Trishna's

IIT JEE Super Course in Chemistry

PHYSICAL CHEMISTRY



ALWAYS LEARNING

PEARSON

Super Course in Chemistry

PHYSICAL CHEMISTRY

for IIT-JEE

Volume 1

Trishna Knowledge Systems

A division of
Triumphant Institute of Management Education Pvt. Ltd

PEARSON

Chandigarh • Delhi • Chennai

T.I.M.E.
Triumphant Institute of Management Education Pvt. Ltd.

T.I.M.E. is the No. 1 institute for the IIT-JEE courses. It is the largest, and the most successful, classroom-based entrance exam training institute in India. For more details, visit *www.time4education.com*

The aim of this publication is to supply information taken from sources believed to be valid and reliable. This is not an attempt to render any type of professional advice or analysis, nor is it to be treated as such. While much care has been taken to ensure the veracity and currency of the information presented within, neither the publisher nor its authors bear any responsibility for any damage arising from inadvertent omissions, negligence or inaccuracies (typographical or factual) that may have found their way into this book.

Copyright © Trishna Knowledge Systems, 2011
Licensees of Pearson Education in South Asia

No part of this eBook may be used or reproduced in any manner whatsoever without the publisher's prior written consent.

This eBook may or may not include all assets that were part of the print version. The publisher reserves the right to remove any material present in this eBook at any time.

ISBN 9788131759776
eISBN 9789332510852

Head Office: A-8(A), Sector 62, Knowledge Boulevard, 7th Floor, NOIDA 201 309, India
Registered Office: 11 Local Shopping Centre, Panchsheel Park, New Delhi 110 017, India

Contents

Preface *v*

Chapter 1 **Basic Concepts of Chemistry** **1.1—1.63**

STUDY MATERIAL
• *Introduction* • *Definition and Description of Matter* • *Dalton's Atomic Theory and Model of the Atom* • *Mass of an Atom* • *Valency of Elements* • *Valency and Formula of Radicals* • *Formulae of Compounds* • *Mole Concept* • *Molar Mass* • *Chemical Formulae* • *Chemical Equation* • *Excess and Limiting Reactants* • *Oxidation Number or Oxidation State* • *Balancing Redox Reactions* • *Disproportionation Reactions* • *Equivalent Weight* • *Strength of Solutions* • *Volumetric Analysis* • *Neutralization Reactions*

Chapter 2 **States of Matter** **2.1—2.67**

STUDY MATERIAL
• *Introduction* • *Gas laws* • *Absolute Density and Relative Density of a Gas* • *Units of Pressure and Volume* • *Value of R in Different Units* • *Dalton's Law of Partial Pressure* • *Graham's Law of Diffusion* • *Kinetic Theory of Gases* • *Molecular Velocity Distribution Among Gas Molecules* • *Real Gases* • *Modification of Ideal Gas Equation: Van Der Waals Equation* • *Critical State of a Gas* • *Relative Humidity (RH)*

Chapter 3 **Atomic Structure** **3.1—3.80**

STUDY MATERIAL
• *Introduction* • *Atomic Models* • *Hydrogen Spectrum* • *Photoelectric Effect* • *Wave–Particle Duality* • *Heisenberg's Uncertainty Principle* • *Quantum Numbers* • *Quantum Mechanical Picture of Hydrogen Atom* • *Probability Distribution Curves* • *Orbitals* • *Electronic Configuration of Elements*

Chapter 4 **Chemical Bonding** **4.1—4.80**

STUDY MATERIAL
• *Introduction* • *Modern Theories of Bonding* • *Hydrogen Bonding* • *Polarization of Ions-Fajan's Rule* • *Dipole Moment* • *Resonance*

Chapter 5 **Chemical Energetics and Thermodynamics** **5.1—5.78**

STUDY MATERIAL
• *Basic Concepts in Thermodynamics* • *Mechanical Work (Pressure–Volume Work)* • *Internal Energy (U)* • *First Law of Thermodynamics* • *Expression for Pressure–Volume Work* • *Enthalpy, H* • *Adiabatic Expansion* • *Work Done in an Isobaric Process* • *Thermochemical Equations* • *Various Forms of Enthalpy of Reaction* • *Bond enthalpy (Bond energy)* • *Laws of Thermochemistry* • *Spontaneous Processes* • *Second law of Thermodynamics and Entropy (S)* • *Entropy Changes During Various Processes* • *Third law of Thermodynamics* • *Gibbs Free Energy (G)* • *Criteria for Equilibrium and Spontaneity* • *Free Energy Change and Useful Work*

Chapter 6 **Chemical and Ionic Equilibria** **6.1—6.90**

STUDY MATERIAL

CHEMICAL EQUILIBRIUM

• *Reversible Reactions* • *Law of mass action* • *Relation between K_p, K_c and K_x* • *Relation between ΔG, $\Delta G°$ and Equilibrium constant* • *Le Chatelier's Principle—Theory of Mobile Equilibrium* • *Homogeneous equilibria in gas phase* • *Effect of the Variation of Total Pressure on a System at Equilibrium* • *Homogeneous Equilibria in Liquid Phase* • *Simultaneous Equilibria Heterogeneous Equilibria* • *Effect of Temperature on Equilibrium Constant* • *Acid- base Concept*

IONIC EQUILIBRIUM

• *pH concept* • *Dissociation Equilibria Involving Weak Acids and Weak Bases* • *Salt Hydrolysis Common ion effect* • *Buffer and Buffer action* • *Solubility Equilibria and Solubility product* • *Application of Solubility Product Principle in Qualitative Analysis* • *Indicators in Neutralisation Reaction*

Chapter 7 **Dilute Solutions and Electrochemistry** **7.1—3.77**

STUDY MATERIAL

DILUTE SOLUTIONS

• *Introduction* • *Colligative Properties*

ELECTROCHEMISTRY

• *General* • *Faraday's Laws of Electrolysis* • *Effect of Electrodes on the Nature of Electrode Reactions During Electrolysis* • *Conductance of Electrolyte Solutions* • *Galvanic Cells* • *Reversible Electrodes* • *Measurement of the EMF of a Galvanic Cell* • *Single Electrode Potentials* • *Reference Electrodes* • *Concentration Cells*

Chapter 8 **Chemical Kinetics and Solid State** **8.1—8.79**

STUDY MATERIAL

CHEMICAL KINETICS

• *Introduction and Importance of the Subject* • *Definition of "Rate"* • *Order of a Reaction* • *Molecularity of a Reaction* • *Evaluation of Rate Constant, k* • *Methods for Following the Progress of a Reaction* • *Methods of Determination of Order of a Reaction* • *Collision Theory* • *Activated Complex Theory of Reaction Rates* • *Kinetics of Consecutive, Parallel and Opposing Reactions*

SOLID STATE

• *Classification of Solids* • *Types of Unit Cells* • *Closest Packing of Constituent Particles (Spheres) in Crystals* • *Number of Atoms Per Unit Cell in a Cubic Lattice* • *Packing in Ionic Compounds* • *Defects in Solids*

Chapter 9 **Surface Chemistry and Nuclear Chemistry** **9.1—9.50**

STUDY MATERIAL

SURFACE CHEMISTRY

• *Adsorption and Absorption* • *Physical Adsorption and Chemisorption* • *Colloids* • *Types of Colloidal Systems* • *Classification of Sols* • *Methods of Preparation of Colloidal Solutions* • *Purification of Colloidal Solutions* • *Properties of Colloidal Solutions* • *Applications of Colloids* • *Emulsifiers*

NUCLEAR CHEMISTRY

• *α, β, γ-Rays and Their Properties* • *Kinetics of Radioactive Decay* • *Units of Radioactivity* • *Isotopes, Isobars and Isotones* • *Nuclear Fission and Fusion Reactions* • *Artificial Disintegration or Transmutation of Elements*

Preface

The IIT-JEE, the most challenging amongst national level engineering entrance examinations, remains on the top of the priority list of several lakhs of students every year. The brand value of the IITs attracts more and more students every year, but the challenge posed by the IIT-JEE ensures that only the *best* of the aspirants get into the IITs. Students require thorough understanding of the fundamental concepts, reasoning skills, ability to comprehend the presented situation and exceptional problem-solving skills to come on top in this highly demanding entrance examination.

The pattern of the IIT-JEE has been changing over the years. Hence an aspiring student requires a step-by-step study plan to master the fundamentals and to get adequate practice in the various types of questions that have appeared in the IIT-JEE over the last several years. Irrespective of the branch of engineering study the student chooses later, it is important to have a sound conceptual grounding in Mathematics, Physics and Chemistry. A lack of proper understanding of these subjects limits the capacity of students to solve complex problems thereby lessening his / her chances of making it to the top-notch institutes which provide quality training.

This series of books serves as a source of learning that goes beyond the school curriculum of Class XI and Class XII and is intended to form the backbone of the preparation of an aspiring student. These books have been designed with the objective of guiding an aspirant to his / her goal in a clearly defined step-by-step approach.

- **Master the Concepts and Concept Strands!**
  This series covers all the concepts in the latest IIT-JEE syllabus by segregating them into appropriate units. The theories are explained in detail and are illustrated using solved examples detailing the different applications of the concepts.
- **Let us First Solve the Examples – Concept Connectors!**
  At the end of the theory content in each unit, a good number of "Solved Examples" are provided and they are designed to give the aspirant a comprehensive exposure to the application of the concepts at the problem-solving level.
- **Do Your Exercise – Daily!**
  Over 200 unsolved problems are presented for practice at the end of every chapter. Hints and solutions for the same are also provided. These problems are designed to sharpen the aspirant's problem-solving skills in a step-by-step manner.
- **Remember, Practice Makes You Perfect!**
  We recommend you work out ALL the problems on your own – both solved and unsolved – to enhance the effectiveness of your preparation.

A distinct feature of this series is that unlike most other reference books in the market, this is not authored by an individual. It is put together by a team of highly qualified faculty members that includes IITians, PhDs etc from some of the best institutes in India and abroad. This team of academic experts has vast experience in teaching the fundamentals and their application and in developing high quality study material for IIT-JEE at T.I.M.E. (Triumphant Institute of Management Education Pvt. Ltd), the number 1 coaching institute in India. The essence of the combined knowledge of such an experienced team is what is presented in this self-preparatory series. While the contents of these books have been organized keeping in mind the specific requirements of IIT-JEE, we are sure that you will find these useful in your preparation for various other engineering entrance exams also.

We wish you the very best!

This page is intentionally left blank.

CHAPTER

1

BASIC CONCEPTS OF CHEMISTRY

CHAPTER OUTLINE

Preview

STUDY MATERIAL

Introduction

Definition and Description of Matter

Dalton's atomic theory and model of the atom

Mass of an Atom

Valency of elements

Valency and Formula of Radicals

Formulae of Compounds

Mole Concept

Molar Mass
- Concept Strands (1-2)

Chemical Formulae
- Concept Strands (3-6)

Chemical Equation
- Concept Strand (7)

Excess and Limiting Reactants
- Concept Strand (8)

Oxidation Number or Oxidation State
- Concept Strands (9-10)

Balancing Redox Reactions
- Concept Strands (11-13)

Disproportionation Reactions
- Concept Strands (14-17)

Equivalent Weight

Strength of Solutions
- Concept Strands (18-32)

Volumetric Analysis
- Concept Strands (33-37)

Neutralization Reactions
- Concept Strands (38-50)

TOPIC GRIP

- Subjective Questions (10)
- Straight Objective Type Questions (5)
- Assertion–Reason Type Questions (5)
- Linked Comprehension Type Questions (6)
- Multiple Correct Objective Type Questions (3)
- Matrix-Match Type Question (1)

IIT ASSIGNMENT EXERCISE

- Straight Objective Type Questions (80)
- Assertion–Reason Type Questions (3)
- Linked Comprehension Type Questions (3)
- Multiple Correct Objective Type Questions (3)
- Matrix-Match Type Question (1)

ADDITIONAL PRACTICE EXERCISE

- Subjective Questions (10)
- Straight Objective Type Questions (40)
- Assertion–Reason Type Questions (10)
- Linked Comprehension Type Questions (9)
- Multiple Correct Objective Type Questions (8)
- Matrix-Match Type Questions (3)

INTRODUCTION

Chemistry is basically an experimental science and chemists are engaged in studying a variety of reactions in the laboratory and many reactions find wide applications in industry. Many industrial processes such as extraction of metals from their ores, refining of petroleum, manufacture of drugs, pharmaceuticals and high polymers are of great practical and commercial value. The success of all such efforts is the result of the strong basic foundation laid by early generations of chemists. This chapter deals with some of the fundamental ideas out of which chemical theory has evolved viz., the atomic theory of matter, mole concept etc.

DEFINITION AND DESCRIPTION OF MATTER

Matter may be defined as anything which has mass and occupies space.

Some common examples of matter (or synonymously also called materials) are salt, sand, water, oxygen etc. Matter may also be classified into the three states of aggregation, namely solid, liquid and gas.

The Alchemists discovered many new substances, developed new types of apparatus and perfected some chemical procedures like crystallization from solution, distillation and separation of mixture etc. Thus, over a period of time, it became evident that matter could be physically separated into different pure substances each having a unique set of properties (crystalline form, colour etc.).

For example, sea water can be separated by distillation into pure water and a residue consisting of a mixture of several salts. This mixture can be further separated into different components, each component having different properties from the others.

The mixtures, consisting of more than one substance, may again be divided into two categories

(i) Homogeneous mixtures
(ii) Heterogeneous mixtures

Homogeneous mixtures

These are mixtures having uniform composition throughout and in which the individual components cannot be physically distinguished from one another.

Air is an example of a homogeneous mixture of several gases (Nitrogen, Oxygen, Carbon dioxide, water vapour etc.). So also are solutions, say, glucose in water.

Heterogeneous mixtures

These are mixtures whose composition is not uniform throughout and in which the individual components can be distinguished from one another.

Some examples are,

(i) an insoluble solid like $BaSO_4$ or AgCl placed in a beaker containing water.
(ii) a gas in contact with a solid in an enclosed vessel, like CO_2 in contact with solid $CaCO_3$.
(iii) A mixture of sand and salt (or Iron powder and Sulphur).

Elements

Elements are defined as fundamentally inseparable elemental materials and consist of only one kind of atom.

Solid substances like carbon, liquids like mercury and gases like helium are elements.

Compounds

The pure substances that are formed by the combination of two or more elements (for example, residue from sea water) are referred to as compounds.

Compounds contain two or more different types of atoms in them, as for example, in compounds like H_2O, NH_3, CO_2 etc. Further, the elements in a compound do not have their individual characteristics.

DALTON'S ATOMIC THEORY AND MODEL OF THE ATOM

Based on the above ideas of matter, elements and compounds, John Dalton put forward his atomic theory in 1803. The three laws of chemical combination, which were already proposed prior to this period, further led Dalton in 1804 to formulate the existence of atoms on the basis of the following postulates.

(i) Matter is discrete (i.e., discontinuous) and is made up of atoms. An atom is the smallest (chemically) particle of an element, which can take part in a chemical change.
(ii) Atoms of a given element are all similar in all respects, size, shape, mass etc., but differ from atoms of other elements.
(iii) An atom of an element has a definite mass.
(iv) Atoms are indestructible i.e., they can neither be created nor destroyed and chemical reactions involve only rearrangement of atoms.

A compound like SO_2 is suggested to be composed of a group of atoms. The group of atoms, which is capable of independent existence, is known as **molecule**. Thus matter was structured in terms of atoms and molecules. Some postulates of Dalton's theory need to be modified in the light of subsequent developments in this area.

(i) Atom is divisible as known from studies of nuclear fission.
(ii) Atoms of an element may have different masses due to existence of isotopes.

MASS OF AN ATOM

Atom cannot be seen or isolated by simple physical/chemical methods. Therefore, the mass of an atom as suggested by Dalton is based on relative terms viz., the average mass of one atom relative to the average mass of another. The method that helped in the development of atomic mass (relative to that of another) is based on Avogadro's hypothesis proposed by Avogadro in 1811.

Avogadro's hypothesis states that equal volumes of all gases at the same temperature and pressure contain equal number of particles.

Considering the formation of water from hydrogen and oxygen, one can write

1 volume of oxygen + 2 volumes of hydrogen = 2 volumes of water vapour — (1)

Assuming that a unit volume of gas contains x particles, equation (1) may be written as

x particles of oxygen + 2x particles of hydrogen = 2x particles of water vapour — (2)

Simplifying equation (2) further, we can write

1 particle of oxygen + 2 particles of hydrogen = 2 particles of water vapour — (3)

Assuming that each particle of oxygen is a diatomic oxygen molecule and each particle of hydrogen is a diatomic hydrogen molecule, equation (3) may be written as

1 molecule of oxygen + 2 molecules of hydrogen = 2 molecules of water vapour

or, $O_{2\,(g)} + 2H_{2\,(g)} = 2H_2O_{\,(g)}$ (the suffix (g) stands for the designation "gas")

Considering that the mass ratio of oxygen to hydrogen is 8:1 (in water) and that in water there are two atoms of hydrogen and one atom of oxygen, it follows that an atom of oxygen is 16 times heavier than a hydrogen atom.

The atomic mass scale is currently based on carbon 12-isotope standard since 1961. It assigns a mass of 12 unified mass (u) to the mass of one atom of ${}_6C^{12}$ isotope.

Atomic mass of an element

$$= \frac{\text{Average mass of 1 atom of the element}}{\frac{1}{12} \times \text{mass of 1 atom of C} - 12} \qquad — (4)$$

Thus atomic mass is a relative mass.

For example, the atomic mass of carbon atom is 12.011 u. This is because the element carbon consists of two isotopes, (i) 98.89 percent of C^{12} isotope whose mass is 12 u and (ii) 1.11 percent of C^{13} isotope whose mass is 13.00335 u.

The atomic mass of an element thus depends on the fractional abundance of the isotopes of that element. The fractional dependence, in turn, represents the fraction of the total number of atoms of the isotope concerned.

1. Relative atomic mass is a number.
2. Gram atomic mass (or) gram-atom is the atomic mass expressed in grams.

Mass number and atomic mass

Mass number refers to the total number of nucleons (protons and neutrons) in an atom. It will be the same as atomic mass if the element does not have an isotope.

^{19}F does not have isotopes. Thus its mass number is equal to its atomic mass.

Molecular mass

$$\text{Molecular mass} = \frac{\text{Mass of 1 molecule}}{\frac{1}{12} \times \text{mass of 1 C} - 12 \text{ atom}}$$

(i) Molecular mass is not absolute but relative and is also known as relative molecular mass.
(ii) Relative molecular mass is a number.
(iii) Molecular mass expressed in grams is the gram-molecular mass (or) gram-mole.

VALENCY OF ELEMENTS

Table 1.1

Elements	Valency
H, Li, Na, K, Rb, Cs, F, Cl, Br, I, Ag	1
Be, Mg, Ca, Sr, Ba, O, S, Ni, Zn, Cd	2
B, Al, N, As, Sb, Bi,	3
C, Si	4

Table 1.2 Elements with variable valency

Element	Valencies	
Fe	2 (Ferrous)	3 (Ferric)
Cu	1 (Cuprous)	2 (Cupric)
Au	1 (Aurous)	3 (Auric)
Hg	1(Mercurous Hg_2^{2+})	2 (Mercuric)
Sn	2 (Stannous)	4 (Stannic)
Pb	2 (Plumbous)	4 (Plumbic)

VALENCY AND FORMULA OF RADICALS

Table 1.3

Valency 1			
Radical	**Formula**	**Radical**	**Formula**
Ammonium	NH_4^+	Hypochlorite	ClO^-
Hydroxide	OH^-	Iodate	IO_3^-
Nitrate	NO_3^-	Periodate	IO_4^-
Nitrite	NO_2^-	Meta aluminate	AlO_2^-
Permanganate	MnO_4^-	Meta borate	BO_2^-
Bisulphate	HSO_4^-	Cyanide	CN^-
Bisulphite	HSO_3^-	Isocyanide	NC^-
Bicarbonate	HCO_3^-	Cyanate	CNO^-
Dihydrogen phosphate	$H_2PO_4^-$	Isocyanate	NCO^-
Chlorate	ClO_3^-	Thiocyanate	CNS^-
Perchlorate	ClO_4^-	Isothiocyanate	NCS^-

Valency 2			
Carbonate	CO_3^{2-}	Dichromate	$Cr_2O_7^{2-}$
Sulphate	SO_4^{2-}	Zincate	ZnO_2^{2-}
Sulphite	SO_3^{2-}	Tetrathionate	$S_4O_6^{2-}$
Thiosulphate	$S_2O_3^{2-}$	Oxalate	$C_2O_4^{2-}$
Hydrogen phosphate	HPO_4^{2-}	Silicate	SiO_3^{2-}
Manganate	MnO_4^{2-}	Tetraborate	$B_4O_7^{2-}$
Chromate	CrO_4^{2-}	Stannate	SnO_3^{2-}
Valency 3			
Phosphate	PO_4^{3-}	Arsenate	AsO_4^{3-}
Borate (Ortho borate)	BO_3^{3-}	Arsenite	AsO_3^{3-}
Aluminate	AlO_3^{3-}		

FORMULAE OF COMPOUNDS

Table 1.4

1.	Calcium chloride	$Ca^{2}Cl^{1}$	$CaCl_2$	6.	Potassium perchlorate	$K^{1}ClO_4^{1}$	$KClO_4$
2.	Magnesium sulphate	$Mg^{2}SO_4^{2}$	$MgSO_4$ (simple ratio)	7.	Sodium meta aluminate	$Na^{1}AlO_2^{1}$	$NaAlO_2$
3.	Stannic sulphide	$Sn^{4}S^{2}$	SnS_2	8.	Sodium zincate	$Na^{1}ZnO_2^{2}$	Na_2ZnO_2
4.	Mercurous chloride	$Hg_2^{2}2Cl^{1}$	Hg_2Cl_2 (dimeric)	9.	Magnesium bicarbonate	$Mg^{2}HCO_3^{1}$	$Mg(HCO_3)_2$
5.	Sodium thiosulphate	$Na^{1}S_2O_3^{2}$	$Na_2S_2O_3$	10.	Ammonium oxalate	$NH_4^{1}C_2O_4^{2}$	$(NH_4)_2C_2O_4$

MOLE CONCEPT

A mole is the amount of any substance containing 6.022×10^{23} particles.

A rigorous definition of mole in SI terminology is given, as *'the mole is the amount of any substance that contains as many elementary entities as there are atoms in 12 grams of carbon-12 isotope.*

Mole is the SI unit of amount of substance

The elementary entities may be atoms, molecules, ions, electrons etc. A mole of any substance always contains the same number of entities irrespective of what the substance is. For example, a mole of nitrogen or a mole of a metal like Li, contains 6.022×10^{23} molecules or atoms respectively. This number is also known as Avogadro's constant and so named in honour of the Italian scientist, A. Avogadro.

MOLAR MASS

The term 'molar mass' refers to the mass of one mole of any substance, be it an element or a compound.

For example, the molar mass of NH_3 represents the mass of 6.022×10^{23} molecules of NH_3 and is equal to 17.064 g mol^{-1}.

It may be noted from equation (4) that the molar mass of any atom (or molecule) is numerically equal to the atomic mass (or molecular mass) in amu.

For example, from equation (4), for carbon-12

$$12\text{amu} = \frac{12.000}{6.022\times10^{23}}\times6.022\times10^{23}\,\text{g} = \text{molar mass in grams of C-12}$$

From Avogadro's hypothesis mentioned earlier, it follows that one mole of any gas at STP (0°C and 1atm) occupies 22.4litres of volume, which is known as the molar volume of the gas at STP.

The number of atoms of an element in a molecule is called its atomicity.

Calculation of Number of Moles

$$\text{No. of moles of atoms} = \frac{\text{weight (gram)}}{\text{atomic mass}}$$

$$\text{No. of moles molecules (or) No. of moles} = \frac{\text{weight (gram)}}{\text{molecular mass}}$$

$$\text{No. of moles} = \frac{\text{No. of molecules}}{6.022\times10^{23}}$$

$$\text{No. of moles atoms} = \frac{\text{No. of atoms}}{6.022\times10^{23}}$$

$$\text{No. of moles of gases} = \frac{\text{Volume at STP(L)}}{22.4}$$

CONCEPT STRANDS

Concept Strand 1

Calculate the number of atoms in

(i) half a mole of nitrogen gas.
(ii) 5.6 litres of helium gas at S.T.P.

Solution

(i) One mole of $N_2 = 6.022 \times 10^{23}$ molecules

$\Rightarrow$ 0.5 moles of $N_2 = \frac{6.022 \times 10^{23}}{2} = 3.015 \times 10^{23}$ molecules.

Since N_2 is diatomic, the number of atoms of N_2 = $3.015 \times 10^{23} \times 2 = 6.022 \times 10^{23}$ atoms

(ii) From Avogadro's hypothesis, 22.4 litres of gas at S.T.P are occupied by one mole.

5.6 litres of gas at S.T.P are occupied by $\frac{5.6}{22.4} = 0.25$ moles

No. of atoms in 0.25moles = $6.022 \times 10^{23} \times 0.25 = 1.51 \times 10^{23}$ atoms.

Since 'He' is monoatomic, number of atoms in 0.25 mole = 1.51×10^{23}

Concept Strand 2

A molecule of a diatomic compound weighs 4.98×10^{-23} g. What is its molecular weight? How many molecules and atoms are present in 30 kg of this compound?

Solution

Molecular weight of this compound = $4.98 \times 10^{-23} \times 6.022 \times 10^{23} = 30$.

Number of moles in 30 kg of this compound = $\frac{30 \times 10^3}{30} = 1000$

No. of molecules in 30 kg = $1000 \times 6.022 \times 10^{23} = 6.022 \times 10^{26}$

No. of atoms in 30 kg = $1000 \times 6.022 \times 10^{23} \times 2 = 1.2046 \times 10^{27}$

CHEMICAL FORMULAE

When dealing with chemical formulae, two types of formulae must be taken into account.

(i) empirical formulae, which give the relative number of different types of atoms present in a molecule of the compound.
(ii) molecular formulae, which give the exact number of atoms in a molecule of the compound.

The empirical formulae of a compound can be obtained by using chemical analytical data on the compound and the atomic masses of various elements present in the compound. The given mass per cent composition of the compound is converted into mole per cent composition by dividing the composition of each element by its atomic mass. The mole per cent composition of each element is next divided by the least value among the composition obtained to get the relative number of atoms of different elements in the compound in the simplest ratio.

The molecular formula is an integral multiple of the empirical formula.

CONCEPT STRANDS

Concept Strand 3

An ore of chromium contains 24.95 weight % of iron, 46.46 weight % chromium and the rest is made up of oxygen. Find the empirical formula of the ore.

(Given: At mass: Fe = 55.85, Cr = 52; O = 16)

Solution

In 100 grams of the ore

Element	Weight	Atomic ratio	Smallest Ratio
Iron	24.95	$\frac{24.95}{55.85} = 0.45$	1
Chromium	46.46	$\frac{46.46}{52.00} = 0.89$	2
Oxygen	28.59	$\frac{28.59}{16.00} = 1.79$	4

Ratio of Fe, Cr, O = 1: 2 : 4 Empirical formula = $FeCr_2O_4$

Concept Strand 4

An inorganic compound, on analysis, gave the following weight percent composition. Mg = 9.86; S = 13.00; O = 71.41, the rest being hydrogen. Assuming that all the hydrogen is in the form of water, calculate its empirical formula. If its molecular weight is 246.5, what is the molecular formula of the compound?

(Atomic mass of Mg = 24.3; S = 32.06; O = 16.00; H = 1)

Solution

Element	Weight%	Atomic ratio	Smallest ratio
Mg	9.86	$\frac{9.86}{24.3} = 0.4$	1
S	13.00	$\frac{13.00}{32.06} = 0.4$	1
O	71.41	$\frac{71.41}{16.00} = 4.4$	11
H	5.73	$\frac{5.73}{1} = 5.7$	14

Empirical formula: $MgSO_{11}H_{14}$.

Since all H atoms are part of water, the above formula can be revised as $MgSO_4.7H_2O$ or $(MgSO_4.7H_2O)_n = 246.5$.

$\therefore$ n = 1

Molecular formula of the compound = $MgSO_4.\ 7H_2O$

Concept Strand 5

A compound contains carbon, hydrogen and nitrogen in the weight ratio 10 : 1 : 2. If the molecular weight of the compound is 93 g mol^{-1}, find its molecular formula.

Solution

Weight of carbon = $\frac{10}{13} \times 93 = 71.5$ g; $\frac{71.5}{12} = 6$ g atom.

Weight of hydrogen = $\frac{1}{13} \times 93 = 7.15$ g; $\frac{7.15}{1} = 7$ g atom.

Weight of nitrogen = $\frac{2}{13} \times 93 = 14.3$ g; $\frac{14.3}{14} = 1$ g atom.

$\therefore$ Molecular formula is C_6H_7N ($C_6H_5NH_2$)

Concept Strand 6

The ratio of weights of mercurous chloride to mercury is 2.354. Find the atomic mass of mercury. (This example illustrates the importance of the use of correct formula for a compound.)

Solution

Mercurous chloride is Hg_2Cl_2. If x is the atomic mass of Hg, then $\frac{2x + (2 \times 35.5)}{x} = 2.354$

Solving for x, we get atomic mass of Hg as 200.6

If we use the formula HgCl, the answer becomes different.

CHEMICAL EQUATION

Chemical equations are a manifestation of chemical reactions both in a qualitative and quantitative manner. The quantitative aspects of mass and volume relations between reactants and products are known as "stoichiometry". The following points must be borne in mind when writing chemical equation.

(i) The equation must be properly balanced as to maintain conservation of mass of all the species participating in the reaction. For example, the reaction

$$CH_4O + O_2 \rightarrow CO_2 + H_2O$$

gives the products correctly but does not show the mass conservation in hydrogen and oxygen. The balanced equation is

$$CH_4O + \frac{3}{2}O_2 \rightarrow CO_2 + 2H_2O$$

multiplying throughout by 2, we get the correct balanced equation

$$2CH_4O + 3O_2 \rightarrow 2CO_2 + 4H_2O \qquad \text{— (5)}$$

(ii) The physical state of all the reactants and products i.e., solid (s), liquid (l) or gas (g) must be indicated along the side of the constituents. Equation (5) is more appropriately written as

$$2CH_4O_{(\ell)} + 3\ O_{2\ (g)} \rightarrow 2\ CO_{2\ (g)} + 4\ H_2O_{\ (\ell)}$$

(iii) In case of thermo chemical equation, it is important to incorporate the ΔH (enthalpy change) of the reaction with the sign. A balanced chemical equation enables calculation of mass of reactants consumed and products formed from the given data.

CONCEPT STRAND

Concept Strand 7

Calculate the weight of hydrochloric acid required to react with 5 g of calcium carbonate completely. How much of CO_2 is formed under the given conditions?

Solution

Balanced equation:

$$CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$$

1mole of $CaCO_3$ react with 2 moles of HCl for complete reaction

i.e., 100 g of $CaCO_3$ react with $2 \times 36.5 = 73.0$ g of HCl

5 g of $CaCO_3$ react with $\frac{5 \times 73.0}{100} = 3.65$ g of HCl

Mass of CO_2 formed:

100 g of $CaCO_3$ on reaction with HCl gives 44 g of CO_2.

5 g of $CaCO_3$ on complete reaction with HCl gives $\frac{5 \times 44}{100} = 2.20$ g of CO_2.

EXCESS AND LIMITING REACTANTS

In the reaction between lead nitrate and potassium iodide to give lead iodide, lead nitrate and potassium iodide in the mole ratio of 1:2 react to give 1 mole lead iodide as per the stoichiometric equation.

$$Pb(NO_3)_2 + 2KI \rightarrow PbI_2 + 2KNO_3 \qquad \text{— (6)}$$

Sometimes, a reactant taken in excess amount, than what is required by stoichiometry, will be left behind after the reaction and it is known as the excess reactant.

In some cases, the amount of reactant taken, limits the amount of the product formed because it is completely consumed when the reaction is complete. As this reactant limits the amount of product formed, it is called a limiting reactant.

If for example, in the above reaction, 2 moles of lead nitrate is allowed to react with 6 moles of potassium iodide, 2 moles of lead iodide are formed and 2 moles of potassium iodide are left behind under these conditions, lead nitrate is the limiting reactant and potassium iodide is the excess reactant.

CONCEPT STRAND

Concept Strand 8

20.0 g of silver nitrate in aqueous solution react with 6.0 g sodium chloride to form silver chloride precipitate. Calculate the amount of silver chloride formed and specify which among the reactants, is an (Atomic mass of Ag = 108, Cl = 35.5, Na = 23).

(i) excess reactant
(ii) limiting reactant.

Solution

The reaction is $AgNO_3 + NaCl \rightarrow AgCl \downarrow + NaNO_3$

1 mole of $AgNO_3$ reacts with one mole of NaCl to give one mole of AgCl.

i.e., 170 g of $AgNO_3$ reacts with 58.5 g of NaCl to give 143.5 g of AgCl.

6.00 g of NaCl react completely with $\frac{6\times170}{58.5} = 17.44$ g of $AgNO_3$.

Since 20.00 – 17.44 = 2.56 g of $AgNO_3$ remains in solution, the excess reactant is $AgNO_3$ and as NaCl limits the amount of AgCl formed, it is the limiting reactant.

$$\text{Amount of AgCl formed} = \frac{6\times143.5}{58.5}$$
$$= 14.72 \text{ g}$$

OXIDATION NUMBER OR OXIDATION STATE

Oxidation number of an atom in a compound is the number of electrons shifting towards or away from that atom. Loss of electron is indicated by a positive sign and gain of electron by a negative sign. In the case of sharing, the more electronegative atom involved in the sharing is given a negative sign and the less electronegative a positive sign.

For example in the ionic compound, $CaCl_2$ the oxidation number of Ca is +2 and that of Cl is –1 and in MgO the oxidation number of Mg is +2 and that of O is –2 and in the covalent compound HCl, the oxidation number of H is +1 and that of Cl is –1.

Rules for assigning oxidation number

(i) The oxidation number of a free element (regardless of whether it exists in mono atomic or poly atomic form. example: Cu, Cl_2, P_4, S_8) is zero.

(ii) Sum of oxidation numbers of all the atoms of a charged species (e.g., SO_4^{2-}, NH_4^+ etc) is equal to the charge on the ion.

(iii) The oxidation number of F in all its compounds is –1, for all the other halogens, oxidation number = –1 except in those with oxygen (e.g., ClO_3^-) and with other halogens (e.g., $BrCl_2^-$).
In the former case it is determined from oxygen, i.e., Oxidation number of Cl = +5 and in the latter case it is determined from the more electronegative halogen i.e., oxidation number of Br is +1.

(iv) Oxidation number of alkali metals = +1 and alkaline earth metals = +2.

(v) Oxidation number of H = +1 except in metal hydrides like CaH_2, $LiAlH_4$ etc., where it is –1.

(vi) Oxidation number of O = –2 except in peroxide and OF_2. In peroxides it is –1 and in OF_2, it is +2.

(vii) Algebraic sum of the oxidation numbers of all the atoms in a neutral molecule is equal to zero and for a radical it is equal to the charge.

CONCEPT STRAND

Concept Strand 9

Calculate the oxidation number of

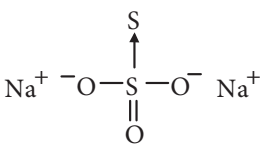
(i) Mn in K_2MnO_4
(ii) Cr in $K_2Cr_2O_7$
(iii) S in $Na_2S_2O_3$
(iv) C in CH_3COOH
(v) N in HN_3
(vi) Cr in CrO_5

Solution

(i) Oxidation number of Mn in K_2MnO_4 is + 2 + x – 8 = 0, x = +6

(ii) Oxidation number of Cr in $K_2Cr_2O_7$ is + 2 + 2x – 14 = 0, x = +6

(iii) Oxidation number of S in $Na_2S_2O_3$ is + 2 + 2x – 6 = 0, x = +2

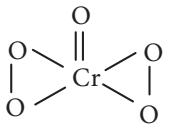


In fact, the two sulphur atoms are in +6 and −2 oxidation states and the average works out to be +2.

(iv) Oxidation number of C in CH_3COOH ($C_2H_4O_2$) is $2x + 4 - 4 = 0$, $x = 0$

(v). Oxidation number of N in HN_3 is $-\frac{1}{3}$.

(vi) Oxidation number of Cr in CrO_5 is +6. (since, $4 \times -1 + x + 1 \times -2 = 0 \Rightarrow x = +6$)



Oxidation and Reduction in terms of oxidation number

An increase in oxidation number indicates oxidation and a decrease in oxidation number indicates reduction.

The species for which there is an increase in oxidation number is said to be oxidized. It is or known as reducing agent (reductant) and the species, which undergoes a decrease in oxidation number, is reduced, and it is or known as oxidizing agent or oxidant.

Oxidants

Table 1.5

Oxidant	Radical or element involved	O.N of effective element	Reduction Product	New O.N	Decrease In O.N	Gain in electrons
$KMnO_4$ (acid medium)	MnO_4^-	Mn = +7	Mn^{2+}	+2	5	5
$KMnO_4$ (alkaline medium)	MnO_4^-	Mn = +7	MnO_4^{2-}	+6	1	1
$KMnO_4$ (neutral medium)	MnO_4^-	Mn = +7	MnO_2	+4	3	3
$K_2Cr_2O_7$ (acid medium)	$Cr_2O_7^{2-}$	Cr = +6	Cr^{3+}	+3	6 (for 2 Cr)	6
Cl_2	Cl	$Cl_2 = 0$	Cl^-	−1	1	1
$FeCl_3$	Fe^{3+}	Fe = +3	Fe^{2+}	+2	1	1
NaOCl	OCl^-	Cl = +1	Cl^-	−1	2	2
KIO_3	IO_3^-	I = +5	I^-	−1	6	6
H_2O_2	O_2^{2-}	O = −1	O^{2-}	−2	2	2

Reductants

Table 1.6

Reductant	Radical or element involved	O.N of the effective element	Oxidation product	New O.N	Increase in O.N	Loss in electrons
$SnCl_2$	Sn^{2+}	Sn = +2	Sn^{4+}	+4	2	2
$FeSO_4$	Fe^{2+}	Fe = +2	Fe^{3+}	+3	1	1
$H_2C_2O_4$	$C_2O_4^{2-}$	C = +3	CO_2	+4	2 (for 2 carbons)	2
H_2SO_3	SO_3^{2-}	S = +4	SO_4^{2-}	+6	2	2

CONCEPT STRAND

Concept Strand 10

In the following reactions, identify the species oxidized, reduced, oxidizing agent and reducing agent.

(i) $H_2SO_3 + 2H_2S \rightarrow 3S + 3\,H_2O$

(ii) $H_2O_2 + H_2O_2 \rightarrow 2H_2O + O_2$

Solution

(i) $\underset{+1\ +4\ -2}{H_2SO_3} + \underset{+1\ -2}{2H_2S} \longrightarrow \underset{0}{3S} + \underset{+1\ -2}{3H_2O}$

Oxidation states of oxygen and hydrogen remain the same. Sulphur in H_2SO_3 is reduced to S ($+4 \rightarrow 0$), while sulphur in H_2S is oxidized to S ($-2 \rightarrow 0$). Hence H_2SO_3 is the oxidizing agent and H_2S is the reducing agent.

(ii) $\underset{+1\ -1}{H_2O_2} + \underset{+1\ -1}{H_2O_2} \longrightarrow \underset{+1\ -2}{2H_2O} + \underset{0}{O_2}$

Oxidation state of hydrogen remains the same. Oxygen in one molecule of H_2O_2 is oxidized to O_2 ($-1 \rightarrow 0$) where as in another molecule it is reduced to H_2O ($-1 \rightarrow -2$). Thus H_2O_2 acts both as an oxidizing and reducing agent.

BALANCING REDOX REACTIONS

Balancing redox equations is based on the fact that the number of electrons gained during reduction must be equal to the number of electrons lost during oxidation. There are two methods for balancing the redox reactions.

I. Ion-electron method

Various steps involved in this method are:

(i) Split up the complete reaction into two halves—oxidation half (change undergone by the reducing agent) and reduction half (change undergone by the oxidizing agent).

(ii) Balance the number of atoms of each element other than oxygen and hydrogen in each half of the reaction.

(iii) For balancing oxygen, H_2O may have to be added, while for balancing hydrogen, H^+ are added. But in alkaline medium, H^+ must be converted to H_2O by adding the same number of OH^-. While adding OH^-, it must be added on both sides of the equation.

(iv) Equalize the charge on both sides by adding electrons to the side where positive charge has to be reduced.

(v) Next is to add the two half reactions, for which the number of electrons on both the reactions must be the same, so that the final equation does not contain any electron. This is achieved by multiplying with suitable numbers.

(vi) Similar terms on either side of the equation can be cancelled.

CONCEPT STRAND

Concept Strand 11

(i) Balance $Fe^{2+} + MnO_4^- \rightarrow Fe^{3+} + Mn^{2+}$ in acid medium.

(ii) Balance $Cr_2O_7^{2-} + Fe^{2+} \rightarrow Cr^{3+} + Fe^{3+}$ in acid medium.

(iii) Balance the equation: $MnO_4^- + C_2O_4^{2-} \rightarrow CO_2 + Mn^{2+}$.

(iv) Balance the equation: $CrO_4^{2-} + SO_3^- \rightarrow CrO_2^- + SO_4^{2-}$ in alkaline medium.

Solution

(i) $Fe^{2+} + MnO_4^- \rightarrow Fe^{3+} + Mn^{2+}$ in acid medium

Oxidation half

$Fe^{2+} \rightarrow Fe^{3+}$

Balancing charge $Fe^{2+} \rightarrow Fe^{3+} + 1e^-$ — (7)

Reduction half

$MnO_4^- \rightarrow Mn^{2+}$

Balancing oxygen, $MnO_4^- \rightarrow Mn^{2+} + 4H_2O$

Balancing hydrogen, $MnO_4^- + 8H^+ \rightarrow Mn^{2+} + 4H_2O$

Balancing charge $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$ — (8)

Equalising number of electrons: Multiply (7) with 5 and then add with (8)

$$5Fe^{2+} \rightarrow 5Fe^{3+} + 5e^-$$
$$MnO_4^{--} + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$$
$$MnO_4^{--} + 5Fe^{2+} + 8H^+ \rightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O$$

(ii) $Cr_2O_7^{2-} + Fe^{2+} \rightarrow Cr^{3+} + Fe^{3+}$ in acid medium

Oxidation half

$Fe^{2+} \rightarrow Fe^{3+}$

$Fe^{2+} \rightarrow Fe^{3+} + 1e^-$ — (9)

Reduction half

$Cr_2O_7^{2-} \rightarrow Cr^{3+}$

$Cr_2O_7^{2-} \rightarrow 2Cr^{3+}$

$Cr_2O_7^{2-} \rightarrow 2Cr^{3+} + 7H_2O$

$Cr_2O_7^{2-} + 14H^+ \rightarrow 2Cr^{3+} + 7H_2O$

$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$ — (10)

(9) × 6 + (10)

$\Rightarrow$ $6Fe^{2+} + Cr_2O_7^{2-} + 14H^+ \rightarrow 6Fe^{3+} + 2Cr^{3+} + 7H_2O$

(iii) $MnO_4^- + C_2O_4^{2-} \rightarrow CO_2 + Mn^{2+}$

Oxidation half

$C_2O_4^{2-} \rightarrow CO_2$

$C_2O_4^{2-} \rightarrow 2CO_2$

$C_2O_4^{2-} \rightarrow 2CO_2 + 2e^-$ — (11)

Reduction half

$MnO_4^- \rightarrow Mn^{2+}$

$MnO_4^- \rightarrow Mn^{2+} + 4H_2O$

$MnO_4^- + 8H^+ \rightarrow Mn^{2+} + 4H_2O$

$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$ — (12)

(11) × 5 + (12) × 2

$\Rightarrow$ $5C_2O_4^{2-} + 2MnO_4^- + 16H^+ \rightarrow 10CO_2 + 2Mn^{2+} + 8H_2O$

(iv) $CrO_4^{2-} + SO_3^- \rightarrow CrO_2^- + SO_4^{2-}$ in alkaline medium.

Reduction half

$CrO_4^{2-} \rightarrow CrO_2^-$

$4OH^- + 4H^+ + CrO_4^{2-} \rightarrow CrO_2^- + 2H_2O + 4OH^-$

$3e^- + CrO_4^{2-} + 2H_2O \rightarrow CrO_2^- + 4OH^-$ — (13)

Oxidation half

$SO_3^- \rightarrow SO_4^{2-}$

$2OH^- + H_2O + SO_3^- \rightarrow SO_4^{2-} + 2H^+ + 2OH^-$

$SO_3^- + 2OH^- \rightarrow SO_4^{2-} + H_2O + e^-$ — (14)

(13) + 3 × (14)

$\Rightarrow$ $CrO_4^{2-} + 3SO_3^- + 2OH^- \rightarrow CrO_2^- + 3SO_4^{2-} + H_2O$

II. Oxidation state method

This involves the following steps.

(i) Identify the elements whose oxidation numbers are changed.

(ii) Select suitable coefficients for the oxidizing and reducing agents so that total increase in oxidation number of the reducing agent becomes equal to the total decrease in the oxidation number of the oxidizing agent.

CONCEPT STRANDS

Concept Strand 12

Balance the equation: $H_2SO_4 + HI \rightarrow H_2S + H_2O + I_2$

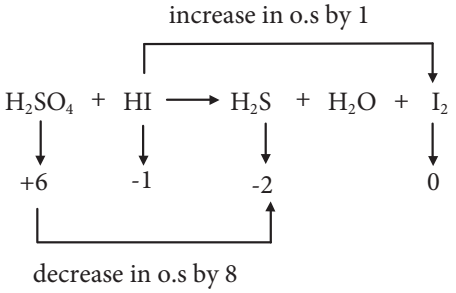
Solution

For making the increase and decrease in the oxidation numbers the same, use the coefficients 8 for HI

i.e., $H_2SO_4 + 8\,HI \rightarrow H_2S + H_2O + I_2$

Balancing the other atoms

$H_2SO_4 + 8\,HI \rightarrow H_2S + 4I_2 + 4H_2O$



Concept Strand 13

Balance the equation

$$K_2Cr_2O_7 + H_2SO_4 + FeSO_4 \rightarrow K_2SO_4 + Cr_2(SO_4)_3 + Fe_2(SO_4)_3 + H_2O$$

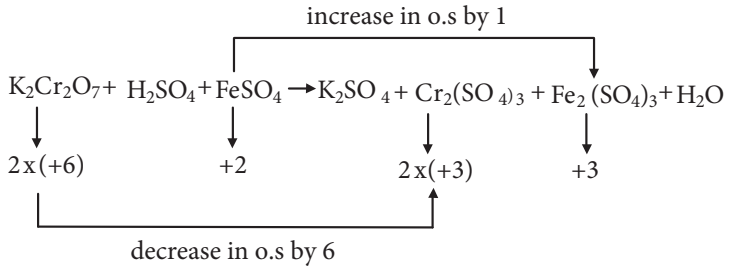
Solution

Using the coefficient 6 for $FeSO_4$

$$K_2Cr_2O_7 + H_2SO_4 + 6FeSO_4 \rightarrow K_2SO_4 + Cr_2(SO_4)_3 + Fe_2(SO_4)_3 + H_2O$$

Balancing all the atoms

$$K_2Cr_2O_7 + 6FeSO_4 + 7H_2SO_4 \rightarrow K_2SO_4 + Cr_2(SO_4)_3 + 3\,Fe_2(SO_4)_3 + 7H_2O$$



DISPROPORTIONATION REACTIONS

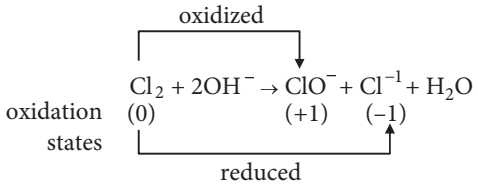
A reaction in which both the oxidizing and reducing agents are the same is called a disproportionation reaction.

CONCEPT STRANDS

Concept Strand 14

Represent the disproportionation reaction of chlorine in alkaline medium.

Solution

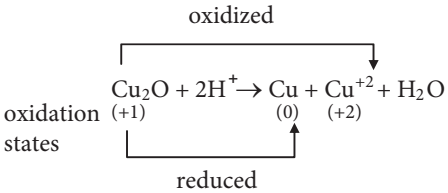


Cl_2 disproportionates to (+1) and (−1) states.

Concept Strand 15

Represent the disproportionation reaction of Cu_2O in acidic medium.

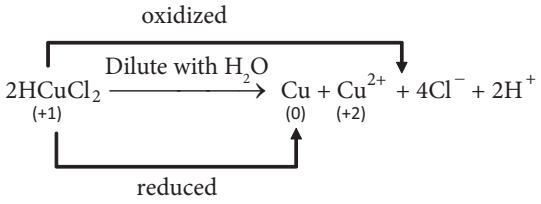
Solution



Concept Strand 16

Represent the disproportionation reaction of $HCuCl_2$.

Solution



Concept Strand 17

In the given reactions, identify the elements whose oxidation numbers are changed.

(i) $2H_2O_2 \rightarrow 2H_2O + O_2$
(ii) $4ClO_3^- \rightarrow 3ClO_4^- + Cl^-$
(iii) $3MnO_4^{2-} + 2H_2O \rightarrow MnO_2 + 2MnO_4^- + 4OH^-$
(iv) $Hg_2^{2+} \rightarrow Hg^{2+} + Hg$

Solution

(i) $\overset{-1}{2H_2O_2} \longrightarrow \overset{-2}{2H_2O} + \overset{0}{O_2}$

(ii) $\overset{+5}{4ClO_3^-} \longrightarrow \overset{+7}{3ClO_4^-} + \overset{-1}{Cl^-}$

(iii) $\overset{+6}{3MnO_4^{2-}} + 2H_2O \longrightarrow \overset{+4}{MnO_2} + \overset{+7}{2MnO_4^-} + 4OH^-$

(iv) $\overset{+1}{Hg_2^{2+}} \longrightarrow \overset{+2}{Hg^{2+}} + \overset{0}{Hg}$

EQUIVALENT WEIGHT

In general equivalent weight is defined as the weight of a substance, which can react with or liberate 1 mole of electrons.

Equivalent weight of an acid

It is defined as the number of parts by weight of the acid that contain one part by weight of replaceable hydrogen or as the weight of the acid which on reaction with a metal produces one part by weight of hydrogen. The number of replaceable hydrogen atoms in a molecule of an acid is known as its basicity. The equivalent weight of an acid is, therefore, related to its molecular weight by the relation.

$$\text{Equivalent weight of an acid} = \frac{\text{molecular weight of acid}}{\text{Basicity}}$$

Basicity of a poly protic acid depends on the nature of the reaction

Acid	Basicity
HCl, HNO_3, H_3BO_3, H_3PO_2, $HClO_4$	1
H_2SO_4, $H_2C_2O_4$, H_3PO_3	2
H_3PO_4	3

Equivalent weight of a base

The equivalent weight of a base (say an alkali) is defined as the number of parts by weight of it that completely neutralizes one gram equivalent of an acid. The 'acidity' of a base is used to calculate the equivalent weight of the base and is defined as the number of –OH groups in the molecule of the base. Alternately, it is the number of gram equivalents of the acid required to neutralize one gram molecule of the base.

$$\text{Equivalent weight of a base} = \frac{\text{molecular weight of base}}{\text{acidity}}$$

Acidity of a polyhydroxy base depends on the nature of the reaction

Base	Acidity
NaOH, KOH	1
$Ca(OH)_2$, $Ba(OH)_2$	2
$Al(OH)_3$	3

Equivalent weight of any other substance

It is defined as the number of parts by weight of one compound that reacts with one gram equivalent weight of another compound.

For example, in the reaction, $Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2$

$$\text{Equivalent weight of } Na_2CO_3 = \frac{\text{Mol.wt}}{2}$$

(because one mole of Na_2CO_3 reacts with 2 gram equivalents of HCl)

In the reaction, $Na_2CO_3 + HCl \rightarrow NaHCO_3 + NaCl$

$$\text{Equivalent weight of } Na_2CO_3 = \frac{\text{Mol.wt}}{1}$$

As seen from above, it is necessary to state the reaction in specifying the equivalent weight of a compound.

Equivalent weights of salts

For a salt reacting in such a manner that no atom of the salt undergoes any change in oxidation state,

$$\text{Equivalent weight} = \frac{\text{Molecular weight}}{\text{Total charge on cation (or) anion}}$$

Example: $2AgNO_3 + MgCl_2 \rightarrow Mg(NO_3)_2 + 2AgCl$

In this reaction it can be seen that the oxidation state of Ag, N, O, Mg and Cl remains the same in reactants and products.

$$\therefore \quad \text{Equivalent weight of } AgNO_3 = \frac{\text{Mol.wt}}{1}$$

$$\text{Equivalent weight of } MgCl_2 = \frac{\text{Mol.wt}}{2}$$

Equivalent weights of oxidizing and reducing agents

$$\text{Equivalent weight of oxidizing agent or reducing agent} = \frac{\text{Molar mass of oxidant or reductant}}{\text{change in oxidation number of oxidant or reductant}}$$

In oxidation-reduction reaction, the equivalent weight of a substance is equal to the molecular weight divided by the number of electrons required in the particular reaction.

Equivalent weights of $KMnO_4$

(i) In acidic medium, oxidation state of Mn changes from +7 to +2.

$$\therefore \quad \text{Eq. wt} = \frac{\text{mol.wt}}{5}, \text{ in acidic medium}$$

(ii) In neutral medium, oxidation state of Mn changes from +7 to +4

$$\therefore \quad \text{Eq. wt} = \frac{\text{mol.wt}}{3}, \text{ in neutral medium}$$

(iii) In basic medium, oxidation state of Mn changes from +7 to +6

$\therefore$ Eq. wt = mol. wt, in basic medium

Equivalent weights of $K_2Cr_2O_7$

In acidic medium oxidation state of Cr changes from $2 \times (+6)$ to $2 \times (+3)$

$$\therefore \quad \text{Eq. wt} = \frac{\text{mol.wt}}{6}$$

It is also possible to obtain the equivalent weight from the stoichiometric reaction representing the redox reaction.

Several oxidizing agents like $KMnO_4$, $K_2Cr_2O_7$, MnO_2, KIO_3 are used in analytical determinations and their equivalent weights have to be calculated on the basis of the change in oxidation state of the oxidizing species or on the basis of the corresponding stoichiometric equation. Some examples are considered below:

(i) Reaction between $KMnO_4$ and $FeSO_4$ in acid solution

$$2KMnO_4 + 8H_2SO_4 + 10\ FeSO_4 \rightarrow K_2SO_4 + 2MnSO_4 + 5Fe_2(SO_4)_3 + 8H_2O \qquad \text{— (15)}$$

In the above reaction, the equivalent weight of $KMnO_4$ may be calculated on the basis of the change in the oxidation state of Mn from +7 (in $KMnO_4$) to + 2 (in $MnSO_4$) i.e., a change of 5 electrons.

$$\therefore \quad \text{Equivalent weight of } KMnO_4 = \frac{\text{Mol.wt}}{5}$$

The equivalent weight of $FeSO_4$ can be calculated by noting that the change in its oxidation state as 1 in the oxidation of Fe^{2+} to Fe^{3+}.

$$\therefore \quad \text{Equivalent weight of } FeSO_4 = \frac{\text{Mol.wt}}{1}$$

Equation (15) may be split into two partial equations as

(i) $2KMnO_4 + 3H_2SO_4 \rightarrow K_2SO_4 + 2MnSO_4 + 3H_2O + 5(O)$ — (16 a)

(ii) $2FeSO_4 + H_2SO_4 + (O) \rightarrow Fe_2(SO_4)_3 + H_2O$ — (16 b)

The quantity of $KMnO_4$ required to liberate 8 g of oxygen (equivalent weight of oxygen), which is the equivalent weight of $KMnO_4$, calculated from equation (16 a) is obtained as

Equivalent weight of $KMnO_4$

$$= \frac{2 \times \text{Mol.wt. of } KMnO_4 \times 8}{80}$$

$$= \frac{2 \times 158 \times 8}{80} = 31.6$$

The equivalent weight of $FeSO_4$ (usually employed in the form of ferrous ammonium sulphate) is calculated from equation (16 b) as

$$\text{Equivalent weight of } FeSO_4 = \frac{\text{Mol. wt} \times 2 \times 8}{16}$$

$$= \frac{151.85 \times 2 \times 8}{16} = 151.85$$

(ii) Oxidation of oxalic acid by $KMnO_4$

The reaction is

$$\begin{matrix} COOH \\ | \\ COOH \end{matrix} + (O) \longrightarrow 2CO_2 + H_2O$$

Equivalent weight of oxalic acid (anhydrous)

$$= \frac{\text{mol.wt} \times 8}{16} = \frac{90 \times 8}{16} = 45$$

Equivalent weight of hydrated oxalic acid ($H_2C_2O_4 . 2H_2O$) $= \frac{\text{mol.wt}}{2} = \frac{126}{2} = 63$

(iii) Reaction between $K_2Cr_2O_7$ and $FeSO_4$ in acid solution

The partial equations are

(i) $K_2Cr_2O_7 + 4H_2SO_4 \rightarrow K_2SO_4 + Cr_2(SO_4)_3 + 4H_2O + 3(O)$

(ii) $2FeSO_4 + H_2SO_4 + (O) \rightarrow Fe_2(SO_4)_3 + H_2O$

The overall equation is

$K_2Cr_2O_7 + 7H_2SO_4 + 6FeSO_4 \rightarrow K_2SO_4 + Cr_2(SO_4)_3 + 3Fe_2(SO_4)_3 + 7H_2O$

From the equation,

Equivalent weight of $K_2Cr_2O_7$

$$= \frac{\text{mol.wt}}{6} = \frac{298 \times 8}{48} = 49.0$$

The equivalent weight of $FeSO_4$ is the same as its molecular weight.

(iv) $H_2O_2 + 2H^+ + 2e^- \rightarrow 2H_2O$

Eq. wt of $H_2O_2 = \frac{\text{mol.wt}}{2}$

(v) $C_2O_4^{2-} \rightarrow CO_2$

Oxidation state of carbon changes from $2 \times (+3)$ to $2 \times (+4)$

Eq. wt $= \frac{\text{mol.wt}}{2}$

(vi) $S_2O_3^{2-} \rightarrow S_4O_6^{2-}$

Oxidation state of S changes from $2 \times (+2)$ to $2 \times (+2.5)$

Eq. wt. of $S_2O_3^{2-}$ = mol. wt

(vii) $Fe_2O_3 \rightarrow FeSO_4$

Oxidation state of Fe changes from $2 \times (+3)$ to $2 \times (+2)$

Eq. wt of $Fe_2O_3 = \frac{\text{mol.wt}}{2}$

n-factor

n-factor is actually the number of moles of electrons gained or lost by one mole of a substance in a reaction. n-factors are used for balancing the chemical reaction between two species by cross multiplying their n-factors.

(i) n-factor of acids

It is the number of H^+ ions replaced by 1 mole of acid in a reaction. In the reaction, $H_2SO_4 + NaOH \rightarrow NaHSO_4 + H_2O$, n-factor of H_2SO_4 is 1

n-factor of H_2SO_4 = 1 or 2
H_2SO_3 = 1 or 2
H_2CO_3 = 1 or 2
H_3PO_3 = 1 or 2
H_3PO_4 = 1, 2 or 3

(ii) n-factor of bases

It is the number of OH^- ions replaced by mole of base in a reaction.

n-factor of NaOH = 1
$Ca(OH)_2$ = 1 or 2
$Al(OH)_3$ = 1, 2 or 3

(iii) n-factor for salts which react such that no atom undergoes change in oxidation state.

The n-factor of such a salt is the total charge on the cation or anion

$$2AgNO_3 + K_2CrO_4 \rightarrow Ag_2CrO_4 + 2KNO_3$$

The oxidation state of Ag, N, O, K and Cr remains the same in reactants and products.

n-factor of $AgNO_3$ = 1
K_2CrO_4 = 2

(iv) n-factor in redox change

For oxidizing or reducing agent n-factor is the change in oxidation number per mole of the substance.

$$\underset{+7}{MnO_4^-} \xrightarrow{H^+} \underset{+2}{Mn^{2+}}$$

n-factor of $KMnO_4$ in acid medium = 5

$$\underset{+7}{MnO_4^-} \xrightarrow{H_2O} \underset{+4}{Mn^{+4}}$$

n-factor of $KMnO_4$ in neutral medium = 3

$$\underset{2x+6}{Cr_2O_7^{2-}} \xrightarrow{H^+} \underset{2x+3}{Cr^{3+}}$$

n-factor = 6

$$\underset{2x+3}{C_2O_4^{2-}} \longrightarrow \underset{2x+4}{CO_2}$$

n-factor = 2

$$\underset{2x+2}{S_2O_3^{2-}} \xrightarrow{OH^-} \underset{2x+6}{SO_4^{2-}}$$

n-factor = 8

$$\underset{2x+2}{S_2O_3^{2-}} \longrightarrow \underset{2\times2.5}{S_4O_6^{2-}}$$

n-factor = 1

$$\underset{-1}{I^-} \longrightarrow \underset{0}{I_2}$$

n-factor = 1

$$\underset{2\times0}{I_2} \longrightarrow \underset{2\times-1}{I^-}$$

n-factor = 2

$$\underset{+2}{Fe}\underset{2\times+3}{C_2O_4} \longrightarrow \underset{+3}{Fe^{3+}} + \underset{2\times+4}{CO_2}$$

n-factor of FeC_2O_4 = (3 − 2) + (8 − 6) = 3

$$\underset{+2}{Fe}\,\underset{2\times+3}{Cr_2\,O_4} \longrightarrow \underset{+3}{Fe^{3+}} + \underset{2\times+6}{Cr^{6+}}$$

n-factor = 1 + 6 = 7

STRENGTH OF SOLUTIONS

Strength of a solution refers to the amount of the solute dissolved in a given amount of solvent/solution. For a binary solution (a solution consisting of two components) the component, which is present in large quantity is the solvent and the component which is in small quantity is called the solute, provided both the components are in the same physical state.

For e.g., in a solution containing 40 mL water and 60 mL ethyl alcohol, water is the solute and alcohol is the solvent.

When the two component is a binary solution are in different physical state, solvent is that component, which is in the same physical state as that of the solution.

For e.g., in a sugar syrup containing 40 % water and 60 % sugar, water is the solvent and not the sugar, even though sugar is present in larger proportion.

Methods of expressing the strength or concentration of a solution

The strength or the concentration of a solution may be expressed in several ways such as percentage by weight (m/m), percentage by volume (m/v), normality (N), molarity (M), molality (m), mole fraction (x) etc.

1. Percentage by weight (m/m)

$$\text{Percentage by weight} = \frac{\text{Weight of solute}}{\text{Weight of solution}} \times 100$$

2. Percentage by volume (m/v)

$$\text{Percentage by volume} = \frac{\text{Weight of solute}}{\text{Volume of solution}} \times 100$$

3. Normality (N)

It gives the number of equivalents of solute in one litre solution.

$$\text{i.e., } N = \frac{\text{weight of solute(g)}}{\text{equivalent weight of solute}} \times \frac{1}{\text{volume of solution(L)}}$$

CONCEPT STRAND

Concept Strand 18

Find the normality of a solution containing 2.45 g H_2SO_4 in 250 mL solution:

Solution

Wt of H_2SO_4 in 1L solution = 2.45 × 4 g

$$\text{No. of equivalents of } H_2SO_4 \text{ in 1L solution} = \frac{2.45\times4}{49} = 0.2$$

$\therefore$ N = 0.2

Weight of solute from the volume and normality of the solution

V (mL) × N = No. of milliequivalents of solute

$\frac{V(mL)\times N}{1000}$ = No. of equivalents of solute (or)

V (L) × N = no. of equivalents of solute

No. of equivalents of solute × equivalent weight = wt of solute (gram)

CONCEPT STRAND

Concept Strand 19

A 2.0 g mixture of potassium carbonate and potassium chloride was completely neutralized by the addition of 100 mL of 0.15 N monobasic acid. Calculate the percentage amount of KCl present in the mixture. (Relative Atomic Weights: K = 39.0; Cl = 35.5; C = 12.00; O = 16.00)

Solution

Milli. eq. of acid required to neutralize $K_2CO_3 = 100 \times 0.15$
$= 15$

$$\frac{\text{mg of } K_2CO_3}{\text{Eq.wt}} = \frac{\text{mg of } K_2CO_3}{69} = 15$$

or, mg of $K_2CO_3 = 69 \times 15$
$= 1035$

weight of KCl in mixture = 2.000 −1.035
= 0.965 g

% KCl in mixture = $\frac{0.965}{2} \times 100 = 48.3$

Normality from density and percentage strength of the solution

$$N = \frac{\text{Density (g mL}^{-1}) \times \text{wt.\% (m/m) of the solute} \times 10}{\text{Eq. wt. of solute}}$$

CONCEPT STRANDS

Concept Strand 20

Calculate the weight of $Ba(OH)_2$ (mol.wt. = 171 g mol^{-1}) required to prepare 300 mL 0.04 N solution?

Solution

Weight of $Ba(OH)_2 = \frac{V \times N \times \text{eq.wt}}{1000} = \frac{300 \times 0.04}{1000} \times \frac{171}{2}$
$= 1.026$ g

Concept Strand 21

Find the density of 32 N sulphuric acid solution having 90% (m/m) strength?

Solution

$$N = \frac{\text{density} \times (\%) \times 10}{49}$$

$$\Rightarrow \quad \text{density} = \frac{32 \times 49}{90 \times 10} = 1.74 \text{ g mL}^{-1}$$

4. Molarity (M)

It is the number of moles of solute in one litre solution. i.e.,

$$M = \frac{\text{weight of solute(g)}}{\text{molecular weight of solute}} \times \frac{1}{\text{volume of solution(L)}}$$

5. Formality (F)

If one formula weight of a solute is dissolved and made up to 1 L of aqueous solution, we defined the solution as 1 formal. This unit is not much used.

CONCEPT STRANDS

Concept Strand 22

If the molarity of a solution containing 1.12 g KOH (Mol. wt = 56 g mol^{-1}) is 0.2, find the volume of the solution?

Solution

0.2 M solution → 0.2 moles

(0.2 × 56 = 11.2 g KOH) in 1000 mL.

When the weight is 1.12 g, volume becomes 100 mL to get the same molarity.

Concept Strand 23

Calculate the number of H^+ ions in 0.5 litres of 0.1 N H_2SO_4.

Solution

Milli equivalents of $H^+ = 500 \times 0.1 = 50$

$$\text{Moles of } H_2SO_4 = \frac{50 \times 10^{-3}}{2} = 0.025$$

Moles of H^+ ion in 0.025 moles of $H_2SO_4 = 0.025 \times 2 = 0.05$

Number of H^+ ion in 0.05 moles = $6.023 \times 10^{23} \times 0.05 = 3.01 \times 10^{22}$

Weight of solute from volume and molarity of the solution

V (mL) × M = No. of millimoles of solute

$\frac{V(mL) \times M}{1000}$ = No. of moles of solute or

V(L) × M = No. of moles of solute

∴ No. of moles of solute × molecular weight = weight of solute (gram)

Molarity from density and percentage strength of the solution

$$M = \frac{\text{Density (g mL}^{-1}) \times \text{weight \% (m / m) of the solute} \times 10}{\text{molecular weight}}$$

CONCEPT STRAND

Concept Strand 24

A 0.5 M solution of Na_2CO_3 (mol.wt. = 106 g mol^{-1}) has a density of 0.96 g mL^{-1}. Find the weight of solvent in 125 g of this solution.

Solution

$$M = \frac{\text{density} \times \text{weight \% of solute (m / m)} \times 10}{\text{Molecular wt.}}$$

$$\text{Weight \% (m/m)} = \frac{0.5 \times 106}{0.96 \times 10} = 5.52$$

i.e., 5.52 g solute in 100 g solution

or, 5.52×1.25 g = 6.9 g solute in 125 g solution.

∴ wt. of solvent in 125g solution = 118.1g

Relation between N and M

$$N \times \text{Eq.wt} = M \times \text{Mol.wt.}$$

For example, mol.wt of $H_2C_2O_4.2H_2O$ (oxalic acid) is 126 and eq.wt is 63.

A 5 N solution of oxalic acid is 2.5M.

Effect of dilution on N and M

When a solution is diluted to 'x' times, then final N or M = $\frac{\text{Initial N(or)M}}{x}$

CONCEPT STRANDS

Concept Strand 25

Calculate the normality of

(i) 50.00 weight % of H_2SO_4 whose density is 1.31 g cm^{-3} at 20°C.
(ii) the volume of acid required to get one litre of 0.2 N acid solution.

Solution

$$N = \frac{1.31 \times 50 \times 10}{49} = 13.37$$

$\Rightarrow$ $13.37 \times V = 0.2 \times 1000$

or $$V = \frac{0.2 \times 1000}{13.37} = 14.96 \text{ mL}$$

i.e., 14.96 mL of 13.37N on dilution to 1 L reduces the normality to 0.2.

Concept Strand 26

250 mL of 0.25 M H_2SO_4 and 250 mL of 0.1N HCl are mixed together. What is the normality of the resulting solution?

Solution

m.eq. of $H_2SO_4 = 250 \times 0.50 = 125$

m.eq. of $HCl = 250 \times 0.10 = 25$

Total m.eq present in 500 mL mixture = 150

$$\text{Normality of mixture} = \frac{150}{0.5} = 300 \times 10^{-3}$$

$$= 0.3$$

Concept Strand 27

50 mL of 6 N HCl is diluted to 200 mL. Find the normality of the resulting solution.

Solution

Dilution factor (x) is 4 as the solution is diluted to 4 times

$\therefore$ $$\text{Final N} = \frac{6}{4} = 1.5$$

6. Molality (m)

It is the number of moles of solute in 1 kg solvent.

$$\text{i.e., } m = \frac{\text{wt. of solute(g)}}{\text{Mol. wt of solute}} \times \frac{1}{\text{wt. of solvent(kg)}}$$

CONCEPT STRAND

Concept Strand 28

Find molality of 0.5 M HNO_3, if its density is 0.92 g mL^{-1}.

Solution

0.5 M $\rightarrow$ 0.5 moles ($0.5 \times 63 = 31.5$ g) in 1000 mL (920 g) solution

$\therefore$ wt of solvent = 888.5 g

$\therefore$ $$m = 0.5 \times \frac{1000}{888.5} = 0.56$$

7. Mole fraction (x)

Let n_1 and n_2 represent the number of moles of solvent and solute respectively.

Then the mole fraction of solvent, $$x_1 = \frac{n_1}{n_1 + n_2}$$

The mole fraction of the solute $$x_2 = \frac{n_2}{n_2 + n_1}$$

$$x_1 + x_2 = 1 \Rightarrow \frac{x_1}{x_2} = \frac{n_1}{n_2}$$

8. Parts per million (ppm)

If 1 g of solute is made up to 1000000 g of solution, then it is 1 ppm.

If 20 g of NaCl is dissolved in 999980 g of water, then we have 20 ppm of NaCl. This is same as 2×10^{-3} % solution.

CONCEPT STRAND

Concept Strand 29

Find the mole fraction of the components in 3.65 M NaOH solution having a density of 0.92 g mL^{-1}.

Solution

3.65 M NaOH, $3.65 \times 40 = 146$ g
wt. of solution = 920 g;
wt. of solvent = 774 g

$$n_1 = \frac{774}{18} = 43 \text{ and}$$

$$n_2 = 3.65 \qquad x_2 = \frac{3.65}{46.65} = 0.078$$

$\therefore \quad x_1 = 0.922$

Inter relation between the concentration terms M, m and x_2

(i) $m = \dfrac{M}{d - MM_2}$ or $M = \dfrac{md}{1 + mM_2}$

(ii) $m = \dfrac{x_2}{x_1.M_1}$ or $x_2 = \dfrac{m.M_1}{1 + mM_1}$

(iii) $M = \dfrac{x_2 d}{x_1.M_1 + x_2 M_2}$ or $x_2 = \dfrac{MM_1}{M(M_1 - M_2) + d}$

subscripts 1 and 2 represent solvent and solute.
d – density of the solution in kgL^{-1}

Working out

(i) Relation between (m) and (M)

Molarity (M) $\rightarrow$ M moles of solute of molar mass M_2 (g mol^{-1}) in 1 L solution.

If density d is in g/mL, then

Mass of solute = MM_2 g

Mass of solution = 1000 d g

Mass of solvent = $(1000\,d - MM_2)$ g

$$\text{Molality (m)} = \frac{M}{1000\,d - MM_2} \times 1000$$

When M_2 is in kg mol^{-1} and d is in kg/L

Mass of solute = MM_2 kg

Mass of solution = d kg

Mass of solvent = $d - MM_2$ kg

$$\therefore \quad m = \frac{M}{d - MM_2}$$

(ii) Relation between (m) and (x_2)

Mass of solvent = W_1 kg
Molar mass of solvent = M_1 kg mol^{-1}

$$x_2 = \frac{n_2}{n_1 + n_2} \qquad m = \frac{n_2}{W_1} = \frac{n_2}{n_1 M_1}$$

$$n_2 = mn_1M_1 \qquad x_2 = \frac{mn_1M_1}{n_1 + mn_1M_1} = \frac{mM_1}{1 + mM_1}$$

(iii) Relation between (M) and (x_2)

M_1 and M_2 in kg mol^{-1}, d in kg/L
Mass of solute = MM_2 kg
Mass of solution = d kg
Mass of solvent = $(d - MM_2)$ kg

$$x_2 = \frac{M}{M + \dfrac{d - MM_2}{M_1}} = \frac{MM_1}{MM_1 + d - MM_2}$$

$$x_2 = \frac{MM_1}{M(M_1 - M_2) + d}$$

Volume strength of H_2O_2

H_2O_2 decomposes according to the equation

$$2H_2O_2 \rightarrow 2H_2O + O_2$$

'V' volume strength of H_2O_2 means 1 litre of H_2O_2 gives 'V' litre of O_2 at STP.

% strength of H_2O_2 solution gives the weight (in grams) of H_2O_2 in 100 mL H_2O_2 solution.

$$\text{Molarity of } H_2O_2 = \frac{V}{11.2}; \quad \text{Normality of } H_2O_2 = \frac{V}{5.6}$$

$$\text{Weight (g/L)} = \frac{V}{5.6} \times 17$$

1% H_2O_2 = 3.3 Volume
1 Volume = 0.303%

CONCEPT STRAND

Concept Strand 30

A sample of H_2O_2 solution is labelled as 28 volume. Find its percentage by volume strength.

Solution

% Strength = V × 0.303
= 28 × 0.303 = 8.484

Degree of hardness of water

Hardness of water is due to the presence of bicarbonates (temporary), chlorides and sulphates (permanent) of calcium and magnesium. The degree of hardness of water is defined as the number of parts of $CaCO_3$ (or equivalent to other calcium and magnesium salts) present in one million (10^6) parts of water. It is usually expressed in ppm.

CONCEPT STRANDS

Concept Strand 31

Find the degree of hardness of a water sample containing 240 ppm $MgSO_4$.

Solution

Equivalent weight of $MgSO_4 = \frac{120}{2} = 60$

Equivalent weight of $CaCO_3 = \frac{100}{2} = 50$

60 g of $MgSO_4 \equiv 50$ g of $CaCO_3$

$\therefore$ Degree of hardness of the water sample = 200 ppm

Concept Strand 32

Calculate the degree of hardness of a water sample containing 0.001 mole of $MgSO_4$ dissolved in 2 litres of water.

Solution

No. of moles of $MgSO_4$ in 1 L = 0.0005

= 0.0005 moles of $MgSO_4 \equiv 0.0005$ moles of $CaCO_3$

= 0.0005 moles of $CaCO_3$ in 1 L

= 50 mg in 1 L

$\therefore$ Degree of hardness of the water sample = 50 ppm

Quantitative analysis

Any quantitative analytical determination depends on the measurement of a property, which is related to the amount of the derived constituent present in the sample under study. Quantitative analytical methods may broadly be divided into three categories. viz., (1) gravimetric methods (2) volumetric methods (3) instrumental methods. Only volumetric methods will be considered below:

VOLUMETRIC ANALYSIS

This method enables a quantitative determination of the concentration or the composition of an unknown solution by studying their reaction with certain standard solutions. From the volume of the standard solution consumed at the end of the reaction, the concentration of the unknown solution is calculated. The completion of the reaction under consideration is observed by use of appropriate indicators. Titrimetric methods can be classified into four categories depending on the nature of the reaction involved:

(i) Neutralization methods
(ii) Precipitation methods
(iii) Oxidation-Reduction methods
(iv) Complex formation

In neutralization methods, the reaction is most often between hydrogen and hydroxyl ions or either of them reacts with another basic or acidic constituent.

In the case of precipitation methods, the desired constituent and the standard solution react to form an insoluble precipitate. It is not necessary to separate out the precipitate to complete the titration using a suitable indicator. The determination of the chloride content of a solution with a standard solution of silver nitrate is a classical example.

In oxidation-reduction methods, the main reaction involves the transfer of electrons between the reduced species and the oxidized species. The standard solution may be either the oxidizing or the reducing agent. The estimation of iron by the reaction of ferrous ions with standard $KMnO_4$ or $K_2Cr_2O_7$ is a good example of this category.

In complexometric or complex formation methods, the formation of a soluble complex ion between the standard solution and the constituent to be estimated is the main reaction. The determination of hardness of water by titrating the Ca^{2+} ions using EDTA as complex forming agent is a good example of this type.

Standard solution

A standard solution is a solution whose concentration is known and it is prepared by taking a known weight of a substance termed as a primary standard and dissolving it in a given volume of water and making up to a definite volume in a standard flask. Potassium hydrogen phthalate, anhydrous oxalic acid, anhydrous sodium carbonate are examples of such standards. Using a suitable standard solution, the strength of a desired solution is obtained by titration.

Primary and secondary standards

Standard solutions are of two categories–primary and secondary.

A primary standard solution can be prepared by directly weighing the required amount of the substance and diluting it to a known volume. Examples of primary standards in acid-base titrations are (i) potassium hydrogen phthalate (ii) sodium carbonate. In redox titrimetry, oxidants like $K_2Cr_2O_7$ can be directly used as primary standard whereas in the case of $KMnO_4$ (being not very stable), substances like oxalic acid are used to estimate the strength of $KMnO_4$ solution.

A secondary standard solution is one whose concentration cannot be determined directly from the weight of the substance and the volume of the solution. Its concentration must be determined by analysis of a portion of the solution.

The chief requirements for a substance to be considered as a primary standard are:

(i) It must be obtainable in a pure and stable form and its solution must be stable too.
(ii) It must not be hygroscopic and must be capable of being dried without decomposition.
(iii) It must be fairly soluble in a suitable solvent.
(iv) It must be capable of entering into a stoichiometric reaction with a substance to be determined.
(v) It must have an adequately high equivalent weight so that fairly large weight may be weighed without causing weighing errors.

Solutions of secondary standard require standardisation against a primary standard.

Note: $Na_2Cr_2O_7$ is not used as a primary standard but $K_2Cr_2O_7$ is used for this purpose. This is because $Na_2Cr_2O_7$ is hygroscopic.

Normality equation used in volumetric analysis

In volumetric analysis, the strength of a solution is most commonly expressed in the unit of normality(N).

Equal volumes of the same normality of an acid and base exactly neutralize each other or in the case of redox reaction, the same statement is applicable for the reduction of an oxidizing species or vice versa. Put in mathematical terms, we can write

$$N_1V_1 = N_2V_2$$

where, V_1 and V_2 represent the volumes of reactants and N_1 and N_2 represent their normalities. In the above reaction, the law of equivalence is valid i.e., milliequivalents of acid = milliequivalents of base. For example, in the reaction,

$$H_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2H_2O,$$

20 mL 0.25N (5 m.eq) of H_2SO_4 is completely neutralised by 50 mL 0.1N (5 m.eq) of NaOH.

But, when the reactants are expressed in moles or milli moles, they react according to the stoichiometry. In this case, 20 mL 0.125M (2.5 m.mols) of H_2SO_4 is completely neutralized by 50 mL 0.1M (5 m.mols) of NaOH, since according to the balanced equation, 1 mole of H_2SO_4 is neutralized by 2 moles of NaOH.

CONCEPT STRANDS

Concept Strand 33

Calculate the normality of a NaOH solution, 30 mL of which is required to neutralize 0.75 g of potassium hydrogen phthalate.

Solution

milli equivalents of NaOH = milli equivalents of potassium hydrogen phthalate

$$N \times 30 = \frac{\text{mg of potassium hydrogen phthalate}}{\text{eq.wt}}$$

$$= \frac{750}{204} = 3.67\ ;$$

$$\therefore \quad N = \frac{3.67}{30} = 0.122$$

Concept Strand 34

To a 25 mL of H_2O_2 solution, excess acidified KI was added. The iodine liberated required 40 mL of 1.5 N sodium thiosulphate solution. Find the volume strength of H_2O_2.

Solution

Using $V_1N_1 = V_2N_2$ and substituting the volumes of H_2O_2, thiosulphate and strength of thiosulphate, we have $N_1 = 2.4$

Volume strength of $H_2O_2 = N \times 5.6$

$= 2.4 \times 5.6 = 13.44$

Concept Strand 35

A sample of Na_2CO_3 weighing 0.53g is added to 50 mL of 0.2N H_2SO_4. Is the resulting solution acidic, basic or neutral?

Solution

$$Na_2CO_3 + H_2SO_4 \rightarrow Na_2SO_4 + H_2O + CO_2$$

$$\text{Eq. Weight of } Na_2CO_3 = \frac{\text{Molecular. wt}}{2} = \frac{106}{2} = 53.0$$

$$\frac{\text{mg of } Na_2CO_3}{\text{Eq.wt}} = \frac{0.53 \times 1000}{53}$$

$$V_{H_2SO_4} \times N_{H_2SO_4} = 50 \times 0.2 = 10$$

Since the milliequivalents of Na_2CO_3 are equal to milliequivalents of H_2SO_4, the solution of acid is exactly neutralized by addition of Na_2CO_3.

Concept Strand 36

A mixture of HCOOH and $H_2C_2O_4$ in 4 : 3 molar ratio is heated with conc.H_2SO_4. The gaseous products are then collected over KOH solution. Calculate the fraction of the volume of the gas collected?

Solution

$$\underset{4\text{ moles}}{HCOOH} \rightarrow \underset{4\text{ moles}}{CO} + H_2O$$

$$\underset{3\text{ moles}}{H_2C_2O_4} \rightarrow \underset{3\text{ moles}}{CO} + \underset{3\text{ moles}}{CO_2} + H_2O$$

Total amount of gaseous product = 10 moles

Fraction of the volume of gas (CO_2) absorbed by KOH $= \frac{3}{10} = 0.3$

Fraction of the volume of the gas (CO) collected over KOH $= \frac{7}{10} = 0.7$

Concept Strand 37

A mixture of KOH and $Ca(OH)_2$ weighing 3.065 g is completely neutralized by 350 mL 0.2 N acid. Find the composition of the mixture.

Solution

Let the wt. of KOH be x g

$$\frac{x}{56} + \frac{3.065 - x}{37} = 0.07$$

$\Rightarrow \quad x = 1.4$ g

i.e., weight of KOH = 1.4 g and weight of $Ca(OH)_2$ = 1.665 g

NEUTRALIZATION REACTIONS

The term 'neutralization' is applied to the reaction of certain equivalent of an acid with same equivalent of a base.

The general nature of this type of reaction may be seen in the various reactions listed below, which are all to be considered as acid-base reactions.

$$H_3O^+ + OH^- \rightarrow H_2O + H_2O$$

$$HCOOH + (Na^+)\ OH^- \rightarrow (Na^+)\ HCOO^- + H_2O$$

$$HCl + NH_3 \rightarrow NH_4^+ + Cl^-$$

$$HCl + (Na^+)\ CH_3COO^- \rightarrow CH_3COOH + (Na^+)\ Cl^-$$

$$NH_4^+(Cl^-) + (Na^+)\ OH^- \rightarrow NH_4OH + (Na^+)(Cl^-)$$

The last two reactions are displacement reactions, where in the first case, a strong acid displaces a weak acid from its salt and in the second case a weak base is displaced from its salt by a strong base.

Calculations in volumetric analysis

In volumetric analysis, calculations are mostly of the stoichiometric type involving a single substance or sometimes a reaction between two substances. Problems involving standardization of solution or percentage calculations of unknowns also belong to this category.

Case I

(a) When applied to a single substance 'X', it is very often necessary to apply an identity as milli equivalents of (x), to arrive at the required result. One meets with such a situation in the calculation of normality of an acid whose strength is given in weight per cent.

(b) When two substances, say (X) and (Y), are titrated against each other, at the equivalence point, the simple relation milli equivalents of (X) = milli equivalents of (Y) holds.

(c) When two substances (X) and (Y) in a given solution, both react quantitatively with another substance (Z), at the equivalence point, the number of milli equivalents of (Z) is numerically equal to the total number of milli equivalents of (X) and (Y)

i.e., milli equivalents of (X) + milli equivalents of (Y) = mili equivalents of (Z)

Case II

In a given solution of a substance (X), the product of a definite volume (V_x) and its normality (N_x) i.e., $N_x \cdot V_x$ = milli equivalents of (X).

Knowing that the normality of a solution is defined as the number of equivalents per litre of the solution, which is the same as stating the number of milli equivalents per mL of the solution, the product of V_x and N_x given in equation is equivalent to milli equivalents of (X).

Case III

The number of milli equivalents of a substance (X) may also be written as

$$\text{Milliequivalents of X} = \frac{\text{milligram of X}}{\text{equivalent weight of X}}$$

The above equations form the basis for solving problems in volumetric analysis.

CONCEPT STRANDS

Concept Strand 38

65.4 milligram of zinc was dissolved in 100 mL of 0.1 N HCl. Calculate (i) the volume of hydrogen displaced from the acid at 27°C and 1 atm (ii) the strength of the remaining acid. (Relative atomic weight of Zn = 65.4)

Solution

$$\underset{1\text{m.mol}}{Zn} + \underset{2\text{m.mols}}{2HCl} \rightarrow ZnCl_2 + \underset{1\text{m.mol}}{H_{2(g)}}$$

65.4 mg Zn = 1 millimole

100 mL 0.1 N HCl = 10 millimoles

Volume of 1 m.mol of H_2 at STP = $22.4 \times 10^{-3} = 0.0224$ L

Volume of H_2 at 27°C and 1 atm = $\frac{0.0224 \times 300}{273} = 0.0246$ L

Millimoles of HCl that remains as unreacted = 10 − 2

$\therefore$ Strength of the unreacted HCl = $\frac{8}{100} = 0.08$ N

Concept Strand 39

A sample of ammonium sulphate was treated with 125 mL of 0.1 N NaOH and the resulting solution was boiled to expel the ammonia completely. The excess alkali remaining in solution required 24.65 mL of 0.2 N HCl for complete neutralization. Calculate the amount of $(NH_4)_2SO_4$ originally present in the solution.

Solution

The displacement of ammonia from $(NH_4)_2SO_4$ by alkali is given by the reaction

$$(NH_4)_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2NH_3 + 2H_2O.$$

Milli.eq of HCl used for titrating excess alkali = 24.65 × 0.2 = 4.93

Milli.eq of alkali originally present in the solution = 125.0 × 0.1 = 12.5

Milli eq. of alkali consumed in the reaction = 12.5 − 4.93 = 7.57;

$\therefore$ Milli. eq. of $(NH_4)_2SO_4$ present = 7.57

eq.weight.of $(NH_4)_2SO_4 = \frac{132}{2} = 66$, since eq.weight. of $(NH_4)_2SO_4 = \frac{mol.wt}{2}$

$\Rightarrow$ Number of moles of $(NH_4)_2SO_4$

$= \frac{7.57 \times 10^{-3}}{2} = 3.785 \times 10^{-3}$

$\therefore$ grams of $(NH_4)_2SO_4 = 3.785 \times 10^{-3} \times 132 = 0.4996$g

Concept Strand 40

Calculate the amount of Mn^{2+} ions present in the solution that requires 40mL of 0.1N $KMnO_4$ for complete conversion of Mn^{2+} into MnO_2.
(Atomic mass of Mn = 54.94)

Solution

$3Mn^{2+} + 2MnO_4^- + 2H_2O \rightleftharpoons 5MnO_2 + 4H^+$.

milli.eq. of $KMnO_4$ used up = $40 \times 0.1 = 4.0$

$\therefore$ Milli eq. of Mn^{2+} in solution = 4

Eq. Weight of $Mn^{2+} = \frac{Atomic.wt}{2}$

$\therefore$ Wt. of $Mn^{2+} = \frac{4 \times 10^{-3} \times 54.94}{2} = 0.11$ g

Concept Strand 41

20mL of a solution of potassium dichromate of unknown strength was treated with excess KI solution and the liberated iodine required 25.8 mL of sodium thiosulphate of 0.1 N strength. Write the relevant reactions. Calculate the normality of $K_2Cr_2O_7$ and also its concentration in g/litre.

Solution

Reactions:

(i) $K_2Cr_2O_7 + 7H_2SO_4 + 6KI \rightarrow 4K_2SO_4 + Cr_2(SO_4)_3 + 7H_2O + 3I_2$

(ii) $2Na_2S_2O_3 + I_2 \rightarrow Na_2S_4O_6 + 2KI$

milli .eq. of $Na_2S_2O_3 = 25.8 \times 0.1 = 2.58$ and m.eq of $Na_2S_2O_3$ = m.eq of $K_2Cr_2O_7$

Strength of $K_2Cr_2O_7 = \frac{2.58}{20} = 0.129$ N

Weight of $K_2Cr_2O_7$ per litre = $\frac{0.129 \times 294.2}{6} = 6.325$ g/L

Concept Strand 42

200 mL of acidified 0.1 N $KMnO_4$ was decolourized by the addition of 25 mL H_2O_2 of certain volume strength: (i) write the redox reaction involved (ii) Calculate the volume of oxygen liberated at S.T.P. (iii) what is the volume strength of H_2O_2? (Mol. weight of $KMnO_4$ = 158.04)

Solution

Reaction: $2KMnO_4 + 3H_2SO_4 + 5H_2O_2 \rightarrow K_2SO_4 + 2MnSO_4 + 8H_2O + 5O_2$

Normality of $H_2O_2 = \frac{200 \times 0.1}{25} = 0.8$ N

Volume strength of $H_2O_2 = N \times 5.6 = 4.48$

Volume of O_2 liberated from 25 mL H_2O_2 at S.T.P = 25×4.48 mL = 112 mL

Concept Strand 43

Copper (II) in 0.5 g of a sample is reduced to copper (I) by KI in acidic medium. 35.4 mL of 0.2 M thiosulphate is consumed by the liberated iodine. Calculate the percentage of copper (II) in the sample. (At. mass of copper = 63.5)

Solution

$2Cu^{2+} + 4I^- \rightarrow Cu_2I_2 + I_2$

$I_2 + 2S_2O_3^{2-} \rightarrow 2I^- + S_4O_6^{2-}$

35.4×0.2 m mols of thio $\equiv 35.4 \times 0.2$ m mols of Cu^{2+}

$\therefore$ Weight of copper = $35.4 \times 0.2 \times 10^{-3} \times 63.5 = 0.45$ g

% of copper = $\frac{0.45 \times 100}{0.5} = 90$

Concept Strand 44

60 mL of a mixture of CO and CO_2 is mixed with required amount of O_2 (x mL) and electrically sparked. The volume collected after the explosion is (35 + x) mL. What would be the residual volume of 84 mL of the original mixture is treated with KOH solution? (All volumes are measured under the same conditions)

Solution

$2CO_{(g)} + O_{2(g)} \rightarrow 2CO_{2(g)}$

Volume of O_2 taken = 60 − 35 = 25 mL

Volume of CO that reacts with 25 mL O_2 = 50 mL

∴ The original mixture contains CO and CO_2 in the ratio 5 : 1

Volume of CO_2 in 84 mL mixture = $\frac{1}{6} \times 84 = 14$ mL

Volume of the residual gas (CO) = 70 mL

Concept Strand 45

A metal forms two oxides. The lower oxide, M_2O_3 contains 30.4 % oxygen. 7.9 g of the lower oxide when converted to the higher oxide becomes 10.3 g. Which is the higher oxide?

Solution

Eq. mass of M in $M_2O_3 = \frac{69.6 \times 8}{30.4} = 18.3$

At. mass of M = 18.3 × 3 = 54.9 (Mn)

Wt. of Mn in 10.3 g of the higher oxide = $7.9 \times \frac{69.6}{100} = 5.5$ g

Eq. mass of Mn in the higher oxide = $\frac{5.5}{4.8} \times 8 = 9.15$

ie, Mn in the higher oxide is in +6 oxidation state, MnO_3

Concept Strand 46

A solution contains 1.35 g $H_2C_2O_4$ and 3.2 g of the salt, KHC_2O_4. Find the volume of 0.5 M acidified $KMnO_4$ required to react completely with the above solution.

Solution

$H_2C_2O_4$ and KHC_2O_4 contain the $C_2O_4^{2-}$ ions.

No. of m. mols of $H_2C_2O_4 = \frac{1.35 \times 10^3}{90} = 15$

No. of m mols of $KHC_2O_4 = \frac{3.2 \times 10^3}{128} = 25$

$$\underset{(n=2)}{5C_2O_4^{2-}} + \underset{(n=5)}{2MnO_4^-} \xrightarrow{H^+} CO_2 + Mn^{2+}$$

40 m mols of $C_2O_4^{2-} \equiv 16$ m mols of MnO_4^-

∴ Volume of $KMnO_4 = \frac{16}{0.5} = 32$ mL

Concept Strand 47

Calculate the minimum volume of 0.5 M H_2SO_4 needed to dissolve 4.94 g of copper (II) carbonate.

Solution

No. of m moles of $CuCO_3 = \frac{4.94 \times 10^3}{123.5} = 40$

$$\underset{(n=2)}{CuCO_3} + \underset{(n=2)}{H_2SO_4} \longrightarrow CuSO_4 + CO_2 + H_2O$$

40 m moles of $CuCO_3 \equiv 40$ m moles of H_2SO_4

∴ Volume of $H_2SO_4 = \frac{40}{0.5} = 80$ mL

Concept Strand 48

3.8 g of a metal oxide, M_2O_3 is completely reduced by 1.35 g Al. Find the oxide.

Solution

$$\underset{(n=6)}{M_2O_3} + \underset{(n=3)}{2Al} \longrightarrow 2M + Al_2O_3$$

Wt. of M_2O_3 that is reduced by 54 g of Al = $\frac{3.8 \times 54}{1.35}$ = 152 g

∴ At. mass of M = $\frac{152 - 48}{2} = 52$ i.e., Cr

∴ Cr_2O_3

Concept Strand 49

Give the equivalent weights of Cu, HNO_3 and N_2 according to the reaction,

$$Cu + HNO_3 \rightarrow CuO + H_2O + N_2.$$

Solution

$$\underset{(n=2)}{5Cu} + \underset{(n=5)}{2HNO_3} \rightarrow CuO + H_2O + N_2$$

Eq. wt. of Cu = $\frac{\text{At. wt.}}{2} = \frac{63.54}{2} = 31.77$

$$\text{Eq. wt. of } HNO_3 = \frac{\text{mol.wt.}}{5}$$

$$= 12.6$$

$$\text{Eq. wt. of } N_2 \left[\underset{2\times 0}{N_2} \rightarrow \underset{\substack{2\times+5 \\ (n=10)}}{HNO_3} \right] = \frac{28}{10}$$

$$= 2.8$$

Concept Strand 50

In a reaction 5 mL 0.25 M of an oxidant, M^{x+} is reduced to M^{2+} by consuming 25 mL 0.2 M Iron (II) sulphate in acid medium. Find the value of 'x'.

Solution

$$\underset{\substack{(n=1) \\ 5\,m.mols}}{Fe^{2+}} + \underset{\substack{(n=?) \\ 1.25\,m.mols}}{M^{x+}} \xrightarrow{H^+} Fe^{3+} + M^{2+}$$

$\therefore$ n-factor for, $M^{x+} \rightarrow M^{2+}$, is 4 $\quad\therefore\quad$ x = 6

SUMMARY

Mole	Amount of any substance containing 6.02×10^{23} particles.
Empirical formula	Simplest ratio of relative number of atoms of different element in the compound.
Molecular formula	Exact number of atoms in a molecule of the compound.
Excess reactant	Reactant taken in excess amount than required by stochiometry.
Limiting reactant	Reactant which is completely consumed in the reaction.
Oxidation number	Number of electrons gained or lost or shared by an atom.
Reductant	Species which undergoes oxidation.
Oxidant	Species which undergoes reduction.
Disproportionation	Reaction in which both the oxidizing and reducing agent are the same.
Equivalent weight	weight of a substance that can react with or liberate one mole of electrons.
Normality	Number of equivalents of solute in one litre solution.
Molarity	Number of moles of solute in one litre solution.
Formality	Number of formula weights of the solute in one litre solution.
Molality	Number of moles of solute in 1 kg solvent.
Mole fraction	Ratio of number of moles of one component to the total number of moles of all the components in a solution.
Degree of hardness of water	Number of parts of $CaCO_3$ equivalent to the total calcium and magnesium salts present in one million (10^6) parts of water.
$\frac{1}{6.02\times10^{23}} = 1.66\times10^{-24}$g	Mass of one atomic mass unit (u).
$M_1 = \frac{W_1}{1.66\times10^{-24}}$	M_1 = atomic mass/molecular mass W_1 = mass of an atom/molecule in gram
$n_1 = \frac{W_1}{M_1}$	n_1 = Number of moles W_1 = mass in grams M_1 = molecular mass

$n_1 = \frac{V}{22.4}$	V = volume in dm^3 at STP
$n_1 = \frac{\text{Number of atoms/molecules}}{N_A}$	$N_A = 6.02 \times 10^{23}$
$E_1 = \frac{M_1}{n}$	E_1 = Equivalent mass (of acid, base, salt, oxidising agent or reducing agent) M_1 = molecular mass n = basicity, acidity, total charge on anion or cation or change in oxidation number per molecule.
Equivalent mass of $KMnO_4 = \frac{M}{n}$ or $\frac{158}{n}$	Where, n is 5 in acid medium 3 in neutral medium and 1 in strongly basic medium.
% by mass (m/m) = $\frac{W_2}{W_1} \times 100$	W_1 = mass of solvent W_2 = mass of solute
% mass by volume(m/v) = $\frac{W_2}{V} \times 100$	V = volume of the solution
$M = \frac{W_2}{M_2} \times \frac{1000}{V}$	M = molarity V = volume in mL M_2 = molar mass of solute
$N = \frac{W_2}{E_2} \times \frac{1000}{V}$	N = Normality E_2 = Equivalent mass of solute
$m = \frac{W_2}{M_2} \times \frac{1000}{W_1}$	W_1 = mass of solvent
$F = \frac{W_2}{F_2} \times \frac{1000}{V}$	F = formality F_2 = formula mass of solute
$V_1M_1 = V_2M_2$	When solution is diluted
$V_1N_1 = V_2N_2$	When solution is diluted or at point of equivalence in chemical reaction.
Normality × Equivalent mass = Molarity × Molar mass	Relation between N and M of the same solution.
$x_2 = \frac{n_2}{n_1 + n_2}$	x_2 = mole fraction of solute n_1 = moles of solvent n_2 = moles of solute
$\frac{W_2}{W_1} \times 10^6$ = ppm	W_2 = mass of solute W_1 = mass of solution

$$N = \frac{d \times \%\left(\frac{m}{m}\right) \text{ of solute} \times 10}{E_2}$$

$$M = \frac{d \times \%\left(\frac{m}{m}\right) \text{ of solute} \times 10}{M_2}$$

$$m = \frac{M}{d - MM_2}$$

$$M = \frac{md}{1 + mM_2}$$

$$m = \frac{x_2}{x_1 M_1}$$

$$x_B = \frac{mM_1}{1 + mM_1}$$

$$M = \frac{x_2 \times d}{x_1 M_1 + x_2 M_2}$$

$$x_B = \frac{MM_1}{M\left(M_1 - M_2\right) + d}$$

d = density of solution

m = molality
M = molarity
d = density of solution
M_1 = molar mass of solvent
M_2 = molar mass of solute
x_1 = mole fraction of solvent
x_2 = mole fraction of solute

$$M = \frac{V}{11.2}$$

$$N = \frac{V}{5.6}$$

$$W = \frac{V}{5.6} \times 1.7$$

$$\% \text{ strength} = V \times 0.303$$

V = volume strength of H_2O_2
M = Molarity of H_2O_2
N = Normality
W = mass per litre of H_2O_2 solution

Degree of hardness of water in ppm $= \frac{x \times 50}{E_2}$

x = ppm of the hardness causing salt
E_2 = Equivalent mass of the salt

TOPIC GRIP

Subjective Questions

1. A hydrated salt, $MCl_2 \cdot 2H_2O$, has 24.9% by weight of water of hydration. Calculate the % by weight of chlorine in it. At weight of chlorine = 35.46.
2. A certain oxide of iron has 72.4% by weight of iron in it (at. weight = 56). If it is regarded as a mixed oxide of FeO and Fe_2O_3, what is the mole ratio of these two compounds in it?
3. One volume of a certain gaseous oxide of carbon requires exactly two volumes of oxygen for oxidation. The product which is a gas is completely absorbed by KOH. Suggest a formula for the oxide.
4. Excess of potassium iodide (in presence of dilute HCl) is added to 0.01 mole of pure Barium chromate suspended in water. Calculate the weight of Iodine (in g) liberated. Write the relevant equations (At. wt of Iodine = 127).
5. In an experiment on the conversion of Oxygen to ozone in an ozonizer, it was observed that 50 ml of O_2 gas yielded 46 ml of ozonized oxygen. If 20 ml of the ozonised oxygen are exposed to an oil like turpentine which can absorb ozone. Find out what may be the residual volume of oxygen. All volume measurements are made at the same temperature and pressure.
6. 1.58 g of a mixture of NaI and KI, were dissolved in water and treated with excess of $AgNO_3$ aq. The total weight of the precipitate (AgI) was 2.35 g. Calculate the weight % of NaI in the original mixture. Atomic weights: Na = 23, K = 39, I = 127, Ag = 108.
7. A mixture of $NaNO_3$ and NaCl is heated till all the $NaNO_3$ in it is converted into $NaNO_2$. If there is a loss of 0.114 g per gram of the original mixture. Determine the weight % of NaCl in it.
8. A mixture of NaBr and $NaBrO_3$ in the mole ratio 2 : 3 is treated with excess of dilute acid. State (i) the componding (reactant) which is limiting and (ii) the number of moles of Bromine liberated. [Consider that actually 2 moles of NaBr and 3 moles of $NaBrO_3$ are present]
9. Four compounds of an element respectively contain 90.3%, 26.4% and 22.8% of the element. The correspoinding vapour densities of these compounds are 31,85.5, 53 and 184. Suggest a probable atomic weight of the element.
10. One litre of a sample of H_2O_2 in solution yields 8.4 litres of (dry) oxygen gas at STP. Calculate the molarity of the H_2O_2 solution.

Straight Objective Type Questions

Directions: **This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

11. 30 g of Nitric oxide were passed slowly over heated copper; if the increase in weight of copper is 10 g. Calculate the weight of NO remaining undecomposed.
    (a) 15.12 g (b) 12.15 g (c) 15.78 g (d) 11.25 g
12. One gram of the oxide of a certain metal is obtained by heating strongly 2.1 g of the carbonate of the metal. The equivalent weight of the metal is
    (a) 16 (b) 12 (c) 10 (d) 24

13. 1.38 g of a hydrated Cu(II) salt of an acid yields on strong heating 0.457 g of CuO. The salt contains 3 molecules of water of hydration per atom of Copper. Calculate the equivalent weight of the acid. Atomic weight of Cu = 63.6
(a) 62.38 (b) 68.32 (c) 38.62 (d) 36.28

14. The aqueous solution of the sulphate of a certain element was treated with excess of Barium chloride solution and yielded 2.05 g of $BaSO_4$ per gram of the sulphate. If the chloride of the element has vapour density ~ 67, suggest the atomic weight of the element [Atomic weights: Ba = 137.4, S = 32, Cl = 35.5]
(a) 18.3 (b) 21.7 (c) 26.8 (d) 16.3

15. An element X forms an acidic oxide which reacts with KOH to form a salt, isomorphous with potassium sulphate (i.e., same crystalline form). Given that the atomic weight of the element is ≃ 79. Calculate the equivalent weight of the element.
(a) 13.17 (b) 17.13 (c) 11.73 (d) 15.1



Assertion–Reason Type Questions

Directions: **Each question contains Statement-1 and Statement-2 and has the following choices (a), (b), (c)and (d), out of which ONLY ONE is correct.**

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False
(d) Statement-1 is False, Statement-2 is True

16. Statement 1
When white phosphorus is heated with NaOH both PH_3 and NaH_2PO_2 are obtained as products.
and

Statement 2
In such reactions eg: Cl_2 + NaOH or S + NaOH etc. the same element suffers both an increase and a decrease of its oxidation state/number.

17. Statement 1
The equivalent weight of H_3PO_3 as an acid is one third of its molecular weight.
and

Statement 2
Equivalent weight of an acid = $\frac{\text{Molecular weight}}{\text{n}}$ where, n is the number of replaceable hydrogen atoms in a molecule of the acid.

18. Statement 1
One gram of tin (atomic weight = 118.7) reduces 337 ml of 0.1 N $K_2Cr_2O_7$ solution in the presence of acid.
and

Statement 2
Oxidation state of tin goes from 0 to 4 in the reaction.

19. Statement 1
In the presence of NaCl and con H_2SO_4, K_2CrO_4 forms CrO_2Cl_2.
and

Statement 2
In this reaction, the Cl^- ion reduces the oxidation state of chromium from 6 to 3.

20. Statement 1

H_2O_2 behaves in different reactions as an oxidizing agent and a reducing agent.

and

Statement 2

The volume strength of a solution of H_2O_2 is the number of litres of O_2 gas at STP obtained by decomposition from one litre of aqueous H_2O_2 solution.

Linked Comprehension Type Questions

Directions: **This section contains 2 paragraphs. Based upon the paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

Passage I

In a reaction where an electron transfer takes place from one species (atom, molecule, ion etc.) to another, the species which gives electrons is termed "reducing agent" and gets oxidized. The species which accepts electrons is called an "oxidizing agent" and gets reduced in the reaction. Each species may be assigned a "so-called" oxidation number, according to certain rules. Increase in this number is oxidation and decrease is reduction.

Thus,

Oxidation number change: negative → zero → positive value

This is oxidation. The changes indicated by reversed arrows is reduction. Equivalent weight of the species

$$= \frac{\text{Mol.weight}}{\text{Change in oxidation number}}$$

21. When $KMnO_4$ reacts with $FeSO_4$ in dilute H_2SO_4 medium it forms $MnSO_4$. What is the decrease in oxidation number for Mn in this reaction?

(a) 5 (b) 3 (c) 6 (d) 2

22. Indicate the equivalent weight of $KMnO_4$ in the above reaction [Atomic weight K = 39, Mn = 55]

(a) 63.2 (b) 31.6

(c) 15.8 (d) 18.5

23. If in the above reaction, instead of $FeSO_4$, Ferrous ammonium sulphate (Mohr's salt) $FeSO_4(NH_4)_2SO_4.6H_2O$ is used, what is its equivalent weight?

(a) 196 (b) 131 (c) 392 (d) 98

Passage II

In the balancing of chemical equations and in stoichiometric calculations, one basic fact is of central importance i.e., one equivalent weight of any species always interacts with one equivalent weight of another species. This presupposes that in any reaction undergone by the species; its equivalent weight must be clearly defined, quite often, in relation to its atomic/molar mass. Depending on the reaction, a chemical species may have different equivalent weights.

24. The equivalent weight of an element is 12.16. This implies that 1.54 g of the oxide of the element when converted into the chloride of the element by a reaction sequence yields the product of weight

(a) 4.36 (b) 4.63 (c) 3.64 (d) 6.34

25. One gram of a metallic chlorate on heating gave 0.685 g of the corresponding chloride. The equivalent weight of the element is

(a) 86.3 (b) 68.89 (c) 63.8 (d) 43.2

26. 2 g of a metallic chloride was quantitatively converted into the corresponding oxide which weighed 0.8 g. The equivalent weight of the element is

(a) 10.33 (b) 13.10 (c) 20.73 (d) 11.39

Multiple Correct Objective Type Questions

Directions: **Each question in this section has four suggested answers out of which ONE OR MORE answers will be correct.**

27. Number of molecules in 7g of nitrogen is the same as the number of

(a) molecules in 4.257 g ammonia' (b) atoms in 5.33 g SO_2

(c) electrons in 0.5 g calcium (d) H_3O^+ ions in 250 ml of 1M H_2SO_4

28. When a solution of KIO_3 is heated with excess of oxalic acid

(a) 3 moles of oxalic acid are consumed per mole of KIO_3 (b) The equivalent weight of KIO_3 is 42.8

(c) In the reaction KIO_3 is reduced to KI (d) 6 moles of CO_2 are produced per mole of KIO_3

[Given atomic weight of Iodine = 127]

29. When hydroxylamine, NH_2OH is boiled with Fe(III) sulphate in H_2SO_4 medium

(a) Hydroxylamine is oxidized to Nitrogen

(b) Both the reactants NH_2OH and Fe^{3+} react in equimolar amounts

(c) The equivalent weight of $Fe_2(SO_4)_3$ is half its molar mass

(d) The SO_4^{2-} ion acts as a catalyst for the reaction

Matrix-Match Type Question

Directions: **Match the elements of Column I to elements of Column II. There can be single or multiple matches.**

30. Match the reactions on the left with the types of reactions on the right.

Column I	**Column II**
(a) $Cl_2 + 2OH^- \rightarrow Cl^- + OCl^- + H_2O$	(p) Redox reaction
(b) $P_4 + 5O_2 \rightarrow P_4O_{10}$	(q) Reaction in acid medium
(c) $MnO_4^- + I^- \rightarrow MnO_2 + IO_3^-$	(r) Disproportionation
(d) $MnO_4^- + Fe^{2+} \rightarrow Mn^{2+} + Fe^{3+}$	(s) Reaction in alkaline medium

IIT ASSIGNMENT EXERCISE

Straight Objective Type Questions

Directions: **This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

31. Nitric oxide combines with oxygen producing nitrogen dioxide. If 50 mL each of nitric oxide and oxygen is mixed, the volume of the gas mixture after reaction is
(a) 100 mL (b) 75 mL (c) 60 mL (d) 50 mL

32. 15 ml of a mixture of C_2H_4 and CH_4 after oxidation with O_2 gave 20 ml of CO_2 gas. Express the composition of the mixture in volume measure. All volume measurements are at the same temperature and pressure.
(a) 10 ml C_2H_4, 5 ml CH_4 (b) 7.5 ml C_2H_4, 7.5 ml CH_4
(c) 5 ml C_2H_4, 10 ml CH_4 (d) 9 ml C_2H_4, 6 ml CH_4

33. 10 ml of a gaseous hydrocarbon were exploded with 70 ml (excess) of oxygen. After explosion the residual gases had a volume of 50 ml which was reduced to 20 ml on exposure to alkali. Identify the hydrocarbon. (All volume measurements were made at STP)
(a) C_2H_4 (b) C_4H_8 (c) C_3H_8 (d) C_4H_{10}

34. A certain metallic carbonate is isomorphous with $MgCO_3$. Given that the atomic weight of the element is ~ 137.4, what is the weight % of the element in the oxalate of the element?
(a) ~ 72 (b) ~ 61 (c) ~ 55 (d) ~ 47

35. The total number of electrons in 3.2 g of oxygen molecule is
(a) 6.022×10^{23} (b) 9.64×10^{23} (c) 6.022×10^{22} (d) 9.64×10^{22}

36. A certain compound containing carbon, hydrogen and iron contains ~ 30% by weight of iron in it. Suggest the minimum molecular weight of the compound. [Atomic weight of Fe = 56]
(a) 168 (b) 195 (c) 159 (d) 187

37. One molecule of a compound contains six carbon atoms, 2×10^{-23} g of hydrogen and 16×10^{-23} g of other atoms. The gram-molecular mass of the compound is?
(a) 180 g (b) 174 g (c) 138 g (d) 114 g

38. 'x' g of pure silver was completely dissolved in dil.HNO_3. The solution was then treated with excess of NaCl and the precipitated AgCl was observed to be 2.87 g. The amount of silver 'x' dissolved initially in nitric acid was:
(a) 1.08 g (b) 2.16 g (c) 2.7 g (d) 1.62 g

39. Atomic weights of two elements (A) and (B) are 32 and 16 respectively. They combine to give two gases AB_2 and AB_4 and the mixture is found to have a vapour density of 38.4. The number of moles of AB_2 in 50 moles of the mixture is
(a) 10 (b) 20 (c) 30 (d) 40

40. A certain hydrate $M.nH_2O$ has 19% by weight of H_2O in it. The molar mass of M = 230 g mol^{-1}. Calculate the value of n
(a) 2 (b) 6 (c) 3 (d) 4

41. A typical pyrex glass contains 12.9% of B_2O_3 and 80.7% of SiO_2 (by weight) besides other oxides (Al_2O_3, Na_2O, K_2O) what is the weight ratio of Silicon to Boron in the glass (Atomic weights Si = 28.1, B = 10.8)
(a) 9.32 : 1 (b) 9.76 : 1 (c) 9.29 : 1 (d) 9.43 : 1

42. Determine the simplest formula of a compound with the following composition: Cr = 26.52%, S = 24.52%, O = 48.96%. Atomic weights: Cr = 52, S = 32

(a) $Cr_2S_3O_8$ (b) $Cr_2S_5O_{16}$ (c) $Cr_2S_3O_{12}$ (d) $Cr_2S_5O_{20}$

43. A non-stoichiometric oxide of Manganese, MnO_x contains 63.7% of Manganese. Atomic weight of Mn = 55. Calculate the value of x

(a) 2 (b) 1 (c) 1.5 (d) 3

44. 50 g of zinc blende (ZnS) containing only silica as impurity is roasted to convert all the ZnS in it into ZnO. This results in decrease in mass by 6 g due to roasting. The weight of the impurity in the sample is [Atomic weight of Zn = 65.4]

(a) 15.3 g (b) 21.0 g (c) 13.5 g (d) 10.7 g

45. A sample of hard water is found to contain 40 mg of Ca^{2+} ions per litre. Amount of washing soda required to soften 4 L of the sample is

(a) 4.24 g (b) 0.424 g (c) 0.0424 g (d) 42.4 g

46. 10 g of an amine hydrochloride, B.HCl is dissolved in a litre (of the solution in water). 50 ml of the solution reacts exactly with 52.4 ml of 0.1 M $AgNO_3$ solution. Calculate the molecular weight of the amine (g mol^{-1}).

(a) 42 (b) 65 (c) 59 (d) 35

47. 15 g of a metal (A) (purity = 95%) was heated with excess oxygen to give its oxide AO_2. If the weight of the oxide collected (assuming no loss) was 17.45 g, what is the atomic weight of the metal?

(a) 14.25 g (b) 142.5 kg (c) 0.1425 kg (d) 1.425 kg

48. Carbon monoxide was passed over heated copper oxide which suffered a loss of weight of 0.5318 g. If the gaseous products are exposed to excess of NaOH, what weight of Na_2CO_3 would be formed?

(a) 3.523 g (b) 2.335 g (c) 3.211 g (d) 2.721 g

49. When $Fe_2(SO_4)_3$ solution reacts with $K_4Fe(CN)_6$ solution, what product is obtained besides K_2SO_4?

(a) $Fe[Fe(CN)_6]_3$ (b) $Fe_2[Fe(CN)_6]$

(c) $K_2Fe[Fe(CN)_6]$ (d) $Fe_3[Fe(CN)_6]_2$

50. In the thermite process, Aluminium is used to reduce metallic oxides to yield metals. Consider the reduction of MnO by Al. If 110 g of Al and 200 g of MnO are used in an experiment, what are the products? Atomic weights: Al = 27, Mn = 55

(a) Al_2O_3 and Mn (b) Al_2O_3, Mn and unreacted MnO

(c) Al_2O_3, Mn and excess Al (d) Al_2O_3, Mn, unreacted Al and unreacted MnO

51. A mixture of gold and copper contains Cu : Au = 2 : 3 in weight ratio. If by combination the compound alloy Cu_3Au is formed, which component is present in excess and what weight of the alloy is Cu_3Au.? Assume a total weight of 100 g of the mixture

Atomic weights: Cu = 63.5, Au = 197.

(a) gold, 91.4 g (b) gold, 81.4 g (c) gold, 71.4 g (d) gold, 61.4 g

52. PbO_2 oxidizes MnO in presence of HNO_3. The products are $HMnO_4$, $Pb(NO_3)_2$ and H_2O. Calculate the number of moles of HNO_3 and PbO_2 consumed per mole of MnO.

(a) 5, 2.5 (b) 6, 3 (c) 4, 2 (d) 5, 3

53. An element forms two oxides containing respectively 24.24% and 34.78% by weight of oxygen. Determine the ratio of the valencies of the element in these two oxides.

(a) 1 : 2 (b) 2 : 3 (c) 1 : 3 (d) 3 : 5

54. 1.5 g of anhydrous cupric salt of an organic acid yields on thermal decomposition 0.392 g of CuO. Calculate the equivalent weight of the acid. Atomic weight of Cu = 63.6

(a) 92.5 (b) 151.5 (c) 101.5 (d) 121.5

55. 0.3660 g of a metallic sulphide was roasted in air to yield the oxide of the metal along with $SO_{2(g)}$ which was passed into Bromine water. The solution was boiled, cooled and treated with Barium chloride solution. 0.875 g of $BaSO_4$ was obtained. Calculate the equivalent weight of the metal. Atomic weight of Ba = 137.4

(a) 23.8 (b) 28.3 (c) ~ 32.8 (d) 38.2

56. 0.431 g of the anhydrous carbonate of a metal evolved 96.5 ml of CO_2 measured at STP (when treated with excess HCl). What is the equivalent weight of the metal?

(a) 30 (b) 35 (c) 25 (d) 20

57. 9.8 g of ferrous ammonium sulphate reacts with 30 mL 0.1 M $KMnO_4$ in acidic medium completely. The percentage purity of the sample is

(a) 26% (b) 12% (c) 60% (d) 80

58. Cr_2O_3 is oxidized by $K_3Fe(CN)_6$ in presence of KOH. The products are K_2CrO_4, $K_4Fe(CN)_6$ and H_2O. How many moles of $[Fe(CN)_6]^{3-}$ are consumed per mole of Cr_2O_3

(a) 6 (b) 4 (c) 2 (d) 8

59. Solid KOH (300 g) which had been inadvertently exposed to moist air is used to prepare 20 litres of 0.25 N solution. Calculate the weight % of the impurities. Atomic weight K = 39

(a) 7.92 (b) 6.67 (c) 5.81 (d) 4.33

60. In the reduction experiment of $KMnO_4$ to MnO_2, the solution has a normality = 1.56. Calculate the molarity.

(a) 0.52 (b) 0.78 (c) 1.04 (d) 0.65

61. 20 ml of 0.1 M potassium nitrate solution and 30 ml of 0.15 M potassium chloride solution are mixed. Calculate the volume in ml of potassium bromide solution (0.5 M) which should be added to make the concentration of K^+ ion equal to 0.3 M

(a) 45.2 (b) 42.5 (c) 24.5 (d) 52.4

62. 11 g of a sample of Zn with a purity of 87% is dissolved by V ml of hydrochloric acid with a density of 1.18 g ml^{-1} containing 35% of HCl by mass. Calculate V ml, Atomic weight of Zn = 65.4.

(a) 25.9 ml (b) 29.5 ml (c) 15.9 ml (d) 20.4 ml

63. A solution of sucrose ($C_{12}H_{22}O_{11}$) contains 13.5 g in 100 ml of the solution. The density = 1.05 kg $litre^{-1}$. Calculate the molality of the solution.

(a) ~ 0.34 (b) ~ 0.43 (c) 0.49 (d) 0.51

64. A certain solution of a non-volatile solute in a solvent has molality m, molarity M and density, d in g ml^{-1}. If the molar mass of the solute is W g mol^{-1}, the relation among d, m, M and W is

(a) $d = \frac{M}{1000}\left(\frac{1}{W}+\frac{1}{m}\right)$ (b) $d = m\left(\frac{1}{M}+\frac{1}{W}\right)$ (c) $d = \frac{mM}{W} \times 1000$ (d) $d = M\left(\frac{W}{1000}+\frac{1}{m}\right)$

65. How many grams of H_2O_2 are present in 250 ml of a 10 volume Hydrogen peroxide solution?

(a) ~7.6 g (b) 8.4 g (c) 5.8 g (d) 6.7 g

66. In what volume ratio should 8 N HCl and 2 N HCl be mixed to yield 5.5 N HCl?

(a) 7 : 3 (b) 5 : 2 (c) 7 : 5 (d) 3 : 1

67. Concentrated H_2SO_4 of density 1.83 g ml^{-1} has 93% of H_2SO_4 by weight. How many ml of this acid should be taken to yield (after addition of water) 200 ml of 2 N acid?

(a) 12.15 ml (b) 15.21 ml (c) 11.52 ml (d) 15.73 ml

68. 7.00 g of a mixture of NaCl and NaBr were dissolved in water and treated with excess of $AgNO_3$ solution. The total precipitate of AgCl and AgBr was found on further analysis to contain 11 g of metallic silver. Determine the composition of the original mixture. Atomic weights: Na = 23, Ag = 108, Br = 80, Cl = 35.5

(a) 55.65% NaCl, 44.35% NaBr (b) 60.56% NaCl, 39.44% NaBr

(c) 50.65% NaCl, 49.35% NaBr (d) 65.56% NaCl, 34.44% NaBr

69. 22 ml of a solution of NaCN of molarity 0.327 M reacts exactly with 16 ml of x M $AgNO_3$ solution. Calculate x.
(a) ~ 0.252 (b) ~ 0.225 (c) 0.202 (d) 0.113

70. A sample (0.518 g) of limestone, dissolved and the calcium in it is precipitated as CaC_2O_4. This after filtering, washing and dissolving in dil. H_2SO_4 needed 40 ml of 0.250 N $KMnO_4$ solution. Calculate % of CaO in the limestone sample.
(a) ~ 44% (b) ~54% (c) ~61% (d) ~ 35%

71. 36 g of Aluminium metal are added to 2 litres of 1.5 M $CuSO_4$ solution. How many gram of copper would be displaced? Atomic weights of Al = 27, Cu = 63.6
(a) 127.2 g (b) 172.2 g (c) 110.2 g (d) 142.2 g

72. The reaction $NH_4Cl + NaOH \rightarrow NaCl + NH_3 + H_2O$ is an example of
(a) acid-base reaction (b) rearrangement (c) oxidation-reduction (d) disproportionation

73. In an experiment of maximal neutralization of an acid by an alkali, a certain oxyacid of phosphorus reacted with NaOH in the weight ratio, acid: alkali = 1.025 g: 1.000 g. Atomic weight of P = 31. The acid is
(a) HPO_3 (b) H_3PO_3 (c) H_3PO_2 (d) H_3PO_4

74. 0.125 g of a solid acid of equivalent weight = 50.0 is neutralized by 80 ml of NaOH. Calculate the weight in gram of NaOH per litre. Atomic weight of Na = 23
(a) 1.50 g (b) 1.00 g (c) 0.75 g (d) 1.25 g

75. By the total hydrolysis of 1.635 g of the chloride of an element, HCl is formed along with the insoluble oxide of the element. The acid thus obtained needed 46.2 ml of 0.56 N NaOH for neutralization. Calculate the equivalent weight of the element.
(a) 24.7 (b) 27.7 (c) 30.4 (d) 20.6

76. A sample of weight 1.632 g, of a metallic carbonate was dissolved in dil H_2SO_4 and the solution was evaporated to crystallization. The crystalline hydrated salt thus obtained was isomorphous with $FeSO_4.7H_2O$. The atomic weight of the metal was ≃ 66. What weight of metallic oxide would be obtained if the sample is strongly heated?
(a) 1.16 g (b) ~1.06 g (c) 1.26 g (d) 0.96 g

77. $NaHSO_3$ reduces $NaIO_3$ to I_2. The products are Na_2SO_4, $NaHSO_4$, I_2 and H_2O. Calculate the weight of $NaHSO_3$ in kg consumed per mole of Iodine formed. Atomic weights of Iodine = 127, S = 32, Na = 23
(a) 0.640 (b) 0.420 (c) 0.720 (d) 0.520

78. SO_2 may be used for production of CS_2 by using its reaction with coke. The products are CO and CS_2. Using one kg of SO_2 with an efficiency of conversion of 80%, calculate the weight of CS_2 produced in grams.
(a) 475 g (b) 525 g (c) 575 g (d) 425 g

79. If equal masses of oxygen and phosphorus react to yield P_4O_{10} and P_4O_6, assuming that the reactants are entirely consumed, calculate approximately the mole ratio of the products P_4O_{10} and P_4O_6. Atomic weight of P = 31
(a) 1 : 2 (b) 2 : 3 (c) 1 : 1 (d) 3 : 5

80. 45 ml of 2 M NaOH and 30 ml H_2SO_4 solution were mixed. A further addition of 12.9 ml of HCl (0.5 M) was required to bring the mixture to the point of neutralization. Calculate the molarity of the H_2SO_4 solution.
(a) 1.932 M (b) 1.239 M (c) 1.392 M (d) 1.229 M

81. A certain ore of metallic oxide contained 15% of moisture, 15% silica and the rest metallic oxide. After partial drying it contained 5% of moisture. What is the % silica in the partially dried sample?
(a) 14.36 (b) 19.26 (c) 16.67 (d) 18.26

82. How many Nitrogen atoms are present in 1 g of $K_2Zn_3[Fe(CN)_6]_2$. Relevant atomic weights are K = 39, Zn = 65.4, Fe = 56
(a) ~ 1.5×10^{21} (b) ~2.00×10^{21} (c) ~1.00×10^{22} (d) ~5×10^{20}

83. The weight of heavy water containing 5×10^5 "D" atoms is
(a) 4.15×10^{-20}g (b) 4.15×10^{-18}g (c) 7.47×10^{-18}g (d) 8.3×10^{-18}g

84. Chlorophyll contains 2% by mass of Mg. The number of Mg atoms in 2 g chlorophyll is
(a) 10^{21} (b) 10^{23} (c) 10^{24} (d) 10^{25}

85. The hydrated chloride of a divalent metal loses 14.73% (by weight) of its mass by dehydration. If one mole of the hydrated salt contains 2 moles of water of hydration, calculate the weight % of chlorine in the hydrated salt. [Atomic weight: Cl = 35.46]
(a) 33% (b) 29% (c) 24% (d) 36%

86. What is the mean molar mass of a mixture (in the weight ratio 3 : 2) of solid $CaCO_3$ and solid $MgCO_3$ (Atomic weights: Ca = 40, Mg = 24]
(a) 78 (b) 81 (c) 85 (d) ~93

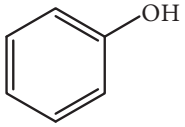
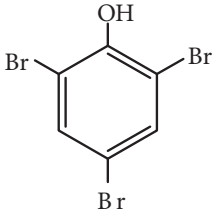
87. One molecule of a compound contains 4 atoms of carbon, 4 atoms of hydrogen, 4 atoms of oxygen and 1.03×10^{-22} g of other elements. The molar mass of the compound in g mol^{-1} would be:
(a) 110.3 (b) 120.6 (c) 162 (d) 178

88. A certain compound containing only iron, oxygen and carbon has very nearly 39% by weight of Fe in it. Suggest a possible formula for it.
(a) $FeCO_3$ (b) $Fe(CO)_5$ (c) $FeO.FeCO_3$ (d) FeC_2O_4

89. A certain compound of Iridium has 39.8% of Iridium by weight in it. If one mole of the compound contains 6 chlorine atoms in it, what are (i) minimum molar mass of the compound and (ii) weight % of chlorine in it? [Atomic weight of Ir = 192.22]
(a) 453, 48% (b) ~483, 44% (c) 453, 41% (d) 435, 41%

90. As_2S_5 is oxidized by con HNO_3 to yield H_3AsO_4, H_2SO_4 and NO_2. Calculate the weight ratio of the reactants. [Atomic weights: As = 75, S = 32]
(a) 1 : 7.6 (b) ~1 : 8.1 (c) 1 : 9.1 (d) 1 : 6.6

91. NaBr and $NaBrO_3$ react in presence of acid to liberate Br_2. This is used to brominate. Calculate the weight of NaBr

phenol,  to yield  required to brominate ultimately 1 g of phenol. (Atomic

weight of Na = 23 and Br = 80)
(a) 4.58 g (b) 5.48 g (c) 4.85 g (d) 8.45 g

92. A mixture of 2 g each of $MgSO_4$ and $CuSO_4$ (anhydrous) is dissolved in water and treated with excess of Barium chloride solution. Calculate the weight of $BaSO_4$ precipitated. [Atomic weights Mg = 24, Cu = 63.6, Ba = 137.4, S = 32]
(a) ~6.82 g (b) ~8.26 g (c) ~8.62 g (d) 6.65 g

93. 32.46 ml of Nitrogen measured at STP is released when 0.1 g of a primary amine is acted upon by excess of nitrous acid. $R\text{-}NH_2 + HNO_2 \rightarrow R\text{-}OH + N_2 + H_2O$. Calculate the molar mass of the amine
(a) 55 (b) 69 (c) 83 (d) 29

94. In Zeisel's method of estimation of methoxy groups (i.e., $-OCH_3$) in an organic compound, the compound $R\text{-}OCH_3$ is heated with constant boiling HI: $R\text{-}OCH_3 + HI \rightarrow R\text{-}OH + CH_3I$. Reaction with alcoholic $AgNO_3$ precipitates AgI which is filtered, washed, dried and weighed.

An organic methoxy compound (Molar mass = 168 g mol^{-1}) gave ~4.2 g of silver iodide per gram of the compound. Molar mass of AgI = 235 g mol^{-1}. Calculate the number of methoxy groups per molecule of the compound.

(a) 3 (b) 5 (c) 2 (d) 1

95. 3.14 g of silver carbonate on being heated strongly yields a residue weighing

(a) 2.16 g (b) 2.45 g (c) 2.3 g (d) 2.64 g

96. A solution of KIO_3 is boiled with excess of oxalic acid; the products are $K_2C_2O_4$, CO_2, H_2O and I_2. Calculate the quantity of oxalic acid in grams used up in the formation of one mole of Iodine.

(a) 540 g (b) 450 g (c) 590 g (d) 420 g

97. Given the equation $2FeSO_4 + K_2S_2O_8 \rightarrow Fe_2(SO_4)_3 + K_2SO_4$. Calculate the weight of $BaSO_4$ in gram when 50 ml of 0.1 N $FeSO_4$ solution is heated with excess of $K_2S_2O_8$ and at the completion of the reaction, cooled and treated with excess of Barium Chloride solution. Molar mass of $BaSO_4$ = 233.4 g mol^{-1}

(a) ~3.52 g (b) ~2.33 g (c) ~3.25 g (d) ~3.38 g

98. Consider the reaction of oxidation of I^- to I_2 by Na_2TeO_3 in presence of HCl to yield Te. Determine the mole ratio Na_2TeO_3 : NaI : HCl in the balanced equation

(a) 1 : 3 : 6 (b) 1 : 4 : 7 (c) 1 : 4 : 6 (d) 1 : 3 : 5

99. Hydroxylamine reduces Ferric Sulphate solution getting oxidized to N_2O. In what mole ratio does NH_2OH and $Fe_2(SO_4)_3$ react?

(a) 2 : 1 (b) 1 : 2 (c) 2 : 3 (d) 1 : 1

100. A certain volume of Ferric sulphate solution is reduced by zinc. The resulting solution could be reoxidized by V_1 ml of 0.1 N $KMnO_4$ solution. When the same volume of Ferric sulphate solution is reduced by another metal, M the resulting solution required V_2 ml of 0.1 N $KMnO_4$ solution for reoxidation. If $V_1 : V_2 = 1 : 1.5$ determine the valencies of the metals Zn and M.

(a) 2, 4 (b) 2, 5 (c) 2,3 (d) 2,6

101. An element X forms two oxides X_2O_3 and X_2O_5. If 0.2 g of X_2O_3 dissolved in H_2SO_4 consumes 43 ml of $KMnO_4$ solution containing 5 g of $KMnO_4$ per litre. Calculate the atomic weight of the element. [Equivalent weight of $KMnO_4$ = 31.6]

(a) ~34.8 (b) 48.3 (c) 43.8 (d) 38.4

102. The molarity and molality of a certain solution of acetic acid in water are 2.03 and 2.27 respectively. Calculate the density.

(a) 1.171 g ml^{-1} (b) 1.017 g ml^{-1} (c) 1.117 g ml^{-1} (d) 1.211 g ml^{-1}

103. Calculate the weight % of O_3 in a sample of ozonized oxygen given that 18.8 ml of 0.1 N thiosulphate solution is consumed by the Iodine liberated from KI in acid medium by 250 ml of the sample at STP.

(a) 12.15 (b) 15.12 (c) 21.51 (d) 25.11

104. Calculate the molar mass of a mono-acid organic base given that 0.334 g of the hydrochloride of the base gave 0.300 g of AgCl (precipitate) with excess of $AgNO_3$ solution.

(a) 132 g mol^{-1} (b) 117 g mol^{-1} (c) 109 g mol^{-1} (d) 123 g mol^{-1}

105. Calculate the volume in ml of 0.1 M $KMnO_4$ in acid medium needed to oxidize one g of FeC_2O_4. [Atomic weight: Fe = 56]

(a) 41.7 ml (b) 47.1 ml (c) 23.6 ml (d) 36.2 ml

106. 0.4 g of silver powder was boiled with an excess of $Fe_2(SO_4)_3$ solution. The products are Ag_2SO_4 and $FeSO_4$. Calculate the volume of 0.1 N $KMnO_4$ in acid medium needed to reoxidise the $FeSO_4$ formed.

(a) ~32ml (b) ~29 ml (c) ~41 ml (d) ~37 ml

107. 2.55 g of saturated solution of $CuSO_4.5H_2O$ was dissolved in water and made up to 200 ml, 50 ml of this solution treated with excess of KI solution liberated Iodine which reacted exactly with 7 ml of 0.092 N sodium thiosulphate solution. Express the solubility of $CuSO_4.5H_2O$ in gram per 100 g of water.

(a) 33.79 (b) 39.73 (c) 30.37 (d) 43.71

108. A mixture of oxalic and sulphuric acid gave the following titrimetric results.

(i) 25 ml of the solution ≡ 16.25 ml of 0.2 N NaOH solution

(ii) 25 ml of the solution warmed ≡17.50 ml of 0.1 N $KMnO_4$ solution in presence of acid. Calculate the concentration of sulphuric acid in g $litre^{-1}$

(a) 5.88 (b) 2.94 (c) 3.92 (d) 1.96

109. 0.2 g of a metal (atomic weight = 51) liberated 87.8 ml of H_2 gas at STP by reaction with dil H_2SO_4. The resulting solution had reducing properties and consumed 58.8 ml of 0.2 N $KMnO_4$. The metal is obviously polyvalent. Calculate the valencies exhibited by the metal.

(a) 2, 3 (b) 3,5 (c) 2,4 (d) 2, 5

110. 50 mL of 0.001 M phosphoric acid are exactly neutralized by the addition of 0.005 N NaOH. The molarity of the resulting salt solution is

(a) 6.25×10^{-4} (b) 6.25×10^{-5} (c) 5×10^{-5} (d) 6.25×10^{-7}



Assertion–Reason Type Questions

Directions: **Each question contains Statement-1 and Statement-2 and has the following choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False
(d) Statement-1 is False, Statement-2 is True

111. Statement 1

The equivalent weight of potassium chromate as an oxidizing agent in acid medium is one third of its molar mass.

and

Statement 2

In acid medium chromate changes to dichromate $2CrO_4^{2-} + 2H^+ \rightarrow Cr_2O_7^{2-} + H_2O$.

112. Statement 1

Molality and molarity of a solution are unaffected by change in temperature.

and

Statement 2

Both molality and molarity involve the number of moles of solutes in solution and the number of moles cannot change with change in temperature.

113. Statement 1

H_2O_2 in solution is oxidized by $KMnO_4$ in presence of dil H_2SO_4.

and

Statement 2

This fact is shown by the equation

$2KMnO_4 + 3H_2SO_4 + 3H_2O_2 \rightarrow K_2SO_4 + 2MnSO_4 + 6H_2O + 4O_2$

Linked Comprehension Type Questions

Directions: **This section contains 1 paragraph. Based upon the paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

Passage I

Stoichiometric calculations can be carried out with great facility on the basis of one principle. i.e., when substances interact chemically, one equivalent of any (chemical) compound/element always reacts with only one equivalent of another compound/element. The valency (a whole number) of an element gives the number of equivalents contained in atomic weight of the element. For elements exhibiting variable valency, the equivalent weight also varies depending on the valency. The equivalent weight of an acid is defined as the ratio of its molar mass to the number of replaceable hydrogen atoms in a molecule. The equivalent weight of a base is the ratio of its molar mass to the number of equivalents of an acid neutralized by one mole of the base. The equivalent of an oxidising or a reducing agent is the ratio of its molar mass to the change of oxidation number of the substance in any particular reaction. This obviously depends on the nature of the reaction.

114. In the reaction $Zn_3Sb_2 + 6H_2O \rightarrow 3Zn(OH)_2 + 2SbH_3$, indicate the element which suffers a change of oxidation state.
(a) Zn (b) Sb (c) both Zn and Sb (d) None

115. Indicate among the following the reaction where the same chemical species suffers both oxidation and reduction.
(a) $I_2 + 5Cl_2 + 6H_2O \rightarrow 2HIO_3 + 10HCl$
(b) $P_4 + 3NaOH + 3H_2O \rightarrow 3NaH_2PO_2 + PH_3$
(c) $I_2 + 2Na_2S_2O_3 \rightarrow 2NaI + Na_2S_4O_6$
(d) $3Cu + 8HNO_3 \rightarrow 3Cu(NO_3)_2 + 4H_2O + 2NO$

116. Indicate among the following the species which reacts as an acid with its molar mass equal to its equivalent weight.
(a) $H_2C_2O_4$
(b) H_3PO_3
(c) H_3PO_2
(d) $C_4H_4O_4$(an organic acid)

Multiple Correct Objective Type Questions

Directions: **Each question in this section has four suggested answers out of which ONE OR MORE answers will be correct.**

117. Indicate the correctly formulated equations
(a) $Fe(CO)_5 + 4NaOH \rightarrow Na_2Fe(CO)_4 + Na_2CO_3 + 2H_2O$
(b) $3HCl\ (conc.) + HNO_3\ (con.) \rightarrow 2H_2O + NOCl + Cl_2$
(c) $3Zn + 8HNO_3(dil.) \rightarrow 3Zn(NO_3)_2 + 4H_2O + 2NO$
(d) $2KMnO_4 + 3H_2SO_4 + 7H_2O_2 \rightarrow K_2SO_4 + 2MnSO_4 + 10H_2O + 6O_2$

118. Indicate among the following the correct alternatives. A mixture of 20 ml of 0.2 M NaOH and 15 ml of 0.1 M $Ba(OH)_2$ has in it the same number of equivalents of acid (or base) as
(a) 14 ml of 0.5 M HCl
(b) 14 ml of 0.5 M H_2SO_4
(c) 56 ml of 0.125 M KOH
(d) 28 ml of 0.25 M $Ba(OH)_2$

119. Consider the (unbalanced) equation
$K_4Fe(CN)_6 + (n_1)H_2SO_4 + (n_2)H_2O \rightarrow$ Products
The products are $FeSO_4$, K_2SO_4, $(NH_4)_2SO_4$ and CO
By balancing the equation determine the values of n_1 and n_2
(a) $n_1 < n_2$ (b) $n_1 = 6$ (c) $n_1 > n_2$ (d) $n_2 = 6$



Matrix-Match Type Question

Directions: **Match the elements of Column I to elements of Column II. There can be single or multiple matches.**

120. Match the units of concentration on the left with statements on the right

Column I	**Column II**
(a) Molarity	(p) not changed by change in temperature
(b) Molefraction	(q) Moles per Kg
(c) Molality	(r) changed by change in temperature
(d) Normality	(s) moles per litre

ADDITIONAL PRACTICE EXERCISE

Subjective Questions

121. When bleaching powder, CaCl(OCl), reacts with CO_2, chlorine gas is evolved. What volume of chlorine is released when excess bleaching powder was added into a 8 L glass container containing moist CO_2? (Purity of CO_2 = 90%) (Assume that all the CO_2 is used.)

122. I^- ion in 3.32 g sample of XI is removed through precipitation. The precipitate is found to contain 2.54 g of iodine. Identify the alkali metal X.

123. When 400 g of 80% SO_3 is dissolved in 54 g H_2O, a mixture of H_2SO_4 and $H_2S_2O_7$ is obtained.
(i) Calculate the number of moles of H_2SO_4 first formed and the number of moles of $H_2S_2O_7$ formed.
(ii) How many moles of H_2SO_4 remain? What are the mole fractions of H_2SO_4 and $H_2S_2O_7$?
(iii) What is the weight % of $H_2S_2O_7$ formed?

124. The percentage by volume of C_3H_8 in a mixture of C_3H_8, CH_4 and CO is 36.5. Calculate the volume of CO_2 liberated when 200 mL of the mixture is burnt in excess of O_2.

125. When 0.8 g of a hydride MH_2 was heated in oxygen it was completely converted into its oxide M_2O_2. Find the equivalent weight of M if M_2O_2 obtained weighs 5.2 g?

126. A 10 molal solution is obtained when 11.6 g of acetone is dissolved in 22.8 mL of benzene. Find the density of benzene in g L^{-1}.

127. If the ratio in the mole fractions of the components in a solution is 3:22, calculate the molefraction of the solute.

128. 0.5 g of an alloy containing 60% by weight of copper is dissolved in conc.H_2SO_4 and then treated with excess KI. The liberated iodine consumed 30 mL of thiosulphate.
(i) Write down the reaction between Cu^{2+} and I^- and that between iodine and thiosulphate?
(ii) How many moles of thiosulphate are consumed in the reaction?
(iii) Calculate the molarity of thiosulphate solution.

129. In a neutralization reaction 60 mL normal perchloric acid is completely reacted with 1.68 g of a base (M.W = 84). Find the acidity of the base.

130. A mixture of 0.4 L of 1 M HCl and 0.3 L of 4 N HCl is used to neutralize a mixture of 20 mL of 0.35 M $Ca(OH)_2$ and 25 mL of 0.4 N $Ca(OH)_2$.
(i) What is the normality of the acid mixture?
(ii) How many milliequivalents of $Ca(OH)_2$ are present in the mixture?
(iii) How many milli litres of acid mixture are required to neutralise the $Ca(OH)_2$ mixture completely?

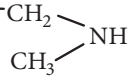
Straight Objective Type Questions

Directions: **This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

131. A certain nitrogen compound contains 12.5% hydrogen and the rest nitrogen. It is a basic compound and combines with H_2SO_4 to form a sulphate containing 24.6% of Sulphur. What may be the formula of the sulphate.
(a) $(NH_4)_2SO_4$ (b) $NH_4.HSO_4$ (c) $N_2H_4.H_2SO_4$ (d) $N_2H_4.2H_2SO_4$

132. An organic compound, A, which is a monoacid base contains wt % ratio C : H : N = 4 : 1 : 1.55. It forms a hydrochloride which reacts with 0.105 eqt of $AgNO_3$ per grm of the hydrochloride. Suggest a possible structure for the base.

(a) $CH_3 - CH_2 - CH_2 - NH_2$ (b) $N(CH_3)_3$

(c)  (d) All the above

133. The hydrated sulphate of a divalent metal loses 51.2% of its weight on heating and thus forming the anhydrous salt. If the molar mass of the hydrated salt is 246 g/mole. Calculate (i) the number of molecules of H_2O per molecule of the hydrated sulphate and (ii) the atomic weight of the metal

(a) 5 , 20 (b) 2, 20 (c) 6, 24 (d) 7, 24

134. How many molecules of oxygen are obtained per gram of $NaNO_{3(s)}$ on heating?

(a) 4.351×10^{22} (b) 4.531×10^{20} (c) 5.341×10^{20} (d) 3.541×10^{21}

135. 2 g of a metallic bromide could be theoretically converted into 1.368 g of the anhydrous sulphate of the metal. If the metal is divalent and if its hydrated sulphate has two molecules of water of hydration per molecule of the sulphate, calculate the molar mass of the hydrated sulphate (in grams/mole)

(a) 128 (b) 192 (c) 172 (d) 148

136. The number of Ca^{2+} and Cl^- ions present in anhydrous $CaCl_2$ is 3.01×10^{23} and 6.023×10^{23} respectively. The weight of the anhydrous sample is

(a) 40 g (b) 55.5 g (c) 222 g (d) 75.5 g

137. A certain compound containing cobalt and other elements has 13.05 wt% of cobalt in it. What may be the minimum molar mass of the compound? Atomic weight of cobalt = 59

(a) 542 g mol^{-1} (b) 524 g mol^{-1} (c) 425 g mol^{-1} (d) 452 g mol^{-1}

138. Ammonium stannic chloride contains tin to a % by weight of 32.28. At. weight of tin 118.7. Calculate the minimum molar mass of the compound.

(a) 267.7 (b) 367.7 (c) 437.7 (d) 307.7

139. An unknown compound X weighing 0.858 g on complete combustion gives 2.63 g of CO_2 and 1.28 g of H_2O. The lowest molecular weight of the compound is

(a) 43 (b) 86 (c) 22 (d) 60

140. An element X forms two different oxides at STP

(i) The first oxide (containing 36.3% of oxygen by weight) has a volume of 505 ml per gram).

(ii) The second oxide (with 53.3 % by weight of oxygen) occupies 735 ml per gram. Calculate the ratio of valencies of X in the two oxides

(a) 1 : 2 (b) 1 : 1 (c) 2 : 1 (d) 2: 3

141. The pair of species having same percentage of carbon is

(a) CH_3COOH and $C_6H_{12}O_6$ (b) CH_3COOH and C_2H_5OH

(c) $HCOOCH_3$ and $C_{12}H_{22}O_{11}$ (d) $C_6H_{12}O_6$ and $C_{12}H_{22}O_{11}$

142. A straight chain saturated hydrocarbon has a vapour density = 71. If 4.26 g of the compound contains 3.6 g of carbon, determine the formula of the compound.

(a) C_8H_{18} (b) $C_{10}H_{22}$ (c) C_9H_{20} (d) $C_{11}H_{24}$

143. Two oxides of a metal contain 50% and 40% metal M, respectively. If the formula of the first oxide is MO_2, the formula of the second oxide will be

(a) MO_2 (b) MO_3 (c) M_2O (d) M_2O_5

144. An alkyl amine contains 65.75 % of carbon, 15.07 % hydrogen and the rest nitrogen. Suggest a formula for the amine.

(a) $C_4H_{11}N$ (b) C_4H_9N (c) $C_5H_{13}N$ (d) $C_5H_{11}N$

145. A certain saturated fuoro chloro hydrocarbon $C_xCl_yF_4$ has a wt% of 29.92 of fluorine. At. wt. of fluorine = 19. Suggest the formula of the compound

(a) $C_3F_6Cl_2$ (b) $C_2Cl_2F_4$ (c) $C_2Cl_4F_2$ (d) $C_3F_4Cl_4$

146. $NaNO_3$ in solution is reduced by Al in presence of alkali (NaOH) to yield NH_3 and $NaAlO_2$. Calculate the weight in grams of $NaNO_3$ reduced by one gram of Al. (At. wieght. of Al = 27)

(a) 1.243 (b) 1.342 (c) 1.432 (d) 1.181

147. The equivalent weight of a certain metal is 68.7. On heating the chlorate of the metal decomposes yielding the chloride of the metal and oxygen. Calculate the weight in grams of the chlorate that forms one gram of the chloride.

(a) 1.641 (b) 1.146 (c) 1.164 (d) 1.461

148. The equivalent weight of a metal is 12. If one gram of the sulphite of the metal is oxidised to sulphate and then treated with excess of $BaCl_2$ to yield a precipitate of $BaSO_4$, what may be the weight of the precipitate? [At. weight of Ba:137.4]

(a) 2.482 g (b) 2.244 g (c) 2.924 g (d) 2.412 g

149. Metallic silver dissolves in NaCN when exposed to air to form $Na[Ag(CN)_2]$ and NaOH. Determine by calculation the weight of NaCN that would dissolve 1 g of Ag.

(At. weight of Ag = 108)

(a) 0.709 g (b) 0.790 g (c) 0.907 g (d) 0.970 g

150. Calculate the number of grams of Borax:$Na_2B_4O_7.10H_2O$ per litre of a solution, 50 ml of which requires 15.6 ml of 0.2 N hydrochloric acid solution for complete neutralization.

Note: Borax in aqueous solution may be assumed to yield by hydrolysis two NaOH and four H_3BO_3. Boric acid is a weak acid and does not react with NaOH under the given condition.

(a) 12.19 (b) 11.92 (c) 19.21 (d) 21.19

151. Which of the following reactions is incorrectly formulated in the equations given

(a) $2HCl + H_2SO_4 \rightarrow Cl_2 + 2H_2O + SO_2$ (b) $I_2 + 2Na_2S_2O_3 \rightarrow 2NaI + Na_2S_4O_6$

(c) $NaCl + NH_3 + CO_2 + H_2O \rightarrow NaHCO_3 + NH_4Cl$ (d) $\underset{(s)}{Na_2B_4O_7.10H_2O} \xrightarrow{heat} \underset{(g)}{10H_2O} + 2NaBO_2 + B_2O_3$

152. 50 litres of methane (CH_4) gas at STP is oxidized by oxygen to yield CO_2 and water. Calculate the number of grams of ethane (C_2H_6) gas, which on oxidation will yield the same number of moles of CO_2.

(a) 43.38 g (b) 48.33 g (c) 38.43 g (d) 33.48 g

153. Calculate the weight of calcium hydride required to produce one kg of hydrogen by the reaction, $CaH_2 + 2H_2O \rightarrow Ca(OH)_2 + 2H_2$. [At. weight of Ca = 40]

(a) 5.25 kg (b) 10.5 kg (c) 15.75 kg (d) 21 kg

154. K_2CrO_4 reacts with NaCl in presence of con.H_2SO_4 and gives a deep red (liquid) compound containing chromium. In the formation of this compound the change of oxidation number of chromium (per atom) is

(a) 3 (b) 4 (c) 2 (d) Zero

155. Number of moles of electrons required to reduce one mole of NO_3^- to NO is

(a) 1.8×10^{23} (b) 1.2×10^{23} (c) 3 (d) 4

156. One mole of N_2H_4 loses 10 mole of electrons to form Y. If Y contains 2 nitrogen atoms and there is no change in oxidation number of H, then oxidation number of N in Y is

(a) −3 (b) +3 (c) +5 (d) +7

157. Sodium hyponitrite $Na_2N_2O_2$ is oxidised by $KMnO_4$ in acid medium (dil H_2SO_4) to yield $NaNO_3$ along with $KHSO_4$, $MnSO_4$ and H_2O. Calculate the mole ratio for the reaction: $Na_2N_2O_2 : KMnO_4$

(a) 3 : 4 (b) 5 : 8 (c) 3 : 8 (d) 3 : 5

158. Sodium thiosulphate ($Na_2S_2O_3$) is oxidized by Cl_2. The reaction (unbalanced, in outline) is
$[S_2O_3]^{2-} + (.....)Cl_2 + (....)H_2O \rightarrow (.....)Na^+ + (.....)Cl^- + (.....)SO_4^{2-} + (....)H^+$
Calculate how many moles of Cl_2 are required to oxidize one gram of thiosulphate.
(a) 0.0253 (b) 0.0325 (c) 0.0352 (d) 0.0176

159. When equimolar amounts of $SnCl_2$ and $HgCl_2$ solutions are mixed, the metal ions present after complete reaction are
(a) Sn^{2+} and Hg^{2+} (b) Sn^{4+} and Hg^{2+} (c) Sn^{4+} only (d) Sn^{4+} and Hg_2^{2+}

160. The equivalent weight of an element = 13.16. It forms an acidic oxide which dissolves in KOH to form a salt isomorphous with potassium sulphate. The element forms another oxide in which it exhibits tetravalency. What is its equivalent weight in this second oxide?
(a) 19.74 (b) 17.49 (c) 14.97 (d) 15.45

161. 5.20 g of a metallic chloride were dissolved in water and treated with an excess of $AgNO_3$ solution. The precipitated AgCl was filtered, washed and dried. It weighed 10 g. Calculate the eqt. weight of the metallic carbonate. At. weight of Ag = 108
(a) 62.19 (b) 69.12 (c) 34.56 (d) 31.1

162. An unknown hydrocarbon weighing 0.858 g on complete combustion gives 2.63 g of CO_2 and 1.28 g of H_2O. The molecular weight of the hydrocarbon is
(a) 86 (b) 100 (c) 128 (d) 78

163. Density of 6 N Na_2CO_3 solution in water = 1.2 g ml^{-1}. The % by weight of Na_2CO_3 in the solution is
(a) 53 (b) 39.75 (c) 13.25 (d) 26.5

164. A solution contains acetic acid is 2 M and has a molality = 2.174. Calculate the density of the solution in g mL^{-1}
(a) 1.02 (b) 1.04 (c) 1.1 (d) 1.07

165. $KClO_{3(s)}$ on heating decomposes to yield oxygen gas + KCl and $KClO_4$. In a different reaction if one mole of $KClO_3$ reacts entirely yielding 0.8 mole of O_2 with KCl and $KClO_4$. The mole fraction of $KClO_4$ is
(a) 0.51 (b) 0.35 (c) 0.42 (d) 0.29

166. 0.517 g of the carbonate of a metal was added to 50 ml of 0.491 N H_2SO_4. After dissolving and boiling to expel carbon dioxide, the residual acid required 32.71 ml of 0.435 N sodium hydroxide for neutralization. Calculate the equivalent weight of the metal
(a) ~20.1 (b) 23.0 (c) 39.0 (d) 12.0

167. A solution of potassium iodide of concentration 2×10^{-3} M was used in an oxidation experiment in presence of con. HCl. The product of oxidation was ICl. Calculate the no. of eqts of oxidizing agent required to oxidize the iodide ion in 100 ml of the solution.
(a) 4×10^{-4} (b) 2×10^{-4} (c) 2×10^{-3} (d) 4×10^{-3}

168. 0.5 g of an alloy containing 60% by weight of copper is dissolved in con. H_2SO_4. The solution is then diluted and treated with excess of KI. The liberated iodine consumed 30 ml of a certain solution of sodium thiosulphate. Calculate the molarity of the thiosulphate solution. At. wt. of Cu = 63.6
(a) 1.752×10^{-4} (b) 1.257×10^{-3} (c) 1.572×10^{-4} (d) 1.752×10^{3}

169. Calculate the volume (in ml) of a mixture of 3 N HCl and 6 N HCl, prepared in the ratio 4 : 1 required to neutralize 10 ml of 6 N NaOH
(a) 17.66 ml (b) 13. 33 ml (c) 16.67 ml (d) 19.67 ml

170. The molar mass of an acid = 40 g mol^{-1}. 50 ml of one normal $Ca(OH)_2$ is neutralized by 0.5 g of the acid. The basicity of the acid is
(a) 4 (b) 3 (c) 2 (d) 1

Assertion–Reason Type Questions

Directions: **Each question contains Statement-1 and Statement-2 and has the following choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False
(d) Statement-1 is False, Statement-2 is True

171. Statement 1
The number of molecules in 16 g of SO_2 is one fourth of the number of molecules of 16 g of methane.
and

Statement 2
One mole of any gas behaving ideally has the same (Avogadro) number of molecules in it.

172. Statement 1
Oxidation state of sulphur is the same in both SO_3 and H_2SO_4.
and

Statement 2
Con. H_2SO_4 dissolves SO_3 to yield $H_2S_2O_7$.

173. Statement 1
Formaldehyde (HCHO) and lactic acid [$CH_3CH(OH)COOH$] have the same percentage composition.
and

Statement 2
Both compounds have the same functional groups.

174. Statement 1
$NH_4Cl_{(s)}$ on heating dissociates into ammonia and HCl.
and

Statement 2
All ammonium salts yield ammonia on heating.

175. Statement 1
The oxidation (state) number of manganese in MnO_4^- is 7.
and

Statement 2
Reduction of MnO_4^- in acid medium is catalysed by Mn^{2+} ion.

176. Statement 1
SO_2 reduces acidified dichromate in acid medium.
and

Statement 2
SO_2 oxidizes H_2S to sulphur.

177. **Statement 1**
Al_2O_3 can be reduced by red hot carbon to metallic Al.
and

Statement 2
Both carbon and CO are reducing agents.

178. **Statement 1**
One gram of hydrogen combines with 80 g of bromine or 20 g of calcium.
and

Statement 2
One equivalent of a substance always reacts with one equivalent of any other substance.

179. **Statement 1**
Normality and molarity of a solution are unaffected by a change in temperature.
and

Statement 2
Both normality and molarity involve the number of moles of solute in solution which does not change with temperature.

180. **Statement 1**
20 ml of 0.05 M Na_2CO_3 requires 20 mL 0.1 M HCl at the equivalence point.
and

Statement 2
Equal volumes of a base and an acid of equal normality react completely at equivalence point.



Linked Comprehension Type Questions

Directions: **This section contains 3 paragraphs. Based upon the paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

Passage I

The balanced chemical equation for a reaction indicates the mole ratio of the reactants that would be exactly right for the reactants to form the products in a total reaction. If in an actual case this mole ratio is not observed then one or more of the reactants may be present in excess and will be left over while the other reactant(s) not present in excess will be completely consumed. These other reactants used up completely are termed limiting reactants.

181. 6 g of NaCl and 20 g of $AgNO_3$ are allowed to react in solution. (At. wt. of Ag = 108)
(a) Excess reactant is NaCl
(b) Neither reactant is limiting
(c) $AgNO_3$ is present in excess
(d) Amount of AgCl formed is 15.64 g

182. If one kg of aluminium is made to react with one kg of MnO_2 and the products are Mn and Al_2O_3 indicate among the following the correct statement. (At. wts. Al = 27, Mn = 55]
(a) The limiting reactant is aluminium
(b) The limiting reactant is manganese dioxide
(c) Unreacted MnO_2 left over is 207 g
(d) Unreacted aluminium left over is 216 g

183. In the reduction of heated Fe_3O_4 by hydrogen to form iron and water vapour, if H_2 should not be the limiting reactant, what volume of H_2 measured at STP would be required to reduce one gram of Fe_3O_4. (At wt. Fe = 56)
(a) 0.386 litre (b) 0.368 litre (c) 0.245 litre (d) 0.490 litre

Passage II

In volumetric analysis the two reactants used in most of the cases are (i) an acid and a base or (ii) an oxidizing agent and a reducing agent, generally both reactants are present in solution. Atleast one of the solutions must have a known concentration, such a solution is called a standard solution. Its concentration is expressed as a normality i.e., number of equivalents present in a litre of the solution or as a molarity i.e., number of moles per litre of the solution.

184. Which among the following cannot be the equivalent weight of $KMnO_4$ in acid, neutral or alkaline medium?
(a) 31.6 (b) 158 (c) 52.7 (d) 39.5

185. Which of the following is not used as a primary standard in acid-alkali or redox titrations?
(a) $KMnO_4$ (b) $K_2Cr_2O_7$
(c) anhydrous Na_2CO_3 (d) Potassium hydrogen phthalate

186. How many ml of 0.4 M NaOH would exactly neutralize a mixture formed from 20 ml of 0.1 M KOH and 30 ml of 0.1 M H_2SO_4?
(a) 15 (b) 10 (c) 5 (d) 7.5

Passage III

Balancing of redox equations may be done by either (i) the ion-electron method or (ii) the oxidation state method. In (i) the total reaction is first written in two halves, the oxidation half and the reduction half. Both include electrons, a certain number in each. The two halves are then added after multiplying each by a suitable number (coefficient), so as to cancel the number of electrons on both sides.

In (ii) suitable coefficients are used for the oxidizing and reducing agents so that the total increase in the oxidation number of the reducing agents equals the total decrease in the oxidation number of the oxidizing agents.

187. $\alpha\ H_2SO_4 + \beta\ HI \rightarrow$ products (H_2S, I_2 and H_2O). What are the numbers α and β?
(a) 1, 6 (b) 2, 9 (c) 1, 8 (d) 2, 7

188. How many grams of Mohr's salt $FeSO_4.(NH_4)_2SO_4.6H_2O$ will react with one gram of $KMnO_4$ in acid medium? At. wt. Fe = 56
(a) 12.4 (b) 14.2 (c) 21.4 (d) 24.1

189. $K_2Cr_2O_7$ in dil. H_2SO_4 functions as an oxidizing agent with an equivalent weight, $\frac{1}{n}$ of its molar mass.
(a) n = 3 (b) n = 6 (c) n = 8 (d) n = 4

Multiple Correct Objective Type Questions

Directions: **Each question in this section has four suggested answers out of which ONE OR MORE answers will be correct.**

190. The different oxidation states, exhibited by chromium in its compounds are
(a) 1 (b) 2 (c) 3 (d) 6

191. Which of the following reactions does not involve oxidation/reduction?
(a) $2K_2CrO_4 + H_2SO_4 \rightarrow K_2Cr_2O_7 + K_2SO_4 + H_2O$ (b) $MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2$
(c) $4FeCl_3 + 3K_4[Fe(CN)_6] \rightarrow Fe_4[Fe(CN)_6]_3 + 12KCl$ (d) $N_2H_4 + HN_3 \rightarrow N_5H_5$

192. By redox reactions I^- ion may be converted into I_2 or I^+. The equivalent weights of iodide ion are (at wt of I = 127)
(a) 127 (b) 254 (c) 63.5 (d) 42.33

193. Which among the following are disproportionation reactions?
(a) $P_4 + 3NaOH + 3H_2O \rightarrow 3NaH_2PO_2 + PH_3$ (b) $P_4 + 8SOCl_2 \rightarrow 4PCl_3 + 4SO_2 + 2S_2Cl_2$
(c) $6NaOH + 3Cl_2 \rightarrow 5NaCl + NaClO_3 + 3H_2O$ (d) $Na_2S_2O_3 + 2HCl \rightarrow 2NaCl + SO_2 + H_2O + S$

194. Which of the following is a reaction where the same element undergoes both oxidation and reduction?
(a) $Ca(OH)_2 + Cl_2 \rightarrow CaOCl_2 + H_2O$
(b) $2H_2S + SO_2 \rightarrow 2H_2O + 3S$
(c) $NaH + H_2O \rightarrow NaOH + H_2$
(d) $3Br_2 + 6NaOH \rightarrow 5NaBr + NaBrO_3 + 3H_2O$

195. The equivalent weights of sulphur in its oxides are
(a) 32 (b) 8 (c) 24 (d) 5.33

196. 2 moles of a solute of molar mass 60 were dissolved in a kilogram of water. The volume of the solution was 1.02 litres.
(a) molality of the solution is 2
(b) molarity of the solution is ~ 1.96
(c) % by wt. of the solute is 12
(d) mole fraction of the solute = 0.0347

197. A mixture of 10 ml of 0.5 N HCl and 20 ml of 0.2 N HCl
(a) has a concentration of 0.3 moles litre^{-1}.
(b) is equivalent to 90 ml of 0.1 N HCl.
(c) may be neutralized exactly by 60 ml of 0.15 N NaOH.
(d) may be neutralized by 45 ml of 0.25 N $Ba(OH)_2$.

Matrix-Match Type Questions

Directions: **Match the data in Column I with data in Column II. There can be single or multiple matches.**

198.

Column I	**Column II**
(a) Disproportionation	(p) sodium carbonate + HCl neutralization using methyl orange
(b) mole ratio of reactants 1 : 2	(q) $I_2 + Na_2S_2O_3$ reaction
(c) used in volumetric estimation	(r) I_2 + hot alkali
(d) Inflammable volatile product	(s) white Phosphorus with hot NaOH

199. Match the quantities on the left with number of particles on the right

Column I	**Column II**
(a) 224 cm^3 helium at STP	(p) 1.807×10^{22} atoms
(b) 180 mg water	(q) 6.022×10^{21} atoms
(c) 224 cm^3 hydrogen at STP	(r) 3.011×10^{21} molecules
(d) 112 cm^3 ethene at STP	(s) 6.022×10^{21} molecules

200.

Column I	**Column II**
(a) $NaH + H_2O \rightarrow NaOH + H_2$	(p) Redox reaction
(b) $2NaOH + Cl_2 \rightarrow NaCl + NaClO + H_2O$	(q) same element act as both oxidizing and reducing agents
(c) $2KClO_3 \rightarrow 2KCl + 3O_2$	(r) disproportion reaction
(d) $Ca(OH)_2 + Cl_2 \rightarrow CaOCl_2 + H_2O$	(s) Metal does not involve oxidation or reduction

SOLUTIONS

ANSWERS KEYS

Topic Grip

1. 49.1
2. 1 : 1
3. C_3O_2
4. 3.81 g of Iodine
5. 16.52 mL
6. 47.3
7. 39.44%
8. 1.2
9. 2.8
10. 0.75
11. (d) **12.** (b) **13.** (a)
14. (c) **15.** (a) **16.** (a)
17. (d) **18.** (a) **19.** (c)
20. (b) **21.** (a) **22.** (b)
23. (c) **24.** (c) **25.** (b)
26. (a)
27. (a), (b), (c)
28. (a), (b)
29. (b), (c)
30. (a) → (p), (r), (s)
(b) → (p)
(c) → (p), (s)
(d) → (p), (q)

IIT Assignment Exercise

31. (b) **32.** (c) **33.** (c)
34. (b) **35.** (b) **36.** (d)
37. (a) **38.** (b) **39.** (c)
40. (c) **41.** (d) **42.** (c)
43. (a) **44.** (c) **45.** (b)
46. (c) **47.** (c) **48.** (a)
49. (a) **50.** (c) **51.** (b)
52. (c) **53.** (d) **54.** (d)
55. (c) **56.** (d) **57.** (c)
58. (a) **59.** (b) **60.** (a)
61. (b) **62.** (a) **63.** (b)
64. (d) **65.** (a) **66.** (c)
67. (c) **68.** (d) **69.** (b)
70. (b) **71.** (a) **72.** (a)
73. (b) **74.** (d) **75.** (b)
76. (b) **77.** (d) **78.** (a)
79. (c) **80.** (c) **81.** (c)
82. (c) **83.** (d) **84.** (a)
85. (b) **86.** (d) **87.** (d)
88. (d) **89.** (b) **90.** (b)
91. (b) **92.** (a) **93.** (b)
94. (a) **95.** (d) **96.** (a)
97. (b) **98.** (c) **99.** (d)
100. (c) **101.** (a) **102.** (b)
103. (a) **104.** (d) **105.** (a)
106. (d) **107.** (a) **108.** (b)
109. (d) **110.** (a) **111.** (b)
112. (d) **113.** (c) **114.** (d)
115. (b)
116. (c)
117. (a), (b)
118. (a), (c)
119. (b), (d)
120. (a) → (s), (r)
(b) → (p)
(c) → (p), (q)
(d) → (r)

Additional Practice Exercise

121. 7.2 L
122. Potassium
123. (i) H_2SO_4 = 3 mol
$H_2S_2O_7$ = 1 mol
(ii) Remaining H_2SO_4 = 2 mol
Mole fraction of H_2SO_4 = 0.667
Mole fraction of $H_2S_2O_7$ = 0.333
(iii) 47.6%
124. 346 ml
125. 20.4 ml
126. 877g L^{-1}
127. 0.12
128. (i) $2Cu^{2+} + 4I^- \rightarrow \underset{\text{insoluble}}{2CuI} + I_2$
$I_2 + 2S_2O_3^{2-} \rightarrow S_4O_6^{2-} + 2I^-$
(ii) 0.0047
(iii) 0.157 M
129. (3)
130. (i) 2.286 N
(ii) 24 milliequivalents
(iii) 10.5 mL
131. (c) **132.** (d) **133.** (d)
134. (d) **135.** (c) **136.** (b)
137. (d) **138.** (b) **139.** (a)
140. (a) **141.** (a) **142.** (b)
143. (b) **144.** (a) **145.** (d)
146. (d) **147.** (d) **148.** (b)
149. (c) **150.** (b) **151.** (a)
152. (d) **153.** (b) **154.** (d)
155. (c) **156.** (b) **157.** (b)
158. (a) **159.** (c) **160.** (a)
161. (b) **162.** (a) **163.** (d)
164. (b) **165.** (d) **166.** (a)
167. (a) **168.** (c) **169.** (c)
170. (a) **171.** (a) **172.** (b)
173. (c) **174.** (c) **175.** (b)
176. (b) **177.** (d) **178.** (a)
179. (d) **180.** (d) **181.** (c)
182. (b) **183.** (a) **184.** (d)
185. (a) **186.** (b) **187.** (c)
188. (a)
189. (b)
190. (b), (c), (d)
191. (a), (c), (d)
192. (a), (c)
193. (a), (c), (d)
194. (b), (c)
195. (b), (d)
196. (a), (b), (d)
197. (a), (b), (c)
198. (a) → (r), (s)
(b) → (p), (q), (r)
(c) → (p), (q)
(d) → (s)
199. (a) → (q), (s)
(b) → (p), (s)
(c) → (s)
(d) → (p), (r)
200. (a) → (p), (q), (s)
(b) → (p), (q), (r), (s)
(c) → (p), (s)
(d) → (p), (q), (r), (s)

HINTS AND EXPLANATIONS

Topic Grip

1. Since one mole of $MCl_2 \cdot 2H_2O$ contains 2 moles i.e., 36 g of water, if the molar mass of $MCl_2 \cdot 2H_2O$ = m g,

then $\frac{36}{m} \times 100 = 24.9$

$\therefore \quad m = \frac{36 \times 100}{24.9}$ g = 144.6 g

$\therefore$ Weight % of Chlorine = $\frac{71}{144.6} \times 100 = 49.1$

2. % ratio of Iron: Oxygen i.e., Fe : O = 72.4 : 27.6

Atom ratio = $\frac{72.4}{56} = \frac{27.6}{16} = 1.293 : 1.725$

i.e., 3 : 4

$\therefore$ the empirical formula = Fe_3O_4 i.e., $FeO.Fe_2O_3$

The mole ratio = 1 : 1

3. $\underset{1\text{ vol}}{C_xO_y} + \underset{2\text{ vol}}{2O_2} \rightarrow xCO_2$

2x = y + 4 Since y exceeds 2x by 4, it must be an even number.

$\therefore \quad C_3O_2$

4. $BaCrO_4 + 8HCl + 3KI \rightarrow BaCl_2 + CrCl_3 + 3KCl + 4H_2O + \frac{3}{2} I_2$. This suffers a change of oxidation state of +6 to +3. Accordingly one mole of $BaCrO_4 \equiv \frac{3}{2}$ mole of I_2. i.e., $\frac{3}{2} \times 254 = 381$ g

$\therefore \quad 0.01$ mole $\equiv 3.81$ g of Iodine

5. Suppose V ml of O_2 is converted into Ozone. Since $3O_2 \rightarrow 2O_3$. V ml $O_2 \equiv \frac{2}{3}$ V ml O_3

$\therefore$ total volume equals $[(50 - V) + \frac{2}{3} V]$ ml

$= \left(50 - \frac{V}{3}\right)$ ml = 46 ml

$\therefore \quad \frac{V}{3} = 4$ ml

$\therefore \quad V = 12$ ml [check $(50 - 12) + \frac{2}{3} \times 12$

$= (38 + 8)$ ml = 46 ml]

$\therefore$ if 46 ml of ozonised oxygen is exposed to oil of turpentine, 38 ml will be the residual volume of oxygen. Proportionately if 20 ml of ozonised oxygen are exposed, residual volume = $\frac{38}{46} \times 20$

= 16.52 ml

6. Suppose 1.58 g of the mixture contains x g of NaI, and (1.58 – x) g of KI. Then

$$\frac{x}{150} + \frac{1.58 - x}{166} = \frac{2.35}{235} = 0.01$$

$\therefore \quad \frac{x}{150} - \frac{x}{166} = 0.01 - 0.00952$

$$\frac{(166 - 150)x}{166 \times 150} = 4.8 \times 10^{-4} = \frac{16x}{166 \times 150}$$

$$= 4.8 \times 10^{-4}$$

$$x = \frac{4.8 \times 10^{-4} \times 166 \times 150}{16}$$

$$= 0.747 \text{ g}$$

$\therefore$ Weight % of NaI = $\frac{0.747}{1.58} \times 100 = 47.3$

7. The original mixture (1 g) contains x g of NaCl (say) and (1 – x) g of $NaNO_3$

$$\underset{M = 85\,g.mol^{-1}}{NaNO_3} \rightarrow \underset{M = 69\,g.mol^{-1}}{NaNO_2} + \underset{16\text{ g}}{\frac{1}{2}O_2}$$

$\therefore \quad \frac{(1 - x)}{85} \times 16 = 0.114$ g

$\therefore \quad (1 - x) = \frac{0.114 \times 85}{16}$ g = 0.6056 g

$\therefore \quad x = 0.3944$ g

% of NaCl = 39.44

8. The equation is $5NaBr + NaBrO_3 + 6HCl \rightarrow 6NaCl + 3Br_2 + 3H_2O$

$\therefore$ 5 moles of NaBr would react with one mole of $NaBrO_3$

$\therefore$ 2 moles of NaBr $\equiv \frac{2}{5}$ mole of $NaBrO_3$

$\therefore$ $NaBrO_3$ is in excess

$\therefore$ NaBr is the limiting reactant

Further according to the equation, 5 moles of NaBr would yield in the reaction 3 moles of Br_2

$\therefore$ 2 moles would yield $\frac{3}{5} \times 2 = \frac{6}{5}$ moles

= 1.2 moles of Br_2

9. Molecular mass of the compounds are 62, 171, 106 and 368. The mass of element present in one molecule of these compounds are 56, 28, 28 and 84. The least value can be taken as the atomic mass of the element.

10. Suppose the molarity is m. Then one litre of the solution contains m moles of H_2O. Then $\underset{1 \text{ mole}}{H_2O_2} \rightarrow H_2O + \underset{0.5 \text{ mole}}{\frac{1}{2}O_2}$

i.e., 11.2 litres at STP

$\therefore$ molarity m = $\frac{8.4}{11.2} = 0.75$

11. $\underset{30\text{ g}}{NO} + Cu \rightarrow \underset{14\text{ g}}{\frac{1}{2}N_2} + CuO.$

Increase in weight of copper is due to oxygen from NO.

10 g oxygen is obtained from $\frac{30}{16} \times 10 = 18.75$ g NO

residual NO = (30 − 18.75)g = 11.25 g

12. Let ε be the equivalent weight of the metal

$$\frac{\varepsilon + 8}{\varepsilon + 30} = \frac{1}{2.1}$$

$\therefore$ $\varepsilon = 12 \left[\text{check } \frac{12+8}{12+30} = \frac{20}{42} = \frac{1}{2.1}\right]$

13. Molar mass of the salt = $\left(\frac{1.38}{0.457} \times 79.6\right)$

= 240.37 g mol^{-1}

Molar mass of anhydrous salt = 240.37 − 54 = 186.37

Equivalent weight = 93.18 g

$\therefore$ Equivalent weight of acid = 93.18 − 31.8 + 1

= 62.38

14. Let the equivalent weight of the element be ε

$$\frac{\varepsilon + 48}{116.7} = \frac{1}{2.05}$$

$\therefore$ $\varepsilon + 48 = \frac{116.7}{2.05} = 56.93$

$\therefore$ $\varepsilon = 8.93$ Equivalent weight of the chloride

= 44.43

Molar mass of the chloride = 134 ≈ 3× 44.43

$\therefore$ Valency = 3

Atomic weight = 8.93 × 3 = 26.79

15. It is obvious that the salt of K_2XO_4 is formed from XO_3

$\therefore$ Equivalent weight = $\frac{79}{6} = 13.17$

16. Statement (1) and (2) are correct. Statement (2) is the correct explanation to statement (1)

17. Statement (1) is false.
Statement (2) is true

18. Statement (1) and (2) are correct. Statement (2) is the correct explanation to statement (1)

19. Statement (1) is true
Statement (2) is false

20. Statement (1) and (2) are correct.
Statement (2) is not the correct explanation to statement (1)

21. MnO_4^- changes to Mn^{2+}

22. Since the change in ON is by five units equivalent weight = 31.6

23. Equivalent weight. of $FeSO_4.(NH_4)_2SO_4.6H_2O$ in the reaction is 392

24. Equivalent weight of chloride = 12.16 + 35.5

= 47.66

Equivalent weight of oxide = 12.6 + 8 = 20.16

Weight of chloride formed from 1.54 g oxide

= $\frac{47.66}{20.16} \times 1.54 = 3.64$

25. 1 equivalent chlorate produces 6 equivalents of oxygen

1 g chlorates liberates 1 − 0.685 = 0.315 g

oxygen = 0.0394 equivalents

Equivalent weight of chlorate = 152.4

Equivalent weight of metal = 68.89

26. $\frac{E+35.5}{E+8}=\frac{2}{0.8}$

Solving E = 10.33

27. No. of molecules in 7 g $N_2 = \frac{7\times6\times10^{23}}{28}$

$= 1.5 \times 10^{23}$

(a) No. of molecules in 4.257 g ammonia

$= \frac{W\times N_A}{M} = \frac{4.257\times6\times10^{23}}{17}$

$= 1.5 \times 10^{23}$

(b) No. of atoms in 5.33 g SO_2

$= \frac{5.33\times6\times10^{23}\times3}{64} = 1.5 \times 10^{23}$

(c) No. of electrons in 0.5 g calcium

$= \frac{0.5\times6\times10^{23}\times20}{40} = 1.5 \times 10^{23}$

(d) No. of H_3O^+ ions in 250 ml of 1M H_2SO_4

$= \frac{1\times250\times6\times10^{23}\times2}{1000} = 3 \times 10^{23}$

28. (a), (b) are correct

The reaction is

$$2KIO_3 + 6H_2C_2O_4 \rightarrow K_2C_2O_4 + I_2 + 6H_2O + 10CO_2$$

29. (b), (c) are correct

The reaction is

$$2NH_2OH + 2Fe_2(SO_4)_3 \rightarrow N_2O + H_2O + 4FeSO_4 + 2H_2SO_4$$

30. (a)→ (p), (r), (s)

(b) → (p)

(c) → (p), (s)

(d) → (p), (q)

IIT Assignment Exercise

31. $2NO + O_2 \rightarrow 2NO_2$

50 50

50 mL NO combines with 25 mL O_2 to produce 50 mL NO_2.

Volume O_2 remaining = 25 mL

∴ Total volume of the gas mixture after the reaction = 50 + 25 = 75 mL

32. $\underset{g}{C_2H_4} \overset{\text{oxidation}}{\equiv} 2\underset{g}{CO_2}$; $\underset{g}{CH_4} \overset{\text{oxidation}}{\equiv} \underset{g}{CO_2}$

Suppose we have V ml of C_2H_4 and (15 – V) ml of CH_4, then

$2V + (15 - V) = 20$

∴ $V + 15 = 20$

∴ $V = 5$ ml

∴ 5 ml C_2H_4, 10 ml CH_4

33. Clearly since 10 ml of the hydrocarbon

$C_xH_y \equiv (50 - 20)$ ml = 30 ml of CO_2 i.e., 1 : 3

∴ $x = 3$.

Further $C_xH_y + \left(x + \frac{y}{4}\right)O_2 \rightarrow xCO_2 + \frac{y}{2}H_2O$.

Since 20 ml of O_2 was left unused in 70 ml of O_2, 50 ml was used for 10 ml of C_xH_y which gives

$x + \frac{y}{4} = 5 \qquad x = 3,\ \frac{y}{4} = 2,\ y = 8$

∴ $C_xH_y = C_3H_8$

34. Clearly the metallic carbonate is MCO_3 and the oxalate is MC_2O_4

∴ weight % $= \frac{137.4}{225.4} \times 100 = 60.96$

35. No. of moles $= \frac{3.2}{32} = 0.1$

No. of molecules of oxygen

$= 0.1\times6.023\times10^{23} = 6.023\times10^{22}$

No. of electrons $= 6.023\times10^{22}\times16 = 9.64\times10^{23}$

36. Minimum molecular weight contains atleast one mole of iron

∴ $M = \frac{100}{30} \times 56 = \frac{560}{3} = 187$

37. Weight of a molecule

$$= \left[\frac{6\times12}{6\times10^{23}}\right] + \left[2\times10^{-23}\right] + \left[16\times10^{-23}\right]$$

$= 30\times10^{-23}$ g

Gram molecular mass = $30 \times 10^{-23} \times 6 \times 10^{23} = 180$ g

38. 2.87 g AgCl contain 2.16 g Ag & 0.71 g Cl (mol. mass of AgCl = 108 + 35.5 = 143.5).

39. Mol. wt of AB_2 = 64 g mol^{-1} and AB_4 = 96 g mol^{-1}

Mol. wt of mixture = 2 x 38.4 = 76.8

Let x be the no. of moles of AB_2 in 50 moles of the mixture.

$\therefore \quad x \times 64 + (50 - x)\, 96 = 50 \times 76.8$

$x + (50 - x)\, 1.5 = 60 \Rightarrow \quad x = 30$

40. % of water $= \dfrac{n \times 18 \times 100}{230 + 18n} = 19$

$n = 3$

41. Mass of Si $= \dfrac{80.7 \times 28.1}{60.1}$

Mass of B $= \dfrac{12.9 \times 21.6}{69.6}$

Si : B = 9.43 :1

42. Cr : S : O $= \dfrac{26.52}{52} : \dfrac{24.52}{32} : \dfrac{48.96}{16}$

$= 2 : 3 : 12$

43. Mn : O $= \dfrac{63.7}{55} : \dfrac{36.3}{16} = 1 : 2$

44. $\underset{97.4\text{ g}}{ZnS} \xrightarrow{\text{Oxidation}} \underset{81.4\text{ g}}{ZnO} + \text{product}$

For a decrease of 16 g (97.4 – 81.4), one mole i.e., 97.4 g of ZnS is present.

For 6 g decrease, $\dfrac{97.4}{16} \times 6 = \dfrac{292.2}{8}$

= 36.5 g of ZnS are present in 50 g

$\therefore$ Impurity = (50 – 36.5)g = 13.5 g

45. The equation is $Ca^{2+} + Na_2CO_3 \rightarrow CaCO_3 + 2Na^+$

Total Ca^{2+} ion in 4 litre of hard water

$= 4 \times 40 = 160$ mg

From the equation, 40 g Ca^{2+} require 106 g washing soda

$\therefore$ 160 mg require washing soda

$= \dfrac{106}{40} \times 160\, 10^{-3} = 0.424$ g

46. $\underset{1\text{ mole}}{B.HCl} + \underset{1\text{ mole}}{AgNO_3} \rightarrow B.HNO_3 + AgCl$

50 ml of the solution ≡ 52.4 ml of 0.1 M $AgNO_3$ solution

$\therefore$ 1 litre of the solution $\underset{(10\text{ g})}{\equiv}$ 1048 ml of 0.1 M $AgNO_3$ solution

$\equiv \dfrac{1048}{1000} \times 0.1$ mole = 0.1048 mol

$\therefore$ 0.1048 mole of $AgNO_3 \equiv 10$ g

$\therefore$ 1 mole $\equiv \dfrac{10}{0.1048}$ g = 95.42 g = molar mass of B.HCl

$\therefore$ molar mass of amine = 95.42 – 36.5

$= 59$ g mol^{-1}

47. Amount of A $= \dfrac{15 \times 95}{100} = 14.25$g

$\underset{1\text{mole}}{A} + \underset{1\text{ mole}}{O_2} \rightarrow AO_2$

Let x be the atomic weight of A

No. of moles of the metal = No. of moles of AO_2

$$\frac{14.25}{x} = \frac{17.45}{x+32} \Rightarrow 14.25x + 14.25 \times 32 = 17.45x$$

$$x = \frac{14.25 \times 32}{3.2} = 142.5\text{g} = 0.1425\text{ kg}$$

48. $\underset{1\text{ mole}}{CO} + \underset{1\text{ mole}}{CuO} \rightarrow Cu + \underset{1\text{ mole}}{CO_2}$; which gives one mole of Na_2CO_3(106 g)

$\therefore$ A loss of 16 g ≡ 106 g

$\therefore$ 0.5318 g $\equiv \dfrac{106}{16} \times 0.5318$ g which is 3.523 g

49. The reaction is

$2K_4[Fe(CN)_6] + 3ZnSO_4 \rightarrow$

$K_2Zn_3[Fe(CN)_6]_2 + 3K_2SO_4$

50. MnO is the limiting reagent.

51.

Mass ratio	Cu	Au
	2 :	3
Wt. in 100 g alloy	40	60
No. of moles	0.63	0.305

Excess amount of Au = 0.095 moles = 18.6 g

Amount of alloy = 81.4

52. 10 mole HNO_3, 2 mole MnO and 5 mol PbO_2 react together to form products.

53. Equivalent weights are $E_1 = \dfrac{75.76}{24.24} \times 8$ and

$$E_2 = \frac{65.22}{34.78} \times 8$$

i.e., 3.125 and 1.875 ≃ 5 : 3, valencies are in the inverse ratio 3 : 5

54. Molar mass of the salt = $\frac{1.500}{0.392} \times 79.6 = 304.6$

$\therefore$ Equivalent weight of the acid = 152.3 − 31.8 + 1 = 121.5

55. Wt. of sulphur in 0.875 g $BaSO_4$

$= \frac{32}{233.3} \times 0.875 = 0.12$ g

Wt. of metal in 0.366 g metal sulphide

= 0.366 − 0.12

= 0.246

Eq. wt . of metal = 32.8

56. One equivalent weight of metallic carbonate ≡ 1 equivalent of CO_2 ≡ 11200 ml at STP.

$\therefore$ 96.5 ml of CO_2 at STP ≡ $\frac{96.5}{11200}$ equivalent

= 0.431 g

$\therefore$ 1 equivalent weight of the carbonate

$= \frac{0.431 \times 11200}{96.5}$ g = 50 g

subtracting the equivalent weight of the "CO_3" radical i.e., 30

Equivalent weight of metal = 20

57. Mohr's salt is $(NH_4)_2SO_4.FeSO_4.6H_2O$

Molecular wt. = 392

$2KMnO_4 + 10FeSO_4 + 8H_2SO_4 \rightarrow K_2SO_4 + 2MnSO_4 + 5Fe_2(SO_4)_3 + 8H_2O$

1 mole of $KMnO_4$ react with 5 moles of $FeSO_4$

3 milli moles of $KMnO_4$ = 15 milli moles of $FeSO_4$

15 milli mole of $FeSO_4$ is present in 15 milli moles of Mohr's salt.

Weight of 15 milli moles of Mohr's salt

$= 15 \times 10^{-3} \times 392 = 5.88$ g

Percentage purity = $\frac{5 \times 10^5 \times 20}{6.023 \times 10^{23} \times 2}$

58. The balanced equation is

$6K_3[Fe(CN)_6] + 10KOH + Cr_2O_3 \rightarrow 6K_4[Fe(CN)_6] + 2K_2CrO_4 + 5H_2O$

59. Wt. of KOH in 20 L 0.25 N solution = 280 g

% impurity = 6.67

60. Change in ON is from +7 to +4

61. $(20 \times 0.1) + (30 \times 0.15) + (V \times 0.5)$

$= (50 + V) \times 0.3$

$2 + 4.5 + (0.5\ V) = 15 + 0.3V$

$\therefore$ $0.2\ V = 8.5$ $\quad V = 42.5$ ml

62. Considering the equation

$Zn + 2HCl \rightarrow ZnCl_2 + H_2$

65.4 g Zn react with 73 g HCl

Wt. of Zn in 11 g sample = 9.57

Wt. of HCl that react with 9.57 g Zn

$= \frac{73}{65.4} \times 9.57 = 10.68$

Let the volume of HCl taken be 'V' ml

$= \frac{V \times 1.18 \times 35}{100} = 10.68$

V = 25.9 mL

63. Molarity of solution = $\frac{135}{342} = 0.395$

Wt. of 1 L solution = 1050 g

Wt. of solvent = 915

Molality = 0.43

64. $d = M\left(\frac{W}{1000} + \frac{1}{m}\right)$

65. $\underset{1\text{ mole}}{H_2O_2} \rightarrow H_2O + \frac{1}{2}\underset{\frac{22.4}{2}=11.2\text{ litres at STP}}{O_2}$

$\therefore$ $\underset{\text{at STP}}{11.2\ \text{litres}} \equiv 34$ g litre^{-1}

$\underset{\text{STP}}{10\ \text{litres}} \equiv \left(\frac{34}{11.2} \times 10\right)$ g litre^{-1}

= 30.36 g

$\therefore$ in 250 ml, $\simeq$ 7.6 g

66. $8V_1 + 2V_2 = 5.5(V_1 + V_2)$

Solving = $V_1 : V_2 = 7 : 5$

67. Let the volume be V mL

Wt. of $H_2SO_4 = \frac{V \times 1.83 \times 93}{100} = 1.702\ V$

Wt. of acid in 200 mL 2 N acid = 19.6 g

$19.6 = 1.702\ V$

V = 11.52 mL

68. No equivalents of Ag = $\frac{11}{108} = 0.102$

7 g NaX = 0.102 equivalent

Solving

$(x \times 58.5) + (0.102 - x)\ 103 = 7$

$x = 0.0788$

Mass of NaCl = 4.61 g

% NaCl = 65.56%

69. $\frac{22 \times 0.327}{2} = 16 \times x$

Solve for x

70. no. of moles of $C_2O_4^{2-}$ = moles of Ca^{2+} = moles of CaO

$= \frac{40 \times 0.250}{1000 \times 2} = 5 \times 10^{-3}$

Wt. of CaO = 0.28

% of CaO = 54

71. 36 g Al = 4 equivalents

hence 127.2 g Cu displaced

72. $\underset{\text{acid}}{NH_4^+} + \underset{\text{base}}{OH^-} \rightarrow \underset{\text{conjugate base}}{NH_3} + \underset{\text{conjugate acid}}{H_2O}$

73. The equivalent wt. of the acid is 41

74. Normality of NaOH = 0.03125

75. No. of equivalent of HCl = 0.0259

equivalent weight of chloride = 63.2 g

Eq. wt. of element = 63.2 – 35.5 = 27.7 g

76. The metal is divalent, hence

$MCO_3 \rightarrow MO + CO_2$

126 g 82 g

1.632 g 1.06 g

77. The equation is

$5NaHSO_3 + 2NaIO_3 \rightarrow 2Na_2SO_4 + 3NaHSO_4 + H_2O + I_2$

hence 520 g $NaHSO_3$ per mole of I_2

78. $2SO_2 + 5C \rightarrow CS_2 + 4CO$

2 moles of SO_2 produce 1 mole CS_2

79. % by mass of P : O= 50 : 50

EF = PO_2

Hence P_4O_{10} and P_4O_6 in the mole ratio = 1 : 1

80. $45 \times 2 - 30 \times N = 12.9 \times 0.5$

Hence molarity = 1.392

81. % silica = $\frac{15}{5+15+70} \times 100$

82. Mol mass of $K_2Zn_3[Fe(CN)_6]_2$ is 698.2

Hence $\frac{1 \times 12 \times 6.02 \times 10^{23}}{698.2}$ atoms

83. 2 moles of D are present in one mole of D_2O

$6.023 \times 10^{23} \times 2$ atoms are present in 20 g of D_2O

Weight of D_2O containing in 5×10^5 atoms of D

$= \frac{5 \times 10^5 \times 20}{6.022 \times 10^{23} \times 2}$

$= \frac{10^7}{6.022 \times 10^{23} \times 2} = 8.3 \times 10^{-18}$ g

84. Weight of Mg in 2 g chlorophyll = $\frac{2 \times 2}{100} = 0.04$ g

No. of gram atoms of Mg = $\frac{0.04}{24} = 1.66 \times 10^{-3}$

No. of atoms of Mg = $1.66 \times 10^{-3} \times 6.02 \times 10^{23} = 10^{21}$

85. $\frac{2 \times 18 \times 100}{M + (2 \times 35.5) + (2 \times 18)} = 14.73$

M = 137.4

% of chlorine in hydrated salt = 29

86. The mass ratio is 3 : 2

Mole ratio is $\frac{3}{100} : \frac{2}{84}$ i.e., 63 : 50

Average molarmass = $\frac{63 \times 100 + 50 \times 84}{113}$

87. 1 atomic mass unit 'u' = 1.66×10^{-24} g

Hence 1.03×10^{22} g = 62 u

The molecular mass

$= (4 \times 12) + (1 \times 4) + (4 \times 16) + 62 = 178.$

88. FeC_2O_4 contain 39% Fe

89. $\frac{192.22 \times 100}{\text{mol mass}} = 39.2$

M = 483

% chlorine = $\frac{6 \times 35.5}{483} \times 100$

90. 40 moles of HNO_3 reacts with 1 mole As_2S_5

91. $C_6H_5OH + 3Br_2 \rightarrow C_6H_2Br_3(OH) + 3HBr$

1 g phenol = $\frac{1}{94}$ mole requires $\frac{3}{94}$ mole Br_2

$1BrO_3^-$ oxidizes $5Br^-$ to $3Br_2$

92. $\frac{2}{120}$ moles of $MgSO_4$ and $\frac{2}{159.6}$ mole $CuSO_4$ are reacting

93. Since 0.1 g amine forms 32.46 mL

i.e., 1.45×10^{-3} mole N_2

hence molar mass = 69

94. 4.2 g AgI ie., 0.0179 moles are produced from 1 g

hence three CH_3O per mole

95. $Ag_2CO_3 \rightarrow 2Ag + CO_2 + {}^1/_2O_2$

276 216

3.14 2.45

96. Six moles of $H_2C_2O_4$ are required to form 1 mole I_2

97. 2 mole $FeSO_4$ produce 4 mole SO_4^{2-} hence the answer

98. The correct set of co-efficients are 1, 4 & 6

99. The balanced equation contain equal no. of moles of $Fe_2(SO_4)_3$ & NH_2OH

100. Since 1.5 times more $KMnO_4$ is required the valency of M should be 3 in addition to 2

101. Normality of $KMnO_4$ = 0.158

No. of equivalents of $KMnO_4$ used = 6.8×10^{-3}

$X_2O_3 \rightarrow X_2O_5$

Change in O.N = 4

Hence no. of moles = $\frac{6.8\times10^{-3}}{4} = 1.7 \times 10^{-3}$

102. $\frac{M\times1000}{1000d - MM'} = m$

103. Moles of O_3 present = $\frac{18.8\times0.1}{1000\times2} = 9.4 \times 10^{-4}$

Mole of mixture = $\frac{250}{22400} = 0.01116$

Moles of $O_2 = 0.01116 - 9.4 \times 10^{-4}$

Wt. of O_3 = 0.04512 g

Wt. of oxygen = 0.3270 g

Total wt. of sample = 0.32718 g

% of $O_3 = \frac{0.04512}{0.32718}\times100$

104. No. of equivalents of AgCl = $\frac{3}{143.5} = 2.1 \times 10^{-3}$

Hence Eq. Wt. of the base = 123

105. 5 mol FeC_2O_4 are oxidized by 3 mol $KMnO_4$

$$\frac{N\times0.1}{1000\times3} = \frac{1}{144\times5}$$

106. $\underset{0.4\,g}{2Ag} + Fe_2(SO_4)_3 \rightarrow Ag_2SO_4 + 2FeSO_4$

3.7×10^{-3} moles	1.85×10^{-3} moles	1.85×10^{-3} moles	3.7×10^{-3} moles

$5FeSO_4^{2+} + MgO_4^- \rightarrow Fe^{3+} + Mn^{2+}$

(Mn: +7)

n = 1 n = 5

3.7×10^{-3} moles 7.4×10^{-4}

$V \times 0.1$ = 0.74 M. moles

= 0.74 × 5 M eq

$\therefore$ V = 37 mL

107. Equivalent weight of $CuSO_4$ in iodometry is its molar mass. Amount of $CuSO_4$ in 2.55 g solution = $7 \times 0.092 \times 10^{-3} \times 4$

Weight of $CuSO_4 5H_2O$ = 0.644 g

Solubility = 33.79

108. Normality of oxalic acid = $\frac{17.5\times0.1}{25}$

Normality of H_2SO_4 = 0.06

weight per litre = 2.94 g

109. Equivalent weight in reaction with acid = 2

As reducing agent ON changes by three hence ON 5

110. 13.50 mL 0.001 M $H_3PO_4 = 50 \times 0.001 = 0.05$ milli moles

Reaction:

$H_3PO_4 + 3NaOH \rightarrow Na_3PO_4 + 3H_2O$

0.05 milli moles H_3PO_4 require $0.05 \times 3 = 0.15$ milli moles of NaOH

$\Rightarrow$ $0.15 = 0.005 \times V$

$\Rightarrow$ $V = \frac{0.15}{0.005} = 30$ mL

0.05 milli moles of H_3PO_4 on reaction gives 0.05 milli moles of Na_3PO_4.

i.e., 5×10^{-5} moles of Na_3PO_4 are formed.

80 mL contains 5×10^{-5} moles of Na_3PO_4

1000 mL contains $\frac{1000\times5\times10^{-5}}{80}$

$= 6.25 \times 10^{-4}$ moles of Na_3PO_4.

111. Statement 1 and 2 are correct
But statement 2 not explanation to 1

112. Statement 1 is wrong since molarity is affected by temperature
Statement 2 is correct

113. Statement 1 is correct
Statement 2 is not balanced

114. There is no change in the oxidation number for any of the species involved

115. It is a disproportionation reaction

116. It is a monobasic acid

117. (a) and (b) are correct

118. (a) and (c)
Both equals 7 milli moles

119. (b) and (d)
The correct co-efficients of the balanced equation are $K_4[Fe(CN)_6] + 6H_2SO_4 + 6H_2O \rightarrow$

120. (a) → (s), (r)
(b) → (p)
(c) → (p), (q)
(d) → (r)

Additional Practice Exercise

121. $CaCl(OCl) + \underset{1\text{ mole}}{CO_2} \rightarrow CaCO_3 + \underset{1\text{ mole}}{Cl_2}$

Volume of $CO_2 = 8\times\frac{90}{100} = 7.2$ L

Volume of $Cl_2 = 7.2$ L

122. No. of moles of $I^- = \frac{2.54}{127} = 0.02$

∴ Formula = XI

If I^- = 0.02 moles, then X = 0.02 moles

Mass of X in sample = XI - I precipitated = 3.32 – 2.54 = 0.78 g of X

Molar mass of X = $\frac{0.78}{0.02} = 39$. It must be potassium.

123. (i) No. of moles of $SO_3 = \frac{400\times0.8}{80} = 4$

($\because$ Mol wt of SO_3 = 80)

No. of moles of $H_2O = \frac{54}{18} = 3$

Reaction:

$SO_3 + H_2O \rightarrow H_2SO_4$

$H_2SO_4 + SO_3 \rightarrow H_2S_2O_7$

3 moles of water react with 3 moles of SO_3

No. of moles of H_2SO_4 formed = 3

No. of moles of SO_3 remaining = 4 – 3 = 1

No. of moles of $H_2S_2O_7$ formed = 1

(ii) No. of moles of H_2SO_4 remaining = 3 – 1 = 2

No. of moles of $H_2S_2O_7$ formed = 1

Mole fraction of $H_2SO_4 = \frac{2}{2+1} = 0.667$

Mole fraction of $H_2S_2O_7 = 0.333$

(iii) Wt% of $H_2S_2O_7$

$= \frac{1\times178}{1\times178+2\times98}\times100 = 47.6\,\%$

124. Let a, b and c be the volume of C_3H_8, CH_4 and CO respectively.

a + b + c = 200 mL

volume of C_3H_8 in 100 mL mixture = 36.5 mL

∴ a = (2 × 36.5) mL

$C_3H_{8(g)} + 5O_{2(g)} \rightarrow 3CO_{2(g)} + 4H_2O_{(\ell)}$

$CH_{4(g)} + 2O_{2(g)} \rightarrow CO_{2(g)} + 2H_2O_{(\ell)}$

$CO_{(g)} + ½\ O_{2(g)} \rightarrow CO_{2(g)}$

CO_2 from three equations = 3a + b + c

= 3 (36.5 × 2) + b + c = 3(73) + (200 – 73)

= 219 + 127 = 346 mL of CO_2

125. $2MH_2 + 2O_2 \rightarrow M_2O_2 + 2H_2O\uparrow$

2 eq. of $MH_2 \rightarrow$ 1 eq. M_2O_2

$2\left(\frac{0.8}{E+2}\right) = \frac{5.2}{2E+32}$

$3.2E + (1.6 \times 32) = 5.2E + 10.4$

$2E = 51.2 - 10.4$

$E = \frac{40.8}{2} = 20.4$

126. 10 molal → 10 moles in 1000 g

$\therefore \quad \frac{11.6}{58} = 0.2$ moles acetone in $\frac{0.2}{10} \times 1000$

= 20 g of solvent ≡ 10 molal

$$\text{Density} = \frac{20(g)}{22.8 \times 10^{-3}(L)} = 877 g\ L^{-1}$$

127. $\frac{x_2}{x_1} = \frac{3}{22} = \frac{n_2}{n_1}$

$$\therefore x_2 = \frac{3}{25} = 0.12$$

128. (i) $2Cu^{2+} + 4I^- \rightarrow \underset{\text{insoluble}}{2CuI} + I_2$

$I_2 + 2S_2O_3^{2-} \rightarrow S_4O_6^{2-} + 2I^-$

(ii) No. of grams of copper in alloy

$$= \frac{0.5 \times 60}{100} = 0.3$$

No. of moles of thiosulphate consumed

$$= \frac{0.3}{63.54} = 0.0047$$

($\because$ one mole Cu^{2+} requires one mole of $S_2O_3^{2-}$)

(iii) 30 mL contain $\frac{0.3}{63.54}$ moles of thio.

1000 mL contain $\frac{1000 \times 0.3}{63.54 \times 30} = 0.157$.

Molarity of thio = 0.157 M

129. No. of m. eq. Of $HClO_4 = 60 \times 1$ = No. of m. eq. of the base

Wt. of the base

$$= 60 \times \frac{\text{Eq. wt}}{1000} = 1.68 \Rightarrow \text{Eq. wt} = 28$$

$$\text{Acidity} = \frac{\text{Mol. wt}}{\text{Eq. wt}} = 3$$

130. (i) Total equivalents of HCl = 0.4 × 1 + 0.3 × 4 = 1.6

$$\text{Normality of acid mixture} = \frac{1.6}{0.4 + 0.3}$$

$$= \frac{1.6}{0.7} = 2.286$$

(ii) Milli equivalents of $Ca(OH)_2$ in mixture

= 20 × 0.7 + 25 × 0.4 = 14 + 10 = 24

(iii) $V \times 2.286 = 24 \Rightarrow V = \frac{24}{2.286} = 10.5$ mL

131. It is $N_2H_4.H_2SO_4$

132. The EF is C_3H_9N

All are monobasic with equivalent weight of hydrochloride 96.5

i.e, 1 g chloride require 0.105 eq $AgNO_3$

133. $\frac{x \times 18}{250} \times 100 = 51.2$

hence number of water molecules 7

Mol. mass of $MSO_4.7H_2O = M + 96 + 126 = 246$

M = 24

134. $2NaNO_3 \rightarrow 2NaNO_2 + O_2$

3.54×10^{21} O_2 molecule per g $NaNO_3$

135. Let the metal sulphate be MSO_4

$$\frac{M + 96}{M + 160} = \frac{1.368}{2}$$

Solving $MSO_4.2H_2O = 172$

136. $CaCl_2 \rightarrow 3.01 \times 10^{23}$ ions and 6.023×10^{23} ions

i.e., 0.5 moles of Ca^{2+} and 1 mole of Cl^-

$\therefore$ 0.5 moles of $CaCl_2$

Weight of $CaCl_2 = 111 \times 0.5 = 55.5$ g

137. Let M be the molar mass

$$\frac{59 \times 100}{M} = 13.05$$

Solving M = 452

138. $\frac{118.7}{M} \times 100 = 32.28$

139. Percentage of C

$$= \frac{12}{44} \times \frac{2.63}{0.858} \times 100 = 83.5\%$$

$\frac{83.5}{12} = 6.96 \quad 1 \times 3 = 3$

Percentage of H = $\frac{2}{18} \times \frac{1.28}{0.858} \times 100 = 16.57\%$

$\frac{16.57}{1} = 16.57 \qquad 2.38 \times 3 = 7$

$\therefore$ EF = C_3H_7

Lowest molecular weight of the compound

= 36 + 7 = 43

140. Equivalent weights are 7 and 14

valency ratio 2 : 1

141. CH_3COOH and $C_6H_{12}O_6$.

24 g C in 60 g of CH_3COOH

$\Rightarrow$ % of C $= \left(\frac{24}{60}\times 100\right) = 40\%$.

72 g C in 180 g (i.e., 72 + 12 + 96) of $C_6H_{12}O_6$.

$\Rightarrow$ % of C $= \frac{72}{180}\times 100 = 40\%$

142. Molar mass = $2 \times 71 = 142$ g mol^{-1}. Hence $C_{10}H_{22}$

143.

Oxide	Metal(%)	Oxygen(%)	Formula
1st	50	50	MO_2
2nd	40	60	$M_{\frac{4}{5}}O_{2\times\frac{6}{5}}$

The formula is MO_3

144. EF = $C_4H_{11}N$

145. Molar mass = 254

146. $8Al + 3NaNO_3 + 5NaOH + 2H_2O \rightarrow 8NaAlO_2 + 3NH_3$

147. Equivalent weight of metal chlorate = 152.2
Equivalent weight of metal chloride = 104.2

148. Let E be the Equivalent weight of metal
$E_2SO_3 \rightarrow E_2SO_4 \rightarrow BaSO_4$
104g E_2SO_3 gives $\rightarrow$ 233.4g $BaSO_4$

149. 2 mol NaCN reacts with every mol of Ag

150. Normality of the basic solutions
$= \frac{15.6\times 2}{50} = 0.0624\,N$
1M Borax gives 2N OH^-
hence weight/litre = 11.92 g

151. HCl is not a reducing agent

152. $\underset{50L}{CH_4} + 2O_2 \rightarrow \underset{50L\ (2.23mol)}{CO_2} + 2H_2O$

$C_2H_6 + 3\frac{1}{2}O_2 \rightarrow 2CO_2 + 3H_2O$

2.23mol CO_2 is produced from $1/2 \times 2.23$ mole C_2H_6.

153. mol mass of CaH_2 = 42
$\frac{42}{4} = 10.5$ kg

154. There is no change in oxidation number during the reaction

155. $\underset{+5}{NO_3^-} \rightarrow \underset{+2}{NO}$
For one mole of NO_3^-, 3 moles of e^- are required.

156. N_2H_4
N_2^{2-} loses 10 electrons. So one N^{2-} loses $5e^-$.
Therefore oxidation number of N in Y should be $(-2) - (-5) = +3$

157. The reactants in balanced equation are
$5Na_2N_2O_2 + 8KMnO_4 + 12H_2SO_4 \rightarrow$

158. 1 mol $Na_2S_2O_3$ reacts with 4 mole Cl_2.
Mol mass of $Na_2S_2O_3$ = 158

159. $2HgCl_2 + SnCl_2 \rightarrow Hg_2Cl_2 + SnCl_4$
$Hg_2Cl_2 + SnCl_2 \rightarrow 2Hg + SnCl_4$
After complete reaction metal ions present are Sn^{4+} only.

160. Since potassium of salt is isomorphous with K_2SO_4 O.S of element is + 6 hence in + 4 State equivalent weight $= 13.16 \times 1.5 = 19.74$

161. $\frac{E+35.5}{143.5} = \frac{5.2}{10}$ where n is equivalent weight of metal
Solving E = 39.12
Equivalent weight of carbonate = 69.12

162. % of carbon = 83.5%
% of hydrogen = 16.5%
Empirical formula = C_3H_7
Molecular formula of possible hydrocarbon is C_6H_{14}
Molecular weight = 86

163. 1200g solution contains 318 g Sodium carbonate, hence % strength = 26.5

164. $d = \frac{M\times 1000}{1000d - MM^1}$

165. $2KClO_3 \rightarrow 2KCl + 3O_2$
0.533 mol KClO3 produces 0.8 mol O2 and 0.533 mol KCl
0.467 mol KClO3 is converted to KClO4 according to the equation
4KClO3 $\rightarrow$ 3$KClO_4$ + KCl
No. of moles of KClO3 formed = 0.35
No. of moles of KCl = 0.1168
Mole fraction of $KClO_4$
$= 0.35 / (0.533 + 0.1168 + 0.35) = 0.35$

166. Number of milli equivalents of acid reacted
$= (50 \times 0.491) - 32.71 \times 0.435 = 10.32$

167. Since I^- is oxidised to ICl
conc. of $I^- = 2 \times 10^{-3}$ M or $2 \times 2 \times 10^{-3}$ N
Number of equivalents of oxidizing agent
$= \frac{4\times 10^{-3}}{1000}\times 100$

168. 1 mol $CuSO_4$ liberate iodine sufficient to react with 1 mol thio

$$\frac{0.3}{63.6}\text{ mol Cu} = 30 \times 1.572 \times 10^{-4}$$

169. Normality of resultant solutions

$$= \frac{(4\times3)+(6\times1)}{5} = 3.6$$

$= 3.6 \times V = 10 \times 6$ hence V

170. Equivalent weight of the acid = 10

171. Statement (1) and statement (2) are correct
Statement (2) is the explanation for 1

172. Statement (1) and (2) are correct

173. Statement (1) is correct and (2) is wrong

174. Statement (1) is correct and 2 is wrong

175. Statement (1) and (2) are correct.
Statement (2) is not the correct explanation.

176. Statement (1) and (2) are correct.
Statement (2) is not the correct explanation.

177. Statement (1) is false and (2) is true.

178. Statement (1) and statement (2) are correct
Statement (2) is the correct explanation for 1

179. Statement (1) false and (2) is true.

180. Statement (1) is false and (2) is true.

181. $AgNO_3$ is the excess reagent

182. MnO_2 is the limiting reactant

183. $Fe_3O_4 + 4H_2 \rightarrow 3Fe + 4H_2O$
232g Fe_3O_4 reacts with 4 mole H_2

184. The equivalent weight of $KMnO_4$ are

$$\frac{158}{1}, \frac{158}{3} \text{ or } \frac{158}{5} \text{ only}$$

185. $KMnO_4$ is not used as a primary standard

186. Normality of mixture $= \frac{(30\times0.2)-(20\times0.1)}{50} = 0.08$

$0.08 \times 50 = .4 \times v$

$v = 10$ ml

187. The co-efficients of balanced equation are
$H_2SO_4 + 8HI \rightarrow$ products

188. Equivalent weight of $KMnO_4 = 31.6$
Equivalent weight of Mohrs salt = 392
Hence 1g $KMnO_4$ = 12.4g Mohr salt

189. ON changes by 6 units per molecule of $K_2Cr_2O_7$

190. (b), (c), (d)
ON 1 is not exhibited by chromium

191. (a), (c), (d)
MnO_2 & HCl forming Cl_2 is a redox reaction.

192. Eq. wt of I^- when converted to

$$I_2 = \frac{\text{atomic weight}}{\text{change in oxidation state}}$$

$$= \frac{127}{1} = 127$$

Eq wt of I^- when converted to

$$I^+ = \frac{\text{atomic weight}}{\text{change in oxidation state}} = \frac{12}{2}$$

$= 63.5$

193. (a) P_4 disproportionates to PH_3 and NaH_2PO_2
(b) SO_2Cl_2 is reduced by P_4 to SO_2 and S_2Cl_2
(c) Cl_2 disproportionates to Cl^- and ClO^-
(d) $S_2O_3^{2-}$ disproportionates to S and SO_2

194. (b), (c)
They are not disproportionation reaction, but the element udnergoing oxidation and reduction is the same

195. (b), (d)
S do not exhibit +1 or +1.33 O.S in its oxides

196. (a), (b), (d)
The % by weight of solute is not 12

197. (a), (b), (c)
45 ml 0.25 N $Ba(OH)_2$ is more than required to neutralize the mixture

198. (a) $\rightarrow$ (r), (s)
(b) $\rightarrow$ (p), (r), (s)
(c) $\rightarrow$ (p), (q)
(d) $\rightarrow$ (s)

199. (a) $\rightarrow$ (q), (s)
(b) $\rightarrow$ (p), (s)
(c) $\rightarrow$ (s)
(d) $\rightarrow$ (p), (r)

200. (a) $\rightarrow$ (p), (r), (s)
(b) $\rightarrow$ (p), (q), (r), (s)
(c) $\rightarrow$ (p), (s)
(d) $\rightarrow$ (p), (q), (r), (s)

This page is intentionally left blank.

CHAPTER

2

STATES OF MATTER

CHAPTER OUTLINE

Preview

STUDY MATERIAL

Introduction

Gas laws
- Concept Strand (1)

Absolute Density and Relative Density of a Gas
- Concept Strand (2)

Units of Pressure and Volume

Value of R in Different Units

Dalton's Law of Partial Pressure
- Concept Strands (3-5)

Graham's Law of Diffusion
- Concept Strands (6-17)

Kinetic Theory of Gases

Molecular Velocity Distribution Among Gas Molecules
- Concept Strands (18-22)

Real Gases

Modification of Ideal Gas Equation: Van Der Waals Equation
- Concept Strands (23-29)

Critical State of a Gas

Relative Humidity (RH)

TOPIC GRIP
- Subjective Questions (10)
- Straight Objective Type Questions (5)
- Assertion–Reason Type Questions (5)
- Linked Comprehension Type Questions (6)
- Multiple Correct Objective Type Questions (3)
- Matrix-Match Type Question (1)

IIT ASSIGNMENT EXERCISE
- Straight Objective Type Questions (80)
- Assertion–Reason Type Questions (3)
- Linked Comprehension Type Questions (3)
- Multiple Correct Objective Type Questions (3)
- Matrix-Match Type Question (1)

ADDITIONAL PRACTICE EXERCISE
- Subjective Questions (10)
- Straight Objective Type Questions (40)
- Assertion–Reason Type Questions (10)
- Linked Comprehension Type Questions (9)
- Multiple Correct Objective Type Questions (8)
- Matrix-Match Type Questions (3)

INTRODUCTION

There exist many different kinds of substances in nature which are generally referred as "matter". These substances may be divided into three important categories, namely, solids, liquids and gases and are known as the three states of matter. Some characteristic features that distinguish the three states of matter are tabulated below.

Table 2.1

Gases	Liquids	Solids
The volume of a gas is very sensitive to changes in temperature and pressure.	Liquids have no definite shape but take the shape of the vessel in which they are placed.	Solids have a definite shape for a given mass/ volume.
Gases tend to occupy the available space completely and have no bounding surface.	Liquids have a surface unlike gases.	Solids have a definite bounding surface.
Gases are highly compressible compared to liquids and solids.	Liquids are more denser than gases.	The change in volume of solids on application of pressure or change of temperature is comparatively much lesser than that for gases

Thus the gaseous state is distinguished from the solid and liquid states by

(i) Indefinite expansivity
(ii) High compressibility
(iii) Very low density under moderate pressures and temperatures
(iv) Easy diffusivity
(v) Homogeneity (in pure mixed states)
(vi) Ability to exert pressure on the walls of the container
(viii) Complete disorder of structure and translational molecular motion.

Thus gas can be defined as, a homogeneous substance whose volume increases continuously at constant temperature as the pressure is decreased.

An important property by gases is their diffusion into each other to form a homogeneous mixture when two or more gases are brought into contact in any proportion.

Intermolecular forces operating among gaseous molecules

"The force of attraction or repulsion operating between molecules is known as intermolecular force". This does not include the electrostatic force of attraction between two oppositely charged ions or the force that holds atoms in a molecule. The different types of intermolecular forces are ionic, covalent, metallic and van der Waal's forces.

van der Waals force: This include dispersion or London forces, dipole-dipole forces and dipole-induced dipole forces.

(a) Dipole–dipole forces

This type of forces occur between *neutral molecules having permanent dipole moment.* When a bond formed between two atoms leaves partial charges on the atoms, it is known as a polar bond. HCl for example, is a polar molecule in which the shared electron pair is closer to "Cl" than "H" because of higher electronegativity of "Cl" atom. This electric dipole interacts with another dipolar molecule. This was studied by Keesom who showed that polar molecules would find to orient themselves so that oppositely charged ends of different molecules were adjacent, leading to electrostatic attraction extending through out the volume of the sample. This orientation effect is opposed by thermal agitation with the result that this (Keesom) interaction energy is inversely dependent on temperature.

The interaction energy is proportional to r^{-6}, for rotating polar molecules and proportional to r^{-3} for stationary polar molecules where "r" is the distance between the two molecules.

(b) Dipole–induced dipole forces

In the presence of a molecule having permanent dipole moment, a non polar molecule in the vicinity gets polarised. The interaction between the permanent dipole and induced dipole is known as dipole – induced dipole forces. The magnitude of the interaction depends on the strength of the permanent dipole and polarisability of non polar molecule and also proportional to r^{-6} where r is distance between the two molecules. This was studied by Debye who showed that this interaction energy is quite small in magnitude compared to the Keesom interaction.

(c) Dispersion or London forces

This type of van der Waals forces exist in all molecules including non polar molecules like O_2, N_2, He, Ar etc. The oscillation of electrons in a molecule with respect to nuclei results in an unsymmetrical distribution of charge in the molecule. Thus a non polar molecule becomes polarized momentarily and induces a dipole moment in neighbouring molecule. The force of attraction between the induced dipoles among polarized molecules are known as **London forces**.

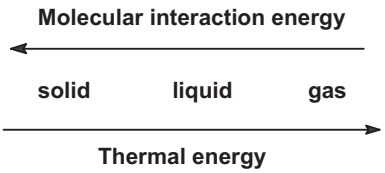
The interaction energy due to this is *inversely proportional to sixth power of distance between interacting particles i.e.,* $\left(\alpha \frac{1}{r^6}\right)$ *where "r" is the distance between two molecules.*

The oscillations of the neighbouring molecular dipoles are in phase and in contrast to the Debye and Keesom interactions, these are additive. Even at the limit of $T \rightarrow$ zero, these oscillations do not cease because of the existence of zero point energies of atoms/molecules.

In addition to these forces there is an interaction of repulsion between atoms/molecules when the interatomic or intermolecular distance is extremely small. This repulsion energy is taken to be r^{-12}

Intermolecular forces and thermal energy

The three states of matter are the result of a competition between the molecular interaction energy, which tends to keep molecules together and thermal energy keeps them away. The competition between the two types of interaction decides whether a substance is a gas, liquid or solid.



Thermal energy is the energy possessed by molecules as a result of their temperature and it is also a measure of the thermal motion. Molecular interaction energies comprise of intermolecular attractive as well as repulsive forces.

GAS LAWS

There are two important laws connecting the temperature, pressure and volume of gases, which are known as Boyle's law and Charle's law or Gay-Lussac's law.

Boyle's law

This law (proposed by R.Boyle in 1662) states that the *volume of a given mass of gas is inversely proportional to pressure at constant temperature.*

Mathematically, it may be expressed as $P \propto \frac{1}{V}$ — (1)

where P and V represent the pressure and volume respectively of a given mass of gas at a definite temperature. Equation (1) may be written as

$PV = \text{constant}$ — (2)

The value of the constant depends on the nature of the gas, its mass and temperature. Equation (2) may also be put in another form

$$P_1V_1 = P_2V_2$$

where V_1 refers to the volume of given mass of gas when its pressure is P_1 and V_2 refers to the volume of the same mass of gas when its pressure is changed to P_2, all these quantities referring to the same temperature.

The pressure-volume relationship at a given temperature (also known as the ***isotherm***) is represented graphically in the Fig. 2.1

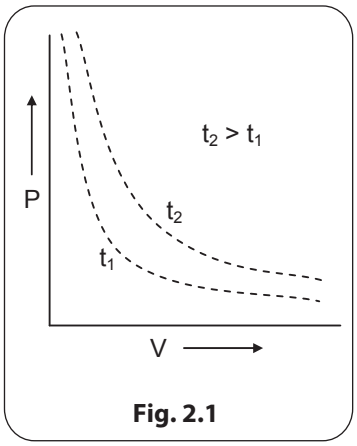


Fig. 2.1

The above drawn isotherm is also known as a ***rectangular hyperbola***.

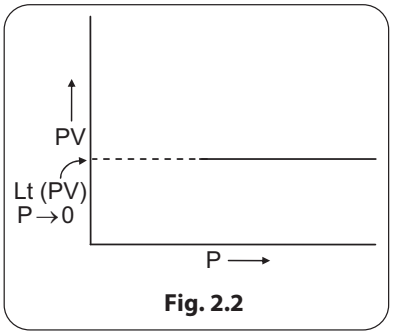


Fig. 2.2

Since PV is a constant for a given mass of a gas at constant temperature, the plot of PV against P isotherm is a straight line parallel to the pressure axis.

Also, $P \propto \frac{1}{V}$ at constant temperature, then the plot of P against $\frac{1}{V}$ isotherm shows linear variation.

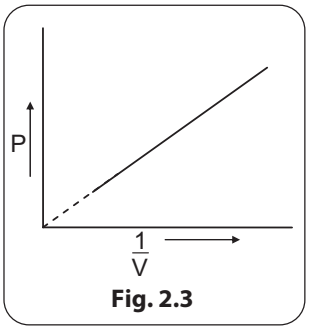


Fig. 2.3

Gay-Lussac's law (also referred to as Charle's law):

This law states that at constant pressure, the volume of a fixed mass of gas expands by the same fraction of its volume at 0°C per unit rise in temperature. If V_o and V_t are the volumes occupied by a definite mass of gas at 0°C and t°C respectively at the same pressure, according to Gay-Lussac's law

$$V_t = V_0(1 + \alpha t)$$

$$\text{or} \quad \alpha = \frac{V_t - V_0}{V_0 t} \qquad \text{— (3)}$$

α is known as the coefficient of expansion and is the same for all gases, the value being found experimentally to be equal to $\frac{1}{273.1}$ per °C. Substituting this value in equation (3), we get,

$$\frac{V_t}{V_0} = \frac{273.1 + t}{273.1} \qquad \text{— (4)}$$

If a new temperature scale is defined such that it starts at –273.1°C, one can write

T (in the new scale) = 273.1+ t

T_0 (in the new scale) = 273.1

The new scale of temperature represented as K is known as absolute temperature scale. The K is named in honour of Lord Kelvin. Equation (4) may now be written as

$$\frac{V_T}{V_0} = \frac{T}{T_0} \qquad \text{— (5)}$$

From equation (5), Charle's law may be stated in another way. At constant pressure, the volume of a given mass of gas is directly proportional to absolute temperature. Equation (5) may thus be written as

$$\frac{V_1}{T_1} = \frac{V_2}{T_2} \quad \text{or} \quad \frac{V}{T} = \text{constant} \qquad \text{— (6)}$$

V_1 and V_2 correspond to the volumes at the absolute temperatures T_1 and T_2 respectively. Rearranging of equation (6) gives

$$\frac{V_1}{V_2} = \frac{T_1}{T_2}$$

If T_1 = 0K, the volume V_1 of the gas should be zero. A plot of V against the absolute temperature T for a given mass of gas at constant pressure (known as ***isobar***) is linear passing through the origin as shown in Fig. 2.4.

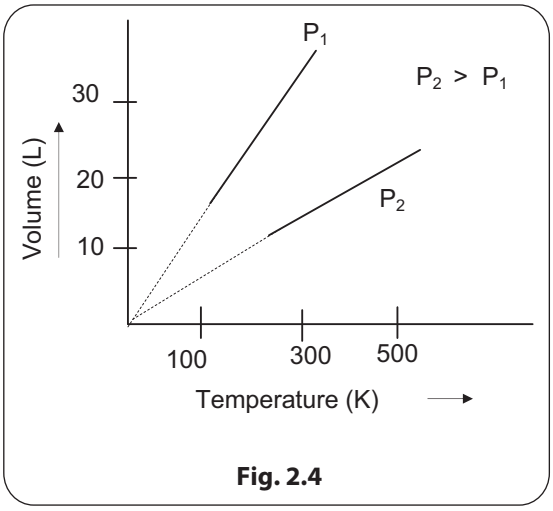


Fig. 2.4

For every isobar, the lower the pressure, greater is the slope and all these lines converge identically toward an intercept of the horizontal axis at 0K i.e., they extrapolate to zero volume at zero Kelvin. (Fig.2.5)

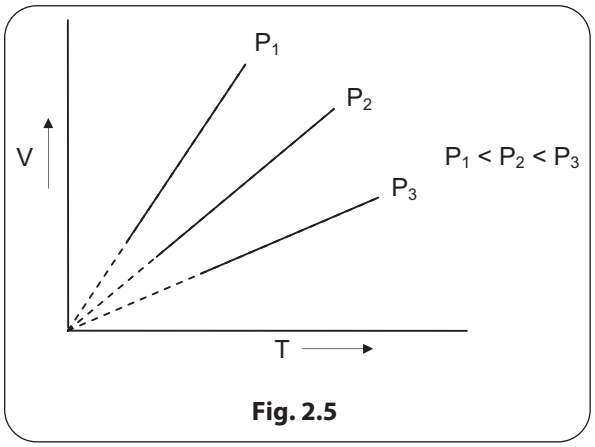


Fig. 2.5

If a plot of V against the temperature in °C is made, the straight line will not pass through the origin but will cut the temperature axis at –273.1°C where V = 0. This variation is shown in Fig. 2.6.

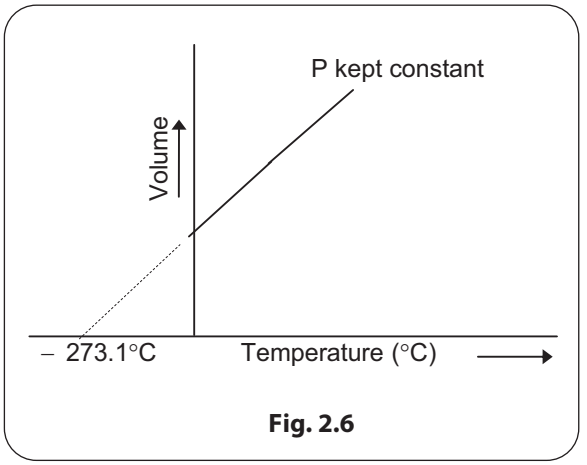


Fig. 2.6

CONCEPT STRAND

Concept Strand 1

An ideal gas initially occupying 20 L at 27^0C and 2 atm pressure is allowed to expand into an evacuated vessel such that the final volume is 50 L. If the pressure were to remain at the same value of 2 atm, to what temperature should the gas be heated?

Solution

Applying Charles' Law, $\frac{V_1}{T_1} = \frac{V_2}{T_2} \Rightarrow \frac{20}{300} = \frac{50}{T_2}$

$\Rightarrow \quad T_2 = \frac{300 \times 50}{20} = 750K$

(or) t (in ^{0}C) = 750 – 273 = 477°C

Combination of Boyle's and Charle's laws

By combining Boyle's and Charle's laws, a new relation is obtained which connects the pressure, volume and temperature of a given mass of gas and it is known as the **equation of state**. Let P, V and T represent the pressure, volume and temperature of gas. Then

By Boyle's law, $V \propto \frac{1}{P}$ (at constant temperature) and by Charle's law $V \propto T$ (at constant pressure)

or $V \propto \frac{T}{P}$ from which $\frac{PV}{T}$ = constant or PV = kT — (7)

where, k is a proportionality constant. Equation (7) may be written in a different form as

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2} = \frac{P_3V_3}{T_3} \quad \text{— (8)}$$

where P_1, V_1 correspond to the temperature T_1 and P_2, V_2 correspond to the temperature T_2 and so on.

Relation between Pressure and Temperature

From equation (8), it is seen that for a definite mass of gas at constant volume, the relation is

$$\frac{P_1}{T_1} = \frac{P_2}{T_2} = \frac{P_3}{T_3} \quad \text{or} \quad \frac{P}{T} = \text{constant}$$

Thus the pressure of a given mass of gas at constant volume is directly proportional to absolute temperature. A plot of the pressure against absolute temperature T (for a given mass of gas) at constant volume is thus linear, such a graph is known as an ***isochore.***

For every isochore, the lower the volume greater is the slope and these lines converge identically toward an

intercept of the horizontal axis at 0K i.e., they interpolate to zero pressure at zero Kelvin.

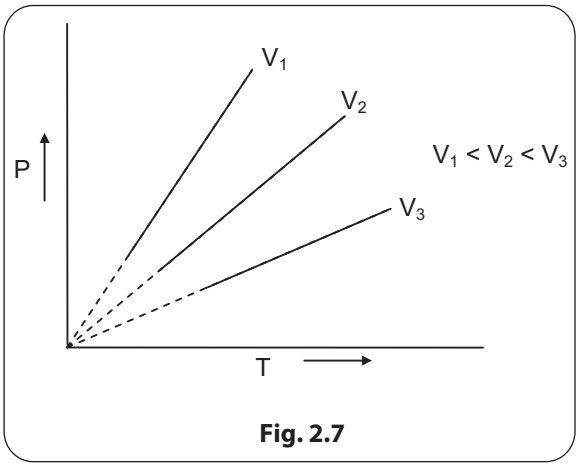


Fig. 2.7

The change of pressure or volume of a given mass of gas, with temperature forms the basis of construction of gas thermometers.

Interpretation and evaluation of the constant k of equation (7)

The volume of a gas V, is proportional to the amount of gas taken at a given temperature and pressure. Since V is proportional to k under the above conditions, one may write that k ∝ amount of gas taken.

Making use of Avogadro's hypothesis which states that one gram molecule of any gas occupies the same volume under conditions of constant pressure and temperature, it may be stated that the value of k is the same for one gram molecule of all gases independent of the conditions under which the gases are studied. The value of k, referred per gram molecule of a gas, is represented by R and is known as the ***molar gas constant*** (it is a universal constant) and it has the same value for all gases. Equation (7) may now be written as

$$PV = RT \qquad \text{— (9)}$$

where, V is the volume occupied by one-gram molecular weight of a gas under a pressure P and temperature T. Equation (9) is known as the ideal gas equation.

Important points regarding equation (9)

(i) It is applicable for one mole of an ideal gas.

(ii) For n moles of an ideal gas, equation (9) is written as

$$PV = nRT \qquad \text{— (10)}$$

where, P and V refer to the pressure and volume of 'n' moles of the gas at the temperature T.

(iii) If there are W g of a gas of molecular weight M, equation (10) may be written as

$$PV = \frac{W}{M}.RT$$

or
$$P = \frac{W}{V}.\frac{RT}{M} = d.\frac{RT}{M} \qquad \text{— (11)}$$

where d is the density of the gas in g/L. From equation (11), it is seen that P ∝ d for a given gas at constant temperature T.

(iv) There are two important temperatures we refer to for standard conditions. If we have 273K and 1 atmosphere pressure, we call it as standard temperature and pressure (STP) and if we use 298K and 1 atmosphere pressure, we call this as standard ambient temperature and pressure (SATP).

Dimensions of the gas constant, R

The gas constant, R, has the dimensions of energy per mole and its numerical value varies with the unit chosen.

ABSOLUTE DENSITY AND RELATIVE DENSITY OF A GAS

The absolute density of a substance is defined as mass per unit volume in g cm^{-3} or g/L. The relative density (or specific gravity) is the ratio of the absolute density of a substance and that of a standard substance. In the case of solids and liquids, water at 4°C is taken as the standard because its density is one at this temperature. In the case of gases and vapours, the relative density is commonly referred as vapour density, the standard of reference being hydrogen gas.

The vapour density of a gas is defined by the equation,

$$\text{Vapour density} = \frac{\text{wt. of a certain volume of a gas or vapour at a given temperature and pressure}}{\text{wt. of the same volume of hydrogen at the same temperature and pressure}}$$

In calculating the relative density of a gas (or vapour), a volume of 11.2 L of hydrogen at STP is taken as the unit as its weight is close to unity at STP. Based on this unit, the vapour density is defined as

$$\text{Vapour density or relative density} = \frac{\text{(wt. of 11.2 L of a gas at the given temperature and pressure)}}{\text{(wt. of 11.2 L of hydrogen gas at the same temperature and pressure)}}$$

Since the weight of 11.2 L of hydrogen at STP is unity, the absolute density of a gas in g L^{-1} is obtained by dividing the relative density of a gas at STP by 11.2.

Since 1 mole of any gas at STP occupies 22.4 L, the mass of half a mole corresponds to 11.2 L. i.e.,

$2 \times$ vapour density = molar mass.

CONCEPT STRAND

Concept Strand 2

The vapour density of partially dissociated iodine vapour at a certain temperature was found to be 80. What is its degree of dissociation? [Molecular weight of $I_2 = 254$]

Solution

Let α be the degree of dissociation under equilibrium.

$$\underset{1-\alpha}{I_{2(g)}} \rightleftharpoons \underset{2\alpha}{2I_{2(g)}}$$

Vapour density of undissociated vapour

$$(d_0) = \frac{254}{2} = 127$$

Number of moles at equilibrium $= 1 - \alpha + 2\alpha = 1 + \alpha$

$$\frac{\text{Vapour density of undissociated } I_2}{\text{Vapour density of equilibrium mixture}} = \frac{d}{d_0} = \frac{1}{1+\alpha}$$

$$\text{Degree of dissociation } \alpha = \frac{d_0 - d}{d} = \frac{127 - 80}{80} = 0.587$$

UNITS OF PRESSURE AND VOLUME

SI and CGS units of pressure are N m^{-2} (or pascals, Pa) and dyne cm^{-2} respectively.

Other units of pressure are atmospheres (atm), centimeters of Hg, millimeters of Hg (torr) and bar.

1 atm = 76 cm of Hg pressure (hdg = $76 \times 13.6 \times 981$ dyne cm^{-2}) = 760 torr

1 atm = 1.013×10^6 dyne cm^{-2} = 1.013×10^5 Pa
1 bar = 10^5 Pa

SI and CGS units of volume are m^3 and cm^3 (cc) respectively.

Another unit of volume is Litre (dm^3)
10^3 dm^3 = $1m^3$ (or) dm^3 = 10^{-3} m^3

VALUE OF R IN DIFFERENT UNITS

(a) R in the units of L-atm K^{-1} mol^{-1}

$$R = \frac{P \times V}{T} = \frac{1 \times 22.4}{273}\left(\frac{\text{atm} \times \text{L mol}^{-1}}{\text{K}}\right)$$

$= 0.0821$ L-atm $K^{-1}mol^{-1}$

(b) R in C.G.S units

$$R = \frac{P \times V}{T} = \frac{76 \times 13.6 \times 981 \times 22400}{273}$$

$$\left[\frac{\text{dynes}}{\text{cm}^2} \times \frac{\text{cm}^3}{\text{mole}} \times \frac{1}{\text{K}}\right] = 8.314 \times 10^7 \text{ erg } K^{-1} \text{ mol}^{-1}$$

(c) R in cal K^{-1} mol^{-1}

1 cal = 4.184 J = 4.184×10^7 erg

Or, $R = \frac{8.314 \times 10^7}{4.184 \times 10^7} = 1.987$ cal $K^{-1} mol^{-1}$

(d) R in SI units

We have, $R = 8.314 \times 10^7$ erg K^{-1} mol^{-1}

But 10^7 ergs = 1 Joule

R = 8.314 Joule K^{-1} mol^{-1}

DALTON'S LAW OF PARTIAL PRESSURE

When we are dealing with a mixture of gases, the relation between the total pressure P of the mixture and the pressure of the individual gases is of great importance. This relation is known as Dalton's law of partial pressure and *it states that the total pressure of a mixture of non reacting gases is equal to the sum of partial pressure of the component gases.*

Definition of partial pressure of gas

The partial pressure of a constituent 'c' in a mixture of gases is defined as the pressure the gas would exert if it alone occupies the whole volume of the mixture at the same temperature.

Consider a vessel of volume V containing a mixture of gases viz. n_1 mole of gas (1), n_2 mole of gas (2) and n_3 moles of gas (3) etc. Let the pressure exerted by gas (1) be P_1 when the vessel is occupied exclusively by it at the same temperature. Under the same conditions, let the pressure exerted by gas (2) be P_2 if it alone occupied the volume V. Extending the same arguments to gas (3), let the pressure exerted by it be P_3 under the conditions stated above. If the total pressure is P in the vessel containing the mixture of all gases, according to Dalton's law

$$P = P_1 + P_2 + P_3$$

If each of the gases in the mixture behaved ideally, one can write

$$P_1V = n_1RT \; ; \; P_2V = n_2RT \; ; \; P_3V = n_3RT \qquad \text{— (12)}$$

From Dalton's law, $(P_1 + P_{2+} P_3)V = (n_1 + n_2 + n_3)RT$

or $PV = nRT$ — (13)

where $P = P_1 + P_{2+} P_3$; $n = n_1 + n_2 + n_3$

Dividing each of the equation (12) by equation (13), we get

$$\frac{P_1}{P} = \frac{n_1}{n} = x_1; \qquad \frac{P_2}{P} = \frac{n_2}{n} = x_2; \qquad \frac{P_3}{P} = \frac{n_3}{n} = x_3$$

$$P_1 = x_1P; \; P_2 = x_2P; P_3 = x_3P \qquad \text{— (14)}$$

The fractions x_1, x_2 and x_3 are known as the mole fractions of the gases 1, 2 and 3 in the mixture. Equations (14) may be written in a general form,

partial pressure P_i of a component in the mixture = mole fraction of i × P

For gases, volume fraction = mole fraction

Usefulness of the law

Pressure of dry gas = Atmospheric pressure – aqueous tension at the same temperature .

CONCEPT STRANDS

Concept Strand 3

One litre of a gas was collected over water at 25^0C at a pressure of 750 torr. Given that the vapour pressure of water at this temperature is 3170 pascals. Calculate the mole fraction of the dry gas.

Solution

Vapour pressure of water in torr $= \frac{3170 \times 760}{1.013 \times 10^5} = 23.78$ torr

Partial vapour pressure of dry gas = 750 – 23.78 = 726.22 torr

Mole fraction of dry gas = $\frac{726.22}{750} = 0.968$

Concept Strand 4

Calculate the density of a gas mixture consisting of 0.20 mole fraction of methane, 0.50 mole fraction of nitrogen and 0.3 mole fraction of ethane at 37^0C and 2 atm pressure.

Solution

Partial pressures of $CH_4 = 0.20 \times 2 = 0.40$ atm, $N_2 = 0.50 \times 2 = 1.00$ atm, $C_2H_6 = 0.60$ atm

Using the relation $P_i = \frac{W_i}{V}.\frac{RT}{M_i}$,

$$P_{CH_4}M_{CH_4} + P_{N_2}M_{N_2} + P_{C_2H_6}M_{C_2H_6}$$

$$= RT\left(\frac{W_{CH_4}}{v} + \frac{W_{N_2}}{v} + \frac{W_{C_2H_6}}{v}\right) = RTd_{mix}$$

or $$d_{mix} = \frac{0.40\times16 + 1.0\times28 + 0.60\times30}{0.0821\times310}$$

$$= \frac{6.4 + 28 + 18}{25.45} = 2.06\,g\,L^{-1}$$

Concept Strand 5

A mixture of 56 g of nitrogen and 44 g of CO_2 occupies a volume of 10 L at 310K. Calculate the partial pressures and mole fractions of each gas in the mixture and the total pressure.

Solution

Moles of $N_2 = \frac{56}{28} = 2$; Moles of $CO_2 = \frac{44}{44} = 1$;

Mole fraction of $N_2 = \frac{2}{3}$;

Mole fraction of $CO_2 = \frac{1}{3}$

Total pressure $= \frac{3\times0.0821\times310}{10} = 7.63$ atm

$\therefore$ Partial pressure of $N_2 = \frac{2}{3}\times7.63 = 5.09$ atm,

Partial pressure of $CO_2 = 2.54$ atm

GRAHAM'S LAW OF DIFFUSION

Diffusion refers to the process of mixing of two or more gases uniformly with each other when they are brought into contact. Strictly, *it may be defined as the tendency of any gas to spread uniformly throughout the available space.* Diffusion occurs in liquids also but diffusion among gases occur more rapidly. Diffusion is often applied to the passage of gases through porous media. If a porous vessel filled with a gas like hydrogen is placed in air, hydrogen diffuses out through the pores and air will diffuse into the vessel from outside.

The process is referred to as *effusion* and is similar to diffusion. It pertains to the passage of gases through fine pores. Streaming of a gas through a pinhole also comes under this category.

According to Graham (1829), the rate of diffusion of a gas is inversely proportional to the square root of the density of the gas. If r_1 and r_2 are the rates of diffusion (ie., the Volumes in mL diffusing over the small time interval under the same conditions of temperature and pressure) of two gases (1) and (2) whose densities are d_1 and d_2 respectively, application of Graham's law yields

$$\frac{r_1}{r_2} = \sqrt{\frac{d_2}{d_1}} \qquad \text{— (15)}$$

Since the density of a gas is directly proportional to the molecular weight of the gas, M, equation (15) may be written as

$$\frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}} \qquad \text{— (16)}$$

where M_1 and M_2 are the molecular weights of the gases, (1) and (2) respectively.

Important conclusion from Graham's law

(i) A lighter gas will diffuse rapidly than a heavier one. This is best illustrated by considering the separation of U^{235} isotope from U^{238} isotope and is important from the point of view of atomic energy. This is achieved by taking advantage of the faster rate of diffusion of $U^{235}F_6$ than $U^{238}F_6$.

(ii) Graham's law is used to determine the densities of gases in the following way. The method involves the determination of the time required for a definite volume of a gas to diffuse through a small hole in a thin metallic plate. Next, the experiment is

repeated with a gas of known density under the same conditions.

If t_1 and t_2 are the times required for the same volume of gases (1) and (2) to diffuse through a small hole, we can write

$$\frac{t_1}{t_2} = \frac{r_2}{r_1} \qquad \text{— (17)}$$

where, x_1 and x_2 are the rates of diffusion of gases (1) and (2) since the time is inversely proportional to the rate of diffusion.

Combining equation (17) with equation (15) and (16),

$$\frac{t_1}{t_2} = \sqrt{\frac{d_1}{d_2}} = \sqrt{\frac{M_1}{M_2}} \qquad \text{— (18)}$$

Equation (18) may be used to determine the density or the molecular weight of a gas if d and M of another gas are known.

It is possible to consider the rates of diffusion of two gases (1) and (2), where in V_1 mL and V_2 mL are the volumes effusing out over the small time interval 't' under the same conditions of temperature and pressure. For the rates of diffusion r_1 and r_2 of these gases, one can write

$$\frac{r_1}{r_2} = \frac{V_1 / t}{V_2 / t} = \frac{V_1}{V_2} = \sqrt{\frac{d_2}{d_1}}$$

Sometimes an equation involving the pressures of the gases is useful in calculating the rates of diffusion. Equation (15) may be written in such case as

$$\frac{r_1}{r_2} = \sqrt{\frac{d_2}{d_1}} = \frac{P_1}{P_2}\sqrt{\frac{M_2}{M_1}}$$

where, P_1 and P_2 are the pressures of gases (1) and (2) and M_1 and M_2 are their molecular weights.

CONCEPT STRANDS

Concept Strand 6

Two 5L flasks were attached to each other through a narrow tube of negligible volume. This is filled with 6 moles of carbon monoxide at room temperature (27°C). In an accident one of the flask breaks. After some time the other flask is sealed. Calculate the amount of carbon monoxide lost.

Solution

The unbroken flask is at 1 atmospheric pressure and 27°C

i.e., 300K V = 5 L

So n_{CO} remaining $= \frac{PV}{RT} = \frac{1 \times 5}{0.082 \times 300} = 0.2$ moles

Amount lost = 6 − 0.2 = 5.8 moles

Weight of CO = 5.8 × 28 = 162.4 g

Concept Strand 7

500 mL of sulphur dioxide effuses through a pinhole in 60 minutes. How much time does the same volume of methane take to effuse, under similar conditions of temperature and pressure? How much volume of helium will effuse in 10 minutes?

Solution

$$\frac{\text{Rate of effusion of } SO_2}{\text{Rate of effusion of } CH_4} = \frac{\left(\frac{500}{60}\right)}{\left(\frac{500}{t}\right)}$$

$$= \frac{t}{60} = \sqrt{\frac{d_{CH_4}}{d_{SO_2}}} = \sqrt{\frac{M_{CH4}}{M_{SO_2}}} \quad \text{(or)}$$

$$\frac{t}{60} = \sqrt{\frac{16}{64}} = \sqrt{\frac{1}{4}} = \frac{1}{2}; \quad \text{(or)} \quad t = 30 \text{ minutes}$$

Volume of Helium $\frac{r_{He}}{r_{CH_4}} = \frac{\frac{V_{He}}{10}}{\frac{500}{30}} = \sqrt{\frac{d_{CH_4}}{d_{He}}} = \sqrt{\frac{M_{CH_4}}{M_{He}}}$

$$\frac{V_{He} \times 3}{500} = \sqrt{\frac{16}{4} = 2}$$

(or) Volume of helium effusing in 10 minutes = 333.4 mL

Concept Strand 8

A molecule X_4 dissociates into X_2 upto 50% at 800°C. If the molecule diffuses 0.62 times as fast as oxygen. Calculate the atomic weight of X.

Solution

$$\frac{r_x}{r_{O_2}} = \sqrt{\frac{M_{O_2}}{M_x}} \quad (0.62)^2 = \frac{M_{O_2}}{M_x}$$

$$M_x = \frac{32}{0.62 \times 0.62} = 82.7 \text{ g mol}^{-1}$$

$$X_4 \rightarrow 2X_2$$

1

$1 - x \qquad 2x \qquad$ let M be molecular weight of X_4

$$82.7 = \frac{2 \times x \times M/2 + (1-x)M}{1+x} = \frac{0.5M + 0.5M}{1.5}$$

$$M = 82.7 \times 1.5 = 124.07 \text{ g}$$

$$\text{Atomic weight of X} = \frac{124}{4} = 31 \text{ g mol}^{-1}$$

Concept Strand 9

Oxygen gas at 27°C is heated in vessel which allows gas to diffuse out only. After 15 min. only $\frac{3}{5}$ of the gas remains in the vessel. Calculate the final temperature.

Solution

In the given case, volume and pressure are constant. Therefore nT is a constant

$$n_1T_1 = n_2T_2$$

$$1 \times 300 = \frac{3}{5} \times T_2$$

$$T_2 = \frac{5}{3} \times 300 = 500\text{K}$$

Concept Strand 10

A mixture of CH_4 and HBr effuses with equal initial rates for both gases at TK. Calculate the mole fractions of the gases in the mixture.

Solution

$$\frac{r_{CH_4}}{r_{HBr}} = \frac{P_{CH_4}}{P_{HBr}}\sqrt{\frac{M_{HBr}}{M_{CH_4}}} \qquad 1 = \frac{P_{CH_4}}{P_{HBr}}\sqrt{\frac{81}{16}} \qquad \frac{P_{CH_4}}{P_{HBr}} = \frac{4}{9}$$

$$\therefore \quad \text{Mole fraction of } CH_4 = \frac{4}{13}$$

$$\text{Mole fraction of HBr} = \frac{9}{13}$$

Concept Strand 11

A gas stream having a composition 75% methane and 25% hydrogen by volume leaks by effusing through a porous section of a pipe through which it is being pumped. Calculate the composition of the gas lost by leakage.

Solution

$$\frac{\text{Rate of effusion of } H_2}{\text{Rate of effusion of } CH_4} = \sqrt{\frac{M_{CH4}}{M_{H2}}} \Rightarrow \sqrt{\frac{16}{2}} = 2.83$$

Since the original ratio of CH_4 to H_2 is 75 : 25

i.e., 3 : 1

The ratio of H_2 to CH_4 coming out = $2.83 \times 1 : 3 \times 1$

$= 2.83 : 3$

$$\%H_2 \text{ leaking out} = \frac{100 \times 2.83}{5.83} = 48.54\,;$$

$\%CH_4$ leaking out = 51.46

Concept Strand 12

Consider a 5 litre vessel containing 10^{20} molecules of oxygen, each moving with velocity 602 m s^{-1}. If all of them are moving towards the wall of the container simultaneously, what is the force exerted on the container walls if each molecule make one collision per second?

Solution

Force = rate of change of momentum

momentum of hitting molecule = mu

momentum of rebounding molecule = –mu

change of momentum = mu – (–mu)

$= 2mu$

total change in momentum due to all the molecules

$$= 2 \times \frac{32 \times 10^{-3} \text{ kg mol}^{-1}}{6.02 \times 10^{23} \text{ mol}^{-1}} \times 602 \text{ m s}^{-1}$$

$$= 6.4 \times 10^{-3} \text{ kg m s}^{-1}$$

$$\text{rate of change of momentum} = \frac{\text{change of momentum}}{\text{time taken}}$$

$$= 6.4 \times 10^{-3} \text{ kg m s}^{-2}$$

$$= 6.4 \times 10^{-3} \text{ N}$$

Concept Strand 13

The reaction between gaseous NH_3 and HBr produces white solid NH_4Br. Suppose a small quantity of NH_3 & HBr are introduced simultaneously into opposite ends of an open tube with one metre long. Calculate the distance of white solid formed from the end which was used to introduce NH_3

Solution

Let 'x' be the distance of white solid from NH_3 end

$$\frac{r_1}{r_2} = \frac{x}{100 - x} = \sqrt{\frac{M_{HBr}}{M_{NH_3}}}$$

since rate of diffusion are proportional to the distance

$$\frac{x}{100 - x} = \sqrt{\frac{81}{17}} = 2.18$$

$$x = 68.55 \text{ cm}$$

Concept Strand 14

At 27°C H_2 is leaked out through a tiny hole in a vessel for 20 minutes. Another unknown gas at the same temperature and pressure also leaked through the same hole for 20 minutes. The effused H_2 and unknown gas collected in a 3 litre vessel at the same temperature exerted a total pressure 6 atmosphere and it was found to contain 0.7 moles of H_2. Calculate the molecular mass of unknown gas

Solution

Let P_{H_2} and P_{un} be the partial pressures of the gases

$$P_{H_2} = \frac{0.7}{3} \times 0.0821 \times 300 = 5.747 \text{ atmosphere}$$

$$P_{un} = 6 - 5.747 = 0.253 \text{ atmosphere}$$

$$\text{Total number of moles of gases} = \frac{PV}{RT} = \frac{6 \times 3}{0.0821 \times 300}$$

$$= 0.73082$$

Number of moles of unknown gas = 0.73082 − 0.7 = 0.03082

$$\text{rate of effusion } \frac{r_{H_2}}{r_{un}} = \sqrt{\frac{M_{(un)}}{M_{H_2}}}$$

$$\text{i.e., } \frac{0.7/20}{0.3082/20} = \sqrt{\frac{M_{(un)}}{M_{H_2}}}$$

$$M = 1032$$

Concept Strand 15

100 cm^3 of NH_3 diffuses through a pin hole in 32.5 seconds. How much time will 60 cm^3 of N_2 take to diffuse under the same conditions?

Solution

$$\frac{r_{NH_3}}{r_{N_2}} = \sqrt{\frac{28}{17}}$$

$$\frac{100/32.5}{60/t} = \sqrt{\frac{28}{17}}$$

$$\frac{100}{32.5} \times \frac{t}{60} = \sqrt{\frac{28}{17}}$$

$$t = 25 \text{ seconds}$$

Concept Strand 16

The ratio of root mean square velocity of 5 moles of nitrogen and 8 moles of methane is 1 : 3. Calculate the ratio between their temperatures

Solution

$$\frac{C_{rms}(N_2)}{C_{rms}(CH_4)} = \frac{1}{3}$$

$$\sqrt[3]{\frac{3RT_{N_2}}{28}} = \sqrt{\frac{3RT_{CH_4}}{16}}$$

$$9 \times \frac{T_{N_2}}{28} = \frac{T_{CH_4}}{16} \quad \frac{T_{N_2}}{T_{CH_4}} = \frac{28}{9 \times 16} = \frac{0.19}{1}$$

$$\cong 0.2 : 1 \text{ i.e., } 1 : 5$$

KINETIC THEORY OF GASES

In the earlier section, the properties of ideal gases viz., gases that obey Boyle's and Charles's laws have been considered in detail. It is now proposed to consider a model or a theory known as 'kinetic theory of gases, which enables us to interpret the gaseous behaviour and coordinate the various gas laws.

There are certain basic ***postulates*** of the theory which are listed below:

(i) The gas consists of a large number of very minute particles (atoms or molecules) moving about in all directions randomly i e., all directions of motions are equally probable.

(ii) Due to continuous movement, the particles (or the molecules) frequently collide with each other. However, no loss of total translational energy results during these collisions. The collisions are, hence, said to be elastic.

(iii) The pressure exerted by the gas is due to the elastic collisions of the gas molecules on the walls of the container.

(iv) There are no forces of attraction or repulsion between gas molecules.

(v) The molecules of the gas are very small and their actual volume is negligible in comparison to the total volume of the gas.

(iv) Gravitational field has no influence on molecular motion.

Using the postulates mentioned above, it is possible to derive an equation for the pressure of the gas in terms of the mass 'm' of the gas molecule, 'n' the number of molecules in a total volume, V, and the mean square velocity of gas molecules, $\overline{c^2}$.

The kinetic theory expression known as kinetic gas equation is

$$P = \frac{1}{3}\frac{mn\overline{c^2}}{V} \qquad \text{— (19)}$$

where, m = mass of gas molecule (g), n = number of molecules in the volume V, $\overline{c^2}$ = mean square velocity of gas molecules ($cm^2\ sec^{-2}$) and V = volume of the vessel (or gas) (cm^3). The units of p from the above equation may be shown to be

$$P = \frac{1}{3}\frac{kg \times no. \times m^2}{s^2 \times m^3} = \frac{kg \times m}{s^2 \times m^2} = \text{Newton } m^{-2} = \text{Pascal}$$

Relation of equation $P = \frac{1}{3}\frac{mn\overline{c^2}}{V}$ to kinetic energy

The velocity of gas molecules increases with increase of temperature. The mean kinetic energy per molecule of a gas is given by,

$$\text{Mean kinetic energy per molecule of a gas} = \frac{1}{2}m\overline{c^2}$$

Total kinetic energy of all the molecules in one mole of the gas,

$$E_K = \frac{1}{2} \times N_A \times m \times \overline{c^2} \qquad \text{— (20)}$$

where, N_A is the Avogadro number.

Modification of equation (19) for one mole of a gas containing N_A molecules and considering V as the volume occupied by N_A molecules of the gas, it is possible to rewrite equation (19) as

$$P \times V = \frac{1}{3}m \times N_A \times \overline{c^2} = \frac{2}{3} \times \frac{1}{2} \times mN_A\overline{c^2}$$

$$\Rightarrow \quad PV = \frac{2}{3}E_K \qquad \text{— (21)}$$

Since PV = RT for one mole of a gas

$$E_K = \frac{3}{2}RT \qquad \text{— (22)}$$

$$\text{Kinetic energy per molecule} = \frac{3}{2}\frac{RT}{N_A} = \frac{3}{2}k_BT$$

where, k_B is the gas constant per molecule known as ***Boltzmann constant***

From equation (21), it is seen that the *kinetic energy of random movement of gas molecules is proportional to temperature.*

It may also be inferred from equation (21) that *at a specified temperature all gases have the same average kinetic energy per mole.*

Root mean square velocity (or RMS velocity)

For two gases (1) and (2), we can write from equation (21)

$$E_{K(1)} = \frac{3}{2} \times \frac{1}{3}\left(mN\overline{c^2}\right)_1$$

$$E_{K(2)} = \frac{3}{2} \times \frac{1}{3}\left(mN\overline{c^2}\right)_2$$

Equating the above two expressions

$$\frac{1}{3}\left(mN\overline{c^2}\right)_1 = \frac{1}{3}\left(mN\overline{c^2}\right)_2$$

$$m_1\overline{c_1^2} = m_2\overline{c_2^2}$$

$$\frac{\overline{c_1}}{\overline{c_2}} = \sqrt{\frac{m_2}{m_1}} = \sqrt{\frac{M_2}{M_1}} \quad (\because M = m \times N) \qquad \text{— (23)}$$

$\overline{c_1}$ and $\overline{c_2}$ in equation (23) are known as root mean square velocities of the molecules of gases (1) and (2) and it may be defined as

$$C_{RMS} = \sqrt{\overline{c^2}} = \sqrt{\frac{c_1^2 + c_2^2 + c_3^2 + ... + c_n^2}{n}}$$

where, 'n' is the number of molecules of the gas in the given system with velocities c_1, c_2, c_3 etc.
From equations (20) and (22)

$$\frac{3}{2}RT = \frac{1}{2}M\overline{c^2} \quad (\text{where } M = m \times N_A) \text{ or } \overline{c^2} = \frac{3RT}{M}$$

$$C_{RMS} = \sqrt{\overline{c^2}} = \sqrt{\frac{3RT}{M}}$$

Using the kinetic theory expression, it is possible to prove Avogadro's hypothesis, Boyle's law and Graham's law in a simple manner as shown below.

Verification of Avogadro's hypothesis

For two gases (1) and (2)

$$P_1V_1 = \frac{1}{3}m_1n_1\overline{c_1^2}$$

$$P_2V_2 = \frac{1}{3}m_2n_2\overline{c_2^2}$$

If these gases are at the same pressure and occupy the same volume $P_1 = P_2$ and $V_1 = V_2$

The above equation may then be simplified by writing

$$\frac{1}{3}m_1n_1\overline{c_1^2} = \frac{1}{3}m_2n_2\overline{c_2^2}$$

Since the mean kinetic energy of all gases is the same at the same temperature, we can write $\frac{1}{2}m_1\overline{c_1^2} = \frac{1}{2}m_2\overline{c_2^2}$

Rearranging the above equation, $\frac{2}{3}n_1 \times \frac{1}{2} \times m_1\overline{c_1^2} = \frac{2}{3} \times n_2\frac{1}{2} \times m_2\overline{c_2^2}$ and cancelling the kinetic energy term on the both sides of the above equation, it is seen that $n_1 = n_2$, i.e., equal volumes of two gases at the same temperature and pressure contain equal number of molecules.

Verification of Boyle's law

For a definite mass of gas, the number of molecules 'n' in the gas is constant. Further, at a given temperature, the mean kinetic energies of all gases are the same and thus are constants.

$$\therefore P \times V = \frac{2}{3} \times \frac{1}{2} \times m \times n \times \overline{c^2} = \frac{2}{3} \times n \times \frac{1}{2} \times m\overline{c^2} = \text{constant}$$

The above equation thus proves Boyle's law under the above stated conditions.

Verification of Graham's law of diffusion

As a result of frequent collisions, the molecules of a gas do not travel far in a straight line. But the rate of diffusion of the gas molecules is proportional to the mean velocity of the gas molecules. Under the same conditions, if r_1 and r_2 are the rates of diffusion of gases (1) and (2), whose mean velocities are $\overline{c_1}$ and $\overline{c_2}$ we can write

$$\frac{r_1}{r_2} = \sqrt{\frac{\overline{c_1}}{\overline{c_2}}} = \sqrt{\frac{M_2}{M_1}}, \text{ in analogy with rms velocity}$$

or $$\frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}} = \sqrt{\frac{d_2}{d_1}}$$

where, d_1 and d_2 are the densities of the two gases. The above equation is a mathematical form of Graham's law derived from kinetic theory expression.

MOLECULAR VELOCITY DISTRIBUTION AMONG GAS MOLECULES

The molecules of a gas do not move with the same speed. This is due to the continuous interchange of momentum due to frequent collisions among themselves. Even assuming that all the molecules of a gas start off with the same velocity in parallel lines, any slight disturbance due to forces results in collisions among themselves and chaotic movement.

The problem about the velocities with which gas molecules move and the distribution of the possible velocities among various gas molecules has been worked out by J. C. Maxwell in 1860 and is known as the Maxwell's law of distribution of molecular velocities.

The following graph represents a typical Maxwell distribution where in the ordinate is the fraction and the abscissa the velocity c

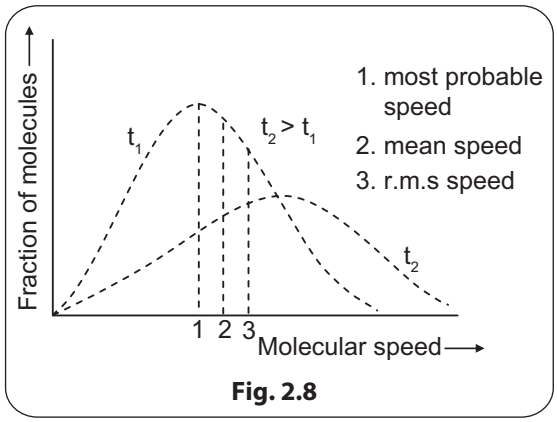


Fig. 2.8

The general nature of the curve is the same (with a maximum) irrespective of the molecular weight and temperature of the gas. *The maximum in the curve (MPV) is known as the* ***most probable velocity*** *i.e., the velocity possessed by maximum number of molecules.*

Equation for MPV is

$$C_{MPV} = \sqrt{\frac{2RT}{M}}$$

Apart from most probable velocity, ***the mean velocity (or average velocity, AV)*** defined by

$$\bar{c} = \frac{c_1 + c_2 + c_3 + ... + c_n}{n}$$

where, c_1, c_2 etc., represent the magnitude of the velocities possessed by gas molecule (1), gas molecule (2), etc., can also be calculated as

$$\bar{c} = \sqrt{\frac{8RT}{\pi M}}$$

the ratio of mean velocity to most probable velocity is obtained as

$$\frac{C_{AV}}{C_{MPV}} = \frac{\bar{c}}{C_{MPV}} = \sqrt{\frac{8RT}{\pi M} \times \frac{M}{2RT}} = \sqrt{\frac{8}{2\pi}} = 1.128$$

The ratio of RMS velocity to mean velocity ($\bar{c}$ or C_{AV}) is given by

$$\frac{C_{RMS}}{C_{AV}} = \frac{C_{RMS}}{\bar{c}} = \sqrt{\frac{3RT}{M} \times \frac{\pi M}{8RT}} = \sqrt{\frac{3\pi}{8}} = 1.085$$

From the above equation and it is concluded that, at a given temperature for a gas, the three velocities decrease in the order.

$$C_{RMS} > C_{AV} > C_{MPV}$$

$$C_{MPV} : C_{AV} : C_{RMS} = 1 : 1.128 : 1.224$$

$$C_{RMS} : C_{AV} : C_{MPV} = 1 : 0.921 : 0.817$$

Effect of temperature on velocity distribution

Curve 2, of Fig 2.8 represents the velocity distribution curve at a higher temperature t_2. The salient features of this curve are (i) the maximum is shifted to the right (compared to curve 1) indicating an increase of molecular velocity with temperature (ii) the maximum is more flattened suggesting a wider distribution of velocities. There is a considerable increase in the number of molecules with velocities greater than the mean velocity.

The marked effect of temperature in increasing the proportion of molecules having higher velocities (or higher kinetic energies) arises from the exponential term in the Maxwell equation.

CONCEPT STRANDS

Concept Strand 17

Calculate the root mean square velocity, average velocity and most probable velocity of nitrogen gas molecule at 310K. (R = 8.314 J deg^{-1} mol^{-1})

Solution

$$\text{RMS velocity} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3 \times 8.314 \times 310}{28 \times 10^{-3}}} = 525.5 \text{ m s}^{-1}$$

$$\text{Average velocity} = \sqrt{\frac{8RT}{\pi M}} = \sqrt{\frac{8 \times 8.314 \times 310}{\frac{22}{7} \times 28 \times 10^{-3}}}$$

$$= 484 \text{ m s}^{-1}$$

$$\text{Most probable velocity} = \sqrt{\frac{2RT}{M}} = \sqrt{\frac{2 \times 8.314 \times 310}{28 \times 10^{-3}}}$$

$$= 429 \text{ m s}^{-1}$$

Alternatively,

$C_{RMS} : C_{AV} : C_{MP} = 1 : 0.921 : 0.817$

$C_{AV} = 525.5 \times 0.921 = 484 \text{ m s}^{-1}$

$C_{MP} = 525.5 \times 0.817 = 429 \text{ m s}^{-1}$

Concept Strand 18

At 27°C oxygen is completely converted to ozone. (Temperature is maintained using insulating material). Calculate the percentage change in root mean square velocity

Solution

$$C_{RMS} \text{ of } O_2 = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3\times8.314\times300}{32\times10^{-3}}}$$

$$C_{RMS} \text{ of } O_3 = \sqrt{\frac{3\times8.314\times300}{48\times10^{-3}}}$$

Percentage change

$$= \left(\frac{\sqrt{\frac{3\times8.314\times300}{32\times10^{-3}}} - \sqrt{\frac{3\times8.314\times300}{48\times10^{-3}}}}{\sqrt{\frac{3\times8.314\times300}{32\times10^{-3}}}} \right) \times 100$$

$$= \sqrt{\frac{3\times8.314\times300}{10^{-3}}} \left(\frac{\sqrt{\frac{1}{32}} - \sqrt{\frac{1}{48}}}{\sqrt{\frac{3\times8.314\times300}{32\times10^{-3}}}} \right) \times 100$$

$$= \frac{\sqrt{\frac{1}{16}}}{\sqrt{\frac{1}{32}}} \left(\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{3}} \right) \times 100$$

$$= \sqrt{2}\,(0.707 - 0.577) \times 100 = 18.38\%$$

Concept Strand 19

A vessel of 1 L capacity contains 10^{20} gas molecules each of mass 10^{-25} g and the mean square velocity of the gas molecule is 10^{10} cm^2/sec^2. Calculate

(i) the pressure of the gas in the units atmospheres and kilo pascals.

(ii) the total kinetic energy of the molecules in ergs.

(iii) the temperature of the gas.

Solution

(i) $PV = \frac{1}{3}mnc^2$

$V = 1 \text{ L} = 1 \times 10^{-3} \text{ m}^3$

$m = 10^{-25} \times 10^{-3} \text{ kg} = 10^{-28} \text{ kg}$

$n = 10^{20}$

$c^2 = 10^{10} \text{ cm s}^{-2} = 10^6 \text{ m}^2 \text{ s}^{-2}$

$$P \times 1 \times 10^{-3} \text{ m}^3 = \frac{1}{3} \times 10^{-28} \text{ kg} \times 10^{20} \times 10^6 \text{ m}^2 \text{ s}^{-2}$$

$$P = \frac{10^{-2} \text{ kg m}^2 \text{ s}^{-2}}{3\times10^{-3}} = 3.33 \text{ kg m}^{-1} \text{ s}^{-2}$$

$= 3.33 \text{ Pa}$

(ii) Total kinetic energy of gas molecules $= E_k = \frac{3}{2} \times PV$

$$= \frac{3}{2} \times 3.33 \text{ kg m}^{-1} \text{ s}^{-2} \times 1 \times 10^{-3} \text{ m}^3 = 5 \times 10^{-3} \text{ kg m}^2 \text{ s}^{-2}$$

$= 5 \times 10^{-3} \text{ J}$

(iii) $E_k \text{ (per molecule)} = \frac{5\times10^{-3} \text{ J}}{10^{20} \text{ molcules}}$

$= 5 \times 10^{-23} \text{ J molecule}^{-1}$

Total KE per mole $= 5 \times 10^{-23} \text{ J molecule}^{-1} \times 6.022 \times 10^{23}$ molecules $= 30.11 \text{ J mol}^{-1}$

But KE $= \frac{3}{2} RT$

$$30.11 \text{ J mol}-1 = \frac{3}{2} \times 8.314 \text{ JK}^{-1} \text{ mol}^{-1} \times T$$

$$T = \frac{30.11 \text{ J mol}^{-1} \times 2}{3\times8.314 \text{ L JK}^{-1} \text{ mol}^{-1}} = 2.4 \text{ K}$$

Concept Strand 20

Calculate the temperature at which the root mean square velocity of SO_3 gas molecules becomes equal to the average velocity of argon gas molecules at 27^{0}C.

($M_{SO_3} = 80 \text{ g mol}^{-1}$; $M_{Ar} = 40 \text{ g mol}^{-1}$)

Solution

RMS velocity of SO_3 gas molecules $= \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3RT}{80}}$

Average velocity of "Ar" gas molecules $= \sqrt{\frac{8R\times300}{\frac{22}{7}\times40}}$

$$\Rightarrow \quad \frac{3RT}{80} = \frac{8R \times 300}{\frac{22}{7} \times 40}$$

$$T = \frac{8 \times 300 \times 7 \times 80}{22 \times 40 \times 3} = 509.1K$$

Concept Strand 21

Calculate the density of helium gas at N.T.P. given that the most probable velocity of helium atoms is 10.62×10^4 cm/sec.

Solution

Most probable velocity $= \sqrt{\frac{2RT}{M}} = \sqrt{\frac{2PV}{M}} = \sqrt{\frac{2P}{d}}$

Most probable velocity $= 10.26 \times 10^2 \text{ m s}^{-2}$

$P = 1.01325 \times 10^5$ Pa

$d = ?$

$$10.62 \times 10^2 \text{ m s}^{-1} = \sqrt{\frac{2 \times 1.01325 \times 10^5 \text{ Pa}}{d}}$$

$$d = \frac{2 \times 1.01325 \times 10^5 \text{ Pa}}{(10.62 \times 10^2 \text{ m s}^{-1})^2} = 0.1797 \text{ kg m}^{-3} = 0.1797 \text{ g L}^{-1}$$

Concept Strand 22

Calculate the density of a gas in g L^{-1} at a pressure of 10^3 dynes cm^{-2}. The mean velocity of the gas molecules is 2.765×10^4 cm sec^{-1}

Solution

Mean velocity 'u' $= \sqrt{\frac{8RT}{\pi M}}$

Considering the gas to be ideal

$$u = \sqrt{\frac{8PV}{\pi M}}$$

$$= \sqrt{\frac{8P}{\pi d}} \qquad \because \frac{M}{V} = d$$

$$u^2 = \frac{8P}{\pi d}$$

$$d = \frac{8P}{\pi u^2} = \frac{8 \times 10^3 \text{ dynes (g cm s}^{-2}) \text{ cm}^{-2}}{3.14 \times (2.765 \times 10^4 \text{ cm s}^{-1})^2}$$

$= 3.33 \times 10^{-6} \text{ g cm}^{-3}$

$= 3.33 \times 10^{-3} \text{ g L}^{-1}$

REAL GASES

The gas laws viz., Boyle's law, Charle's law and Avogadro's hypothesis discussed earlier hold well for many gases over a limited range of temperatures and pressures. Most gases show deviation from that expected for an ideal gas when the range of temperature and pressure are extended during experimental studies. For example, from the ideal gas equation, $V = \frac{RT}{P}$, the volume of a gas at constant temperature should approach zero as the pressure is increased to large value. Similarly, from Charle's law, the volume of a gas at constant pressure should be equal to zero at absolute zero.

These deductions from the ideal gas law do not correspond to the observed behaviour of most gases at high pressure and low temperature. For this reason, all gases are strictly to be considered as real gases.

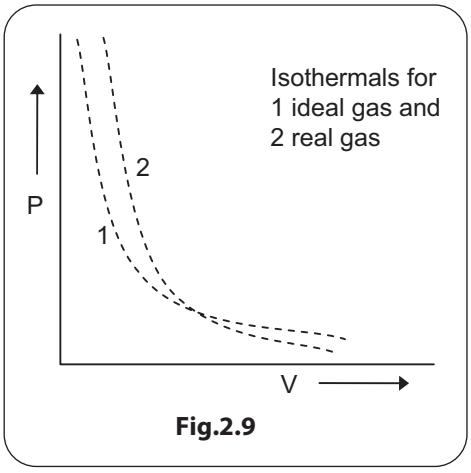


Fig.2.9

It is well known that as a real gas is cooled continuously under constant pressure its volume decreases and the gas liquefies at some definite temperature. There is however, not much further decreased on decrease of temperature after liquefaction.

Similarly, liquefaction of a gas occurs when a gas is subjected to high pressure under certain isothermal conditions. Once it is liquefied, further increase of pressure produces little change in volume.

To illustrate the deviation from ideal gas law, the P–V isotherm of hydrogen is shown in Fig. 2.9.

Causes for deviation from ideal behaviour

Boyle's law and Charle's law were derived from the kinetic theory on the basis of two important assumption (i) the volume of gas molecules is negligible in comparison to the total volume of the gas (ii) the molecules of the gas do not exert any attraction on one another. i.e., there is no inter molecular attraction among the gas molecules.

Since, neither of these assumptions is applicable for real gases, they show deviation from ideal behaviour . The fact that gases can be liquefied and also solidified at sufficiently low temperatures (than occupying a finite volume) under suitable pressure conditions proves that the above assumptions are invalid. Further, the existence of intermolecular forces among gas molecules is proved by the Joule-Thomson effect, which results in cooling of gases and ultimately results in liquefaction.

Compressibility factor 'Z'

For one mole of an ideal gas PV = RT i.e., the 'PV' product of one mole of an ideal gas at T Kelvin temperature is always equal to RT. But for real gases the product of volume and pressure at which the volume is measured are not found to be exactly equal to 'RT' i.e., they deviate from ideal behaviour. The extent of deviation is considered in terms of the difference in measured 'PV' value of the real gas from the 'PV' value of ideal gas at the same temperature. Actually the ratio $\frac{(PV)_{real}}{(PV)_{ideal}}$ is taken as the measure of deviation of a real gas from ideal behaviour. Since $PV_{(ideal)} = RT$, the ratio becomes $\frac{PV_{(real)}}{RT}$. When the measured volume of the gas is less than expected for the real gas at the same temperature the gas is considered it to be more compressed than ideal gas. Hence the ratio $\frac{PV}{RT}$ gives measure of how much compressible the gas is. So it is known as compressibility factor Z. When Z < 1 the gas is more compressible and when Z > 1 the gas is less compressible than ideal gas, the compressibility is related to he pressure to which the gas is applied. How the compressibility of a gas change with pressure can be obtained from the plot of 'Z' vs P at constant temperature.

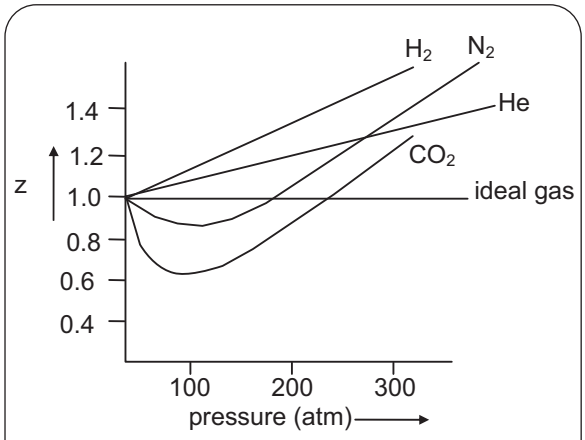


Fig. 2.10

From the graph it is evident that at extremely low pressures all the gases have 'Z' close to unity, which means that the gases behave almost ideally. At very high pressures all gases have 'Z' more than unity indicating that the gases are less compressible than ideal gases.

At moderately low pressure CO_2 and N_2 are more compressible than ideal gas. This is due to the fact that the intermolecular attractive forces are dominant and favour compression. The 'Z' value decreases with pressure and reaches a minimum. When the repulsive force between molecules becomes dominant, further decrease of pressure results in increase in repulsive force between molecules and Z increases after reaching a minimum. It can be observed that N_2exhibits remarkable deviation from ideal behaviour only at high pressures, but CO_2 shows large deviation even at low pressures

Hydrogen and Helium are found to be less compressible than the ideal gas at all pressures i.e., Z > 1. However, if the temperature is sufficiently low there gases also have Z – P plot as shown by CO_2 or N_2 at 0°C.

When, z > 1, repulsive forces dominate and implies that the gas is less compressible.

When, z < 1, attractive forces dominate and implies that the gas is more compressible.

The deviation from ideal behavior may also be depicted as a graph of the product P × V versus P (also known as Amagat curves).

MODIFICATION OF IDEAL GAS EQUATION: VAN DER WAALS EQUATION

Corrections to the ideal gas equation due to the invalidity of the two assumptions stated earlier are, therefore, necessary.

(i) Correction for the intermolecular attraction among gas molecules

A molecule (A) in the interior of a gas, [Fig.2.11(a)] is surrounded by other molecules distributed uniformly in all direction. There is no resultant attractive force on the molecule in any direction.

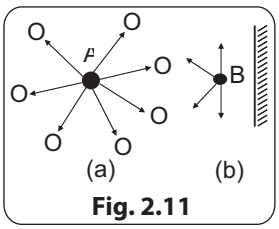


Fig. 2.11

On the other hand, consider a gas molecule (B) near the wall of the vessel [Fig 2.11(b)], experiences

(a) Forces of attraction on a molecule in the interior of a gas.
(b) Forces of attraction on a molecule near the wall.

In this case, there is no uniform distribution of gas molecules around it and therefore other molecules tend to exert an attractive force pulling it inwards. Thus at the time when the gas molecule tends to hit the wall and constitute its part to the total pressure, the molecules inside the bulk pull it away from the wall. The result is that the measured pressure of the gas, P_i, is less than the ideal pressure as required by the kinetic theory. A correction term is, therefore, to be added to the observed pressure to get the ideal pressure as shown below.

Attractive force exerted on a molecule which is about to strike the wall of the vessel μ n, where, n is the number of molecules per unit volume in the bulk of gas and

$\therefore$ The number of molecule striking the wall of the vessel $\propto$ n.

$\therefore$ Total attractive force acting on all the molecules near the wall $\propto n^2$.

If V is the volume of one mole of gas, $n \propto \frac{1}{V}$ or $n^2 \propto \frac{1}{V^2}$ or $n^2 = \frac{a}{V^2}$,

where, a is proportionality constant.,

$\therefore$ Corrected (ideal) pressure $= \left(P + \frac{a}{V^2}\right)$

P in equation is the actual or observed pressure of the gas.

(ii) Correction for finite size of gas molecules

If V is the (actually) measured volume of the gas (at the given T and P), the actual space available for movement of gas molecules is less than V. To obtain the ideal volume, it is necessary to subtract a term from the total volume.

This correction term, b, is referred as the co-volume.

$\therefore$ Corrected (ideal) volume = V – b

The corrected equation may be written as $P_{ideal} \times V_{ideal} = RT$ (for one mole of gas)

$$\left(P + \frac{a}{V^2}\right)(V - b) = RT \text{ (for one mole of gas)}$$

The above equation is known as van der Waals equation and the constants a and b are known as van der Waals constants. The constants a and b depend on the nature of the gas and also on temperature. For n moles of a gas in a total volume V, the van der Waals equation takes the form

$$\left(P + \frac{n^2 a}{V^2}\right)(V - nb) = nRT$$

'a' has the units litre^2 atm mole^{-2}'b' has the units litre mole^{-1}

The above two equations are known as the van der Waals equation of state for a real gas.

Relation between observed pressure (P) and ideal pressure (P_i) and between observed volume (V) and ideal volume (V_i) are

$$P_i = P + \frac{a}{V^2} \quad \therefore \quad P_i > P$$

$$V_i = V - b \quad \therefore \quad V > V_i$$

van der Waals constants

Physical significances of 'a' and 'b'

'a' is the pressure correction term and gives a measure of the attraction between gas molecules. 'b' is the volume correction term and gives a measure of the size of the gas molecules.

For example when we compare the 'a' and 'b' values of NH_3 and N_2, due to the inter molecular hydrogen bonding in ammonia molecules, 'a' is greater for NH_3 than N_2. For comparing the values of 'b' let us assume that the volume occupied by hydrogen atoms is very small, then 'b' is slightly higher than half that of N_2.

Explanation based on Van der Waals equation

For one mole of a real gas the Van der Waals equation is

$$\left(P + \frac{a}{V^2}\right)(V - b) = RT$$

When the pressure is not too high the volume correction 'b' can be neglected in comparison to V and the equation becomes $\left(P + \frac{a}{V^2}\right)V = RT$

i.e., $PV + \frac{a}{V} = RT$ or $PV = RT - \frac{a}{V}$

Then the PV observed will be less than RT by an amount $\frac{a}{V}$ as pressure increases V decreases and $\frac{a}{V}$ increases. The measured 'PV' becomes lesser and less than RT. This explain the dip in the isotherms of gases CO_2 and N_2 in the figure.

As a typical example, the PV – P plot for nitrogen is shown at different temperatures in Fig.2.12. In this figure, the product P × V is arbitrarily taken as 1.0 at one atmosphere at a given temperature for an ideal gas.

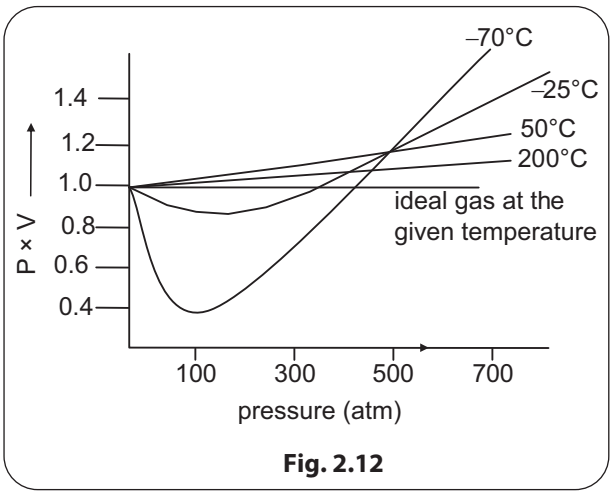


Fig. 2.12

It is seen from the figure that the minimum in the graph vanishes as the temperature is increased above a certain value. Also, the general nature of Amagat curves markedly varies with temperature.

When pressure is too high

At high pressure the volume 'V' will be small hence 'b' cannot be neglected but 'P' is high and hence $\frac{a}{V^2}$ becomes negligible compared to P and the equation becomes

$$P(V - b) = RT$$

$$PV = RT + Pb$$

Then PV is greater than RT, since as pressure increase Pb also increases. Hence there is a continuous increase in Z-P graph at high pressures.

For most of the real gases at ordinary temperature the effect of the term $\frac{a}{V^2}$ is predominant at low pressures and that of 'Pb' term at high pressures. Here at some intermediate pressure the two effects should balance each other. In this range of pressure they should exhibit ideal behaviour. But how much is this range depends on the gas and temperature. For gases like CO and CH_4 there is a small horizontal portion close to that of the ideal gas exists at intermediate pressures in the Z vs P curve. But for most of the gases the magnitudes of $\frac{a}{V^2}$ and Pb do not remain similar at an appreciable range. Hence there is only a point at a given pressure. When temperature increases the volume also increases and $\frac{a}{V}$ becomes less and less significant and finally it becomes negligible. The term 'b' also becomes less significant and finally negligible. So the gas behaves ideally at a high temperatures so long as the pressure is not too high. This explains the decrease in depression in isotherms of N_2 at different temperatures.

For H_2 and He the value of $\frac{a}{V^2}$ remains negligible even at ordinary temperature due to very less value of intermolecular attraction hence the Van der Waals equation is

$$PV = RT + Pb$$

and so there is continuous increase of PV with pressure. Both a and b are temperature dependent. For every real gas there is a temperature at which the magnitude of $\frac{a}{V^2}$ and Pb are nearly equal over an appreciable range of pressure. This temperature which is characteristic of a gas is known as its **Boyle temperature**. It is given by the relation $\mathbf{T_B = \frac{a}{Rb}}$

At other temperatures the effect due to 'a' and 'b' becomes equal only at certain specific pressures. Hence ideal behaviour is not observed over a range. At zero pressure all real gases behave ideally. At every temperature there is a particular pressure at which the gas can exhibit ideal behaviour. Above that pressure it exhibits positive deviation and below that negative deviation.

Other equations of state

In order to account for the behaviour of real gases several other equations have been proposed from time to time.

DIETERICI equation involves an exponential factor to account for intermolecular attraction.
It is $P(V - b) = RT\ e^{-a/RTV}$

Clausius equation

This equation accounts for variation of 'a' with temperature

It is $\left[P + \frac{a}{T(V-c)^2}\right](V - b) = RT$

Where 'c' is a constant

Virial equation

It was proposed by Kammerligh Onnes. He expressed PV as a power series of $\frac{1}{V}$ as

$$PV = RT\left(1 + \frac{B_2(T)}{V} + \frac{B_3(T)}{V^2} + ...\right)$$

where, $B_2(T)$ $B_3(T)$ ..... are temperature dependent Virial Co-efficients. They are calculated from the value of intermolecular potential energy. The equation can also be expressed in terms of a power series is pressure as

$$PV = RT(1 + A_2(T)P + A_3(T)P^2 + ...)$$

the temperature at which the second virial co-efficient vanishes is known as Boyle temperature.

CONCEPT STRAND

Concept Strand 23

Calculate the pressure exerted by 5 moles of oxygen gas at 37^0C occupying a volume of 10 L using (a) ideal gas equation (b) van der Waals equation. The van der Waals constants for the gas are a = 1.36 L^2 atm mol^{-2}, b = 0.0318 L mol^{-1}.

Solution

Using ideal gas equation,

$$P = \frac{nRT}{V} = \frac{5 \times 0.0821 \times 310}{10} = 12.72\,atm$$

Using van der Waals Equation,

$$\left(P + \frac{n^2a}{V^2}\right)(V - nb) = nRT$$

$$\left(P + \frac{25 \times 1.36}{100}\right)(10 - 5 \times 0.0318) = 5 \times 0.0821 \times 310$$

$$(P+0.34)9.841 = 127.25$$

$$\Rightarrow P = \frac{127.25}{9.841} - 0.34 = 12.93 - 0.34 = 12.59\ atm$$

van der Waals constant 'b' and radius of molecules

The van der Waals constant b is related to volume of gas molecules by

$b = 4\,N_o V = 4\,N_0(\frac{4}{3}\pi r^3)$
constants 'a' and 'b' assuming that it obeys van der Waals equation.

CONCEPT STRANDS

Concept Strand 24

A gas cylinder of capacity 50 L containing CO_2 gas can withstand a maximum pressure of 100 atm. If the cylinder contains 6.6×10^3 g of the gas, at what temperature will the cylinder break? Assume that carbon dioxide obeys van der Waals equation with the van der Waals constants a and b being 3.60 L^2 atm mol^{-2}, 4.30×10^{-2} L mol^{-1} respectively.

Solution

$V = 50\ L,\ P = 100\ atm,\ n = \frac{6.6\times10^3}{44} = 150\,moles$

Substituting the data in van der Waals equation, $\left(P+\frac{an^2}{V^2}\right)(V - nb) = nRT$

$$\left(100+\frac{150^2\times3.6}{50^2}\right)(50\text{-}150\times0.043)$$

$$=150\times0.0821\times T$$

$$(100+32.4)(50 - 6.45) = 12.31T$$

$$T=\frac{132.4\times43.55}{12.31} = 468.4K$$

Concept Strand 25

The van der Waals constants 'a' and 'b' for nitrogen are 0.816 L^2 atm mol^{-2} and 0.03 L mol^{-1} respectively. Calculate the compressibility factor, z, of nitrogen at a temperature of 150K and 100 atm pressure.

Solution

Expanding the van der Waals equation and simplifying will give an equation for z given by $z = 1+\frac{1}{V}\left(b-\frac{a}{RT}\right)$ where, V is the molar volume of the gas, a and b are van der Waals constants and T is the temperature in Kelvin

Molar volume, $V = \frac{nRT}{P} = \frac{1\times0.0821\times150}{100} = 0.123$ L

$$z = 1+\frac{1}{0.123}\left[0.03-\frac{0.816}{0.0821\times150}\right]$$

$$= 1+\frac{1}{0.123}\left[0.03-0.066\right]$$

$$= 1 + (8.13 \times -0.036) = 1 - 0.293 = 0.707$$

Since the value of z is less than 1, it shows that the attractive forces between the gas molecules dominate.

Concept Strand 26

The van der Waals constant b for methane gas is 6.9×10^{-5} m^3 per mole. Find the radius of methane molecule.

Solution

$$b = 4\,N_0.V = 4\,N_0\left(\frac{4}{3}\pi r^3\right)$$

$$r^3 = \frac{6.9\times10^{-5}\left(m^3\right)}{4\times6.02\times10^{23}}\times\frac{3}{4}\times\frac{7}{22}$$

$\Rightarrow$ $r = 1.9 \times 10^{-10}$ m.

Concept Strand 27

Calculate the temperature at which a sample of ideal gas exhibit a pressure of one atmosphere at a concentration 1 mol dm^{-3}

Solution

Since the gas is ideal PV = nRT

$$P = \frac{n}{V}.RT$$

i.e., 1 atm = 1 mol L^{-1} × 0.0821 L atm K^{-1} mol^{-1} × T

$$T = \frac{1}{0.0821} = 12.18\ K$$

Isotherm of a real gas

Let us consider the pressure-volume isotherm of a real gas measured at various temperatures. The results of T. Andrews on carbon dioxide are taken as an example. A certain mass of gas was taken in a closed tube at constant temperature and the volume of the gas was measured at different pressures and the results are shown in Fig.2.13

At the lowest temperature, 13°C, CO_2 is a gas at low pressure and as the pressure is increased, volume decreases as shown in the curve ABCD. At B, liquefaction commences and hence V decreases rapidly. At C the liquefaction of the gas is complete and a further increase of pressure along CD does not decrease the volume appreciably and curve rises steeply.

The transition from the gaseous state in curve (63) is AB – only gas, BC – gas + liquid, CD – only liquid

The region BC represents the vapour pressure of liquid CO_2 at the given temperature. At a higher temperature, 25°C, the curve is similar, but the horizontal

portion is reduced. As the temperature is increased, the horizontal portion decreases further and above 31.1°C the horizontal portion disappears practically. This implies that CO_2 cannot be liquefied above 31.1°C even on increasing the pressure to high values. It is the limit of temperature above which the gas cannot be liquefied, no matter what the pressure is and this is known as critical temperature, T_c. This is a general conclusion applicable to all gases but the critical temperature varies with the nature of the gas.

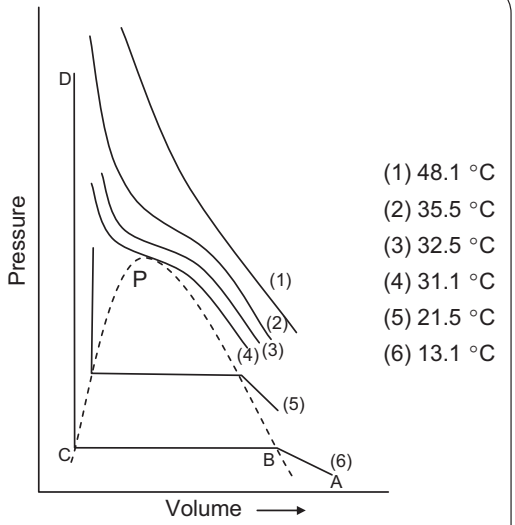


Fig 2.13

The pressure required to cause liquefaction of the gas at the critical temperature (T_c) is known as the critical pressure (P_c). The term vapour is used to describe a gaseous substance when its temperature is below the critical temperature. It may be mentioned that a gas shows marked deviations from the P-V isotherm expected for an ideal gas just above the critical temperature. At higher temperatures, the deviation from ideal behaviour is less.

Continuity of states

It may be noted from Fig.2.13, that out side the bounded region EFG, point A on the curve represents the gaseous state of CO_2. Within the bounded region both liquid and vapour are in equilibrium. Consider a point A^1 in the figure, which is also outside the bounded region EFG. It is possible to go from state A to state A^1 along the dashed lines without passing through the two-phase region. The fact that it is not possible to distinguish between liquid and gas (by following a procedure as mentioned above) is known as the *principle of continuity of states.*

CRITICAL STATE OF A GAS

van der Waals equation, being a cubic equation, has three roots for certain values of temperature and pressure. At the critical point i.e., at the critical temperature T_c and critical pressure P_c, all the three roots of the van der Waals equation merge and give one value of V i.e., V_c. By suitable mathematical procedure, it is possible to relate the critical constants P_c, V_c and T_c to the van der Waals constants a and b and the relevant equations may be given as

$$V_c = 3b \;;\; P_c = \frac{a}{27b^2} \;;\; T_c = \frac{8a}{27Rb}$$

If the values of the van der Waals constants a and b are known, it is possible to calculate P_c, V_c and T_c from above equation.

The critical compressibility factor, $Z = \frac{P_c V_c}{RT_c} = \frac{3}{8}$

We can also evaluate a, b and R in terms of P_c, V_c and T_c from above equation as

$$a = P_c \times 3 \times 9b^2 = 3P_c V_c^2 \;;$$

$$b = \frac{V_c}{3} \text{ and } R = \frac{8a}{27T_c b} = \frac{8P_c V_c}{3T_c} \qquad \text{— (24)}$$

substituting , $V_c = \frac{3RT_c}{8P_c}$ from the above equation we get,

$$a = \frac{27(RT_c)^2}{64P_c} \;;\; b = \frac{RT_c}{8P_c}$$

The law of corresponding states

Using the values of a, b and R from equation (24), the van der Waals equation may be written as

$$\left(P + \frac{3P_c V_c^2}{V^2}\right)\left(V - \frac{V_c}{3}\right) = \left(\frac{8P_c V_c}{3T_c} \times T\right)$$

Rearranging the above equation and substituting

$$\frac{P}{P_c} = \alpha, \frac{V}{V_c} = \beta \text{ and } \frac{T}{T_c} = \gamma,$$

we get $$\left(\alpha + \frac{3}{\beta^2}\right)(3\beta - 1) = 8\gamma$$

α, β and γ are known as reduced variables of state and above equation is known as reduced equation of state. It does not contain any constants particular to an individual gas. The above equation expresses one variable, say γ, in terms of two other variables α and β and it may be noted that two gases at the same reduced temperature and pressure have the same reduced volume. This statement is known as the *law of corresponding states*.

Volume of van der Waals gas by successive approximation method

The van der Waals equation for one mole is

$$\left(P + \frac{a}{V^2}\right)(V - b) = RT$$

This is rewritten as $V - b = \dfrac{RT}{P + \left(\dfrac{a}{V^2}\right)}$

$$\Rightarrow V = \frac{RT}{P + \left(\frac{a}{V^2}\right)} + b$$

Since this equation in V is a cubic equation, we can solve it by successive approximation method.

At first, we use the ideal volume RT/P and introduce it in the right hand side of equation for V given above. We will get a new value of V, which is introduced into the right hand side for V and the procedure is repeated till two successive values of V become quite close.

Let us consider methane gas whose van der Waals constants a and b are 2.29 L^2 bar mol^{-1} and 0.043 L mol^{-1}. (1 bar = 10^5 Pa, hence 1.013 bar = 1 atm)

$$\frac{RT}{P} = 0.248$$

$$V = \frac{0.08314 \times 298}{100 + \frac{2.29}{(0.248)^2}} + 0.043 = \frac{24.8}{100 + 37.2} + 0.043$$

$$= 0.181 + 0.043 = 0.224 \text{ L mol}^{-1}.$$

If we repeat this once more, we get V as 0.215 L mol^{-1}. Thus we can calculate volume of a van der Waals gas.

Boyle's Temperature

If a plot of z for a real gas against P is made at various temperatures, the resulting graph is similar to Fig. 2.10.

The slope of two curves may be shown to be equal to

$$\left(\frac{\partial z}{\partial P}\right)_T = \frac{1}{RT}\left(b - \frac{a}{RT}\right)$$

At P = 0, the L.H.S. of the equation corresponds to the initial slope of the curve (Fig. 2.10) and it is seen that when $b > \dfrac{a}{RT}$, the slope is + ve when, $b < \dfrac{a}{RT}$, the slope is – ve.

At some temperature, T_B, the slope vanishes and becomes zero, At this temperature, known as Boyles Temperature (T_B),

$$b = \frac{a}{RT_B} \text{ or } T_B = \frac{a}{R \times b}$$

At this temperature, any real gas behaves ideally over a considerable range of pressure due to the effect of size and intermolecular forces roughly compensating each other.

CONCEPT STRAND

Concept Strand 28

Boyle temperature for H_2 is 108K. Derive the expression for compressibility factor 'z' for H_2 above 108K

Solution

Above Boyle's temperature, Z is greater then 1
When, Z > 1 the relevant equation is

$$P(V - b) = RT$$

i.e., $$\frac{PV}{RT} = 1 + \frac{Pb}{RT}$$

i.e., $$Z = 1 + \frac{Pb}{RT}$$

RELATIVE HUMIDITY (RH)

At a given temperature, RH is given by the equation,

$$RH = \frac{\text{Partial pressure of water vapour in air}}{\text{Vapour pressure of water}}$$

SUMMARY

Boyle's Law	$PV = \text{constant}$
	$P_1V_1 = P_2V_2$
Gay–Lussac's or Charle's law	$V_t = V_0(1 - \alpha t)$
α = Coefficient of expansion	$\alpha = \frac{V_t - V_0}{V_0 t}$
V_t = Volume of gas at t°C	
V_0 = Volume of gas at 0°C	$\frac{V_1}{T_1} = \frac{V_2}{T_2}$
T = Temperature	
Gas law	$PV = nRT$
N = number of moles of an ideal gas	
Value of R in different units	$R = 0.0821 \text{ L atm } K^{-1} \text{ mol}^{-1}$
	$8.314 \text{ JK}^{-1} \text{ mol}^{-1}$
	$8.314 \times 10^7 \text{ ergs } K^{-1} \text{ mol}^{-1}$
	$1.987 \text{ cal } K^{-1} \text{ mol}^{-1}$
Dalton's law of partial pressures	$P = P_1 + P_2 + P_3 + \ldots\ldots$
x_1 = mole fraction of gas	$P_1 = x_1 \times P$
P = total gas pressure	
P_1, P_2, P_3…..are the partial pressure of gases in the mixture	
Grahams law of diffusion	$\frac{r_1}{r_2} = \sqrt{\frac{d_2}{d_1}}$
r = rate of diffusion	
d = density of the gas	
M = molecular mass of the gas	$\frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}}$
t = time for diffusion of equal volume of gases	$\frac{r_1}{r_2} = \frac{t_2}{t_1}$
P_1 and P_2 are the pressure of gases	
	$\frac{r_1}{r_2} = \frac{P_1}{P_2}\sqrt{\frac{M_2}{M_1}}$
Kinetic gas equation	$P = \frac{1}{3}\frac{mn\overline{c^2}}{V}$
m = mass of gas molecule	
N = number of molecules in the volume V	
$\overline{c^2}$ = mean square velocity	

R = gas constant KE = Kinetic energy	$KE(\text{per mole}) = \frac{3}{2}RT$
k = Boltzman constant	$KE \text{ per molecule} = \frac{3}{2}kT$
Different types of molecular velocities	$C_{RMS} = \sqrt{\frac{C_1^2 + C_2^2 + ... + C_n^2}{n}}$
	$C_{RMS} = \sqrt{\frac{3RT}{M}}$; $C_{MPV} = \sqrt{\frac{2RT}{M}}$
	$C_{AV} = \frac{C_1 + C_2 + ... + C_n}{n}$; $C_{AV} = \sqrt{\frac{8RT}{\pi M}}$
Relation between different types of molecular velocities	$C_{MPV} : C_{AV} : C_{RMS} = 1 : 1.128 : 1.224$ $C_{RMS} : C_{AV} : C_{MPV} = 1 : 0.921 : 0.817$
Z = compressibility factor	$Z = \frac{PV}{nRT}$
van der waals equation	$\left(P + \frac{n^2a}{V^2}\right)\left(V - nb\right) = nRT$
	$\left(P + \frac{a}{V^2}\right)(V - b) = RT$
van der waals constants a and b relates to critical temperature, pressure and volume	$V_C = 3b$; $P_C = \frac{a}{27b^2}$; $T_C = \frac{8a}{27Rb}$
$\alpha = \frac{P}{P_C}, \beta = \frac{V}{V_C}, \gamma = \frac{T}{T_C}$ reduced equation of state	$\left(\alpha + \frac{3}{\beta^2}\right)(3\beta - 1) = 8\gamma$
Boyle temperature	$T_B = \frac{a}{Rb}$
Dieterici equation	$P(V - b) = RT\ e^{-a/RTV}$
Clausius equation	$\left[P + \frac{a}{T\left(V - c\right)^2}\right](V - b) = RT$
Virial equation	$PV = RT\left(1 + \frac{B_{2(T)}}{V} + \frac{B_{3(T)}}{V^2} + ...\right)$
	$PV = RT(1 + A_{2(T)}P + A_{3(T)}P^2 + \ldots)$
Relative humidity	$R_H = \frac{\text{partial pressure of water vapour in air}}{\text{V.P of water}}$

TOPIC GRIP

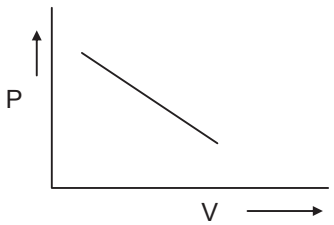
Subjective Questions

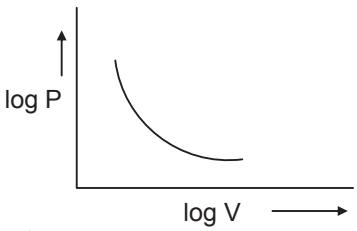
1. The mean molar mass of a mixture of CO and CO_2 is 32 g mol^{-1}. What is the mole fraction of CO in it?
2. At 80°C the density of N_2O_4 is found to be 25.8. Calculate the percentage by weight of NO_2 molecules.
3. 1.78×10^{22} molecules of a gas trapped in a 200 ml flask exerted a pressure of 3.66×10^5 Nm^{-2} at 298 K. Identify the nature of the gas(ideal or real) and justify.
4. On heating a mixture of chlorine and an oxide of chlorine in equal volumes (30ml + 30ml) and cooling to room temperature at the same pressure, the volume observed was 75 ml of which 60 ml were absorbed by alkali. Suggest the formula of the oxide.
5. The fraction of partial pressure exerted by hydrogen in a mixture of equal masses of two gases (including hydrogen) is 8/9 at 25°C. Identify the other gas.
6. The gaseous fluoride of an element contains 27% of the element. 50 ml of the gas diffuses under certain conditions in 25 seconds. Under the same conditions 100 ml O_2 the gas diffuses in 27.7 seconds. What may be the probable atomic weight of the element.
7. Calculate the rms speed, mean speed and most probable speed of N_2 molecules at 27°C.
8. Calculate the compressibility factor of CO_2, given its density = 1.96 g L^{-1} at 0°C and 1 atmospheric pressure.
9. The van der Waals constants, b for methane gas is 6.9×10^{-5} m^3 mol^{-1}. Find the molecular radius.
10. 2 litres of a gas at 40 atm and 27°C is compressed to 60 atm and 17°C. The compressibility factors are found to be 1.12 and 1.68, respectively. Calculate the final volume?

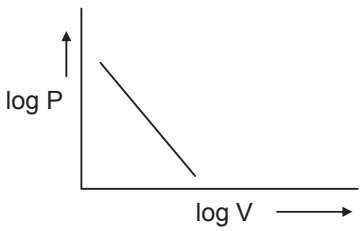
Straight Objective Type Questions

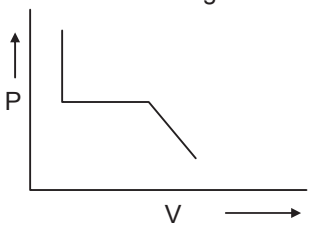
Directions: **This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

11. Which of the following graphical plots represent Boyle's law for a given mass of gas (assuming ideal behaviour)?

(a)


(b)


(c)


(d)


12. The ratio of mean velocity of molecules of a gas to its most probable velocity at a given temperature is
(a) 1.182 (b) 1.128 (c) 1.144 (d) 1.133

13. Among real gases largest deviation from ideal behaviour is shown by the gas with
(a) large values of a and b
(b) low values of a and b
(c) large value for a with low/moderate value of b
(d) low value of a with moderate value of b

14. Indicate the incorrect statement.
(a) The mean free path of gas molecules is inversely proportional to their collision diameter.
(b) The larger the pressure of a gas at a given temperature the shorter the mean free path.
(c) The van der Waals constant b for a real gas is related to the radius of gas molecules.
(d) For a van der Waals gas the constant 'a' has the dimensions : atm L^2 mol^{-2}.

15. For a real gas (one mole) conforming in behaviour to the van der Waals equation, the Boyle temperature is
(a) $\frac{2a}{Rb}$ (b) $\frac{8a}{27Rb}$ (c) $\frac{3a}{Rb}$ (d) $\frac{a}{Rb}$



Assertion–Reason Type Questions

Directions: **Each question contains Statement-1 and Statement-2 and has the following choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False
(d) Statement-1 is False, Statement-2 is True

16. Statement 1
Dalton's law of partial pressures holds for mixtures of non interacting gases at low/moderate pressures at not-too-low values of temperature.
and

Statement 2
Under these conditions intermolecular interactions may be neglected.

17. Statement 1
In a container with a mixture of CH_4 and SO_2 at a certain temperature, it is observed that the initial rates of effusion through a pinhole are equal.
and

Statement 2
The partial pressures of CH_4 and SO_2 in the mixture are 1 : 2.

18. Statement 1
van der Waals constant, b for one mole of a gas is the product of volume per molecule and the Avogadro number.
and

Statement 2
At moderately low pressures and high temperatures the total volume of all the molecules in a given mass of the gas is quite negligible in comparison with the volume of the gas.

19. Statement 1
For (one mole of)an ideal gas, the Joule-Thomson effect is zero.
and

Statement 2
For any gas (real or ideal) Joule-Thomson expansion is also termed as adiabatic expansion.

20. Statement 1
van der Waals equation can be expressed in the so-called reduced form applicable to all van der Waals gases by taking $P = \pi P_c$, $V = \phi V_c$, $T = \theta T_c$.
and

Statement 2
The result $\frac{RT_c}{P_c V_c} = \frac{8}{3} = 2.67$ is not observed for most of the real gases. This shows the limitations of the van der Waals equation.

Linked Comprehension Type Questions

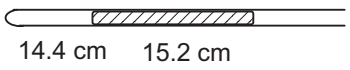
Directions: **This section contains 2 paragraphs. Based upon the paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

Passage I

The ideal gas equation for n moles is PV = nRT which may also be expressed in the form $PV = \frac{W}{M}RT$ where W = mass of the gas in grams and M is the molar mass of the gas in g mol^{-1}. Another formulation of the equation is PM = dRT where, $d = \frac{W}{V}$ i.e., density in g L^{-1}.

21. Calculate the weight of Helium in grams that must be added to a 2.24 litre container containing 1.6 g of O_2 at 0°C in order to have a total pressure of 2 atm
(a) 0.6 g (b) 0.5 g (c) 0.7 g (d) 0.4 g

22. An air column of length 14.4 cm is trapped by a mercury column 15.2 cm long in a capillary tube of uniform bore when the tube kept horizontally. At the open end the pressure is atmospheric, what would be the length of the air column when the tube is kept vertically with its open end up?



(a) 11 cm (b) 13 cm (c) 12 cm (d) 10.5cm

23. A gas is heated in a cylinder fitted with a nozzle for 20 min. The initial temperature is 27°C. It is observed that $\frac{1}{3}$ of the original volume escapes when the temperature is maintained steadily at the higher temperature t°C. Calculate t. The pressure is maintained at 1 atm throughout
(a) 150°C (b) 177°C (c) 160°C (d) 190°C

Passage II

For a given mass of ideal gas at a given temperature, the pressure and volume are inversely related. i.e., (PV) is constant at all pressures for a given mass of gas at a given temperature. Real gases approximately conform to this behaviour at not too high values of pressure and not too low values of temperature but deviations from this 'ideal' behaviour are observed especially at low temperatures and at moderate to high pressures. The ideal gas equation has to be modify to take into consideration these deviations. One of the most useful equations for real gas behaviour is the van der Waals equation for one mole of a real gas. This is $\left(P + \frac{a}{V^2}\right)(V - b) = RT$ where a, b are van der Waals constants.

24. The units in which the van der Waals constant a may be expressed are
(a) atm L mol^{-1} (b) atm L^2mol^{-2} (c) L atm deg^{-1} mol^{-1} (d) atm mol L^{-1}

25. It may be shown that the van der Waals constant, b is related to the molecular radius $b = 4N_0\left(\frac{4}{3}\pi r^3\right)$ where $N_0 = 6.02 \times 10^{23}$. The molecular radius of the methane molecule is 1.9 Å. Calculate the value of b in litre per mole
(a) $\sim 6.9 \times 10^{-2}$ (b) $\sim 5.9 \times 10^{-2}$ (c) $\sim 7.9 \times 10^{-2}$ (d) $\sim 4.9 \times 10^{-2}$

26. The van der Waals equation can be expanded as a series in $\left(\frac{1}{V}\right)$ (virial equation) An approximate form of the equation for 1 mole of gas is $Z = \frac{PV}{RT} = 1 + \frac{\left(b - \frac{a}{RT}\right)}{V}$. The value of Z for T = 150 K, V = 0.123 litres [a = 0.816 and b = 0.03 in the usual litre-atm-units] is
(a) 0.770 (b) 0.670 (c) 0.660 (d) 0.707

Multiple Correct Objective Type Questions

Directions: **Each question in this section has four suggested answers of which ONE OR MORE answers will be correct.**

27. For a mixture of 0.5 mole of Helium and 0.3 mole of N_2 in a container of 0.82 litre at 27°C
(a) the total pressure is 24 atm (b) the total pressure is 2.4 atm
(c) the partial pressure of Helium is 15 atm (d) the partial pressure of Nitrogen is 0.9 atm

28. Which of the following gases have the same mean velocity at the same temperature?
(a) Cyclopropane (b) Ethene (c) Nitrogen (d) Carbon monoxide

29. Which of the following is true for a real gas?
(a) Application of the equation PM = dRT gives precise value for the molar mass M only at the limit $p \to 0$ [d = density g L^{-1}].
(b) For a van der Waals gas $Z = \frac{PV}{RT}$ (for one mole) is almost 1 at the Boyle temperature.
(c) The ratio $\left(\frac{RT_c}{P_cV_c}\right) = \frac{8}{3}$ for almost all real gases.
(d) For temperatures below T_c, van der Waals equation is a cubic in V.

Matrix-Match Type Question

Directions: **Match the elements of Column I to elements of Column II. There can be single or multiple matches.**

30. For one mole of each of the following

Column I	**Column II**
(a) CH_4 at 100 atm, 300 K	(p) $Z < 1$
(b) O_2 at 1000 atm, 300 K	(q) $Z > 1$
(c) H_2 at the Boyle temperature	(r) $Z = 1$
(d) He at 300 K, 200 atm	(s) $V_{actual} > V_{ideal}$

IIT ASSIGNMENT EXERCISE

Straight Objective Type Questions

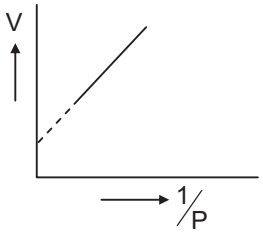
Directions: **This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

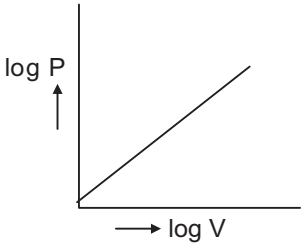
31. Which of the following properties can be seen exclusively in gases only?
(a) They intermix spontaneously.
(b) They fill the whole vessel in which they are placed.
(c) They exert pressure on the walls of the container.
(d) Their molecules are constantly in a state of motion.

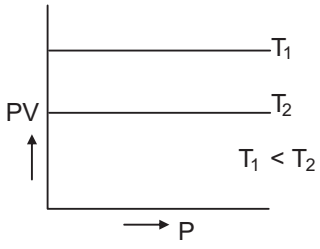
32. 4 L of a gas at 2 atm is filled in a 5 L flask at constant temperature. After some time the flask is evacuated, 90% of the gas is removed and pressure drops to 0.2 atm. The final volume of gas remaining is
(a) 5 L (b) 0.4 L (c) 0.39 L (d) 2 L

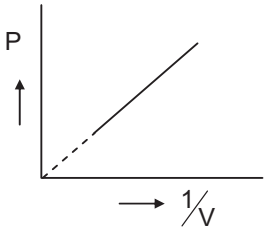
33. Consider a vessel filled with carbondioxide at 27°C and 5 atmospheric pressure. A part of the gas is removed at 27°C and it fills a 3L container at 1 atm and the pressure drops to 3.5 atm in the vessel. The volume of the vessel is
(a) 1 L (b) 5.8 L (c) 3.5 L (d) 2 L

34. Boyle's law can be represented as

(a) 

(b) 

(c) 

(d) 

35. If equal weights of oxygen and methane are enclosed in two identical vessels at 1 atm, then
(a) both the gases are at same temperature
(b) oxygen is at a lower temperature
(c) oxygen is at twice the Kelvin temperature as methane
(d) methane is at twice the Kelvin temperature as oxygen

36. A gas mixture contain 1 mole of H_2. 3.5 moles of He, 1.5 mole of N_2 and 3 mole of Ar. The partial pressure of the Argon gas in the mixture is 4 atmosphere. The total pressure of the gas is
(a) 18 atm (b) 36 atm (c) 12 atm (d) 9 atm

37. Considering ideal behaviour, the volume occupied by one chlorine molecule and one hydrogen molecule at STP can be compared as

(a) $V_{Cl_2} = V_{H_2}$ (b) $V_{Cl_2} = 2V_{H_2}$ (c) $V_{Cl_2} = 35.5V_{H_2}$ (d) $V_{Cl_2} = \frac{1}{2}V_{H_2}$

38. 2 moles of a gas was filled in a closed vessel at 2 atm pressure. The gas shows a 20% increase in pressure for every 10° rise in temperature. The volume of the vessel is

(a) 0.41 L (b) 8.2 L (c) 4.1 L (d) 0.2 L

39. The initial pressure of a mixture of methane and ethane is P_1. The mixture undergoes complete combustion and the final pressure at 25°C is P_2. The molar ratio of methane and ethane is (volume is constant)

(a) $P_1 : 3P_2$ (b) $P_1 - P_2 : 3P_2$ (c) $\frac{P_1 - 2P_2}{P_1 - P_2}$ (d) $\frac{2P_1 - P_2}{P_2 - P_1}$

40. The volume occupied by one molecule of nitrogen dioxide at STP is (in cm^3)

(a) 22400 (b) $22400/N_A$ (c) $\frac{22400}{N_A} \times 23$ (d) $\frac{22400}{N_A} \times 46$

41. If equal volumes of oxygen and nitrogen are mixed together, the vapour density of the mixture, at constant temperature and pressure is

(a) 8.2 (b) 7.5 (c) 15 (d) 23

42. If carbon dioxide and oxygen are mixed in the ratio 1 : 9, the density (g/ L) of the gas at STP is approximately

(a) 0.98 (b) 0.93 (c) 2.62 (d) 1.5

43. At 1 atm, if, in the mercury column, mercury was replaced by water, the height of column would be

(a) 760 mm (b) 1013 mm (c) 10336 mm (d) 1520 mm

44. An open vessel kept at 27°C was cooled to 7°C. The percentage increase in weight of the vessel is (take weight of air = 29.2 g mol^{-1})

(a) 7% (b) 17% (c) 9.3% (d) 93%

45. The density of Krypton gas at STP is (atomic wt = 84 g/mol)

(a) 2.5 gL^{-1} (b) 3.75 gL^{-1} (c) 4.2 gL^{-1} (d) 5.0 gL^{-1}

46. 10 L of a gas was collected over 1000 ml of water at 27°C. After collection weight of water decreased by 0.24 g. The vapour pressure of water is

(a) 0.08 atm (b) 30 mm Hg (c) 0.033 atm (d) 0.6 cm Hg

47. A container has a partition exactly at the centre. One half contains bromine at 0.5 atm and the other hydrogen at 1 atm. At 400K, the partition is removed, the gases react completely. The partial pressure of HBr formed is

(a) 0.75 atm (b) 0.92 atm (c) 0.4 atm (d) 0.5 atm

48. Ammonia burns in oxygen with a greenish yellow flame $4NH_3 + 3O_2 \rightarrow 2N_2 + 6H_2O$. At 127°C and 2 atm pressure, 30 mL each of NH_3 and O_2 are mixed to undergo complete reaction. The amount of N_2 produced is

(a) 16 g (b) 0.016 g (c) 0.0255 g (d) 12 g

49. The numerical value of R in eV $mol^{-1}K^{-1}$ is

(a) 1.33×10^{-19} (b) 5.19×10^{-20} (c) 1.33×10^{-18} (d) 5.19×10^{19}

50. Equal weights of O_2 and SO_2 are enclosed in a container when O_2 exerts a partial pressure of 0.67 atm. This will remain the same even on

(a) adding equal weights of O_2 and SO_2
(b) removing equal weights each of O_2 and SO_2
(c) removing equal moles of O_2 and SO_2
(d) doubling both volume and temperature

51. Consider the reaction $C_{(s)} + H_2O_{(l)} \rightarrow CO_{(g)} + H_{2(g)}$. If 72 g carbon reacts at high temperature and products formed are brought back to STP, the partial pressure of CO in the product is
(a) 0.5 atm (b) 0.067 atm (c) 0.166 atm (d) 0.8 atm

52. A divers cylinder contains 90% O_2 by weight, the other gas being helium. The percentage of helium by volume is
(a) 40 (b) 10 (c) 47 (d) 53

53. Work done by one mole of an ideal gas per degree temperature at STP is approximately
(a) 1 J (b) $\frac{1}{273}$ J (c) 8.314 J (d) 0.08 J

54. Equal weights of fluorine and Argon gas are enclosed in a 10 L vessel at 2 atm and 373K. If partial pressure of fluorine and argon are p_F and p_{Ar}, we can say
(a) $p_F \cong p_{Ar}$ (b) $p_F \cong 2p_{Ar}$ (c) $p_F \cong 0.5p_{Ar}$ (d) $p_F < p_{Ar}$

55. 3 moles each of SO_2 and O_2 are mixed at 10 atm in a closed vessel to form 1.0 mole of SO_3 at constant temperature. The partial pressure of SO_3 is
(a) 2.98 atm (b) 2.6 atm (c) 1.66 atm (d) 1.81 atm

56. An observatory balloon is filled with hydrogen and nitrogen. If this mixture effuses 1.97 times faster than nitrogen, the mole percentage of hydrogen is
(a) 40% (b) 80% (c) 20% (d) 92%

57. The ratio of time taken for diffusion of 5 g each of helium and argon at constant temperature and pressure from two different vessels is
(a) 4 : 1 (b) 10 : 1 (c) 1 : 3.16 (d) 3.16 : 1

58. At a certain temperature, 1 mole of chlorine at 1.2 atm takes 40 sec to diffuse while 1 mole of its oxide at 2 atm takes 26.5 sec. The oxide is
(a) Cl_2O (b) ClO_2
(c) Cl_2O_6 (d) Cl_2O_7

59. The average kinetic energy of a molecule of oxygen at 27°C is
(a) 6.21×10^{-23} J (b) 6.21×10^{-21} J
(c) 3.7 kJ (d) 6.2 kJ

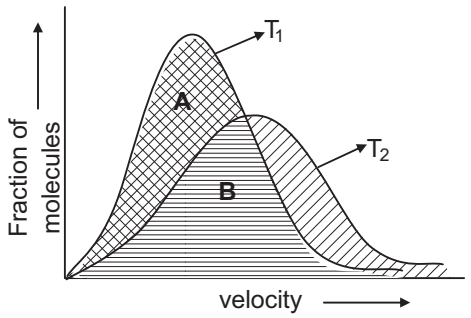
60. When 5.6 g of nitrogen and 6.4 g of oxygen are filled in a evacuated flask, the one with larger kinetic energy is
(a) oxygen (b) nitrogen
(c) both have same velocity (d) temperature dependent

61. The true statement for a gas that obeys the kinetic theory of ideal gases is
(a) Total volume of gas = no. of molecules × volume of each molecule
(b) For a given gas, molecules have same mass but differ slightly in size
(c) Energy is transferred between molecules on collision
(d) All molecules of a given gas have same kinetic energy at a given temperature.

62. The rms velocity of oxygen at a high temperature is u. If at this temperature oxygen dissociates to its atoms, the new rms velocity is
(a) u (b) $\sqrt{2}$ u
(c) 2u (d) $u/\sqrt{2}$

63. The root mean square speed of nitrogen molecules at 27°C and 700 torr pressure is
(a) 700 ms^{-1} (b) 515 ms^{-1} (c) 135 ms^{-1} (d) 162 ms^{-1}

64. Let A and B be the area under the curve for a given sample of gas. If $T_2 > T_1$ then



(a) A = B (b) A > B (c) A < B (d) A ≠ B

65. If f_1 and f_2 be the fraction of molecules moving with speed = average speed and speed = root mean square speed respectively. The correct relation between f_1 and f_2 is

(a) $f_1 > f_2$ (b) $f_1 < f_2$

(c) $f_1 \leq f_2$ (d) f_1 and f_2 cannot be compared

66. Chlorine molecule decomposes to chlorine atoms at 1200°C. The percentage dissociation of chlorine molecules, if it has an average velocity of 674.4 m s^{-1} is

(a) 3.5% (b) 0.80% (c) 2.8% (d) 6.7%

67. When the compressibility factor of a gas is greater than one, then

(a) it is difficult to compress the gas (b) molar volume is less than 1 L at STP

(c) molar volume is less than 22.4 L at STP (d) it is easy to compress at low pressure

68. At STP, a real gas has a molar volume of 20.8 L. The compressibility factor of the gas is

(a) 1.03 (b) 0.93 (c) 0.75 (d) 1.08

69. If the compressibility factor of 5 moles of methane, at 0°C and 50 atm is 0.8, then its volume is

(a) 2.25 L (b) 1.8 L (c) 1.2 L (d) 3.8 L

70. On plotting a curve between PV and P for hydrogen, only positive deviation is observed because

(a) intermolecular forces of attraction are negligible

(b) hydrogen does not have neutrons

(c) hydrogen is a mixture of ortho and para isomers

(d) hydrogen is easily liquefiable

71. A correction using the term an^2/v^2 is made in the ideal gas equation to account for

(a) inward pull experienced by the molecule at the surface

(b) random motion of gas molecules

(c) pressure exerted on molecule inside the gas

(d) volume occupied by gas molecule

72. For CS_2, the true statement among the following is

(a) low Z value, due to greater van der Waals force of attraction

(b) high Z value, due to large sulphur atom

(c) low Z value, due to symmetric structure

(d) low Z value, due to triatomic linear nature

73. Consider gases 1 and 2 such that they have vander Waals constant a_1, b_1 and a_2, b_2 respectively. The condition when '1' is more compressible then '2' is

(a) $a_1 = a_2$ and $b_1 > b_2$ (b) $a_1 < a_2$ and $b_1 > b_2$ (c) $a_1 < a_2$ and $b_1 = b_2$ (d) $a_1 > a_2$ and $b_1 < b_2$

74. The radius of a molecule which has a vander Waals constant value of 0.03 L mol^{-1} is

(a) 7.2A° (b) 2.6A° (c) 1.43A° (d) 0.72A°

75. If an isobar is plotted for one mole of a real gas at high pressure, the straight line obtained will cut the Y axis (volume) at

(a) b (b) 22.4 cm^3 (c) 0.082 cm^3 (d) the origin

76. If the pressure of a gas is increased 10 times, at constant temperature, the mean free path of the gas molecules

(a) remains unchanged (b) increases 10 times (c) decreases 10 times (d) decreases $\sqrt{10}$ times

77. For a certain gas, a = 6.71 atm L^2 mol^{-2} and b = 0.056 L mol^{-1}. The gas

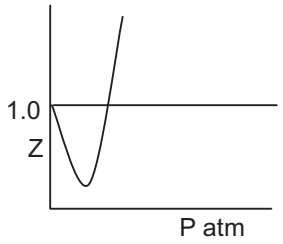
(a) can be liquefied at room temperature on applying pressure
(b) cannot be liquefied at 0°C
(c) gas can be liquefied only on cooling
(d) exists as gas at 0K

78. In the case of nitrogen, at 50°C, the value of Z remains close to 1 up to nearly 100 atm. This temperature can be derived using the formula

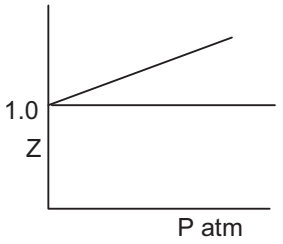
(a) $\frac{a}{27b^2}$ (b) $\frac{a}{Rb}$ (c) $\frac{8a}{27Rb}$ (d) $\frac{a}{27Rb}$

79. Let the Boyle temperature of a gas be T. The plot of graph between Z and P for this gas at Temperature T', such that T' > T will be

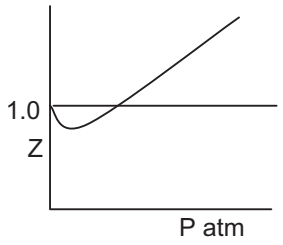
(a)



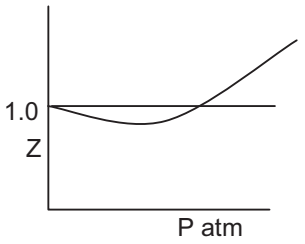
(b)



(c)



(d)



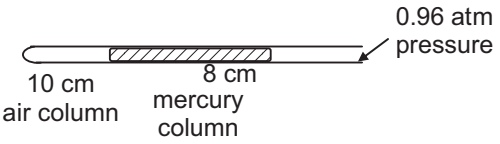
80. The vapour pressure of water at 20°C is 18 mm Hg. If the relative humidity is 35%, the weight of water per litre of air at 20°C is (in gL^{-1})

(a) 6.3×10^{-2} (b) 6.3×10^{-3}
(c) 4.7×10^{-3} (d) 6.3×10^{-4}

81. Which among the following gases exert a pressure of 5 atm at 0°C when placed in a 1L container?

(a) 0.223 moles of CH_4 (b) 0.30 moles of cyclopropane
(c) 0.277 moles of PH_3 (d) 0.24 moles of NH_3

82. The figure shows a narrow tube kept horizontally with air column of 10 cm in length trapped by 8 cm of mercury column. The pressure of air outside is 0.96 atm. [1 atm = 76 cm of Hg]. If the tube is kept in a slanting position at 45° to the vertical with the open end up, Calculate the length of the trapped column of air



(a) 8.28 cm (b) 7.28 cm (c) 9.28 cm (d) 7.82 cm

83. A mixture of $N_2O_{4(g)}$ and $NO_{2(g)}$ under dissociation equilibrium $N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$ has a degree of dissociation = 0.6 at temperature TK. The mean molar mass of the mixture is

(a) 75.5 g mol^{-1} (b) 67.5 g mol^{-1} (c) 50.7 g mol^{-1} (d) 57.5 g mol^{-1}

84. Two cylinders A and B contain the same gas at the same temperature. The pressure and volume of A are both twice those of B. Then the ratio of the number of molecules of A and B is

(a) 1 : 4 (b) 4 : 1 (c) 2 : 1 (d) 1 : 2

85. An electronic tube of volume 200 ml was sealed off during manufacture at a pressure 1.8×10^{-5} torr. At T = 300 K, calculate the number of molecules contained in it

(a) 5.8×10^{13} (b) 1.16×10^{14} (c) 2.9×10^{14} (d) 2.9×10^{13}

86. A 10 litre cylinder of oxygen at 4 atm pressure at 17°C developed a leak. When the leak was repaired the remaining oxygen had a pressure of 3 atm. Calculate the number of moles of oxygen escaped

(a) 0.63 (b) 0.515 (c) 0.42 (d) 0.71

87. The density of oxygen gas at 293 K and 3 atm pressure is nearly

(a) 3.5 g L^{-1} (b) 2.9 g L^{-1} (c) 4 g L^{-1} (d) 4.5 g L^{-1}

88. A stony mass believed to be a meteorite contained, as revealed by analysis, $_{36}Ar$ atoms to the extent of 2×10^{-7} litre at STP per kg of the mass. Calculate the number of atoms of Ar, per kg of the mass

(a) 5.4×10^{15} (b) 4.5×10^{14} (c) 4.4×10^{16} (d) 6.5×10^{16}

89. A gas cylinder contains 11.6 moles of O_2 gas at 298 K. The cylinder has a volume of 9.43 litres. The cylinder is now heated to 75°C and then a valve is opened to allow the oxygen to escape. The final temperature and pressure are 75°C and 1 atm. Calculate the mass of oxygen that has escaped

(a) ~342 g (b) ~360 g (c) ~412 g (d) ~312 g

90. Krypton at 500 torr in a 250 ml container and helium at 950 torr in a 450 ml flask both at the same temperature are mixed by opening a stop cock in a narrow tube connecting the containers. Calculate the total final pressure.

(a) 690 torr (b) 670 torr (c) 720 torr (d) 790 torr

91. Calculate the pressure of H_2S gas in atm at 27°C if the density is 2.76 g L^{-1} (At.wt. of S = 32) is

(a) 2 atm (b) 1.5 atm (c) 2.25 atm (d) 2.5 atm

92. At a certain altitude above the surface of the earth (mean sea level) the temperature and pressure of air are −100°C and 5×10^{-7} torr. Calculate the density assuming the density of air at STP = 1. Assume also that the composition of the air is uniform throughout

(a) 10^{-8} (b) ~10^{-9} (c) 5×10^{-8} (d) 9×10^{-9}

93. Three volatile (gaseous) compounds of a certain elements have densities of 6.75 g L^{-1}, 9.56 g L^{-1} and 10.08 g L^{-1} respectively at STP. The three compounds contain 96%, 33.9% and 96.4% by weight of the element (Assume ideal behaviour of the gases) Suggest an atomic weight for the element

(a) 145 (b) 218 (c) 54.4 (d) 72.5

94. The approximate the volume in litres of CO_2 gas at 298 K and a pressure of 765 mm Hg that would produce 0.300 kg of calcium bicarbonate by reaction with $Ca(OH)_2$. Atomic weight of Ca = 40

(a) 75 L (b) 110 L (c) 90 L (d) 80 L

95. A mixture of CH_4 and C_4H_{10} has a pressure of 400 torr at temperature TK and volume V litre. The mixture is burnt in excess of oxygen. The partial pressure of CO_2 formed is 600 torr for the same temperature and volume. Calculate the mole fractions of CH_4 and C_4H_{10}

(a) 0.17, 0.83 (b) 0.34, 0.66 (c) 0.66, 0.34 (d) 0.83, 0.17

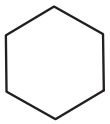
96. 5 litres of N_2 gas is collected over water at 27°C at a total pressure of 750 torr. Aqueous tension = 27 torr. If the volume (5 litres of moist gas) is dried over anhydrous $CuSO_4$ yielding $CuSO_4.5H_2O$, how many moles of $CuSO_4$ would be needed?

(a) 1.64×10^{-3} (b) 1.64×10^{-2} (c) 1.44×10^{-3} (d) 1.44×10^{-2}

97. In a gaseous mixture at 20°C, the total pressure = 775 torr. If the sum of the partial pressures of H_2 and CO_2 is 350 torr and the partial volume of hydrogen 19.4%, what is the partial pressure of CO_2

(a) 225 torr (b) 300 torr (c) 275 torr (d) 200 torr

98. A certain hydrocarbon (as a gas) has a rate of effusion about one sixth that of hydrogen under the same conditions of temperature and pressure. Which of the following structures would be most appropriate for it?

(a)  (b) C_4H_{10} (c) $C(CH_3)_4$ (d) 

99. The approximate temperature, T Kelvin at which the mean translational kinetic energy per mole of H_2 molecules would equal 104 k cal mol^{-1} (the dissociation energy of the molecules to yield atoms)

(a) 34900 K (b) 49300 K (c) 43900 K (d) 39400 K

100. The rms speed of CH_4 is 3.8×10^4 cm s^{-1}. The total kinetic energy of a mixture of 0.8 moles of CH_4 and 0.2 moles of C_4H_{10} is

(a) 289 J (b) 578 J (c) 1156 J (d) 2890 J

101. The rms speed of CH_4 at T_1 K and SO_2 at T_2 K are the same. The ratio T_1/T_2 is

(a) 0.25 (b) 0.30 (c) 0.20 (d) 0.35

102. The most probable speed of atoms of Argon (at weight = 40) at 27°C (in m s^{-1}) is

(a) ~420 m s^{-1} (b) ~350 m s^{-1} (c) ~320 m s^{-1} (d) ~400 m s^{-1}

103. At what temperature would the most probable speed of CO_2 molecules be twice that at 323 K?

(a) 911°C (b) 1091°C (c) 1272 K (d) 1019°C

104. The ratio of temperatures (K), $\frac{T_A}{T_B}$ at which the mean speed of molecules of gas A equals to rms speed of molecules of gas B(given $\frac{M_A}{M_B} = 16$) is

(a) ~20 (b) ~17 (c) ~19 (d) ~18

105. 10 moles of a real gas has a pressure of 11 atm in a container of volume 16 litres at 310K. Calculate the value of PV/RT where, V = volume per mole, and the magnitude of the second virial coefficient $\left(b - a/RT\right)$ in L mol^{-1}.

(a) −0.308 (b) +0.308 (c) +0.439 (d) −0.493

106. The vapour pressure values at the room temperature for liquids A, B, C and D are respectively 14.6 mm, 10 mm, 38 mm and 48 mm. Suggest which of these liquids has the largest value of the ratio $\left(\frac{a}{b}\right)$ of van der Waals constants

(a) A (b) B (c) C (d) D

107. The molecular radius for a certain gas is 2 Å. The van der Waals constant, b (in L mol^{-1}) is

(a) 0.0708 (b) 0.0805 (c) 0.0404 (d) 0.0440

108. The ratio of van der Waals constants $\frac{b}{a}$ for a certain gas is ~ 3.7×10^{-2}(in the accepted litre atm mol^{-1} units). Calculate T_c, the critical temperature

(a) 75.9 K (b) 79.5 K

(c) 59.7 K (d) 97.5 K

109. The critical temperature T_c and the critical pressure P_C for a certain gas have the values −118°C and 50 atm. Calculate the van der Waals constant, a in atm L^2 mol^{-2}

(a) 1.366 (b) 1.636

(c) 1.663 (d) 1.991

110. The total moisture in terms of kg of water vapour present in a cubical enclosure of side length 4.0 m at 27°C if the relative humidity 50%, aqueous tension at 27°C = 26.7 mm (relative humidity corresponds to actual vapour pressure expressed as a fraction of the saturated vapour pressure (aqueous tension)

(a) 0.82 kg (b) 0.72 kg

(c) 0.89 kg (d) 0.92 kg



Assertion–Reason Type Questions

Directions: **Each question contains Statement-1 and Statement-2 and has the following choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False
(d) Statement-1 is False, Statement-2 is True

111. **Statement 1**

In the graph of Maxwellian distribution of molecular speeds, the maximum of the graph shifts to the right at a higher temperature, but the graph itself becomes flatter.

and

Statement 2

The total (integrated) probability for all molecular speeds, (0 to ∞) represented by the area under the curve is the same for all temperatures.

112. **Statement 1**

In the plot of pressure, P against volume V, for one mole of an ideal gas, one isothermal and one adiabatic passes through every finite point.

and

Statement 2

For one mole of a van der Waals gas $Z = \frac{PV}{RT}$ is almost equal to 1 at the Boyle temperature.

113. Statement 1

For one mole of a van der Waals gas $\frac{RT_c}{P_cV_c}=\frac{8}{3}$

and

Statement 2

For most of the known gases the observed value of $\frac{RT_c}{P_cV_c}$ is quite close to $\frac{8}{3}$

Linked Comprehension Type Questions

Directions: **This section contains 1 paragraph. Based upon the paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

The Van der Waals equation for n moles is formulated in the form $\left(P+\frac{n^2a}{V^2}\right)\left(V-nb\right)=nRT$ where, a and b are known as Van der Waals constants. Each real gas is characterized by its own values of a and b. The van der Waals equation for one mole, of the form $P=\left(\frac{RT}{V-b}-\frac{a}{V^2}\right)$ can be expressed as a cubic equation in V for given values of P and V. The roots of the equation are all real numbers (3 roots) at lower temperature but only one root is real at high temperature. The temperature at which the cubic equation has three equal roots is known as the critical temperature, T_c, the corresponding pressure and volume are termed critical pressure P_c and critical volume V_c one may express P_c, V_c and T_c in terms of a and b(and R); V_c = 3b, $P_c=\frac{a}{27b^2}$ and $T_c=\frac{8a}{27Rb}$

114. The van der Waals constants for Nitrogen gas are a = 1.39 atm litre^2 mol^{-2} and b = 0.0391 litre mol^{-1}. For 5 moles of N_2 at 0°C and a volume of 0.560 litre, calculate the pressure

(a) 196.64 (b) 169.4 (c) 146.9 (d) 144.9

115. Considering van der Waals equation for one mole of Nitrogen in the form, $\frac{PV}{RT}=1+\left(\frac{b-\frac{a}{RT}}{V}\right)+$ higher order terms, the calculated the value of the term $\left[b-\frac{a}{RT}\right]$, also known as the second virial coefficient at 300 K (in L mol^{-1}), is

(a) 0.0137 (b) 0.0317 (c) –0.0173 (d) 0.0371

116. Using a = 1.39 atm (L mol^{-1})^2 and b = 0.0391 L mol^{-1} for Nitrogen. Calculate the magnitudes of P_C, V_C, and T_C.

(a) 33.67, 0.3117, 128.3 (b) 33.67, 0.1173, 128.3 (c) 37.63, 0.3117, 182.3 (d) 36.37, 0.3711, 182.3

Multiple Correct Objective Type Questions

Directions: **Each question in this section has four suggested answers of which ONE OR MORE answers will be correct.**

117. The density of a certain gas is 1.19 g litre^{-1} at 50°C and 730 mm pressure

(a) Molar mass of the gas = 32.85 g mol^{-1} (b) Mean kinetic energy mol^{-1} = 4028 J

(c) rms speed of molecules = 495 m s^{-1} (d) Most probable speed of molecules = 342 m s^{-1}

118. The value of the ratio of the van der Waal's constants $\frac{b}{a}$ for a certain gas is 0.029 in magnitude

(a) The Joule-Thomson coefficient changes sign at the temperature 840 K according to van der Waals equation.

(b The gas cannot be liquefied by the application of pressure above the temperature −149°C.

(c) The value of $\frac{RT_c}{P_cV_c}$ is $\frac{8}{3}$ for many gases.

(d) The molecular radius of the gas is ~ 1.58 A° (given b = 0.04 L mol^{-1}).

119. Which among the following statements is/are correct?

(a) At the critical temperature the densities of the liquid and vapour phases are equal.

(b) The square of the mean velocity of gas molecules is less in magnitude than the mean squared velocity of gas molecules.

(c) The van der Waals equation is a cubic in V for given magnitudes of P and T and the corresponding isothermal of P vs V at temperature < T_c exhibits a part which is a non equilibrium variation of P vs V.

(d) The van der Waals constant a and b for any gas are independent of temperature.



Matrix-Match Type Question

Directions: **Match the elements of Column I to elements of Column II. There can be single or multiple matches.**

120.

Column I	**Column II**
(a) T is doubled, P is halved, same number of moles	(p) V decreases to one fourth the previous value
(b) T is halved, P is doubled, same number of moles	(q) V increases to twice the preceding value
(c) T is halved, P is halved, number of moles is doubled	(r) V remains constant
(d) T is increased to 4 times its previous value, P is doubled, number of moles is halved	(s) V increases to 4 times the previous value

ADDITIONAL PRACTICE EXERCISE

Subjective Questions

121. A gas cylinder (I) contains 2 L of N_2 gas having a pressure of 4.8 atm at 27°C. This cylinder was then connected to another cylinder (II) of capacity 1 L filled with N_2, which is kept at 1.92 atm and at 27°C. Find the number of moles of N_2 transferred from cylinder (I) to (II). ($R = 0.08$ L atm mol^{-1} K^{-1}).

122. Calculate the weight of He that must be added to a 2.24 L cylinder containing 1.6 g O_2 at 0°C, in order to establish a total pressure of 2 atm.

123. A balloon has a volume of 600 mL at 15°C, when it is blown to 4/5 th of its maximum volume, will it burst at 27°C? What will be its bursting temperature?

124. A mixture of N_2 and O_2 at 1 atm pressure is stored in a 5 L iron container at constant temperature. If the entire oxygen reacts to form a solid iron oxide of negligible volume and if the final pressure is 456 torr, find the weight percent of O_2 in the original gaseous mixture.

125. An equimolar mixture of He and Ar gases exerts a pressure of 2.47×10^5 N m^{-2} in a 2 L glass bulb at 25°C. Find the weights of gases in the mixture. (Relative atomic mass of Ar = 40)

126. A mixture of chlorine gas and chlorine atoms at 1473K effuse through a pin hole 1.2 times as fast as krypton under similar conditions. Calculate the degree of dissociation of chlorine gas at this temperature (Mol.Wt.of Kr = 84).

127. An ideal gas is stored in a container at 27°C. Calculate the average kinetic energy per molecule of this gas in SI unit.

128. A certain mass of oxygen gas is maintained at 25°C and 800 torr.
(i) What is its volume at STP?
(ii) Calculate root mean square velocity of the gas in m s^{-1}.
(iii) What is the most probable velocity of the gas?

129. Mercury vapour has P_c = 3550 atm and Tc = 1900 K. Calculate the van der Waals constants a and b

130. 5 moles of a van der Waal's gas contained in a 8 L flask exerts a pressure of 11 atm at 37°C.
(i) What is the pressure of the gas if it behaves ideally?
(ii) Calculate the value of 'a'. Given, b = 0.05 dm^3 mol^{-1}. ('a' and 'b' are van der Waals constants).
(iii) What is the Boyle temperature of the gas?
(iv) Calculate the critical pressure P_c, volume V_c and temperature T_c of the gas.

Straight Objective Type Questions

Directions: **This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

131. According to Charles law V = KT where K is a constant. The unit of K is
(a) m^3 K^{-1} (b) m^{-3} K (c) m^3 K^2 (d) m^{-3} K^{-2}

132. An ideal gas initially having a volume of 20 litres at 27°C and a pressure of 2 atm, is allowed to expand into an evacuated vessel such that the final volume = 50 litres. Calculate the final pressure and the final temperature
(a) $0.8\text{atm}_{27^\circ C}$ (b) $0.8\text{atm}_{22^\circ C}$ (c) $0.7\text{atm}_{22^\circ C}$ (d) $0.65\text{atm}_{21^\circ C}$

133. A commercial gas cylinder contains 100 litres of Helium at a pressure of 11 atm. How many 2.5 litre-balloons at 1 atm may be filled by the gas in the cylinder?

(a) 350 (b) 450 (c) 500 (d) 400

134. At STP density of Nitrogen is 1.25 kg m^{-3}. At 0°C when a sample of the gas is compressed to 575 atm, the volume decreased from 1.5 litres to 4 cm^3. Clearly this is not in conformity with Boyle's law. By how much has the measured density deviated from Boyle's law?

(a) 250 g L^{-1} (b) 200 g L^{-1} (c) 300 g L^{-1} (d) 350 g L^{-1}

135. A container of volume 500 ml contains at 298 K a mixture of $N_2O_{4(g)}$ and $NO_{2(g)}$ under dissociation equilibrium : $N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$. The pressure is 1 atm. If initially 0.01726 mole of (only) $N_2O_{4(g)}$ was introduced into the vessel, what are (i) the degree of dissociation and (ii) the mean molar mass in g mol^{-1}

(a) 18.4%, 77.7 (b) 14.8%, 87.7 (c) 15.8%, 82.5 (d) 17.0%, 70.0

136. At 70°C the density of N_2O_4 is found to be 27.6. The percentage by volume of NO_2 molecules is

(a) 60 (b) 70 (c) 80 (d) 90

137. A certain container has 5×10^{23} molecules of a gas at a pressure of 900 torr. Partial photochemical dimerization is achieved at constant temperature and volume by allowing light to fall on the gas. The pressure then drops to 500 torr. How many monomer molecules are present at this stage?

(a) 5.56×10^{22} (b) 6.71×10^{22} (c) 9.73×10^{21} (d) 9.73×10^{22}

138. Ammonium carbamate decomposes as $NH_2COONH_4 \rightleftharpoons CO_2 + 2NH_3$ when 5 g of ammonium carbamate is completely vapourized at 200°C, it has a volume of 7.66 litres at a pressure of 740 m Hg:

The approximate the degree of dissociation is

(a) 0.85 (b) ~0.95 (c) ~1 (d) 0.75

139. The vapour density of nickel carbonyl, $Ni(CO)_4$ is 83.3 at 63°C and 70.8 at 100° C. Calculate the % dissociation at these temperatures. Atomic weight of Ni = 58

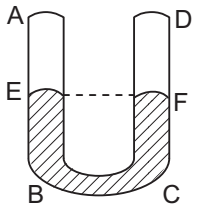
(a) 0.6%, 5.5% (b) 0.8%. 7.3% (c) 0.7%, 6.7% (d) 0.5%, 7.3%

140. A non volatile oil has a density of 0.9 g ml^{-1}. What height of a column of oil will represent an atmospheric pressure. value of 740 torr (of Hg) [density of Hg = 13.6 g ml^{-1}(assume T = 298 K)]

(a) 1811 cm (b) 1118 cm

(c) 811 cm (d) 1081 cm

141. In a tube ABCD closed at A(see fig) mercury has been poured. The levels of mercury E and F in both limbs are the same. The pressure is 1 atm. T = 298 K, AB = DC = 100 cm. AE = DF = 50 cm. Mercury is now poured into the open limb right upto the top (i.e., up to D) To what height will mercury rise above the previous level in the limb AB



(a) ~15.5 cm (b) ~17.5 cm

(c) ~13.5 cm (d) ~12.5 cm

142. A flask (of volume of 1103 ml) containing Nitrogen at a pressure of 710.5 mm is connected to an evacuated flask of volume V litre. The gas expands to occupy both flask. The final pressure is 583 mm. Calculate V

(a) 0.214 L (b) 0.412 L (c) 0.421 L (d) 0.241 L

143. A mixture of NH_3 and N_2 had a pressure of 1.50 kPa initially which decreased to 1.00 kPa after the NH_3 was absorbed from the mixture (at a certain temperature and volume). If, instead of absorbing, the NH_3 had been completely decomposed into N_2 and H_2 (at the same T and V) what would have been the pressure?

(a) 1.75 kPa (b) 2.5 kPa

(c) 2.25 kPa (d) 2 kPa

144. An open vessel at 10°C is heated to 400°C. What fraction of the weight of air originally contained in the vessel is expelled? [P = 1 atm, Volume remains constant]
(a) ~0.58 (b) 0.45 (c) 0.63 (d) 0.39

145. The vapour pressure of liquid water at 25°C is 24 torr. By what factor does the volume increase when one mole of water (density : 1 g m L^{-1}) vapourizes. Assume ideal gas behaviour for the vapour.
(a) 43042 (b) 34024 (c) 24043 (d) 23404

146. Three gases A, B and C have molar masses of 64 g mol^{-1}, 28 g mol^{-1} and 44 g mol^{-1} respectively. A mixture of A and B have the same density as C for the same temperature and pressure. Calculate the composition (mole ratio) of the mixture of A and B
(a) 3 : 7 (b) 4 : 5 (c) 3 : 5 (d) 4 : 7

147. Mercury vapour condensed in the cold trap of mercury diffusion pumps has a vapour pressure of 10^{-16} torr at 153 K. Calculate in the vapour the number of mercury atoms per ml
(a) 7.3 (b) 6.3 (c) 8.3 (d) 5.3

148. At a certain temperature and 760 mm pressure a gas mixture consists of N_2, O_2 and CO_2 at partial pressures of 420 mm, 190 mm and 150 mm respectively. Calculate the number of moles L^{-1} of each taking T = 300 K
(a) 0.02424, 0.01105, 0.00810 (b) 0.02442, 0.01510, 0.00810
(c) 0.02244, 0.01015, 0.00801 (d) 0.04224, 0.05110, 0.00810

149. The density at 2 atm and 310K of a gas mixture of CH_4, C_2H_6 and N_2 at mole fractions of 0.2, 0.3 and 0.5 is
(a) 1.56 g L^{-1} (b) 2.06 g L^{-1} (c) 2.26 g L^{-1} (d) 1.26 g L^{-1}

150. A mixture of Na_2CO_3 and $NaHCO_3$ of weight = m g is heated to decompose the bicarbonate in it. Among the products, the CO_2 gas obtained was n_1 moles. Next m g of the same mixture were treated with excess of HCl. In this case the CO_2 gas was n_2 moles. Given m = 10 g and $\frac{n_1}{n_2} = 0.3$, the volumes at STP of n_1 moles and n_2 moles are respectively
(a) 4.142 L, 0.742 L (b) 1.424 L, 0.472 L (c) 2.414 L, 0.724 L (d) 4.412 L, 0.247 L

151. One mole of solid $KClO_3$ is heated and decomposes along two reaction channels: $2KClO_{3(s)} \rightarrow 2KCl(s) + 3O_{2(g)}$; $4KClO_{3(s)} \rightarrow 3KClO_{4(s)} + KCl(s)$. In the experiment 0.25 mole of $KClO_4$ was obtained per mole of $KClO_3$. Assuming that no $KClO_3$ was left undecomposed. The volume of oxygen obtained at 740 mm and 27°C is
(a) 25.3 L (b) 23.5 L (c) 22.8 L (d) 26.6 L

152. An impure sample of Zinc blende (ZnS) containing Silica as impurity undergoes partial oxidation by roasting and suffers a loss of weight of 6 g
$ZnS_{(s)} + \frac{3}{2}O_{2(g)} \rightarrow ZnO_{(s)} + SO_{2(g)}$. Calculate the volume of $SO_{2(g)}$ obtained as measured at STP
(a) 4.2 L (b) 6.3 L (c) 8.4 L (d) 10.5 L

153. Density of neon will be highest at
(a) STP (b) 0^0C and 2 atm (c) 273^0C and 1 atm (d) 273^0C and 2 atm

154. A certain container holds 8 g of Argon at STP. What mass of Helium will it hold if it contains a mixture of Helium and Argon in the mole ratio 2 : 1 at 100°C and 100 atm? (Atomic weight of Ar = 40)
(a) 52 g (b) 47 g (c) 39 g (d) 29 g

155. Two flasks A and B of volumes 2 litres and 3 litres respectively are connected by a narrow tube carrying a stop cock 'A' contains oxygen at pressure, p_1 torr and B contains Nitrogen at pressure, p_2 torr. The stop cock kept closed at first is now opened and the gases allowed to mix. Assuming no change in temperature, The final pressure is (given $p_1 + p_2$ = 320 torr and $\frac{p_1}{p_2} = 2.2$).
(a) 128 torr (b) 138 torr (c) 158 torr (d) 148 torr

156. To which of the following mixtures, Dalton's law may not be applicable?
(a) CO_2 and CO at room temperature. (b) Helium and hydrogen at 27°C.
(c) Ammonia and steam at 100°C. (d) Nitrogen and CO_2 at room temperature.

157. A container of capacity 200 litres contains 3 moles of hydrogen gas and 1 mole of oxygen gas at 27°C (initially). The temperature is raised to 127°C and the gases react. What is the pressure(partial) of water vapour at this temperature. What is the total pressure?
(a) 0.3842 atm, 0.5226 atm (b) 0.4283 atm, 0.5225 atm (c) 0.3284 atm, 0.4926 atm (d) 0.2843atm,0.3126atm

158. If 4 g of oxygen diffuses through a very narrow hole, how much hydrogen would have diffused under identical conditions?
(a) 16 g (b) 1 g (c) $\frac{1}{4}$ g (d) 64 g

159. If E is the kinetic energy per unit volume of a gas, the equation for pressure of gas in terms of E can be given as
(a) $P = \frac{2}{3}E$ (b) $P = \frac{3}{2}E$ (c) $P = \frac{2}{3} \times \frac{E}{T}$ (d) $P = \frac{1}{3}E$

160. If a monoatomic ideal gas is heated through 10K, at constant volume, the increase in kinetic energy per mole of the gas is
(a) 10 cal $mol^{-!}$ (b) 10 J mol^{-1} (c) 30J mol^{-1} (d) 30 cal mol^{-1}

161. What is the ratio of kinetic energy per mole of Argon at 27°C and Helium at 127°C?
(a) 0.75 : 1 (b) 1 : 1 (c) 1 : 0.67 (d) 1 : 1.25

162. The rms speed of hydrogen molecules at a certain temperature t°C is 2.15 km s^{-1}. Calculate the temperature t°C.
(a) 25°C (b) 0°C (c) 250°C (d) 100°C

163. Assuming ideal behaviour. Calculate the most probable speed of methane molecules at 88°C in m s^{-1}.
(a) 651.2 (b) 561.2 (c) 612.5 (d) 526.2

164. Calculate for helium (at STP) the magnitude of the density in g L^{-1} given that the most probable speed = 10.62×10^4 cm s^{-1}.
(a) 0.1987 (b) 0.1798 (c) 0.1879 (d) 0.2179

165. Two gases A and B have the same magnitude of most probable speed at 298 K for A and 150 K for B. Calculate the ratio of their molar masses $\left(M_A / M_B\right)$.
(a) 2 : 1 (b) 1 : 0.75 (c) 1 : 2 (d) 3 : 1

166. Which of the following gases will show positive deviation from ideal gas behaviour at all pressures?
(a) N_2 (b) H_2 (c) CH_4 (d) CO_2

167. The compressibility factor of a certain gas is < 1 (per mole) at STP. The molar volume V_m, is
(a) $V_m = 22.4$ L (b) $V_m > 22.4$ L (c) $V_m < 22.4$ L (d) $V_m \geq 22.4$ L

168. A 50 litre gas cylinder is capable of withstanding a maximum pressure of 100 atm. It holds 6.6 kg of CO_2 van der Waals constants for CO_2 are a = 3.60 $atmL^2\ mol^{-2}$ and b = 4.30×10^{-2} L mol^{-1} respectively. Calculate the maximum temperature to which the gas can be heated with the cylinder remaining intact.
(a) 486.4 K (b) 648.4 K (c) 864.4 K (d) 468.4 K

169. The internal pressure for one mole of a van der Waals gas is
(a) $\frac{RT}{V-b}$ (b) $\frac{a}{V^2}$ (c) $\frac{a}{V}$ (d) $\frac{a}{b^2}$

170. A sample of Helium gas is maintained at STP in a container. The atomic radius of Helium = 176 pm. Calculate the fraction of the total volume occupied by the helium atoms
(a) 6.12×10^{-4} (b) 6.21×10^{-3} (c) 2.45×10^{-3} (d) 3.11×10^{-4}

Assertion–Reason Type Questions

Directions: **Each question contains Statement-1 and Statement-2 and has the following choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False
(d) Statement-1 is False, Statement-2 is True

171. Statement 1

Intermolecular attraction is due to dispersion forces for species such as noble gases.

and

Statement 2

Repulsive intermolecular forces are of shorter range compared to attractive forces.

172. Statement 1

The rate of effusion of a gas depends on the number of molecules per unit volume, $T^{1/2}$ and $M^{-1/2}$.

and

Statement 2

Number of collisions of gas molecules on the container wall per unit area per unit time is $\left(n\bar{c}/4\right)$.

173. Statement 1

Pressure of a given mass of gas is inversely proportional to volume at constant temperature.

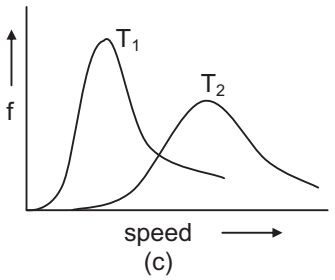
and

Statement 2

When volume is reduced the number of molecules per unit volume and the collisions per unit area of the wall of the container are increased.

174. Statement 1

In the Maxwellian graph of f(c) i.e., fractional probability for a given speed, C vs the molecular speed, the area under the graph is the same at all temperatures.



and

Statement 2

As in the plots shown above $T_1 < T_2$.

175. Statement 1

When an ideal gas expands into vacuum, there is no change in temperature.

and

Statement 2

At the temperature, $T_i = \left(2a/Rb\right)$ for a van der Waals gas there is a temperature change (rise or fall) in a Joule-Thomson experiment i.e., streaming of the gas through a porous plug.

176. Statement 1

At low pressures compressibility factor of O_2 is less than 1 and at high pressures it is greater than 1.

and

Statement 2

At low pressures molecules are far apart and repulsive forces are less than attractive forces and at high pressures there is more of repulsive forces than attractive forces between molecules.

177. Statement 1

The Van der Waals equation for one mole of a gas shown that $\left(PV/RT\right) \lesseqgtr 1$ depending on the temperature.

and

Statement 2

Most of the known gases which are easily liquefied have graphical plots showing $Z = \frac{PV}{RT}$ as a continuously increasing function of P at all pressures and temperatures.

178. Statement 1

An ideal gas has a low value of T_c(> 0 K) compared to any real gas.

and

Statement 2

The van der Waals equation for one mole of gas, has (at temperature < T_c) a graphical plot with a maximum and minimum at finite values of V.

179. Statement 1

For most of the known gases of low molar mass $\left(\frac{RT_c}{p_c V_c}\right)$ is very nearly equal to 2.67 experimentally.

and

Statement 2

van der Waals equation for one mole i.e. $P = \left(\frac{RT}{V-b} - \frac{a}{V^2}\right)$ may be (after suitable manipulation) expanded as a series in powers of $\left(\frac{1}{V}\right)$.

180. Statement 1

The Boyle temperature of a van der Waals gas depends on the ratio $\left(a/b\right)$ of the van der Waals constants.

and

Statement 2

The van der Waals constant, b, may be related to the molecular radius.

Linked Comprehension Type Questions

Directions: **This section contains 3 paragraphs. Based upon the paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

Passage I

The kinetic theory of ideal gases introduced theoretical (and mathematical) simplification by neglecting two important features of real gas behaviour viz., intermolecular attraction and molecular size. This procedure is (in a sense) justified even in the case of real gases at low pressures and high temperature. Under these conditions, the volume per mole of the gas is so large that the consequent large mean distances of molecules (except in the case of intermolecular collisions) resulted in totally negligible magnitudes of the (short range) attractions and (the shorter range) repulsions of molecules.

181. Calculate for oxygen gas at 27°C and a pressure of 800 torr, the density in g L.

(a) 1.683 (b) 1.368 (c) 1.638 (d) 1.836

182. The most probable molecular speed C_{mp} is given by the expression $C_{m.p} = \left(\frac{2RT}{M}\right)^{1/2}$. Calculate this for oxygen at 27°C in m s^{-1}.

(a) 395 (b) 539 (c) 359 (d) 593

183. Calculate for oxygen at 800 torr pressure and 27°C the number of molecules per ml and the mean kinetic energy per molecule in Joule.

(a) $2.97 \times 10^{18}, 6.52 \times 10^{-21}$ (b) $2.97 \times 10^{20}, 6.73 \times 10^{-20}$ (c) $2.57 \times 10^{19}, 6.215 \times 10^{-21}$ (d) $2.79 \times 10^{20}, 6.37 \times 10^{-20}$

Passage II

The kinetic theory of ideal gases ignores both intermolecular attractions and repulsions. The latter is encapsulated in the assumptions that (i) molecules are rigid spheres and (ii) molecules have, each a definite radius. A rather intuitive approach to a formulation of an equation which takes into consideration both intermolecular attractions and molecular size was that of van der Waals. The equation per mole is $p = \left(\frac{RT}{V-b} - \frac{a}{V^2}\right)$ where, a and b are known as van der Waals constants, with characteristic values for each gas, the constant b is related to molecular radius: $b = 4 \times N_0 \times \left(\frac{4}{3}\pi r^3\right)$; N_0 = Avogadro number $\left(\frac{a}{V^2}\right)$ stands for intermolecular attractions and is called internal pressure

184. The value of b for methane gas is 0.043 litre mol^{-1}. Calculate the radius of the molecule.

(a) 1.26A° (b) 2.16 A° (c) 2.161A° (d) 1.62A°

185. A certain container of volume 10 litre contains 2 moles of a gas A at 310 K and a pressure of 7.00 atm. By how much does the compressibility factor per mole, $\frac{PV}{RT}$ deviate from the ideal gas value?

(a) 0.357 (b) –0.357 (c) 0.375 (d) +0.186

186. For one mole of CO_2 gas at a pressure 101.32 kPa T = 273.15 K and $V = 22.263 \times 10^{-3}$ m^3 mol^{-1} and a = 0.3637 m^6 Pa mol^{-2}. Calculate $\left(\frac{\partial E}{\partial V}\right)_T = \frac{a}{V^2}$ (internal pressure) assuming CO_2 to be a van der Waals gas.

(a) 733.79 Pa (b) 937.37 Pa (c) 973.73 Pa (d) 779.33 Pa

Passage III

The van der Waals equation for one mole = $\left(P + \frac{a}{V^2}\right)(V - b) = RT\right)$. On expanding with clearing of denominators one gets the cubic $PV^3 - (Pb + RT)V^2 + aV - ab = 0$. This is a cubic equation in V and must have 3 roots for given values of p and T viz.,(i) three real roots (ii) three identical roots or (iii) one real root with the other two roots as complex conjugates. The situation of identical roots is easy to investigate. viz.,$(V - V_C)^3 = V^3 - (3V_C)V^2 + (3V_C^2)V - V_C^3 = 0$ where, V_C is the root. Compare $V^3 - \frac{(Pb + RT)}{P}V^2 + \left(\frac{a}{P}\right)V - \frac{ab}{P} = 0$. Equating corresponding coefficients $3V_C^2 = \frac{a}{P}$, $V_C^3 = \frac{ab}{P}$, $\frac{V_C}{3} = b$, $V_C = 3b$ With a little more manipulation one may show $P = P_C = \left(a/27b^2\right)$ and $T_C = T = \frac{8a}{27Rb}$. The three quantities P_C, V_C and T_C are called critical constants and have characteristic values for each gas.

187. P_C and T_c for certain gas are 3550 atm and 1900K. The van der Waals constant a is

(a) 3.90 atm L^2 mol^{-2} (b) 3.60 atm L^2 mol^{-2} (c) 2.90 atm L^2 mol^{-2} (d) 2.50 atm L^2 mol^{-2}

188. Using the equation $b = 4 \times 6.02 \times 10^{23} \times \left(\frac{4}{3}\pi r^3\right)$, Calculate the molecular radius of the gas.

(a) 0.871 A° (b) 0.817A° (c) 0.718 A° (d) 0.781 A°

189. Using the values of a and b calculated from the data given above, Calculate the second virial coefficient $\left(b - \frac{a}{RT}\right)$ at 27°C.

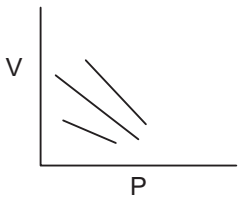
(a) -0.2211 L mol^{-1} (b) -0.3322 L mol^{-1}

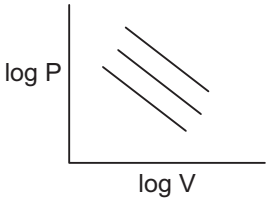
(c) -0.2233 L mol^{-1} (d) -0.1122 L mol^{-1}

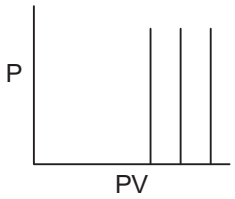
Multiple Correct Objective Type Questions

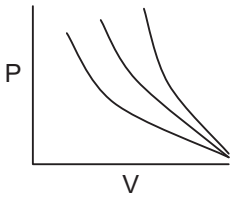
Directions: **Each question in this section has four suggested answers of which ONE OR MORE answers will be correct.**

190. Which of the family of isotherms at different temperatures correctly represent Boyle's law?

(a) 

(b) 

(c) 

(d) 

191. In the light of the kinetic theory of gases

(a) Pressure of a gas is proportional to the mean squared velocity at a given number density of molecules.

(b) Mean kinetic energy of gas molecules is directly proportional to the Kelvin temperature.

(c) Since molecular collisions are perfectly elastic the kinetic energy of a colliding molecule is the same before and after a collision.

(d) The actual total volume of all the gas molecules per mole is equal to the van der Waals constant, b of the gas.

192. Which of the following gases/vapour have the same rms speed at a particular temperature?

(a) H–CHO (b) CH_4

(c) C_2H_6 (d) NO

193. According to the Maxwellian law for distribution of molecular speeds

(a) As temperature increases the fraction of the total number of molecules having high speeds increases.

(b) The ratio of the most probable velocity to the mean velocity of molecules depends on temperature.

(c) The fraction of the total number of molecules with speeds less than the most probable speed equals the corresponding fraction of molecules with speeds greater than the most probable speed.

(d) At a particular temperature and pressure the most probable velocity of a lighter gas is greater than the most probable velocity of a heavier gas.

194. The mean velocity of molecules of a gas (assuming ideal gas behaviour) becomes half, when

(a) The temperature is decreased to half its initial value.

(b) Pressure is decreased to half its initial value.

(c) Both P and T are reduced to one fourth their initial values.

(d) T is decreased to one fourth the initial value.

195. Identify the correct statement/samong the following
(a) Theoretically speaking an ideal gas cannot undergo Joule Thomson cooling by expansion through a porous plug.
(b) Theoretically speaking an ideal gas cannot be liquefied by adiabatic expansion and cooling.
(c) According to van der Waals equation molecules of a real gas are hard rigid spheres with definite radii.
(d) When any gas at any temperature and pressure expands through a porous plug there is always a fall in temperature.

196. Which of the following statements are correct?
(a) For an ideal gas, the compressibility factor = 1 at all temperatures.
(b) For a van der Waals gas, compressibility factor is nearly equal to 1 at the Boyle temperature.
(c) For Helium gas the compressibility factor is less than 1, at all pressures.
(d) For a real gas deviation from ideality is zero if its liquefaction temperature is low.

197. A comparison of N_2 and NH_3 shows that unlike N_2, NH_3 is quite easily condensable i.e., liquefiable. The reason is
(a) NH_3 has a larger value of van der Waals constants a.
(b) NH_3 has a pyramidal structure.
(c) N_2 does not have a dipole moment while NH_3 molecule has a large value of dipole moment.
(d) NH_3 is not a diatomic molecule like N_2.



Matrix-Match Type Question

Directions: **Match the elements of Column I to elements of Column II. There can be single or multiple matches.**

198.

Column I	**Column II**
(a) Carbon monoxide	(p) Diffuses faster than C_2H_6
(b) Formaldehyde(vapour)	(q) Diffuses as fast as NO
(c) Hydrogen	(r) Diffuses as fast as N_2
(d) Deuterium	(s) Diffuses faster than all other gases given

199.

Column I	**Column II**
(a) Same rms speed	(p) He at 300K and 200 atm
(b) $Z > 1$	(q) CH_4 at 27°C and O_2 at 327°C
(c) Ratio of magnitudes of translational kinetic energy per mole = 1 : 2	(r) N_2 above the Boyle temperature
(d) $V_{real} > V_{ideal}$	(s) Same mean speed of molecules

200.

Column I	**Column II**
(a) $\left(a/b\right)$ ratio of van der Waals equation related to	(p) $\left(b - a/RT\right)$
(b) Second virial coefficient	(q) Large intermolecular attractive forces
(c) Easy liquefaction of the gas	(r) Boyle temperature
(d) $Z = 1$	(s) Critical temperature

SOLUTIONS

ANSWER KEYS

Topic Grip

1. 0.75
2. 78.3
3. Ideal gas
4. Cl_2O
5. CH_4
6. 28.12
7. $C_{RSM} = 517\ ms^{-1}$
$C_{MPV} = 422\ ms^{-1}$
$C_{AV} = 476\ ms^{-1}$
8. 1
9. 1.9×10^{-10} m
10. 1.932

11. (c)	**12.** (b)	**13.** (c)
14. (a)	**15.** (d)	**16.** (a)
17. (a)	**18.** (d)	**19.** (c)
20. (b)	**21.** (c)	**22.** (b)
23. (b)	**24.** (b)	**25.** (a)

26. (d)
27. (a), (c)
28. (b), (c), (d)
29. (a), (b), (d)
30. (a) → (p)
(b) → (q), (s)
(c) → (r)
(d) → (q), (s)

IIT Assignment Exercise

31. (b)	**32.** (a)	**33.** (d)
34. (d)	**35.** (c)	**36.** (c)
37. (a)	**38.** (c)	**39.** (d)
40. (b)	**41.** (c)	**42.** (d)
43. (c)	**44.** (a)	**45.** (b)
46. (c)	**47.** (d)	**48.** (c)
49. (d)	**50.** (d)	**51.** (a)
52. (d)	**53.** (c)	**54.** (a)
55. (c)	**56.** (b)	**57.** (c)
58. (a)	**59.** (b)	**60.** (c)

61. (c)	**62.** (b)	**63.** (b)
64. (a)	**65.** (a)	**66.** (a)
67. (a)	**68.** (b)	**69.** (b)
70. (a)	**71.** (a)	**72.** (a)
73. (d)	**74.** (c)	**75.** (a)
76. (c)	**77.** (a)	**78.** (b)
79. (b)	**80.** (b)	**81.** (a)
82. (c)	**83.** (d)	**84.** (b)
85. (b)	**86.** (c)	**87.** (c)
88. (a)	**89.** (b)	**90.** (d)
91. (a)	**92.** (b)	**93.** (d)
94. (c)	**95.** (d)	**96.** (c)
97. (d)	**98.** (c)	**99.** (a)
100. (c)	**101.** (a)	**102.** (b)
103. (d)	**104.** (c)	**105.** (d)
106. (b)	**107.** (b)	**108.** (d)
109. (a)	**110.** (a)	**111.** (a)
112. (b)	**113.** (c)	**114.** (a)
115. (c)	**116.** (b)	

117. (a), (b), (c)
118. (a), (b), (d)
119. (a), (b), (c)
120. (a) → (s)
(b) → (p)
(c) → (q)
(d) → (r)

Additional Practice Exercise

121. 0.08
122. 0.6 g
123. 360 K
124. 43%
125. Weight of Al = 4 g
Weight of He = 0.4 g
126. 0.218
127. 6.21×10^{-21} J
128. (i) 23.22 L
(ii) 482 ms^{-1}
(iii) 393.5 ms^{-1}
129. a = 2.889 atm $L^2\ mol^{-2}$
b = 0.00549 L mol^{-1}
130. (i) 15.89 atm
(ii) 13.82 atm $L^2\ mol^{-2}$
(iii) 3371 K

131. (a)	**132.** (a)	**133.** (d)
134. (a)	**135.** (a)	**136.** (c)
137. (a)	**138.** (c)	**139.** (c)
140. (b)	**141.** (a)	**142.** (d)
143. (d)	**144.** (a)	**145.** (a)
146. (b)	**147.** (b)	**148.** (c)
149. (b)	**150.** (c)	**151.** (a)
152. (c)	**153.** (b)	**154.** (c)
155. (d)	**156.** (c)	**157.** (c)
158. (b)	**159.** (a)	**160.** (d)
161. (a)	**162.** (d)	**163.** (c)
164. (b)	**165.** (a)	**166.** (b)
167. (c)	**168.** (d)	**169.** (b)
170. (c)	**171.** (b)	**172.** (a)
173. (a)	**174.** (b)	**175.** (c)
176. (a)	**177.** (c)	**178.** (d)
179. (d)	**180.** (b)	**181.** (b)
182. (a)	**183.** (c)	**184.** (d)
185. (c)	**186.** (a)	**187.** (c)

188. (b)
189. (d)
190. (b), (c)
191. (a), (b)
192. (a), (c), (d)
193. (a), (d)
194. (c), (d)
195. (a), (b), (c)
196. (a), (b)
197. (a), (c)
198. (a) → (p), (r)
(b) → (q)
(c) → (p), (s)
(d) → (p)
199. (a) → (q), (s)
(b) → (p), (r)
(c) → (q)
(d) → (p), (r)
200. (a) → (q), (r)
(b) → (p), (r)
(c) → (q), (s)
(d) → (r)

HINTS AND EXPLANATIONS

Topic Grip

1. Suppose the mole ratio of CO : CO_2 = x(1 – x)

$28x + 44(1 - x) = 32 \;\therefore\; (44 - 32) = (44 - 28)\,x$

$\therefore \quad x = \frac{12}{16} = 0.175$ = mole fraction of CO

2. $N_2O_4 \rightleftharpoons 2NO_2$

$$\alpha = \frac{d_0 - d}{d} = \frac{46 - 25.8}{25.8} = 78.3$$

% by weight of NO_2 = 78.3

3. Since 1 atm = 1.013×10^5 Pa = 1.013×10^5 Nm^{-2}

$$\frac{3.66 \times 10^5}{1.013 \times 10^5} = 3.613 \text{ atm}$$

Since $\frac{PV}{RT}$ is $\frac{3.613 \times 0.2}{0.0821 \times 298} = 0.02954$

$$\frac{1.78 \times 10^{22}}{6.02 \times 10^{23}} = 0.02957$$

In a real gas $\frac{PV}{RT}$ = n(1 + a factor)

In an ideal gas $\frac{PV}{RT}$ = n.

In the above case n = 0.02954. The agreement by two methods shows that the gas behaves ideally.

4. It is clear that 30 ml of the oxide of chlorine were decomposed into 30 ml of chlorine and 15 ml of O_2. Since for the same pressure and temperature partial volumes are proportional to numbers of moles, the oxide of chlorine contains Cl_2 and O_2 in the ratio 30 : 15 i.e., 2 : 1. Thus the formula is suggested to be Cl_2O.

5. Partial pressure = mole fraction × P

Let w be the weight of the gases and m be the molecular weight of unknown gas.

Then $$\frac{\frac{w}{2}}{\frac{w}{2} + \frac{w}{M}} = \frac{8}{9}$$

$$\frac{\frac{1}{2}}{\frac{1}{2} + \frac{1}{M}} = \frac{8}{9}$$

$$1 + \frac{2}{M} = \frac{9}{8} \;\Rightarrow\; \frac{2}{M} = \frac{9}{8} - 1 = \frac{1}{8}$$

M = 16

$\therefore$ The gas is methane.

6. $$\frac{r_g}{r_{O_2}} = \frac{50/25}{100/27.7} = \frac{\sqrt{32}}{\sqrt{M}}$$

$\therefore$ M = 104

Eq.wt. = $\frac{27}{73} \times 19 = 7.03$

Equivalent weight of the fluoride = 19 + 7.03

$$= 26.03 = \frac{104}{4}$$

$\therefore$ Atomic weight = 4 × 7.03 = 28.12

7. $$\left(\frac{3RT}{M}\right)^{1/2} = \left(\frac{3 \times 8.314 \times 10^7 \times 300}{28}\right)^{1/2} \text{ cm s}^{-1}$$

$= 5.169 \times 10^4$ cm s^{-1} = 517 m s^{-1}

$\left(\frac{2RT}{M}\right)^{1/2} = 4.22 \times 10^4$ cm s^{-1} = 422 m s^{-1}

$\left(\frac{8RT}{\pi M}\right)^{1/2} = 4.76 \times 10^4$ cm s^{-1} = 476 m s^{-1}

8. Volume per mole = 22.45 litre

$\therefore$ compressibility factor

$$= \frac{PV}{RT} = \frac{1 \times 22.45}{0.0821 \times 273} = 1$$

9. $b = 4N_0\left(\frac{4}{3}\pi r^3\right)$

Straight substitution with $N_0 = 6.02 \times 10^{23}$ gives

$r = 1.9 \times 10^{-10}$m

10. $P_1V_1 = Z_1nRT_1 \qquad P_2V_2 = Z_2nRT_2$

$$V_2 = \frac{Z_2}{Z_1} \times \frac{T_2}{Z_1} \times \frac{P_1V_1}{P_2}$$

$$= \frac{1.68}{1.12} \times \frac{290}{300} \times \frac{40 \times 2}{60} = 1.93 \text{ L}$$

11. Acc to Boyle's law PV should be constant for a given mass of gas at a given temperature

$\therefore$ log P + log V = constant

or log P = –log V + constant.

The plot has a slope = –1

(descending at 45° to the vertical)

12. Mean velocity = $\sqrt{\frac{8}{\pi}} \times \sqrt{\frac{RT}{M}}$.

Most probable velocity = $\sqrt{2}\ \sqrt{\frac{RT}{M}}$

Ratio = $\sqrt{\frac{8}{2\pi}} = \sqrt{\frac{4}{\pi}} = 1.128$

13. Large a and small b indicates large intermolecular attraction, easy liquifiability and large deviation from ideal behaviour.

14. Mean free path $\ell = \frac{1}{\sqrt{2}\,\pi n\sigma^2}$. Inverse proportionality to the square of the collision diameter.

15. Boyle temperature $T_B = \frac{a}{Rb}$,

because $b - \frac{a}{RT_B} = 0$

16. Both statements are true and are certainly linked.

17. If the rates of effusion are equal then $\frac{r_{CH_4}}{r_{SO_2}} = 1$

$$= \frac{p_{CH_4}}{p_{SO_2}} \sqrt{\frac{64}{16}}$$

$$\therefore \quad \frac{p_{CH_4}}{p_{SO_2}} = \sqrt{\frac{16}{64}} = \frac{1}{2}$$

18. Statement 1 is incorrect. $b = 4 \times \left(N_0 \times \frac{4}{3}\pi r^3 \right)$

19. Statement 2 is incorrect

20. Both statements are true but not linked

21. $2 \times 2.24 = n \times 0.0821 \times 273$

$\therefore$ n = 0.2 mol

1.6 g of $O_2 \equiv \frac{1.6}{32}$ mole = 0.05 mol

$\therefore$ number of moles of He = (0.2 – 0.05)mol

= 0.15 mole

= 0.15 × 4 = 0.6 g

22. $14.4 \times 76 = \ell \times (76 + 15.2)$ (Boyle's law)

$\therefore \quad \ell = 12$ cm

23. Obviously $PV = n_1RT_1 = n_2RT_2$

$\therefore \quad n_1T_1 = n_2T_2$

$$n_1 \times 300 = \frac{2}{3} n_1 \times T_2$$

$$\therefore \quad T_2 = \frac{3}{2} \times 300 = 450 \text{ K} = 177°\text{C}$$

24. The unit of "a" is atm L^2 mol^{-2}

25. $b \approx 6.9 \times 10^{-2}$ L mol^{-1}

26. Z = 0.707

27. $P \times 0.82 = 0.8 \times 0.0821 \times 300$

$\therefore \quad P = 24$ atm

$$pHe = \frac{0.5}{0.8} \times 24 = 15 \text{ atm}$$

28. C_2H_4, N_2, CO have the same molar masses = 28

(a) is incorrect

29. Obviously (c) is wrong

30. (a) → p

(b) → q, s

(c) → r

(d) → q, s

IIT Assignment Exercise

31. Gases fill the whole container, seen only in gaseous state.

a, c and d may be observed in liquids also.

32. Volume of gas = Volume of the container.
Only number of moles will change.

33. Let volume of vessel be VL
Then pressure of gas removed is 5 – 3.5
= 1.5 atm
$1.5 \times V = 3 \times 1$
$V = \frac{3 \times 1}{1.5} = 2$ L

34. PV = constant P = constant $\frac{1}{V}$ (y = mx type)
So line should pass through origin
log P = –log V + constant . So it is a negative slope (m = –1)
In graph (c), $T_1 > T_2$
So only (d) is the right option

35. Both are occupying equal volume and are at same pressure.
$n_1T_1 = n_2T_2$
$\frac{w}{32}T_{O_2} = \frac{w}{16}T_{CH_4}$
$T_{O_2} = 2T_{CH_4}$

36. PV is a constant at constant T
$P_fV_f = P_i(V_A + V_B)$
$1 \times 10 = 2(2 + V_B)$
$V_B = \frac{10-4}{2} = 3$ L

37. At STP all gas have a molar volume = 22.4 L
∴ One molecule of all gases will occupy
$\frac{22.4}{6.02\times10^{23}}$ L

38. Two equations with two unknowns can be written
Initially, $2 \times V = 2 \times R \times T_1$ — (1)
Finally, $2 + \left(\frac{20}{100}\times2\right)$.
$V = 2 \times R \times (T_1 + 10)$
Or $2.4\ V = 2RT_1 + 20R$ — (2)
(2) – (1)
$0.4\ V = 20R$ $V = \frac{20}{0.4} \times 0.082 \cong 4.1$ L

39. Let molar ratio be $n_1 : n_2$
So after combustion, at 25°C only CO_2 is gas so the ratio of CO_2 from CH_4 and C_2H_6 is $n_1 : 2n_2$
At constant temperature and Volume. $P \propto n$
So $\frac{P_1}{P_2} = \frac{n_1 + n_2}{n_1 + 2n_2}$
$\Rightarrow P_1(n_1 + 2n_2) = P_2(n_1 + n_2)$
$n_1[P_1 - P_2] = n_2[P_2 - 2P_1]$
$\Rightarrow \frac{n_1}{n_2} = \frac{2P_1 - P_2}{P_2 - P_1}$

40. 1 mole occupies 22400 cm^3
1 molecule will occupy $\frac{22400}{6.02\times10^{23}}$ cm^3

41. Vapour density of O_2 = 16 (V.d = $\frac{\text{mol wt}}{2}$)
$N_2 = 14$
$V.d = \frac{1}{2} \times 16 + \frac{1}{2} \times 14 = 15$

42. Density = $\frac{\text{mass}}{\text{volume}}$
Volume of gas at STP = 22.4 L
Mass = molar mass = $\frac{1}{10} \times 44 + \frac{9}{10} \times 32$
= 4.4 + 28.8 = 33.2 g
$d = \frac{33.2}{22.4} = 1.48 \cong 1.5\ gL^{-1}$

43. $P = hg\rho$
Pressure = atmospheric pressure = 1 atm
$h_1g\rho_1 = h_2g\rho_2$ 1 – mercury
$760 \times 13.6 = h_2 \times 1$ 2 – water
$h_2 = 760 \times 13.6 = 10336$ mm

44. Pressure and volume are constant
so $n_1T_1 = n_2T_2$
$\frac{n_1}{n_2} = \frac{T_2}{T_1} = \frac{280}{300} = 0.93$
$1 - \frac{n_1}{n_2} = 1 - 0.93$
$\frac{n_2 - n_1}{n_2} = 0.07$
percentage increase = $\frac{n_2 - n_1}{n_2} \times 100 = 7\%$

45. $d = \frac{PM}{RT} = \frac{1\times84}{0.082\times273} = 3.75\ gL^{-1}$

or $\frac{84}{22.4} = 3.75\ gL^{-1}$

46. Volume of water vapour = 10 L i.e., volume of the gas

$PV = nRT$

$P = \frac{0.24}{18} \times 0.082 \times 300 \times \frac{1}{10}$

$= 0.0328 \cong 0.033$ atm

47.

	H_2 +	Br_2 →	2HBr
Initial	$\frac{1\times V}{0.082\times400}$	$\frac{0.5V}{0.082\times400}$	
Final	$\frac{0.5V}{0.082\times400}$	0	$\frac{2\times0.5V}{0.082\times400}$

Total final pressure P

$$= \frac{3\times\left(\frac{0.5V}{0.082\times400}\right)\times0.082\times400}{2V} = 0.75 \text{ atm}$$

Partial pressure of HBr $= \frac{2\times\frac{0.5V}{RT}}{3\times\frac{0.5V}{RT}}\times0.75$

$= 0.5$ atm

48.

	$4NH_3$ +	$3O_2$ →	$2N_2$ +	$6H_2O$
Initial	30 ml	30 ml		
Final	0	7.5 ml	15 ml	45 ml

NH_3 is the limiting reagent

$PV = \frac{w}{M}RT$

$w = \frac{PVM}{RT} = \frac{2\times15\times10^{-3}\times28}{0.082\times400} = 0.0255$ g

49. $R = 8.314\ J\ mol^{-1}\ K^{-1}$

$1eV = 1.6 \times 10^{-19}\ J$

$R = \frac{8.314}{1.6\times10^{-19}}\ eV\ mol^{-1}\ K^{-1}$

$= 5.19 \times 10^{19}\ eV\ mol^{-1}\ K^{-1}$

50. In the case of a, b, c there is a change in total pressure. So irrespective of change in mole fraction, partial pressure will change.

Only in the case of 'd' both total pressure and mole fraction do not change.

51. Total pressure = 1 atm

Irrespective of amount of carbon, equal moles of CO and H_2 are formed.

So $X_{CO} = 0.5$

So partial pressure $= 0.5 \times 1 = 0.5$ atm

52. At STP, 32 g of O_2 occupies 22.4 L

So $\frac{V_{He}}{V_{He}+V_{O_2}} = \frac{22.4\times\frac{10}{4}}{22.4\times\frac{10}{4}+22.4\times\frac{90}{32}}$

$= \frac{\frac{10}{4}}{\frac{10}{4}+\frac{90}{32}} = \frac{10}{4}\times\frac{32}{170}$

$= \frac{8}{17} = 0.47$

53. The constant R is work done/degree/mole.

At STP R = 8.314 J/K/mole

54. Molecular weight of fluorine $F_2 = 38\ g\ mol^{-1}$ and since Argon is monoatomic At. Weight $= 39.9\ g\ mol^{-1}$. Since both are almost the same weight, there are equal number of moles of each. Hence both will have same partial pressure.

55.

	$2SO_2$ +	O_2 →	$2SO_3$
Initial	3	3	0
Final	3–1	3 – 0.5	1.0
	= 2	= 2.5	

Mole fraction $= \frac{1}{5.5}$

At constant temperature and Volume $\frac{P_i}{P_f} = \frac{n_i}{n_f}$

$P_f = P_i \times \frac{n_f}{n_i} = 10 \times \frac{5.5}{6.0}$

Partial pressure of $SO_3 = \frac{1}{5.5} \times \frac{10 \times 5.5}{6}$

$= 1.66$ atm

56. $\frac{r_{mix}}{r_{N_2}} = 1.97 = \sqrt{\frac{M_{N_2}}{M_{mix}}}$

$1.97 \times 1.97 = \frac{28}{M_{mix}}$

$M_{mix} = 7.2 \text{ g mol}^{-1}$

$(x \times M_{H_2}) + (1 - x)\ M_{N_2} = 7.2$

$2x + 28 - 28x = 7.2$

$26x = 20.8$

$x = \frac{20.8}{26} = 0.8$

57. $\frac{w_1}{t_1} \times \frac{t_2}{w_2} = \sqrt{\frac{M_1}{M_2}}$

$\frac{t_{He}}{t_{Ar}} = \sqrt{\frac{M_{Ar}}{M_{He}}} = \sqrt{\frac{40}{4}} = \sqrt{10} = 3.16$

58. $\frac{r_{Cl_2}}{r_{ClO_x}} = \frac{P_{Cl_2}}{P_{ClO_x}} \sqrt{\frac{M_{ClO_x}}{M_{Cl_2}}}$

$\frac{n_{Cl_2}}{n_{ClO_x}} \times \frac{t_{ClO_x}}{t_{Cl_2}} = \frac{P_{Cl_2}}{P_{ClO_x}} \cdot \sqrt{\frac{M_{ClO_x}}{M_{Cl_2}}}$

$\frac{26.5}{40} = \frac{1.2}{2} \sqrt{\frac{M_{ClO_x}}{71}}$

$\sqrt{\frac{M_{ClO_x}}{71}} = 1.11$

$M_{ClO_x} = 71 \times 1.11 \times 1.11 = 87.4$ g

This is approximately molecular weight of Cl_2O – (71 + 16).

59. K.E $= \frac{3}{2}$ nRT for n moles

For one molecule K.E $= \frac{3}{2} \times \frac{8.314 \times 300}{6.02 \times 10^{23}}$

$= 6.21 \times 10^{-21}$

60. K.E $= \frac{3}{2}$ nRT T is constant

n is 0.2 for both N_2 and O_2

61. Energy is conserved during collision.

Volume of a gas is much greater than volume of the molecule which is negligible. Both mass and size are equal. Molecules move with different velocities hence kinetic energy is different for different molecules. Only average kinetic energy

$= \frac{3}{2}$ nRT.

62. $u_{rms} = \sqrt{\frac{3RT}{M}}$

$u = \sqrt{\frac{3RT}{M}}$ $M = 32$

$u' = \sqrt{\frac{3RT}{16}}$

$\frac{u'}{u} = \sqrt{\frac{32}{16}}$

$u' = \sqrt{2}\ u$

63. $d = \frac{PM}{RT} = \frac{\frac{700}{760} \times 28 \times 10^{-3}}{0.082 \times 300}$

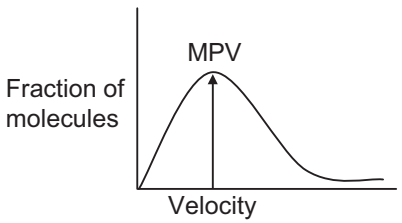
$= 1.05 \times 10^{-3} \text{ kg L}^{-1} = 1.05 \text{ kg m}^{-3}$

$C_{rms} = \sqrt{\frac{3P}{d}} = \sqrt{\frac{3 \times \frac{700}{760} \times 1.01 \times 10^5}{1.05}}$

$= \sqrt{2.65 \times 10^5} = 515.5 \text{ ms}^{-1}$

64. Area under the curve represents number of molecules. For a given sample it is a constant so A = B.

65. The curve for the plot of fraction of molecules Vs speed is



Since MPV < Av.V < C_{rms}. We are looking at that part of curve where fraction of molecules decreases with increase in speed.

Since average speed < C_{rms}, $f_1 > f_2$

66. $V_{Avg} = \sqrt{\frac{8RT}{\pi M}}$

$M = \frac{8\times8.314\times1473}{3.14\times674.4\times674.4}$

$= 0.0686\ kg = 68.6\ g$

$Cl_2 \rightarrow 2Cl$

Initial 1

Final $1-x$ $2x$

So $68.6 = \frac{2x\times35.5+(1-x)71}{1+x}$

$68.6 + 68.6\,x = 71\,x + 71 - 71x$

$x = \frac{2.4}{71} = 0.035$

% dissociation = 3.5%

67. $V_m > 22.4$

It is difficult to compress gas.

68. $Z = \frac{V_{real\ gas}}{V_{ideal\ gas}} = \frac{20.8}{22.4} = 0.93$

69. $Z = \frac{PV}{nRT}$

$V = \frac{Z.nRT}{P} = \frac{0.8\times5\times0.082\times273}{50} = 1.8\ L$

70. Since intermolecular forces are negligible

$P(V-b) = RT$

$PV = RT + Pb$

So Z is always greater than 1

71. $\frac{n^2a}{v^2}$ is included for pressure correction due to the force of attraction between molecules, the molecule near the wall experiences an inward pull and so the pressure exerted is less than the ideal pressure.

72. CS_2 being a polar molecule, has greater Van der Waals forces of attraction, hence easily liquefiable, so low Z value.

73. $a_1 > a_2$ – gas 1 has greater intermolecular forces of attraction.

$b_1 < b_2$ – molecules of gas 2 occupy less space.

74. $b = 4V$

$b = 0.03\ L\ mol^{-1}$

$= \frac{0.03\times10^{-3}\ m^3 molecule^{-1}}{6\times10^{23}}$

$= 5\times10^{-29}\ m^3\ molecule^{-1}$

$4\times\frac{4}{3}\pi r^3 = 5\times10^{-29}$

$r = \sqrt[3]{\frac{5\times10^{-29}\times3}{4\times4\times3.14}}$

$= \sqrt[3]{2.98\times10^{-30}}$

$= 1.43\times10^{-10}\ m = 1.43A°$

75. At high pressure $P >> a/V^2$

$\therefore$ Eqn is $P(V-b) = RT$

$PV = RT + Pb$

$V = \frac{R}{V}T + b$

$y = mx + c$

So intercept for this line is 'b'

76. $\lambda \propto \frac{T}{P}$

At constant temperature $\lambda \propto \frac{1}{P}$

So as pressure increases mean free path decreases

77. $a = 6.71\ atm\ L^2\ mol^{-2}$

$b = 0.056\ L\ mol^{-1}$

$T_C = \frac{8a}{27Rb} = \frac{8\times6.71}{27\times0.082\times0.056} = 432.9$

Therefore, T_C is above room temperature, so gas can be liquefied on applying pressure

78. 50°C is Boyle temperature, when Z = 1, gas is behaving ideally over a range of temperature

Boyle temperature $T_B = \frac{a}{Rb}$

79. Above Boyle temperature, a gas will always show positive deviation.

80. $\%RH = \dfrac{\text{partial pressure of } H_2O}{\text{Vap pr}} \times 100$

$p_{H_2O} = \dfrac{35 \times 18}{100} = 6.3 \text{ mm Hg}$

$PV = \dfrac{w}{M}RT$

$w = \dfrac{P \times V \times M}{RT} = \dfrac{\frac{6.3}{760} \times 1 \times 18}{0.082 \times 293}$

$= 0.0062 = 6.2 \times 10^{-3} \text{ gL}^{-1}$

81. 0.223 moles of CH4 has a pressure

$= \dfrac{0.223 \times 0.0821 \times 273}{1}$

$= 5 \text{ atm}$

82. 0.96 atm ≡ 0.96 × 76 cm of Hg = 72.96 cm

At 45° inclination effective height of the mercury column = $8 \times \cos 45° = 8 \times \dfrac{1}{\sqrt{2}} = 5.656$ cm

$\therefore \quad 10 \times 72.96 = \ell \times (72.96 + 5.656)$ Boyle's law

$\therefore \quad \ell = \dfrac{10 \times 72.96}{78.616} \simeq 9.28 \text{ cm}$

83. $\underset{(1-\alpha)}{N_2O_{4(g)}} \rightleftharpoons \underset{2\alpha}{2NO_{2(g)}} \Big] \equiv \text{total } 1 + \alpha$

$\therefore \quad \text{mean molar mass} = \dfrac{\text{Molar mass of } N_2O_4}{1 + \alpha}$

$= \dfrac{92}{1.6} = 57.5 \text{ g mol}^{-1}$

84. Obviously in B, $p_1V_1 = n_1RT$

In A $p_2V_2 = n_2RT$

$\therefore \quad \dfrac{p_2V_2}{p_1V_1} = 4 = \dfrac{n_2}{n_1}$

Ratio = 4 : 1

85. Number of moles is

$\dfrac{1.8 \times 10^{-5}}{760} \times \dfrac{200/1000}{0.0821 \times 300}$

$= 1.923 \times 10^{-10}$

Number of molecules

$= 1.923 \times 10^{-10} \times 6.02 \times 10^{23}$

$= 11.576 \times 10^{13}$

$= 1.16 \times 10^{14}$

86. Initial number of moles = $\dfrac{10 \times 4}{0.0821 \times 290}$

$= 1.68$ mols

After the leak was repaired, the number of moles

$= \dfrac{10 \times 3}{0.0821 \times 290} = 1.26$ mol

$\therefore \quad$ Number of moles escaped = (1.68 − 1.26)

$= 0.42$

87. PM = dRT, $3 \times 32 = d \times 0.0821 \times 293$

$d \simeq 4 \text{ g L}^{-1}$

88. Number of atoms per kg

$= \dfrac{2 \times 10^{-7}}{22.4} \times 6.02 \times 10^{23}$

$= 0.54 \times 10^{16} = 5.4 \times 10^{15}$

89. Initially the number of moles = 11.6

Finally number of moles = $\dfrac{1 \times 9.43}{0.0821 \times 348}$

$= 0.33$ mol

$\therefore \quad O_2$ escaped = 11 27 moles

$= (11.27 \times 32) \text{ g} \simeq 360 \text{ g}$

90. Final pressure = $\dfrac{(500 \times 250) + (950 \times 450)}{700}$

$\simeq 790$ torr

91. $p \times M = dRT \quad p \times 34 = 2.76 \times 0.0821 \times 300$

$\therefore \quad p = \dfrac{2.76 \times 0.0821 \times 300}{34} \text{ atm} \simeq 2 \text{ atm}$

92. At STP $760 \times M = d_1 \times R \times 273$

M = mean molar mass of air assuming constant composition

At the higher altitude $5 \times 10^{-7} \times M = d_2 \times R \times 173$

$\therefore \quad \dfrac{2.76 \times 0.0821 \times 300}{34}$

$\therefore \quad \dfrac{d_2}{d_1} = \dfrac{5 \times 10^{-7}}{760} \times \dfrac{273}{173}$

$= 1.038 \times 10^{-9} \simeq 1 \times 10^{-9}$

93. Molar masses are 6.75×22.4, 9.56×22.4, 10.08×22.4,

i.e., 151.2, 214.144, 225.792 g mol^{-1} respectively

% weight of the element (g mol^{-1}) = 645.152, 72.59 and 217.66.

These are nearly in the ratio 2 : 1 : 3. One may suggest that these correspond to 2 atoms, one atom and 3 atoms.

$\therefore$ The suggested atomic weight is $\simeq$ 72.5.

94. $Ca(OH)_2 + 2CO_2 \rightarrow Ca(HCO_3)_2$

2×44 162 g

(2 moles)g

$\therefore$ for 0.3 kg = 300 g

number of moles of $CO_2 = \left(\frac{2}{162} \times 300\right) = 3.704$

$\therefore \quad \frac{765}{760} \times V = 3.704 \times 0.0821 \times 298$

$\therefore \quad V = \frac{3.704 \times 0.0821 \times 298 \times 760}{765}$ L $\simeq$ 90 L

95. Since V and T are the same, partial pressures of the components are proportional to the number of moles.

$\therefore$ indicating CH_4 and C_4H_{10} by the suffixes 1 and 2

$p_1 + p_2 = 400$; $p_1 + 4p_2 = 600 \equiv p_1 = 333$ torr

$p_2 = 66$ torr

$\therefore$ mole fractions are $\frac{333}{400}$ and $\frac{66}{400}$

$\simeq$ 0.83 and 0.17

96. Number of moles of water vapour

$$= \frac{\left(5 \times 27/760\right)}{0.0821 \times 300}$$

$= 7.2 \times 10^{-3}$ mols

since one mole of anhydrous $CuSO_4$ removes 5 moles of water in the formation of $CuSO_4.5H_2O$, number of moles of $CuSO_4$ needed $= \frac{7.2}{5} \times 10^{-3} = 1.44 \times 10^{-3}$

97. $pH_2 + pCO_2 = 350$ torr; $pH_2 = \frac{19.4}{100} \times 775$

$= 150.35$ torr

$\therefore \quad pCO_2 = (350 - 150.35)$torr $\simeq$ 200 torr

98. $\frac{r_{H_2}}{r_{CxHy}} = 6 = \sqrt{\frac{M_{CxHy}}{2}}$

$\therefore \quad M_{CxHy} = 72$

99. Mean translational kinetic energy per mole

$$= \frac{3}{2}RT$$

$$= \frac{3}{2} \times 2 \times T = 104 \times 10^3$$

$\therefore \quad T = \frac{104 \times 10^3}{3} = 34700$ K

A more accurate value $\simeq$ 34900 K

100. Total = 0.2 + 0.8 = 1 mole Kinetic energy = $\frac{3}{2}$ RT

But $\frac{3RT}{M} = (3.8 \times 10^4)^2$

$\therefore \quad \frac{3}{2}RT = \left[\frac{1}{2} \times \left(3.8 \times 10^4\right)^2 \times 16\right]$ ergs mol^{-1}

$= 1155$ J mol

101. Taking CH_4 as indicated by suffice 1 and SO_2 by suffix 2

$$\frac{3RT_1}{M_1} = \frac{3RT_2}{M_2}$$

$\therefore \quad \frac{T_1}{T_2} = \frac{M_1}{M_2} = \frac{16}{64} = \frac{1}{4} = 0.25$

102. $\left(\frac{2RT}{M}\right)^{1/2} = \left(\frac{2 \times 8.314 \times 10^7 \times 300}{40}\right)^{1/2}$

$= 3.53 \times 10^4$ cm s^{-1} = 350 m s^{-1}

103. $\left(\frac{2RT}{M}\right)^{1/2} = 2\left(\frac{2 \times R \times 323}{M}\right)^{1/2}$

$\therefore \quad T = 4 \times 323$ K $\simeq$ 1292 K $\simeq$ 1019°C

104. $\frac{RT_A}{\pi M_A} = \frac{3RT_B}{M_B}$

$\therefore \quad \frac{8T_A}{3\pi T_B} = \frac{M_A}{M_B} = 16$

$\frac{T_A}{T_B} = 16 \times \frac{3\pi}{8} \simeq 19$

105. Volume per mole $= \frac{16}{10} = 1.6$ L mol^{-1}

$\frac{pV}{RT} = \frac{11 \times 1.6}{0.0821 \times 310} = 0.692$

Equating $\frac{pV}{RT}$ to $\left[1+\frac{\left(b-\frac{a}{RT}\right)}{V}\right]$ one gets

$$\left(b-{}^{a}\!/_{RT}\right)\Big/_{V} = 0.308$$

$$\therefore \quad \left(b-{}^{a}\!/_{RT}\right) = -0.308 \times 1.6 = -0.493 \text{ L mol}^{-1}$$

106. When $\frac{a}{b}$ is large attractive interaction among molecules is large

$\therefore$ Vapour pressure will be low
(i.e., non volatile) liquid B

107. $b = 4 \times 6.02 \times 10^{23} \times \left(\frac{4}{3}\pi r^3\right)$

$= 4 \times 6.02 \times 10^{23} \times \frac{4}{3} \times 3.14 \times 8 \times 10^{-24}$

$= 80.45 \text{ cm}^3 \text{ mol}^{-1} = 0.0805 \text{ L mol}^{-1}$

108. $T_c = \frac{8a}{27Rb} = \frac{8}{27} \times \frac{1}{3.7 \times 10^{-2} \times 0.0821}$

$= 97.5 \text{ K}$

109. $T_c = \frac{8a}{27Rb} = 155 \text{ K}$

$p_c = \frac{a}{27b^2} = 50 \text{ atm}$

$\frac{RT_c}{P_c V_c} = \frac{8}{3}$

Calculating $\frac{V_c}{3} = b$, one gets $b = 0.0318 \text{ L mol}^{-1}$

$a = \frac{155 \times 27Rb}{8} = \frac{155 \times 27 \times 0.0821 \times 0.0138}{8}$

$= 1.366 \text{ atm L}^2 \text{ mol}^{-2}$

110. Number of moles of water vapour in the enclosure

$= \frac{26.7}{2 \times 760} \times \frac{64 \times 10^3}{0.0821 \times 300}$

$= 45.644$ moles

$\therefore$ Number of kg of water vapour $= \frac{45.644 \times 18}{1000}$

$= 0.82 \text{ kg}$

111. Statements (1) and (2) are correct and (2) is the correct explanation to (1).

112. Statements (1) and (2) are correct but (2) is not the correct explanation to (1).

113. Statement (1) is true (2) is false.

114. $P = \frac{nRT}{V-nb} - \frac{n^2a}{V^2}$.

Substitute $R = 0.0821 \text{ litre atm deg}^{-1} \text{ mol}^{-1}$

$T = 273, \quad V = 0.560 \text{ litre},$

$n = 5, \quad a = 1.39 \text{ atm L}^2\text{mol}^{-2}$

$b = 0.0391 \text{ L mol}^{-1}$. After detailed calculation (not difficult) the answer is 196.64 atm

115. $P = \frac{RT}{V-b} - \frac{a}{V^2}$; $\frac{RTV}{V-b} - \frac{a}{V}$

$= PV = \frac{RT}{1-\frac{b}{V}} - \frac{a}{V}$

$\therefore \quad PV = RT\left(1+\frac{b}{V}+\frac{b^2}{V^2}+.....\right) - \frac{a}{V}$

$= RT + \frac{RTb-a}{V}$ + higher order terms

$\therefore \quad \frac{PV}{RT} = 1 + \frac{b-\frac{a}{RT}}{V}$ + higher order terms

Also by substitution,

$b - \frac{a}{RT} = \left(0.0391 - \frac{1.39}{0.0821 \times 300}\right)$

$= -0.0173 \text{ L mol}^{-1}$

116. $V_c = 3b = (3 \times 0.0391) \text{ L mol}^{-1}$

$= 0.1173 \text{ L mol}^{-1}$

$P_c = \left(\frac{a}{27b^2}\right) = \left[\frac{1.39}{27 \times (0.0391)^2}\right]$

$= 33.67 \text{ atm}$

$T_c = \left(\frac{8a}{27Rb}\right) = \left[\frac{8 \times 1.39}{27 \times 0.0821 \times 0.0391}\right]$

$= 128.3 \text{ K}$

Additional Practice Exercise

121. PV = nRT

$$n_1 = \frac{4.8 \times 2}{0.08 \times 300} = 0.4 ;$$

$$n_2 = \frac{1.92 \text{ x } 1}{0.08 \text{ x } 300} = 0.08$$

Total number of moles = 0.4 + 0.08 = 0.48

After connecting two cylinders the pressure and temperature are identical through out the system.

Number of moles distributed is based on the volume $n \propto V$.

Total volume = 2 + 1 = 3 L

$\therefore$ Number of moles in cylinder I after connecting

$$0.48 \times \frac{2}{3} = 0.32$$

$\therefore$ Number of moles transferred from cylinder I to II = 0.4 - 0.32 = 0.08

122. 1 mole - 0°C - 1 atm - 22.4 L

0.1 mole - 0°C - 1 atm - 2.24 L

0.2 mole - 0°C - 2 atm - 2.24 L

0.05 moles of O_2 is there in the cylinder and an additional 0.15 moles is required.

The weight of He that has to be added

= 0.15 × 4 = 0.6 g

123. Balloon is blown to $\frac{4}{5}$th of the original volume, then the maximum capacity of the balloon will be $\frac{5}{4}$th of its volume i.e., $600 \times \frac{5}{4} = 750$ mL

From Charles Law:

$$\frac{V_1}{T_1} = \frac{V_2}{T_2} \qquad \frac{600}{288} = \frac{V_2}{300}$$

$\therefore V_2 = 625$ mL

Balloon will not burst because its capacity is 750 mL.

If the volume exceeds 750 ml, it will burst. By Charles Law,

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

$$\frac{600}{288} = \frac{750}{T_2}$$

$$T_2 = 360K$$

At temperatures above 360K (or) 87°C the balloon bursts.

124. Volume of remaining gas (N_2) at 1atm pressure

$$= \frac{P_1 V_1}{P_2} = \frac{456 \times 5}{760} = 3L$$

Volume of O_2 consumed = 5 – 3 = 2 L

Weight percentage of oxygen

$$\frac{2 \times 32}{2 \times 32 + 3 \times 28} = 43\%$$

125. Number of moles of gas = $\frac{PV}{RT}$

$$= \frac{2.47 \times 10^5 \times 2 \times 10^{-3}}{8.314 \times 298} = 0.2 \text{ mole}$$

i.e., 0.1 mole each

Weight of Ar = 4 g

Weight of He = 0.4 g

126. $\frac{r_{mixture}}{r_{Kr}} = 1.2\sqrt{\frac{M_{Kr}}{M_{mix}}}$

$$= \sqrt{\frac{84}{M_{mix}}} ;$$

$$\frac{84}{M_{mix}} = 1.44$$

$$M_{mix} = \frac{84}{1.44} = 58.3$$

Considering the equation $Cl_2 \rightleftharpoons 2Cl$

If α is degree of dissociation

At equilibrium $[Cl_2] = 1 - \alpha$; $[Cl] = 2\alpha$

Total moles at equilibrium = $1 - \alpha + 2\alpha = 1 + \alpha$

$$\frac{(1-\alpha)71 + 2\alpha \times 35.5}{1 + \alpha} = 58.3$$

$$71 - 71\alpha + 71\alpha = 58.3 + 58.3\alpha$$

$$\alpha = 0.218$$

127. Kinetic energy per molecule = $\frac{3}{2}\frac{RT}{N_A}$

$$= \frac{3}{2} \times \frac{8.314 \times 300}{6.023 \times 10^{23}}$$

$$= 6.21 \times 10^{-21} \text{ J}$$

128. (i) $\frac{P_1V_1}{T_1}=\frac{P_2V_2}{T_2}$

$$\frac{1\times22.4}{273}=\frac{\frac{800}{760}\times V_2}{298}$$

$$V_2=\frac{22.4}{273}\times\frac{298}{1.053}=23.22\ L$$

(ii) $C_{RMS}=\sqrt{\frac{3RT}{M}}=\sqrt{\frac{3\times8.314\times298}{32\times10^{-3}}}=482\ m\ sec^{-1}$

(iii) $C_{MPV}=\sqrt{\frac{2RT}{M}}=\sqrt{\frac{2\times8.314\times298}{32\times10^{-3}}}$

$= 393.5\ m\ s^{-1}$

129. $P_c=\frac{a}{27b^2}=3550\ atm$

$T_c=\frac{8a}{27\ Rb}=1900\ K$

using $\frac{RT_c}{P_cV_c}=\frac{8}{3}=\frac{0.0821\times1900}{3550\times V_c}$

we get $V_c=\frac{3}{8}\times\frac{0.0821\times1900}{3550}\ L\ mol^{-1}$

$V_c = 0.01648\ L\ mol^{-1} = 3b$

$\therefore$ $b = 0.00549\ L\ mol^{-1}$

Substituting in $Pc = 3550=\frac{a}{27\times(5.49\times10^{-3})^2}$

we get a

$= 3550 \times 27 \times (5.49)^2 \times 10–6\ atm\ L^2\ mol^{-2}$

$a = 2.889\ atm\ L^2\ mol^{-2}$

130. (i) $P=\frac{nRT}{V}=\frac{5\times0.082\times310}{8}=15.89\ atm$

(ii) $\left[P+\frac{n^2a}{V^2}\right]\left[V-nb\right]=nRT$

$$\left(11+\frac{(5)^2a}{8^2}\right)\left(8-5\times0.05\right)=5\times0.0821\times310$$

$\Rightarrow$ $11+\frac{25a}{64}=\frac{5\times0.082\times310}{7.75}=16.4$

$\frac{25a}{64}=5.4$

$\Rightarrow$ $a = 13.82\ atm\ L^2\ mol^{-2}$

(iii) $T_b=\frac{a}{Rb}$

$$=\frac{13.82\ atm\ L^2\ mol^{-2}}{0.082\ L atm\ deg^{-1}\ mol^{-1}\times0.05\ L mol^{-1}}$$

$= 3371K.$

(iv) $V_c = 3b = 3 \times 0.05 = 0.15\ L\ mol^{-1}$

$P_c=\frac{a}{27b^2}=\frac{13.82\ L^2\ atm\ mol^{-2}}{27\times0.0025\ L^{-2}\ mol^{-2}}$

$= 204.7\ atm$

$T_c=\frac{8a}{27Rb}=\frac{3371\times8}{27}=998.8K$

131. Unit of K = unit of $\frac{V}{T}=m^3K^{-1}$.

132. Expansion for the ideal gas occurs without any work being done

$\therefore$ temperature does not change

pressure $= 2\times\frac{20}{50}\ atm = 0.8\ atm$

133. Initial volume = 100 L at 11 atm. This would be equivalent to $\frac{100\times11}{1}=1100$ L at 1 atm. Subtracting 100 L and dividing by 2.5, one gets $\frac{1100-100}{2.5}=\frac{1000}{2.5}$ $= 400$

134. $1.25\ kg\ m^{-3} \equiv 1.25\ g\ L^{-1}$ at STP

Suppose we have n moles of Nitrogen

$$\frac{n\times28}{1.5}=1.25,\ \frac{n\times28}{4}\times1000=x$$

$\therefore$ $x=\frac{1000\times1.5\times1.25}{4}=468.75\ g\ L^{-1}$

Suppose we use the ideal gas formula $pM = dRT$.

At same $T\ \frac{p_1}{d_1}=\frac{p_2}{d_2}$

i.e., $\frac{1}{1.25}=\frac{575}{d_2}$

$\therefore$ $d_2 = 575 \times 1.25 = 718.75\ g\ L^{-1}$

deviation $=\left(\underset{\text{1 deal}}{718.75}-\underset{\text{ac}}{468.75}\right)\ g\ L^{-1} = 250\ g\ L^{-1}$

135. If x mole of $N_2O_{4(g)}$ has dissociated the total number of moles under equilibrium

$= (0.01726 – x) + 2x$

$= (0.01726 + x)$. Thus 1×0.5

$= (0.01726 + x) \times 0.0821 \times 298$

$\therefore \quad (0.01726 + x) = \dfrac{0.5}{0.0821 \times 298} = 0.02044$

$x = 3.1767 \times 10^{-3}$

$\therefore$ Degree of dissociation $= \dfrac{3.1767 \times 10^{-3}}{1.726 \times 10^{-2}} \times 100$

$= 18.40\%$

mean molar mass $= \dfrac{0.01726 \times 92}{0.02044}$

$= 77.7 \text{ g mol}^{-1}$

136. $\alpha = \dfrac{d_0 - d}{d} = \dfrac{46 - 27.6}{27.6} = 66.67\%$

% by wt. of $NO_2 = 1.32$ and $N_2O_4 = 0.33$

No. of moles of $NO_2 = 1.45$ and $N_2O4 = 0.36$

$\therefore$ Molefraction of $NO_2 = \dfrac{1.32}{1.65} = 0.80$

% by volume = 80

137. Suppose n molecules of monomer dimerise then total number of molecules $= (5 \times 10^{23} - n) + \dfrac{n}{2}$

$= 5 \times 10^{23} - \dfrac{n}{2}$

$\therefore \quad \dfrac{500}{900} = \dfrac{5 \times 10^{23} - \frac{n}{2}}{5 \times 10^{23}} = 1 - \dfrac{n}{10^{24}}$

$\therefore \quad n = \dfrac{4}{9} \times 10^{24} = 4.444 \times 10^{23}$

$\therefore$ Number of monomer molecules

$= (5 - 4.444) \times 10^{23}$

$= 0.556 \times 10^{23} = 5.56 \times 10^{22}$

138. Using PV = nRT, $\dfrac{740}{760} \times 7.66 = n \times 0.0821 \times 473$

$n \simeq 0.1921$ mol

5g of $NH_2 - COONH_4 = \dfrac{5}{78}$ mol

i.e., 0.0641 mol. We observe that 0.0641×3 equals 0.1923 mole $\simeq$ n moles

since $\underset{\text{one mole solid}}{NH_2 - COONH_4} \rightarrow \underset{\text{3mols(gas)}}{2NH_3 + CO_2}$ the degree of dissociation is very nearly 100%

139. $\underset{\substack{1 \text{ mol} \\ 1 - x}}{Ni(CO)_{4(g)}} \rightarrow Ni_{(s)} + \underset{\substack{4 \text{ mol} \\ 4x}}{4CO_{(g)}}$

one, mole by dissociation forms $(1 - x + 4x)$

$= 1 + 3x$ m

Molar mass of $Ni(CO)_4 = 58 + 4 \times 28$

$= 170 \text{ g mol}^{-1}$

V.d = 83.3

$\therefore \quad M = 166.6 \text{ g mol}^{-1}$

$\therefore \quad 170 \times 1 = 166.6 \times (1 + 3x)$

$\therefore \quad (170 - 166.6) = 166.6 \times 3x$

$\therefore \quad x = 6.8 \times 10^{-3} \simeq 0.7\%$

Similarly, for V.d = 70.8, M = 141.6 g mol^{-1}

$(170 - 141.6) = 141.6 \times 3x \qquad x = 0.06685$

$\simeq 6.68\%$

140. By using high $h_1\rho_1 g = h_2 g \rho_2$: $\dfrac{h_2}{h_1} = \dfrac{\rho_1}{\rho_2} = \dfrac{13.6}{0.9}$

$\therefore \quad H_2 = \left(\dfrac{13.6}{0.9} \times 74\right) \text{cm} = 1118 \text{ cm}$

141. The length of the trapped air column may be taken as proportional to the volume of the trapped air, Using Boyle's law. Let the length of the air column after pouring mercury be ℓ cm. Before pouring mercury it is 50 cm

$\therefore \quad \ell(76 + \ell) = (50 \times 76) \qquad 76\ell + \ell^2 = 3800$

$\ell^2 + 76\ell - 3800 = 0$

$\therefore \quad \ell \text{ cm} = \dfrac{-76 \pm \sqrt{5776 + 15200}}{2}$

$\therefore \quad \ell = 34.5$ cm

$\therefore$ difference in pt = (50 − 34.5) cm = 15.5 cm

142. $V_1 = 1.103$ litres, $V_2 = (1.103 + V)$

$(710.5) \times 1.103 = 583(1.103 + V)$ on calculation

$V = 0.241$ L

143. Clearly $p_{N_2} = 1 \text{ KP}_a$ and $p_{NH_3} = 0.5 \text{ KP}_a$

$\underbrace{2NH_3}_{2\text{moles}} \rightarrow \underbrace{N_2 + 3H_2}_{4\text{mole}}$

These p_{NH_3} yields $(N_2 + 3H_2)$ and 1 KP_a

$\therefore$ Total pressure $= (1 + 1)KP_A = 2KP_A$

144. PV = nRT. Both before and after heating P and V remain the same

$\therefore \quad n_1T_1 = n_2T_2$

$$\frac{n_1}{n_2} = \frac{T_2}{T_1} = \frac{400+273}{10+273} = 2.378$$

$$\frac{n_2}{n_1} = 0.42.$$

It follows that a fraction$(1 - 0.42) = 0.58$ of the original weight has been expelled

145. At 24 torr and 298 K, using pV = RT for one mole

$$\frac{24}{760} \times V = 0.0821 \times 298$$

$$V = \frac{0.0821 \times 298 \times 760}{24} \text{ L}$$

$$V = 774.75 \text{ L}$$

Volume of one mole of liquid water $= \frac{18}{1000}$ L

$\therefore$ The required ratio $= \frac{774.75 \times 1000}{18}$

$= 43042$ (at 298 K)

146. PM = dRT. Same density for the same pressure and temperature implies that the mean molar mass of the mixture of A and B equals the molar mass of C. Thus $64x + 28(1 - x) = 44$

$64x - 28x = (44 - 28)$ $36x = 16$

$\therefore \quad x = \frac{16}{36} = \frac{4}{9} \quad (1 - x) = \frac{5}{9}$

$\therefore$ the mixture has the mole ratio A : B = 4 : 5

147. Using pV = RT mol^{-1} $\frac{10^{-16}}{760} \times V$ L

$= 0.0821 \times 153$

$$V = \frac{0.0821 \times 153}{10^{-16}} \times 760 \text{ L mol}^{-1}$$

$= 9547 \times 10^{16}$ L mol^{-1}

$\therefore$ Number of atoms mol^{-1} $= \frac{6.02 \times 10^{23}}{9547 \times 10^{19}}$

$= 6.3$ atom mol^{-1}

148. Consider one litre of the mixture at a temperature 300 K

pV = nRT i.e., $1 \times 1 = x \times 0.0821 \times 300$

$\therefore \quad n = 1/0.0821 \times 300$

n = 0.0406 mole(total) Breaking this up into components

Number of moles of $N_2 = 0.0406 \times \frac{420}{760}$

$= 0.02244$ mol

Number of moles of $O_2 = 0.0406 \times \frac{190}{760}$

$= 0.01015$ mol

Number of moles of $CO_2 = 0.0406 \times \frac{150}{760}$

$= 0.008013$ mol

149. Mean molar mass

$= (0.2 \times 16) + (0.3 \times 30) + (0.5 \times 28)$ g mol^{-1}

$= 3.2 + 9 + 14 = 26.2$ g mol^{-1}

Treating all of them as ideal gases,

PM = dRT; $2 \times 26.2 = d \times 0.0821 \times 310$

$\therefore \quad d = \left(52.4/0.0821 \times 310\right)$ g mol^{-1}

$= 2.0589$ g L^{-1}

150. Suppose we have x moles of Na_2CO_3 and y moles of $NaHCO_3$ $(x \times 106) + (y \times 84) = 10$ g. By stoichiometry

$(x + y) = n_2$ and $\frac{y}{2} = n_1 \frac{y}{2(x+y)} = \frac{3}{10}$

$\therefore \quad 10y = 6x + 6y$

$\therefore \quad 4y = 6x \quad 2y = 3x$

This gives the ratio $\frac{y}{x} = \frac{3}{2}$.

Put y = 3K and x = 2K

Then $(2K \times 106) + (3K \times 84)$

$= 10$ g$(212 + 252)K = 10$

$\therefore \quad K = \frac{10}{464}$g

$\therefore \quad x = \frac{20}{464}$mol $y = \frac{30}{464}$ mol

$\therefore \quad n_1 = \frac{y}{2}$ mole $= \frac{15}{464}$mol $n_2 = \frac{50}{464}$mol

volumes at STP are $\left(\frac{15}{464}\times 22.4\right)$ litres

$= 0.724$ L and $\left(\frac{50}{464}\times 22.4\right)$ L $= 2.414$ L

151. Since $4KClO_3$ mole yields $3KClO_4$ mole, $\frac{1}{4}$ mole of $KClO_4$ is obtained from $\frac{4}{3}\times\frac{1}{4}=\frac{1}{3}$ mole of $KClO_3$. Remaining $\frac{2}{3}$ mole of $KClO_3$ yields oxygen. $2KClO_3$ moles $\equiv 3O_2$ mole

i.e., $= 1$ mole of O_2; $\frac{740}{760}\times V = 1\times 0.0821\times 300$

$\therefore \quad V=\left(0.0821\times 300\times\frac{760}{740}\right)$ L $= 25.3$ L

152. When one mole of ZnS is oxidized to form one mole of ZnO, loss in weight $= (32-16) = 16$ g and one mole i.e., 22.4 L of SO_2 at STP are obtained.

$\therefore$ for a loss of 6 g, $\left(\frac{6}{16}\times 22.4\right)$ L

i.e., 8.4 L of SO_2

153. $PV = nRT$

$$P=\frac{dRT}{M}\Rightarrow d=\frac{PM}{RT}$$

So density will be the maximum at higher pressure and lower temperature.

154. $pV = nRT \quad 1\times V=\frac{8}{40}\times 0.821\times 273$. A mixture of He and Ar in the mole ratio 2 : 1 has a mean molar mass,

$$\frac{(2\times 4)+(1\times 40)}{3}=16 \text{ g}$$

$$100\times V=\frac{\omega}{16}\times 0.0821\times 373$$

$$100\times\frac{8}{40}\times 0.0821\times 273$$

$$=\frac{\omega}{16}\times 0.0821\times 373;\ 20\times 273$$

$$=\frac{\omega}{16}\times 373$$

$$\therefore \quad \omega=\frac{20\times 273\times 16}{373}\text{ g}\simeq 234\text{ g}$$

$$\simeq 14.625 \text{ moles}$$

Number of moles of Helium $= 14.625\times\frac{2}{3}=9.75$

$\therefore$ Mass of He $= (9.75\times 4)$ g $= 39$ g

155. Since $(p_1 + p_2) = 320$ torr and $\frac{p_1}{p_2}$

$= 2.2$, $p_1 = 220$ torr and $p_2 = 100$ torr

Oxygen will have a partial pressure

$$=\frac{2}{5}\times 220\text{ torr}=\frac{440}{5}=88\text{ torr.}$$

Nitrogen will have a partial pressure of $\frac{3}{5}\times 100$

$$=\frac{300}{5}=60\text{ torr}$$

$\therefore$ total final pressure $88 + 60 = 148$ torr

156. Daltons law is applicable only to a mixture of gases, that are non reacting.

157. $2H_2 + O_2 \rightarrow 2H_2O$

Oxygen is the limiting reactant. It is completely consumed. 2 moles of H_2 are consumed

$\therefore$ 1 mole of H_2 and 2 moles of water vapour $\equiv$ 3 moles are the product.

Using $pV = nRT$, $p\times 200 = 3\times 0.0821\times 400$

$$\therefore \quad p_{total}=\frac{3\times 0.0821\times 400}{200}=0.4926\text{ atm}$$

$$p_{H_2O}=\left(\frac{2}{3}\times 0.4926\right)\text{atm}=0.3284\text{ atm}$$

158. $$\frac{V_{(H_2)}/t_{(H_2)}}{V_{(O_2)}/t_{(O_2)}}=\sqrt{\frac{M_{O_2}}{M_{H_2}}}$$

$t_{H_2} = t_{O_2}$ and $V \propto n$

$$\frac{V_{(H_2)}}{V_{(O_2)}}=\frac{m_{(H_2)}/2}{m_{(O_2)}/32}$$

$$\frac{m_{H_2}/2}{4/32} = \sqrt{\frac{32}{2}}$$

$$m_{H_2} = 4 \times \frac{4}{32} \times 2 = 1 \text{ g}$$

159. $PV = \frac{1}{3}M\bar{c}^2$ — Kinetic theory of gases

$$PV = \frac{2}{3} \times \frac{1}{2}M\bar{c}^2$$

$$\Rightarrow \quad P = \frac{2}{3} \times \left[\frac{1}{2}\frac{M\bar{c}^2}{V}\right]$$

Since, $K.E = \frac{1}{2}M\bar{c}^2$

$$P = \frac{K.E}{V} \times \frac{2}{3} = \frac{2}{3} \times E$$

Since $= \frac{K.E}{V} = E$

160. $K.E = \frac{3}{2}RT$,

$R = 1.987 \text{ cal mol}^{-1} \text{ K}^{-1} \cong 2$

$$K.E = \frac{3}{2} \times 2 \times 10 = 30 \text{ cal mol}^{-1}$$

161. For ideal behaviour kinetic energy per mole + $\frac{3}{2}RT$

$\therefore$ ratio of kinetic energy per mole

$$= \left(T_{argon}/T_{Helium}\right) = \frac{273+27}{273+127}$$

$$= 300/400 = 0.75 : 1$$

162. $C_{rms} = \left(\frac{3RT}{M}\right)^{1/2}$,

$$\therefore \quad \frac{3RT}{M} = \left\{\left[3 \times 8.314 \times 10^7 \times T\right]/2\right\}$$

$$= (2.15 \times 10^5)^2$$

$\therefore$ $T = 371$ K; $t = (T - 273)°C \simeq 100°C$

163. Most probable speed =

$$\left(\frac{2RT}{M}\right)^{1/2} = \left[\frac{2 \times 8.314 \times 10^7 \times 361}{16}\right]^{1/2} \text{ cm s}^{-1}$$

$$= 6.125 \times 10^4 \text{ cm s}^{-1}$$

$$= 612.5 \text{ m s}^{-1}$$

164. $\left(\frac{2RT}{M}\right)^{1/2} = 10.62 \times 10^4 \text{ cm s}^{-1}$

$\therefore$ $\frac{2RT}{M} \simeq 112.8 \times 10^8 \text{ cm}^2 \text{ s}^{-2}$

$PM = dRT$

$\therefore$ $d = \frac{pM}{RT}$

$$= \left[\left[76 \times 13.6 \times 981\right]/56.4 \times 10^8\right] \times 10^3 \text{ L mol}^{-1}$$

$$= 0.1798 \text{ g L}^{-1}$$

165. Most probable speed = $\left(\frac{2RT}{M}\right)^{1/2}$

Acc to the problem $\frac{T_A}{M_A} = \frac{T_B}{M_B}$

$\therefore$ $\frac{M_A}{M_B} = \frac{T_A}{T_B} = \frac{298}{150} \simeq 2$

$\therefore$ $M_A : M_B \simeq 2 : 1$

166. For H_2, the mass is very small and the intermolecular forces of attractions are small. Hence the pressure correction factor $\frac{a}{V^2}$ is negligible at all pressures. Hence van der Waal's equation is reduced to

$P(V-b) = RT$

$PV = RT + Pb$

This explains why H_2 shows positive deviation only.

167. For a molar volume = 22.4 L mol^{-1} at STP then, compressibility factor = 1. Clearly in this case $V_m < 22.4$ L mol^{-1}

168. 50 litres = V P = 100 atm

$$n = \frac{6.6 \times 10^3}{44} = 150 \text{ moles}$$

Substituting $\left(100 + \frac{150^2 \times 3.6}{50^2}\right)(50–150\times 0.043)$

$= 150 \times 0.0821 \times T$

Calculating T = 468.4 K

169. The internal pressure per mole = $\frac{a}{V^2}$

170. Volume of one molecule = $\frac{4}{3}\pi r^3$

Actual volume occupied = $4 \times \frac{4}{3}\pi r^3$

Volume of one mole = $6.022 \times 10^{23} \times 4 \times \frac{4}{3}\pi r^3$

$= 6.022 \times 10^{23} \times 4 \times \frac{4}{3}\pi \times (176 \times 10^{-10})^3$

Fraction of the total volume occupied

$= \frac{55}{22400} = 2.45 \times 10^{-3}$

171. Statements (1) and (2) are correct. Statement (2) is not an explanation to (1).

172. Statements (1) and (2) are correct. Statement (2) is the correct explanation to (1).

173. Statement (1) is correct

Statement (2) is correct and is the correct explanation of statement 1

174. Statements (1) and (2) are correct. Statement (2) is not an explanation to (1).

175. Statement (1) is true (2) is false.

176. Statement (1) is correct

Statement (2) is correct and is the explanation for statement (1)

177. Statement (1) is true (2) is false.

178. Statement (1) is false (2) is true.

179. Statement (1) is false (2) is true.

180. Statements (1) and (2) are true. Statement (2) is not the correct explanation to (1).

181. $pM = dRT;\ \frac{800}{760} \times 32 = d \times 0.0821 \times 300$

$\therefore \quad d = \frac{800}{760} \times \frac{32}{0.0821 \times 300}\ g\ L^{-1} = 1.3676\ g\ L^{-1}$

182. $\left[\frac{2 \times 8.314 \times 10^7 \times 300}{32}\right]^{1/2}$ cm s^{-1}

$= 3.948 \times 10^4$ cm $s^{-1} \simeq 395$ m s^{-1}

183. (Using pV = nRT and taking V as one litre

$\frac{800}{760} \times 1 = n \times 0.0821 \times 300$

$\therefore \quad n = \frac{800}{760 \times 0.021 \times 300}$ moles

$\therefore$ Number of molecules ml^{-1}

$= \frac{800 \times 6.02 \times 10^{23}}{760 \times 0.0821 \times 300 \times 1000}$

$= 2.5728 \times 10^{19}$

mean kinetic energy per mole = $\frac{3}{2}RT$ which is

$\left[\frac{3}{2} \times 8.314 \times 10^7 \times 300\right] \Big/ \left(6.02 \times 10^{23}\right)$ ergs mol^{-1}

which equals 6.215×10^{-21} J mol^{-1}

184. $4 \times 6.02 \times 10^{23} \times \left(\frac{4}{3}\pi r^3\right)$

$= (0.043 \times 10^3 \times 10^{24})(A°)^3$

$\therefore \quad r^3 = 4.263(A°)^3 \qquad r \simeq 1.62\ A°$

185. Volume per mole = 5L

$\therefore \quad \frac{pV}{RT} = \frac{7 \times 5}{0.0821 \times 310} = 1.375$

The ideal gas value = 1

$\therefore$ deviation = (1.375 − 1) = 0.375

186. By direct substitution, $\frac{0.3637 \times 10^6}{\left(22.263\right)^2}$

$\simeq$ 733.79 P_a

187. $p_c = \dfrac{a}{27b^2} = 3550 \text{ atm}$

$T_C = \dfrac{8a}{27Rb} = 1900 \text{ K}$

$\therefore \quad T_C \big/ P_c = \dfrac{8a}{27Rb} \times \dfrac{27b^2}{a} = \dfrac{8b}{R} = \dfrac{1900}{3550}$

$\therefore \quad b = \dfrac{1900}{3550} \times \dfrac{R}{8} = \dfrac{1900}{3550} \times \dfrac{0.0821}{8}$

$= 5.493 \times 10^{-3} \text{ L mol}^{-1}$

$\therefore \quad a = 3550 \times 27 \times b^2 = 2.90 \text{ atm L}^2 \text{ mol}^{-2}$

188. As seen above $b = 5.493 \times 10^{-3} \text{ L mol}^{-1}$

$\therefore \quad 4 \times 6.02 \times 10^{23} \times \left(\dfrac{4}{3}\pi r^3\right) = (b \times 10^3 \times 10^{24})A^{\circ 3}$

$\therefore \quad r^3 = \left(5.493 \times 10^{24}\right) \big/ 4 \times 6.02 \times 1.33 \times 3.14 \times 10^{23}$

$= 0.5462A^{\circ 3}$

$\therefore \quad r = 0.817A^{\circ}$

189. By direct substitution, using T = 300 K

$$\left(b - \frac{a}{RT}\right) = \left(5.493 \times 10^{-3} - \frac{2.90}{0.0821 \times 300}\right)$$

$= -0.1122 \text{ L mol}^{-1}$

190. (b), (c)

191. (a), (b)

192. (a), (c), (d)

193. (a), (d)

194. (c), (d)

195. (a), (b), (c)

196. (a), (b)

197. (a), (c)

198. (a) → (p), (r)
(b) → (q)
(c) → (p), (s)
(d) → (p)

199. (a) → (q), (s)
(b) → (p), (r)
(c) → (q)
(d) → (p), (r)

200. (a) → (q), (r)
(b) → (p), (r)
(c) → (q), (s)
(d) → (r)

This page is intentionally left blank.

CHAPTER

3

ATOMIC STRUCTURE

CHAPTER OUTLINE

Preview

STUDY MATERIAL

Introduction
- Concept Strands (1-2)

Atomic Models
- Concept Strands (3-10)

Hydrogen Spectrum
- Concept Strands (11-37)

Photoelectric Effect
- Concept Strands (38-39)

Wave–Particle Duality
- Concept Strands (40-43)

Heisenberg's Uncertainty Principle
- Concept Strands (44-47)

Quantum Numbers
- Concept Strand 48

Quantum Mechanical Picture of Hydrogen Atom

Probability Distribution Curves

Orbitals
- Concept Strands (49-58)

Electronic Configuration of Elements
- Concept Strands (59-60)

TOPIC GRIP
- Subjective Questions (10)
- Straight Objective Type Questions (5)
- Assertion–Reason Type Questions (5)
- Linked Comprehension Type Questions (6)
- Multiple Correct Objective Type Questions (3)
- Matrix-Match Type Question (1)

IIT ASSIGNMENT EXERCISE
- Straight Objective Type Questions (80)
- Assertion–Reason Type Questions (3)
- Linked Comprehension Type Questions (3)
- Multiple Correct Objective Type Questions (3)
- Matrix-Match Type Question (1)

ADDITIONAL PRACTICE EXERCISE
- Subjective Questions (10)
- Straight Objective Type Questions (40)
- Assertion–Reason Type Questions (10)
- Linked Comprehension Type Questions (9)
- Multiple Correct Objective Type Questions (8)
- Matrix-Match Type Questions (3)

INTRODUCTION

The concept of atoms and molecules

Matter in the universe is made up of about 92 elements. Elements are made from simple basic units called atoms. Democritus proposed the earliest theory on atoms that *all matter is made up of tiny particles called atoms.*

Dalton's atomic theory

Early in the nineteenth century, John Dalton enunciated the atomic theory. Some significant principles of the atomic theory proposed by Dalton, though not valid in the light of present day concepts, are summarized below.

1. *Matter is made up of small and indivisible particles known as atoms.*
2. *Atoms can neither be created nor destroyed.*
3. Atoms of the same element are exactly alike in all the properties such as mass, volume etc. Atoms of different elements differ in their properties.
4. When elements combine to form compounds, atoms of these elements combine in small whole number ratios.

Atoms of two or more elements can combine in small whole number ratios to form compound atoms.

Berzelius determined the atomic weights of elements and invented the system of giving a symbol to the elements using one or two letters. William Prout believed that atomic weights of elements were whole number multiples of that of hydrogen. Experiments using Field ionization microscope and X-ray diffraction pattern of crystals have proved beyond doubt the existence of atoms.

Sub-atomic particles

(i) Cathode rays–discovery of electrons

The cathode rays were produced when a potential of above 1000 volts was applied between electrodes in a discharge tube containing a gas at very low pressures of about 10^{-4} atmospheres. Cathode rays consist of a stream of negatively charged material particles known as electrons emitted from cathode.

The value of charge to mass ratio (e/m) also known as specific charge of an electron was determined by J.J. Thomson through the deflection of the cathode rays under the influence of electric and magnetic fields and was found to be 1.76×10^{11} coulombs per kilogram.

The actual charge on an electron as found out by the Millikan's oil drop method was 1.602×10^{-19} coulombs. The mass of the electron was found out from the values of $\frac{e}{m}$ and the charge 'e'. Mass of the electron is 9.1×10^{-31} kg.

(ii) Prdotons

When cathode rays strike the molecules of a gas at low pressures, neutral molecules and atoms lose electrons to produce positive ions.

$$M + e^- \rightarrow M^+ + 2\,e^-$$

The positive ions thus produced can pass through a perforated cathode and come out as positive rays. Goldstein (1886) discovered that the simplest positive particle was produced from hydrogen in the discharge tube. This fundamental particle is called the proton.

The actual mass of a proton is 1.67×10^{-27} kg. The electrical charge on a proton is equal in magnitude and opposite in sign to that of an electron. A proton is a subatomic particle, which has a mass of one atomic mass unit and possesses one unit positive charge.

(iii) Neutrons

In 1932, James Chadwick discovered a third subatomic particle, when a stream of alpha particles (${}^4_2He$) was allowed to fall on beryllium metal.

$${}^4_2He + {}^9_4Be \rightarrow {}^{12}_6C + {}^1_0n$$

He named it as neutron. A neutron has a mass almost equal to (slightly higher than) that of a proton and is electrically neutral.

So the structure of the atom is based on the three subatomic particles electrons, protons and neutrons and these are the three main subatomic particles.

Many other subatomic particles such as positrons, mesons, neutrinos, pions are also known thereby making the picture of an atom more complex.

Characteristics of fundamental particles

Table 3.1

Symbol	Electron e or e^-	Proton p or p^+	Neutron n or n°
Relative mass (approx)	$\frac{1}{1836}$	1	1
Relative charge	–1	+1	No charge
Mass in kg	9.109×10^{-31}	1.673×10^{-27}	1.675×10^{-27}
Mass in amu	0.000549	1.007	1.008
Charge in coulomb	-1.6×10^{-19}	$+1.6 \times 10^{-19}$	Zero
Charge in esu	-4.8×10^{-10}	$+4.8 \times 10^{-10}$	Zero
Charge in emu	-1.6×10^{-20}	$+1.6 \times 10^{-20}$	Zero
Discovered by	J.J. Thomson	Goldstein	J. Chadwick

Atomic number (Z) of an element

Atomic number of an element is defined as the number of protons in the nucleus. As an atom is electrically neutral, the number of protons in the nucleus is equal to the number of electrons revolving in orbits outside the nucleus.

Z = Number of protons in the nucleus = Number of electrons in a neutral atom

Moseley's work on Atomic numbers

Roentgen in 1895 found that when cathode rays were allowed to fall on a metal, which acted as an anticathode or target material, a new type of radiation known as X-rays were produced.

Moseley in 1913 found that the wavelengths of these X-rays decreased in a regular manner in passing from one element to another used as anticathode. Moseley plotted the atomic number of the target material against the square root of the frequency of the X-rays emitted and obtained a straight line thereby indicating that the wavelength of the emitted X-rays were related to the atomic numbers of the elements or the number of positive charges or protons in the nucleus.

Moseley's equation is $\sqrt{\upsilon} = a\,(Z - b)$, where 'a' and 'b' are constants, υ is the frequency of X-ray and Z is the atomic number of the anticathode.

CONCEPT STRAND

Concept Strand 1

When made an anticathode in an X-ray tube a metal emits X-rays of wavelength 2.285 Å. If the values of Moseley's constants a and b are 5×10^7 and 1.375 respectively for this line, calculate the atomic number of the element and identify it.

Solution

$$\sqrt{\upsilon} = a(Z - b)$$

$$\sqrt{\frac{3\times10^8}{2.285\times10^{-10}}} = 5 \times 10^7(Z - 1.375)$$

$$\frac{1.145\times10^9}{5\times10^7} = Z - 1.375$$

$$Z = 24.275 \approx 24$$

The element with atomic number 24 is chromium.

Mass number and atomic weight

The total number of protons and neutrons in the nucleus of an atom is called the mass number of the atom. The mass number of an atom should be a whole number. Since electrons have practically no mass, the entire mass of an atom is almost due to the mass of protons and neutrons, each of which has a mass of approximately one mass unit. Generally the mass number of an atom can be obtained by rounding off the actual value of the atomic mass or atomic weight to the nearest whole number. The actual atomic mass of sodium is 22.9898 and its mass number is 23 and the atomic mass of aluminium is 26.9815 and its mass number is 27.

CONCEPT STRAND

Concept Strand 2

Among the following radiations, which is most easily stopped by air, why?

(i) X-ray (ii) α-ray
(iii) β-ray (iv) γ-ray

Solution

Alpha particle tracks (in length) are restricted in air because of the considerable ionizing power of the rays for molecules of air.

Atomic terms

(i) Atomic number (Z)

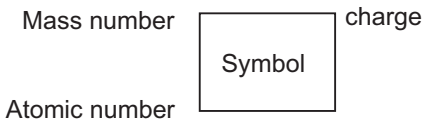
(ii) Mass number (A)

(iii) Nucleus

The centre of an atom where the mass and positive charge of the atom are concentrated is called the nucleus.

(iv) IUPAC notation of an atom

The atomic number, the mass number, charge and the symbol of the element are to be written as



For example, ${}^{4}_{2}He$, ${}^{19}_{9}F$, ${}^{40}_{18}Ar$, ${}^{4}_{2}He^{2+}$, ${}^{16}_{8}O^{2-}$ etc

In a molecule, atomicity may be written at the right hand bottom corner and charge on the right hand top corner.

For example, H_2, O_2, O_2^{2-}, H_2^{+} etc.

(v) Nuclide

Various species of atoms or nuclei with a definite atomic number and mass number are known as nuclides.

(vi) Isotopes

These are atoms of the same element with the same atomic number but different mass numbers i.e., isotopes are atoms, which differ only in mass but have same electrical charge on their nucleus and same chemical properties but different physical properties.

Example, (a) The three isotopes of hydrogen are ${}^{1}_{1}H$ (ordinary hydrogen or protium) ${}^{2}_{1}H$ (heavy hydrogen or deuterium – represented also as D) and ${}^{3}_{1}H$ (radioactive hydrogen or tritium – represented also as T)

Example, (b) The two isotopes of chlorine are ${}^{35}_{17}Cl$ and ${}^{37}_{17}Cl$

(vii) Isobars

These are atoms of different elements having the same mass number but different atomic numbers.

Example,

(a) ${}^{14}_{6}C$ and ${}^{14}_{7}N$

(b) ${}^{40}_{18}Ar$, ${}^{40}_{19}K$ and ${}^{40}_{20}Ca$

(viii) Isotones

Atoms of different elements having the same number of neutrons but different number of protons or mass number.

Example,

(a) ${}^{40}_{18}Ar$ and ${}^{41}_{19}K$

(b) ${}^{14}_{6}C$, ${}^{15}_{7}N$ and ${}^{16}_{8}O$

(ix) Isosters

Molecules having the same number of atoms and also same number of electrons are known as isosters.

Example,

(a) HCl and F_2
(b) CO and N_2
(c) CO_2 and N_2O

(x) Isodiaphers

Atoms having the same difference in the number of neutrons and protons or same isotopic number are known as isodiaphers.

Example,

(a) $^{19}_{9}F$ and $^{35}_{17}Cl$ (b) $^{14}_{6}C$ and $^{40}_{19}K$

Atom	No. of neutrons	No. of protons	Difference
(a) $^{19}_{9}F$	10	9	1
$^{35}_{17}Cl$	18	17	1
(b) $^{14}_{6}C$	8	6	2
$^{40}_{19}Cl$	21	19	2

(xi) Isoelectronic species

Atoms, molecules or ions having the same number of total electrons are known as isoelectronic species.

Example,

(a) HF, Ne and F^-

(b) Ca, Sc^+ and SiC

ATOMIC MODELS

Plum pudding model of atom

Thomson proposed a plum pudding model for an atom in which electrons were embedded in a ball of positive charge.

This model was rejected due to several reasons, including the instability of the atom.

Rutherford's nuclear model of the atom

Rutherford's experiment using α-particles as projectiles and a gold foil as a target, gave an insight on the nuclear model of the atom. Majority of the α-particles went through the foil without getting deflected. This shows that most of the atoms is empty space. The striking feature was that one in about 20,000 particles retraced their path. Such intense deflection is possible only if a strong electrical field is present inside the atom. Calculations revealed that a centre known as nucleus where the entire mass and positive charge of the atom are concentrated exists in the atom, spread over a sphere of 10^{-15} m radius. The electrons are outside the nucleus at a radius of about 10^{-10} m. The positive charge on the nucleus is due to the protons, which are positively charged particles in the nucleus. The radius of a nucleus can be roughly expressed as,

$$r\ (m) = 1.3 \times 10^{-15}\ A^{1/3}$$

where, A is the mass number.

CONCEPT STRANDS

Concept Strand 3

Calculate the radius of helium nucleus (A = 4). Nucleus of which element has a radius double that of He.

Solution

$$X = 1.3 \times 10^{-15}\ \sqrt[3]{A}\ \text{m}$$

$$r_{He} = 1.3 \times 10^{-15}\ \sqrt[3]{4} = 2.06 \times 10^{-15}\ \text{m}$$

Radius of the second atom = $2 \times 2.06 \times 10^{-15}$

Its mass number (A) = $\left(\frac{2 \times 2.06 \times 10^{-15}}{1.3 \times 10^{-15}}\right)^3 = 31.83 \approx 32$

The element with mass number 32 is sulphur.

Concept Strand 4

The radius of the nucleus of an atom of an element is found to be 3.9×10^{-15} m. Identify the element.

Solution

Radius (nucleus) $= 3.9 \times 10^{-15}$ m

$$= 1.3 \times 10^{-15} \times A^{1/3}\ \text{m}$$

$\Rightarrow$ A = 27

$\therefore$ The element is Al.

Rutherford's Model of the atom

The model of the atom proposed by Rutherford was known as the nuclear or planetary model of the atom.

The postulates of this model are as follows.

(i) Atom has a small dense central core known as nucleus, which holds the entire mass of the atom, known as nucleus and the remaining space surrounding the nucleus is almost free or empty.
(ii) The entire positive charge of the atom is centered on the nucleus and the negatively charged electrons are distributed in the vacant space around the nucleus.
(iii) The negatively charged electrons are moving in circular paths called orbits around the nucleus similar to the planets revolving round the sun.
(iv) The centrifugal force resulting from the circular movement of the electron is exactly balanced by the centripetal force arising from the electrostatic attraction between the nucleus and electrons. Therefore, the electrons remain in the orbits.
(v) The positive charge of the nucleus is equal to the negative charge of the electrons. Therefore, the atom is electrically neutral.

Drawbacks of Rutherford's model

(i) According to classical electromagnetic theory, when a charged particle (in this case electrons) revolves around an oppositely charged particle (a positively charged nucleus), the revolving particle (in this case an electron) continuously radiates energy, decreases in speed and travels in a spiral path eventually falling into the nucleus. Then the atom or nucleus becomes unstable and destroys itself into energy. In reality, an atom is very stable and Rutherford's model of atom should not have a long life-time.
(ii) As the energy of the electron continuously decreases, the frequency of radiation emitted by the electron continuously changes. This produces a continuous spectrum. But in reality, the atom produces line spectrum, and not continuous spectrum.
(iii) Rutherford's model does not say anything about the electronic structure of the atom.

Wave parameters

A wave has the following measurable properties

(i) Wavelength (λ)

The distance between successive crests or troughs is known as wavelength. The units in which wavelength is expressed are cm, m, nm (1nm = 10^{-9} m) and Å (1Å = 10^{-10} m).

(ii) Velocity (c)

It is the distance travelled by the wave in one second and is expressed in m s^{-1} or cm s^{-1}.

(iii) Frequency (υ)

It is the number of waves that pass a given point in a medium in one second. It is expressed in cycles per second (cps) or Hertz (Hz) or s^{-1}

1Hz = 1cps.The frequency of a wave is related to its wavelength as $\upsilon = \frac{c}{\lambda}$

(iv) Wave number ($\bar{\upsilon}$)

It is the number of waves per unit length (or) it is the reciprocal of wavelength and is expressed in m^{-1} or cm^{-1}. i.e., $\bar{\upsilon} = \frac{1}{\lambda}$ (or) $\bar{\upsilon} = \frac{\upsilon}{c}$

(v) Amplitude (a)

It is the height of the crest or the depth of the trough of a wave.

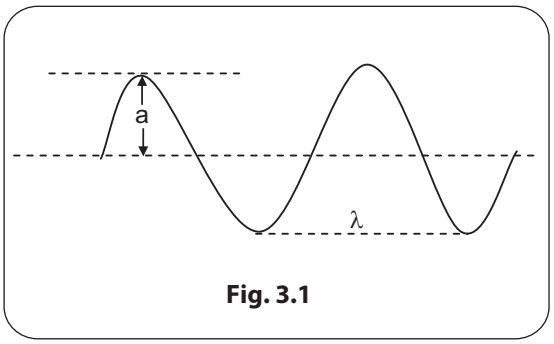


Fig. 3.1

CONCEPT STRAND

Concept Strand 5

Identify the shortest wavelength of radiation among the following.

(i) 740 nm (ii) 6.3×10^{-5} cm
(iii) 1.05 μm (iv) 3.5×10^{-6} m

Convert all of them into metres and then compare.

Solution

(i) $740\ \text{nm} \times \dfrac{10^{-9}\ \text{m}}{1\text{nm}} = 7.40 \times 10^{-7}\ \text{m}$

(ii) $6.3 \times 10^{-5}\ \text{cm} \times \dfrac{1 \times 10^{-2}\ \text{m}}{1\text{cm}} = 6.3 \times 10^{-7}\ \text{m}$

(iii) $1.05\ \mu\text{m} \times \dfrac{1 \times 10^{-6}\ \text{m}}{1\ \mu\text{m}} = 1.05 \times 10^{-6}\ \text{m}$

(iv) $3.5 \times 10^{-6}\ \text{m}$

6.3×10^{-7} m is the shortest wave length of radiation.

Electromagnetic radiations

Radiation is the emission and transmission of energy through space in the form of waves. Waves consisting of oscillating electric and magnetic fields are known as electromagnetic radiations and they require no medium to pass through.

The field components viz., electric and magnetic fields have the same wavelength, frequency, speed and amplitude and travel in planes perpendicular to each other and also perpendicular to the direction of propagation of the electromagnetic wave. In vacuum, all the electromagnetic radiations travel at the same speed of 3×10^8 m s^{-1}.

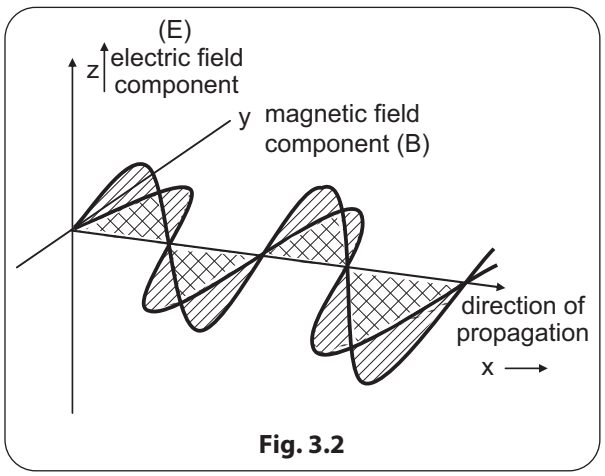


Fig. 3.2

Spectrum

When a ray of light passes through a prism or a denser medium, it gets refracted. The wave with a shorter wavelength bends more than the one with a longer wavelength. Ordinary white light consisting of waves with wavelengths in the visible range, when passed through a prism gets dispersed into seven colours to produce a spectrum on screen. Thus spectrum is an arrangement of waves (or radiations) in the decreasing or increasing order of wavelength or frequency.

In the above spectrum, red light with the longest wavelength suffers minimum deviation and the violet light with the shortest wavelength suffers maximum deviation. The spectrum of white light is a continuous spectrum or continuum as the blue merges into green, green into yellow and so on.

There are spectra consisting of well-defined lines and they are known as line spectra. A line spectrum contains only discrete frequencies (or wavelengths) whereas a continuous spectrum contains a continuous range of frequencies.

Electromagnetic spectrum

It is the spectrum of electromagnetic radiation.

Table 3.2

Radiations	Wavelength (Å)
Cosmic rays	0.00 – 0.01
Gamma rays	0.01 – 0.1
X-rays	0.1 – 150
Ultra violet (UV)	150 – 4000
Visible	4000 – 7000
Infrared (IR)	7000 – 10^6
Micro waves	10^6 – 10^9
Radio waves	10^9 – 10^{14}

Atomic spectrum

When an atom is exposed to electromagnetic radiation, the electric field of the radiation exerts a time varying force on the electrical charges (electrons and nucleus) of each atom and the electrons pass from lower energy levels to higher levels by absorbing certain specific frequencies. Consequently, waves of the corresponding frequencies are lost from the incident radiation and appear as dark lines in the spectrum. These dark lines constitute the absorption spectrum. When the excited atoms lose energy, the electrons

pass from higher energy levels to lower levels and waves of definite frequencies are emitted and appear as bright lines. These bright lines constitute the emission spectrum. To produce an emission spectrum, energy is supplied to an atom by heating it or irradiating it and the frequency (or wavelength) of the radiation emitted, as the sample gives up the absorbed energy, is recorded. An absorption spectrum of an atom is like the photographic negative of its emission spectrum. It is important to note that atomic spectrum is line spectrum.

Quantum theory and Bohr's model of atom

Rutherford's theory of the atom failed to explain the line spectrum produced by atoms. Also, Rutherford model could not explain how the spectral lines are produced by a simple atom like hydrogen when an electron jumps from one orbit to another.

Planck's Quantum Theory

Energy can be transmitted through space by electromagnetic radiation, which is made up of waves, having electrical and magnetic properties. The wave theory of transmission of radiant energy proposed that energy was absorbed or emitted in the form of continuous waves.

Max Planck in 1900 proposed that radiation of light emitted by a hot body was produced discontinuously by the molecules, each of which was vibrating with a particular frequency and the frequency of vibration increases with increase of temperature. According to Planck, when a hot body radiates energy, it does not give out continuous waves, but it gives small units of waves. The 'unit wave' or 'pulse of energy' is called a quantum. The quanta (plural of quantum) of light are called photons.

According to quantum theory of radiation, when atoms or molecules absorb or emit energy, they do so in discrete units or quanta or photons.

The energy 'E' of a quantum is given by the relation,

$$E = h\upsilon$$

where, E is the energy, h is Planck's constant which is 6.626×10^{-27} erg sec or 6.626×10^{-34} joules sec and υ is the frequency of radiation.

Since, $\upsilon = \frac{c}{\lambda}$, E can also be written as

$$E = \frac{hc}{\lambda}$$

The magnitude of a quantum or photon of energy is directly proportional to the frequency of radiation (υ) and inversely proportional to wavelength (λ). An atom or molecule can emit or absorb one quantum of energy ($h\upsilon$) or a simple multiple of it. So the energy E can be in terms of $h\upsilon$, $2h\upsilon$, $3h\upsilon$ and so on, but not in terms of fractions such as $1.25h\upsilon$ or $3.6h\upsilon$.

Besides, according to the theory of relativity, the effective mass 'm' of a particle moving with a velocity 'v' is given by, $m = \frac{m_0}{\sqrt{1 - v^2/c^2}}$, where m_0 is the so called rest mass i.e., mass of particle at rest and c is the velocity of light.

According to this theory, if a particle moves with the speed of light, it should not have mass. Otherwise its effective mass during motion would be infinite.

$\therefore \quad m_0 = 0$

$\therefore$ Energy of a photon is given by $E = h\upsilon = pc$, where p is the momentum associated with the particle. $E = pc$ is valid for all particles travelling with the speed of light (massless). Hence it is to be noted that a photon is always absorbed or destroyed completely or remains intact.

CONCEPT STRANDS

Concept Strand 6

Two wavelengths are in the ratio 1:2. What will be the ratio of their energies?

Solution

Since $E = h\upsilon$

$\Rightarrow \quad E = \frac{hc}{\lambda} \quad \therefore E \propto \frac{1}{\lambda}$

Ratio of energies = 2:1

Concept Strand 7

In an experiment, energy required is 3.6 kJ min^{-1} from a light source. If the scientist uses a monochromatic light bulb, what wattage should he choose?

Solution

$1\ J\ sec^{-1} = 1$ watt; $1\ J\ min^{-1} = \frac{1}{60}$ watts

$$3.6\times10^3\ J\,min^{-1} = \frac{3.6\times10^3}{60} = 60\ watts$$

Concept Strand 8

The energy required to remove an electron from a metal is 1.655×10^{-20}J. Calculate the maximum wavelength of light that can eject an electron from the metal.

Solution

$$\lambda = \frac{hc}{E} = \frac{6.63\times10^{-34}\,\text{Js}\times\left(3\times10^{8}\,\text{m s}^{-1}\right)}{1.655\times10^{-20}\,\text{J}} = 12\times10^{-6}\,\text{m}$$

Concept Strand 9

Calculate the number of photons of light having a wavelength of 4000 Å to provide 2 J of energy.

Solution

$$E_{photon} = h\upsilon = \frac{hc}{\lambda} = \frac{\left(6.63\times10^{-34}\,\text{Js}\right)\times\left(3\times10^{8}\,\text{m s}^{-1}\right)}{4000\times10^{-10}\,\text{m}}$$

$$= 4.97\times10^{-19}\,\text{J}$$

For 2 J energy, the no. of photons

$$= \frac{2\,\text{J}}{4.97\times10^{-19}\,\text{J photon}^{-1}} = 4.02\times10^{18}$$

Concept Strand 10

Iodine dissociates by absorbing light of wavelength 450 nm. If one quantum of radiation is absorbed by each molecule, calculate the kinetic energy of iodine atom? (Bond energy of iodine = 240 kJ mol^{-1})

Solution

Bond energy per molecule = $\frac{240\times10^{3}}{6.022\times10^{23}} = 3.98\times10^{-19}\,\text{J}$

Energy of one photon,

$$E = h\frac{c}{\lambda} = \frac{6.626\times10^{-34}\times3\times10^{8}}{450\times10^{-9}} = 4.42\times10^{-19}\,\text{J}$$

Kinetic energy per atom =

$$\frac{1}{2}(4.42-3.98)10^{-19} = 2.2\times10^{-20}\,\text{J}$$

Bohr's model of the atom

A Danish physicist, Neils Bohr in 1913 put forward a new model for the hydrogen atom based on Rutherford's nuclear model and Planck's quantum theory.

The postulates of this model of atom are as follows.

(i) The electron in the hydrogen atom revolves round the nucleus only in certain fixed, concentric, circular orbits, or stationary states.

(ii) Each orbit is associated with a definite amount of energy or with a definite quantum of energy. Therefore, orbits are also called energy levels.

(iii) Greater the distance of the orbit from the nucleus, greater will be the energy of the electron in that orbit.

(iv) The energy of an electron cannot change continuously. An electron 'jumps' from one energy level to another, but does not 'flow' from one level to another.

(v) The angular momentum of an electron moving around the nucleus in an allowed orbit is given by, angular momentum = mvr = n (h/2π) where, m = mass; v = linear velocity of the electron; r = radius of the orbit; n = an integer (n = 1, 2, 3, etc).
Thus an electron can move only in those orbits for which its angular momentum is an intergral multiple of $\frac{h}{2\pi}$. That is why only certain orbits are allowed.

(vi) As long as an electron remains in a particular orbit, it neither emits nor absorbs energy. These energy levels are therefore called stationary states. This concept of stationary states explains the stability of the atom, since an electron cannot lose energy gradually and fall into the nucleus.

(vii) When the electron absorbs a quantum of energy, it jumps to an outer orbit and when it emits a quantum of energy, it jumps to an inner orbit.

(viii) When the electron jumps from a lower orbit of energy E_1 to a higher orbit of energy E_2, the difference in energy, $\Delta E = E_2 - E_1$ will be absorbed as a quantum of electromagnetic radiation of definite frequency υ. The energy difference ΔE and frequency υ are related according to Planck's quantum theory as

$$\Delta E = E_2 - E_1 = h\upsilon.$$

When the electron jumps from the higher orbit of energy E_2 to the lower orbit of energy E_1, the same quantum of electromagnetic radiation of frequency υ is emitted.

HYDROGEN SPECTRUM

Application of Bohr's theory to Hydrogen atom

The single electron in a hydrogen atom at ordinary temperatures resides in the ground state or lowest energy state. When energy is supplied to hydrogen gas in a discharge tube, hydrogen molecules split into atoms and the electrons in different atoms move from the ground state to discrete higher energy levels depending on the quantum of energy absorbed. Now the hydrogen atom is said to be in an excited state. Atoms cannot remain in the excited state for long. The electron in a hydrogen atom after reaching higher level returns to lower energy level emitting the excess of energy in the form of one or more photons. This gives rise to a one or more spectral lines. Depending on the region of appearance in the electromagnetic spectrum, these lines are grouped into the following spectral series.

(i) **Lyman series** is obtained when the electron returns to the ground state (i.e., $n_1 = 1$) from higher levels (i.e., $n_2 = 2, 3, 4, 5$ etc.)

(ii) **Balmer series** results when the electron returns to the second level ($n_1 = 2$) from higher levels ($n_2 = 3, 4, 5, 6$ etc.)

(iii) **Paschen series** is observed when the electron returns to third level ($n_1 = 3$) from higher levels ($n_2 = 4, 5, 6$ etc.)

(iv) **Brackett series** results when the electron returns to level 4 ($n_1 = 4$) from higher levels ($n_2 = 5, 6, 7$ etc.)

(v) **Pfund series** is the series of lines observed when the electron returns to level 5 ($n_1 = 5$) from higher levels ($n_2 = 6, 7$ etc.)

(vi) **Humphrey series** is the series of lines observed when the electron returns to level 6 ($n_1 = 6$) from higher levels ($n_2 = 7, 8, 9$ etc.)

Table 3.3

Series	Region of appearance	n_1	n_2
Lyman	UV	1	2, 3, 4, 5 etc
Balmer	visible	2	3, 4, 5, 6 etc
Paschen	Near I.R	3	4, 5, 6, 7 etc
Brackett	I.R	4	5, 6, 7, 8 etc
Pfund	Far I.R	5	6, 7, 8, 9 etc
Humphrey			7, 8, 9, 10. etc

So the spectral lines in the hydrogen spectrum are due to emission of energy acquired by electronic transitions. Working on the above postulates, Bohr calculated the energy of an electron remaining in any orbit, from the equation,

$$E_n = \frac{-2\pi^2 Z^2 me^4}{n^2 h^2} \text{ (in C.G.S.)}$$

$$E_n = \frac{-k^2 2\pi^2 Z^2 me^4}{n^2 h^2} \text{ (in S.I.)}$$

Where,

E_n = Energy of electron in the nth orbit

Z = Atomic number

m = Mass of an electron

e = Charge of an electron

n = Principal quantum number

h = Planck's constant

k = Coulomb's constant = $\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9 \text{ N m}^2 \text{C}^{-2}$.

This is known as the Bohr equation

Refinement of Bohr's Equation

We cannot however call this E_n precisely the energy of the electron, because in its derivation we have assumed that the electron moves about a stationary nucleus, in fact, the nucleus cannot be at rest. According to the laws of dynamics, whenever a particle moves in a circle about another particle of finite mass, the latter also must move in a circle about the common centre of mass. But the motion of one particle may be neglected, if the mass of the other particle is replaced with the reduced mass of the two particles, $\frac{mM}{m+M}$ or $\frac{m}{1+\frac{m}{M}}$, where, m and M are the masses of electron and nucleus respectively.

$$\therefore E_n = \frac{-k^2 2\pi^2 Z^2 me^4}{n^2 h^2 \left(1+\frac{m}{M}\right)}$$, where, k is Coulombs constant,

$k = 9 \times 10^9 \text{ J m C}^{-2}$

Bohr equation is applicable to hydrogen, hydrogen like atoms and uni–electronic species like He^+, Li^{2+}, Be^{3+} (Z = 1 for H, Z = 2 for He^+, and Z = 3 for Li^{2+})

For H atom, in the ground state, $E_1 = -13.6 \text{ eV atom}^{-1}$

Energy of any other uni–electronic species in the nth orbit is

$$E_n = \frac{-13.6Z^2}{n^2} \text{ eV atom}^{-1}$$

$$= \frac{-2.18 \times 10^{-18}}{n^2} \text{ J atom}^{-1}$$

$$= \frac{-1312}{n^2} \text{ kJ mol}^{-1}$$

According to the Bohr's equation, the energy of the electron in orbit n_1 of hydrogen atom is $E_{n_1} = \frac{-2\pi^2 me^4}{n_1^2 h^2}$ (Z = 1) and the energy of the electron in orbit n_2 is $E_{n_2} = \frac{-2\pi^2 me^4}{n_2^2 h^2}$

The difference in energy between the levels n_1 and n_2, i.e., $\Delta E = E_{n_2} - E_{n_1}$

$$\Delta E = \frac{2\pi^2 me^4}{h^2}\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right).$$

But, $\Delta E = \frac{hc}{\lambda}$

$$\text{or } \frac{hc}{\lambda} = \frac{2\pi^2 me^4}{h^2}\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

$$\Rightarrow \frac{1}{\lambda} = \frac{2\pi^2 me^4}{h^3 c}\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

Taking $\frac{2\pi^2 me^4}{ch^3}$ as a constant, R_H

$$\bar{\upsilon} = \frac{1}{\lambda} = R_H\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right] \text{ for hydrogen atom}$$

$$\bar{\upsilon} = R_H Z^2\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right] \text{ for any other uni–electronic}$$

species where, R_H is known as Rydberg constant for hydrogen. The value of R_H can be calculated as the values of e, m, h, and c are all constants. The value of R_H is 109677 cm^{-1} (or) 1.32×10^6 J mol^{-1} (or) 2.18×10^{-18} J $atom^{-1}$

Maximum number of emission lines produced from a uni–electronic species, when a single electron falls from the nth level = $\frac{n(n-1)}{2}$ i.e., when the electron falls from the 5th level, maximum number of emission lines = 10

CONCEPT STRAND

Concept Strand 11

In the Bohr spectrum, a line was observed at a frequency of 4.3×10^{14} s^{-1}. To which series would it correspond in the hydrogen spectrum?

Solution

$$\upsilon = 4.3 \times 10^{14} \text{ s}^{-1}$$

$$\Rightarrow \lambda = \frac{c}{\upsilon} = \frac{3 \times 10^8 \text{ ms}^{-1}}{4.3 \times 10^{14} \text{ s}^{-1}} = 6.976 \times 10^{-7} \text{ m} = 6976 \text{ Å}$$

This is in the visible region. Hence Balmer series.

Calculation of radius 'r' of orbits

Making use of the value of angular momentum of the revolving electron, $mvr = \frac{nh}{2\pi}$, one can arrive at the radius of the orbit which is given by the expression,

$$r_n = \frac{n^2 h^2}{4\pi^2 mZe^2} \text{ (in C.G.S)} \qquad \text{or}$$

$$r_n = \frac{n^2 h^2}{4\pi^2 kmZe^2} \text{ (in S.I.)}$$

The values of h, m, and e are determined experimentally and the value of 'r' gets reduced to

$$r_n = \frac{n^2 \times 0.529 \times 10^{-8}}{Z} \text{ cm}$$

When, n = 1, r = 0.529×10^{-8} cm and it is the Bohr radius of first orbit in a hydrogen atom. When the value of n changes to 2, 3, 4, 5 etc. 'r' becomes

$r_n = 2^2 \times 0.529 \times 10^{-8} = 2.12 \times 10^{-8}$ cm (when n = 2)

$r_n = 3^2 \times 0.529 \times 10^{-8} = 4.76 \times 10^{-8}$ cm (when n = 3)

$r_n = 4^2 \times 0.529 \times 10^{-8} = 8.46 \times 10^{-8}$ cm (when n = 4)

For hydrogen like atoms, $r_n = \frac{n^2}{Z} \times 0.529 \times 10^{-8}$ cm

$$= \frac{n^2}{Z} \times 0.529 \text{ Å}.$$

CONCEPT STRANDS

Concept Strand 12

Find out the hydrogen like ion having the wavelength difference between the first lines in the Paschen and Balmer series equal to 48.4 nm.

Solution

$$\frac{1}{\lambda} = R_H Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

For a first Paschen series line,

$$\frac{1}{\lambda_P} = R_H Z^2 \left(\frac{1}{3^2} - \frac{1}{4^2}\right)$$

$$= R_H Z^2 (0.1111 - 0.0625)$$

$$\Rightarrow \quad \lambda_P = \frac{20.57}{R_H Z^2} \text{ cm}$$

For a first Balmer series $\frac{1}{\lambda_B} = R_H Z^2 \left(\frac{1}{2^2} - \frac{1}{3^2}\right)$

$$\Rightarrow \quad \lambda_B = \frac{7.2}{R_H Z^2} \text{ cm} \qquad \therefore \ \lambda_P - \lambda_B = 48.4 \times 10^{-7} \text{ cm}$$

$$\Rightarrow \quad \frac{20.57 - 7.2}{R_H Z^2} = 48.4 \times 10^{-7} \text{ cm}$$

$$\Rightarrow \quad Z^2 = 25.188$$

$$\Rightarrow \quad Z = 5$$

It is boron, and the ion is B^{4+}

Concept Strand 13

If the Bohr's radius of the 4th orbit is 848 pm, find that of the 5th.

Solution

$$\text{Radius} = \frac{848 \times 5^2}{4^2} \times 10^{-12} \text{ m} = 1.32 \times 10^{-9} \text{ m}$$

Concept Strand 14

Find the ratio of the radius of the 2nd Bohr orbit of Hydrogen atom (at. no. = 1) and the 4th Bohr orbit of He^+ ion (at. no. = 2).

Solution

$$r_0 = \frac{n^2 a_0}{Z}$$

$$r_{2(H)} = \frac{4a_0}{1}; \qquad r_{4_{(He^+)}} = \frac{16a_0}{2}$$

$$r_{2(H)} : r_{4_{(He^+)}} = 4a_0 : 8a_0 = 1 : 2$$

Concept Strand 15

The energy level of triply ionized beryllium, which will have same radius as ground state of hydrogen atom is

(a) 2 (b) 1 (c) 3 (d) none

Solution

(a)

$$r_n = \frac{n^2}{Z} . r_{1,H} \quad r_{n,Be^{3+}} = \frac{n^2}{4} r_{1,H}$$

$$r_{1,H} = \frac{n^2}{4} r_{1,H} \Rightarrow n^2 = 4 \text{ or } n = 2$$

Energy of an electron

The total energy, E of an electron is the sum of potential and kinetic energies.

$$\text{Kinetic energy} = \frac{1}{2} mv^2$$

$$\text{Potential energy} = \frac{-Ze \times e}{r^2} \times r = \frac{-Ze^2}{r} \text{(C.G.S.) (or)}$$

$$\frac{-k Ze \times e}{r^2} \times r = \frac{-kZe^2}{r} \text{(S.I.)}$$

(negative sign indicates attraction).

From Coulomb's law, the electrostatic attraction between the electron and nucleus

$$= \frac{Ze \times e}{r^2} = \frac{Ze^2}{r^2} \text{(C.G.S.)} \quad \text{(or)} \quad \frac{k Ze^2}{r^2} \text{(S.I.)}$$

$$\text{The centrifugal force of an electron} = \frac{mv^2}{r}$$

An electron can remain in an orbit when the centrifugal force is balanced exactly by the electrostatic force of attraction.

i.e., $\frac{mv^2}{r} = \frac{Ze^2}{r^2}$

$\Rightarrow \quad mv^2 = \frac{Ze^2}{r}$ (C.G.S.) (or) $mv^2 = \frac{kZe^2}{r}$ (S.I.)

$\therefore$ Kinetic energy,

$\frac{1}{2}mv^2 = \frac{Ze^2}{2r}$ (C.G.S.) (or) $\frac{1}{2}mv^2 = \frac{k\,Ze^2}{2r}$ (S.I.)

$\therefore$ Total energy of the electron, $E = \frac{Ze^2}{2r} - \frac{Ze^2}{r}$

$= -\frac{Ze^2}{2r}$ (C.G.S.) (or) $\frac{-kZe^2}{2r}$ (S.I.)

Substituting for 'r', the above equations become

$E = \frac{-2\pi^2 m Z^2 e^4}{n^2 h^2}$ (C.G.S.) (or)

$E = \frac{-2\pi^2 k^2 m Z^2 e^4}{n^2 h^2}$ (S.I.)

CONCEPT STRAND

Concept Strand 16

An α particle moving with a velocity of 2×10^7 m s^{-1} is collided with the nucleus of a metal atom and stopped at a distance of 10^{-14} m from the nucleus. Which is that metal atom? (Mass of hydrogen atom = 1.67×10^{-27} kg)

Solution

K.E = P.E

$\frac{1}{2}mv^2 = \frac{2Ze^2}{4\pi\varepsilon_0 r_0}$

$\Rightarrow \quad Z = \frac{mv^2(4\pi\varepsilon_0)r_0}{4e^2}$

$\frac{1}{4\pi\varepsilon_0} = 9\times10^9$

$\Rightarrow \quad 4\pi\varepsilon_0 = 1.1 \times 10^{-10}$

$$Z = \frac{4\times1.67\times10^{-27}\text{ kg}\times\left(2\times10^7\right)^2\text{ m}^2\text{ s}^{-2}\times 10^{-14}\text{ m}\times1.1\times10^{-10}\text{ C}^2\text{ N}^{-1}\text{ m}^{-2}}{4\times\left(1.6\times10^{-19}\right)^2\text{ C}^2} = 29$$

The metal atom is copper.

Energy associated with the first five energy levels in hydrogen

Maximum number of emission lines produced from a uni–electronic species, when the single electron falls from the nth level = $\frac{n(n-1)}{2}$

i.e., when the electron falls from the 5th level, maximum number of emission lines = 10

$$E = = \frac{-2\pi^2 me^4}{n_1^2 h^2}$$

Substituting the values of constants, where

m = mass of electron = 9.1×10^{-28} g;

e = charge of an electron = 1.60×10^{-19} coulomb

h = Planck's constant = 6.626×10^{-34} J s.

$E = \frac{-2.18\times10^{-11}}{n^2}$ erg $atom^{-1}$

$= \frac{-2.18\times10^{-18}}{n^2}$ joule $atom^{-1}$

$= \frac{-2.18\times10^{-18}\times6.02\times10^{23}}{n^2}$ joule mol^{-1}

$= \frac{-1311.8}{n^2}$ kJ mol^{-1}

$= \frac{-313.3}{n^2}$ kcal mol^{-1}

$= -\frac{13.6}{n^2}$ eV $atom^{-1}$

$= -\frac{109700}{n^2}$ cm^{-1} $atom^{-1}$

Energy at the five levels are,

$n = 1, \quad E_1 = \dfrac{-2.18 \times 10^{-11}}{1^2}$

$= -2.179 \times 10^{-11}$ erg atom^{-1} = – 1311.8 kJ mol^{-1}

$n = 2, \quad E_2 = -0.5448 \times 10^{-11}$ erg atom^{-1} = –327.9 kJ mol^{-1}

$n = 3, \quad E_3 = -0.2421 \times 10^{-11}$ erg atom^{-1} = – 145.8 kJ mol^{-1}

$n = 4, \quad E_4 = -0.1362 \times 10^{-11}$erg atom^{-1} = – 82.0 kJ mol^{-1}

$n = 5, \quad E_5 = -0.0872 \times 10^{-11}$ erg atom^{-1} = – 52.49 kJ mol^{-1}

Hydrogen like ion

All systems with (Z – 1) positive charges become hydrogen like ions since all of them have only one electron. He^+, Li^{2+}, Be^{3+} etc are called hydrogen like ions. The Rydberg equation for these ions is, $\dfrac{1}{\lambda} = \bar{\upsilon} = R_H Z^2 \left(\dfrac{1}{n_1^2} - \dfrac{1}{n_2^2} \right)$ where, $\bar{\upsilon}$ = wave number of the spectral line, R_H = Rydberg constant for 'H' atom and Z = the atomic number of the element. Ionization potential of the hydrogen like ion is given by $E = hc\bar{\upsilon} = hcR_H Z^2 \left(\dfrac{1}{1^2} - \dfrac{1}{\infty^2} \right)$

The ionization potential of hydrogen is 13.56 eV and the hydrogen like ions will have ionisation potential equal to 13.56 Z^2, where Z refers to the atomic number of the element.

Significance of negative value of energy

The energy of an electron at rest or at infinity is assumed to be zero. This is called zero–energy state. When an electron comes closer to the nucleus and under the influence of the nucleus, it spends some of its energy and does some work in this process. Therefore, the energy of an electron decreases from the zero value and it assumes a negative value. In other words, an amount of energy equivalent to the value of the energy of the electron has to be put in, in order to pull the electron from the influence of the nucleus to the zero energy state at infinite distance from the nucleus.

Electron volt (eV)

An electron volt is the energy gained by an electron when it is accelerated through a potential difference of 1 volt.

1 eV = 1.6×10^{-19} J = 23 kcal mol^{-1} = 96.4 kJ mol^{-1}

CONCEPT STRANDS

Concept Strand 17

A diatomic molecule undergoes photochemical dissociation into two atoms of which one is having the energy of 3.2 eV and the other 2.6 eV. What is the effective frequency for this dissociation? (Given: Dissociation energy of the given molecule = 500 kJ mol^{-1})

Solution

1 mole $\rightarrow 6.023 \times 10^{23}$ diatomic molecules

6.023×10^{23} molecules $\rightarrow 500 \times 10^3$ J

$\therefore$ for 1 diatomic molecule,

$$\frac{500 \times 10^3}{6.023 \times 10^{23}} = 8.30 \times 10^{-19} \text{ J/molecule}$$

For 1 excited atom, 3.2 eV $\quad$ (1eV = 1.6×10^{-19}J)

Energy = $3.2 \times 1.6 \times 10^{-19} = 5.12 \times 10^{-19}$ J

For the other excited atom $\rightarrow$ 2.6 eV

Energy = $2.6 \times 1.6 \times 10^{-19} = 4.16 \times 10^{-19}$J

$\therefore$ Total energy = $5.12 \times 10^{-19} + 4.16 \times 10^{-19} + 8.30 \times 10^{-19}$
$= 17.58 \times 10^{-19}$ J

$$E = h\upsilon \Rightarrow \upsilon = \frac{E}{h} = \frac{17.58 \times 10^{-19} \text{ J}}{6.6 \times 10^{-34} \text{ Js}} = 2.6 \times 10^{15} \text{ Hz}$$

Concept Strand 18

The ionization potential of Li^{2+} corresponds to light of wavelength 101.3Å. Calculate (i) R_H and (ii) the wave number of the first line in Lyman series of hydrogen atom.

Solution

$$\bar{\upsilon} = \frac{1}{\lambda} = R_H Z^2 \left[\frac{1}{1^2} - \frac{1}{\infty^2} \right] = R_H Z^2$$

$$\frac{1}{\lambda} = \frac{1}{101.3 \times 10^{-8}} = R_H \times 3 \times 3$$

$9R_H = 987166.8 \text{ cm}^{-1} \Rightarrow R_H = 109685 \text{ cm}^{-1}$

$$\bar{\upsilon} \text{ (Lyman)} = 109685 \left(1 - \frac{1}{4} \right) = 82263.7 \text{ cm}^{-1}$$

Concept Strand 19

If the longest wavelength of helium ion (He^+) in the Lyman series is x, then the shortest wavelength would be

(a) $\frac{3}{4x}$ (b) $\frac{4}{3x}$ (c) $\frac{3x}{4}$ (d) $\frac{4x}{3}$

Solution

$$\frac{1}{x} = R_H \times 2^2\left[\frac{1}{1^2}-\frac{1}{2^2}\right] \qquad \frac{1}{x} = 3R_H \text{ or } R_H = \frac{1}{3x}$$

For shortest wavelength

$$\frac{1}{\lambda} = R_H \times 2^2\left[\frac{1}{1^2}-\frac{1}{\infty^2}\right] = 4R_H = \frac{4}{3x}$$

$$\lambda = \frac{3x}{4}$$

Concept Strand 20

Calculate the energy required for the removal of the electron from the first orbit of hydrogen atom.

Solution

$$\frac{1}{\lambda} = 109678\left(\frac{1}{1^2}-\frac{1}{\infty}\right) = 109678 \text{ cm}^{-1}$$

$$= 10967800 \text{ m}^{-1}$$

$$= 1.09678 \times 10^7 \text{ m}^{-1}$$

$$E = \frac{hc}{\lambda} = 6.634\times10^{-34}\times3\times10^8\times1.09678\times10^7$$

$$= 2.18 \times 10^{-18} \text{ J atom}^{-1}.$$

Concept Strand 21

Calculate the value of Rydberg constant, R_H using Bohr equation.

Solution

$$R_H = \frac{2\pi^2k^2me^4}{ch^3}, \text{ where } k = \frac{1}{4\pi\varepsilon_0} = 9\times10^9 \text{ N m}^2 \text{ C}^{-2}$$

On substituting the values,

$$R_H = \frac{2\times(3.14)^2\times(9\times10^9 \text{ J m C}^{-2})^2\times}{9.1\times10^{-31} \text{ kg } (1.6\times10^{-19} \text{ C})^4}$$

$$(6.626\times10^{-34} \text{ Js})^3\times3\times10^8 \text{ ms}^{-1}$$

$$= 1.0926 \times 10^7 \text{ m}^{-1}.$$

Concept Strand 22

Which electronic transition in Li^{2+} can produce the radiation of same wavelength as the red line in hydrogen spectrum?

Solution

$n_2 = 9$ to $n_1 = 6$. For red line in hydrogen spectrum, $n_2 = 3$ to $n_1 = 2$

The energy of the 2 and 3 orbits of hydrogen is same as that of 6th and 9th orbits of Li^{2+}.

Concept Strand 23

Convert 4.8×10^{-20} coulomb to esu.

Solution

$$1C = 3 \times 10^9 \text{ esu}$$

$$\Rightarrow \quad \frac{3\times10^9}{1}\times4.8\times10^{-20} = 1.44\times10^{-10}$$

Concept Strand 24

What are the two successive energy levels for which the energy difference is the maximum?

Solution

First and second energy levels. Energy gap decreases as the distance of the orbit from the nucleus increases.

Concept Strand 25

Given below are the orbits between which the electronic transition takes place. Arrange the radiations corresponding to these transitions in the decreasing order of their wavelengths.

(i) n = 10 to n = 4
(ii) n = 9 to n = 3
(iii) n = 8 to n = 2
(iv) n = 7 to n = 1

Solution

(i) > (ii) > (iii) > (iv).
Falling to a higher level radiates lesser energy (or) the resulting radiation will have longer wavelength.

Concept Strand 26

Which of the following transitions produce a spectral line in hydrogen atom with the maximum wave length?

(a) n = 2 to n = 1
(b) n = 3 to n = 2
(c) n = 10 to n = 3
(d) n = 5 to n = 1

Solution

(c)
Spectral line of maximum wavelength has the least energy. The least energy transition is produced when electron falls from higher levels to the level with highest principal quantum number

Concept Strand 27

A certain transition emits a radiation of wavelength λ in a Li^{2+} atom. The same transition in Be^{3+} ion will have a wavelength

(a) λ (b) $\frac{9\lambda}{16}$

(c) $\frac{3\lambda}{4}$ (d) $\frac{3\lambda}{8}$

Solution

(b)

$$\frac{1}{\lambda} = R_H Z^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

$$\frac{\lambda_1}{\lambda_2} = \frac{Z_2^2}{Z_1^2} \qquad \frac{\lambda_1}{\lambda_2} = \frac{16}{9} \text{ so } \lambda_2 = \frac{9\lambda_1}{16}$$

Concept Strand 28

Calculate the ratio of the difference in energy between first and second orbits to that between second and third in H-atom.

Solution

$$E = -13.6Z^2\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

Here Z is same; $E \propto \left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$

$$\frac{E_{1-2}}{E_{2-3}} = \frac{\left(\frac{1}{1^2} - \frac{1}{2^2}\right)}{\left(\frac{1}{2^2} - \frac{1}{3^2}\right)} = \frac{1 - \frac{1}{4}}{\frac{1}{4} - \frac{1}{9}} = \frac{\frac{4-1}{4}}{\frac{9-4}{36}}$$

$$= \frac{3/4}{5/36} = \frac{3}{4} \times \frac{36}{5} = \frac{27}{5}$$

Concept Strand 29

In the excited state of a hydrogen atom the electron is in the 4th orbit. It falls to the ground state in two steps by falling from 4th to the second and then from second to the first orbit. Calculate the wave number and wavelength of the two lines produced by these transitions.

Solution

Wave number of the line produced by the first transition,

$$\bar{\upsilon} = 109679\left[\frac{1}{2^2} - \frac{1}{4^2}\right] = 20564 \text{ cm}^{-1}$$

Wave number of the first transition = $\lambda = \frac{1}{\bar{\nu}} = \frac{1}{20564}$

$$= 4.8626 \times 10^{-5} \text{ cm}$$

Wave number of the second transition

$$= \bar{\upsilon} = 109677\left[\frac{1}{1^2} - \frac{1}{2^2}\right] = 82258 \text{ cm}^{-1}$$

Wavelength = $\frac{1}{\bar{\upsilon}}$

$$= \frac{1}{82259} = 1.216 \times 10^{-5} \text{ cm}$$

Concept Strand 30

Calculate the wave number of the first line in the Lyman series of hydrogen atom.

Solution

$$\bar{\upsilon} = \frac{1}{\lambda} = R_H Z^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

$$= \frac{1}{101.3 \times 10^{-8} \times 9} = 109685 \text{ cm}^{-1}$$

Wave number of first line in Lyman series

$$= 109685\left[\frac{1}{1^2} - \frac{1}{2^2}\right] = 82263 \text{ cm}^{-1}$$

Velocity of an electron in a Bohr orbit of hydrogen

$$v = \frac{nh}{2\pi mr}$$

Substituting for r,

$$v = \frac{2\pi Ze^2}{nh} \text{ (in C.G.S.)}$$

(or) $$v = \frac{2\pi kZe^2}{nh} \text{ (in S.I.)}$$

Velocity of an electron in the first orbit of hydrogen

$$= \frac{1}{137} \times \text{velocity of light.}$$

$$= 2.19 \times 10^6 \times \frac{Z}{n} \text{ ms}^{-1}$$

Calculation of number of revolution of an electron in an orbit in one second

No. of revolutions per second

$$= \frac{v}{2\pi r} = \frac{2\pi mvZe^2}{n^2h^2} \qquad \left(\because r = \frac{n^2h^2}{4\pi^2 mZe^2}\right)$$

Calculation of number of waves in any orbit

No. of waves in an orbit =

$$\frac{\text{circumference of that orbit}}{\text{wave length}} = \frac{2\pi r}{\lambda} = \frac{2\pi r}{\left(h/mv\right)}$$

$$= \frac{2\pi}{h}(mvr) = \frac{2\pi}{h} \times \frac{nh}{2\pi} \qquad \left(\because mvr = \frac{nh}{2\pi}\right) = n$$

No. of waves in an orbit = No. of the orbit

CONCEPT STRANDS

Concept Strand 31

Which statement about hydrogen spectrum is not true?

(a) The line corresponding to the transition of $n_1 = 4$ to $n_2 = 3$ occurs in the IR region.
(b) The line with $n_1 = 3$ and $n_2 = 2$ has a higher wave number than the line with $n_1 = 4$ and $n_2 = 3$.
(c) The energy of the line corresponding to $n_1 = \infty$ and $n_2 = 1$ is the ionization energy of the hydrogen atom.
(d) He^+ has more lines in the spectrum than H.

Solution

(d)

He^+ and H are one-electron species with similar number of energy levels. Therefore, have the same number of transitions to each level

Concept Strand 32

An electron is at a distance of 4.77 Å from the nucleus in a hydrogen atom. The wavelength of this electron is (Radius of 1st orbit = 0.53 Å)

(a) 9.98 Å
(b) 0.68 Å
(c) 3.4 Å
(d) 6.8 Å

Solution

(a)

Waves are in phase $2\pi r = n\lambda$

$r_n = n^2 a_0$

$4.77 \text{ Å} = n^2 \times 0.53 \text{Å}$

$n^2 = 4.77/0.53 = 9$ so $n = 3$

$$\lambda = \frac{2\pi r}{n} = \frac{2 \times 3.14 \times 4.77 \text{ Å}}{3} = 9.98 \text{ Å}$$

Concept Strand 33

Calculate the number of revolutions/sec of an electron in the fifth orbit of the hydrogen like lithium ion.

Solution

$$r_n = \frac{0.529n^2}{Z} \text{Å} = \frac{0.529 \times (5)^2}{3} = 4.408 \text{ Å}$$

$$= 4.408 \times 10^{-10} \text{ m}$$

$$v_n = \frac{2.182 \times 10^6}{n} \times Z \text{ m s}^{-1}$$

$$= \frac{2.182 \times 10^6 \times 3}{5} = 1.31 \times 10^6 \text{ m s}^{-1}$$

The number of revolutions per second

$$= \frac{\text{velocity of electron}}{\text{circumference of orbit}}$$

$$= \frac{v_n}{2\pi r_n} = \frac{1.31\times10^6\ \text{m s}^{-1}}{2\times3.14\times4.408\times10^{-10}\ \text{m}} = 4.732\times10^{14}.$$

Concept Strand 34

Show that the number of waves made by a Bohr electron in one complete revolution in any orbit is the same as the principal quantum number of the electron.

Solution

$$mvr = \frac{nh}{2\pi} \text{—} (1) \qquad \lambda = \frac{h}{mv}$$

$$\text{or } mv = \frac{h}{\lambda} \text{—} (2)$$

$$(2) \text{ in } (1) \Rightarrow \frac{r}{\lambda} = \frac{n}{2\pi} \Rightarrow 2\pi r = n\lambda$$

$2\pi r$ is the circumference of the circular orbit. This indicates that the circumference of an orbit is equal to n times the wavelength of the electron. Hence the electron makes 'n' waves around the nth orbit.

Concept Strand 35

The circumference of the 4th Bohr orbit is 53 Å. Find the velocity of electron using de Broglie relation.

Solution

According to Bohr model, $mvr = \frac{nh}{2\pi} \Rightarrow 2\pi r = \frac{nh}{mv} = n\lambda.$

$$4\lambda = 2\pi r = 53\ \text{Å} = \frac{4h}{mv}$$

$$\Rightarrow \quad v = \frac{4h}{53\times10^{-10}\times m}$$

$$\Rightarrow \quad v = \frac{4\times6.6\times10^{-34}}{53\times10^{-10}\times9.1\times10^{-31}} = 5.47\times10^5\ \text{m s}^{-1}$$

Concept Strand 36

The radius of the nth orbit is given by the relation $r_n = 0.53 \times n^2$ Å. What is the wavelength of the electron in the 3rd shell?

(a) 4.77Å (b) 10Å
(c) 1.59Å (d) 0.53Å

Solution

(b)

$r_3 = 0.53 \times 3^2$

$2\pi r = 0.53 \times 3^2 2\pi$

In the third orbit there are three standing waves of electrons

$$\therefore \quad \lambda = \frac{0.53\times3^2\times2\pi}{3} = 9.99\text{Å} = 10\text{Å}$$

Defects of Bohr's theory of hydrogen spectrum

(i) No explanation for the fine structure of the spectrum was offered. Only the principal energy levels or principal quantum numbers were explained. Existence of additional quantum numbers was not accounted.
(ii) Multi-electron spectrum could not be explained on the basis of Bohr theory. It was applicable only to one electron system.
(iii) Movement of electrons around the nucleus occurs in three dimensional planes whereas Bohr theory limited the motion of electron in a single plane.
(iv) The Zeeman effect (splitting of spectral lines under the influence of the magnetic field) could not be explained.
(v) The Stark effect (the splitting of lines under the influence of electric field) could not be explained by Bohr's theory.

CONCEPT STRAND

Concept Strand 37

Bohr's theory can explain the spectra of which of the following, Li^{2+}, H^+, H^- and H_2, justify.

Solution

Only Li^{2+} is a one electron system.

PHOTOELECTRIC EFFECT

When light radiation of suitable wavelength shines on a metal surface, electrons are ejected from the metal surface. This is photoelectric effect and the ejected electrons are called photoelectrons. The minimum energy required for the ejection of electrons is called threshold energy ($h\upsilon_0$). When a metal is irradiated with a light of frequency υ, the kinetic energy of the photoelectrons is given by the relation,

KE = $h\upsilon - h\upsilon_0$, where, υ_0 is the threshold frequency

$\frac{1}{2}mv^2 = h(\upsilon - \upsilon_0)$. {Since, actually the KE of the emitted electron from a continuous range, the equation refers to the maximum KE}

$h\upsilon_0$ is also known as the work function (ϕ) of the metal, and is equal to the ionisation energy of the metal. Intensity of incident light $\propto$ number of photoelectrons. If the frequency of the incident light is equal to the threshold frequency, the electrons will be ejected without any kinetic energy. Frequency of incident light $\propto$ KE of photoelectrons.

CONCEPT STRANDS

Concept Strand 38

In a photoelectric experiment a metal was irradiated with two different frequencies υ_1 and υ_2 which are in the ratio 1 : 2. If the velocity v_2 of the photoelectrons is two times that of v_1 calculate the two frequencies. Threshold frequency of the metal is 3.0×10^{15} Hz.

Solution

$$h\upsilon = h\upsilon_0 + \frac{1}{2}mv^2 \Rightarrow h(\upsilon - \upsilon_0) = \frac{1}{2}mv^2$$

$$\frac{\upsilon_1 - \upsilon_0}{\upsilon_2 - \upsilon_0} = \left(\frac{v_1}{v_2}\right)^2 \Rightarrow \frac{\upsilon_1 - \upsilon_0}{2\upsilon_1 - \upsilon_0} = \left(\frac{v_1}{2v_1}\right)^2$$

$$\Rightarrow \quad \frac{\upsilon_1 - 3\times10^{15}}{2\upsilon_1 - 3\times10^{15}} = \frac{1}{4}$$

$$\Rightarrow \quad (\upsilon_1 - 3\times10^{15})\,4 = 2\upsilon_1 - 3\times10^{15}$$

$$\Rightarrow \quad 2\upsilon_1 = -3\times10^{15} + 4\times3\times10^{15}$$

$$\upsilon_1 = \frac{9\times10^{15}}{2} = 4.5\times10^{15}\ \text{Hz}$$

$$\upsilon_2 = 9\times10^{15}\ \text{Hz}$$

The photoelectric effect can be used to evaluate Planck's constant. We have the photoelectric equation, $\frac{1}{2}mv^2 = h\upsilon - h\upsilon_0$. A plot of kinetic energy against υ, the incident frequency, gives intercepts on x and y axes as $\frac{\phi}{h}$ and $-\phi$ where, ϕ is work function $h\upsilon_0$. From the two intercepts, we can get the value of Planck's constant.

Concept Strand 39

Calculate the maximum wavelength of the radiation that will cause photoelectric emission from potassium metal, if the kinetic energy of the photoelectron is 6.8×10^4 J mol^{-1} when an electromagnetic radiation of wavelength 500 nm falls on the metal surface.

Solution

Energy of the radiation, $E = h\frac{c}{\lambda}$

$$= \frac{6.626\times10^{-34}\times3\times10^{8}}{500\times10^{-9}} = 3.97\times10^{-19}\ \text{J}$$

The energy of 1 mole of photons = $3.97 \times 10^{-19} \times 6.022 \times 10^{23}$

= 2.39×10^5 J mol^{-1}

The minimum energy required for photo electric emission = $(2.39 - 0.68)\ 10^5$

= 1.71×10^5 J mol^{-1}

= 2.84×10^{-19} J $electron^{-1}$

Maximum wavelength of this radiation for the same,

$$\lambda = \frac{hc}{E} = \frac{6.626\times10^{-34}\times3\times10^{8}}{2.84\times10^{-19}} = 700\,\text{nm}$$

The radiation is red light.

WAVE–PARTICLE DUALITY

Bohr assumed that an electron is a particle of extremely small mass. It was established by de Broglie that an electron has a dual nature as a material particle and as a wave. An electron as a particle was assumed to revolve round the nucleus at a definite velocity. But in reality any exact measurement of the position of an electron which also behaved as a wave creates uncertainty because it is impossible to measure simultaneously and precisely the exact position and exact velocity of an electron. So Bohr's theory is contradictory to Heisenberg's uncertainty principle.

Sommerfeld's modification of Bohr's atomic model

The existence of close multiple spectral lines observed in the spectrum could not be explained by Bohr. While circular electronic orbits were considered by Bohr, the presence of elliptical orbits also was proposed by Sommerfeld. An ellipse has a major and minor axis. The angular momentum of an electron moving in an elliptical orbit is also quantized. Angular momentum can be an integral part of $\frac{h}{2\pi}$ units, or Angular momentum = k $(\frac{h}{2\pi})$ where, k is called the azimuthal quantum number. It can have values equal to 1, 2, 3…..n. The quantum number considered by Bohr was 'n' which is principal quantum number. The two quantum numbers can be related as

$$\frac{n}{k} = \frac{\text{length of major axis}}{\text{length of minor axis}}$$

Thus for any given value of n (except 1), the value of k can be more than one. If k and n are equal, then the orbit is circular. If k becomes smaller, the orbit assumes elliptical shape, and k cannot assume a value of zero because the minor axis in that instance would imply a linear motion. According to Sommerfeld, the energy of an electron depends both on the principal quantum number and to some extent on the azimuthal quantum number of an electron. This would seem to explain the fine structure of lines in the spectrum. Later calculations showed that the circular and elliptical orbits having the same value of 'n' have the same energy and thus Sommerfield's model became unable to explain the fine structure of H atom.

de-Broglie hypothesis

In contemplating the dual nature of light supported by Planck's expression (E = hυ), Einstein's mass-energy relationship ($E = mc^2$) and Compton's work on X-ray scattering, de Broglie had the novel idea that the dual nature of light should be considered in parallel with the dual nature of matter. According to him, a moving material particle of mass 'm' and velocity 'v' has wave properties associated with it and its wavelength is given by the expression,

$$\lambda = \frac{h}{mv}$$

where, λ denotes de Broglie's wave length and h is Planck's constant. However this value of λ should be too small as h is extremely small. To illustrate, a stone of mass 100 g, moving with a velocity of 100 cm sec^{-1}, should have a de Broglie wavelength of 6.6×10^{-31} cm which is too small to be detected and can be ignored. If the same calculation is extended to an electron, traveling with a velocity of 5.94×10^8 cm s^{-1} λ becomes,

$$\lambda = \frac{6.626 \times 10^{-27} \text{ erg sec}}{9.1 \times 10^{-28} \text{ g} \times 5.94 \times 10^{8} \text{ cm s}^{-1}}$$

$= 1.22 \times 10^{-8}$ cm = 1.22 Å.

Thus the wavelength in the case of an electron cannot be ignored. Thus the de-Broglie's hypothesis is applicable only for sub-atomic particles like electrons, protons, neutrons etc. It was found that electrons while passing through a crystal produced a diffraction pattern on a photographic plate similar to X-rays. Apart from electrons, beams of neutrons, and whole atoms like hydrogen and helium also produce diffraction patterns when scattered by crystals.

Since diffraction is an inherent property of waves, the above observation proved the wave property of electrons beyond doubt.

Distinction between photon and sub-atomic particles

Table 3.4

Photon	Sub-atomic particles
Energy = hυ	Energy = $\frac{1}{2}mv^2$
Wavelength = $\frac{c}{\upsilon}$	Wavelength = $\frac{h}{mv}$

Relationship between the accelerating potential and de Broglie wavelength

If P is the accelerating potential,

$$P \times e = \frac{1}{2}mv^2 \Rightarrow Pe = \frac{(mv)^2}{2m}$$

$$\Rightarrow \quad (mv)^2 = 2mPe$$

$$\lambda = \frac{h}{mv} = \frac{h}{\sqrt{2mPe}}$$

CONCEPT STRANDS

Concept Strand 40

How is the kinetic energy (KE) of an electron related to its wavelength (λ)?

Solution

$$K.E = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{\frac{2K.E}{m}}$$

$$\lambda = \frac{h}{mv} = \frac{h}{m\sqrt{\frac{2KE}{m}}} = \frac{h}{\sqrt{2mK.E}}$$

Concept Strand 41

Find the de Broglie wavelength of an electron accelerated through a potential of 1000 V.

Solution

$$\lambda = \frac{h}{\sqrt{2mPe}} = \frac{6.6\times10^{-34}}{\sqrt{2\times9.1\times10^{-31}\times1000\times1.6\times10^{-19}}}$$

$= 38.676 \times 10^{-12}$ m = 38.676 pm.

Concept Strand 42

Find the ratio of wavelengths of two particles A and B moving with the same velocity with masses in the ratio 1:3.

Solution

$$\lambda = \frac{h}{mv} \quad \therefore \lambda \propto \frac{1}{m} \text{ at constant velocity}$$

$$\therefore \quad \lambda_A : \lambda_B = 3:1$$

Concept Strand 43

A scientist needs to produce an electron beam and a proton beam of same wavelength for his experiment. If he has to accelerate these beams through a potential difference, compare the potential difference applied?

(Mass of electron = 9.1×10^{-31} kg, mass of proton = 1.67×10^{-27} kg)

Solution

$$PE = eV = \frac{1}{2}mv^2 \Rightarrow \frac{1}{2}\frac{(mv)^2}{m} = \frac{1}{2m}\frac{h^2}{\lambda^2}$$

since $\lambda = \frac{h}{mv}$

$$mv = \frac{h}{\lambda}$$

When, $\lambda_1 = \lambda_2$ for electron and proton,

$$\frac{V_1}{V_2} = \frac{m_2}{m_1} = \frac{1.67\times10^{-27}}{9.1\times10^{-31}} = 1835.$$

HEISENBERG'S UNCERTAINTY PRINCIPLE

This principle discusses the relationship between a pair of conjugate properties. According to this principle, *it is impossible to determine simultaneously the position and momentum of a moving particle with absolute certainty.* If the momentum (or velocity) be measured very accurately, a measurement of the position of the particle at the same

time becomes less precise. If the position is determined accurately, momentum becomes less precisely known or becomes uncertain. Thus determination of one property introduces uncertainty in the determination of another.

$$\Delta x \times \Delta p \geq \frac{h}{4\pi}$$

or $$\Delta x \times m\Delta v \geq \frac{h}{4\pi}$$

or $$\Delta x \times \Delta v \geq \frac{h}{4\pi m}$$

where, Δx is the uncertainty in the determination of position and (Δp) or ($m\Delta v$), is the uncertainty in determination of momentum. This is the expression of Heisenberg's uncertainty principle.

The uncertainty principle does not hold good in the case of heavy objects, like a ball or a big stone.

To illustrate, for a moving ball of 500 g weight, the uncertainty assumes the form

$$\Delta x \times m\Delta v \geq \frac{h}{4\pi}$$

$$\Rightarrow \quad \Delta x \times \Delta v \geq \frac{h}{4\pi m} \geq \frac{6.626\times10^{-27}}{4\times3.14\times500}$$

$= 1.05 \times 10^{-30}$ cm^2 s^{-1}

This is very small and can be ignored. Thus for very large objects, the equation is not quite significant.

Take the case of an electron, m = 9.1×10^{-28} g

$$\Delta x \times \Delta v \geq \frac{h}{4\pi m} \geq \frac{6.626\times10^{-27}\ \text{erg sec}}{4\times3.14\times9.1\times10^{-28}\ \text{g}}$$

≈ 0.6 cm^2 s^{-1}

The latter value for an electron is significant compared to the value for a stone or a ball.

The uncertainty principle is applicable only for small particles like electrons or neutrons and for larger particles, it does not have much of significance. It also explains the origin of lines within a spectrum. The probability concept towards the position of electron in an atom was introduced by the uncertainty principle.

CONCEPT STRANDS

Concept Strand 44

For a particle of mass m, the uncertainty in position was found to be zero. Find the uncertainty in its momentum.

Solution

$\Delta x.\Delta p = \frac{h}{4\pi}$. If $\Delta x = 0$, then $\Delta p = \infty$

Concept Strand 45

The velocity of an electron moving around an orbit is 2.2×10^6 m s^{-1}. If the momentum can be measured with an accuracy of 1%, then calculate the minimum uncertainty in its position.

Solution

Momentum = Mass × velocity = 9.1×10^{-31} kg × $2.2 \times 10^{+6}$ ms^{-1} = 2×10^{-24} kg m s^{-1}

$$\frac{\Delta p}{p} = .01$$

Δp (1% uncertainty) = 2×10^{-26}

$$\Delta x = \frac{h}{4\pi\Delta p} = \frac{6.6\times10^{-34}\ \text{Js}}{4\pi\times2\times10^{-26}\ \text{kg m s}^{-1}}$$

$= 2.63 \times 10^{-9}$

Concept Strand 46

Using Heisenberg's uncertainty principle show that electron cannot stay inside the nucleus.

Radius of nucleus = 5×10^{-15} m, Mass of electron = 9.1×10^{-31}kg.

Solution

Radius of nucleus = 5×10^{-15} m

$$\Delta x \times \Delta p = \frac{h}{4\pi}$$

$$\Delta v = \frac{h}{4\pi.\ m.\Delta x}$$

$$= \frac{6.626\times10^{-34}\ \text{Js}}{4\times3.14\times9.1\times10^{-31}\ \text{kg}\times5\times10^{-15}\ \text{m}}$$

$= 1.16 \times 10^{10}$ ms^{-1}

This speed is more than that of light, therefore it is impossible, to retain an electron in the nucleus.

Concept Strand 47

The uncertainty in the velocity of a moving particle is two times the uncertainty in its position. Calculate the maximum uncertainty in its momentum.

Solution

$$\Delta x.\Delta v = \frac{h}{4\pi m} \qquad \text{since } \Delta x = \frac{\Delta v}{2}$$

$$\frac{\Delta v}{2}.\Delta v = \frac{h}{4\pi m}$$

$$(\Delta v)^2 = \frac{2h}{4\pi m}$$

$$\Delta v = \sqrt{\frac{h}{2\pi m}}$$

$$\therefore\ m\Delta v = \Delta p = \sqrt{\frac{mh}{2\pi}}$$

Bohr's theory and Heisenberg's Uncertainty principle

According to Heisenberg's principle we cannot describe the exact path of an electron, due to the wave nature of the electron. Hence Bohr's theory according to which the electrons are having fixed positions in orbits of definite energy is not correct. In order to describe the position of an electron, we can only predict the probability or relative chance of finding or locating an electron with a probable velocity in a particular space or region around the nucleus. The Bohr orbits which are circular paths in which the electron revolve round the nucleus are replaced in the modern wave mechanical model by orbitals which are three dimensional space around the nucleus where there is a greater probability of finding an electron.

QUANTUM NUMBERS

The concept of orbit or energy level designated by principal quantum number 'n' proposed by Bohr could not adequately explain the hydrogen spectrum in all aspects, particularly the fine structure of lines and Zeeman effect.

Spectroscopes of high resolving power revealed the presence of closely spaced lines. It has become necessary to allow more possible energy changes of an electron within an atom or the number of possible energy states where an electron can be considered to exist.

Wave mechanics permits three more possible states in addition to the one already proposed by Bohr. These states identified by quantum numbers fully describe the position and energy of an electron. These are called quantum numbers and there are in all four such quantum numbers and each one describes one particular characteristic of an electron.

The principal quantum number 'n' is the most important factor that decides the energy of an electron. However, in the presence of a magnetic field, the energy of an electron depends on all four quantum numbers so that a given electronic energy is specified by a set of four quantum numbers (n, ℓ, m and s).The four quantum numbers which are used to describe an electron completely are

(i) Principal quantum number 'n'
(ii) Azimuthal quantum number 'ℓ'
(iii) Magnetic quantum number 'm' (or) m_ℓ
(iv) Spin quantum number 's' (or) m_s

Principal quantum number (n)

It is the serial number of the shells or main energy levels starting from the innermost shell and is denoted by n. It can have integral values with n = 1, 2, 3, ...∞. The shells with these n-values are designated as K, L, M, N, O, P etc. respectively. As the value of n increases the size of the shells and that of the orbitals in these shells and the energy of the electron in these orbitals increases.

The energy E_n of an electron in the nth orbit is given by

$$E_n = -\frac{2\pi^2 me^4}{n^2h^2} \text{ (in CGS units)}$$

The only variable in the equation is n and as the value of n increases the value of energy becomes less negative and when the value of n is close to infinity then E_∞ becomes zero, which means, it is a free electron.

Significance of n

(1) Principal quantum number is a measure of the size of the orbital.
(2) It is a rough measure of the energy of an electron in an orbital.
(3) The number of orbitals in a shell is given by n^2 and the maximum number of electrons in a shell is given by $2n^2$.

Table 3.5

n	Shell	Orbitals(n^2)	Max. no. of e^- s ($2n^2$)
1	K	1	2
2	L	4	8
3	M	9	18
4	N	16	32
5	O	25	50

Subsidiary or azimuthal or orbital angular momentum quantum number.

The main energy levels or shells in an atom are divided into subshells. The subshells are denoted by the subsidiary quantum number, ℓ. ℓ can have n values starting from zero upto (n − 1) in each shell. ℓ = 0, 1, 2, 3,....(n − 1). The subshells corresponding to ℓ = 0, 1, 2, 3, ...etc are denoted by small letters s, p, d, f, g, h etc.

Table 3.6

n	Shell	ℓ values	subshells	No. of subshells
1	K	0	1s	1
2	L	0, 1	2s, 2p	2
3	M	0, 1, 2	3s, 3p, 3d	3
4	N	0, 1, 2, 3	4s, 4p, 4d, 4f	4

Orbitals of a subshell of a particular ℓ - value have a particular shape. When, ℓ = 0 the subshell is s and the orbital is spherical. When ℓ = 1 the subshell is p and the orbitals are dumb-bell shaped. When, ℓ = 2 the subshell is d and the orbitals in it are double dumb-bell shaped and so on.

Significance of ℓ

(1) Prinicipal and subsidiary quantum numbers together give the exact energy of the electron in an orbital in a many electron atom.
(2) Subsidiary quantum number indicates the subshell of the electron.
(3) It defines the shapes of the orbital.
(4) It is a measure of the orbital angular momentum of the electron

Orbital angular momentum = $\sqrt{\ell(\ell+1)}\,\frac{h}{2\pi}$.

Magnetic quantum number (m or m_ℓ)

The subshells are further divided into orbitals. An orbital can contain a maximum of two electrons.

The different orbitals in a subshell are denoted by magnetic quanum number m (or m_ℓ). It can have values ranging form $-\ell$ through 0 to $+\ell$. For every ℓ – value there will be $(2\ell + 1)$ values of m. Each m-value indicates an orbital. Therefore, there are $(2\ell + 1)$ orbitals in a subshell with subsidiary quantum number ℓ.

Orbitals are named after the subshell to which they belong. Thus orbital in s-subshell is an s-orbital. Orbitals in p-subshell are called p-orbitals and so on.

Table 3.7

ℓ	m – values $-\ell$ ....0.....$+\ell$	No. of orbitals $(2\ell + 1)$
0	0	one s – orbital
1	−1, 0, 1	three p – orbitals
2	−2, −1, 0, 1, 2	five d – orbitals
3	−3, −2, −1, 0, 1, 2, 3	seven f – orbitals

All the orbitals in a subshell have the same energy. That is, they are said to be degenerate. But they have different orientations in space. In a strong magnetic field they acquire different energies depending on the orientation and the lines in the hydrogen spectrum split into a number of lines. This effect is called Zeeman effect. This effect can be explained on the basis of magnetic quantum number.

Significance of magnetic quantum number

(1) Magnetic quantum number denotes the orientation of orbitals of a subshell in three dimensional space.
(2) Magnetic quantum number indicates the orbitals in a subshell.

Spin quantum number (s or m_s)

The above three quantum numbers n, l and m are derived from Schroedinger wave equation. They are enough to define an orbital in an atom.

A fourth quantum number was necessary to explain the fine structure of atomic spectrum. This is the spin quantum number.

An electron is spinning (rotating) on its own axis at the same time as it is revolving around the nucleus. It may spin in the clockwise or in the anticlockwise direction indicated by the spin quantum number s (or m_s) which can have values $+\frac{1}{2}$ or $-\frac{1}{2}$ respectively. Thus for each value of m (i.e., in each orbital) there will be two values of s namely $+\frac{1}{2}$ and $-\frac{1}{2}$ and are indicated sometimes by upward and downward arrows. (↑ and ↓). The spin angular momentum corresponding to these two values of s are calculated using the formula.

Spin angular momentum = $\sqrt{s(s+1)}\frac{h}{2\pi}$

Significance of s

(1) s value gives the direction of spin of the electron.
(2) It gives the spin angular momentum of the electron.

Summary of quantum numbers

Table 3.8

n	ℓ	m	s	No. of electrons	
				In subshell	In a shell
1	0	0	$\pm\frac{1}{2}$	2	2
2	0	0	$\pm\frac{1}{2}$	2	
	1	−1	$\pm\frac{1}{2}$	6	8
		0	$\pm\frac{1}{2}$		
		+1	$\pm\frac{1}{2}$		
3	0	0	$\pm\frac{1}{2}$	2	18
	1	−1	$\pm\frac{1}{2}$	6	
		0	$\pm\frac{1}{2}$		
		+1	$\pm\frac{1}{2}$		
	2	−2	$\pm\frac{1}{2}$	10	
		−1	$\pm\frac{1}{2}$		
		0	$\pm\frac{1}{2}$		
		1	$\pm\frac{1}{2}$		
		2	$\pm\frac{1}{2}$		

CONCEPT STRAND

Concept Strand 48

The number of electrons with orbital angular momentum equal to zero in sodium atom is

(a) 1 (b) 3 (c) 5 (d) 11

Solution

(c) s-electrons have orbital angular momentum equal to zero. There are five's' electrons ($1s^2$, $2s^2$, $3s^1$).

QUANTUM MECHANICAL PICTURE OF HYDROGEN ATOM

de Broglie's concept of the dual nature of wave and particle, for an electron, followed by the uncertainty principle proposed by Heisenberg and the failure of Bohr's concept to adequately explain the exact structure of the atom, a new model of the atom based on wave mechanical concept was developed.

The wave mechanical picture of the atom proposed by Erwin Schroedinger in 1926 took into account three major factors. They are

(i) the wave nature of the electron.
(ii) uncertainty in the position of the electron in an atom.
(iii) the idea of fixed energy levels for electrons revolving round the nucleus, based on Bohr's theory.

Every object in motion has a wave character. Classical mechanics deals with objects only with particle behaviour and it fails with sub microscopic particles like electrons. The branch of quantum mechanics takes into account the dual behaviour of matter as a wave and particle.

According to the quantum mechanical model developed by Schroedinger, an atom is considered as a positively charged nucleus surrounded by standing or stationary electron waves which move round the nucleus in orbits. This is a three dimensional wave.

Schroedinger Wave equation

The wave motion of an electron wave propagating in three dimensions in space, along any of the three axes x, y and z can be described by the simplified term of a complex equation

$$\frac{\partial^2\psi}{\partial x^2}+\frac{\partial^2\psi}{\partial y^2}+\frac{\partial^2\psi}{\partial z^2}+\frac{8\pi^2 m}{h^2}(E-V)\psi = 0,$$

where h is Planck's constant, m, E and V are mass, total energy and potential energy of the electron respectively and (E – V) gives the kinetic energy of the electron, ψ is a mathematical function, which is called wave function. ψ represents the amplitude of the electron wave at various points surrounding the nucleus. Schroedinger equation is a partial differential equation of the second order. The same equation can also be written in the form

$$\nabla^2\Psi+\frac{8\pi^2 m}{h^2}(E-V)\Psi = 0, \text{ where } \nabla^2 = \frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}$$

∇^2 is known as the Laplacian operator. It is to be implied that $\nabla^2\psi$ is not a product of ∇^2 and ψ and it means that the operator ∇^2 operates on the mathematical function of ψ.

Hydrogen atom and Schroedinger Wave equation

When this equation is solved for hydrogen atom, the solution gives the possible energy states that the electron can occupy and the corresponding wave functions (ψ) or also called atomic orbitals. The wave function corresponding to that energy state contains all information about the electron. Some of the results are,

(i) The energy of electrons in an atom are quantized.
(ii) The various quantized or (step wise) electronic energy levels reflect the wave properties of electrons.
(iii) The precise path of an electron in any atom cannot be determined or known and therefore the term probability of finding an electrons at different points in an atom is introduced.
(iv) An atomic orbital is the wave function ψ for an electron in an atom.
(v) Whenever an electron is described by a wave function, an electron occupies a specific orbital. In the case of hydrogen it is a spherically symmetrical orbital for the ground state.
(vi) A large number of orbitals are possible in an atom. These orbitals can be distinguished by their size, shape and orientation.
(vii) The solution of Schroedinger equation gives the quantum numbers n, ℓ and m.

PROBABILITY DISTRIBUTION CURVES

The probability of finding an electron at various points in space can be obtained by two types of probability distribution curves. They are

(i) Radial probability distribution curves
(ii) Angular probability distribution curves

Radial Probability Distribution Curves

In the wave model of an atom, the discrete energy levels of electrons as developed by Bohr are replaced by mathematical functions ψ, which are related to the probability of finding an electron in a specific area around the nucleus.

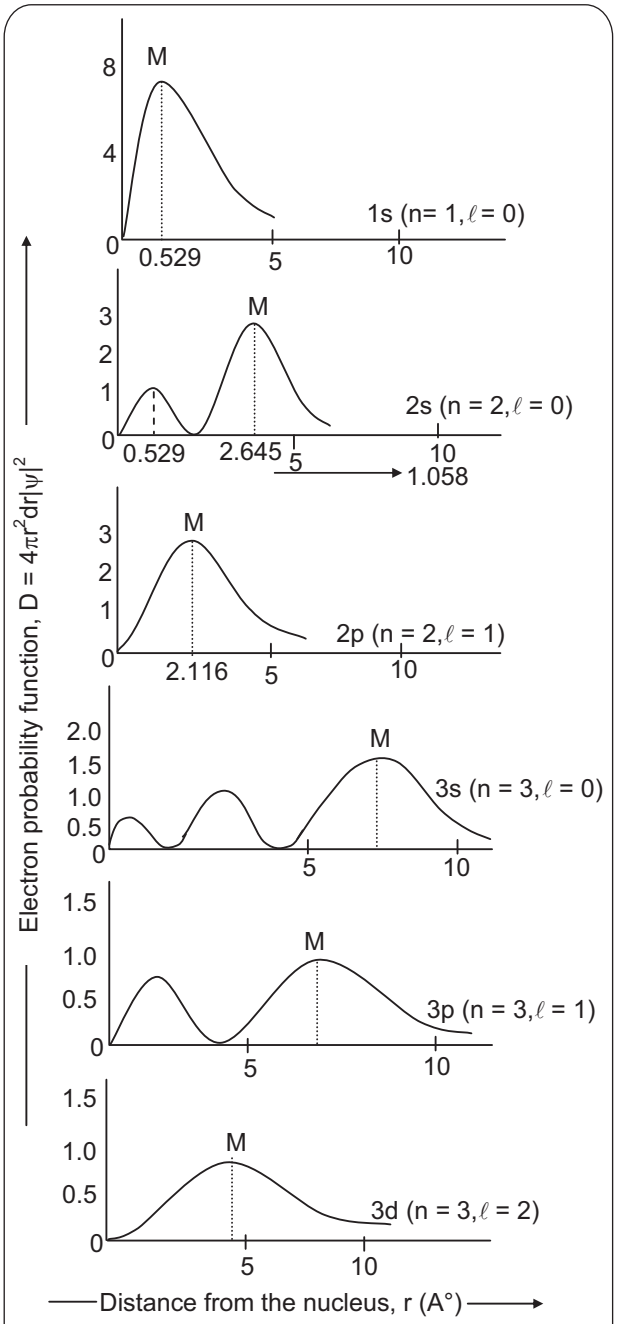
These curves are obtained by plotting the electron probability distribution function D (in the y-axis) against the distance 'r' (in the x-axis) from the nucleus. In a simplified form of the equation, D is given a value of $|\psi|^2 4\pi r^2 dr$

The probability of finding an electron at a distance 'r' from the nucleus is obtained from these curves for 1s, 2s, 2p, 3s, 3p and 3d orbital electrons.

Now the radial probability is the product of two terms namely $4\pi r^2 dr$ and $|\psi|^2$. At the nucleus, for s orbital $|\psi|^2$ is

large, but since r = 0, the radial probability is zero. As 'r' increases the corresponding value of $4\pi r^2$ increases but the remaining part of D decreases.

Therefore, we get a curve starting from zero, rising to a maximum and then decreasing again to zero (for 1s orbital).



Radial probability distribution curves for 1s, 2s, 2p, 3s, 3p and 3d orbitals. The point M in each curve represents the maximum electron density.

Fig. 3.3

It may be noted that the number of nodes i.e., the regions at which the radial probability is n – ℓ – 1 = 0, helps us to draw the curves.

For a hydrogen atom wave function, of principal quantum number, n there are

(i) n – ℓ –1 radial nodes
(ii) ℓ angular nodes
(iii) (n – 1)total nodes

Therefore for 1s (n = 1, ℓ = 0) orbital there is no node.
for 2s (n = 2, ℓ = 0) orbital, will have only one radial node.
for 2p (n = 2, ℓ = 1) orbital, there will be one angular node.
for 3s (n = 3, ℓ = 0) orbital, there will be two radial nodes.
for 3p(n = 3, ℓ = 1) orbital, there will be one radial and one angular node.
for 3d (n = 3, ℓ = 2) orbital, there will be two angular nodes.

The number of nodal planes will be equal to ℓ.

The following points are significant from the radial probability distribution curves.

(a) In every case, the probability of finding an electron at the nucleus or at the origin is zero. This leads to the conclusion that an electron is never found at the nucleus.
(b) The distance for the maximum probability of finding an electron in an orbital increases with the increase in the principal quantum number 'n'.
(c) The total number of peaks for the various curves for s, p & d orbitals is equal to n, (n-1) and (n-2) in the same order where, n is the principal quantum number.
(d) According to wave mechanical model, a 1s electron is most likely to be found at a distance of 0.53Å from the nucleus. This distance (0.53 Å) is obtained from the time-independent Schroedinger equation (by maximizing the probability)

The radial probability is the total probability of finding an *electron* in a radial shell between spheres of radii 'r' and 'r+dr' where, dr is an infinitely small distance.

The point at which the probability of finding an electron is zero is called a nodal point.

Angular Probability Distribution Curves

These types of curves give us the probability of finding the electrons varying with the direction from the nucleus without any reference to the distance from the nucleus.

The angular function is a function of two angle variables.

The complete wave function represents an orbital in which the radial part depends on quantum numbers n and ℓ and give us an idea of the size of the orbital whereas, the angular part depends on the quantum numbers ℓ and m and provides information about the shape of the orbital.

ORBITALS

There are regions where the chances of finding the electron are relatively greater. The specific location of the electrons in space is not known but such regions are expressed in terms of electron charge cloud or charge density regions or high probability regions.

The three dimensional region where there is a high probability of finding an electron of a certain energy is called an orbital. Orbital is the spatial description of the movement of an electron corresponding to a particular energy level.

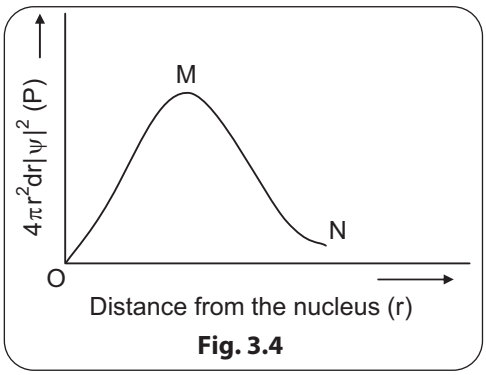


Fig. 3.4

The earlier part of the curve OM essentially signifies the r^2 dependence of P. From M to N and onwards it reflects the exponential (decaying) part.

On the basis of the pictorial description, the probability of finding an electron increases gradually to a maximum value at 0.53 Å and then drops (Fig. 3.5).

The Fig. 3.6 indicates the cross sectional view of the electron cloud. The circle represents the boundary which encompasses 90% of the electron density belonging to that orbital.

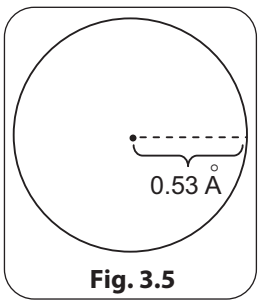


Fig. 3.5

The quantum numbers n, ℓ and m obtained by solving this equation for the energy of an electron in a hydrogen atom, are related as follows.

(i) $(n + \ell)$ and $(2\ell + 1)$ are whole numbers.
(ii) equal to $(n - 1)$ or less than $(n - 1)$
(iii) m has values from $-\ell$ to $+\ell$.

Spin quantum number does not appear in the solution of a wave equation.

Difference between orbits and orbitals

An orbit is a circular path at a definite distance from the nucleus. According to Bohr's theory an electron of hydrogen is found at a distance of 0.53Å from the nucleus. Orbits are circular or elliptical in shape according to Sommerfield.

An orbital is postulated on the wave concept of electron. It is a three dimensional region where there is a non zero probability of finding an electron. An orbital cannot accommodate more than two electrons. These two electrons must have opposite spins. Orbitals have different shapes.

The complete wave function of a one electron atom or hydrogen like atom is equal to the product of the radial wave function and the angular wave functions.

Eigen values and Eigen functions

The Schroedinger wave equation is a second degree partial differential equation. If the potential energy term (V) is known, the total energy E and the wave function corresponding to that can be arrived at. The wave function is zero at infinite distance from the nucleus, but it is finite, and continuous and single valued at other instances. Certain characteristic values given for the E to meet the requirements are called Eigen values. Corresponding to the values of E, we have several characteristic forms for the wave function ψ and they are called Eigen functions. In Schroedinger wave equation φ represents the amplitude of the spherical wave. The square of the amplitude, $|\psi|^2$, is proportional to the density of the wave. The probability distribution is maximum at a distance of 0.53 Å for hydrogen and is spherically symmetrical. This distance corresponds to Bohr's first radius when the electron gets excited and goes to higher

energy levels like n = 2 or 3, the solution of wave equation gives a set of values of $|\psi|^2$ which gives different shapes to the spatial distribution of the electrons.

Shapes of s, p, d and f orbitals

For a given value of ℓ, there can be a total of $(2\ell + 1)$ values for m. Different values of m give the total number of different ways in which orbitals of s, p, d and f sub shells can be oriented in space. The spatial orientations of the s, p, d and f orbitals are thus determined by the value of 'm'.

s-orbital

For an s-orbital, ℓ should be 0. If $\ell = 0$, m can only be 0: There is no special orientation for the s-orbital. The shape of s-orbital is spherically symmetrical and is usually represented by a circle or a cut sphere.

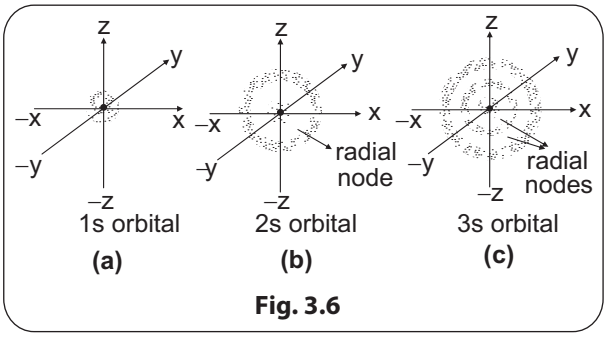


Fig. 3.6

An s-orbital does not have directional characteristics. As the values of the principal quantum number increases from n = 1 to higher values like n = 2, 3, 4, 5 etc, the size of the s-orbital increases. An electron in a 2s orbital can go farther away from the nucleus and has greater energy than an electron in a 1s orbital where the electron is nearer to the nucleus and has low energy. Similarly, 3s orbital is larger in size and a 3s electron has more energy than a 1s or 2s electron. The number of spherical nodal surfaces in the 1s-orbital where the probability of finding an electron is zero. For any given main energy level with the principal quantum number 'n' and azimuthal quantum number 'ℓ', the number of spherical or radial nodes is equal to $(n - \ell - 1)$.

p-orbital

For a p-orbital, the sub shell value is $\ell = 1$. Now when $\ell = 1$, m can take the values –1, 0 and +1 (three values). Three values of m imply that the real part of p-orbitals have three spatial orientations along the x, y and z axis. The three orientations of the p-orbitals are designated as p_x, p_y and p_z (m = 0). The three p orbitals, since the three axes are perpendicular to each other, have orientations perpendicular to each other.

All the three p orbitals have directional characteristics and are dumb bell shaped. Each p-orbital, has two egg shaped lobes and are disposed symmetrically along the orbital axis. The two lobes of the p orbitals extend outwards and away from the nucleus along the axis.

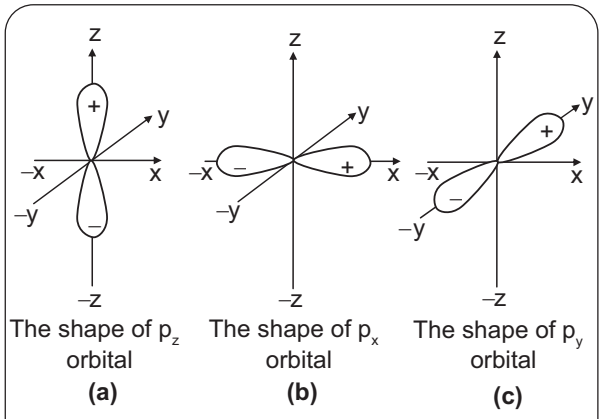


Fig. 3.7

The p–orbitals of all the energy levels with n = 2, 3, 4 etc have similar shapes, but with the increase in the value of n, the size and energy of the p-orbitals also increase. The two lobes of a p-orbital can be divided by a plane. This plane known as nodal plane contains the nucleus and it is perpendicular to the axis of the p-orbital. The probability of finding an electron at the nucleus and the nodal plane is zero.

For each one of the three p orbitals, there is one nodal plane, for example the two lobes of the p_x, orbital are separated by the yz plane in which the wave function has zero value. i.e., the nodal planes of p_x, p_y and p_z orbitals are in the yz, xz and xy planes respectively. Also, a 2p orbital has no radial node, 3p has one and 4p has two radial nodes as given by the formula,

No. of radial nodes = $n - \ell - 1$

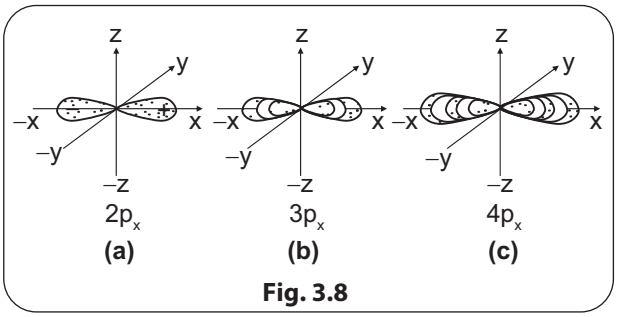


Fig. 3.8

All the three p-orbitals are of equal energy, in the absence of magnetic and electrical fields. Orbitals having the same energy are said to be degenerate. In the absence of the external fields these three p-orbitals are said to be triply degenerate. The signs of the angular wave function changes as the nodal plane is crossed and therefore the two lobes of the p-orbitals are of opposite signs.

The three dimensional representation of the p-orbitals will be better obtained by rotating the two dimensional form about the appropriate axis like x, y or z.

d-orbitals

For the d-orbitals, $\ell = 2$ and m can take values −2, −1, 0, +1 and +2. The five orbitals are named as d_{xy}, d_{yz}, d_{xz}, $d_{x^2-y^2}$ and d_{z^2}. These are shown in the figure. The d_{xy}, d_{yz}, d_{xz} and $d_{x^2-y^2}$ orbitals have four lobes each.

In the absence of the magnetic and electrical fields they are all equivalent in energy and are said to be five fold degenerate. The shapes and features of these d-orbitals can be described as follows.

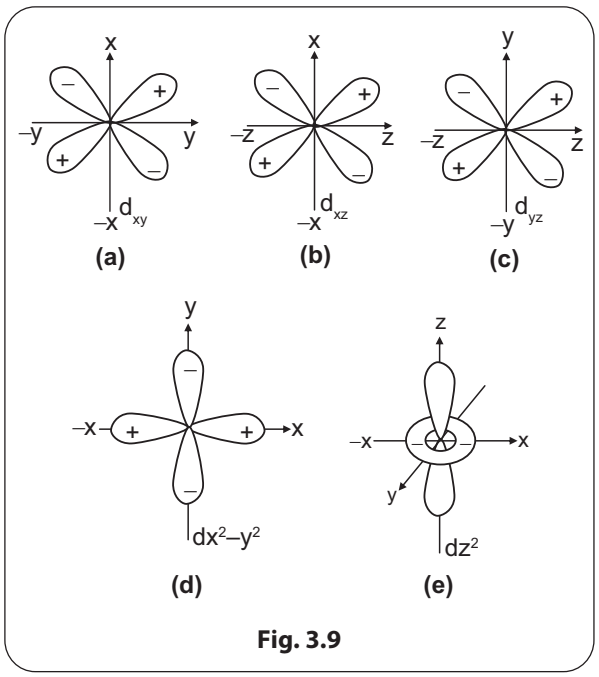


Fig. 3.9

The set of three orbitals, viz., d_{xy}, d_{yz} and d_{xz} have their lobes lying symmetrically between the coordinate axes. For example, the lobes of d_{xy} orbital are lying between x and y-axes. The $d_{x^2-y^2}$ and d_{z^2} orbitals have their lobes along the axes. For example the lobes of $d_{x^2-y^2}$ orbital are along the x and y axes and those of d_{z^2} are along the z-axis. Besides, the d_{z^2} orbital has an annular ring in the xy plane.

Each of the d_{xy}, d_{yz} and d_{xz} have double dumb bell shape. In the case of d_{xy}, the lobes of xy component will be positive when both x and y are positive and xy will be negative when both x and y are negative. If one breaks the orbital shapes in the form of four quadrants, the sign will be either positive or negative in the opposite quadrants. Similarly, the other two d_{yz} and d_{xz} have the same description for the shape and signs.

The $d_{x^2-y^2}$ orbital is also double dumb bell shaped and their lobes lie along the axes 'x' and 'y'. The sign of the lobes along the x-axis will be positive by convention and those on the lobes along y-axis will be negative. A nodal plane or sphere separates the positive and negative lobes.

The d_{z^2} orbital has dumb bell shaped lobes symmetric about the z-axis. In addition, there is a ring like collar along the xy plane. The lobes have positive sign where as the collar around has a negative sign.

A 5d orbital has two radial node while a 4d orbital has one and a 3d orbital has no radial nodes.

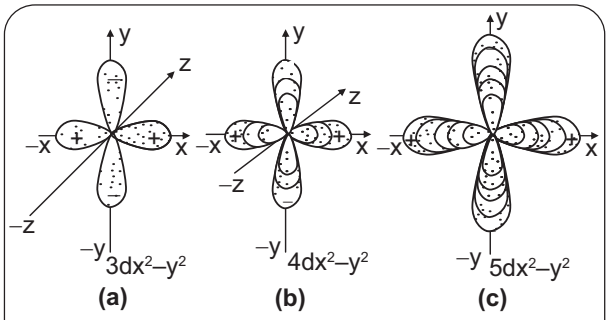


Fig. 3.10

The d-orbitals are, in general, more complicated in shape than the p-orbitals. With the exception of d_{z^2} orbital, all the d-orbitals have four lobes alternating in sign. While p-orbitals have one nodal plane, the d-orbitals have two nodal planes. The energy of an orbital increases as the number of nodal planes increases.

Another important feature is that all the five d orbitals, though degenerate in the absence of any external field are not mutually equivalent. This is in contrast to the p-orbitals which are equivalent.

The d-orbitals can thus be divided into two different types. The $d_{x^2-y^2}$ and d_{z^2} belong to one type and these orbitals lie along the axial directions, while the other type namely d_{xy}, d_{yz} and d_{xz} point along 45° angle to the axial directions. Because of this difference in orientation, the energies of the two types of orbitals are differently affected by the applied electric fields.

CONCEPT STRANDS

Concept Strand 49

The electron of Orbital in the 4th energy level goes nearest to nucleus?

Solution

s electron cloud starts from the nucleus. Hence 4s electron goes nearest to nucleus.

Concept Strand 50

How many radial nodes and angular nodes are there in $3d_{z^2}$ orbital?

Solution

Radial nodes = 3 – 2 – 1 = 0
Angular nodes = $\ell = 2$

Concept Strand 51

From the given set of orbitals identify those for which the orbital angular momentum of an electron would be zero.

(i) 2s and 3s (ii) 2p and 3p
(iii) 3p and 4s (iv) 3d and 4d

Solution

2s and 3s
The value of ℓ for s orbital is 0

Hence $\sqrt{\ell(\ell+1)}\frac{h}{2\pi} = 0$

Concept Strand 52

Which of the following sets of quantum numbers are allowed?

(i) n = 2; ℓ = 2; m = –1 (ii) n = 2; ℓ = –1; m = –1
(iii) n = 2; ℓ = 3; m = –2 (iv) n = 2; ℓ = 1; m = 0

Solution

(i) is not allowed because ℓ cannot have a value of 2 for n = 2.
(ii) is not allowed because ℓ cannot have negative values.
(iii) is not allowed because ℓ cannot exceed n.

Concept Strand 53

Which of the following set of quantum numbers is allowed for '5d' electron?

(i) n = 5, ℓ = 5, m = 0, s = $-\frac{1}{2}$

(ii) n = 5, ℓ = 2, m = 0, s = $+\frac{1}{2}$

(iii) n = 5, ℓ = 2, m = –5 , s = $-\frac{1}{2}$

(iv) n = 5, ℓ = 3, m = 4, s = $+\frac{1}{2}$

Solution

(i) is not allowed because ℓ cannot be equal to n.
(iii) is not allowed because m cannot be numerically greater than ℓ.
(iv) is not allowed because m can be from – 3 to +3 only.

Concept Strand 54

An electron has an angular momentum of $\frac{\sqrt{3}h}{\pi}$. Which orbital does it belong to?

Solution

Angular momentum = $\sqrt{\ell(\ell+1)}\frac{h}{2\pi} = \sqrt{3}\frac{h}{\pi}$

$= \sqrt{12}\frac{h}{2\pi}$

$\therefore \quad \ell(\ell + 1) = 12$

$\ell = 3$ i.e., f-orbital

Concept Strand 55

For an electron, n = 3 and m = -2. Which among the following statements is not applicable for this electron?

(i) the electron is in a p-orbital
(ii) the electron is in a d-orbital
(iii) the electron is in the third orbit
(iv) the spin quantum number of the electron is $\frac{1}{2}$

Solution

The electron is in a d-orbital.
When, m = –2, ℓ = 2 (d-orbital).

Concept Strand 56

Given below are statements pertaining to atomic orbitals. Identify the one which is not true and how it can be corrected.

(i) describes three dimensional probability distribution of electron density
(ii) have directional characteristics
(iiI) have different shapes
(iv) maximum electron accommodating capacity is two

Solution

Have directional characteristics. (All orbitals except 's' have directional characteristics).

Concept Strand 57

How will you distinguish between s, p and d orbitals, based on the electron distribution?

Solution

Electron distribution in the p-orbital is antisymmetrical whereas in s and d-orbitals it is symmetrical.

Concept Strand 58

Find the number of orbitals present in the level like lithium ion Li^{2+} having the energy of –30.6 eV in a hydrogen.

Solution

$$E_n = \frac{-13.6Z^2}{n^2}eV$$

$$n^2 = \frac{13.6 \times 9eV}{30.6eV} = 4$$

n = 2
ℓ = 0
m = 0
ℓ = 1,
m = –1, 0, +1
$\therefore$ Number of orbitals is 4

ELECTRONIC CONFIGURATION OF ELEMENTS

The distribution of electrons in various shells, subshells and orbitals is called electronic configuration of an atom.

Distribution of electrons in various orbitals

The filling of electrons in the ground state of the orbitals of atoms is decided by the following rules.

(i) Aufbau principle
(ii) Hund's rule of maximum multiplicity
(iii) Pauli's exclusion principle

Aufbau Principle

In the ground state of an atom, electrons occupy the available orbitals in the order of increasing energy, the orbitals of lower energy being filled up first. This building up principle (Aufbau is a German word meaning building up) is known as aufbau principle.

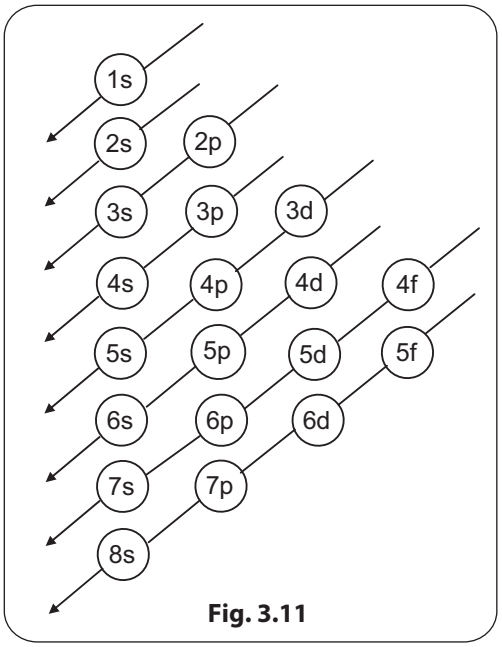


Fig. 3.11

The increasing order of energy of orbitals starting with the lowest energy at the 1s level will be 1s < 2s < 2p < 3s < 3p < 4s < 3d < 4p < 5s < 4d < 5p < 6s < 4f < 5d < 6p < 7s < 5f < 6d < 7p < 8s and so on.

This order of increasing energy is given by a simple step wise ladder.

The orbitals of lowest energy are better seats for electrons and therefore, they are occupied first. Electrons occupy orbitals of minimum energy first and progressively they occupy orbitals of higher energy.

(n + ℓ) rule

The orbital having the lowest value of (n + ℓ) has the lowest energy and hence it is filled up first with electrons and then in increasing order.

When two or more orbitals have the same value of (n + ℓ), the orbital with the lower value of 'n' is lower in energy and therefore is filled up first.

A few deviations or limitations of aufbau principle are observed in the case of configuration of lanthanum (atomic no.57), actinium (atomic no.89) and thorium (atomic no.90).

CONCEPT STRAND

Concept Strand 59

4 orbitals are considered at random. If we have to fill one electron each into these orbitals, what would be the order of priority? The quantum numbers of the orbitals are

(i) n = 5 ℓ = 0
(ii) n = 3 ℓ = 2
(iii) n = 4 ℓ = 2 and
(iv) n = 2 ℓ = 1

Solution

(n + ℓ) for (i) = 5 (n = 5), (ii) = 5(n = 3), (iii) = 6 and (iv) = 3
The order of priority for filling electrons is

(iv) > (ii) > (i) > (iii)

Hund's rule of maximum multiplicity

Hund's rule states that when electrons are added to a set of degenerate orbitals of a subshell, they are added singly with parallel spin and when all the orbitals are half filled, pairing will take place with opposite spin.

According to Hund's rule, electrons are distributed among the orbitals of a sub shell in such a way as to give the maximum number of unpaired electrons.

In other words, pairing of the electrons takes place only when all the orbitals of the same energy are half filled. For example, in nitrogen, the electronic configuration is $1s^2 2s^2 2p^3$. The three p electrons occupy three separate orbitals $2p_x$, $2p_y$ and $2p_z$, all the three electrons remain unpaired and have parallel spins. This exactly half filled p orbitals of nitrogen imparts extra stability for the nitrogen atom. In the case of carbon, the electronic configuration is $1s^2 2s^2 2p^2$. There are three p orbitals.

When we draw box diagrams, it can be illustrated.



$1s^2$ $2s^2$ $2p^2$

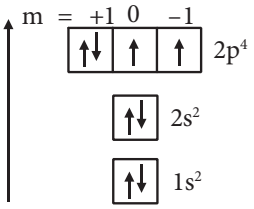
The two electrons in the 2p orbital remain unpaired and one of the three orbitals at the 2p level remains vacant in the ground state of carbon.

In the degenerate orbitals i.e., orbitals with the same energy, if two electrons with opposite spins are placed in the same orbital, the electrostatic repulsion would be greater than when they are placed in separate orbitals. For least repulsion, electrons in degenerate orbitals should be placed separately or singly until it becomes necessary to place two of them in the same orbital. This happens when the number of electrons exceeds the number of degenerate orbitals.

By Hund's rule, there will be a maximum number of unpaired electrons. By entering the orbitals in this manner minimizes the inter electron repulsion energy and a more stable system is obtained.

Electronic configuration of oxygen is another illustration of Hund's rule.

Oxygen has 8 electrons with the configuration $1s^2 2s^2 2p^4$. They are arranged as shown.



There are four electrons in the 2p level and pairing of electrons commences only when all the three p orbitals are singly occupied first. This results in two single unpaired electrons in an oxygen atom, in the ground state.

The concept known as 'multiplicity' is derived from the number of lines in the spectrum.

The term 'multiplicity' is related to the number of unpaired electrons and is expressed as 2S + 1. If S = 0, then the value is 1 and it is called a singlet. If S = $\frac{1}{2}$, the value is 2, the multiplicity is 2 and the state is a doublet. If S = 1, then it is a triplet state.

According to Hund's rule of maximum multiplicity, in the ground state of an atom, there will be the greatest multiplicity. In the case of carbon, the two 2p electrons may be paired in which case its S = 0, or the two electrons may be unpaired in which they have the same (parallel) spins, when they are in two different orbitals in which case S = 1. Hund's rule predicts that it is a triplet state.

Hund's rule has been explained by assuming that there is less repulsion between electrons in the high spin state, thereby stabilizing it. It is already established that electrons having same spin are highly correlated and actually repel each other much more than the electrons having anti - parallel spins.

Since electrons of parallel spins avoid each other, they shield or protect each other from the nucleus much less and as a result the attraction between nucleus and electron is greater and it predominates. The net result is that overall energy is lowered.

Pauli's Exclusion Principle

This rule states that in any particular atom no two electrons can have all the four quantum numbers identical.

Obeying Pauli's exclusion principle, limits the maximum number of electrons in each orbital to two. At the most only two electrons of opposite spins can occupy the same or identical energy state.

In accordance with this principle, if two electrons in an atom have identical values of the three quantum numbers namely n, ℓ and m, it necessarily follows that these two electrons differ in their spins. Their spins are anti-parallel, represented as $\uparrow\downarrow$ or spin quantum number of one electron is $+\frac{1}{2}$ while that of the other is $-\frac{1}{2}$.

The net result of Pauli's exclusion principle is that it throws considerable light on the electron build up in an atom and the stability of an atom. This principle is of great value in fixing the maximum number of electrons to be accommodated in any shell or orbit.

Another way of statement of this rule is that no two electrons residing in the same orbital of an atom can be described by an identical set of four quantum numbers.

Illustration of Pauli's exclusion principle

If we consider helium (atomic no. 2), the element has 2 electrons in 1s orbital. We know as per the electronic configuration, n = 1 and ℓ = 0 and m = 0.

The exact quantum numbers for these two electrons of helium are described in the table.

Table 3.9

Electron	Quantum numbers			
	n	**ℓ**	**m**	**s**
1st	1	0	0	$+\frac{1}{2}$ ($\uparrow$)
2nd	1	0	0	$-\frac{1}{2}$ ($\downarrow$)

A close look at the quantum numbers of the two electrons of helium will reveal that out of the four quantum numbers, principal, azimuthal, magnetic and spin, the first three are identical for both electrons, since the two electrons are in the same orbital.

But the two electrons differ only in their spin ($+\frac{1}{2}$ for one and $-\frac{1}{2}$ for the other) or spin quantum number (s).

The possible orbitals

On the basis of the three quantum numbers, n, ℓ and m, a particular value for these numbers result in the orbitals as shown

Table 3.10

n = 1	ℓ = 0	m = 0	1s orbital
n = 2	ℓ = 0	m = 0	2s orbital
n = 2	ℓ = 1	m = −1, 1, 0	$2p_x$, $2p_y$, $2p_z$ orbitals
n = 3	ℓ = 0	m = 0	3s orbital

n = 3	$\ell = 1$	m = −1, 1, 0	$3p_x, 3p_y, 3p_z$ orbitals
n = 3	$\ell = 2$	m = 0, −2, −1, +1, +2	$3d_{z^2}$, $3d_{x^2-y^2}$, $3d_{xy}$, $3d_{yz}$, $3d_{xz}$ orbitals
n = 4	$\ell = 0$	m = 0	4s orbital

The relationship between quantum numbers, electron distribution and the three rules governing electronic configuration can be summarized in the following set of conclusions.

(i) The type of orbital is determined by the quantum number 'ℓ'.

$\ell = 0$'s' orbitals
$\ell = 1$ 'p' orbitals
$\ell = 2$ 'd' orbitals
$\ell = 3$ 'f' orbitals

(ii) There are $(2\ell + 1)$ orbitals of each type, that is one s, three p, five d, seven f, per set.

(iii) There are 'n' type of orbitals in the nth energy level. For example, when n = 3, this energy level has three different types of orbitals namely s, p and d types.

(iv) There are $n - \ell - 1$ nodes in the radial distribution function of all orbitals. For example, 3s orbital has two nodes while the 4d orbitals have one each. These nodes are called radial nodes.

(v) There are ℓ nodal plane surfaces in the angular distribution functions of all orbitals, for example, s orbitals have none, d orbitals have two. These nodes are called angular nodes.

(vi) Each orbital can contain two electrons, corresponding to the two permitted values of spin $+\frac{1}{2}$ and $-\frac{1}{2}$.

(vii) Atoms having only paired electrons (S = 0) are repelled in a magnetic field and they are known as diamagnetic. Atoms having one or more unpaired electrons ($S \neq 0$) are strongly attracted by a magnetic field and they are called paramagnetic.

Salient features of the electronic configuration of elements

The inert gases possess the completely filled orbitals and hence they have the most stable electronic configuration. When electrons are filled in orbits on the basis of Bohr-Bury scheme, the maximum number of electrons each orbit can hold as given by $2n^2$ formula will be 2 for the first orbit, 8 for the second, 18 for the third and 32 for the fourth and so on.

The maximum numbers of electrons in the outer most orbit of any element (irrespective of its atomic number and its position in the periodic table) will be 8 and that in the penultimate orbit (last but one), will be18 electrons. It is not necessary for an inner orbit to be completed before the outer orbit starts getting filled up.

The outer most orbit cannot have more than 2 electrons and the penultimate orbit cannot have more than 8 electrons as long as the next inner orbit, in each case, has not received the maximum electrons.

For example, element, gallium (atomic no. 31) has an electronic configuration of $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^1$. The outer most orbital has 3 electrons (more than 2) and in this case both the inner orbits are completely filled up and they have the maximum number of electrons.

Another example is potassium (atomic no.19), the outer most orbital is not completely filled. The penultimate orbit also is incomplete with 8 electrons ($3d^0$) where it can actually hold 18 electrons including the 3d electrons.

There are a few elements with a slight variation in the filling of electrons from the patterns according to rules. This is for the extra stability of the element on account of electronic configuration.

A few examples are given below

(i) Chromium
The atomic number is 24. The expected electronic configuration is $1s^2 2s^2 2p^6 3s^2 3p^6 3d^4 4s^2$. But the actual configuration observed is $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^1$. This is due to the fact that by attaining a half filled $3d^5$ configuration, chromium, attains extra stability.

(ii) Copper
The atomic number is 29. Expected configuration is $1s^2 2s^2 2p^6 3s^2 3p^6 3d^9 4s^2$, but the actual configuration $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^1$, which results in a completely filled 3d subshell with 10 electrons providing greater stability for Cu.

Half filled or completely filled orbitals

When orbitals get exactly half filled or completely filled up, such orbitals are tend to be more stable compared to other orbitals (partially filled) and these half filled and fully filled orbits give a more stable and energetically favourable configuration to the atom.

On the basis of the rules followed, the electronic configurations of elements with atomic numbers 1 to 36 are given in the form of the following table.

Table 3.11

No	Period in the Periodic Table	Symbol of the element	Atomic Number	Electronic Configuration			
				K Shell 1 s	L Shell 2s 2p	M Shell 3s 3p 3d	N Shell 4s 4p 4d 4f
1	1	H	1	1			
2	1	He	2	2			
3	2	Li	3	2	1		
4	2	Be	4	2	2		
5	2	B	5	2	2, 1		
6	2	C	6	2	2, 2		
7	2	N	7	2	2, 3		
8	2	O	8	2	2, 4		
9	2	F	9	2	2, 5		
10	2	Ne	10	2	2, 6		
11	3	Na	11	2	2, 6	1	
12	3	Mg	12	2	2, 6	2	
13	3	Al	13	2	2, 6	2, 1	
14	3	Si	14	2	2, 6	2, 2	
15	3	P	15	2	2, 6	2, 3	
16	3	S	16	2	2, 6	2, 4	
17	3	Cl	17	2	2, 6	2, 5	
18	3	Ar	18	2	2, 6	2, 6	
19	4	K	19	2	2, 6	2, 6	1
20	4	Ca	20	2	2, 6	2, 6	2
21	4	Sc	21	2	2, 6	2, 6, 1	2
22	4	Ti	22	2	2, 6	2, 6, 2	2
23	4	V	23	2	2, 6	2, 6, 3	2
24	4	Cr	24	2	2, 6	2, 6, 5	1
25	4	Mn	25	2	2, 6	2, 6, 5	2
26	4	Fe	26	2	2, 6	2, 6, 6	2
27	4	Co	27	2	2, 6	2, 6, 7	2
28	4	Ni	28	2	2, 6	2, 6, 8	2
29	4	Cu	29	2	2, 6	2, 6, 10	1
30	4	Zn	30	2	2, 6	2, 6, 10	2
31	4	Ga	31	2	2, 6	2, 6, 10	2, 1
32	4	Ge	32	2	2, 6	2, 6, 10	2, 2
33	4	As	33	2	2, 6	2, 6, 10	2, 3
34	4	Se	34	2	2, 6	2, 6, 10	2, 4
35	4	Br	35	2	2, 6	2, 6, 10	2, 5
36	4	Kr	36	2	2, 6	2, 6, 10	2, 6

Extra stability of half-filled and completely filled orbitals

For a p_x^1 configuration, the electronic charge is concentrated in the x-direction. For a p_x^1, p_y^1 configuration, the electronic charge is concentrated in the xy plane. For a p_x^2, p_y^2, p_z^1 configuration, the electronic charge is more concentrated along xy plane.

But for p_x^1, p_y^1, p_z^1 and p_x^2, p_y^2, p_z^2 configurations, electronic charges are uniformly distributed in space or the distribution of electronic charge is symmetrical. Such symmetrical systems are more stable than unsymmetrical ones.

Similarly, for d^5 and d^{10} configurations also have symmetrical distribution of electronic charges in space and hence they are also more stable compared to other partially filled d orbitals. Thus half filled and completely filled orbitals are more stable than partially filled orbitals.

Exchange and pairing energies

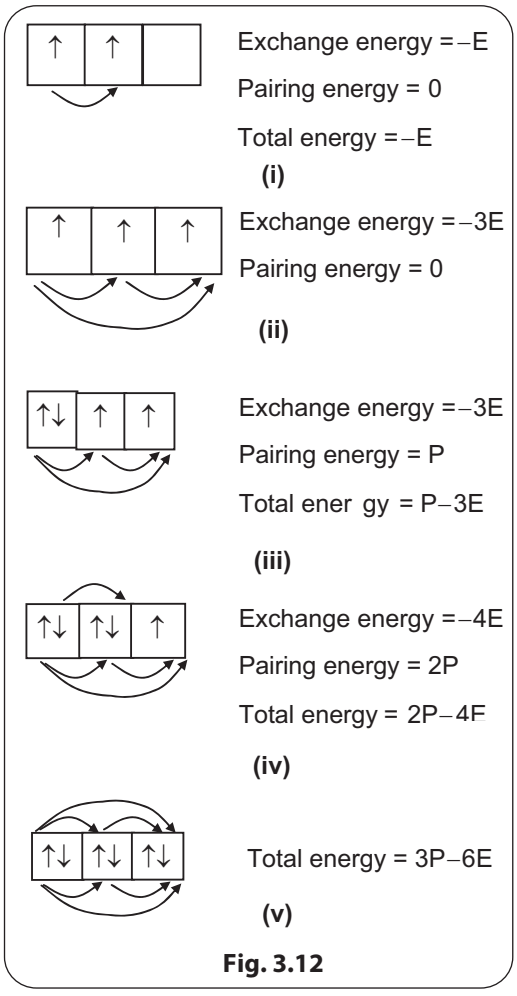


Fig. 3.12

On exchanging the position of two electrons in space with parallel spins, there is no change in the electronic arrangement, but it would lead to decrease in energy.

The energy decrease per exchange pair of electrons is known as exchange energy.

Placing of two electrons in the same orbital results in repulsion and *the energy required for placing two electrons together in the same orbital is known as pairing energy.*

Thus pairing energy destabilizes the system where as exchange energy stabilizes.

The overall stability of a system depends on the total of exchange energy (E) and pairing energy (P).

The stability of the electronic arrangements given in the illustrations (i) – (v) cannot be explained only in terms of the total energy.

Rather it is explained by the symmetrical distribution of electronic charge and hence the electronic arrangements given in the illustrations (ii) and (v) are the most stable among (i) – (v).

Electronic configuration of transition elements (3d elements)

The ten elements starting from Scandium (Atomic number.21) to Zinc (Atomic number.30) have a different electronic arrangement. They are known as 3d elements or transition elements as the filling of electrons occur in the inner 3d level. On account of the vacant or partially filled 3d orbitals, these transition elements exhibit some characteristic properties like colour of their salts, formation of coordination compounds, variable oxidation states, magnetic nature and so on.

Electronic configuration of lanthanides and actinides (4f and 5f elements)

In the series of elements, lanthanides with atomic number 57 to 71 (15 elements) and actinides with atomic numbers 89 to 103 (15 elements), the filling of electrons occurs in 4f and 5f levels respectively. They are called inner transition elements and they have some unique and similar properties. In addition, the 14 elements of the lanthanides except lanthanum are placed in one and the same position of the periodic table while the 14 elements of the actinides except actinium too are kept in one and the same position in the periodic table.

Electronic Configuration of Lanthanides

$_{57}$La: [Xe] $6s^2$ $5d^1$ (and not $4f^1$)

$_{58}$Ce: [Xe] $6s^2$ $5d^1$ $4f^1$

$_{59}$Pr: [Xe] $6s^2$ $5d^1$ $4f^2$
$_{63}$Eu: [Xe] $6s^2$ $5d^0$ $4f^7$
$_{64}$Gd: [Xe] $6s^2$ $5d^1$ $4f^7$

Electronic Configuration of Actinides

$_{89}$Ac: [Rn] $7s^2$ $6d^1$ (and not $5f^1$)
$_{90}$Th: [Rn] $7s^2$ $6d^2$
$_{91}$Pa: [Rn] $7s^2$ $6d^1$ $5f^2$
$_{95}$Am: [Rn] $7s^2$ $6d^0$ $5f^7$
$_{96}$Cm: [Rn] $7s^2$ $6d^1$ $5f^7$

Electronic Configuration of elements with atomic number beyond 103

$_{104}$Rf: [Rn] $5f^{14}$ $6d^2$ $7s^2$
$_{105}$Db: [Rn] $5f^{14}$ $6d^3$ $7s^2$
$_{106}$Sg: [Rn] $5f^{14}$ $6d^4$ $7s^2$
$_{112}$Uub: [Rn] $5f^{14}$ $6d^{10}$ $7s^2$

CONCEPT STRAND

Concept Strand 60

Write down the electronic configurations of elements with atomic elements with atomic numbers 24, 29, 42, 46, 74. also mention the reasons for deviation from expected configuration if any.

Solution

Z	Element	Configuration
24	Chromium	[Ar] $3d^5$ $4s^1$
29	Copper	[Ar] $3d^{10}$ $4s^1$
42	Molybdenum	[Kr] $4d^5$ $5s^1$
46	Palladium	[Kr] $4d^{10}$
74	Tungsten	[Xe $4f^{14}$, $5d^4$ $6s^2$

Elements with atomic numbers 24, 29, 42 and 46 have exceptional electronic configurations. The reason is the special stability of half filled and completely filled d subshells. This, in turn, is due to greater exchange energy of these configurations.

SUMMARY

Particles in atom

Electron	$e/m = 1.76 \times 10^{11}$ C kg^{-1} $e = 1.602 \times 10^{-19}$ C $m = 9.11 \times 10^{-31}$ kg
Proton	$m = 1.673 \times 10^{-27}$ kg $e = 1.602 \times 10^{-19}$ C
Neutron	$m = 1.675 \times 10^{-27}$ kg
Moseley's equation	$\sqrt{\upsilon} = a(Z - b)$ υ = frequency of X-ray Z = atomic number
Relation between atomic no., mass no. & no. of neutrons	A = Z + N
Isotopes	Same atomic no. different mass no.
Isobar	Same mass no.
Isotones	Same number of neutrons

Isosters	Same no. of atoms Same no. of electrons
Isodiaphers	Same difference between neutrons & protons
Isoelectronic	Same no. of total electrons

Atom models

Plum pudding model of atom	Proposed by Thomson
Discovery of atomic nucleus	Rutherford's α-scattering by gold foil experiment.
$\frac{\text{Radius of atom}}{\text{Radius of atomic nucleus}}$	$\frac{10^{-10}\ \text{m}}{10^{-15}\ \text{m}}$
Radius of atomic nucleus	$R = 1.3 \times 10^{-15}\ A^{1/3}$ A is the mass no.

Electromagnetic radiation

Relation between wavelength λ, wave no. $\bar{\upsilon}$, frequency υ & velocity c	$\bar{\upsilon} = \frac{1}{\lambda} \quad \upsilon = \frac{c}{\lambda} = c\bar{\upsilon}$
Velocity of electromagnetic radiation	$3 \times 10^{10}\ ms^{-1}$
Unit of λ	m or cm
Unit of $\bar{\upsilon}$	m^{-1} or cm^{-1}
Unit of υ	s^{-1}
Amplitude of the wave	Height of the crest or the depth of the trough

Electromagnetic spectrum

Regions in electromagnetic spectrum	Cosmic, γ-rays, X-rays, UV-rays, Visible, IR, radio waves
Absorption spectrum	Spectrum of absorbed radiations.
Emission spectrum	Spectrum of emitted radiation.
Atomic spectrum	Spectrum of radiations emitted by excited atoms.
Molecular spectrum	Spectrum of radiations emitted by excited molecules.
Line spectrum	Spectrum consisting of discrete lines only
Band spectrum	Spectrum consisting of broad bands
Hydrogen spectrum	Spectrum of radiations emitted by excited hydrogen atoms.
Planck's equation	$E = h\upsilon$
Planck's constant	$h = 6.626 \times 10^{-34}$ J s
Effective atomic mass of a particle of rest mass 'm_0' moving with velocity 'v'	$m = \frac{m_0}{\sqrt{1 - v^2/c^2}}$
Quantization of angular momentum	$mvr = \frac{nh}{2\pi}$

Energy of electromagnetic radiation for transition between two energy levels that differ by ΔE	$\Delta E = h\upsilon = \frac{hc}{\lambda} = \bar{\upsilon}$

Hydrogen spectrum

Rydberg equation	$\bar{\upsilon} = R_H\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$
Lyman series	u.v region $n_1 = 1$ $n_2 = 2, 3, ...$
Balmer series	Visible $n_1 = 2$ $n_2 = 3, 4, ...$
Paschen series	IR region $n_1 = 3$ $n_2 = 4, 5, ...$
Brackett series	IR region $n_1 = 4$ $n_2 = 5, 6, ...$
Pfund series	IR region $n_1 = 5$ $n_2 = 6, 7, ...$
Rydberg constant	$R_H = 109677\ cm^{-1}$ $2.18 \times 10^{-18}\ J\ atom^{-1}$ $1.32 \times 10^{6}\ J\ mol^{-1}$
Energy of electron in hydrogen like species also known as Bohr equation	$E = \frac{-2\pi^2 mZ^2 e^4}{n^2 h^2}$ in (CGS) $E = \frac{-k^2 2\pi^2 mZ^2 e^4}{n^2 h^2}$ in (SI) Where $k = 9 \times 10^9\ Nm^2C^{-2}$
Refined Bohr equation	$E = \frac{-k^2 2\pi^2 mZ^2 e^4}{n^2 h^2\left(1+\frac{m}{M}\right)}$ where, 'm' is mass of electron and 'M' is mass of nucleus
Energy of electron transition in hydrogen atom	$\Delta E = h\upsilon = \frac{2\pi^2 me^4}{h^2}\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$ where, n_1 is characteristic of the series and n_2 characterising the line.

Parameters of hydrogen atom and Bohr theory

Radius of orbit in hydrogen atom	$r_n = \frac{n^2 h^2}{4\pi^2 me^2}$ (in C.G.S) $r_n = \frac{n^2 h^2}{4\pi^2 mke^2}$ (in S.I.)
Radius of orbit of hydrogen like species	$r_n = \frac{n^2}{Z} \times 0.529 \times 10^{-8}$ cm

Velocity of electron	$v = \frac{2\pi Ze^2}{nh}$ (in C.G.S.) $v = \frac{2\pi kZe^2}{nh}$ (in S.I.) $= 2.19 \times 10^6 \times \frac{Z}{n}\ ms^{-1}$ or $\frac{1}{137}$ times velocity of light
K.E of electron in hydrogen atom	$\frac{1}{2}mv^2 = \frac{kZe^2}{2r}$ (SI) ; $\frac{Ze^2}{2r}$ (CGS)
PE of electron in H-atom	$PE = \frac{-kZe^2}{r}$ (SI) ; $\frac{-Ze^2}{r}$ (CGS)
Total energy of electron in hydrogen atom	$PE + KE = \frac{-Ze^2}{r} + \frac{Ze^2}{2r}$ $= \frac{-Ze^2}{r}$ in (CGS) unit or $\frac{-kZe^2}{2r}$ (in SI)
I E of hydrogen atom	13.56 eV
I E of hydrogen like atom	$13.56 \times Z^2$ eV
1 eV energy	1 eV = 23 k cal mol^{-1} or 96.53 kJ mol^{-1}
No. of revolutions per second in hydrogen like species	$n = \frac{v}{2\pi r} = \frac{2\pi mvZe^2}{n^2h^2}$
No. of electron waves in an orbit	$2\pi r = n\lambda$ $n = \frac{2\pi r}{\lambda}$ where λ is the length of electron wave

Photoelectric effect

K.E of photoelectron	$KE = h(\upsilon - \upsilon_0)$
Threshold frequency	υ_0
Work function w_0	$W_0 = h\upsilon_0$

Modifications of Bohr model of atom

Elliptical orbits by Sommerfield	Angular momentum = $k\left(\frac{h}{2\pi}\right)$ k-azimuthal quantum number $\frac{n}{k} = \frac{\text{length of major axis}}{\text{length of minor axis}}$
de-Broglie equation	$\lambda = \frac{h}{mv} = \frac{h}{p}$
Relation between de-Broglie wavelength λ and accelerating potential P	$\lambda = \frac{h}{\sqrt{2mPe}}$

Heisenberg uncertainty principle	$\Delta x \times \Delta p \geq \frac{h}{4\pi}$ $\Delta x \times m\Delta v \geq \frac{h}{4\pi}$ $\Delta x \times \Delta v \geq \frac{h}{4\pi m}$

Quantum numbers

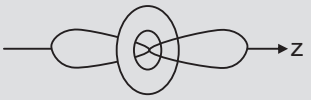
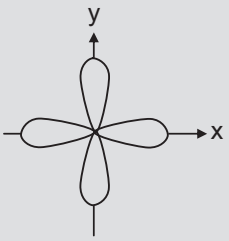
Principal quantum no. n	$E_n = \frac{-2\pi^2 me^4}{n^2h^2}$ n is principal quantum no. It has non-zero integral values 1, 2, ... decide energy of electron shell
Azimuthal or orbital angular momentum quantum number 'ℓ'	Corresponding to every values of 'n' there are 'n' values for ℓ they are 0, 1, 2,...(n – 1) or s, p, d, f.....
Orbital angular momentum	$\frac{h}{2\pi}\sqrt{\ell(\ell+1)}$ 'ℓ' is azimuthal quantum no.
Magnetic quantum number 'm_ℓ'	Orientation of orbitals in space each value of 'ℓ' has $(2\ell + 1)$ values of m_ℓ they are $-\ell$ ....0.....+ℓ
Spin quantum number 's'	Denote spin of electrons in an orbital . It can have values $+\frac{1}{2}$ or $-\frac{1}{2}$

Quantum mechanical picture of Hydrogen atom

Wave function ψ	Amplitude of electron wave in an area obtained by solving Schroedinger wave equation.
Schroedinger wave equation	$\nabla^2\Psi + \frac{8\pi^2 m}{h^2}(E - V)\psi = 0$ or $\hat{H}\Psi = \hat{E}\Psi$
Radial probability distribution curve	Plot of radius 'r' against $4\pi r^2 dr\psi^2$ The curve is characteristic of a sublevel
Significance of ψ^2	It is a measure of the probability of finding an electron in space within an atom
Angular probability distribution	It provides information about shape of orbitals
Orbitals	Three dimensional region in an atom where there is maximum probability of finding electron.
Eigen values and Eigen functions	The accepted solutions of Schroedinger wave equation corresponding to characteristic values of energy E

Shape of orbitals

's' orbitals	Spherical
'p' orbitals	Dumb bell shaped, three degenerate orbitals p_x, p_y and p_z in mutually perpendicular direction.

'd' orbitals	Double dumb bell shaped Five degenerate orbitals
d_{z^2}	
$d_{x^2-y^2}$	along x and y axis 
d_{xy}, d_{xz}, d_{yz}	All identical in shape to that of $d_{x^2-y^2}$ but directed in between two axis

Electronic configuration of atoms

Pauli's exclusion principle	No two electron in an atom can have the same set of all the four quantum numbers
Hund's rule of maximum multiplicity	When electrons occupy degenerate orbitals all the orbitals are singly occupied before any pairing takes place.
Aufbau principle – $(n + \ell)$ rule	Electrons occupy orbitals in the increasing order of $(n + \ell)$ value . In orbitals with same $(n + \ell)$ value filling occurs in the increasing order of 'n'
Aufbau order of atomic orbitals	1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, 4d, 5p, 6s, 4f, 5d, 6p, 7s, 5f, 6d, 7p, 8s,.....
Exchange energy	The decrease in energy for a pair of electrons due to mutual exchange of electrons with parallel spin in identical orbitals.
Pairing energy	Energy required to place two electrons with anti-parallel spins together in an orbital by over-coming repulsion between them.

Electronic configuration which are exceptions

Chromium (Z = 24)	$[_{18}Ar]3d^5 4s^1$
Copper (Z = 29)	$[_{18}Ar]3d^{10} 4s^1$
Palladium (Z = 46)	$[_{36}Kr]4d^{10} 5s^0$
Lanthanum (Z = 57)	$[_{54}Xe]4f^0 5d^1 6s^2$
Actinium (Z = 89)	$[_{86}Rn]7s^2\ 5f^0 6d^1$
Europium (Z = 63)	$[_{54}Xe]6s^2 5d^0 4f^7$
Thorium (Z = 90)	$[_{86}Rn]7s^2\ 5f^0 6d^2$
Americium (Z = 95)	$[_{86}Rn]7s^2\ 5f^7 6d^0$

TOPIC GRIP

Subjective Questions

1. By what % does the mass of electron change when it has been accelerated from a very low speed to a speed of 5×10^7 m s^{-1}?
2. A certain electronic transition in the spectrum of singly ionized helium (He^+) ion has λ = 1086 Å. Identify the transition.
3. In a certain atomic spectrum, the principal series is given by $\upsilon_n(cm^{-1}) = \upsilon_\infty(cm^{-1}) - \left[\frac{R}{(n+\mu)^2}\right]$ where n = quantum no., μ = 0.9596, R = 109722 cm^{-1}. Given υ_2 = 30952 cm^{-1}, calculate υ_∞.
4. Identify the transitions in the atomic spectrum of hydrogen corresponding to λ values 3971 Å and 3890 Å.
5. Calculate, according to the Bohr model the radius of the Be^{3+} ion.
6. For the Bohr orbits of quantum numbers n_1 and n_2 of the hydrogen atom, the difference in the ionization energy = 2.55 eV. Calculate the values of (i) n_1, n_2 (ii) the corresponding ionization energies
7. The angular momentum of an electron in a Bohr orbit of hydrogen atom is 5.281×10^{-34} kg $m^2 s^{-1}$. Calculate the energy of the radiation emitted (in eV) when that electron falls from this level to the next successive lower level.
8. Calculate the mass of the photon in kg corresponding to the series limit of the Lyman transition of the H-atom.
9. An electron initially accelerated through 100 volts is decelerated till its de Broglie wavelength is 50% greater. Calculate the magnitude of the retarding potential.
10. An electron is trapped in a region of extension 1 Å along the X-axis. Calculate the momentum uncertainty in this direction approximately.

Straight Objective Type Questions

Directions: **This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

11. For an ion, hydrogen-like, with only one electron, the radius of the nth Bohr orbit is not proportional to

    (a) $\frac{n}{Z^2}$ (b) (reduced mass)$^{-1}$

    (c) $\frac{n^2}{Z}$ (d) v^{-2} (v = speed of the electron)

12. If the ionization potential of Hydrogen is 13.6 eV, then the ionization potentials of He^+ and Li^{2+} are respectively.
    (a) 4.35×10^{-18} J & 6.53×10^{-18} J
    (b) 8.7×10^{-18} J & 1.96×10^{-17} J
    (c) 27.2 eV and 40.8 eV
    (d) 54.4 eV and 122.4 eV

13. Electromagnetic radiation of $\lambda = 100$ nm falling on a surface causes photoelectric emission of electrons of de Broglie wavelength 4.2 Å. Calculate the magnitude of υ_0, the threshold frequency.
(a) $9.475 \times 10^{14}\ s^{-1}$ (b) $7.495 \times 10^{14}\ s^{-1}$ (c) $4.795 \times 10^{13}\ s^{-1}$ (d) $5.947 \times 10^{13}\ s^{-1}$

14. The mass of a neutron is 1.009 u. Calculate the de Broglie wavelength of the neutron in Å with energy = 1 MeV.
(a) 2.478×10^{-3} (b) 4.278×10^{-3}
(c) 2.874×10^{-4} (d) 4.872×10^{-2}

15. The ratio of de Broglie wavelengths of α particles accelerated through potentials of 100 V and 10000 V is
(a) 5 (b) 20 (c) 10 (d) 15



Assertion–Reason Type Questions

Directions: **Each question contains Statement-1 and Statement-2 and has the following choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False
(d) Statement-1 is False, Statement-2 is True

16. Statement 1
The emission spectrum of sodium vapour shows a doublet i.e two very close spectral lines in the yellow region of the spectrum.
and

Statement 2
Atomic sodium contains one unpaired electron with two spin quantum states.

17. Statement 1
According to Bohr's theory of the spectrum of atomic hydrogen the angular momentum of the electron at the second quantum level is $\frac{h}{\pi}$.
and

Statement 2
Bohr's theory implied the concept of "selection rules" for spectral transitions.

18. Statement 1
In a Bohr orbit of quantum no. 'n' of a hydrogen like species, the circumference corresponds to 'n' de Broglie wavelengths.
and

Statement 2
In the above case the radius, r is proportional to n^2.

19. Statement 1
When a moving electron is decelerated, its de Broglie wavelength increases.
and

Statement 2
When monochromatic radiation of sufficiently high frequency falls on a metallic surface the emitted photoelectrons have different de Broglie wavelengths.

20. Statement 1

The orbitals $2p_x$, $2p_y$ and $2p_z$ of a hydrogen-like atomic species correspond to the magnetic quantum numbers +1, −1 and 0

and

Statement 2

The 3s atomic orbital has one radial node and one angular node.

Linked Comprehension Type Questions

Directions: **This section contains 2 paragraphs. Based upon the paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

Passage I

According to the Bohr theory for the atomic spectrum of hydrogen the energy levels of the proton-electron system depends on the quantum number n. In an electron transition from a higher quantum level, n_2 to the lower level n_1, radiation is emitted. The frequency, υ of the radiation is given by $h\upsilon = \left(E_{n_2} - E_{n_1}\right)$ where, h is Planck's constant and E_{n_2}, E_{n_1} are the energy level values for the quantum numbers n_2 and n_1. A useful formula for the wavelength, $\lambda = c/\upsilon$ is given by $\lambda(\text{Å}) = \frac{912}{Z^2} \times \left(\frac{n_2^2 n_1^2}{n_2^2 - n_1^2}\right)$ where, Z = atomic number of any one electron species viz H, He^+, Li^{2+}, Be^{3+}……. For hydrogen Z = 1

21. In the above formula, when $n_2 \to \infty$, we have $\lambda = \frac{912}{1^2} \times n_1^2$. This value of λ is known as the series limit for the given value of n_1. Calculate the value of n_1 for the series limit λ = 8208Å.

(a) 5 (b) 2 (c) 3 (d) 4

22. Calculating n_2 for the series limit = 22800Å. Calculate λ for the transition $(n_2 + 1)$ to $(n_2 - 1)$ (q-levels).

(a) 26266Å (b) 22665Å (c) 25266Å (d) 22566Å

23. A certain spectral line of singly ionized helium, He^+ has a wavelength = 1642 Å. Identifying the quantum numbers n_2 and n_1 $(n_2 > n_1)$ in this transition, calculate λ for the transition $(n_2 + 1)$ to $(n_1 - 1)$.

(a) 423Å (b) 243Å (c) 432Å (d) 324Å

Passage II

The equation of Schroedinger for the hydrogen atom in the time-independent, non-relativistic form is a partial differential equation involving the position coordinates(x, y and z). The potential energy term for the proton-electron system is spherically symmetric of the form $\frac{-1}{4\pi\varepsilon_0} \times \left(e^2/r\right)$. Thus it is advantageous to change over from the cartesian coordinates (x, y and z) to the spherical polar coordinates, (r, θ and ϕ). In this form the equation becomes separable into the radial part involving r and the angular part involving θ and ϕ. The probability of locating the electron within a volume element $d\tau = 4\pi r^2 dr$ is then given by $|\psi|^2(4\pi r^2 dr)$ where, ψ is a function of r, θ and ϕ. With proper conditions imposed on ψ, the treatment yields certain functions, ψ, known as atomic orbitals which are solutions of the equations. Each function ψ corresponds to quantum numbers n, ℓ and m, the principal, the azimuthal and the magnetic quantum numbers respectively, n has values 1, 2, 3,...., ℓ has values 0, 1, 2,…..(n − 1) for each value of n and m(or m_ℓ) has values $+\ell$, + (ℓ − 1),….1, 0, −1, −2….$-\ell$ i.e., (2ℓ + 1) values for each value of ℓ. In addition a further quantum number called spin had to be introduced with values $\pm \frac{1}{2}$. Any set of four values for n, ℓ, m and s characterizes a spin orbital. Pauli's exclusion principle states that a given spin orbital can accommodate not more than one electron. Further the values $\ell = 0$, $\ell = 1$, $\ell = 2$, $\ell = 3$ are designated s, p, d and f orbitals respectively.

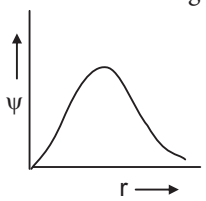
24. How many spin orbitals are there corresponding to n = 3?

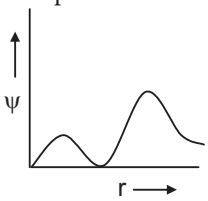
(a) 9 (b) 18 (c) 3 (d) 6

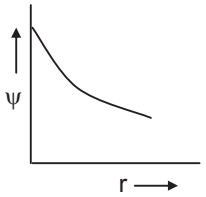
25. It is a basic fact that any two electrons are indistinguishable. 3 electrons are to be accommodated in the spin orbitals included under the designation 2p, conforming to the Pauli principle. Calculate the number of ways in which this may be done.

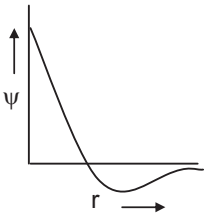
(a) 20 (b) 30 (c) 60 (d) 120

26. Which of the following diagrams corresponds to the 2s orbital?

(a) 

(b) 

(c) 

(d) 

Multiple Correct Objective Type Questions

Directions: **Each question in this section has four suggested answers out of which ONE OR MORE answers will be correct.**

27. The "Rydberg constant" for a one-electron species such as He^{+}, Li^{2+} etc.

(a) varies slightly from species to species
(b) depends on the nuclear charge
(c) does not depend on the quantum number
(d) is inversely proportional to the Planck's constant h

28. Identify the correct Statements among the following

(a) The number of nodes in a hydrogen like (one electron) species is (n − 1) for the quantum number = n.
(b) The angular momentum of the electron in the 3p orbital is $1.5\,h/\pi$ (since $mvr = n\,h/2\pi$).
(c) The $3d_{z^2}$ orbital has characteristic nodal surfaces.
(d) The 1s orbital of the hydrogen atom has a non zero magnitude at $r \to 0$ i.e., at the nucleus.

29. Atomic orbitals have.

(a) physically measurable/observable magnitudes.
(b) different symmetries and energy values. No two orbitals have the same quantum numbers or the same energies.
(c) electrons are accommodated in them in the increasing order of their energies in the ground state of an atom.
(d) have directional properties except the s-orbitals.

Matrix-Match Type Question

Directions: **Match the elements of Column I to elements of Column II. There can be single or multiple matches.**

30.

Column I	**Column II**
(a) Angular momentum of electron determines	(p) magnetic quantum number m
(b) Zeeman effect	(q) principal quantum number n
(c) Energy levels of electron in the theory of Bohr	(r) quantum number ℓ
(d) Number of nodal planes in an atomic orbital	(s) magnetic moment of the electron

IIT ASSIGNMENT EXERCISE

Straight Objective Type Questions

Directions: **This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

31. Of the following Statements, the one, not proposed by Sir John Dalton in the atomic theory is
(a) Atoms are indivisible
(b) Atoms of same element are identical in all respects
(c) The mass of an atom can be changed into energy
(d) Atoms are smallest particle that take part in a chemical reaction

32. Consider an atom of atomic number Z. If energy required to remove a proton and electron are E_p and E_e respectively then
(a) $E_p = ZE_e$ (b) $E_p < E_e$ (c) $E_p > E_e$ (d) $E_p = 1837\ E_e$

33. The ratio of mass of a hydrogen ion and the electron that is removed from it, is approximately
(a) 1837 : 1 (b) 1 : 13.6 (c) $1 : 2.18 \times 10^{-18}$ (d) 1 : 1

34. The charge by mass ratio of an ion M^+ is 4.2×10^6 C kg^{-1}. The ion may be
(a) Ag^+ (b) Cu^+ (c) Cs^+ (d) Na^+

35. The frequency of X rays produced by striking cathode rays on lithium metal will be (Given a = b = 1 in the Moseley's equation)
(a) 4 sec^{-1} (b) 16 Hz (c) 2 sec^{-1} (d) 40 sec^{-1}

36. The atoms $^{19}_{9}F$ and $^{35}_{17}Cl$ are
(a) isotones (b) isodiaphers (c) isoelectronic (d) isosters

37. The number of neutrons present in a phosphonium ion is $\left[{}^{31}_{15}P,\ {}^{1}_{1}H\right]$
(a) 16 (b) 17 (c) 18 (d) 19

38. Which of the following is not a limitation to the Rutherford's model of atom
(a) It could not explain based on Maxwell's law of electrodynamics
(b) It could not explain spectral lines
(c) It could not account for discontinuity in spectrum
(d) It could not explain electrical neutrality of atom

39. Two photons of wavelength 400 Å and 800 Å are absorbed by a gas and emitted as a single radiation. Assuming no loss, wavelength of emitted radiation is approximately
(a) 600 Å (b) 800 Å (c) 267 Å (d) 1200 Å

40. The wrong Statement about matter waves is
(a) They move with speed of light.
(b) They cannot be radiated into empty space.
(c) They are associated with the particle under consideration.
(d) They generally have small wavelength.

41. Two monochromatic radiations with wavelengths of 4×10^{-6} m and 8×10^{-4} m, differ by an energy of
(a) 5×10^{-10} J (b) 5×10^{-20} J (c) 5×10^{-32} J (d) 5×10^{-8} J

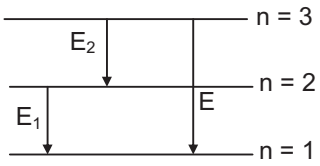
42. An electron in the 5s orbital, has an orbital angular momentum of

(a) zero (b) $\frac{5h}{2\pi}$ (c) $\frac{\sqrt{5}h}{2\pi}$ (d) $\frac{\sqrt{30}h}{2\pi}$

43. Consider a 60 W lamp giving out light radiations of wavelength 500 nm. It will give 1 mole of photons in

(a) 4000 sec (b) 82 sec (c) 4×10^5 sec (d) 8×10^5 sec

44. Consider the transition



The true Statement is

(a) $E_1 > E_2$ (b) $E \leq E_1 + E_2$ (c) $E \geq E_1 + E_2$ (d) $\frac{1}{E} = \frac{1}{E_1} + \frac{1}{E_2}$

45. An electron falls from the 8th orbit in a hydrogen atom. The spectral line of longest wavelength in the Brackett series is from

(a) 5th orbit (b) 6th orbit (c) 4th orbit (d) 7th orbit

46. The radiation of smallest wavelength in the Balmer series of helium ion (He^+) is the same as that for

(a) Lyman series of Be^{3+} ion (b) Balmer series of Li^{2+} ion
(c) Lyman series of hydrogen atom (d) Brackett series of Li^{2+} ion

47. Given that shortest wavelength in Lyman series of Li^{2+} ion is x . Then

(a) longest wavelength in Balmer series is $\frac{5}{36x}$

(b) longest wavelength in Lyman series of He^+ is 3x

(c) shortest wavelength in Balmer series is $\frac{4}{3}x$

(d) shortest wavelength in Lyman series of He^+ is $\frac{4}{9}x$

48. If a hydrogen atom is supplied with 15.8 eV of energy, the kinetic energy of the escaped electron is (Energy of 1st orbit of hydrogen = 13.6 eV)

(a) 1.75×10^{-19}J (b) 29.4 eV (c) 1.1 eV (d) 3.5×10^{-19} J

49. Consider an electron in the second orbit of Be^{3+}. The ratio of kinetic energy to potential energy for this electron is

(a) −2 : 1 (b) 2 : −1 (c) 1 : −2 (d) 2 : 1

50. The difference between the potential energy of an electron in the first and second bohr orbit of hydrogen atom is

(a) $4 \times 2.18 \times 10^{-18}$ J (b) $2 \times 2.18 \times 10^{-18}$ J (c) 4×10^{-28} J (d) 3.24×10^{-18} J

51. A H_2 molecule is exposed to a radiation of wavelength 250 nm. If dissociation energy of H_2 is 7.15×10^{-19} J/molecule, the kinetic energy of each atom is

(a) 7.7×10^{-23} J (b) 7.7×10^{-20} J (c) 7.9×10^{-16} J (d) 7.9×10^{-20} J

52. An electron moves a distance equal to its wavelength in 5 sec. Its speed is

(a) $\sqrt{5h}$ (b) $\sqrt{\frac{h}{5m}}$ (c) $5\sqrt{\frac{h}{p}}$ (d) $\sqrt{\frac{5h}{m}}$

53. For an electron moving in the 3rd orbit, the number of waves made in one complete revolution is

(a) 27 (b) 9 (c) 3 (d) 6

54. The number of revolutions made by an electron per second in the third orbit of hydrogen atom is (In the first orbit of hydrogen atom, velocity = 2.18×10^6 m/s and radius = 0.53 Å)
(a) 2.42×10^{14} (b) 2.42×10^6 (c) 2.42×10^{10} (d) 2.42×10^4

55. The velocity of an electron in a Bohr orbit is directly proportional to
(a) atomic number of the atomic species
(b) quantum number of the orbit
(c) square of the quantum number of the orbit
(d) mass of the electron

56. Photoelectric effect is observed only in metals and not in non metals because
(a) Non-metals have very high ionization energy
(b) Non-metals occur only in combined state
(c) In non metals last electron enters 'p' orbital
(d) Non-metals generally have low atomic weight

57. Radiation of energy 8 eV is incident on a metal surface. If its work function is 3.5eV, the stopping potential is
(a) −4.5 V (b) +3.5 V (c) −11.5 V (d) −3.5 V

58. A particle, for which mass and velocity are numerically equal, moves with a de Broglie wavelength of 165 Å. Its kinetic energy is
(a) 6.25×10^{37} (b) 4×10^{-39} J (c) 6.25×10^{-37} J (d) 4×10^{-26}J

59. An electron and proton are travelling such that their wavelengths are in the ratio 1 : 2. Their momentum are in the ratio
(a) 1836 : 1 (b) 2 × 1836 : 1 (c) 1 : 1836 (d) 2 : 1

60. The de Broglie wavelength of a proton at 27°C moving with rms velocity is (mass of proton = 1.67×10^{-27} kg)
(a) 45.68 Å (b) 145 Å (c) 1.45 Å (d) 2413 Å

61. The kinetic energy of a moving particle is 2.5×10^{-28} J. The frequency (in s^{-1}) of this particle is
(a) 1.25×10^5 (b) 7.5×10^5 (c) 1.25×10^6 (d) 5.2×10^6

62. The Heisenberg's uncertainity principle was experimentally verified by
(a) Davison Germer experiment (b) Compton effect
(c) Zeeman effect (d) Photoelectric effect

63. It is not possible to build an instrumental device that can defy Heisenberg uncertainty principle by reducing the error because
(a) position and momentum are conjugate pair of variables
(b) velocity can be measured only along one axis
(c) to observe the electron it has to be illuminated
(d) as mass of particle increases, uncertainty decreases

64. We can talk only in terms of probability of finding an electron around the nucleus because
(a) position has to be determined along three axes
(b) both exact position and exact velocity cannot be determined simultaneously
(c) electron has both wave like and matter like behaviour
(d) the electrons have a spin

65. Consider the set of quantum numbers 3, 2, −2, $+\frac{1}{2}$, if the given subshell is completely filled. The next electron will enter orbital with n and l value
(a) n = 3, ℓ = 3 (b) n = 4, ℓ = 1 (c) n = 4, ℓ = 0 (d) n = 2, ℓ = 1

66. The quantum number that explains Stark effect is
(a) Principal quantum number (b) Azimuthal quantum number
(c) Angular momentum quantum number (d) Magnetic quantum number

67. Given that an orbital is symmetric about the nucleus, then the value of azimuthal quantum number and magnetic quantum number are respectively
(a) −1, +1 (b) +1, +1 (c) 0, 0 (d) 1, 0

68. The correct quantum number values for the 19th electron of titanium ion is
(a) n = 3, $\ell = 0$ (b) n = 4, $\ell = 1$ (c) n = 3, $\ell = 2$ (d) n = 4, $\ell = 0$

69. Consider an orbital of the shape  . The number of such orbitals present in subshell with n = 2 is
(a) 1 (b) zero (c) 2 (d) 3

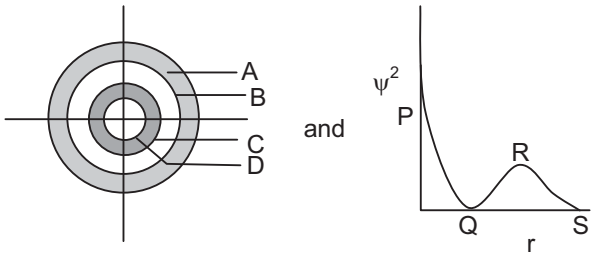
70. The quantum number that cannot be derived from Schroedinger equation
(a) can take only two value for an electron
(b) determines the magnetic moment
(c) takes odd number of values for an electron
(d) determines the energy of the orbital

71. The radius of maximum probability in the case of a hydrogen atom coincides with
(a) wavelength of electron in the first orbit
(b) uncertainty in position, Δx with respect to Heisenberg
(c) the Bohr radius
(d) distance between two nodes

72. Nodes are regions around the nucleus where,
(a) $\psi^2 = 1$ (b) $|\psi^2| = 1$ (c) $\psi = \infty$ (d) $\psi^2 = 0$

73. The number of radial nodes in a 4d orbital is
(a) 1 (b) 2 (c) 3 (d) 4

74. Consider the two graphs for 2s orbital

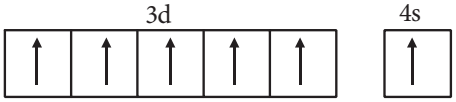


In the two, similar points are
(a) A and P (b) B and Q (c) B and R (d) D and S

75. As the total number of nodes increases, the parameter that will not show a regular increase is
(a) energy (b) distance from nucleus
(c) principal quantum number (d) angular nodes

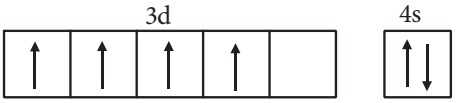
76. For an orbital with value $\ell = 1$, it will be symmetric about
(a) either x, y or z axis (b) the x axis only (c) the nucleus (d) it is unsymmetric

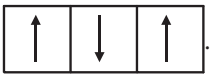
77. The electronic configuration where there is a violation of Aufbau principle is

(a) 

(b) 

(c) 

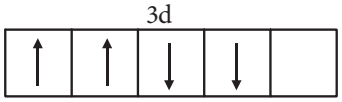
(d) 

78. The configuration $2p^3$ cannot be written as  . This is based on

(a) Pauli's exclusion principle
(b) $n + \ell$ rule
(c) Hund's Rule
(d) Aufbau's Rule

79. Which of the following results have not been obtained from Pauli's exclusion principle

(a) Maximum number of electrons in a shell = $2n^2$
(b) A subshell tends to attain exactly half filled configuration to attain stability
(c) Maximum number of electrons in p subshell = 6
(d) Maximum number of electrons in an orbital = 2

80. The arrangement 

is not allowed for $3d^4$ because

(a) the electrons should be paired
(b) 4s orbital should be filled before 3d
(c) single electrons with same spin in a given subshell are stable
(d) the arrangement is not symmetric

81. Moseley's equation for the Kα series of characteristic X rays is $\upsilon^{\frac{1}{2}} = a\,(z - b)$ where,
υ = frequency of the X ray line, a and b are constants;
Z = atomic number of the element. Given $\lambda_1 = 9.87$Å and $\lambda_2 = 2.29$Å for z-values
$z_1 = 12$ and $z_2 = 24$, calculate the magnitude of b.
(a) 0.50 (b) 0.67 (c) 0.85 (d) 0.91

82. Dissociation of Iodine molecules to yield atoms i.e., $I_{2(g)} \rightarrow 2I_{(g)}$ may be brought about by the absorption of light of frequency above a certain value. If the minimum dissociation energy is 57.2 kcal/mol. Calculate the critical wavelength, λ above which dissociation is not possible.
(a) ~ 5000Å (b) ~ 6000Å (c) ~ 4000Å (d) ~ 6500Å

83. Calculate the momentum (in kg m s^{-1}) of a photon of wave length $\lambda = 5000$ Å.
(a) 1.523×10^{-28} (b) 1.325×10^{-27} (c) 1.253×10^{-28} (d) 2.153×10^{-28}

84. A hydrogen atom in its ground state absorbs a photon and goes into the first excited state. It then absorbs a second photon which just ionizes it. What is the ratio of the wave lengths of the first photon and the second photon?
(a) 0.5 (b) 0.25 (c) 0.33 (d) 0.67

85. Given that 2.00×10^{18} photons of a certain wavelength, λ in the blue violet region of the spectrum have a total energy of one joule, calculate an approximate value of λ.
(a) ~ 4500Å (b) ~ 4000Å (c) ~ 4700Å (d) ~ 4300Å

86. The energy (relativistic), E of a certain particle and its momentum, p, are related by the equation, $E = p \times c$ where c = speed of light in vacuum. The particle is
(a) an electron (b) a neutron (c) a photon (d) a deuteron

87. The frequency of a 1 kilo watt radio transmitter is 880 k Hz s^{-1}. Calculate the number of photons emitted per sec. $h = 6.626 \times 10^{-34}$ J.s.
(a) 2.01×10^{29} (b) 1.86×10^{30} (c) 1.5×10^{29} (d) 1.72×10^{30}

88. Values of wave length λ in the spectrum of singly ionized helium (He^+) are suggested. Identify the wrong value.
(a) 1642 Å (b) 4690 Å (c) 304 Å (d) 1126Å

89. According to the Bohr model the radius of the electron orbit in the first excited state of the Li^{2+} ion is
(a) 0.705Å (b) 0.751Å (c) 0.925Å (d) 0.952Å

90. Three successive lines in the atomic spectrum of hydrogen have wave lengths 18746Å, 12815Å and 10935Å. Calculate the series limit for the spectral series to which the above lines belong.
(a) 8820Å (b) 8208Å (c) 9280Å (d) 9028Å

91. Calculate the ratio of the Rydberg constants for atomic hydrogen and $He^{(+)}$ ion. The mass of $He^{(+)}$ may be taken to be four times the mass of the Hydrogen atom. Take the proton mass to be 1840 × electron mass.
(a) 1.0000 exactly (b) very slightly > 1 (c) very slightly < 1 (d) 1.01 exactly

92. Which of the following hydrogen like, one electron species have the same Bohr radius?
(i) first orbit of hydrogen atom (ii) first orbit of He^+
(iii) second orbit of Li^{2+} (iv) second orbit of Be^{3+}
Which pair has the same magnitude?
(a) (ii) and (iii) (b) (i) and (iii) (c) (ii) and (iv) (d) (i) and (iv)

93. In a certain electronic transition in the hydrogen atom from a higher quantum number n to the ground state in several steps, only one line of the Paschen series is observed. Calculate the series limit corresponding to the quantum number n.
(a) 14592Å (b) 15924Å (c) 12594Å (d) 19542Å

94. The first line in the Paschen series for a certain hydrogen like species (i.e., an ion with only one electron) has a wave length λ = 2085Å. Identify the ion
(a) Li^{2+} (b) Be^{3+} (c) He^+ (d) Ne^{9+}

95. Which of the following energy transitions is possible in the hydrogen atom according to Bohr's theory?
(a) 6.03 eV (b) 1.56 eV (c) 9.2 eV (d) 0.66 eV

96. Calculate the speed of the electron approximately in the nth Bohr orbit of the hydrogen atom in cm/sec (electronic charge = -4.802×10^{-10} esu)
(a) $\frac{3.2\times10^8}{n}$ (b) $\frac{2.2\times10^7}{n}$ (c) $\frac{2.2\times10^8}{n}$ (d) $3.2\times\frac{10^7}{n}$

97. Calculate the ratio of the velocity of light in vacuum to the speed of the electron in the first Bohr orbit of the He^+ ion.
(a) 56.8 (b) 86.5 (c) 137 (d) 68.5

98. The time, T taken by an electron in the hydrogen atom for one revolution in the Bohr–orbit of quantum number = n is proportional to
(a) $\frac{1}{n}$ (b) n^3 (c) n^{-2} (d) constant for all n

99. Consider the photo chemical equation $K.E_{max} = h(\upsilon - \upsilon_0)$ In two different experiments using radiations of frequencies υ_1 and υ_2, if $\upsilon_1 : \upsilon_2 : \upsilon_0 = 8 : 5 : 2$. Calculate the ratio of the kinetic energy (max) values of the emitted photo electrons $\frac{(KE)_1}{(KE)_2}$.
(a) 1.5 (b) 1.25 (c) 1.75 (d) 2

100. Indicate among the following the incorrect alternative.
Radiation has a particle nature (i.e., consists of photons) This is implied by
(a) The photoelectric equation
(b) The (double-slit) interference experiment
(c) de Broglie equation
(d) The Compton formula

101. Calculate the de Broglie wavelength of the electron (in the spirit of the Bohr theory) revolving around the nucleus in the third orbit of the hydrogen atom.
(a) 7.92Å (b) 9.97Å (c) 5.79Å (d) 6.55Å

102. An electron is accelerated from a very low velocity (~ zero speed) by the application of a potential difference of V volts. If the de Broglie wavelength should change (i.e., decrease) by 1.0% what percent increase in V causes it
(a) 0.5% (b) 1% (c) 1.5% (d) 2%

103. A projectile of mass 1 kg is shot at a target. At the moment of hitting the target its de Broglie wavelength is 6.626×10^{-36}m. If during its passage through air it slowed down to 95% of its initial speed, calculate its initial speed in m/s.
(a) ~110 m/s (b) ~ 107 m/s. (c) ~105 m/s. (d) ~ 102 m/s

104. The 1s electron in the hydrogen atom is ejected from the atom after absorbing a total amount of energy = 20.4 eV. Calculate its de Broglie wavelength in Å (mass of electron = 9.1×10^{-31} kg)
(a) 4.7 (b) 3.7 (c) 5.7 (d) 2.7

105. A charged particle, accelerated through a potential drop has a velocity = 7.92×10^4 m/s and a de Broglie wavelength of 5.00 pm. Suggest the identity of the particle. Electron mass (m_e) 9.1×10^{-31} kg, masses of the proton, deuteron and α–particle are respectively (2000 me), (4000 me) and (8000 me) approximately.
(a) electron (b) proton (c) deuteron (d) alpha particle

106. An electron first accelerated through 100 volts suffers successively two retardations (i) through 19 volts and then (ii) through 32 volts. Its de Broglie wavelengths in the three situations are λ_1, λ_2 and λ_3. Calculate $\frac{\lambda_3 - \lambda_2}{\lambda_1}$.

(a) $\frac{20}{41}$ (b) $\frac{10}{63}$ (c) $\frac{20}{63}$ (d) $\frac{10}{41}$

107. Calculate approximately the uncertainty in momentum of an electron, supposed to be trapped within a nucleus of radius = 10^{-12} cm. h = 6.626×10^{-27} erg.sec.
(a) 2.57×10^{-15} g.cm.s^{-1} (b) 2.75×10^{-14} g.cm.s^{-1} (c) 2.64×10^{-16} g.cm.s^{-1} (d) 7.25×10^{-13} g.cm.s^{-1}

108. The position along the x-axis and the corresponding momentum p_x are determined with uncertainties of 1Å in the position and Δp_x in the momentum for a 1 keV electron. Express $\left(\frac{\Delta p_x}{p_x} \times 100\right)$ i.e., as a %.

(a) 7.3% (b) 3.1% (c) 5.5% (d) 4.7%

109. Indicate among the following the incorrect Statement
(a) The s-orbital Ψ_{ns}, has a non zero probability at the nucleus i.e., r = 0.
(b) The 2s, $2p_x$, $2p_y$ and $2p_z$ orbitals of the hydrogen atom are degenerate i.e., have the same energy.
(c) The Bohr theory of the hydrogen atom did not explain the fine structure of spectral lines.
(d) In an electronic transition in a hydrogen-like (one electron) species, from higher to lower and yet lower (energy) quantum levels [ie higher to lower value – quantum numbers) The orbital speed of the electron in the circular (Bohr) orbit decreases.

110. Assume that any two electrons are indistinguishable. An atomic orbital is to be specified by four quantum numbers n, ℓ, m_ℓ, m_s. Calculate the number of ways of accommodating two electrons in the 3d orbitals of an atom.
(a) 45 ways (b) 90 ways (c) 10 ways (d) 5 ways

Assertion–Reason Type Questions

Directions: **Each question contains Statement-1 and Statement-2 and has the following choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False
(d) Statement-1 is False, Statement-2 is True

111. Statement 1
According to the Bohr theory of the spectrum of the hydrogen atom successive energy levels get closer and closer at higher and higher quantum numbers.
and

Statement 2
The Sommerfield extension of Bohr's theory introduced selection rules for electronic transitions.

112. Statement 1
None of the p-orbitals of the hydrogen atom has spherical nodes.
and

Statement 2
The 3 d-orbitals, except $3d_z^2$ orbital have shapes exhibiting 2 nodal planes in each orbital.

113. Statement 1
Although the 2s and 2p orbitals are degenerate in the hydrogen atom, they are not so in polyelectronic species.
and

Statement 2
In poly electronic species, interelectron repulsions change the energy levels of orbitals.

Linked Comprehension Type Questions

Directions: **This section contains 1 paragraph. Based upon the paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

Passage I

Even in the Rutherford's α - particle scattering experiments the existence of the atomic nucleus and its charge were clearly indicated by the experimental verification of the Rutherford's theoretical formula. Further to the work, Moseley studied the X-radiation (characteristic) emitted by metallic anticathodes struck by fast-moving electrons. His formula is $\sqrt{\upsilon} = a(Z - b)$ where, a and b are approximately constant and υ is the frequency of the characteristic X-radiation and Z is the atomic number.

114. Wave lengths of the characteristic X-ray lines are 2.892Å and 2.775Å for cesium (At.no.= 55) and Barium (At. no. = 56) calculate the values of a and b from these data.
(a) 1.32×10^8, 7.81 (b) 1.32×10^6, 8.71 (c) 2.31×10^6, 8.87 (d) 2.13×10^7, 7.18

115. The characteristic X-radiation λ values for three elements are λ_1, λ_2 and λ_3. The atomic numbers are 57, 72 and 74 calculate $\left(\upsilon_3^{1/2} - \upsilon_1^{1/2}\right)$.
(a) 0.085 (b) 0.15 (c) 0.12 (d) 0.17

116. For the characteristic wavelength $\lambda = 1.522$ Å, calculate the energy of the X-ray photon in joule.
(a) 1.0361×10^{-16} (b) 1.3061×10^{-15} (c) 1.6310×10^{-16} (d) 1.6130×10^{-14}

Multiple Correct Objective Type Questions

Directions: **Each question in this section has four suggested answers out of which ONE OR MORE answers will be correct.**

117. Identify the correct Statements among the following
(a) The photon has zero rest mass and zero spin.
(b) The deuteron has both non-zero rest mass and a non-zero spin.
(c) In a Bohr orbit the speed of the electron is less, the lower the quantum number.
(d) The angular momentum of a 2s electron has a zero value according to the Schroedinger theory.

118. Identify the correct Statements from the following
(a) Interelectron repulsion have a destabilizing effect on the (energy of an) orbital with paired electrons.
(b) The spin–angular momentum of an electron is $\frac{\sqrt{3}}{2}$ in terms of $\frac{h}{2\pi}$.
(c) In an atom, orbital angular momentum and spin angular momentum couple vectorially.
(d) Unlike the electron, the proton has a spin not equal to $\frac{1}{2}$, last equals to zero.

119. The 4 p_x orbital has
(a) 3 nodes
(b) one nodal plane
(c) zero probability for finding the electron at the centre r = 0
(d) has lower energy than the 3d orbital

Matrix-Match Type Question

Directions: **Match the elements of Column I to elements of Column II. There can be single or multiple matches.**

120.

Column I	**Column II**
(a) Presence of electrons with parallel spins	(p) The photoelectric equation of Einstein
(b) Particle nature of radiation	(q) Hund's rules
(c) Concept of work function	(r) Compton effect
(d) Partial filling of orbitals by electrons	(s) Paramagnetism

ADDITIONAL PRACTICE EXERCISE

Subjective Type Questions

121. Moseley's equation for atomic number is $\sqrt{\upsilon} = a(Z - b)$, where, υ is the frequency of X-rays, 'a' and 'b' are constants and Z is the atomic number. The wavelength of X-ray lines of Fe and Co are 0.1956 nm and 0.1789 nm respectively. Find the constant a and b of Moseley's equation.

122. (A) is an inert gas. It has equal number of protons, electrons and neutrons. A^+ is a hydrogen like ion. If A^+ emits two photons in succession to the ground state with λ = 108.5 nm and 30.4 nm, find n, the excited state quantum number.

123. In the hydrogen like lithium ion spectrum, a line in Balmer series was traced at 48 nm. What is the transition corresponding to this line and what are the wavelengths of the other intermediate transitions in the same series?

124. When Li^{2+} was exposed to light of wavelength 1550 Å it emits electrons of kinetic energy 3.2 eV. Find out the Bohr's orbit from which the electronic emission takes place.

125. An electron in hydrogen atom is excited from first to the sixth Bohr orbit. Calculate the increase in distance from the nucleus (in Å).

126. If the energy of the electron in the 4th Bohr orbit of hydrogen is – 0.85 eV, find the velocity of the electron in the ground state of Li^{2+}.

127. An electron moves with a speed of v_1 in the 4th orbit. It absorbs energy of 1.6×10^{16} J and jumps to the 6th orbit. How the new velocity v_2 of the electron in the 6th orbit is related to v_1?

128. For an electron in the third orbit of hydrogen atom, calculate its velocity.

129. When a silver plate was exposed to a light of frequency 2.1×10^{14} Hz, the emitted photo-electrons have thrice the kinetic energy as the photoelectrons emitted when the same metal was exposed to another light of frequency 1.5×10^{14} Hz. Calculate the threshold frequency for the silver metal.

130. The kinetic energies of electrons emitted are 1.95 and 0.98 eV for wavelengths 250 nm and 333 nm respectively. Find the approximate value of Planck's constant.

Straight Objective Type Questions

Directions: **This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

131. Which of the following Statements is incorrect?

(a) X-rays from an anticathode (struck by fast moving electrons) as well as radiation from an incandescent source provides a continuous spectrum.

(b) Fast moving neutrons have wave properties and can be used in diffraction experiments.

(c) The relativistic equation for the energy $E = mc^2$ of a particle and its momentum p is $E^2 = p^2c^2 + m_0{}^2c^4$ where, m_0 is the rest mass of the particle and c is the velocity of light in vacuum.

(d) Very high energy photons (say- of cosmic rays) striking a heavy metal block, like lead produce positron electron pairs.

132. The X-ray frequencies of Sn(Z = 38) and Cl (Z = 17) are respectively 3.33×10^{18} Hz and 6.34×10^{17} Hz. Calculate a and b in the equation $\upsilon^{1/2} = a(Z - b)$ given by Moseley.

(a) $a = 4.90 \times 10^7$; b = 0.75 (b) $a = 4.72 \times 10^6$; b = 0.70 (c) $a = 4.27 \times 10^8$; b = 0.65 (d) $a = 4.40 \times 10^8$; b = 0.8

133. Calculate the energy per mole of photons in joules corresponding to the violet and the red ends of the visible spectrum, viz., 4000 Å and 7000 Å respectively. $h = 6.626 \times 10^{-34}$ J s, $c = 3 \times 10^8$ m s^{-1}

(a) 2.79×10^4 , 1.61×10^4 (b) 2.85×10^4, 1.91×10^4 (c) 2.02×10^5, 1.42×10^5 (d) 2.99×10^5, 1.71×10^5

134. A certain F.M station broadcasts at wavelength equal to 3.5 m. How many photons per second correspond to transmission of one kilowatt?

(a) 2.24×10^{27} (b) 1.76×10^{28} (c) 2.26×10^{28} (d) 1.43×10^{26}

135. The mass of a photon is 5.52×10^{-32} kg. Calculate the wavelength.

(a) 0.4 Å (b) 0.7 Å (c) 1 Å (d) 0.2 Å

136. The wave length of a spectral line for an electronic transition is inversely related to

(a) the difference in energy of the levels involved in the transition.
(b) the number of electrons undergoing transition.
(c) atomic number of the element.
(d) mass number of the element.

137. An electromagnetic radiation of wavelength 242 nm is sufficient to ionize sodium atom. The ionization energy (in kJ mol^{-1}) of sodium is

(a) 495 (b) 408 (c) 506 (d) 686

138. Calculate the ratio (r) of the Rydberg constants for Li^{2+} ion (Hydrogen like) and the hydrogen atom. Take the mass ratio of Li^{2+} to H as 6.89. Mass of the hydrogen atom $\simeq 1840 \times m_e$; [i.e., electron mass]

(a) r = 1 (b) r < 1 slightly (c) r > 1 slightly (d) r = 1.1

139. In the spirit of Bohr's theory calculate the radius of B^{4+} ion.

(a) 0.1587 Å (b) 0.2645 Å (c) 0.1323 Å (d) 0.1058 Å

140. In a certain transition belonging to one of the spectral series of the hydrogen like species He^+, the following wavelengths are observed; 1642 Å, 1216 Å, ..., 1026 Å

Calculate the missing wavelength.

(a) 1086 Å (b) 1076 Å (c) 1096 Å (d) 1066 Å

141. A wavelength of 1642 Å is observed in an electronic (Balmer) transition in a hydrogen like species. Among the following identify the species.

(a) Li^{2+} (b) Be^{3+} (c) He^+ (d) B^{4+}

142. Calculate the speed of an electron, the mass (m) of which is 1.42% more than the rest mass, m_0 of the electron.

(a) $\sim 2.5 \times 10^6$ ms^{-1} (b) $\sim 1.75 \times 10^7$ ms^{-1} (c) $\sim 1.75 \times 10^6$ ms^{-1} (d) $\sim 5 \times 10^7$ ms^{-1}

143. A Bohr orbit in a hydrogen atom has a radius of 8.464 Å. How many electron i.e., transitions may occur from this orbit to the ground state?

(a) 10 (b) 3 (c) 6 (d) 15

144. The radii of two of the Bohr orbits of the hydrogen atom are 13.225 Å and 4.761 Å. Calculate the wavelength of the photon emitted when an electron "jumps" from the orbit of higher energy to that of lower energy of these two orbits.

(a) 12825 Å (b) 18525 Å (c) 15825 Å (d) 17255 Å

145. The minimum magnitude of the wavelength in the absorption spectrum of deuterium is 911.52 Å. Calculate the Rydberg constant.

(a) 109707 cm^{-1} (b) 109678 cm^{-1} (c) 109825 cm^{-1} (d) 109556 cm^{-1}

146. The wave no. values for the quantum state transition, n_2 to n_1 in the Hydrogen atom and another one electron species are 15200 cm^{-1} and 243000 cm^{-1}. Identify the species.

(a) Li^{2+} (b) He^+ (c) B^{4+} (d) Be^{3+}

147. A certain electronic transition in the hydrogen atom corresponds to photon energy of 1.89 eV. Identify the transition. Calculate the wavelength in Å for the series limit of the transition.

(a) $3 \rightarrow 1$, 912 Å (b) $2 \rightarrow 1$, 912 Å

(c) $3 \rightarrow 2$, 3648 Å (d) $4 \rightarrow 3$, 8208 Å

148. The ionization potential in eV of the outermost electron of the sodium atom is ≃ 5.44 eV. Take the sodium atom to be hydrogen like with an inner core (a nucleus plus the inner 10 electrons) and just one outermost electron. Calculate the effective nuclear charge seen by the electron.

(a) 3.8 (b) 2.6 (c) 1.9 (d) 1.2

149. The energy emitted when electrons of 2.2 g hydrogen like lithium ion undergoes 4 to 2 transition [Relative atomic weight of Li = 7] is

(a) 77 kJ (b) 696 kJ (c) 23 kJ (d) 231 kJ

150. The uncertainty in position of a particle is $\sqrt{\frac{h}{4\pi m}}$. We can say that

(a) Uncertainty in position is twice that of momentum.

(b) Only position of particle is altered by impact of light.

(c) Uncertainty in momentum is very large compared to uncertainty in position.

(d) Uncertainty in position and velocity are equal.

151. Show by what factor the speed of the electron in a Bohr orbit will change if the value of the principal quantum number, n is doubled

(a) no change (b) 2 (c) $\frac{1}{\sqrt{2}}$ (d) $\frac{1}{2}$

152. Calculate the number of revolutions per second in the third orbit of the Li^{2+} ion.

(a) 1.29×10^{14} (b) 1.92×10^{16} (c) 2.19×10^{15} (d) 1.22×10^{14}

153. Calculate $\frac{\upsilon}{c}$ where c is the velocity of light in vacuum and υ is the speed of the electron in the second Bohr orbit of the Hydrogen atom.

(a) $\sim 3.35 \times 10^{-4}$ (b) $\sim 3.85 \times 10^{-4}$ (c) $\sim 5.33 \times 10^{-3}$ (d) $\sim 3.65 \times 10^{-3}$

154. Photo electrons with maximum kinetic energy 5×10^{-18} J are emitted in a certain experiment when radiation of frequency γ is used. If $\gamma = 4\upsilon_0$, calculate the value of υ_0. ($h = 6.626 \times 10^{-34}$ J sec)

(a) $\sim 1.25 \times 10^{16}$ (b) $\sim 1.5 \times 10^{14}$ (c) $\sim 2.5 \times 10^{15}\ s^{-1}$ (d) $\sim 1.75 \times 10^{14}$

155. The work function of a certain metal is 4.3 eV. Calculate the threshold frequency υ_0 for photoelectric emission.

(a) 1.308×10^{14} (b) 1.380×10^{16} (c) 1.830×10^{16} (d) 1.038×10^{15}

156. Calculate the momentum of the photon in kg ms^{-1} with wavelength $\lambda = 5890$ Å.

(a) 1.125×10^{-27} (b) 2.115×10^{-26} (c) 2.511×10^{-27} (d) 1.512×10^{-26}

157. Express in Å, the wave length λ of a photon with energy equal to 10.2 eV.

(a) 1128 (b) 1218 (c) 2118 (d) 1812

158. An electron initially accelerated through 100 volts is then retarded through 56 volts. If before and after retardation its de Broglie wavelengths are λ_1 and λ_2, calculate the ratio, $\frac{\lambda_2}{\lambda_1}$ approximately

(a) 1.25 (b) 1.75 (c) 1.5 (d) 1.7

159. If the de Broglie wavelength of an electron is 6 Å, what is its speed?
$m_e = 9.11 \times 10^{-31}$ kg

(a) 1.212×10^6 m s^{-1} (b) 2.112×10^5 m s^{-1} (c) 2.211×10^6 m s^{-1} (d) 1.102×10^5 m s^{-1}

160. Calculate the wavelength, λ of monochromatic radiation, given that 2×10^{18} photons provide one Joule of energy.
(a) ~4000 Å (b) ~5000 Å (c) ~5500 Å (d) ~6000 Å

161. Calculate the de Broglie wavelength of a He atom at 27°C. Assume the molar mass to be 4.00×10^{-3} kg.
(a) $\sim 7.92 \times 10^{-10}$ m (b) $\sim 7.24 \times 10^{-11}$ m (c) $\sim 7.62 \times 10^{-12}$ m (d) $\sim 7.82 \times 10^{-12}$ m

162. If the uncertainty in the velocity of an electron in the 3rd orbit of He^+ ion is 0.02%, the uncertainty in its position is
(a) 88 nm (b) 132 nm (c) 176 nm (d) 198 nm

163. A certain excited state in an atomic species with an energy E relative to the ground state has a mean life time $\Delta t \simeq 10^{-8}$ sec. Using $\Delta E.\, \Delta t \geq \frac{h}{2\pi}$ calculate approximately the minimum uncertainty, ΔE (actually an error) in the measurement of the energy, E ,of the excited state.
(a) 1.5064×10^{-34} J (b) 1.0546×10^{-26} J (c) 1.5640×10^{-28} J (d) 1.5640×10^{-34} J

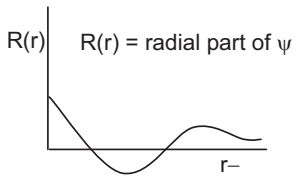
164. A particle is confined to a one dimensional trap (X axis) of extension 10 Å. Calculate the minimum uncertainty in its momentum along the X axis. $h = 6.626 \times 10^{-34}$ J s (answer in kg m s^{-1}).
(a) 1.165×10^{-24} (b) 1.156×10^{-26} (c) 5.275×10^{-26} (d) 1.166×10^{-26}

165. What is the angular momentum of the electron in the 4f orbital of a one-electron species according to wave mechanics?
(a) $\sqrt{3} \times \frac{h}{\pi}$ (b) $2\frac{h}{\pi}$ (c) $\sqrt{\frac{3}{2}}\frac{h}{\pi}$ (d) $\frac{1}{\sqrt{2}}\frac{h}{\pi}$

166. What is the maximum angle made by the angular momentum vector of a 3d- electron with the positive direction of the Z axis. The angle given by cos θ is
(a) $\sqrt{\frac{2}{3}}$ (b) $-\sqrt{\frac{2}{3}}$ (c) $\frac{1}{\sqrt{2}}$ (d) $-\frac{1}{\sqrt{2}}$

167. The radial distribution plot: $4\pi r^2 dr|\psi(r)|^2$ (on Y axis) against r (on the axis of X) has a single maximum. Which of the following orbitals does not fit into this description?
(a) 3d (b) 3p (c) 1s (d) 2p

168. The following is the plot of the radial part of an atomic orbital of the Hydrogen atom. Suggest the designation of the orbital



(a) 2s (b) 3p (c) 4p (d) 4d

169. Indicate among the following the incorrect Statement.
(a) The $3d_{z^2}$ orbital has nodes but not nodal planes
(b) The $2p_x$, $2p_y$ and $2p_z$ orbitals correspond to the magnetic quantum numbers, m = +1, m = −1 and m =0.
(c) In the hydrogen atom the 2s, $2p_x$, $2p_y$ and $2p_z$ orbitals are all degenerate.
(d) The orbital angular momentum of an s electron is zero according to wave mechanics (Schroedinger).

170. A transition metal X has configuration $[Ar]3d^4$ in its +3 oxidation state. Its atomic number is
(a) 19 (b) 27 (c) 25 (d) 26

Assertion–Reason Type Questions

Directions: **Each question contains Statement-1 and Statement-2 and has the following choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False
(d) Statement-1 is False, Statement-2 is True

171. Statement 1
The significant observation in Rutherford's experiment on the Scattering of alpha particles by gold foil was that quite a few alpha particles were scattered through unexpectedly large angles.
and

Statement 2
According to the classical electromagnetic theory, Rutherford's atom model could not "exist".

172. Statement 1
In a polyelectronic atom any electron is shielded from the attraction of the nucleus by every other electron.
and

Statement 2
Bohr's theory could not be extended to polyelectronic atoms.

173. Statement 1
The radius of a Bohr orbit of the hydrogen atom increases as the square of its quantum number.
and

Statement 2
Both Bohr's theory and its extension by Sommerfeld did not imply the wave nature of the electron.

174. Statement 1
The ionization energy values for the atomic orbitals of the hydrogen atom get closer and closer for higher and higher values of the quantum numbers in Bohr's theory.
and

Statement 2
The intensities of spectral lines formed an integral part of Bohr's original theory.

175. Statement 1
Every excited atomic state has a mean life Δt, of the order of 10^{-8} sec or less.
and

Statement 2
This is a consequence of the uncertainty principle.

176. Statement 1
The Schroedinger equation , $\nabla^2\psi + \frac{2\mu}{\hbar^2}(E - V)\psi = 0$ is a non-relativistic equation.

and

Statement 2
The Schroedinger equation given above provides the four quantum numbers, n, l, m_ℓ and m_s for each electron orbital.

177. Statement 1
A beam of neutrons (monoenergetic) does not exhibit wave properties because (unlike electrons) neutrons have no charge.
and

Statement 2
The photon has a nonzero spin but has neither charge nor rest mass.

178. Statement 1
The 1s orbital of hydrogen atom has no radial node.
and

Statement 2
The 1s orbital of hydrogen atom is smaller in size than its 2s orbital.

179. Statement 1
The atomic orbital $\psi(r, \theta, \phi)$ of an electron is physically measurable/observable, just like its energy value.
and

Statement 2
Energy levels of orbitals are changed by interelectronic repulsions.

180. Statement 1
The $2p_x$, $2p_y$ and $2p_z$ atomic orbitals are necessarily degenerate but the 2s orbital is accidentally degenerate with them in the hydrogen atom.
and

Statement 2
The symmetry of the 2s orbital is different from that of the 2p orbital and hence it is differently affected by the potential.



Linked Comprehension Type Questions

Directions: **This section contains 3 paragraphs. Based upon the paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

Passage I

According to Bohr's theory successive energy levels in the Hydrogen like (one electron) species vary according to $\left(-13.6\times\frac{Z^2}{n^2}\right)$ eV where, Z is the atomic number and n is the quantum number of the Bohr orbit. The radius of the orbit = $\left(0.529\times\frac{n^2}{Z}\right)$ Å. Bohr's formula for the wave number, $\left(\frac{1}{\lambda}\right)(cm^{-1})$ of a spectral line is $\frac{1}{\lambda}=RZ^2\left(\frac{1}{n_1^{\,2}}-\frac{1}{n_2^{\,2}}\right)$ where, R is known as the Rydberg constant and an electron transition occurs from the higher quantum state n_2 to the lower quantum state n_1. For n_1 = 1, 2, 3,.....we have a series of spectral lines: Lyman, Balmer, Paschen, Brackett etc. When $n_2 \to \infty$ we have the series limit. We may use the simplified formula $\lambda(Å) = \frac{912}{Z^2}\times\frac{n_2^{\,2}n_1^{\,2}}{n_2^{\,2}-n_1^{\,2}}$ and for the series limit $n_2 \to \infty$, $\lambda(Å) = \frac{912}{Z^2}\times n_1^{\,2}$.

181 Three successive lines in the spectrum of the hydrogen atom have λ values 18746 Å, 12815 Å, 10935 Å. Identify the series by its n_1 value.
(a) 2 (b) 4 (c) 1 (d) 3

182. For the Bohr orbits of the hydrogen atom of quantum numbers, n_1 and n_2, the difference in ionization energy = 2.55 eV. Identify n_1 and n_2. ($n_2 > n_1$).

(a) 2, 4 (b) 2, 3 (c) 1, 3 (d) 1, 4

183. What according to the Bohr model is the radius of the C^{5+} ion?

(a) 0.0529 Å (b) 0.1058 Å (c) 0.0882 Å (d) 0.0353

Passage II

The wave nature of the electron was proposed by de Broglie in his celebrated equation, $\lambda = \frac{h}{mv}$ where, λ is the wavelength of the electron, m is the electron mass an v is its speed. The equation was experimentally verified by Davisson and Germer and independently by G.P.Thomson. After substitution of relevant values and simplifying it turned out that $\lambda(\text{Å}) = \frac{12.225}{V^{1/2}}$ where V in volts is the potential difference through which the free electron is accelerated.

184. What is the de Broglie wavelength ratio for an electron in the second orbit and the third orbit of the Hydrogen atom in the spirit of the Bohr theory?

(a) $\frac{2}{3}$ (b) $\frac{4}{9}$ (c) $\frac{1}{2}$ (d) $\frac{3}{2}$

185. An electron initially accelerated through 81 volts is retarded then by 32 volts. Calculate the ratio of the de Broglie wavelengths in the two situations before and after retardation.

(a) $\frac{49}{81}$ (b) $\frac{32}{49}$ (c) $\frac{32}{81}$ (d) $\frac{7}{9}$

186. Suppose an electron is confined to the interior of an atomic nucleus ~10^{-12} cm

(i) What may be its momentum according to the de Broglie theory?

(ii) Can we assume that the mass of the electron = 9.11×10^{-28} g = m_0 (rest mass)

(a) (i) 6.626×10^{-12} kg m s^{-1}, (ii) No (b) (i) 6.626×10^{-15} g cm s^{-1} (ii) No

(c) (i) 6.626×10^{-15} g cm s^{-1}, (ii) Yes (d) (i) 6.626×10^{-12} kg m s^{-1} (ii) Yes

Passage III

The equation of Schroedinger for the hydrogen atom was $\frac{\partial^2\Psi}{\partial x^2}+\frac{\partial^2\Psi}{\partial y^2}+\frac{\partial^2\Psi}{\partial z^2}+\frac{8\pi^2\mu}{h^2}\left(E+\frac{e^2}{r}\right)\Psi = 0$ [in cgs unit] $r = \left(x^2+y^2+z^2\right)^{1/2}$

On solving the equation one finds the functional form of ψ known as the atomic orbital. The solution is usually done in terms of the spherical polar coordinates r, θ and ϕ, with the relationship $x = r \sin\theta \cos\phi$, $y = r \sin\theta \sin\phi$, $z = r \cos\theta$. Each atomic orbital function ψ corresponds to a value E (total energy) of the system. The same energy may correspond to several orbital functions which are then said to be degenerate. Each orbital has its characteristic symmetry exhibited by nodes (planes, spheres etc). Each (Schroedinger) orbital also has three quantum numbers.

n : the principal quantum number signifying the energy

ℓ: the second (azimuthal) quantum number for the angular momentum

m_ℓ : the magnetic quantum number standing for the Z-component of the angular momentum. The angular momentum is a vector having the magnitude $\sqrt{\ell(\ell+1)}\frac{h}{2\pi}$ .The magnetic quantum number m_ℓ is an integer with values $-\ell$, ...0, $+\ell$ for each value of ℓ.

187. Which of the following orbitals has neither nodal planes nor nodal spheres?

(a) $3p_x$ (b) 2s (c) $3d_{z^2}$ (d) $2p_z$

188. Which orbital may be assigned the value $m_\ell = 0$?

(a) $2p_y$ (b) $2p_z$

(c) $2p_x$ (d) all the orbitals ($2p_x$, $2p_y$, $2p_z$)

189. Point out the incorrect Statement

(a) Any s orbital (for any principal quantum number) is independent of θ and ϕ.

(b) The 3s orbital has two nodes both spherical.

(c) ℓ and m_ℓ are related as $\left[\sqrt{\ell(\ell+1)}\right](\cos\theta)\frac{h}{2\pi} = m_\ell \frac{h}{2\pi}$.

(d) A free electron also has the so called spin quantum state.

Multiple Correct Objective Type Questions

Directions: **Each question in this section has four suggested answers out of which ONE OR MORE answers will be correct.**

190. In the Bohr theory of the hydrogen atom the electron transition from n = 5 to lower levels

(a) can theoretically yield 20 wavelength.

(b) may yield 10 spectral lines.

(c) may yield 3 lines in the visible part of the spectrum.

(d) may yield 3 lines in the infrared region of the spectrum.

191. Identify the correct Statement/s among the following.

(a) Bohr's theory introduced only one quantum number because it dealt with only circular orbits.

(b) Bohr's theory did not consider intensities of spectral lines.

(c) Bohr's theory (and Sommerfield's extension of it) both accounted for the fine structure of spectral lines.

(d) According to Bohr's theory all Hydrogen like (one electron) species have the same Rydberg constant

192. The particle nature of radiation is brought out

(a) in the photoelectric effect.

(b) in the Compton Scattering experiment.

(c) in Young's double slit interference experiment.

(d) in using unpolarized monochromatic light and a nicol prism thus obtaining plane polarized light.

193. The extension of Bohr's theory by Sommerfeld

(a) introduced elliptic orbits.

(b) quantized both the radial momentum and the angular momentum.

(c) empirically introduced selection rules for electronic transitions.

(d) introduced the azimuthal quantum number k which had the values 0, 1,....(n − 1) for the principal quantum number n.

194. The correct formulations of the uncertainty relations are

(a) $\Delta z \times \Delta p_z \geq \frac{h}{4\pi}$ (b) $\Delta E \times \Delta t \geq \frac{h}{4\pi}$ (c) $\Delta x \times \Delta p_y \geq \frac{h}{4\pi}$ (d) $\Delta p_y \times \Delta p_z \geq \frac{h}{4\pi}$

195. In the orbits of the Bohr theory a lower quantum number is associated with

(a) a lower energy .

(b) a lower angular momentum for the electron.

(c) closer energy levels.

(d) lower orbital speed of the electron.

196. The orbital angular momentum vector for a single electron in a d-orbital may have the component value

(a) $\sqrt{6}\times\frac{h}{2\pi}$ along the z axis

(b) $\frac{h}{\pi}$ along the z axis

(c) zero along the z axis

(d) $-\sqrt{6}\times\frac{h}{2\pi}$ along the z axis

197. The p-orbitals for any principal quantum number n (as obtained by solving the wave equation for a one electron species)

(a) are always three in number.
(b) always have only one nodal plane.
(c) are (always) stabilized only by the presence of unpaired electrons in them.
(d) have in different situations magnetic moments exceeding 4 BM (Bohr magnetons)

Matrix-Match Type Questions

Directions: **Match the elements of Column I to elements of Column II. There can be single or multiple matches.**

198.

Column I	**Column II**
(a) Electron transition from a quantum number n > 1 to n = 1 in Bohr's theory of the hydrogen spectrum	(p) $\frac{n(n-1)}{2}$ possible wavelength
(b) Time, T for one revolution of the electron in a circular Bohr orbit	(q) Proportional to n^3
(c) wavelength anticipated in the visible part of the spectrum when the excited electron is in 7th or higher level.	(r) Lyman series
(d) ultra violet part of the spectrum	(s) $\lambda < 4000$ Å

199.

Column I	**Column II**
(a) elliptic orbits	(p) Sommerfield's extension of Bohr's theory
(b) azimuthal quantum number	(q) electron treated as a particle
(c) the ψ function	(r) Schroedinger's equation for the H–atom
(d) concept of orbital degeneracy	(s) product of both radial and angular functions

200.

Column I	**Column II**
(a) 1s orbital of the H-atom	(p) plot of radial probability function vs r (distance from the centre) is maximum at $r = a_0 = 0.529$ Å
(b) $(4\pi r^2) \exp\left(\frac{-2r}{a_0}\right)$	(q) no radial node
(c) $2p_x$, $2p_y$, $2p_z$ orbitals	(r) degenerate
(d) $3d_{z^2}$ orbital	(s) nodal cones

SOLUTIONS

ANSWER KEYS

1. 1.42%
2. $n_1 = 2; n_2 = 5$
3. 43478 cm^{-1}
4. $7 \rightarrow 2$ and $8 \rightarrow 2$
5. 0.132 Å
6. (i) $n_1 = 2; n_2 = 4$
(ii) 13.6, 3.4, 1.51, 0.85 (eV)….
7. 0.306 eV
8. 2.42×10^{-35} kg
9. 55.56 volts
10. 5.273×10^{-25} kgms^{-1}

11.	(a)	**12.**	(b)	**13.**	(a)
14.	(c)	**15.**	(c)	**16.**	(a)
17.	(c)	**18.**	(b)	**19.**	(b)
20.	(c)	**21.**	(c)	**22.**	(a)
23.	(b)	**24.**	(b)	**25.**	(a)

26. (d)
27. (a), (b), (c)
28. (a), (b), (d)
29. (c), (d)
30. (a) → (p), (s)
(b) → (p), (s)
(c) → (q)
(d) → (r)

31.	(c)	**32.**	(c)	**33.**	(a)
34.	(d)	**35.**	(a)	**36.**	(b)
37.	(a)	**38.**	(d)	**39.**	(c)
40.	(a)	**41.**	(b)	**42.**	(a)
43.	(a)	**44.**	(a)	**45.**	(a)
46.	(c)	**47.**	(b)	**48.**	(d)
49.	(c)	**50.**	(d)	**51.**	(b)
52.	(b)	**53.**	(c)	**54.**	(a)
55.	(a)	**56.**	(a)	**57.**	(a)
58.	(b)	**59.**	(d)	**60.**	(c)
61.	(b)	**62.**	(b)	**63.**	(a)
64.	(b)	**65.**	(b)	**66.**	(d)
67.	(c)	**68.**	(d)	**69.**	(b)

70.	(a)	**71.**	(c)	**72.**	(d)
73.	(a)	**74**	(b)	**75.**	(d)
76.	(a)	**77.**	(c)	**78.**	(c)
79.	(b)	**80.**	(c)	**81.**	(c)
82.	(a)	**83.**	(b)	**84.**	(c)
85.	(b)	**86.**	(c)	**87.**	(d)
88.	(d)	**89.**	(a)	**90.**	(b)
91.	(c)	**92.**	(d)	**93.**	(a)
94.	(a)	**95.**	(d)	**96.**	(c)
97.	(d)	**98.**	(b)	**99.**	(d)
100.	(b)	**101.**	(b)	**102.**	(d)
103.	(c)	**104.**	(a)	**105.**	(b)
106.	(c)	**107.**	(c)	**108.**	(b)
109.	(d)	**110.**	(a)	**111.**	(b)
112.	(d)	**113.**	(a)	**114.**	(d)
115.	(c)	**116.**	(b)		

117. (b), (d)
118. (a), (b), (c)
119. (a), (b), (c)
120. (a) → (q), (s)
(b) → (p), (r)
(c) → (p)
(d) → (q), (s)
121. $a = 5.6 \times 10^7$, b = 3.893
122. n = 5
123. $5 \rightarrow 2$, $\lambda_{3 \rightarrow 2} = 73$ nm, $\lambda_{4 \rightarrow 2} = 54$ nm
124. 5th orbit
125. 18.55 Å
126. 6.6×10^6 ms^{-1}
127. $V_2 = \frac{2}{3} V_1$
128. 7.3×10^5 ms^{-1}
129. 1.2×10^{14} Hz
130. 5.17×10^{-34} Js

131.	(a)	**132.**	(a)	**133.**	(d)
134.	(b)	**135.**	(a)	**136.**	(a)
137.	(a)	**138.**	(c)	**139.**	(d)

140.	(a)	**141.**	(c)	**142.**	(d)
143.	(c)	**144.**	(a)	**145.**	(a)
146.	(d)	**147.**	(c)	**148.**	(c)
149.	(b)	**150.**	(d)	**151.**	(d)
152.	(c)	**153.**	(d)	**154.**	(c)
155.	(d)	**156.**	(a)	**157.**	(b)
158.	(c)	**159.**	(a)	**160.**	(a)
161.	(b)	**162.**	(d)	**163.**	(b)
164.	(c)	**165.**	(a)	**166.**	(a)
167.	(b)	**168.**	(c)	**169.**	(b)
170.	(c)	**171.**	(b)	**172.**	(d)
173.	(b)	**174.**	(c)	**175.**	(a)
176.	(c)	**177.**	(d)	**178.**	(b)
179.	(d)	**180.**	(a)	**181.**	(d)
182.	(a)	**183.**	(c)	**184.**	(a)
185.	(d)	**186.**	(b)	**187.**	(c)

188. (b)
189. (d)
190. (b), (c), (d)
191. (a), (b)
192. (a), (b), (c)
193. (a), (b), (c)
194. (a), (b)
195. (a), (b)
196. (b), (c)
197. (a), (b)
198. (a) →(p), (r), (s)
(b) → (q)
(c) → (p), (s)
(d) → (r), (s)
199. (a) → (p), (q)
(b) → (p), (r)
(c) → (r), (s)
(d) → (r)
200. (a) → (p), (q)
(b) → (p), (q)
(c) → (q), (r)
(d) → (q), (s)

HINTS AND EXPLANATIONS

Topic Grip

1. At very low speed $m = m_0$. For the greater speed one may use the formula $m = \dfrac{m_0}{\sqrt{1 - v^2/c^2}}$

i.e., $\left(\dfrac{m}{m_0}\right) = \dfrac{1}{\left(1 - v^2/c^2\right)^{1/2}}$

$= \dfrac{1}{\left(1 - \dfrac{25 \times 10^{14}}{9 \times 10^{16}}\right)^{1/2}}$

$= \dfrac{1}{0.9860} = 1.014185$

$\therefore \quad \left(\dfrac{m - m_0}{m_0}\right) \times 100 = 1.4185\ \% \simeq 1.42\%$

2. $1086\text{Å} = \dfrac{912}{2^2} \times \dfrac{n_2^2 n_1^2}{n_2^2 - n_1^2}$ Å

$\therefore \quad \dfrac{n_2^2 n_1^2}{n_2^2 - n_1^2} = 4.763$

obviously the corresponding transition in the hydrogen atom for the same n_2 and n_1 would have occurred at 4344Å

$\left[4344 = \dfrac{912}{1^2} \times \dfrac{n_2^2 n_1^2}{n_2^2 - n_1^2}\right.$

$\therefore \quad \dfrac{n_2^2 n_1^2}{n_2^2 - n_1^2} = \dfrac{4343}{912} = 4.763\Big]$

Obviously too it is a Balmer transition i.e., $n_1 = 2$

$\therefore \quad 4.763 = \dfrac{n_2^2 \times 4}{n_2^2 - 4}$

By trial and error $n_2 = 5$

i.e., $\dfrac{5^2 \times 4}{5^2 - 4} = \dfrac{100}{21} = 4.762$

The transition is from $n_2 = 5$ to $n_1 = 2$

3. $30952\ \text{cm}^{-1} = \upsilon_\infty - \dfrac{109722}{(2.9596)^2} = \upsilon_\infty - 12526.44$

$\therefore \quad \upsilon_\infty = 43478\ \text{cm}^{-1}$

4. The λ values given are almost equal to 4000Å. i.e., the violet end of the visible spectrum

$\therefore$ one may try the formula for a Balmer transition $(n_1 = 2)$

i.e., $3971 = \dfrac{912}{1^2} \times \dfrac{n_2^2 \times 4}{n_2^2 - 4}$ (Å)

i.e., $4.354 = \dfrac{4n_2^2}{n_2^2 - 4}$

solving $4.354\, n_2^2 - 4n_2^2 = 4 \times 4.354$

$\therefore \quad n_2^2 = \dfrac{4 \times 4.354}{0.354}$

$n_2^2 \simeq 49$

$\therefore \quad n_2 = 7$ similarly, $\dfrac{3890}{912} = 4.264$

$\therefore \quad n_3^2 = \dfrac{4 \times 4.264}{0.264} = 64.6$

$\therefore \quad n_3 \approx 8$

The transitions are $(7 \rightarrow 2)$ and $(8 \rightarrow 2)$ quantum transition

5. One uses the formula, radius, $r = r_0 \times \dfrac{n^2}{Z}$ where r_0 = 0.529 Å. Be^{3+} ion is a one electron species and the electron is in the 1s orbital

i.e., $n = 1$

Be^{3+} ion has $Z = 4$

$\therefore \quad r = 0.529 \times \dfrac{1}{4}\text{Å} \simeq 0.132\text{Å}$

6. The ionization energies (eV) are $\dfrac{13.6}{n^2}$

i.e., 13.6, 3.4, 1.51, 0.85, 0.544,….

For getting 2.55 eV as the difference we observe that (3.4 − 0.85)eV = 2.55 eV

i.e., $n_1 = 2$, $n_2 = 4$

7. $mvr = \dfrac{nh}{2\pi}$

$n = \dfrac{5.281 \times 10^{-34}\ \text{kg m}^2\ \text{s}^{-1} \times 2 \times 3.14}{6.626 \times 10^{-34}\ \text{Js}} = 5$

The transition is from 5th to 4th level.

$$E = \frac{-13.6Z^2}{n^2} = 13.6\left(\frac{1}{4^2} - \frac{1}{5^2}\right) = 0.306\,eV$$

8. The series limit has the wave length,

$\lambda = \frac{912}{1^2} \times 1^2 Å$

Since $h\upsilon = \frac{hc}{\lambda} = mc^2$ $\qquad \frac{h}{\lambda} = mc$

$\therefore \quad m = h/\lambda c$

$\therefore \quad m(kg) = \dfrac{6.626 \times 10^{-34}}{\left(3 \times 10^8\right)\left(912 \times 10^{-10}\right)}$

$= 2.4218 \times 10^{-3} \times 10^{-32}$

$= 2.4218 \times 10^{-35}$ kg

Alternatively, the series limit corresponds to 13.6 eV

i.e., $13.6 \times 1.6 \times 10^{-19}$ Joule = Equating this to mc^2

$$m = \frac{13.6 \times 1.6 \times 10^{-19}}{\left(3 \times 10^8\right)^2} = 2.4178 \times 10^{-35}\ kg$$

$\therefore \quad 2.42 \times 10^{-35}$ kg

9. Initially the de Broglie wavelength

$$\lambda(Å) = \frac{12.225}{\sqrt{100}} = \frac{12.225}{10}$$

After deceleration the wavelength is

$$\frac{12.225}{10} \times \frac{3}{2} = \frac{12.225}{\sqrt{v}}$$

$$\therefore \quad \frac{3}{20} = \frac{1}{v^{1/2}} \text{ or } v^{1/2} = \frac{20}{3}$$

$$\therefore \quad v = \frac{400}{9} = 44.44v$$

$\therefore$ Decelerating voltage = (100 − 44.44) volts

= 55.56 volts

10. One may use the formula $\Delta x.\Delta p_x \simeq h/4\pi$.

In cgs units $\Delta x = (1Å) = 10^{-8}$ cm

$h = 6.626 \times 10^{-27}$ ergs

$$\therefore \quad 10^{-8} \times \Delta p_x = \frac{6.626 \times 10^{-27}}{4 \times 3.1416} \text{ ergs}$$

$$\therefore \quad \Delta p_x = \frac{6.626 \times 10^{-27}}{4 \times 3.1416 \times 10^{-8}}$$

$= 5.273 \times 10^{-25}$ kg m s^{-1}

11. (a)

12. $IP_{He^+} = Z^2 \times 13.6$ eV

$= 4 \times 13.6$ eV $= 54.4$ eV $= 8.7 \times 10^{-18}$ J

$IP_{Li^{2+}} = (3)^2 \times 13.6$ eV

$= 9 \times 13.6$ eV $= 122.4$ eV $= 1.96 \times 10^{-17}$ J

13. For $\lambda = 100$ nm, energy $h\upsilon = \frac{hc}{\lambda}$

$$= \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{100 \times 10^{-9}} \text{ J which equals}$$

19.878×10^{-19} J

For the de Broglie wave length of 4.2 Å the equivalent accelerating voltage

$$V = \left(\frac{12.225}{4.2}\right)^2 \text{ volts} \equiv \frac{150}{17.64} \times 1.6 \times 10^{-19}\ J$$

which equals 13.6×10^{-19} J

using K.E (J) = 13.6×10^{-19}

$= 19.878 \times 10^{-19} - h\upsilon_0$

we get $h\upsilon_0 = 6.278 \times 10^{-19}$ J

$\therefore \quad 6.626 \times 10^{-34}$ (J s) $\times \upsilon_0$ (s^{-1}) $= 6.278 \times 10^{-19}$ J

$$\therefore \quad \upsilon_0 = \frac{6.278 \times 10^{-19}}{6.626 \times 10^{-34}}\ s^{-1}$$

which is $0.9475 \times 10^{15} s^{-1} = 9.475 \times 10^{14}\ s^{-1}$

14. Energy 1 MeV = 10^6 eV = $10^6 \times 1.6 \times 10^{-19}$ J

$= 1.6 \times 10^{-13}$ J

mass 1.009u = 1.661×10^{-27} kg $\qquad E = \frac{p^2}{2m}$

$\therefore \quad p^2 = 2mE$

$= (2 \times 1.661 \times 10^{-27} \times 1.6 \times 10^{-13})\left(\text{kg m s}^{-1}\right)^2$

$= 5.3152 \times 10^{-40}\left(\text{kg m s}^{-1}\right)^2$

$\therefore \quad p = 2.3055 \times 10^{-20}$ kg m s^{-1}

de Broglie wave length $\lambda = h/p$

$$= \frac{6.626 \times 10^{-34}}{2.3055 \times 10^{-20}}\ m = 2.874 \times 10^{-14}\ m$$

$= 2.874 \times 10^{-14} \times 10^{10}$ Å

$= 2.874 \times 10^{-4}$ Å

15. $PE_{(\text{at cathode})} = KE_{(\text{at anode})}$

$$V_q = \frac{1}{2}mv^2 \quad v = \sqrt{\frac{2V_q}{m}}$$

$$\lambda = \frac{h}{mv} = \frac{h}{m\sqrt{2\frac{V_q}{m}}} = \frac{h}{\sqrt{2mV_q}}$$

$$\frac{\lambda_1}{\lambda_2} = \sqrt{\frac{V_2}{V_1}} = \sqrt{\frac{10000}{100}} = 10$$

16. (a)

17. (c)

18. (b)

19. (b)

20. (c)

21. $\frac{8208}{912} = 9 = 3^2 \qquad \therefore\ n_1 = 3$

22. $\frac{22800}{912} = 25 = 5^2 \qquad \therefore\ n_2 = 5$

$(n_2 + 1) = 6(n_2 - 1) = 4$

$\therefore \quad \lambda Å = 912 \times \frac{6^2 \times 4^2}{6^2 - 4^2} = 912 \times \frac{36 \times 16}{20}$

$= 26266Å$

23. $1642Å = \frac{912}{4} \times \frac{n_2^2 n_1^2}{n_2^2 - n_1^2}$

$\therefore \quad 7.202 = \frac{n_2^2 n_1^2}{n_2^2 - n_1^2}$

By trial and error, it is easy to get $n_2 = 3$ and $n_1 = 2$

$(n_2 + 1) = 4, (n_1 - 1) = 1$

$\therefore \quad \lambda Å = \frac{912}{4} \times \frac{4^2 \times 1^2}{4^2 - 1^2} = 228 \times \frac{16}{15} \simeq 243Å$

24. $2n^2 = 18$

25. $\frac{6 \times 5 \times 4}{3!} = 5 \times 4 = 20$

26. (d) showing one radial node

27. (a), (b), (c) are correct

28. (a), (b), (d) are correct

29. (c), (d) are correct

30. (a) $\rightarrow$ p, s
(b) $\rightarrow$ p, s
(c) $\rightarrow$ q
(d) $\rightarrow$ r

IIT Assignment Exercise

31. By Daltons theory atoms are indestructible, changing mass into energy was later added based on Einstein's theory.

32. It is very difficult to remove a proton from the nucleus compared to an electron from an atom.

33. This is ratio of mass of proton:electron. m_e is $1/1837$ times mass of proton.

34. $\frac{e}{m} = 4.2 \times 10^6 C\ kg^{-1} \qquad e = 1.6 \times 10^{-19} C$

$m = \frac{e}{e/m} = \frac{1.6 \times 10^{-19}}{4.2 \times 10^6} = 3.8 \times 10^{-26} kg$

mass of one mole $= 3.8 \times 10^{-26} \times 6.02 \times 10^{23}$

$= 0.023\ kg = 23\ g$

$\therefore$ The metal ion may be Na^+

35. Moseley's equation $\sqrt{\upsilon} = a(z - b)$

$a = 1, b = 1$ and $z = 3$

$\sqrt{\upsilon} = 1(3 - 1)$

$\upsilon = 2^2 = 4\ s^{-1}$

36. difference between number of neutrons and protons is the same

$^{19}_{9}F \quad 10 - 9 = 1$

$^{35}_{17}Cl \quad 18 - 17 = 1$

37. PH_4^+

neutrons in phosphorus $= 31 - 15 = 16$

neutrons in hydrogen $= 0$

38. Rutherford's atom was neutral

number of protons in the nucleus was equal to number of electrons

39. $\frac{1}{\lambda} = \frac{1}{\lambda_1} + \frac{1}{\lambda_2}$

$= \frac{1}{400} + \frac{1}{800} = \frac{3}{800} = \frac{1}{267}$

$\lambda = 267Å$

40. They travel with speed much less than that of light

41. $E_1 = \frac{hc}{4 \times 10^{-6}} \qquad E_2 = \frac{hc}{8 \times 10^{-4}}$

$E_1 - E_2 = \frac{hc}{4 \times 10^{-6}}\left[1 - \frac{1}{2 \times 10^2}\right]$

$$= \frac{6.6\times10^{-34}\times3\times10^{8}}{4\times10^{-6}}\times\frac{199}{200}$$

$$= 4.9\times10^{-20}\ \text{J}$$

42. Orbital angular momentum = $\sqrt{\ell(\ell+1)}\,\frac{h}{2\pi}$

since $\ell = 0$, orbital angular momentum is also zero

43. $E = \frac{nhc}{\lambda}$

$E = 60\ \text{J s}^{-1}$

So $60 = \frac{n\times6.6\times10^{-34}\times3\times10^{8}}{500\times10^{-9}}$

$n = \frac{60\times500\times10^{-9}}{6.6\times10^{-34}\times3\times10^{8}} = 1.5\times10^{20}$ photons

1.5×10^{20} photons $\rightarrow$ 1 s

$\therefore\quad 6\times10^{23}$ photons $\rightarrow \frac{6.0\times10^{23}}{1.5\times10^{20}}\times1$

$= 4000$ s

44. $E_1 > E_2$ and $E = E_1 + E_2$

45. For Brackett series $n_1 = 4$

least energy line will have longest wavelength

so $n_2 = 5$

46. Smallest wavelength–highest energy so for Balmer series $n_1 = 2$ and $n_2 = \infty$

$$R_H\times Z^2\left[\frac{1}{2^2}-\frac{1}{\infty}\right] = R_H\times Z_1{}^2\left[\frac{1}{n_1{}^2}-\frac{1}{\infty}\right]$$

Comparing Balmer series of helium and Lymann series of hydrogen

$$R_H\times2^2\left[\frac{1}{2^2}\right] = R_H\times1^2\left[\frac{1}{1^2}-\frac{1}{\infty^2}\right]$$

47. $\frac{1}{\lambda} = R_H.3^2\left[\frac{1}{1^2}-\frac{1}{\infty^2}\right]$ $\qquad R_H = \frac{1}{9x}$

shortest wavelength in Lymann series of Li^{2+}

(a) $\frac{1}{\lambda} = R_H\times3^2\left[\frac{1}{2^2}-\frac{1}{3^2}\right]$

$\frac{1}{\lambda} = \frac{1}{9x}\times9\times\frac{5}{36} = \frac{5}{36x}$ $\quad \lambda = \frac{36x}{5}$

(b) $\frac{1}{\lambda} = R_H\times2^2\left[\frac{1}{1^2}-\frac{1}{2^2}\right]$ $\quad \frac{1}{\lambda} = \frac{1}{9x}\times4\times\frac{3}{4} = \frac{1}{3x}$

$\lambda = 3x$

(c) $\frac{1}{\lambda} = R_H\times3^2\left[\frac{1}{2^2}-\frac{1}{\infty^2}\right]$

$\frac{1}{\lambda} = \frac{1}{9x}\times9\times\frac{1}{4} = \frac{1}{4x}$ $\quad \lambda = 4x$

(d) $\frac{1}{\lambda} = R_H\times2^2\left[\frac{1}{1^2}-\frac{1}{\infty^2}\right]$ $\quad \frac{1}{\lambda} = \frac{1}{9x}\times4\times1 = \frac{4}{9x}$

$\lambda = \frac{9x}{4}$

48. K.E = E − I.E

$= 15.8 - 13.6$

$= 2.2$ eV

$1\text{eV} = 1.6\times10^{-19}\text{J}$

$2.2\text{eV} = 2.2\times1.6\times10^{-19}\ \text{J} = 3.5\times10^{-19}\ \text{J}$

49. $\text{K.E} = \frac{1}{2}\times\frac{Ze^2}{r_n}$ and $\text{P.E} = -\frac{Ze^2}{r_n}$

$\text{K.E : P.E} = \frac{1}{2}\times\frac{Ze^2}{r_n} : \frac{-Ze^2}{r_n} = 1:-2$

50. $\text{P.E} = \frac{-e^2}{r_n}$

1st orbit $\text{P.E}_1 = \frac{-e^2}{r}$ $\qquad r = 0.53$ Å

2nd orbit $\text{P.E}_2 = \frac{-e^2}{n^2.r} = \frac{-e^2}{4r}$

$$\Delta\text{P.E} = \frac{-e^2}{4r} - \left(\frac{-e^2}{r}\right) = \frac{e^2}{r}\left[-\tfrac{1}{4}+1\right] = \frac{3e^2}{4r}$$

$$= \frac{3\times\left(1.6\times10^{-19}\right)^2}{4\times0.53\times10^{-10}} = 3.6\times10^{-28}\text{C}^2\text{m}^{-1}$$

Conversion into J

$\Delta E = 3.6\times10^{-28}\text{C}^2\ \text{m}^{-1}\times9\times10^{9}\ \text{Nm}^2\ \text{C}^{-2}$

$K = \frac{1}{4\pi\varepsilon_0} = 9\times10^{9}\ \text{Nm}^2\ \text{C}^{-2}$

$= 3.24\times10^{-18}$ J

51. E of radiation $= \frac{hc}{\lambda} = \frac{6.6\times10^{-34}\times3\times10^{8}}{250\times10^{-9}}$

$= 7.92\times10^{-19}$ J

K.E = E − dissociation energy

$= 7.92\times10^{-19}\ \text{J} - 7.15\times10^{-19}\text{J}$

$= 7.7\times10^{-20}$ J

52. Given $v = \frac{\lambda}{5}$

$$\lambda = \frac{h}{mv}$$

$$5v.v = \frac{h}{m} \quad v = \sqrt{\frac{h}{5m}}$$

53. $2\pi r = n\lambda$ quantized

so $n = 3$

54. Time taken for one revolution $= \frac{\text{distance}}{\text{velocity}}$

$\therefore$ Number of revolution in 1 s $= \frac{v}{d}$

$$v_n = \frac{v_1}{n} \Rightarrow v_3 = \frac{2.18 \times 10^6}{3} \text{ and}$$

$d = 2\pi r = 2\pi(3^2 \times 0.53)$

Number of revolutions

$$= \frac{2.18 \times 10^6}{3} \times \frac{1}{2\pi \times 3 \times 3 \times 0.53}$$

$$= 2.42 \times 10^{14}$$

55. $v = \dfrac{2\pi\left(Ze^2/4\pi\varepsilon_0\right)}{nh}$

56. Due to high ionization energy, ejecting an electron is difficult. Threshold energy is very high

57. $K.E = h\upsilon - w$

$= 8 - 3.5 = 4.5$ eV

so stopping potential is -4.5 V

58. $m = v = \sqrt{\frac{h}{\lambda}} = \sqrt{\frac{6.6 \times 10^{-34}}{165 \times 10^{-10}}} = \sqrt{4 \times 10^{-26}}$

$$= 2 \times 10^{-13}$$

$$K.E = \frac{1}{2}mv^2 = \frac{1}{2} \times \left(2 \times 10^{-13}\right)^3 = 4 \times 10^{-39} \text{ J}$$

59. $\lambda = \frac{h}{mv} = \frac{h}{p}$ (p = momentum)

$$\frac{\lambda_1}{\lambda_2} = \frac{p_2}{p_1} \quad \frac{\lambda_e}{\lambda_p} = \frac{1}{2}$$

so $\frac{p_e}{p_p} = \frac{2}{1}$

60. $C_{rms} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3 \times 8.314 \times 300}{1 \times 10^{-3}}} = 2735 \text{ ms}^{-1}$

$$\lambda = \frac{h}{mv} = \frac{6.6 \times 10^{-34}}{1.67 \times 10^{-27} \times 2735}$$

$$= 1.45 \times 10^{-10} \text{m} = 1.45 \text{Å}$$

61. $\lambda = \frac{h}{mv}$ and $\upsilon = \frac{v}{\lambda}$

$$\upsilon = \frac{v.mv}{h} = \frac{2.K.E}{h}$$

$$\upsilon = \frac{2 \times 2.5 \times 10^{-28}}{6.6 \times 10^{-34}} = 7.5 \times 10^5 \text{ s}^{-1}$$

62. In Compton effect, the wavelength of incident X-ray increases due to interaction with electron. This energy is transferred to the electron, so momentum increases during the process.

63. Error cannot be reduced in measurement below the $h/4\pi$ limit because $\Delta x.\Delta p \geq \frac{h}{4\pi}$ but because position and momentum is a conjugate pair of variables, it cannot be done.

64. Heisenberg's uncertainty principle

65. In the question 3d is given
so next 4p will be filled

66. This specifies the orientation of atomic orbital in a magnetic field or electric field. Stark effect is splitting of lines in an electric field

67. The orbital is symmetric about the nucleus. So it is s orbital. So it is $\ell = 0$ and $m = 0$

68. 19th electron enters the 4s orbital hence $n = 4$ and $\ell = 0$

69. The orbital represents a d orbital. When $n = 2$, the shell does not contain d orbital

70. The quantum number not described by the Schroedinger equation is the spin quantum number.
b, c and d are not true for s

71. On plotting probability against distance, the distance at which there is maximum probability of finding an electron is equal to the Bohr radius i.e., 0.529Å

72. Nodes are region where probability of finding an electron is zero.

73. Number of radial nodes = $n - \ell - 1$

For 4d $n = 4, \ell = 2$

$n - \ell - 1 = 4 - 2 - 1 = 1$ radial node

74. Both B and Q represent nodes where probability of finding an electron is zero.

75. Total number of nodes = $n - 1$

so which means n increases, so energy increases and so does distance from nucleus

Angular nodes = ℓ that will depend on shape of the orbital.

76. $\ell = 1$ refers to the p orbital

p orbital is dumb bell shaped and will be symmetric about the respective axis.

77. Electrons should be filled in the 4s orbital before filling 3d orbital.

78. By Hunds' rule, same spin of unpaired electrons in a subshell gives rise to stability.

79. Pauli's exclusion principle restricts the number of electrons and determines their spin.

80. Two electrons with same spin can exchange positions giving rise to decrease in energy called exchange energy, hence they are more stable.

81. $\left.\begin{aligned}\upsilon_1^{1/2} &= a(Z_1 - b)\\ \upsilon_2^{1/2} &= a(Z_2 - b)\end{aligned}\right\} \left(\frac{\upsilon_2}{\upsilon_1}\right)^{\frac{1}{2}} = \frac{Z_2 - b}{Z_1 - b}$ i.e., $\left(\frac{\lambda_1}{\lambda_2}\right)^{\frac{1}{2}}$

$$= \left(\frac{9.87}{2.29}\right)^{\frac{1}{2}} = \left(\frac{24 - b}{12 - b}\right)$$

$$2.076 = \frac{24 - b}{12 - b}$$

$$(12 - b)\ 2.076 = (24 - b)$$

$$(12 \times 2.076) - 24 = (2.076 - 1)b$$

$$b = 0.848$$

82. $57.2 \times 10^3 \times 4.18$

$$= \left[6.02 \times 10^{23} \times 6.626 \times 10^{-34} \times 3 \times 10^8\right] \Big/ \lambda$$

$$\therefore \quad \lambda \text{ (meters)} = \left[119.67 \times 10^{-3}\right] \Big/ \left[239.1 \times 10^3\right]$$

$$= 0.5 \times 10^{-6} \text{ m}$$

$$\lambda \text{ Å} = 0.5 \times 10^{-6} \times 10^{10} = 0.5 \times 10^4 \text{ Å}$$

$$= 5000 \text{ Å}$$

83. $mc = \frac{h}{\lambda} = \frac{6.626 \times 10^{-34}}{5000 \times 10^{-10}}$

$$= 1.3252\ 10^{-27} \text{ kg m/s}$$

84. Between n = 1 and n = 2 (first excited state) the energy difference = (13.6 − 3.4)

$$= 10.2 \text{ eV} = \text{Energy of photon } \frac{hc}{\lambda_1}$$

From the first excited state to ionization, energy absorbed = 3.4 eV = $\frac{hc}{\lambda_2}$

$$\therefore \quad \frac{3.4}{10.2} = \frac{hc}{\lambda_2} \times \frac{\lambda_1}{hc} = \frac{\lambda_1}{\lambda_2} = 0.33$$

85. $2 \times 10^{18} \times h\upsilon$

$$= \left[2 \times 10^{18} \times 6.626 \times 10^{-34} \times 3 \times 10^8\right] \Big/ \lambda = 1 \text{ J}$$

$\therefore \quad \lambda = 39.756 \times 10^{-8} \text{ m} = 3975.6 \text{Å} \approx 4000 \text{Å}$

86. The energy (ε) – momentum (p) relation is

$\varepsilon^2 = p^2c^2 + m_0^2c^4$.

Where c is the speed of light, m is mass.

For a body at rest, the momentum is zero, so the equation simplifies to

$\varepsilon = m_0c^2$.

If the object is massless then the equation reduces to

$\varepsilon = pc$, as is the case for a photon.

87. 1 Kilo watt = 1000 J/s $\upsilon = 880 \times 10^3$/ sec

$\therefore \quad h\upsilon = 6.626 \times 10^{-34} \times 880 \times 10^3$ J

$$= 5830.88 \times 10^{-31} \text{J} \qquad = 5.831 \times 10^{-28} \text{ J}$$

$\therefore$ number of photons / s = $\frac{1000}{5.831 \times 10^{-28}}$

$$= 171.5 \times 10^{28} = 1.72 \times 10^{30}$$

88. Calculating λ values; $\frac{912}{4} \times \frac{3^2 \times 2^2}{3^2 - 2^2} = 1642 \text{Å}$

$$\frac{912}{4} \times \frac{4^2 \times 3^2}{4^2 - 3^2} = 4690 \text{Å}; \quad \frac{912}{4} \times \frac{2^2 \times 1^2}{2^2 - 1^2} = 304 \text{Å}$$

$$\frac{912}{4} \times \frac{4^2 \times 2^2}{4^2 - 2^2} \neq (4690 \text{Å}) \text{ the value is 1216 Å}$$

89. $0.529 \times \frac{n^2}{z} = \text{radius} = 0.529 \times \frac{2^2}{3} = 0.705$ Å

90. The λ values (beyond the 4000 Å to 7000 Å ranges suggest the Paschen series)

$\therefore$ by trial, $18746 = \frac{912}{1}\times\frac{n_2^2\times 9}{n_2^2-9}$

$\therefore$ $n_2^2 \simeq 4^2$. $n_2 = 4$

Again $12815 = \frac{912}{1}\times\frac{n_3^2\times 9}{n_3^2-9}$ $\quad n_3 = 5$

$10935 = \frac{912}{1}\times\frac{n_4^2\times 9}{n_4^2-9}$ $\quad n_4 = 6$

$\therefore$ It is the Paschen series; series limit

$\frac{912}{1}\times 9 = 8208\,\text{Å}$

91. Rydberg constants are proportional to the reduced mass of the system. Taking the masses of He^+ and H as M_2 and M_1 and m = electron mass

$R_{He^+} = \frac{mM_2}{m+M_2}$ and $R_H = \frac{mM_1}{m+M_1}$

The ratio required is $\frac{mM_1}{m+M_1}\times\frac{m+M_2}{mM_2}$

$= \frac{1+\frac{m}{M_2}}{1+\frac{m}{M_1}} = \left(1+\frac{1}{4\times 1840}\right) \Big/ \left(1+\frac{1}{1840}\right)$

$\approx \left(1+\frac{1}{4\times 1840}\right)\left(1-\frac{1}{1840}\right)$, which is nearly

$1+\frac{1}{4\times 1840}-\frac{1}{1840} = 1-\frac{3}{4\times 1840}$ which is a little less than 1.

92. radii are for (i) $r_0 = 0.529$ Å. for (ii) $\frac{n^2}{z}\times r_0 = \frac{1}{2}r_0$

(iii) $\frac{2^2}{3}r_0 = \frac{4}{3}r_0$ $\quad$ (iv) $\frac{2^2}{4}r_0 = r_0$

Ans (i) and (iv)

93. Clearly the transition is downwards from n = 4 to the lower levels. The series limit is 912 × 16

= 14592 Å

94. $2085 = \frac{912}{Z^2}\times\frac{4^2\times 3^2}{\left(4^2-3^2\right)}$

$\therefore$ $Z^2 = \frac{912\times 144}{2085\times 7} \simeq 9$ $\quad\therefore$ Z = 3

Li^{2+} ion

95. Energy levels are in eV −13.6, n =1

−3.4, n = 2

− 1.51, n = 3

− 0.85, n = 4…..

Clearly the energy difference

(−0.85 eV) − (−1.51eV) = 0.66 ev

96. In cgs unit, $v = \frac{2\pi e^2}{nh}$

$= \left(\frac{2\times 3.1416\times\left(4.802\times 10^{-10}\right)^2}{n\times 6.626\times 10^{-27}}\right)\text{cm/s}$

$\therefore$ $v = \frac{21.866}{n}\times 10^7\ \text{cm/s} = \frac{2.2\times 10^8}{n}\text{cm/s}$

97. In cgs units speed of the electron = $\frac{2\pi Ze^2}{nh}$

$= \frac{2\times 3.1416\times 2\times\left(4.802\times 10^{-10}\right)^2}{1\times 6.626\times 10^{-27}}\text{cm/s}$

$= \frac{289.77\times 10^{-20}}{6.626\times 10^{-27}}\text{cm/s} = 43.732\times 10^7\ \text{cm/s}$

$c = 3\times 10^{10}$ cm/s

$\therefore$ $\frac{c}{\upsilon} = \frac{3\times 10^{10}}{4.3732\times 10^8} = 6.86\times 10 \approx 68.6$

98. $T = \frac{2\pi r}{\upsilon} = \left[\frac{(2\pi r_0)\times n^2}{2\pi e^2}\times nh\times(\text{constant})\right]$

proportional to n^3

99. $KE_1\ \alpha\ (8-2)$; $KE\ \alpha\ (5-2)$ $\quad\therefore$ $\frac{KE_1}{KE_2} = \frac{6}{3} = 2$

100. Interference experiment brings out the wave nature not the particle nature.

101. $2\pi r = n\lambda$; i.e., $2\pi r_0\times n^2 = n\lambda$

$\therefore$ $\lambda = 2\pi r_0\times n = (2\times 3.14\times 0.529\ \text{x}\ 3)$ Å

$\simeq 9.97$ Å

102. $\lambda = \frac{12.225}{V^{\frac{1}{2}}}$ $\therefore$ $\lambda V^{\frac{1}{2}} = \text{constant}$

$\ln\lambda + \frac{1}{2}\ln V = \text{constant}$ $\therefore$ $\frac{d\lambda}{\lambda}+\frac{1}{2}\frac{dV}{V} = 0$

if $\frac{d\lambda}{\lambda} = -\frac{1}{100}$ then $\frac{dV}{V} = \frac{2}{100}$. i.e., 2%

103. $\lambda = \dfrac{h}{mv} \qquad \therefore mv = \dfrac{h}{\lambda} = \dfrac{6.626\times10^{-34}}{6.626\times10^{-36}} kg\,ms^{-1}$

$= 100\ kg\ ms^{-1}$

$m = 1\ kg\ \therefore$ speed at the instant of hitting

$= 100 m/s = \dfrac{95}{100} \times$ initial speed

$\therefore$ initial speed = 105 m/s

104. In ejection 13.6 eV are lost in ionization

$\therefore$ kinetic energy = (20.4 – 13.6) eV = 6.8 eV

$$= \frac{p^2}{2m} = \frac{h^2}{2m\lambda^2}$$

i.e., $6.8 \times 1.6 \times 10^{-19}\ J = \dfrac{\left(6.626\times10^{-34}\right)^2}{2\times9.1\times10^{-31}\times\lambda^2}$

$\therefore \quad \lambda^2 = \dfrac{43.9\times10^{-68}}{18.2\times10^{-31}\times10.88\times10^{-19}}$

$= 0.2217 \times 10^{-18}\ m^2$

$\therefore \quad \lambda = 0.4708 \times 10^{-9}\ m = 4.708\ Å$

105. $\lambda = 5 \times 10^{-12}\ m$

$\therefore \quad mv = \dfrac{h}{\lambda} = \dfrac{6.626\times10^{-34}}{5\times10^{-12}} kg\,m/s$

$mv = 1.325 \times 10^{-22}\ kg.ms^{-1}\ v = 7.94 \times 10^4\ ms^{-1}$

$\therefore \quad m = \dfrac{1.325\times10^{-22}}{7.94\times10^4} kg = 0.1669 \times 10^{-26} kg$

Dividing by 9.1×10^{-31}, we get the ratio 1.834×10^3 which is nearly 2000. The particle is a proton.

106. $\lambda_1 = \dfrac{12.225}{10},\ \lambda_2 = \dfrac{12.225}{9},\ \lambda_3 = \dfrac{12.225}{7}$

$$\therefore \quad \frac{\lambda_3 - \lambda_2}{\lambda_1} = \frac{\left(\frac{1}{7} - \frac{1}{9}\right)}{\left(\frac{1}{10}\right)}$$

$$= \frac{2}{63} \times 10 = \frac{20}{63}$$

107. Diameter $\simeq 2 \times 10^{-12}$ cm = Δx

$$\therefore \quad \Delta px = \frac{h}{4\pi\Delta x} = \frac{6.626\times10^{-27}}{4\times3.14\times2\times10^{-12}}$$

$\simeq 2.64 \times 10^{-16}\ g\ cm\ s^{-1}$

108. $1\ k\ eV = 1 \times 10^3 \times 1.6 \times 10^{-19} J$;

$\Delta x = 1Å = 1\times10^{-10} m$

$$\therefore \quad \Delta p_x = \frac{6.626\times10^{-34}}{4\times3.1416\times10^{-10}}$$

$= 5.273 \times 10^{-25}\ kg\ ms^{-1}$

$1\ kev = 1.6 \times 10^{-16}\ J = \dfrac{px^2}{2m}$

$\therefore \quad p_x^2 = 1.6 \times 10^{-16} \times 2 \times 9.1 \times 10^{-31}\ (kg\ ms^{-1})^2$

$p_x^2 = [29.12 \times 10^{-47}\ (kg\ ms^{-1})^2$

$\therefore \quad p_x = 17.06 \times 10^{-24} kg\ ms^{-1}$

$$\therefore \quad \frac{\Delta p_x}{p_x} \times 100 = \frac{5.273\times10^{-25}}{17.06\times10^{-24}} \times 100 = 3.091$$

Ans : 3.1%

109. The wrong Statement is (d)

110.

+2	+1	0	–1	–2	
					$+\frac{1}{2}$
					$-\frac{1}{2}$

There are 10 spin orbitals

$\therefore$ number of ways = $\dfrac{10\times9}{1\times2} = 45$

111. (b)

112. (d)

113. (a)

114. $\lambda_1 = 2.892 \times 10^{-10}\ m\ \therefore\ \upsilon_1 = \dfrac{3\times10^8}{2.892\times10^{-10}}$

$= 1.0373 \times 10^{18},\ \upsilon_1^{\frac{1}{2}} = 1.0185\times10^9$

$\lambda_2 = 2.775 \times 10^{-10}\ m$

$\therefore \quad \upsilon_2 = \dfrac{3\times10^8}{2.775\times10^{-10}}$

$= 1.0811 \times 10^{18}\ \upsilon_2^{\frac{1}{2}} = 1.0398\times10^9$

$$\left.\begin{aligned} 1.0185\times10^9 &= a(55-b) \\ 1.0398\times10^9 &= a(56-b) \end{aligned}\right\} 0.0213\times10^9$$

$= a = 2.13\times10^7 \qquad b = 7.1831$

115. $\upsilon_1^{\frac{1}{2}} = a(57-b),\ \upsilon_2^{\frac{1}{2}} = (72-b),$

$\upsilon_3^{\frac{1}{2}} = a(74-b)$

$$\upsilon_3^{\frac{1}{2}} - \upsilon_2^{\frac{1}{2}} = a(74-72) = 2a$$

$$\upsilon_3^{\frac{1}{2}} - \upsilon_1^{\frac{1}{2}} = a(74-57) = a \times 17$$

$$\therefore \quad \frac{\upsilon_3^{\frac{1}{2}} - \upsilon_2^{\frac{1}{2}}}{\upsilon_3^{\frac{1}{2}} - \upsilon_1^{\frac{1}{2}}} = \frac{2}{17} = 0.117 \simeq 0.12$$

116. $h\upsilon = \frac{hc}{\lambda} = \left(\frac{6.626\times10^{-34}\times3\times10^{8}}{1.522\times10^{-10}}\right)$ J

$= 13.061\times 10^{-16}$ J $= 1.3061 \times 10^{-15}$ J

117. (b), (d) are correct

118. (a), (b), (c) are correct

119. (a), (b), (c) are correct

120. (a)→ (q), (s)
(b)→ (p), (r)
(c)→ (p)
(d)→ (q), (s)

Additional Practice Exercise

121. (i) $\lambda = \frac{c}{\upsilon} \Rightarrow \upsilon = \frac{c}{\lambda}$

$$\sqrt{\frac{c}{\lambda}} = a(26 - b)$$

$$\sqrt{\frac{3\times10^{8}}{0.1956\times10^{-9}}} = a(26-b)$$

$1.238 \times 10^9 = a(26 - b)$ —(1)

$$\sqrt{\frac{3\times10^{8}}{0.1789\times10^{-9}}} = a\,(27-b)$$

$1.294 \times 10^9 = a(27 - b)$ —(2)

$$\frac{\text{equation(1)}}{\text{equation(2)}} \Rightarrow \frac{1.238}{1.294} = \frac{26-b}{27-b}$$

$1.238\,(27 - b) = 1.294\,(26 - b)$

$33.426 - 1.238b = 33.644 - 1.294\,b$

$\Rightarrow$ $b = 3.893$ $a = 5.6 \times 10^7$

122. (A) is He and A^+ is He^+

$$\bar{\upsilon} = R_H \times 2^2 \left(\frac{1}{1^2} - \frac{1}{n^2}\right)$$

$$\bar{\upsilon}_{total} = \bar{\upsilon}_1 + \bar{\upsilon}_2$$

$$= \frac{1}{108.5}\times10^9 + \frac{1}{30.4}\times10^9$$

$$= 10^9\left[\frac{1}{108.5} + \frac{1}{30.4}\right]$$

$= 0.0422 \times 10^9 = 4.22 \times 10^7$

$$4.22 \times 10^7 = 1.09679 \times 10^7 \times 2^2 \left(1 - \frac{1}{n^2}\right)$$

$$1 - \frac{1}{n^2} = \frac{4.22\times10^7}{1.09679\times4\times10^7} = 0.962$$

$$\frac{1}{n^2} = 0.038$$

$$n^2 = \frac{1}{0.038} = 26.3 \Rightarrow n \approx 5$$

123. For Balmer series the Rydberg equation is

$$\frac{1}{\lambda} = R_H Z^2 \left[\frac{1}{n_f^2} - \frac{1}{n_i^2}\right]$$

$$\frac{1}{48\times10^{-9}} = 1.09\times10^7 \times 9\left(\frac{1}{4} - \frac{1}{n_i^2}\right)$$

$$\frac{1}{n_i^2} = 0.04 \; ; \Rightarrow \; n_i = 5$$

$\therefore$ The transition is $5 \rightarrow 2$.

The other intermediate transitions are $3 \rightarrow 2$ and $4 \rightarrow 2$.

The corresponding wavelengths are

$$\frac{1}{\lambda_{3\to2}} = 1.09\times10^7 \times 9\left(\frac{1}{2^2} - \frac{1}{3^2}\right)$$

$\lambda = 73$ nm

$$\frac{1}{\lambda_{4\to2}} = 1.09\times10^7 \times 9\left(\frac{1}{2^2} - \frac{1}{4^2}\right)$$

$\Rightarrow$ $\lambda = 54$ nm

124. $h\upsilon$ = kinetic energy + ionization energy

$$h\upsilon = \frac{6.626\times10^{-34}\times3\times10^{8}}{1550\times10^{-10}\times1.6\times10^{-19}} = 8.015\text{eV}$$

$\therefore$ I.E = 8.015 – 3.2 = 4.815 eV

For Li^{2+}, I.E $= 13.6\times\frac{Z^2}{n^2}$eV

$$4.815 = 13.6\times\frac{9}{n^2} \Rightarrow n = 5$$

$\therefore$ The electronic emission takes place from the 5th orbit.

125. $r_6 = a_0(6)^2 \quad r_1 = a_0(1)^2$

$r_6 - r_1 = a_0\left[(6)^2 - (1)^2\right] = a_0[36-1] = 35a_0$

$= 35 \times 0.53\,\text{Å} = 18.55\,\text{Å}$

126. Energy of the electron in the 1st Bohr orbit

$= -0.85 \times 4^2 = -13.6$ eV

$= -13.6 \times 1.6 \times 10^{-19}\,\text{J} = -2.18 \times 10^{-18}\,\text{J}$

Energy of the electron in the 1st orbit of Li^{2+}

$= -2.18 \times 10^{-18} \times 3^2 = -1.96 \times 10^{-17}\,\text{J}$

$$\text{Velocity} = \sqrt{\frac{2E}{m}} = \sqrt{\frac{2 \times 1.96 \times 10^{-17}}{9.1 \times 10^{-31}}}$$

$$= 6.6 \times 10^6\,\text{m sec}^{-1}$$

127. $mvr = \dfrac{nh}{2\pi}$; $mvr = \dfrac{nh}{2\pi}$

From Bohr's equation, $r \propto \dfrac{n^2}{Z}$

$$v \propto \frac{n}{\left(\frac{n^2}{Z}\right)} \propto \frac{Z}{n} \propto \frac{1}{n}$$

$$\frac{v_1}{v_2} = \frac{n_2}{n_1} = \frac{3}{2}$$

$$\Rightarrow \quad v_2 = \frac{2}{3}v_1$$

128. Velocity: $v_n = \dfrac{c}{137} \times \dfrac{Z}{n}$

$$V_3 = \frac{3 \times 10^8}{137} \times \frac{1}{3} = 7.3 \times 10^5\,\text{m sec}^{-1}$$

129. $KE = h\upsilon - h\upsilon_0 = h(\upsilon - \upsilon_0)$

$KE_1 = h(\upsilon_1 - \upsilon_0)$ —(1)

$KE_2 = h(\upsilon_2 - \upsilon_0)$ —(2)

Dividing equation (1) by (2),

$$\frac{KE_1}{KE_2} = \frac{\upsilon_1 - \upsilon_0}{\upsilon_2 - \upsilon_0}$$

$$3 = \frac{2.1 \times 10^{14} - \upsilon_0}{1.5 \times 10^{14} - \upsilon_0}$$

$$\Rightarrow \quad \upsilon_0 = 1.2 \times 10^{14}\,\text{Hz}$$

130. $$\frac{h \times c}{250 \times 10^{-9}} = h\upsilon_o + 1.95 \times 1.6 \times 10^{-19}$$

$$\frac{h \times c}{333 \times 10^{-9}} = h\upsilon_0 + 0.98 \times 1.6 \times 10^{-19}$$

$$hc\left[\frac{10^9}{250} - \frac{10^9}{333}\right] = 1.6 \times 10^{-19}\,(1.95 - 0.98)$$

$$h \times 3 \times 10^8 \times 10^9 \times 10^{-3} = 1.6 \times 10^{-19}\,(0.97)$$

$$h = \frac{1.6 \times 10^{-19} \times 0.97}{3 \times 10^{14}} = 5.17 \times 10^{-34}\,\text{J s}$$

131. (a)

X-rays from an anticathode consists of characteristic wavelength only.

132. This can be done by direct substitution and solving

133. Energy per mole of photons $= N_0 h\upsilon = N_0 h\,c/\lambda$

$$\therefore \quad \frac{\left[6.02 \times 10^{23} \times 6.626 \times 10^{-34} \times 3 \times 10^8\right]}{\left[4000 \times 10^{-10}\right]}$$

$= 2.99 \times 10^5$ J similarly,

$$\frac{\left[6.02 \times 10^{23} \times 6.626 \times 10^{-34} \times 3 \times 10^8\right]}{\left[7000 \times 10^{-10}\right]}$$

$= 17.1 \times 10^4\,\text{J} = 1.71 \times 10^5\,\text{J}$

134. One kilowatt = 1000 J s^{-1}

$\lambda = 3.5$ m

$$\therefore \quad h\upsilon = \frac{hc}{\lambda} = \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{3.5}$$

$= 5.679 \times 10^{-26}$ J

$\therefore$ Number of photons per second

$$= \frac{1000}{5.679 \times 10^{-26}}$$

$\simeq 0\ 176 \times 10^{26} = 1.76 \times 10^{28}$

135. $E = h\upsilon = \dfrac{hc}{\lambda} = mc^2 = 5.52 \times 10^{-32} \times 9 \times 10^{16}$ J

$$\therefore \quad \lambda = \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{5.52 \times 9 \times 10^{-32} \times 10^{16}}$$

$$= \frac{6.626 \times 10^{-34}}{5.52 \times 10^{-32} \times 3 \times 10^8}\,\text{m}$$

$\lambda = 0.4 \times 10^{-10}\text{m} = 0.4$ Å

136. The difference in energy of the levels involved in the transition. i.e., $\Delta E = h\upsilon = h\dfrac{c}{\lambda}$.

137. $E = Nh\upsilon = Nh\dfrac{c}{\lambda}$

$= 6.023\times10^{23}\times6.626\times10^{-34}\times\dfrac{3\times10^{8}}{242\times10^{-9}}$

$= 495\ \text{kJ mol}^{-1}$

138. The ratio required is the ratio of the reduced masses

$$\frac{\mu_2}{\mu_1} = \left(\frac{m_e \times M_{Li^{2+}}}{m_e + M_{Li^{2+}}}\right)\times\frac{\left(m_e + M_H\right)}{\left(m_e \times M_H\right)}$$

$$= \frac{\left(1+\dfrac{m_e}{M_H}\right)}{\left(1+\dfrac{m_e}{M_{Li^{2+}}}\right)} \simeq \frac{(1+\alpha)}{\left(1+\dfrac{\alpha}{7}\right)}$$

Since the numerator is slightly larger than the denominator the answer is (c)

139. Acc. to Bohr' theory $r = r_0 \times \dfrac{n^2}{z}$

$= (0.529 \times \dfrac{1^2}{5})\ \text{Å} = 0.1058\ \text{Å}$

140. $\lambda' = \dfrac{912}{4}\times\dfrac{n_2^2n_1^2}{n_2^2-n_1^2} = \dfrac{1642}{228} = 7.2$

$= \dfrac{3^2\times2^2}{3^2-2^2}$ by trial and error

i.e., first Balmer transition

$\lambda''' = 1216 = \dfrac{n_2^2n_1^2}{n_2^2-n_1^2}\times228$

$\dfrac{1216}{228} = 5.33 = \dfrac{4^2\times2^2}{4^2-2^2}$ second Balmer transition

$\lambda'' = 1026 = \dfrac{n_2^2n_1^2}{n_2^2-n_1^2}\times228$

$\dfrac{1026}{228} = 4.5 = \dfrac{6^2\times2^2}{6^2-2^2}$ fourth Balmer transition

Missing wavelength λ(Å)

$= \dfrac{5^2\times2^2}{5^2-2^2}\times228 = 1086\ \text{Å}$

141. $1642 = \dfrac{912}{Z^2}\times\dfrac{n_2^2n_1^2}{n_2^2-n_1^2} = 1.8\times Z^2 = \dfrac{n_2^2n_1^2}{n_2^2-n_1^2}$

By trial and error if we take Z = 2. Then $7.2 = \dfrac{3^2\times2^2}{3^2-2^2}$.

Clearly, Z = 2 and it is the first Balmer transition

142. $\dfrac{m}{m_0} = 1.0142 = \left(1-\dfrac{v^2}{c^2}\right)^{-1/2}$

$\therefore\quad \left(1-\dfrac{v^2}{c^2}\right) = 0.972$

$\therefore\quad \dfrac{v^2}{c^2} = 0.028$

$\therefore\quad v^2 = 0.028\times9\times10^{16}\ \text{m}^2\ \text{s}^{-2}$

$= 0.252\times10^{16}\ \text{m}^2\ \text{s}^{-2}$

$\therefore\quad v = \left(25.2\times10^{14}\right)^{1/2} \simeq 5\times10^7\ \text{m s}^{-1}$

143. $r = r_0\times\dfrac{n^2}{Z}$ $8.464\ \text{Å} = 0.529\times\dfrac{n^2}{1}$

$\therefore\quad n^2 = \dfrac{8.464}{0.529} = 16 \qquad \therefore\ n = 4$

$\therefore\quad$ Number of transitions $= \dfrac{n(n-1)}{2} = \dfrac{4\times3}{2} = 6$

144. $r_2 = 0.529\times n_2^2 = 13.225\ \text{Å}$

$\therefore\quad n_2^2 = 25,\ n_2 = 5$

$r_1 = 0.529\times n_1^2 = 4.761\ \text{Å}$

$\therefore\quad n_1^2 = 9;\ n = 3$

$\therefore\quad$ It is a$(5\rightarrow3)$ transition

$\lambda(\text{Å}) = \dfrac{912}{1^2}\times\dfrac{5^2\times3^2}{5^2-3^2} = \left(912\times\dfrac{225}{16}\right)\ \text{Å}$

$= 12825\ \text{Å}$

145. $\dfrac{1}{\lambda} = R\left(\dfrac{1}{1^2}-\dfrac{1}{n^2}\right)$ λ is minimum when $\dfrac{1}{\lambda}$ is maximum

i.e., $\dfrac{1}{\lambda}$ = R(Rydberg constant)

$= \dfrac{1}{911.52\times10^{-8}}$

$= 1.097069\times10^5 \simeq 109707\ \text{cm}^{-1}$

146. $\frac{1}{\lambda_1} = R_H \times \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) = 15200 \text{ cm}$

$$\frac{1}{\lambda_2} = R_x \times (Z^2)\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) = 243000$$

$$\text{ratio} = \frac{243000}{15200}$$

$= Z^2 \times \frac{R_x}{R_H}$ since $\frac{R_x}{R_H}$ is nearly = 1

$Z^2 \simeq 16 \qquad Z = 4 \qquad \therefore$ Ans Be^{3+}

147. Energy levels (eV): −13.6, −3.4, −1.51, −0.85, −0.544…

The value 1.89 eV corresponds to

[(−1.51) − (−3.4)eV

$\therefore$ It is transition from $n_2 = 3$ to $n_1 = 2$

i.e., first Balmer transition

$\therefore$ series limit $= \frac{912}{1^2} \times 2^2 Å = 3648 Å$

148. If Z^* is the effective charge $\frac{(Z^*)^2}{9} \times 13.6 = 5.44$

$$\therefore \quad Z^* = \left(\frac{5.44 \times 9}{13.6}\right)^{1/2} \simeq 1.9$$

149. $E = -2.18 \times 10^{-18} \times 3^2 \left(\frac{1}{2^2} - \frac{1}{4^2}\right)$

$= 3.68 \times 10^{-18} \text{ J ion}^{-1}$

$\text{Total } E = \frac{6.02 \times 10^{23}}{7} \times 2.2 \times 3.68 \times 10^{-18}$

$= 696 \text{ kJ}$

150. $\Delta x.\Delta p \geq h/4\pi$

$$\Delta x = \sqrt{\frac{h}{4\pi m}} \text{ so } \sqrt{\frac{h}{4\pi m}}\, m\Delta v = \frac{h}{4\pi}$$

$$\therefore \quad \Delta v = \sqrt{\frac{h}{4\pi m}}$$

151. According to the Bohr theory, $v = \frac{2\pi e^2}{nh}$(cgs) . Thus v $\propto \frac{1}{n}$. Hence, when n is doubled, v is halved

152. Calculating $v/2\pi r = \frac{Z^2 e^2}{n^3 h r_0}$

$$= \frac{9 \times \left(4.802 \times 10^{-10}\right)^2}{27 \times 6.626 \times 10^{-27} \times 0.529 \times 10^{-8}} \text{ in cgs}$$

which gives 2.1929×10^{15} revolutions per sec.

153. For the H-atom $V = \frac{2\pi e^2}{nh}$(in cgs)

$$= \frac{2 \times 3.1416 \times \left(4.802 \times 10^{-10}\right)^2}{2 \times 6.626 \times 10^{-27}} \text{ cm s}^{-1}$$

$= 10.933 \times 10^7 \text{ cm s}^{-1}$

$$\therefore \quad \frac{V}{c} = \frac{10.933 \times 10^7}{3 \times 10^{10}} = 3.644 \times 10^{-3}$$

154. $K.E_{max} = 5 \times 10^{-18} \text{ J} = 6.626 \times 10^{-34}(4\upsilon_0 - \upsilon_0)$

$= 6.626 \times 10^{-34} \times 3\upsilon_0$

$$\therefore \quad \upsilon_0 = \left(5 \times 10^{-18}\right) / \left(6.626 \times 10^{-34} \times 3\right) \text{ s}^{-1}$$

$= 0.2515 \times 10^{16} \text{ s}^{-1}$

$= 2.515 \times 10^{15} \text{ s}^{-1}$

155. $h\upsilon_0 = 4.3 \times 1.6 \times 10^{-19} \text{ J}$

$$\therefore \quad \upsilon_0 = \frac{4.3 \times 1.6 \times 10^{-19}}{6.626 \times 10^{-34}} = 1.038 \times 10^{15} \text{ s}^{-1}$$

156. Momentum,(mc) $= \left(h/\lambda\right)$

$$= \left[6.626 \times 10^{-34}\right] / \left[5890 \times 10^{-10}\right]$$

$= 1.125 \times 10^{-27} \text{ kg m s}^{-1}$

157. $\frac{hc}{\lambda} = \frac{\left[6.626 \times 10^{-34} \times 3 \times 10^8\right]}{\left[\lambda \times 10^{-10}\right]}$

$= 10.2 \times 1.6 \times 10^{-19} \text{ J}$

$\therefore \quad \lambda(\text{in Å})$

$$= \left[6.626 \times 3 \times 10^{-26}\right] / \left[10.2 \times 1.6 \times 10^{-29}\right]$$

$= 1218Å$

158. $\lambda_1 = \dfrac{12.225}{10}$ $\qquad \lambda_2 = \dfrac{12.225}{6.63}$

$\therefore \quad \dfrac{\lambda_2}{\lambda_1} = \dfrac{10}{6.63} \simeq 1.5$

159. $\lambda = \dfrac{h}{mV}$; 6×10^{-10}m

$= \dfrac{6.626 \times 10^{-34}}{9.11 \times 10^{-31} \times v}$

$\therefore \quad v\left(\text{m}/\text{s}\right)$

$= \left(6.626 \times 10^{-34}\right) \Big/ \left(9.11 \times 10^{-31}\right) \times \left(6 \times 10^{-10}\right)$

$= 1.212 \times 10^6$ m s^{-1}

160. $2 \times 10^{18} \times \dfrac{hc}{\lambda} = 1$ J

i.e., $\dfrac{2 \times 10^{18} \times 6.626 \times 10^{-34} \times 3 \times 10^8}{\lambda} = 1$

$\therefore \quad \lambda = 39.756 \times 10^{-8}$ m $= 3975.6$Å $\simeq 4000$ Å

161. The rms speed

$= \left(\dfrac{3RT}{M}\right)^{1/2} = \left(\dfrac{3 \times 8.314 \times 10^7 \times 300}{4}\right)^{1/2}$

which equals $\left(1.871 \times 10^{10}\right)^{1/2}$ cm s^{-1}

$= 1.3677 \times 10^5$ cm $s^{-1} \simeq 1368$ m s^{-1}

calculate $\lambda = \dfrac{h}{mv}$

$= \dfrac{6.626 \times 10^{-34} \times 6.02 \times 10^{23}}{4 \times 10^{-3} \times 1368}$ m

$= (7.289 \times 10^{-3}) \times 10^{-8}$

$= 7.289 \times 10^{-11}$ m

162. $\Delta V = 2.19 \times 10^6 \times \dfrac{2}{3} \times \dfrac{0.02}{100} = 2.92 \times 10^2$ ms^{-1}

$\therefore \quad \Delta x = \dfrac{h}{4\pi m \times \Delta V}$

$= \dfrac{6.625 \times 10^{-34}\,(Js)}{4 \times 3.14 \times 9.1 \times 10^{-31}\,(kg) \times 2.92 \times 10^2\,(ms^{-1})}$

$= 198$ nm

163. $\Delta E \times \Delta t = \dfrac{h}{2\pi}$ where ΔE is minimum

$\therefore \quad \Delta E = \dfrac{6.626 \times 10^{-34}}{2 \times 3.1416 \times 10^{-8}}$

$\Delta E = 1.0546 \times 10^{-26}$ J

164. $\Delta x.\Delta p_x = h/4\pi$

$10 \times 10^{-10} \times \Delta p_x = \dfrac{6.626 \times 10^{-34}}{4 \times 3.1416}$

$\therefore \quad \Delta p_x = \dfrac{6.626 \times 10^{-34}}{4 \times 3.1416 \times 10^{-9}}$ kg m s^{-1}

$= 5.275 \times 10^{-26}$ kg m s^{-1}

165. For the f orbital $\ell = 3$

$\therefore$ angular momentum is $\sqrt{3 \times 4} \times \dfrac{h}{2\pi}$

$= \sqrt{3} \times \dfrac{h}{\pi}$

166. $\dfrac{h}{2\pi} \cos\theta \sqrt{\ell(\ell+1)} = m_\ell . \dfrac{h}{2\pi}$

When $m_\ell = 2$, $\cos\theta = \sqrt{\dfrac{2}{3}}$

167. If the plot has a single maximum like the 1s orbital plot it has no radial node

$\therefore \quad (n - \ell - 1) = 0$

3d : $3 - 2 - 1 = 0$, 1s : $1 - 0 - 1 = 0$,

2p : $2 - 1 - 1 = 0$ but for 3p : $3 - 1 - 1 = 1$ (non zero)

$\therefore$ 3p orbital does not fit into the description

168. There are two radial nodes

i.e., $n - 1 - \ell = 2$

In 4p, $n = 4$, $\ell = 1$

$\therefore \quad n - 1 - \ell = 4 - 1 - 1 = 2$

169. (b)

m = 0 corresponds to the orbital arranged along Z – axis, but for other values of 'm' there is no specified orientation.

170. $[Ar]3d^4 \quad \rightarrow +3$ state

$[Ar]3d^5\,4s^2 \rightarrow$ ground state

Total number of electrons = 25

Atomic number = 25

171. (b)

172 (d)

173. (b)

174. (c)

175. (a)

176. (c)

177. (d)

178. (b)

179. (d)

180. (a)

181. (d)

182. (a)

183. (c)

184. (a)

185. (d)

186. (b) Since the electron is confined to the nucleus of radius 10^{-12} cm.

Dimension, $n\lambda = 10^{-12}$ cm

Taking $n = 1$, $\lambda = 10^{-12}$ cm

$\therefore \quad \lambda = \frac{h}{p}, p = \frac{h}{\lambda} = 6.626 \times 10^{-15}$ g cm s^{-1}

Since the velocity of electron is much greater than that of light, the mass of electron cannot be the rest mass.

187. (c)

188. (b)

189. (d) A free electron is not quantized. Only the electrons in an atomic orbital are quantized.

190. (b), (c), (d)

191. (a), (b)

192. (a), (b)

193. (a), (b), (c)

194. (a), (b)

195. (a), (b)

196. (b), (c)

197. (a), (b)

198. (a) $\rightarrow$ (p), (r), (s)
(b) $\rightarrow$ (q)
(c) $\rightarrow$ (p), (s)
(d) $\rightarrow$ (r), (s)
(c) : when $n_1 = 2$ and $n_2 = 6$, $\lambda = 4104$ Å
$n_1 = 2$ and $n_2 = 7$, $\lambda = 3972$ Å

199. (a) $\rightarrow$ (p), (q)
(b) $\rightarrow$ (p), (r)
(c) $\rightarrow$ (r), (s)
(d) $\rightarrow$ (r)

200. (a) $\rightarrow$ (p), (q)
(b) $\rightarrow$ (p), (q)
(c) $\rightarrow$ (q), (r)
(d) $\rightarrow$ (q), (s)

CHAPTER

4

CHEMICAL BONDING

CHAPTER OUTLINE

Preview

STUDY MATERIAL

Introduction
- Concept Strands (1-8)

Modern Theories of Bonding
- Concept Strands (9-28)

Hydrogen Bonding
- Concept Strands (29-34)

Polarization of Ions- Fajan's Rule
- Concept Strands (35-38)

Dipole Moment
- Concept Strands (39-47)

Resonance
- Concept Strands (48-50)

TOPIC GRIP
- Subjective Questions (10)
- Straight Objective Type Questions (5)
- Assertion–Reason Type Questions (5)
- Linked Comprehension Type Questions (6)
- Multiple Correct Objective Type Questions (3)
- Matrix-Match Type Question (1)

IIT ASSIGNMENT EXERCISE
- Straight Objective Type Questions (80)
- Assertion–Reason Type Questions (3)
- Linked Comprehension Type Questions (3)
- Multiple Correct Objective Type Questions (3)
- Matrix-Match Type Question (1)

ADDITIONAL PRACTICE EXERCISE
- Subjective Questions (10)
- Straight Objective Type Questions (40)
- Assertion–Reason Type Questions (10)
- Linked Comprehension Type Questions (9)
- Multiple Correct Objective Type Questions (8)
- Matrix-Match Type Questions (3)

INTRODUCTION

A chemical bond is the attractive force that holds two or more atoms together as a stable molecule. Valency is the capacity or potential of an element to combine with other elements. To be precise, valency is the number of bonds formed by an atom in a molecule. The outermost electrons in an atom are the ones that take part in chemical bonding. Therefore, these electrons are named as valence electrons. The electrons involved in bonding are called bonding electrons and those, which are not involved in bonding, are called non-bonding electrons.

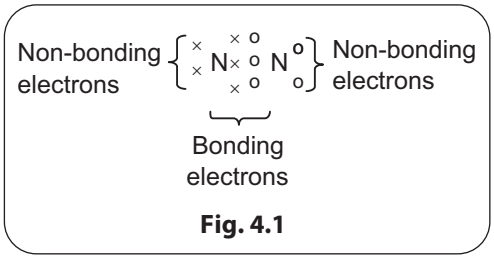


Fig. 4.1

Lewis structures or electron dot structures

This is a simple representation of molecules and ions with the valence electrons represented by dots. It shows an appropriate shape of the molecules and appropriate arrangement of atoms. The valence electrons are distributed as either bonding or lone pairs. Lewis structures can be quickly written using the following guidelines.

1. The total number of electrons in the structure can be obtained by adding together the valence electrons of all the atoms in the molecules or ion.
2. For each negative charge on an ion one electron is added and for each positive charge on electron is subtracted from the above total.

CONCEPT STRANDS

Concept Strand 1

Write down the Lewis electron dot structure of SO_3.

Solution

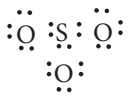
1. The total number of electrons = 6 + (6 × 3) = 24
2. Sulphur is the central atom and Oxygen atoms surround it.

   O S O
   O

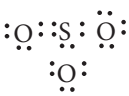
3. Bonding pairs are added

   O : S : O
   O

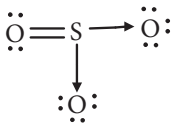
4. Octet of oxygens are completed

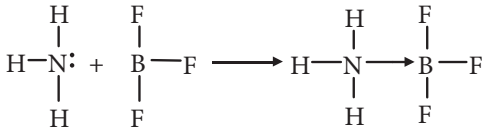


5. Complete the octet of Sulphur by shifting a lone pair of one of the oxygen atoms as a bond pair.



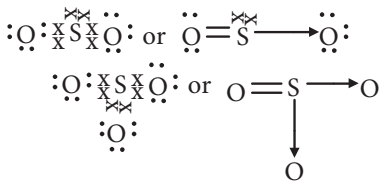
This electron dot structure may also be written as





ammonia-boron trifluoride adduct

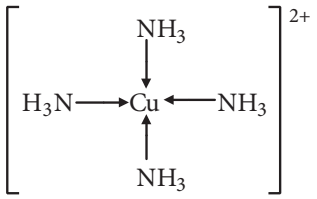
Other examples are SO_2, SO_3, $[Cu(NH_3)_4]^{2+}$



As shown above a coordinate bond is represented by an arrow mark (→), the arrow head shows the direction of donation of electrons. It is also represented by a line with + and – signs on the two ends. + charge is near the donor of electron pair and – charge is near the acceptor of electron pair SO_2 molecule is thus represented as

$$O = S^+ - {}^-O$$

Coordinate bond is also called donor–acceptor bond or dative bond or semi-polar bond.

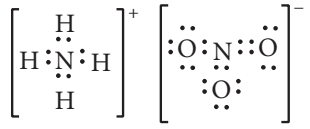


Concept Strand 2

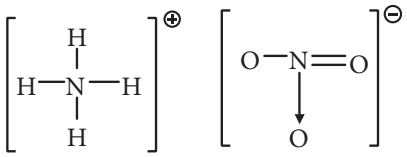
Write down the electron dot structure of (i) ammonium nitrate and (ii) sodium carbonate.

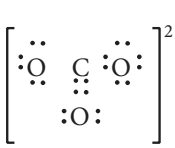
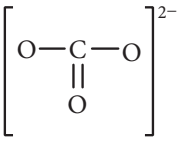
Solution

(i)



or



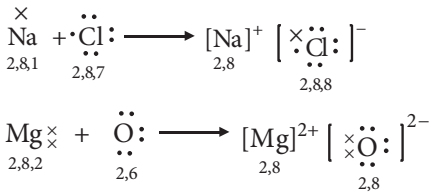
(ii) $2Na^+$  or $2Na^+$ 

Lewis and kossel theory of bonding

The valency and bonding of atoms was first explained on the basis of electronic structure of atoms by G.N. Lewis and W. Kossel (1916). This theory is also called electronic theory of valency.

They noted that the noble gases do not combine with other elements or with themselves to form molecules. This is because they have already a stable electronic configuration in their atoms with an octet (group of eight) of electrons. Atoms of other elements combined to form compounds in an attempt to attain a stable octet structure in their outermost shell.

Octet can be completed by losing, gaining or sharing of electrons. Losing and gaining of electrons by two atoms results in transfer of electrons from one atom to another producing a positive ion called cation and a negative ion called anion. The two ions are bounded by electrostatic attraction or an electrovalent or ionic bond.



Lattice energy

The lattice energy (u) of a crystal is the energy liberated when one gram-mole of the crystal is formed from its gaseous ions.

$$Na_{(g)}^{+} + Cl_{(g)}^{-} \rightarrow NaCl_{(crystal)} \qquad u = -782 \text{ kJ mol}^{-1}$$

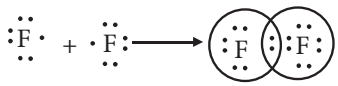
Higher the lattice energy, greater will be the ease of forming an ionic compound.

1. Lattice energy increases as the inter-ionic distance decreases.
2. Lattice energy increases as the charge on the ions increases.

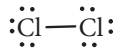
An ionic compound dissolves in a solvent if the value of solvation energy is higher than the lattice energy. Solvation energy is the energy released when the ions are surrounded by the solvent molecules when placed in that solvent.

It is to be noted that the highly electropositive elements (gp 1 and 2) combine with highly electronegative elements (gp 16 and 17) to form ionic bonds.

On the other hand when two fluorine atoms combine to form an F_2 molecule transfer of electrons will not produce octet on both the atoms. Instead, sharing of one electron from each of the fluorine atoms forming an electron pair forms what is called a covalent bond.



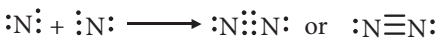
Now both the atoms can get an octet around it. Sharing of a pair of electrons produces one covalent bond represented by a line between the two atoms.



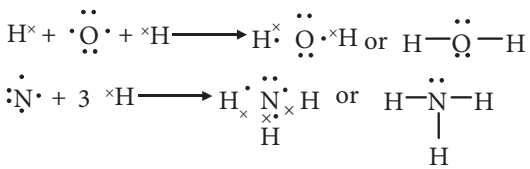
O_2 molecule is formed by the sharing of two electrons producing a double bond.



N_2 molecule has a triple bond in it.



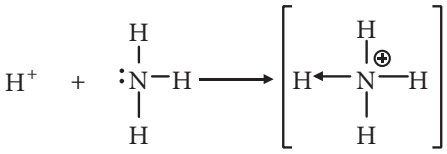
H_2O and NH_3 molecules are represented as follows.



The electron pairs in molecules which are not used for bonding are called unshared pairs or lone pairs of electrons.

Coordinate covalent bonds are special covalent bonds produced by the sharing of electron pairs between two atoms where both the electrons come from one of the bonded atoms.

Usually a stable molecule or ion will share its electron pair of its already completed octet with another atom, ion or molecule which has not yet completed an octet.



Hydrogen ion from an acid | ammonia molecule | ammonium ion

Generally a simple and symmetrical arrangement of atoms is the structure of the molecule. The atom which is present only once in the molecule is usually the central atom and atoms present in larger numbers will surround the central atom. The central atom will usually be the least electronegative atom in the molecule. The C, P and N are the central atoms CO_2, PCl_5 and N_2O respectively. Prior knowledge of the structure or an educated guess about the structure will help immensely.

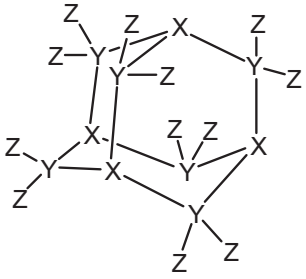
Place an electron pair between all neighbouring atoms and complete the octet of surrounding atoms with the available electrons.

If the octet of the central atom is not complete make multiple bonds by shifting the lone pairs of the peripheral atoms between that atom and the central atom.

CONCEPT STRANDS

Concept Strand 3

A compound formed from the atoms of elements, C, H and N is given below:



Identify the atoms X, Y and Z.

Solution

Valency of X is three.

Valency of Y is four.

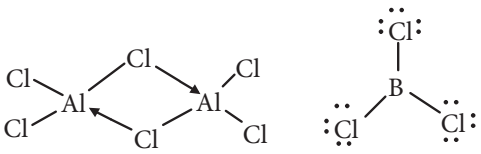
Valency of Z is one.

Hence X is 'N', Y is C and Z is H.

Concept Strand 4

Why anhydrous $AlCl_3$ exists as a dimer where as BCl_3 does not?

Solution



Due to small size of B – atom, the vacant 2p orbitals of Boron is not able to accept the 3p electrons of Cl – atom. In $AlCl_3$ due to large size of Al – atom, the vacant 3p orbital of Al can accept 3p electron pairs of Cl – atom.

Concept Strand 5

Arrange the following in the increasing order of their lattice energy, KI, KBr, KCl and KF.

Solution

KI > KBr > KCl > KF

Lattice energy increases with the increase in covalent character.

Concept Strand 6

Considering the three ionic compounds AP, BQ and CR having the same lattice type, and the radius of cation and anion from which the compounds are formed, arrange them in the decreasing order of lattice energy?

Ions	A^+	B^+	C^{2+}	P^-	Q^-	R^{2-}
Radius A°	1.4	1.8	1.5	1.4	1.8	1.5

Solution

CR > AP> BQ

Lattice enthalpy depends on size of the ions and their charges

The ionic charge is highest in CR

Similarly, the ionic radii are small

Concept Strand 7

The observed lattice energy of NaF and MgO are –915 and –3933 kJ mol^{-1} respectively. Give reasons for this observation.

Solution

In Mg – O the interionic distance is smaller than that in NaF and charge on anion and cation are twice that in NaF.

Concept Strand 8

Why does anhydrous aluminium chloride fumes in moist air?

Solution

Aluminium chloride fumes in moist air due to hydrolysis to form HCl as one of the products.

MODERN THEORIES OF BONDING

The modern theories of bonding are based on the principles of quantum mechanics. There are two important theories of bonding.

(i) Valence bond theory (VBT) and
(ii) Molecular orbital theory (MOT)

Valence bond theory

In valence bond theory, it is assumed that atoms with all their bonding electrons approach each other to form a molecule. This theory was first applied by Heitler and London in 1927 to the formation of the hydrogen molecule. It was further extended by Slater and Pauling.

The valence bond theory is a quantum mechanical extension of the ideas of electron pairing given by Lewis and others. A discussion on Valence bond theory is based on atomic orbitals, electronic configuration and orbital overlap.

Orbital overlap and covalent bond

Hydrogen molecule formation

According to the valence bond theory, a covalent bond is formed between two hydrogen atoms, H – H, by the

overlap of the 1s orbitals of the two hydrogen atoms containing one electron each with opposite spins. When the two 1s orbitals approach each other, they overlap, which means that the two orbitals share a common region in space.

Let us consider two hydrogen atoms H_A and H_B lying far apart from each other, so that no interaction occurs between them. It is assumed that when the two atoms are far away from each other, that is the distance between H_A and H_B is infinity, the potential energy of the system is zero.

When the two hydrogen atoms combine together to form a hydrogen molecule, forming a chemical bond between the two atoms, there is an overall decrease in the potential energy of the combining atoms. Or, the system having the bonded atoms has a lower energy than the system having free hydrogen atoms. A system results out of bonding, which has lower energy and therefore greater stability.

Potential energy curve

When the two hydrogen atoms approach each other, repulsive forces are produced between the two positively charged nuclei and between the two negatively charged electrons. At the same time, attractive forces are produced between the positive nucleus of one hydrogen atom and the negative electron of the other hydrogen atom (Fig 4.2).

When two hydrogen atoms approach each other, the long - range attractive forces begin to appear between the nucleus of one atom and the electron of the other atoms. This results in the lowering of the potential energy. The two hydrogen atoms may approach each other with electronic spins in the same direction (↑↑) or in opposite directions (↑↓). So there will be two curves, shown as (a) and (b) in the figure.

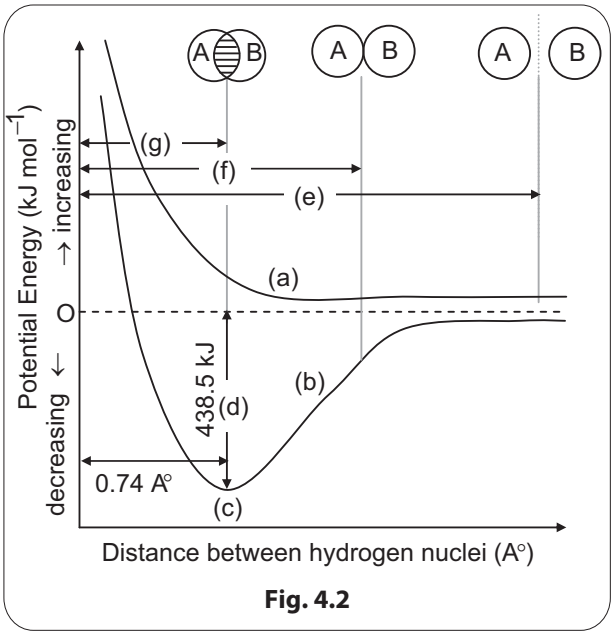


Fig. 4.2

(a) Curve with parallel spins of electrons
(b) Curve with opposite spins of electrons
(c) Minimum in potential energy of the system
(d) Bond energy
(e) Infinite distance between atoms
(f) Atoms coming closer as the energy starts decreasing (curve (b))
(g) Distance between H atoms for maximum decrease in potential energy

In the curve (b), where the electrons are paired with opposite spins, potential energy continuously decreases and there is an energy state (c) corresponding to minimum potential energy, which represents the optimum distance (g) between the atoms and the maximum interaction leading to bond formation. At this point corresponding to (c) there is maximum overlap of the 1s orbitals of the two hydrogen atoms, there is maximum electron charge density in the region between the two nuclei, which leads to a stable H_2 molecule.

Any further decrease in the distance between atoms results in a steep increase in the potential energy due to increased internuclear and interelectronic repulsions.

The decrease in potential energy is accompanied by a release of energy when the H_2 molecule is formed from the two H atoms. That means, heat is given out when a bond is formed. If a H_2 molecule (H – H) is to be broken, the process is endothermic since the same amount of energy must be supplied.

When the distance becomes close, the two electrons of the H_A and H_B are no longer identified with their original nuclei. That is each electron has an equal probability of being a part of either atom and each nucleus may be associated with both the electrons. Then we can say that the electrons are paired and shared. Thus VBT is a quantum mechanical extension of Lewis concept of localized electron pair bonds.

When two electrons come with opposite spin with in the limits of atomic distance for bond formation, the mutual neutralization of their spin moments result in an attractive force between the two atoms. If the spins are parallel, the two atoms provide a repulsive force. Potential energy does not show a minimum (represented by curve 'a'). There would be anti-bonding conditions if spins are parallel and bonding conditions if the spins are opposite.

Covalent bond

In the covalent bonding there is a change in electron densities in the combining atoms. Accumulation of electron densities between the two nuclei results in bond formation.

When two atoms approach, there is overlapping of electron waves and maximum overlapping would be at the mid point between the two atoms.

The covalent bond arises from the attraction of the two positive nuclei with the negatively charged electron cloud of the shared electron pairs in between.

Types of covalent bonds–sigma (σ) and pi (π) bonds

The strength of the bond depends upon the extent of the overlap. The strongest bonding occurs when the orbitals overlap the most. *The bond that is formed between two atoms by the linear overlap of their atomic orbitals along the inter-nuclear axis is a strong bond due to maximum overlap and is known as the σ bond. The bond that is formed by lateral overlapping of p-orbitals in a direction at right angles to the inter-nuclear axis is called π bond and is weak due to partial overlap of the orbitals.* All overlaps involving s-orbitals are σ. The first overlap between two p-orbitals of two atoms is σ and the next overlaps are π.

Comparison of σ and π bonds

Table 4.1

σ bond	π bond
It is formed by linear overlap of s-s or s-p or p-p orbitals of two atoms.	It is formed by lateral overlap of p-p and p-d orbitals.
The extent of overlap is quite large and hence σ bond is a strong bond.	The extent of overlap is small and hence π bond is usually weaker than σ.
There can be only one σ bond between two atoms.	There can be one or two π bonds between two atoms.
Electron cloud is cylindrically symmetrical about the line joining the two nuclei.	Electron cloud of π bond is above and below the line joining the two nuclei
Free rotation of atoms about a σ bond is possible. Rotation means rotation about the axis with respect to other atoms attached to it. Thus if we rotate CH_3–CH_3, we rotate the whole –CH_3 group with respect to the other.	Free rotation of atoms with respect to each other about a π bond is not possible because on rotation, overlapping vanishes and so the bond breaks. The bond breaking requires energy.
σ bonds involve the overlapping of hybrid orbitals. They determine the shape of the molecule.	π bond usually involve the overlapping of unhybridised orbitals. They do not determine the shape of molecules.

Results of valence bond theory

(i) The atoms, which unite to form a molecule, retain their identities in the resulting molecule.

(ii) The greater the overlap between atomic orbitals, the greater is the strength of the bond formed.

(iii) Electrons, which are already paired in the valence shell usually cannot participate in bond formation, i.e., only half filled orbitals (singly occupied) can overlap with each other. One exception is the coordinate bond where, a filled orbital overlaps with an empty orbital to form a coordinate covalent bond.

Similarly, a filled p-orbital may laterally overlap with an empty p-orbital to form a π bond.

Example, back bonding in BF_3 molecule.

Overlapping of s–s orbitals

Hydrogen molecule (H_2) formation is an example of 1s-1s overlap between two hydrogen atoms, resulting in the formation of a covalent bond.

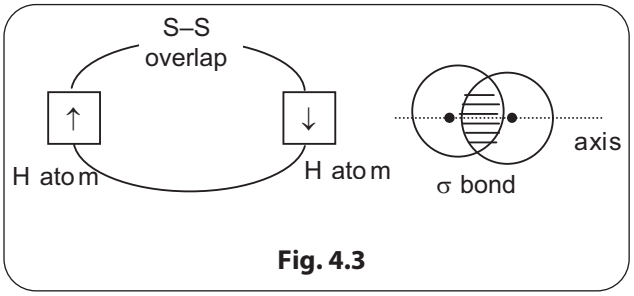


Fig. 4.3

Overlapping of s–p orbitals

The half filled s-orbital of one atom overlaps with the half-filled p orbital of another atom resulting in the formation of a chemical bond. Examples of this type of s-p overlap are the formation of compounds HF, H_2O, NH_3, HCl etc.

The general representation of an s-p orbital overlap can be made as shown.

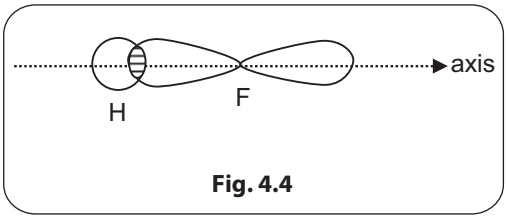


Fig. 4.4

Formation of H-Cl molecule

The 1s orbital of hydrogen overlaps with the $3p_z$ orbital of chlorine to form H-Cl bond. The atomic number of chlorine is 17. ($1s^2 2s^2 2p^6 3s^2 3p_x^2 3p_y^2 3p_z^1$) The $3p_z$ orbital contains a single electron.

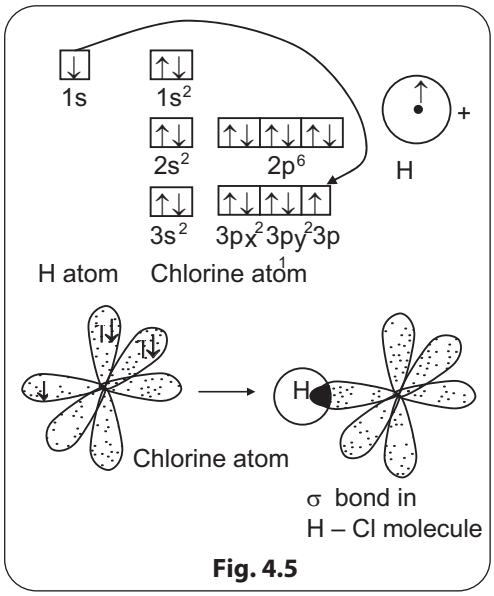


Fig. 4.5

Formation of water molecule

Water is a molecule in which there is a central oxygen atom, which is bonded to two hydrogen atoms. The electronic configuration of oxygen in the ground state is $1s^2 2s^2 2p_x^2 2p_y^1 2p_z^1$. Oxygen has two unpaired electrons, which can form two bonds with the two hydrogen atoms. The p orbitals are aligned at right angles (90°) to each other and from the direction of the two p orbitals $2p_y^1$ and $2p_z^1$ involved in bonding, the bond angle of H – O – H in water should be 90^0.

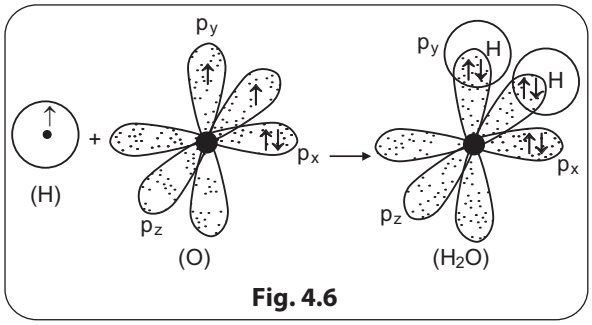


Fig. 4.6

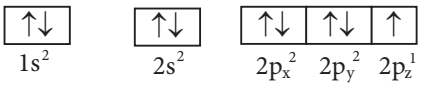
Actually it is found that the angle H – O – H in water is 104°, which is far higher than the anticipated. This large difference is due to the fact that oxygen does not use pure p orbitals for bonding with hydrogen but uses hybridized orbitals.

Overlap of p-p orbitals

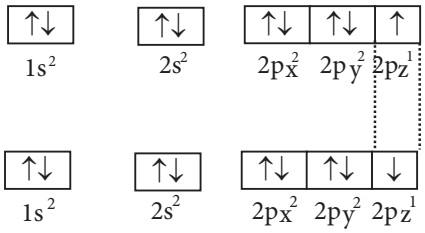
Two p-orbitals can overlap in two different ways forming two types of bonds.

(i) Linear overlap or head to head overlap

Fluorine molecule (F_2) is formed by the linear (length wise) overlap of the two p orbitals of fluorine atoms. Each fluorine atom has one $2p_z^1$ electron in the p orbital. The electronic configuration of fluorine is



Overlap between two $2p_z^1$ orbitals of fluorine atoms is as shown



The linear overlap of two p-orbitals may also be represented as shown below

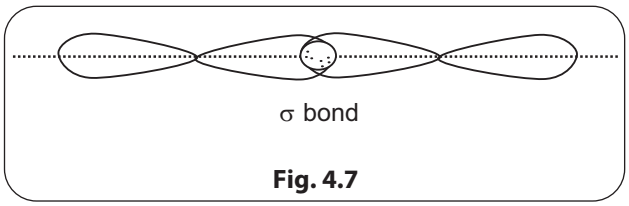


Fig. 4.7

The strength of the σ bond resulting from the linear overlap is in the order

$$s - s > s - p > p - p.$$

Lateral or parallel or sde to side overlap of p-p orbitals

In this type of bonding, the overlapping p orbitals are parallel to each other and perpendicular to the bond axis or they overlap side to side.

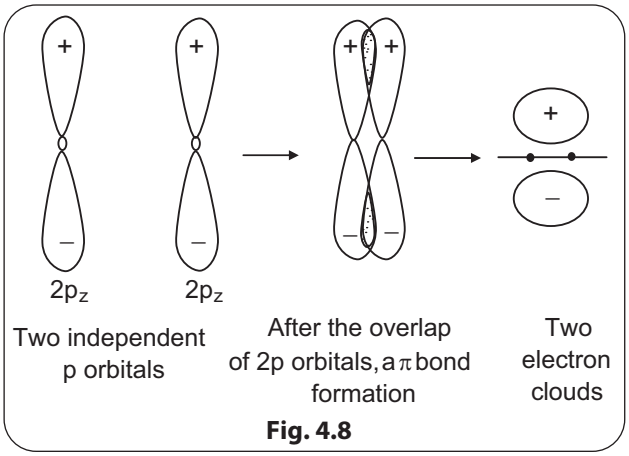
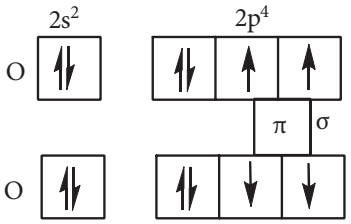


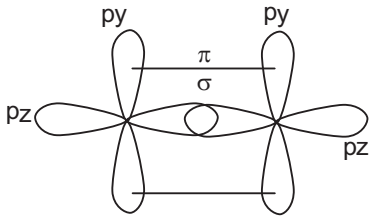
Fig. 4.8

Formation of oxygen and nitrogen molecules

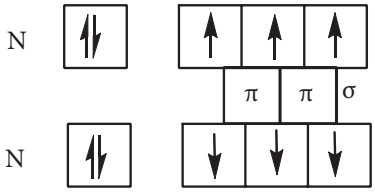
Oxygen atom contains two unpaired electrons in p_y and p_z orbitals. When O_2 molecule is formed the p_z orbitals linearly overlap to form a σ bond. The p_y orbitals overlap laterally to form a π bond.



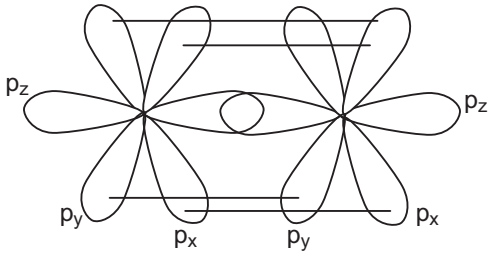
Overlapping may also be represented as follows.



Similarly, a nitrogen atom has three unpaired electrons in p_x, p_y and p_z orbitals. They form a σ bond and two π-bonds.



Overlapping may also be represented as follows:



Bond length

Bond length or bond distance is the average distance between the nuclei of two bonded atoms in a molecule. For example, bond length of H – H = 0.74 Å, Cl – Cl = 1.99 Å H – Cl = 1.27 Å. Bond length decreases as the electro negativity difference between the two bonded atoms increases. For example, bond lengths of H – F = 0.92 Å, H – Cl = 1.27 Å, H – Br = 1.41 Å and H – I = 1.61 Å. Also, the bond length decreases with the multiplicity of the bond formed between the two atoms. Bond length of C – C = 1.54 Å, C = C is 1.34 Å and C ≡ C is 1.20 Å. Bond length is the sum of the atomic (covalent) radii of the two atoms. Thus smaller atoms form shorter bonds and larger atoms form longer bonds.

CONCEPT STRANDS

Concept Strand 9

Compare the silicon–silicon bond length with that of carbon–carbon and justify?

Solution

C–C bond length is less than that of Si–Si. Covalent bond energy for non-metals decreases down the group. C–C bond length = 0.77 Å and Si–Si bond length is equal to 1.17 Å.

C–C bond is shorter and stronger than Si–Si bond.

Concept Strand 10

In which of the following molecules is N to N bond distance expected to be the shortest?

(i) N_2H_4 (ii) N_2
(iii) N_2O_4 (iv) N_2O

Solution

N ≡ N has the highest multiple bond character and hence the shortest bond distance.

Bond angle

Bond angle is defined as the internal angle between the orbitals containing bonding electron pairs in the valence shell of the atoms in a molecule. For example, in H_2O molecule, the angles, 1, 2 and 4 are not considered as bond angles, but angle 3 is the bond angle since it is between two bonded atoms. *Bond angle may also be defined as the angle subtended by two atoms on a third atom, which is bonded to the first two atoms.*

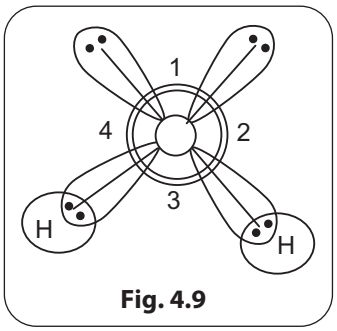


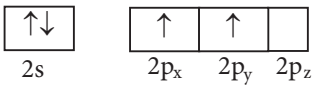
Fig. 4.9

Hybridization

According to V.B theory, valency of an element is given by the number of half filled orbitals it has when it combines. For example valency of H = 1 ($1s^1$), oxygen = 2 ($1s^2$, $2s^2$, $2p_x^2$ p_y^1 p_z^1), Nitrogen = 3 ($1s^2$, $2s^2$, $2p_x^1$ p_y^1 p_z^1) etc.

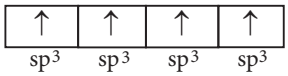
In the ground state, a carbon atom has two electrons in two half filled 2p orbitals ($2p_x$ and $2p_y$) and one 2p orbital ($2p_z$) is vacant. On this basis carbon should form compounds with only two single bonds. However, all known compounds of carbon are derived from tetra valent carbon. It is also well known that all the four bonds are equivalent and these four bonds are tetrahedrally oriented in shape.

In carbon the 2s orbital is fully occupied and the vacant $2p_z$ orbital has a slightly higher energy than the 2s level.



Electrons in the ground state of carbon

When one of the paired 2s electrons is promoted to the vacant $2p_z$ orbital, then the four electrons will have a new configuration viz $2s^1 2p_x^1 2p_y^1 2p_z^1$ and this represents an excited state. In the valence bond approach, carbon forms compounds when it is in the excited state. Of the four electrons in the excited state, all of them are not equivalent. All the four orbitals are mixed or hybridized or merged together so that the energy is redistributed to give a set of new four equivalent orbitals with the same energy. The resulting orbitals are known as sp^3 hybridized orbitals, since one s and three p orbitals are mixed.



(after hybridization)

Hybridization is only for molecules or ions having three or more atoms. The central atom of the molecule or ion creates the required number of hybridized orbitals, with which the half–filled orbitals of surrounding atoms overlap. All overlaps involving hybridized orbitals are sigma.

The hybridization of the central atom or ion is decided by the total number of sigma bonds and lone pairs around it. The total number of sigma bonds and lone pairs around the central atom or ion is equal to the total number of hybridised orbitals. π bonds have no contribution towards hybridization. A coordinate bond is a sigma bond and is treated as a covalent bond in hybridization.

Table 4.2

Total number of σ bonds and lone pairs around the central atom/ion	Hybridization
2	sp
3	sp^2
4	sp^3 or dsp^2
5	sp^3d or dsp^3
6	sp^3d^2 or d^2sp^3
7	sp^3d^3

As the s–character in a hybrid orbital increases, its electronegativity increases, since the s-electrons are more close to the nucleus.

Hybridisation	sp^3	sp^2	sp
% s-character	25%	33.3%	50%
Electronegativity order	$sp^3 < sp^2 < sp$		

Formula for finding the hybridization

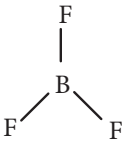
No. of hybrid orbitals = $\frac{1}{2}[V + M - C + A]$

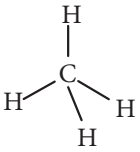
where, V = the total no. of valence electrons on the central atom

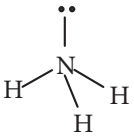
M = the no. of monovalent atoms attached to the central atom.

C = the charge on cation, A = the charge on anion } in the case of radicals

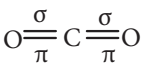
(i) Cl – Be – Cl It has two σ bonds around Be. Hybridization of Be is sp in $BeCl_2$.

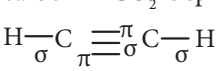
(ii)  Three σ bonds around B. Hybridization of B in BF_3 is sp^2.

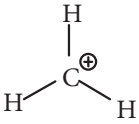
(iii)  Four sigma bonds and hybridization of C in CH_4 is sp^3.

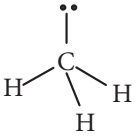
(iv)  Three σ bonds + one lone pair. Hybridization of N in NH_3 is sp^3

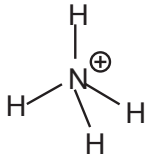
(v)  Two σ bonds + two lone pair. Hybridization of O in H_2O is sp^3

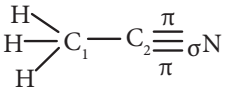
(vi)  Only two σ bonds. Hybridization of carbon in CO_2 is sp

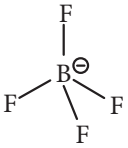
(vii)  There are two σ bonds around each carbon. Both the carbons are central atoms and they are sp hybridized.

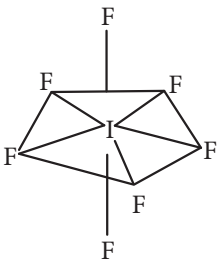
(viii)  Carbon is sp^2 hybridized in $^{\oplus}CH_3$.

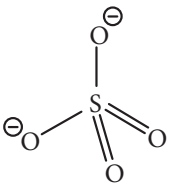
(ix)  Carbon is sp^3 hybridized in $^{\ominus}CH_3$.

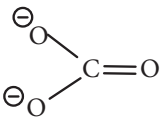
(x)  There are four σ bonds around N and there is no lone pair on N. The hybridization of N in NH_4^+ is sp^3

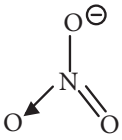
(xi)  There are four σ bonds around C_1 and hence it is sp^3 hybridized whereas there are only two σ bonds around C_2 and it is sp hybridized.

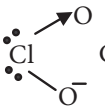
(xii)  B in BF_4^- is sp^3 hybridized.

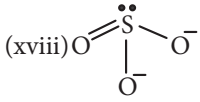
(xiii)  I in IF_7 is sp^3d^3 hybridized.

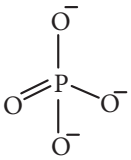
(xiv)  Sulphur in SO_4^{2-} is sp^3 hybridized.

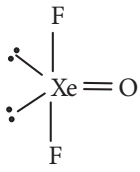
(xv)  Carbon in CO_3^{2-} is sp^2 hybridized.

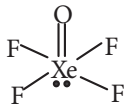
(xvi)  N in NO_3^- is sp^2 hybridized.

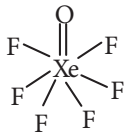
(xvii)  Cl in ClO_2^- is sp^3 hybridized ($2\sigma + 2\ell p$); ℓp = Lone pair

(xviii)  S in SO_3^{2-} is sp^3 hybridized ($3\sigma + 1\ell p$)

(xix)  P in PO_4^{3-} is sp^3 hybridized (4σ bonds only)

(xx)  Xe in $XeOF_2$ is sp^3d hybridized ($3\sigma + 2\ell p$)

(xxi)  Xe in $XeOF_4$ is sp^3d^2 hybridized ($5\sigma + 1\ell p$)

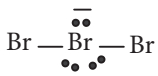
(xxii)  Xe in $XeOF_6$ is sp^3d^3 hybridized (7σ bonds only)

CONCEPT STRANDS

Concept Strand 11

What is the type of hybridisation involved in Br_3^-?

Solution



Br_3^- = 3 lone pairs + 2 bond pairs = 5

$5 \equiv sp^3d$

Concept Strand 12

Which is more stable, BeF_2 or BF_3 why?

Solution

BeF_2. BeF_2 involves sp-p overlap where as BF_3 involves sp^2-p overlap. Greater the s-character of the hybrid orbital, stronger are the bonds formed.

VSEPR (valence shell electron pair repulsion) theory

In a molecule consisting of several atoms and bonds, the direction of bonds around the central atom and the overall shape of the molecule depends on how the electron pairs-bond pairs and the lone pairs are arranged spatially around the central atom. The electron pairs are arranged in space as far apart as possible to minimize the electrostatic repulsion between the electrons.

The decreasing order of repulsion among the lone pair and bond pair of electrons is lone pair – lone pair >> lone pair – bond pair > bond pair – bond pair.

The lone pairs occupy a greater angular volume than the bonding pairs and hence the repulsion created by such orbitals are the largest.

When a molecule has a lone pair of electrons, the bond pairs are pushed closer to each other so that the bond angle becomes lesser than the expected value. Molecular shapes or geometry of molecules may be linear, angular, pyramidal, tetrahedral or any other determined shape. Multiple bonds behave as a single bond pair for the purpose of VSEPR. Double bonds exert a repulsion almost equal to that of a lone pair. Depending on the nature of repulsion, the following shapes of molecules are recognized.

CONCEPT STRAND

Concept Strand 13

The repulsion between lone pairs are greater than that between bond pairs, why?

Solution

Non-bonded orbitals occupy relatively more space compared to bonded orbitals and thus repulsion is greater. Bonded pair is attracted by two nuclei and thus occupy less space.

sp³ hybridization

sp³ hybrid orbitals are formed by the interaction of one s and three p orbitals. Four hybridized or equivalent or identical orbitals are formed by this process.

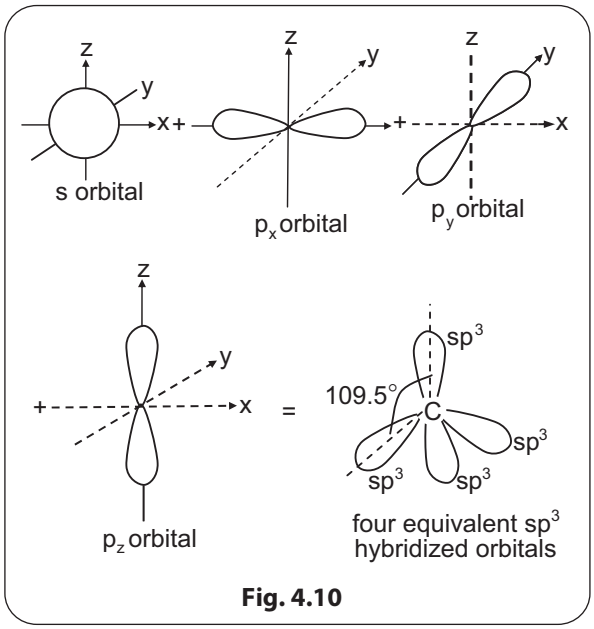
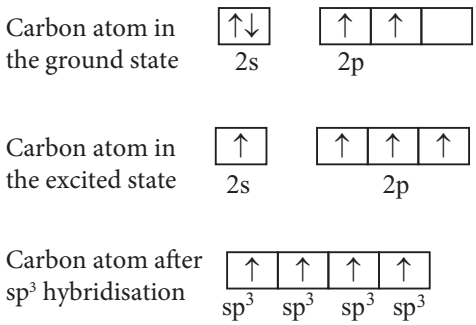


Fig. 4.10

Structure of methane



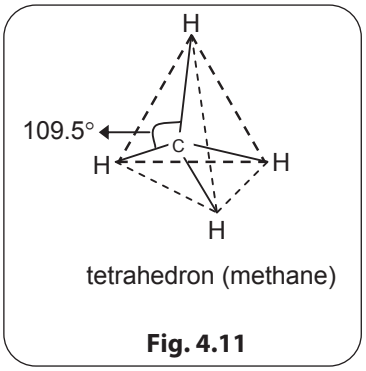


Fig. 4.11

In methane, the four sp³ hybridized orbitals of carbon are directed towards the four corners of a regular tetrahedron with the carbon atom located at the centre and four hydrogen atoms at the corners.

Each of the four sp³ hybrid orbitals on carbon is singly filled. In the formation of methane, each of these hybrid orbitals overlaps with the half filled 1s orbital of hydrogen. This results in four C – H bonds and these single bonds are known as sigma (σ) bonds. The axes of the sp³ orbitals are directed towards the four corners of a regular

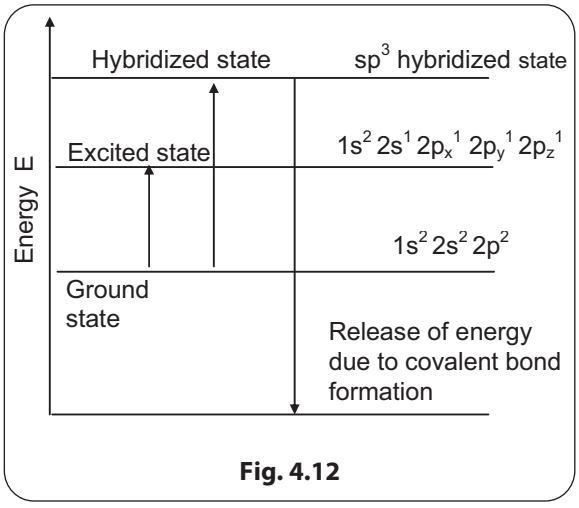


Fig. 4.12

tetrahedron, with the carbon at the centre with H – C – H bond angle 109.5^0.

Energy is required for the promotion of a 2s electron to the 2p orbital and for the hybridization of the orbitals to give equivalent orbitals, but this is compensated by the release of energy in the formation of four covalent bonds involving the sp^3 hybrid orbitals. In an sp^3 hybrid orbital, there is 25% s character and 75% p character.

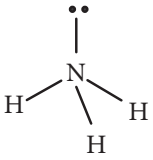
So the entire input of energy is more than recovered by the release of energy when the hybrid orbitals combine with orbitals of other atoms to result in covalent bonds.

Ground energy state $\xrightarrow{+\text{ Energy}}$ Excited energy state $\xrightarrow{+\text{ Energy}}$ sp^3 hybridized carbon

Energy released $\rightarrow$ Overlap of sp^3 hybrid orbitals with half–filled orbitals of hydrogen

Ammonia molecule

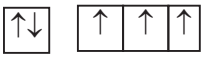
In ammonia molecule, central nitrogen atom is bound to three hydrogen atoms.



The nitrogen atom uses only three electrons for bonding.

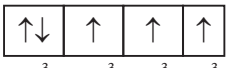
Three sigma bonds are formed with three H atoms while the two free electrons remain as non-bonding electrons on nitrogen. Nitrogen atom in ammonia molecule is sp^3 hybridized.

Nitrogen atom in the ground state $2s^2$, $2p^3$



Nitrogen atom after sp^3 hybridization

$(sp^3)^2$ $(sp^3)^1$ $(sp^3)^1$ $(sp^3)^1$



Although the arrangement of the sp^3 hybridized orbitals around the central nitrogen is tetrahedral, due to the presence of lone pair of electrons in one of the orbitals, the shape of the NH_3 molecule gets distorted and becomes trigonal pyramidal and the H – N – H bond angle decreases from the expected value of 109.5° to 107°.

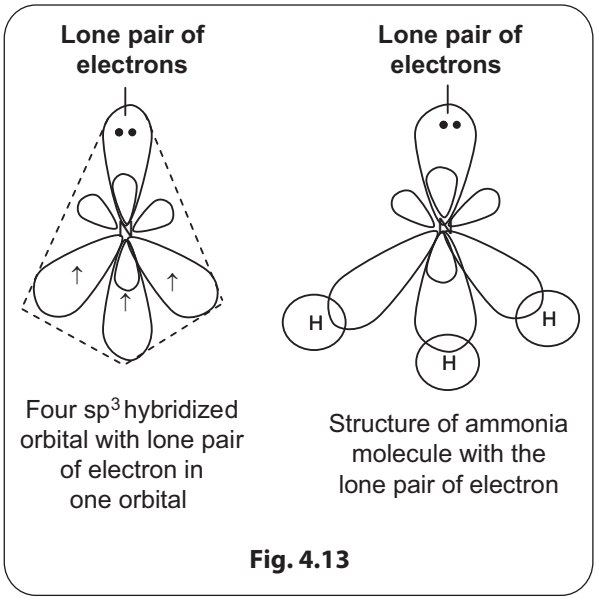
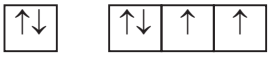


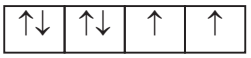
Fig. 4.13

Water molecule

Oxygen atom in the ground state



Oxygen atom in sp^3 hybridization



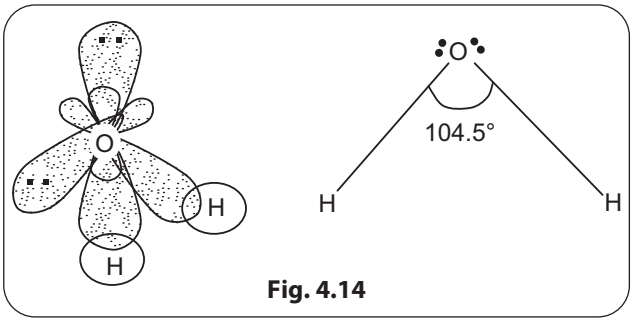


Fig. 4.14

Oxygen uses its sp^3 hybridized orbital in the formation of water. There are two bond pairs and two lone pairs in the hybridized orbitals. Two of the four sp^3 hybridized orbitals of oxygen form σ bonds with 1s orbitals of the hydrogen atom.

The bond angle H – O – H in water is found to be 104.5° much less than the expected tetrahedral angle of 109° in an sp^3 hybridized structure. This difference has been explained due to the greater lone pair-lone pair repulsion than lone pair-bond pair or bond pair-bond pair repulsion.

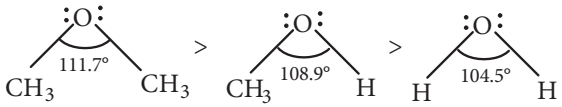
CONCEPT STRANDS

Concept Strand 14

CH_3–O–CH_3 (A), CH_3–O–H (B) and H – O – H (C) have the same hybridization on oxygen. Compare the bond angle involving oxygen in (A), (B) and (C) and justify.

Solution

Oxygen atom is sp^3 hybridized in all cases.



Lone pair repulsion is counter balanced by the steric repulsion of bulky CH_3 groups and hence the bond angle in CH_3–O–CH_3 is maximum.

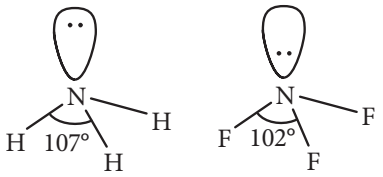
Concept Strand 15

Arrange the compounds NF_3, NH_3, PH_3, and AsH_3 in the decreasing order of their bond angle and justify.

Solution

NH_3 > NF_3 > PH_3 > AsH_3 (107°, 102°, 94° and 92° respectively)

In N, P and As, the p-orbitals involved in hybridization are 2p, 3p and 4p respectively. As the size of the orbital increases, distance of the bonded electron from the central atom increases as well as the repulsion between the bonded orbitals decreases. Hence the bond angle decreases from NH_3 to PH_3 to AsH_3.In NH_3 and NF_3 – N atoms are sp^3 hybridized. In NF_3, due to high electronegativity of fluorine, the shared pair of electrons are pulled towards fluorine hence reducing the bond pair-bond pair repulsion. This brings the bonds together hence reducing the bond angle.



Concept Strand 16

Arrange the following in the decreasing order of single bond dissociation energy and justify.

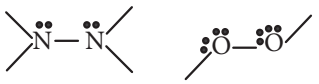
(i) O – O
(ii) F – F
(iii) N – N and
(iv) C–C

Solution

C — C > N — N > F — F > O — O
81.6 39 38 34.9 kcal mol^{-1}

Bond strength depends on the extent of overlap. Extent of overlap is more for sp^3 – sp^3 (C – C) than for p – p (F – F).

In the case of N – N and O – O bonds, even though the overlaps are sp^3 – sp^3, the presence of lone pairs on the sp^3 hybridized atoms decreases the extent of overlap and hence the bond strength.



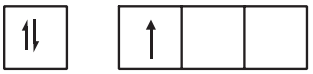
sp^2 hybridization

sp^2 hybrid orbitals result from a combination of one s and two p orbitals. Out of the four orbitals (one s and three p) available for hybridization, only one s and two p orbitals take part in this sp^2 hybridization.

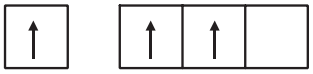
There are three new equivalent sp^2 orbitals formed with 33% s character and 67% p character. The three equivalent orbitals have a symmetrical distribution and are directed towards the corners of a planar trigonal structure at an angle of 120° from one another.

Structure of borontrifluoride

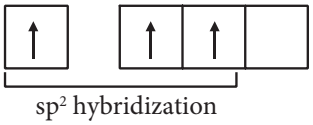
Ground state of Boron atom is

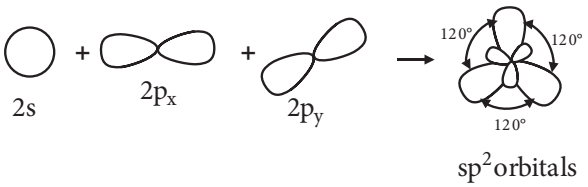


To form BF_3 it must form three covalent bonds. For that it requires three unpaired electrons. It is achieved by unpairing and exciting an electron from 2s to 2p.



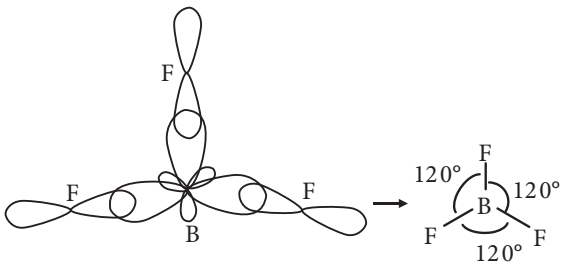
Three orbitals are hybridized in order to produce three σ bonds





sp^2 orbitals

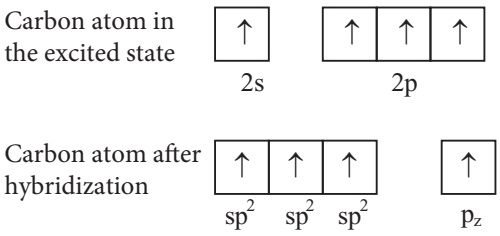
The three sp^2 hybridized orbitals are directed towards the three corners of an equilateral triangle. They overlap with the p_z orbitals of fluorine atoms to form BF_3.



The shape of the molecule is trigonal planar.

Structure of ethylene $CH_2 = CH_2$

Both the carbons in ethylene are sp^2 hybridized.



For every carbon with sp^2 hybridized orbitals, mixing occurs as shown below.

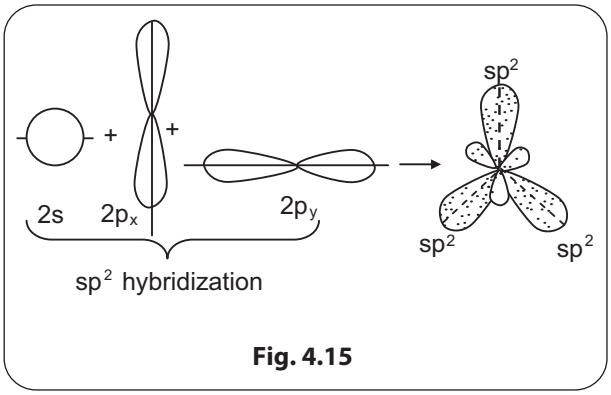


Fig. 4.15

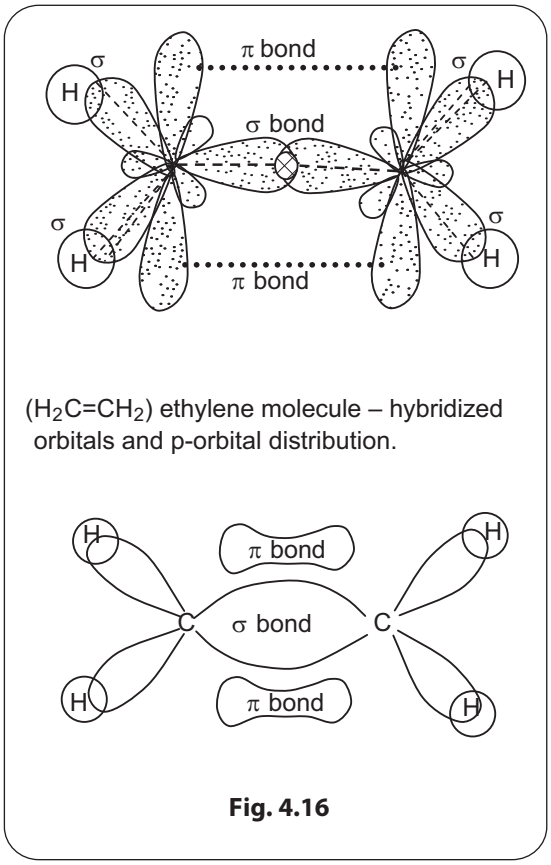


Fig. 4.16

In ethylene molecule, the sp^2 hybrid orbitals form σ-bonds by linear overlap whereas the unhybridized pure p orbitals which are half filled, overlap by their sides to form a π bond.

Structure of $BeCl_2$

In $BeCl_2$, Cl – Be – Cl , the central Be atom uses both its valence electrons ($2s^2$) – atomic number $4(1s^2 2s^2)$ – in

forming 2σ bonds with the two chlorine atoms. There are three vacant p orbitals in the ground state.

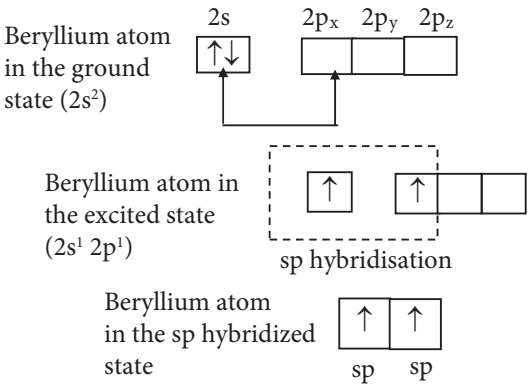


Fig. 4.17

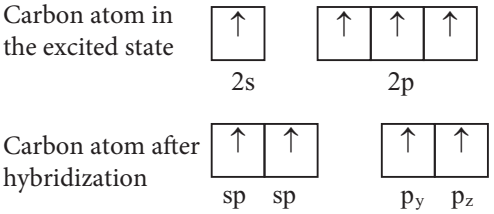
Few other compounds which exhibit sp hybridization are C_2H_2, BeF_2, CO_2, CH_3CN, HCN etc.

sp hybridization

One s and one p orbital take part in hybridization to result in two equivalent sp hybridized orbitals. The sp hybrid orbitals have 50% s and 50% p character. The sp hybrid orbitals are linear and lie in the same line at an angle of 180^0 from each other.

Structure of acetylene, H – C ≡ C – H

Both the carbons in acetylene are sp hybridized.



Each carbon has two pure 2p orbitals which can overlap laterally or in a parallel way to form two separate π bonds to give triple bonded structure. The sp hybridized orbitals overlap coaxially or linearly, to form σ bonds to hydrogen atoms and between carbon atoms.

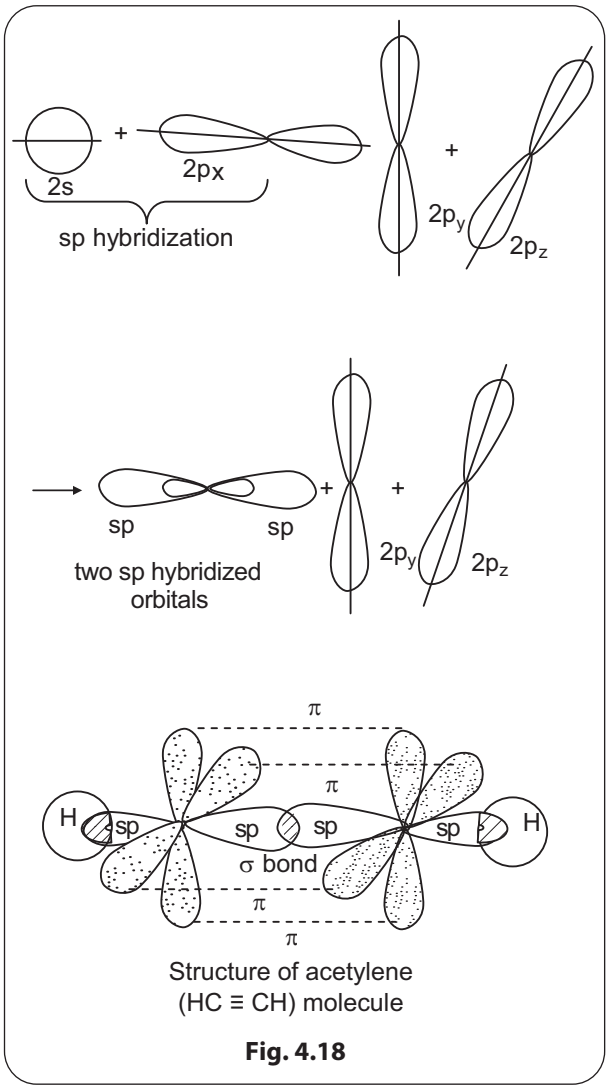


Fig. 4.18

CONCEPT STRANDS

Concept Strand 17

Find the hybridization of central carbon in

(i) O = C = C = C = O

(ii) $CH_2 = C = CH_2$

(iii) $CH_2 = C = O$

Solution

In all these cases, carbon forms 2 double bonds and hence hybridization is sp.

Concept Strand 18

Arrange the hybrid orbitals involving s and p orbitals in the increasing order of their size.

Solution

$sp^3 > sp^2 > sp$. Size increases with increase in p-character

sp^3d hybridization

The d-orbital used in sp^3d hybridization may be either d_{z^2} or $d_{x^2-y^2}$. If the orbital used is d_{z^2}, then the geometry is trigonal bipyramidal, otherwise it will be square pyramidal.

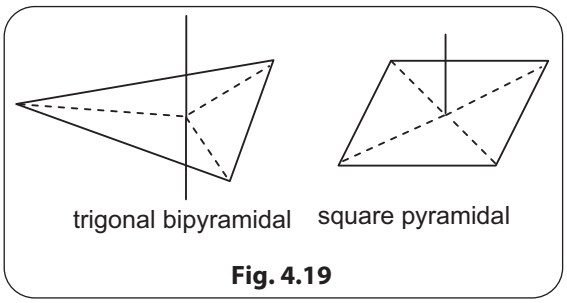
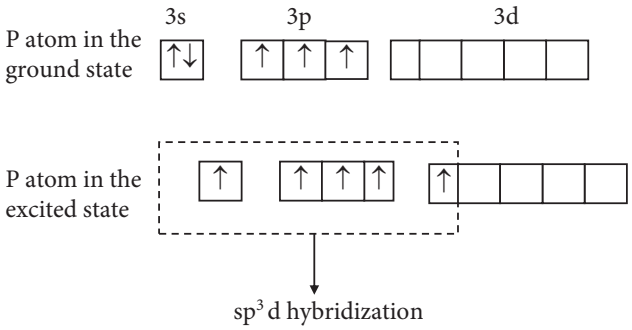


Fig. 4.19

In trigonal bipyramidal geometry, the axial bonds are slightly longer than the equatorial bonds. This is because the axial bonds experience greater repulsion from other bonds than the equatorial bonds. The electron pair repulsions decrease sharply with increase in bond angles. Moreover of lone pairs are present in sp^3d hybridized molecules they occupy the equatorial positions because here they experience less repulsion from other bonds. Note that an equatorial bond has only two bonds at 90° to it (the two axial bonds) while an axial bond has three bonds at 90° to it (the three equatorial bonds).

Structure of PCl_5

The structure of PCl_5 shows that the central atom P uses all its five electrons from its valence shell ($3s^23p^3$) in forming the 5σ bonds with five chlorine atoms. There are five bond pairs and no lone pair of electrons.



The result is that there are five non-equivalent sp^3d hybrid orbitals with 2 axial orientations and 3 equatorial (lateral) orientations. Hence, there are five sp^3d hybrid orbitals which are singly occupied. They form five σ bonds with five p orbitals of five chlorine atoms.

PCl_5 has a trigonal bipyramidal shape. Of the five bonds, three equatorial bonds are equal in length and two axial bonds are longer than the equatorial bonds. Axial bonds are less strong than equatorial bonds.

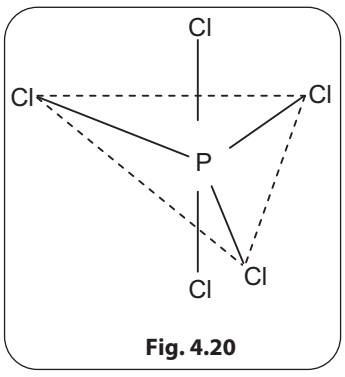


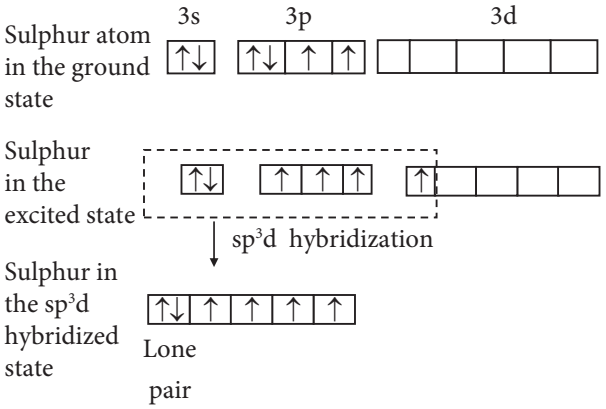
Fig. 4.20

Structure of SF_4

In SF_4 molecule, the central sulphur atom is linked to four fluorine atoms and in this process, sulphur uses four of its six electrons in the valence shell. Four bonds are formed with 4 electrons and a lone pair remains. The valence shell electrons are involved in sp^3d hybridization.

The electronic configuration of sulphur (Atomic number:16) is $1s^2 2s^2 2p^6 3s^2 3p^4 3d^o$. Sulphur has five vacant d orbitals in the third level.

The sp^3d hybridization results in five equivalent orbitals. Four of the hybrid orbitals are used for bonding with 4 orbitals from four fluorine atoms while one hybridized orbital contains the lone pair of electrons.



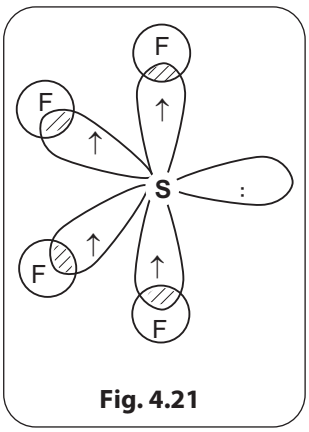
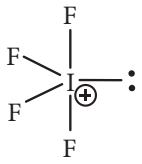


Fig. 4.21

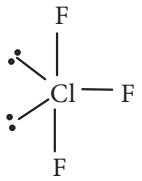
The spatial arrangement of five electron pairs (4 bond pairs and one lone pair) around the central sulphur atom is trigonal bipyramidal. However, due to the presence of the lone pair of electrons in the equatorial hybrid orbital, the shape of SF_4 molecule is see-saw. The distortion in the shape of the molecule is due to the greater lone pair–bond pair repulsion.

Another species in which the central atom exhibits sp^3d hybridization is IF_4^+ (a cation).

This species also has a see saw structure.



ClF_3 molecule also shows sp^3d hybridization at the central chlorine atom. It has T-shape.



XeF_2 has a structure with three lone pairs and two bond pairs with two fluorine atoms. The molecule has a linear structure as shown.

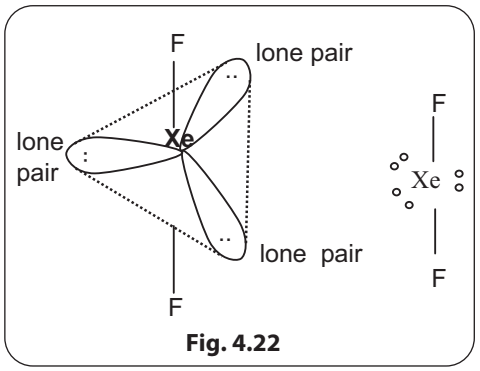


Fig. 4.22

ICl_2^- and I_3^- ions also have linear structures with sp^3d hybridization

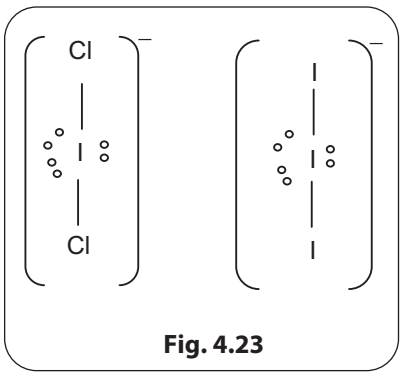


Fig. 4.23

CONCEPT STRAND

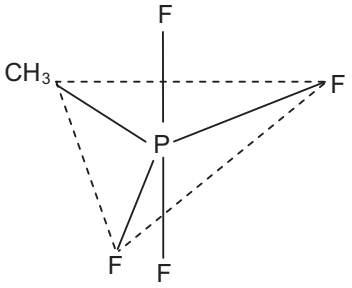
Concept Strand 19

Choose the correct statement
In PF_4CH_3

(i) all the four P – F bond lengths are equal
(ii) three P – F bond length are shorter than the other P – F bond lengths
(iii) one P – F bond length is shorter than the other three P – F bond lengths
(iv) two P – F bond lengths are shorter than the other two P – F bond lengths

Solution

Two P – F bond lengths are shorter than the other two P – F bond lengths



PCl_2Br_3

Hybridization of P in PCl_2Br_3 is sp^3d. The two chlorine atoms are arranged along the axial positions because larger Br atoms prefer equatorial positions.

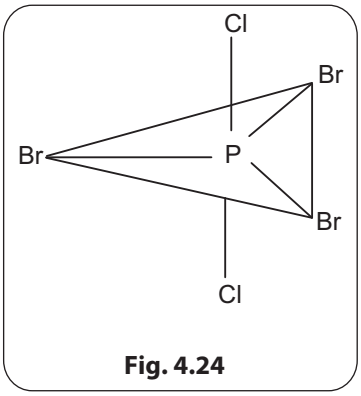


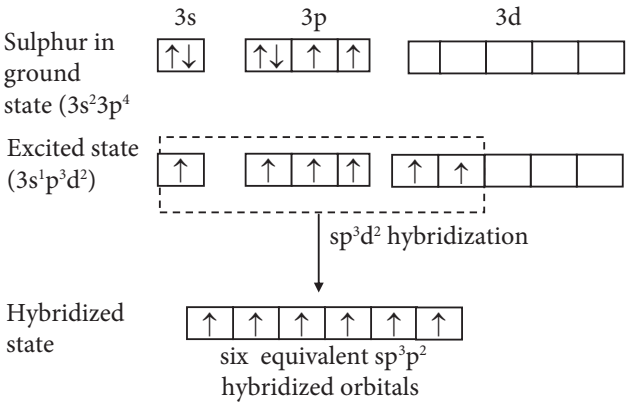
Fig. 4.24

sp^3d^2 hybridization

The d-orbitals used in sp^3d^2 hybridisation are $d_{x^2-y^2}$ and d_{z^2}

Sulphur hexafluoride (SF_6) is an example of a molecule where sulphur shows sp^3d^2 hybridization.

In SF_6, all the six valence electrons of sulphur are used up. There are six sp^3d^2 equivalent orbitals formed after hybridization with no lone pair of electrons. Each of the six hybridized sp^3d^2 orbitals are singly filled before bonding.



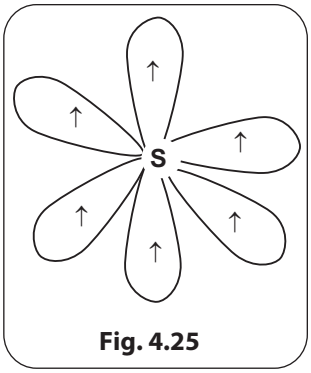
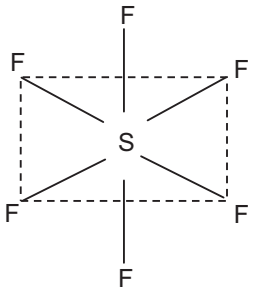


Fig. 4.25

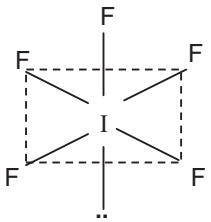
Each one of these sp^3d^2 hybridized orbitals favourably overlaps with p orbitals of six fluorine atoms to form SF_6. The molecule has an octahedral shape.



Four fluorine atoms and sulphur atom are in a plane while two fluorine atoms are one above the plane and the other below the plane.

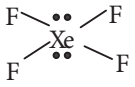
Structure of IF_5

Iodine is in group 17 of the periodic table. Atomic number of iodine is 53 and its electronic configuration is $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^2 4d^{10} 5p^5$. The valence shell electrons are $5s^2 5p^5$. Five of these seven electrons form five iodine–fluorine sigma bonds and the remaining two electrons form the lone pair. One 5s three 5p and two of the d-orbitals which are singly occupied mix and hybridise to form six sp^3d^2 orbitals. The lone pair of electrons occupy one of the corners. It is to be remembered that all six corners of an octahedron are equivalent. The molecule has square pyramidal shape.



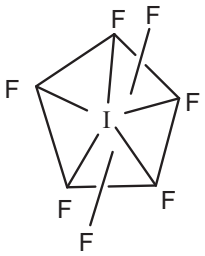
Structure of XeF_4

In xenon atom, the 5s and 5p orbitals are full and five 5d orbitals are vacant. Hence in XeF_4 there are two lone pairs and four singly occupied hybrid orbitals in the sp^3d^2 hybridization. XeF_4 assumes a square planar shape.



Other species like ICl_4^-, BrF_4^- have the hybridization of the central atom as sp^3d^2. They are isostructural with XeF_4.

sp^3d^3 hybridization



Iodine in IF_7 is sp^3d^3 hybridized. Molecule is pentagonal bipyramidal.

Linear molecules

There is a central atom with two sigma (σ) bonding pairs and with no lone pair of electrons. These molecules have a linear arrangement.

Example, $BeCl_2$ (Beryllium chloride)

Cl : Be : Cl. The electrons are far apart and the bond angle is 180°

Other examples of molecules which have linear geometry are $ZnCl_2$, $HgCl_2$, BeF_2, BeH_2, CO_2

Trigonal planar geometry

In trigonal planar geometry, there are three bonds around the central atom-that means three bonding pairs and no lone pair of electrons. The only way by which these three pairs of electrons can be spread out is at an angle of 120° on a plane to be as far apart as possible. Such a type of arrangement is known as trigonal planar.

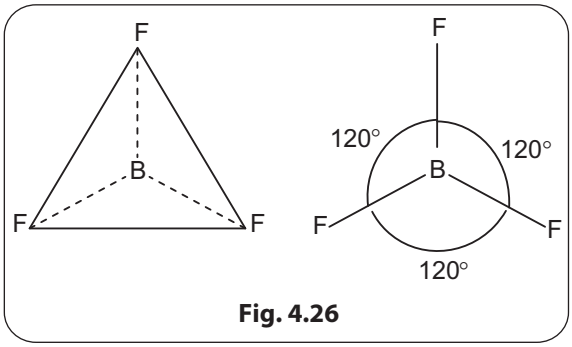


Fig. 4.26

Molecules, which have similar geometry are BCl_3 and $GaCl_3$. Some molecules like $PbCl_2$, $SnCl_2$ have angular or V-shape because of the presence of lone pair, since while

naming the shape (geometry) of a molecule, lone pairs are ignored and only atoms are considered.

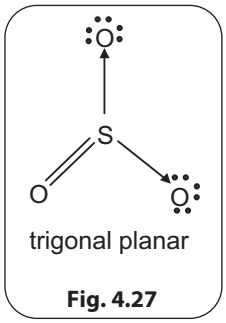


Fig. 4.27

In sulphur trioxide (SO_3), the central sulphur atom is linked to two oxygen atoms by coordinate bonds and to the third oxygen by a double covalent bond. Since there is no lone pair on the central sulphur atom, SO_3 has trigonal planar shape.

Trigonal pyramidal geometry

Ammonia (NH_3)

In ammonia central nitrogen is surrounded by3 bond pairs and a lone pair. The electron pairs are arranged tetrahedrally. Since the lone pair-bond pair repulsion is greater than the bond pair- bond pair repulsion, the molecule gets distorted and the bond angle is reduced from the expected 109.5° to 107°. Again due to the presence of a lone pair, the shape of the molecule cannot be regular tetrahedron: By ignoring the lone pair, the molecule assumes a trigonal pyramidal shape.

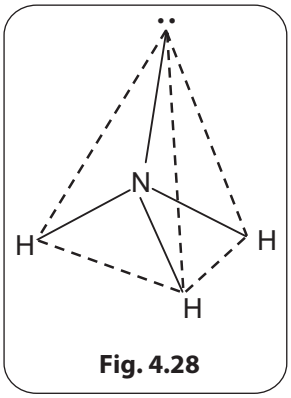


Fig. 4.28

Angular geometry

Water (H_2O)

There are two bond pairs and two lone pairs around oxygen. VSEPR theory predicts a tetrahedral geometry for water. However, the bond angle in water is found to be 104.5° as against the anticipated 109.5°. This distortion is due to repulsive interaction of the lone pair–lone pair (there are two lone pairs in oxygen), which is greater compared to the lone pair–bond pair and bond pair–bond pair repulsions. Ignoring the two lone pairs on oxygen, the geometry of water molecule is bent or angular or V-shaped.

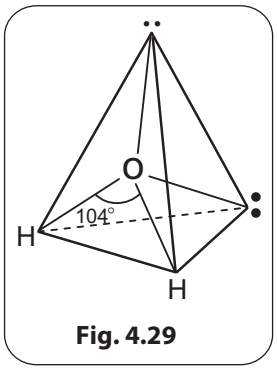


Fig. 4.29

Stannous chloride ($SnCl_2$)

$SnCl_2$ molecule assumes a V shaped geometry due to the presence of a lone pair. Because of the lone pair–bond pair repulsion, the bond angle is less than 120°.

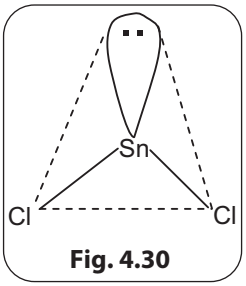


Fig. 4.30

SO_2 molecule

Sulphur atom forms two double bonds to two oxygen atoms and has a lone pair of electrons. Therefore the molecule is V-shaped or bent. The hybridization theory also leads to the same result as follows.

The central sulphur atom makes use of sp^2 hybridized orbitals for sigma bonding. In SO_2, two oxygens are bound to sulphur by sp^2–p overlap resulting in σ bonds. In addition to the orbitals used by sulphur and oxygen for σ bonding, sulphur has one p and one d orbital which are half filled as well as unhybridized and each oxygen atom has one half filled p orbital. So a pπ-pπ bond is formed by the overlap of the p orbitals of sulphur and oxygen. Similarly, a pπ-dπ bond is formed by the overlap of the d-orbital of sulphur with the p-orbital of oxygen.

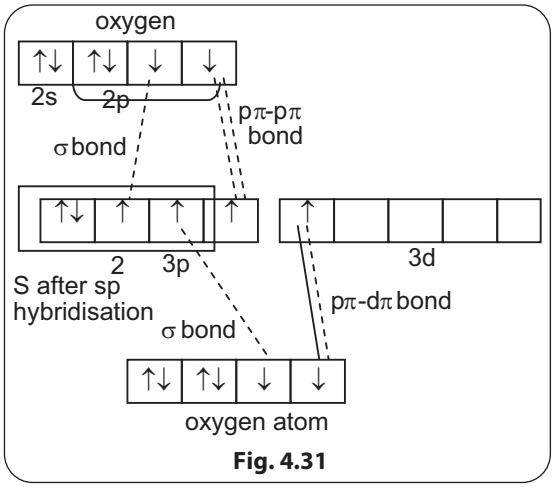


Fig. 4.31

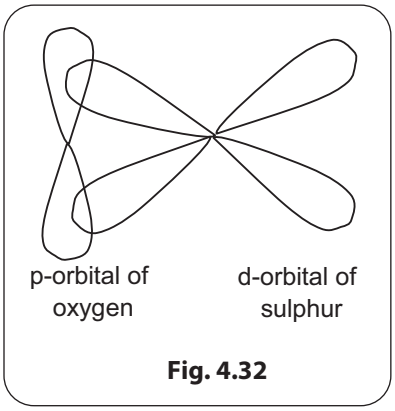


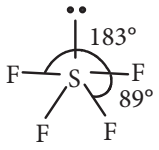
Fig. 4.32

See–saw geometry

SF_4 molecule

There are four bond pairs and one lone pair around the central sulphur atom.

SF_4 molecule assumes a see–saw geometry with bond angles of 89° and 177° instead of the expected 90° and 180°. There are two types of bonding pairs of electrons in this molecule–the axial and equatorial.



T–shaped geometry

ClF_3 molecule

There are three bond pairs and two lone pairs. They are arranged at the corners of a trigonal bipyramid. This molecule is T–shaped with a bond angle of 87.6° instead of 90°.

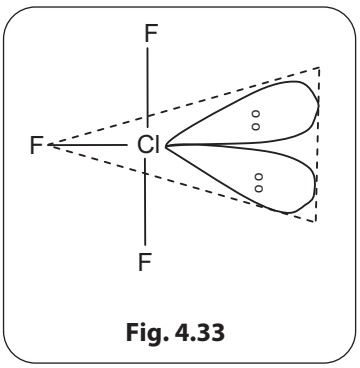


Fig. 4.33

Tetrahedral geometry

When there are four electron pairs around a central atom they are arranged at the corners of a tetrahedron. Examples are methane CH_4, ethane, $SiCl_4$, Sulphate ion (SO_4^{-2}), ClO_4^{-1}.

Structure of sulphate ion (SO_4^{2-})

By VSEPR theory the four electron pairs are arranged tetrahedrally.

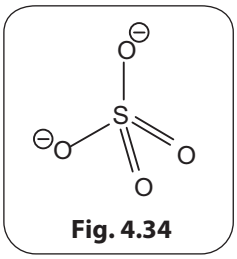


Fig. 4.34

Structure of perchlorate ion (ClO_4^-)

By VSEPR theory it has tetrahedral structure. Also by hybridization theory. ClO_4^- is sp^3 hybridized and the shape is tetrahedral.

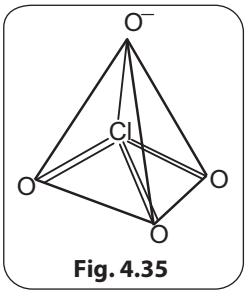


Fig. 4.35

Geometry of the electron pairs and the molecule

Table 4.3 (a)

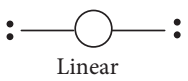
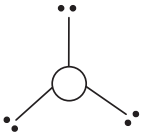
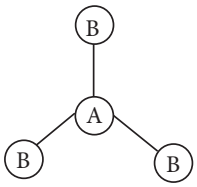
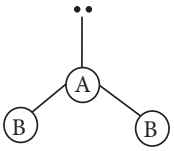
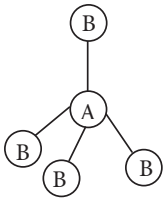
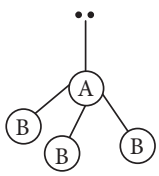
Total No. of electron pairs ($\sigma + \ell p$)	Type of Hybridization	Geometry of the electron pairs	Bond pairs	Lone pairs	Geometry of the molecule with examples
2	sp	 Linear	2	0	 Linear $BeCl_2$, CO_2
3	sp^2	 Trigonal planar	3	0	 Trigonal planar BF_3, SO_3
			2	1	 Bent or angular or V–shaped $SnCl_2$, : CH_2 (carbene)

Table 4.3 (b)

Total No. of electron pairs ($\sigma + \ell p$)	Type of Hybridization	Geometry of the electron pairs	Bond pairs	Lone pairs	Geometry of the molecule
			4	0	 Tetrahedral CH_4
			3	1	 Trigonal pyramidal NH_3

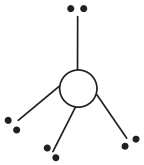
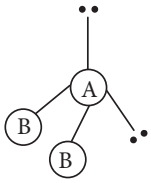
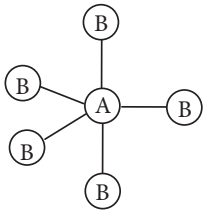
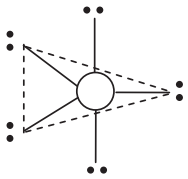
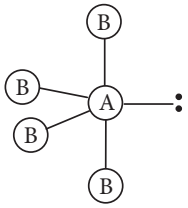
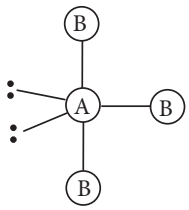
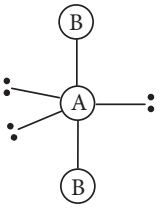
Total No. of electron pairs ($\sigma + \ell p$)	Type of Hybridization	Geometry of the electron pairs	Bond pairs	Lone pairs	Geometry of the molecule
4	sp^3	 Tetrahedral	2	2	 Bent or angular or V–shaped H_2O

Table 4.3 (c)

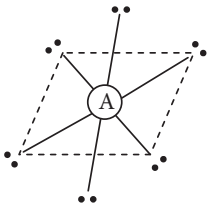
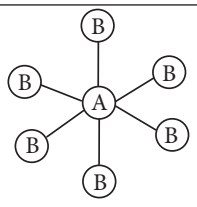
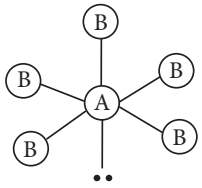
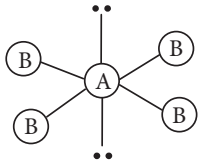
Total No. of electron pairs ($\sigma + \ell p$)	Type of Hybridization	Geometry of the electron pairs	Bond pairs	Lone pairs	Geometry of the molecule
			5	0	 Trigonal bipyramidal PCl_5
5	sp^3d	 Trigonal bipyramidal	4	1	 See saw XeO_2F_2, SF_4
			3	2	 T–shaped $BrCl_3$

(Contd.)

Total No. of electron pairs ($\sigma + \ell p$)	Type of Hybridization	Geometry of the electron pairs	Bond pairs	Lone pairs	Geometry of the molecule
			2	3	 Linear XeF_2

Note: Whenever there are lone pairs in trigonal bipyramid geometry, they are arranged in the equatorial position, so as to have minimum repulsion.

Table 4.3 (d)

Total No. of electron pairs ($\sigma + \ell p$)	Type of Hybridization	Geometry of the electron pairs	Bond pairs	Lone pairs	Geometry of the molecule
6	sp^3d^2	 Octahedral	6	0	 Octahedral SF_6
			5	1	 Square pyramidal BrF_5, $XeOF_4$
			4	2	 Square planar XeF_4
7	sp^3d^3		7	0	IF_7
			6	1	XeF_6

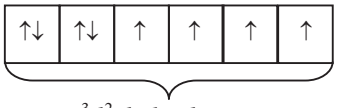
CONCEPT STRANDS

Concept Strand 20

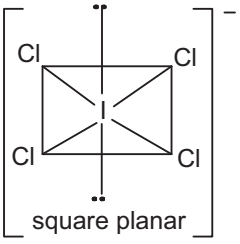
Predict the shape of ICl_4^-.

Solution

I: $5s^2, 5p^5, 5d^0$
I^- (ground state): $5s^2, 5p^6$
I^- (excited state): $5s^2, 5p^4, 5d^2$



sp^3d^2 hybridization

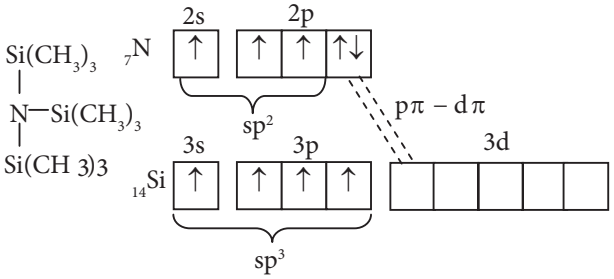


Concept Strand 21

Why $[Si(CH_3)_3]_3N$ is a weak Lewis base?

Solution

Due to pπ-dπ back bonding, lone pair on nitrogen is not available for donation.

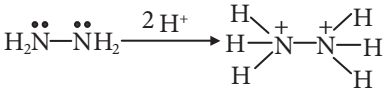


Concept Strand 22

Explain why?

Nitrogen–nitrogen bond distance in hydrazine ($NH_2 - NH_2$) is reduced from 1.47 to 1.4 Å by protonation

Solution



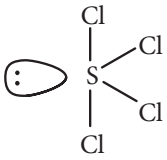
By protonation, the lone pairs are removed, the repulsion is reduced and hence the bond length is decreased.

Concept Strand 23

Among SO_4^{2-}, SCl_4, NH_4^+ and PO_4^{3-} which one is not isostructural with $SiCl_4$?

Solution

'S' in SCl_4 possess See–saw structure having sp^3d hybridization whereas $SiCl_4$ is tetrahedral.



Molecular orbital theory

In the Valence bond theory (VBT), a molecule is assumed to be made up of atoms. Formation of a bond between two atoms involves the overlapping of the orbitals of the two atoms, which are half filled.

These orbitals may or may not be hybridized. The atomic orbitals retain their identity even after the atom is chemically bonded to another atom in a molecule.

According to molecular orbital theory (MOT), all the atomic orbitals (AO) of the individual atoms participating in the formation of a molecule come close to each other and get mixed up to give an equal number of new orbitals that now totally belongs to the molecule. These new orbitals formed are known as molecular orbitals (MO).

When nuclei of two atoms come close to each other, the individual atomic orbitals interact leading to the for-

mation of molecular orbitals. Once the molecular orbital is formed, the individual atomic orbitals lose their identity. Each electron in a molecular orbital belongs to all the nuclei in the molecule.

LCAO (linear combination of atomic orbitals)

Molecular orbitals are obtained by solving the Schroedinger wave equation for the molecule. This is a very difficult task. To overcome this problem an approximate method called Linear combination of atomic orbitals (LCAO) method is used.

Linear combination of two atomic orbitals leads to the formation of two molecular orbitals. The energy of one molecular orbital is lower than the energies of the atomic orbitals. Such a molecular orbital is designated as bonding molecular orbital. The energy of the other molecular orbital formed is higher than the energies of the individual atomic orbitals. Such a molecular orbital is designated as anti bonding molecular orbital.

Mathematically speaking, the bonding molecular orbital ψ_b is formed by the addition combination of two atomic orbitals, ψ_1 and ψ_2 and the antibonding molecular orbital ψ_a is formed by the subtraction combination of the two atomic orbitals.

i.e., $\psi_b = \psi_1 + \psi_2$

$\psi_a = \psi_1 - \psi_2$

Greater the overlap of the two combining atomic orbitals, the lower would be the energy of the bonding molecular orbital and higher will be the energy of the anti-bonding molecular orbital. In a bonding MO the electron charge density between the two nuclei is high and therefore the repulsion between the nuclei is very low. Bonding molecular orbital, therefore, gets stabilized and bond formation is favoured.

In an anti-bonding MO the electron charge density between the two nuclei is low and therefore the repulsion between the nuclei is high and hence the anti-bonding molecular orbital is destabilized. Bond formation is not favoured in the antibonding molecular orbital.

Combination of s–s atomic orbitals

Bonding molecular orbital is designated as σ and the anti bonding molecular orbital (ABMO) as σ*.

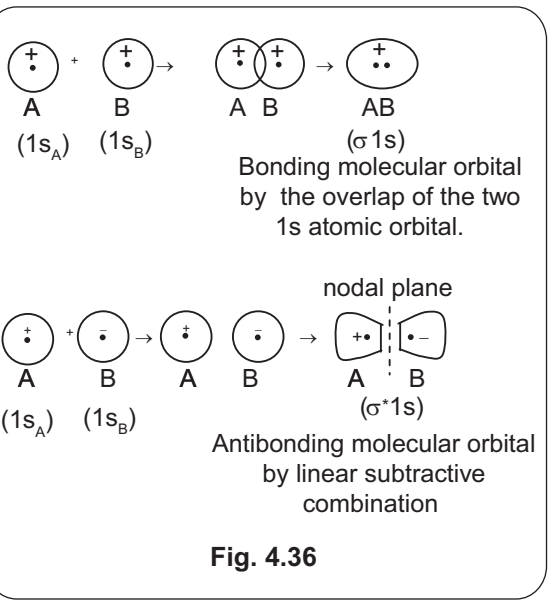


Fig. 4.36

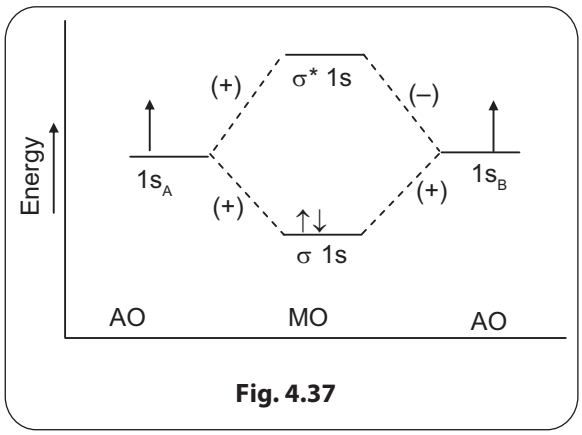


Fig. 4.37

Energy level diagram for σ* 1s bonding and σ* 1s antibonding molecular orbitals formed by the linear combination of 1s atomic orbitals of the two hydrogen atoms in a hydrogen molecule is shown above. It is important to note that (+) and (–) signs given for the atomic and molecular orbitals are only algebraic signs of wave functions and not to be confused as (+) or (–) electrical charges.

It is to be noted that the lowering of energy of bonding M.O w.r. to the A.O's is equal to the raising of energy of the anti-bonding M.O.

Combination of s-p orbitals

In the diagram shown below, the p-orbital does not favourably combine with the s-orbital of A to form a bond. This is

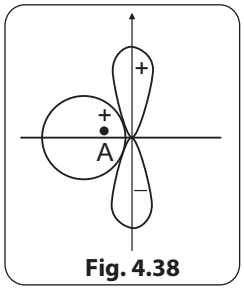


Fig. 4.38

because the axis of the p-orbital is oriented at right angles to the axis of the s-orbital. No combination is possible and so they may produce a non-bonding interaction.

A p-orbital can combine with an s-orbital to form molecular orbitals if there is linear overlapping.

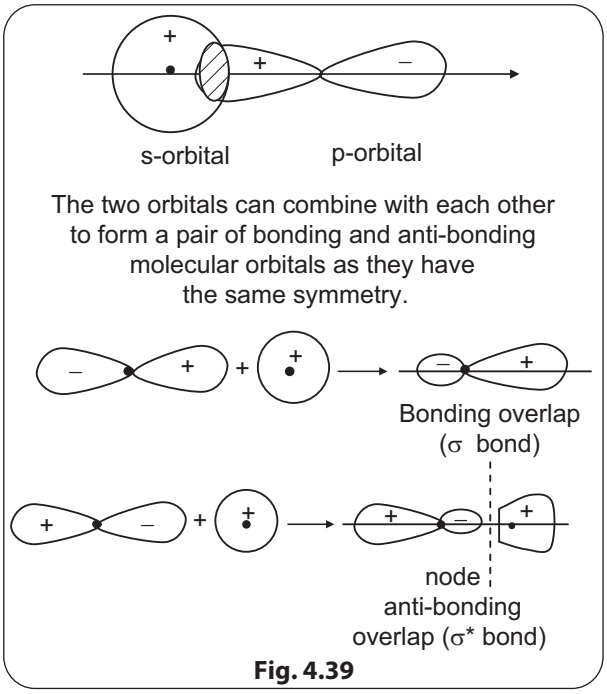


Fig. 4.39

Representations of bonding and anti-bonding overlaps of s and p-orbitals are given above.

When the overlapping lobes have the opposite signs an antibonding molecular orbital is formed with reduced electron density between the nuclei.

Combination of p – p orbitals

When two p-orbitals combine with the two lobes (+) and (–) lying along the axis joining the two nuclei, two different molecular orbitals can be produced namely a bonding σ orbital and an antibonding σ* orbital.

They are represented as shown below.

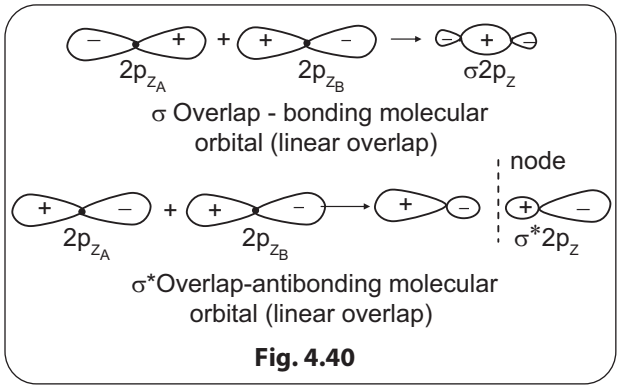


Fig. 4.40

Lateral overlap (parallel overlap)

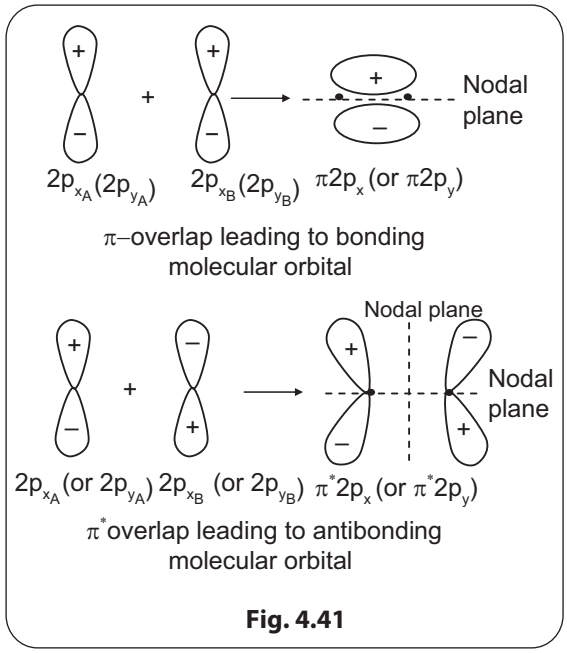


Fig. 4.41

When two p orbitals combine in which the lobes are perpendicular to the axis joining the nuclei, two sets of orbitals, bonding π orbital and anti-bonding π* orbital are obtained.

$$1s_A + 1s_B = \sigma 1s$$

$$1s_A - 1s_B = \sigma^* 1s$$

$$2s_A + 2s_B = \sigma 2s$$

$$2s_A - 2s_B = \sigma^* 2s$$

$$2p_{z_A} + 2p_{z_B} = \sigma 2p_z$$

$$2p_{z_A} - 2p_{z_B} = \sigma^* 2p_z$$

$$2p_{y_A} + 2p_{y_B} = \pi 2p_y$$

$$2p_{x_A} + 2p_{x_B} = \pi 2p_x$$

$$2p_{y_A} - 2p_{y_B} = \pi^* 2p_y$$

$$2p_{x_A} - 2p_{x_B} = \pi^* 2p_x$$

CONCEPT STRAND

Concept Strand 24

How many nodal planes are there for the following molecular orbitals?

(i) $\sigma^* 2p_z$

(ii) $\pi^* 2p_y$

(iii) $\sigma 2s$

(iv) $\pi 2p_y$

Solution

Molecular orbital	Nodal plane(s)
$\sigma^* 2p_z$	1
$\pi^* 2p_y$	2
$\sigma 2s$	0
$\pi 2p_y$	1

Rules for LCAO (linear combination of atomic orbitals)

(i) The atomic orbitals must have the same or nearly the same energy.

(ii) Orbitals must overlap as much as possible. The radial distribution functions of the two atoms must be similar at this close distance.

(iii) In order to produce bonding and antibonding molecular orbitals, both atomic orbitals must change symmetry in an identical manner.

Bond order

An outcome of the electronic configuration of molecules in discussing the bonding and anti-bonding orbitals is the bond order.

$$\text{Bond order} = \frac{1}{2}[N_b - N_a]$$

where, N_b = No. of e^-s in bonding molecular orbitals and

N_a = No. of e^-s in antibonding molecular orbitals

If the number of electrons in the bonding and antibonding orbitals is the same, the bond order is zero and hence no bond formation. Bond orders 1, 2 and 3 represent single, double and triple bonds respectively. Higher the bond order, stronger as well as shorter is the bond formed.

Molecular orbital configuration for Homonuclear Diatomic molecules

(i) Hydrogen molecule (H_2)

There is one electron from each atom.

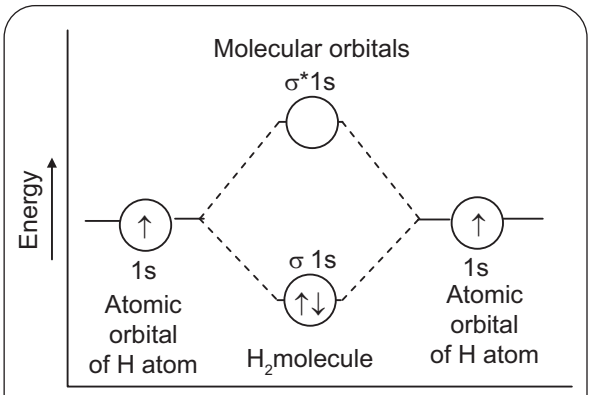


Fig. 4.42

The bonding σ 1s molecular orbital is full. A σ bond is formed between two hydrogen atoms to form a hydrogen molecule. H_2 molecule exists and is known. The orbitals in the increasing order of energy is σ 1s, σ* 1s. Molecular orbital configuration of H_2 is $(\sigma 1s)^2$

$$\text{Bond order} = \frac{1}{2}(2 - 0) = 1.$$

This means the two hydrogen atoms are bonded together by a single bond.

(ii) Hydrogen molecule ion (H_2^+)

The molecule has only one electron and the electronic configuration of H_2^+ is $(\sigma 1s)^1$.

$\text{Bond order} = \frac{1}{2}(1 - 0) = \frac{1}{2}$. The value $+\frac{1}{2}$ means that the bond is formed and the bond in H_2^+ ion is weaker than in a neutral hydrogen molecule. H_2^+ is unstable and exists transiently in electric discharge tubes containing hydrogen gas.

(iii) Diatomic helium molecule (He_2)

Four electrons have to be accommodated in a diatomic helium. A molecular orbital can accommodate two electrons of opposite spins. So one pair of electrons can be accommodated in the bonding molecular orbital, σ(1s) bonding, and the other two electrons will be in the anti-bonding orbital, σ*1s antibonding. So $He_2 = \sigma(1s)^2\, \sigma^*(1s)^2$

Bond order = $\frac{1}{2}(2-2) = 0$.

Bond order zero means that the diatomic helium molecule, He_2, does not exist.

(iv) Diatomic lithium molecule (Li_2)

There are six electrons (3+3) in a diatomic lithium molecule. The molecular orbitals can be arranged as $\sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2$. The electrons in the inner shell do not contribute to bonding.

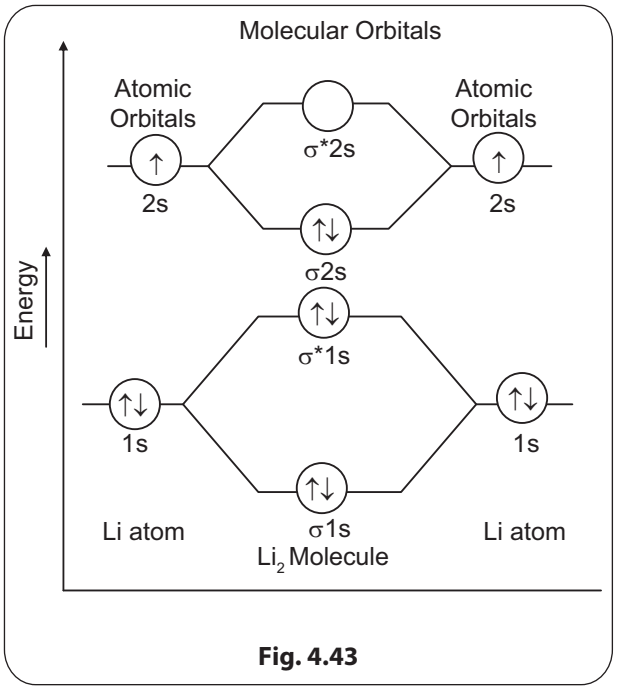


Fig. 4.43

The bonding occurs from the electrons of the σ2s orbital. $Li_2 = (KK)\sigma(2s)^2$

The two Ks represent K shells of two lithium atoms carrying two electrons each and the electrons in the KK levels do not produce any bonding. Bond order = $\frac{1}{2}(4-2) = 1$

There is only Li – Li single bond. Since s-s overlap is inefficient, bond energy is very small and Li_2 molecule is not very stable. In lithium vapour only about 1% atoms combine to form Li_2 molecule.

(v) Diatomic beryllium molecule (Be_2)

$Be_2 = KK\, \sigma(2s)^2, \sigma^*(2s)^2$

Bond order = $\frac{1}{2}(4-4) = 0$.

Diatomic Be_2 molecule does not exist.

(vi) Nitrogen molecule (N_2)

A nitrogen atom has a total of 7 electrons. Therefore, a nitrogen molecule has (7+7) = 14 electrons.

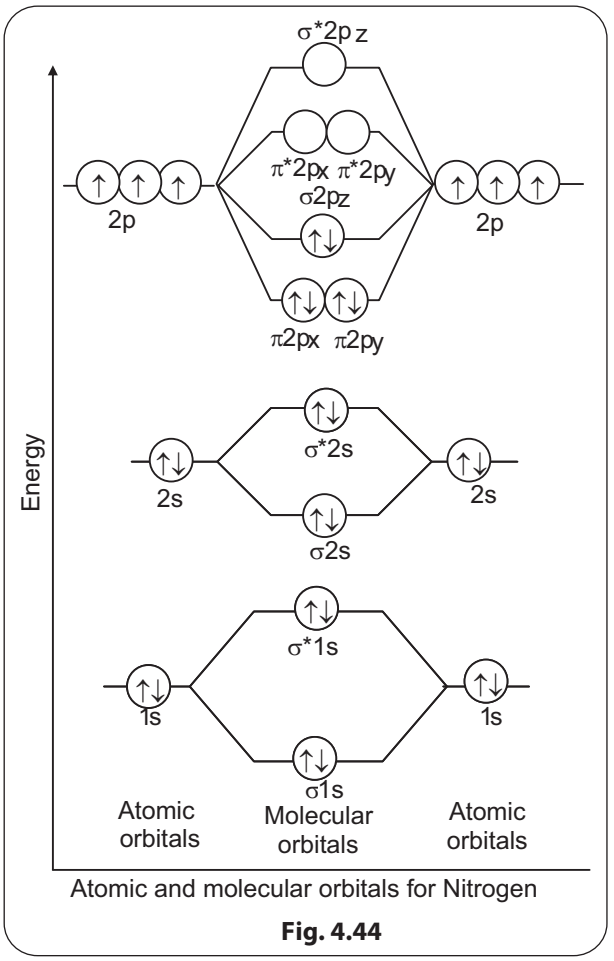


Fig. 4.44

The molecular orbitals formed by a nitrogen molecule.

$$(KK)\, \sigma 2s^2, \sigma^* 2s^2, \pi 2p_x^2, \pi 2p_y^2 \sigma 2p_z^2.$$

The inner shell electrons (KK) do not produce any bonding. Bond order = $\frac{1}{2}(10-4) = 3$.

Bond order is 3, there is one σ bond and two π bonds in the nitrogen molecule. The increasing order of energy of molecular orbitals for simple homonuclear diatomic molecules, is

$$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \sigma 2p_z < \begin{cases} \pi 2p_x \\ \pi 2p_y \end{cases} < \begin{cases} \pi^* 2p_x \\ \pi^* 2p_y \end{cases} < \sigma^* 2p_z$$

It is important to note that the π bonding and π* antibonding molecular orbitals are given by $2p_x$ and $2p_y$ atomic orbitals. So these two orbitals are doubly degenerate. The order shown is observed to be correct for oxygen and heavier elements, but for elements like boron, carbon and nitrogen, a slightly different order is being followed. The increasing order is

$$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \begin{cases} \pi 2p_x \\ \pi 2p_y \end{cases} < \sigma 2p_z < \begin{cases} \pi^* 2p_y \\ \pi^* 2p_x \end{cases} < \sigma^* 2p_z$$

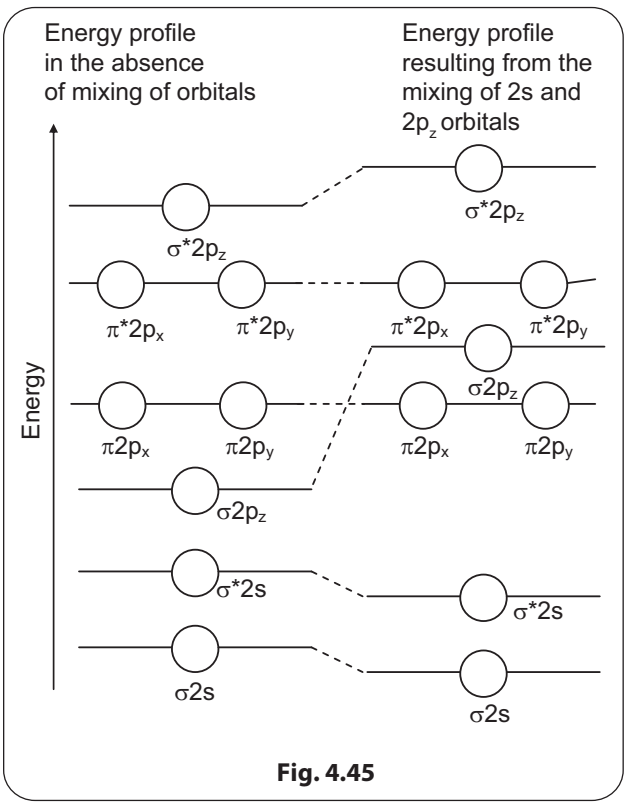


Fig. 4.45

In the case of second row elements upto nitrogen, mixing of 2s and $2p_z$ orbitals occurs as the energy difference between these orbitals is lesser (Table 4.4). Thus σ2s and σ*2s do not retain pure s-character and σ2p_z and σ*2p_z do not retain pure p-character. All the four M.Os acquire a mixed sp character.

Due to this s- p mixing, the energies of all the four M.Os change in such a way that σ2s and σ*2s which also contain some p-character become more stable and are thus lowered in their energy content. where as σ2p_z and σ*2p_z which also contain some s-character now become less stable and thus raised in their energy content. Since the $2p_x$ and $2p_y$ orbitals are not involved in mixing, their energies remain the same.

Table 4.4

Difference in energies between 2s and $2p_z$ orbitals							
Atoms	Li	Be	B	C	N	O	F
Energy difference between 2s and $2p_z$ (kJ mol^{-1})	178	262	449	510	570	1430	1970

(*vii) Boron molecule (B_2)*

Boron has a total of 5 electrons, with two in the 1st level and three in the 2nd level. So a boron molecule should contain (5 + 5) = 10 electrons.

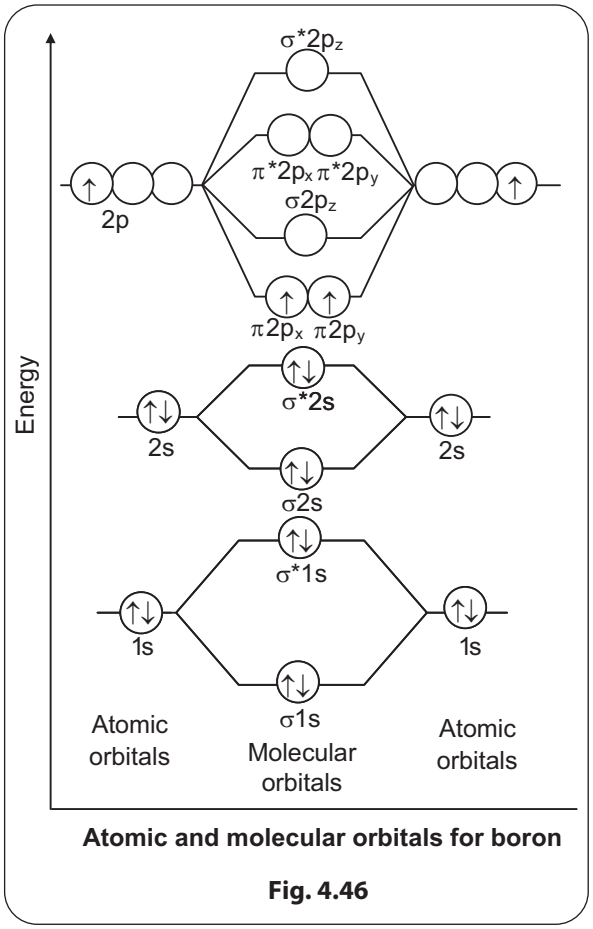
The arrangement of electrons in the molecular orbitals will be

$$KK, \sigma 2s^2, \sigma^* 2x^2, \begin{cases} \pi 2p_x^1 \\ \pi 2p_y^1 \end{cases}.$$

Boron is a light atom and the order of energies of molecular orbitals is different from the conventional arrangement. It is observed that π2p orbitals are of lower energy than the σ2p_z orbitals. Since π2p_x and π2p_y orbitals are of the same energy (degenerate), each of the π2p_x and π2p_y orbitals is singly occupied.

The effects of bonding and antibonding σ2s orbitals cancel, however, it leads to stabilization from the filling of the π2p orbitals and a bond exists in B_2. Bond order of B_2 molecule is equal to $\frac{1}{2}(6-4) = 1$.

A weak π bond exists between the two boron atoms, it has two unpaired electrons in the π orbitals and hence B_2 molecule is paramagnetic.



Atomic and molecular orbitals for boron

Fig. 4.46

(viii) Oxygen molecule (O_2)

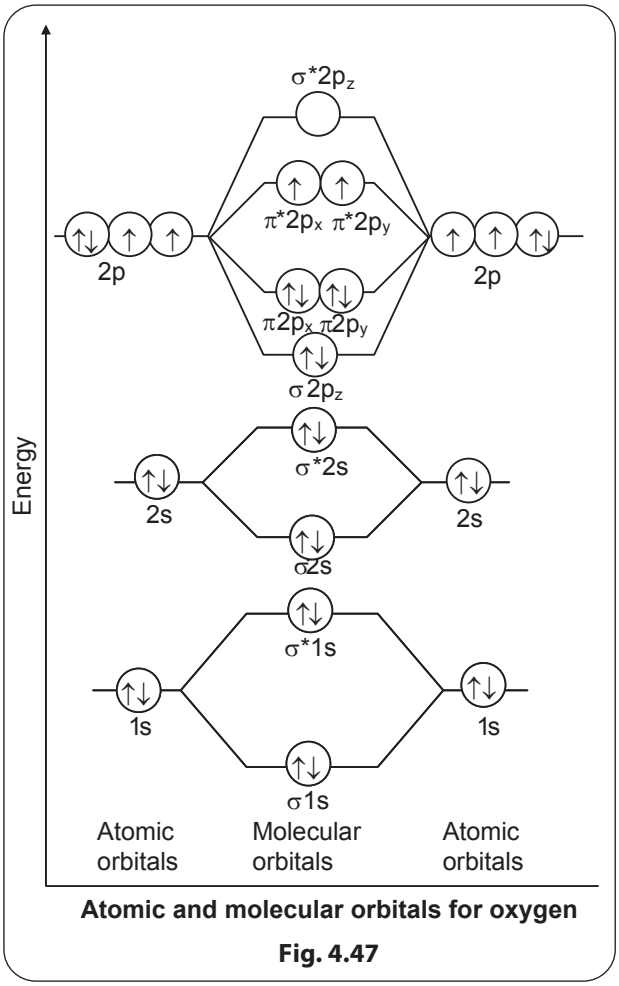
Oxygen atom has 8 electrons and a molecule contains 16 electrons. The arrangement of electrons in a molecule of oxygen is

$$KK\sigma 2s^2\sigma^* 2s^2\sigma 2p_z^{\,2}\begin{cases}\pi 2p_x^{\,2} \\ \pi 2p_y^{\,2}\end{cases}\begin{cases}\pi^* 2p_x^{\,1} \\ \pi^* 2p_y^{\,1}\end{cases}$$

The antibonding π^*2p_x and π^*2p_y orbitals are singly occupied. So there are two unpaired electrons with parallel spins and oxygen molecule is paramagnetic. A σ bond results from the filling of $\sigma 2p_z^{\,2}$. Since $\pi^*2p_y^{\,1}$ is half filled and therefore cancels one half of the effect of the $\pi 2p_y^{\,2}$ which is completely filled. So, half of a π bond results. Similarly, another half of a π bond results from the bonding $\pi 2p_x^{\,2}$ and the antibonding $\pi^*2p_x^{\,1}$. So the total number of bonds = $1+\frac{1}{2}+\frac{1}{2}$ = 2 bonds (one σ and one π).

$$\text{Bond order} = \frac{1}{2}(10-6) = 2$$

The bond that exits between two oxygen atoms in an oxygen molecule is a double bond consisting of a σ and π bonds. Since there are two unpaired electrons in O_2 molecule, the molecule is paramagnetic. Simple VBT predicts diamagnetic nature for O_2. This is one of the earliest triumphs of MOT over VBT.



Atomic and molecular orbitals for oxygen

Fig. 4.47

(ix) Super oxide ion (O_2^-)

O_2^- ion is formed by the combination of an oxygen atom with an oxygen ion O^-. There are 13 electrons in the molecular orbital of O_2^-, exclusive of the K shell.

$$O_2^- = KK\,\sigma 2s^2\sigma^* 2s^2\sigma 2p_z^{\,2}\pi 2p_x^{\,2}\pi 2p_y^{\,2}\pi^* 2p_x^{\,2}\pi^* 2p_y^{\,1}$$

$$\text{Bond order} = \frac{1}{2}(10-7) = 1\frac{1}{2}$$

There is one unpaired electron and O_2^- is paramagnetic.

(x) Peroxide ion (O_2^{2-})

This has 18 electrons including the K shell. They are arranged as

$$KK\sigma 2s^2\sigma^* 2s^2\sigma 2pz^2 \begin{cases} \pi 2p_x^{\ 2} \begin{cases} \pi^* 2p_x^{\ 2} \\ \pi^* 2p_y^{\ 2} \end{cases} \\ \pi 2p_y^{\ 2} \end{cases}$$

The inner shell does not take part in bonding. The bond order is $\frac{1}{2}(10 - 8) = 1$. O_2^{2-} ion has a bond order of 1 and O_2^- ion has a bond order of $1\frac{1}{2}$ and O_2 has a bond order of 2. Among the species N_2 and N_2^+, the bond orders are 3 and 2.5 respectively. The electronic configurations are

$$N_2 : \sigma 1s^2\sigma^* 1s^2\sigma 2s^2\sigma^* 2s^2\pi 2p_x^{\ 2}\pi 2p_y^{\ 2}\sigma 2p_z^{\ 2}$$

$$N_2^+ : \sigma 1s^2\sigma^* 1s^2\sigma 2s^2\sigma^* 2s^2\pi 2p_x^{\ 2}\pi 2p_y^{\ 2}\sigma 2p_z^{\ 1}$$

Properties of O_2, N_2 and their ions related with bond order

Table 4.5

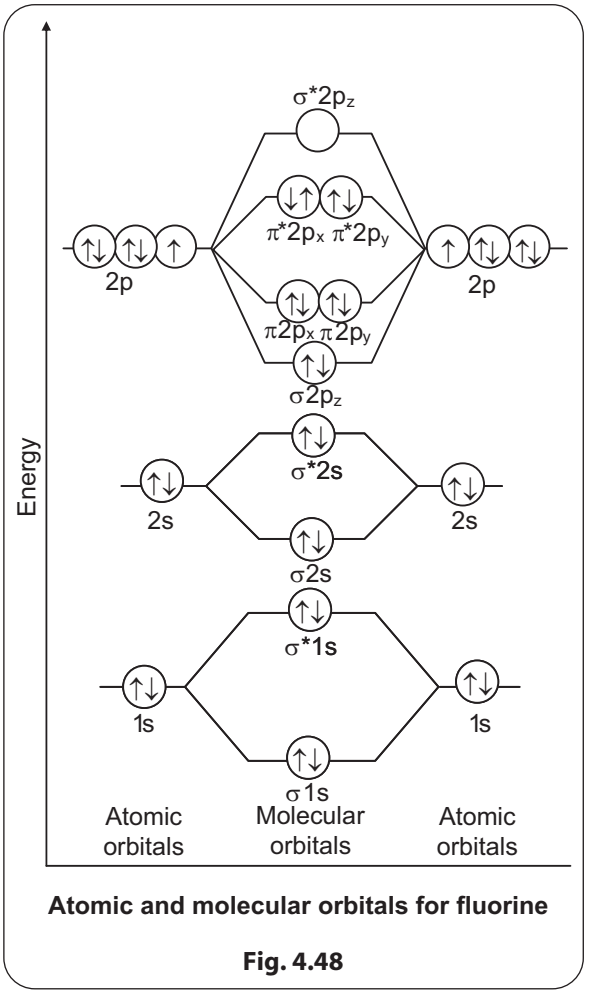
Species	Bond order	Unpaired Electrons	Magnetic Property	Variation of Characteristic Properties
O_2	2	2	Paramagnetic	Bond length: $O_2 < O_2^- < O_2^{2-}$
O_2^-	1.5	1	Paramagnetic	Bond strength: $O_2 > O_2^- > O_2^{2-}$
O_2^{2-}	1	0	Diamagnetic	Magnetic moment: $O_2 > O_2^- > O_2^{2-}$
N_2	3	0	Diamagnetic	Bond length: $N_2 < N_2^+$
N_2^+	2.5	1	Paramagnetic	Bond strength: $N_2 > N_2^+$ Magnetic moment: $N_2 < N_2^+$

(xi) Fluorine molecule (F_2)

The electronic configuration of fluorine atom is $1s^2 2s^2 2p^5$. There are 14 electrons in a fluorine molecule, excluding the four in the two K shells. The electronic configuration of the molecular orbitals of fluorine is

$$KK\ \sigma(2s)^2\sigma^*(2s^2)\ \sigma(2p_z)^2,$$
$$\pi(2p_x)^2\,\pi(2p_y)^2\,\pi^*(2p_x)^2\,\pi^*(2p_y)^2$$

Bond order = $\frac{1}{2}(10 - 8) = 1$



Atomic and molecular orbitals for fluorine

Fig. 4.48

CONCEPT STRANDS

Concept Strand 25

Explain why, one of the following species gets stabilized by losing an electron (Assume all other interactions are same).

(i) N_2

(ii) O_2^{2+}

(iii) N_2^+

(iv) O_2

Solution

In all species, except O_2, the electron is lost from bonding orbitals. Only in O_2, the electron is lost from antibonding orbital.

Concept Strand 26

(i) Two ions of O_2 have the bond orders in the ratio 5:3. Identify the ions. Comment on their magnetic properties.

(ii) Comment on the stability of N_2 and F_2 When
   (a) an electron is added
   (b) an electron is removed

Solution

(i) O_2 - $\sigma 1s^2\ \sigma^* 1s^2\ \sigma 2s^2\ \sigma^* 2s^2\ \sigma 2p_z^2\ \pi 2p_x^2\ \pi 2p_y^2\ \pi^* 2p_x^1\ \pi^* 2p_y^1$

O_2^+ - $\sigma 1s^2\ \sigma^* 1s^2\ \sigma 2s^2\ \sigma^* 2s^2\ \sigma 2p_z^2\ \pi 2p_x^2\ \pi 2p_y^2\ \pi^* 2p_x^1$

Bond order $= \dfrac{10-5}{2} = 2.5$

O_2^- - $\sigma 1s^2\ \sigma^* 1s^2\ \sigma 2s^2\ \sigma^* 2s^2\ \sigma 2p_z^2\ \pi 2p_x^2\ \pi 2p_y^2\ \pi^* 2p_x^2\ \pi^* 2p_y^1$

Bond order $= \dfrac{10-7}{2} = \dfrac{3}{2} = 1.5$

$O_2^+ : O_2^- = 2.5 : 1.5 = 5 : 3$

Both are paramagnetic having one unpaired electron each.

(ii) N_2: $\sigma 1s^2\ \sigma^* 1s^2\ \sigma 2s^2\ \sigma^* 2s^2\ \pi 2p_x^2\ \pi 2p_y^2\ \sigma 2p_z^2$

Bond order $= \dfrac{10-4}{2} = 3$

(a) By adding electron, bond order $= \dfrac{10-5}{2} = 2.5$ (destabilized).

(b) On removing electron, bond order $= \dfrac{9-4}{2} = 2.5$ (destabilized).

F_2: $\sigma 1s^2\ \sigma^* 1s^2\ \sigma 2s^2\ \sigma^* 2s^2\ \sigma 2p_z^2\ \pi 2p_x^2\ \pi 2p_y^2\ \pi^* 2p_x^2\ \pi^* 2p_y^2$

Bond order $= \dfrac{8-6}{2} = 1$

(a) On adding electron, bond order $= \dfrac{10-9}{2} = 0.5$ (destabilized).

(b) On removing electron, bond order $= \dfrac{10-7}{2} = 1.5$ (more stable).

Concept Strand 27

Arrange the following in the increasing order of their bond dissociation energy O_2, O_2^+, O_2^- and O_2^{2-}.

Solution

$O_2^{2-} < O_2^- < O_2 < O_2^+$. Bond order increases in this order.

As bond order increases, bond length decreases and bond strength or bond dissociation energy increases.

Concept Strand 28

Even though the bond orders of N_2^+ and N_2^- are the same, N_2^+ is more stable than N_2^-. Why?

Solution

For N_2^+ B.O $= \dfrac{9-4}{2} = 2.5$

For N_2^- B.O $= \dfrac{10-5}{2} = 2.5$

N_2^+ is more stable than N_2^- as it contains less no. of antibonding electrons which contribute to instability.

Molecular orbital configuration for heteronuclear diatomic molecules

(i) Carbon monoxide (CO)

The electronic configuration of carbon is $1s^2 2s^2 2p^2$ that of oxygen is $1s^2 2s^2 2p^4$

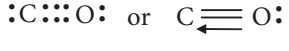
There are four electrons in the valence shell of carbon and six electrons in the valence shell of oxygen. So there are in all ten electrons to be filled up in the molecular orbitals of CO.

The distribution of electrons in the molecular orbital of CO molecule is

$$CO : KK\ \sigma 2s^2\ \sigma^*2s^2\ \pi 2p_x^{\ 2}\ \pi 2p_y^{\ 2}\ \sigma 2p_z^{\ 2}$$

$$\text{Bond order} = \frac{10-4}{2} = 3$$

The bond order for CO is 3. It means that it has one σ-bond and two π-bonds. The Lewis structure of CO is



Resonance forms in VBT are



(ii) Nitric oxide (NO)

There are totally 15 electrons in the nitric oxide molecule. The arrangement of electrons in the molecular orbital of NO is

$$KK\ \sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^{\ 2} \begin{cases} \pi 2p_x^{\ 2} \\ \pi 2p_y^{\ 2} \end{cases} \begin{cases} \pi^* 2p_x^{\ 1} \\ \pi^* 2p_y^{\ 0} \end{cases}$$

The electrons in the bonding and non bonding molecular orbitals in the 2s orbital cancel each other. A σ bond is formed from $2p_z^{\ 2}$ orbital. The $2p_x^{\ 2}$ electrons form the π orbital. The $\pi 2p_y^{\ 2}$ orbital is matched with the half filled $\pi 2p_y^{\ 1}$ orbital.

$$\text{The bond order is } \frac{10-5}{2} = 2.5$$

The molecule has an odd electron and therefore paramagnetic. The molecule has a σ bond and two π bonds, minus the anti bonding effect of one unpaired electron in the $\pi 2p_y^{\ 1}$ orbital. In this case too the bonding orbital has the greater characteristics of atomic orbitals of nitrogen.



(iii) Nitrosonium ion (NO^+)

The configuration of electrons in this species can be represented as

$$NO^+ : KK\ \sigma 2s^2 \sigma^* 2s^2 \pi 2p_x^{\ 2}\ \pi 2p_y^{\ 2}\ \sigma 2p_z^{\ 2}$$

$$\text{Bond order} = \frac{10-4}{2} = 3$$

There is one σ bond and two π bonds. (NO^+ is isoelectronic with N_2)

(iv) Nitrosyl anion (NO^-)

The electronic configuration of this anion is

$$NO^- : KK\ \sigma 2s^2\ \sigma^* 2s^2\ \pi 2p_x^{\ 2}\ \pi 2p_y^{\ 2}\ \sigma 2p_z^{\ 2}\ \pi^* 2p_x^{\ 1}\ \pi^* 2p_y^{\ 1}$$

$$\text{Bond order} = \frac{10-6}{2} = 2$$

Bond order of NO^- is less than that of NO^+.

Total number of electrons 16. (It is isoelectronic with O_2)

HYDROGEN BONDING

Molecules in which hydrogen is bonded covalently to highly electronegative elements like nitrogen, oxygen, fluorine have certain typical properties. Covalent bonds in these compounds are more polar than the bonds observed between hydrogen and any other element. The charge separation between hydrogen and the electronegative element is greater with a greater positive charge residing with hydrogen.

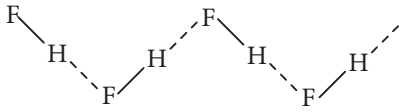
It was also found that boiling points of compounds like H_2O, HF and NH_3 were quite high, which led to belief that some attractive force operates between the molecules in these compounds.

As a result of fairly high partial charges and small sizes of the atoms, dipole–dipole interactions between molecules with hydrogen–fluorine bond, hydrogen–oxygen bond and hydrogen–nitrogen bond are very strong. This type of weak electrostatic attraction between hydrogen attached to a strongly electronegative atom such as F, O or N and another electronegative atom like Cl, N, O or F is called hydrogen

bonding. It is represented as X – H - - - - -Y where, X = F, O, N and Y = F, N, O, Cl

Hydrogen bond is a dipole-dipole attraction.

In a hydrogen bonded structure such as for example, HF, hydrogen is attached to one fluorine by a covalent bond and on the other side attached to the other fluorine by a hydrogen bond or a weak force.



The three atoms joined by a hydrogen bond are usually in a straight line.

Hydrogen bond is a weak bond compared to a usual covalent bond.

Covalent bond energy	200 – 400 kJ mol^{-1}
Hydrogen bond energy	10 – 40 kJ mol^{-1}
van der Waals force energy	less than 2 kJ mol^{-1}

Water, which is a liquid, will remain as a gas if there were no hydrogen bonding. Water is a liquid because of hydrogen bonding, due to intermolecular attractions.

Each hydrogen in a water molecule can form hydrogen bonds to oxygen in another water molecule and oxygen in each water molecule can form two hydrogen bonds with two other water molecules.

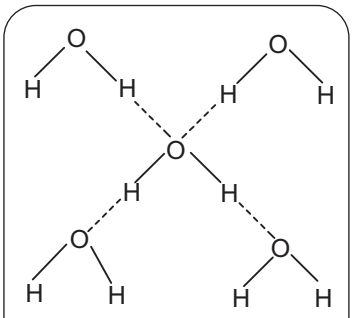


Fig. 4.49

Hydrogen bond is regarded as a weak electrostatic attraction between lone pair of electrons on an electronegative atom and a covalently bonded hydrogen that carries a small positive charge. Hydrogen bonds are formed only with most electronegative atoms like F, O and N. Despite the low bond energy of a hydrogen bond, they are of great significance in biological systems like linking of polypeptide chains in proteins and linking in nucleic acid containing units.

Types of hydrogen bonds

Hydrogen bonds are of two types.

(i) Intermolecular hydrogen bonding and
(ii) Intramolecular hydrogen bonding.

Intermolecular hydrogen bonding

When two different molecules are connected by a hydrogen bonding, it is known as intermolecular.

Examples are

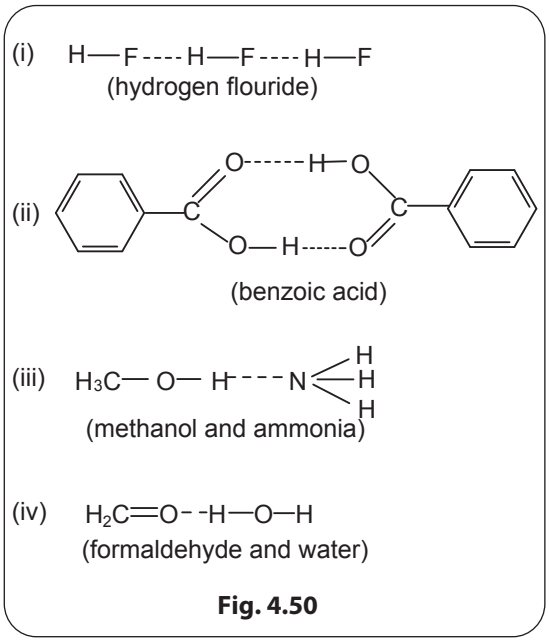


Fig. 4.50

CONCEPT STRANDS

Concept Strand 29

What is the constitution of the compound KHF_2?

Solution

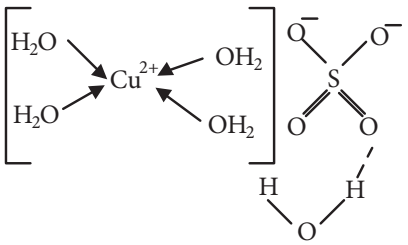
K^+ and HF_2^-. HF_2^- is a resonance hybrid of $F - H - - - F^- \leftrightarrow F^- - - - H - F$. This is the strongest hydrogen bond with an energy of 209 kJ mol^{-1}.

Concept Strand 30

Describe the different types of bonds present in $CuSO_4.5H_2O$.

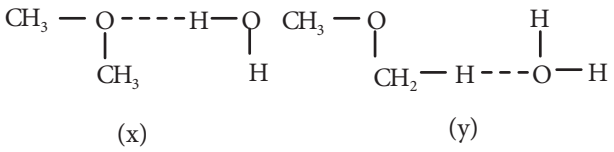
Solution

Electrovalent, covalent, co-ordinate and hydrogen bonds.



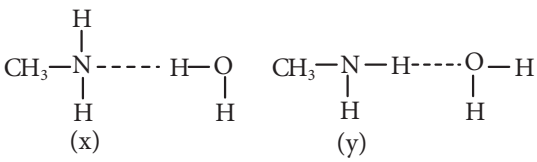
Concept Strand 31

(i) Ethyl ether dissolves in water to the extent of 8 g per 100 mL. Is the reason for this, the presence of hydrogen bonding? If so which representation is correct and why?



(x) (y)

(ii) Which is the correct representation of hydrogen bonding in methyl amine in water?



Solution

(i) (x) is the right representation. Hydrogen should be covalently bonded to oxygen atom for hydrogen bonding.

(ii) Representation (x) is correct. Oxygen is more electronegative than nitrogen, so hydrogen attached to oxygen will show stronger hydrogen bonding.

Concept Strand 32

Give the decreasing order of boiling points of Methyl alcohol, water and a solution of Methyl alcohol and water?

Solution

$$H_2O > CH_3OH + H_2O > CH_3OH$$

Strength of H-bonding decreases in the above order.

Concept Strand 33

Glucose dissolves in water due to the formation of hydrogen bonds. The initially formed hydrogen bonds are broken when it is more diluted with water. Represent the temperature change with the addition of water to glucose?

Solution

On adding water heat is released initially due to formation of H-bonds. Later heat is absorbed due to breaking of H-bonds.

Concept Strand 34

At 27°C and one atm pressure the observed density of HF is 3.25 g L^{-1}. In which associated state does HF existing as per data?

Solution

$$\text{The calculated molar mass } M = \frac{dRT}{P}$$

$$= \frac{3.25 \times 0.0821 \times 300}{1}$$

$$= 80.05$$

$$\text{Extent of association} = \frac{\text{observed molar mass}}{\text{mass of single molecule}}$$

$$= \frac{80.05}{20} = 4$$

This means that the molecule form clusters. This can happen through hydrogen bonding.

Intramolecular hydrogen bonding.

When a hydrogen bonding occurs within a simple molecule, by linking two groups to form a cyclic structure, it is known as intramolecular hydrogen bonding. Examples of such type of bonding are

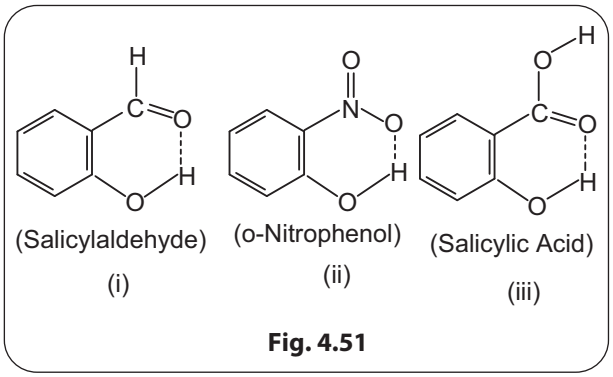


Fig. 4.51

Intermolecular hydrogen bonding has a profound influence on the physical properties such as melting points, boiling points, enthalpies of sublimation, vaporization. The high boiling point of NH_3 compared to PH_3, SbH_3 and AsH_3 is on account of hydrogen bonding. Intramolecular hydrogen bonding increases the stability of the molecule due to the formation of the chelated structure.

Steam distillation

When compounds are involved in intramolecular hydrogen bonding, their boiling points get reduced and they can be steam distilled. Among ortho and para nitrophenols, the ortho compound can be steam distilled due to intramolecular hydrogen bonding.

Electronegativity and bond polarity

When a covalent bond is formed, the electrons, which are used for bonding may not be shared equally by the two bonding atoms. Small atoms attract electrons strongly than the larger atoms. The power of an atom in a molecule to attract the electrons to itself is termed as electronegativity. The term electronegativity is used in a more qualitative way and the concept of this term can predict whether a chemical bond between two similar or dissimilar atoms is ionic or polar covalent bond or totally non-polar covalent bond.

If two atoms have the same electro negativities or close values of electronegativities, the tendency of the two atoms to draw the bonding electrons towards themselves will be the same.

Example, H – H or Cl – Cl

Therefore, the bond will be predominantly covalent. If there is a large electronegativity difference between the two atoms, then the bond will have a polar character. The ionic character of a bond varies with the difference in electronegativity.

Example H – Cl

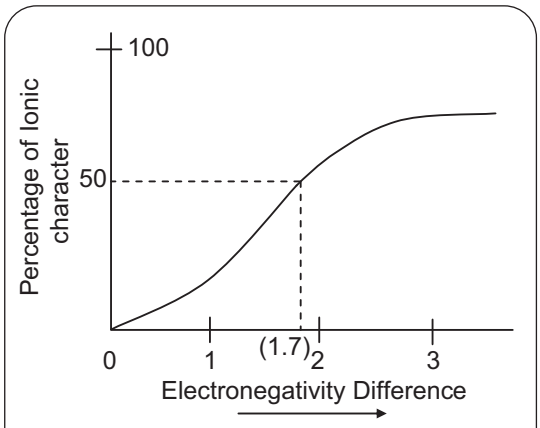


Fig. 4.52

Fifty per cent ionic character is observed if the electronegativity difference is 1.7. There is greater polar character if the electronegativity difference is greater than 1.7. If it is less than 1.7, the bond is more covalent than ionic.

POLARIZATION OF IONS- FAJAN'S RULE

Although the bond in a compound like A^+B^- is considered to be 100% ionic, actually it has partial covalent character. Partial covalent character evolving in ionic compounds is explained by *Fajan's rule.* When two oppositely charged ions (say A^+ and B^-) approach each other, the positive ion attracts electron on the outer most shell of the anion. This results in the polarization (distortion or deformation) of the anion. If the polarization is very small, an ionic bond is

favoured, while if the extent of polarisation is large, a covalent bond is favoured. Thus *the ability of the cation to distort the anion is known as its polarizing power and the tendency of the anion to get polarized is known as its polarizability.* Greater the polarisation power or polarizability of an ion, greater will be its tendency to form a covalent bond. Covalent character of an ionic bond is large under the following conditions.

(i) Small cation

Due to greater concentration of positive charge in a small volume (high charge density), a small cation has high polarizing power. That is why LiCl is more covalent than NaCl.

(ii) Large anion

The larger the anion, the greater is its polarizability and hence greater is the tendency to form covalent bond. This explains why the covalent character of silver halides increases in the order AgCl < AgBr < AgI

(iii) Higher charge on either of the two ions

As the charge on the ion increases, the electrostatic attraction of the cation for the valency electrons of the anions also increases, with the result its ability for forming covalent bond increases. Thus covalency increases from Na^+ to Mg^{2+} to Al^{3+} and also from Cl^- to S^{2-} to P^{3-}.

(iv) Electronic configuration of the cation

For the two ions of the same size and charge, one with a pseudo noble gas configuration (i.e., 18 electrons in outermost shell) will be more polarizing than a cation with noble gas configuration (i.e., 8 electrons in outermost shell).

Thus copper (I) chloride is more covalent than NaCl though Cu^+ (0.96A°) and Na^+ (0.95A°) have same size and charge. Thus it can be seen that, greater the possibility of polarization, greater is the tendency to form covalent bonds and consequently lower is the melting point and heat of sublimation and greater is the solubility in non-polar solvents.

CONCEPT STRANDS

Concept Strand 35

Arrange the silver halides in the increasing order of their water solubility and justify?

Solution

AgI < AgBr < AgCl < AgF. Ionic character increases from AgI to AgF.

Concept Strand 36

Alkaline earth metals have higher melting point than alkali metals. Why?

Solution

The small inter atomic distance and stronger metallic bonding due to two valency electrons are the reason for group 2 metals to have higher melting point than alkali metals.

Concept Strand 37

The boiling point of ICl is much higher than that of Br_2. Why?

Solution

This is due to the fact that ICl is polar and Br_2 is non-polar.

Concept Strand 38

Among NaCl, $PbCl_2$, $AlCl_3$ and $SnCl_4$. Which one is having the lowest melting point? Explain.

Solution

$SnCl_4$ has more covalent character as the polarization of Cl^- ions by Sn^{4+} ion is greater as cation has higher charge. Hence melting point of $SnCl_4$ is lowest.

DIPOLE MOMENT

The degree of polar character in a polar covalent bond is given in terms of dipole moment. This is expressed as a product of the magnitude of the electric charge (e) in electrostatic units and the distance (d) in Angstrom units between the positive and negative ends of the molecule.

Dipole moment is represented as μ.

$$\mu = e \times d$$

where, μ = dipole moment

e = charge on an atom;

d = distance between the nuclei of the two atoms

If the charge 'e' is 10^{-10} esu and distance is 10^{-8} cm, then

$$\mu = 10^{-10} \times 10^{-8} = 10^{-18} \text{ esu} - \text{cm or 1D}$$

where, D is called Debye

$$1D = 10^{-18} \text{ esu-cm}$$

SI unit of dipole moment is meter – Coulomb (m C)

$$1D = 3.33 \times 10^{-30} \text{ m. c}$$

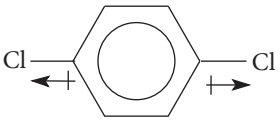
CONCEPT STRAND

Concept Strand 39

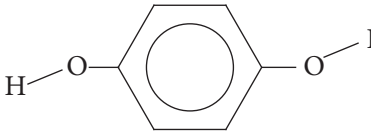
Among p-dichlorobenzene and p-dihydroxybenzene (quinol), which is polar, explain?

Solution

p-Dichloro benzene is non-polar where as quinol is polar.

In p-dichloro benzene, 

the two bond moments are acting exactly in the opposite direction. Hence the molecule as a whole is non polar. Whereas in quinol,



the bond moments are not exactly canceling with each other and hence the molecule has a resultant dipole moment or quinol is polar.

Dipole moment and percentage of ionic character

The percentage of the ionic character is the ratio of the observed dipole moment to the dipole moment after the complete electron transfer.

$$\% \text{ ionic character} = \frac{\text{Actual dipole moment of the bond}}{\text{Dipole moment of a pure ionic bond}} \times 100$$

For the hydrogen halides, the relationship between dipole moment and percentage ionic character is given as follows.

Table 4.6

Compound	Dipole Moment	% ionic character
H–Cl	1.03	17
H – Br	0.87	12
H – I	0.38	5
H – F	1.92	43

Calculations of ionic character of HCl bond

Electronic charge = 4.8×10^{-10} esu

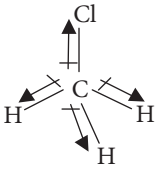
H – Cl bond length = 1.27 Å

The dipole moment of 100% ionic bond = $4.8 \times 10^{-10} \times 1.27 \times 10^{-8} = 6.10$ D

The observed dipole moment of HCl = 1.03 D

% ionic character = $(1.03/6.1) \times 100 = 17$

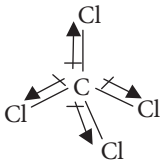
In a polyatomic molecule containing two or more bonds, the net dipole moment of the molecule is obtained by the vectorial addition of the dipole moments of the constituent bonds. For example, dipole moment of CH_3Cl is the vectorial addition of the dipole moments of the three C – H bonds and one C – Cl bond.



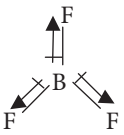
μ = 1.86 D

In linear molecules like CO_2, CS_2, $HgCl_2$ etc. the individual bond moments exactly cancel each other on vectorial addition and the dipole moments are zero.

Symmetrical molecules like CCl_4, BF_3 etc also have zero dipole moment.



$\mu = 0$



$\mu = 0$

The magnitude of dipole moment of a polar molecule depends on the difference in electro negativities of the bonded atoms. Greater the difference in the electro negativities, greater is the dipole moment of a polar molecule. For example, dipole moments of hydrogen halides decrease in the order

$$HF > HCl > HBr > HI$$

CO_2 and H_2O are triatomic compounds. The dipole moment of CO_2 is zero, while that of H_2O is 1.84D. This is because CO_2 has a linear structure, while water has a bent structure. The dipole moment of CH_4 and CCl_4 are both zero, since both the molecules are symmetrical.

CONCEPT STRANDS

Concept Strand 40

A diatomic molecule has a dipole moment of 0.48 D. If its interatomic distance is 0.5 Å, calculate the fraction of the charge carried by each atom?

Solution

0.48 D = 0.48×10^{-10} esu Å

$$\mu = e \times D \quad e = \frac{0.48 \times 3.33 \times 10^{-30}}{0.5 \times 10^{-10}} \text{ mC} \quad \text{For 100%}$$

ionic character, $\mu = 4.8 \times 10^{-10}$(esu) × 0.5 Å

Fractional charge

$$= \frac{0.48 \times 10^{-10}(\text{esu Å}) \times 100}{4.8 \times 10^{-10}(\text{esu}) \times 0.5(\text{Å})} \times 100 = 20\%$$

Concept Strand 41

The dipole moment of two bonds A – B and A – C were found to be 4.0×10^{-30} m C and 4.67×10^{-30} m C. If their bond lengths were 145 pm and 139 pm respectively, calculate the percentage ionic character of the two bonds A – B and A – C. Comment on the electronegativities of B and C.

Solution

For molecule A – B

Calculated dipole moment = 4.8×10^{-10}(esu) × 1.45 Å (for 100% ionic character)

$$\% \text{ Ionic character} = \frac{\mu_{\text{observed}}}{\mu_{\text{calculated}}} \times 100$$

$$= \frac{\frac{4}{3.33} \times 10^{-10}(\text{esu Å}) \times 100}{4.8 \times 10^{-10}(\text{esu}) \times 1.45(\text{Å})}$$

$$= 17.2\%$$

For molecule A – C

Calculated dipole moment = 4.8×10^{-10}(esu) × 1.39 Å (for 100% ionic character)

Percentage Ionic character

$$= \frac{\frac{4.67}{3.33} \times 10^{-10}(\text{esu Å})}{4.8 \times 10^{-10}(\text{esu}) \times 1.39(\text{Å})} \times 100 = 21\%$$

Since the percentage ionic character of A – C is greater than A – B, A – C is more ionic than A – B. Therefore C is more electronegative than B.

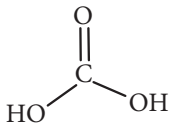
Concept Strand 42

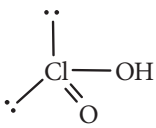
Among the following compounds, the one that is polar and in which the central atom has sp^2 hybridization is

(i) BF_3 (ii) PCl_3
(iii) $HClO_2$ and (iv) H_2CO_3

Solution

H_2CO_3



$HClO_2$ is 

where chlorine is sp^3 hybridized.

Concept Strand 43

From the given data, calculate the bond distances of HCl and HF?

Compound	Dipole moment(D)	% ionic character
HCl	1.03	17
HF	1.92	43

Solution

$$\% \text{ ionic character} = \frac{\mu(\text{experimental})}{\mu(100\% \text{ ionic})} \times 100$$

For $$HCl, 17 = \frac{1.03 \times 10^{-18}}{d \times 4.8 \times 10^{-10}} \times 100$$

$$d = 1.26\text{Å}$$

Similarly for HF, d = 0.93Å

Concept Strand 44

Arrange the following compounds in the decreasing order of polarity of bonds NH_3, PH_3, AsH_3, SbH_3. Electronegativities of N = 3, P = 2.1, As=2.0, Sb = 1.9 and H = 2.1

Solution

Decreasing order of electronegativity difference is

$$NH_3 > SbH_3 > AsH_3 > PH_3$$

Concept Strand 45

Dipole moment of CO_2=0 whereas $SO_2 = 5.37 \times 10^{-30}$ m C. What indication does it give about the geometry of the molecules?

Solution

CO_2 is a linear molecule and SO_2 is a bent molecule.

Concept Strand 46

Arrange the following C – C bonds in the increasing order of bond moments.

$C(sp^3) - C(sp^2)$, $C(sp^3) - C(sp)$, $C(sp^2) - C(sp)$?

Solution

Since the electronegativity of a carbon depends on its hybridization,
i.e., as the s-character increases electronegativity increases.

$\therefore$ Bond moments increase in the order

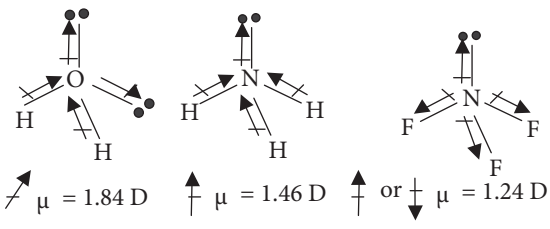
$$C(sp^3) - C(sp^2) < C(sp^2) - C(sp) < C(sp^3) - C(sp)$$

Concept Strand 47

Arrange NH_3, NF_3 and H_2O in the decreasing order of their dipole moments and justify.

Solution

Considering the contribution of dipole moment of lone pair, dipole moments of these molecules decrease in the order



RESONANCE

When a molecule cannot be represented by a single Lewis structure but can be represented by two or more different structures, then the molecule is said to be the resonance hybrid of the various structures. As a consequence of resonance, the energy of the molecule becomes lower than the energy of any of the resonating structures. The molecule with a lower energy has a higher stability. Therefore resonance gives additional stability to the molecule. Resonance energy is the difference between the actual energy of the molecule and the energy of the most stable among the resonance contributors.

Few examples of resonance

Carbonate ion can be represented as a resonance hybrid of the following structures

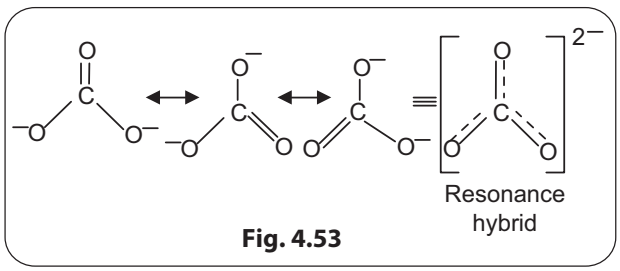


Fig. 4.53

Sulphate ion can be represented as a hybrid

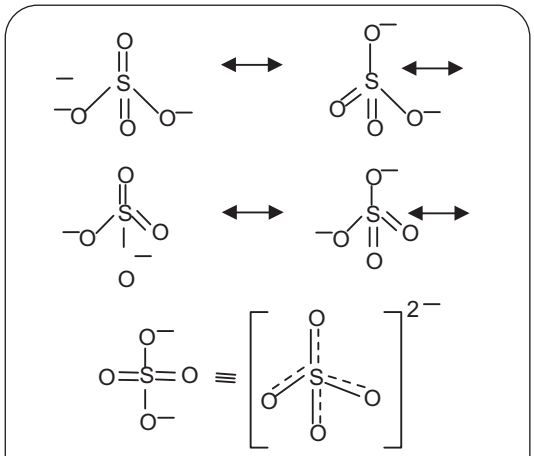


Fig. 4.54

CONCEPT STRANDS

Concept Strand 48

Among the following pairs, identify the one having the same bond order.

(i) CO_3^{2-} and NO_3^-

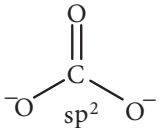
(ii) N_3^- and NO_3^-

(iii) NO_2^- and CO_3^{2-}

(iv) NO_2^- and NO_3^-

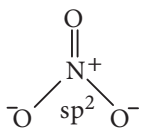
Solution

CO_3^{2-} and NO_3^-

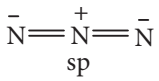


$$\text{Bond order of C–O} = \frac{\text{No. of bonds present}}{\text{No. of atoms around the central atom resonating}}$$

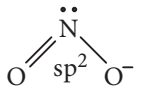
$$= \frac{4}{3} = 1.33$$



Bond order $= \frac{4}{3} = 1.33$



Bond order $= \frac{4}{2} = 2$



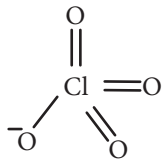
Bond order $= \frac{3}{2} = 1.5$

Note: Bond order is calculated for resonating bonds using the above formulae.

Concept Strand 49

What is the Cl–O bond order in perchlorate ion?

Solution



Bond order = $\frac{7}{4}$ = 1.75

Concept Strand 50

Arrange the following in the increasing order of Cl – O bond lengths ClO_2^-, ClO_3^- and ClO_4^-?

Solution

$$ClO_4^- < ClO_3^- < ClO_2^-$$

Bond order increases from ClO_2^- to ClO_4^-. i.e, ClO_2^- has bond order 1.5 which is the lowest bond order. Hence it has highest bond length.

SUMMARY

Electronic theory of valency	G.N Lewis and W. Kossel
Octet rule	Stability of outer inert gas electronic configuration Lewis structure
Ionic Bond	Gain and loss of electrons
Co-valent bond	Sharing of electrons: simple, double and triple bonds
Co-ordinate bond	Donating and sharing of electron pairs
Lattice enthalpy	Solubility in polar solvents
Potential energy of co-valent bond formation	Bond formed corresponding to the energy minimum
Sigma and π bonds	Head on overlap and lateral overlap of orbitals
Different types of σ bonds	s–s, s–p, p–p…sp^3–s….overlaps
Bond length	Sum of co-valent radii of bonded atoms
Bond angle	Characteristic of a molecule CH_4, NH_3, H_2O….
Hybridization	Orbital mixing
Types of hybridization	sp^3–CH_4, NH_3, H_2O sp^2–BF_3,C_2H_4 sp–$BeCl_2$, C_2H_2 sp^3d–PCl_5 sp^3d^2–SF_6
Additional examples of different molecules involving sp^3 d and sp^3d^2 hybridization	SF_4, ClF_3, ICl_2^-, BrF_5, XeF_4, XeF_2
Involvement of lone pair in molecular geometry	Distortion of molecule due to presence of lone pairs
VSEPR theory	Decides molecular geometry from the number of bp and ℓp electrons. $SnCl_2$, SO_2 and SO_3

Geometry of certain common species	SO_4^{2-}, ClO_4^-, ICl_4^-
Geometry and different number of electron pairs	Shapes of commonly encountered molecules
Some unique species	$[Si(CH_3)_3]_3N$, N_2H_4
Molecular orbitals	Formation from atomic orbitals
Combination of AO	s–s, σs and σ*s s–p, p–p σp, σ*p , πp, π*p
Aufbau order of MO	$\sigma 1s\ \sigma^*1s\ \sigma 2s\ \sigma^*2s\ \sigma 2p_z \begin{matrix}\pi 2p_y\ \pi^*2p_y \\ \pi 2p_x\ \pi^*2p_x\end{matrix} \sigma^*2p_z$ for elements upto Be_2 and beyond N_2 $\sigma 1s\ \sigma^*1s\ \sigma 2s\ \sigma^*2s \begin{matrix}\pi 2p_y \\ \pi 2p_x\end{matrix} \sigma 2p_z \begin{matrix}\pi^*2p_y \\ \pi^*2p_x\end{matrix} \sigma^*2p_z$ for B_2, C_2, N_2
Node	Regions between probable regions where there is no probability of finding electron
Bond order	Relation to bonding and anti-bonding electrons $B.O = \frac{1}{2}(N_b - N_a)$ N_b = bonding electrons N_a = antibonding electron
Different diatomic species	H_2, H_2^+, He_2,….He_2^{++}, Li_2, Be_2, B_2, C_2, N_2, O_2, F_2 and Ne_2
Their B.O and magnetic nature	
Non formation of certain species	He_2, Be_2, Ne_2
Energy diagram for different homoatomic diatomic species	(a) H_2, He_2, Li_2, Be_2 (b) B_2, C_2, N_2 (c) Cl_2, F_2, Ne_2
Heteroatomic diatomic species	CO, NO, NO^+, NO^-
Bond order in poly atomic species	CO_3^{2-}, NO_3^-, NO_2, NO_2^-, NO_2^+
Hydrogen bonding	H_2O, HF, benzoic acid, $CuSO_4.5H_2O$
Intramolecular and intermolecular hydrogen bonding	o-Nitro and p-nitro phenols Salicylic acid, salicyl aldehyde
Electronegativity and bond polarity	Dipole moment, Fajan's rules
Dipole moment	$\mu = e \times d$
% ionic character	$\frac{\mu_{observed}}{\mu_{calculated}} \times 100$ $= \frac{\mu_{observed}}{4.8 \times x} \times 100$ where, x is bond length in Å ID = 10^{-18} esu cm $= 3.3 \times 10^{-30}$ mC
Resonance	To account for stability of poly atomic species

TOPIC GRIP

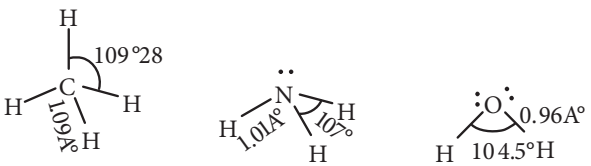
Subjective Questions

1. PCl_5 has trigonal bipyramidal shape, but IF_5 has square pyramidal shape. Explain.
2. A compound AB_3 is triangular planar. When B^- is added it becomes AB_4^-. When B^- is removed from AB_3, it becomes AB_2^+. What are the geometries of AB_4^- and AB_2^+?
3. Hydrogen is diatomic but He is monatomic. Explain using M.O. theory.
4. Explain the following.
   (i) $SnCl_2$ is ionic where as $SnCl_4$ is covalent.
5. Explain, why the experimentally determined P – O bond length in $POCl_3$ is greater than the sum of double bond covalent radii of P and O?
6. Using the physical properties, how will you distinguish the following?
   (i) BaO and MgO
   (ii) Br_2 and ICl
7. How will you distinguish between the following molecules in terms of dipole moment?
   (i) PCl_3 (ii) NH_3 (iii) BrF_5 (iv) BF_3
8. Na_2SO_4 is readily soluble in water whereas $BaSO_4$ is only sparingly soluble, why?
9. NH_4Cl is highly soluble in water than NaCl. Explain.
10. Even though $BaSO_4$ is an electrovalent compound, it is insoluble in water. Explain.

Straight Objective Type Questions

Directions: **This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

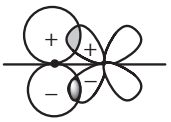
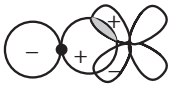
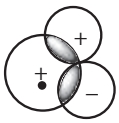
11. Calcium forms $CaCl_2$ rather than CaCl because
   (a) the removal of second electron from calcium requires only less energy.
   (b) the electron gain enthalpy value is greater than the ionization energy of calcium to Ca^{2+}.
   (c) more energy is released during the formation of $CaCl_{2(s)}$ than formation of CaCl.
   (d) CaCl is a covalent molecule while $CaCl_2$ is ionic.
12. Identify the correct statement regarding the trio.



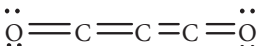
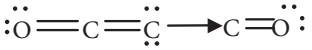
   (a) Bond length decreases with increase in s character of hybridized orbitals.
   (b) Increase in repulsion between hydrogen atoms decreases the bond length.

(c) Decreasing b.p –b.p repulsion decreases the bond angle.
(d) Increase in non bonding electron pair increases the bond angle.

13. Considering the molecular orbitals represented below, the bonding combination is

(a) 
(b) 
(c) 
(d) 

14. Of the following represented Lewis structures which one is the most appropriate to carbon suboxide C_3O_2

(a) 
(b) 
(c) 
(d) 

15. BF_3 forms the adduct $(CH_3)_3N \rightarrow BF_3$ with $(CH_3)_3N$. The B – F bond length in BF_3 is 1.30 A°. Which among the following statements are true about the B – F bond length in the adduct.

(a) It is equal to 1.30 A°
(b) It is greater than 1.30 A°
(c) It is less than 1.30 A°
(d) B – F bond becomes ionic and hence the bond length becomes very large.



Assertion–Reason Type Questions

Directions: **Each question contains Statement-1 and Statement-2 and has the following choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False
(d) Statement-1 is False, Statement-2 is True

16. Statement 1
LiCl is soluble in alcohol.
and

Statement 2
LiCl is a covalent molecule.

17. Statement 1
CH_4 molecule is non-polar.
and

Statement 2
C – H bonds are non-polar.

18. Statement 1
Bonding MO has no nodal plane perpendicular to the line joining the two nuclei while antibonding MO has such a node.
and

Statement 2
A bonding MO is formed by addition and an antibonding MO is formed by subtraction combination of atomic orbitals.

19. Statement 1
Fe_2O_3 is more acidic than FeO.
and

Statement 2
The electronegativity of central atom and correspondingly its non metallic character increases with increase in oxidation number.

20. Statement 1
CO_2 is nonpolar while SO_2 is polar.
and

Statement 2
C – O bond is non polar while S–O bond is polar.

Linked Comprehension Type Questions

Directions: **This section contains 2 paragraphs. Based upon the paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

Passage I

There is a very natural correlation between the orientation of the bonds to an atom and the spatial requirements for the bondings and non bonding (lone pair) electrons that reside at and hence occupy the space surrounding that atom. Electrons will tend to stay as far apart from one another to minimize repulsion. This dictates the geometry of the molecule

21. The largest bond angle in the molecule of acetone is
(a) 109°48' (b) 120° (c) 90° (d) 107°

22. The number of bond pairs along the nuclear axis and lonepairs in carbon monoxide is
(a) 1, 1 (b) 2, 1 (c) 1, 0 (d) 2, 0

23. The number of lone pairs on oxygen in water is same as that in
(a) SF_4 (b) $P(CH_3)_3$ (c) ClF_3 (d) XeF_2

Passage II

Covalent Bonding is explained by Valence Bond Theory (VBT) and Molecular Orbital Theory(MOT). According to MOT, the nuclei of the two atoms lie at appropriate distance and all the atomic orbitals of one atom overlap with all the atomic orbitals of other atom provided the overlapping orbitals are of similar energy and same symmetry.

24. The total energy of the molecule can be given as
(a) sum of energies of occupied molecular orbitals
(b) sum of energies of all the molecular orbitals
(c) sum of energies of bonding molecular orbitals
(d) sum of energies of occupied bonding molecular orbitals

25. Antibonding molecular orbitals are of higher energy because
(a) it does not form a stable bond
(b) electron charge density between the nucleus is reduced
(c) it is formed by substractive combination
(d) it is formed from p orbitals of the combining atom

26. If the z axis is an internuclear axis, which of the following will not form a σ bond

(a) s orbital + s orbital
(b) p_z orbital + p_z orbital
(c) s orbital and p_x orbital
(d) s orbital and p_z orbital



Multiple Correct Objective Type Questions

Directions: **Each question in this section has four suggested answers out of which ONE OR MORE answers will be correct.**

27. Which of the following molecules have all the bonds equal in length?

(a) PCl_5 (b) SF_4 (c) SiF_4 (d) NCl_3

28. Which of the following species have identical bond order?

(a) CO (b) NO (c) O_2^{2+} (d) CN^-

29. Molecules which are nonpolar are

(a) SCl_4 (b) BCl_3 (c) NO_2 (d) SO_3



Matrix-Match Type Question

Directions: **Match the elements of Column I to elements of Column II. There can be single or multiple matches.**

30. Match the compounds with shapes on the right

Column I	**Column II**
(a) XeO_2F_2	(p) seesaw
(b) I_3^-	(q) square pyramid
(c) BrF_5	(r) trigonal pyramid
(d) SO_3^{2-}	(s) linear

IIT ASSIGNMENT EXERCISE

Straight Objective Type Questions

Directions: **This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

31. The formation of ionic bond is favoured by
(a) high heat of hydration (b) low ionization energy (c) low electron affinity (d) low lattice energy

32. A stable ion with ns^2 configuration is
(a) Cs^+ (b) Ba^{2+} (c) Fe^{2+} (d) Pb^{2+}

33. Among the following, the compound having both electrovalent and covalent bond is
(a) $H_2S_{(g)}$ (b) $AlF_{3(s)}$ (c) $NaOH_{(aq)}$ (d) $HNO_{3(l)}$

34. The maximum covalency of an atom is equal to the total number of s and p electrons present in valence shell. This is true for
(a) All elements (b) Elements containing d orbitals
(c) Elements containing vacant d orbitals (d) Elements of large size

35. The number of shared electron pairs present in nitrogen molecule is
(a) one (b) two (c) three (d) None of these

36. On decreasing the internuclear distance below the optimum distance (where potential energy is minimum), there is steep increase in potential energy due to
(a) increase in force of attraction between electrons and nucleus
(b) loss of symmetry
(c) equal probability of finding the electron in either nuclei
(d) increase in internuclear and interelectronic repulsions

37. The strength of a covalent bond increases when there is
(a) an increase in extent of overlap
(b) a lateral overlap only
(c) overlap of orbitals that are not directionally concentrated
(d) presence of half filled stable gas configuration

38. During the formation of a fluorine molecule from the atom, there is a
(a) lateral overlap of p orbital that is singly occupied
(b) axial overlap of p orbitals that is singly occupied
(c) lateral overlap of any two p orbitals one from each atom
(d) axial overlap between a s orbital and p orbital

39. Even though triple bond is a stronger bond, it is more reactive than a single bond because
(a) triple bond is a shorter bond (b) a single bond has no unpaired electrons
(c) a triple bond exists only along with a σ bond (d) a triple bond has mobile π-electrons

40. The number of sigma and π bonds in $CH_3 - COCOOH$ is
(a) 9σ and 1 π bond (b) 8σ and 1 π bond (c) 9σ and 2π bond (d) 8σ and 2π bonds

41. The overlap leading to a bond about which free rotation may not be possible is
(a) s – s overlap (b) p – p overlap (c) s – p overlap (d) p – p and s – p overlap

42. For a given molecule, lateral overlap of orbitals
(a) determines the shape of molecule
(b) increases reactivity of molecule
(c) can take place independently
(d) weakens the sigma bond

43. In the molecule of SO_2, there are two π bonds. These π bonds are formed by
(a) pπ – pπ overlap only
(b) pπ – dπ overlap only
(c) pπ – pπ and pπ – dπ overlap
(d) pπ – pπ and dπ – dπ overlap

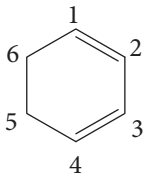
44. The bond that has maximum s character is
(a) sp (b) sp^2 (c) sp^3 (d) sp and sp^2

45. Boron contains only one unpaired electron. But it is trivalent. This anomaly can be explained by
(a) VSEPR theory
(b) Hybridization
(c) Fajan's rule
(d) Linear combination of atomic orbitals

46. The wrong statement regarding hybridization is
(a) The energy required for hybridization is recovered from formation of covalent bond.
(b) Hybridization occurs for orbitals of same atom only.
(c) Hybrid orbitals form sigma and pi bonds or remain as a lone pair of electron.
(d) The hybrid orbitals have greater energy than the atomic orbitals.

47. The compound in which all the bond lengths are not equal is
(a) PCl_5 (b) SF_6 (c) XeF_4 (d) BCl_3

48. The hybridization of central atom in ClO_3^- is
(a) sp^2 (b) sp^3 (c) sp^3d (d) sp^3d^2

49. The hybridization of each carbon in the molecule 

in the order marked is

(a) all sp^2
(b) 1, 2, 3, 4 – sp^2, 5, 6 – sp^3
(c) 1, 3 sp^2 2, 4, 5, 6 – sp^3
(d) 1, 2, 3 sp^2 4, 5, 6 sp^3

50. Two orbitals at 180° to each other can be obtained by hybridization of one s orbital with
(a) two p orbitals
(b) one p and one d orbital
(c) two p and one d orbital
(d) one p orbital

51. The triatomic molecule which has sp^3d hybridization is
(a) $SnCl_2$ (b) XeF_2 (c) O_3 (d) $AgCl_2^-$

52. For an sp^3d hybridization, irrespective of the number of lone pairs, the lone pairs always preferably occupy
(a) the equatorial position
(b) the axial position
(c) either equatorial or axial
(d) first axial, when both axial are occupied, equatorial is occupied

53. In $SnCl_2$, Sn undergoes sp^2 hybridization, and not direct formation of bond by pairing of two unpaired electrons of Sn with that of Cl. This can be proved by
(a) by X-ray analysis Cl –Sn – Cl angle is close to 120°
(b) $SnCl_2$ is found to be paramagnetic
(c) $SnCl_2$ has a net dipole moment
(d) $SnCl_2$ is a good reducing agent

54. The repulsion between two lone pairs is greater than that between two bond pairs because
(a) shape of molecule determines the repulsion
(b) lone pair is attracted to only one nucleus
(c) lone pair causes distortion of bond angles
(d) lone pairs are formed by hybridized orbitals

55. The magnitude of repulsions between bond pairs depends on
(a) number of lone pairs on the central atom
(b) shape of the molecule
(c) hybridization of central atom
(d) electronegativity of central atom

56. The correct order of increasing bond angles is
(a) $NCl_3 < PCl_3 < NF_3$ (b) $PF_3 < PCl_3 < NCl_3$ (c) $NF_3 < NH_3 < PH_3$ (d) $NF_3 < PH_3 < NH_3$

57. The bond dissociation energy of a molecule is inversely proportional to
(a) extent of overlap
(b) bond length
(c) bond order
(d) presence of unpaired electrons

58. The total number of bonding molecular orbitals formed by LCAO is equal to
(a) total number of atomic orbitals
(b) twice the atomic orbitals with nodal plane
(c) twice the number of antibonding molecular orbitals
(d) half the number of atomic orbitals

59. For B_2, B_2^+, B_2^- the correct order of increasing bond order is
(a) $B_2 < B_2^+ < B_2^-$ (b) $B_2 < B_2^- < B_2^+$ (c) $B_2^+ < B_2 < B_2^-$ (d) $B_2^- < B_2 < B_2^+$

60. The species having same magnetic moment as B_2 is
(a) C_2^+ (b) Li_2 (c) He_2^+ (d) O_2

61. In the following CN^+, CN^-, CN, the one with the shortest bond length is
(a) CN^- (b) CN (c) CN^+ (d) both CN and CN^-

62. The molecule that may be destabilized by adding an electron is
(a) Nitrogen (b) H_2^+ (c) CN (d) C_2

63. The trend in boiling points of hydrides of group 15 elements shows a regular increase, while that of group 15 is irregular because
(a) corresponding group 15 elements have small size
(b) first element of group 15 has higher electronegativity
(c) first element of group 15 has lower ionization energy
(d) group 15 shows irregular increase in size

64. The compounds which will not show hydrogen bonding is
(a) H_3BO_3 (b) H_3PO_4 (c) $CH_3 - CHO$ (d) HCOOH

65. Even though both glycerol and 1 pentanol are both alcohols and have same molecular weight, glycerol is more viscous than pentanol because
(a) glycerol had three O – H covalent bonds
(b) pentanol is a straight chain molecule
(c) glycerol is more polar than pentanol
(d) there are two lone pairs present in oxygen of R – OH

66. Even though fluorine is more electronegative than oxygen, HF has lower melting point than H_2O because
(a) HF has slightly higher molecular weight than H_2O
(b) fluorine is a much smaller atom than oxygen
(c) fluorine has very low electron affinity
(d) H_2O is associated with 4 hydrogen bonds per molecule

67. The electrostatic force of attraction between Na^+ and Cl^- is greater than that of Cs^+ and Cl^- because
(a) Na has higher ionization energy
(b) Na does not have filled d orbitals
(c) Na^+ has smaller size than Cs^+
(d) Cs^+ is more reactive than Na^+

68. On moving down the group the solubility of aluminium halides in aqueous solution
(a) is almost a constant
(b) increases and then remains a constant
(c) increases
(d) decreases

69. Identify the correct statement
(a) $SnCl_2$ is ionic while $SnCl_4$ is covalent
(b) $PbCl_2$ has a lower melting point than $PbCl_4$
(c) $MgCl_2$ and $AlCl_3$ are equally soluble in water
(d) Sulphides of a metal are more soluble than its oxide

70. CuCl has a much lower melting point of 442°C while NaCl melts at 800°C. This difference is because
(a) Cu^+ is a larger cation than Na^+
(b) Na^+ is a larger cation than Cu^+
(c) Cu^+ has d electrons in its shell
(d) the ionization energy of $Na \rightarrow Na^+$ is greater than $Cu \rightarrow Cu^+$

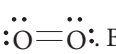
71. Which of the following will have least conductivity?
(a) LiBr (b) $MgCl_2$ (c) NaF (d) RbBr

72. Symmetric molecules will have zero dipole moment only if
(a) bond angles are equal
(b) all bond lengths are of equal length
(c) they have no lone pair of electrons on the central atom
(d) they are planar

73. Which of the following cannot be found from dipole moment measurements?
(a) Geometry
(b) Hybridization
(c) Ionic character
(d) Comparable boiling points

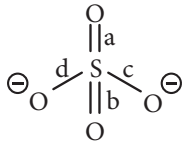
74. A molecule of the type AX_4 with sp^3d^2 hybridization will have
(a) $\mu = 0$
(b) $\mu = 1$ D
(c) μ will depend on electronegativity of A and X
(d) μ will depend on direction of lone pair of electrons

75. The dipole moment of BCl_3 and NCl_3 are not equal because
(a) Nitrogen is more electronegative than Boron
(b) Nitrogen and chlorine have almost same electronegativity
(c) Nitrogen and boron have different hybridization
(d) Bond angles in BCl_3 are equal but NCl_3 has two different bond angles

76. Oxygen molecule is represented as  . By resonance it can be written as  . These representations fail to explain
(a) paramagnetic property in oxygen
(b) π bonding in oxygen
(c) covalent nature of oxygen
(d) Octet rule in oxygen

77. The factor that changes when a molecule undergoes resonance
(a) No. of unpaired electrons
(b) Symmetry of molecule
(c) Structure of molecule
(d) Number of paired electrons

78. Consider SO_4^{2-}, the four S – O bonds are marked as 'a', 'b', 'c', 'd'. We can say that



(a) a = b < c = d
(b) a = b ≠ c = d
(c) a ≤ b < c ≤ d
(d) a = b = c= d

79. The bond order of the Cl – O bond in ClO_4^- is
(a) 1
(b) 1 and 2
(c) 1.75
(d) 1.5

80. Compare the carbon carbon single bond length in $^{+}CH_2 - CH_2 - CH_3$(I) and $^{+}CH_2 - CH = CH_2$(II). We can say bond length of single bond in
(a) I > II (b) II > I (c) I = II (d) I = ½ II

81. If a bond is formed by sharing of a pair of electrons, equally, then the bond is between
(a) two identical metals (b) two identical non metals
(c) two atoms with opposite spins (d) a metal and a non metal

82. The compound which is not isomorphous with NaF is
(a) NaCl (b) KCl (c) MgO (d) LiF

83. For which of the following, the magnitude of lattice enthalpy is the highest?
(a) LiF (b) LiI (c) RbF (d) CsI

84. During overlap of atomic orbitals, the two bonding electrons are of opposite spin. This is in accordance with
(a) Valence Bond Theory (b) Hund's Rule
(c) Pauli's Exclusion principle (d) Exchange energy rule

85. According to Valence Bond Theory, Helium cannot exist as He_2 because
(a) Both bonding and antibonding orbitals contain equal number of electrons
(b) The repulsive forces are dominant over attractive forces
(c) Helium has noble gas configuration
(d) Helium has small atomic size

86. The probability of finding the electron after a sigma bond formation is
(a) maximum in between the bonding nuclei (b) distributed in either atom
(c) far away from each other to reduce repulsion (d) each electron in their respective atom

87. Among the following, the compound without a π bond is
(a) $SO_{3(g)}$ (b) $SiO_{2(s)}$ (c) $F_2O_{2(g)}$ (d) $SO_{2(g)}$

88. Consider ethane, ethene and ethyne. The strongest C – H bond is formed in
(a) Ethane
(b) Ethene
(c) Ethyne
(d) All the three will have approximately same bond energy.

89. The total number of hybrid orbitals present in a given plane for a molecule of benzene is
(a) 6 (b) 12 (c) 18 (d) 24

90. In which of the following hybridizations, angle between the hybrid orbitals formed are not equivalent?
(a) sp (b) sp^3d^2 (c) sp^3d (d) dsp^2

91. The wrong statement about hybrid orbitals is
(a) hybrid orbitals are of higher energy than atomic orbitals.
(b) a hybrid orbital overlaps more effectively than atomic orbital.
(c) Only electrons of nearly same energy belonging to same atom undergo hybridization.
(d) hybrid orbitals determine geometry based on energy considerations.

92. Which of the following molecules have a bond angle of 180°?
(a) NO_2^+ (b) $SnCl_2$ (c) NO_2^- (d) $PbCl_2$

93. The structure and hybridization of ICl_2^- is
(a) linear, sp^3d^2 (b) angular, sp^2 (c) linear, sp^3d (d) angular, sp^3d

94. The common aspect about SO_2 and SCl_2 are
(a) Both have two lone pair of electrons
(b) Both contain one pπ-dπ bond
(c) Both have angular shape
(d) Both have sp^2 hybridised central atom

95. Consider, π-Bonding molecular orbital, π*. Anti-bonding molecular orbital and n-non bonding molecular orbital. The increasing order of energy is
(a) $\pi < \pi^* < n$ (b) $n < \pi < \pi^*$ (c) $\pi^* < \pi < n$ (d) $\pi < n < \pi^*$

96. Considering bond order the most stable species among the following is
(a) Li_2 (b) C_2^+ (c) Be_2^+ (d) B_2^{2+}

97. The formation of CN^- from CN involves
(a) destabilization of the species
(b) decrease in bond order
(c) increase of electron in ABMO
(d) change in magnetic moment

98. The set of compounds that show a lesser boiling point than the expected value due to hydrogen bonding is
(a) ortho nitrophenol and salicylic acid
(b) phenol and nitrobenzene
(c) pentane 1, 2 diol and pentane 1, 3 diol
(d) all compounds show increase in boiling point due to hydrogen bonding

99. Which of the following is not a consequence of hydrogen bonding?
(a) Sucrose, an organic molecule of high molecular weight dissolves in water
(b) Formation of long zig zag chains of HF molecule
(c) Acidic nature of benzoic acid
(d) Solubility of aldehydes in water

100. Among all elements, only hydrogen is capable of forming hydrogen bonds because
(a) hydrogen gives a high electronegativity difference with F, O, N
(b) of small size of hydrogen and presence of one electron
(c) hydrogen has minimum atomic mass
(d) H–H bond enthalpy is appreciable

101. The compound with maximum covalent character among the following is
(a) LiCl (b) LiF (c) $BeCl_2$ (d) BCl_3

102. The correct order of melting points of $CaCl_2$, $SrCl_2$ and $BaCl_2$ can be given as
(a) $CaCl_2 < BaCl_2 < SrCl_2$
(b) $CaCl_2 < SrCl_2 < BaCl_2$
(c) $CaCl_2 < SrCl_2 \cong BaCl_2$
(d) $BaCl_2 < SrCl_2 < CaCl_2$

103. Among the halides of aluminium, the one with maximum solubility in water is
(a) AlF_3 (b) $AlCl_3$ (c) $AlBr_3$ (d) AlI_3

104. A bond X–Y of percentage ionic character 80% has a bond length of 200 pm. Its dipole moment is
(a) 2.56 D (b) 3.2 D (c) 7.68 D (d) 4.8 D

105. Among the following, the molecule with zero dipole moment is
(a) NF_3 (b) SO_3 (c) PCl_5 (d) CH_2Cl_2

106. Consider a molecule PBr_3Cl_2. This molecule will show zero dipole moment if
(a) All the bromine occupy equatorial position
(b) If both Cl are at 90° to each other
(c) If Br and Cl are at 60° to each other
(d) due to presence of lone pair in the molecule bond moment can never be zero

107. Identify the incorrect statement with respect to resonance?
(a) The actual molecule cannot be depicted by a single lewis structure
(b) Resonance energy is the difference in energy between the resonance hybrid and most stable resonating form
(c) greater the resonance energy, more stable the resonance hybrid
(d) Resonating structures do not have real existence

108. The bond order in NO_3^- is
(a) 1.66 (b) 2 (c) 1.0 (d) 1.33

109. The O–O bond length in ozone could be considered as an intermediate between the O–O bond lengths in
(a) H_2O_2 and O_2 (b) H_2O_2 and KO_2
(c) F_2O_2 and H_2O_2 (d) F_2O_2 and KO_2

110. Even though Br_2 and ICl have the same molecular weight, ICl has a higher boiling point than Br_2 because
(a) ICl molecule has larger size than Br_2
(b) Cl in ICl is $2\frac{1}{2}$ times heavier than air
(c) ICl is polar while Br_2 is non polar
(d) Iodine at room temperature exists as a solid

Assertion–Reason Type Questions

Directions: **Each question contains Statement-1 and Statement-2 and has the following choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False
(d) Statement-1 is False, Statement-2 is True

111. Statement 1
A sharp increase in inter-ionic distance does not produce a sharp decrease in lattice enthalpy.
and

Statement 2
Change in inter ionic distance affects structure of the compounds and the modified structure compensates the expected change in lattice enthalpy.

112. Statement 1
XeO_2F_2 has a planar structure.
and

Statement 2
Xe atom is in sp^3d hybridization.

113. Statement 1
NF_3 has very small dipole moment while NH_3 has a large dipole moment.
and

Statement 2
The electronegativity between N and F is very small and that between N and H is very large.

Linked Comprehension Type Questions

Directions: **This section contains 1 paragraph. Based upon the paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

Hybridized orbitals are formed by mixing of atomic orbitals of similar energy and redistributing. Each hybrid orbital has a positive lobe concentrated in a particular direction and is therefore able to overlap very strongly with an orbital on another atom located at an appropriate distance in that direction. The extent of overlap obtained is greater than that by pure atomicorbital.

114. The hybridization of sulphur is SO_2 and SO_3 are

(a) sp^2, sp^2 (b) sp^2, sp^3 (c) sp, sp^2 (d) sp, sp^3

115. Hybrid orbitals formed from atomic orbitals are

(a) of lower energy, hence stable
(b) of equivalent energies and identical shape
(c) orbitals with equal distribution of electrons on either side of nucleus
(d) directed similarly in space or have same orientation

116. The hybridization of the central atom in $HgCl_2$ is

(a) sp^3d (b) sp (c) sp^2 (d) dsp^2

Multiple Correct Objective Type Questions

Directions: **Each question in this section has four suggested answers out of which ONE OR MORE answers will be correct.**

117. Species with linear structure is/are

(a) NO_2^+ (b) I_3^- (c) HCN (d) CS_2

118. Which among the following are the conditions for stability of a diatomic molecule?

(a) Bond order is positive and large (b) Bond length is small
(c) Bond energy is small (d) There is ionic covalent resonance

119. Which of the following molecules exhibit intermolecular hydrogen bonding?

(a) O-nitro phenol (b) p-nitrophenol (c) chloral hydrate (d) Boric acid

Matrix-Match Type Question

Directions: **Match the elements of Column I to elements of Column II. There can be single or multiple matches.**

120.

Column I	**Column II**
(a) C_2H_2	(p) contains at least one nonpolar bond
(b) CH_2Cl_2	(q) contains at least one polar bond
(c) S_8	(r) molecule is polar
(d) CO_2	(s) molecule is nonpolar

ADDITIONAL PRACTICE EXERCISE

Subjective Questions

121. Which is having a greater bond angle PF_3 or PCl_3,. Why?

122. What happens to the bond lengths in the following changes
$N_2 \rightarrow N_2^+$ and $O_2 \rightarrow O_2^+$?

123. Using the aspect of polarisation of ions, explain why AgI is coloured, but NaI is colourless.

124. The percentage ionic character for HF is 43. The experimental dipole moment is 1.92 D. Calculate the internuclear distance in HF?

125. Dipole moment for CO_2 is zero, but SO_2 has a non-zero dipole moment.

126. Give the hybridisation of the central atoms of the following oxy acids of chlorine. Also give the bond orders.
(i) $HClO_4$ and ClO_4^- (ii) $HClO_3$ and ClO_3^- (iii) $HClO_2$ and ClO_2^-

127. Explain, why the Si–F bond length in SiF_4 is shorter than that calculated from the covalent radii?

128. Explain the following:
Atomic radius decrease along the period. However, inert gases, which are at right extreme have maximum atomic radius.

129. HCl gas is covalent where as hydrochloric acid is ionic. Why?

130. The crystalline salts of alkaline earth metals contain more water of crystallization than corresponding alkali metals. Why?

Straight Objective Type Questions

Directions: **This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

131. Magnesium oxide has a much higher lattice energy than sodium fluoride because
(a) Na^+ is larger than Mg^{2+} (b) MgO has partial covalent character
(c) Both Mg and O have higher charge than Na and F (d) Number of neutrons in Mg^{2+} is higher

132. Formation of sodium fluoride is easier compared to formation of sodium oxide because
(a) F^- is a smaller ion than O^{2-} (b) formation of O^{2-} from O^- is endothermic
(c) oxide requires two sodium atoms (d) F^- has higher effective nuclear charge

133. According to the Born Haber cycle, the enthalpy of formation of an ionic bond AB (A^+B^-) is

(a) $\Delta H_f^\circ = \Delta H_{vap}^\circ(A) + \frac{1}{2}\Delta H_{diss}^\circ(B_2) + \Delta H_{EA}^\circ(B) + \Delta H_{ion}^\circ(A) + U$

(b) $\Delta H_f^\circ = \Delta H_{vap}^\circ(A) + \frac{1}{2}\Delta H_{diss}^\circ(B_2) - \Delta H_{EA}^\circ(B) + \Delta H_{ion}^\circ(A) - U$

(c) $\Delta H_f^\circ = \Delta H_{vap}^\circ(B) + \frac{1}{2}\Delta H_{diss}^\circ(A) + \Delta H_{EA}^\circ(A) + \Delta H_{ion}^\circ(B) + U$

(d) $\Delta H_f^\circ = \Delta H_{vap}^\circ(B) + \frac{1}{2}\Delta H_{diss}^\circ(A) - \Delta H_{EA}^\circ(A) + \Delta H_{ion}^\circ(B) - U$

134. The formal charge on nitrogen bonded to oxygen in the molecule N_2O is

(a) +1 (b) −1 (c) −2 (d) −0.5

135. Halides i.e., Cl^-, Br^- and I^- can be quantitatively estimated by using $AgNO_3$ solution. This cannot be applied for F^- because

(a) Fluorine has high electronegativity and so AgF is insoluble in water
(b) Fluorine is a small anion compared to others and so AgF is soluble in water
(c) Fluorine does not have vacant d orbitals and so AgF is insoluble in water
(d) Fluorine is capable of hydrogen bonding and hence sparingly soluble in water.

136. The total number of π bonds present in the complex $Na_2[Zn(CN)_4]$ is

(a) 4 (b) 2 (c) 0 (d) 8

137. The total number of σ and π bonds in $C_2H_2Cl_2$ is

(a) 5σ and 1π or 4σ and 1π (b) 5σ and 1π
(c) 5σ and 0π (d) 6σ and 0π

138. Consider a bond formed between

(i) $sp^2 - p_y$ (ii) $p_x - p_x$ (lateral overlap
(iii) $dsp^2 - p_x$ (iv) $p_x - d_{x^2-y^2}$ (lateral overlap)

The bond in which electron density is absent in between the nuclei of combining atom is

(a) (i) and (iii) (b) (iii) only (c) (ii) and (iv) (d) (iii) and (iv)

139. BCl_3 being a lewis acid, accepts lone pair of electrons on oxygen in ether. During the process

(a) B–Cl bond length increases (b) Bond angle increases
(c) Bond polarity increases (d) Oxygen of ether is repelled by chlorine

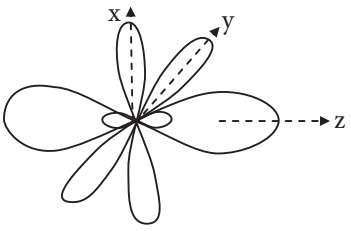
140. The true statement about aluminium chloride is

(a) $AlCl_3$ is a strong lewis base
(b) Thermal decomposition of hydrated $AlCl_3$ is accompanied by decrease in density
(c) Hydrated $AlCl_3$ is more soluble in ether than anhydrous $AlCl_3$ due to hydrogen bonding
(d) Aluminium in $AlCl_3$ is sp^3 hybridised

141. A molecule AB_4 is formed such that all the bonding hybrid orbitals are in one plane. An example for AB_4 is

(a) SF_4 (b) CH_4 (c) $TeBr_4$ (d) XeF_4

142.



If the central atom is carbon, and the orbitals are represented as given in the figure, the molecule formed may be

(a) C_6H_6 (b) C_2H_4 (c) C_2H_2 (d) CO_3^{2-}

143. In SF_6 even though sulphur is surrounded by 6 fluorine atoms it is more stable than PF_5 because
(a) Fluorine is an electronegative atom
(b) Bonds in PF_5 are not equivalent
(c) Bond angle in PF_5 is greater than SF_6
(d) Phosphorus has larger size and lesser charge than SF_6

144. IF_4^+ will have the same structure as
(a) NH_4^+ (b) BH_4^+ (c) XeF_4 (d) SF_4

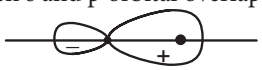
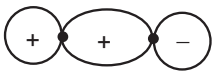
145. The hybridization, geometry of NH_2^- are
(a) sp^3, linear (b) sp^3, angular (c) sp^3d, linear (d) sp^2, angular

146. In the molecule of ClF_3, the lone pair of electrons are always present on the equatorial orbitals because
(a) axial bonds are elongated
(b) lone pairs are kept 120° apart which is maximum
(c) it is the only possibility where $\ell p - \ell p$ repulsion is zero
(d) extent of repulsion is minimum

147. In which of the following cases, the lone pair should be placed taking into consideration the minimum repulsion?
(a) sp^3d^2, two lone pairs (b) sp^3, two lone pairs
(c) sp^3d^2, one lone pair (d) sp^2, one lone pair

148. In H_2O, if two singly filled p orbitals in oxygen overlaps with singly filled s orbital of two hydrogens, the bond angle would have been 90°. But the bond angle is 104° 5' . This is explained by
(a) Presence of one unpaired electrons
(b) Presence of two unpaired electrons
(c) Hybridisation in oxygen atom
(d) Pauli's exclusion principle

149. If a double bond is involved in the formation of an ion,
(a) it does not influence geometry of the ion at all
(b) it may or may not involve hybrid orbitals
(c) it always occurs between similar atomic orbitals
(d) it cannot be polarized easily

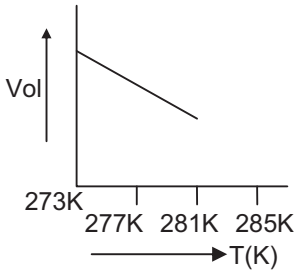
150. When s and p orbital overlap to form antibonding molecular orbitals, they may be represented as
(a) 
(b) 
(c) 
(d) 

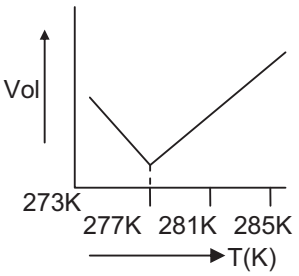
151. The molecule that does not exist according to molecular orbital theory is
(a) F_2^{2-} (b) He_2^+ (c) Be_2^+ (d) Li_2^+

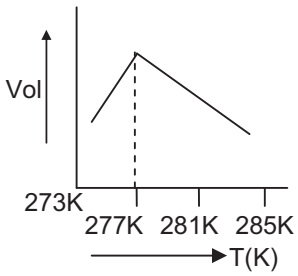
152. Oxygen is paramagnetic by MOT because filling up of molecular orbitals is governed by
(a) Pauli's exclusion principle (b) LCAO Rules
(c) Hund's Rule (d) Aufbau principle

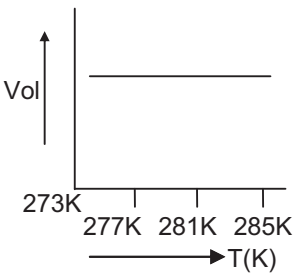
153. The concepts that is encountered both in VBT as well as MOT is
(a) equal weightage to both covalent and ionic bond
(b) stabilization due to delocalization
(c) bond formation by overlap of valence shell orbitals
(d) bond formation by overlap of orbitals of identical symmetry

154. Solid ice is melted and converted to water. The change in volume with respect to temperature can be represented as

(a) 

(b) 

(c) 

(d) 

155. Even though oxygen in H_2O is sp^3 hybridized, the structure of solid ice is open cage like because
(a) oxygen is very big compared to hydrogen
(b) the lone pair electron of hydrogen is shared
(c) In solid the structure is rigid
(d) covalent bonds are shorter and are stronger than hydrogen bonds

156. Which of the following natural phenomena are not based on hydrogen bonds formation?
(a) Colour of natural dye
(b) Structure of proteins
(c) Rigidity and tensile strength of silk
(d) Stickiness of honey

157. Even though electronegativity of nitrogen and chlorine are almost equal, HCl does not show hydrogen bonding because
(a) Cl^- in HCl attains noble gas configuration
(b) Cl has reduced electrostatic forces operating for a hydrogen bond
(c) Cl has high electron affinity
(d) Cl contains empty d orbitals

158. The compound with least ionic characteristics among the following is
(a) HF
(b) HCl
(c) HBr
(d) HI

159. Consider two ions of the same size and charge. The one with more polarizing power will have
(a) $(n-1)d^x ns^0$ configuration
(b) $(n-1)s^2(n-1)p^6 ns^0$ configuration
(c) $(n-1)d^{10} ns^2 np^x$ configuration
(d) $(n-1)d^x ns^2$ configuration

160. Which of the following is not a consequence of polarization of the bond?
(a) higher lattice energy of iodides e.g., LiI
(b) solubility of polar compounds in organic solvents
(c) insolubility nature of $BaSO_4$ in water
(d) decrease in hardness of ionic compounds

161. If the measured dipole moment of water is 1.84D, the O–H bond moment is ($\cos 52° 25' = 0.612$)
(a) 0.613 D
(b) 0.92 D
(c) 1.85 D
(d) 1.5 D

162. If a diatomic molecule has a dipole moment of 2.4D and a bond distance of 1.2A°, then the fraction of electronic charge on each atom is

(a) 42% (b) 58% (c) 20% (d) 21%

163. Arrange H_2O, CH_3OCH_3, $C_2H_5OC_2H_5$ in increasing order of dipole moment

(a) $H_2O < CH_3OCH_3 < C_2H_5OC_2H_5$
(b) $H_2O < C_2H_5OC_2H_5 < CH_3OCH_3$
(c) $CH_3OCH_3 < C_2H_5OC_2H_5 < H_2O$
(d) $C_2H_5OC_2H_5 < CH_3OCH_3 < H_2O$

164. The compound among the following that has zero dipole moment is

(a) Trans 1 chloro propene (b) cyclopropane
(c) SF_4 (d) XeF_6

165. The energy of an induced dipole interaction is given as $E = \frac{-\mu^2\alpha}{r^6}$ where μ is moment of inducing dipole, α is its polarizibility and r is the distance. We can say that these are

(a) applicable for noble gases (b) induced if charged species is large
(c) short range forces (d) very weak forces

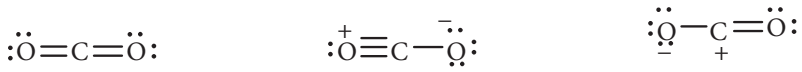
166. The weakest chemical interaction among the following is

(a) Ion-dipole (b) dipole-dipole (c) dipole-induced dipole (d) London forces

167. In VBT, the ionic nature of a H_2 molecule is highlighted by considering that

(a) only one electron can be associated with a given nucleus at a given time
(b) electrons tend to avoid each other due to mutual repulsion
(c) electrons are allowed to exchange places
(d) there is a probability of finding both electrons in the same atom

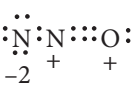
168. Resonating forms of carbon dioxide are



The most unstable form is

(a) I (b) II (c) III (d) II and III

169. Among the following structure of Nitrous oxide, the most significant structure is

(a)  (b)  (c)  (d) 

170. The atom in CO_3^{2-} which has a non zero formal charge is

(a) carbon (b) single bonded oxygen (c) double bonded oxygen (d) both oxygen atoms

Assertion–Reason Type Questions

Directions: **Each question contains Statement-1 and Statement-2 and has the following choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1

(c) Statement-1 is True, Statement-2 is False
(d) Statement-1 is False, Statement-2 is True

171. Statement 1
Co-valency of carbon is four in the ground state.
and

Statement 2
In the excited state carbon can form sigma bonds utilizing hybridized orbitals only.

172. Statement 1
SO_3^{2-} is pyramidal while NO_3^- is planar.
and

Statement 2
SO_3^{2-} has a lone pair on sulphur while NO_3^- has none on nitrogen.

173. Statement 1
A π-molecular orbital has always a nodal plane containing the line joining the two nuclei.
and

Statement 2
There is negative overlap above and below this nodal plane.

174. Statement 1
Monomeric BN molecule is diamagnetic and has a double bond consisting of two π-bonds without a σ-bond.
and

Statement 2
BN molecule with 12 electrons is isoelectronic with C_2 molecule.

175. Statement 1
The species HF_2 exists in solid state and liquid state but not in aqueous solution.
and

Statement 2
Hydrogen bonding between HF and H_2O is stronger than that between HF and HF.

176. Statement 1
Polarising power of Cr^{2+} is more than that of Mg^{2+}.
and

Statement 2
Inert gas configuration is the best shielding configuration for nuclear charge.

177. Statement 1
Covalent character of CH_3Na is greater than that of CH_3ONa.
and

Statement 2
Electron distribution over carbon is more polarizable than that over oxygen.

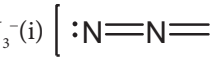
178. Statement 1
During resonance canonical forms are in a dynamic equilibrium in which a definite percentage of each canonical form exists.

and

Statement 2

The percentage contribution of a canonical structure towards the resonance hybrid is proportional to its stability.

179. Statement 1

Among the canonical structures of N_3^-(i)  and (ii)  , structure (ii) has greater contribution towards the hybrid.

and

Statement 2

Structure (ii) is more stable.

180. Statement 1

The central carbon atom in $F_2C = C = CF_2$ and both the carbons in $F_2B - C \equiv C - BF_2$ are sp hybridized.

and

Statement 2

Because they are planar molecules

Linked Comprehension Type Questions

Directions: **This section contains 3 paragraphs. Based upon the paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

Passage I

The electronic configuration of Lithium (Z = 3) is $1s^2\ 2s^1$. There are six electrons in the lithium molecule. The four electrons are present in K shell and there are only two electrons to be accommodated in the molecular orbital. These two electrons go into σ2s BMO which are of lower energy and σ*2s ABMO remains empty

181. Among the hypothetical ions, the non existent one according to MOT is

(a) Li_2^+ (b) Li_2^- (c) Li_2^{2+} (d) Li_2^{3+}

182. Even though bond order of both H_2 and Li_2 molecules are 1, the stronger bond is

(a) H_2, because of small size of 1s orbital
(b) H_2, because of presence of one electron
(c) Li_2, because of large size of lithium atom
(d) Li_2, presence of more number of electrons

183. In the gas phase, according to MOT, we can expect to find

(a) diatomic Li and diatomic Be
(b) monatomic Li and monatomic Be
(c) diatomic Li and monatomic Be
(d) triatomic Li and monatomic Be

Passage II

The hydrogen bond is a class in itself. It arises from electrostatic forces between positive end of one molecule and negative end of another molecule, generally of the same substance. The strength of bond has been found to vary between 10 – 40 kJ mol^{-1}.

184. In alcohols, hydrogen bonding extends over several molecules whereas in acetic acid it lead to

(a) polymerization (b) increase in acidity (c) decrease in acidity (d) dimerisation

185. For a bond H – A…..H – A. The bond A…..H is expected to be

(a) as long as H–A bond
(b) equal to the sum of vander waal radii
(c) much less than sum of vander wall radii
(d) a little shorter than H–A bond length

186. Which of the molecules are stabilized by ring formation in the molecule due to hydrogen bonding?
(a) H_3BO_3 (b) Phenol (c) $CuSO_4.5H_2O$ (d) HCN

Passage III

A covalent bond between two atoms acquires a partial polar character if the values of electronegativity of the bonded atoms differ. The two charged ends of the bond behave as electrical dipole and the degree of polarity is measured in terms of dipole moment

187. Even though SO_2 and SO_3 have same type of polar bond, SO_2 has a dipole moment, but SO_3 has zero dipole moment because
(a) sulphur is surrounded by 3 oxygen atoms in SO_3 (b) there are two $p\pi-d\pi$ bond is SO_3
(c) SO_3 is symmetric while SO_2 is angular (d) SO_2 is sp^2 and SO_3 is sp^3 hybridised

188. The dipole moment of a bond AB is 5D and its bond distance is 150 pm. The percentage ionic character of this bond is
(a) 78.5% (b) 83% (c) 24% (d) 69.5%

189. On moving down the group, the dipole moment of the hydrides of group V elements
(a) increases (b) decreases
(c) increases then decreases due to hydrogen bonding (d) does not show a regular trend

Multiple Correct Objective Type Questions

Directions: **Each question in this section has four suggested answers out of which ONE OR MORE answers will be correct.**

190. Which of the following facts can be explained by the statement 'oxygen is a smaller atom than sulphur'?
(a) H_2O boils at a much higher temperature than H_2S.
(b) H_2O undergoes intermolecular hydrogen bonding.
(c) O – H bond is stronger than S – H bond.
(d) S – H bond is longer than O – H bond.

191. PCl_5 is highly reactive and in solid state it splits into ions. The shape of these ions is
(a) square pyramidal (b) octahedral
(c) tetrahedral (d) trigonal bipyramidal

192. The d orbitals involved in sp^3d^2 hybridization are (internuclear axis is the Z axis)
(a) $d_{x^2-y^2}$ (b) d_{xy} (c) d_{z^2} (d) d_{yz}

193. Among the sp^3d^2 hybridized compounds, SF_6, BrF_5, XeF_4 and ClF_5, those with bond angle of 90° is/are
(a) SF_6 (b) BrF_5 (c) XeF_4 (d) ClF_5

194. By isoelectronic principle, isoelectronic species and species with same number of valence electrons have the same structure. This is shown by
(a) NH_4^+, BF_4^- (b) O_3, ONCl (c) ONCl, NO_2 (d) N_3^-, NO_2^+

195. The change(s) observed on substituting hydrogen in NH_3 by SiH_3 is/are
(a) hybridization of central atom (b) geometry of molecule
(c) charge on central metal atom (d) type of bonding

196. We know that N_2 and O_2 have different bond order. We can say that the species with identical bond order is
(a) N_2^- and O_2^- (b) N_2^- and O_2^+ (c) N_2^+ and O_2^+ (d) N_2^+ and N_2^-

197. By Resonance, hydrogen molecule can be written as $H-H$ (I) or as H^+H^- (II). Both are acceptable but structure (II) is of higher energy because

(a) It is in ionic form
(b) hydrogen has high ionization energy
(c) hydrogen is a no neutron system
(d) hydrogen has low electron affinity

Matrix-Match Type Questions

Directions: **Match the elements of Column I to elements of Column II. There can be single or multiple matches.**

198.

Column I	**Column II**
(a) Covalent	(p) HCN
(b) Ionic	(q) H_3PO_4
(c) Coordinate	(r) $K_4[Fe(CN)_6]$
(d) Hydrogen Bonding	(s) fluoro benzene

199.

Column I	**Column II**
(a) SF_4	(p) Two lone pairs
(b) NH_2^-	(q) Angular
(c) SO_2	(r) One lone pair
(d) ClF_3	(s) Presence of π bonds

200.

Column I	**Column II**
(a) Paramagnetic	(p) CO
(b) Fractional Bond order	(q) C_2^-
(c) destabilized on adding one electron	(r) Li_2^+
(d) mixing of orbitals	(s) F_2

SOLUTIONS

ANSWER KEYS

Topic Grip

1. PCl_5 – 5 bond pairs – trigonal bipyramidal
IF_5 – 5 bond pairs and 1 lone pair – square pyramidal

2. AB_4^- is tetrahedral
AB_2^+ is linear

3. H_2, bond order = 1
He_2, bond order = 0

4. Sn^{4+} – more polarization than Sn^{2+}
– Fajan' s rule

5. P – O bond has partial ionic character

6. (i) BaO – water soluble
MgO – water insoluble
(ii) ICl – High boiling point
Br_2 – Low boiling point
PCl_3 – trigonal planar

7. NH_3 – Tetrahedral – $\mu \neq 0$
BrF_5 – Square pyramidal – $\mu \neq 0$
BF_3 – trigonal planar $\mu = 0$

8. $BaSO_4$, Lattice energy > Hydration energy

9. NH_4Cl – ion-dipole and hydrogen bonding
NaCl – ion-dipole interaction only

10. $BaSO_4$ – Hydration energy < Lattice energy

11. (c) **12.** (a) **13.** (a)
14. (b) **15.** (b) **16.** (c)
17. (c) **18.** (a) **19.** (a)
20. (c) **21.** (b) **22.** (a)
23. (c) **24.** (a) **25.** (b)
26. (c)
27. (c), (d)
28. (a), (c), (d)
29. (b), (d)
30. (a) → (p)
(b) → (s)
(c) → (q)
(d) → (r)

IIT Assignment Exercise

31. (b) **32.** (d) **33.** (c)
34. (c) **35.** (c) **36.** (d)
37. (a) **38.** (b) **39.** (d)
40. (c) **41.** (b) **42.** (b)
43. (c) **44.** (a) **45.** (b)
46. (c) **47.** (a) **48.** (b)
49. (b) **50.** (d) **51.** (b)
52. (a) **53.** (a) **54.** (b)
55. (d) **56.** (b) **57.** (b)
58. (d) **59.** (c) **60.** (d)
61. (a) **62.** (a) **63.** (b)
64. (c) **65.** (a) **66.** (d)
67. (c) **68.** (d) **69.** (a)
70. (c) **71.** (a) **72.** (c)
73. (d) **74.** (a) **75.** (c)
76. (a) **77.** (b) **78.** (d)
79. (c) **80.** (a) **81.** (b)
82. (d) **83.** (a) **84.** (c)
85. (b) **86.** (a) **87.** (c)
88. (c) **89.** (c) **90.** (c)
91. (c) **92.** (a) **93.** (c)
94. (c) **95.** (d) **96.** (b)
97. (d) **98.** (a) **99.** (c)
100. (b) **101.** (d) **102.** (b)
103. (a) **104.** (c) **105.** (b)
106. (a) **107.** (b) **108.** (d)
109. (a) **110.** (c) **111.** (a)
112. (d) **113.** (c) **114.** (a)
115. (b)
116. (b)
117. (a), (b), (c), (d)
118. (a), (b), (d)
119. (b), (d)
120. (a) → (p), (q), (s)
(b)→ (q), (r)
(c) → (p), (s)
(d) → (q), (s)

Additional Practice Exercise

121. $PCl_3 > PF_3$

122. $N_2 \rightarrow N_2^+$
$\therefore$ bond length increases
$O_2 \rightarrow O_2^+$
$\therefore$ bond length decreases

123. AgI – more polarisation

124. 0.93 Å

125. CO_2 – linear
SO_2 – angular

126. (i) sp^3 & 1.75
(ii) sp^3 & 1.66
(iii) sp^3 & 1.5

127. $p\pi - d\pi$ back bonding

128. Inert gases – van der Waals radii
Rest of atoms–Covalent radii

129. High hydration energy of HCl

130. High charge density of alkaline earth metals

131. (c) **132.** (b) **133.** (a)
134. (a) **135.** (b) **136.** (d)
137. (b) **138.** (c) **139.** (a)
140. (b) **141.** (d) **142.** (c)
143. (b) **144.** (d) **145.** (b)
146. (d) **147.** (a) **148.** (c)
149. (a) **150.** (c) **151.** (a)
152. (c) **153.** (d) **154.** (b)
155. (d) **156.** (a) **157.** (b)

158. (d) **159.** (a) **160.** (c)
161. (d) **162.** (a) **163.** (d)
164. (b) **165.** (c) **166.** (d)
167. (d) **168.** (c) **169.** (a)
170. (b) **171.** (d) **172.** (a)
173. (c) **174.** (a) **175.** (a)
176. (a) **177.** (a) **178.** (d)
179. (a) **180.** (c) **181.** (c)
182. (a) **183.** (c) **184.** (d)
185. (c) **186.** (a) **187.** (c)
188. (d)
189. (b)

190. (c), (d)
191. (b), (c)
192. (a), (c)
193. (a), (c)
194. (a), (b), (d)
195. (a), (b)
196. (b), (c), (d)
197. (b), (d)
198. (a) → (p), (q), (r), (s)
(b)→ (r)
(c) → (r)
(d) → (p), (q)

199. (a) → (r)
(b)→ (p), (q)
(c) → (q), (r), (s)
(d) → (p)
200. (a) → (q), (r)
(b)→ (q), (r)
(c) → (p), (s)
(d) → (p), (q), (r), (s)

HINTS AND EXPLANATIONS

Topic Grip

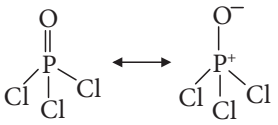
1. In PCl_5, there are 5 bond pairs around P. Hence the shape is trigonal bipyramidal. In IF_5, there are 5 bond pairs and one lone pair. Hence the shape is square pyramidal.

2. AB_4^- is tetrahedral (4 bond pairs).
AB_2^+ is linear (two bond pairs)

3. In H_2, there are 2 electrons in bonding molecular orbital. There are no electrons in antibonding molecular orbital. Hence bond order = $\frac{2-0}{2}=1$
For He_2, there are 2 electrons in bonding molecular orbital and 2 electrons in antibonding molecular orbital.
Bond order = $\frac{2-2}{2}=0$. Bond order zero means that diatomic He does not exist, i.e., He is monoatomic.

4. (i) According to Fajan's rule, smaller cationic size causes more polarization and hence forms covalent compounds. Sn^{4+} is smaller in size than Sn^{2+} and therefore it causes more polarization to Cl^- ions and forms covalent compound.

5. Due to the partial ionic character of the P – O bond, it has only partial double bond character.



6. (i) BaO is soluble in water whereas MgO is insoluble. Since BaO is more ionic than MgO
(ii) ICl it is having higher boiling point than Br_2 because ICl is polar while Br_2 is non polar.

7. Geometry is related to dipole moment. Molecules of the type, AB_4 tetrahedral, AB_2 linear and centro symmetric, AB_3 trigonal planar, AB_4 square planar and AB_6 octahedral have zero dipole moment. BF_3 is trigonal planar and hence has zero dipole moment.

8. For $BaSO_4$, Lattice energy > Hydration energy

9. Na^+ ion is solvated only by ion–dipole interaction, where as NH_4^+ ion is solvated both by ion–dipole and H – bonding and hence more hydration energy is released which results in greater solubility of NH_4Cl.

10. The hydration energy of $BaSO_4$ is lesser than the lattice energy and hence it will not dissociate in water into the ions.

11. The high value of lattice enthalpy stabilizes $CaCl_2$ and favours its formation.

12. Increase in s character in hybridized orbitals decreases the bond length. Bond angle is decreased by increase in no. of lone pair electrons.

13. Equal stabilization due to ++ overlap and destabilization due to + – overlap.

14. The formal charge on all the bonded atoms are zero, hence it is considered to be the most appropriate structure.

15. In BF_3 because of back bonding B – F bond is shorter than expected single bond length . But in the adduct there is no back bonding and hence the bond length increases.

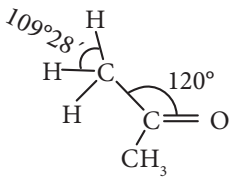
16. Statement 1 is correct
Statement 2 is wrong

17. Statement 1 is correct
Statement 2 is wrong

18. Statement 1 is correct
Statement 2 is correct and is the correct explanation for statement 1.

19. As oxidation number increases, electronegative character increases.
Statements 1 and 2 are correct and 2 is the reason for 1

20. Statement 1 is correct
Statement 2 is wrong

21. 

carbonyl carbon is sp^2 hybridized hence bond angle is 120°

22. carbon is sp hybridized two hybrid orbitals one forms a bond and other remains as lone pair

23. $SF_4 - AB_4E$ $ClF_3 - AB_3E_2$ and $H_2O - AB_2E_2$
$P(CH_3)_3 - AB_3E$ $XeF_2 - AB_2E_3$

24. Each molecular orbital wave function ψ is associated with definite energy. The sum of these energies will be the energy of the molecule

25. As electron charge density in between the nucleus decreases, the two nuclei are not screened enough, so they repel to a greater extent, hence there is increase in energy

26. Since s and p_x do not have same symmetry overlap may not be sufficient for bond formation

27. SiF_4 is tetrahedral and NCl_3 is trigonal pyramidal. Therefore all the bonds in them have equal length.

28. CO, O_2^{2+} and CN^- have bond order, 3

29. BCl_3 and SO_3 molecules have no resultant dipole moment.

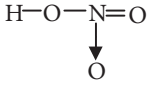
30. (a) → (p)
(b) → (s)
(c) → (q)
(d) → (r)

IIT Assignment Exercise

31. When ionization energy is low, cation is formed easily.

32. Cs^+ and Ba^{2+} has ns^2np^6 configuration. Fe^{2+} has $3d^64s^0$ configuration. Pb^{2+} has ns^2 configuration and is stable due to inert pair effect.

33. H_2S – only covalent
AlF_3 – only electrovalent
NaOH – covalent and electrovalent
$Na^+[O - H]^-$
HNO_3 – covalent and coordinate



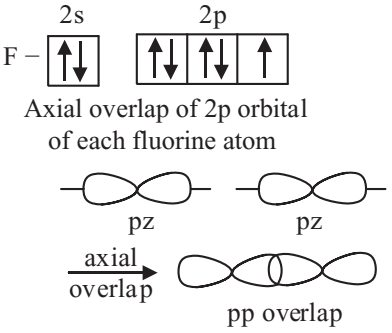
34. On bonding with more electronegative elements, s and p electrons unpair to exhibit variable valency.

35. Nitrogen has three unpaired electrons
$1s^2\ 2s^2\ 2p^3$
Pairing between these three unpaired electrons from each nitrogen atom leads to a triple bond.

36. As the distance decreases, repulsive forces dominate attractive forces leading to increase in potential energy.

37. Greater the extent of overlap, stronger the bond

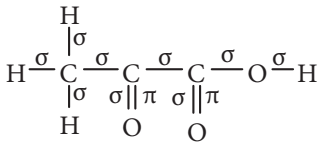
38.



Note: Correct key is (b) and not (a)

39. The presence of mobile π electrons allows a point of attack for the attacking reagent, making it more reactive.

40.

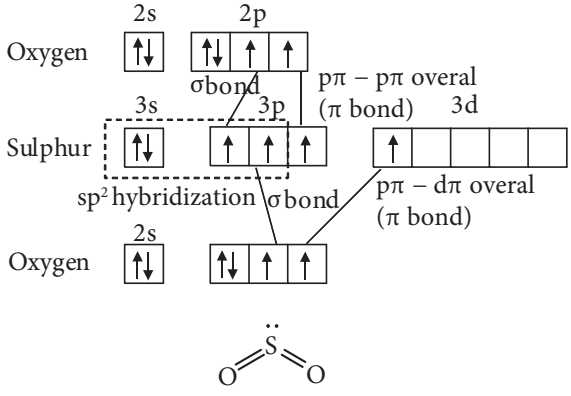


9σ and 2 π bonds

41. Among ss, pp and sp
Only pp overlap can have a lateral overlap. So it may posess a restricted rotation about the bond.

42. Lateral overlap leads to a pi bond which increases reactivity of molecule due to presence of electrons.

43. Sulphur undergoes sp^2 hybridization

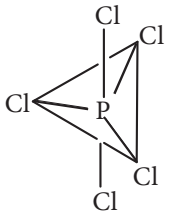


44. s character sp = 50%
sp^2 = 33%
sp^3 = 25%

45. Based on hybridization, electrons in the 2s and 2p orbitals pool up their energy and redistribute. So three electrons are available for bonding.

46. Hybrid orbitals do not form π bonds usually

47. PCl_5 – trigonal bipyramidal



Axial bonds are longer than equatorial bond

48. $X = SA + \frac{1}{2}[G - V + a]$

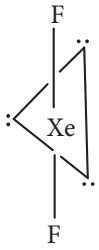
$= 3 + \frac{1}{2}[7 - 6 + 1] = 4$

Since X = 4, hybridization is sp^3

49. 1, 2, 3 and 4 are sp^2 hybridized and 5 and 6 sp^3 hybridized.

50. sp hybridization gives a linear structure

51. Xe in XeF_2 is sp^3d hybridization. It has three lone pairs of electron and therefore has linear structure.



52. When equatorial position is occupied by the lone pair, in this way they are placed farthest away and experience minimum repulsion.
Example, SF_4, ClF_3, XeF_2

53. Only when Sn undergoes sp^2 hybridization, the bond angle can be 120°. Otherwise bond angle will be equal to the angle between two p orbitals i.e., 90°.

54. Lone pair is attracted to one nucleus while the bond pair is attracted to two nuclei, so the lone pair takes up more space around the central atom hence shows greater repulsion.

55. The repulsion between bond pairs depends on the electronegativity difference between the central atom and the other atom.

If the central atom is more electronegative the electrons are closer to the central atom hence bond pair repulsion is greater.

56. $\underset{98°}{PF_3} < \underset{100°}{PCl_3}$ due to high electronegativity of F electron pairs are farther away from central atom, so they can be compressed.

$\underset{100°}{PCl_3} < \underset{107°}{NCl_3}$ Nitrogen is more electronegative, so lone pair is close to central metal atom lone pair can also back bond with empty d orbitals of Cl atom.

57. Shorter the bond, stronger the bond

58. No. of atomic orbitals combining = No. of molecular orbitals formed
And, BMO = ABMO

59. $B_2 - \sigma 1s^2\ \sigma^* 1s^2\ \sigma 2s^2\ \sigma^* 2s^2\ \pi 2py^1\ \pi 2px^1$

Bond order of $B_2 = \frac{6-4}{2} = 1$

For $B_2^+ = \frac{5-4}{2} = 0.5$

For $B_2^- = \frac{7-4}{2} = 1.5$

60. B_2 has two unpaired electrons, O_2 also has two unpaired electrons.

61. Bond order $= \frac{BMO - ABMO}{2}$

For CN, B.O $= \frac{9-4}{2} = 2.5$

For CN^+, B.O $= \frac{8-4}{2} = 2$

For CN^-, B.O $= \frac{10-4}{2} = 3$

Bond order is inversely proportional to bond length.

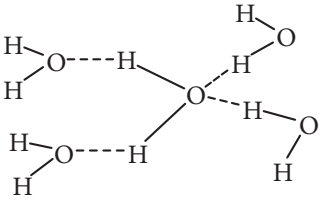
62. Only in the case of nitrogen electron enters the antibonding molecular orbital. Hence the molecule is destabilized.
In H_2^+, CN, C_2 electron enters the bonding molecular orbital.

63. Nitrogen being electronegative, has hydrogen bonds in NH_3. Hence there is a decrease in boiling point followed by a regular increase in size of group 15.

64. There are no covalent bonds between hydrogen and electronegative atom.

65. Due to high electronegativity of oxygen, hydrogen bonding due to three covalent O – H bonds are possible. This makes glycerol viscous. Pentanol has only – O – H bond. Hence there is only one hydrogen bond.

66.



H— F---H—F

67. For the same anion, cation with a smaller size forms stronger ionic bond. As charge/size ratio is greater.

68. On moving down the group, anion size increases, so covalent character increases, so solubility in aqueous solution decreases.

69. Based on Fajans rule, greater the polarization, greater the covalent character. So lesser the solubility in aqueous solution and lesser the melting point.

70. Cu^+ – 0.96 A° and Na^+ 0.95 A°.
Both cation has same size but due to the presence of low shielding d electrons in Cu^+ it has high effective nuclear charge, hence it has greater polarizing power.

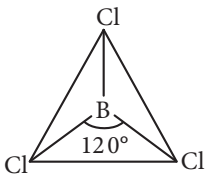
71. LiBr
Small cation hence more covalent. Greater covalent nature, low conductivity in solid state.

72. Symmetric molecules without lone pair of electrons will have $\mu = 0$

73. Molecules such as XeF_4 containing two lone pairs also have $\mu = 0$

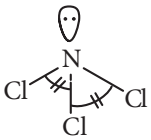
74. The given molecule will have square planar structure with lone pair of electrons symmetrically above and below the plane . So $\mu = 0$

75. Boron is sp^2 hybridized symmetric, planar molecule.



So $\mu = 0$

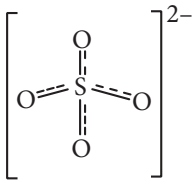
But in NCl_3, N is sp^3 hybridized, non planar, pyramidal and has a lone pair of electrons.



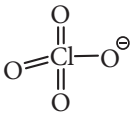
76. Oxygen is paramagnetic, but according to the structure there are no unpaired electrons.

77. Due to change in bonds in the molecule, symmetry of the molecule changes. The structure of molecule, number of electrons (paired and unpaired electrons) does not change.

78. Sulphate ion exists as a resonance hybrid and can be represented as



79.



The πe^- are delocalized, hence bond order

$$= \frac{7}{4} = 1.75$$

80.



Single bond in the II compound has partial double bond character. Hence bond length decreases.

81. Two identical non metals share their electrons to form a covalent bond. Since electronegativity is equal, they are shared equally.

82. NaF, NaCl, KCl and MgO are isomorphous.

83. This refers to bond with highest lattice energy

Lattice energy $\propto \frac{q_1 q_2}{d^2}$

Since all have similar charge, smaller ion will have greater lattice energy.

84. By Pauli's exclusion principle, no two electrons in an atom can have same value for all four quantum

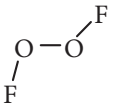
numbers. So electrons in the same orbitals have opposite spin.

85. In He_2 the electron-electron and nuclei-nuclei repulsive forces operating are greater than the attractive forces. So He_2 does not exist.

86. The electron density is maximum in between the two atoms. A large part of bonding force is electrostatic force of attraction between the nuclei and accumulated electron cloud.

87. F_2O_2 has a peroxide linkage

All others have π bonds



SiO_2 has three dimensional co-valently bonded network structure.

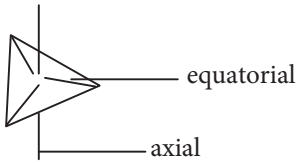
88. C–H bond energy is greater in ethyne. Relative overlap of hybrid orbitals decreases in the order

$sp > sp^2 > sp^3 >> p$

89. Carbon is sp^2 hybridized in benzene. All hybrid orbitals are in one plane.

$\therefore$ Hybrid orbitals in benzene = $6 \times 3 = 18$

90. In sp^3d the hybrid orbitals are disposed as



The equatorial orbitals are not exactly similar to axial orbitals

91. Only orbitals undergo hybridization and not electrons.

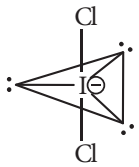
92. NO_2^+ $\quad X = 2 + \frac{1}{2}[5 - 4 - 1] = 2$

Hence sp hybridized–linear

$SnCl_2$, $PbCl_2$ and NO_2^- are sp^2 hybridized and angular.

93. $ICl_2^- X = 2 + \frac{1}{2}[7 - 2 + 1] = 5$

Hybridization is sp^3d. Hence it is AB_2L_3 type like I_3^-



94. SO_2 – sp^2, angular, AB_2L type

SCl_2 – sp^3, angular, AB_2L_2 type

95. Non bonding orbitals, formed have no overall change in energy

BMO – energy is lowered

ABMO – energy is higher

96. C_2^+ has higher bond order

Li_2 $\quad B.O = \frac{4-2}{2} = 1$

C_2^+ $\quad B.O = \frac{7-4}{2} = 1.5$

Be_2^+ $\quad B.O = \frac{4-3}{2} = 0.5$

B_2^{2+} $\quad B.O = \frac{4-4}{2} = 0$

97. CN

$\sigma 1s^2\ \sigma^* 1s^2\ \sigma 2s^2\ \sigma^* 2s^2\ \sigma 2p_z^2\ \pi 2p_x^2\ \pi 2p_y^1$

Formation of CN^- from CN involves addition of one electron to the species

which adds on to BMO is stabilized and Bond order increases from 2.5 to 3

paramagnetic species becomes diamagnetic

98. Ortho nitrophenol and salicylic acid have intramolecular hydrogen bonding. So there are no interactions between the molecules. Hence presence of hydrogen bond does not help to increase the boiling point.

99. acidic nature of benzoic acid is due to resonance stabilization of $C_6H_5-C(=O)-O^{\ominus}$

100. as there is only one electron, the small H^+ formed is capable of keeping two electronegative atoms close together due to electrostatic force of attraction

larger the size lesser will be the electrostatic force of attraction

101. Apply Fajan's rules

since F^- is smaller than Cl^-

comparing LiCl, $BeCl_2$, BCl_3

B^{3+} has the highest charge

Hence BCl_3 will be most covalent

102. Cl^- is the common anion

Ca^{2+}, Ba^{2+}, Sr^{2+} have same charge

Smaller the size, more polarizing the cation

hence greater covalent character

$\therefore$ Melting point will be less

$Ca^{2+} < Sr^{2+} < Ba^{2+}$

$772^\circ C \quad 872^\circ C \quad 960^\circ C$

103. As anion size increases covalent character increases, hence solubility in water decreases

104. $\mu_{ionic} = e^- \times d = 1.6 \times 10^{-19} \times 200 \times 10^{-12}$

$= 3.2 \times 10^{-29}$ cm

$$\frac{\mu_{obsr}}{\mu_{ionic}} \times 100 = \% \text{ ionic character}$$

$$80 = \frac{\mu}{3.2 \times 10^{-29}} \times 100$$

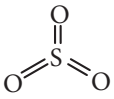
$$\mu = \frac{80 \times 3.2 \times 10^{-29}}{100}$$

$$= 2.56 \times 10^{-29} \text{ cm}$$

$$= \frac{2.56 \times 10^{-29}}{3.33 \times 10^{-30}}$$

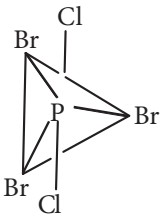
$$= 7.68 \text{ D}$$

105.



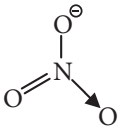
due to planar and symmetric nature, it has zero dipole moment

106. PBr_3Cl_2 has trigonal bipyramidal structure. Therefore the structure with zero dipole moment is



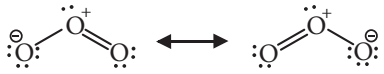
107. Resonance energy is the energy difference between actual energy of the molecule and energy of the most stable resonating form

108.

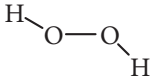


Bond order $= \frac{4}{3} = 1.33$

109. Ozone exists as



So the bond length is intermediate between O – O and O = O

H_2O_2 is  and O_2 is O = O

110. Due to polar nature of ICl, it has higher boiling point.

111. Statement 1 and statement 2 are correct. Statement 2 is the correct explanation for statement 1.

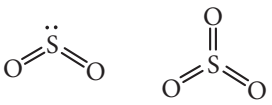
112. Statement 1 is wrong

Statement 2 is correct

113. Statement 1 is correct

Statement 2 is wrong. In fact the difference in electronegativity between N and F and between N and H are almost the same.

114.



115. Hybrid orbitals are of higher energy

electron density is concentrated on one side

hybrid orbitals are directed in preferred direction to minimize repulsion

116. **$HgCl_2$** is linear in shape

117. All are linear.

118. Large bond order, small bond length and resonance increase stability of a diatonic molecule.

119. p-nitrophenol and solid boric acid exhibit inter molecular hydrogen bonding.

Additional Practice Exercise

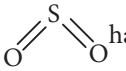
121. Bond angle of $PF_3 < PCl_3$
In PF_3, due to high electronegativity of fluorine the shared pair of electrons is pulled towards fluorine. This pushes the bond pair to minimize lone pair-bond pair repulsion thereby reducing the bond angle.

122. $N_2 \rightarrow N_2^+$
Bond order $3 \rightarrow 2.5$
$\therefore$ bond length increases
$O_2 \rightarrow O_2^+$
Bond order $2 \rightarrow 2.5$
$\therefore$ bond length decreases

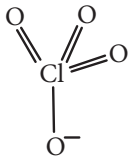
123. Ag^+ is a small cation and I^- is a large anion. Hence AgI is polarized to a large extent. The polarization leads to colour in AgI.

124. $\dfrac{\text{experimental dipole moment}}{\text{charge} \times \text{distance}} \times 100 = 43$

Experimental dipole moment
$= 1.92 \times 10^{-10}$ esu Å
Charge $= 4.8 \times 10^{-10}$ esu
distance $\dfrac{1.92 \times 100}{43 \times 4.8} = 0.93$ Å

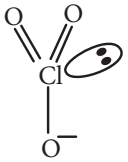
125. O = C = O is a linear molecule with centre of symmetry. Individual C – O bond moments get cancelled.
But  has bent structure and will have a net dipole moment.

126. (i) ClO_4^-



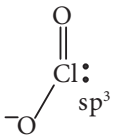
chlorine atom is sp^3 hybridized
B.O $= \dfrac{7}{4} = 1.75$

(ii) ClO_3^-



Due to resonance all Cl-O bonds are of equal length. B.O $= \dfrac{5}{3}$

(iii) ClO_2^-



Due to resonance the two Cl-O bonds are of equal length. B.O $= \dfrac{3}{2} = 1.5$

127. Due to $p\pi - d\pi$ back bonding

128. Atomic radii of inert gases are van der Waals radii where as for rest all atoms, atomic radii is determined experimentally are either covalent radius or metallic radius

129 HCl undergoes hydration to form H_3O^+ ion and $Cl^-_{(aq)}$. The large hydration energy is sufficient to ionise covalent HCl.

130. Due to high charge density of alkaline earth metal cations, the extent of hydration will be more.

131. Lattice energy $\propto \dfrac{q_1 q_2}{d^2}$

Charge in MgO is double that in NaF hence Lattice energy is almost 4 times.

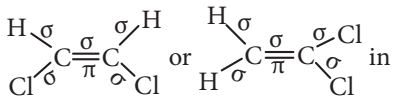
132. Fluorine forms F^- and bonds with Na^+ while oxygen forms O^{2-}, addition of an electron into a negative species is difficult hence energy has to be supplied. Since formation of O^{2-} is difficult, NaF is formed more readily than Na_2O.

133. For the reaction $A_{(s)} + \dfrac{1}{2}B_{2(g)} \rightarrow AB_{(s)}$ the enthalpy of formation is given as (a)

134. 
formal charge = Number of e^- in valence shell – $\dfrac{1}{2}$ (Number of bonding e^-) – lone electrons

135. Due to small size F^- is not polarizable like the other halides.
So AgF formed is ionic and therefore water soluble.
So a precipitate of AgF is not formed

136. CN^-, C ≡ N has 2π bonds
There are 4 such moieties. So total of 8π bonds.

137. The structure is  or in either case there are 5σ and 1π bond

138. Only π bonds are perpendicular to nuclear axis (ii) and (iv) are π bonds

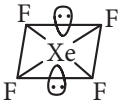
139. B – Cl bond in BCl_3 has partial double bond character due to pπ – pπ back bonding

On bonding with ether, this back bonding is lost and B – Cl is a pure single bond. Therefore bond length increases

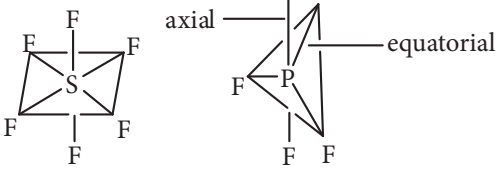
140. $AlCl_3$ is a strong lewis acid

$$Al_2Cl_6.xH_2O \xrightarrow{190^\circ C} Al_2Cl_6 + \text{water}$$

hydrated Al_2Cl_6 is ionic $[AlCl_2(H_2O)_4]^+ [AlCl_4(H_2O)_2]^-$ so are placed closer due to electrostatic attraction while Al_2Cl_6 is neutral so the interionc distance is greater hence density decreases. Hydrated $AlCl_3$ is less soluble in ether because it is ionic. Al is sp^2 hybridized in $AlCl_3$

141. XeF_4 is sp^3d^2 hybridized  square planar structure

142. These orbitals represent sp hybridization
so it is H – C ≡ C – H

143. 

PF_5 is sp^3d hybridized, axial bonds are longer than the equatorial bonds hence they are more reactive.

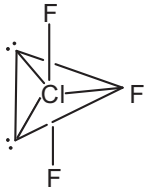
SF_6 is sp^3d^2 hybridized. All bonds are equivalent hence more stable

144. IF_4^+ $X = 4 + \frac{1}{2}\left[7 - 4 - 1\right] = 5$

AB_4L type hence sp^3d hybridized which is seen in SF_4

145. NH_2^- $\quad X = 2 + \frac{1}{2}\left[5 - 2 + 1\right] = 4$

sp^3 hybridized AB_2L_2 type molecule like water

146. 

Axial positions are at 90° where as equitorial positions are at 120°. Greater the angle lesser is the repulsion and greater is the stability.

147. In (b), (c), (d) placement of lone pair is not an issue, all points are equivalent

sp^3d^2, two lone pairs are placed opposite one another as it is more stable

148. Due to hybridization bonding orbitals are equivalent and these hybrid orbitals overlap with orbitals of other atoms to form molecules.

149. π bond does not involve hybrid orbitals, it can form from pπ –dπ overlap and can be easily polarized

150. There is a node in between the nuclei.

151. F_2^{2-} has bond order = 0

152. By Hund's Rule, electron pairing cannot take place until all the orbitals of same energy are singly filled. So O_2 configuration is $\pi^*2p_y^1\ \pi^*2p_x^1$ giving rise to two unpaired electrons

153. (a) is based on MOT

(b), (c) are based on VBT

only (d) is seen in both MOT and VBT

154. On heating volume decreases upto 4°C and then increases due to hydrogen bonding.

155. Oxygen in ice is tetrahedrally surrounded by 4 hydrogen atoms 2 covalent and 2 hydrogen bonded. Since covalent bonds are shorter, it attains elongation leading to open cage like structure.

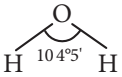
156. Colours are due to presence of chromophores.

157. As Cl has a larger size than nitrogen, inspite of high electronegativity it does not show hydrogen bonding as the electrostatic forces reduce with increase in distance.

158. HI $\quad I^-$ is the largest anion.
larger the anion, greater the covalent character.

159. For two ions of same charge and size, the one which has one or more d electrons which shield the nucleus poorly has a better polarizing power.

160. This is due to high lattice energy of $BaSO_4$

161. 

Dipole moment $\mu_R = \sqrt{\mu_1^2 + \mu_2^2 + 2\mu_1\mu_2\cos\theta}$

$\mu_1 = \mu_2 = \mu_{O-H}$

$$\mu_R = \sqrt{2\mu_1^2 + 2\mu_1^2\cos\theta}$$

$$= \sqrt{2}\,\mu_1\sqrt{1+\cos\theta}$$

$$\left[\left(1+\cos\theta\right) = 2\cos^2\theta/2\right]$$

$$= \sqrt{2}\,\mu_1 \times \sqrt{2}\cos\theta/2 = 2\mu_1\cos\theta/2$$

$$1.85 = 2\mu_1\cos\frac{104.5^\circ}{2}$$

$$\mu_{O-H} = \frac{1.85}{2\times\cos 52.25} = 1.5D$$

162. $\mu = e \times d$

$$\text{partial charge} = \frac{\mu}{d} = \frac{2.4\times10^{-18}\text{ esu cm}}{1.2\times10^{-8}\text{ cm}}$$

$$= 2\times10^{-10}\text{ esu}$$

$$\text{fraction of charge} = \frac{2\times10^{-10}}{4.8\times10^{-10}}$$

$$= 0.416 \cong 0.42$$

% fraction = 42%

163. As the size of group attached to 'O' increases bond angle increases

As θ increases, cosθ decreases in 0 to π range

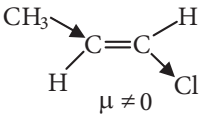
∴ dipole moment decreases

$\mu_{H_2O} = 1.85D$

$\mu_{CH_3OCH_3} = 1.30D$

$\mu_{C_2H_5OC_2H_5} = 1.15D$

164. Cyclopropane symmetric structure $\mu = 0$

(a) 

(c) SF_4 see saw structure $\mu = 0$

(d) XeF_6 distorted octahedron $\mu \neq 0$

165. We can see that $E \propto \frac{1}{r^6}$

so lesser the distance greater the energy

166. London forces are instantaneous

dipole – Induced dipole interactions

167. Inspite of the mutual repulsion, the electrons may tend to stay in one atom though this probability is less than one electron being associated with a given nucleus at a given time. So H_2 can exist as

$$\underset{\text{covalent}}{H-H} \leftrightarrow \underbrace{H^+ - H^- \leftrightarrow H^- - H^+}_{\text{ionic}}$$

168. It has higher energy due to loss of stabilization by π bonding

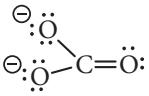
169. Most important 

Others are unfavourable because

(b) there is a change in position

(c) positive charge on electronegative atom

(d) lesser number of covalent bonds

170. 

Formal charge on carbon $= 4 - 0 - \frac{1}{2}(8) = 0$

on double bonded oxygen $= 6 - 4 - \frac{1}{2}(4) = 0$

on single bonded oxygen $= 6 - 6 - \frac{1}{2}(2) = -1$

[formal charge = Number of valence electrons – Number of lone electrons – $\frac{1}{2}$ (Number of bonded electrons)

171. Statement 1 is wrong and statement 2 is correct

172. Statement 1 is correct

Statement 2 is correct and is the explanation for Statement 1

173. Statement 1 is correct

Statement 2 is wrong

174. Statement 1 is correct

Statement 2 is correct and is the correct explanation of Statement 1

175. In aqueous medium, H_3O^+ and F^- are the species that exist because HF is more acidic than H_2O.

176. Statement 1 and 2 are correct. Statement 2 is the correct explanation for statement 1.

177. Due to lesser electronegative nature of carbon Na^+ can polarize carbon more effectively than oxygen.

178. Statement 1 is wrong

Statement 2 is correct but not the explanation of Statement 1

179. Structure II is more stable and it contributes more towards the resonance structure.

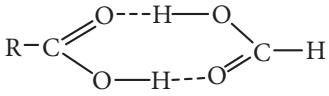
180. Statement 1 is correct but statement 2 is wrong $F_2B-C\equiv C-BF_2$ molecule is planar.
In $F_2C=C=CF_2$, the two $-CF_2$ groups are planar but in mutually perpendicular planes.

181. Li_2^{2+} Bond order $= \dfrac{2-2}{2} = 0$

182. Due to small size of 1s orbital, overlap is efficient

183. By MOT Bond order of $Li_2 = 1$
$Be_2 = 0$

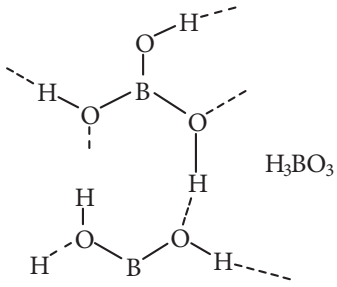
184. Acetic acid exists as



carboxylic acid hydrogen bond is limited to application of two molecules

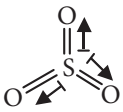
185. The A....H bond is shorter than expected because hydrogen penetrates the electron cloud of atom A
Example, in water O...H both is expected to be 260 pm but observed length is 170 pm.

186.



H_3BO_3

187.



Bond moments cancel each other $\mu = 0$



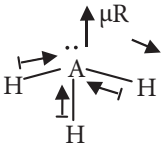
188. $\mu_{cal} = e^- \times d$
$= 1.6 \times 10^{-19} \times 150 \times 10^{-12} = 2.4 \times 10^{-29}$ cm

$$\text{percentage ionic character} = \frac{\mu_{obs}}{\mu_{cal}} \times 10$$

$$= \frac{5 \times 3.3 \times 10^{-29}}{2.4 \times 10^{-29}} \times 100$$

$= 69.47\% \cong 69.5\%$

189.



As electronegativity of A decreases, dipole moment decreases. On moving down the group electro negativity decreases hence dipole moment decreases

190. a, b are consequences of O being more electronegative. Due to larger size of S, S – H bond is weaker. Due to the weaker bond, H^+ is given out easily hence acidic.

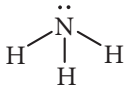
191. Due to unsymmetric nature of PCl_5, it is unstable and splits into $[PCl_4]^+$ and $[PCl_6]^-$
$[PCl_4]^+$ – tetrahedral
$[PCl_6]^-$ – octahedral

192. The d orbitals involved in sp^3d^2 hybridization is $d_{x^2-y^2}$ and d_{z^2}

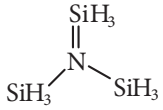
193. SF_6 is octahedral, no lone pairs hence bond angle is 90°
BrF_5 and ClF_5 have one lone pair and angle is reduced
BrF_5 – 84°30'
In XeF_4 there are two lone pairs. Ultimately structure is square planar, hence bond angle is 90°.

194. NH_4^+, BH_4^- are tetrahedral
N_3^- and NO_2^+ are linear
O_3 and ONCl are angular

195.



sp^3 hybridized-pyramidal



sp^2 hybridization-trigonal planar

196. Bond order

$$N_2 = \frac{10-4}{2}, N_2^+ = \frac{9-4}{2}, N_2^- = \frac{10-5}{2}$$

$$O_2 = \frac{10-6}{2}, O_2^+ = \frac{10-5}{2}, O_2^- = \frac{10-7}{2}$$

197. Since formation of $H^+.H^-$ is very difficult it has high energy considerations.

198. (p) HCN has covalent and hydrogen bonding
(q) H_3PO_4 has both covalent and hydrogen bonding
(r) $K_4[Fe(CN)_6]$ covalent, ionic and coordinate
(s) Fluorobenzene only covalent
(a) → (p), (q), (r), (s)
(b) → (r)
(c) → (r)
(d) → (p), (q)

199. (a) SF_4 – sp^3d, AB_4L type – see saw structure
(b) NH_2^- – sp^3, AB_2L_2 type – angular
(c) SO_2 – sp^2, AB_2L with two double bonds angular
(d) ClF_3 – sp^3d, AB_3L_2 T shaped angle is about 87° hence not planar
(a) → (r)
(b) → (p), (q)
(c) → (q), (r), (s)
(d) → (p)

200. (p) CO – $\sigma 2s^2\ \sigma^*2s^2\ \pi 2p_x^2\ \pi 2p_y^2\ \sigma 2p_z^2$ mixing of orbitals occurs
e^- enters antibonding orbital, diamagnetic
(q) C_2^- $\sigma 1s^2\ \sigma^*1s^2\ \sigma 2s^2\ \sigma^*2s^2\ \pi 2p_x^2\ \pi 2p_y^2\ \sigma 2p_z^1$
mixing of orbitals, e^- enters BMO, paramagnetic
Bond order = $\frac{9-4}{2} = 2.5$
(r) Li_2^+ $\sigma 1s^2\ \sigma^*1s^2\ \sigma 2s^1$
e^- enters BMO, paramagnetic,
bond order = $\frac{3-2}{2} = 0.5$
(s) F_2 – KK $\sigma 2s^2\ \sigma^*2s^2\ \sigma 2p_z^2\ \pi 2p_x^2\ \pi 2p_y^2\ \pi^*2p_x^2\ \pi^*2p_y^2$
e^- enter ABMO, paramagnetic
(a) → q, r
(b) → q, r
(c) → p, s
(d) → p, q, r, s

CHAPTER

5

CHEMICAL ENERGETICS AND THERMODYNAMICS

CHAPTER OUTLINE

Preview

STUDY MATERIAL

Basic Concepts in Thermodynamics

Mechanical Work (Pressure–Volume Work)
- Concept Strands (1-2)

Internal Energy (U)

First Law of Thermodynamics

Expression for Pressure – Volume Work

Enthalpy, H
- Concept Strands (3-7)

Adiabatic Expansion
- Concept Strands (8-14)

Work Done in an Isobaric Process
- Concept Strand 15

Thermochemical Equations

Various Forms of Enthalpy of Reaction
- Concept Strands (16-20)

Bond enthalpy (Bond energy)
- Concept Strands (21-24)

Laws of Thermochemistry
- Concept Strand (25)

Spontaneous Processes

Second law of Thermodynamics and Entropy (S)

Entropy Changes During Various Processes
- Concept Strands (26-35)

Third law of Thermodynamics

Gibbs Free Energy (G)

Criteria for Equilibrium and Spontaneity

Free Energy Change and Useful Work
- Concept Strands (36-45)

TOPIC GRIP
- Subjective Questions (10)
- Straight Objective Type Questions (5)
- Assertion–Reason Type Questions (5)
- Linked Comprehension Type Questions (6)
- Multiple Correct Objective Type Questions (3)
- Matrix-Match Type Question (1)

IIT ASSIGNMENT EXERCISE
- Straight Objective Type Questions (80)
- Assertion–Reason Type Questions (3)
- Linked Comprehension Type Questions (3)
- Multiple Correct Objective Type Questions (3)
- Matrix-Match Type Question (1)

ADDITIONAL PRACTICE EXERCISE
- Subjective Questions (10)
- Straight Objective Type Questions (40)
- Assertion–Reason Type Questions (10)
- Linked Comprehension Type Questions (9)
- Multiple Correct Objective Type Questions (8)
- Matrix-Match Type Questions (3)

BASIC CONCEPTS IN THERMODYNAMICS

Energy is the most precious commodity we have. Energy gives us the power to work. In every country, people's living standards are closely related to the availability of energy.

The transfer of energy to or from chemicals plays a crucial part in the chemical processes in industry and in living things.

Consequently the study of these energy changes is as important as the study of the changes in the materials themselves.

System and surroundings

Part of the universe selected for investigation, which has definite boundaries, is called the system.

The rest of the universe outside the system is called the surroundings.

Usually it is restricted to the immediate vicinity of the system. There are complex systems as a human body and simple as a mixture consisting of a drop of acid or a drop of base.

System + Surroundings = Universe

Types of systems

Systems may be classified into three types.

Open System

A system is said to be open if it can exchange heat and matter with surroundings. For e.g., hot coffee in an open flask.

Closed System

A system is said to be closed if it can exchange heat but not matter with the surroundings. For e.g., hot coffee in a stainless steel air tight flask.

Isolated system

A system is said to be isolated if it can exchange neither heat nor matter with surroundings. No example is available within the universe, but the entire universe can be taken as an isolated system as there is no surroundings.

Macroscopic properties of the system

A system is said to be macroscopic when it consists of a large number of molecules, atoms or ions. Properties associated with a macroscopic system are called macroscopic properties.

E.g., pressure, volume, temperature, mass, composition, surface area etc.

Homogeneous and heterogeneous systems

A system is homogeneous when its physical properties and chemical composition are uniform throughout the system. E.g., a system with one phase–pure gas, pure liquid or pure solid.

A system is heterogeneous when it is not uniform throughout i.e., it consists of parts, each of which has different physical and chemical properties. E.g., a system with two or more phases such as water and its vapour in equilibrium.

Note:

Gaseous mixture is always one phase system. Every pure solid is a separate phase. Completely miscible liquids form one phase, whereas immiscible liquids form separate phases.

Illustration

$2NH_{3(g)} \rightleftharpoons 3H_{2(g)} + N_{2(g)}$ One phase system

$CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$ Three phase system (Two solid phases and one gaseous phase)

State functions

Any property of a system that depends only on the present state of the system and that does not depend on how the system gets to the state exhibiting that property can be described as a state function.

The change in a state function depends only on the initial and final values of that state function in the initial and final states of the system and not on the manner or path adopted to bring about that change.

State variables

The state of the system changes when one or more of macroscopic properties change. Hence these properties are also called state variables. E.g., for 1 mole of an ideal gas PV = RT (R is gas constant and P, V and T are state variables). If P and T are known then V can be calculated. The two variables generally specified are temperature and pressure and are called independent variables. The third variable, volume is called dependent variable.

Intensive and extensive properties

The properties of a system can be classified into two types.

Intensive properties

The property of the system whose value is independent of the amount of the substance present in the system is called intensive property. E.g., temperature, pressure, concentration (molarity, normality) density, viscosity, pH, refractive index, dipole moment, gas constant, surface tension, specific heat capacity, molar volume, e.m.f of a galvanic cell, vapour pressure, specific gravity, boiling point, freezing point etc.

Extensive properties

The property of the system whose value depends upon the amount of the substance present in the system is called extensive property. E.g., volume, energy, mass, heat capacity, enthalpy, entropy, free energy etc.

Types of processes

The operation by which a system changes from one state to another is called a process. A process is accompanied by change in energy (and matter also if system is open). Some common types of processes are

Isothermal process

If a process is taking place at constant temperature it is said to be an isothermal [iso – same, therm – heat (temperature)] process.

Adiabatic process

If no heat enters or leaves the system during the process, it is said to be an adiabatic.

Isobaric process

A process in which the pressure of the system remains constant. (If a process is taking place at constant pressure, it is said to be an isobaric process).

Isochoric process

If a process is taking place at constant volume, it is said to be an isochoric process.

Reversible process

A process in which the change is carried out so slowly that the system and surroundings are always in equilibrium with each other is said to be thermodynamically reversible.

A characteristic of such a process is that a state function like temperature or pressure of the system never differ from those of the surroundings by more than an infinitesimal amount.

Thermodynamically reversible processes are idealized processes and can never be realized in practice. They are thus conceptual in nature.

Irreversible process

Irreversible processes are those that proceed at finite rate and the condition of equilibrium between the system and the surroundings is no longer applicable in such cases. Thus they are practical in nature. For e.g., the reaction between nitrogen and hydrogen to give ammonia.

Cyclic process

A process in which the system returns to its original state after undergoing a number of different process.

When the system returns to the original state all the thermodynamic properties assume their initial values and hence the difference between the initial and final values (Δ) of all these properties become zero.

Thermodynamic equilibrium

A system in which the macroscopic properties such as temperature and pressure do not change with time is said to be in thermodynamic equilibrium. The condition of thermodynamic equilibrium in a system is very important from the standpoint of the reversible processes.

Sign conventions of work and heat

Heat absorbed by the system is regarded as a positive quantity whereas heat released by the system is regarded as a negative

quantity. In the case of work, the convention is to take the *work done on the system as a positive quantity whereas the work done by the system is taken as a negative quantity* i.e., in a compression process, work done is positive and in an expansion process it is negative.

The heat and work are not state functions because their magnitude for a process depends upon the path by which the change is accomplished. However, when work is done adiabatically at constant pressure it becomes a state function.

MECHANICAL WORK (PRESSURE–VOLUME WORK)

This is the common type of work met with in the expansion or compression of gases. It is the work done when the gas expands or contracts against the external pressure, usually atmospheric pressure.

Mechanical work is related to heat produced (H) as,

$$W = JH$$

where, J is mechanical equivalent of heat (J = 4.184 joule)

Mechanical work is defined as $w = -\int P_{ext}\, dV$

When, P_{ext} is constant $w = -P_{ext} \int dV$

The work done in a finite stage is $W = -P_{ext} \int_{V_1}^{V_2} dV$

$$= -P_{ext} (V_2 - V_1).$$

$$W = P_{ext} \Delta V$$

Note:
To convert one energy unit into another, use different values of R (gas constant)

CONCEPT STRANDS

Concept Strand 1

A gas expands from 10 L to 20 L against an external pressure of 2.0 atm. Calculate the work done in kJ.

Solution

$$W_{exp} = -P_{ext} (\Delta V) = -2 (20 - 10) \text{ L atm}$$

$$= -20 \text{ L atm}$$

$$20 \text{ L atm} = \frac{8.314}{0.0821} \times 20 = 2.02 \text{ kJ} \quad (\because \text{gas constant},$$

$$R = 0.0821 \text{ L atm mol}^{-1} \text{ K}^{-1}$$

$$= 8.314 \text{ J mol}^{-1} \text{ K}^{-1})$$

$$W_{exp} = -2.02 \text{ kJ}$$

The negative sign indicates that the expanding system loses energy and does work on the surroundings.

Concept Strand 2

654 g of zinc reacts with dilute hydrochloric acid in an open vessel at 37°C liberating hydrogen. (i) Calculate the work done by the hydrogen gas in (a) litre atmosphere (b) joules (c) calories (ii). What is the work done by the gas if the same reaction were carried in a sealed vessel? (Relative at.wt. of Zn = 65.4)

Solution

Reaction: $Zn + 2HCl \rightarrow ZnCl_2 + H_2$

(i) (a) 654 g = $\frac{654}{65.4}$ = 10 moles of Zn react to give 10 moles of H_2.

Volume occupied by 10 moles H_2 at S.T.P. = $10 \times 22.4 = 224$ L

Volume of H_2 at 37°C and 1 atm

$$= \frac{224 \times 310}{273} = 254.36 \text{ L}$$

Work done by the gas = $-P\Delta V$

$$= -254.36 \times 1 = -254.36 \text{ L atm.}$$

(b) In joules: 0.0821 L atm = 8.314 J

$$\text{Work done} = \frac{-254.36 \times 8.314}{0.0821} = -2.58 \times 10^4 \text{ J}$$

(c) In calories: Work done

$$= \frac{-2.58 \times 10^4}{4.184} = -6.16 \times 10^3 \text{ cal}.$$

(ii) No work is done by the gas against external pressure. i.e., W = 0, as there is no volume change.

INTERNAL ENERGY (U)

Every substance is associated with a definite amount of energy known as internal energy, which depends on its chemical nature as well as on its temperature, pressure and volume. *Internal energy is made up of kinetic and potential energies of the constituent particles.* The kinetic energy arises due to the motions of all the particles which comprise their translation, rotation and vibration energies. The potential energy arises due to different types of interaction between the particles. These are the interactions between the electrons and the nuclei in atoms and molecules, and also the interactions between molecules. Internal energy is represented by the symbol U. We cannot measure the exact internal energy of a system. We can only determine the change in internal energy (ΔU) that accompanies a change in the state of the system.

Illustration

$$H_{2(g)} \rightarrow 2H_{(g)}$$

$$U_{1(\text{unknown})} \; U_1 + 436 \text{ kJ mol}^{-1}$$

$$\Delta U = U_{\text{final}} - U_{\text{initial}} = (U_1 + 436) - U_1 = 436 \text{ kJ mol}^{-1}$$

The value of ΔU is positive when energy is transferred from the surroundings to the system and it is negative when energy is transferred from the system to the surroundings.

Thermodynamics is essentially based on three main generalizations known as the first, second and third laws of thermodynamics which came into existence based on the observations.

FIRST LAW OF THERMODYNAMICS

It is same as the law of conservation of energy

(i) Energy can neither be created nor destroyed, but it can be converted from one form to an equivalent amount of another form.
(ii) The total energy of the universe is constant.
(iii) The mass and energy of an isolated system remain constant.
(iv) Heat lost from the system = Heat gained by the surroundings or vice versa.

Mathematical statement of 1st law

Let us consider a system of internal energy U_1. If 'Q' amount of heat is supplied to the system then its internal energy increases to $U_1 + Q$. If W is the work done by the system, the internal energy in the final state of the system U_2 is given by

$$U_2 = U_1 + Q - (-W) \Rightarrow U_2 - U_1 = Q + W$$

$$\Delta U = Q + W$$

This relationship is the mathematical statement of the first law.

While using the above equation we have to follow the sign convention given below.

If heat is absorbed by the system, Q is positive.

If heat is lost from the system, Q is negative.

If work is done on the system, W is positive

If work is done by the system, W is negative

Mechanical work is specially important in systems that contain gases.

EXPRESSION FOR PRESSURE – VOLUME WORK

(i) Work of expansion against constant pressure P_{ext} under isothermal conditions

$$W = -P_{ext}.\Delta V$$

where, $\Delta V \rightarrow$ change in volume, $(V_2 - V_1)$

(ii) Work of reversible expansion under isothermal conditions

$$W = -2.303\ nRT \log \frac{V_2}{V_1} \text{ (or)}$$

$$W = -2.303\ nRT \log \frac{P_1}{P_2}$$

Work is maximum if it is done under reversible conditions

Derivation of the expression for work done during isothermal reversible expansion

When a gas is expanding reversibly, the gas pressure differs from external pressure, P_{ext}, only by an infinitesimal amount.

$$\therefore \quad w = -\int P_{ext}\, dV = -\int P_{gas}\, dV$$

For a finite change of state involving a volume change in the initial state V_1, to a final state V_2 of n moles of an ideal gas, we can write

$$W = -\int_{V_1}^{V_2} P_{gas}\, dV = -\int_{V_1}^{V_2} \frac{nRT}{V} dV$$

$$= -nRT \int_{V_1}^{V_2} \frac{dV}{V} = -nRT \ln \frac{V_2}{V_1}$$

We can also write,

$$W = -nRT \ln \frac{P_1}{P_2}$$

where, P_1 and P_2 are the initial and final pressures of the gas, since $V \propto \frac{1}{P}$ for an isothermal process.

ENTHALPY, H

Most reactions in the laboratory are carried out in open vessels under conditions of constant pressure, which is often the atmospheric pressure (and also is the pressure of the system).

Heat of reaction at constant pressure

At constant pressure, the volume of the system changes. The thermal changes at constant pressure not only involves the change in internal energy but also the work done either in expansion or in contraction of the system.

$\therefore$ from the 1st law, $Q_p = \Delta U - W$

$$Q_P = \Sigma U_P - \Sigma U_R + P(V_P - V_R)$$

$$= \Sigma U_P + PV_P - (\Sigma U_R + PV_R)$$

U + PV is taken as a new state function 'H', known as the enthalpy of the system.

$$\therefore \quad Q_P = \Sigma H_P - \Sigma H_R = \Delta H$$

where, ΣH_P is the total enthalpy of the products and ΣH_R that of the reactants.

$$\Delta H = Q_P$$

If $\Sigma H_P < \Sigma H_R$, the reaction is exothermic, i.e., ΔH = –ve

If $\Sigma H_P > \Sigma H_R$, the reaction is endothermic, i.e., ΔH = +ve

Units of work are ergs (in C.G.S.) and joules (in S.I.).

Heat of reaction at constant volume.

A process at constant volume means that it is carried out in a closed container and no work is involved in the process.

$\therefore \quad \Delta U = Q + W$ becomes $\Delta U = Q_V$

$\therefore \quad \Delta U = Q_V$

In other words heat absorbed or released by a system at constant volume is equal to the change in internal energy of the system.

Zeroth law of thermodynamics

This law, though discovered after first law, is more fundamental and therefore, named as zeroth law.

According to this law, two systems A and B are separately in thermal equilibrium with a third system C, then A and B will also be in thermal equilibrium with each other.

CONCEPT STRANDS

Concept Strand 3

Calculate the work done in joules when five moles of an ideal gas expand isothermally and reversibly at 27°C until its volume becomes ten fold of the initial. What will be the work done if the same expansion were carried out irreversibly against a constant external pressure of 2 atm assuming that the initial volume of the gas is 5 L.

Solution

During reversible expansion,

$$W = -nRT \ln \frac{V_2}{V_1}$$

$$= -5 \times 8.314 \times 300 \times 2.303 \times \log 10$$

$$\Rightarrow \quad W = -2.872 \times 10^4 \text{ J}.$$

During irreversible expansion,

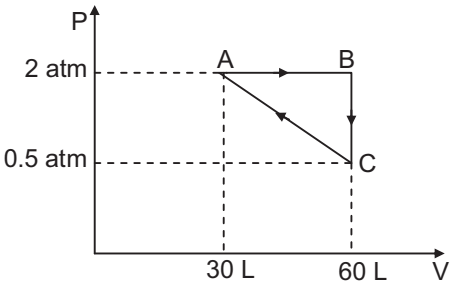
$$W = -P_{ext} \times \Delta V = -2 \text{ atm} \times (50 - 5) \text{ L} = -90 \text{ L atm}.$$

$$\Rightarrow \quad W = \frac{-90 \times 8.314}{0.0821} = -9.114 \times 10^3 \text{ J}.$$

By a system thus work done in a reversible process is greater than that in an irreversible process.

Concept Strand 4

Calculate the total work done in Joule for a system for the isothermal process given below:



Solution

$\therefore$ Cyclic process

$$\Delta U = 0, \Delta H = 0, \Delta S = 0, \Delta U = Q + W$$

$$0 = Q + W \Rightarrow Q = -W$$

Total work $= W_{A \to B} + W_{B \to C} + W_{C \to A}$

$$= -P(V_B - V_A) + 0 + 2.303nRT \log \frac{V_C}{V_A}$$

$$= -P(V_B - V_A) + 0 + 2.303 \times P_A \times V_A \log \frac{V_C}{V_A}$$

$$= -2(60 - 30) + 0 + 2.303 \times 2 \times 30 \times \log \frac{60}{30}$$

$$= -18.41 \text{ L atm}$$

$$W = -1864.32 \text{ Joule}$$

Relation between ΔH and ΔU

From $H = U + PV$

We can write $\Delta H = \Delta U + P\Delta V$ at constant pressure

The change in volume, ΔV occurs due to change in the number of moles of gases.

Hence $P\Delta V = \Delta n_g RT$

$\therefore \quad \Delta H = \Delta U + \Delta n_g RT.$

This equation is applicable in reactions involving only gases or those involving solids or liquids and gases as well. If

$\Delta n = 0, \quad \Delta H = \Delta U$

$\Delta n > 0, \quad \Delta H > \Delta U$

$\Delta n < 0, \quad \Delta H < \Delta U$

CONCEPT STRANDS

Concept Strand 5

Determine ΔU at 620K for the following reaction

$$2CO + O_2 \rightarrow 2CO_2\ \Delta H = -424\ kJ$$

The pressure in the 1 litre vessel changes from initial pressure of 85 atm to 20 atm. The gases deviate appreciably from ideal behaviour.(1 L atm = 0.1 kJ)

Solution

$$\Delta H = \Delta U + \Delta(PV)$$
$$= \Delta U + P_2V_2 - P_1V_1 \quad \text{— (i)}$$

We cannot use here $\Delta H = \Delta U + \Delta nRT$ since, gases deviate from ideal gas behaviour,

$$\text{Thus , } \Delta(PV) = P_2V_2 - P_1V_1 = 20 \times 1 - 85 \times 1$$
$$= -65\ L\ atm = -6.5\ kJ$$

$\therefore$ Substituting the Δ(PV) in equation (i)

$-424 = \Delta U - 6.5 \qquad \Delta U = -417.5\ kJ$

Concept Strand 6

Two moles of an ideal gas are compressed isothermally and reversibly from an initial pressure of 5 atmospheres to a final pressure of 40 atmospheres at 30°C. Calculate

(i) the work done in joules (ii) ΔU
(iii) ΔH (iv) Q

Solution

(i) $W = nRT\ln\frac{P_2}{P_1}$

$$= 2\times8.314\times303\times2.303\times\log\frac{40}{5} = 10478.7\,J$$

(ii) For an ideal gas at constant temperature, the energy is independent of volume.

$\therefore$ $\Delta U = 0$.

(iii) $\Delta H = \Delta U + \Delta(PV) = 0 + 0 = 0$

(iv) $Q = \Delta U - W = 0 - 10478.7 = -10478.7$ J. This is the thermal energy rejected to the environment.

Concept Strand 7

One mole of ammonia gas is condensed to the liquid at its boiling point – 33.4°C under reversible condition. Calculate

(i) work done, W
(ii) heat changes for the process, Q
(iii) ΔU
(iv) ΔH.

The latent heat of vapourisation of liquid ammonia is 1368.4 J g^{-1} at its boiling point.

Solution

(i) $W = -P_{ext}\,\Delta V = -P_{ext}(V_I - V_g) \approx P_{ext} \times V_v = nRT$

(V_l and V_g are the volumes of the liquid and vapour respectively)

$$W = 1 \times 8.314 \times 239.6 = 1992.0\ J$$

($\because$ at boiling point $P_{ext} = P_{gas} = 1$ atm)

(ii) $Q_p = \Delta H = -1368.4\ (J\ g^{-1}) \times 17\ (g\ mol^{-1})$
$= -23263.0\ J\ mol^{-1}$

(iii) $\Delta U = Q + W = \Delta H + W$
$= -23263.0 + 1992.0 = -21271$ J.

Heat Capacity

Heat capacity is the tendency to store heat for gases. There are two types of heat capacities, one measured at constant volume (C_v) and the other measured at constant pressure (C_p).

Heat capacity, $C = \left(\partial q/\partial T\right)$

Heat capacity at constant volume

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V$$

Heat capacity at constant pressure

$$C_P = \left(\frac{\partial H}{\partial T}\right)_P$$

For an ideal gas $C_P - C_V = R$, where, R is the gas constant. This can be illustrated as follows:

The difference between C_P and C_V is equal to the work done by 1 mole of a gas during expansion when heated through 1^0C.

Work done by the gas at constant pressure = $P\Delta V$

For 1 mole of gas $PV = RT$

When temperature is raised by 1^0C, the volume becomes $V + \Delta V$.

So $P(V + \Delta V) = R(T + 1)$

$\Rightarrow$ $P\Delta V = R$

Hence

$$C_P - C_V = P\Delta V = R$$

Ratio of heat capacities $(\gamma) = \frac{C_P}{C_V}$

For a monatomic gas, $\gamma = 1.66$

For a diatomic gas, $\gamma = 1.40$

For a triatomic gas, $\gamma = 1.33$

Kirchhoff's equation

Effect of temperature on heats of reaction is given by Kirchhoff's equation

$$\frac{\Delta H_2 - \Delta H_1}{T_2 - T_1} = \Delta C_P$$

and $$\frac{\Delta U_2 - \Delta U_1}{T_2 - T_1} = \Delta C_V$$

ADIABATIC EXPANSION

From first law, $\Delta U = Q + W$

In adiabatic expansion, no heat is allowed to enter or leave the system. Hence $Q = 0$. $\therefore \Delta U = W$

In expansion, work is done by the system on the surroundings. Hence W is negative.

Accordingly ΔU is also negative. i.e., internal energy decreases and therefore the temperature of the system falls.

In case of compression, ΔU is positive i.e., internal energy increases and therefore the temperature of the system rises. The molar heat capacity at constant volume of an ideal gas is given by

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V \quad \Rightarrow dU = C_V dT$$

and for a finite change,

$$\Delta U = C_V \Delta T \quad \text{(for 1 mole)}$$

$$\Delta U = nC_V \Delta T \text{ (for n moles)}$$

Similarly,

$$\Delta H = C_P \Delta T \quad \text{(for 1 mole)}$$

$$\Delta H = nC_P \Delta T \text{ (for n moles)}$$

The value of ΔT depends upon the process, whether it is reversible or irreversible.

CONCEPT STRANDS

Concept Strand 8

Five moles of helium at 25°C and 1 atm are heated at constant pressure until its volume is doubled. Calculate ΔU and ΔH for this process. Assume helium to be an ideal gas. ($C_V = 3.0$ cal deg^{-1} mol^{-1})

Solution

The final temperature T_2 of the gas may be calculated from the equation,

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

$$\frac{V_1}{298} = \frac{2V_1}{T_2} \Rightarrow T_2 = 596\,K$$

$$\Delta U = nC_V\Delta T = 5 \times 3 \times (596 - 298) \times 4.184$$
$$= 18702 \text{ J.}$$

$$\Delta H = nC_P\Delta T = 5 \times 5 \times (596 - 298) \times 4.184$$
$$= 31171 \text{ J.}$$

Concept Strand 9

195.0 g of solid benzene (melting point 278.6K) is converted into vapour (boiling point 353.2K) under a constant pressure of 1 atm. Calculate (i) ΔH and (ii) ΔU for the overall process. The heat capacity, C_P of benzene may be taken as 0.136 kJ K^{-1} $mole^{-1}$ in the given temperature range. The enthalpy of fusion at the melting point and the enthalpy of vapourisation of benzene at its boiling point are 10.6 kJ mol^{-1} and 30.8 kJ mol^{-1} respectively.

Solution

(i) The enthalpy change may be calculated by the three stages indicated below:

$$\text{No of moles of benzene} = \frac{195}{78} = 2.5$$

Enthalpy change for the conversion of 2.5 moles of solid benzene to liquid at 278.6K is $2.5 \times 10.6 = 26.5$ kJ.

Enthalpy change for the heating of 2.5 moles of benzene liquid from 278.6K to 353.2K is $nCp\ (T_2 - T_1) = 2.5 \times 0.136\ (353.2 - 278.6) = 25.36$ kJ.

Enthalpy change for the conversion of 2.5 moles of benzene liquid at 353.2 K into vapour is $2.5 \times 30.8 = 77.0$ kJ.

Total enthalpy change = 26.5 + 25.36 + 77.0 = 128.9 kJ.

(ii) The change in volume, ΔV for the first two steps is negligible.

$$\therefore \Delta H \approx \Delta U \text{ for these steps.}$$

For the third step, $\Delta H = \Delta U + P\Delta V$
$= \Delta U + P(V_g - V_l) \approx \Delta U + nRT$

Since

$V_g - V_l = V_g$ and $PV_g = nRT$

$$\Delta U = \Delta H - nRT = 77.0 - 2.5 \times 8.314 \times 353.2 \times 10^{-3}$$
$$= 77.0 - 7.34 = 69.66 \text{ kJ.}$$

$\therefore$ Total $\Delta U = 26.50 + 25.36 + 69.66 = 121.5$ kJ.

Concept Strand 10

36 g of supercooled water at −10°C is converted to ice at the same temperature under 1 atm pressure. Calculate ΔH for the process. Given, the latent heat of fusion of ice at 0°C is 6008 J mol^{-1}. The heat capacity of water is 75.3 J deg^{-1} mol^{-1} and that of ice is 37.6 J deg^{-1} mol^{-1}. (Assume that latent heat of fusion and the heat capacities of water and ice to be independent of temperature).

Solution

ΔH can be calculated by considering the following steps.

$$\text{Water } (-10°C) \xrightarrow{\Delta H_I} \text{water } (-0°C) \xrightarrow{\Delta H_{II}}$$
$$\text{Ice } (0°C) \xrightarrow{\Delta H_{III}} \text{Ice } (-10°C)$$

I Step:

ΔH_I in heating supercooled water (2 moles) at −10°C to water at 0°C.

$$\Delta H_I = 2 \times 75.3 \times (273 - 263) = 1506 \text{ J}$$

II Step:

ΔH_{II} required for freezing water (2 moles) at 0°C to ice at 0°C.

$$\Delta H_{II} = -2 \times 6008 = -12016.0 \text{ J.}$$

III Step:

ΔH_{III} required for cooling 2 moles of ice at 0°C to ice at −10°C.

$$\Delta H_{III} = 2 \times 37.6\ (263 - 273) = -752 \text{ J.}$$

Total $\Delta H = \Delta H_I + \Delta H_{II} + \Delta H_{III}$
$= 1506 - 12016 - 752 = -11262$ J

Heat liberated or absorbed

When there is no phase change,

$$\left.\begin{array}{c}\text{The amount of heat liberated} \\ \text{(or)} \\ \text{The amount of heat absorbed}\end{array}\right\} = ms(t_2 - t_1)$$

where, m is the mass, s is the specific heat, t_2 is the final temperature and t_1 is the initial temperature

Important relationships in reversible adiabatic expansion

(i) $PV^{\gamma} = \text{constant}$

(ii) $\left(\frac{T_1}{T_2}\right)^{\gamma} = \left(\frac{P_1}{P_2}\right)^{\gamma-1} = \left(\frac{P_2}{P_2}\right)^{1-\gamma}$

(iii) $TV^{\gamma-1} = \text{constant}$

(iv) $\frac{T_1}{T_2} = \left(\frac{V_2}{V_1}\right)^{\gamma-1} = \left(\frac{V_1}{V_2}\right)^{1-\gamma}$

CONCEPT STRANDS

Concept Strand 11

Three moles of nitrogen initially at 47°C expand adiabatically and reversibly from an initial volume of ten litres to a final volume of 200 L. Calculate

(i) final temperature of the gas
(ii) the temperature if five moles of the gas are taken for the same expansion
(iii) ΔU and ΔH for step (i)
(iv) the final pressure of the gas
(C_V for nitrogen = 20.9 J deg^{-1} mol^{-1}).

Solution

(i) $\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1}$

$T_1 = 320K$, $V_1 = 10$ L, $V_2 = 200$ L,

$\gamma = \frac{C_P}{C_V} = \frac{29.2}{20.9} J\ deg^{-1}\ mol^{-1}$

$\frac{T_2}{320} = \left(\frac{10}{200}\right)^{\frac{29.2}{20.9}-1}$

$T_2 = 320 \times (0.05)^{\frac{8.3}{20.9}}$

$\Rightarrow \log T_2 = \log 320 + 0.397 \log 0.05$
$\Rightarrow T_2 = 97.41K.$

(ii) Final temperature will be the same as in (i), as temperature is an intensive property.

(iii) $\Delta U = nC_V (T_2 - T_1) = 3 \times 20.9 (97.41 - 320)$
$= -13.96$ kJ.
$\Delta H = nC_P (T_2 - T_1) = 3 \times 29.2 (97.41 - 320) = -19.5$ kJ.

(iv) $P \times 200 = 3 \times 0.0821 \times 97.41;$
$P = 0.12$ atm

Concept Strand 12

Two moles of argon gas initially at one atmosphere pressure and occupying a volume of 50 L, are compressed adiabatically and reversibly until the pressure becomes ten atmospheres. Calculate

(i) initial and final temperatures of the gas
(ii) ΔU
(iii) W
(iv) ΔH.

(C_V of argon may be taken as12.55 J deg^{-1} mol^{-1})

Solution

(i) Initial temperature of the gas
Using $PV = nRT_1$; $1 \times 50 = 2 \times 0.0821 \times T_1$
$T_1 = 304.5K$

$\gamma = \frac{C_P}{C_V} = \frac{12.55 + 8.314}{12.55} = 1.66$

$\left(\frac{T_1}{T_2}\right)^{\gamma} = \left(\frac{P_1}{P_2}\right)^{\gamma-1} \Rightarrow \left(\frac{304.5}{T_2}\right)^{1.66} = \left(\frac{1}{10}\right)^{0.66}$

$\Rightarrow 1.66 \log \frac{304.5}{T_2} = -0.66 \Rightarrow \left(\frac{304.5}{T_2}\right)^{1.66} = \left(\frac{1}{10}\right)^{0.66}$

$\Rightarrow 1.66 \log \frac{304.5}{T_2} = -0.66$

$\Rightarrow \frac{304.5}{T_2} = 0.4 \Rightarrow T_2 = 761K$

(ii) $\Delta U = nC_V (T_2 - T_1) = 2 \times 12.55 (761 - 304.5) = 11.46$ kJ

(iii) For an adiabatic compression process, ΔU = W = 11.46 kJ

(iv) $\Delta H = nC_P (T_2 - T_1) = 2 \times 20.86 (761 - 304.5) = 19.05$ kJ

Concept Strand 13

A sample of argon at 1 atm pressure and 27°C expands reversibly and adiabatically from 3.25 dm^3 to 7.25 dm^3. Calculate the enthalpy change in this process. C_V for argon is 12.48 J K^{-1} mol^{-1}.

Solution

Number of moles of argon present in the sample

$n = \frac{PV}{RT} = \frac{3.25 \times 1}{0.0821 \times 300} = 0.1319$

For adiabatic expansion

$\frac{T_1}{T_2} = \left(\frac{V_2}{V_1}\right)^{\gamma-1}$

$\frac{300}{T_2} = \left(\frac{7.25}{3.25}\right)^{\gamma-1}$ [$\because \gamma = 1.66$ for a monoatomic gas]

$\frac{300}{T_2} = \left(\frac{7.25}{3.25}\right)^{1.66-1}$

$T_2 = 176.67K$
$C_P = C_V + R$
$= 12.48 + 8.314 = 20.794$ J K^{-1} mol^{-1}

$\Delta H = n \times C_p \times \Delta T$

$\Delta H = 0.1319 \times 20.794 \times (300 - 176.67)$

$\Delta H = 338.2$ J

Concept Strand 14

One mole of Helium at 27°C and 1 atmosphere is expanded adiabatically and reversibly until its pressure falls to half the original value. Calculate

(i) Final temperature of the gas
(ii) ΔU
(iii) ΔH
(iv) W

(C_V of Helium= 12.55 J deg^{-1} mol^{-1})

Solution

(i) Using the relation, $\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}}$,

$$\gamma = \frac{20.86}{12.55} = 1.66$$

$$\frac{T_2}{300} = \left(\frac{0.5}{1}\right)^{0.4} \Rightarrow T_2 = 300 \times (0.5)^{0.4} \Rightarrow T_2 = 227.3\text{K}.$$

(ii) $\Delta U = nC_V (T_2 - T_1) = 1 \times 12.55\ (227.3 - 300)$
$= -912.4$ J.

(iii) $\Delta H = nC_P (T_2 - T_1) = 1 \times 20.86\ (227.3 - 300)$
$= -1516.5$ J

(iv) ΔU = W for a reversible adiabatic expansion process.

$\therefore$ $W = \Delta U = -912.4$ J.

Reversible adiabatic expansion

Work done $= C_V \Delta T = \frac{R}{\gamma - 1}(T_2 - T_1)$ for 1 mole

For 'n' moles, work done $= \frac{nR}{\gamma - 1}(T_2 - T_1)$

Irreversible adiabatic expansion

In free expansion $\Delta T = 0$, $\Delta U = 0$, $W = 0$ and $\Delta H = 0$

In irreversible expansion, the volume changes from V_1 to V_2 against external pressure, $P_{ext.}$

$$W = -P_{ext}(V_2 - V_1) = -P_{ext}\left(\frac{RT_2}{P_2} - \frac{RT_1}{P_1}\right)$$

$$= -P_{ext}\left(\frac{T_2P_1 - T_1P_2}{P_1P_2}\right) \times R$$

$$W = -P_{ext}\left(\frac{T_2P_1 - T_1P_2}{P_1P_2}\right) \times R$$

WORK DONE IN AN ISOBARIC PROCESS

In an isobaric process the pressure is constant and volume changes. The work done by the system is

$$W = -\int_{V_1}^{V_2} PdV = -P\int_{V_1}^{V_2} dV = -P(V_2 - V_1) = -P\Delta V$$

CONCEPT STRAND

Concept Strand 15

(i) Derive an equation for the work done when a gas obeying van der Waals equation expands isothermally and reversibly from an initial volume V_1 to a final volume V_2 at temperature T kelvin.

(ii) Calculate the work done for the isothermal reversible expansion of one mole of carbon dioxide from an initial volume of 10 L to a final volume of 50 L at 300K. Compare it with the work done if CO_2 is an ideal gas. The van der Waals constants of the gas are, a = 3.59 L^2 atm mol^{-2} and b = 0.0427 L mol^{-1}.

Solution

(i) The van der Waals equation for one mole of a gas is

$$\left(P + \frac{a}{V^2}\right)(V - b) = RT$$

Rearranging the above equation, $P = \frac{RT}{V-b} - \frac{a}{V^2}$

The amount of work done in an infinitesimal process = $-PdV$

i.e., the total work, W is obtained by integrating the above equation between proper limits

$$= \int_{V_1}^{V_2} -PdV = \int_{V_1}^{V_2} -\left(\frac{RT}{V-b} - \frac{a}{V^2}\right)dV$$

$$W = -RT\int_{V_1}^{V_2}\frac{dv}{(V-b)} + a\int_{V_1}^{V_2}\frac{dV}{V^2}$$

$$= -RT\Big[\ln(V-b)\Big]_{V_1}^{V_2} - a\left[\frac{1}{V}\right]_{V_1}^{V_2}$$

$$W = RT\ln\frac{(V_1-b)}{(V_2-b)} - a\left(\frac{1}{V_2} - \frac{1}{V_1}\right)$$

(ii) $W = 0.0821 \times 300 \times 2.303\log\frac{10-0.0427}{50-0.0427} - 3.59\left(\frac{1}{50} - \frac{1}{10}\right)$

$= -39.73 + 0.287$

$= -39.443$ L atm

$= \frac{-39.443 \times 8.314}{0.0821} = -3.99$ kJ

If CO_2 is an ideal gas,

$$W = nRT\ln\frac{V_1}{V_2} = 1 \times 8.314 \times 300 \times 2.303\log\frac{10}{50}$$

$= 8.314 \times 300 \times 2.303 \times -0.70$

$= -4.015$ kJ

Work done in an isochoric process

In an isochoric process the volume remains constant and no work is done by the system.

THERMOCHEMICAL EQUATIONS

A thermochemical equation is a chemical equation, which indicates the physical states of the reactants and products and also the heat change accompanying the reaction.
E.g.,

$$C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)}; \Delta H = -393.0 \text{ kJ}$$

$$2H_{2(g)} + O_{2(g)} \rightarrow 2H_2O_{(\ell)}; \Delta H = -572 \text{ kJ}$$

Standard state of a substance

It is thermodynamically the most stable state of a substance at 1 atm and at a specified temperature (usually 25°C). For example, the standard state of carbon is graphite, water is liquid and CO_2 is gas.

Standard state of carbon

$$C_{(\text{graphite})} + O_{2(g)} \rightarrow CO_{2(g)} \quad \Delta_f H° = -393.5 \text{ kJ}$$

$$C_{(\text{diamond})} + O_{2(g)} \rightarrow CO_{2(g)} \quad \Delta_f H° = -395.41 \text{ kJ}$$

Standard state is the thermodynamically more stable state, which is graphite for carbon.

VARIOUS FORMS OF ENTHALPY OF REACTION

Enthalpy of formation

It is the enthalpy change when 1 mole of a compound is obtained from its constituent elements. It is denoted by $\Delta_f H$.

E.g.,

$$C_{(s)} + 2H_{2(g)} \rightarrow CH_{4(g)}\ \Delta_f H = -74.8 \text{ kJ}$$

Enthalpy of formation expressed under standard conditions is called standard enthalpy of formation $\left(\Delta_f H^\circ\right)$. E.g.,

$$C_{(graphite)} + 2H_{2(g)} \rightarrow CH_{4(g)}$$

The enthalpy of free elements in their standard states is taken to be zero.

The value of $\Delta_f H°$ of the compound is related to its chemical stability.

Enthalpy or heat of formation of a compound is sometimes referred to as the enthalpy of the compound.

Enthalpy of combustion

It is the enthalpy change when one mole of a compound undergoes complete combustion in excess of oxygen.
E.g.,

$$CH_{4(g)} + 2O_{2(g)} \rightarrow CO_{2(g)} + 2H_2O_{(l)}; \Delta H = -890 \text{ kJ}$$

Enthalpy of combustion is negative since combustion reactions are exothermic.

Enthalpy of solution

Integral enthalpy of solution is the enthalpy change when one mole of a substance is dissolved in n_1 moles of solvent so that on further dilution no appreciable heat change occurs.

Enthalpy of dilution

It is the enthalpy change when a solution containing one mole of a solute in n_1 moles of a solvent is further diluted by adding, say, n_2 moles of solvent.

Enthalpy of hydration

It is the enthalpy change when one mole of a solute is hydrated.
E.g.,

$$CuSO_{4(s)} + 5H_2O_{(\ell)} \rightarrow CuSO_4.5H_2O_{(s)}; \Delta H_{Hydration} = -78.21 \text{ kJ}$$

Hydration is an exothermic process because it involves bond formation between water molecules and central metal ion.

Enthalpy of fusion

It is the enthalpy required to change one mole of the solid completely into liquid at its melting point.
E.g.,

$$H_2O_{(s)} \rightarrow H_2O_{(\ell)}; \Delta H = 6.0 \text{ kJ}$$

Enthalpy of vaporization

It is the enthalpy change when one mole of the liquid substance is completely converted to vapour at its boiling point.
E.g.,

$$H_2O_{(\ell)} \rightarrow H_2O_{(g)}; \Delta H = 40.6 \text{ kJ}$$

Enthalpy of neutralization

It is the enthalpy change when 1 gram equivalent of an acid is neutralized by 1 gram equivalent of a base in dilute solutions.
The enthalpy of neutralisation of a strong acid with a strong base in a dilute solution is always a constant and is –57.1 kJ mol^{-1} (–13.6 kcal mol^{-1}). E.g.,

$$HCl_{(aq)} + NaOH_{(aq)} \rightarrow NaCl_{(aq)} + H_2O_{(\ell)};$$
$$\Delta H = -57.1 \text{ kJ mol}^{-1}$$

In case of strong acids and bases the net reaction is the formation of one mole of water from hydrogen ions (H^+) and hydroxide ions (OH^-).

$$H^+_{(aq)} + OH^-_{(aq)} \rightarrow H_2O_{(\ell)} \Delta H = -57.1 \text{ kJ mol}^{-1}$$

Hence the neutralization enthalpy is the change in enthalpy when one mole of H_2O is formed. Neutralization enthalpies are quoted for infinitely dilute solution where there is no interaction between ions.

In case of neutralization of a weak acid with a strong base or vice versa, the enthalpy of neutralization will be different from –57.1 kJ mol^{-1} as some amount of heat is utilized for dissociation of the weak acid or base molecules.

Heat of neutralization of a strong acid with a strong base + heat of ionization of a weak acid (or weak base) = heat of neutralization of a weak acid with a strong base (or strong acid with a weak base).

Enthalpy of hydrogenation

It is the enthalpy change when 1 mole of an unsaturated organic compound is fully hydrogenated.

Enthalpy of transition

It is the enthalpy change when 1 mole of the substance undergoes transition from one crystalline form to another.
E.g.,

$$S_{(r)} \rightarrow S_{(m)} \Delta H = +2.5 \text{ kJ};$$

where, $S_{(r)}$ = rhombic sulphur and $S_{(m)}$ = monoclinic sulphur

CONCEPT STRANDS

Concept Strand 16

Calculate the standard enthalpy of formation of carbon tetrachloride vapour using the following data.

(i) $CCl_{4(g)} + 2H_2O_{(g)} \rightarrow CO_{2(g)} + 4HCl_{(g)}$; $\Delta H = -173.2$ kJ
(ii) $H_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow H_2O_{(l)}$; $\Delta H^\circ = -286.0$ kJ.
(iii) $C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)}$; $\Delta H^\circ = -394.2$ kJ.
(iv) $\frac{1}{2}H_{2(g)} + \frac{1}{2}Cl_{2(g)} + aq \rightarrow HCl_{(aq)}$; $\Delta H^\circ = -165.5$ kJ
(v) $H_2O_{(\ell)} \rightarrow H_2O_{(g)}$ $\Delta H^\circ = 40.7$ kJ
(vi) $HCl_{(aq)} \rightarrow HCl_{(g)} + aq$; $\Delta H^\circ = 73.2$ kJ

Solution

Required equation is

$$C_{(s)} + 2Cl_{2(g)} \rightarrow CCl_{4(g)}$$

By adding equations (ii) and (v), we get

$$H_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow H_2O_{(g)};\ \Delta H^\circ = -245.3 \text{ kJ}$$

By adding equations (iv) and (vi), we get

$$\frac{1}{2}H_{2(g)} + \frac{1}{2}Cl_{2(g)} \rightarrow HCl_{(g)};\ \Delta H^\circ = -92.3 \text{ kJ}$$

ΔH° of reaction (i) $= -173.2 = \Delta_f H^\circ (CO_2) + 4\Delta_f H^\circ (HCl_{(g)}) - \Delta_f H^\circ (CCl_{4(g)}) - 2\Delta_f H^\circ (H_2O_{(g)})$

or $\Delta_f H^\circ (CCl_{4(g)}) = -394.2 + 4 \times -92.3 - (-173.2) - 2 \times (-245.3)$
$= -394.2 - 369.2 + 173.2 + 490.6$
$= -99.6$ kJ.

Concept Strand 17

The standard enthalpy of combustion of aniline is –3390 kJ at 298K. Calculate its standard enthalpy of formation. Given, the standard enthalpies of formation of $CO_{2(g)}$, $H_2O_{(\ell)}$ and $NO_{2(g)}$ are –394.2 kJ, –286.0 kJ and 33.2 kJ respectively. Give the reaction for which the result has been calculated.

Solution

The stoichiometric equation for the combustion of aniline is given by

$$4C_6H_5NH_{2(\ell)} + 35O_{2(g)} \rightarrow 24CO_{2(g)} + 14H_2O_{(\ell)} + 4NO_{2(g)};\ \Delta H^\circ = -13560 \text{ kJ.}$$

$\Rightarrow -13560 = 24\,\Delta_f H^\circ (CO_2) + 14\Delta_f H^\circ (H_2O_{(\ell)}) + 4\Delta_f H^\circ (NO_2) - 4\Delta_f H^\circ (C_6H_5NH_{2(\ell)})$

$\Rightarrow 4\Delta_f H^\circ (C_6H_5NH_{2(\ell)}) = 24 \times -394.2 + 14 \times -286.0 + 4 \times 33.2 + 13560.$
$= -9460.8 - 4004.0 + 132.8 + 13560$

$\Rightarrow \Delta_f H^\circ = \frac{228}{4} = 57$ kJ

Reaction is

$$6C_{(\text{graphite})} + \frac{7}{2}H_{2(g)} + \frac{1}{2}N_{2(g)} \rightarrow C_6H_5NH_{2(\ell)}$$

Concept Strand 18

Calculate the calorific value of C_3H_6 in kJ g^{-1} if

$\Delta H_f\, C_3H_6 = 71.6$ kJ mol^{-1}
$\Delta H_f\, CO_2 = -291$ kJ mol^{-1}
$\Delta H_f\, H_2O_{(l)} = -186$ kJ mol^{-1} respectively

Solution

$$C_3H_{6(g)} + \frac{9}{2}O_{2(g)} \rightarrow 3CO_{2(g)} + 3H_2O_{(l)}$$

$\Delta H = \sum \Delta H_f$ products $- \sum \Delta H_f$ reactants
$= [3 \times (-291 \text{ kJ}) + 3(-186 \text{ kJ})] - [71.6 \text{ kJ}]$

$\Delta H = -873$ kJ $- 558$ kJ $- 71.6$ kJ

$\Delta H = -1502.6$ kJ mol^{-1}

Calorific value = Fuel efficiency $= \frac{1502.6}{42} = 35.77$ kJ g^{-1}

Enthalpy of combustion for 1 g fuel $\simeq$ 35.8 kJ

Concept Strand 19

The enthalpy change for the following reaction $2C_6H_{6(\ell)} + 15O_{2(g)} \rightarrow 12CO_{2(g)} + 6H_2O_{(\ell)}$ at 298K is –6536.0 kJ. Calculate, Q_V for the combustion of benzene per mole at the same temperature. Also, calculate its standard heat of formation, given the standard heats of formation of $CO_{2(g)}$ and liquid water are –394.6 kJ and –286.2 kJ respectively.

Solution

Standard enthalpy of combustion of benzene

$$= Q_p = \frac{-6536.0}{2} = -3268.0 \text{ kJ mol}^{-1}$$

$$Q_p = Q_v + RT\Delta n;$$

$$Q_v = -3268 - 8.314 \times 10^{-3} \times 298 \times \left(-\frac{3}{2}\right)$$

$$\left(\because \Delta n = -\frac{3}{2}\right)$$

$$= -3268 + 3.7 = -3264.3 \text{ kJ}.$$

Calculation of standard heat of formation of benzene:

$$C_6H_{6\,(\ell)} + 7½O_{2(g)} \rightarrow 6CO_{2(g)} + 3H_2O_{(\ell)};\ \Delta H° = -3268.0 \text{ kJ}$$

$$\Delta H° = -3268.0$$
$$= 6\Delta_f H°\,(CO_2) + 3\Delta_f H°\,(H_2O_{(\ell)}) - \Delta_f H°\ \ (C_6H_{6(\ell)})$$
$$= 6 \times -394.6 + 3 \times -286.2 - \Delta_f H°\ \ (C_6H_{6(\ell)})$$

$$\Rightarrow\ \ \Delta_f H°\ \ (C_6H_{6(\ell)}) = -2367.6 - 858.6 + 3268$$
$$= 41.8 \text{ kJ}$$

Concept Strand 20

From the enthalpies of neutralisation given below, calculate the enthalpy of neutralization of nitrous acid by sodium hydroxide in dilute solution.

(i) $NH_4OH_{(aq)} + HNO_{2(aq)} \rightarrow NH^+_{4(aq)} + NO^-_{2(aq)} + H_2O;$ $\Delta H_1 = -38.0$ kJ.

(ii) $NH_4OH_{(aq)} + HCl_{(aq)} \rightarrow NH^+_{4(aq)} + Cl^-_{(aq)} + H_2O;$ $\Delta H_2 = -51.5$ kJ.

(iii) $HCl_{(aq)} + NaOH_{(aq)} \rightarrow Na^+_{(aq)} + Cl^-_{(aq)} + H_2O;\ \Delta H_3 = -57.3$ kJ.

Also, calculate the enthalpy of ionization of nitrous acid in dilute solution.

Solution

To get the enthalpy of neutralization of nitrous acid in NaOH add equations (i) and (iii) and subtract (ii)

$$HNO_{2(aq)} + NaOH_{(aq)} \rightarrow$$
$$Na^+_{(aq)} + NO^-_{2(aq)} + H_2O;\ \Delta H = -43.8 \text{ kJ} \quad \text{— (iv)}$$

Enthalpy of ionization of nitrous acid is given by the equation,

$$HNO_{2(aq)} \rightarrow H^+_{(aq)} + NO^-_{2(aq)};\ \Delta H = ?$$

As ionic reactions,

(iii) $\Rightarrow H^+_{(aq)} + Cl^-_{(aq)} + Na^+_{(aq)} + OH^-_{(aq)} \rightarrow Na^+_{(aq)} + Cl^-_{(aq)} + H_2O;\ \Delta H = -57.3$ kJ

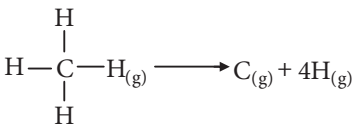
(iv) $\Rightarrow HNO_{2(aq)} + Na^+_{(aq)} + OH^-_{(aq)} \rightarrow Na^+_{(aq)} + NO^-_{2(aq)} + H_2O;\ \Delta H = -43.8$ kJ

(iv) – (iii) gives, $HNO_{2(aq)} \rightarrow H^+_{(aq)} + NO^-_{2(aq)};$ $\Delta H = +13.5$ kJ.

BOND ENTHALPY (BOND ENERGY)

The amount of energy required to separate an atom by breaking a bond is the dissociation energy of that bond. Bond energy is the average energy required to break all the bonds of one type present in one mole of the substance so as to get the gaseous atoms. For diatomic molecules, the term bond dissociation energy and bond energy are the same. For example, H–Cl, H–H, O=O, N≡N etc have bond dissociation energy and bond energy the same. Bond energy is the average value of the dissociation energies of a given bond.

For example,

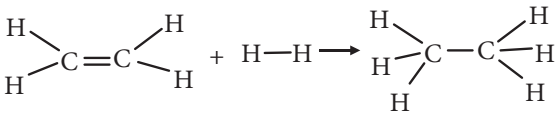


CH_4 has four bond dissociation energies and they are the energies required to break the four C–H bonds one by one. The first bond dissociation energy is the highest as CH_4 is a stable molecule and as the stability decreases by breaking the bonds one by one, bond dissociation energies goes on decreasing.

The average of these four dissociation energies gives the bond energy of C–H in CH_4. It has been found that average bond enthalpies differ from compound to compound. For e.g., the average C - H bond enthalpy in methane is 416 kJ mol^{-1}, but differs slightly in CH_3CH_2Cl, CH_3NO_2 etc.

ΔH from bond energy data

ΔH= sum of the bond energies of all the bonds broken in the reactants – sum of the bond energies of all the bonds formed in the products.



$$H_{H-H} + H_{C=C} - \left[2\,H_{C-H} + H_{C-C}\right]$$

where, H_{H-H} is the bond energy of H – H or enthalpy of H – H bond and so on.

CONCEPT STRANDS

Concept Strand 21

The standard heat of formation of $HI_{(g)}$ is 26.5 kJ. The bond dissociation energy of H – H and H – I are 436.0 kJ and 295 kJ respectively. From these data, calculate the bond dissociation energy of I – I bond?

Solution

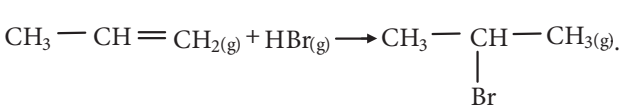
Given, $\frac{1}{2}H_2 + \frac{1}{2}I_2 \rightarrow HI$, $\Delta H = 26.5$ kJ

$$26.5 = \frac{1}{2} \times 436 + \frac{1}{2} \times H_{I-I} - 295$$

$$H_{I-I} = 207 \text{ kJ}$$

Concept Strand 22

From the given data calculate the enthalpy change of the reaction



The bond energies of C–C and C=C bonds are 348 kJ and 612.0 kJ respectively. The bond energies of C–Br, C–H and H–Br bonds are 276.0 kJ, 412.0 kJ and 366.0 kJ respectively.

Solution

$$\Delta H = H_{C=C} + H_{H-Br} - (H_{C-H} + H_{C-C} + H_{C-Br})$$

$$= 612 + 366 - (412 + 348 + 276) = -58 \text{ kJ}$$

Concept Strand 23

Determine the average S–F bond energy in SF_6. The standard heat of formation values of $SF_{6(g)}$, $S_{(g)}$ and $F_{(g)}$ are: –2250, 425 and 95 kJ mol^{-1} respectively.

Solution

$$SF_{6(g)} \rightarrow S_{(g)} + 6F_{(g)}$$

$$\Delta H_{reaction} = [\Delta_f H°S_{(g)}] + 6\,[\Delta_f H°F_{(g)}] - [\Delta_f H°SF_{6(g)}]$$

$$\Delta H_{reaction} = 425 + 6\,(95) - (-2250) = 3245 \text{ kJ mol}^{-1}$$

$$\therefore \quad \text{Average S-F bond energy} = \frac{3245}{6} = 540.83 \text{ kJ mol}^{-1}$$

Concept Strand 24

Calculate the bond energy of O–H bond in ethanol from the following data.

(i) $2C_{(graphite)} + 3H_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow C_2H_5OH_{(g)}$; $\Delta H = -234.5$ kJ

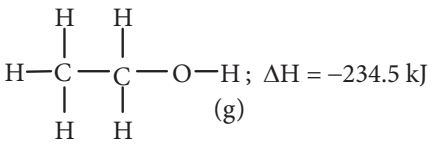
(ii) $C_{(graphite)} \rightarrow C_{(g)}$; $\Delta H = 715.0$ kJ

(iii) $3H_{2(g)} \rightarrow 6H_{(g)}$; $\Delta H = 1308.0$ kJ

The bond energies of C–C, C–H, C–O bonds are 348.0 kJ, 412.0 kJ, 335.0 kJ respectively. The O=O bond energy is 498.0 kJ

Solution

Given, $2C_{(graphite)} + 3H_{2\,(g)} + \frac{1}{2}O{=}O_{(g)} \rightarrow$



$\Delta H = -234.5$ kJ

$$-234.5 = 2H_{C_{(graphite)}-C_{(g)}} + 3H_{H-H} + \frac{1}{2}H_{O=O} - \left[5H_{C-H} + H_{C-C} + H_{C-O} + H_{O-H}\right]$$

$$-234.5 = 2 \times 715 + 1308 + \frac{1}{2} \times 498 - \left[5 \times 412 + 348 + 335 + H_{O-H}\right]$$

$$H_{O-H} = 478.5 \text{ kJ}$$

LAWS OF THERMOCHEMISTRY

Lavosier and Laplace's Law

The enthalpy change taking place during the reaction is equal in magnitude but opposite in sign to the enthalpy change occurring in the reverse process.

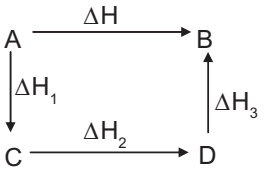
E.g., $H_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow H_2O_{(\ell)}$; $\Delta H = -286$kJ

$H_2O_{(\ell)} \rightarrow H_{2(g)} + \frac{1}{2}O_{2(g)}$; $\Delta H = +286$ kJ

Hess's law of constant heat summation

It states that enthalpy change for a reaction is the same whether it occurs in one step or in a series of steps. This supports the fact that enthalpy is a state function. Hence its change is independent of the path by which a reaction occurs. Hess's law is one form of 1st law of thermodynamics.

According to Hess's law, thermo-chemical equations can be added, subtracted and multiplied just like mathematical equations. It also helps in calculation of enthalpy changes for reactions, which cannot be experimentally determined. Representation of Hess's law:



Then according to Hess's law, $\Delta H = \Delta H_1 + \Delta H_2 + \Delta H_3$

Illustration

Let us look at an example involving butane (C_4H_{10}). Carbon and hydrogen will not react directly to produce butane. But its formation can be represented by the equation.

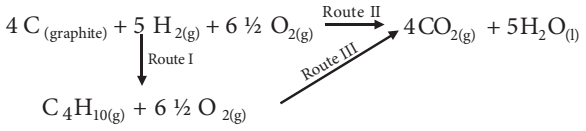
$$4C_{(graphite)} + 5H_{2(g)} \rightarrow C_4H_{10(g)}$$

This cannot be carried out directly in a calorimeter. However, the $\Delta_f H^\circ$ of butane can be obtained by measuring the standard heats of combustion of butane, carbon and hydrogen.

$$4\,C_{(graphite)} + 4\,O_{2(g)} \rightarrow 4\,CO_{2(g)} \rightarrow (1)$$

$$5H_{2(g)} + 2\,½\,O_{2(g)} \rightarrow 5\,H_2O_{(l)} \rightarrow (2)$$

(1) + (2) ⇒



By Hess's law,

Enthalpy change for route I = Enthalpy change for route II − Enthalpy change for route (III)

$$= 4 \times \Delta_c H^\circ_{(graphite)} + 5 \times \Delta_c H^\circ_{H_2(g)} - \Delta_c H^\circ_{C_4H_{10}(g)}$$

$$= 4 \times -393 + 5 \times -286 - (-2877) = -125 \text{ kJ mol}^{-1}$$

Limitations of first law of thermodynamics

The first law says that whenever one form of energy disappears, an equivalent amount of energy in another form appears. But it does not tell us about the extent and convertibility of one form of energy into another and the direction of flow of heat energy. For example, when two bodies, which are capable of exchanging heat energy are brought together, the first law states that the heat gained by one must be equal to that lost by the other. The first law, however, does not tell us which of the two would lose or gain heat energy. It also does not tell how much heat energy would be transferred from one body to the other. From experience, we know that heat must flow from a hotter body to a colder one.

CONCEPT STRAND

Concept Strand 25

Calculate the electron affinity of the element, X

$$\frac{1}{3}X^{2+}_{(g)} + \frac{2}{3}X^{-}_{(g)} \rightarrow X_{(g)};\ \Delta H_1 = -750\ \text{kJ} \qquad \text{—(i)}$$

$$X^{2+}_{(g)} + 2e^{-} \rightarrow X_{(g)};\ \Delta H_2 = -4150\ \text{kJ} \qquad \text{— (ii)}$$

Solution

Reversing the equation (i) $\times \frac{3}{2}$

$$\frac{3}{2}X_{(g)} \rightarrow \frac{1}{2}X^{2+}_{(g)} + X^{-}_{(g)} \qquad \text{— (iii)}$$

$$\Delta H_1 = 750 \times \frac{3}{2} = 1125\ \text{kJ mol}^{-1}$$

Multiplying equation (ii) by $\frac{1}{2}$

$$\frac{1}{2}X^{2+}_{(g)} + e^{-} \rightarrow \frac{1}{2}X_{(g)} \qquad \text{— (iv)}$$

$$\Delta H_2 = \frac{-4150}{2} = -2075\ \text{kJ mol}^{-1}$$

Adding equations (iii) and (iv)

$$X_{(g)} + e^{-} \rightarrow X^{-}_{(g)}$$

$$\Delta H = -2075 + 1125 = -950\ \text{kJ mol}^{-1}$$

$\therefore$ $\Delta H = -950\ \text{kJ mol}^{-1}$

SPONTANEOUS PROCESSES

Physical or chemical processes, which occur on their own direction under a given set of conditions are referred to as spontaneous processes. All natural processes take place spontaneously and they cannot be reversed without the aid of an external agency. Some examples of such processes are the evaporation of water from an open vessel, the flow of gas from a region of higher pressure into a region of lower pressure when they are connected by a tube etc.

Note: The spontaneity of a process is not at all related to its speed i.e., it may be slow or fast process.

All spontaneous processes proceed to a state of final equilibrium and during their passage can provide energy for useful work. They are thus important in this context and it is necessary to study such process and to know under what conditions they occur.

Spontaneous processes are exothermic as well as endothermic, hence ΔH is not the sole factor to conclude whether the process is spontaneous or not.

SECOND LAW OF THERMODYNAMICS AND ENTROPY (S)

An important aspect of second law concerns with the irreversibility of spontaneous processes. This law is usually stated in the form, *"all spontaneous processes are thermodynamically irreversible"*. In another form, the second law may be stated as *"it is impossible for cyclic processes to transfer heat from a system at lower temperature to one at higher temperature without applying some work on this system"*.

An important thermodynamic property, which enables us to ascertain the feasibility or spontaneity of a process is known as entropy (denoted by 'S'). It is defined as follows:

$$dS = \frac{\delta Q_{rev}}{T}$$

where, δQ_{rev} is the amount of heat absorbed reversibly in an infinitesimal process at temperature T. For a finite process,

$$\Delta S = S_2 - S_1 = \frac{\delta Q_{rev}}{T}$$

Entropy is a state function and is also an extensive property. It has the unit of J K^{-1}. Molar entropy has the unit of J K^{-1} $mol^{-1.}$

Physical significance of entropy

Physically, entropy may be associated with the 'disorder of the system'. It is a measure of the disorder or randomness of the molecules of a system. As disorder of a system increases, entropy increases. Entropy is also a function of the thermodynamic probability of a system. As the state of equilibrium is the state of maximum probability, entropy is maximum in this state.

Conditions for spontaneity or equilibrium

For a general reversible system A $\rightleftharpoons$ B

$$(S_B - S_A)_{system} = \frac{Q_1}{T_1} \text{ and } (S_B - S_A)_{surroundings} = \frac{-Q_1}{T_1}$$

Hence the entropy change of the universe during a reversible isothermal process is zero.

If the process is irreversible

$$\Delta S_{irr} = \frac{Q}{T_2} - \frac{Q}{T_1} = Q\left[\frac{1}{T_2} - \frac{1}{T_1}\right].$$

As conduction of heat always taking place from a higher to a lower temperature,

$$T_1 > T_2 \text{ or } \frac{1}{T_2} > \frac{1}{T_1} \text{ or } \frac{Q}{T_2} > \frac{Q}{T_1} \text{ or } \frac{Q}{T_2} - \frac{Q}{T_1} > 0$$

i.e., $\Delta S_{sorr} > 0$

Hence entropy of the universe increases in an irreversible process.

For a process occurring under reversible conditions, (as for example, isothermal reversible expansion of an ideal gas).

$$\Delta S_{system} + \Delta S_{surroundings} = \Delta S_{total} = 0$$

For a process occurring spontaneously (or irreversibly) (as for expansion of an ideal gas from a finite pressure into vacuum)

$$\Delta S_{sys} + \Delta S_{surr} = \Delta S_{total} > 0$$

ENTROPY CHANGES DURING VARIOUS PROCESSES

$$\Delta S = C_V \ln \frac{T_2}{T_1} + R \ln \frac{V_2}{V_1}$$

$$\Delta S = C_p \ln \frac{T_2}{T_1} - R \ln \frac{P_2}{P_1}$$

Entropy change during isothermal reversible expansion of an ideal gas

$$\Delta S_T = 2.303\ nR \log \frac{P_1}{P_2}$$

$$= 2.303\ nR \log \frac{V_2}{V_1}$$

Entropy change during an adiabatic process

$$\Delta S = 0 \qquad (\because Q = 0)$$

Hence adiabatic processes are also known as isentropic processes.

Entropy change during phase transition

$$\Delta S_{fusion} = \frac{\Delta H_{fusion}}{T} \qquad T \rightarrow \text{melting point in K}$$

$$\Delta S_{vap} = \frac{\Delta H_{vap}}{T} \qquad T \rightarrow \text{Boiling point in K}$$

$$\Delta S_{Transition} = \frac{\Delta H_{Transition}}{T} \qquad T \rightarrow \text{Transition temperature in K}$$

Trouton's rule

The ratio of the molar heat of vaporization of a liquid (given in joules) to its normal boiling point of the liquid on the absolute scale is nearly equal to 88 J K^{-1} mol^{-1}. This is known as Trouton's rule.

Thus,

$$\frac{\Delta H_{vap}}{T_b} = 88 \text{ J } K^{-1} \text{ mol}^{-1}$$

where, ΔH_{vap} is enthalpy of vaporization and T_b is boiling point in kelvin.

The equation is only approximate as the values $\frac{\Delta H_{vap}}{T_b}$ are seen to vary over a wide range, viz., from about 60 to 110 J mol^{-1} K^{-1}. The deviations from Trouton's rule, thus occur in both directions. The quantity $\frac{\Delta H_{vap}}{T_b}$ is also called entropy of vaporization, ΔS_{vap}. Trouton's rule is useful for estimating the heat of vaporization of a liquid, if its boiling point is known.

Entropy change without phase change

$$\Delta S = ms \ln \frac{T_2}{T_1}$$

where, m is the mass and s is the specific heat.

For an isochoric change,

$$\Delta S_v = C_v \ln \frac{T_2}{T_1} \text{ (for 1 mole of a perfect gas)}$$

For an isobaric change,

$$\Delta S_p = C_p \ln \frac{T_2}{T_1} \text{ (for 1 mole of a perfect gas)}$$

For an isothermal change,

$$\Delta S_T = R \ln \frac{V_2}{V_1} \text{ (for 1 mole of a perfect gas)}$$

$$\Rightarrow \Delta S_T = R \ln \frac{P_1}{P_2} \text{ (for 1 mole of a perfect gas)}$$

The second law of thermodynamics suggests that all kinds of energy of a given substance cannot be completely converted into useful work. Thus heat of a reaction is also not fully convertible into work.

The unavailable energy = Heat of reaction − Energy available for useful work

Second law also suggests that a part of energy to be lost is proportional to the absolute temperature at which the reaction takes place.

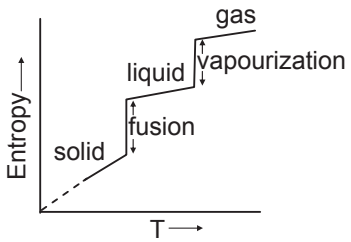
$\therefore$ Unavailable energy $\propto$ T

Unavailable energy = $\Delta S.T$

$$\Delta S = \frac{\text{unavailable energy}}{T}$$

Tephigraph

Graph of entropy of a substance against temperature



Heat and work–carnot cycle

The machine used for the conversion of heat into work is called heat engine. Carnot designed an imaginary reversible cycle with maximum conversion of heat into work. Here the working system consists of one mole of an ideal gas enclosed in a cylinder fitted with a frictionless piston. The heat engine takes heat Q_2 from the reservoir kept at a higher temperature T_2 (source) converts some heat into work and returns the remaining heat Q_1 to another reservoir kept at a lower temperature T_1 (sink).

$$\text{Efficiency of heat engine, } \eta = \frac{W}{Q_2} = \frac{Q_2 - Q_1}{Q_2} = \frac{T_2 - T_1}{T_2}$$

This shows that the efficiency of the engine depends only on the temperatures at which the source and sink are kept and does not depend on the nature of the working system or on the mode of operation.

CONCEPT STRANDS

Concept Strand 26

Five moles of an ideal gas expand isothermally and reversibly from an initial volume of 20 L to a final volume of 250 L at 37°C. Calculate

(i) enthalpy change undergone by the gas
(ii) work done by the gas
(iii) entropy change of the surroundings (thermostat)
(iv) $\Delta S_{sys} + \Delta S_{surr}$

Solution

(i) $\Delta S_{(gas)}$

$$= \frac{Q_{rev}}{T} = nR \ln \frac{V_2}{V_1}$$

$$= 5 \times 8.314 \times 2.303 \times \log \frac{250}{20}$$

$$= 105 \text{ J K}^{-1}$$

(ii) Work done by the gas = W = Q_{rev} (Since $\Delta U = 0$)

$$\therefore \quad W = nRT\ln\frac{V_1}{V_2} = 5 \times 8.314 \times 310 \times 2.303 \times \log\frac{20}{250}$$

$$= -32554 \text{ J.}$$

(iii) $\Delta S_{surr} = \dfrac{-Q_{rev}}{T} = -105 \text{ J K}^{-1}$

(iv) $\Delta S_{sys} + \Delta S_{surr} = 0$ (Since the process is reversible)

Concept Strand 27

Five moles of an ideal gas is allowed to expand irreversibly against vacuum at constant temperature from an initial volume of 20 L to a final volume of 250 L.

Calculate (i) ΔS of the gas (ii) ΔS_{surr} (iii) $\Delta S_{sys} + \Delta S_{surr}$ (iv) work done by the gas.

Solution

(i) $\Delta S_{gas} =$

$$nR\ln\frac{V_2}{V_1} = 5 \times 8.314 \times 2.303\log\frac{250}{20} = 105 \text{ J K}^{-1}$$

(ΔS_{sys} does not undergo any change because it is a state function).

(ii) $\Delta S_{surr} = 0$ (because in the isothermal expansion of an ideal gas, $\Delta U = 0$, $\therefore q = W$ and $W = 0$ because gas expanded against vacuum.

$\therefore$ $Q = 0$ i.e., no heat is absorbed by the gas).

(iii) $\Delta S_{sys} + \Delta S_{surr} = 105 + 0 = 105$ J

($\Delta S_{sys} + \Delta S_{surr}$ is > 0 and thus positive).

(iv) $W = 0$ ($\because$ the gas expands against vacuum).

Concept Strand 28

Two moles of carbon dioxide at a pressure of 5 atm and 25°C are heated to 125°C and compressed to 25 atm. Given C_p of the gas is 37.1 J K^{-1} $mole^{-1}$ and assuming it to be independent of temperature, calculate ΔS for the process. Also, calculate ΔS considering the volume changes under the given conditions. What is the conclusion?

Solution

$$\Delta S = n\left(C_p \ln\frac{T_2}{T_1} + R\ln\frac{P_1}{P_2}\right)$$

$$= 2 \times 2.303\left(37.1\log\frac{398}{298} + 8.314\log\frac{5}{25}\right)$$

$$= 4.606\left[4.66 - 5.81\right] = -5.30 \text{ J K}^{-1}$$

$$\frac{V_2}{V_1} = \frac{P_1T_2}{P_2T_1} = \frac{5 \times 398}{25 \times 298} = 0.267$$

$$\Delta S = n\left(C_v \ln\frac{T_2}{T_1} + R\ln\frac{V_2}{V_1}\right)$$

$$= 2 \times 2.303\left[28.8\log\frac{398}{298} + 8.314 \times \log 0.267\right]$$

(Using calculated C_v from $C_p = C_v + R$)

$$= 4.606\ [3.62 - 4.77] = -5.30 \text{ J K}^{-1}$$

ΔS, calculated by the two methods, are the same as expected.

Concept Strand 29

Calculate the entropy change of the surroundings for the rusting of iron at 298K. If the standard enthalpy of formation of $Fe_2O_{3(s)}$ is -908.6 kJ mol^{-1} and the standard entropy change for the reaction is -750 J K^{-1} mol^{-1}.

Solution

$$4Fe_{(s)} + 3O_{2(g)} \rightarrow 2Fe_2O_{3(s)}$$

Heat released during the reaction

$$= 2 \times (-908.6)$$

$$= -1817.2 \text{ kJ}$$

This heat is absorbed by surroundings at 298K

$$\Delta S_{surroundings} = \frac{q}{T} = \frac{1817.2 \text{ kJ}}{298} = 6097.9 \text{ J K}^{-1}$$

Entropy change of the given reaction $\Delta S_{system} = -750$ J K^{-1} mol^{-1}

$$\therefore \quad \Delta S_{total} = \Delta S_{system} + \Delta S_{surr}$$

$$= -750 + 6097.9 \text{ J K}^{-1}$$

$$= 5347.9 \text{ J K}^{-1}$$

$\therefore$ Total entropy change is +ve the rusting is spontaneous at 298K.

Concept Strand 30

Two moles of CO_2 are heated from 25°C to 125°C (C_p of CO_2 = 37.1 J K^{-1} mol^{-1})

(i) What is ΔS at constant pressure?
(ii) What is ΔS at constant volume?
(iii) Assuming that the gas is compressed from 5 atm to 25 atm at a constant temperature of 25°C, what is its ΔS value?

Solution

(i) $\Delta S_p = nC_p \ln\frac{T_2}{T_1}$

$= 2 \times 37.1 \times 2.303 \log\frac{398}{298} = 21.47 \text{ J K}^{-1}$

(ii) $\Delta S_v = nC_V \ln\frac{T_2}{T_1}$

$= 2 \times 2.303 \times 28.8 \times \log\frac{398}{298} = 16.67 \text{ J K}^{-1}$

(iii) $\Delta S_T = nR \ln\frac{P_1}{P_2}$

$= 2 \times 8.314 \times 2.303 \log\frac{5}{25} = -26.77 \text{ J K}^{-1}$

Concept Strand 31

78 g of solid benzene at its freezing point 5.6°C are heated to benzene vapours at 100°C at constant pressure of 1 atm. What is the total ΔS for the process?

Given C_p for liquid benzene = 136.1 J deg^{-1} mol^{-1}; C_p for benzene vapour = 81.7 J deg^{-1} mol^{-1}; latent heat of fusion of benzene at its freezing point = 10.6 kJ mol^{-1}, latent heat of vapourization of benzene at its boiling point of 80°C = 30.8 kJ mol^{-1}. (Assume C_p's to be independent of temperature).

Solution

I step:

To calculate ΔS_1 for the melting of solid to liquid benzene at 5.6°C (phase change)

$$\Delta S_1 = \frac{10600}{278.6} = 38.05 \text{ J K}^{-1}$$

II step:

To calculate ΔS_2 for heating liquid benzene from 278.6K to 353.0K (no phase change)

$$\Delta S_2 = nC_p \ln\frac{T_2}{T_1} = 1 \times 136.1 \times 2.303 \log\frac{353.0}{278.6}$$

$= 32.22 \text{ J K}^{-1}$.

III step:

To calculate ΔS_3 for vapourising liquid benzene at 353.0K to vapour at the same temperature.

$$\Delta S_3 = \frac{30800}{353} = 87.25 \text{ J K}^{-1}$$

IV step:

To calculate ΔS_4 for heating benzene vapour from 353.0K to 373.0K.

$$\Delta S_4 = nC_p \ln\frac{T_2}{T_1}$$

$$= 1 \times 81.7 \times 2.303 \log\frac{373}{353} = 4.50 \text{ J K}^{-1}$$

$$\Delta S_{total} = 38.05 + 32.22 + 87.25 + 4.50 = 162.02 \text{ J K}^{-1}$$

Concept Strand 32

64 g of methanol at a temperature of 60°C are poured into 128 g of methanol kept at 20°C in an insulated vessel. Calculate the net change in entropy that occurs.
(C_p of liquid methanol = 81.6 J deg^{-1} mol^{-1})

Solution

Let the resultant temperature of the mixture be T kelvin.

Heat lost from the added methanol

$$= \frac{64}{32} \times 81.6 \times (333 - T)$$

Heat gained by the methanol in the vessel

$$= \frac{128}{32} \times 81.6 \times (T - 293)$$

Heat lost = Heat gained

$2 \times 81.6 \times (333 - T) = 4 \times 81.6 (T - 293)$

$\Rightarrow$ T = 306.3K.

ΔS due to cooling of added methanol = $nC_p \ln\frac{T_2}{T_1}$

$$= 2 \times 81.6 \times 2.303 \log\frac{306.3}{333} = -13.64 \text{ J K}^{-1}$$

ΔS gained by the heating of methanol in the vessel

$$= 4 \times 81.6 \times 2.303 \times \log\frac{306.3}{293} = 14.49 \text{ J K}^{-1}.$$

Net $\Delta S = 14.49 - 13.64 = 0.85 \text{ J K}^{-1}$.

Concept Strand 33

Two moles of water at 50°C are kept in contact with a large heat reservoir at 100°C until the water reaches the same

temperature. Find the entropy change of water and the hot reservoir and $\Delta S_{total.}$ Explain the significance of ΔS_{total}.(C_p of water = 75.6 J deg^{-1} mol^{-1}).

Solution

$$\Delta S \text{ due to heating of water } = nC_p \ln\frac{T_2}{T_1}$$

$$= 2 \times 75.6 \times 2.303 \log\frac{373}{323}$$

$$= 21.76 \text{ J K}^{-1}$$

Heat lost from hot reservoir = $2 \times 75.6\ (323 - 373)$

$$= -7560 \text{ J.}$$

$$\Delta S \text{ of heat reservoir } = \frac{-7560}{373} = -20.27 \text{ J K}^{-1}$$

$\Delta S_{total} = 21.76 - 20.27 = 1.49 \text{ J K}^{-1}$.

ΔS_{total} is positive because of the spontaneous nature of the process.

Concept Strand 34

Calculate the change in entropy associated with the oxidation of β- D-fructose by oxygen according to $\underset{(\beta-D\,fructose)}{C_6H_{12}O_{6(s)}} + 6O_{2(g)} \rightarrow 6CO_{2(g)} + 6H_2O_{(\ell)}$ using the following standard entropies of the various compounds at 298K and 1 atm.

S° (J K^{-1} mol^{-1})	$C_6H_{12}O_{6(s)}$	$O_{2(g)}$	$CO_{2(g)}$	$H_2O_{(\ell)}$
	212.0	205.1	213.7	109.6

Solution

$\Delta S°$ of the reaction = $6 \times 213.7 + 6 \times 109.6 - (212.0 + 6 \times 205.1)$

$$= 497.2 \text{ J K}^{-1} \text{ mol}^{-1}$$

Concept Strand 35

Five moles of an ideal gas expand isothermally at 37°C from an initial volume of 31 L to a final volume of 155 L against a constant external pressure of 0.5 atm.

Calculate

(i) ΔS_{syste}
(ii) ΔS_{surr}
(iii) ΔS_{total}.

Solution

(i) $\Delta S_{system} = nR \ln\frac{V_2}{V_1} = 5 \times 8.314 \times 2.303 \log\frac{155}{31}$

$$= 5 \times 8.314 \times 2.303 \times 0.7$$

$$= 67.0 \text{ J K}^{-1}.$$

(ii) For an isothermal expansion process involving an ideal gas, $\Delta U = 0$; $\therefore q = W$.

$$W = -P_{ext}\,\Delta V = -0.5 \times (155 - 31)$$

$$= -0.5 \times 124 \text{ L atm}$$

$$= \frac{-0.5 \times 124 \times 8.314}{0.0821} = -6278.5 \text{J}$$

Assuming that the heat is lost by thermostat reversibly,

$$\Delta S_{surr} = \frac{-6278.5}{310} = -20.25 \text{ J K}^{-1}$$

$\Delta S_{sys} + \Delta S_{surr} = 67.0 - 20.25 = +46.75$ J

THIRD LAW OF THERMODYNAMICS

At absolute zero (zero Kelvin), the entropy of a perfectly crystalline substance approaches zero.

$$\underset{T \rightarrow 0K}{Lt}\ S \rightarrow 0$$

For a solid at T K, $S_T - S_0 = \int_0^T C_P \frac{dT}{T} = C_P \ln T$

From 3rd law, $S_0 = 0$

$\therefore$ Absolute entropy, $S_T = 2.303\ C_p \log T$

GIBBS FREE ENERGY (G)

Gibbs free energy is a thermodynamic parameter, which can be used as criterion of spontaneity of the system by studying changes in systems alone.
The Gibbs free energy is defined as

$$G = H - TS,$$

where, H is enthalpy, T, the temperature in kelvin scale and S the entropy. For a process taking place at constant temperature the free energy change,

$$\Delta G = \Delta H - T\,\Delta S.$$

If energy decreases and entropy increases, i.e., ΔH is negative and ΔS is positive, the process is always spontaneous. Thus for a spontaneous change ΔG is negative.

G is called the free energy

If we consider TS as the part of the system's energy that is already disordered, then H – TS is the part of the system's energy still in ordered form and therefore available (free) to cause spontaneous change by becoming disordered.

CRITERIA FOR EQUILIBRIUM AND SPONTANEITY

The condition for equilibrium is $\Delta G_{P,T} = 0$

The condition for spontaneity is $\Delta G_{P,T} < 0$

Any physical or chemical transformation under constant temperature and pressure tends to a state of equilibrium. As it proceeds, the free energy of the system decreases and at equilibrium it becomes zero.

Thermodynamic criteria of spontaneity using Gibbs Helmholtz equation

Thermodynamic criteria of spontaneity can be arrived at using Gibbs Helmholtz equation, $\Delta G = \Delta H - T.\Delta S$.

(i) If ΔH is negative and ΔS is positive, ΔG is negative and hence such a process is spontaneous at all temperatures.

(ii) If ΔH is positive and ΔS is negative, ΔG is always positive and hence such a process is non spontaneous at all temperatures.

(iii) If ΔH and ΔS are positive, ΔG is negative only if $T.\Delta S > \Delta H$. Hence such processes are spontaneous only at higher temperatures, i.e., when, $T > \dfrac{\Delta H}{\Delta S}$

(iv) If ΔH and ΔS are negative, ΔG is negative only when $|\Delta H| > T|\Delta S|$

Such processes are spontaneous only at low temperature.

i.e., when, $T < \dfrac{\Delta H}{\Delta S}$

FREE ENERGY CHANGE AND USEFUL WORK

The useful work, i.e., the work other than work of expansion done by the system is equal to the Gibbs free energy change. Thus if ΔG is the free energy change and ΔH the enthalpy change, the efficiency of the system to do useful work is given by $\dfrac{\Delta G}{\Delta H}$

Standard Gibbs free energy of formation

$\Delta_f G°$ is defined as the free energy change associated with the formation of one mole of the compound from its constituent elements in their standard states.

$\Delta_f G°$ of free elements in their standard states is taken to be zero.

$$\Delta G^\circ = \sum G^\circ_{f(Products)} - \sum G^\circ_{f(Reactants)}$$

ΔG^o is related to ΔH^o and ΔS^o, as

$$\Delta G^o = \Delta H^o - T.\Delta S^o$$

ΔG^o is related to the equilibrium constant (K) of a reaction as

$$\Delta G^o = -2.303\ RT\ \log K$$

Also, $\Delta G = \Delta G^0 + RT\ \ln Q$, where Q is the reaction quotient.

$\Rightarrow$ $\Delta G = -RT \ln K + RT \ln Q$

i.e.,

$$\Delta G = RT \ln\left(\frac{Q}{K}\right)$$

Relation of ΔG and ΔG° to the EMF of a reversible cell

ΔG and ΔG^o are related to the EMF of a galvanic cell operating under reversible conditions according to $\Delta G = -nFE$

$$\Delta G^\circ = -nFE^\circ$$

Variation of Free energy change with pressure for a perfect gas

For n moles of ideal gas, at constant temperature,

$$\Delta G = nRT \ln \frac{P_2}{P_1}$$

The above equation may also be written in terms of volume as,

$$\Delta G = nRT \ln \frac{V_1}{V_2}$$

Variation of free energy with temperature

$$\left[\frac{\partial(\Delta G)}{\partial T}\right]_P = -\Delta S$$

$\left[\frac{\partial(\Delta G)}{\partial T}\right]_P$ is also known as the temperature coefficient of free energy change. It gives the variation of free energy change with temperature at constant pressure.

$\therefore$ $\Delta G = \Delta H - T\Delta S$ becomes

$$\Delta G = \Delta H + T\left[\frac{\partial(\Delta G)}{\partial T}\right]_P$$

CONCEPT STRANDS

Concept Strand 36

Calculate the change in Gibbs free energy (ΔG) associated with the isothermal reversible expansion of 3 moles of an ideal gas from an initial volume of 10 L to a final volume of 200 L at 37°C.

Solution

$\Delta G_T = nRT \ln \frac{V_1}{V_2} = 3 \times 8.314 \times 310 \times 2.303 \times \log \frac{10}{200} = -23.17$ kJ.

Concept Strand 37

Calculate ΔG for the following reaction at 298K and predict whether the reactions is spontaneous or not?

$$N_{2(g)} + 3H_{2(g)} \rightarrow 2NH_{3(g)}$$

$\Delta U = -43.5\ kJ\ mol^{-1}$, $\Delta S = -95\ J\ mol^{-1}$

Solution

$$\Delta H = \Delta U + \Delta(PV) = \Delta U + \Delta(nRT)$$

$$= \Delta U + (\Delta n)RT$$

$$= 2 \times -43.5 - 2 \times 8.314 \times 10^{-3} \times 298$$

$$= -91.96\ kJ$$

$$\Delta G = \Delta H - T\Delta S$$

$$\Delta G = -91.96 - 298 \times 2 \times -95 \times 10^{-3}$$

$\therefore$ $\Delta G = -35.34$ kJ

$\therefore$ ΔG = –ve, the reaction is spontaneous

Concept Strand 38

Compute the standard free energy of a reaction at 27°C for the combustion of propane using the following data.

Species	C_3H_8	O_2	CO_2	H_2O
$\Delta H_f°$ (kJ mol^{-1})	–91.2	–	–413.5	–331.2
S°/ (J K^{-1} mol^{-1})	197	231	301	86

Solution

$$C_3H_{8(g)} + \frac{10}{2}O_2 \rightarrow 3CO_{2(g)} + 4H_2O_{(l)}$$

$$\Delta H° = 3\Delta H_f°(CO_2) + 4\Delta H_f°(H_2O) - \Delta H_f°(C_3H_8)$$

$$= 3(-413.5) + 4(-331.2) - (-91.2)$$

$$= -2474.1 \text{ kJ mol}^{-1}$$

$$\Delta S° = 3S°(CO_2) + 4S°(H_2O) - S°(C_3H_8) - \frac{10}{2}S°(O_2)$$

$$= -105 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$\Delta G° = \Delta H° - T\Delta S°$$

$$= -2474.1 - 300 \times (-105 \times 10^{-3})$$

$$= -2442.6 \text{ kJ mol}^{-1}$$

Concept Strand 39

Calculate ΔG of the reaction at 300K, $N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$. When 2 moles of each are taken and temperature is kept at 300K, the total pressure was found to be 10 bar.

$$\Delta_f G°_{N_2O_4} = 220 \text{ kJ}, \ \Delta_f G°_{NO_2} = 110 \text{ kJ}$$

Solution

The reaction is $N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$. The no. of moles of both N_2O_4 and NO_2 are same, hence their partial pressures will also be same.

$$pN_2O_4 = pNO_2 = \frac{10}{2} = 5 \text{ bar}$$

$$\therefore \quad Q_p = \frac{[pNO_2]^2}{[pN_2O_4]} = \frac{(5)^2}{5} = 5 \text{ bar}$$

$$\Delta G°_{\text{reaction}} = 2\Delta G°_{f_{NO_2}} - \Delta G°_{f_{N_2O_4}}$$

$$= 2 \times 110 - 220 = 0$$

$$\therefore \quad \Delta G = \Delta G° - 2.303 \text{ RT log Q}$$

$$= 0 - 2.303 \times 8.314 \times 300 \times \log 5$$

$$\Delta G = -4015 \text{ J}$$

Concept Strand 40

Calculate the standard Gibbs free energy change ($\Delta G°$) and the equilibrium constant Kp of the reaction, $4NH_{3(g)} + 5O_{2(g)} \rightarrow 4NO_{(g)} + 6H_2O_{(\ell)}$ using the following data.

	$NH_{3(g)}$	$O_{2(g)}$	$NO_{(g)}$	$H_2O_{(\ell)}$
S° (J deg^{-1} mol^{-1})	192.4	205.1	210.8	70
$\Delta_f H°$ (kJ mol^{-1})	–46.1	0	90.2	–285.8

Solution

$\Delta S°$ of the reaction = ($4 \times 210.8 + 6 \times 70$) – ($4 \times 192.4 + 5 \times 205.1$) = -531.9 J K^{-1}.

$\Delta H°$ of the reaction = $(4 \times 90.2 + 6 \times -285.8) - (4 \times -46.1 + 0)$ = –1169.6 kJ.

$\Delta G°$ of reaction $= \Delta H° - T\Delta S°$

$$= -1169.6 - 298 \times -531.9 \times 10^{-3}$$

$$= -1169.6 + 158.5 = -1011.1 \text{ kJ}.$$

$$\Delta G° = -RT \ln Kp$$

$$= -8.314 \times 10^{-3} \times 298 \times 2.303 \times \log Kp$$

$$\log K_p = \frac{1011.1}{8.314 \times 10^{-3} \times 298 \times 2.303} = 177.2$$

$$\Rightarrow \quad K_p = 1.58 \times 10^{177}.$$

Concept Strand 41

The standard Gibbs free energy change, ΔG, associated with the reaction, $2C_2H_{2(g)} + 5O_{2(g)} \rightarrow 4CO_{2(g)} + 2H_2O_{(\ell)}$ at 298 K and 1 atm is –2470.2 kJ. Given that the standard free energy of formation of $CO_{2(g)}$ and $H_2O_{(\ell)}$ are –394.4 kJ and –237.1 kJ respectively, calculate the standard free energy of formation of acetylene gas.

Solution

ΔG of the reaction = $-2470.2 = 4\Delta_f G°(CO_2) + 2\Delta_f G°(H_2O_{(\ell)}) - 2\Delta_f G°(C_2H_{2(g)})$

$$\Rightarrow \quad -2470.2 = 4 \times -394.4 - 2 \times 237.1 - 2\Delta_f G°(C_2H_{2(g)})$$

$$2\Delta_f G°(C_2H_{2(g)}) = -1577.6 - 474.2 + 2470.2$$

$$= 418.4 \text{ kJ}$$

$$\Rightarrow \quad \Delta_f G°(C_2H_{2(g)}) = \frac{418.4}{2} = 209.2 \text{ kJ}$$

Concept Strand 42

The temperature coefficient of the free energy change of a reaction is +4.80 J deg^{-1}. If the enthalpy change of the reaction is – 100 kJ, calculate ΔG of the reaction at 303K.

Solution

$$\left(\frac{\partial \Delta G}{\partial T}\right)_P = -\Delta S = +4.80 \text{ J deg}^{-1}$$

$\Rightarrow \quad \Delta S = -4.80 \text{ J deg}^{-1}$.

$$\Delta G = \Delta H - T\Delta S = -100 - (303 \times -4.80 \times 10^{-3})$$

$$= -100 + 1.45 = -98.55 \text{ kJ}.$$

Concept Strand 43

A sample of air consisting of N_2 and O_2 is heated to 2000°C to establish the equilibrium, $N_{2(g)} + O_{2(g)} \rightleftharpoons 2\ NO_{(g)}$, with an equilibrium constant, $K_c = 2 \times 10^{-3}$. If the mole percentage of NO at equilibrium is 2%, calculate the initial mole percentage of N_2 in air.

Solution

Number of moles	$N_{2(g)}$ +	$O_{2(g)}$ ⇌	$2\ NO_{(g)}$
Initial	(a)	100 – a	0
Eqbm.	a – x	100 – a – x	2x
	2x = 2	x = 1	

$$K_c = \frac{[NO]^2}{[N_2][O_2]}$$

$$2\times 10^{-3} = \frac{4x1}{(a-1)(99-a)}$$

a = 70%

i.e., mole percentage of N_2 = 70 and mole percentage of O_2 = 30

Concept Strand 44

Acetic acid CH_3COOH can form a dimer $(CH_3COOH)_2$ in the gas phase. The dimer is held together by H–bonds with a total strength of 88.5 kJ per mol of dimer. The equilibrium constant for the dimerization is 2.3×10^3 at 25°C. Calculate ΔS° of the reaction.

Solution

$$2CH_3COOH_{(g)} \rightleftharpoons (CH_3COOH)_{2(g)}$$

$\Delta G° = -2.303\ RT \log K$

$= -2.303 \times 8.314 \times 298 \times \log (2.3 \times 10^3)$

$= -19181.51 \text{ J} = -19.181 \text{ kJ}$

$\Delta G° = \Delta H° - T\Delta S°$

$-19.181 = -88.5 - 298 \times \Delta S°$

$$\Delta S° = \frac{-88.5+19.181}{298}$$

$$\Delta S° = \frac{-69.319}{298} = -0.232 \text{ kJ}$$

$\therefore \quad \Delta S° = -0.232$ kJ

Concept Strand 45

Ice and water are at equilibrium at 0°C. ΔH = 4 k J mol^{-1} for the process

$H_2O_{(s)} \rightarrow H_2O_{(l)}$ calculate ΔS and ΔG for the conversion of ice to liquid water.

Solution

ice ⇌ water

ΔG = 0 at equilibrium

$\Delta G = \Delta H - T\Delta S$

$\Delta H - T\Delta S = 0$

$T\Delta S = \Delta H$

$$\Delta S = \frac{\Delta H}{T}$$

$\Delta H = 4$ k J $mol^{-1} = 4000$ J mol^{-1} and T = 0°C = 273K

$$\therefore \quad \Delta S = \frac{4000 \text{ J mol}^{-1}}{273}$$

$= 14.65 \text{ J K}^{-1} \text{ mol}^{-1}$

SUMMARY

Terms used	System, surroundings
	Open, closed and isolated systems.
	Homogeneous and Heterogeneous system.
State of a system	Condition where the properties have constant value.
	State variables and state functions
Thermodynamic properties	Intensive and Extensive properties
Process and types of process	Process is an operation in which change of state is brought about.
	Isothermal, isobaric , isochoric , cyclic, adiabatic, reversible and irreversible
Thermodynamically irreversible process	Spontaneous process
Thermodynamic equilibrium	Macroscopic properties do not change with time.
Sign of heat and work	Heat gained by system = +Q
	Work done on the system = +W
	Heat lost by the system = –Q
	Work done by the system = –W
Relation between mechanical work W and heat produced H	$W = J \times H$
	When J = 4.184 Joules
Mechanical work W	$W = P_{(ext)} \Delta V$
Internal energy	Represented by U
	State function and extensive property. It is the sum of different forms of energy associated with the system.
First law of thermodynamics	$\Delta U = Q + W$
Work of isothermal reversible expansion	$W = -2.303\ nRT \log \frac{V_2}{V_1}$ or
	$W = -2.303\ nRT \log \frac{P_1}{P_2}$
Enthalpy	Heat constant of the system
	Represented by H
	Given by the relation H = U + PV
	State function & extensive property
Relation between ΔH, ΔU, Q_P and Q_V	$\Delta H = Q_P$; $\Delta U = Q_V$
Heat of reaction at constant volume	$Q_V = \Delta U$
	$\Delta U = \Sigma U_{products} - \Sigma U_{reactants}$
Heat of reaction at constant pressure	It is enthalpy or reaction
	$\Delta H = Q_P$
	$\Delta H = \Sigma H_{Products} - \Sigma H_{reactants}$
Kirchoff's equation	$\frac{\Delta H_2 - \Delta H_1}{T_2 - T_1} = \Delta C_P$; $\frac{\Delta U_2 - \Delta U_1}{T_2 - T_1} = \Delta C_V$

Zeroth law of thermodynamics	Law of thermal equilibrium
Relation between ΔH and ΔU	$\Delta H = \Delta U + P\Delta V$
Heat capacity	$C = \left(\frac{\partial q}{\partial T}\right)$; $C_V = \left(\frac{\partial U}{\partial T}\right)_V$; $C_P = \left(\frac{\partial H}{\partial T}\right)_P$
	$C_P - C_V = P\Delta V = R$; $\frac{C_P}{C_V} = \gamma$
Adiabatic expansion	$\Delta U = W$; $C_V\Delta T = \Delta U$ $\Delta U = nC_V\Delta T$ for n moles $\Delta H = C_P\Delta T$ for 1 mole $\Delta H = nC_P\Delta T$ for n moles
Heat liberated or absorbed	$= ms(t_2 - t_1)$ m = mass ; s = specific heat t_2 = final temperature; t_1 = initial temperature
Adiabatic expansion	PV^γ = constant $\left(\frac{T_1}{T_2}\right)^\gamma = \left(\frac{P_1}{P_2}\right)^{\gamma-1} = \left(\frac{P_2}{P_1}\right)^{1-\gamma}$ $TV^{\gamma-1}$ = constant $\frac{T_1}{T_2} = \left(\frac{V_2}{V_1}\right)^{\gamma-1} = \left(\frac{V_1}{V_2}\right)^{1-\gamma}$
Reversible adiabatic expansion	$W = C_V(T_2 - T_1) = -P_{ext}\left(\frac{T_2P_1 - T_1P_2}{P_1P_2}\right) \times R$
Work done in isobaric process	$W = -P(V_2 - V_1) = -P\Delta V$
Work done by real gases	$W = RT\ln\frac{(V_1 - b)}{(V_2 - b)} - a\left(\frac{1}{V_2} - \frac{1}{V_1}\right)$
Thermochemical equations	Balanced equation indicating the state of the involved substances and the energy changes involved.
Standard state	Most stable state at 298K and 1 atmosphere pressure
Various forms of enthalpy of reaction	Enthalpy of formation Enthalpy of combustion Enthalpy of solution Enthalpy of dilution Enthalpy of hydration Enthalpy of fusion Enthalpy of vaporization Enthalpy of neutralization Enthalpy of hydrogenation Enthalpy of transition

Enthalpy of neutralization	For strong acid and strong base $\Delta H = -57.1$ kJ mol^{-1}
Bond enthalpy	Bond dissociation enthalpy corresponds to breaking of a particular bond Average of dissociation enthalpies of a particular type of bonds in a molecule is known as bond enthalpy
Relation between ΔH reaction and bond enthalpy	$\Delta H = \sum$Bond enthalpies of reactants $- \sum$Bond enthalpies of products
Laws of thermochemistry	1. Lavosier and Laplace's Law 2. Hess's law of constant heat summation
Limitations of first law of thermodynamics	No explanation for direction of process
Spontaneous process	Process that takes place on its own whether it occurs slowly or fast does not matter
Second law of thermodynamics	Statement of second law
Entropy	Thermodynamic function given as $\Delta S = \frac{\delta Q_{rev}}{T}$ State function and extensive property Unit J K^{-1} mol^{-1}
Physical significance of entropy	It is a measure of disorder of system
Condition for spontaneity	$\Delta S_{total} = \Delta S_{(system)} + \Delta S_{(surroundings)} > 0$
Entropy changes during various processes	$\Delta S = C_V \ln \frac{T_2}{T_1} + R \ln \frac{V_2}{V_1}$ $\Delta S = C_p \ln \frac{T_2}{T_1} - R \ln \frac{P_2}{P_1}$
Isothermal reversible expansion of gas	$\Delta S_T = 2.303\ nR \log \frac{P_1}{P_2} = 2.303\ nR \log \frac{V_2}{V_1}$
Adiabatic process	$\Delta S = 0$
Phase transition	$\Delta S_{fusion} = \frac{\Delta H_{fusion}}{T_{(mp)}}$ $\Delta S_{vap} = \frac{\Delta H_{vap}}{T_{(bp)}}$ $\Delta S_{Transition} = \frac{\Delta H_{(transition)}}{T_{(transition)}}$
Trouton's Rule	$\frac{\Delta H_{vap}}{T_b} = 88$ J K^{-1} mol^{-1}
Entropy change without phase change	$\Delta S = ms \ln \frac{T_2}{T_1}$

Isochoric change	$\Delta S_v = C_v \ln \frac{T_2}{T_1}$
Isobaric change	$\Delta S_p = C_p \ln \frac{T_2}{T_1}$
Isothermal change	$\Delta S_T = R \ln \frac{V_2}{V_1} = R \ln \frac{P_1}{P_2}$
Tephigraph	Graph of entropy of a substance against temperature
Heat and work – Carnot cycle	$\eta = \frac{W}{Q_2} = \frac{Q_2 - Q_1}{Q_2} = \frac{T_2 - T_1}{T_2}$
Third law of thermodynamics	Statement of third law
Absolute entropy	$S_T = 2.303\ C_P \log T$
Gibbs free energy (G)	State function and extensive property Given as $G = H - TS$
Gibbs-Helmholtz equation	$(\Delta G)_{T,P} = \Delta H - T\Delta S$
Criteria for spontaneity	$(\Delta G)_{T,P} = 0$ for equilibrium $(\Delta G)_{T,P} < 0$ for spontaneous process
Thermodynamic criteria for spontaneity using Gibbs Helmholtz equation	$\Delta G = \Delta H - T.\Delta S \leq 0$

ΔH	ΔS	
–	+	Spontaneous at all temperatures
+	–	Non-spontaneous at all temperatures
+	+	Spontaneous when $T > \frac{\Delta H}{\Delta S}$
–	–	Spontaneous only when $T < \frac{\Delta H}{\Delta S}$

Standard free energy of formation	$\Delta G^\circ = \sum G^\circ_{f(\text{Products})} - \sum G^\circ_{f(\text{Reactants})}$
Relation involving ΔG, ΔG° and E°	$\Delta G^\circ = -nFE^\circ$ $\Delta G^\circ = -RT \ln K$; $\Delta G = \Delta G^\circ + RT \ln Q$ $\Delta G = RT \ln\left(\frac{Q}{K}\right)$
ΔG and pressure	$\Delta G = nRT \ln\left(\frac{P_2}{P_1}\right)$

ΔG and volume	$\Delta G = nRT \ln \left(\frac{V_1}{V_2} \right)$
Temperature co-efficient of free energy change	$\left[\frac{\partial(\Delta G)}{\partial T} \right]_P$
Variation of ΔG with temperature	$\left[\frac{\partial(\Delta G)}{\partial T} \right]_P = -\Delta S$
ΔG in terms of temperature co-efficient	$\Delta G = \Delta H + T\left[\frac{\partial(\Delta G)}{\partial T} \right]_P$

TOPIC GRIP

Subjective Questions

1. Calculate the work done in calories in the isothermal expansion of a system by one litre at the constant pressure of 740 mm T = 298 K.
2. ΔH_f values for $C_2H_{2(g)}$ and $C_2H_{6(g)}$ are + 47.0 k cal mol^{-1} and –29.3 k cal mol^{-1} respectively. Calculate for the reaction $C_2H_{2(g)} + 2H_{2(g)} \rightarrow C_2H_{6(g)}$ both ΔH and ΔU; T = 298 K.
3. 28 g of oxygen at 0°C and 5 atm pressure expand adiabatically so that the final pressure is one atm. Calculate the final temperature (γ for oxygen = 1.4).
4. 5 litres of oxygen at 2 atm expand adiabatically to 20 litres. Calculate the work done in litre atm $\gamma = \frac{C_p}{C_v} = 1.4$ for oxygen.
5. The enthalpy of combustion of ethylene, $C_2H_{4(g)}$ is –333.00 k cal mol^{-1}
   Enthalpies of formation of $CO_{2(g)}$ and $H_2O(\ell)$ are –97.3 and –68.4 k cal mol^{-1} respectively. T = 298 K. Calculate ΔH_f of $C_2H_{4(g)}$.
6. The enthalpy of neutralization of a weak monoprotic acid is –13.385 k cal eqt^{-1}. Assuming that the acid is 14% ionized in solution, Calculate the enthalpy of ionization of the acid. ΔH for the reaction $H_3O^+ + OH^- \rightarrow 2H_2O$ in solution is –13.7 k cal eqt^{-1}.
7. The enthalpies of neutralization of two weak monoprotic acids, HA and HB are –13350 cal eqt^{-1} and –11200 cal eqt^{-1} respectively. When one equivalent of NaOH in dilute solution is added to a mixture of 1 equivalent of HA and 1 equivalent of HB, 12900 calories are evolved. Find the number of moles of HA and HB that remain as unreacted in the mixture.
8. Calculate ΔH_f for CO(g) given ΔU_f = –29.60 k cal mol^{-1} given ΔH_f for $CO_{2(g)}$ = –97.30 k cal. Calculate ΔU for the reaction $CO_{(g)} + \frac{1}{2}O_{2(g)} \rightarrow CO_{2(g)}$. T = 298K.
9. Calculate ΔH : $C_2H_{2(g)} + H_2O_{(l)} \rightarrow CH_3–CHO_{(l)}$ given
   (i) $2C_{(s)} + H_{2(g)} \rightarrow C_2H_{2(g)}$ ΔH_f = 46.8 k cal mol^{-1}
   (ii) $H_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow H_2O_{(\ell)}$ ΔH = –68.4 k cal mol^{-1}
   (iii) $C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)}$; ΔH = –97.3 k cal mol^{-1}
   (iv) $CH_3–CHO_{(l)} + \frac{5}{2}O_{2(g)} \rightarrow 2CO_{2(g)} + 2H_2O_{(l)}$; ΔH = –282.0 k cal mol^{-1}
10. The enthalpies of neutralization of the two acids HA and HB are –13.350 and –13.7 k cal respectively. Calculate the enthalpy change of the reaction $NaA + HB \rightleftharpoons HA + NaB$.

Straight Objective Type Questions

Directions: **This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

11. 5 g of zinc(Atomic weight ≃ 65 assumed) react with excess of dilute acid at 15°C at the atmospheric pressure = 750 mm. Calculate in calories the work done against the atmosphere by the evolving hydrogen gas.
    (a) 22.3 (b) 32.3 (c) 44.0 (d) 34.4

12. ΔH_f values for $CO_{2(g)}$ and $N_2O_{(g)}$ are –94.00 k cal mol^{-1} and +17.70 k cal mol^{-1} respectively. Calculate enthalpy of combustion of carbon in Nitrous oxide.
(a) –129.4 k cal (b) –149.2 k cal (c) –214.9 k cal (d) +149.2 k cal

13. Given $CH_2Cl_{2(g)} + O_{2(g)} \rightarrow CO_{2(g)} + 2HCl_{(g)}$; $\Delta H = -106{,}800$ cal

ΔH_f values of $CO_{2(g)}$ and $HCl_{(g)}$ are –94,400 cal mol^{-1} and –22000 cal mol^{-1}

Calculate ΔH_f of $CH_2Cl_{2(g)}$
(a) –36100 cal mol^{-1} (b) –31600 cal mol^{-1} (c) +16300 cal mol^{-1} (d)+13600 cal mol^{-1}

14. A certain monoprotic acid shows enthalpy of neutralization = –13200 cal. Calculate its enthalpy of ionization assuming its degree of ionization to be 0.25. For HCl–NaOH neutralization, enthalpy value = –13700 cal eqt^{-1}.
(a) 550 cal (b) –550 cal (c) –766 cal (d) –667 cal

15. Given $Au(OH)_3 + 4HCl \rightarrow HAuCl_4 + 3H_2O$; $\Delta H = -23000$ cal

$Au(OH)_3 + 4HBr \rightarrow HAuBr_4 + 3H_2O$; $\Delta H = -36800$ cal

On mixing 1 mole of $HAuBr_4$ with 4 moles of HCl, 510 cal are absorbed. what % of $HAuBr_4$ is transformed into $HAuCl_4$ in this process?
(a) 3.7% (b) 4.5% (c) 2.1% (d) 5.4%

Assertion–Reason Type Questions

Directions: **Each question contains Statement-1 and Statement-2 and has the following choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False
(d) Statement-1 is False, Statement-2 is True

16. Statement 1
The I law of thermodynamics is often considered equivalent to the principle of conservation of energy.
and

Statement 2
In an isolated thermodynamic system the entropy either remains constant or increases but never decrease.

17. Statement 1
For one mole of an ideal gas $(C_p - C_v) = R$

i.e., $\left(\frac{\partial H}{\partial T}\right)_p - \left(\frac{\partial U}{\partial T}\right)_v = R$

and

Statement 2

For an ideal gas $C_p = \frac{5}{2}R + x$ and $C_V = \frac{3}{2}R + x$ where 'x' is a constant that depends on the atomicity of the gas.

18. Statement 1
The relation between ΔH and ΔU for a reaction (at T) is $\Delta H = \Delta U + (\Delta n)RT$ where, (Δn) = [Number of moles of products – number of moles of reactants] in the balanced equation.

and

Statement 2

In an equilibrium involving gases as (say) reactants or products or both, K_p does not change as the pressure changes but changes as the temperature changes.

19. Statement 1

When a gas undergoes an irreversible expansion the absorbed heat Q (when divided by T) does not yield the full entropy change, ΔS.

and

Statement 2

When ΔH has a negative value (ie, exothermic) in a reaction, the products are in every case more stable than the reactants.

20. Statement 1

Two liquids, like phenol and water, which are only partially miscible at lower temperatures become completely miscible and form a homogeneous solution on suitably raising the temperature.

and

Statement 2

In this case, change over from a 2-phase system to a one-phase homogeneous system is accompanied by an increase in entropy(i.e., positive ΔS). Hence since $\Delta G = \Delta H - T\Delta S$, ΔG may certainly become (and does become) negative above a certain temperature.



Linked Comprehension Type Questions

Directions: **This section contains 2 paragraphs. Based upon the paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

Passage I

Enthalpies of neutralization of strong acids and strong bases (say HCl and NaOH) is nearly a constant = –13700 cal eqt^{-1} for any strong acid and strong base. But for a weak acid and/or a weak base the value is different from –13700 cal. This is because the weak acid or base is ionized only to a small extent in dilute solution (unlike a strong acid/base). Salts formed from weak acids/bases are however fully ionized barring the (not-too-significant) complication of hydrolysis. The difference between the observed enthalpy of neutralization and 13700 is often taken as a measure of the enthalpy of ionization of the weak acid/base.

21. Enthalpies of neutralization of HCl and $ClCH_2–COOH$ by NaOH are –13700 cal eqt^{-1} and –13200 cal eqt^{-1} respectively. Calculate the enthalpy of ionization of $Cl–CH_2–COOH$ (in cal mol^{-1})

(a) –550 (b) +550 (c) +450 (d) +500

22. Enthalpies of neutralization of NaOH and NH_4OH by HCl are –13780 cal eqt^{-1} and –12,270 cal eqt^{-1} respectively. Calculate the enthalpy of ionization of NH_4OH.

(a) +1150 cal (b) +1510 cal (c) –1150 cal (d) –1015 cal

23. Enthalpies of neutralization for (i) BOH and HCl (ii) NaOH and HA and (iii) NaOH and HCl are –51.46, –50.63 and –55.90 kJ eqt^{-1} respectively. Calculate the neutralization enthalpy in kJ eqt^{-1} of BOH and HA.

(a) –46.19 (b) –41.96 (c) –49.16 (d) –41.69

Passage II

The bond dissociation energy is the energy needed to break a given bond in a compound. If there are several equivalent bonds in a compound, one may define a property called bond energy or bond enthalpy which is an average value for all the equivalent bonds. Example, CH_4 has four equivalent bonds C–H.

Now $CH_4 \longrightarrow \dot{C}H_3 + \dot{H}$ (homolytic fission) energy needed ΔH (bond dissociation energy)

$\dot{C}H_3 \longrightarrow \dot{C}H_2 + \dot{H}$; ΔH_2

$\dot{C}H_2 \longrightarrow \dot{C}H + \dot{H}$; ΔH_3

$\dot{C}H \longrightarrow C + \dot{H}$; ΔH_4

ΔH_1, ΔH_2, ΔH_3 and ΔH_4 are bond dissociation energies. Their magnitudes are different. But their average value, $\Delta H = \left[\frac{\Delta H_1 + \Delta H_2 + \Delta H_3 + \Delta H_4}{4}\right]$ is called bond energy (i.e., bond enthalpy). Bond energy for the same bond, (say C–H) is assumed to have the same magnitude in all the compounds wherein the bond occurs. Obviously, this is not quite correct and therefore the accepted bond energy values are approximate. A consequence of the above assumption is that bond energies are additive.This helps in the approximate calculation of many thermodynamic-thermochemical-properties of compounds for which the structures are either known or assumed. One very useful quantity is the enthalpy of sublimation of C(graphite) which is 712.96 kJ $mol^{-1} \simeq$ 170 k cal mol^{-1}

24. The enthalpy of formation ΔH_f for $CH_{4(g)} = -74.84$ kJ mol^{-1}

Given (i) $C_{(graphite)} \rightarrow C_{(g)}$; $\Delta H = +712.96$ kJ mol^{-1} and $H_{2(g)} \rightarrow 2\dot{H}_{(g)} = 436$ kJ mol^{-1}. Calculate the bond enthalpy of the C–H bond (energy needed per mole to break the bond)

(a) 541 kJ mol^{-1} (b) 451 kJ mol^{-1} (c) 415 kJ mol^{-1} (d) 514 kJ mol^{-1}

25. Given $CH_{4(g)} + 2O_{2(g)} \rightarrow CO_{2(g)} + 2H_2O_{(g)}$; $\Delta H = -802.24$ kJ.

The total energy absorbed in breaking the reactants into atoms.

i.e., into C + 4H + 4O = 2648.8 kJ. The bond enthalpy values for C = O and O–H bonds are in the ratio 1.713 : 1. The bond enthalpy values in kJ mol^{-1} of the C = O and O–H bonds are

(a) ~ 796.1 kJ mol^{-1}, ~464.7 kJ mol^{-1} (b) 967.1 kJ mol^{-1}, 446.7 kJ mol^{-1}

(c) 976.1 kJ mol^{-1}, 644.7 kJ mol^{-1} (d) 679.1 kJ mol^{-1}, 674.4 kJ mol^{-1}

26. Given : (i) $C_{(Graphite)} + 2H_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow CH_3{-}OH_{(g)}$; $\Delta H = -226.05$ kJ

(ii) $C_{(graphite)} \rightarrow C_{(g)}$; $\Delta H = 712.96$ kJ mol^{-1}

(iii) $H_{2(g)} \rightarrow 2H_{(g)}$, $\Delta H = 436$ kJ mol^{-1}

(iv) $O_{2(g)} \rightarrow 2O_{(g)}$; $\Delta H = 495$ kJ mol^{-1}

Calculate the total energy change per mole of the product formed when C, H and O atom combine and form the required bonds to yield methanol

(a) 2580.5 kJ (b) 2058.5 kJ (c) 5028.5 kJ (d) 5280.5 kJ

Multiple Correct Objective Type Questions

Directions: **Each question in this section has four suggested answers of which ONE OR MORE answers will be correct.**

27. The following thermodynamical quantities always have negative values

(a) enthalpy of formation of a definite pure compound

(b) enthalpy of neutralization of an acid and a base in aqueous solution

(c) enthalpy of solution of a solute in the solvent say water

(d) enthalpy of combustion of a definite pure compound

28. When a thermodynamic system undergoes a transition from one equilibrium state to another, the change of entropy, ΔS

(a) depends only on the initial and final states

(b) equals $\frac{Q}{T}$ when Q is the net thermal energy absorbed or thrown out, when the transition is reversible

(c) is never negative in an isolated system

(d) may have a net non zero value in a cyclic reversible process

29. The change in the Gibbs free energy, ΔG in a system, is

(a) zero once the system is in equilibrium

(b) equal to $\Delta H + T\left(\frac{\partial \Delta G}{\partial T}\right)_p$ in a general way

(c) never negative when ΔH and ΔS are both positive or both negative

(d) never positive when ΔH and ΔS have opposite signs



Matrix-Match Type Question

Directions: **Match the elements of Column I to elements of Column II. There can be single or multiple matches.**

30.

Column I	**Column II**
(a) enthalpy of formation of a component is related to	(p) component elements in standard states
(b) concept of additivity of component energy terms	(q) bond energy calculations
(c) enthalpy of neutralization	(r) Hess' law
(d) Ionization constants	(s) strengths of acids/bases

IIT ASSIGNMENT EXERCISE

Straight Objective Questions

Directions: **This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

31. A gas behaving ideally expands against a constant external pressure of P atm, from an initial volume V_1 litre to the final volume V_2 litres. Given $V_1 = 8.2$ litre and $V_2 = 16.4$ litres. If the work done per mole is 16.4 litre atm. Calculate the value of P.

(a) 1.95 atm (b) 2 atm (c) 2.15 atm (d) 1.75 atm

32. 40 g of magnesium metal are added to dil H_2SO_4 kept in an open beaker. Temperature = 27°C. Atmospheric pressure is 740 mm. Calculate the work done (in litre atm) against the atmosphere by the evolving hydrogen gas. [At. wt. of Mg = 24]

(a) 43.11 litre atm (b) 41.13 litre atm (c) 31.14 litre atm (d) 34.11 litre atm

33. Calculate W, the work done for the isothermal, reversible expansion of one mole of $CO_{2(g)}$ from an initial volume of 10 litres to the final volume of 50 litres at 300K. van der Waals constants for CO_2 are (i) a = 3.59 atm L^2 mol^{-2} and (ii) b = 0.0427 L mol^{-1}

(a) +3.92 kJ (b) −4.72 kJ (c) −3.99 kJ (d) +4.72 kJ

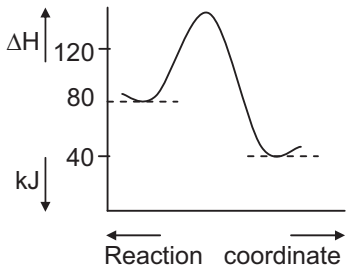
34. At 17°C $C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)}$; $\Delta H = -96.96$ k cal mol^{-1}

$$CO_{(g)} + \frac{1}{2} O_{2(g)} \rightarrow CO_{2(g)}; \quad \Delta H = -67.96 \text{ k cal}$$

Calculate in k cal ΔU_f for the formation of $CO_{(g)}$. i.e., $C_{(s)} + \frac{1}{2}O_{2(g)} \rightarrow CO_{(g)}$

(a) +22.90 (b) −22.90 (c) −29.29 (d) +19.29

35. Consider the diagram for the progress of a reaction ΔH for the reaction is (in kJ)



(a) −40 (b) +40 (c) −120 (d) +120

36. The difference between dQ_P and dQ_V is equal to (where the subscripts indicate constant P or V)

(a) $dU_P - dU_V$ (b) $dU_P - dU_V - PdV_P$ (c) $PdP_V - dU_V + VdP_V$ (d) $dU_P - dU_V + PdV_P$

37. Calculate ΔH° for the reaction : $CaC_{2(s)} + 2H_2O_{(l)} \rightarrow Ca(OH)_{2(s)} + C_2H_{2(g)}$ given ΔH_f values in kJ mol^{-1} for $CaC_{2(s)}$, $H_2O_{(l)}$, $Ca(OH)_2$ and $C_2H_{2(g)}$ are −62.76, −285.84, −986.59 and +226.75 respectively

(a) −125.4 (b) −154.2 (c) +142.5 (d) +214.5

38. In which of the following reactions is $\Delta H = \Delta U$
(a) Decomposition of $CaCO_{3(s)}$ to give $CaO_{(s)}$ and $CO_{2(g)}$
(b) Dissociating of $PCl_{5(g)}$ to give $PCl_{3(g)}$ and $Cl_{2(g)}$
(c) Combination of $SO_{2(g)}$ and Oxygen gas to yield $SO_{3(g)}$
(d) Formation of ethyne (g) from carbon (graphite) and $H_{2(g)}$

39. Calculate ΔH_f for diethyl ether, given its enthalpy of combustion is $-660.000\,k\,cal\,mol^{-1}$. ΔH_f values for $CO_{2(g)}$ and $H_2O_{(l)}$ are –97.000 and –68.400 k cal mol^{-1}.
(a) +76.0 k cal (b) –55.0 k cal (c) –70.0 k cal (d) +55.0 k cal

40. The molar heat of vapourization of $CH_3 - OH$ = 8890 cal mol^{-1} at 40°C. Take the vapour pressure = 700 mm. Calculate ΔU for vapourization per mol.
(a) 8426 cal (b) 8642 cal (c) 8264 cal (d) 8024 cal

41. An ideal gas undergoes reversible adiabatic compression from an initial pressure p_1 atm to a final pressure of p_2 atm. Given that $\frac{p_1}{p_2} = 3$ and the initial temperature is 300K, calculate the final temperature $\gamma = \frac{C_p}{C_v} = \frac{5}{3}$
(a) 173K (b) 213K (c) 203K (d) 193K

42. The work done by 'n' moles of an ideal gas in an irreversible adiabatic expansion is given by
(a) $W = \frac{nR}{\gamma - 1}(T_2 - T_1)$ (b) $W = nRP_{ext}\frac{(T_1P_2 - T_2P_1)}{P_1P_2}$
(c) $W = nC_p(T_2 - T_1)$ (d) $W = \frac{nR}{\gamma - 1}\frac{T_2 - T_1}{T_1T_2}$

43. The enthalpy of formation of $CO_{2(g)}$, $H_2O_{(l)}$ and propene are –193.5, –235.8 and 12.34 kJ mol^{-1} respectively. The enthalpy of isomerisation of cyclopropane to propene is –24 kJ mol^{-1}. The enthalpy change for the combustion of cyclopropane at 298K in kJ mol^{-1} is
(a) –2563.16 (b) –1300.24 (c) –1335.90 (d) –1324.24

44. Given $C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)}$; $\Delta H = -96960$ cal

$H_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow H_2O_{(l)}$; $\Delta H = -68360$ cal

Enthalpies of combustion of $C_6H_{6(l)}$ and $C_2H_{2(g)}$ are –799350 cal mol^{-1} and –310050 cal mol^{-1}. Calculate ΔH_f values for $C_6H_{6(l)}$ and $C_2H_{2(g)}$ in cal mol^{-1}
(a) 11250, 46700 (b) 11520, 45120 (c) 15210, 44210 (d) 12510, 47770

45. The enthalpy of hydration: $MgSO_4$ to $MgSO_4.H_2O$ is –6.98 k cal. Enthalpy of solution of anhydrous $MgSO_4$ = –20.28 k cal. Calculate in k cal the enthalpy of solution of $MgSO_4.H_2O$.
(a) +11.3 (b) –13.3 (c) –16.3 (d) –11.3

46. Enthalpy of solution of $BaCl_{2(s)}$ in excess of water is –2070 cal. Enthalpy of hydration: $BaCl_{2(s)} + 2H_2O \rightarrow BaCl_2.2H_2O$ is –6970 cal. What is enthalpy of solution of $BaCl_2.2H_2O$ in excess of water?
(a) –4700 cal (b) +4900 cal (c) –5200 cal (d) +5200 cal

47. The enthalpies of solution of anhydrous $MgSO_4$ and $MgSO_4.7H_2O$ in water are –20.28 k cal mol^{-1} and +3.80 k cal mol^{-1}. Calculate the enthalpy of hydration of $MgSO_4$ to $MgSO_4.7H_2O$ in k cal mol^{-1}.
(a) –24.08 (b) – 28.40 (c) +28.40 (d) +20.48

48. $CCl_{4(l)}$ and $CHCl_{3(l)}$ have boiling points = 76.7 °C, 61.5°C ΔH vapourization of CCl_4 = 30.71 kJ mol^{-1}. Calculate ΔH vapourization of $CHCl_3$.

(a) 29.38 kJ mol^{-1} (b) 23.98 kJ mol^{-1} (c) 32.89 kJ mol^{-1} (d) 38.92 kJ mol^{-1}

49. Calculate enthalpy of ionization of the weak base NH_4OH, assuming it to be unionized: Given enthalpies of neutralization of NH_4OH and NaOH by HCl are respectively −12.270 and −13.680 k cal eqt^{-1}

(a) +1.410 k cal (b) −1.140 k cal (c) +1.014 k cal (d) −1.014 k cal

50. The basic equation for acid base neutralization is $H^+_{(aq)} + OH^-_{(aq)} \rightarrow H_2O_{(l)}$; ΔH = −13.3 k cal. Given that −68.3 k cal mol^{-1} is ΔH_f for $H_2O_{(l)}$. Calculate the standard enthalpy of formation of $^-OH_{(aq)}$ ion taking the standard enthalpy of formation of $H^+_{(aq)}$ ion as zero

(a) −55.0 k cal mol^{-1} (b) −81.6 k cal mol^{-1} (c) +81.6 k cal mol^{-1} (d) −41.7 k cal mol^{-1}

51. Consider the equations

(i) $HCN_{(aq)} \rightarrow H^+_{(aq)} + CN^-_{(aq)}$ ΔH_1 = x k cal

(ii) $H^+_{(aq)} + O\,H^-_{(aq)} \rightarrow H_2O_{(l)}$ ΔH_2 = −13.3 k cal eqt^{-1}

(iii) $HCN_{(aq)} + OH^-_{(aq)} \rightarrow H_2O_{(l)} + CN^-_{(aq)}$; ΔH_3 = y k cal

Given that $\left(\frac{x}{y}\right)$ = −3.6 and y has a negative value

Calculate y and x in k cal

(a) −1.9, +6.5 (b) −1.9, +7.2 (c) −2.9, + 10.4 (d) −3.1, +11.5

52. Acids HA and HB show enthalpies of neutralization of −13550 cal eqt^{-1} and −14700 cal eqt^{-1}. When 1 eqt of NaOH was completely neutralized by a mixture of HA and HB, 14350 calories of thermal energy were evolved. Compare the ratio HA : HB in terms of equivalence?

(a) 1 : 1.19 (b) 1 : 2.11 (c) 1 : 2.45 (d) 1 : 2.29

53. $\Delta H°_f$ for $HI_{(g)}$ = 26.5 kJ. Bond dissociation energies of H_2 and HI are 436 kJ mol^{-1} and 295 kJ mol^{-1} respectively. Calculate the bond energy of the I – I bond

(a) 207 kJ (b) −270 kJ (c) +270 kJ (d) −315 kJ

54. Calculate enthalpy change for the reaction

$CH_3 - CH = CH_{2(g)} + HBr_{(g)} \rightarrow CH_3 - CHBr - CH_{3(g)}$ given the bond energy values in kJ : "C – C" : 348 ; "C = C" : 612; "C – Br" : 276; "C – H" : 412 and "H – Br" : 366

(a) −58 kJ (b) +48 kJ (c) −48 kJ (d) +64 kJ

55. Given ΔH_f for $CO_{2(g)}$ = −94.3 k cal mol^{-1} and the enthalpy of combustion of $CO_{(g)}$ = −67.4 k cal mol^{-1}. Given the enthalpy of sublimation of $C_{(graphite)}$ = 170 k cal mol^{-1}. Taking the energy for the dissociation $O_{2(g)} \rightarrow 2O$ as 117.4 k cal. Calculate the dissociation energy for $CO \rightarrow C + O$ (atom).

(a) +255.6 k cal (b) −275.5 k cal (c) +235.5 k cal (d) −225.6 k cal

56. $CH_{4(g)} + 2O_{2(g)} \rightarrow CO_{2(g)} + 2H_2O_{(l)}$ ΔH = −213 k cal mol^{-1}. The dissociation processes

(i) $CO_2 \rightarrow C + O_2$ and (ii) $H_2O \rightarrow 2H + \frac{1}{2}O_2$ absorb 265 and 160 k cal mol^{-1} respectively. Calculate the bond energy of the "C – H" bond.

(a) −73 k cal (b) +93 k cal (c) +73 k cal (d) −81 k cal

57. The solution of metallic sodium in excess of water releases 1880 calories per gram of sodium. Similarly, 1g of Na_2O on dissolving in excess of water releases 1020 calories. Calculate ΔH_f for Na_2O; Given $H_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow H_2O_{(l)}$; ΔH_f = −68.00 k cal mol^{-1}.

(a) −91240 cal (b) +94120 cal (c) −90412 cal (d) +90412 cal

58. Given $K + (H_2O + aq) \rightarrow KOH_{(aq)} + \frac{1}{2}H_{2(g)}$. $\Delta H = -48.10$ k cal

$H_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow H_2O_{(l)}$. $\Delta H = -68.40$ k cal mol^{-1}

$KOH_{(s)} + (aq) \rightarrow KOH_{(aq)}$ $\Delta H = -13.30$ k cal

Calculate ΔH_f for $K_{(s)} + \frac{1}{2}O_{2(g)} + \frac{1}{2}H_{2(g)} \rightarrow KOH_{(s)}$ (in k cal)

(a) +120.30 (b) +132.00 (c) −103.20 (d) −132.00

59. Calculate ΔH_f for the compound Al_2Cl_6 (anhydrous).

(i) $2Al_{(s)} + 6HCl_{(aq)} \rightarrow Al_2Cl_{6(aq)} + 3H_{2(g)}$; $\Delta H = -239.76$ k cal

(ii) $H_{2(g)} + Cl_{2(g)} \rightarrow 2HCl$ $\Delta H = -44.00$ k cal

(iii) $HCl + (aq) \rightarrow HCl_{(aq)}$ $\Delta H = -17.315$ k cal

(iv) $Al_2Cl_6 + (aq) \rightarrow Al_2Cl_{6(aq)}$ $\Delta H = -153.690$ k cal

(a) −321.960 k cal mol^{-1} (b) +123.690 k cal mol^{-1} (c) +132.960 k cal mol^{-1} (d) −162.390 k cal mol^{-1}

60. Given : $2C_6H_{6(l)} + 15O_{2(g)} \rightarrow 12CO_{2(g)} + 6H_2O_{(l)}$; $\Delta H = -1598700$ cal and $2C_2H_{2(g)} + 5O_{2(g)} \rightarrow 4CO_{2(g)} + 2H_2O_{(l)}$; $\Delta H = -620100$ cal

Calculate at 17°C, ΔU for the reaction, $3C_2H_{2(g)} \rightarrow C_6H_{6(l)}$

(a) +210917 cal (b) −129071 cal (c) −201917 cal (d) +197120 cal

61. Calculate ΔH_f for $As_2O_{3(s)}$, given the following data

(i) $As_2O_{3(s)} + (3H_2O + aq) \rightarrow 2H_3AsO_{3(aq)}$; $\Delta H = +7550$ cal

(ii) $As_{(s)} + \frac{3}{2}Cl_{2(g)} \rightarrow AsCl_{3(l)}$; $\Delta H = -71390$ cal

(iii) $AsCl_{3(l)} + (3H_2O + aq) \rightarrow H_3AsO_{3(aq)} + 3HCl_{(aq)}$ $\Delta H = -17580$ cal

(iv) $H_{2(g)} + Cl_{2(g)} \rightarrow 2HCl_{(g)}$ $\Delta H = -44000$ cal

(v) $HCl_{(g)} + [aq] \rightarrow HCl_{(aq)}$ $\Delta H = -17315$ cal

(vi) $H_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow H_2O_{(l)}$ $\Delta H = -68360$ cal

(a) +105468 cal (b) −145860 cal (c) −154680 cal (d) +164508 cal

62. Calculate formation enthalpy of $HNO_{2(aq)}$ [i.e., $\frac{1}{2}H_{2(g)} + \frac{1}{2}N_{2(g)} + O_{2(g)} + aq \rightarrow HNO_{2(aq)}$ $\Delta H = ?$)

(i) $(NH_4)NO_{2(s)} \rightarrow N_2 + 2H_2O$; $\Delta H = -71770$ cal

(ii) $H_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow H_2O_{(l)}$; $\Delta H = -68360$ cal

(iii) $N_{2(g)} + 3H_{2(g)} + (aq) \rightarrow 2NH_{3(aq)}$; $\Delta H = -40640$ cal

(iv) $NH_{3(aq)} + HNO_{2(aq)} \rightarrow (NH_4)NO_{2(aq)}$; $\Delta H = -9110$ cal

(v) $(NH_4)NO_2 + (aq) \rightarrow (NH_4)NO_{2(aq)}$; $\Delta H = +4750$ cal

(a) −30770 cal (b) +37700 cal (c) −37700 cal (d) +30070 cal

63. Calculate $\Delta H°_f$ for $CCl_{4(g)}$ given

(i) $CCl_{4(g)} + 2H_2O_{(g)} \rightarrow CO_{2(g)} + 4HCl_{(g)}$; $\Delta H = -173.2$ kJ

(ii) $H_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow H_2O_{(l)}$; $\Delta H = -286.0$ kJ

(iii) $C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)}$; $\Delta H = -394.2$ kJ

(iv) $\frac{1}{2}H_{2(g)} + \frac{1}{2}Cl_{2(g)} + (aq) \rightarrow HCl_{(aq)}$; $\Delta H = -165.5$ kJ

(v) $H_2O_{(l)} \rightarrow H_2O_{(g)} + (aq)$ $\Delta H = 40.7$ kJ

(vi) $HCl_{(aq)} \rightarrow HCl_{(g)} + (aq)$ $\Delta H = 73.2$ kJ

(a) +92.6 kJ (b) −72.6 kJ (c) +72.6 kJ (d) −99.6 kJ

64. Given (i) $S_{(s)} + O_{2(g)} + Cl_{2(g)} \rightarrow SO_2Cl_{2(g)}$; $\Delta H = -89800$ cal

(ii) $\frac{1}{2}H_{2(g)} + \frac{1}{2}Cl_{2(g)} + (aq) \rightarrow HCl_{(aq)}$; $\Delta H = -39300$ cal

(iii) $H_{2(g)} + S_{(s)} + 2O_{2(g)} + (aq) \rightarrow H_2SO_{4(aq)}$; $\Delta H = -210000$ cal

(iv) $H_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow = H_2O_{(l)}$; $\Delta H = -68000$ cal

Calculate ΔH for the reaction $SO_2Cl_{2(g)} + 2H_2O + (aq) \rightarrow H_2SO_{4(aq)} + 2HCl_{(aq)}$

(a) +31000 cal (b) −31000 cal (c) −62000 cal (d) +72000 cal

65. Given that the enthalpies of combustion of $CH_{4(g)}$ and $C_2H_{4(g)}$ are −212.000 k calmol^{-1}and −333.000 k cal mol^{-1} respectively and ΔH_f for H_2O = −68.00 k cal mol^1. Calculate ΔH for the reaction $2CH_{4(g)} \rightarrow C_2H_{4(g)} + 2H_{2(g)}$.

(a) +45.0 k cal (b) +54.0 k cal (c) −54.0 k cal (d) −65.0 k cal

66. In a reversible Carnot cycle, W = 150 kJ; Q(supplied) at T_1 = 225 kJ . Calculate (i) efficiency and (ii) the lower temperature, T_2K . (T_1 = 227°C)

(a) 0.5; 150 (b) 0.75; 170 (c) 0.75 ; 180 (d) 0.67 ; 299

67. An ideal gas at 3 atm pressure and 300K is expanded isothermally and reversibly to double its initial volume. Calculate in cal K^{-1} the entropy change for 10 moles of the gas

(a) 17.37 (b) 11.73 (c) 13.77 (d) 17.11

68. For the reaction $2Cl_{(g)} \rightarrow Cl_{2(g)}$, What are the signs of ΔH and ΔS?

(a) Both ΔH and ΔS are positive (b) ΔH negative, ΔS positive

(c) ΔH positive, ΔS negative (d) ΔH negative, ΔS negative

69. Which of the following changes has the sign of ΔS (+ve or −ve) opposite to that of the other three?

(a) $C_{(graphite)} \rightarrow C_{(diamond)}$ (b) $Br_{2(l)} \rightarrow Br_{2(g)}$

(c) $N_{2(g,\ 10\ atm)} \rightarrow N_{2(g,\ 1\ atm)}$ (d) $C_{(s)} + H_2O_{(g)} \rightarrow CO_{(g)} + H_{2(g)}$

70. Two moles of $CO_{2(g)}$ at a pressure of 5 atm and 25°C are heated to 125°C and compressed to 25 atm. C_p = 37.1 J K^{-1} mol^{-1} (assumed to be independent of temperature). Calculate ΔS

(a) +5.03 J K^{-1} (b) −5.30 J K^{-1} (c) −3.50 J K^{-1} (d) +3.50 J K^{-1}

71. Which of the following processes is accompanied by a decrease in entropy?

(a) Isothermal compression of an ideal gas

(b) A liquid heated from 298K to its boiling point at constant pressure

(c) Urea dissolved in water

(d) The fusion of a solid at atmospheric pressure

72. Indicate among the following the correct statement.

(a) The feasibility of a reaction is determined by the magnitude of the entropy.

(b) At 0°C the entropy of a perfectly crystalline substance is taken to be zero.

(c) For a reversible adiabatic volume change of an ideal gas pV^γ = constant for one mole.

(d) n-Hexane and 2,2-dimethyl butane have exactly the same magnitude of ΔH_f.

73. Given $2C_2H_{2(g)} + 5O_{2(g)} \rightarrow 4CO_{2(g)} + 2H_2O_{(l)}$, $\Delta G° = -2470.2$ kJ (T = 298K, p = 1 atm)
$\Delta G°_f$ of $CO_{2(g)}$ and $H_2O_{(l)}$ are -394.4 kJ mol^{-1} and -237.1 kJ mol^{-1} respectively. Calculate $\Delta G°_f$ of $C_2H_{2(g)}$
(a) -22.09 kJ mol^{-1} (b) 209.2 kJ mol^{-1} (c) $+22.09$ kJ mol^{-1} (d) -220.9 kJ mol^{-1}

74. Under the equilibrium conditions for the chemical reaction system: Reactants $\rightleftharpoons$ products.
(a) $\Delta H > T\Delta S$ (b) $\Delta H = T\Delta S$ (c) $\Delta H < T\Delta S$ (d) $T\Delta S > 0$

75. For the reaction A + 2B $\rightarrow$ C, calculate ΔS and the ratio $\left(\frac{\Delta S}{\Delta H}\right)$, given that both are positive and $\Delta H = 100$ k cal and that the forward reaction becomes spontaneous above 2000K
(a) 0.005 kcal K^{-1} ; 5×10^{-3} (b) 0.05 k cal K^{-1} ; 5×10^{-4}
(c) 0.005 k cal K^{-1}; 5×10^{-5} (d) 0.01 k cal K^{-}; 5×10^{-5}

76. For the reaction; $2A_{(g)} + 3B_{(g)} \rightarrow 2C_{(g)}$, $\Delta U°_{300K} = -5$ k cal, $\Delta S°_{300K} = -20$ cal K^{-1}. Predict the feasibility of the reaction by calculating $\Delta G°$
(a) +700 cal; not feasible (b) -800 cal ; feasible
(c) -650 cal ; feasible (d) +650 cal ; not feasible

77. Calculate the phase-transition temperature for $Sn_{(grey)} \rightleftharpoons Sn_{(white)}$.
$\Delta H° = 2.1$ kJ mol^{-1}. Entropy (abs) values : $Sn_{(white)} = 51.5$ J K^{-1} mol^{-1};
$Sn_{(grey)} = 44.1$ J K^{-1} mol^{-1}
(a) ~ 260K (b) ~ 290K (c) ~ 230K (d) ~ 280 K

78. $K_p = 1.6 \times 10^{-3}$ at 2025°C for the conversion $N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}$. If $N_{2(g)}$ and $O_{2(g)}$ are to be passed into a reaction chamber at 50 atm for each and NO gas is to be withdrawn at 30 atm from the chamber; What is ΔG for the process? Is the process feasible?
(a) +153.0 kJ ; not feasible (b) -153.0 kJ ; feasible (c) +130.5 kJ ; not feasible (d) +103.5 kJ ; not feasible

79. $Ag_2O_{(s)} \rightleftharpoons 2Ag_{(s)} + \frac{1}{2}O_{2(g)}$. $\Delta G° = +11.21$ kJ, T = 298K
Calculate partial pressure of O_2 under equilibrium.
(a) 1.61×10^{-3} atm (b) 1.06×10^{-3} atm (c) 1.16×10^{-4} atm (d) 1.26×10^{-5} atm

80. The temperature coefficient of the free energy change of a reaction at constant pressure i.e., $\left(\frac{\partial \Delta G}{\partial T}\right)_P = +4.8$ J deg^{-1}. $\Delta H = -100$ kJ. Calculate ΔG for the reaction at 303K
(a) +89.55 kJ (b) -89.55 kJ (c) -98.54 kJ (d) +58.95 kJ

81. A steel cylinder of capacity = 10 litres holds 20 moles of oxygen gas at 27°C. An imperfection in the valve of the cylinder allows the gas to leak out so slowly that the temperature does not change. External pressure = 1 atm. The cylinder finally contains the gas at 1 atm. Calculate in calories the total work done by the gas.
(a) ~12300 cal (b) ~10100 cal (c) ~13700 cal (d) ~11700 cal

82. One mole of an ideal gas at t°C expands (isothermally) and reversibly to double its initial volume. If the maximum work done by the gas is 1.719×10^{10} ergs mol^{-1}. Calculate t °C.
(a) 25 (b) 30 (c) 20 (d) 35

83. 3 moles of an ideal monoatomic gas ($\gamma = 1.67$) at 27°C and 1 atm pressure are subjected to a reversible adiabatic compression to half the initial volume. Calculate Q, W and ΔU.
(a) Q = 0, W = +1757 cal, $\Delta U = -7.35$ kJ (b) Q= 0, W = 1775 cal, $\Delta U = -7.43$ kJ
(c) Q =0, W = +1585 cal, $\Delta U = -6.63$ kJ (d) Q = 0, W = 775 cal, $\Delta U = -3.24$ kJ

84. 220 g of CO_2 are (i) heated from 27°C to 527°C at a constant pressure of 1 atm and then (ii) compressed isothermally to 100 atm. Calculate energy absorbed in k cal. Assume ideal behaviour (C_p = 5 cal deg^{-1} mol^{-1}).
(a) 19.35 k cal (b) 15.39 k cal (c) 13.95 k cal (d) 11.95 k cal

85. ΔH_f values for $C_2H_{4(g)}$ and $C_2H_{6(g)}$ are respectively +12.50 and −20.24 k cal mol^{-1}. Calculate ΔH for the reaction $C_2H_{4(g)} + H_{2(g)} \rightarrow C_2H_{6(g).}$
(a) −37.42 k cal (b) −42.37 k cal (c) −32.74 k cal (d) −43.72 k cal

86. Consider the (hypothetical) reaction
$2CH_3OH_{(\ell)} + C_2H_{4(g)} \rightarrow 2C_2H_5{-}OH_{(\ell)}$ Given ΔH_f values of $CH_3{-}OH_{(\ell)}$, $C_2H_{4(g)}$ and $C_2H_5{-}OH_{(\ell)}$ are −57.02 k cal mol^{-1}, +12.50 k cal mol^{-1} and −66.36 k cal mol^{-1} respectively. Calculate ΔH for the reaction.
(a) −38.11 k cal (b) −18.31 k cal (c) −13.81 k cal (d) −31.18 k cal

87. Reduction of $Al_2Cl_{6(s)}$ by metallic sodium to yield $NaCl_{(s)}$ and $Al_{(s)}$ is accompanied by the evolution of 256.8 k cal per mole of $Al_2Cl_{6(s)}$. ΔH_f of $NaCl_{(s)}$ is −98.2 k cal mol^{-1}. Calculate ΔH_f of $Al_2Cl_{6(s)}$.
(a) −233.4 k cal mol^{-1} (b) −332.4 k cal mol^{-1} (c) −243.3 k cal mol^{-1} (d) −432.2 k cal mol^{-1}

88. Enthalpy of combustion of $C_6H_{6(\ell)}$ at 298K to yield $CO_{2(g)}$ and $H_2O_{(\ell)}$ is ΔH = −781.0 k cal mol^{-1}. Calculate ΔU.
(a) −708.1 k cal mol^{-1} (b) −780.1 k cal mol^{-1}
(c) −801.7 k cal mol^{-1} (d) −810.7 k cal mol^{-1}

89. ΔH of vaporization of a liquid at 500 K and one atm pressure is 10 k cal mol^{-1}. Calculate ΔU for the vaporization of 3 moles of liquid at this temperature.
(a) 13 k cal (b) +11 k cal (c) +7 k cal (d) +9 k cal

90. The combustion of Benzoic acid at constant volume is accompanied by an evolution of 771.5 k cal mol^{-1}. 1.247 g of Benzoic acid at 25°C were burnt in a bomb calorimeter and showed a temperature rise = 2.870°. Calculate the total heat capacity of the calorimeter [Molar mass = 122.12 g mol^{-1}].
(a) 2400 cal deg^{-1} (b) 2745 cal deg^{-1} (c) 2975 cal deg^{-1} (d) 2225 cal deg^{-1}

91. Calculate the final temperature T_2 when n moles of an ideal mono atomic gas are compressed adiabatically and reversibly from 1 atm to 10 atm. Initial temperature T_1 = 25 °C
(a) 715 K (b) 517 K (c) 571 K (d) 748 K

92. According to Trouton's law, the molar entropy of vapourization of normal liquids ≃ 21 cal deg^{-1} mol^{-1}. The normal boiling point of chloroform = 61.2 °C. Estimate latent heat of vapourization per gram
(a) ~59 cal g^{-1} (b) ~64 cal g^{-1} (c) 49 cal g^{-1} (d) 69 cal g^{-1}

93. Given (i) $H_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow H_2O_{(g)}$; ΔH = −241.8 kJ mol^{-1}

(ii) $CO_{(g)} + \frac{1}{2}O_{2(g)} \rightarrow CO_{2(g)}$; ΔH = −283 kJ mol^{-1}. The heat evolved in the combustion of 112 Litres of water gas (at STP) containing equimolar mixture of $CO_{(g)}$ and $H_{2(g)}$ is

(a) 1132 kJ (b) 1312 kJ (c) 524.8 kJ (d) 283 kJ

94. The integral heat of solution of $CaCl_{2(anhydrous)}$ in 400 moles of water to yield $CaCl_{2(aq.)}$ is −19.3 k cal mol^{-1}. The corresponding value for $CaCl_2.6H_2O$ in 394 moles of water is ΔH = +3.9 k cal mol^{-1}. What is ΔH for the hydration?
$CaCl_{2(s)(anhydrous)} + 6H_2O_{(\ell)} \rightarrow CaCl_2.6H_2O_{(s)}$
(a) −23.2 k cal (b) −32.2 k cal (c) −20.3 k cal (d) −30.2 k cal

95. The enthalpy of hydration of $CuSO_{4(anhydrous)}$ to $CuSO_4.5H_2O$ has enthalpy change, ΔH_1 mol^{-1}(negative sign). Enthalpy change $CuSO_4.5H_2O$ to $CuSO_4$ aq is ΔH_2 mol^{-1} $\left(\Delta H_1 / \Delta H_2\right) = -7.79$. The ΔH value for $CuSO_{4(anhydrous)} \xrightarrow{aq}$ $CuSO_{4(aq)}$ is -16.43 k cal mol^{-1}. Calculate ΔH_1 and ΔH_2 in that order

(a) -17.85 k cal mol^{-1}, $+1.42$ k cal mol^{-1} (b) -19.85 k cal mol^{-1}, $+3.42$ k cal mol^{-1}

(c) -20.85 k cal mol^{-1}, $+4.42$ k cal mol^{-1} (d) -18.85 k cal mol^{-1}, $+2.42$ k cal mol^{-1}

96. Under identical conditions how many ml of 1 M KOH and 1 M H_2SO_4 solutions when mixed for a total volume of 100 mL produce the highest rise in temperature

(a) 50 mL KOH and 50 mL H_2SO_4 (b) 75 ml KOH and 25 ml H_2SO_4

(c) 67 ml KOH and 33 mL H_2SO_4 (d) 33 mL KOH and 67 ml H_2SO_4

97. ΔH for the ionization of NH_4OH = 5.52 kJ mol^{-1}. A dilute solution of sulphuric acid containing 98 g of it is completely neutralized with a dilute solution of ammonium hydroxide. Calculate the amount of heat liberated, given that the enthalpy of neutralization of a strong acid by a strong base = -57 kJ eqt^{-1}.

(a) -125 kJ (b) -93 kJ (c) ~(-103) kJ (d) -112 kJ

98. Given (i) $CaO_{(s)} + H_2O_{(\ell)} \rightarrow Ca(OH)_{2(s)}$ $\Delta H = -63$ kJ

Dilution of slaked lime (ii) $Ca(OH)_{2(s)} + aq \rightarrow Ca(OH)_{2(aq)}$ $\Delta H = -12.5$ kJ

Neutralizing (iii) $CaO_{(s)} + 2HCl_{(aq)} \rightarrow CaCl_{2(aq)} + H_2O$. $\Delta H = -192.5$ kJ

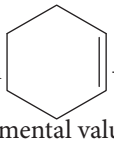
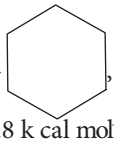
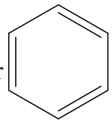
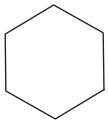
Calculate the enthalpy of neutralization of $Ca(OH)_{2(aq)}$ with dil HCl (per eqt)

(a) -117 kJ eqt^{-1} (b) -85.5 kJ eqt^{-1} (c) -73.5 kJ eqt^{-1} (d) -58.5 kJ eqt^{-1}

99. Calculate ΔH for the reaction : $CH_{4(g)} + Cl_{2(g)} \rightarrow CH_3Cl_{(g)} + HCl_{(g)}$. T = 298 K. Bond energies are

(i) "C–H" : 98 k cal mol^{-1} (ii) "Cl–Cl" : 57 k cal mol^{-1} (iii) "C–Cl" : 78 k cal mol^{-1} (iv) "H–Cl" : 102 k cal mol^{-1}

(a) -29 k cal (b) -25 k cal (c) -21 k cal (d) $+27$ k cal

100. For the hydrogenation  $+ H_{2(g)} \rightarrow$  , $\Delta H = -28.6$ k cal mol^{-1} at 82 °C. For  $+ 3H_2 \rightarrow$  (at 82 °C). The experimental value = -49.8 k cal mol^{-1}.

The calculated value of ΔH for the process & the resonance energy are respectively

(a) -57.2 k cal, differ 7.4 k cal (b) -85.8 k cal, differ 36 k cal

(c) -85.8 k cal, differ 135.6 (d) -85.8 k cal, differ 121.8 k cal

101. If the bond enthalpies of X_2, Y_2 and XY are in the ratio 1 : 2 : 4 and ΔH_f of XY = -200 kJ, find the bond enthalpy of Y_2

(a) 160 kJ (b) 80 kJ (c) 240 kJ (d) 200 kJ

102. If the mean bond enthalpies of "C = C", "C – H", "O = O", "C = O" and "O – H" are 615, 413, 493, 707 and 463 kJ mol^{-1} respectively, calculate ΔH for the combustion of ethene

(a) -934 kJ (b) -488 kJ (c) -1406 kJ (d) -1050 kJ

103. Given $H_2O_{(\ell)} \rightarrow H_{(g)} + OH_{(g)}$ $\Delta H = 129.22$ k cal mol^{-1}

$OH_{(g)} \rightarrow H_{(g)} + O_{(g)}$ $\Delta H = 101.5$ k cal mol^{-1}

$H_2O_{(\ell)} \rightarrow H_2O_{(g)}$ $\Delta H = 9.72$ k cal mol^{-1}

Find the bond energy of O–H in water

(a) 101.5 k cal mol^{-1} (b) 110.5 k cal mol^{-1} (c) 95.5 k cal mol^{-1} (d) 120.5 k cal mol^{-1}

104. Calculate from the following data the enthalpy of sublimation of carbon (graphite) in k cal mol^{-1}. ΔH_f for $C_2H_{6(g)} = -20.2$ k cal mol^{-1} at 298 K. Bond energy values are H–H : 103k cal mol^{-1}, C–C: 80 k cal mol^{-1} C–H : 98 k cal mol^{-1}

(a) ~185 (b) ~165 (c) ~170 (d) ~160

105. Enthalpies of combustion at 25 °C for n-butane and isobutane are –688 k cal mol^{-1} and –686.35 k cal mol^{-1} respectively. Calculate ΔH_f values of the compounds and the enthalpy of isomerisation of n-butane to isobutane. $\Delta H_{f(CO_2)} \simeq -96.0$ k cal mol^{-1}, $\Delta H_{f(H_2O)} \simeq -68.0$ k cal

(a) –32 k cal mol^{-1}, –34.65 k cal mol^{-1}, –2.65 k cal mol^{-1}
(b) –36 k cal mol^{-1}, –37.65 k cal mol^{-1}, –1.65 k cal mol^{-1}
(c) –33 k cal mol^{-1}, –35.65 k cal mol^{-1}, –2.65 k cal mol^{-1}
(d) –31 k cal mol^{-1}, –35.65 k cal mol^{-1}, –4.65 k cal mol^{-1}

106. Calculate ΔS when 2 moles of $SO_{2(g)}$ undergo a change of state from 25 °C and 1 atmospheric pressure to 323°C and 20 atmospheric pressure. Take C_p to have a mean value of 6.1 cal deg^{-1} mol^{-1}(hypothetical).

(a) +4.3 cal deg^{-1} (b) –3.5 cal deg^{-1} (c) –4.3 cal deg^{-1} (d) +2.2 cal deg^{-1}

107. A certain forward reaction becomes spontaneous above a certain temperature. Which of the following possibilities is/are implied. (i) ΔH and ΔS are both positive (ii) ΔH and ΔS are both negative (iii) ΔH is positive and ΔS is negative (iv) ΔH is negative and ΔS is positive

(a) both (i) and (ii) (b) (i), (ii) and (iii) (c) only (i) (d) (ii) and (iv)

108. $\Delta H°_f$ values of Benzene liquid and Benzene vapour are 49.0 and 82.9 kJ mol^{-1}. The corresponding entropy values are 173.3 J K^{-1} mol^{-1} and 269.3 J K^{-1} mol^{-1}. ΔG for the vapourisation of benzene at its normal boiling point = 80°C is

(a) 1.5 kJ (b) 0 (c) 67 kJ (d) 192 kJ

109. $2H_2O_{(\ell)} + 2SO_{2(g)} \rightarrow 2H_2S_{(g)} + 3O_{2(g)}$(hypothetical). ΔH and ΔS are 1053 kJ and 389 J K^{-1} at 300 K. The reaction as formulated above is

(a) spontaneous at 300 K
(b) at equilibrium at 300 K
(c) non spontaneous at 300 K
(d) spontaneous only much below 300 K

110.

Given	$\Delta H°_f$(kJ mol^{-1})	$\Delta_f G°$(kJ mol^{-1})	
$NH_{3(g)}$	–48.2	–16.6	
$HCl_{(g)}$	–95.3	–93.9	
$NH_4Cl_{(s)}$	–311.4	–203.9	T = 298 K

$\Delta S°$ for the reaction $NH_{3(g)} + HCl_{(g)} \rightarrow NH_4Cl_{(s)}$ is

(a) –0.2 kJ mol^{-1} K^{-1} (b) –0.248 kJ $mol^{-1}K^{-1}$ (c) –0.25 kJ mol^{-1} K^{-1} (d) –0.273 kJ mol^{-1} K^{-1}



Assertion–Reason Type Questions

Directions: **Each question contains Statement-1 and Statement-2 and has the following choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False
(d) Statement-1 is False, Statement-2 is True

111. Statement 1

Enthalpies of ionization of weak acids/weak bases may be positive or negative.

and

Statement 2

Ionization in aqueous solution involves several energy factors such as separation of ions with energy influenced by mean dielectric constant, hydration, hydrogen bonding ion pair formation, entropic(steric) factors etc.

112. Statement 1

Calculation of ΔH values for reactions is facilitated by a knowledge of mean bond enthalpy values of the relevant bonds and the principle of additivity.

and

Statement 2

The second law of thermodynamics enables a calculation of ΔS and ΔG values for reactions and implies that the entropy of all perfectly crystalline substances should tend to the same value at Lt $T \to 0$.

113. Statement 1

In a cyclic adiabatic process $\Delta S \geq 0$.

and

Statement 2

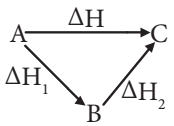
Principle of entropy conservation does not apply to irreversible processes.

Linked Comprehension Type Questions

Directions: **This section contains 1 paragraph. Based upon the paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

According to the Hess' law, when a certain chemical reaction may occur in either in one step or more than one step, the total enthalpy change is the same.

i.e.,



$\Delta H = \Delta H_1 + \Delta H_2$. Two principles are involved in this (i) additivity of enthalpy values (ii) constancy of ΔH for each step whatever the other steps may be. Inspite of its approximate nature the law has been of immense use in many situations wherein the above principles may be accepted as reasonable assumptions.

114. Given $P_{4(s)} + 6Cl_{2(g)} \to 4PCl_{3(\ell)}$; $\Delta H = -132.2$ kJ

$PCl_{3(\ell)} + Cl_{2(g)} \to PCl_{5(s)}$ $\Delta H = -141.4$ kJ

Calculate the enthalpy of formation of $PCl_{5(s)}$ per mole from elemental phosphorus and $Cl_{2(g)}$.

(a) −174.5 kJ (b) −154.7 kJ

(c) −147.5 kJ (d) +147.5 kJ

115. The dissociation energy of $CH_{4(g)}$ = 360 k cal mol^{-1} and that of $C_2H_{6(g)}$ is 620 k cal mol^{-1}. Calculate the bond enthalpy of the C–C bond mol^{-1} (i.e., energy to break the C–C bond per mole) in k cal.

(a) 80 (b) 260

(c) 130 (d) 180

116. One mole of a mixture of CH_4 and C_2H_6 on complete combustion evolved 1292 kJ. One mole of CH_4 on combustion evolves 890 kJ mol^{-1} and one mole of C_2H_6 yields on combustion 1560 kJ mol^{-1}. Calculate the ratio of CH_4 and C_2H_6 in the mixture.

(a) 3 : 4 (b) 2 : 3

(c) 2 : 1 (d) 1 : 2

Multiple Correct Objective Type Questions

Directions: **Each question in this section has four suggested answers of which ONE OR MORE answers will be correct.**

117. Identify the correct statements among the following
(a) Carbon (graphite) has enthalpy of sublimation approximately equal to 171 k cal mol^{-1}.
(b) Carbon (diamond), is thermodynamically more stable than carbon (graphite)since it is harder.
(c) Bond enthalpy of "C=C" bond is not double the bond enthalpy of C–C bond.
(d) The $\gamma = \left(C_p / C_v\right)$ ratio for a gas depends on the atomicity of the molecule.

118. Identify the correct statements among the following
(a) If a reaction has a negative value of ΔH i.e., exothermic it is spontaneous
(b) At the limit $T \to 0$ [where T is the Kelvin temperature] the entropy of a perfectly crystalline substance may be taken to be zero.
(c) For a reversible adiabatic expansion of one mole of an ideal gas $\left[P^{\gamma-1} / T^{\gamma}\right]$ is constant.
(d) n-Hexane and 2, 2-dimethyl butane have somewhat different enthalpies of formation ΔH_f.

119. Identify the correct statements among the following
(a) In the vapourization at 298 K of one mole of liquid into vapour, ΔH and ΔU both have the same sign.
(b) Enthalpy of combustion, ΔH of a compound always has a negative value.
(c) If ΔH and ΔS for a reaction both have negative values the reaction is spontaneous at all temperatures.
(d) In a system in chemical equilibrium at constant pressure and temperature change(infinitesimal) of dG = 0.

Matrix-Match Type Question

Directions: **Match the elements of Column I to elements of Column II. There can be single or multiple matches.**

120.

Column I	**Column II**
(a) Reversible adiabatic expansion of a gas related to	(p) Trouton's law
(b) $\left(\frac{\partial H}{\partial T}\right)_P - \left(\frac{\partial U}{\partial T}\right)_V$	(q) $\Delta S = \frac{Q_{rev}}{T}$
(c) Entropy change for the vaporization of a liquid	(r) $(C_p - C_v)$
(d) Always positive	(s) $W = \frac{nR}{(\gamma - 1)}\left(T_1 - T_2\right)$ for an ideal gas

ADDITIONAL PRACTICE EXERCISE

Subjective Questions

121. Calculate the work done in the conversion of 2 moles of liquid water to water vapour at its boiling point and 1 atm (Assuming ideal behaviour for water vapour).

122. Find the change in internal energy, when 0.5 mole of Ar having a specific heat at constant pressure of 20.814 J g^{-1} deg^{-1} is heated from 27°C to 31°C at constant volume.

123. One mole of methane occupying a volume of 15 L at 310K. What is the ratio of the work done by the gas under isothermal reversible and irreversible conditions such that the final volume of the gas is 20 L? The constant external pressure under which the irreversible expansion occurs is 0.5 atm.

124. The standard enthalpies of formation of $CO_{2(g)}$ and $H_2O_{(\ell)}$ are –393.5 kJ mol^{-1} and –285.8 kJ mol^{-1} respectively. The standard enthalpy of combustion of cyclopropane is –2091.3 kJ mol^{-1}. Calculate $\Delta_f H°$ of cyclopropane.

125. Diborane is a potential rocket fuel which undergoes combustion as follows.

$$B_2H_{6(g)} + 3O_{2(g)} \rightarrow B_2O_{3(s)} + 3H_2O_{(g)}$$

Calculate the enthalpy change for the combustion of diborane from the following data.

$2B_{(s)} + \frac{3}{2}O_{2(g)} \rightarrow B_2O_{3(s)}$ — (i) ($\Delta H = -1643$ kJ mol^{-1})

$H_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow H_2O_{(l)}$ — (ii) ($\Delta H = -420$ kJ mol^{-1})

$H_2O_{(l)} \rightarrow H_2O_{(g)}$ — (iii) ($\Delta H = 32$ kJ mol^{-1})

$2B_{(s)} + 3H_{2(g)} \rightarrow B_2H_{6(g)}$ — (iv) ($\Delta H = 16$ kJ mol^{-1})

126. What would be the change in entropy, when 40.0 g of helium is expanded to 1/10th of the initial pressure at 27°C? (assuming ideal behaviour).

127. The standard entropies of the various species in the reduction of Fe_2O_3 by Al are as follows.

Species	$Fe_2O_{3\,(s)}$	$Al_{(s)}$	$Al_2O_{3\,(s)}$	$Fe_{(s)}$
S° (J K^{-1} mol^{-1})	87.4	28.3	50.9	27.3

What is the standard entropy change of the reaction?

128. Two moles of liquid benzene at its freezing point 5.5°C are kept in contact with a water bath kept at 0°C until it freezes completely. Calculate the entropy change of benzene, of water bath and the ΔS_{net}. What is the conclusion? (Heat of fusion of benzene at freezing point = 127 J g^{-1}).

129. For the reaction $2A_{(g)} + 3B_{(g)} \rightarrow 2C_{(g)}$, $\Delta U^0_{300K} = -5$ kcal and $\Delta S^0_{300K} = -20$ cal K^{-1}.
Predict the feasibility of the reaction at the experimental condition.

130. The standard Gibbs free energy of formation, $\Delta_f G°$ of Fe_2O_3 and Al_2O_3 are –742 kJ and –1582 kJ respectively.
(i) Will the reduction of Fe_2O_3 by Al take place under these conditions?
(ii) Using the above data, find the temperature coefficient of $\Delta G°$ of the same reaction.
(iii) What is the $\Delta H°$ of the reaction?

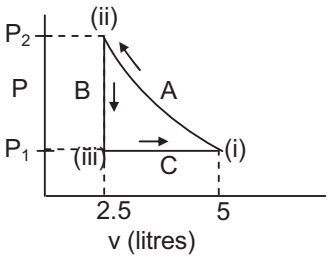
Straight Objective Type Questions

Directions: **This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

131. Metallic zinc dissolves in dil. H_2SO_4 liberating H_2 gas in an open vessel at 37°C. Calculate the work done by the liberated gas against the constant external pressure of 1 atm per mole of metal dissolved.

(a) 54.42 litre atm (b) 45.24 litre atm (c) 25.44 litre atm (d) 12.72 litre atm

132. Examine the adjoining diagram (cyclic process 1 mole ideal gas)
Process A: isothermal compression at 298K
Process B : cooling at constant volume
Process C: heating at constant pressure.
Calculate the total work done 'w' over the cycle



(a) 125 cal (b) 215 cal
(c) ~115 cal (d) 152 cal

133. $\Delta U°$ for the combustion of the hydrocarbon $C_4H_{8(g)}$ is $-x$ kJ mol^{-1}. $\Delta H°$ for the process is $-y$ k J mol^{-1}. Then (T = 298K, given x and y are +ve)

(a) x= y (b) x > y (c) x < y (d) x is exactly $\frac{y}{2}$

134. ΔH for the vapourization of a liquid at its normal boiling point 80°C is 8 k cal mol^{-1}. Calculate ΔU for the vapourization of 5 moles of the liquid under these conditions.

(a) ~4.47 k cal (b) ~7.46 k cal (c) ~3.76 k cal (d) ~5.76 k cal

135. The reaction of cyanamide ($H_2N - CN$) with oxygen was conducted in a bomb calorimeter (at constant volume) $\Delta U = -742.7$ kJ mol^{-1} at 298K. Calculate $\Delta H°_{298}$ for the reaction $NH_2 - CN_{(s)} + \frac{3}{2} O_{2(g)} \rightarrow N_{2(g)} + CO_{2(g)} + H_2O_{(l)}$.

(a) −714.5 kJ (b) −741.5 kJ (c) −571.5 kJ (d) −471.5 kJ

136. The specific heats of a gas at constant volume and constant pressure are respectively 0.033 cal deg^{-1} g^{-1} and 0.062 cal deg^{-1} g^{-1}. Calculate the molar mass in g mol^{-1}.

(a) 43 (b) 52 (c) 68.5 (d) 84

137. When 12.7 g of copper is heated from 27°C to 137°C, 128.7 cal is required. The value of C_p for copper in cal mol^{-1} K^{-1} is

(a) 5.85 (b) 11.7 (c) 23.4 (d) 58.5

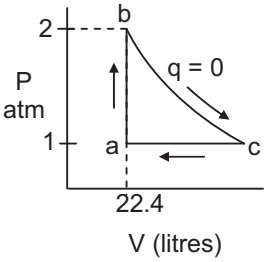
138. n moles of a gas are compressed reversibly and adiabatically to 10 times the initial pressure. $\gamma = 1.67$. Calculate the ratio $\left(\frac{T_2}{T_1}\right)$ of the final to the initial temperature.

(a) 2.512 (b) 2.215 (c) 1.522 (d) 1.255

139. Two moles of a gas of volume 50 litres and one atm pressure are compressed adiabatically and reversibly to 10 atm . Given $\left(\frac{T_1}{T_2}\right) = 0.4$. What is the atomicity of the gas.

(a) 1 (b) 2 (c) 3 (d) 4

140. Consider the (P.V) diagram for one mole of an ideal mono atomic gas. The work done over the entire cycle is close to



(a) 115 cal
(b) 125 cal
(c) 105 cal
(d) 120 cal

141. Consider the reaction $CH_{4(g,1atm)} + 4CuO_{(s)} \rightarrow CO_{2(1atm)} + 2H_2O_{(l)} + 4Cu_{(s)}$

ΔH_f values are : CO_2 : −94.05 k cal mol^{-1}; H_2O : −68.32 k cal mol^{-1}; $CH_{4(g,\ 1\ atm)}$: −17.89 k cal mol^{-1}. ΔH for the reaction = −62.4 k cal. Calculate the standard enthalpy of formation of CuO at 298K.

(a) −73.6 k cal mol^{-1} (b) −76.3 k cal mol^{-1} (c) −63.7 k cal mol^{-1} (d) −37.6 k cal mol^{-1}

142. The normal boiling point of acetone = 56°C. Calculate the specific enthalpy of vapourisation of acetone per gram . Molar mass = 58 g mol^{-1}. Assume the validity of Trouton's law: Molar entropy of vaporization for normal liquids at their normal boiling points is ≃ 88 J mol^{-1} K^{-1}.

(a) 359 J g^{-1} (b) 499 J g^{-1} (c) 529 J g^{-1} (d) 295 J g^{-1}

143. The "Heat of total cracking", H_{TC}, for a hydrocarbon is defined for the purpose of this question as ΔH_{298} for a reaction of the type: $C_nH_{m(g)} + \left(2n - \frac{m}{2}\right)H_2 \rightarrow nCH_{4(g)}$

Given (i) H_{TC} = −15.6 k cal for $C_2H_{6(g)}$ and −20.9 k cal for $C_3H_{8(g)}$ and
(ii) ΔH_f of $CH_{4(g)}$ = −17.9 k cal (all at 298K).

Calculate ΔH_{298} for $CH_{4(g)} + C_3H_{8(g)} \rightarrow 2C_2H_{6(g)}$

(a) 13.3 k cal (b) 12.3 k cal (c) 10.3 k cal (d) 11.3 k cal

144. Given (i) $HS^- \rightarrow H^+ + S^{2-}$; ΔH = 14.22 k cal and
(ii) $OH^- + HS^- \rightarrow S^{2-} + H_2O$; ΔH = + 0.86 k cal

Calculate ΔH for $H^+ + OH^- \rightarrow H_2O$

(a) −13.36 k cal (b) −13.93 k cal (c) −13.13 k cal (d) −13.85 k cal

145. Calculate the enthalpy ΔH°_f of the ion $X^-_{(aq)}$. Given

$H_{2(g)} + X_{2(g)} \rightarrow 2HX_{(g)}$; ΔH = −44.2 k cal

$HX + (aq) \rightarrow H^+_{(aq)} + X^+_{(aq)}$; ΔH = −17.9 k cal

Note : ΔH°_f of the ion $H^+_{(aq)}$ is taken to be zero.

(a) −35 k cal (b) +35 k cal (c) −40 k cal (d) +45 k cal

146. ΔH_f values for $C_2H_{4(g)}$, $CO_{2(g)}$ and $H_2O_{(l)}$ at 17°C are +2.710 k cal mol^{-1}, −96.960 k cal mol^{-1} and −68.360 k cal mol^{-1} respectively. Calculate the enthalpy of combustion of $C_2H_{4(g)}$ at 17°C at constant volume.

(a) −333.350 k cal mol^{-1} (b) −233.910 k cal mol^{-1}
(c) −293.310 k cal mol^{-1} (d) −392.130 k cal mol^{-1}

147. Enthalpy of solution of anhydrous $MgSO_{4(s)}$ in a large amount of water is −20.280 k cal mol^{-1}. For the processes $MgSO_{4(s)}$ (anhydrous) + $H_2O \rightarrow MgSO_4.H_2O_{(s)}$; ΔH_1 k cal mol^{-1}

$MgSO_4.H_2O_{(s)} \xrightarrow{aq} MgSO_{4(aq)}$; ΔH_2 k cal mol^{-1}

Given that $\Delta H_1 : \Delta H_2$ = 1 : 1.9. Calculate ΔH_1. (k cal mol^{-1})

(a) −6.99 (b) −7.32 (c) −5.87 (d) −8.57

148. The enthalpy change of a reaction in which 140 g of ethylene changes into polythene is (Given that bond energies of 'C = C' and 'C – C' are 540 and 322 kJ mol^{-1} at 298K respectively)
(a) –600 kJ (b) –620 kJ (c) –640 kJ (d) –660 kJ

149. The enthalpies of neutralization of (i) HCl with NaOH (both in dilute solution) and (ii) $CH_3 – COOH_{(aq)}$ with $NaOH_{(aq)}$ are –13680 cal eqt^{-1} and –13400 cal eqt^{-1}. Calculate the enthalpy of ionization of acetic acid per mole.
(a) –280 cal (b) –560 cal (c) +280 cal (d) +560 cal

150. The enthalpies of neutralization of (i) $HNO_{3(aq)}$ and (ii) $CHCl_2 – COOH_{(aq)}$ by $NaOH_{(aq)}$ are –13680 cal eqt^{-1} and –14830 cal eqt^{-1} respectively. When one eqt of $NaOH_{(aq)}$ is added to a dilute solution containing one eqt of HNO_3 and one eqt of $CHCl_2$ – COOH, 13960 cal are liberated. In what ratio [i.e., HNO_3 : $CHCl_2$ – COOH] is the base distributed between the acids.
(a) 2.7 : 1 (b) 3.5 : 1 (c) 2.5 : 1 (d) 3.1 : 1

151. Enthalpies of neutralization of acids HA and HB are –13.780 k cal eqt^{-1} and –14.280 k cal eqt^{-1} respectively. When one eqt of acid HA is added to one eqt of the salt NaB in dilute aqueous solution there is an absorption of 455 cal. How much of NaB is decomposed according to the equation NaB + HA → NaA + HB?
(a) 0.91 eqt (b) 0.75 eqt (c) 0.85 eqt (d) 0.81 eqt

152. Average bond enthalpies of H –H, Cl – Cl and HCl are 435, 243 and 431 kJ mol^{-1} respectively. Calculate ΔH_f of HCl.
(a)–24.7 kJ mol^{-1} (b) –92 kJ mol^{-1} (c) –123 kJ mol^{-1} (d) –49 kJ mol^{-1}

153. Calculate the resonance energy of benzene from the following data
(i) $\Delta H°_f$ of $C_6H_{6(g)}$ = 19.8 k cal mol^{-1}
(ii) $C_{(s)} \rightarrow C_{(g)}$ ΔH = 171.4 k cal mol^{-1}. Bond enthalpies are
(iii) H – H : 104.2
(iv) C – H : 99.4
(v) C = C : 143.6
(vi) C – C : 83.2 (all in k cal mol^{-1})
(a) 64.2 k cal (b) 84.2 k cal (c) 19.8 k cal (d) 44.4 k cal

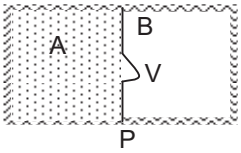
154. ΔH_f for $C_2H_5 – OH_{(l)}$ = –66 k cal mol^{-1}. ΔH_f of $CO_{2(g)}$ = –94 k cal mol^{-1}
ΔH_f of $H_2O_{(l)}$ = –68 k cal mol^{-1}. ΔH for combustion of $CH_3 – O – CH_{3(g)}$ to form $CO_{2(g)}$ and $H_2O_{(l)}$ is –348 k cal mol^{-1}. Calculate ΔH for the isomerization
$C_2H_5 – OH_{(l)} \rightarrow CH_3 – O – CH_{3(g)}$
(a) 27 k cal (b) 32 k cal (c) 18 k cal (d) 22 k cal

155. ΔH for the combustion of cyclopropane = –500 k cal mol^{-1}. ΔH_f for $CO_{2(g)}$ and $H_2O_{(l)}$ are –94 and –68 k cal mol^{-1} respectively. ΔH_f for propene is 4.9 k cal mol^{-1}. Calculate the enthalpy of isomerization of cyclopropane to propene.
(a) +12.3 k cal (b) +9.1 k cal (c) –12.3 k cal (d) +13.2 k cal

156. Given (i) ΔH for combustion of $C_2H_5 – OH_{(l)}$ = –341.800 k cal mol^{-1}
(ii) ΔH_f for $CO_{2(g)}$ = –96.000 k cal and (iii) ΔH_f for $H_2O_{(l)}$ is –68.000 k cal. Calculate ΔH_f for $C_2H_5 – OH_{(l)}$.
(a) –45.2 k cal mol^{-1} (b) –25.4 k cal mol^{-1} (c) +25.4 k cal mol^{-1} (d) –54.2 k cal mol^{-1}

157. A certain Carnot's cycle operates between the temperatures T_1 and T_2. $T_1 > T_2$. The efficiency i.e., $\left(\frac{T_1 - T_2}{T_1}\right) = \in$. If T_2 is raised by 100 degrees the efficiency becomes 0.86. Instead if T_1 is raised by 400 degrees. The efficiency becomes 1.2 $\in$. Calculate T_1 and T_2
(a) T_1 = 900K; T_2 = 400K (b) T_1 = 900K; T_2 = 300K
(c) T_1 = 800K ; T_2 = 300K (d) T_1 = 1000K ; T_2 = 600K

158. Box with two chambers of the same volume v litres totally adiabatically enclosed (see fig). Chamber A contains one mole of ideal gas. Chamber B is vacuous. The partition P and the walls are thermal insulators. The value v is kept closed at first. On opening v, the gas spreads out and finally occupies both chambers with uniform density. The initial temperature = 300K. Calculate (i) the final temperature (ii) change ΔU (iii) change ΔS



(a) less than 300K; ΔU is –ve; ΔS = R
(b) more than 300K; ΔU is +ve; ΔS = 2R
(c) 300K; ΔU = zero; ΔS = 0.693R
(d) more than 300K; ΔU is –ve; ΔS = R

159. Calculate the work done in joules in the isothermal and reversible expansion at 27°C of 5 moles of an ideal gas from an initial volume of V litres to a final volume of 10 V litres. What are the values of ΔU, ΔH and ΔS? [convention work done by the gas is taken to be negative).

(a) -2.271×10^3 J, zero, zero , 75.9 increase
(b) -2.531×10^5 J, zero, zero, 59.7 increase
(c) -2.872×10^4 J, zero, zero, 95.7 J deg^{-1} increase
(d) -2.011×10^3J, zero, zero, 57.9 increase

160. Estimate the normal boiling point of $PCl_{3(l)}$, given the free energy difference ΔG° between liquid and vapour = 4.6 kJ mol^{-1}; ΔS° = 94.6 J K^{-1}. Calculate ΔH° at 298K assuming $\Delta H°_{298}$ and ΔS° are independent of temperature.

(a) 74°C; ~32.76 kJ
(b) 78°C; ~23.67 kJ
(c) 64°C; ~26.73 kJ
(d) 84°C; ~16.38 kJ

161. For which of the following processes is the sign of ΔS different from the other three

(a) $C_{(s)} + H_2O_{(g)} \rightarrow CO_{(g)} + H_{2(g)}$
(b) $N_{2(g)} + 3H_{2(g)} \rightarrow 2NH_{3(g)}$
(c) $N_{2(g,10\ atm)} \rightarrow N_{2(g,\ 1\ atm)}$
(d) $Br_{2(l)} \rightarrow Br_{2(g)}$

162. An ideal gas expands isothermally at 25°C but somewhat irreversibly, producing 1000 cal of work. The entropy change is 10 cal deg^{-1}. Calculate the degree of irreversibility i.e., ratio $\frac{w_{actual}}{w_{rev}}$ where, w_{rev} is the reversible work for the isothermal expansion of the same gas to the same final volume.

(a) 0.68 (b) 0.34 (c) 0.17 (d) 0.51

163. ΔS for the vapourization of ether at its normal boiling point is 88.31 J K^{-1}. Enthalpy of vaporization = 27.2 kJ mol^{-1}. Calculate the normal boiling point of ether.

(a) 329K (b) 319K (c) 300K (d) 308K

164. One mole of an ideal monoatomic gas initially at STP expands isothermally and irreversibly to 44.8 litres. Calculate ΔS.

(a) 2.63 cal deg^{-1} (b) zero (c) –2.63 cal deg^{-1} (d) 1.37 cal deg^{-1}

165. The value of ΔS for a certain reaction = –4.80 J deg^{-1}. ΔG for the reaction at 303K = –98.55 kJ. Calculate ΔH for the reaction.

(a) –110 kJ (b) –100 kJ (c) –85 kJ (d) –120 kJ

166. For the reaction : $2A_{(g)} + B_{(g)} \rightarrow 2D_{(g)}$, $\Delta U°_{298} = -2.50$ k cal. $\Delta S°_{298} = -10.5$ cal K^{-1}. Calculate $\Delta G°_{298}$. Assume ideal gas behaviour

(a) –0.024 k cal (b) +0.024 k cal (c) +0.047 k cal (d) +0.033 k cal

167. Paraffin wax is heated and kept at its melting point, at 1 atm pressure. The system is subject to

(a) spontaneous change
(b) non-spontaneous change
(c) equilibrium situation
(d) decomposition

168. The equilibrium constant K_p for the gaseous system, $N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}$ at 2025°C is 1.6×10^{-3}. Calculate the value of ΔG°. (in k cal).

(a) 22.69 (b) 26.29 (c) 29.62 (d) 22.96

169. Consider $A_{2(g)} + B_{2(g)} \rightleftharpoons 2AB_{(g)}$. Given that ΔG° = –10.10 kJ at 500K. Calculate the value of K_P.

(a) 8.20 (b) 11.20 (c) 9.20 (d) 10.20

170. For the equilibrium $Ag_2O_{(s)} \rightleftharpoons 2Ag_{(s)} + \frac{1}{2}O_{2(g)}$, The partial pressure p_{O_2} of oxygen = 0.089 torr. Calculate $\Delta G°_{298}$

(a) 10.02 kJ (b) 12.25 kJ (c) 12.52 kJ (d) 11.21 kJ



Assertion–Reason Type Questions

Directions: **Each question contains Statement-1 and Statement-2 and has the following choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement- 2 is NOT a correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False
(d) Statement-1 is False, Statement-2 is True

171. Statement 1
When a system (gas) expands, the work done in expansion is always calculated as $\int_i^f PdV$ when i and f indicate initial and final states of the system.
and

Statement 2
In an isolated system, the change in entropy is never negative.

172. Statement 1
For an ideal gas (1 mole), the internal energy U, is independent of the volume V at a given temperature T.
and

Statement 2
For an ideal gas $\left(\frac{\partial U}{\partial V}\right)_T = 0$ and $dU = C_V dT$

173. Statement 1
For a monoatomic gas, the ratio $\gamma = \frac{C_P}{C_V}$ equals $\frac{5}{3} = 1.67$.
and

Statement 2
For a monoatomic gas, the molecules have only three translational degrees of freedom and $C_V = \frac{3}{2}R$ while $C_P - C_V = R$.

174. Statement 1
Heat of combustion is always negative for compounds.
and

Statement 2
In exothermic process, products are more stable than reactants.

175. Statement 1
Enthalpies of ionization of weak acids and weak bases are always positive in aqueous solution.
and

Statement 2
Separation of opposite charges (i.e., ions) involves absorption of energy.

176. Statement 1
The molar entropy of vaporization of many (normal) liquids at their normal boiling point is nearly 88 J deg^{-1} mol^{-1}.
and

Statement 2
By the above statement, one finds that water is not a normal liquid due to hydrogen bonding.

177. Statement 1
The bond enthalpy of a bond say C – Cl has exactly the same value in all compounds containing that bond.
and

Statement 2
In calculations of ΔH_f values of organic compounds of assumed structures, the enthalpy of sublimation of carbon (graphite) is taken to be ~171 k cal mol^{-1}.

178. Statement 1
Heat absorbed by a system at constant volume is used to increase the internal energy and not to perform work.
and

Statement 2
At constant volume, mechanical work done is zero.

179. Statement 1
The Kelvin temperature T of a crystalline solid tends to zero, one may reasonably assume that all perfectly crystalline substances tend to have the same constant value of entropy which may be taken as zero.
and

Statement 2
Entropy of a system may be interpreted as a measure of randomness of the components in a system, thermal, configurational etc.

180. Statement 1
Using G = H – TS and H = U + PV one may derive the equation dG = VdP – SdT [TdS = dU + PdV for a reversible infinitesimal process]
and

Statement 2

$$\left(\frac{\partial \Delta G}{\partial T}\right)_P = -\Delta S \text{ and } \Delta G = \Delta H - T\Delta S$$



Linked Comprehension Type Questions

Directions: **This section contains 3 paragraphs. Based upon the paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

Passage I
The enthalpy of formation of a compound is the enthalpy change, ΔH_f when one mole of a compound at its standard state is formed from its elements in their standard states. For elements in their standard states ΔH_f is taken to be zero.

Enthalpy of combustion of a compound is the enthalpy change, ΔH when one mole of a compound is burnt in excess of oxygen at constant pressure.

Enthalpy of a reaction, ΔH is similarly defined. The basis of such calculations is Hess's law. If a reaction occurs in several steps, then the total enthalpy change $\Delta H = \sum_i \Delta H_i$ where, ΔH_i is the enthalpy change in the ith step.

181. Consider the two reactions undergone by carbon:

(i) Water gas production : $C_{(s)} + H_2O_{(g)} \rightarrow \underbrace{CO_{(g)} + H_{2(g)}}_{\text{water gas}}$ $\Delta H = 131.3$ k J mol^{-1}

(ii) $C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)}$ $\Delta H = -393.5$ k J mol^{-1}

Assume that both reactions are conducted in such a way that the endothermicity of (i) is just balanced by the exothermicity of (ii). Neglecting heat losses and all other possible reactions such as $C_{(s)} + \frac{1}{2} O_{2(g)} \rightarrow CO_{(g)}$ or $CO_{(g)} + \frac{1}{2} O_{2(g)} \rightarrow CO_{2(g)}$ etc, calculate the ratio in which one mole of carbon is consumed in reactions (i) and (ii)

(a) 3 : 2 (b) 3 : 1 (c) 1 : 1 (d) 2 : 1

182. Given $C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)}$ $\Delta H = -94$ k cal mol^{-1} and $C_{(s)} + \frac{1}{2} O_{2(g)} \rightarrow CO_{(g)}$ $\Delta H = -26.4$ k cal mol^{-1}; in a certain experiment utilizing a limited quantity of oxygen $\Delta H = -57.5$ k cal per mole of carbon oxidized: Calculate the mole fraction of CO_2 formed.

(a) 0.41 (b) 0.51 (c) 0.50 (d) 0.46

183. Given $\Delta_f H$ values of $CO_{(g)}$ and $CO_{2(g)}$ are respectively -26.4 k cal mol^{-1} and -94 k cal mol^{-1} at 298K, calculate the ratio $\frac{-\Delta H}{n_{O_2}}$ i. e., $\frac{\text{heat liberated}}{\text{no of moles of } O_2 \text{ consumed}}$ when one mole of carbon is oxidized and 0.28 mole of carbon forms $CO_{(g)}$

(a) 87.3 k cal mol^{-1} of O_2 (b) 73.8 k cal mol^{-1} of O_2

(c) 63.7 k cal mol^{-1} of O_2 (d) 67.3 k cal mol^{-1} of O_2

Passage II

The enthalpy of neutralization of aqueous strong acids (eg $HCl_{(aq)}$) and strong bases (e.g., $NaOH_{(aq)}$) is always very nearly a constant $\simeq -13.7$ k cal eqt^{-1}. This is because, water is a leveling solvent and all strong acids by "total" ionization yield H_3O^+ ions and strong bases yield OH^- ions. Thus in all such cases, the neutralization reaction is $H_3O^+ + OH^- \rightarrow 2H_2O$. In weak acids or weak bases however, the enthalpy of neutralization differs from -13.7 k cal eqt^{-1} because in such cases neutralization entails ionization and $\Delta H_{(ionization)}$ gets added to -13.7 k cal eqt^{-1}.

184. The enthalpy of neutralization of a weak monoprotic acid HA is -13680 cal eqt^{-1}. Assume that the acid HA is 20% ionized in aqueous solution. Calculate the enthalpy of ionization of HA.

(a) 5 cal mol^{-1} (b) 15 cal eqt^{-1} (c) 25 cal mol^{-1} (d) 100 cal mol^{-1}

185. Two acids HA and HB in a mixture are neutralized by NaOH. The mixture contains 1 eqt of each of HA and HB in dilute solution . So this one eqt of NaOH in dil solution is added. Enthalpies of neutralization of HA and HB are -13680 cal eqt^{-1} and -13280 cal eqt^{-1}. If the base distributes itself in the ratio 9 : 1 between HA and HB, how much of heat is evolved in the experiment.

(a) -13680 cal (b) -13640 cal (c) -13600 cal (d) -13000 cal

186. As in the preceding problem, HA and HB have enthalpies of neutralization of -13680 cal eqt^{-1} and -13280 cal eqt^{-1}. One eqt of HA is added to a dilute solution of 1eqt of NaB (the salt) in dilute solution. What thermal change is observed–absorption or evolution of heat and by how many calories?

(a) 450 cal absorbed (b) 400 cal evolved (c) 550 cal absorbed (d) 450 cal evolved

Passage III

The first law of thermodynamics has often been described as a restatement of the principle of conservation of energy. It admits the interconversion of different forms of energy such as mechanical, thermal, electrical, etc but has nothing to say about the question whether a given amount of thermal energy, absorbed from a source, can be totally converted into mechanical energy (with no changes elsewhere). It is the II law of thermodynamics that asserts that 100% conversion of other forms of energy (say mechanical energy) into thermal energy is achievable, the opposite process 100% conversion of thermal into mechanical energy is not possible without leaving changes elsewhere. This is done through the concept of entropy change, $\Delta S = \frac{Q}{T}$ where, Q is the thermal energy change reversibly made. Associated concepts are ΔG i.e., Gibbs free energy change and ΔH i.e., Helmholtz free energy change. For a system in eqm. $\Delta G = 0$. Further, the general equation is $\Delta G = \Delta H - T\Delta S$. As a system approaches the equilibrium state from some other initial state free energy decreases i.e., ΔG is negative. Free energy reaches a minimum value at equilibrium under given conditions while entropy tends to reach a maximum value.

187. Calculate ΔG for the reaction $C_6H_{12}O_{6(s)} + 6O_{2(g)} = 6CO_{2(g)} + 6H_2O_{(l)}$ at 25°C. Given $\Delta H = -673$ k cal mol^{-1} and $\Delta S = 60.4$ cal deg^{-1}.

(a) −619 k cal (b) −916 k cal (c) −961 k cal (d) −691 k cal

188. Given the information that a reaction with $\Delta H = +37$ k cal is spontaneous only above 97°C, calculate ΔG and ΔS at this temperature.

(a) $\Delta S = +100$ cal K^{-1}; $\Delta G = 0$ (b) $\Delta S = -100$ cal K^{-1}; ΔG is +ve

(c) $\Delta S = -100$ cal K^{-1}; ΔG is −ve (d) $\Delta S = 0$; ΔG is +ve

189. Calculate the molar entropy of vapourization of liquid water at its normal boiling point, latent heat of vapourization is 540 cal g^{-1}.

(a) ~23 cal deg^{-1} mol^{-1} (b) ~29 cal deg^{-1} mol^{-1} (c) ~26 cal deg^{-1} mol^{-1} (d) ~20 cal deg^{-1} mol^{-1}

Multiple Correct Objective Type Questions

Directions: **Each question in this section has four suggested answers of which ONE OR MORE answers will be correct.**

190. The specific heats of a gas at constant volume and constant pressure are 0.038 cal deg^{-1} g^{-1} and 0.062 cal deg^{-1} g^{-1}.

(a) The atomicity of the gas molecule is 2.

(b) The molar mass of the gas is 83 g mol^{-1}.

(c) C_p for the gas is 5 cal deg^{-1} mol^{-1}.

(d) The gas has such a low critical temperature that it cannot be condensed to form a liquid above 10K.

191. Identify the correct statement(s)

(a) Work done in the isothermal, reversible expansion of one mole of a van der Waals gas from V_1 litres to V_2 litres is $RT \ln \left(\frac{V_2 - b}{V_1 - b}\right)$.

(b) For every chemical reaction at equilibrium, Gibbs free energy of reaction is zero.

(c) At constant temperature and pressure chemical reactions are spontaneous in the direction of decreasing free energy.

(d) The enthalpy change accompanying the reaction, $C_{(diamond)} + O_{2(g)} \rightarrow CO_{2(g)}$ cannot be considered as the standard enthalpy of formation of CO_2.

192. Identify the correct statement(s).

(a) In the combustion of (i) butene-1 and (ii) butene-2 (T = 298K), both $\Delta H°$ and $\Delta U°$ are negative.

(b) In the above combustion $|\Delta H°|$ is numerically greater than $\Delta U°$.

(c) In the above combustion $(\Delta H° - \Delta U°) \simeq +1.200$ k cal.

(d) In the above combustion $\Delta H° = \Delta U° - 3RT$.

193. Identify the correct statement(s).

(a) ΔH for the neutralization of $KOH_{(aq)}$ and $HNO_{3(aq)} \simeq -13.7$ k cal eqt^{-1}.

(b) ΔH for the neutralization of a weak acid and $Ba(OH)_{2(aq)}$ is different from -13.7 k cal eqt^{-1}.

(c) When both ΔH and ΔS are positive, a reaction becomes spontaneous above a certain temperature.

(d) When ΔH is negative for any reaction, ΔG is also negative at all temperatures.

194. A gas is allowed to expand adiabatically.

(a) In the above process the internal energy of the gas decreases always.

(b) In all such expansions the differential element $TdS \geq dU + PdV$.

(c) The efficiency of a Carnot's cycle is always < 1.

(d) Any reversible cycle to which an ideal gas is subjected may be regarded as a superposition of very small Carnot's cycles.

195. Identify the correct statement(s).

(a) The vapourization of a liquid at temperature, T, is associated with work done, RT

(b) The internal energy change per mole when liquid water changes to steam at its normal boiling point is 8.97 k cal.

(c) The molar entropy of vapourization of many normal liquids is ~88 J deg^{-1} mol^{-1}.

(d) Since heat and work are path properties, the sum of heat + work is also a path property.

196. Identify the correct statement(s).

(a) The ratio $\frac{C_P}{C_V}$ for CO_2 gas is 1.67.

(b) The work done in isothermal (W_1), isobaric (W_2), isochoric (W_3) and adiabatic (W_4) processes follows the order, $W_2 > W_1 > W_4 > W_3$.

(c) It is given that, in a process for an ideal gas dW = 0 and dq = –ve . Then the process is accompanied by decrease in internal energy.

(d) For a system in equilibrium $\Delta G^\circ = -RT \ln K_P$.

197. Identify the correct statement(s).

(a) Relation of ΔG to ΔG° in a reaction system is $\Delta G = \Delta G^\circ - RT \ln Q$ where, Q is the reaction quotient.

(b) For an ideal gas $\Delta G = nRT \ln \left(\frac{P_2}{P_1}\right)$ for n moles at constant temperature.

(c) $\left(\frac{\partial \Delta G}{\partial P}\right)_T = V$

(d) The entropy change during an isothermal reversible expansion of an ideal gas is $R \ln \left(\frac{V_2}{V_1}\right)^{1/n}$ where, n is the no. of moles of the gas.

Matrix-Match Type Questions

Directions: **Match the elements of Column I to elements of Column II. There can be single or multiple matches.**

198.

Column I	**Column II**
(a) Enthalpy of formation	(p) nearly constant for strong acids and strong bases
(b) Additivity of enthalpy values	(q) Enthalpy of ionization
(c) Principle of energy conservation applied	(r) bond energy values
(d) Enthalpy of neutralization	(s) Hess's law

199.

Column I	**Column II**
(a) Adiabatic expansion of a gas	(p) $\left(\frac{\partial G}{\partial T}\right)_P = -S$
(b) $\left(\frac{\partial H}{\partial T}\right)_P$	(q) Isoentropic
(c) $\Delta G = \Delta H + T\left[\frac{\partial \Delta G}{\partial T}\right]_P$	(r) C_P
(d) $\frac{5}{2}R$ for monoatomic ideal gas	(s) $TV^{\gamma - 1}$ = constant

200.

Column I	**Column II**
(a) $-\left(\frac{\partial \Delta G}{\partial T}\right)_P$	(p) (ΔS)
(b) $\frac{\Delta H - \Delta G}{T}$	(q) $-T\Delta S_{Total}$
(c) nearly constant (values) for the vapourization of normal liquids per mole at their normal boiling points	(r) $RT \ln Q + \Delta G^\circ$
(d) ΔG	(s) Trouton's law

SOLUTIONS

ANSWERS KEYS

Topic Grip

1. 23.565 cal

2. $\Delta H = -76.3$ k cal mol^{-1}
$\Delta U = -75.12$ k cal mol^{-1}

3. 172.4K

4. 10.64 L atm

5. 1600 cal

6. 366 cal mol^{-1}

7. Unreacted HA = 0.2 moles
Unreacted HB = 0.79 moles

8. –68.3 k cal

9. –27.8 k cal

10. 0.35 k cal

11. (c) **12.** (a) **13.** (b)
14. (d) **15.** (a) **16.** (b)
17. (a) **18.** (d) **19.** (c)
20. (a) **21.** (d) **22.** (b)
23. (a) **24.** (c) **25.** (a)
26. (b) **27.** (b), (d)
28. (a), (b), (c)
29. (a), (b)
30. (a) → (p), (q), (r)
(b) → (q), (r)
(c) → (s)
(d) → (s)

IIT Assignment Exercise

31. (b) **32.** (b) **33.** (c)
34. (c) **35.** (a) **36.** (d)
37. (a) **38.** (d) **39.** (c)
40. (c) **41.** (d) **42.** (b)
43. (d) **44.** (d) **45.** (b)
46. (b) **47.** (a) **48.** (a)
49. (a) **50.** (a) **51.** (c)
52. (d) **53.** (a) **54.** (a)
55. (a) **56.** (b) **57.** (a)
58. (c) **59.** (a) **60.** (b)
61. (c) **62.** (a) **63.** (d)
64. (c) **65.** (a) **66.** (d)
67. (c) **68.** (d) **69.** (a)
70. (b) **71.** (a) **72.** (c)
73. (b) **74** (b) **75.** (b)
76. (b) **77.** (d) **78.** (d)
79. (c) **80.** (c) **81.** (d)
82. (a) **83.** (c) **84.** (a)
85. (c) **86.** (d) **87.** (b)
88. (b) **89.** (c) **90.** (b)
91. (d) **92.** (a) **93.** (b)
94. (a) **95.** (d) **96.** (c)
97. (c) **98.** (d) **99.** (b)
100. (b) **101.** (a) **102.** (a)
103. (b) **104.** (c) **105.** (b)
106. (b) **107.** (c) **108.** (b)
109. (c) **110.** (b) **111.** (a)
112. (c) **113.** (a) **114.** (a)
115. (a) **116.** (b)
117. (a), (c), (d)
118. (b), (c), (d)
119. (a), (b), (d)
120. (a) → (s)
(b) → (r)
(c) → (p), (q)
(d) → (r)

Additional Practice Exercise

121. –6.2 kJ

122. 1.6 kJ

123. $\frac{W_{rev}}{W_{irrev}} = 2.93$

124. 53.4 kJ

125. –2823 kJ mol^{-1}

126. 191.5 J K^{-1}

127. –38.5 J K^{-1}

128. $\Delta S_{benzene} = -71.1$ J K^{-1}
$\Delta S_{water\ bath} = 72.6$ J K^{-1}
$\Delta S_{net} = 1.5$ J K^{-1}

129. Spontaneous at 300K

130. (i) ΔG = –ve, feasible
(ii) 38.5 J K^{-1}
(iii) –828.5 kJ

131. (c) **132.** (c) **133.** (c)
134. (a) **135.** (b) **136.** (c)
137. (a) **138.** (a) **139.** (a)
140. (a) **141.** (d) **142.** (b)
143. (c) **144.** (a) **145.** (c)
146. (a) **147.** (a) **148.** (b)
149. (c) **150.** (d) **151.** (a)
152. (b) **153.** (d) **154.** (d)
155. (b) **156.** (d) **157.** (c)
158. (c) **159.** (c) **160.** (a)
161. (b) **162.** (b) **163.** (d)
164. (d) **165.** (b) **166.** (d)
167. (c) **168.** (c) **169.** (b)
170. (d) **171.** (b) **172.** (a)
173. (a) **174.** (a) **175.** (d)
176. (a) **177.** (d) **178.** (a)
179. (b) **180.** (a) **181.** (b)
182. (d) **183.** (a) **184.** (c)
185. (b) **186.** (b) **187.** (d)
188. (a)
189. (c)
190. (b), (c)
191. (b), (c), (d)
192. (a), (b), (d)
193. (a), (b), (c)
194. (b), (c), (d)
195. (a), (b), (c)
196. (b), (c), (d)
197. (b), (c)
198. (a) → (r), (s)
(b) → (q), (r), (s)
(c) → (q), (r), (s)
(d) → (p), (q)
199. (a) → (q), (s)
(b) → (r)
(c) → (p)
(d) → (r)
200. (a) → (p)
(b) → (p)
(c) → (p), (s)
(d) → (q), (r)

HINTS AND EXPLANATIONS

Topic Grip

1. work done $= -P\Delta V = -\left(\frac{740}{760} \times 1\right)$ litre atm

$= -\left(\frac{740}{760} \times \frac{1.987}{0.0821}\right)$ cal

$= -23.565$ cal

2. $C_2H_{2(g)} + 2H_{2(g)} = C_2H_{6(g)}$

$+47.0 \quad$ zero $\quad -29.3$ k cal

$\therefore \quad \Delta H = (-29.3 - 47.0)$ k cal $\quad = -76.3$ k cal

$\Delta n = 1 - 3 = -2$

$\Delta H = \Delta U + RT\Delta n$

$-76300 = \Delta U - 2 \times 1.987 \times 298$

$\therefore \quad \Delta U = -75116$ cal $= -75.12$ k cal mol^{-1}

3. $\left.\begin{matrix} PV^{\gamma} = \text{const}_1 \\ TV^{(\gamma-1)} = \text{const}_2 \end{matrix}\right\} \Leftrightarrow \left(P^{\gamma-1}/T^{\gamma}\right) = \text{constant}$

[Both P and I are intensive variables]

$\therefore \quad \left(P_1^{\gamma-1}/T_1^{\gamma}\right) = \left(P_2^{\gamma-1}/T_2^{\gamma}\right)$

$\therefore \quad \left(\frac{T_2}{T_1}\right)^{\gamma} = \left(\frac{P_2}{P_1}\right)^{\gamma-1}$

$\left(T_2/273\right)^{1.4} = \left(1/5\right)^{0.4}$

$\therefore \quad 1.4 \log\left(T_2/273\right) = 0.4 \log(0.2)$

calculation (taking finally antilog) gives $T_2 = 172.4$ K which is -100.6 °C

4. $p_1V_1 = nRT_1 = 2 \times 5 = 10$ litre atm

Also $T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1}$

$\therefore \quad \frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1} = \left(\frac{5}{20}\right)^{0.4} = 0.5743$

$= \left(nRT_2/nRT_1\right)$

$nRT_2 = 0.5743 \times 10 = 5.743$ litre atm

work done $= \left(\frac{nRT_1 - nRT_2}{\gamma - 1}\right)$

$= \left(\frac{10 - 5.743}{0.4}\right)$ litre atm

$= 10.6425$ litre atm

5. $C_2H_{4(g)} + 3O_{2(g)} = 2CO_{2(g)} + 2H_2O_{(l)}$

$x \quad$ zero $\quad -2 \times 97.3 \quad -2 \times 68.4$(in k cal)

$\Delta H = -333.0$ k cal

$\therefore \quad -194.6 - 136.8 - x = -333.0$

$\therefore \quad x = +1.6$ k cal

i.e., $\quad +1600$ cal

6. Enthalpy of ionization for the partially ionized acid = (13700 − 13385) cal = 315 cal (for the 86% ionization of the unionized part)

$\therefore$ enthalpy of ionization (total)

$= \left[315 \times \frac{100}{86}\right]$ cal $= 366$ cal mol^{-1}

7. $13350x + 11200(1 - x) = 12900$ cal

$\therefore \quad x = 0.79$

$\therefore \quad (1 - x) = 0.21$

No. of moles of HA that remains as unreacted $= 0.21$

No. of moles of HB that remains as unreacted $= 1 - 0.21 = 0.79$

8. $C_{(s)} + \frac{1}{2}O_{2(g)} = CO_{(g)}$

$\Delta U_f = -29{,}600$ cal

$\Delta n = 1 - \frac{1}{2} = \frac{1}{2}$

$\Delta H_f = \Delta U_f + RT\Delta n$

$= -29600 + [(1.987) \times 298 \times 0.50]$

On calculation $\Delta H_f \simeq -29304$ cal mol^{-1}

$\simeq -29.3$ k cal mol^{-1}

$C_{(s)} + O_{2(g)} = CO_{2(g)}$ $\Delta H_f = -97300$ cal —(1)

$C_{(s)} + \frac{1}{2} O_{2(g)} = CO_{(g)}$ $\Delta H_f = -29300$ cal —(2)

Reversing the second equation and adding to the first,

$CO_{(g)} + \frac{1}{2} O_{2(g)} = CO_{2(g)}$ $\Delta H = -68000$ cal

$\Delta H_f = \Delta U + \frac{1}{2} RT$; $-68000 = \Delta U + 296$ cal

$\therefore$ $\Delta U = (-68000 - 296)$ cal $= -68296$ cal

$= -68.3$ k cal

9. $CH_3{-}CHO_{(\ell)} + \frac{5}{2} O_{2(g)} = 2CO_{2(g)} + 2H_2O_{(l)}$;

x zero -2×97.3 -2×68.4 k cal

$\Delta H = -282$ k cal

[x = ΔH_f of $CH_3{-}CHO$]

$-331.4 - x = -282$ k cal

$\therefore$ $x = -49.4$ k cal

Now consider $C_2H_{2(g)} + H_2O_{(l)} \rightarrow CH_3{-}CHO_{(l)}$;

46.8 -68.4 -49.4

$\Delta H = (-49.4 - 46.8 + 68.4)$ k cal

$= -27.8$ k cal

10. (i) $HA + NaOH \rightarrow NaA + H_2O$;

$\Delta H = -13.35$ k cal

(ii) $HB + NaOH \rightarrow NaB + H_2O$;

$\Delta H = -13.7$ k cal

(ii) – (i) $\Rightarrow$ $NaA + HB \rightarrow NaB + HA$

$\therefore$ $\Delta H = -13.7 + 13.35 = 0.35$ k cal

11. Work done = $P\Delta V = nRT$

$= \left[\left(\frac{5}{65}\right) \times 1.987 \times 288\right]$ cal

$= 44.0$ cal

12. $C_{(s)} + 2N_2O_{(g)} \rightarrow CO_{2(g)} + 2N_{2(g)}$

zero 2×17.7 -94 zero

k cal k cal

$\Leftrightarrow$ $\Delta H = (-94 - 35.4)$ k cal $= -129.4$ k cal

13. $CH_2Cl_{2(g)} + O_{2(g)} \rightarrow CO_{2(g)} + 2HCl_{(g)}$

x 0 $-94{,}400$ -2×22000 cal

$\Delta H = -106{,}800$ cal

$\therefore$ $[-94{,}400 - 44{,}000 - x]$ cal $= -106{,}800$ cal

$x = -31600$ cal

14. $[-13200 = -13700 + \Delta H]$ cal

$\therefore$ $\Delta H = -500$ cal for 75% ionization

$\therefore$ for 100% ionization

$\Delta H = \left(-500 \times \frac{4}{3}\right)$ cal $= -667$ cal

15. $x \times [36800 - 23000] = 510$

$\therefore$ $x = \frac{510}{13800} = 0.03696$

$\therefore$ % conversion $\simeq 3.7$

21. -13200 cal $= -13700$ cal + ΔH ionization cal

$\therefore$ $\Delta H_{ioni} = +500$ cal

22. -12270 cal $= -13780$ cal + ΔH ionization cal

$\therefore$ ΔH ionization $= +1510$ cal

23. (i) $BOH + HCl \rightarrow B^+ + Cl^- + H_2O$

$\Delta H = -51.46$ k J eqt^{-1}

(ii) $Na^+OH^- + HA \rightarrow Na^+ + A^- + H_2O$

$\Delta H = -50.63$ k J eqt^{-1}

(iii) $H^+ + OH^- \rightarrow H_2O$; $\Delta H = -55.90$ k J

Add (i) and (ii) and subtract (iii) cancel common terms we get

$BOH + HA \rightarrow B^+ + A^- + H_2O$

$\Delta H = -46.19$ k J

24. $C_{(graphite)} + 2H_{2(g)} = CH_{4(g)}$

712.96 872

$\{C_{(g)} + 4H\} - 4x$

$\Delta H_f = -74.84$ k J mol^{-1}

$\therefore$ $712.96 + 872 - 4x = -74.84$ k J

$\therefore$ $x = 415$ k J mol^{-1}

[Energy mol^{-1} to break the C–H bond]

25.

$CH_{4(g)} + 2O_{2(g)}$	$CO_{2(g)}$	$2H_2O_{(g)}$
$\downarrow$ 2648.8 k J	$\downarrow$	$\downarrow$
$C + 4H + 4O$(atoms)	$C + 2O$	$4H + 2O$
	$[2 \times 1.713x]$	$4x$

$\Delta H = -802.24$ k J

$\therefore$ $2648.8 - (2 \times 1.713x) - (4x) = -802.24$,

$x = 464.7$ k J mol^{-1}

$1.713x = 796.07$ k J mol^{-1}

26. $C_{(graphite)} + 2H_{2(g)} + \frac{1}{2}O_{2(g)} = \underset{\Delta H}{CH_3 - OH_{(g)}}$

$\Delta H = -226.05$ k J

$\downarrow 712.96$ k J $\downarrow 872$ k J $\downarrow \frac{495}{2}$ k J

$[C_{(g)} + 4H + O]$

$$\left[712.96 + 872 + \frac{495}{2}\right] - \Delta H = -226.05 \text{ k J}$$

$\therefore$ $\Delta H = 2058.5$ k J

IIT Assignment Exercise

31. Work done $= p(\Delta V) = [p \times (16.4 - 8.2)]$ litre atm

$= p \times 8.2 = 16.4$ litre atm

$\therefore$ $p = \frac{16.4}{8.2} = 2$ atm

32. 40 g of Mg $\equiv \frac{40}{24} = 1.67$ mol.

Work done $= p \times \Delta V = (\Delta n)RT$

This equals $(1.67 \times 0.0821 \times 300)$ litre atm

$= 41.13$ litre atm

33. (c)

$$W = RT \ln\left(\frac{V_1 - b}{V_2 - b}\right) - a\left(\frac{1}{V_2} - \frac{1}{V_1}\right)$$

$$= \left[0.0821 \times 300 \times 2.303 \log\left(\frac{10 - 0.0427}{50 - 0.0427}\right)\right]$$

$$- 3.59\left(\frac{1}{50} - \frac{1}{10}\right)$$

Calculation gives $W = -39.44$ L atm $= -3.99$ kJ

34. Using the two given equations, reversing the second equation and adding to the first, $C_{(s)} + \frac{1}{2}O_{2(g)} = CO_{(g)}$.

For this $\Delta H_f = [-96.96 + 67.96]$ k cal

$= -29.00$ k cal

$\Delta H = \Delta U + (\Delta n)RT$. $\Delta n = \frac{1}{2}$

$\therefore$ -29.00×10^3 cal $= \Delta U + \left[\frac{1}{2} \times 1.987 \times 290\right]$ cal

$\therefore$ $\Delta U = -29290$ cal $= -29.29$ k cal

35. $\Delta H = -40$ kJ (exothermic)

36. From the first law, $dU = dQ + dW$

$dQ = dU - dW$

$= dU + PdV$ for a closed system with only P – V work

$dQ_P = dU_P + PdV_P$ and $dQ_V = dU_V$

$\therefore$ $dQ_P - dQ_V = dU_P - dU_V + PdV_P$

37. $\underset{-62.76}{CaC_{2(s)}} + \underset{-2\times285.84}{2H_2O_{(l)}} = \underset{-986.59}{Ca(OH)_{2(s)}} + \underset{+226.75}{C_2H_{2(g)}}$

$[226.75 - 986.59 + 62.76 + 2 \times 285.84]$

$= -125.4$ k cal

38. Since $\Delta n = 0$

39. $C_2H_5 - O_{(l)} - C_2H_5 + 6O_{2(g)} \rightarrow 4CO_{2(g)} + 5H_2O_{(l)}$

x	zero	$-4 \times$	$-5 \times$
		97.000	68.4000

$\Delta H = -660.000$ k cal mol^{-1}

$\therefore$ $-388 - 342 - x = -660.000$

$\therefore$ $x = -70.000$ k cal mol^{-1}

40. $\Delta H = \Delta U + (\Delta n \times RT)$

$CH_{3(l)} - OH = CH_3 - OH_{(g)}$ $\Delta n = 1$

$\therefore$ 8890 cal mol^{-1} $= \Delta U + 1 \times 2 \times 313$ cal mol^{-1}

$= \Delta U + 626$ cal

$\therefore$ $\Delta U = 8264$ cal mol^1

41. Since pV^γ = constant (1) and

$TV^{\gamma - 1}$ = constant (2), eliminating V, we get

$\left(\frac{p^{\gamma-1}}{T^\gamma}\right)$ = constant. Thus $\frac{p_1^{2/3}}{T_1^{5/3}} = \frac{p_{21}^{2/3}}{T_2^{5/3}}$.

Thus $\left(\frac{p_1}{p_2}\right)^{2/3} = \left(\frac{T_1}{T_2}\right)^{5/3}$

$$\frac{2}{3}\log\left(\frac{p_1}{p_2}\right) = \frac{5}{3}\log\left(\frac{T_1}{T_2}\right)$$

i.e., $\frac{2}{3}\log 3 = \frac{5}{3}\log\left(\frac{300}{T_2}\right)$

Calculation gives $\frac{300}{T_2} = 1.5518$

$T_2 \simeq 193K$

42. $W = nC_V(T_2 - T_1)$

$W = -P_{ext}(V_2 - V_1)$

$= -P_{ext}\left(\frac{RT_2}{P_2} - \frac{RT_1}{P_1}\right)$

$= -P_{ext}\, R\left(\frac{T_2}{P_2} - \frac{T_1}{P_1}\right)$

$W = -RP_{ext}\,\frac{T_2P_1 - T_1P_2}{P_1P_2}$

for 1 mole

43. Sol. $C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)}$ — (i)

$\Delta H = -193.5$ kJ mol^{-1}

$H_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow H_2O_{(\ell)}$ — (ii)

$\Delta H = -235.8$ kJ mol^{-1}

$3C_{(s)} + 3H_{2(g)} \rightarrow C_3H_{6(g)}$ — (iii)

$\Delta H = 12.34$ kJ mol^{-1}

$\triangle_{(g)} \rightarrow C_3H_{6(g)}$ — (iv)

$\Delta H = -24$ kJ mol^{-1}

$\triangle + \frac{9}{2}O_{2(g)} \rightarrow 3CO_{2(g)} + 3H_2O_{(l)}$

(i) × 3 + (ii) × 3 + (iv) − (iii)

$\Delta H = -1324.24$ kJ mol^{-1}

44. $C_6H_{6(l)} + \frac{15}{2}O_{2(g)} = 6CO_{2(g)} + 3H_2O_{(l)}$;

$\Delta H = -799350$ cal

$C_{(graphite)} + O_{2(g)} = CO_{2(g)}$; $\Delta H_f = -96960$ cal

$H_{2(g)} + \frac{1}{2}O_{2(g)} = H_2O_{(g)}$; $\Delta H_f = -68360$ cal

One follows the usual procedure as in problem (7) above

The answer is ΔH_f of $C_6H_{6(l)} = +12510$ cal mol^{-1}

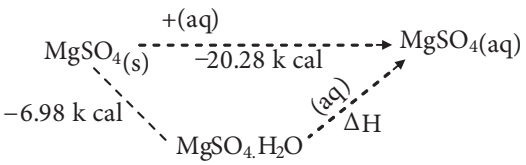
The same method may be used for $C_2H_{2(g)}$

$C_2H_{2(g)} + \frac{5}{2}O_{2(g)} = 2CO_{2(g)} + H_2O_{(l)}$;

$\Delta H = -310050$ cal

The answer is : ΔH_f of $C_2H_2 = +47770$ cal mol^{-1}

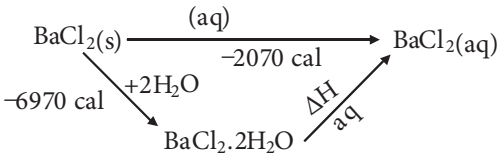
45.



$-20.28 = -6.98 + \Delta H$ k cal mol^{-1}

$\Delta H = -13.30$ k cal mol^{-1}

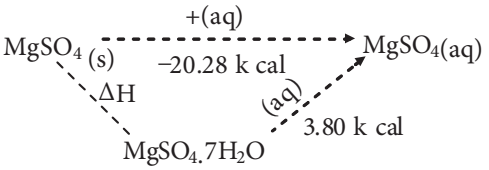
46.



$-2070 = -6970 + \Delta H$

$\therefore \quad \Delta H = +4900$ cal

47.



$-20.28 = \Delta H + 3.80$ k cal

$\Delta H = [-20.28 - 3.80]$ k cal mol^{-1}

$= -24.08$ k cal mol^{-1}

48. By Trouton's law, the enthalpy of vapourization per mole ΔH_V divided by Boiling point (normal) in Kelvin is very nearly constant for normal liquids. Thus

$$\frac{30.71\times10^3}{(273+76.7)}\ \text{J mol}^{-1} = \frac{(\Delta H_{vap})\text{of } CHCl_3}{273+61.5}$$

$\therefore \quad (\Delta H_V)$ of $CHCl_3 = 30.71 \times 10^3 \times \frac{334.5}{349.7}$

Which give $(\Delta H_v)_{CHCl_3} = 29.375 \times 10^3$ J

$\simeq 29.38$ k J

49. -12.270 k cal $= -13.680$ k cal $+ \Delta H_{ionization}$

$\therefore \quad \Delta H_{ionization} = +1.410$ k cal

50. $\underset{zero}{H^+_{(aq)}} + \underset{x}{^-OH_{(aq)}} = \underset{-68.3}{H_2O_{(l)}}$; $\Delta H = -13.3$ k cal

$-68.3 - x = -13.3$ k cal

$\therefore \quad x = -55.0$ k cal

51. (i) x k cal : enthalpy of ionization

(iii) y k cal : Enthalpy of neutralization

$\therefore \quad y = -13.3 + x$ k cal

$\therefore \quad y - x = -13.3$ k cal

Since $\frac{x}{y} = -3.6$

We have $y + 3.6\ y = 4.6\ y = -13.3$

$$y = \frac{-13.3}{4.6} \quad y \simeq -2.9 \text{ k cal}$$

$x = -3.6\ y = -3.6 \times (-2.9) \simeq 10.4$ k cal

52. $(x \times 13550) + (1 - x)\ 14700 = 14350$ cal

$\therefore \quad (14700 - 14350) = (14700 - 13550)x$

$\therefore \quad x = \frac{350}{1150} = 0.304$

$(1 - x) = 0.696$

$\therefore \quad$ HA : HB (equivalence) $= 0.304 : 0.696$

$= 1 : 2.29$

53. $\frac{1}{2}H_{2(g)} + \frac{1}{2}I_{2(g)} = HI_{(g)}\ \Delta H = 26.5$ k J

$\therefore \quad 26.5 = \frac{436}{2} + \frac{1}{2} \times x - 295$ k J

$\therefore \quad x = 207$ k J

54. $\Delta H = 612 + 366 - (412 + 276 + 348) = -58$ k J

55. $C_{(s)} + O_{2(g)} = CO_{2(g)} \qquad \Delta H = -94.3$ k cal mol^{-1}

$CO_{(g)} + \frac{1}{2}O_2 = CO_{2(g)} \quad \Delta H = -67.4$ k cal mol^{-1}

Subtracting

$$\underset{+170}{C_{(s)}} + \frac{1}{2}O_{2(g)} = \underset{+58.7,\ -x}{CO_{(g)}} = -26.9 \text{ k cal}$$

$$C_{(g)} + O_{(g)}$$

$170 + 58.7 - x = -26.9$ k cal

$\therefore \quad x =$ bond energy $= 255.6$ k cal

$\therefore \quad$ dissociation energy $= +255.6$ k cal

56.

$$\underset{+265}{CO_{2(g)}} + \underset{+320}{2H_2O_{(l)}} = \underset{-4x}{CH_{4(g)}} + 2O_2$$

$$C_{(s)} + O_{2(g)} \quad 4H + O_2$$

$\Delta H = +213$ k cal

$\therefore \quad 265 + 320 - 4x = 213$ k cal

$\therefore \quad x = 93$ k cal mol^{-1}

57. $2Na_{(s)} + \frac{1}{2}O_{2(g)} = Na_2O_{(s)} \qquad \Delta H_f = ?$

(i) $Na_{(s)} \xrightarrow{[(aq) + H_2O]} NaOH_{(aq)} + \frac{1}{2}H_{2(g)}$

$\Delta H = [-23 \times 1880]$ cal $= -43240$ cal

(ii) $Na_2O_{(s)} \xrightarrow[+H_2O]{(aq)} 2\ NaOH_{(aq)}$

$\Delta H = [-62 \times 1020]$ cal $= -63240$ cal

Multiply the first equation by 2, reverse the second equation and add; and finally adding

$H_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow H_2O$; $\Delta H = -68000$ cal; One gets after some manipulation,

$\Delta H_f = -91240$ cal.

58. $K_{(s)} + (H_2O + aq) = KOH_{(aq)} + \frac{1}{2}H_{2(g)}$;

$\Delta H = -48.10$ k cal

$KOH_{(aq)} = KOH_{(s)} + aq$; $\Delta H = +13.30$ k cal

$H_{2(g)} + \frac{1}{2}O_{2(g)} = H_2O_{(l)}$; $\Delta H = -68.40$ k cal

Adding we get the required equation

$$K_{(s)} + \frac{1}{2}O_{2(g)} + \frac{1}{2}H_{2(g)} = KOH_{(s)}$$

$\Delta H = (-48.10 + 13.30 - 68.40)$ k cal

$= -103.20$ k cal

59. The step by step method is

Eqn(i) − eqn (iv) + 6 eqn (iii) + 3 eqn (ii)

60. Multiply the second equation by 3; Then subtract equation (1)

$6C_2H_{2(g)} + 15O_{2(g)} = 12CO_{2g)} + 6H_2O_{(l)}$

$\Delta H = -3 \times 620100$ cal

$12CO_{2(g)} + 6H_2O_{(l)} = 2C_6H_{6(l)} + 15O_{2(g)}$

$\Delta H = +1598700$ cal

$6C_2H_{2(g)} = 2C_6H_{6(l)}$ $\Delta H = -261600$ cal

Divide by 2

$3C_2H_{2(g)} = C_6H_{6(l)}$ $\Delta H = -130800$ cal

$\Delta H = \Delta U + \Delta n \times RT$;

$-130800 = \Delta U + (-3) \times 1.987 \times 290$ cal

$\therefore$ $\Delta U = -130800 + 3 \times 1.987 \times 290$

$= -130800 + 1728.69 \simeq -129071$ cal

61. By the usual method of working:

Equations $[-(i) + 2(iii) + 2(ii) - 6(v) - 3(iv) + 3(vi)]$

One gets $[-7550 - 35160 - 142780 + 103890 +$

$132000 - 205080]$ cal

$= -154680$ cal

62. By the usual method of working:

Equations $[-(iv) + (v) - (i) + \frac{1}{2} \times (iii) + 2 \times (ii)]$ give

$[9110 + 4750 + 71770 + 20320 - 136720]$ cal

$= -30770$ cal

63. The usual method of working gives

$(-394.2 - 369.2 + 173.2 + 490.6)$ k J $= -99.6$ k J

64. $SO_2Cl_{2(g)} + 2H_2O_{(l)} + (aq) = H_2SO_{4(aq)} + 2HCl_{(aq)}$

-89800 -136000 -210000 -78600

$\Delta H = [-210000 - 78600 + 89800 + 136000]$ cal

$= -62000$ cal

65. (i) $2CH_{4(g)} + 4O_{2(g)} = 2CO_{2(g)} + 4H_2O_{(l)}$

$\Delta H = -424.000$ k cal

(ii) $C_2H_{4(g)} + 3O_{2(g)} = 2CO_{2(g)} + 2H_2O_{(l))}$

$\Delta H = -333.000$ k cal

(iii) $2H_{2(g)} + O_{2(g)} = 2H_2O_{(l)}$

$\Delta H = -136.000$ k cal

Subtracting equations (2) and (3) from eqn (1) and rearranging $2CH_{4(g)} = C_2H_{4(g)} + 2H_{2(g)}$

$\Delta H = 45.000$ k cal

66. Efficiency $= \frac{150}{225} = 0.67$

$\frac{T_1 - T_2}{T_2} = 0.67$, $T_1 = 500$K; $T_2 = 299$K

67. Isothermal change for an ideal gas. Thus, no change in internal energy. Q = work done

$$= \int PdV = nRT \ln\left(\frac{V_2}{V_1}\right)$$

$$\frac{Q}{T} = \Delta S = nR \ln\left(\frac{V_2}{V_1}\right)$$

$= 10 \times 1.987 \times 2.303 \log 2 = 13.77$ cal K^{-1}

68. The change is exothermic. Product has less randomness than reactant,.

69. In (a) ΔS is negative

In (b), (c), (d) ΔS is +ve

70. $$\Delta S = n\left[C_P \ln\frac{T_2}{T_1} + R\ln\frac{P_1}{P_2}\right]$$

$$= 2 \times 2.303\left[37.1 \log\left(\frac{398}{298}\right) + 8.314 \log\left(\frac{5}{25}\right)\right]$$

Calculation gives -5.30 J K^{-1}

71. Since $\Delta S = nR\ln\left(\frac{V_2}{V_1}\right)$ and $V_2 < V_1$,

ΔS is negative

73. $\Delta G = -2470.2 = 4 \times (-394.4) + 2 \times (-237.1) - 2x$

Where, $x = \Delta G^\circ_f$ of $C_2H_{2(g)}$. Calculation gives

$x = 209.2$ k J

74. Since $\Delta G = \Delta H - T\Delta S = 0$

75. $\Delta H = T\Delta S$ for spontaneity, T = 2000K

$\therefore$ $\frac{\Delta S}{\Delta H} = \frac{1}{2000} = 5 \times 10^{-4}$

$\Delta H = 100$ k cal

$\Delta S = 5 \times 10^{-4} \times 100$

$= 5 \times 10^{-2}$ k cal K^{-1}

76. $\Delta H = \Delta U + (\Delta n)RT = [-5000 + (-3 \times 2 \times 300]$ cal

$= -6800$ cal

$\Delta G = \Delta H - T\Delta S = -6800 \text{ cal} - [300 \times (-20)]$

$= [-6800 + 6000] \text{ cal}$

$= -800$ cal (negative), feasible

77. Under equilibrium, $\Delta G = \Delta H - T\Delta S$

$\Delta S = (51.5 - 44.1) \text{ J K}^{-1} \text{ mol}^{-1} = 7.4 \text{ J K}^{-1} \text{ mol}^{-1}$

$\therefore \quad T = \frac{\Delta H}{\Delta S} = \left(\frac{2100}{7.4}\right) \simeq 280\text{K}$

78. $\Delta G = \Delta G^\circ + RT\ln Q = -RT\ln K + RT \ln Q$

$= RT \ln \left(\frac{Q}{K}\right)$

$K = 1.6 \times 10^{-3} \qquad Q = \frac{30^2}{50^2} = 0.36$

$T = 2298 \text{ K} \qquad R = 8.314 \text{ J}$

Direct substitution gives ΔG

$= 2.303 \times R \times T \log \left(\frac{Q}{K}\right)$

$= 103.5$ k J (i.e., +ve), not feasible.

79. $Ag_2O_{(s)} \rightleftharpoons 2Ag_{(s)} + \frac{1}{2}O_{2(g)} \quad \Delta G^\circ = 11.21 \text{ k J}$

$\Delta G^\circ = -2.303 \text{ RT} \log\left[P_{O_2}\right]^{1/2};$

$11210 \text{ J} = -2.303 \times (8.314) \times 298 \times \log\left[P_{O_2}\right]^{1/2}$

By calculation, $PO_2 = 0.000116$ atm

$= 1.16 \times 10^{-4}$ atm

80. $\Delta G = \Delta H + T\left(\frac{\partial \Delta G}{\partial T}\right)_P$

$= \left[-100 + \frac{303 \times 4.8}{1000}\right]$

$= -98.5456$ k J

81. The number of moles left as residue in the cylinder = PV/RT which is $(10 \times 1)/(0.0821 \times 300) \simeq 0.4$

$\therefore$ $(20 - 0.4) = 19.6$ moles escaped against and external pressure of 1 atm, work done, $W = PdV = nRT \simeq (19.6 \times 1.987 \times 300)$ cal $\simeq$ 11700 cal

82. Work done by the gas(taken to be positive for the present calculation = $nRT \ln\left(\frac{V_2}{V_1}\right)$

$1 \text{ J} = 10^7$ org

1.719×10^{10} orgs = 1719 J

$1719 = 1 \times 8.314 \times T \times 2.303 \log 2$

$= 25°C$

83. Since the process is adiabatic Q = 0. Work done in an adiabatic compression

But $TV^{\gamma - 1}$ = a constant

$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma - 1}$

$\left(\frac{T_2}{300}\right) = \left(\frac{1}{0.5}\right)^{1.67-1}$

$T_2 = 300(2)^{0.67} = 477\text{K}$

$W = \frac{nR}{\gamma - 1}\left(T_2 - T_1\right)$

$\therefore \quad W = \frac{3 \times 2 \times 177}{0.67} = 1585$ cal

$\Delta U = -W$

$= -6.63$ kJ

84. 220 g of $CO_2 \equiv 5$ moles

For heating from 27°C to 527°C absorbs

$C_p \Delta T$ per mole = [5 × 500] cal mol^{-1}

Work done $P\Delta V$ absorbs $R\Delta T$ = [2 × 500] cal per mole

Total = [7 × 500] cal = 3500 cal mol^{-1}

Work done by isothermal compression on the gas ≡ heat rejected = $RT\ell n\frac{P_2}{P_1}$

$= -2 \times 800 \times 2.303 \times 2 = -7369.6$ cal

net energy change per mole = −3.87 k cal

$\therefore$ for 5 moles energy change = −19.35 k cal

i.e., + 19.35 by reversed conversion

85. $C_2H_{4(g)} + H_{2(g)} \rightarrow C_2H_{6(g)}$

+12.50 zero −20.24 k cal

$\therefore \quad \Delta H = (-20.24 - 12.50) \text{ k cal} \qquad = -32.74 \text{ k cal}$

86. $2CH_3OH_{(\ell)} + C_2H_{4(g)} \rightarrow 2C_2H_5{-}OH_{(\ell)}$

-2×57.02 k cal $+12.50$ k cal -2×66.36 k cal

$\Delta H = -31.18$ k cal(analogous method : prob 3 proceeding)

87. $Al_2Cl_{6(g)} + 6Na_{(s)} \rightarrow 6NaCl_{(s)} + 2Al_{(s)}$;

x zero -6×98.2 zero k cal

$\Delta H = -256.8$ k cal

$\therefore$ -589.2 k cal $- x = -256.8$ k cal

$\therefore$ $x = -332.4$ k cal mol^{-1}

88. $C_6H_{6(\ell)} + \frac{15}{2} O_{2(g)} \rightarrow 6CO_{2(g)} + 3H_2O_{(\ell)}$

$\Delta H = -781.0$ k cal mol^{-1}

$\Delta n = 6 - 7.5 = -1.5$

$\Delta H = \Delta U + \Delta n \times RT$

i.e., $-781.0 = \Delta U - 1.5 \times \frac{1.987 \times 298}{1000}$ k cal

$\therefore$ $\Delta U = (-781 + 0.888)$ k cal

$\therefore$ $\Delta U = -780.112$ k cal mol^{-1}

89. $\Delta H = \Delta U + p\Delta U \rightarrow \Delta U + \Delta n \times RT$

$= \Delta U + \frac{3 \times 2 \times 500}{1000}$

10 k cal $= \Delta U + 3$ k cal

$\therefore$ $\Delta U = 10 - 3 = 7$ k cal mol^{-1}

90. If the heat capacity = x : mol mass of benzoic acid = 122.12

then $2.87 \times x = \frac{1.247}{122.12} \times 771.5 \times 1000$

$\therefore$ $x = 2745$ cal deg^{-1}

91. $\frac{P^{\gamma-1}}{T^{\gamma}}$ = constant $\gamma = 1.67$

$\therefore$ $\left(\frac{P_2}{P_1}\right)^{0.67} = \left(\frac{T_2}{T_1}\right)^{\gamma}$

$(10)^{0.67} = \left(\frac{T_2}{298}\right)^{1.67}$

$\therefore$ $0.67 = 1.67 \log\left(\frac{T_2}{298}\right)$

$\therefore$ $\frac{T_2}{298} = 2.512$

$\therefore$ $T_2 = 748$K

92. $\frac{\Delta H}{T} = 21$ cal deg^{-1} mol^{-1}

$T = (273 + 61.2)$K $= 334.2$ K

$\therefore$ $\Delta H = 7018.2$ cal mol^{-1}

$\therefore$ $\ell = \frac{7018}{119.5} = 58.73$ cal g^{-1} ≈ 59 cal g^{-1}

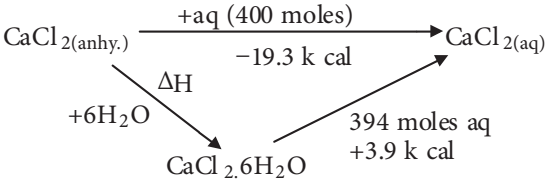
93. Since it is an equimolar mixture, mean ΔH of combustion $= -262.4$; 112 litres at STP $= \frac{112}{22.4}$

$= 5$ moles

$\therefore$ heat evolved $= 5 \times +262.4 = 1312$ k J

$\Delta H = -1312$ k J

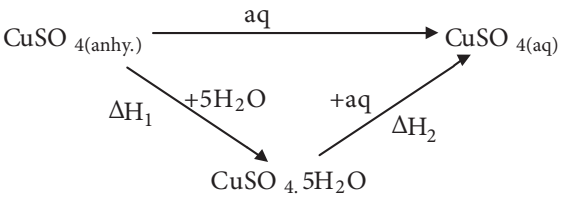
94.



$\Delta H + 3.9 = -19.3$ k cal

$\therefore$ $\Delta H = -23.2$ k cal

95.



$\Delta H = -16.43$ k cal

$\Delta H_1 + \Delta H_2 = -16.43$ k cal

$\therefore$ $-7.79\ \Delta H_2 + \Delta H_2 = -16.43$ k cal

$\therefore$ $\Delta H_2 = \frac{-16.43}{-6.79}$ k cal $\simeq 2.42$ k cal

$\therefore$ $\Delta H_1 = -18.85$ k cal

96. since maximum neutralization is achieved

97. 98 g of H_2SO_4 = 1 mol = 2 eqts

For 1 eqt ΔH neutralization

$= \Delta H_{ionization} + (-57)$ k J eqt^{-1}

$=(5.52 - 57)$k J eqt^{-1} = −51.48 k J

∴ for 2 eqts ΔH = −102.96 k J

98. $CaO_{(s)} + H_2O_{(\ell)} \rightarrow Ca(OH)_{2(s)}$ ΔH = −63 k J

$Ca(OH)_{2(s)} + aq \rightarrow Ca(OH)_{2(aq)}$ ΔH = −12.5 k J

(i) $CaO_{(s)} + (H_2O + aq) \rightarrow Ca(OH)_2aq$

ΔH = −75.5 k J

(ii) $CaO_{(s)} + 2HCl_{aq} \rightarrow CaCl_{2(aq)} + (H_2O)$

ΔH = −192.5 k J

Reverse (i) and add (ii)

$Ca(OH)_{2(aq)} + CaO_{(s)} + 2HCl_{(aq)}$

$\rightarrow CaO_{(s)} + (H_2O + aq) + CaCl_{2(aq)} + H_2O$

−192.5 + 75.5 k J = −117.0 k J

$Ca(OH)_{2(aq)} + 2HCl_{(aq)} \rightarrow CaCl_{2(aq)} + 2H_2O$

This is for 2 eqts ∴ ΔH for one eqt

= −58.5 k J eqt^{-1}

99. $CH_{4(g)} + Cl_{2(g)} \rightarrow CH_3Cl_{(g)} + HCl_{(g)}$

In the overall reaction one C–H bond and one Cl–Cl bond are broken and one "C–Cl" bond and one HCl bond are formed. Thus

ΔH =(98 + 57 − 78 − 102) k cal = −25 k cal

100. It is obvious that for the reduction of three double bonds, ΔH = 3 × (−28.6) k cal = −85.8 k cal mol^{-1} for C_6H_6. The discrepancy is the difference between −49.8 k cal and −85.8 k cal = 36 k cal mol^{-1} which is an extra stabilization per mole for benzene. This is resonance stabilization energy

101. $X_2 + Y_2 \rightarrow 2XY$

ΔH 2ΔH 2 × 4ΔH

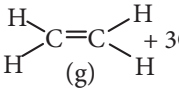
∴ Total energy change (release) for 2 moles

formation of XY = (8 − 3)ΔH = 5ΔH

∴ $\Delta H_f XY = -2.5\ \Delta H = -200$ k J

∴ $\Delta H = \frac{200}{2.5}$ k J = 80 k J;

∴ 2ΔH = 160 k J

102. 

$+ 3O_{2(g)} \rightarrow 2CO_{2(g)} + 2H_2O_{(1)}$

ΔH = [(615 + 4 × 413 + 3 × 493)] k J

− [4 × 707 + 4 × 463] k J

= −934 k J

103. $H_2O_{(l)} \longrightarrow \dot{H}_{(g)} + \dot{O}H_{(g)}$; ΔH = 129.22 k cal

$H_2O_{(g)} \rightarrow H_2O_{(\ell)}$ ΔH = −9.72 k cal

Add

$H_2O_{(g)} \longrightarrow \dot{H}_{(g)} + \dot{O}H_{(g)}$ ΔH = 119.50 k cal

$\dot{O}H_{(g)} \longrightarrow \dot{H}_{(g)} + \ddot{O}_{(g)}$ ΔH = 101.50 k cal

$H_2O_{(g)} \longrightarrow 2\dot{H}_{(g)} + \ddot{O}_{(g)}$ ΔH = 221 k cal

= 2 × bond enthalpy

∴ bond enthalpy = $\frac{221}{2}$ k cal = 110.5 k cal mol^{-1}

104. $2C_{(Graphite)} + 3H_{2(g)} \rightarrow C_2H_{6(g)}$

ΔH_f = −20.2 k cal mol^{-1}

consider (i) $2C_{(s)} \rightarrow 2C_{(g)}$; 2x

(ii) $3H_{2(g)} \rightarrow 6H$; 3 × 103 k cal

one C–C bond : 80 k cal

six C–H bonds; 6 × 98 k cal

∴ 2x + 309 − 80 − 588 = −20.2

∴ 2x = 338.8 k cal

∴ x = 169.4 k cal mol^{-1} ≃ 170 k cal mol^{-1}

105. (i) $C_4H_{10(n\ butane)} + \frac{13}{2}O_{2(g)} \rightarrow 4CO_{2(g)} + 5H_2O_{(\ell)}$;

ΔH = −688 k cal mol^{-1}

(ii) $C_4H_{10(isobutane)} + \frac{13}{2}O_{2(g)} \rightarrow 4CO_{2(g)} + 5H_2O_{(\ell)}$

ΔH = −686.35 k cal mol^{-1}

subtracting (ii) from (i) and rearranging

$$C_4H_{10(\text{n-butane})} \rightarrow C_4H_{10(\text{isobutane})}$$

$$\Delta H = -1.65 \text{ k cal mol}^{-1}$$

Also by the usual procedure

$$\underset{x}{C_4H_{10}}(n-) + \frac{13}{2}\underset{\text{zero}}{O_{2(g)}} \rightarrow \underset{-384}{4CO_{2(g)}} + \underset{-340 \text{ k cal}}{5H_2O_{(\ell)}}$$

$$\Delta H = -688 \text{ k cal}$$

$$-724 - x = -688 \text{ k cal}$$

$\therefore$ $x = -36$ kcal mol^{-1}

By the same procedure,

$$-724 - x = -683.35 \text{ k cal}$$

$\therefore$ $x = -37.65$ k cal mol^{-1}

106. ΔS per mole $= C_p \ell n \frac{T_2}{T_1} + R\ell n \frac{P_1}{P_2}$

$= [6.1 \times 0.693 + 2 \times (-2.99)]$cal mol^{-1} deg^{-1}

$= -1.75$ cal deg^{-1} mol^{-1}

$\therefore$ for 2 moles the answer is -3.5 cal deg^{-1}

107. ΔH and $T\Delta S$ are positive

$\therefore$ for T greater than the value $\frac{\Delta H}{\Delta S}$, $T\Delta S$ is negative so as to make ΔG, negative

108. clearly at the normal boiling point, liquid and vapour are in equilibrium

$\therefore$ $\Delta G = 0$

109. ΔG is positive, hence non spontaneous.

110. $NH_{3(g)} + HCl_{(g)} = NH_4Cl_{(s)}$

$-16.6 \quad -93.9 \quad -203.9(\Delta G°_f)$ k J mol^{-1}

$\therefore$ net $\Delta G = -93.4$ k J

$-48.2 \quad -95.3 \quad -311.4(\Delta H°_f)$k J mol^{-1}

$\therefore$ net $\Delta H = -167.9$ k J

$\Delta G° = \Delta H - T\Delta S \quad -93.4 = -167.9 - T\Delta S$

$\therefore$ $T\Delta S = (-167.9 + 93.4) \times 10^3 J = -74.5 \times 10^3$ J

$\therefore$ $\Delta S = \frac{-74.5}{298}$ k J deg^{-1}

$\simeq -0.248$ k J deg^{-1}

113. If A $\rightarrow$ B process is atleast partially irreversible $\frac{q}{T}$ < dS since randomness may increase without a corresponding absorption of thermal energy

114. $P_{4(s)} + 6Cl_{2(g)} \rightarrow 4PCl_{3(\ell)} \quad \Delta H = -132.2$ k J

$4PCl_{3(\ell)} + 4Cl_{2(g)} \rightarrow 4PCl_{5(s)} \quad \Delta H = -565.6$ k J

Add

$P_{4(s)} + 10Cl_{2(g)} \rightarrow 4PCl_{5(s)} \quad \Delta H = -697.8$ k J

$\therefore$ for ΔH_f for 1 mole $PCl_{5(s)}$ $\Delta H = -174.5$ k J

115. $CH_{4(g)} \rightarrow C_{(g)} + 4H_{(g)} \quad \Delta H = 360$ k cal mol^{-1}

$\therefore$ bond energy of C–H = 90 k cal mol^{-1}

$C_2H_{6(g)} \rightarrow 2C_{(g)} + 6H = 620$ k cal mol^{-1} which includes one C–C bond and 6 CH bonds

$\therefore$ for the C–C bond, enthalpy

$= (620 - 540)$ k cal mol^{-1}

$= 80$ k cal mol^{-1}

116. $[x \times 890 + (1 - x)1560]$ k J mol^{-1} = 1292 k J mol^{-1}

$\therefore (1560 - 1292) = (1560 - 890)x$

$268 = 670x$

$\therefore$ $x = \frac{268}{670} = 0.4; (1 - x) = 0.6$

ratio $CH_4 : C_2H_6 = 0.4 : 0.6 = 2 : 3$

Additional Practice Exercise

121. $W = -P\Delta V = -PV_v$ where V_v is the volume of the vapour (volume of liquid water is negligible)

$= -nRT = -2 \times 8.314 \times 373 = -6.2$ kJ

122. At constant volume $\Delta U = Q_v$, $W = 0$

$Q_v = nC_v\Delta T = 0.5 \times (40 \times 20.814 - 8.314) \times 4$

$= 1.6$ kJ

123. Work done under reversible conditions

$= -nRT \ln \frac{V_2}{V_1}$

$= -1 \times 8.314 \times 310 \times 2.303 \log \frac{20}{15}$

$= 5935.6 \times 0.125 = -742$ J

Work done under irreversible conditions

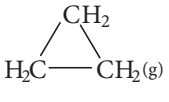
$$= -p_{ext}(V_2 - V_1)$$

$$= -0.5(20 - 15) = -2.5 \text{ L atm}$$

$$= \frac{-2.5 \times 8.314}{0.0821} = -253 \text{ J}$$

$$\text{Ratio } \frac{W_{rev}}{W_{in}} = \frac{-742}{-253} = 2.93$$

124. The enthalpy of combustion of cyclopropane is given by equation



$$+ \frac{9}{2}O_{2(g)} \rightarrow 3CO_{2(g)} + 3H_2O_{(\ell)};$$

$$\Delta_C H^\circ = -2091.3 \text{ kJ}$$

$$\Delta_C H^\circ = 3\Delta_f H^\circ_{(CO_2)_{(g)}} + 3\Delta_f H^\circ_{(H_2O)_{(\ell)}} - \Delta_f H^\circ_{(C_3H_6)(g)}$$

$$-2091.3 = 3 \times -393.5 - 3 \times 285.8 - \Delta_f H^\circ(C_3H_6)$$

$$\Rightarrow \quad \Delta_f H^\circ(C_3H_6) = -1180.5 - 857.4 + 2091.3$$

$$= 53.4 \text{ kJ.}$$

125. The required equation is

$$B_2H_{6(g)} + 3O_{2(g)} \rightarrow B_2O_{3(s)} + 3H_2O_{(g)}$$

ΔH can be calculated by rearranging the equations.

Eq (i) + 3eq (ii) + 3eq (iii) − eq

(iv) = −2823 kJ mol^{-1}

126. $$\Delta S_T = nR \ln \frac{V_2}{V_1} = nR \ln \frac{P_1}{P_2}$$

$$= 2.303 \times 10 \times 8.314 \times \log 10 = 191.5 \text{ J K}^{-1}$$

127. Reaction:

$$Fe_2O_3 + 2Al \rightarrow Al_2O_3 + 2Fe$$

$$\Delta S^\circ_r = 50.9 + 2 \times 27.3 - (87.4 + 2 \times 28.3)$$

$$= (105.5 - 144.0) = -38.5 \text{ J K}^{-1}$$

128. ΔS of benzene (system) $= -\frac{127 \times 2 \times 78}{278.5}$

$$= -71.1 \text{ J K}^{-1}$$

ΔS of water bath (surr) $= \frac{127 \times 2 \times 78}{273}$

$$= 72.6 \text{ J K}^{-1}$$

$$\Delta S_{net} = -71.1 + 72.6 = 1.5 \text{ J K}^{-1}$$

129. $\Delta H^\circ = \Delta U^\circ + P\Delta V = \Delta U^\circ + \Delta nRT$

$$= -5000 + (-3 \times 2 \times 300) = -6800 \text{ cal}$$

$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ = -6800 - (300 \times -20)$

$$= -800 \text{ cal}$$

i.e., Reaction is spontaneous at 300K

130. (i) Reaction:

$$Fe_2O_3 + 2Al \rightarrow Al_2O_3 + 2Fe$$

$$\Delta G^\circ_r = -1582 - (-742) = -840 \text{ kJ}$$

The reaction is feasible, since ΔG is negative, under standard conditions.

(ii) $$\left(\frac{\partial \Delta G^\circ}{\partial T}\right)_P = -\Delta S^\circ$$

$$\Rightarrow \quad \frac{\partial}{\partial T}(\Delta G^\circ)_p = 38.5 \text{J K}^{-1}$$

(iii) $\Delta H^\circ_r = \Delta G^\circ + T\Delta S^\circ$

$$= -840 + 298 \times 38.5 \times 10^{-3}$$

$$= -840 + 11.5 = -828.5 \text{ kJ.}$$

131. One mole of Zn liberates one mole of H_2.

For which work done = $P\Delta V = RT$ for one mole

i.e., (0.0821×310) litre atm

i.e., 25.45 litre atm

132. For one mole of gas at 298K, isothermal compression from 5 litres to 2.5 litres has work done on the gas = $RT \ln_2 = 1.987 \times 298 \times 0.693$ calories (step A)

During step B, volume remains constant

$\therefore$ no work is done. During step C, the gas expands at constant pressure.

Work done is by the gas $P\Delta V = P(5 - 2.5)$ litre atm $= P \times 2.5$

But the initial pressure P

$$= \frac{RT}{V} = \frac{R \times 298}{5} = \frac{0.0821 \times 298}{5} \text{ atm}$$

$\therefore \quad P\Delta V = \left[\frac{0.0821\times298}{5}\times2.5\right]$ litre atm

$= \frac{0.0821\times298}{2}\times\frac{1.987}{0.0821}$ cal

Which is $\simeq$ 298 cal

$\therefore$ net work done on the gas is

$[1.386 \times 298 - 298]$ cal = $[0.386 \times 298]$ cal

$\simeq$ 115 cal

133. $C_4H_{8(g)} + 6O_{2(g)} = 4CO_{2(g)} + 4H_2O_{(l)}$

$\Delta n = (4 - 7) = -3$

$\therefore \quad \Delta H^\circ = \Delta U^\circ + \Delta n(RT)$

$-y = \frac{-x - 3\times8.314\times298}{1000}$

$\therefore$ y = x + positive quantity. (both y and x are +ve)

$\therefore$ y > x or x < y

134. For the vapourization of 5 moles of liquid.

$\Delta H = \Delta U + 5 \times RT = \Delta U + 5 \times \frac{1.987\times353}{1000}$

$= \Delta U + 3.53$ k cal

$\therefore \quad \Delta U = 8 - 3.53 \simeq 4.47$ k cal

135. $NH_2 - CN_{(s)} + \frac{3}{2}O_{2(g)} = N_{2(g)} + CO_{2(g)} + H_2O_{(l)}$

$\Delta n = 2 - 1.5 = 0.5; \Delta H = \Delta U + (\Delta n)RT$

$\Delta H = -742.7$ k J $+ \frac{0.5\times8.314\times298}{1000}$

$= (-742.7 + 1.239)$ k J

$= \sim 741.5$ k J mol^{-1}

136. $(0.062 - 0.033)$ cal g^{-1} = 0.029 cal g^{-1} is the specific heat difference.

For one mole $(C_p - C_v)$ = 1.987 cal mol^{-1}

$\therefore$ the molar mass = $\left(\frac{1.987}{0.029}\right)$ g mol^{-1}

= 68.5 g mol^{-1}

137. Heat $= nC_p\Delta T$

$128.7 = \frac{12.7}{63.5} \times C_p \times 110$

C_p = 5.85 cal mol^{-1} K^{-1}

138 $\left(\frac{P_2}{P_1}\right)^{\gamma-1} = \left(\frac{T_2}{T_1}\right)^{\gamma} \qquad 10^{0.67} = \left(\frac{T_2}{T_1}\right)^{1.67}$

$0.67 \log 10 = 1.67 \log\left(\frac{T_2}{T_1}\right)$

$\log\left(\frac{T_2}{T_1}\right) = \frac{0.67}{1.67} = 0.4$

$\therefore \quad \frac{T_2}{T_1} = 2.512$

139. $\left(\frac{P_1}{P_2}\right)^{\gamma-1} = \left(\frac{T_1}{T_2}\right)^{\gamma} \cdot \left(\frac{1}{10}\right)^{\gamma-1}$

$= (0.4)^{\gamma}$ or $10^{\gamma-1} = (2.5)^{\gamma}$

Taking log $(\gamma - 1) = \gamma \times 0.3979$

$\therefore \quad \frac{\gamma-1}{\gamma} \simeq 0.4 \qquad \gamma = 1.67$

atomicity = 1

140. At a: P = 1 atm ,

V = 22.4 litre for 1 mole of ideal gas .

$\therefore \quad$ T = 273K

At b: P = 2 atm, V = 22.4 litre, T = 546K

At c: P = 1 atm b $\rightarrow$ c is adiabatic.

Work done is $\int_b^c PdV = \frac{R}{\gamma - 1} \times \Delta T$

But $\left(\frac{P_2}{P_1}\right)^{\gamma-1} = \left(\frac{T_2}{T_1}\right)^{\gamma} \cdot \left(\frac{1}{2}\right)^{0.67} = \left(\frac{T_2}{546}\right)^{1.67}$

$0.67 \times \log\left(\frac{1}{2}\right) = 1.67 \times \log\left(\frac{T_2}{546}\right)$

$\therefore \quad \frac{0.67}{1.67} \times 0.3010 = \log\left(\frac{546}{T_2}\right)$

$0.1208 = \log\frac{546}{72}$

$\therefore \quad 1.321 = \frac{546}{T_2}$

$\therefore \quad T_2 = 413.3$K

$\therefore \quad$ Work done $= \frac{R}{\gamma - 1}(546 - 413.3)$

$= \left(\frac{1.987}{0.67}\times132.7\right)$ cal = 393.5 cal

This is work done in adiabatic expansion

From c → a we have isobaric compression (PΔV) at constant pressure i.e., as volume decreases at constant pressure temperature falls. For one mole it is RΔT = 1.987(413.3 – 213) cal which is 278.8 cal. For a to b no work is done since volume is constant (PΔV = 0). Total work done = (393.5 – 278.8) cal, which is 114.7 cal [Note a → b is a heating process, b to c, adiabatic cooling, c to a isobaric cooling heat absorbed externally.

$a \rightarrow b = C_V \Delta T = 3 \times 273 = 819$ cal

b → c no heat absorbed or rejected.

C → a heat rejected

$= C_p \Delta T = (5 \times 140.3)$ cal = 701.5 cal

Balance = 819 – 701.5 = 117.5 cal which practically agrees with the total work done i.e., 114.7 cal barring small errors due to approximations).

141. $CH_{4(g)} + 4CuO \rightarrow CO_{2(g)} + 2H_2O_{(l)} + 4Cu_{(s)}$

$-17.89 \quad 4 \times x \quad -94.05 \quad -2 \times 68.32 \quad$ zero

$\Delta H = -62.4$ k cal

$-230.69 + 17.89 - 4x = -62.4$

$\therefore \quad 4x = -150.4$ k cal

$\therefore \quad x = -37.6$ k cal mol^{-1}

142. ΔH per mole = TΔS = (329 × 88) J mol^{-1}

= 28952 J

$\therefore$ Specific enthalpy of vapourization i.e., latent heat of vapourization per gram = $\frac{28952}{58}$ J g^{-1}

= 499 J g^{-1}

143. $C_2H_{6(g)} + H_2 = 2CH_4 \qquad H_{TC} = -15.6$ k cal

$C_3H_{8(g)} + 2H_{2(g)} = 3CH_4 \qquad H_{TC} = -20.9$ k cal

Subtract 2 × eqn (1) from eqn (2) and rearrange

$C_3H_{8(g)} + CH_4 = 2C_2H_6$

$-20.9 - 2(-15.6) = (-20.9 + 31.2)$ k cal

= +10.3 k cal

144. Reverse eqn (i), add to eqn (ii), cancel common terms.

The answer is (–14.22 + 0.86) k cal

= –13.36 k cal

145. $\frac{1}{2} H_{2(g)} + \frac{1}{2} X_{2(g)} = HX_{(g)} \qquad \Delta H = -22.1$ k cal

$HX \quad + aq \quad = \quad H_{(aq)}^{+} \quad + \quad X_{(aq)}^{-}$

$-22.1 \qquad\qquad$ zero $\qquad\quad \Delta H_f$

$\Delta H = -17.9$ k cal

$\therefore \quad \Delta H_f - (-22.1) = -17.9$ k cal

$\therefore \quad \Delta H_f = (-22.1 - 17.9)$ k cal $= -40$ k cal

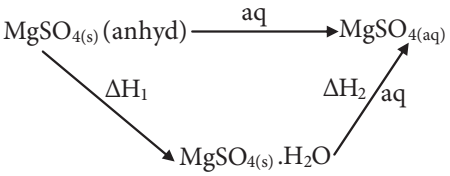
146. $C_2H_{4(g)} + 3O_{2(g)} = 2CO_{2(g)} + 2H_2O_{(l)}$

$2.710 \quad$ zero $\quad 2 \times (-96.960) \quad 2 \times (-68.360)$

ΔH = ?

ΔH = –136.72 – 193.92 – 2.710 = –333.35 k cal

147.

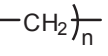


$\Delta H = -20.280$ k cal mol^{-1}

$\Delta H_1 + \Delta H_2 = -20.280$ K

1 : 1.9

$\Delta H_1 = \left(-20.28 \times \frac{1}{2.9}\right)$ k cal $= -6.99$ k cal mol^{-1}

148. $nCH_2 = CH_2 \rightarrow$ 

$\Delta H = E_{C=C} - 2E_{C-C}$

$= +540 - 2 \times 332 = -124$ kJ mol^{-1}

$\therefore$ 140 g of ethylene = $\frac{140}{28} = 5$ mol

$\therefore$ Total heat change = 5 × 124 = –620 kJ

149. $-13400 = -13680 + \Delta H_{ionization}$ cal

$\therefore \quad \Delta H_{ionization} = (13860 - 13400)$ cal = +280 cal

150. $x \times (-13680) + (1 - x) \times (-14830) = -13960$ cal

$\therefore \quad 13680\,x + 14830(1 - x) = 13960$

$(14830 - 13680)\,x = (14830 - 13960)$

$1150\,x = 870$

$\therefore \quad x = 0.757(1 - x) = 0.243$

ratio = 0.757 : 0.243 = 3.115 : 1

151. One may consider that x eqt of NaB decomposes to yield HB and x eqt of NaA are formed

$\therefore$ $x(14280 - 13780) = 455$ cal

$\therefore$ $x = \dfrac{455}{500} = 0.91$

152.

$$H_{2(g)} + Cl_{2(g)} = 2HCl_{(g)}$$

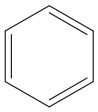
$+435$ $+243$ -2×431

$2H + 2Cl$

$\Delta H = (435 + 243 - 862)$ k J for 2 moles

$= -184$ k J

$\therefore$ for one mole -92 k J mol^{-1}

153. Assuming a cyclic triene structure  (kekule)

$$6C_{(s)} + 3H_{2(g)} = C_6H_{6(g)}$$

$$6C_{(g)} + 6H$$

ΔH by calculation is [in k cal]

$(6 \times 171.4) + (3 \times 104.2) - (6 \times 99.4) -$

$(3 \times 143.6) - (3 \times 83.2)$

Thus ΔH (by calculation) $= 64.2$ k cal mol^{-1}

The observed value $= 19.8$ k cal

$\therefore$ resonance energy $= 44.4$ k cal

154.

$$C_2H_5 - OH_{(l)} + 3O_{2(g)} \equiv 2CO_{2(g)} + 3H_2O_{(l)}$$

-66 zero -2×94 -3×68

$\Delta H = ?$

$\Delta H = (-2 \times 94) + (-3 \times 68) - (-66) = -326$ k cal

$CH_3 - O - CH_{3(g)} + 3O_{2(g)} = 2CO_{2(g)} + 3H_2O_{(l)}$

$\Delta H = -348$

Subtracting $-326 + 348$ k cal $= +22$ k cal

155.

$$C_3H_6(\text{propene}) + \frac{9}{2}O_2 = 3CO_{2(g)} + 3H_2O_{(l)}$$

4.9 zero -282 -204

$\Delta H = -400.9$ k cal

$\therefore$ cyclopropane $\rightarrow$ propene

-500 -400.9

$\Delta H = +9.1$ k cal

156.

$$C_2H_5{-}OH_{(l)} + 3O_{2(g)} \rightarrow 2CO_{2(g)} + 3H_2O_{(l)}$$

x zero $2 \times (-96.0)$ $3 \times (-68.0)$

$\Delta H = \pm 341.800$ k cal

$\therefore$ $-192 - 204 - x = -341.8$

$\therefore$ $x = [-396 + 341.8]$ k cal mol^{-1}

$x = -54.2$ k cal mol^{-1}

157. $\dfrac{T_1 - T_2}{T_1} = \eta$ $\dfrac{T_1 - T_2 - 100}{T_1} = 0.8\,\eta$

$$= \frac{T_1 - T_2}{T_1} - \frac{100}{T_1} = 0.8\,\eta$$

$\therefore$ $\dfrac{100}{T_1} = 0.2\,\eta$ $\dfrac{T_1 + 400 - T_2}{T_1 + 400} = 1.2\,\eta$

$$\frac{T_1 + 400 - T_2}{T_1} \times \frac{T_1}{T_1 + 400} = 1.2\,\eta$$

$$\left(\frac{T_1 - T_2}{T_1} + \frac{400}{T_1}\right)\left(\frac{T_1}{T_1 + 400}\right) = 1.2\,\eta;$$

$$(\eta + 0.8\,\eta)\left(\frac{T_1}{T_1 + 400}\right) = 1.2\,\eta$$

$\therefore$ $\dfrac{T_1 + 400}{T_1} = \dfrac{1.8}{1.2} = \dfrac{3}{2}$ $\quad \therefore T_1 = 800$K

$$\frac{100}{800} = 0.2\,\eta$$

$\therefore$ $\eta = \dfrac{100}{800 \times 0.2} = \dfrac{100}{160} = \dfrac{5}{8}$

$\dfrac{800 - T_2}{800} = \dfrac{5}{8} = \dfrac{500}{800}$ $\quad \therefore T_2 = 300$K

158. Since no work is done in the expansion. There is no change in the internal energy,

$\therefore$ no change in temperature. Thus $T_{final} = 300$K,

$$\Delta U = 0,\ \Delta S = R\ln\left(\frac{V_2}{V_1}\right) = R.\ln 2 = 0.693\,R$$

159. Work done $= 5 \times 8.314 \times 300 \ln\left(\dfrac{1}{10}\right)$ J

$= -2.872 \times 10^4$ J (convention work done by the gas is negative]

Isotheral process for ideal gas : $\Delta U =$ zero

PV does not change in the process

$\therefore$ $\Delta H =$ zero $\Delta S = \dfrac{Q}{T} = 95.7$ J deg^{-1} increase since heat [Q] is absorbed.

160. $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$

$$\Delta G^\circ + T\Delta S^\circ = \Delta H^\circ = \left[4.6 + \frac{298\times 94.6}{1000}\right] \text{k J}$$

$\Delta H^\circ = 32.76$ k J

$$\text{Normal boiling point} = \frac{\Delta H}{\Delta S} = \frac{32.76\times 10^3}{94.6}$$

$= 346.6\text{K} \quad = 73.3^\circ\text{C}$

161. For the reaction $N_{2(g)} + 3H_{2(g)} \rightarrow 2NH_{2(g)}$ ΔS is –ve. For the other three, ΔS is positive.

162. In this case the amount of gas is not specified.

$$\Delta S = nR \ln\left(\frac{V_2}{V_1}\right)$$

$$T\Delta S = W_{rev} = nRT \ln\left(\frac{V_2}{V_1}\right)$$

$= 298 \times 10 = 2980$ cal

$$\therefore \frac{W_{(actual)}}{W_{(reversible)}} = \frac{1000}{2980} \simeq 0.34$$

163. This is an application of Trouton's law

$$T = \frac{27.2\times 1000}{88.31}\text{K} = 308\text{K}$$

164. $\therefore \Delta S = R\ln\left(\frac{V_2}{V_1}\right) = 1.987 \times 2.303 \times \log 2$

$\simeq 1.37$ cal deg^{-1}

165. $\Delta G = \Delta H - T\Delta S$; $-98.55 = \Delta H - \frac{303(-4.80)}{1000}$

$= (\Delta H + 1.4544)$ k J $\qquad \therefore \Delta H = -100$ k J

166. $\Delta n = (2 - 3) = -1$

$$\Delta H = \Delta U + \Delta n\, RT = \left[-2.5 + \frac{(-1)\times 2\times 298}{1000}\right] \text{k cal}$$

$\Delta H = [-2.5 - 0.596]$ k cal $= -3.096$ k cal

$$\Delta G^\circ = \Delta H^\circ - T\Delta S = -3.096 - \frac{(298)\times(-10.5)}{1000}$$

$= (-3.096 + 3.129)$ k cal

$= 0.033$ k cal

167. At the melting point solid $\rightleftharpoons$ liquid. We have an equilibrium situation.

168. $\Delta G^\circ = -RT \ln K$

$$= -1.987 \times \frac{2298}{1000} \times 2.303 \times \log(1.6\times 10^{-3}) \text{ k cal}$$

$\simeq -2 \times 2.3 \times 2.303 \times (-2.7959)$ k cal $= 29.62$ k cal

169. Using the given equation as such,

$\Delta G^\circ = -RT \ln K_p$

$= -2.030\ RT \log K_p$

$$\frac{-10.1\times 10^3}{-2.303\times 8.314\times 500} = \log K_p$$

$K_p = \text{Antilog } 1.055 = 11.2$

170. $\Delta G^\circ = -RT \ln K_p$ and $K_p = P_{O_2}^{1/2}$

$$= \left(\frac{0.089}{760}\right)^{1/2} = 0.01082$$

$$\therefore \quad \Delta G^\circ = -8.314 \times \frac{298}{1000} \times \ln(0.01082)$$

$= +11.214$ k J

171. Any process in an isolated system has $\Delta S > o$ if it is irreversible $q = 0$ for an isolated system.

$\therefore \quad \frac{q}{T} = 0$ for reversible change. In an irreversible process $\Delta S > \frac{q}{T}$. i.e., $\Delta S > 0$

172. Obviously U depends only on T for an ideal gas since intermolecular interactions (attractive or repulsive) are absent.

$$\left(\frac{\partial U}{\partial V}\right)_T = 0 \text{ and } dU = C_V\, dT$$

show that internal energy of a perfect gas does not vary with temperature.

173. $C_V = \frac{3}{2}R$, $C_P = \frac{5}{2}R$.

$$\therefore \quad r = \frac{C_P}{C_V} = 1.67$$

For 3 translational degrees of freedom U

$$= 3\times\left(\frac{1}{2}RT\right) \text{ acc. To the principle of equipartion}$$

of energy. $\therefore C_V = \left(\frac{\partial U}{\partial T}\right)_V = \frac{3}{2}R$.

174. Both statement –1 and statement –2 are correct and statement –2 is the correct reason for statement –1.

175. Ionization in aqueous solution involves not only charge separation but other factors like hydration of ions reorientation of water molecules in the proximity of ions modified dielectric constants etc. Hence ΔH

ionization may be +ve or –ve. E.g., for acetic acid ΔH_{ion}: + 280 cal mol^{-1} for butyric acid $\Delta H_{ion} = -120$ cal mol^{-1}.

176. Statement –1 is an approximate law due to Trouton . Statement –2 is true.

177. Statement –1 is wrong. Bond enthalpy values are only approximate but reasonably constant to allow a certain measure of additivity in calculations.

178. $\Delta U = Q_V$
If there is no volume change there is no mechanical work.

179. Both statements are true (standard) (well known).

180. $G = H - TS \qquad dG = dH - TdS - SdT,$
$dH = dU + PdV + VdP$
$\therefore \quad dG = dU + PdV + VdP - TdS - SdT,$
$TdS = dU + PdV$
$\therefore \quad dG = VdP - SdT$

It follows that $\left[\left(\frac{\partial G}{\partial T}\right)_P = -S\right]$

$\Rightarrow \quad \left(\frac{\partial(\Delta G)}{\partial T}\right)_P = -\Delta S \ \therefore$ From $\Delta G = \Delta H - T\Delta S$

$\Delta G = \Delta H + T\left(\frac{\partial \Delta G}{\partial T}\right)_P$

181. The ratio is the inverse of 131.3 and 393.5.
Thus 3 : 1

182. $-(94 \times x) + [-26.4)(1 - x)] = -57.5;$
$(94 - 26.4)x = (57.5 - 26.4)$

$x = \frac{31.1}{67.6} = 0.46$

183. $\frac{-(94 \times 0.72) + (-26.4)(0.28)}{(0.72 + 0.14)}$ k cal

$= \frac{-67.68 - 7.392}{0.86}$ k cal $\qquad = -87.29$ k cal

$\therefore \quad \frac{-\Delta H}{n} = 87.3$ k cal per mole of O_2 consumed.

184. ΔH of ionization $= (13700 - 13680) \times \left(\frac{100}{80}\right)$

$= 25$ cal mol^{-1}

185. $\left(-13680 \times \frac{9}{10}\right) + \left(-13280 \times \frac{1}{10}\right) = -13640$ cal

186. $NaB + HA = NaA + HB$
$HA + NaOH \rightarrow NaA + H_2O \quad \Delta H = -13680$ cal
$HB + NaOH \rightarrow NaB + H_2O \quad \Delta H = -13280$ cal
$\therefore \quad NaB + HA \rightarrow NaA + HB$
$\Delta H = -13680 + 13280 = -400$ cal

187. $\Delta G = \Delta H - T\Delta S$

$= \left[-673 - \frac{298 \times 60.4}{1000}\right]$ k cal mol^{-1}

$= -691$ k cal mol^{-1}

188. $97°C = (273 + 97) = 370K$
At eqm. At 370K $\Delta G = 0$
$\therefore \quad \Delta H = T\Delta S$
$37000 = 370 \times \Delta S \qquad \therefore \quad \Delta S = 100$ cal deg^{-1}

189. $\Delta S = \left(\frac{18 \times 540}{373}\right)$ cal deg^{-1} $mol^{-1} = 26$

190. The gas is monatomic $\therefore$ (a) is wrong
(d) has no real basis

191. $P = \frac{RT}{V - b} - \frac{a}{V^2}$

$\therefore \quad \int PdV = RT \ln\left(\frac{V_2 - b}{V_1 - b}\right) + a\left(\frac{1}{V_2} - \frac{1}{V_1}\right)$

192. Simple calculation shows that (c) is incorrect.

193. $\Delta G = \Delta H(+ve) - T\Delta S(-ve)$

At a certain $T = \frac{\Delta H}{\Delta S}$, $\Delta G = 0$

Above this temperature ΔG becomes –ve (spontaneous reaction)

$\Delta G = \Delta H - T\Delta S$. ΔH is –ve. But ΔS may be –ve

In such a case $T\Delta S$ is +ve

$\therefore \quad \Delta G$ may be +ve

194. If we have an ideal gas undergoing free expansion (against vacuum) $\Delta U = 0$

195. (a), (b), (c)

Heat + work = Δu, which is a state property.

196. CO_2 is not monatomic, it is also easily condensable non-ideal in behaviour.
On comparing the work done

$W_{isobaric} > W_{isothermal} > W_{adiabatic} > W_{isochoric}$

$\Delta u = -q$

Temperature falls and hence internal energy decreases.

200. (a) $\rightarrow$ (p)

$\Delta S_{Total} = \Delta S_{System} + \Delta S_{Saroundings}$

$\Delta S_{Total}\ \Delta S - \dfrac{\Delta H}{T}$

Multiplying by $-T$

$-T\Delta S_{Total} = -T\Delta S + \Delta H = \Delta G$

CHAPTER

6

CHEMICAL AND IONIC EQUILIBRIA

CHAPTER OUTLINE

Preview

STUDY MATERIAL

CHEMICAL EQUILIBRIUM

Reversible Reactions

Law of Mass Action

Relation Between K_p, K_c and K_x
- Concept Strand (1)

Relation Between ΔG, $\Delta G°$ and Equilibrium Constant
- Concept Strands (2-3)

Le Chatelier's Principle—Theory of Mobile Equilibrium

Homogeneous Equilibria in Gas Phase
- Concept Strands (4-8)

Effect of the Variation of Total Pressure on a System at Equilibrium
- Concept Strands (9-10)

Homogeneous Equilibria in Liquid Phase
- Concept Strand (11)

Simultaneous Equilibria

Heterogeneous Equilibria
- Concept Strands (12-14)

Effect of Temperature on Equilibrium Constant
- Concept Strand (15)

Acid-base Concept

IONIC EQUILIBRIUM

pH Concept
- Concept Strands (16-24)

Dissociation Equilibria Involving Weak Acids and Weak Bases
- Concept Strands (25-29)

Salt Hydrolysis
- Concept Strands (30-34)

Common ion effect
- Concept Strand (35)

Buffer and Buffer Action
- Concept Strands (36-39)

Solubility Equilibria and Solubility Product
- Concept Strands (40-45)

Application of Solubility Product Principle in Qualitative Analysis
- Concept Strands (46-48)

Indicators in Neutralization Reaction
- Concept Strand (49)

TOPIC GRIP
- Subjective Questions (10)
- Straight Objective Type Questions (5)
- Assertion–Reason Type Questions (5)
- Linked Comprehension Type Questions (6)
- Multiple Correct Objective Type Questions (3)
- Matrix-Match Type Question (1)

IIT ASSIGNMENT EXERCISE
- Straight Objective Type Questions (80)
- Assertion–Reason Type Questions (3)
- Linked Comprehension Type Questions (3)
- Multiple Correct Objective Type Questions (3)
- Matrix-Match Type Question (1)

ADDITIONAL PRACTICE EXERCISE
- Subjective Questions (10)
- Straight Objective Type Questions (40)
- Assertion–Reason Type Questions (10)
- Linked Comprehension Type Questions (9)
- Multiple Correct Objective Type Questions (8)
- Matrix-Match Type Questions (3)

CHEMICAL EQUILIBRIUM

There are two important aspects in the study of chemical reactions (i) to what extent the given reaction will proceed towards products (ii) at what rate will it approach the equilibrium state. While the second aspect deals with the subject of chemical kinetics, the first aspect pertains to the study of chemical equilibria and is considered in detail in this chapter.

REVERSIBLE REACTIONS

Generally, all chemical reactions occur to some extent both in the forward and reverse directions, like for example the reaction between hydrogen and iodine at 700K according to

$$H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$$

But in many cases, the reaction takes place nearly completely in one direction with the reverse reaction being negligible. An example of such a reaction is the neutralization of a strong acid by a strong base in dilute solution given by

$$HCl_{(aq)} + NaOH_{(aq)} \rightarrow NaCl_{(aq)} + H_2O$$

To show that a reaction occurs in both directions, one can consider the decomposition of HI in a closed vessel at a certain temperature, say 450°C.

$$2HI_{(g)} \rightleftharpoons H_{2(g)} + I_{2(g)}$$

After a certain time, the reaction attains an equilibrium state with both hydrogen and iodine together with some undissociated hydrogen iodide being present in the system.

Any reaction, in general, can be considered as a reversible reaction, i.e., a reaction which takes place in both directions. When the experimental conditions are such that both forward and reverse reactions occur to an appreciable extent, the reaction is said to be a reversible reaction.

Several such reactions can be considered as examples.

(i) The reaction between hydrogen gas and oxygen gas in the required proportion in a closed vessel at 200°C given by

$$2H_{2(g)} + O_{2(g)} \rightleftharpoons 2H_2O_{(g)}$$

(ii) The esterification of ethyl alcohol by acetic acid in liquid phase in a sealed tube at 100°C according to the reaction.

$$C_2H_5OH_{(\ell)} + CH_3COOH_{(\ell)} \rightleftharpoons CH_3COOC_2H_{5(\ell)} + H_2O_{(\ell)}$$

(iii) The decomposition of calcium carbonate at high temperatures as per the reaction

$$CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$$

Characteristics of Reversible Reactions

There are two important observations which characterize a reversible reaction.

(i) All such reactions reach a state of chemical equilibrium after the lapse of a sufficient interval of time, i.e., a state in which no further change in the composition of the system is observed with time, assuming that the temperature and pressure are maintained constant.

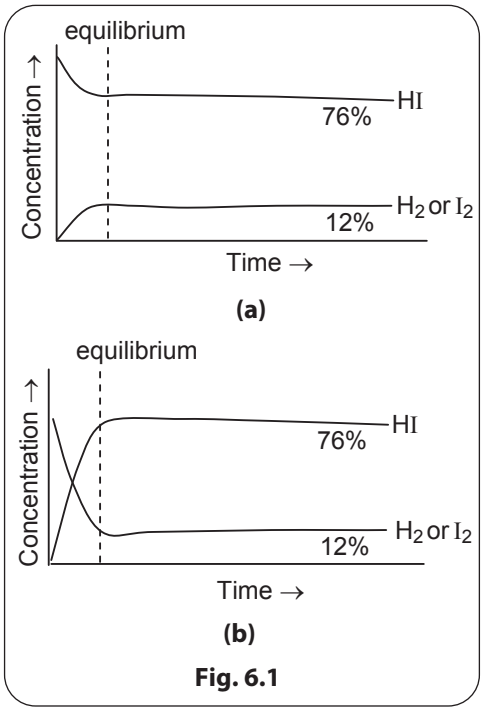


Fig. 6.1

(ii) The same state of equilibrium is obtained starting either from the reactants or products under properly chosen condition.

For example, $HI_{(g)}$ decomposes at 425°C to give an equilibrium mixture consisting of 12 mole% H_2, 12 mole% I_2 and 76 mole% HI (Figure 6.1(a)). The same equilibrium composition is reached when $H_{2(g)}$ and iodine vapour in equimolar proportion are heated in a sealed tube at a temperature of 698K (Fig.6.1(b)).

The constancy of equilibrium composition with time in a given reaction under chosen experimental condition is due to the fact that both forward and reverse reactions are proceeding at the same rate. The reacting system is then said to be in a state of dynamic equilibrium, which can be proved by isotopic labeling methods.

LAW OF MASS ACTION

The concept of dynamic equilibrium can be used to derive a relation between the concentrations of reactants and products at equilibrium. Such a relation is based on the "law of mass action" proposed by two Norwegian chemists C.M. Guldberg and P. Waage in 1864.

The law states that the rate of a chemical reaction depends on and in many cases is proportional to the product of the "active masses" of the reacting substances. The active mass in modern terminology is the activity or the "effective" concentration of a species and for all practical purposes it is taken as the molar concentration of the reacting species.

Consider a simple general reaction involving reactants A, B and products C, D given by

$$A + B \underset{k'}{\overset{k}{\rightleftharpoons}} C + D$$

Applying the mass law gives,

r_f = rate of forward reaction = $k[A][B]$

r_b = rate of the backward reaction = $k'[C][D]$

When the system attains equilibrium, the rate of forward reaction, r_f is equal to the rate of the backward reaction, r_b. k and k' are proportionality constants and are known as velocity constants or rate constants or specific reaction rates of the forward and backward reactions respectively and the formulae of the species in square brackets represent the concentrations of the various reacting species in mol L^{-1}.

Thus at equilibrium, $r_{f(equilibrium)} = r_{b(equilibrium)}$

$$\Rightarrow k[A]_{eq}[B]_{eq} = k'[C]_{eq}[D]_{eq}$$

Rearranging the above equation, we get,

$$\frac{[C]_{eq}[D]_{eq}}{[A]_{eq}[B]_{eq}} = \frac{k}{k'} = K_c$$

The subscript "eq" on various species indicates that they are equilibrium concentrations and K_c is called the equilibrium constant of the reaction. The subscript "c" on the equilibrium constant signifies that it is a constant based on the concentration of the reactants and products, expressed in moles $litre^{-1}$.

Extending the above equation to a more general reaction, $aA + bB + \ldots \rightleftharpoons 1L + mM + \ldots$, the equation for K_c may be written as

$$K_c \frac{[L]^{\ell}_{eq} \times [M]^{m}_{eq} \times .....}{[A]^{a}_{eq} \times [B]^{b}_{eq} \times .....}$$

where, the letters ℓ, m, a, b indicate the stoichiometric coefficients of the species L, M, A, B, respectively.

Test of equilibrium condition

An important test to verify whether a reaction like that given by the above equation is in equilibrium or not, is as follows. Addition of the reactant A or B must shift the equilibrium in the direction of the products C and D. Alternatively, addition of the product C or D results in a shift of the equilibrium in the direction of the reactants. If a system is not in a state of true equilibrium, either of these shifts will not take place or they will not occur in a manner predicted by equilibrium expression.

Convention in expressing equilibrium constant

In expressing the equilibrium constant, it is conventional to use the concentration of the products in the numerator and those of the reactants in the denominator. The inverse of K_C, $\left(\frac{1}{K_c} = K'_c\right)$ is also a constant.

The equilibrium constant of a reaction strongly varies with temperature.

It is pertinent to mention that the rate equations do not have general validity, i.e., they are not applicable in all cases because they depend on the mechanism of the reaction. However, the state of equilibrium is independent of any mechanism and equilibrium expressions are correct. Further consideration of this aspect is undertaken in the study of chemical kinetics. Some examples on the use of equation are given below.

(i) $N_{2(g)}+3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$

For which $K_c = \dfrac{[NH_3]^2}{[N_2][H_2]^3}$

(ii) $3O_{2(g)} \rightleftharpoons 2O_{3(g)} \quad K_c = \dfrac{[O_3]^2}{[O_2]^3}$

(iii) $2CH_3COOH \rightleftharpoons (CH_3COOH)_2$

For dimerization occurring in solvent like benzene

$$K_c = \frac{[(CH_3COOH)_2]}{[CH_3COOH]^2}$$

Equilibria involving gases

Considering the general equilibrium as a reaction occurring between ideal gases, the equation for the equilibrium constant, K_p, may be expressed in terms of the partial pressures of the various gaseous species as

$$K_p = \frac{p_L^{\ell} \times p_M^{m} \times .....}{p_A^{a} \times p_B^{b} \times .....}$$

The partial pressures of the various species L, M, A, B, etc., are those at equilibrium and the subscript "p" of K indicates that it is a constant based on partial pressures of reactants and products.

RELATION BETWEEN K_p, K_c AND K_x

The partial pressure p_i, of a gas i, in a mixture of ideal gases is given by

$$p_i \frac{n_i RT}{V} = C_i RT$$

Where, n_i is the number of moles of the constituent i in a total volume V (= concentration, C_i), R is the gas constant, and T is the temperature in kelvin. Substituting $p_i = C_i RT$ in the above K_p expression, we get

$$K_p = \frac{C_L^{\ell} \times C_M^{m} \times .....}{C_A^{a} \times C_B^{b} \times .....} \times \frac{(RT)^{\ell+m+....}}{(RT)^{a+b+....}} \Rightarrow$$

$$K_p = K_c (RT)^{\Delta n}$$

Where, Δn = total no. of moles of gaseous products – total no. of moles of gaseous reactants.
For example, for equilibrium reaction,

$$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$$

$\Delta n = 2 - 4 = -2.$

For equilibrium reaction,

$$2HI_{(g)} \rightleftharpoons H_{2(g)} + I_{2(g)},$$

$\Delta n = 2 - 2 = 0.$

It is also possible to express the equilibrium constant in terms of mole fraction units. The mole fraction, x_i of a constituent i in a mixture of ideal gas is given by

$$x_i = \frac{p_i}{P} \Rightarrow p_i = x_i P$$

where, p_i is the partial pressure of i and P is the total pressure. Substituting $p_i = x_i P$ in the K_p expression, we have

$$K_p = \frac{x_L^{\ell} \times x_M^{m} \times .....}{x_A^{a} \times x_B^{b} \times .....} \times P^{(\ell+m+...)-(a+b+...)} \Rightarrow$$

$$K_p = K_x . P^{\Delta n}$$

where, $\Delta n = (\ell + m + \ldots) - (a + b + \ldots)$ and K_x is the equilibrium constant in terms of mole fraction of various species.

CONCEPT STRAND

Concept Strand 1

The equilibrium constant, K_p for the reaction, $2CO_{(g)} + 6H_{2(g)} \rightleftharpoons 2CH_{4(g)} + 2H_2O_{(g)}$ is 1.52×10^{-3} at 727°C. Calculate K_c.

What will be the value of K_x if the total pressure of the system is 5 atm?

Also calculate K_p for the reaction, $CO_{(g)} + 3H_{2(g)} \rightleftharpoons CH_{4(g)} + H_2O_{(g)}$ at 727°C?

(R= 0.0821 L atm deg^{-1} mol^{-1})

Solution

$K_p = K_c (RT)^{\Delta n}$ $\Delta n = (2+2) - (2+6) = -4$

$$\Rightarrow \quad K_c = \frac{K_p}{(RT)^{\Delta n}} = \frac{1.52 \times 10^{-3}}{(0.0821 \times 1000)^{-4}} = 6.91 \times 10^4$$

$$K_x = \frac{K_p}{(P)^{\Delta n}} = \frac{1.52 \times 10^{-3}}{(5)^{-4}} = 0.95$$

For the reaction,

$CO_{(g)} + 3H_{2(g)} \rightleftharpoons CH_{4(g)} + H_2O_{(g)}$

$K_p = \sqrt{1.52 \times 10^{-3}} = 3.9 \times 10^{-2}$

RELATION BETWEEN ΔG, $\Delta G°$ AND EQUILIBRIUM CONSTANT

It is pertinent to mention at this stage that the relation

$$\Delta G = \Delta G° + RT \ln Q$$

$$\Delta G = -RT \ln K + RT \ln Q$$

$$\Rightarrow \quad \Delta G = RT \ln \frac{Q}{K}$$

is applicable to all cases where, K may be expressed in any concentration unit, viz. partial pressures, moles litre^{-1}. Correspondingly, Q also has to be expressed in the same unit. Q is known as the reaction quotient and K is the equilibrium constant of the reaction.

The relations are

$$\Delta G^\circ = -RT \ln K_p$$

$$\Delta G^\circ = -RT \ln K_c$$

$$\Delta G^\circ = -RT \ln K_x$$

where, $\Delta G°$ represents the standard Gibbs free energy change in the desired unit. Strictly, K_p, K_c, K_x are considered to be dimensionless quantities as each partial pressure, concentration or mole fraction belonging to a reactant or product is a ratio of the actual value to that in a chosen standard state.

CONCEPT STRANDS

Concept Strand 2

The standard Gibbs free energies of formation of $SO_{2(g)}$ and $SO_{3(g)}$ are –300.2 kJ and –371.1 kJ respectively at 298K. Calculate K_p, K_c and K_x of the reaction.

$2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)}$ at this temperature. The total pressure of the reaction mixture may be taken as 5 atm.

Solution

Reaction

$2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)}$

$\Delta G°$ of this reaction $= -2 \times 371.1 - (2 \times -300.2)$

$= -742.2 + 600.4 = -141.8$ kJ

$$-141.8 = -8.314 \times 10^{-3} \times 298 \times 2.303 \log K_p$$

$$\log K_p = \frac{141.8}{8.314 \times 10^{-3} \times 298 \times 2.303}$$

$= 24.85$; $K_p = 7.08 \times 10^{24}$

$$K_c = \frac{K_p}{(RT)^{\Delta n}} = \frac{7.08 \times 10^{24}}{(0.0821 \times 298)^{-1}} = 1.73 \times 10^{26}$$

$$K_x = \frac{7.08 \times 10^{24}}{(P)^{\Delta n}} = \frac{7.08 \times 10^{24}}{5^{-1}} = 3.54 \times 10^{25}$$

Concept Strand 3

Calculate the equilibrium constant of the reaction, $4HCl_{(g)} + O_{2(g)} \rightleftharpoons 2H_2O_{(g)} + 2Cl_{2(g)}$. Given that the standard enthalpy and entropy changes of the reaction are –116 kJ and –109 J K^{-1}. From these data, discuss the possibility of preparing pure chlorine by the oxidation of HCl with O_2.

Solution

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ = -116 - 298 \times -109 \times 10^{-3} = -83.52 \text{ kJ}$$

Since ΔG° is negative, the reaction is feasible

$$\Delta G^\circ = -RT \ln K$$

$$\log K = \frac{83.52 \times 10^3}{2.303 \times 8.314 \times 298} = 14.64$$

$$K = 4.36 \times 10^{14}$$

The very high value of K also proves that the forward reaction is highly favourable.

Magnitude of the equilibrium constant

The magnitude or the numerical value of the equilibrium constant of a reaction depends on the manner in which the reaction is written.

For example, the equilibrium constant, K_p of the reaction,

$$N_{2(g)}\, 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$$

at 673K is 1.64×10^{-4} with the partial pressures being expressed in atmosphere.

$$K_p = \frac{p^2_{NH_3}}{p_{N_2} \times p^3_{H_2}} = 1.64 \times 10^{-4}$$

If the above equation is written as

$$\frac{1}{2}N_{2(g)} + \frac{3}{2}H_{2(g)} \rightleftharpoons NH_{3(g)}$$

$$K'_p = \frac{p_{NH_3}}{p_{N_2}^{1/2} \times p_{H_2}^{3/2}} = \sqrt{K_p}$$

$$= \sqrt{1.64 \times 10^{-4}} = 1.28 \times 10^{-2}$$

Although, stoichiometrically both equations are correct, the equilibrium constant values differ due to different ways in which the equations have been written.

LE CHATELIER'S PRINCIPLE—THEORY OF MOBILE EQUILIBRIUM

For any given reaction, the position of equilibrium remains unaltered so long as the temperature and pressure are kept constant. However, when the temperature or pressure of a system is altered, the position of the equilibrium changes and this change is governed by the principle, referred to as the Le Chatelier's principle.

This principle formulated by French chemist H.L. Le Chatelier states that *"if a change occurs in any one of the external parameters such as the temperature or pressure of a system at equilibrium, the system will tend to adjust itself in such a way as to cancel the effect of this change"*.

Effect of change of pressure

Let us consider the effect of pressure on the equilibrium reaction involving the formation of $NH_{3(g)}$ from nitrogen and hydrogen according to

$$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$$

Increase of pressure on this reaction (at equilibrium) will tend to shift the reaction in the direction in which smaller number of molecules are formed i.e., towards a position where the pressure will decrease. In this case, the reaction is shifted towards the formation of more of ammonia.

As a second example, let us consider the decomposition of phosphorus pentachloride vapour into the trichloride and chlorine

$$PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$$

In this case, the equilibrium is shifted towards the formation of more of PCl_5 on increasing the pressure on the system, i.e., the position of the equilibrium is shifted backwards.

Effect of change in temperature

Consider the synthesis of ammonia from nitrogen and hydrogen as an example, it is an exothermic reaction with the

evolution of heat of about 47.7 kJ mol^{-1}. Increase of temperature of the system, therefore, shifts the position of equilibrium such that the rise of temperature is compensated i.e., the reaction proceeds in a direction involving absorption of heat i.e., the equilibrium is shifted backwards, i.e., in the direction of the reactants. This implies that higher temperatures are not favourable for formation of NH_3. Thus *all exothermic reactions are favoured at lower temperatures while endothermic reactions are favoured at higher temperatures.*

HOMOGENEOUS EQUILIBRIA IN GAS PHASE

Some examples of homogeneous equilibria in gas phase are considered below and relevant equations for the equilibrium constant will be derived.

Gaseous reactions in which $\Delta n = 0$

As a typical example, the reaction between hydrogen and iodine to give hydrogen iodide will be considered.

$$H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$$

In this case, $\Delta n = 2 - (1 + 1) = 0 \therefore K_p = K_c = K_x = K$.

If n_{H_2}, n_{I_2}, n_{HI} are the numbers of moles of H_2, I_2 and HI at equilibrium in a volume V litres, the concentration of each species = $\frac{n_i}{V}$

$$\therefore K = \frac{[HI]^2}{[H_2][I_2]} = \frac{\left(\frac{n_{HI}}{V}\right)^2}{\left(\frac{n_{H_2}}{V}\right)\left(\frac{n_{I_2}}{V}\right)} = \frac{n_{HI}^2}{n_{H_2} \times n_{I_2}}$$

The above equation may also be obtained by writing the expression for equilibrium constant in terms of partial pressures of various species.

$$K_p = K = \frac{p_{HI}^2}{p_{H_2} \times p_{I_2}} = \frac{\frac{n_{HI}^2}{N^2} \times P^2}{\frac{n_{H_2}}{N} \times P \times \frac{n_{I_2}}{N} \times P}$$

$$\Rightarrow K = \frac{n_{HI}^2}{n_{H_2} \times n_{I_2}}$$

Here, $p_i = \frac{n_i}{N} \times P$ where, n_i = number of moles of i, N = total number of moles of the reaction system and P = total pressure

From the above two equations, it is seen that the equation for the equilibrium constant does not contain the term V or P as they get cancelled. Thus it is independent of these parameters. This observation is applicable for all reactions for which $\Delta n = 0$. The hydrogen – iodine reaction has a K value of 48.7 at 457.6°C.

CONCEPT STRANDS

Concept Strand 4

The equilibrium constant of the reaction, $H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$ at 427°C is 54.6. Calculate the mole% equilibrium composition of the reaction mixture when a mole each of hydrogen and iodine are heated at this temperature in a 5 L vessel till equilibrium is attained. What are the concentrations of various species at equilibrium?

Solution

Let x moles of H_2 and I_2 react at equilibrium

Moles of H_2 left at equilibrium = moles of I_2 left at equilibrium = (1 – x)

Moles of HI formed = 2x

$$\Rightarrow \frac{2x}{1 - x} = \sqrt{54.6} = 7.389; \; x = 0.787$$

Mole percentage of $H_2 = \frac{(1 - 0.787)}{2} \times 100 = 10.65$

Mole percentage of I_2 = 10.65; Mole % of HI = 78.7

Concentration of H_2 = Concentration of I_2 = $\frac{1 - 0.787}{5} = 0.0426$ mol L^{-1}

Concentration of HI = $\frac{2 \times 0.787}{5} = 0.3148$ mol L^{-1}

Concept Strand 5

One litre of dinitrogen tetroxide weighs 3 g at 27°C and 742 mm pressure. Calculate the observed molar mass and the degree of dissociation.

Solution

$M_{N_2O_4}$ = molar mass of undissociated N_2O_4

$M_{(eq)}$ = molar mass of equilibrium mixture

Calculation of M_{eq}

Density = 3 g L^{-1}

Pressure = 742 mm Hg

$$= \frac{742}{760} \text{ atm}$$

T = 300K

$$\rho = \frac{PM_{(eq)}}{RT} \qquad M_{(eq)} = \frac{\rho RT}{P}$$

$$= \frac{3 \times 0.0821 \times 300}{\frac{742}{760}} = 75.68$$

$$\alpha = \frac{M_{N_2O_4} - M_{eq}}{M_{eq}} = \frac{92 - 75.68}{75.68} = 0.216$$

Determination of K and composition of the reaction mixture at equilibrium

From the knowledge of equilibrium constant, K, it is possible to determine the composition of the reaction mixture for any initial concentrations (or moles of the component in this case) of the reactants. For example, in the reaction,

$$H_2 + I_2 \rightleftharpoons 2HI$$

Let the number of moles of H_2 taken initially = a

Number of moles of I_2 taken initially = b

Number of moles of H_2 and I_2 reacting at equilibrium = x

Then, number of moles H_2 remaining at equilibrium = (a–x)

Number of moles I_2 remaining at equilibrium = (b–x)

Number of moles of HI formed at equilibrium = 2x

$$\Rightarrow \quad K = \frac{n_{HI}^2}{n_{H_2} \times n_{I_2}} = \frac{(2x)^2}{(a-x)(b-x)} = \frac{4x^2}{(a-x)(b-x)}$$

Since the above expression for the equilibrium constant does not contain pressure and volume terms, altering the pressure (or volume) of the system does not alter the position of equilibrium of this reaction. The same conclusion is applicable for all reactions for which $\Delta n = 0$. Application of Le Chatelier's principle leads to the same conclusion.

Other reactions in this category are

(i) Water gas reaction given by

$$CO_{2(g)} + H_{2(g)} \rightleftharpoons CO_{(g)} + H_2O_{(g)}$$

(ii) $N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}$

Gaseous reactions in which $\Delta n \neq 0$

For gaseous reactions in which the total number of moles between the reactants and products are not the same i.e., $\Delta n \neq 0$, the position of equilibrium depends on the total pressure.

It should be emphasized that any change in the total pressure of a system merely alters the composition of the system while not affecting the equilibrium constant at the same time. In other words, the concentration or partial pressures of the various constituents adjust themselves in such a way as to keep the equilibrium constant unchanged when the total pressure of the system is altered. Some examples of this category are considered below

Decomposition of N_2O_4

$$N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$$

In this case $\Delta n = 2 - 1$, $\therefore K_p = K_c.RT$; also $K_p = K_x.P$

For this reaction, $K_p = \frac{p_{NO_2}^2}{p_{N_2O_4}}$, where the partial pressure terms represents the values of the given species at equilibrium. We can write,

$$p_{NO_2} = \frac{n_{NO_2}}{N} \times P; \qquad p_{N_2O_4} = \frac{n_{N_2O_4}}{N} \times P$$

where, n represents the number of moles of the given species at equilibrium, P the total pressure of the system and N the total number of moles

$$K_p = \frac{n_{NO_2}^2}{N^2} \times P^2 \times \frac{N}{n_{N_2O_4} \times P} = \frac{n_{NO_2}^2}{n_{N_2O_4}} \times \frac{P}{N}$$

It is seen that the above equation contains the total pressure P and hence altering the same will affect the position of equilibrium. If P is increased (at constant N) at a

given temperature, n_{NO_2} must decrease and $n_{N_2O_4}$ must increase to keep K_p constant. Thus the equilibrium is shifted in the direction of N_2O_4, i.e., the reaction takes place in the backward direction.

The composition of the system under the new equilibrium conditions (i.e., when the pressure is increased) can be calculated by a different procedure.

Let the degree of dissociation of N_2O_4 be 'α' at equilibrium, we have

Number of moles of NO_2 formed = 2α

Number of moles of N_2O_4 remaining = 1 − α

Total number of moles at equilibrium = 2α + 1 − α = 1 + α

$$p_{NO_2} = \frac{2\alpha}{1+\alpha}.P \; ; \; p_{N_2O_4} = \frac{1-\alpha}{1+\alpha}.P$$

where, P is the total pressure of the system

$$K_p = \frac{P_{NO_2}^2}{P_{N_2O_4}} = \frac{4\alpha^2}{(1+\alpha)^2} \times P^2 \times \frac{(1+\alpha)}{(1-\alpha)p}$$

$$\frac{4\alpha^2}{(1+\alpha)(1-\alpha)} \times P$$

$$K_p = \frac{4\alpha^2}{1-\alpha^2} \times P$$

From the above equation, knowledge of K_p and P, will enable an evaluation of α. The above equation on rearrangement gives,

$$\alpha = \left(\frac{K_p}{4P + K_p}\right)^{1/2}$$

It is seen from the above equation that if P increases (at constant temperature), α must decrease to keep K_p constant. Both these conclusions are in accordance with the conclusions drawn on the basis of Le chatelier's principle.

Decomposition of PCl_5

$$PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$$

The expression for the equilibrium constant, K_p, in terms of degree of dissociation, α, of PCl_5 may be obtained in the following way.

At equilibrium

Number of moles of PCl_5 remaining = (1 − α)

Number of moles of PCl_3 = Number of moles of Cl_2 formed = α

Total number of moles at equilibrium = 1 − α + α + α

$$K_p = \frac{p_{PCl_3} \times p_{Cl_2}}{p_{PCl_5}}$$

$$= \frac{\alpha}{(1+\alpha)} \times P \times \frac{\alpha}{(1+\alpha)} \times P \times \frac{(1+\alpha)}{(1-\alpha)p}$$

$$K_p = \frac{\alpha^2}{1-\alpha^2} \times P$$

where, P is the total pressure of the system

$$\alpha = \left(\frac{K_p}{K_p + P}\right)^{1/2}$$

It is seen from the above equation that as P increases (at constant temperature) α decreases i.e., the degree of dissociation of PCl_5 decreases as the pressure of the system is increased. Thus the equilibrium is shifted in the direction of PCl_5. This conclusion is in accordance with the Le chatelier's principle.

CONCEPT STRANDS

Concept Strand 6

For the reaction $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$, K_p at 250°C is 1.78. Calculate the number of grams of PCl_5 to be taken in a five litre vessel to get an equilibrium concentration of chlorine of 0.05 mol L^{-1}.

Solution

$K_p = K_c(RT)^{\Delta n} \Rightarrow 1.78 = K_c(0.0821 \times 523)$

$$\Rightarrow \quad K_c = \frac{1.78}{0.0821 \times 523} = 4.145 \times 10^{-2}$$

Let x moles of PCl_5 be required to be taken in a 5 L vessel

$$\text{Concentration of } PCl_5 \text{ at equilibrium} = \left(\frac{x}{5} - 0.05\right)$$

$$K_c = 4.145 \times 10^{-2} = \frac{0.05 \times 0.05}{\left(\frac{x}{5} - 0.05\right)} \Rightarrow \left(\frac{x}{5} - 0.05\right) = 0.0603$$

$$\frac{x}{5} = 0.1103; \; x = 0.5515 \text{ moles}$$

$$= 0.5515 \times 208.5 \text{ g} = 115 \text{ g}$$

Concept Strand 7

A partially dissociated phosphorous pentachloride gas in a 10 L vessel weighed 50.0 g at a pressure of 2 atm and a temperature of 300°C. Calculate the degree of dissociation and K_p.

Solution

$$PV = \frac{W}{M}RT;\ 2 \times 10 = \frac{50}{M} \times 0.0821 \times 573$$

$$M = \frac{50 \times 0.0821 \times 573}{20} = 117.6$$

Total mass before dissociation = Total mass after dissociation

$$PCl_5 \rightleftharpoons PCl_3 + Cl_2$$

Eqm $1 - \alpha$ α α

Total number of moles after dissociation = $1 + \alpha$

$\therefore$ $208.5 = (1 + \alpha)\ 117.6$

$\alpha = 0.773$

$$K_p = \frac{\alpha^2}{1 - \alpha^2} \times P = \frac{(0.773)^2}{1 - (0.773)^2} \times 2 = \frac{0.597 \times 2}{0.403}$$

$= 2.963$

Concept Strand 8

The standard free energy change for the reaction $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$ is −2.51 kJ at 523K. Calculate the degree of dissociation, α, when 0.2 moles of PCl_5 are placed in a 5 L vessel containing 0.2 moles of chlorine.

Solution

$$\Delta G^\circ = -RT \ln K_p$$

$$= -8.314 \times 10^{-3} \times 523 \times 2.303 \log K_p - 2.51$$

$$= -10.014 \log K_p$$

(or) $\log K_p = +0.2506 \Rightarrow K_p = 1.781$

Let x moles of PCl_3 be formed at equilibrium

$$p_{PCl_5} \text{ at equilibrium} = \frac{(0.2 - x) \times RT}{5};$$

$$P_{PCl_3} \text{ at equilibrium} = \frac{x \times RT}{5};$$

$$p_{Cl_2} \text{ at equilibrium} = \left(\frac{x + 0.2}{5}\right)RT$$

$$K_p = \frac{x.RT}{5} \times \left(\frac{x + 0.2}{5}\right)RT \times \frac{5}{(0.2 - x)RT}$$

$$1.781 = \frac{x^2RT + 0.2xRT}{5(0.2 - x)}$$

$$8.905\ (0.2 - x) = (x^2 + 0.2x)\ RT$$

$$= (x^2 + 0.2x)\ 0.0821 \times 523$$

$$42.94x^2 + 17.49\ x - 1.78 = 0$$

Solving, $x = 0.0844$

(or) $\alpha = \frac{x}{0.2} = \frac{0.0844}{0.2} = 0.422$

Determination of degree of dissociation from vapour density measurements

Density measurements in gaseous reactions provide a simple method of determination of the degree of dissociation of gases. When a gas dissociates producing more molecules, the volume of the system increases at constant temperature and pressure. The density, defined as weight per litre at 1 atm pressure (at constant pressure) decreases when volume increases (at a given temperature) and thus the difference between the density of the undissociated gas and the partially dissociated gas provides a method of calculating the degree of dissociation.

Consider the reaction

$$A \rightleftharpoons xB$$

Initially, let one mole of A be taken and α be its degree of dissociation.

At equilibrium,

Number of moles of A = $(1 - \alpha)$

Number of moles of B = $x\alpha$

Total number of moles = $1 - \alpha + x\alpha = 1 + \alpha\ (x - 1)$

The density of a given weight of gas at constant pressure is inversely proportional to the number of moles ($\because$ $\frac{\text{weight}}{\text{volume}}$ decreases as volume increases), the ratio d_1 of the undissociated gas to the density, d_2 of the dissociated gas is given by

$$\frac{d_1}{d_2} = \frac{1 + \alpha(x - 1)}{1}$$

$$\Rightarrow \quad \alpha = \frac{d_1 - d_2}{d_2(x - 1)}$$

The correctness of the above equation may be checked in the following way.

If there is no dissociation, $\alpha = 0$, $\therefore$ $d_1 = d_2$

If dissociation is complete, $\alpha = 1$, $\therefore$ $d_1 = x\ d_2$

EFFECT OF THE VARIATION OF TOTAL PRESSURE ON A SYSTEM AT EQUILIBRIUM

The total pressure on a system at equilibrium may be altered by the addition of a reactant or a product or even an inert gas. The effect of such an addition on the position of equilibrium will be briefly discussed.

By adding a reactant gas or gaseous product keeping the total volume of the system constant

We first write the equilibrium constant, K_p, for the general reaction,

$$aA + bB + \ldots \rightleftharpoons 1L + mM + \ldots \text{ as}$$

$$K_p = \frac{n_L^{\ell} \times n_M^{m} \times \ldots}{n_A^{a} \times n_B^{b} \times \ldots} \times \left(\frac{P}{N}\right)^{(\ell+m+\ldots)-(a+b+\ldots)} = K_n \left(\frac{P}{N}\right)^{\Delta n}$$

Under the given conditions, both P and N increase and the ratio $\frac{P}{N}$ is constant (at constant temperature). If a reactant is added (i.e., A or B….), the denominator of the above equation increases and to keep K_p constant, the numerator must also increase.

This implies addition of a reactant shifts the position of equilibrium in the direction of products. The addition of a product in turn, shifts the equilibrium in the direction of reactants.

By adding an inert gas keeping the total volume of the system constant

In this case, the pressure P as well as N, the total number of moles, increase but the ratio $\frac{P}{N}$ is constant. Also, K_n is not affected.

The addition of inert gas at constant volume has no effect on the position of equilibrium irrespective of the sign of Δn (i.e., whether it is positive or negative)

Effect of addition of an inert gas at constant pressure

In the above equation P is constant and N increases. Now considering a reaction with positive Δn, $\left(\frac{P}{N}\right)^{\Delta n}$ decreases. To keep K_p constant, K_n must increase i.e., n_L, n_M,…. must increase and n_A, n_B decreases. Thus the equilibrium shifts in the direction of products.

If Δn is negative, $\left(\frac{P}{N}\right)^{\Delta n}$ increases and to keep K_p constant K_n must decrease. Thus n_L, n_M must decrease and n_A, n_B must increase. Thus the equilibrium is shifted in the direction of reactants.

CONCEPT STRANDS

Concept Strand 9

The equilibrium constant K_p for the reaction, $N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$ is 1.354 at 328K.

(i) Calculate K_c.
(ii) What is the degree of dissociation at a total pressure of 1 atm?
(iii) What is the effect of decreasing the pressure to 0.5 atm on the degree of dissociation?
(iv) What is the effect of adding an inert gas on the position of equilibrium (a) at constant volume (b) at constant pressure?

Solution

(i) $K_c = \frac{K_p}{(RT)^{\Delta n}} = \frac{K_p}{RT} \quad (\because \Delta n = 1)$

$\Rightarrow \quad K_c = \frac{1.354}{0.0821 \times 328} = 5.03 \times 10^{-2}$

(ii) $\alpha = \sqrt{\frac{K_p}{K_p + 4P}} = \sqrt{\frac{1.354}{1.354 + 4}} = 0.503$

(iii) $\alpha \text{ at } 0.5 \text{atm} = \sqrt{\frac{K_p}{K_p + 2}} = \sqrt{\frac{1.354}{3.354}} = 0.635$

(iv) (a) There is no effect on the position of equilibrium when an inert gas is added at constant volume.

(b) From the equation $K_p = \frac{n^2_{NO_2}}{n_{N_2O_4}} \times \frac{P}{N}$

Addition of an inert gas at constant pressure, results in a decrease of $\frac{P}{N}$.

$\therefore$ n_{NO_2} must increase and $n_{N_2O_4}$ must decrease to keep K_p constant. The equilibrium shifts in the direction of NO_2 i.e., α increases.

Concept Strand 10

The equilibrium constant, K_p for the reaction, $SO_2Cl_{2(g)} \rightleftharpoons SO_{2(g)} + Cl_{2(g)}$ is 0.042 at 298K. Calculate its degree of dissociation at a total pressure of 5 atm. Will the degree of dissociation change if nitrogen is introduced into the system at a partial pressure of 0.5 atm keeping the total pressure same as above?

Solution

$$SO_2Cl_{2(g)} \rightleftharpoons SO_{2(g)} + Cl_{2(g)}$$

$$K_p = \frac{\alpha^2}{1-\alpha^2} \times P = \frac{\alpha^2}{1-\alpha^2} \times 5$$

$$\frac{\alpha^2}{1-\alpha^2} = \frac{0.042}{5} = 0.0084;\ \alpha = 0.091$$

In presence of N_2 at 0.5 atm,

$$p_{SO_2} + p_{Cl_2} + p_{SO_2Cl_2} = 5.0 - 0.5 = 4.5 \text{ atm}$$

When N_2 is introduced at constant P, $\frac{P}{N}$ decreases and K_n must increase. $\therefore \alpha$ changes

Using, $K_p = \frac{\alpha^2}{1-\alpha^2} \times P = \frac{\alpha^2}{1-\alpha^2} \times 4.5$

$$\frac{\alpha^2}{1-\alpha^2} = \frac{0.042}{4.5} = 0.0093; \frac{\alpha^2}{1-\alpha^2} = 0.0093; \alpha = 0.096$$

α increases from 0.091 to 0.096

HOMOGENEOUS EQUILIBRIA IN LIQUID PHASE

For a reaction taking place in liquid phase, the equilibrium constant may be expressed either in terms of mole fraction, x_i, or concentration units, C_i. The equilibrium constant of a general reaction of the type of equation, $aA + bB + \rightleftharpoons 1L + mM + \ldots$, may be expressed in terms of mole fraction unit as, $K_x = \frac{x_L^{\ell} \times x_M^m \times .....}{x_A^a \times x_B^b \times .....}$

An equation, similar to this may be written for K_c i.e., the equilibrium constant in terms of concentration of various reacting species.

One of the most commonly studied reaction of this category is the reaction between acetic acid and ethyl alcohol to give the ester and water according to

$$CH_3COOH_{(aq)} + C_2H_5OH_{(aq)} \rightleftharpoons CH_3COOC_2H_{5(aq)} + H_2O_{(\ell)}$$

K_x for this reaction may be written as

$K_x = \frac{x_{ester} \times x_{water}}{x_{alcohol} \times x_{acid}}$, where the x's are the mole fractions of the respective species.

In dilute solution, $x_i = \frac{n_i}{n_s}$ where, n_i is the number of moles of species, i(solute) and n_s is the number of moles of solvent. It may be noted that in the equation for x_i, it has been assumed that $n_i << n_s$ and hence $\frac{n_i}{n_i + n_s} \approx \frac{n_i}{n_s}$. The equilibrium constant may thus be written as

$$K_x = \frac{n_{ester} \times n_{H_2O}}{n_{alcohol} \times n_{acid}} \ (\because n_s \text{ term cancel out})$$

$$K_c = \frac{\frac{n_{ester}}{V} \times \frac{n_{water}}{V}}{\frac{n_{alcohol}}{V} \times \frac{n_{acid}}{V}} = \frac{n_{ester} \times n_{H_2O}}{n_{alcohol} \times n_{acid}}$$

The "V" terms cancel out due to the fact that there are same number of molecules in the reactants and product side. The equilibrium constant of the esterification is found to be 4.0 at 100°C. From a knowledge of the equilibrium constant, it is possible to calculate the composition of the reaction mixture at equilibrium for any given initial concentration of the reactants in the following way.

Let "a" and "b" be the number of moles of acetic acid and alcohol taken initially. Let y moles each of ester and water be formed at equilibrium.

$\therefore$ Number of moles of acetic acid left at equilibrium = (a – y)

Number of moles of alcohol left at equilibrium = (b – y)

The equilibrium constant, K, for the reaction is thus given by

$$K = \frac{y \times y}{(a-y)(b-y)} = \frac{y^2}{(a-y)(b-y)}$$

y can be evaluated from the above equation.

CONCEPT STRAND

Concept Strand 11

Two moles of acetic acid and one mole of ethanol are heated in a sealed tube at 100°C. Calculate the percentage composition in mole percentage of the reaction mixture at equilibrium. The equilibrium constant of the reaction, K at this temperature is 4. What is the mole percentage of ester formed if the same reaction was carried in the presence of one mole of water?

Solution

Let y moles of ester be formed at equilibrium

$C_2H_5OH + CH_3COOH \rightleftharpoons CH_3COOC_2H_5 + H_2O$

$\therefore$ At equilibrium, we have $n_{ester} = n_{water} = y$

At equilibrium, we have $n_{acid} = (2 - y)$

At equilibrium, we have $n_{alcohol} = (1 - y)$

$$\Rightarrow \quad 4 = \frac{y^2}{(2-y)(1-y)} \Rightarrow 3y^2 - 12y + 8 = 0$$

Solving, y = 3.15 and 0.845. But 'y' cannot be greater than the number of moles of acetic acid or ethanol

$\therefore$ y = 0.845

$\therefore$ $n_{ester} = n_{water} = 0.845$; $n_{acid} = 1.155$; $n_{alcohol} = 0.155$

Total number of moles = 3

$$\text{Mole\% of ester} = \text{Mol\% of water} = \frac{0.845}{3} \times 100 = 28.17$$

$$\text{Mol \% of acid} = \frac{1.155}{3} \times 100 = 38.5$$

Mol % of alcohol = 100 – (2 × 28.17 + 38.5) = 5.16

In presence of water

Let z moles of ester be formed in presence of added water of one mole.

$\therefore$ $n_{ester} = z$, $n_{water} = (1 + z)$,

$n_{acid} = (2 - z)$; $n_{alcohol} = (1 - z)$

$$\Rightarrow \quad \frac{z(z+1)}{(1-z)(2-z)} = 4$$

$3z^2 - 13z + 8 = 0$, z = 0.743 (cannot be 3.59)

Total number of moles = 0.743+1.743+1.257+0.257 = 4

$\therefore$ % of ester formed $= \frac{0.743}{4} \times 100 = 18.6$

SIMULTANEOUS EQUILIBRIA

In some equilibria, two or more reversible reactions with some common reactants and products occur simultaneously,

The equilibrium reaction, $CO_{2\,(g)} + H_{2\,(g)} \overset{K_p}{\rightleftharpoons} CO_{(g)} + H_2O_{(g)}$

is an example of this type. Two other equilibria occurring in the system are

$$H_2O_{(g)} \overset{K'_p}{\rightleftharpoons} H_{2(g)} + \frac{1}{2}O_{2(g)}$$

$$CO_{2(g)} \overset{K''_p}{\rightleftharpoons} CO_{(g)} + \frac{1}{2}O_{2(g)}$$

The equilibrium constant K_p is related to K'_p and K''_p by the equation, $K_p = \frac{K''_p}{K'_p}$

HETEROGENEOUS EQUILIBRIA

Any system consisting of more than one phase is called a heterogeneous system. Equilibria in systems consisting of different phases are called heterogeneous equilibria. As examples of such equilibria, we have

$$Fe_{(s)} + CO_{2(g)} \rightleftharpoons FeO_{(s)} + CO_{(g)}$$
$$Fe_{(s)} + H_2O_{(g)} \rightleftharpoons FeO_{(s)} + H_{2(g)}$$
$$CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$$

which consists of gaseous and solid phases.

In the application of law of mass action to heterogeneous equilibria, Guldberg and Waage stated that whenever a solid is involved in a reversible reaction its "active mass" should be regarded as constant at a given temperature irrespective of the amount of solid present. This implies that the equation for the equilibrium constant need not contain any term for the substances present as solids at equilibrium.

For pure solids and pure liquids in their standard state, active mass = 1 by convention.

As an example, let us consider the dissociation of calcium carbonate into calcium oxide and CO_2 according to

$$CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$$

The equilibrium constant of the reaction is given by

$$K_p = p_{CO_2}$$

Other examples of this type are

$$Ag_2O_{(s)} \rightleftharpoons 2Ag_{(s)} + \frac{1}{2}O_{2(g)}$$

$$HgO_{(s)} \rightleftharpoons Hg_{(\ell)} + \frac{1}{2}O_{2(g)}$$

Heterogeneous equilibria involving one solid and two gaseous phases

A good example of equilibrium of this type is the decomposition of ammonium hydrosulphide.

$$NH_4HS_{(s)} \rightleftharpoons NH_{3\,(g)} + H_2S_{(g)}$$

For which $K_p = p_{NH_3} \times p_{H_2S}$

If 'P' is the total pressure of the system at equilibrium and assuming that no product is present initially,

$$p_{NH_3} = p_{H_2S} = \frac{P}{2}$$

$$\Rightarrow K_p = \left(\frac{P}{2}\right)^2 = \frac{P^2}{4}$$

Equation for K_p in presence of a product

If the decomposition is taking place in presence of a product, say, ammonia at a partial pressure of x mm Hg, the equation for equilibrium constant must be derived in a different manner. Let the partial pressure of H_2S obtained on dissociation of NH_4HS be P' at equilibrium.

$\therefore$ Partial pressure of H_2S = P'

Partial pressure of NH_3 = (P' +x)

$\Rightarrow$ K_p = (P' + x)P'

P = total pressure of the system = P'+ P' + x =(2P' + x)

P' can be evaluated from a knowledge of K_p and x.

CONCEPT STRANDS

Concept Strand 12

The equilibrium constant K_p for the reaction, $CO_{2(g)} + C_{(s)} \rightleftharpoons 2CO_{(g)}$, is 122.0 at 1000°C. Calculate the mole% composition of the reacting gases and their partial pressures at a total pressure of 12.2 atm.

Solution

Let α be the degree of conversion of CO_2 to CO
At equilibrium

Moles of CO formed = 2α
Moles of CO_2 remaining =$(1 - \alpha)$
Total number of moles = $1 - \alpha + 2\alpha = 1 + \alpha$

Partial pressure of CO at equilibrium = $\frac{2\alpha}{1+\alpha} \times P$

Partial pressure of CO_2 at equilibrium = $\frac{1-\alpha}{1+\alpha} \times P$

$$K_p = \frac{p^2_{CO}}{p_{CO_2}} = \frac{4\alpha^2}{(1+\alpha)^2} \times P^2 \times \frac{(1+\alpha)}{(1-\alpha)P} = \frac{4\alpha^2}{1-\alpha^2} \times P$$

$$\frac{4\alpha^2}{1-\alpha^2} \times 12.2 = 122; \quad \frac{\alpha^2}{1-\alpha^2} = 2.5; \quad \alpha = 0.845$$

$$p_{CO} = \frac{2 \times 0.845}{1.845} \times 12.2 = 11.17 \text{ atm}$$

$$p_{CO_2} = \frac{0.155}{1.845} \times 12.2 = 1.02 \text{ atm}$$

$$\text{Mole \% of CO} = \frac{2 \times 0.845}{1.845} \times 100 = 91.6$$

$$\text{Mole \% of } CO_2 = \frac{0.155}{1.845} \times 100 = 8.4$$

Concept Strand 13

0.1 moles of ammonium hydrogen sulphide are vapourized in a 2 L vessel and the total pressure of the system at equilibrium was observed to be 360.0 mm of Hg The reaction is given by, $NH_4HS_{(s)} \rightleftharpoons NH_{3(g)} + H_2S_{(g)}$

(i) Calculate K_p (ii) What are the partial pressures of the gases if H_2S is introduced at a partial pressure of 0.05 atm into the system at equilibrium?

Solution

$$K_p = p_{H_2S} \times p_{NH_3}$$

$$\text{Total pressure} = \frac{360}{760} = 0.474 \text{atm}$$

$$\Rightarrow \quad p_{H_2S} = p_{NH_3} = 0.237 \text{ atm}$$

$$\therefore \quad K_p = (0.237)^2 = 0.0562 \text{ atm}^2$$

Let p'_{NH_3} be the partial pressure of NH_3 due to dissociation of NH_4HS

Partial pressure of H_2S at equilibrium = $(p'_{NH_3} + 0.05)$

$$\Rightarrow \quad (p'_{NH_3} + 0.05)\ p'_{NH_3} = 0.0562$$

$$\left(p'_{NH_3}\right)^2 + 0.05\ p'_{NH_3} - 0.0562 = 0$$

$$p'_{NH_3} = \frac{-0.05 \pm \sqrt{0.0025 + 0.2248}}{2}$$

$$= \frac{-0.05 \pm 0.4767}{2} = 0.213 \text{ atm}$$

Concept Strand 14

The dissociation pressure of mercuric oxide at 693K is 385mm Hg. Calculate the equilibrium constant K_p for the reaction, $2HgO_{(s)} \rightleftharpoons 2Hg_{(g)} + O_{2(g)}$.

Solution

Dissociation pressure = Total pressure P exerted on dissociation of HgO

P = 385 mm. The partial pressures of Hg vapour and oxygen are in the ratio 2:1

$$\therefore \ p_{Hg} = \frac{385 \times 2}{3} = 256.7\text{mm} = 0.338 \text{ atm}$$

$$p_{O_2} = 128.3\text{mm} = 0.169 \text{ atm}$$

$$K_p = (p_{Hg})^2 \times p_{O_2}$$

$$= (0.338)^2 \times 0.169 = 1.93 \times 10^{-2} \text{ atm}^3$$

EFFECT OF TEMPERATURE ON EQUILIBRIUM CONSTANT

The equilibrium constant of a reaction varies with temperature and this variation may be derived in a quantitative manner as shown below:

We know,

$$\Delta G^\circ = -RT \ln K_p$$

Differentiating the above equation with respect to temperature at constant pressure

$$\left[\frac{\partial}{\partial T}\left(\Delta G^0\right)\right]_P = -R \ln K_p - RT\left(\frac{\partial \ln k_p}{\partial T}\right)_P$$

Multiplying throughout by T

$$T\left[\frac{\partial}{\partial T}\left(\Delta G^0\right)\right]_P = -RT \ln K_p - RT^2\left(\frac{\partial \ln K_p}{\partial T}\right)_P$$

$$-T\Delta S^0 = \Delta G^0 - RT^2\left(\frac{\partial \ln K_p}{\partial T}\right)_P$$

On rearrangement, we get $\left(\frac{\partial \ln K_p}{\partial T}\right)_P = \frac{\Delta H^0}{RT^2}$

$\Delta H°$ in the above equation is the standard enthalpy change of the reaction.

The integrated form of the above equation may be obtained in the following way.

Integrating the above equation between the two temperature limits T_1 and T_2, T_1 being the lower and T_2 being the higher temperature, we get

$$\int_{(K_p)_1}^{(K_p)_2} \partial \ln K_p = \int_{T_1}^{T_2} \frac{\Delta H^0}{RT^2} dT$$

$$\Rightarrow \quad \ln \frac{(K_p)_2}{(K_p)_1} = -\frac{\Delta H^0}{R}\left[\frac{1}{T_2} - \frac{1}{T_1}\right] = \frac{\Delta H^0}{R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$$

$$= \frac{\Delta H^0}{R}\left[\frac{T_2 - T_1}{T_1 T_2}\right]$$

$$\Rightarrow \quad \log \frac{(K_p)_2}{(K_p)_1} = \frac{\Delta H^0}{2.303R}\left[\frac{T_2 - T_1}{T_1 T_2}\right]$$

In the above equation, $(K_p)_2$ and $(K_p)_1$ are the equilibrium constants at the temperatures T_1 and T_2, $\Delta H°$ is the standard enthalpy change of the reaction and is assumed to be independent of temperature between the temperature limits T_1 and T_2. For endothermic reaction ($\Delta H°$ is positive), the equilibrium constant increases with increase of temperature while for exothermic reaction, $\Delta H°$ being negative, the equilibrium constant decreases with increase of temperature.

A similar equation for the variation of K_c with temperature may be obtained as

$$\left(\frac{\partial \ln K_c}{\partial T}\right) = \frac{\Delta U^\circ}{RT^2}$$

where, $\Delta U°$ = standard internal energy change associated with the reaction.

CONCEPT STRAND

Concept Strand 15

The equilibrium constant K_p for the synthesis of ammonia given by $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$ at 673K is 2.13×10^{-4}. In a certain experiment N_2 and H_2 are mixed in the ratio of 1:3 and it was observed that 0.5 moles of NH_3 were formed at a total pressure of 100 atm. At what temperature would this experiment have been performed given the standard enthalpy of formation of ammonia is –46.1 kJ mol^{-1}.

Solution

$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$

When 0.5 moles of ammonia are formed at equilibrium

Moles of N_2 consumed = 0.25

Moles of H_2 consumed = $\frac{0.5 \times 3}{2} = 0.75$

Number of moles of N_2 at equilibrium =1 – 0.25 = 0.75

Number of moles of H_2 at equilibrium =3 – 0.75 = 2.25

Total number of moles at equilibrium = 0.75 + 2.25 + 0.5 = 3.5

$$p_{N_2} = \frac{0.75}{3.5} \times 100 = 21.43 \text{ atm};$$

$$p_{H_2} = \frac{2.25}{3.5} \times 100 = 64.28 \text{ atm};$$

$$p_{NH_3} = \frac{0.5}{3.5} \text{x} 100 = 14.29 \text{ atm}$$

$$K_p = \frac{p^2_{NH_3}}{p_{N_2} \times p^3_{H_2}} = \frac{(14.29)^2}{21.43 \times (64.28)^3} = 3.59 \times 10^{-5}$$

$$\log \frac{(K_p)_2}{(K_p)_1} = \frac{\Delta H^\circ}{2.303R}\left(\frac{T_2 - T_1}{T_1 T_2}\right)$$

$$\log\left(\frac{2.13 \times 10^{-4}}{3.59 \times 10^{-5}}\right) = -\frac{2 \times -46.1}{2.303 \times 8.314 \times 10^{-3}}\left[\frac{673 - T_1}{673 T_1}\right]$$

$$0.7733 = -4815.34\left[\frac{673 - T_1}{673 T_1}\right]$$

$$\Rightarrow \quad T_1 = 754.5\text{K}$$

IONIC EQUILIBRIUM

Equilibria between ions in solution may broadly be divided into the following categories:

(i) Equilibria dealing with the dissociation of weak acids or bases.
(ii) Equilibria associated with the solubility of sparingly soluble salts.
(iii) Equilibria involving the hydrolysis of salts formed from the neutralization of
  (a) weak acid by a strong base.
  (b) weak base by a strong acid.
  (c) weak acid by a weak base in aqueous solution.

These equilibria will be considered in detail in this chapter.

ACID-BASE CONCEPT

Introduction

The development of the theory of acids and bases has played a historical and significant part in the early stages of the development of physical chemistry of solutions. The pioneering work of Ostwald and Arrhenius on the dissociation behaviour of electrolytes and the catalytic influence of some substances in enhancing the rates of reactions like the inversion of sucrose in water led to the present day concepts about acids and bases.

Although acids and bases have been defined in several ways, the impetus to further developments in this area may be said to have begun with the definition of "acids" and "bases' by S. Arrhenius in 1884.

Arrhenius theory

According to Arrhenius an acid is a substance that gives out hydrogen ions in aqueous solution and a base is a substance that releases OH^- ions in aqueous solution. Thus, the following reactions are typical of Arrhenius theory

$$HNO_{3(aq)} \rightarrow H^+_{(aq)} + NO^-_{3(aq)}$$

$$KOH_{(aq)} \rightarrow K^+_{(aq)} + OH^-_{(aq)}$$

Other acid–base reactions according to Arrhenius theory may be represented as

$$HSO^-_{4(aq)} + H^+_{(aq)} \rightarrow H_2SO_{4(aq)}$$

$$OH^-_{(aq)} + HClO_{4(aq)} \rightarrow H_2O + ClO^-_{4(aq)}$$

Arrhenius theory suffers from the following drawbacks.

(i) H^+ and OH^- do not exist freely in water and are associated with molecules of water. For example, H^+ in aqueous solution exists at least with one molecule of water and is known as the hydronium ion. This is not made explicit in the theory.
(ii) The theory is applicable only to aqueous solutions. It doesnot take into account acid–base behaviour in other solvents like alcohols, glacial acetic acid, etc.
(iii) There are compounds like ammonia, hydrazine which behave as bases but do not contain OH^- ions. These are not recognized as bases under Arrhenius theory.

Bronsted–lowry theory

A major development in this area is the more general definition of acids and bases proposed by J.N Bronsted and T.M Lowry independently in 1923. It is known as the protonic theory of acids and bases.

According to them, *an acid is any substance which has a tendency to donate a proton and a base is any substance which has a tendency to accept a proton through coordination.*

An acid is thus a proton donor and a base is a proton acceptor. An acid-base equilibrium may be represented as

$$\underset{\text{acid}}{HA} \rightleftharpoons \underset{\text{base}}{A^-} + \underset{\text{proton}}{H^+}$$

The acid and base which differ by a proton are referred as a conjugate acid–base pair. Every acid therefore, has a conjugate base and every base, likewise, has a conjugate acid.

The following are some examples of conjugate acid–base pairs.

Acid	Conjugate base
HNO_3	NO_3^-
HSO_4^-	SO_4^{2-}
NH_4^+	NH_3

From the above examples, it is seen that acids and bases may be neutral molecules or ions.

In order that an acid may exhibit its acidic properties, a base, i.e., a substance capable of accepting the proton, must be present. In many cases, solvents like alcohols, liquid etc., NH_3 fulfill this role and the ionisation of acids and bases may therefore be ascribed to transfer of proton from an acid to the solvent or to the acceptance of a proton from the solvent by the base.

For example,

$$CH_3COO^- + H_2O \rightarrow CH_3COOH + OH^-$$

$$NH_4^+ + H_2O \rightarrow NH_3 + H_3O^+$$

Acid–base equilibria of this type can be written in a general form as

$$\text{Acid}_1 + \text{Base}_2 \rightleftharpoons \text{Acid}_2 + \text{Base}_1$$

An acid–base equilibrium may be written as reaction between two conjugate acid–base pairs.

For example, in the equation,

$$CH_3COO^- + H_2O \rightarrow CH_3COOH + OH^-,$$

acid_1 = water, $\text{base}_2 = CH_3COO^-$, $\text{acid}_2 = CH_3COOH$, $\text{base}_1 = OH^-$

The position of the equilibrium in a reaction like the one given above depends on the relative strengths of the acids and bases involved in the equilibrium reaction.

Generally, there exists a reciprocal relationship between the strength of an acid and its conjugate base, irrespective of whether they are neutral molecules or ion. The reverse of this statement is also true, i.e., the conjugate base of a weak acid is also strong (e.g., acetic acid–acetate ion).

Classification of solvents

Solvents like water which can act both as an acid and a base are called amphiprotic solvents. Alcohols like methanol, ethanol also belong to this category.

On the other hand, there are solvents, which have predominantly acidic or basic character. The former are called acidic solvents, examples being glacial acetic acid, conc. H_2SO_4 etc. Examples of basic solvents, i.e., the latter category, are liquid NH_3, ethylene diammine, etc.

Solvents which have poor proton donating or poor proton accepting tendencies are known as aprotic solvents, examples being benzene, cyclohexane etc.

Advantages and disadvantages of bronsted–lowry theory

The following are the advantages of Bronsted–Lowry Theory over the Arrhenius theory.

(i) The theory is not restricted to aqueous solutions alone but can be extended to other solvents as well.

(ii) The theory is more general as neutral, positively charged or negatively charged species can behave as acids or bases. No restriction is placed on a species behaving as a base like for example, in Arrhenius theory where a species is a base if it gives rise to OH^- ions in aqueous solution.

Drawbacks

A major drawback of Bronsted–Lowry theory is that it does not recognize substances not having a proton as acid or base.

For example, substances like BF_3 and $AlCl_3$ which are known to exhibit typical acid properties are not acids in the Bronsted–Lowry category.

Lewis acid–base theory

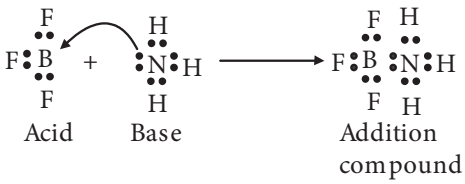
G.N Lewis proposed a new theory of acids and bases around 1930 and it is much more broad based compared to Bronsted–Lowry theory.

According to Lewis concept, *an acid is a substance which can accept a pair of electrons and base is any substance which can donate a pair of electrons in coordination.*

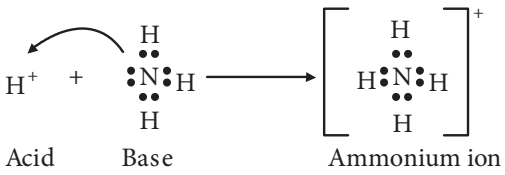
An acid–base reaction in Lewis theory involves a sharing of an electron pair through a covalent bond (or coordinate covalent bond) formation.

Some examples of Lewis acid–base reactions are given below.

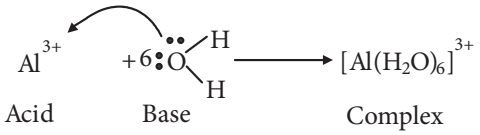
(i) Reaction between BF_3 and NH_3



(ii) Reaction between H^+ and NH_3



(iii) Reaction between Al3+ and H2O



Advantages of Lewis Theory

(i) Reactions, which do not involve the transfer of a proton, like the above given examples, are typical Lewis acid–base reactions.

(ii) All Bronsted–Lowry acid–base reactions also are typical Lewis acid–base reactions because the loss or gain of a proton is accompanied by a gain or loss of an electron pair.

Despite the more general nature of Lewis theory, it has a disadvantage in that this theory is not found to be useful in the quantitative comparison of acid–base strengths. For this reason, Bronsted–Lowry theory is more commonly used.

pH CONCEPT

An aqueous solution of a strong acid like HCl, $HClO_4$ or HBr contains hydronium ions (H_3O^+) at a concentration equal to the stoichiometric concentration of the acid. This implies that such acids are completely dissociated in solution and it is so even up to moderately high concentrations.

In aqueous solutions, the H^+ ion concentration can vary over a wide range, i.e., from a concentration of one mole per litre (at 1.0 M concentration of a strong acid) to 10^{-14} g ions L^{-1} (because of the ionic product equilibrium). In order to bring such a wide variation in H^+ concentration over 14 powers of ten, into a compact easily realizable numbers, S.P.L Sorensen introduced the concept of pH in 1909. The pH is defined as

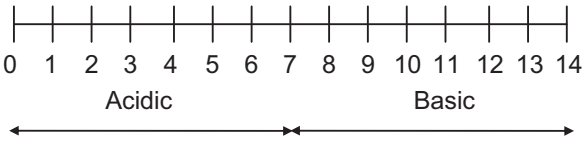
$pH = -\log_{10} a_{H^+}$ where, a_{H^+} is the activity of H^+ in solution

For dilute solutions, activity can be taken as concentration, then

$$pH = -\log_{10} [H^+]$$

where, $[H^+]$ is in gram–ions per litre.

From this definition, it is seen that smaller is the H^+ ion concentration, the greater will be the pH of the solution. For the above range of concentrations, the pH varies from 0 to 14. The following diagram shows the range of pH values in aqueous solution.



In calculating the pH of solutions of weak acids or bases, the degree of dissociation of the acid or base must be known.

The concentration of OH^- ions in solutions may be expressed in terms of pOH, a quantity analogous to pH and is given by the equation

$$pOH = -\log [OH^-]$$

The relation between pH and pOH is given via the ionic product relation, viz.,

$[H_3O^+]\,[OH^-] = K_w \Rightarrow -\log[H_3O^+] - \log [OH^-] = -\log K_w$

i.e.,

$$pH + pOH = pK_w$$

CONCEPT STRANDS

Concept Strand 16

Calculate the pH of the following solutions (i) 0.02 M H_2SO_4 (ii) 0.001 M $Ba(OH)_2$ at 298 K. Assume complete dissociation of the acid and base respectively.

Solution

(i) 0.02 M H_2SO_4
After complete dissociation, $[H^+] = 0.04$ M
$pH = -\log [H^+] = -\log [0.04] = -[\bar{2}.6] = 1.4$

(ii) 0.001 M $Ba(OH)_2$
After complete dissociation, $[OH^-] = 0.002$ M
$pOH = -\log[OH^-] = -\log [0.002] = -[\bar{3}.30] = 2.7$
$pH = pK_w - pOH = 14 - 2.7 = 11.3$

Concept Strand 17

Calculate the H^+ ion concentration of (i) a solution whose pH is 3.5 (ii) a solution whose pOH = 11.5.

Solution

(i) $pH = 3.5$; $-\log [H^+] = 3.5$; $\log [H^+] = \bar{4}.5$
$[H^+] = 3.16 \times 10^{-4}$

(ii) pOH = 11.5
$pH + pOH = 14$; $pH = 14 - 11.5 = 2.5$
$-\log [H^+] = 2.5$, $\log [H^+] = -2.5 = \bar{3}.5$; $[H^+] = 3.16 \times 10^{-3}$

Concept Strand 18

Calculate the pH of the following:

(i) 2.8 g of potassium hydroxide are dissolved in 100 mL of carbondioxide free water at 25°C.

(ii) A solution of H_2SO_4 having density of 1.02 g cm^{-3} and containing 3.24 wt per cent of acid at 20°C. Assume complete dissociation.

Solution

(i) 1000 mL of water contains $\frac{28}{56} = 0.5$ moles of KOH.

$[OH^-] = 0.5$ $[H^+] = \frac{10^{-14}}{0.5} = 2 \times 10^{-14}$

$pH = -\log[2 \times 10^{-14}] = -[\overline{14}.3] = 13.7$

(ii) Normality of the acid solution

$$= \frac{1.02 \times 3.24 \times 10}{49} = 0.674$$

$[H^+] = 0.674$
$pH = -\log [0.674] = 0.171$

Concept Strand 19

100 mL of a solution of HCl containing 3.65 g L^{-1} are mixed with 100 mL of a solution of NaOH containing 2.0 g L^{-1}. What is the pH of the resulting solution?

Solution

3.65 g L^{-1} of HCl = 0.1 M HCl

100 mL 0.1 M HCl = 10 milli moles

2.0 g L^{-1} of NaOH = $\frac{2}{40}$ = 0.05 M NaOH

100 mL of 0.05 M NaOH = 5 milli moles

when equal volume are mixed, (10 – 5) milli moles of H^+ is present in 200 mL

Final concentration = $\frac{5 \times 10^{-3}}{200} \times 1000 = 0.025 = [H^+]$

$pH = -\log [H^+] = 1.61$

Concept Strand 20

Calculate the pH of a solution which contains 5×10^{-8} moles of HCl in one litre of water at 25°C.

Solution

Since the concentration of acid in water is of the same order of H^+ ions obtained from pure water, the H^+ ions existing in water must also be taken into account. Equilibria to be considered are

$[H_3O^+]\,[OH^-] = 10^{-14}$,

Charge balance condition $[H_3O^+] = [OH^-] + [Cl^-]$

$[Cl^-]$ from dissociation of HCl = 5×10^{-8}

$\therefore$ $[H_3O^+] = [OH^-] + 5 \times 10^{-8}$

Substituting the above in ionic product equilibrium,

$[H_3O^+]\,[H_3O^+ - 5 \times 10^{-8}] = 10^{-14}$

$\Rightarrow$ $[H_3O^+]^2 - 5 \times 10^{-8}\,[H_3O^+] - 10^{-14} = 0$

On solving, $[H_3O^+] = 12.81 \times 10^{-8}$ from which pH = 6.89

Concept Strand 21

Calculate the number of H^+ and OH^- ions in 500 mL of a solution whose pH is 10.0.

Solution

$[H^+] = 10^{-10}$ mol L^{-1}

In 500 mL of solution, we have $0.5 \times 10^{-10} = 5 \times 10^{-11}$ g ions.

Number of ions = $5 \times 10^{-11} \times 6.023 \times 10^{23} = 3 \times 10^{13}$

$$[OH^-] = \frac{K_w}{[H^+]} = \frac{10^{-14}}{10^{-10}} = 1 \times 10^{-4}\ \text{g} - \text{ions/L}$$

In 500 mL of solution, we have $0.5 \times 10^{-4} = 5 \times 10^{-5}$ g ions.

∴ Number of OH^- ions = $5 \times 10^{-5} \times 6.023 \times 10^{23} = 3 \times 10^{19}$

Concept Strand 22

Calculate the pH of a mixture made up of (i) one gram of HCl in 750 mL of water (ii) one gram of NaOH in 250 mL of water.

Solution

(i) 0.0274 moles of HCl in 750 ml of water

(ii) 0.025 moles of NaOH in 250 mL of water

The mixture contains an excess of 0.0274 − 0.025 = 0.0024 moles of H^+ in 1000 mL.

$[H^+] = 0.0024$ M;

pH = -log [0.0024] = 2.62

Concept Strand 23

Assuming that the pH concept is applicable to methanol as solvent ($K_{autoprotolysis} = 10^{-19}$ at 298K), what is the pH range available in methanol? Would a solution containing 10^{-8} M $CH_3OH_2^+$ ions in methanol be considered as acidic or basic?

Solution

On analogy with water, pH range in methanol is 0 – 19. The neutral point in methanol corresponds to $p[CH_3OH_2^+] = 9.5$ and since the $p[CH_3O^+H_2]$ in 10^{-8} M solution is 8, the solution is acidic in this solvent.

Concept Strand 24

Ethanol can act as an acid and a base and the self ionization or auto protolysis constant of ethanol at 25°C is 8×10^{-20}.

(i) Write the relevant equation for self ionization of ethanol.

(ii) What is the $C_2H_5OH_2^+$ ion concentration in pure ethanol and what is the $P_{(C_2H_5OH_2^+)}$ (corresponding to pH in water) value?

(iii) What are the neutral pH, acidic and basic regions in this solvent?

Solution

(i) $2C_2H_5OH \rightleftharpoons \underset{\text{ethoxide ion}}{C_2H_5O^-} + \underset{\text{ethoxonium ion}}{C_2H_5OH_2^+}$

(ii) $\left[C_2H_5OH_2^+\right] = [C_2H_5O^-] = \sqrt{8 \times 10^{-20}}$

$= 2.83 \times 10^{-1}$

$pC_2H_5OH_2^+ = -\log\left[C_2H_5OH_2^+\right]$

$= -\log [2.83 \times 10^{-10}] = 9.55$

(iii) The neutral pH in this solvent is 9.55, values less than 9.55 pertain to acidic solution and values greater than 9.55 corresponds to basic solutions.

DISSOCIATION EQUILIBRIA INVOLVING WEAK ACIDS AND WEAK BASES

Electrolytes are classified as strong or weak on the basis of their conducting power in solution. Solutions of electrolytes like NaCl and KCl have a high conductivity and the equivalent conductivity of such salts decreases relatively little with increasing concentration and they are practically completely dissociated in solution. On the other hand, solution of organic acids like acetic acid and benzoic acid and organic bases like aniline have low conductivity and their equivalent conductivity increases greatly with dilution. This increase is attributed to a continuous in-

crease in the number of ions with dilution or decrease of concentration.

Consider a weak acid HA dissociating partially into H^+ and A^- ions in water according to

$$HA + H_2O \rightleftharpoons H_3O^+ + A^-$$

The equilibrium constant, K_a, of the reaction, which may also be designated as the dissociation constant of the weak acid is given by

$$K_a = \frac{[H_3O^+][A^-]}{[HA]} = \frac{[H^+][A^-]}{[HA]}$$

For simplicity, $[H_3O^+]$ in the above equation is represented as $[H^+]$. The term involving the concentration of the solvent, water, is omitted from the equilibrium expression, since it remains virtually constant (the change in its concentration being negligibly small).

Let "n" moles of a weak acid HA be dissolved in V litres of solvent (say water) and let a fraction, α, of the acid dissociate into free ions according to

$$HA + H_2O \rightleftharpoons H_3O^+ + A^-$$

Number of moles at equilibrium	$n(1-\alpha)$	$n\alpha$	$n\alpha$
Conc. at Equilibrium	$\frac{n(1-\alpha)}{V}$	$\frac{n\alpha}{V}$	$\frac{n\alpha}{V}$

The equilibrium constant of the reaction may be written as

$$K_a = \frac{\frac{n\alpha}{V} \times \frac{n\alpha}{V}}{\frac{n(1-\alpha)}{V}} = \frac{n\alpha^2}{V(1-\alpha)} \approx \frac{n\alpha^2}{V} \text{ (when } \alpha << 1)$$

Since $\frac{n}{V} = c$ (mol L^{-1}), the above equation may be written as

$$K_a = \frac{\alpha^2 c}{(1-\alpha)} \approx \alpha^2 c \text{ (when } \alpha << 1)$$

From the above equations, it is seen that as V increases or as the solution is made more dilute, α increases (at a given temperature) or α increases as c decreases to keep K_a constant.

The above conclusions are known as *Ostwald's dilution law* and it may be stated as follows: *The degree of dissociation of a weak acid increases with dilution.* The same conclusion is also applicable for weak bases.

Equation for H^+ concentration (–pH)

From the above equations, $\alpha = \sqrt{\frac{K_a}{c}}$

And, $[H^+] = \alpha c = \sqrt{K_a c}$

$\Rightarrow$ $pH = -\log[H^+]$

$$= -\frac{1}{2}\log K_a - \frac{1}{2}\log c = \frac{1}{2}pK_a - \frac{1}{2}\log c$$

$$pH = \frac{1}{2}\left[pK_a - \log C\right]$$

The dissociation constants of weak acids or weak bases can be determined by a variety of experimental methods such as conductance, EMF or colorimetric methods.

Consider the dissociation of a weak base like ammonium hydroxide

$$NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$$

If c is the concentration of the base in mol L^{-1}, taken originally,

$[NH_4^+]$ at equilibrium = αc

$[OH^-]$ at equilibrium = αc

Concentration of undissociated acid = (1 – α)c

$\therefore$ Dissociation constant,

$$K_b = \frac{[NH_4^+][OH^-]}{[NH_4OH]} = \frac{\alpha c \times \alpha c}{(1-\alpha)c} = \frac{\alpha^2 c}{1-\alpha};$$

$K_b \approx \alpha^2 c$ (when $\alpha << 1$)

The $[OH^-]$ concentration is given by

$[OH^-] = \alpha c = \sqrt{K_b c}$

$$pOH = -\log[OH^-] = \frac{1}{2}pK_b - \frac{1}{2}\log c$$

Using the relation $pH + pOH = pK_w$ (K_w = ionic product of water)

$\Rightarrow$ $pH = pK_w - pOH$

$$= pK_w - \frac{1}{2}pK_b + \frac{1}{2}\log c$$

$$pH = pK_w + \frac{1}{2}[\log c - pK_b]$$

CONCEPT STRANDS

Concept Strand 25

The dissociation constant of nitrous acid at 25°C is 5.1×10^{-4}. Calculate the degree of dissociation of a 0.02 M acid under two conditions (i) assuming $\alpha << 1$ (ii) without any assumption. On the basis of results, justify whether the assumption (i) is acceptable. Calculate H^+ concentration using the latter value (case (ii)) and pH of the solution.

Solution

$$HNO_2 + H_2O \rightleftharpoons H_3O^+ + NO_2^-$$

At eqm $(1-\alpha)c$ αc αc

$$K_a = \frac{\alpha^2 c}{1-\alpha}$$

assumption (i) $K_a = \alpha^2 c$

$$\alpha = \sqrt{\frac{K_a}{c}} = \sqrt{\frac{5.1\times10^{-4}}{0.02}} = \sqrt{2.55\times10^{-2}} = 0.160$$

assumption (ii)

$$\alpha^2 \times 0.02 = 5.1 \times 10^{-4} - 5.1 \times 10^{-4}\,\alpha$$

On simplification, we get

$$\alpha^2 + 2.55 \times 10^{-2}\alpha - 2.55 \times 10^{-2} = 0$$

$$\alpha = 0.147$$

Assumption (i) is not justifiable as the difference between the two values is nearly 9%.

$$[H^+] = \alpha c = 0.02 \times 0.147 = 2.94 \times 10^{-3}$$

$$pH = -\log [H^+] = -\log [2.94 \times 10^{-3}] = 2.53$$

Concept Strand 26

The pH of a 0.1 M solution of ammonia is found to be 11.10 at 25°C. Calculate the dissociation constant, K_b of the base.

Solution

$$pH + pOH = 14.0$$

$\Rightarrow$ pOH of the solution $= 14 - 11.10 = 2.90$

$\Rightarrow$ $[OH^-] = 1.26 \times 10^{-3}$

$$K_b = \frac{\alpha c \times \alpha c}{(1-\alpha)c} = \frac{(1.26\times10^{-3})^2}{0.10 - 0.00126}$$

$$\Rightarrow \quad K_b = \frac{1.588\times10^{-6}}{0.09874} = 1.608\times10^{-5}$$

Concept Strand 27

At a concentration of 1.23×10^{-2} M, trimethyl ammonium ion is 0.01 % dissociated. Calculate its pH. Also calculate the dissociation constant, K_b of trimethyl amine and the pH of 0.0813 M solution of the base.

Solution

$$\alpha \text{ of } (CH_3)_3{}^+NH \text{ at } 1.23 \times 10^{-2}\text{M} = \frac{0.01}{100} = 10^{-4}$$

$$K_a = \alpha^2 c = (10^{-4})^2 \times 1.23 \times 10^{-2}$$

$$= 1.23 \times 10^{-10} \text{ (assume } \alpha << 1 \text{ as } K_a \text{ is small)}$$

$$pK_a = -\log [1.23 \times 10^{-10}] = 9.91$$

K_b of trimethylamine

$$K_b = \frac{K_w}{K_a} = \frac{10^{-14}}{1.23\times10^{-10}} = 8.13\times10^{-5}$$

$$[OH^-] = \sqrt{K_b \times c} = \sqrt{8.13\times10^{-5}\times1.23\times10^{-2}}$$

$$= \sqrt{10^{-6}} = 10^{-3}$$

pOH = 3, pH = 11

Concept Strand 28

A two colour indicator, having yellow colour in anionic form and red colour in the acid form, has a $K_a = 1.2 \times 10^{-5}$ (i) What should be the pH of the solution if 80% of the indicator were to be in acid form? (ii) What will be the change in pH if the ratio of the acid form to anionic form changes from 10 to 0.1.

Solution

$$HIn + H_2O \rightleftharpoons H_3O^+ + In^-$$

acid form anionic formt

(red) (yellow)

$$K_a = \frac{[H_3O^+][In^-]}{[HIn]}$$

$$-\log K_a = -\log [H_3O^+] - \log [In^-] + \log [HIn]$$

$$pK_a = pH + \log\frac{[HIn]}{[In^-]}$$

$$pK_a = -\log [1.2 \times 10^{-5}] = 4.92$$

(i) $pH = 4.92 - \log \frac{80}{20}$,

$pH = 4.92 - 0.60 = 4.32$

(ii) pH when the ratio of the acid form to anionic form is 10

$pH = 4.92 - \log 10 = 3.92$

pH when the ratio of the acid form to anionic is 0.1

$pH = 4.92 - \log 0.1 = 4.92 + 1 = 5.92$

$\Delta pH = 5.92 - 3.92 = 2$

Concept Strand 29

What is the molarity of a solution of benzoic acid having a pH of 3.1 at 25°C? The dissociation constant, K_a, of the acid is 6.3×10^{-5} at this temperature.

Solution

$-\log [H_3O^+] = 3.1; \quad \log [H_3O^+] = -(3.1) = \bar{4}.9$

$[H_3O^+] = 7.94 \times 10^{-4}$

$\Rightarrow \quad \alpha c = 7.94 \times 10^{-4}$

$$\frac{\alpha c \times \alpha c}{(1-\alpha)c} = 6.3 \times 10^{-5}$$

$$\Rightarrow \quad \frac{(7.94 \times 10^{-4})^2}{c - 7.94 \times 10^{-4}} = 6.3 \times 10^{-5}$$

$$c = \frac{(7.94 \times 10^{-4})^2}{6.3 \times 10^{-5}} + 7.94 \times 10^{-4}$$

$$= \frac{63.04 \times 10^{-8}}{6.3 \times 10^{-5}} + 7.94 \times 10^{-4}$$

$\Rightarrow \quad c = 0.01 + 0.000794 = 0.0108$

SALT HYDROLYSIS

When an equivalent amount of a strong acid, like HCl, is neutralized by an equivalent amount of strong base, like NaOH, the resulting solution will be neutral and its pH is equal to 7.0 at 25°C. This is because the ions formed from the salt, viz., Na^+ and Cl^- ions have no tendency to react with water.

On the other hand, when an equivalent amount of a weak acid like acetic acid is neutralized by an equivalent amount of a strong base, the resulting solution has a pH greater than 7.0, i.e., it is alkaline. In a similar manner, when a weak base, like ammonium hydroxide, is exactly neutralized by a strong acid like HCl, the solution formed has a pH lower than 7.0 i.e., it is acidic. In both cases, the alkaline or the acidic nature of the solutions is due to the interactions of the ions of the salts with water as shown below.

For example, in the first case, the sodium acetate reacts with water according to $CH_3COO^- + (Na^+) + H_2O \rightleftharpoons CH_3COOH + (Na^+) + OH^-$
resulting in the formation of OH^- ions.

In the latter case, the ammonium chloride reacts with water according to

$$NH_4^+ + (Cl^-) + H_2O \rightleftharpoons NH_3 + H_3O^+ + (Cl^-)$$

which results in the formation of H_3O^+ ions.

In both the cases, a partial reversal of neutralization occurs, resulting in the regeneration of acetic acid in the first case and ammonia in the second case. This partial reversal of neutralization by interaction of the ions of the salt with water is known as hydrolysis. Hydrolysis also occurs in the case of salts formed by neutralisation of a weak acid by a weak base, as for example, in the neutralization of acetic acid by ammonium hydroxide, which results in the formation of ammonium acetate. The extent of hydrolysis in a reaction also depends on the acid-base nature of the solvent.

It is convenient to discuss hydrolysis under the following three categories:

(i) Hydrolysis of salts of weak acids and strong bases.
(ii) Hydrolysis of salts of weak bases and strong acids.
(iii) ydrolysis of salts of weak acids and weak bases.

(i) Hydrolysis of salts of weak acids and strong bases

The hydrolysis of a salt like sodium acetate is a typical example of this category. Representing the anion of the salt (CH_3COO^-), which enters into reaction with the solvent water in a general way on A^-, the hydrolysis reaction may be written as

$$A^- + H_2O \rightleftharpoons HA + OH^-$$

The equilibrium constant of the reaction, which is also known as the hydrolysis constant, K_h, is given by

$$K_h = \frac{[HA][OH^-]}{[A^-]}$$

If 'h' is the degree of hydrolysis of the salt and 'c' is the stoichiometric initial concentration of the salt, at equilibrium we have,

$$[HA] = hc;\ [OH^-] = hc;\ [A^-] = (1 - h)c$$

$$K_h = \frac{hc \times hc}{(1-h)c} = \frac{h^2c}{(1-h)}$$

When the degree of hydrolysis is small, $h << 1$,

$$K_h = h^2c \quad \Rightarrow h = \sqrt{\frac{K_h}{c}}$$

The hydrolysis constant, K_h, is related to the ionic product, K_w, and the dissociation constant of the weak acid, K_a and the relationship may be arrived at in the following way.

Multiplying and dividing the expression for K_h by $[H_3O^+]$, we get,

$$K_h = \frac{[HA][H_3O^+][OH^-]}{[H_3O^+][A^-]} \Rightarrow K_h = \frac{K_w}{K_a}$$

The pH at a stoichiometric concentration of the salt, c, can be calculated by writing

$$[H_3O^+] = \frac{K_w}{[OH^-]} = \frac{K_w}{hc} = \frac{K_w}{\left(\frac{K_h}{c}\right)^{1/2} \times c} = \frac{K_w}{(K_h c)^{1/2}}$$

$$\Rightarrow \quad pH = -\log [H_3O^+] = pK_w + \frac{1}{2}\log K_h + \frac{1}{2}\log c$$

$$= pK_w + \frac{1}{2}\log K_w - \frac{1}{2}\log K_a + \frac{1}{2}\log c$$

$$\Rightarrow \quad pH = \frac{1}{2}\left[\, pK_w + pK_a + \log c\right]$$

The above equation may be used to calculate the pH of the salts of weak acids and strong bases from the known values of K_w, K_a and the concentration of the salt.

(ii) Hydrolysis of salts of weak bases and strong acids

The hydrolysis of a salt like NH_4Cl is a typical example belonging to this class. Representing the cation of the salt (NH_4^+), which enters into reaction with the solvent water as BH^+, the hydrolysis reaction may be represented as

$$BH^+ + H_2O \rightleftharpoons B + H_3O^+$$

The hydrolysis constant, K_h, of the reaction may be written as

$$K_h = \frac{[B][H_3O^+]}{[BH^+]}$$

If 'h' is the degree of hydrolysis of the salt and 'c' is the stoichiometric initial concentration of the salt,

$$K_h = \frac{hc \times hc}{(1-h)c} = \frac{h^2c}{(1-h)}$$

$\Rightarrow \quad K_h = h^2c$ (assuming $h < 1$)

$\Rightarrow \quad h = \sqrt{K_h/c}$

The hydrolysis constant, K_h, is related to the ionic product, K_w, and the dissociation constant of the weak base, K_b, in the following way.

Multiplying and dividing equation for K_h by $[OH^-]$, we get,

$$K_h = \frac{[B][H_3O^+][OH^-]}{[BH^+][OH^-]} \Rightarrow K_h = \frac{K_w}{K_b}$$

The pH of the solution at a stoichiometric concentration 'C' of the salt may be calculated as follows.

$$[H_3O^+] = hc = \sqrt{\frac{K_h}{c}} \times c = \sqrt{K_h c} = \left(\frac{K_w c}{K_b}\right)^{1/2}$$

$$\Rightarrow \quad pH = -\log [H_3O^+] = -\frac{1}{2}\log K_w - \frac{1}{2}\log c + \frac{1}{2}\log$$

$$K_b = \frac{1}{2}pK_w - \frac{1}{2}pK_b - \frac{1}{2}\log c$$

$$\Rightarrow \quad pH = \frac{1}{2}\left[\, pK_w - pK_b - \log c\right]$$

The above equation may be used to calculate pH of the salts of weak bases and strong acids from known values of K_w, K_b and the concentration 'c' of the salt.

(iii) Salts of weak acids and weak bases

Salts like ammonium acetate and anilinium acetate are examples of this category. Representing such a salt in the general equation BH^+A^-, its hydrolysis reaction may be represented as

$$BH^+ + A^- + H_2O + H_2O \rightleftharpoons B + H_3O^+ + HA^+OH^-$$

Neglecting the ionic product equilibrium, which is in any case present in water, the hydrolysis constant, K_h, is given by

$$K_h = \frac{[B][HA]}{[BH^+][A^-]}$$

If c is the stoichiometric initial concentration of the salt and h is the degree of hydrolysis, at equilibrium, we have,

$[B] = hc; [HA] = hc; [BH^+] = (1 - h)c; [A^-] = (1 - h)c$

$$\Rightarrow \quad K_h = \frac{hc \times hc}{(1-h)c \times (1-h)c} = \frac{h_2}{(1-h^2)}$$

$$\Rightarrow \quad \frac{h}{1-h} = \sqrt{K_h} \text{ and } h = \frac{\sqrt{K_h}}{1+\sqrt{K_h}}$$

K_h can be related to the dissociation constant of the weak acid K_a and the dissociation constant of the weak base K_b in the following way.

Multiplying and dividing the above equation by the term $[H_3O^+]\,[OH^-]$, we get,

$$K_h = \frac{[B][H_3O^+][HA][OH^-]}{[BH^+][H_3O^+][A^-][OH^-]} \Rightarrow K_h = \frac{K_w}{K_a \times K_b}$$

To calculate the pH at a stoichiometric initial concentration of the salt, C, the equation for the dissociation constant of the weak acid HA will be written in the form

$$[H_3O^+] = K_a \frac{[HA]}{[A^-]}$$

$$= K_a \frac{hc}{(1-h)c} = K_a \frac{h}{(1-h)} = K_a \sqrt{K_h}$$

$$= K_a \sqrt{\frac{K_w}{K_a K_b}} = \sqrt{\frac{K_a K_w}{K_b}}$$

$$\Rightarrow \quad pH = \frac{1}{2} pK_w + \frac{1}{2} pK_a - \frac{1}{2} pK_b$$

$$\Rightarrow \quad pH = \frac{1}{2}\left[pK_w + pK_a - pK_b \right]$$

It is seen from the equation that the pH will depend on the pK_a and pK_b values of the acid and base.

If $pK_a < pK_b$, pH will be less than 7.
If $pK_a > pK_b$, pH will be greater than 7.
If $pK_a = pK_b$, pH will be equal to 7.

It may be noted from the equation that the pH is independent of salt concentration.

CONCEPT STRANDS

Concept Strand 30

Calculate the degree of hydrolysis of 0.1 N solution of potassium cyanide at 25°C. What is the pH of the solution? ($K_{a(HCN)} = 7.2 \times 10^{-10}$ at 25°C)

Solution

The hydrolysis reaction is

$$CN^- + H_2O \rightleftharpoons HCN + OH^-$$

$$K_h = \frac{K_w}{K_a} = \frac{10^{-14}}{7.2 \times 10^{-10}} = 1.39 \times 10^{-5}$$

$$K_h = 1.39 \times 10^{-5} = h^2 c$$

$$\Rightarrow \quad h = \sqrt{\frac{1.39 \times 10^{-5}}{0.1}} = 0.0118$$

$[OH^-] = hc = 0.0118 \times 0.1 = 0.00118$

$pOH = -\log [0.00118] = +2.93$

$\Rightarrow$ $pH = 14 - pOH = 14.00 - 2.93 = 11.07$

Alternatively, $pH = \frac{1}{2} pK_w + \frac{1}{2} pK_a + \frac{1}{2} \log c$

$$= 7 + \frac{9.14}{2} + \left(\frac{-1.0}{2}\right) = 7 + 4.57 - 0.5 = 11.07$$

Concept Strand 31

Calculate the degree of hydrolysis and hydrolysis constant of a 0.01 M solution of ammonium chloride at 25°C. Given the pH of the solution is 5.63. What is the pK_b of ammonia?

Solution

Hydrolysis reaction:

$$NH_4^+ + H_2O \rightleftharpoons NH_3 + H_3O^+$$

$-\log [H_3O^+] = 5.63$; $\log [H_3O^+] = -5.63 = \bar{6}.37$

$\Rightarrow$ $[H_3O^+] = 2.344 \times 10^{-6}$

$\Rightarrow$ $h \times c = [H_3O^+] = 2.344 \times 10^{-6}$

$$\Rightarrow \quad h = \frac{2.344 \times 10^{-6}}{0.01} = 2.344 \times 10^{-4}$$

$$K_h \approx h^2c = (2.344 \times 10^{-4})^2 \times 0.01 = 5.49 \times 10^{-10}$$

$$K_h = \frac{K_w}{K_b}$$

$$\Rightarrow \quad K_b = \frac{10^{-14}}{5.49 \times 10^{-10}} = 1.82 \times 10^{-5}$$

$$\Rightarrow \quad pK_b = 4.74$$

Concept Strand 32

The hydroxyl ion concentration of 0.055 M solution of sodium acetate was found to be 5.5×10^{-6} at 25°C. Calculate (i) degree of hydrolysis (ii) hydrolysis constant of the salt. What is the dissociation constant of acetic acid?

Solution

$$CH_3COO^- + H_2O \rightleftharpoons CH_3COOH + OH^-$$

Conc. of $OH^- = h \times c = h \times 0.055$

$$\Rightarrow \quad h = \frac{5.5 \times 10^{-6}}{0.055} = 10^{-4}$$

$$K_h \approx h^2c = (10^{-4})^2 \times 0.055 = 5.5 \times 10^{-10}$$

$$K_a = \frac{K_w}{K_h} = \frac{10^{-14}}{5.5 \times 10^{-10}} = 1.8 \times 10^{-5}$$

Concept Strand 33

Calculate the degree of hydrolysis, hydrolysis constant and the pH of a 0.02 M ammonium acetate solution at 25°C given that K_a of acetic acid is 1.8×10^{-5} and K_b of ammonium hydroxide has also the same value ($K_w = 10^{-14}$).

Solution

Hydrolysis reaction:

$$NH_4^+ + Ac^- + H_2O + H_2O \rightleftharpoons NH_3 + H_3O^+ + HAc + OH^-$$

$$K_h = \frac{K_w}{K_a \times K_b} = \frac{10^{-14}}{1.8 \times 1.8 \times (10^{-5})^2} = \frac{10^{-14}}{3.24 \times 10^{-10}}$$

$$= 3.086 \times 10^{-5}$$

$$= \frac{\sqrt{K_h}}{1 + \sqrt{K_h}} = \frac{\sqrt{3.086 \times 10^{-5}}}{1 + \sqrt{3.086 \times 10^{-5}}}$$

$$= \frac{5.56 \times 10^{-3}}{1.0056} = 5.53 \times 10^{-3}$$

$$pH = \frac{1}{2}pK_w + \frac{1}{2}pK_a - \frac{1}{2}pK_b$$

$$= 7.0 + 2.37 - 2.37 = 7.0.$$

It is noteworthy that the pH is independent of the concentration of the salt.

Concept Strand 34

Calculate the hydrolysis constant, degree of hydrolysis and the pH of a 0.01 M solution of ammonium cyanide at 25°C. (K_a(HCN) = 7.2×10^{-10}; $K_b(NH_3) = 1.8 \times 10^{-5}$).

Solution

$$K_h = \frac{K_w}{K_a \times K_b} = \frac{10^{-14}}{7.2 \times 10^{-10} \times 1.8 \times 10^{-5}} = 0.772$$

$$h = \frac{\sqrt{K_h}}{1 + \sqrt{K_h}} = \frac{\sqrt{0.772}}{1 + \sqrt{0.772}} = \frac{0.878}{1.878} = 0.467$$

$$pH = \frac{1}{2}pK_w + \frac{1}{2}pK_a - \frac{1}{2}pK_b = 7.0 + \frac{9.14}{2} - \frac{4.74}{2}$$

$$= 9.20.$$

COMMON ION EFFECT

Addition of an ion, common to the ions participating in a dissociation equilibrium (For example, acetate ions or H^+ ions in the dissociation of acetic acid) suppresses the degree of dissociation of the acid in accordance with the mass law.

Similarly, addition of NH_4^+ or OH^-, common to the ions in dissociation equilibrium of NH_4OH, results in a lowering of degree of dissociation (α) of the base, i.e., the equilibrium is shifted to the left.

CONCEPT STRAND

Concept Strand 35

Calculate the H^+ ion concentration and the percentage of dissociation of 0.01 M solution of propionic acid containing 0.005 M sodium propionate. [pK_a of propionic acid = 4.87). Compare this value with that in the absence of added sodium salt.

Solution

K_a of acid = antilog (−4.87) = 1.35×10^{-5}

$\alpha^2 c = 1.35 \times 10^{-5}$;

$$\alpha^2 = \frac{1.35 \times 10^{-5}}{0.01} = 1.35 \times 10^{-3}\ ;\ \alpha = 0.0367$$

$[H^+] = 0.0367 \times 0.01 = 3.67 \times 10^{-4}$;

percentage of dissociation = 3.67

Let α' be the degree of dissociation in presence of 0.005 M salt

Concentration of undissociated acid = $(1 - \alpha')\ c$

Concentration of propionate ion = $(\alpha' c + 0.005)$

Concentration of $H_3O^+ = \alpha' c$

$$\therefore \quad \frac{\alpha' c(\alpha' c + 0.005)}{(1 - \alpha')c} = 1.35 \times 10^{-5}$$

$\alpha' (\alpha' c + 0.005) = 1.35 \times 10^{-5}$ considering $(1 - \alpha^1) \approx 1$

(or) $(\alpha')^2 \times 0.01 + 0.005\alpha' - 1.35 \times 10^{-5} = 0$

$\alpha' = 0.0027$

percentage dissociation = 0.27

$[H^+] = \alpha' c = 0.0027 \times 0.01 = 2.7 \times 10^{-5}$

Degree of dissociation is decreased by the presence of common ion.

Self ionization equilibria

Solvents like water, methanol, etc can act both as an acid or base and thereby undergo self ionization which may be represented as

$$H_2O + H_2O \rightleftharpoons H_3O^+ + OH^-$$

$$CH_3OH + CH_3OH \rightleftharpoons CH_3OH_2^+ + CH_3O^-$$

The above reactions are known as auto protolysis reactions and the equilibria are referred as self ionization equilibria.

The equilibrium constant for reaction involving water may be written as

$$K'_w = \frac{[H_3O^+][OH^-]}{[H_2O]^2}$$

Since water is in large excess, the above equation may be rewritten as

$$K'_w[H_2O]^2 = K_w = [H_3O^+][OH^-]$$

The constant K_w is known as the ionic product of water and has a value of 1×10^{-14} at 25°C. It increases with increase of temperature. Thus the above process is an endothermic with mean heat of ionization of water being 56.7 kJ mol^{-1}. It may be recalled that this value with a negative sign is equal to the heat of neutralization of a strong acid by a strong base in dilute solution. In pure water, $[H_3O^+] = [OH^-] = 10^{-7}$ mol L^{-1} (or) g – ions L^{-1}.

The self ionization constant of methanol and ethanol are about 10^{-19} and 8×10^{-20} respectively at 25°C.

BUFFER AND BUFFER ACTION

In physicochemical measurements, analytical work and in industrial practice, it is often necessary to control the pH at a certain value to avoid undesirable reactions and decompositions. We use buffer solutions for this purpose.

Definition of buffer

A buffer solution or a buffer is defined as a solution which resists changes in its pH upon the addition of small amounts of strong acid or strong base or on dilution or on keeping

for a long time. Buffers usually consist of a mixture of weak acid and its salt with a strong base (anion) or a weak base and its salt with a strong acid. Salts formed by neutralization of a weak acid and weak base also have some buffer action.

Reactions involved in buffer action

Let us consider a buffer made up of acetic acid and sodium acetate. Suppose a few mL of a strong acid are added to the buffer. The H^+ ions of the strong acid react with the acetate ions to form undissociated acid thereby resisting any lowering of pH. The reaction may be written as

$$CH_3COO^- + H_3O^+ \rightleftharpoons CH_3COOH + H_2O$$

If a few mL of a strong base are added to the buffer, the OH^- ions react with the undissociated acid to give acetate ion and water according to

$$CH_3COOH + OH^- \rightleftharpoons CH_3COO^- + H_2O$$

In a similar way, one can explain the buffer action on other types of buffers, viz., buffers made up of NH_4Cl and NH_4OH or boric acid and borax, etc.

Equation for calculating the pH of a Buffer

We consider two types of buffers

(i) consisting of a mixture of weak acid HA and its salt, NaA

(ii) consisting of a mixture of weak base B and its salt, BHX

(i) Mixture of HA and NaA (Acidic Buffer)

On the basis of the reaction,

$$HA + H_2O \rightleftharpoons H_3O^+ + A^-$$

the equation for the dissociation constant of the weak acid is given as

$$K_a = \frac{[H_3O^+][A^-]}{[HA]}$$

$$\Rightarrow \quad [H_3O^+] = K_a \frac{[HA]}{[A^-]}$$

For a weak acid which is only slightly dissociated, the addition of the salt represses its dissociation further and so, the equilibrium concentration of HA may be taken as equal to the stoichiometric concentration of the acid.

The above equation may be written as

$$[H_3O^+] = K_a \frac{[\text{Acid}]}{[\text{Salt}]}$$

where, [HA] = concentration of the acid taken, [Salt] = concentration of salt added in the buffer

$$pH = pK_a + \log\frac{[\text{Salt}]}{[\text{Acid}]}$$

This equation is known as Henderson–Hasselbalch equation or in short Henderson equation.

(ii) Mixture of B and BH^+

The equation for the dissociation constant of the weak base is given on the basis of the reaction

$$B + H_2O \rightleftharpoons BH^+ + OH^-$$

$$\text{as } K_b = \frac{[BH^+][OH^-]}{[B]}$$

$$\Rightarrow \quad [OH^-] = K_b \frac{[B]}{[BH^+]}$$

Applying arguments similar to the previous case, the equilibrium concentration of B may be taken as equal to the stoichiometric concentration of the base taken and the concentration of the salt BH^+ is that present in the mixture. Thus

$$pOH = pK_b + \log\frac{[BH^+]}{[B]}$$

$$pOH = pK_b + \log\frac{[\text{Salt}]}{[\text{Base}]}$$

Once pOH is known, pH can be calculated from, pH = 14 – pOH

The buffer is most effective when the concentration of weak acid (or weak base) and their salt are nearly equal, i.e., in the region $pH \approx pK_a$ or pK_b

Buffer Capacity

The capacity of a solution to resist a change in pH is known as its buffer capacity, indicated as β.

Buffer capacity (β) is defined by, $\beta = \frac{dc}{dpH}$ *where, dc is the number of moles of a strong base added to one litre of the buffer solution which results in an increase of pH by dpH.* The addition of a strong acid produces the opposite effect i.e., it is equivalent to –dc i.e., a negative increment of base

leading to a decrease of pH i.e., –dpH. Thus β remains a positive quantity.

The efficacy of a buffer becomes clear by considering the following example. Suppose we add 10 mL of 0.1 M HCl to 100 mL of water. The pH of the resulting solution is 2.0, a difference of five units from that of pure water. If the same amount of 0.1 M HCl is added to 100 mL of an acetic acid–odium acetate buffer where the concentrations of acetic acid and sodium acetate are 0.2 M each, the change in pH is only about 0.04 units.

CONCEPT STRANDS

Concept Strand 36

Calculate the pH of a buffer solution containing 0.01 M boric acid and 0.02 M sodium borate. (The dissociation constant of boric acid is 7.3×10^{-10} at 25°C).

Solution

$$pH = pK_a + \log\frac{[salt]}{[acid]}$$

$$pK_a = -\log(7.3 \times 10^{-10}) = 9.14$$

$$\Rightarrow \quad pH = 9.14 + \log\frac{0.02}{0.01} = 9.14 + 0.30 = 9.44$$

Concept Strand 37

Calculate the pH of a buffer solution, which contains 0.02 M sodium acetate and 0.05 M acetic acid.

(i) What will be the change in pH when 5 mL of 0.2 N HCl are added to one litre of the buffer?

(ii) What is the change in pH when 5 mL of 0.2 N NaOH are added to one litre of the buffer? (The dissociation constant of acetic acid at 25°C = 1.8×10^{-5}).

Solution

(i) pK_a of the acid = 4.74

$$pH = 4.74 + \log\frac{[salt]}{[acid]} = 4.74 + \log\frac{0.02}{0.05} = 4.34$$

(ii) Milli moles of HCl added = $5 \times 0.2 = 1.0$

The acid added will react with CH_3COO^- ions to give CH_3COOH.

$\therefore$ Acid concentration will increase by 1.0 milli moles and acetate concentration will decrease by the same amount.

$\therefore$ $[CH_3COOH] = 0.05 + 1.0 \times 10^{-3} = 0.051$ M

$[CH_3COO^-] = 0.02 - 0.001 = 0.019$ M

$$pH = 4.74 + \log\frac{0.019}{0.051} = 4.74 - 0.43 = 4.31$$

(iii) Milli moles of OH^- added = $5 \times 0.2 = 1.0$

The OH^- ions added will react with acetic acid to give acetate ions. Therefore the acetate ion concentration increases by 1.0 milli mole and the acid concentration decreases by the same amount.

$[CH_3COOH] = 0.05 - 1.0 \times 10^{-3} = 0.049$ M

$[CH_3COO^-] = 0.02 + 1.0 \times 10^{-3} = 0.021$ M

$$pH = 4.74 + \log\frac{0.021}{0.049} = 4.74 - 0.37 = 4.37$$

Concept Strand 38

A buffer solution consisting of a mixture of lactic acid and sodium lactate having a pH 4.30 is to be prepared. Calculate the number of grams of sodium lactate that is necessary to be added to two litres of 0.1 M lactic acid in order that the resulting solution may have the above pH. (K_a of lactic acid at 25°C = 2.5×10^{-4})

Solution

pK_a of lactic acid = $-\log[2.5 \times 10^{-4}]$

$= 3.60$

$$4.30 = 3.60 + \log\frac{[salt]}{[0.1]}; \log\frac{[salt]}{[0.1]} = 0.70$$

$[salt] = 5.01 \times 0.1 = 0.501$

Weight of sodium lactate required for 2 litres = $0.501 \times 112 \times 2 = 112.2$ g.

Concept Strand 39

Calculate the amount of ammonium chloride to be added to one litre of 0.01 M ammonia solution in order to have a buffer of pH = 9.50

(K_b of ammonium hydroxide at 25°C = 1.78×10^{-5})

Solution

pOH of the solution = 14.0 − 9.50 = 4.50

$$pOH = pK_b + \log\frac{[salt]}{[base]}$$

$$pK_b = -\log K_b = -\log[1.78 \times 10^{-5}]$$
$$= -[-4.75] = 4.75$$

$$4.5 = 4.75 + \log\frac{[salt]}{[0.01]}$$

$$\log\frac{[salt]}{0.01} = -0.25; \frac{[salt]}{[0.01]} = 0.562$$

[salt] = 0.00562 M; Amount of salt = 0.00562 × 53.5
= 0.300 g of salt

SOLUBILITY EQUILIBRIA AND SOLUBILITY PRODUCT

Salts like sodium chloride, potassium nitrate, potassium chloride are highly soluble in water and also are practically completely dissociated in aqueous solution.

On the other hand, salts like silver chloride, barium sulphate, lead sulphate, calcium fluoride are sparingly soluble in water and the equilibria obtained in saturated solutions of such salts are of considerable interest due to their effect on solubility.

Let us consider a salt like silver chloride added to a certain amount of pure water taken in a bottle, with a tight fitting stopper and placed in a thermostat at a definite temperature. The bottle is occasionally shaken and left in the thermostat for a few days to attain equilibrium. The equilibrium existing in the saturated solution of the salt under the given conditions may be represented as

$$AgCl_{(solid)} \rightleftharpoons AgCl \text{ (in saturated solution)}$$
$$\rightleftharpoons Ag^+_{(aq)} + Cl^-_{(aq)}$$

The concentration of silver chloride in the saturated solution remains a constant and, assuming that whatever salt has gone into solution is completely dissociated, the equilibrium constant may be expressed as

$$K = \frac{[Ag^+][Cl^-]}{[AgCl]_{solid}} \Rightarrow K_{sp} = K\,[AgCl]_{solid} = [Ag^+]\,[Cl^-]$$

K_{sp}, which is equal to the product of the concentrations of the ions in the saturated solution is a constant at a given temperature and is known as the solubility product of the salt.

The solubility product, K_{sp}, may be expressed in terms of the solubility of the salt in the following way. If "S" represents the solubility of the salt in mol L^{-1} at the given temperature, at equilibrium

$[Ag^+]$ = S g ions L^{-1}; $[Cl^-]$ = S g ions L^{-1} and

$K_{sp} = [Ag^+]\,[Cl^-] = S^2$

The equation may be generalized by extending it to salts of other valence types or multivalent electrolytes as well.

Let us consider a salt of the type M_mA_n, which dissociates in solution according to

$$M_mA_n \rightleftharpoons mM^{n+} + nA^{m-}$$

where, m and n are stoichiometric numbers of M and A,

If S is the solubility of the salt in gram mol L^{-1},

$[M^{n+}] = mS$; $[A^{m-}] = nS$

then, $K_{sp} = (mS)^m\,(nS)^n$

$K_{sp} = m^m\,n^n\,(S)^{m+n}$

Definition of K_{sp}

The solubility product of a salt may be defined as the product of the concentrations of the ions of the salt in its saturated solution (expressed as gram ions L^{-1}), each concentration term being raised to a power representing the number of ions of that type formed from the dissociation of one molecule of the salt.

The solubility product is a constant at a given temperature but is altered when the temperature is changed. The variation of the solubility product with temperature is given by

$$\left(\frac{\partial \ln K_{sp}}{\partial T}\right)_P = \frac{\Delta H^\circ_{soln}}{RT^2}$$

where, $\Delta H^\circ_{solution}$ is the standard enthalpy change for the dissolution process.

In calculating the solubility product of a sparingly soluble salt, it is important to note that the solubility must be expressed in terms of mol L^{-1} of the salt.

CONCEPT STRANDS

Concept Strand 40

The solubility products of silver iodide and silver arsenite at 20°C are 8.3×10^{-17} and 4.5×10^{-19}, respectively. Which of their saturated solutions will have a higher concentration of silver ions and which one has lower Ag^+ concentration?

Solution

Ag^+ concentration from saturated solution of

$$AgI = \sqrt{K_{sp}}$$

$$= \sqrt{8.3 \times 10^{-17}} = 9.11 \times 10^{-9} \text{ mol } L^{-1}$$

For Ag_3AsO_3 (Silver arsenite)

If S is the solubility of silver arsenite,

$$Ag_3AsO_{3(s)} \rightleftharpoons 3Ag^+ + AsO_3^{3-}$$

$$K_{sp} = 27S^4$$

$$S^4 = \frac{4.5 \times 10^{-19}}{27} = 1.67 \times 10^{-20}$$

$S = \sqrt[4]{1.67 \times 10^{-20}} = 1.137 \times 10^{-5}$ mol L^{-1}. The silver ion concentration from a saturated solution of Ag_3AsO_3 is higher than that from a saturated solution of AgI. Thus Ag_3AsO_3 is more soluble.

Concept Strand 41

The solubility product of silver chloride in water at 298K is 1.78×10^{-10}. Calculate its solubility in water at 373K given that the enthalpy of solution of the salt is 174.2 kJ mol^{-1}

Solution

$$\log \frac{(K_{sp})_2}{(K_{sp})_1} = \frac{\Delta H^\circ}{2.303R}\left[\frac{T_2 - T_1}{T_1 T_2}\right]$$

where, $(K_{sp})_2$ is the solubility product at 373K and $(K_{sp})_1$ is the solubility product at 298K.

$$\log \frac{(K_{sp})_2}{1.78 \times 10^{-10}}$$

$$= \frac{174.2}{2.303 \times 8.314 \times 10^{-3}} \times \frac{75}{298 \times 373} = 6.139$$

$$\frac{(K_{sp})_2}{1.78 \times 10^{-10}} = 1.377 \times 10^6$$

$\Rightarrow$ $(K_{sp})_2$ at 373K $= 2.45 \times 10^{-4}$

Solubility $= \sqrt{K_{sp}} = 1.56 \times 10^{-2}$ mol L^{-1}

Effect of common ion on solubility equilibrium

The addition of an ion common to the ions of the salt participating in the solubility equilibrium profoundly affects the solubility of the salt, viz., the solubility of the salt decreases due to common ion effect. For example, addition of silver ion as silver nitrate or chloride ion as NaCl or KCl decreases the solubility of silver chloride. It is possible to calculate the solubility in presence of an added common ion in the following way. Let S' be the solubility of silver chloride in water in presence of a concentration of x g L^{-1} of chloride ion added in the form of sodium chloride.

At saturation equilibrium,

$$[Ag^+] = S'$$

$$[Cl^-] = (S' + x)$$

$$K_{sp} = S'(S' + x)$$

Since $S' << x$ (S' is also less than s, where, s is the solubility of silver chloride in the absence of any Cl^- ion), the above equation may be simplified as

$$S'x = K_{sp}$$

$$\Rightarrow \quad S' = \frac{K_{sp}}{x}$$

The principle of solubility product finds extensive application in qualitative analysis, viz., in the identification of various groups of metal ions.

CONCEPT STRANDS

Concept Strand 42

The solubility product of $Mg(OH)_2$ at 298K is 1.2×10^{-11}. Calculate its solubility in g L^{-1} (i) in pure water (ii) in 0.01 M $MgCl_2$ solution [Molecular weight of $Mg(OH)_2$ = 58.3].

Solution

If S is the solubility in g mol L^{-1}, from the solubility equilibrium

$$Mg(OH)_{2(s)} \rightleftharpoons Mg^{2+} + 2OH^-$$

$$K_{sp} = 4S^3$$

(i) Solubility in pure water

$4S^3 = 1.2 \times 10^{-11}$

$$S = \sqrt[3]{0.3 \times 10^{-11}} = 1.442 \times 10^{-4} \text{ mol } L^{-1}$$

$= 1.442 \times 10^{-4} \times 58.3 = 8.41 \times 10^{-3}$ g L^{-1}

(ii) Solubility in 0.01 M $MgCl_2$ solution

Let S' be the solubility in presence of 0.01 M $MgCl_2$

Concentration of OH^- = 2S'

Concentration of Mg^{2+} = (S' + 0.01)

$K_{sp} = (2S')^2 (S' + 0.01) = 4(S')^2 (S' + 0.01)$

Assuming S' < < 0.01,

we write $4(S')^2 \times 0.01 = 1.2 \times 10^{-11}$

$$\Rightarrow \quad S' = \sqrt{\frac{1.2 \times 10^{-11}}{0.04}} = \sqrt{\frac{1.2}{4} \times 10^{-9}}$$

$$= \sqrt{3 \times 10^{-10}} = 1.732 \times 10^{-5} \text{ mol } L^{-1}$$

S' (in g L^{-1}) = 1.01×10^{-3}

Concept Strand 43

The concentration of Pb^{2+} ions in a saturated solution of lead chromate at 25°C is 1.34×10^{-7} g ions L^{-1}. Calculate the solubility product of the salt. What is the solubility in g L^{-1} in a solution of 0.02 M sodium chromate (Molecular weight of lead chromate = 323.2).

Solution

$$PbCrO_{4(s)} \rightleftharpoons Pb^{2+} + CrO_4^{2-}$$

$K_{sp} = S^2$ (where, S = solubility in mol L^{-1})

$= (1.34 \times 10^{-7})^2 = 1.8 \times 10^{-14}$

Solubility in presence of 0.02 M Na_2CrO_4

Let S' be the solubility in presence of 0.02 M Na_2CrO_4

$\therefore$ Concentration of Pb^{2+} = S'

Concentration of CrO_4^{2-} = (S' + 0.02)

$\Rightarrow$ $S'(S' + 0.02) = 1.8 \times 10^{-14}$

Assuming S' < < 0.02, we can write

$S' \times 0.02 = 1.8 \times 10^{-14}$

$$\Rightarrow \quad S' = \frac{1.8 \times 10^{-14}}{2 \times 10^{-2}} = 0.9 \times 10^{-12}$$

S' (in g L^{-1}) = $9 \times 10^{-13} \times 323.2 = 2.91 \times 10^{-10}$ g L^{-1}

Concept Strand 44

The pK_{sp} values of calcium fluoride and lead fluoride at 25°C are 10.40 and 7.43 respectively. Calculate the concentration of Ca^{2+} and Pb^{2+} ions in a saturated solution of the two salts.

Solution

$pK_{sp}(CaF_2) = 10.40$

K_{sp} of $CaF_2 = 4.0 \times 10^{-11}$

$pK_{sp}(PbF_2) = 7.43$ $\quad$ K_{sp} of $PbF_2 = 3.71 \times 10^{-8}$

In each saturated solution $\frac{1}{2}[F^-] = [M^{2+}]$

$$\Rightarrow \quad \frac{1}{2}[F^-]_{Total} = [Ca^{2+}] + [Pb^{2+}]$$

$$\Rightarrow \quad 0.5[F^-] = \frac{K_{sp}(CaF_2)}{[F^-]^2} + \frac{K_{sp}(PbF_2)}{[F^-]^2}$$

$$= \frac{4.0 \times 10^{-11}}{[F^-]^2} + \frac{3.71 \times 10^{-8}}{[F^-]^2}$$

$$[F^-]^3 = \frac{3.714 \times 10^{-8}}{0.5} = 7.43 \times 10^{-8}$$

$$[F^-] = \sqrt[3]{7.43 \times 10^{-8}}$$

$[F^-] = 4.2 \times 10^{-3}$

Concentration of Ca^{2+}

$$= \frac{4.0 \times 10^{-11}}{17.64 \times 10^{-6}} = 2.27 \times 10^{-6} \text{ g ions } L^{-1}$$

Concentration of Pb^{2+}

$$= \frac{3.71 \times 10^{-8}}{17.64 \times 10^{-6}} = 2.1 \times 10^{-3} \text{ g ions } L^{-1}$$

Concept Strand 45

Silver nitrate solution of a certain concentration is added dropwise to a solution mixture containing chloride ions and chromate ions. Calculate the optimum concentration of chloride ions at which silver chromate begins to precipitate if the solution contains 0.01 mol L^{-1} of the chromate ions. (K_{sp} (AgCl) at 25°C = 1.78×10^{-10}; K_{sp} (Ag_2CrO_4) at 25°C = 1.2×10^{-12})

Solution

Let us first calculate the solubility of AgCl and Ag_2CrO_4 in water.

$$S_{(AgCl)} = \sqrt{1.78 \times 10^{-10}} = 1.33 \times 10^{-5} \text{ mols } L^{-1}$$

$$S_{(Ag_2CrO_4)} = \sqrt[3]{\frac{1.2 \times 10^{-12}}{4}} = \sqrt{0.3 \times 10^{-12}}$$

$$= 0.669 \times 10^{-4} = 6.69 \times 10^{-5} \text{ mol } L^{-1}$$

Since the solubility of silver chloride is smaller, it begins to precipitate out. The precipitation will continue until the remaining concentrations satisfy the following equations.

$[Ag^+]\,[Cl^-] = K_{sp}$ (AgCl);

$[Ag^+]^2\,[CrO_4^{2-}] = K_{sp}$ (Ag_2CrO_4)

$$\Rightarrow \quad [Ag^+] = \frac{K_{sp}(AgCl)}{[Cl^-]} = \sqrt{\frac{K_{sp}(Ag_2CrO_4)}{[CrO_4^{2-}]}}$$

$$\Rightarrow \quad \frac{[Cl^-]}{\sqrt{[CrO_4^{2-}]}} = \frac{K_{sp}(AgCl)}{\sqrt{K_{sp}(Ag_2CrO_4)}}$$

$$= \frac{1.78 \times 10^{-10}}{\sqrt{1.2 \times 10^{-12}}} = \frac{1.78 \times 10^{-10}}{1.09 \times 10^{-6}} = 1.63 \times 10^{-4}$$

When CrO_4^{2-} concentration is 0.01 mol L^{-1}

$[Cl^-] = 1.63 \times 10^{-4} \times \sqrt{0.01} = 1.63 \times 10^{-4} \times 0.1 = 1.63 \times 10^{-5}$ g ion L^{-1}

APPLICATION OF SOLUBILITY PRODUCT PRINCIPLE IN QUALITATIVE ANALYSIS

The technique of qualitative analysis is mainly based on the application of solubility product principle to the various sparingly soluble salts of the metals involved in the analysis. By decreasing or increasing the concentration of one of the ions of the salt under study by the common ion effect or by changing the acidity of solution, one can adjust the product of the concentrations of the ions (IP) in the solution appropriately. If this product is greater than the solubility product of the particular salt, it is precipitated out of the solution. However, if this product (IP) is less than the solubility product of the salt, it will remain in solution.

When IP > K_{sp}, IP is reduced by precipitation, i.e., precipitation takes place and when IP < K_{sp}, IP is increased by dissolution, i. e., dissolution takes place or no precipitation takes place.

Since the solubility products of various salts of metals differ widely, it is possible to precipitate out one metal ion while retaining the others in solution. The selective precipitation of some metal ions keeping some other in solution is discussed at some length below.

Analysis of the precipitation of group I metals in qualitative analysis as their chlorides

Silver, lead and mercurous (I) salts belong to this group and their chlorides are sufficiently insoluble to be precipitated whereas, chlorides of other metals are soluble. Among the chlorides of these metal ions, K_{sp} of mercurous chloride is the lowest ($K_{sp} = 1.1 \times 10^{-18}$).

In a solution containing all the three ions, Hg_2^{2+} ions will be precipitated first, followed by silver ions and then lead (II) ions on the addition of dilute HCl. Since the solubility product of lead (II) chloride is quite high ($K_{sp} = 1.6 \times 10^{-5}$) it is often difficult to precipitate Pb^{2+} ions completely as chloride in the 1st group.

Metals belonging to Groups II and IV

Group II (Pb, Cu,Cd and Hg(II)) and group IV metals (Zn, Co, Ni and Mn) are precipitated as their sulphides using hydrogen sulphide for supplying S^{2-} ions but under different experimental conditions. By initially maintaining a low sulphide ion concentration, which is achieved by increasing the acidity of the solutions to about 1 N, the maximum

product of the concentration of the ions (IP) that can be obtained for any of the group IV metal ions can be made to lie well below the solubility products of their sulphides. Thus these metal ions remain in solution while the group II metal ion sulphides, which have very low solubility products, are precipitated out. This is because the low sulphide ion concentration available under these acidic conditions is sufficient to exceed the solubility products of the group II metal ion sulphides. As an illustration of the sulphide ion concentration available under these conditions, we consider the dissociation equilibrium of H_2S in water given by

$$H_2S \overset{K_1}{\rightleftharpoons} HS^- + H^+ \quad SH^- \overset{K_2}{\rightleftharpoons} S^{2-} + H^+$$

$$\text{For which } K_1 = \frac{[HS^-][H^+]}{[H_2S]} = 1 \times 10^{-7}$$

$$K_2 = \frac{[S^{2-}][H^+]}{[HS^-]} = 1 \times 10^{-14}$$

$$\Rightarrow \quad K_1 \times K_2 = \frac{[H^+]^2[S^{2-}]}{[H_2S]} = 1 \times 10^{-21}$$

Assuming that the concentration of H_2S in water at 25°C to be about 0.1 M, it follows that $[H^+]^2\,[S^{2-}] = 1 \times 10^{-21} \times 0.1 = 1 \times 10^{-22}$

It is clear from the above equation that the sulphide ion concentration can be varied by adjusting the H^+ ion concentration of the medium.

Calculations for CuS and MnS

We now calculate the sulphide ion concentration necessary for precipitating Cu^{2+} (a group II metal ion) and Mn^{2+} (a group IV metal ion) when they are present at 0.1 M concentration in a mixture.

(i) CuS

Since K_{sp} of CuS, i.e., $[Cu^{2+}]\,[S^{2-}] = 8.5 \times 10^{-36}$

The sulphide ion required to just precipitate $[Cu^{2+}]$ is

$$\frac{8.5 \times 10^{-36}}{0.1} = 8.5 \times 10^{-35}$$

Assuming the acidity of the solution to be about 1 M, the highest S^{2-} concentration obtainable is

$$\frac{1.0 \times 10^{-22}}{1^2} = 1.0 \times 10^{-22}$$

$\therefore$ Maximum product of the concentration of ions in CuS $= [Cu^{2+}][S^{2-}] = 0.1 \times 1.0 \times 10^{-22} = 1.0 \times 10^{-23}$

Since the above value exceeds the solubility product of CuS by a wide margin, copper (II) ion is completely precipitated under these acid (1 M) conditions employed.

(ii) MnS

Maximum product of Mn^{2+} and S^{2-} ions under 1 M acid conditions $= 0.1 \times 1.0 \times 10^{-22}$

$= 1 \times 10^{-23}$

Solubility product of MnS $= [Mn^{2+}]\,[S^{2-}] = 2.5 \times 10^{-10}$

Since the maximum product of Mn^{2+} and S^{2-} under the acid conditions employed (1 M) is much less than the solubility product of MnS, Mn^{2+} ions will remain in solution. Extending the above arguments to other group IV metal ions, Zn, Co, Ni, the solubility products of ZnS, CoS and NiS at 25°C are 1.2×10^{-22}, 5.0×10^{-22} and 2×10^{-21}, respectively. Thus, in 0.1 M solutions of their salts, the S^{2-} ion concentration required for precipitation as their sulphides is 1.2×10^{-21} M, 5.0×10^{-21} M and 2×10^{-20} M, respectively. However, since the maximum S^{2-} ion concentration in 1 M acid solution is only 1.0×10^{-22}, it is clear that these ions are not precipitated under the existing acid conditions (1 M).

Metals belonging to group III

The metal ions Al^{3+}, Fe^{3+} and Cr^{3+} belong to this group and they are precipitated as their hydroxides using ammonium hydroxide and ammonium chloride as group reagent. After the group II metal cations are precipitated out, the H_2S in the solution is boiled off completely, and ammonium chloride and ammonium hydroxide in the required concentrations are added to the solution. The solution at this stage is alkaline.

Calculation of $[OH^-]$ in presence of added NH_4Cl

The dissociation constant, K_b of ammonium hydroxide is given by

$$K_b = \frac{[NH_4^+][OH^-]}{[NH_4OH]} = 1.8 \times 10^{-5}$$

Considering that the solubility of ammonia is about 0.5 mole L^{-1}

$$[NH_4^+]\,[OH^-] = 1.8 \times 10^{-5} \times 0.5 = 9 \times 10^{-6}$$

$\Rightarrow$ $[OH^-] = [NH_4^+] = 3 \times 10^{-3}$ (under the given conditions)

When ammonium chloride is added at a concentration of one mol litre^{-1}, $[OH^-] = 9 \times 10^{-6}$

i.e., a decrease of more than 300 times due to common ion effect.

Calculation for $Fe(OH)_3$

Considering a solution, which is 0.1 M in Fe^{3+} (a group III metal ion) and 0.1 M Mg^{2+}, one can verify which of the ions will precipitate under the given conditions.

(i.e., $[NH_4Cl] = 1.0$ M; $[NH_3] = 0.5$ M)

K_{sp} of $Fe(OH)_3 = 1.1 \times 10^{-36}$

$$\Rightarrow \quad [OH^-], \text{ when } [Fe^{3+}] \text{ is } 0.1, = \sqrt[3]{\frac{1.1 \times 10^{-36}}{0.1}} = 2.2 \times 10^{-12}$$

Calculation for $Mg(OH)_2$

In the case of $Mg(OH)_2$, $K_{sp} = 1.2 \times 10^{-11}$

$$[OH^-], \text{ when } [Mg^{2+}] \text{ is } 0.1 = \sqrt{\frac{1.2 \times 10^{-11}}{0.1}} = 1.095 \times 10^{-5}$$

Since $[OH^-]$ is only 9×10^{-6} under these conditions and is below that required for precipitating $Mg(OH)_2$, Mg^{2+} ions will remain in solution. However, Fe^{3+} gets precipitated as $Fe(OH)_3$.

Zinc and Nickel ions are not precipitated under these conditions as they form soluble amino complexes, $\left[Zn(NH_3)_4\right]^{2+}$ and $[Ni(NH_3)_6]^{2+}$ respectively. Because of the formation of these complexes, their free metal ion concentrations are reduced to low values. Under these low metal ion concentrations, fairly large $[OH^-]$ are required to precipitate them as their hydroxides. Such large OH^- concentrations are not reached in mixtures of NH_4Cl and NH_4OH. If NH_4Cl is not added prior to the addition of ammonia to these metal salt solutions, the corresponding hydroxides of these metals are initially precipitated but go into solution when excess NH_4OH is added to form complexes.

Metals belonging to Group IV

Zinc, Cobalt, Nickel and Manganese belong to this group and they are precipitated as their sulphides in alkaline medium. After separating the hydroxides (of group III), H_2S is passed through the ammoniacal medium containing NH_4Cl. Ammonium sulphide is formed, which produces a large concentration of sulphide ion due to the fact that it is a salt. The S^{2-} concentration available under these conditions is adequate for the precipitation of Zn, Co, Ni and Mn as sulphides. As a first step towards understanding the precipitation of these sulphides, we first calculate the sulphide ion concentration available under the ammoniacal conditions prevailing when fourth group metal ions are analysed.

Calculation of S^{2-} ion concentration under ammoniacal conditions

Conc. of $NH_4OH = 0.5$ M; Conc. of $NH_4Cl = 1.0$ M

Under the above conditions, conc. of OH^- in solution = 9.0×10^{-6}

$$[H^+] = \frac{K_w}{[OH^-]} = \frac{10^{-14}}{9.0 \times 10^{-6}} \approx 10^{-9}$$

S^{2-} conc., when H_2S is passed under these conditions =

$$\frac{1.0 \times 10^{-22}}{[H^+]^2} = \frac{1 \times 10^{-22}}{\left(10^{-9}\right)^2} = 1 \times 10^{-4}$$

Assuming that the metal ions (Zn^{2+}, Ni^{2+} and Mn^{2+}) are present at 0.1 M concentration, the maximum product attainable is $0.1 \times 1 \times 10^{-4} = 1 \times 10^{-5}$.

Since this product is greater than the solubility products of all group IV sulphides like those of Zn, Co, Ni, Mn are precipitated out as their sulphides.

Metals belonging to Group V

Calcium, Strontium and Barium ions belong to this group and they are precipitated as their carbonates by using ammonium carbonate as the group reagent in presence of NH_4Cl and NH_4OH.

In presence of NH_4^+ ions, the carbonate ion concentration is reduced to a low value when using $(NH_4)_2CO_3$ and this low concentration is inadequate for precipitation of Mg^{2+} ion as $MgCO_3$. This is because the solubility product of $MgCO_3$ is relatively high ($K_{sp} = 2.6 \times 10^{-5}$) compared to that of $CaCO_3$ ($K_{sp} = 1 \times 10^{-8}$) and $BaCO_3$ ($K_{sp} = 5 \times 10^{-9}$). The latter two carbonates are however precipitated because of their low solubility products.

CONCEPT STRAND

Concept Strand 46

The solubility product of $Mg(OH)_2$ at 25°C is 1.2×10^{-11}. 250 mL of 0.1 M magnesium sulphate and 250 mL of 0.1 M ammonium hydroxide are mixed and ammonium chloride is added to this mixture such that its concentration is 0.1 M. Will $Mg(OH)_2$ precipitate out under these conditions? What happens if ammonium chloride is not added?

(K_b of $NH_4OH = 1.8 \times 10^{-5}$)

Solution

$$\frac{[NH_4^+][OH^-]}{[NH_4OH]} = 1.8 \times 10^{-5}$$

Let the concentration of OH^- be x

$[NH_4^+] = 0.1$ M

$$\therefore \quad \frac{0.1 \times x}{0.05} = 1.8 \times 10^{-5} \Rightarrow x = 9 \times 10^{-6}\,M$$

Concentration of $Mg^{2+} = 0.05$ M

$\therefore$ Ionic product of Mg^{2+} and OH^- = $[Mg^{2+}]\,[OH^-]^2 = 0.05 \times (9 \times 10^{-6})^2 = 4.05 \times 10^{-12}$

Since this is less than K_{sp}, $Mg(OH)_2$ is not precipitated.

In absence of NH_4Cl

$[NH_4^+] = [OH^-] = x$

$$\Rightarrow \quad \frac{x \times x}{0.05} = 1.8 \times 10^{-5}\,;\, x^2 = 9 \times 10^{-7} \Rightarrow x = 9.5 \times 10^{-4}$$

Product of ions of Mg^{2+} and $OH^- = 0.05 \times (9.5 \times 10^{-4})^2 = 4.51 \times 10^{-8}$

Since this product is greater than K_{sp}, $Mg(OH)_2$ will precipitate out in the absence of added ammonium salt.

CONCEPT STRANDS

Concept Strand 47

A solution containing 0.1 M Zn^{2+} is saturated with hydrogen sulphide

(i) in presence of 1 M HCl.
(ii) in presence of 0.01 M HCl.

Will precipitation of ZnS occur in either of the cases? Explain (K_{sp} of ZnS = 1.2×10^{-23}); Concentration of H_2S in saturated solution = 0.1 M; $\dfrac{[H^+]^2[S^{2-}]}{[H_2S^-]} = 9.0 \times 10^{-23}$

Solution

(i) $[H^+] = 1$ M

$[S^{2-}]$ available under the given condition

$$= \frac{9 \times 10^{-23} \times [H_2S]}{[H^+]^2} = 9 \times 10^{-24}$$

Product of Zn^{2+} and S^{2-} concentration $= 0.1 \times 9 \times 10^{-24} = 9 \times 10^{-25}$

Since this product is lesser than K_{sp}, ZnS will not precipitate out.

(ii) $[H^+] = 0.01$ M

$$[S^{2-}] = \frac{9 \times 10^{-23} \times 0.1}{(0.01)^2} = \frac{9 \times 10^{-24}}{(0.01)^2} = 9 \times 10^{-20}$$

Since the product of Zn^{2+} and S^{2-} is 9.1×10^{-21} and is greater than K_{sp}, ZnS precipitates out.

Concept Strand 48

50 mL of 0.001 M barium chloride solution and 200 mL of 0.01 M sodium sulphate solution are mixed at 25°C. Will barium sulphate precipitate under these conditions? How much of barium ions are left behind in solution. (K_{sp} of $BaSO_4$ at 25°C = 1.08×10^{-10})

Solution

Milli moles of $Ba^{2+} = 50 \times 0.001 = 0.05$

Milli moles of $SO_4^{2-} = 200 \times 0.01 = 2.00$

$$[Ba^{2+}] = \frac{0.05 \times 10^{-3}}{(200 + 50)10^{-3}}\,\text{mol L}^{-1}$$

$$[SO_4^{2-}] = \frac{2 \times 10^{-3}}{250 \times 10^{-3}}\,\text{mol L}^{-1}$$

Product of the concentration of ions

$$= \frac{5 \times 10^{-5} \times 2 \times 10^{-3}}{0.25 \times 0.25} = \frac{10^{-7}}{0.0625} = 1.6 \times 10^{-6}$$

Since this product is greater than K_{sp}, $BaSO_4$ will precipitate out.

Let x g ion L^{-1} be left behind after precipitation.

Milli moles of $[Ba^{2+}]$ used up $= (5 \times 10^{-5} - x) \approx 5 \times 10^{-5}$

$K_{sp} = [Ba^{2+}]\,[SO_4^{2-}]$

$$= [Ba^{2+}]\left[\frac{2 \times 10^{-3} - 5 \times 10^{-5}}{0.25}\right] = 1.08 \times 10^{-10}$$

$$[Ba^{2+}] = \frac{1.08 \times 10^{-10} \times 0.25}{1.95 \times 10^{-3}} = 1.38 \times 10^{-8}\,\text{g ion L}^{-1}$$

Side reactions, which affect the application of solubility product principle

Other ionic equilibria such as hydrolysis and complex formation, which are simultaneously present alongside the solubility equilibrium complicate the application of solubility product principle. For example, the actual solubility product of lead chloride is much lower than that calculated from the corresponding solubility value i.e., K_{sp}(actual) < $4S^3$ where, S = observed solubility of $PbCl_2$. This is due to the fact that a considerable portion of Pb^{2+} ions in solution exist as complex ions $PbOH^+$ owing to hydrolysis and these ions do not contribute to the solubility product. This is because only simple Pb^{2+} ions are considered in the calculation of K_{sp}. Sometimes complex ions, like $AgCl_2^-$ in the case of silver chloride in presence of excess Cl^- ions complicate the application of this principle. Thus it is necessary to consider the possible side reactions in applying this principle in the analysis of solubility data.

INDICATORS IN NEUTRALIZATION REACTION

A neutralization indicator is an organic dye and is usually a weak acid or a weak base. It is partially ionised in solution. The ionized and non-ionized forms of the indicator have different colours and the form in which they exist depends on the pH of the medium.

The change of pH required to change the indicator from one colour form to the other, is two pH units, one unit on either side of pK_{Ind}. Since the useful range of an indicator is restricted to about two pH units, it is clear that a number of indicators are required to cover the pH range 0 to 14. Some important indicators and their useful pH range are given below.

Indicator	pK	Colour change		pH range
		acid	alkaline	
Methyl orange	3.7	Red	Yellow	3.1 – 4.4
Methyl red	5.1	Red	Yellow	4.2 – 6.3
Phenol red	7.9	Yellow	Red	6.8 – 8.4
Phenolphthalein	9.4	Colourless	Pink	8.3– 10.0

Double titration

The two indicators, viz., phenolphthalein (Hph) and methyl orange (MOH) show the end point of neutralization reactions at different positions. In the presence of Hph, one mole of H^+ per mole of base is consumed.

Thus in the presence of Hph, strong base will be completely neutralized by acid, but weak diacidic bases will not be completely neutralized.

When a mixture of NaOH and Na_2CO_3 is titrated with HCl using Hph as the indicator in the initial stage and MOH as the indicator in the latter stage of the titration.

$$\text{m.eq. of NaOH} + \frac{1}{2}\text{m.eq. of } Na_2CO_3 = \text{m.eq. of } HCl_{(II)}$$

The neutralization reactions in the former stage are

$$NaOH + HCl_{(I)} \rightarrow NaCl + H_2O$$
$$Na_2CO_3 + HCl_{(I)} \rightarrow NaHCO_3 + NaCl$$

In the latter stage,

$$\text{m.eq. of } NaHCO_3 = \text{m.eq. of } (HCl)_{11}$$

Reaction is, $NaHCO_3 + HCl_{(II)} \rightarrow NaCl + H_2O + CO_2$

$HCl_{(I)}$ and $HCl_{(II)}$ are the volumes of HCl consumed by using Hph an MOH, respectively.

When a mixture of NaOH, $NaHCO_3$ and Na_2CO_3 is titrated against H_2SO_4, the basis of calculations remain the same.

In the presence of Hph,

$$\text{m.eq. of NaOH} + \frac{1}{2}\text{m.eq. of } Na_2CO_3$$
$$= \text{m.eq. of } (H_2SO_4)_{(I)}$$

In the presence of MOH,

m.eq. of $NaHCO_3$ (produced from Na_2CO_3) + m.eq. of $NaHCO_3$ (originally present)

$$= \text{m.eq. of } (H_2SO_4)_{(II)}$$

If we use only MOH as the indicator, then

m.eq. of NaOH + m.eq. of Na_2CO_3 + m.eq. of $NaHCO_3$ = m.eq. of H_2SO_4.

CONCEPT STRAND

Concept Strand 49

A mixture of $NaHCO_3$ and Na_2CO_3 of unknown composition was dissolved in sufficient amount of water and the volume was made upto 250 mL. 25 mL of an aliquot of this mixture was titrated to a phenolphthalein end point that requires 2.8 mL of 0.01 N HCl. To this solution, methyl orange was added and the titration was continued and a further 11.2 mL of HCl was required for complete neutralization. Calculate the normalities of the two carbonates and their percentage composition in the original mixture.

Solution

$$CO_3^{2-} + H^+ \xrightarrow{\text{phenolphthalein}} HCO_3^-$$

$$HCO_3^- + H^+ \xrightarrow{\text{Methyl orange}} CO_2 + H_2O$$

Analysis of data involving phenolphthalein as indicator:

Milli eq of HCl required = $2.8 \times 0.01 = 0.028$

$Na_2CO_3 + HCl \rightarrow NaHCO_3 + NaCl$

0.028 m.eq of HCl reacted with 0.028 m.eq. of Na_2CO_3

$\Rightarrow$ weight of HCl = $0.028 \times 10^{-3} \times 36.5 = 0.01022$ g

Analysis of data involving methyl orange:

Milli .eq of HCl required = $11.2 \times 0.01 = 0.112$

$NaHCO_3 + HCl \rightarrow NaCl + H_2O + CO_2$

0.112 m.eq of HCl react with 0.112 m.eq of $NaHCO_3$. This amount also includes $NaHCO_3$ formed in the earlier titration.

$\therefore$ m.eq of $NaHCO_3$ latter in original mixture = $0.112 - 0.028 = 0.084$ m.eq

$$\text{N of } NaHCO_3 = \frac{0.084 \times 10^{-3}}{0.025} = 0.0034 \text{ ; N of } Na_2CO_3$$

$$= \frac{0.028 \times 10^{-3}}{0.025} = 0.0011$$

Composition of original mixture:

Weight of $NaHCO_3$ present in original mixture = $0.084 \times 10^{-3} \times 84 \times 10 = 0.070$ g.

Weight of Na_2CO_3 present in original mixture = $0.028 \times 10^{-3} \times 106 \times 10 = 0.03$ g

$$\%NaHCO_3 = \frac{0.07}{0.1} \times 100 = 70 \text{ ; } \% Na_2CO_3$$

$$= \frac{0.03}{0.1} \times 100 = 30$$

SUMMARY

Equilibrium	Definition Physical and chemical equilibrium
Reversible reactions	Kinetically reversible
Characteristic of reversible reactions	Attainment of the same equilibrium either from reactants or from the product.
Law of mass action	Proposed by C.M. Guldberg and P. Waage
Application of law of mass action to the equation $aA + bB + \ldots \rightleftharpoons \ell L + mM + \ldots$	Rate of forward reaction $r_f = k_f [A]^a [B]^b \ldots$ where, k_f is rate constant of forward reaction Rate of backward reaction $r_b = k_b [L]^\ell [M]^m \ldots$ where, k_b is the rate constant of backward reaction. At equilibrium $r_f = r_b$ i.e., $k_f[A]^a[B]^b \ldots = k_b [L]^\ell [M]^m \ldots$
Equilibrium constant	$K_C = \frac{k_f}{k_b} = \frac{[L]^\ell \times [M]^m \times \ldots}{[A]^a \times [B]^b \times \ldots}$

Test for equilibrium condition	Adding one of the reactant and observing the direction of the process proceeding
K_C and K_C'	$K_C' = \frac{1}{K_C}$ where, K_C' is the equilibrium constant of the reverse reaction
Equilibria in gases	$K_P = \frac{P_L^{\ell} . P_M^{m} .....}{P_A^{a} . P_B^{b} .....}$ where, $P_{L.....}$ is the partial pressure of the gaseous reactant and products.
Relation between K_P, K_C and K_X	$K_P = K_C(RT)^{\Delta n}$ Δn is the difference between gaseous products and gaseous reactants $K_P = K_X P^{\Delta n}$
Relation between ΔG , $\Delta G°$ and equilibrium constant	$\Delta G = \Delta G° + RT \ln Q$ $\Delta G° = - RT \ln K$ [$K = K_P$, K_C or K_X]
Le Chatelier's principle	Statement Effect of change in pressure and temperature
Homogeneous equilibria in gas phase	When, $\Delta n = 0$ $K_P = K_C = K_X = K$
For the reaction $N_2O_4 \rightleftharpoons 2NO_2$	$K_P = \frac{4\alpha^2 P}{1 - \alpha^2}$ $\alpha = \left(\frac{K_P}{4P + K_P}\right)^{1/2}$
For the reaction $PCl_5 \rightleftharpoons PCl_3 + Cl_2$	$K_P = \frac{\alpha^2 P}{1 - \alpha^2}$ $\alpha = \left(\frac{K_P}{K_P + P}\right)^{1/2}$
Degree of dissociation from vapour density for the dissociation $A \rightleftharpoons xB$	$\alpha = \frac{d_1 - d_2}{d_2(x - 1)}$ where, d_1 = density of undissociated gas , d_2 is density of dissociated gas.
Addition of inert gas at constant volume	No effect on the position of equilibrium
Addition of inert gas at constant pressure	When $\Delta n > 0$, equilibrium shifts in the direction of products. When $\Delta n < 0$, equilibrium shifts in the direction of reactants
Homogeneous equilibria in liquid phase	K_C and K_X only
Simultaneous equilibria	$A \rightleftharpoons B + C$ K_1 $P \rightleftharpoons Q + C$ K_2 $A + Q \rightleftharpoons B + P$ $K = \frac{K_1}{K_2}$
Heterogeneous equilibria	$CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$ $K_P = p_{CO_2}$

Effect of temperature on equilibrium constant	$\left(\frac{\partial(\ell n K_p)}{\partial T}\right)_P = \frac{\Delta H^0}{RT^2}$
	$\log\frac{(K_p)_2}{(K_p)_1} = \frac{\Delta H^0}{2.303R}\left[\frac{T_2 - T_1}{T_1 T_2}\right]$
Variation of K_C with temperature	$\left(\frac{\partial(\ln K_c)}{\partial T}\right)_v = \frac{\Delta U^\circ}{RT^2}$
	Ionic equilibrium
Arrhenius Theory	Acid – H^+ ion supplier
	Base – OH^- ion supplier
Bronsted-Lewis concept	Acid gives H^+ ions
	Bases accept H^+ ions
Conjugate acid	Base on accepting H^+ forms conjugate acid
Conjugate base	Acid on losing H^+ form conjugate base
Conjugate pair	Pair of species differing by an H^+
Classification of solvents	Acidic e.g., acetic acid
	Basic e.g., liquid NH_3
	Amphoteric Eg., water, alcohol etc.
Lewis Theory	Acid-electron pair acceptor
	Base – electron pair donor
pH	$-\log_{10}[H^+]$
Relation between pH and pOH	$pH + pOH = pK_w$
Dissociation constant of weak acid HA	$K_a = \frac{[H^+][A^-]}{[HA]}$
Relation between K_a and degree of dissociation α	$K_a = \frac{\alpha^2 C}{1-\alpha^2} \approx \alpha^2 C \; \alpha = \sqrt{\frac{K_a}{C}}$
pH of a weak acid	$pH = \frac{1}{2}[pK_a - \log C]$
pH of a weak base	$pH = pK_W + \frac{1}{2}[\log C - pK_b]$
Hydrolysis of salts of weak acids and strong bases (A^-B^+)	Anion undergoes hydrolysis
	$K_h = \frac{[HA][OH^-]}{[A^-]}$
	$K_h = \frac{h^2 C}{1-h^2} = Ch^2$
Degree of hydrolysis	$h = \sqrt{\frac{K_h}{C}}$

pH of the solution	$pH = \frac{1}{2}[pK_w + pK_a + \log C]$
Hydrolysis of salt of weak base and strong acid	Cation undergoes hydrolysis
	$K_h = \frac{[BOH][H^+]}{[B^-]}$
	$K_h = \frac{h^2C}{1-h^2} \approx Ch^2$
Degree of hydrolysis	$h = \sqrt{\frac{K_h}{C}}$
pH of solution	$pH = \frac{1}{2}[pK_w - pK_b - \log C]$
Hydrolysis of salt of weak acid and weak base	Both anion and cation hydrolysis
	$K_h = \frac{h^2}{(1-h)^2} \approx h^2$
Degree of hydrolysis	$h = \sqrt{K_h}$
pH of solution	$pH = \frac{1}{2}[pK_w + pK_a - pK_b]$
Common ion effect	Definition
Self ionization of water	$K_w = [H^+]\,[OH^-] = 10^{-7} \times 10^{-7} = 10^{-14}$ at 298K
$[H^+]$ concentration of water	Depends on temperature; It increases with temperature correspondingly $[OH^-]$ and K_w both increase
Buffer	Definition
pH of a buffer	Calculated using Henderson–Hasselbalch equation
Henderson equation	$pH = pK_a + \log \frac{[salt]}{[acid]}$ for acidic buffer
	$pOH = pK_b + \log \frac{[salt]}{[base]}$ for basic buffer
Buffer capacity	$\beta = \frac{dc}{dpH}$
	dc is no. of moles of acid or base added per litre buffer.
	dpH is change in pH
Ionic product	Product of concentrations of ions in solution
Solubility product	Product of molar concentrations of ions in a saturated solution
Solubility product for the salt M_mA_n	$K_{SP} = m^m n^n (S)^{m+n}$ where, 'S' is the solubility in mol L^{-1}
Variation of solubility product with temperature	$\frac{\partial(\ln K_{sp})}{\partial T} = \frac{\Delta H^\circ_{sol n}}{RT^2}$

Relation between solubility S and K_{SP} for certain salts	Salt	Solubility 'S' in mol L^{-1}
	AB	$\sqrt{K_{SP}}$
	AB_2	$\left(\frac{K_{SP}}{4}\right)^{1/3}$
	AB_3	$\left(\frac{K_{SP}}{27}\right)^{\frac{1}{4}}$

Application of K_{SP} in qualitative analysis

Group I	Low K_{SP} of chlorides of Pb^{2+}, Hg_2^{2+}, Ag^+
Group II and IV	Low in K_{SP} values of group II sulphides compared to the high K_{SP} values of group IV sulphides. The $[S^{2-}]$ suppressed by adding HCl in group II and $[S^{2-}]$ increased by adding NH_4OH in group IV
Group III	Low K_{SP} values of group III hydroxides. The $[OH^-]$ is kept low by adding NH_4Cl to NH_4OH
Side reactions affecting solubility product principle	Complex formation such as $[PbOH]^+$ and $AgCl_2^-$ which increases the solubility
Indicators in neutralization reaction	Methyl orange Phenolphthalein Methyl red, phenol red
Double titrations involving Na_2CO_3 and $NaHCO_3$	Phenolphthalein indicate CO_3^{2-} to HCO_3^- change. Methyl orange can indicate HCO_3^- to CO_2 change also.

TOPIC GRIP

Subjective Questions

1. The forward reaction in the system $2NH_{3(g)} \rightleftharpoons N_{2(g)} + 3H_{2(g)}$ has proceeded to 98% under equilibrium. The equilibrium pressure = 10 atm. Temperature = 400°C. Calculate K_P and K_C.
2. 3 moles of Iodine and 8 moles of hydrogen are kept at 444°C and at constant volume till equilibrium is established at which 5.65 moles of $HI_{(g)}$ are formed $H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$. If one starts with 5.3 moles of $I_{2(g)}$ and 8 moles of hydrogen, how many moles of $HI_{(g)}$ will be formed at equilibrium (at the same temperature). Give an approximate estimate.
3. For the equilibrium : $N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}$ att 2675°C $K_P = 3.5 \times 10^{-3}$. Calculate $\Delta G°$.
4. When the equilibrium : $N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$ is set up at 65°C and at a total (equilibrium) pressure of one atm, the degree of dissociation = 0.6. If keeping the temperature at the same value, but the pressure is adjusted so that the volume is reduced to one half of the initial volume, what would be the degree of dissociation?
5. At 2000°C and at 1 atm pressure, degree of dissociation of $CO_{2(g)}$ is only 1.8% :
   $2CO_{2(g)} \rightleftharpoons 2CO_{(g)} + O_{2(g)}$. Calculate the equilibrium constant.
6. 3.6 g of PCl_5 vapour occupies a volume of one litre at an equilibrium pressure of 1 atm at 200°C; $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$. Calculate the degree of dissociation and the equilibrium constant. Molar mass of PCl_5 = 208 g mol^{-1}.
7. Calculate the $[H^+]$ in a solution containing a mixture of one mole of acetic acid and one mole of cyano acetic acid in a litre. K_a for acetic acid = 1.8×10^{-5} and K_a for cyano acetic acid = 3.7×10^{-3}.
8. Derive an expression for the hydrogen ion concentration in a very dilute solution of acetic acid. Use it to calculate $[H^+]$ in 10^{-4} M acetic acid. Ionic product of water $K_w = 10^{-14}$ at 25°C; K_a for acetic acid = 1.85×10^{-5} at 25°C.
9. K_a for acetic acid at 18°C = 1.8×10^{-5}. Calculate α the degree of dissociation and the hydrogen ion concentration in (i) 0.25 N acetic acid solution and (ii) a solution which is 0.25 N w.r.t sodium acetate and 0.25 N w.r.t acetic acid assuming that sodium acetate is completely ionized in solution.
10. At a certain temperature T, solubility of $AgBrO_3$ in water = 8.1×10^{-3} M. To one litre of the solution 8.5×10^{-3} mole of $AgNO_{3(s)}$ is added. What is the (new) solubility of $AgBrO_3$. Assume that both salts are completely ionized in solution.

Straight Objective Type Questions

Directions: **This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

11. Consider the equilibrium: $2A_{(g)} \rightleftharpoons 2B_{(g)} + C_{(g)}$ at temperature T. It is set up with only A as the starting material. The total equilibrium pressure = P atm. $P_C = \frac{P}{n}$. If $\frac{K_P}{P} = \frac{1}{81}$. Calculate the value of n (a whole no.)

    (a) 3 (b) 6 (c) 9 (d) 12

12. To the equilibrium system: $PCl_{5(g)} = PCl_{3(g)} + Cl_{2(g)}$, gaseous chlorine is added, increasing the volume at the same time in such a manner that the partial pressure of chlorine, pCl_2 in the system remains unchanged. The temperature is kept constant. What happens to the equilibrium?

(a) shifted backward (b) remains unaffected

(b) shifted forward (d) depends on the magnitude f pCl_2

13. A few drops of acetic acid are added to a large excess of water at 25°C. In the "limit of infinite dilution", what % of the acid remains unionized? $K_a = 1.85 \times 10^{-5}$ at 25°C.

(a) 0.54 (b) 0.38 (c) 0.83 (d) 0.45

14. Calculate the change in pH when a 0.1 M solution of acetic acid ($K_a = 1.85 \times 10^{-5}$) in water at 25°C is diluted to a final concentration of 0.01 M.

(a) ~0.5 (b) ~0.4 (c) ~0.7 (d) ~0.6

15. 0.1 M KCl solution, containing 0.0001 M [i.e., 10^{-4} M] K_2CrO_4 is titrated against one M $AgNO_3$. The concentration of Cl^- decreases by precipitation of AgCl. At what (approximate) concentration of Cl^- in the solution will the precipitation of Ag_2CrO_4 commence? K_{SP} of AgCl = 1.7×10^{-10}. K_{sp} of $Ag_2CrO_4 = 2 \times 10^{-12}$.

(a) 2.167×10^{-5} M (b) 2.716×10^{-5} M (c) 1.762×10^{-6} M (d) 1.202×10^{-6}M

Assertion–Reason Type Questions

Directions: **Each question contains Statement-1 and Statement-2 and has the following choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False
(d) Statement-1 is False, Statement-2 is True

16. Statement 1

For the equilibrium : $CuSO_4.5H_2O_{(s)} \rightleftharpoons CuSO_4.3H_2O_{(s)} + 2H_2O_{(g)}$

$\Delta G^\circ = -RT \ln \left(p_{H_2O}\right)$ where p_{H_2O} is the partial pressure of water vapour.

and

Statement 2

In general $\Delta G^\circ = -RT \ln K_p$.

17. Statement 1

Addition of $Cl_{2(g)}$ at constant volume and at constant temperature to the equilibrium system: $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$ reduces the equilibrium concentration of $PCl_{3(g)}$.

and

Statement 2

In the equilibrium system: $H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$. The yield of HI is increased by either raising the pressure or adding $H_{2(g)}$ to the system.

18. Statement 1

Addition of only a few drops of $KI_{(aq)}$ to $HgCl_{2(aq)}$ yields a scarlet precipitate of HgI_2, which dissolves in excess of KI solution

and

Statement 2

$HgI_2 + 2KI \rightarrow \underset{\text{soluble}}{K_2\left(HgI_4\right)}$ (complex compound).

19. Statement 1
The solution of borax is alkaline.
and

Statement 2
Borax is the salt of the strong base NaOH and the weak acid, $H_2B_4O_7$. There is hydrolysis.

20. Statement 1
Addition of solid sodium acetate to aqueous $CH_3 - COOH$ raises the pH of the solution.
and

Statement 2
$CH_3 - COONa$ and $CH_3 - COOH$ have the common ion, $CH_3 - COO^-$. By common ion effect, the ionization of acetic acid is suppressed and pH is raised.

Linked Comprehension Type Questions

Directions: **This section contains 2 paragraphs. Based upon the paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

Passage I

The law of mass action states that the speed of a chemical reaction depends on the activities of the reactants. For pure solids and liquids, activities are taken to be constant and for practical reasons taken to be 1. For gases and solutions, activity is taken to be the same as concentration expressed as moles/litre. If one considers the equilibrium $n_1A + n_2B \rightleftharpoons n_3C + n_4D$, one may, then derive, (by the use of the law of mass action) the expression K, the equilibrium constant = $\left[(a_C)^{n_3} \times (a_D)^{n_4}\right] \Big/ \left[(a_A)^{n_1} (a_B)^{n_2}\right]$ where the a's are the activities.

In more convenient form, one writes $K_C = \dfrac{c_C^{n_3} \times c_D^{n_4}}{c_A^{n_1} \times c_B^{n_2}}$ where, c's are the molar concentrations (in moles/litre). For the case of gases one may use the partial pressure 'p', in the place of the concentration C, since $p = \left(\dfrac{n}{V}\right)RT$ ($\dfrac{n}{V}$ = no. of moles/litre) i.e., p = CRT. Thus one may define an equilibrium constant, $K_P = \dfrac{p_C^{n_3} \times p_D^{n_4}}{p_A^{n_1} \times p_B^{n_2}}$ (assuming that A, B, C and D are gases). One then derives the equation : $K_p = K_C(RT)^{\Delta n}$ where, $\Delta n = (n_2 - n_1)$ where, n_2 = no. of moles of gaseous products and n_1 = no. of moles of gaseous reactants. In other words, solids and liquids are left out in counting for n_2 and n_1.

When considering a single reactant say one mole forming a number of products by dissociation, the fraction of one mole, dissociated is known as the degree of dissociation, α. Both K_P and K_C are related to α, through equations, different in different cases. Both K_P and K_C do not depend on the reactant or product concentrations.

21. One mole of ethanol and one mole of acetic acid are mixed:
$CH_3 - COOH + C_2H_5 - OH \rightleftharpoons CH_3 - COOC_2H_5 + H_2O$. under equilibrium $\frac{2}{3}$ mole of ester and $\frac{2}{3}$ mole of water are formed. If one starts with (a) one mole of acid and 2 moles of alcohol, (b) one mole of ester and 3 moles of water, how much ester is present under equilibrium?
(a) 0.548 mol, 0.546 mol (b) 0.845 mol, 0.465 mol (c) 0.584 mol, 0.564 mol (d) 0.485 mol, 0.282 mol

22. Consider the equilibrium: $N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$. If α is the degree of dissociation, and P(atm) is the total equilibrium pressure, express K_P in terms of α and P (or α in terms of K_P and P)

(a) $K_P = \left(\frac{\alpha^2}{1+\alpha^2}\right)P$ (b) $\alpha = \left(\frac{P}{K_P + P}\right)$

(c) $\alpha = \left(\frac{K_P}{4P + K_P}\right)^{1/2}$ (d) $K_P = \left(\frac{\alpha}{1-\alpha}\right)P$

23. The vapour pressure of solid $(NH_4)HS$ at 25.1°C is 501 mm. [Assume for the sake of calculation that the vapour is practically completely dissociated into $NH_{3(g)}$ and $H_2S_{(g)}$] If solid $(NH_4)HS$ is introduced into a vessel already containing $NH_{3(g)}$ at $p_{NH_3} = 320$ mm at 25.1°C. What is the total pressure at equilibrium?

(a) 595 mm (b) 495 mm (c) 695 mm (d) 395 mm

Passage II

Thermodynamics yields the derivation that the standard free energy change $\Delta G° = -RT \ln K_P$. Further $\Delta G° = \Delta H° - T\Delta S°$, where, T = 298K and , $\Delta H°$ and $\Delta S°$ are standard enthalpy change and entropy change.

24. Calculate K_P for a reaction involving gaseous reactants and products for which $\Delta n = -4$, $K_C = 6.91 \times 10^4$ at 727°C.

(a) 1.25×10^{-4} (b) 2.15×10^{-3} (c) 1.52×10^{-3} (d) 2.51×10^{-4}

25. The magnitude of K_P for a certain gaseous reaction at 298K = 7.08×10^{24}. Calculate $\Delta G°$ for the reaction.

(a) −141.8 kJ (b) −184.1 kJ (c) −92.2 kJ (d) −70.9 kJ

26. $\Delta H°$ and $\Delta S°$ for a certain gaseous reaction are −116 kJ and −109 J K^{-1} respectively. T = 298K. Calculate K_P.

(a) 3.46×10^{12} (b) 4.36×10^{14} (c) 3.64×10^{13} (d) 6.43×10^{12}

Multiple Correct Objective Type Questions

Directions: **Each question in this section has four suggested answers out of which ONE OR MORE answers will be correct.**

27. For the dissociation: $A_{(g)} \rightleftharpoons B_{(g)} + C_{(g)}$; $\Delta H° = 70$ kJ mol^{-1} $\Delta S° = 140$ J K^{-1} at T = 298K

(a) Dissociation is spontaneous above 227°C.

(b) $\Delta G° = 28.73$ kJ mol^{-1}.

(c) K_P for the reverse reaction = 1.867×10^5.

(d) K_P is a function of both pressure and temperature.

28. $SO_2Cl_{2(g)}$ undergoes 40% dissociation at 30°C and 760 mm pressure

$SO_2Cl_{2(g)} \rightleftharpoons SO_{2(g)} + Cl_{2(g)}$.

(a) Partial pressures of $SO_2Cl_{2(g)}$ and $SO_{2(g)}$ under equilibrium are 325.7 mm, 217.15 mm.

(b) K_P for the reaction is 0.19.

(c) Degree of dissociation is 0.525 when the total equilibrium pressure is 380 mm.

(d) The reaction is spontaneous at all temperatures.

29. Choose the incorrect statement(s) among the following.

(a) When the buffer solution of sodium acetate + acetic acid is diluted with water the pH of the solution decreases.

(b) No hydrolysis occurs when NaCl is dissolved in water.

(c) The ionic product of water is temperature dependent.

(d) The commonly used acid–base indicator, phenolphthalein suffers a structural change when in an aqueous solution there is a change of pH from 3 to 7.

Matrix-Match Type Question

Directions: **Match the elements of Column I to elements of Column II. There can be single or multiple matches.**

30.

Column I	**Column II**
(a) $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$ forward reaction favoured by	(p) adding the product of the forward reaction
(b) $PCl_{5(g)}$ in a closed vessel at TK. $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$ dissociation suppressed by	(q) increase of pressure
(c) Ionic product of water is lowered in magnitude by	(r) shows a pH dependent on salt to acid ratio.
(d) Mixture of CH_3–COOH and CH_3 – COONa	(s) lowering of temperature

IIT ASSIGNMENT EXERCISE

Straight Objective Type Questions

Directions: **This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

31. $SO_2Cl_{2(g)} \rightleftharpoons SO_{2(g)} + Cl_{2(g)}$. If a catalyst is added to the system under equilibrium.
(a) The rate of the forward reaction will increase.
(b) System shifts to the left i.e, exothermic change.
(c) There is no effect on the position of equilibrium.
(d) What happens depends on the equilibrium pressure and temperature.

32. $N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}$. Equilibrium constant K_1

$NO_{(g)} \rightleftharpoons \frac{1}{2}N_{2(g)} + \frac{1}{2}O_{2(g)}$ equilibrium constant K_2. Then

(a) $K_1K_2 = 1$ (b) $K_1K_2^2 = 1$ (c) $K_1^2K_2 = 1$ (d) $K_1 = K_2^2$

33. At 40°C and a total equilibrium pressure of 1 atm, $N_2O_{4(g)}$ is dissociated to the extent of 60%: $N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$. Calculate the % dissociation at an equilibrium pressure of 5 atm
(a) ~42% (b) ~32% (c) ~48% (d) ~21%

34. For a certain gaseous equilibrium, $n_1A_{(g)} \rightleftharpoons n_2B_{(g)} + n_3C_{(g)}$ considering pressure in atmosphere $K_P = 9.92 \times 10^4$ in magnitude. $K_C = 32.55$ (in magnitude) at 400°C. What is the value of Δn in the equation $K_P = K_C(RT)^{\Delta n}$?
(a) +2 (b) −2 (c) +1 (d) −1

35. K_C for the equilibrium system: $CH_3 - COOH + C_2H_5 - OH \rightleftharpoons H_2O + CH_3 - COOC_2H_5$ equals 4 at the room temperature. Calculate the weight in grams of ester formed when 92 g of ethanol and 180 g of $CH_3 - COOH$ are initially taken and allowed to come to equilibrium.
(a) 69 g (b) 104 g (c) 138 g (d) 158 g

36. One mole of $HCl_{(g)}$ + 0.5 mole of $O_{2(g)}$ mixed and allowed to come to equilibrium : $4HCl_{(g)} + O_{2(g)} \rightleftharpoons 2Cl_{2(g)} + 2H_2O_{(g)}$ at 400°C. Equilibrium pressure = 1 atm. 0.39 mole of $Cl_{2(g)}$ is formed under equilibrium. Calculate K_P.
(a) 46.22 atm^{-1} (b) 42.26 atm^{-1} (c) 64.22 atm^{-1} (d) 22.64 atm^{-1}

37. For the equilibrium of selenium vapour in dissociation $Se_{6(g)} \rightleftharpoons 3Se_{2(g)}$, $K_P = 0.2$ at 700°C. Calculate approximate % dissociation of the vapour at 700°C and at an equilibrium pressure of 600 mm.
(a) 35% (b) 17% (c) 27% (d) 8%

38. Consider the equilibrium system: $N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}$. At T = 2675K; $K_C = K_P = 3.5 \times 10^{-3}$. Calculate the yield of NO in percentage by volume from air at normal pressure (1 atm) (with 79.2% N_2 and 20.8% O_2)
(a) 2.32% (b) 1.33% (c) 1.56% (d) 3.67%

39. Consider the equilibrium system: $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$ K_P (pressure in atmosphere) = 1.781 at 250°C. Calculate no. of grams of $PCl_{5(g)}$ to be taken in a 5 litre vessel to get an equilibrium concentration of 0.05 mol $litre^{-1}$ of PCl_5.
(a) 87.66 g (b) 99.6 g (c) 105.51 g (d) 120.15 g

40. For the system $[PCl_5 - PCl_3 - Cl_2]$ in equilibrium, K_P (calculated in atmosphere) = 1.781 at 250°C. If into a 5 litre vessel, already containing 0.2 mole of $Cl_{2(g)}$, 0.2 mole of $PCl_{5(g)}$ is introduced what may be the degree of dissociation α
(a) 0.242 (b) 0.224 (c) 0.422 (d) 0.122

41. K_p for the equilibrium: $CO_{2(g)} + C_{(s)} \rightleftharpoons 2CO_{(g)}$ is 122.0 at 1000°C. Calculate mole % of $CO_{(g)}$ and $CO_{2(g)}$ at a total pressure of 12.2 atm.

(a) 82.6% , 17.4% (b) 72.7%, 27.3% (c) 75.2%, 24.8% (d) 91.6%, 8.4%

42. Consider the equilibrium system: $SO_{3(g)} \rightleftharpoons SO_{2(g)} + \frac{1}{2} O_{2(g)}$ At 400K, a 20 g sample of SO_3 exhibited decomposition of 45% at the equilibrium pressure of 1 atm. Calculate K_p. At. wt. of S = 32

(a) 0.5360 $atm^{1/2}$ (b) 0.3506 $atm^{1/2}$ (c) 0.5603 $atm^{1/2}$ (d) 0.6530 $atm^{1/2}$

43. When 2.55 g of $H_{2(g)}$ are mixed with 44.66 g of $N_{2(g)}$ in a one litre flask, 4.25 g of $NH_{3(g)}$ are formed at equilibrium at 325K. Calculate the equilibrium constant.

(a) 1.8×10^{-2} (b) 3.2×10^{-2} (c) 4.3×10^{-2} (d) 5.8×10^{-2}

44. Given $K_p = 1.781$ for the $[PCl_5 - PCl_3 - Cl_2]$ equilibrium system at 250°C. Calculate $\Delta G°$.

(a) −2.51 kJ (b) −2.15 kJ (c) +1.52 kJ (d) −1.52 kJ

45. Given (i) $TiO_{2(s)} + 2Cl_{2(g)} \rightarrow TiCl_{4(l)} + O_{2(g)}$; $\Delta G° = 162$ kJ
(ii) $S_{(s)} + O_{2(g)} \rightarrow SO_{2(g)}$; $\Delta G° = -300$ kJ
Reactions

(a) (i) and (ii) are spontaneous
(b) (i) and (ii) are non-spontaneous
(c) the reaction (iii) $S_{(s)} + TiO_{2(s)} + 2Cl_{2(g)} \rightarrow TiCl_{4(l)} + SO_{2(g)}$ is spontaneous
(d) reactions (i), (ii) and (iii) are non-spontaneous

46. 3.5 g of $CO_{(g)}$ reacts with a certain amount of chlorine in a 2.5 litre flask at 27°C to form phosgene , $(COCl_2)$, $CO_{(g)} + Cl_{2(g)} \rightleftharpoons COCl_{2(g)}$. $\Delta G°$ for the reaction = 718 J. Calculate K_p and K_C

(a) 0.57 atm^{-1}, 15.47 M^{-1} (b) 0.47 atm^{-1}, 13.75 M^{-1} (c) 0.47 atm^{-1}, 14.75 M^{-1} (d) 0.75 atm^{-1}, 18.47 M^{-1}

47. Consider the equilibrium system: $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$ at a certain temperature the vapour density = 65. Calculate the degree of dissociation.

(a) 0.6 (b) 0.7 (c) 0.75 (d) 0.65

48. For the equilibrium system $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$ at 200°C, in a volume = 1 litre, $\alpha = 0.49$. Calculate K_C (mol $litre^{-1}$) [The total mass of the gases = 3.6 g. This is the mass of $PCl_{5(g)}$ initially taken in an empty vessel to establish the equilibrium at 200°C]

(a) 5.81×10^{-3} (b) 5.18×10^{-4} (c) 1.85×10^{-4} (d) 8.15×10^{-3}

49. For the equilibrium: $N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$, the degree of dissociation α is related to K_p and P : $\alpha = \left[\frac{K_P}{K_P + 4P}\right]^{1/2}$.
Express % change in α, for a 10% change in the pressure P. Take $\alpha^2 << 1$

(a) 5.46 % (b) 6.54 % (c) 4.65% (d) 6.45%

50. Consider the system: $(NH_4)HS_{(s)} \rightleftharpoons NH_{3(g)} + H_2S_{(g)}$ in equilibrium. 0.1 mole of solid ammonium hydrogen sulphide was vaporized in a 2 litre vessel; at a certain temperature T, the equilibrium pressure was 360 mm. Calculate K_p

(a) 0.0651 atm^2 (b) 0.0561 atm^2 (c) 0.0615 atm^2 (d) 0.0165 atm^2

51. $2HgO_{(s)} \rightleftharpoons 2Hg_{(g)} + O_{2(g)}$. Given that the dissociation pressure is P mm at 693K and $K_p = 1.93 \times 10^{-2}$ atm^3, calculate P.

(a) 538 mm (b) 435 mm (c) 226 mm (d) 385 mm

52. For a certain gaseous equilibrium, the equilibrium constant increases from 0.0068 to 0.816 when the temperature increases from 27°C to 111°C. Calculate ΔH.

(a) 12.045 k cal (b) +13.047 k cal (c) 11.033 k cal (d) 14.722 k cal

53. For the system $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$ the mean value of $\Delta H°$ may be taken to be -46.1 kJ mol^{-1} over the temperature range 650K to 800K. If K_{P_1} and K_{P_2} are the K_P values at temperatures 673K and 755K, calculate the ratio $\left(\frac{K_{P_2}}{K_{P_1}}\right)$ approximately.

(a) 0.2174 (b) 0.1742 (c) 0.1472 (d) 0.4087

54. At 29°C the vapour pressure of the equilibrium system: $BaCl_2.2H_2O$, $BaCl_2.H_2O$, $H_2O_{(g)}$ = 7.125 mm. At 32°C the vapour pressure = 8.945 mm. Calculate approximately the enthalpy of hydration of the monohydrate salt to the dihydrate salt by water vapour.

(a) −13.97 k cal (b) −17.93 k cal (c) −19.37 k cal (d) +14.22 k cal

55. pH of a solution = 11. Calculate the no. of hydrogen ions present per ml of the solution.

(a) 6.02×10^9 (b) 6.02×10^{11} (c) 6.02×10^{20} (d) 6.02×10^{15}

56. The pH of an aqueous solution of a dilute acid is 5. Calculate the number of hydroxide ion per ml of the solution T = 298K.

(a) 6.02×10^{11} (b) 6.02×10^9 (c) 6.02×10^8 (d) 6.02×10^7

57. Calculate the pH of a solution which contains HCl 5×10^{-7} mol $litre^{-1}$ [of water] at 25°C.

(a) 6.825 (b) 6.582 (c) 6.125 (d) 6.285

58. Calculate the pH of a solution of H_2SO_4 with density of 1.02 g cm^{-3} and containing 3.24 wt percent of acid at 20°C. Assume complete dissociation.

(a) 0.171 (b) 0.271 (c) 0.217 (d) 0.127

59. pH of a solution = 7. To one litre of this solution sufficient solid NaOH is added to raise the pH to 12. Calculate the weight of NaOH added. (T = 298K)

(a) 0.4 g (b) 4 g (c) 0.04 g (d) 0.004 g

60. At 65°C, pH of neutral water = 6.77. Calculate the ionic product of water at this temperature.

(a) 2.67×10^{-14} (b) 2.53×10^{-14} (c) 2.18×10^{-14} (d) 2.88×10^{-14}

61. A weak, monoprotic acid is dissociated to the extent of 3% in 0.02 M solution. Calculate the extent of its dissociation in 0.05 M solution. T = 298K

(a) 1.39% (b) 2.19% (c) 2.32% (d) 1.93%

62. Consider the equilibria (in aq). $H_2S_{(0.1\,M)} \rightleftharpoons H^+_{(aq)} + HS^-_{(aq)}$; $K_a = 5.8 \times 10^{-8}$ and $H_2S_{(0.1\,M)} \rightleftharpoons 2H^+_{(aq)} + S^{2-}_{(aq)}$; $K_a = 1.1 \times 10^{-22}$ (both at 291K). Calculate the hydrogen ion concentration in a saturated solution (i.e., 0.1 M) of H_2S in water.

(a) 7.6×10^{-5} M (b) 6.7×10^{-5} M (c) 6.7×10^{-4} M (d) 7.6×10^{-6} M

63. The solubility product of CdS in water at 18°C = 4×10^{-29}. Calculate the maximum concentration of Cd^{2+} ions which can remain in a solution of 0.1 M HCl saturated with H_2S(0.1M) given $H_2S_{(0.1\,M)} \rightleftharpoons 2H^+_{(aq)} + S^{2-}_{(aq)}$; $K_a = 1.1 \times 10^{-22}$

(a) 6.36×10^{-9} M (b) 3.636×10^{-8} M (c) 6.63×10^{-7} M (d) 3.036×10^{-7} M

64. The dissociation constants of acetic acid and monochloroacetic acid are 1.8×10^{-5} and 1.4×10^{-4} respectively at 298 K. The pH of a mixture which is 0.02 M with respect to acetic acid and 0.01 M with respect to monochloroacetic acid is

(a) 1.33 (b) 2.88 (c) 1.76 (d) 6

65. At a concentration of 1.23×10^{-2} M, trimethyl ammonium ion, $-(CH_3)_3NH^+$ is 0.01% dissociated: Calculate the pH.

(a) 5.19 (b) 5.91 (c) 4.91 (d) 4.19

66. K_a for a certain monoprotic organic acid HA is 1.8×10^{-5}. The pH of a solution of NaA, its sodium salt of molarity 0.1 is [$K_w = 10^{-14}$, T = 298K]

(a) 5.13 (b) 8.87 (c) 3.26 (d) 10.75

67. Which among the following will not undergo hydrolysis?
(a) Na_2CO_3 (b) NH_4Cl (c) $CH_3-COONH_4$ (d) KNO_3

68. The dissociation constants of aniline and acetic acid are 4.8×10^{-10} and 1.8×10^{-5} respectively at 25°C. $K_w = 10^{-14}$. Calculate the $[H^+]$ in 0.01 M and 0.05 M aniline acetate solutions at 25°C.
(a) 2.136×10^{-5} M (b) 1.936×10^{-5} M (c) 1.736×10^{-5} M (d) 1.736×10^{-4} M

69. The hydrolytic constant of urea hydrochloride is 0.786 at 25°C [$K_w = 10^{-14}$]. Calculate K_b for urea.
(a) 1.72×10^{-14} (b) 2.17×10^{-14} (c) 1.27×10^{-14} (d) 1.56×10^{-14}

70. Calculate the hydrolysis constant K_h and degree of hydrolysis h in 0.01 M solution of a salt, BA where in both B – OH (base) and HA(acid) are weak. Calculate also the pH
T = 298K; $K_w = 10^{-14}$ [Given K_b of BOH = 1.8×10^{-5}, K_a for HA = 7.2×10^{-10}
(a) 0.711, 0.392, 8.2 (b) 0.872, 0.402, 11.2 (c) 0.772, 0.4676, 9.2 (d) 0.611, 0.237, 7.2

71. Given $K_a = 1.785 \times 10^{-5}$ for acetic acid at 298K. Calculate the pH of the solution obtained by mixing equal volumes of 0.2 M HCl and 0.6 M sodium acetate
(a) 4.0 (b) 4.4 (c) 5.05 (d) 6.18

72. Which among the following is not a buffer?
(a) $NH_4Cl_{(aq)} + NH_4OH_{(aq)}$ (b) $Ba(OH)_2 + HCOOH_{(excess)}$
(c) $HCl_{(aq)}$ + excess of $CH_3.COONa$ (d) $MgSO_{4(aq)(excess)} + dilH_2SO_4$

73. The solubility product of silver oxalate at 298K is 1.21×10^{-11}. Calculate its solubility in g/litre. Atomic wt of Ag = 108
(a) 7.87×10^{-2} (b) 4.39×10^{-2} (c) 7.12×10^{-2} (d) 9.12×10^{-2}

74. The solubility product of CaF_2 in water at 18°C is 2.8×10^{-11}. Calculate the solubility in g/litre. [At. wt. of Ca =40, F = 19]
(a) 0.025 g litre^{-1} (b) 0.022 g litre^{-1} (c) 0.32 g litre^{-1} (d) 0.015 g litre^{-1}

75. A solution of $KBr_{(0.1\ M)}$ was slowly added to 0.005 M. $AgNO_3$ solution to precipitate AgBr ($K_{SP} = 5 \times 10^{-13}$) until $[Br^-]$ becomes 0.01 M. Calculate the concentration of Ag^+ ion.
(a) 5×10^{-15}M (b) 5×10^{-12} M (c) 1×10^{-3} M (d) 5×10^{-11} M

76. The solubility product values of AgI and Ag_3AsO_3 at 20°C are 8.3×10^{-17} and 4.5×10^{-19} respectively. Calculate the ratio of $[Ag^+]$ in two separate solutions (i) saturated with respect to $AgI_{(s)}$ and (ii) saturated with respect to $Ag_3AsO_{3(s)}$.
(a) 1.335×10^{-3} (b) 1.335×10^{-5} (c) 2.67×10^{-4} (d) 3.92×10^{-5}

77. K_{SP} values of $AgCl_{(s)}$ in water at 25°C = 2.25×10^{-10} and that of $AgBr_{(s)}$ in water = 4.9×10^{-13} at the same temperature. Calculate $[Ag^+]$ in a solution saturated with respect to both $AgCl_{(s)}$ and $AgBr_{(s)}$.
(a) 1.205×10^{-5} (b) 1.025×10^{-5} (c) 2.105×10^{-5} (d) 1.502×10^{-5}

78. K_{SP} of $Fe(OH)_2 = 7.68 \times 10^{-16}$. Calculate the concentration of (OH^-) ion in a saturated solution of $Fe(OH)_2$ in water T = 298K.
(a) 1.450×10^{-4} M (b) 1.541×10^{-4} M (c) 1.154×10^{-5} M (d) 1.245×10^{-3} M

79. K_{SP} of $Ag_2C_2O_4$ at 25°C = 1.21×10^{-11} [At. wt of Ag = 108]. Calculate approximately its solubility in water and its solubility in 0.1 M $AgNO_3$ solution.
(a) 1.446×10^{-4} M, 1.21×10^{-9}M (b) 1.632×10^{-3} M, 1.12×10^{-8} M
(c) 1.623×10^{-4} M, 1.12×10^{-6} M (d) 1.263×10^{-3}, 2.11×10^{-6} M

80. K_{SP} of AgCl = 1.7×10^{-10} at 25°C. If 0.2 M $CaCl_2$ solution is added to 0.01M $AgNO_3$, ignoring the effect of dilution the minimum concentration of $CaCl_2$ above which precipitation of AgCl takes place is
(a) 1.7×10^{-8} M (b) 1.7×10^{-9} M (c) 8.5×10^{-9} M (d) 8.5×10^{-7} M

81. Consider the system $2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)}$. The forward reaction is exothermic. Product yield may be increased if

(a) Temperature is increased while pressure is kept constant
(b) Temperature is reduced and pressure is increased
(c) A catalyst is added to the system in equilibrium
(d) By all the above means

82. The vapour pressure of water at 25°C = 23.8 torr. K_p for the equilibrium: $CuSO_4.5H_2O_{(s)} \rightleftharpoons CuSO_4.3H_2O_{(s)} + 2H_2O_{(g)}$ is 1.085×10^{-4} atm^2. Calculate the relative humidity of the atmosphere below which $CuSO_4.5H_2O$ will effloresce.

(a) 33% (b) 50% (c) 42% (d) 63%

83. Consider the equilibrium: $2A_{(g)} \rightleftharpoons A_{2(g)}$ $K_p = 3.72$ atm^{-1} at some temperature: TK. The total equilibrium pressure = 1.5 atm. Calculate the partial pressure of the dimer.

(a) 0.755 atm (b) 0.815 atm (c) 0.985 atm (d) 0.715 atm

84. Consider the equilibrium system: $A + nB \rightleftharpoons mC$. The equilibrium concentrations are 0.32 mol litre^{-1} for [A], [B] = 0.40 mol litre^{-1}, [C] = 0.35 mol litre^{-1}. K_C at 25°C = 2.4 (in magnitude). Suggest the magnitudes of m and n which are small whole numbers.

(a) m = n = 2 (b) m = 2, n = 1 (c) m = 1, n = 2 (d) m = n = 1

85. $H_{2(g)}$ and $I_{2(g)}$ are taken in the molar ratio 2 : 1 in a closed vessel and the equilibrium: $H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$ is established at a certain temperature. If then, 20% of I_2 has been consumed in the forward reaction, calculate the value of K_p

(a) 0.22 (b) 0.11 (c) 0.33 (d) 0.44

86. For the equilibrium system, $N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$ $p_{N_2O_4} = 0.28$ atm, $p_{NO_2} = 1.1$ atm. If the volume is doubled what will be the equilibrium pressures of NO_2, N_2O_4

(a) $p_{NO_2} = 0.642$ atm, $p_{N_2O_4} = 0.095$ atm (b) $p_{NO_2} = 0.462$ atm, $p_{N_2O_4} = 0.115$ atm
(c) $p_{NO_2} = 0.426$ atm, $p_{N_2O_4} = 0.151$atm (d) $p_{NO_2} = 0.264$ atm, $p_{N_2O_4} = 0.276$ atm

87. $K_w = 1 \times 10^{-14}$ at 25°C. Calculate $\Delta G°$ for the ionization of water.

(a) 19981 cal mol^{-1} (b) 18287 cal mol^{-1}
(c) 18329 cal mol^{-1} (d) 19091 cal mol^{-1}

88. A mixture of $CO_{2(g)}$, $CO_{(g)}$ and $C_{(s)}$ are maintained at equilibrium at 1000°C in a one litre flask. If the mean molar mass of the gaseous mixture is 36 g mol^{-1}. What is the composition of the gas mixture.

(a) 1 : 1 (b) 1: 2 (c) 2 : 1 (d) 3 : 2

89. $ZnO_{(s)}$ is exposed to pure CO at 1300K and the equilibrium: $ZnO_{(s)} + CO_{(g)} \rightleftharpoons Zn_{(g)} + CO_{2(g)}$ is then established at 1 atm pressure. The density of the gas mixture is 0.344 g litre^{-1} at the same temperature. Calculate the partial pressure, pCO, under equilibrium. At. wt. of Zn = 65.4

(a) 0.476 atm (b) 0.467 atm (c) 0.674 atm (d) 0.798 atm

90. Calculate the vapour pressure of water at 95°C, given that the specific latent heat of vaporization is 540 cal g^{-1} at its normal boiling point.

(a) 736 mm (b) 763 mm (c) 636 mm (d) 693 mm

91. Which among the following cannot act both as a Bronsted acid and base?

(a) NH_3 (b) H_2O (c) $H_2PO_4^-$ (d) SO_4^{2-}

92. How many mL of 0.05 M H_2SO_4 must be diluted to exactly one litre so that the pH of the solution will be equal to that of 0.08 M acetic acid. $K_a = 1.8 \times 10^{-5}$ at 298K

(a) 15 mL (b) 10 mL (c) 12 mL (d) 9 mL

93. Calculate the concentration of formate ion in a solution which is 0.15 M in H –COOH and 0.2 M in HCl. $K_a = 1.8 \times 10^{-4}$ for formic acid.
(a) 1.830×10^{-4} M (b) 1.535×10^{-4} M (c) 1.350×10^{-4} M (d) 1.912×10^{-4} M

94. A solution of $Cl - CH_2 - COOH$ containing 9.45 g in 500 mL has a pH = 1.78. Calculate (i) degree of dissociation and (ii) K_a
(a) 7.5%, 1.305×10^{-3} (b) 8.3%, 1.503×10^{-3} (c) 8.9%, 1.705×10^{-3} (d) 7.3%, 1.750×10^{-3}

95. 20 mL of a solution of NaCN (0.1 M) is titrated against 0.1 M HCl added from a burette. Calculate, approximately the pH at the start of the titration and after addition of 20 mL of HCl. Assuming that no product is escaping from the solution. K_a for HCN = 6.2×10^{-10}. T = 298K
(a) 9.2, 6.25 (b) 11.1, 5.25 (c) 13.2, 7.25 (d) 12.8, 8.25

96. K_a for phenol = 1.31×10^{-10} at 25°C. Calculate degree of hydrolysis of the salt, $C_6H_5 - ONa$ in 0.05 M solution in water .$K_w = 1 \times 10^{-14}$ at 25°C.
(a) 3.19% (b) 3.91% (c) 2.19% (d) 2.23%

97. Calculate the pH at the isoelectric point of a dilute solution of α-Alanine at 25°C given that the classical acidic and basic dissociation constants are 1.35×10^{-10} and 2.21×10^{-12} respectively.

Hint:- If $K_1 = \dfrac{[NH_2 - R - COOH][H_3O^+]}{[NH_3^+ - R - COOH]}$ and $K_2 = \dfrac{[NH_2 - R - COO^-][H_3O^+]}{[NH_2 - R - COOH]}$ at the isoelectric point, $[NH_3^+ - R - COOH] = [NH_2 - R - COO^-]$
(a) 6.11 (b) 5.90 (c) 5.75 (d) 6.90

98. Calculate the pH at 25°C of 0.1 M ammonium acetate solution. K_a for acetic acid ≡ K_b for NH_4OH
(a) 6.5 (b) 7 (c) 7.5 (d) 8.0

99. For the hydrolysis of the salt BA wherein both the base B – OH and the acid HA are weak.

(a) $K_h = K_w K_a K_b$ (b) h (degree of hydrolysis) $= \dfrac{\sqrt{K_h}}{1 + \sqrt{K_h}}$

(c) $h = \left(\dfrac{K_h}{1 + K_h}\right)^{1/2}$ (d) $pH = \dfrac{1}{2} pK_w + \dfrac{1}{2}(pK_a + psK_b)$

100. K_b for $C_6H_5 - NH_2 = 4 \times 10^{-10}$. Calculate the pH of 0.04 M $C_6H_5 - \overset{+}{N}H_3Cl^-$ in water at 25°C.
(a) 3 (b) 3.5 (c) 4.0 (d) 4.5

101. Calculate pH of the buffer solution obtained by mixing 100 mL of 0.2 M NaOH and 150 mL of 0.4 M $CH_3 - COOH$. $pK_a = 4.74$. T = 298K
(a) 4.12 (b) 3.92 (c) 4.64 (d) 4.44

102. A certain weak, monoprotic organic acid is titrated against 0.1 M NaOH solution. At the stage of one-third neutralization of the acid, the pH = 4.449. What is the dissociation constant of the acid? (T = 298K)
(a) 1.778×10^{-5} (b) 1.987×10^{-5} (c) 1.987×10^{-4} (d) 1.325×10^{-5}

103. Two buffers each having $CH_3 - COOH$ and $CH_3 - COONa$ as constituents have $[CH_3 - COOH] + [CH_3 - COO^-] = 0.1$ M. Their pH values are 4 and 5, $pK_a = 4.75$. If equal volumes of the two buffers are mixed, what is the pH of the resultant mixture?
(a) 4.11 (b) 3.98 (c) 4.58 (d) 5.84

104. To a solution of $CH_3 - COOH$, solid $CH_3 - COONa$ is added. When x grams are added the pH = α, when y grams are added the pH = β. Given that (β – α) = 0.6, calculate the ratio x : y.
(a) 1 : 3 (b) 1 : 4 (c) 2 : 3 (d) 3 : 5

105. 0.1 M aqueous solution of the sodium salt NaX, (of the weak monoprotic acid HX, $K_a \simeq 1 \times 10^{-10}$) is titrated against 0.1 M HCl. At the theoretical equivalence point, what is the approximate pH? T = 298K

(a) 4.56 (b) 6.54 (c) 4.12 (d) 5.65

106. The solubility of PbI_2 (molar mass = 461.2 g mol^{-1}) is 6.8×10^{-2} g per 100 g of water at 25°C. Calculate its solubility in 0.01 M KI solution.

(a) 1.822×10^{-4} M (b) 1.822×10^{-3} M

(c) 1.282×10^{-4} M (d) 2.122×10^{-4} M

107. K_{SP} for a certain sparingly soluble salt is 1.1×10^{-11}. If the solubility is $\simeq 1.4 \times 10^{-4}$ M and the salt has the formula MX_n (X : univalent), determine n

(a) 1 (b) 3 (c) 4 (d) 2

108. Solid $SrSO_4$ and solid $BaSO_4$ are both shaken up with water until saturation equilibrium is reached. K_{SP} for $SrSO_4 = 7.6 \times 10^{-7}$. K_{SP} for $BaSO_4 = 1.5 \times 10^{-9}$. Calculate $[Ba^{2+}]$ and $[SO_4^{2-}]$.

(a) 1.92×10^{-6}, 7.268×10^{-4} (b) 1.719×10^{-6}, 8.726×10^{-4}

(c) 1.92×10^{-5}, 7.268×10^{-3} (d) 1.53×10^{-5}, 6.873×10^{-4}

109. The solubility, 's' and solubility product K_{SP} for $PbCl_2$ are related as

(a) $S = \left[\frac{4}{K_{SP}}\right]^{1/3}$ (b) $S = \left(\frac{K_{SP}}{4}\right)^{1/2}$ (c) $S = \left[\frac{K_{SP}}{4}\right]^{3}$ (d) $S = \left[\frac{K_{SP}}{4}\right]^{1/3}$

110. A very small amount of phenolphthalein is added to a decinormal solution of sodium acetate at 25°C. What fraction of the indicator would exist in the coloured (i.e., ionized) form. Indicator is regarded as a weak acid: $\underset{\text{colourless}}{HA} \rightleftharpoons H^+ + \underset{\text{coloured}}{A^-}$ $K_a = 3.16 \times 10^{-10}$. K_a for acetic acid = 1.80×10^{-5}, $K_w = 10^{-14}$

(a) 0.24 (b) 0.32 (c) 0.43 (d) 0.211

Assertion–Reason Type Questions

Directions: **Each question contains Statement-1 and Statement-2 and has the following choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False
(d) Statement-1 is False, Statement-2 is True

111. **Statement 1**

For an equilibrium $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$ K_P depends on the pressure and the temperature at equilibrium (P and T).

and

Statement 2

$K_P = K_C \times (RT)$ and at a given temperature $K_P = \frac{\alpha^2 P}{1-\alpha^2}$.

112. **Statement 1**

For an endothermic forward reaction in an equilibrium system, K_p is raised by raising the temperature.

and

Statement 2

For the equilibrium system $MCl_2.2H_2O_{(s)} \rightleftharpoons MCl_{2(s)} + 2H_2O_{(g)}$. $K_P = p_{H_2O}$.

113. Statement 1

The solubility s and solubility product K_{SP} of $Ca_3(PO_4)_2$ are related as $K_{SP} = 108\ s^5$.

and

Statement 2

The solubility of $AgCl_{(s)}$ in water is decreased by the addition of small quantities of solid KCl and is significantly increased by the addition of even small quantities of $Na_2S_2O_3.5H_2O_{(s)}$.

Linked Comprehension Type Questions

Directions: **This section contains 1 paragraph. Based upon the paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

Buffer solutions and buffer action

A buffer solution is a solution which resists changes in its pH by the addition of small amounts of a strong acid or strong base or on dilution. Buffers usually consist of a mixture of a weak acid and its salt with a strong base or a weak base and its salt with a strong acid. Salts formed by neutralization of a weak acid and a weak base also have some buffer action. For a mixture of a weak acid HA and its soldium salt: $pH = pK_a + \log\frac{[salt]}{[acid]}$. For a mixture of a weak base B and its salt BHX, $pOH = pK_b + \log\frac{[salt]}{[base]}$ $pH + pOH = 14$

114. A buffer solution consisting of a mixture of lactic acid and sodium lactate of pH = 4.30 has to be prepared. Calculate the no. of grams of sodium lactate, to be added to 2 litres of 0.1 M lactic acid to get the pH = 4.30 K_a for lactic acid = 2.5×10^{-4} T = 298K. Molar mass of lactic acid = 90 g mol^{-1}.

(a) 112.2 g (b) 122.0 g (c) 56.1 g (d) 65.1 g

115. A buffer is prepared which contains 0.02 M sodium acetate and 0.05 M acetic acid. Calculate its pH . $pK_a = 4.74$.

(a) 3.44 (b) 4.34 (c) 3.85 (d) 4.85

116. A buffer solution of pH = 9.5, contains ammonia and ammonium chloride. K_b for ammonium hydroxide is 1.78×10^{-5}. Calculate the molar ratio of the salt to the base.

(a) 0.652 (b) 0.625 (c) 0.562 (d) 0.732

Multiple Correct Objective Type

Directions: **Each question in this section has four suggested answers out of which ONE OR MORE answers will be correct.**

117. Identify the correct statement(s) from the following.

(a) For the equilibrium $(NH_4)HS_{(s)} \rightleftharpoons NH_{3(g)} + H_2S_{(g)}$ established in a closed vessel with the solid as the starting material $K_P = \left(p_{NH_3}\right)^2 = \frac{P^2}{4}$ where, P = equilibrium pressure.

(b) For the equilibria (i) $SO_{2(g)} + \frac{1}{2}O_{2(g)} \rightleftharpoons SO_{3(g)}$ and $\frac{1}{2}N_{2(g)} + \frac{3}{2}H_{2(g)} \rightleftharpoons NH_{3(g)}$, the ratio $\frac{K_P}{K_C}$ is the same

(c) For the equilibrium $N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}$ K_P increases with temperature

(d) $K_P = e^{-\Delta H^\circ/RT} \times e^{\Delta S^\circ/R}$

118. Identify the correct statement(s) from the following:

(a) When a Bronsted acid/base is weak its conjugate base/acid is strong

(b) When equal volumes of solutions of HCl of pH values 3, 4 and 5 are mixed the resulting solution has pH = 4

(c) For the salt , BA of a weak base B – OH and a weak acid HA, the degree of hydrolysis, $h = \frac{\sqrt{K_h}}{1+\sqrt{K_h}}$ where K_h = hydrolytic constant.

(d) The salt NaCl in aqueous solution is present as ions $Na^+_{(aq)}$, $Cl^+_{(aq)}$ but shows no hydrolysis since the conjugate base and conjugate acid (NaOH and HCl) are strong.

119. Identify the correct statement(s) from the following.

(a) A mixture of (i) NaCl + HCl or (ii) KNO_3 + KOH can behave like a buffer solution.

(b) The ionic product of water has a value dependent on temperature.

(c) The equilibrium : $2H_2O \rightleftharpoons H_3O^+ + OH^-$ has $\Delta G° \cong$ 19 k cal.

(d) The solubility product of $Bi_2S_3 = K_{SP} = 108\ s^5$ where s is the solubility in moles $litre^{-1}$.

Matrix-Match Type Question

Directions: **Match the elements of Column I to elements of Column II. There can be single or multiple matches.**

120.

Column I	**Column II**
(a) Equilibrium : $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$	(p) $K_C = \frac{\alpha^2 C}{(1-\alpha)}$
(b) Solution of borax in water	(q) $K_P = K_C(RT)$
(c) $NaOH_{(aq)}$ + excess of NH_4Cl	(r) pH > 7 (at 25°C)
(d) Equilibrium : $N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$	(s) Colour deepens on heating

ADDITIONAL PRACTICE EXERCISE

Subjective Questions

121. In alkaline solution S^{2-} reacts with solid sulphur to form polysulphide ions like $S_2^{2-}, S_3^{2-}, S_4^{2-}$ etc. Given that,

$S + S^{2-} \rightleftharpoons S_2^{2-}$; $K = 12$

$2S + S^{2-} \rightleftharpoons S_3^{2-}$; $K = 132$

$3S + S^{2-} \rightleftharpoons S_4^{2-}$; $K = 528$

Find the equilibrium constant for the formation of S_4^{2-} from S_3^{2-}, S_2^{2-} and S?

122. Aqueous solution of formaldehyde, HCHO, containing a little H_2SO_4 on distillation polymerises to trioxane, $C_3H_6O_3$, with a theoretical value of 5.4×10^7 for the equilibrium constant.

$3HCHO \rightleftharpoons C_3H_6O_3$

If a 2.0 M solution of trioxane has to reach dissociation equilibrium with respect to the above polymerization, what would be the concentration of HCHO in solution?

123. At 27°C, 5.0 moles of N_2O_4 kept in a container is found to have a degree of dissociation of 0.4. If the total equilibrium pressure of N_2O_4 and NO_2 is 2.8 atm, find K_p for the dissociation.

124. A 5 L container contains 1.5 g of NO, 2.0 g of O_2 and 1.15 g of NO_2, at 300K. The equilibrium constant K_c of the reaction, $2NO_{(g)} + O_{2(g)} \rightleftharpoons 2NO_{2(g)}$ at this temperature is 2.1×10^2

(i) Calculate $\Delta G°$ of the reaction. (ii) What is ΔG of the reaction?

125. In the decomposition study of $SO_{3(g)}$ at 400K, according to the reaction, $SO_{3(g)} \rightleftharpoons SO_{2(g)} + \frac{1}{2} O_{2(g)}$; it was observed that a 20 g sample of SO_3 decomposed by 45% and the equilibrium pressure was 1 atm.

Find the total number of moles formed at equilibrium.

126. Equilibrium constants for different salt hydrates at 27°C are as given below

$A_2X.5H_2O_{(s)} \rightleftharpoons A_2X.2H_2O_{(s)} + 3H_2O_{(g)}$; $K_p = 6.4 \times 10^{-11}$ atm

$BY_3.8H_2O_{(s)} \rightleftharpoons BY_3.3H_2O_{(s)} + 5H_2O_{(g)}$; $K_p = 3.2 \times 10^{-29}$ atm

$CZ_2.6H_2O_{(s)} \rightleftharpoons CZ_{2(s)} + 6H_2O_{(g)}$; $K_p = 6.4 \times 10^{-29}$ atm

$D_2X_3.7H_2O_{(s)} \rightleftharpoons D_2X_{3(s)} + 7H_2O_{(g)}$; $K_p = 1.28 \times 10^{-33}$ atm

Which is the most effective drying agent at 27°C?

127. K_a for benzoic acid is 3×10^{-6}. Find the number of H^+ ions in 50 mL of 0.12 M benzoic acid.

128. A sample of acetic acid ($pK_a = 4.76$) prepared by the oxidation of 108 g of 85% ethanol is diluted to 400 L. What is the H^+ ion concentration of the solution?

129. A solution is prepared by mixing 50 mL 0.1 M HCl with 50 mL 2.9 M CH_3CH_2COOH and 100 mL 0.2 M CH_3CH_2COONa. Find the pH of the resulting solution.

(K_a for CH_3CH_2COOH is 1×10^{-5})

130. The solubility product of silver oxalate at 25°C is 1.21×10^{-11}.

[at.wt : Ag = 108, C = 12, O = 16]

Calculate the solubility of silver oxalate in 0.1 M ammonium oxalate solution?

Straight Objective Type Questions

Directions: **This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

131. $A + B \rightleftharpoons AB$; $K'_C = K_1$; $3A + B \rightleftharpoons A_3B$; $K''_C = K_2$
The equilibrium constant for $AB + 2A \rightleftharpoons A_3B$ is
(a) $\frac{K_2}{K_1}$ (b) $K_2 - K_1$ (c) $K_1 \times K_2$ (d) $\frac{K_1}{K_2}$

132. Gaseous Iodine monobromide is introduced into a container of volume V at 25°C and dissociates: $2IBr_{(g)} \rightleftharpoons I_{2(g)} + Br_{2(g)}$. The quantity of $IBr_{(g)}$, taken is such that; if it had not dissociated its pressure would be 1 atm. At equilibrium, however, $p_{Br_2} = p_{I_2} = 0.35$ atm. Calculate K_P
(a) 1.361 (b) 1.631 (c) 1.613 (d) 1.163

133. For the equilibrium system: $H_{2(g)} + CO_{2(g)} \rightleftharpoons H_2O_{(g)} + CO_{(g)}$ $K_P = 0.012$ at some temperature, T. If the mixture of $H_{2(g)}$ and $CO_{2(g)}$ initially taken has mole % values 33% of $H_{2(g)}$ and 67% of $CO_{2(g)}$, and then the equilibrium is established calculate the mole ratio of $H_{2(g)}$ and $CO_{2(g)}$ at equilibrium.
(a) 0.32, 0.58 (b) 0.28, 0.62 (c) 0.37, 0.67 (d) 0.35, 0.70

134. 4 g of $N_2O_{4(g)}$ initially taken in a volume 1.5 litres dissociates: $N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$. At T = 320K, the equilibrium pressure was 800 mm. Calculate K_P
(a)0.652 (b) 0.795 (c) 0.721 (d) 0.625

135. $PCl_{3(g)} + Cl_{2(g)} \rightleftharpoons PCl_{5(g)}$ $K_P = 2.93$ at 400K. The equilibrium is established with one mole of $Cl_{2(g)}$ and 2 moles of $PCl_{3(g)}$ initially. The equilibrium pressure = 1atm. Calculate the % conversion of the reactants to the product
(a) 57% (b)72% (c) 62% (d)47%

136. Bromine vapour is taken in a flask, initially, at a concentration of 0.1 mol litre^{-1} and heated/maintained at 1750K. $[Br_2]_{(g)} \rightleftharpoons 2Br_{(g)}$. α, the degree of dissociation = 3%. Calculate K_C
(a) 1.37×10^{-4} (b) 3.71×10^{-4} (c) 1.37×10^{-3} (d) 7.13×10^{-5}

137. For the hydrogenation $C_6H_{6(g)} + 3H_{2(g)} = C_6H_{12(g)}$ $\Delta H_{100°C} = -46$ k cal. Partial pressure at equilibrium (100°C) are $p_{H_2} = 1$ mm, $p_{C_6H_6} = 69$ mm, $p_{C_6H_{12}} = 690$ mm. Total = 760 mm. Calculate ΔS.
(a) -64.7 cal deg^{-1} (b) -46.7 cal deg^{-1}
(c) -74.6 cal deg^{-1} (d) -37.3 cal deg^{-1}

138. $2SO_{3(g)} \rightleftharpoons 2SO_{2(g)} + O_{2(g)}$; $K_P = 20$ atm. If mole fraction of $O_{2(g)} = \frac{1}{5}$, what is the equilibrium pressure.
(a) 40 atm (b) 50 atm (c) 80 atm (d) 100 atm

139. When 0.2 mole of carbon and 0.52 mole of oxygen are heated in a flask of one litre capacity at 77°C, the following equilibrium is established. $C_{(s)} + O_{2(g)} \rightleftharpoons CO_{2(g)}$
$K_C = 4 \times 10^{-2}$. Calculate p_{O_2} at equilibrium.
(a) 13.74 atm (b) 14.37 atm (c) 17.34 atm (d) 15.74 atm

140. In a closed container of 8.0 litre capacity, 64.0 g of $CaCO_{3(s)}$ are heated and maintained at 727°C. $K_P = 1.642$ atm at this temperature. Calculate % of $CaCO_3$ decomposed at this temperature.
(a) 35% (b) 25% (c) 20% (d) 32%

141. For the equilibrium system: $2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)}$ $\Delta G°$ (cal) = –45200 + 42.82 T. Calculate K_p at 27°C for the equilibrium : $SO_{3(g)} \rightleftharpoons SO_{2(g)} + \frac{1}{2} O_{2(g)}$ at this temperature.

(a) 1.370×10^{-11} (b) 1.731×10^{-11} (c) 1.312×10^{-12} (d) 1.637×10^{-12}

142. For the reaction system $2Ag_{(s)} + Hg_2Cl_{2(s)} \rightleftharpoons 2AgCl_{(s)} + 2Hg$, T = 298K $\Delta G° = -2.16$ k cal (for the forward reaction). Calculate the equilibrium constant.

(a) 38.4 (b) 48.3 (c) 43.8 (d) 21.9

143. At 750 °C and at a total pressure of 10^5 Pa, iodine vapours contain 20% by volume of iodine atoms. Find K_p for the reaction $I_{2(g)} \rightleftharpoons 2I_{(g)}$.

(a) 2.5×10^3 Pa (b) 3.4×10^3 Pa (c) 5.0×10^3 Pa (d) 6.2×10^3 Pa

144. Given the following data state whether the reactions are thermodynamically feasible. Reactions: (i) $2C_{(s)} + 2H_{2(g)} = C_2H_{4(g)}$ (ii) $2C_{(s)} + 3H_{2(g)} = C_2H_{6(g)}$

Standard entropies S° of $C_{(s)}$, $H_{2(g)}$, $C_2H_{4(g)}$ and $C_2H_{6(g)}$ are (in cal deg^{-1}), 1.4, 31.2, 52.2, 55.0 in that order enthalpies of combustion (ΔH) are in (k cal) : –94.2, –68.4, –333.2 and 372.9 in that order . T = 298K

(a) both are feasible (b) (i) alone is feasible
(c) (ii) alone is feasible (d) both are nonspontaneous

145. Calculate K_C for the equilibrium system: $N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$ given $\Delta G°$ of formation of $NO_{2(g)}$ and $N_2O_{4(g)}$ at 25°C are 12.27 k cal mol^{-1} and 23.44 k cal mol^{-1} respectively.

(a) 6.112×10^{-3} M (b) 4.663×10^{-4} M (c) 4.366×10^{-4} M (d) 6.463×10^{-3} M $litre^{-1}$

146. $Fe_2O_{3(s)} + H_{2(g)} \rightleftharpoons 2FeO_{(s)} + H_2O_{(g)}$, T = 298K $\Delta G°$ of formation for $Fe_2O_{3(s)}$, $FeO_{(s)}$ and $H_2O_{(g)}$ are respectively –177.0, –58.5 and –54.6 (all in k cal mol^{-1}). Calculate the equilibrium constant.

(a) $1.814 \times 10^{+4}$ (b) $1.014 \times 10^{+4}$ (c) 1.1618×10^{-4} (d) 1.841×10^{-3}

147. Calculate from the data given below the pressure of zinc vapour in equilibrium with $ZnO_{(s)}$ and $H_{2(g)}$ at 1 atm pressure and 1000°C. Assume equality of vapour pressures of $Zn_{(g)}$ and $H_2O_{(g)}$: $ZnO_{(s)} + H_{2(g)} \rightarrow Zn_{(g)} + H_2O_{(g)}$ $\Delta G°$ (in k cal) = 55.64 – 0.038277 × T

(a) 0.2559 atm (b) 0.2470 atm (c) 0.3522 atm (d) 0.3255 atm

148. When increasing pressure is applied to the equilibrium system , ice $\rightleftharpoons$ water which of the following will happen?

(a) More ice will be formed
(b) Water will evaporate
(c) More water will be formed
(d) Any of the above depending on the temperature of the system

149. For the dissociation of phosgene, $COCl_{2(g)} \rightleftharpoons CO_{(g)} + Cl_{2(g)}$, K_p at 1500K is 8×10^{-2}. Find the percentage of phosgene dissociated when the partial pressure of Cl_2 at equilibrium is 1 atm.

(a) 2.72 (b) 2.83 (c) 7.4 (d) 8.0

150. One mole of $SO_{3(g)}$ is partially dissociated in the equilibrium $SO_{3(g)} \rightleftharpoons SO_{2(g)} + \frac{1}{2} O_{2(g)}$. If the degree of dissociation i.e., fraction dissociated is 0.5 (or 50%) then the observed molar mass is

(a) 56 (b) 64 (c) 48 (d) 60

151. Consider the equilibrium system: $H_2N - COONH_{4(s)} \rightleftharpoons 2NH_{3(g)} + CO_{2(g)}$. How is K_p related to the total equilibrium pressure, P?

(a) $\frac{4}{9} P^3$ (b) $\frac{4}{27} P^3$ (c) $\frac{2}{27} P^3$ (d) $\frac{2}{9} P^3$

152. An aqueous solution is twice as acidic as pure water (T = 298K). The pH of the solution is
(a) 7.3 (b) 7 (c) 0 (d) 6.7

153. Calculate the ratio of $[CH_3 - COO^-] / [CH_3 - COOH]$ in a solution of acetic acid $[K_a = 1.85 \times 10^{-5}]$ at 25°C when a very large quantity of pure water has been added to the solution.
(a) 9.9×10^2 (b) 1.85×10^2 (c) 8.9 (d) 1.25×10^2

154. What is the pH for 10^{-7} M HCl. T = 298. $K_w = 10^{-14}$?
(a) 6.651 (b) 6.951 (c) 6.791 (d) 6.891

155. Consider the following aqueous solutions
(i) $Na_2CO_{3(aq)}$ (ii) $FeSO_{4(aq)}$ (iii) $C_6H_5 - \overset{\oplus}{N}H_3Cl^-_{(aq)}$ (iv) $KCl_{(aq)}$ (v) Borax. Which of these solutions may have pH > 7 at 25°C
(a) (ii) and (iv) (b) (ii), (iii) and (iv) (c) (ii), (iii) and (v) (d) (i) and (v)

156. 0.1 M solution of the sodium salt, NaA is hydrolysed to the extent of 3% at 25°C. (HA is a weak monoprotic acid) Calculate K_a $[K_w \simeq 1.2 \times 10^{-14}]$.
(a) $\sim 2.3 \times 10^{-10}$ (b) $\sim 2.3 \times 10^{-9}$ (c) $\sim 3.2 \times 10^{-7}$ (d) $\sim 1.3 \times 10^{-10}$

157. Given K_a for HA (weak acid) $\simeq 2 \times 10^{-5}$ and K_b for BOH $\simeq 4 \times 10^{-10}$. $(K_w = 1 \times 10^{-14})$ at 298K, calculate the degree of hydrolysis of a solution of BA
(a) 0.35 (b) 0.53 (c) 0.45 (d) 0.29

158. 50 mL of a weak monoprotic organic acid $(K_a = 1.75 \times 10^{-5})$ of molarity, 0.02 is titrated against 0.2 M NaOH. Calculate the pH after the addition of 3 mL of the alkali.
(a) 3.493 (b) 3.943 (c) 2.845 (d) 4.933

159. K_b for ammonia = 1.6×10^{-5}. Calculate the pH in a solution which is 0.5 M in ammonia and 0.4 M in NH_4Cl.
(a) 8.3 (b) 7.3 (c) 9.3 (d) 10.3

160. Equal volume of 0.5 M NaA and 0.2 M HA (weak monoprotic acid) are mixed. The measured pH = 5.14. Calculate K_a.
(a) 1.65×10^{-5} (b) 1.65×10^{-4} (c) 1.95×10^{-6} (d) 1.8×10^{-5}

161. Calculate the volume of 0.1 N, NaOH that should be added to 500 mL of 0.1N acetic acid to produce a buffer solution with $[H^+] = 2 \times 10^{-6}$ M, $K_a = 1.8 \times 10^{-5}$.
(a) 470 mL (b) 430 mL (c) ~450 mL (d) 400 mL

162. In which of the following media would $Ag_2CrO_{4(s)}$ have the least solubility?
$K_{SP} = 1.9 \times 10^{-12}$
(a) water (b) 0.1 M $AgNO_{3(aq)}$ (c) 0.1 M $NaNO_3$ (d) 0.1 M Na_2CrO_4

163. Pure water at 25°C is saturated with both $AgBr_{(s)}$ $[K_{SP} = 5 \times 10^{-13}]$ and AgSCN $[K_{SP} = 1.1 \times 10^{-12}]$. Calculate the concentration of Ag^+ ion in solution
(a) 1.265×10^{-6} (b) 1.650×10^{-5} (c) 1.652×10^{-4} (d) 1.526×10^{-5}

164. A solution of $ZnCl_2$ (0.01 M) is acidified and saturated with H_2S gas. The concentration of H_2S in the solution is 0.1 M. The first and second ionization constants of H_2S, as an acid are $\sim 10^{-7}$ and $\sim 10^{-14}$ respectively. K_{SP} of ZnS = 1.6×10^{-23}. [T = 298K]. Calculate the (critical) magnitude of pH below which $ZnS_{(s)}$ will not be precipitated.
(a) 0.6021 (b) 0.2016 (c) 0.2610 (d) 0.1620

165. Solid pure $BaSO_4$ is added to 0.1 M H_2SO_4 and shaken to reach saturation point. Calculate the weight in gram of $BaSO_4$ that would remain in solution per litre T = 298K $K_{SP} \simeq 10^{-10}$. Molar mass of $BaSO_4$ = 233.4 g $mole^{-1}$. Assume that the acid is completely ionized.
(a) 2.334×10^{-7} (b) 3.243×10^{-7} (c) 3.243×10^{-8} (d) 4.233×10^{-8}

166. An acidified solution of $MnSO_4$ (10^{-3} M) is saturated with H_2S gas. $[H_2S] = 0.1$ M. Given $H_2S \rightleftharpoons H^+ + HS^-$ (K_1), $K_1 = 1 \times 10^{-7}$; $HS^- \rightleftharpoons H^+ + S^{2-}$ (K_2), $K_2 = 1 \times 10^{-14}$. Calculate $[H^+]$ above which $MnS_{(s)}$ will not be precipitated. K_{SP} of $MnS = 8 \times 10^{-14}$

(a) 2.12×10^{-5} (b) 2.01×10^{-5} (c) 1.53×10^{-7} (d) 1.12×10^{-6}

167. The ionization constant of the indicator , $HI_n = 4 \times 10^{-9}$. Calculate the % of the acid form, HI_n at pH = 9.4.

(a) 12.3% (b) 11.35 (c) 9.1% (d) 10.2%

168. A solution contains NaOH and Na_2CO_3. 25 ml of this solution needs 25.13 ml of 0.0972 N HCl for phenolphthalein end point and 35.10 ml of the same HCl for methyl orange end point. Calculate the number of grams of NaOH per litre of the solution.

(a) 2.36 (b) 3.26 (c) 3.62 (d) 1.81

169. 10 ml of a solution of Na_2CO_3 required 26.32 ml of 1.5 N HCl for titration using methyl orange as indicator. The density of the solution is 1.25 g ml^{-1}. Calculate the weight % of Na_2CO_3.

(a) 11.8 (b) 13.8 (c) 19.8 (d) 16.7

170. A very little quality of phenolphthalein is added to a decinormal solution of CH_3 –COONa. Take $K_a = 1.80 \times 10^{-5}$. $K_w \simeq 1.0 \times 10^{-14}$ (T = 298K). Assume that phenolphthalein is a weak acid, HA. $\underset{\text{colourless}}{HA} \rightleftharpoons H^+ + \underset{\text{coloured}}{A^-}$. If 19% of the indicator is in the coloured form, calculate K_a of HA ($K_w = 1 \times 10^{-14}$).

(a) $\sim 3.15 \times 10^{-10}$ (b) $\sim 1.35 \times 10^{-10}$ (c) $\sim 1.53 \times 10^{-10}$ (d) $\sim 1.53 \times 10^{-9}$

Assertion–Reason Type Questions

Directions: **Each question contains Statement-1 and Statement-2 and has the following choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(b) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(c) Statement-1 is True, Statement-2 is False
(d) Statement-1 is False, Statement-2 is True

171. Statement 1

For the equilibrium $CuSO_4.3H_2O_{(s)} + 2H_2O_{(g)} \rightleftharpoons CuSO_4.5H_2O_{(s)}$ $K_P = \left(p_{H_2O}\right)^2$ where p_{H_2O} is the partial pressure of water vapour.

and

Statement 2

For the equilibria (i) $SO_{3(g)} \rightleftharpoons SO_{2(g)} + \frac{1}{2} O_2$ ($K_P = K_1$)

and for (ii) $2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)}$ ($K_P = K_2$) we have $K_1^2 K_2 = 1$

172. Statement 1

For an equilibrium system $\Delta G° = -RT \ln K_P$

and

Statement 2

For an equilibrium system $\left[\frac{\partial}{\partial T}\left(\frac{\Delta G°}{T}\right)\right]_P = \frac{\Delta H°}{T^2}$

173. Statement 1

For a reaction system [Reactants $\rightleftharpoons$ Products] approaching (but not yet at) equilibrium, $\Delta G = RT \ln\left(\frac{Q}{K}\right)$.

and

Statement 2

Under equilibrium, $\Delta S = \frac{\Delta H^\circ}{T}$ and $Q = K$

174. Statement 1

In (every case of) a simple liquid-solid equilibrium, a small increase of the pressure or the temperature leads to the formation of liquid.

and

Statement 2

$K_p = \exp\frac{-\Delta H^\circ}{RT} \times \exp\frac{\Delta S^\circ}{R}$

175. Statement 1

Consider the equilibrium system: Reactants $\rightleftharpoons$ products. If ΔH is positive for the forward reaction, then K_p increases with T.

and

Statement 2

Addition of a catalyst to a reaction system generally enhances the reaction rates and yields more of the products, as compared to the uncatalysed reaction.

176. Statement 1

The reaction $Cl^- + HOH \rightarrow HCl + OH^-$ does not occur in aqueous solution whereas, $CN^- + HOH \rightarrow HCN + OH^-$ does occur to a significant extent.

and

Statement 2

Hydrolytic reactions occur in such a way that the products i.e., conjugate acids/bases are on the whole weaker than the reactants bases/acids.

177. Statement 1

For the equilibrium, $N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$, α (degree of dissociation) $= \left(\frac{\sqrt{K_p}}{\sqrt{K_p} + 4P}\right)$

and

Statement 2

For the hydrolytic equilibrium of a salt BA of a weak base BOH and a weak acid HA, the pH of the solution does not change even by dilution with some quantity of water.

178. Statement 1

When solid AgCl is shaken up with KI solution AgI is formed whereas, when $AgI_{(solid)}$ is shaken up with KCl solution, nothing significant takes place.

and

Statement 2

The solubility product of AgI is about a millionth in magnitude of the solubility product of AgCl in water (at 25°C).

179. Statement 1

$Ca_3(PO_4)_2$ and Bi_2S_3 have the same relationship between solubility, S(M) and solubility product K_{SP}.

and

Statement 2

In a poly basic weak acid in aqueous solution, the first ionization constant is much larger in magnitude than the subsequent ionization constants.

180. Statement 1

When a certain volume (V ml) of Na_2CO_3 solution is titrated against a standard solution of HCl using (i) phenolphthalein as indicator and (ii) using methyl orange as indicator (for V ml of the same Na_2CO_3 and the same HCl solution), the titre value in the second case is double the value observed in the first case

and

Statement 2

Phenolphthalein shows its end point at the bicarbonate stage.

$Na_2CO_3 + HCl \rightarrow NaHCO_3 + NaCl$

Linked Comprehension Type Questions

Directions: **This section contains 3 paragraphs. Based upon the paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

Passage I

According to thermodynamics the dependence of the equilibrium constant K_p on the Kelvin temperature T is given by $\left(\frac{\partial \ln K_P}{\partial T}\right)_P = \frac{\Delta H^\circ}{RT^2}$, where, ΔH° is the standard enthalpy change in the forward reaction. This equation can be integrated between two different temperatures T_1 and T_2. If T_1 and T_2 are not too widely different then, it is reasonable to assume that ΔH° has the same value over the temperature range. We have $\ln\left(\frac{K_{P_2}}{K_{P_1}}\right) = \frac{-\Delta H^\circ}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$ or $\log\left(\frac{K_{P_2}}{K_{P_1}}\right) = \frac{\Delta H^\circ}{2.303R} \times \left(\frac{T_2 - T_1}{T_1 T_2}\right)$.

One also has the basic relationship: $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ = -RT \ln K_P$:

181. For the equilibrium system: $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$ at 673K, $K_P = 2.13 \times 10^{-4}$ atm^{-2}. Calculate ΔG°

(a) 11.31 k cal (b) 13.13 k cal (c) 31.13 k cal (d) 33.11 k cal

182. Given that $\frac{K_{P_2}}{K_{P_1}} \simeq 6$. $T_2 = 755K$, $T_1 = 673K$, calculate the value of ΔH° per mole NH_3.

(a) -13.10 k cal mol^{-1} (b) -11.03 k cal mol^{-1} (c) -31.10 k cal mol^{-1} (d) $+13.10$ k cal mol^{-1}

183. Calculate the % change of K_p per degree rise of temperature at the (mean) temperature value of 714K.

(a) -1.82% (b) -1.28% (c) -2.18% (d) $+1.28\%$

Passage II

In aqueous solution the concentration of H_3O^+ ion (also written as H^+) can vary over a wide range of values . At 25°C, pure water has a concentration of H^+ ion 10^{-7} M. The very convenient pH scale is defined for a solution as $pH = -\log_{10} a_{H^+}$ where, a_{H^+} is the activity of the (H^+) ion in the solution. For approximate calculation a_{H^+} may be assumed to be the same as concentration of (H^+) ion i.e., $[H^+]$ in moles per litre. Thus $pH = -\log[H^+]$ and for pure water $pH = -\log(10^{-7}) = 7$ at 25°C. In this case $[OH^-]$ is also 10^{-7} $\therefore$ Analogously $p_{OH} = 7$ $\therefore p_H + p_{OH} = 14 = -\log K_w$ where, K_w = ionic product of water = 10^{-14}. In an acidic medium (at 25°C) $[H^+] > 10^{-7}$ $p_H < 7$ and $p_{OH} > 7$. In a basic medium (at this temperature) $p_H > 7$, $p_{OH} < 7$.

184. Calculate the $[H^+]$ in a solution with pH = 3.5 and the $[OH^-]$ in a solution with pH = 8.5. T = 25°C.
(a) $1.63 \times 10^{-4}, 3.16 \times 10^{-5}$ (b) $3.16 \times 10^{-4}, 3.16 \times 10^{-6}$
(c) $1.63 \times 10^{-5}, 3.16 \times 10^{-7}$ (d) $3.16 \times 10^{-3}, 3.16 \times 10^{-9}$

185. A solution of H_2SO_4 has density = 1.05 g mL^{-1} and pH of 0.125 at 20°C. (Assuming complete dissociation of the acid). The weight percentage of the acid in the solution is
(a) 1.75 (b) 3.50 (c) 5.25 (d) 7.0

186. Calculate the number of H^+ and OH^- ions in 500 mL of a solution with pH = 10.
(a) $3 \times 10^{13}, 3 \times 10^{19}$ (b) $3 \times 10^{12}, 3 \times 10^{20}$ (c) $3 \times 10^{11}, 3 \times 10^{18}$ (d) $3 \times 10^{12}, 3 \times 10^{18}$

Passage III

When one dissolves NaCl in water, the solution contains ions of Na^+ and Cl^- but not the hydrolytic products NaOH and HCl. This is because hydrolysis does not yield a stronger acid from a weaker conjugate base, HCl from Cl^-. But the salt $CH_3-COO^-Na^+$ in solution does undergo hydrolysis to a small extent.

$CH_3 - COO^- + HOH \rightleftharpoons CH_3 - COOH + OH^-$ salts of weak acids with strong bases e.g., $CH_3 - COONa$ or of strong acids with weak bases e.g., NH_4Cl or of weak acids with weak bases e.g., $CH_3 - COONH_4$ undergo hydrolysis in aqueous solution to regenerate to a limited extent the weak acids and/or weak bases. One then defines a hydrolytic constant, K_h and a degree of hydrolysis, h.

187. Calculate the degree of hydrolysis and the pH of a solution of NaCN (0.1 M) $K_a = 7.2 \times 10^{-10}$ at 25°C for HCN.
(a) 2.10×10^{-2}, 10.5 (b) 1.92×10^{-2}, 10.5 (c) 1.29×10^{-2}, 9.5 (d) 1.12×10^{-2}, 11.1

188. The hydroxide ion concentration of 0.055 M solution of sodium acetate is 5.5×10^{-6} at 25°C. Calculate the degree of hydrolysis and the hydrolytic constant of the salt.
(a) $10^{-4}, 5.5 \times 10^{-10}$ (b) $10^{-6}, 5.5 \times 10^{-8}$ (c) $10^{-5}, 5.5 \times 10^{-9}$ (d) $10^{-4}, 5.5 \times 10^{-7}$

189. Calculate the degree of hydrolysis, the hydrolytic constant and the pH of 0.02 M ammonium acetate solution at 25°C. $K_a = K_b = 1.8 \times 10^{-5}$. $K_w = 10^{-14}$.
(a) $5.01 \times 10^{-3}, 3.1 \times 10^{-6}, 7$ (b) $5.53 \times 10^{-3}, 3.086 \times 10^{-5}, 7$
(c) $6.01 \times 10^{-3}, 3.21 \times 10^{-6}, 7$ (d) $6.32 \times 10^{-3}, 3.12 \times 10^{-6}, 7$

Multiple Correct Objective Type Questions

Directions: **Each question in this section has four suggested answers out of which ONE OR MORE answers will be correct.**

190. Examples of liquid (phase) $\rightleftharpoons$ gas (phase) equilibria are
(a) vapourization of liquid benzene in a closed vessel.
(b) Mixture of (CO_2 + water vapour) in equilibrium with a solution of CO_2 in water.
(c) sugar solution in equilibrium with water vapour.
(d) ice-water equilibrium at an external pressure of 1 atm and 0°C.

191. The characteristics of chemical equilibrium are
(a) Dynamic nature.
(b) approach from either side, reactants or products.
(c) Homogeneous/heterogeneous in different cases.
(d) Not affected by temperature or pressure changes.

192. $K_P = K_C$ in which of the following cases? Indicate the correct units of choices
(a) $N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}$ (b) $CO_{(g)} + H_2O_{(g)} \rightleftharpoons CO_{2(g)} + H_{2(g)}$
(c) $Fe_2O_{3(s)} + H_{2(g)} \rightarrow 2FeO_{(s)} + H_2O_{(g)}$ (d) $CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$

193. Consider (i) $N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}$; $K_C = K_1$

(ii) $2NO_{(g)} + O_{2(g)} \rightleftharpoons 2NO_{2(g)}$; $K_C = K_2$

(iii) $2NO_{2(g)} \rightleftharpoons N_2O_{4(g)}$; $K_C = K_3$

(a) For all the above equilibria , ΔH (forward step) is positive

(b) In (ii) and (iii), ΔS is negative

(c) $[N_2O_4] = (K_1K_2K_3)[N_2][O_2]^2$

(d) In (iii), colour deepens on cooling the equilibrium system

[**Note:** These equilibria are assumed to be simultaneous for the sake of this question]

194. Identify homogeneous equilibria among the following:

(a) $CH_3 - COOC_2H_5 + H_2O \rightleftharpoons CH_3 - COOH + C_2H_5OH$

(b) $2O_{3(g)} \rightleftharpoons 3O_{2(g)}$

(c) $CO_{2(g)} + C_{(s)} \rightleftharpoons 2CO_{(g)}$

(d) $CO_{(g)} + 2H_{2(g)} \rightleftharpoons CH_3 - OH_{(g)}$

195. Given K_a for acetic acid $\approx 1.8 \times 10^{-5} = K_b$ for ammonia at 25°C

(a) A solution of sodium acetate has pH = 7

(b) A solution of ammonium acetate has pH = 7

(c) Addition of sodium acetate to a solution of acetic acid raises its pH

(d) Solutions of sodium acetate, ammonium chloride and ammonium acetate, all have the same pH at the same concentration.

196. The solubility of $BaSO_4$ will be almost same in.

(a) 0.1M H_2SO_4 (b) 0.1 M $Ba(OH)_2$ (c) 0.1 M $Ba(NO_3)_2$ (d) 0.2 M HCl

197. Identify the correct statements.

(a) $\Delta G°$ for the ionization of water. $2H_2O \rightleftharpoons H_3O^+ + OH^-$ at 298K is nearly 19 k cal.

(b) calcium fluoride, silver chloride and copper oxalate all have the same relationship between their solubility values (SM) and solubility products.

(c) The change of colour red $\rightarrow$ yellow for methyl orange with change of pH occurs below pH = 7.

(d) Boric acid, H_3BO_3 can be satisfactorily titrated as a monoprotic acid against NaOH in the presence of mannitol.

Matrix-Match Type Questions

Directions: **Match the elements of Column I to elements of Column II. There can be single or multiple matches.**

198.

Column I	**Column II**
(a) $CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$ $K_p = 2$ atm	(p) $K_p = 32$
(b) $P_{4(s)} + 6Cl_{2(g)} \rightleftharpoons 4PCl_{3(l)}$ Equilibrium pressure = 0.5 atm	(q) $K_p = 64$
(c) $NH_4HS_{(s)} \rightleftharpoons NH_{3(g)} + H_2S_{(g)}$ Equilibrium pressure = 16 atms	(r) $P_{CO_2} = 2$ atm
(d) $NH_4CO_2NH_{2(s)} \rightleftharpoons 2NH_{3(g)} + CO_{2(g)}$ Equilibrium pressure = 6 atms	(s) $P_{NH_3} = 4$ atm

199.

	Column I		**Column II**
(a)	Equilibrium constant K	(p)	1.8×10^{-9}
(b)	Ratio of dissociated water to undissociated water at 298K	(q)	9.9×10^{-8}
(c)	Hydrogen ion concentration of 0.1 M ammonium acetate having pH 7.005 for 1M solution	(r)	$10^{\frac{nE_0}{0.059}}$
(d)	The ratio $\frac{[X^{n+}]}{[Y^{n+}]}$ for the redox reaction $X_{(s)} + Y^{n+}_{(aq)} \rightleftharpoons Y_{(s)} + X^{n+}_{(aq)}$ at equilibrium at 298K	(s)	$e^{\frac{-\Delta G_0}{RT}}$

200.

	Column I		**Column II**
(a)	Sparingly soluble salt $M^{\ell}A^{m}$	(p)	$\left(\frac{K_w}{K_a}\right)$ or $\left(\frac{K_w}{K_b}\right)$
(b)	$pH = \frac{1}{2}pK_w + \frac{1}{2}pK_a + \frac{1}{2}\log c$	(q)	$C_6H_5\overset{\oplus}{N}H_3\ Cl^-$
(c)	acidic buffer (used in)	(r)	$K_{SP} = \ell^{\ell} m^m s^{\ell+m}$
(d)	$K_h = \left(\frac{h^2C}{1-h}\right)$	(s)	$CH_3 - COONa_{(aq)}$

SOLUTIONS

ANSWER KEYS

Topic Grip

1. $K_P = 9.927 \times 10^4 atm^2$
$K_C = 32.52$
2. 9.256 moles
3. 33.35 k cal
4. 48.3%
5. 2.997×10^{-6} atm
6. $\alpha = 0.49$
$K_C = 8.03 \times 10^{-3}$
7. 6.1×10^{-2} M
8. 4.30×10^{-5} M
9. (i) $\alpha = 8.485 \times 10^{-3}$
$[H^+] = 2.12 \times 10^{-3}$ M
(ii) $\alpha = 7.2 \times 10^{-5}$
$[H^+] = 1.8 \times 10^{-5}$ M
10. 4.9×10^{-3}

11. (c)	**12.** (b)	**13.** (a)
14. (a)	**15.** (d)	**16.** (d)
17. (c)	**18.** (a)	**19.** (a)
20. (a)	**21.** (b)	**22.** (c)
23. (a)	**24.** (c)	**25.** (a)

26. (b)
27. (a), (b), (c)
28. (a), (b), (c)
29. (b), (c), (d)
30. (a) → (q), (s)
(b) → (p), (q), (s)
(c) → (s)
(d) → (r)

IIT Assignment Exercise

31. (c)	**32.** (b)	**33.** (b)
34. (a)	**35.** (c)	**36.** (b)
37. (c)	**38.** (a)	**39.** (b)
40. (c)	**41.** (d)	**42.** (b)
43. (d)	**44.** (a)	**45.** (c)
46. (d)	**47.** (a)	**48.** (d)
49. (c)	**50.** (b)	**51.** (d)
52. (b)	**53.** (d)	**54.** (a)

55. (a)	**56.** (a)	**57.** (d)
58. (a)	**59.** (a)	**60.** (d)
61. (d)	**62.** (a)	**63.** (b)
64. (b)	**65.** (b)	**66.** (b)
67. (d)	**68.** (b)	**69.** (c)
70. (c)	**71.** (c)	**72.** (d)
73. (b)	**74** d	**75.** (d)
76. (c)	**77.** (d)	**78.** (c)
79. (a)	**80.** (c)	**81.** (b)
82. (a)	**83.** (c)	**84.** (a)
85. (b)	**86.** (a)	**87.** (d)
88. (a)	**89.** (c)	**90.** (c)
91. (d)	**92.** (c)	**93.** (c)
94. (b)	**95.** (b)	**96.** (b)
97. (a)	**98.** (b)	**99.** (b)
100. (a)	**101.** (d)	**102.** (a)
103. (c)	**104.** (b)	**105.** (d)
106. (c)	**107.** (d)	**108.** (b)
109. (d)	**110.** (d)	**111.** (d)
112. (c)	**113.** (b)	**114.** (a)

115. (b)
116. (c)
117. (a), (c), (d)
118. (a), (c), (d)
119. (b), (c), (d)
120. (a) → (p), (q)
(b)→ (r)
(c) → (r)
(d) → (q), (s)

Additional Pracitce Exercise

121. 176
122. 3.33×10^{-3} M
123. 2.13 atm
124. (i) $\Delta G^\circ = -13.34$ kJ
(ii) $\Delta G = -5.87$ kJ
125. 0.306
126. (B)$Y_3.3H_2O$
127. 1.8×10^{19} H^+ in 50 ml
128. 2.95×10^{-4}
129. 4
130. 5.5×10^{-6} mol L^{-1}

131. (a)	**132.** (a)	**133.** (b)
134. (c)	**135.** (c)	**136.** (b)
137. (c)	**138.** (d)	**139.** (b)
140. (b)	**141.** (d)	**142.** (a)
143. (c)	**144.** (c)	**145.** (d)
146. (c)	**147.** (b)	**148.** (c)
149. (c)	**150.** (b)	**151.** (b)
152. (d)	**153.** (b)	**154.** (c)
155. (d)	**156.** (d)	**157.** (b)
158. (d)	**159.** (c)	**160.** (d)
161. (c)	**162.** (b)	**163.** (a)
164. (a)	**165.** (a)	**166.** (d)
167. (c)	**168.** (a)	**169.** (d)
170. (a)	**171.** (d)	**172.** (c)
173. (a)	**174.** (d)	**175.** (c)
176. (a)	**177.** (d)	**178.** (a)
179. (b)	**180.** (a)	**181.** (a)
182. (b)	**183.** (c)	**184.** (b)
185. (b)	**186.** (a)	**187.** (d)

188. (a)
189. (b)
190. (a), (b), (c)
191. (a), (b), (c)
192. (a), (b), (c)
193. (b), (c)
194. (a), (b), (d)
195. (b), (c)
196. (a), (b), (c)
197. (a), (c), (d)
198. (a) → (r)
(b)→ (q)
(c) → (q)
(d) → (p), (r), (s)
199. (a) → (r), (s)
(b)→ (p)
(c) → (q)
(d) → (r), (s)
200. (a) → (r)
(b)→ (s)
(c) → (s)
(d) → (p), (q), (s)

HINTS AND EXPLANATIONS

Topic Grip

1. $2NH_{3(g)} \rightleftharpoons N_{2(g)} + 3H_{2(g)}$

2×0.02 (= 0.04 mol) 0.98 mol 2.94 mol

Total = 3.96 mol

T = 673K

$$p_{NH_3} = \left[\frac{0.04}{3.96} \times 10\right] \text{ atm} = 0.1010 \text{ atm}$$

$$P_{N_2} = \left[\frac{0.98}{3.96} \times 10\right] \text{atm} = 2.4747 \text{ atm}$$

$$p_{H_2} = \left[\frac{2.94}{3.96} \times 10\right] \text{ atm} = 7.4242 \text{ atm}$$

$$\therefore \quad K_P = \frac{p_{N_2} \times p_{H_2}^{\ 3}}{p_{NH_3}^{\ 2}} = \frac{2.475 \times (7.424)^3}{(0.101)^2} \text{ atm}^2$$

$$= 9.927 \times 10^4 \text{ atm}^2$$

$$K_C = \frac{K_P}{(RT)^2} = \frac{9.927 \times 10^4}{(0.0821 \times 673)^2} = 32.52$$

2. Only an approximate estimate is needed

$$\underset{\frac{5.175}{V}}{H_{2(g)}} + \underset{\frac{0.175}{V}}{I_{2(g)}} \rightleftharpoons \underset{\frac{5.65}{V}}{2HI} \quad \text{Vol} = V \text{ litres}$$

moles/litre at eqm

$$K_C = \frac{(5.65)^2}{(5.175 \times 0.175)} \simeq 35.25$$

II expt

$$\underset{\frac{8-x}{V}}{H_{2(g)}} + \underset{\frac{5.3-x}{V}}{I_{2(g)}} \rightleftharpoons \underset{\frac{2x}{V}}{2HI_{(g)}}$$

$$K_C = \frac{4x^2}{(8-x)(5.3-x)} = 35.25$$

After simplification

$31.25\ x^2 - 468.83x + 1495 = 0$

$\simeq x^2 - 15x + 48 = 0$

Thus $x \simeq 4.628$

$\therefore$ no. of moles of HI = $2x = 9.256$

3. $\Delta G° = -RT \ln K_P = [2 \times 10^{-3} \times 2948 \times 2.303 \times \log(3.5 \times 10^{-3})]$ litre atm

= 33.35 k cal

4. Initially P = 1 atm. Vol. : V litres

T = 338K

The total no. of moles = (1 –0.6) = 0.4 mole of $N_2O_{4(g)}$ and $(2 \times 0.6) = 1.2$ mole of $NO_{2(g)}$

which is 1.6 mole

$\therefore$ PV = nRT gives $1 \times V = 1.6\ RT$ — (1)

After volume reduction, $P \times \frac{V}{2} = n \times RT$ — (ii)

From (i) and (ii)

$$P \times \frac{1}{2} = \frac{n}{1.6} \text{ or } P = \frac{2n}{1.6}$$

$$P = \frac{4\alpha^2}{1-\alpha^2} \times P = \left[\frac{4 \times 0.36}{(1-0.36)} \times 1\right] \text{ atm}$$

$$= \frac{1.44}{0.64} = \frac{9}{4}$$

After volume reduction $n = (1 - \alpha) + 2\alpha = (1 + \alpha)$

$$\frac{4\alpha^2}{1-\alpha^2} \times P = \frac{4\alpha^2}{1-\alpha^2} \times \frac{2(1+\alpha)}{1.6} = K_p = \frac{9}{4};$$

$$\left(\frac{8\alpha^2}{1-\alpha}\right) = 1.6 \times \frac{9}{4} = 3.6$$

Solving $\alpha = 0.483 = 48.3\%$

5. $\underset{1.964 \text{ mol}}{2CO_{2(g)}} \rightleftharpoons \underset{0.036 \text{ mol}}{2CO_{(g)}} + \underset{0.018 \text{ mol}}{O_{2(g)}}$

Total = 2.018 mol

Pressure = 1 atm

$$\text{Thus } K_p = \left(\frac{0.036}{2.018}\right)^2 \times \left(\frac{0.018}{2.018}\right) \times \left(\frac{2.018}{1.964}\right)^2 \text{ atm}$$

$$= 2.997 \times 10^{-6} \text{ atm}$$

6. Density of vapour d = 3.6 g L^-

Molar mass of the dissociated mixture M

$$= \frac{dRT}{P} = \frac{3.6 \times 0.0821 \times 473}{1} = 139.8$$

$$\alpha = \frac{M - M_{(eq)}}{M_{eq}} = \frac{208 - 139.8}{139.8} = 0.4876 \simeq 0.49$$

$$K_C = \frac{[PCl_3][Cl_2]}{[PCl_5]}$$

Volume = 1 litre

$$\therefore \quad K_C = \frac{(0.00844)^2}{(0.01731 - 0.00844)}$$

$$K_C = \frac{(0.00844)^2}{(0.00887)} = 8.031 \times 10^{-3}$$

7. Both acids contribute to the total $[H^+]$ in the solution say xM from acetic acid and y M from cyanoacetic acid

$\therefore \quad [H^+] = (x + y) \qquad \therefore (x + y)x = (1.8 \times 10^{-5} \times 1);$

$(x + y)y = 3.7 \times 10^{-3} \times 1$

$\therefore \quad (x + y)^2 = (1.8 \times 10^{-5}) + (3.7 \times 10^{-3})$

$= 371.8 \times 10^{-5}$

$\therefore \quad (x + y) = 6.0975 \times 10^{-2}$ M

i.e., $[H^+] \simeq 6.1 \times 10^{-2}$ M

8. The solution contains $[H_3O^+]$ ions, acetate ions and hydroxide ions.

Principle of electroneutrality gives

$[H_3O^+] = [CH_3 - COO^-] + [OH^-]$

(write $[H^+]$ for $[H_3O^+]$ for simplicity and multiply through out by $[H^+]$)

$[H^+]^2 = [H^+][CH_3 - COO^-] + [H^+][OH^-]$

$= (K_a \times C) + K_w$ (Reqd eqn.)

For $C = 10^{-4}$M, $[H^+]^2$

$= (1.85 \times 10^{-5} \times 10^{-4}) + 10^{-14}$

$[H^+]^2 \simeq 1.85 \times 10^{-9}$

$\therefore \quad [H^+] \simeq 4.30 \times 10^{-5}$M

9. $K_a = \frac{\alpha^2 C}{1 - \alpha} = 1.8 \times 10^{-5}$

Put C = 0.25

$\therefore \quad \frac{\alpha^2 \times 0.25}{1 - \alpha} = 1.8 \times 10^{-5}$

$\frac{\alpha^2}{1 - \alpha} = 7.2 \times 10^{-5} \simeq \alpha^2$

$\therefore \quad \alpha \simeq 8.485 \times 10^{-3}$

$[H^+] \simeq \sqrt{K_a \times C} = \sqrt{1.8 \times 10^{-5} \times 0.25}$

$= 2.12 \times 10^{-3}$ M

In the second case, we have buffer $pH = pK_a$

$-\log[H^+] = -\log(1.8 \times 10^{-5})$

$\therefore \quad [H^+] = 1.8 \times 10^{-5}$M

If we take it as αC

Then $\alpha = \frac{1.8 \times 10^{-5}}{0.25} = 7.2 \times 10^{-5}$

10. $S = 8.1 \times 10^{-3}$ M

$\therefore \quad K_{sp} = 6.561 \times 10^{-5}$ (constant)

After adding $AgNO_3$, $[Ag^+] = 8.5 \times 10^{-3} + x$

and $[BrO_3^-] = x$

$\therefore \quad [8.5 \times 10^{-3} + x]\,[x] = 6.561 \times 10^{-5}$

An approximate calculation of x gives

$x = \frac{6.561 \times 10^{-5}}{8.5 \times 10^{-3}}$ i.e., $x = 7.719 \times 10^{-3}$ M

This is of the same order as C_{AgNO_3}

$\therefore \quad$ One may solve the quadratic

$x^2 + 8.5 \times 10^{-3}x - 6.561 \times 10^{-5} = 0$

$\therefore \quad x = \frac{-8.5 \times 10^{-3} \pm \sqrt{7.225 \times 10^{-5} + 26.244 \times 10^{-5}}}{2}$

$= \frac{-8.5 \times 10^{-3} + 1.829 \times 10^{-2}}{2}$

$x = 0.4897 \times 10^{-2} \simeq 4.9 \times 10^{-3}$

11. $P_C = \frac{P}{n} \quad \therefore P_B = \frac{2P}{n} \quad P_A = P - \frac{3P}{n}$

$$K_P = \frac{P_B^2 \times P_C}{P_A^2} = \frac{4p^2}{n^2} \times \frac{P}{n} \times \frac{1}{P^2\left(1 - \frac{3}{n}\right)^2}$$

$$= \frac{4P}{n^3} \times \frac{n^2}{(n-3)^2}$$

$$\frac{K_P}{P} = \frac{4}{n(n-3)^2} = \frac{1}{81}$$

$\therefore \quad n(n-3)^2 = 324$

Clearly n = 9

12. $\frac{p_{PCl_3} \times p_{Cl_2}}{p_{PCl_5}} = K_P$ = constant at a given temperature

In the given problem, p_{Cl_2} is constant

$\therefore \quad \frac{p_{PCl_3}}{p_{PCl_5}}$ = constant.

13. $K_a = 1.85 \times 10^{-5} = \frac{[H^+][CH_3 - COO^-]}{[CH_3 - COOH]}$

In a large excess of water $[H^+] = 10^{-7}$ at 25°C

$\therefore \quad \frac{[CH_3 - COO^-]}{[CH_3 - COOH]} = \frac{1.85 \times 10^{-5}}{10^{-7}} = 185$

$$\therefore \quad \frac{[CH_3-COO^-]}{[CH_3-COOH]+[CH_3-COO^-]}=\frac{185}{186}$$

Denominator on the LHS = total concentration of acetic acid.

The numerator is the ionized part.

$\therefore$ the nonionised fraction = $\frac{1}{186}$ or $\frac{100}{186}$ %

$\simeq$ 0.54%

14. Initially, $1.85\times10^{-5}=\frac{\alpha^2C}{1-\alpha}\simeq\alpha^2C=0.1\times\alpha^2$

$\therefore \quad \alpha^2=1.85\times10^{-4}$; $\alpha=1.36\times10^{-2}$

$\therefore \quad \bar{\alpha}C=[H^+]=1.36\times10^{-3}$

$pH=-\log[1.36\times10^{-3}]=2.866$

After dilution, $1.85\times10^{-5}=\alpha^2\times10^{-2}$

$\therefore \quad \alpha^2=1.85\times10^{-3}$

$\therefore \quad \alpha=4.301\times10^{-2}$

$\alpha C=[H^+]=4.301\times10^{-4}$

$pH=-\log[4.301\times10^{-4}]=3.366$

Change in pH = 0.5

15. At the commencement of precipitation, the solution is saturated with both AgCl and Ag_2CrO_4.

$\therefore \quad [Ag^+]\,[Cl^-]=1.7\times10^{-10}$;

$[Ag^+]^2\,[CrO_4^{--}]=2\times10^{-12}$

Eliminating $[Ag^+]$ from both equations

$$\frac{[Cl^-]^2}{[CrO_4^{--}]}=\frac{\left[1.7\times10^{-10}\right]^2}{2\times10^{-12}}=\frac{2.89\times10^{-20}}{2\times10^{-12}}$$

$$=1.445\times10^{-8}$$

$\therefore \quad [Cl^-]^2=1.445\times10^{-8}$

$[CrO_4^{--}]=1.445\times10^{-12}$

$\therefore \quad [Cl^-]=1.202\times10^{-6}$ M

[Note:-Actually the addition of $AgNO_3$ solution reults in dilution and a lowering of $[CrO_4^{--}]$. But since $AgNO_3$ solution is 10 times as strong as KCl (and further since only an approximate value is required the dilution factor can be ignored.]

16. In statement –1, $K_P=\left(p_{H_2O}\right)^2$.

Hence statement –1 is wrong

17. The equilibrium system :

$H_{2(g)}+I_{2(g)} \rightleftharpoons 2HI_{(g)}$ is independent of pressure.

20. The two statements support each other.
No separate explanation needed.

21. $K_C=\dfrac{\left(\frac{2}{3}\times\frac{2}{3}\right)}{\left(\frac{1}{3}\times\frac{1}{3}\right)}=4$ (volume cancels out)

In (a), $K_C=\dfrac{(x\times x)}{(1-x)(2-x)}=4=\dfrac{x^2}{x^2-3x+2}$

$\therefore \quad 4x^2-12x+8=x^2$

$\therefore \quad 3x^2-12x+8=0$

$$x=\frac{12\pm\sqrt{144-96}}{6}=\frac{12\pm\sqrt{48}}{6}\text{ mol}$$

$$=0.845\text{ mol}$$

In (b)

$$\frac{(1-x)(3-x)}{x^2}=4$$

$4x^2=x^2-4x+3$

$\therefore \quad 3x^2+4x-3=0$

$$x=\frac{-4\pm\sqrt{16+36}}{6}=\frac{-4\pm\sqrt{52}}{6}=0.535\text{ mol}$$

$\therefore \quad 1-x=0.465$ mol

22. $\underset{(1-\alpha)}{N_2O_{4(g)}} \rightleftharpoons \underset{2\alpha}{2NO_{2(g)}}$ (at eqm)

$$p_{N_2O_4}=\left(\frac{1-\alpha}{1+\alpha}\right)\times P;$$

$$p_{NO_2}=\left(\frac{2\alpha}{1+\alpha}\right)P$$

$$\therefore \quad K_P=\left[\frac{p_{NO_2}^2}{p_{N_2O_4}}\right]$$

$$=\frac{4\alpha^2}{(1+\alpha)^2}\times P^2\times\left(\frac{1+\alpha}{1-\alpha}\right)\times\frac{1}{P}=\frac{4\alpha^2P}{1-\alpha^2}$$

$$\therefore \quad \frac{K_P}{4P}=\left(\frac{\alpha^2}{1-\alpha^2}\right)$$

$$\therefore \quad \left(\frac{K_P}{K_P+4P}\right)=\alpha^2$$

$$\therefore \quad \alpha=\left[\frac{K_P}{4P+K_P}\right]^{1/2}$$

23. $(NH_4)HS_{(s)} \rightleftharpoons NH_{3(g)} + H_2S_{(g)}$

Total vapour pressure = 501 mm

$p_{NH_3} = \frac{501}{2} = p_{H_2S}$ mm

$\therefore \quad K_P = \left[\frac{250.5}{760} \times \frac{250.5}{760}\right]$ atm^2.

If the vessel already contains $NH_{3(g)}$ at 320 mm (pr) then $p_{H_2S} = p$ mm

$p_{NH_3} = (320 + p)$ mm

$\therefore \quad \frac{p}{760} \times \frac{320+p}{760} = \left(\frac{250.5}{760}\right)^2$

$\therefore \quad p^2 + 320p = 62750$

$p^2 + 320\,p - 62750 = 0$

$$p = \frac{-320 \pm \sqrt{102400 + 251000}}{2}$$

$$= \frac{-320 + 594.5}{2} = 137 \text{ mm}$$

$\therefore$ Total pressure = p + (320 + p) = 595 mm

24. $K_P = K_C \times (RT)^{\Delta n} = 6.91 \times 10^4 \times (RT)^{-4}$

T = 1000K

Sub R = 0.0821 litre atm deg^{-1} mol^{-1}

$K_P = 6.91 \times 10^4 \times (0.0821 \times 1000)^{-4}$

$$= \frac{6.91 \times 10^4}{(82.1)^4} = 1.521 \times 10^{-3}$$

25. $\Delta G° = -RT \ln K_P$

$= -8.314 \times 298 \times 2.303 \times \log(7.08 \times 10^{24})]$

$\Delta G° = 141790$ J $\simeq$ 141.8 kJ

26. $\Delta G° = \Delta H° - T\Delta S = \left[-116 - \frac{298 \times (-109)}{1000}\right]$ kJ

$= -83.52$ kJ

$$\Delta G° = -RT \ln K_P = \frac{-8.314 \times 298}{1000} \ln K_P$$

$= -2.4775 \ln K_P$

$\therefore \quad \ln K_P = \frac{-83.52}{-2.4775} = 33.71$

$\therefore \quad K_P = 4.36 \times 10^{14}$

27. Calculation shows that (a), (b), (c) are correct

28. Calculation shows that (a), (b), (c) are correct

29. (b), (c), (d) are correct

IIT Assignment Exercise

33. $K_P = \frac{4\alpha^2}{1-\alpha^2} \times P = \frac{4 \times (0.6)^2 \times 1}{1-(0.6)^2}$

$$= \frac{4 \times 0.36}{0.64} \text{ atm} = 2.25 \text{ atm}$$

At 5 atm pressure: $2.25 = \left(\frac{4\alpha^2}{1-\alpha^2}\right) \times 5$

$\therefore \quad 0.1125 = \frac{\alpha^2}{1-\alpha^2}$

$\therefore \quad \alpha = 0.318$

i.e., 31.8%

34. $\frac{K_P}{K_C} = (RT)^{\Delta n} \frac{9.92 \times 10^4}{32.55} = (0.0821 \times 673)^{\Delta n}$

$(3.0476 \times 10^3) = (55.253)^{\Delta n}$

By calculation [and even by rough inspection i.e., $55^2 \approx 3025$] $\Delta n = +2$

35. 92 g of ethanol $\equiv$ 2 moles

180 g of CH_3– COOH $\equiv$ 3 moles

[ester] = x = [water]

$\therefore \quad \frac{10^{-21} \times 10^{-1}}{[H^+]^2} = 4$

$x^2 = 4(x^2 - 5x + 6)$

$\therefore \quad 3x^2 - 20x + 24 = 0$

$\therefore \quad x = \frac{[H^+]^2[S^{--}]}{[H_2S]} = 10^{-21}$

$$= \frac{20 \pm 10.58}{6}$$

Accepted value x = 1.57 moles.

Molar mass of ester = 88 g mol^{-1}

$\therefore$ mass of ester in gram $\simeq$ 138 g

36. $\underset{0.22\,\text{mol}}{4HCl_{(g)}} + \underset{0.305\,\text{mol}}{O_{2(g)}} \rightleftharpoons \underset{0.39\,\text{mol}}{2Cl_{2(g)}} + \underset{0.39\,\text{mol}}{2H_2O_{(g)}}$ under eqm

Total no. of moles = 1.305

Pressure = 1 atm

$\therefore$ Calculating K_P, we get

$$\left[\frac{\left(\frac{0.39}{1.305}\right)^4}{\left(\frac{0.22}{1.305}\right)^4\left(\frac{0.305}{1.305}\right)}\right] = K_P$$

$$\therefore \quad K_P = \frac{(0.39)^4(1.305)}{(0.22)^4(0.305)} = 42.26 \text{ atm}^{-1}$$

37. $\underset{(1-x)\text{mol}}{Se_{6(g)}} \rightleftharpoons \underset{3x \text{ mol}}{3Se_{2(g)}}$ eqm

Total (1 + 2x) mol

Using partial pressures,

$$\frac{\left(p_{Se_2}\right)^3}{\left(p_{Se_6}\right)} = K_p = \left[\frac{\left(\frac{3x}{1+2x}\right)^3 \times p^3}{\frac{(1-x)}{(1+2x)} \times p}\right]$$

where, $p = \frac{600}{760}$

$$\therefore \quad \left[\frac{27x^3}{(1-x)(1+2x)^2}\right] \times \left(\frac{600}{760}\right)^2 = 0.2$$

This is a cubic equation and can be solved approxiately

$$\left[\frac{27x^3}{(1-x)(1+2x)^2}\right] = 0.321$$

By trial and error

$x \simeq 0.27$

$\therefore \quad x = 27\%$

38. $N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}$

In air $p_{N_2} = 0.792$ atm

$p_{O_2} = 0.208$ atm

But under equilibrium

$p_{N_2} = (0.792 - p)$, $p_{O_2} = (0.208 - p)$

$p_{NO} = 2p$

$$\therefore \quad \frac{4p^2}{(0.792-p)(0.208-p)} = 3.5 \times 10^{-3}$$

Since p is likely to be small, one can attempt an aproximate solution and improve it, if necessary.

$$\therefore \quad p^2 \text{ (approx)} = 3.5 \times 10^{-3} \times \frac{0.792 \times 0.208}{4}$$

$$\simeq 0.1441 \times 10^{-3}$$

$\therefore \quad p \simeq 1.2 \times 10^{-2}$ atm. This is quite small. One may attempt the following improvement, substitute $p = 1.2 \times 10^{-2}$ atm in the denominator and recalculate the numerator

$$\therefore \quad p^2 = 3.5 \times 10^{-3} \times \frac{0.78 \times 0.196}{4} = 0.1338 \times 10^{-3},$$

$p = 1.157 \times 10^{-2}$ atm

If we repeat the procedure, the value

$= P = 1.158 \times 10^{-2}$ atm

The improvement is small . Thus no further improvement will be significant.

Accepting $p = 1.158 \times 10^{-2}$ atm

$$2p = P_{NO} = 2.316 \times 10^{-2} \text{ atm}$$

As a %, it is 2.316%.

39. Under equilibrium suppose x moles of PCl_3 are present in 5 litres.

Concentration of $PCl_3 = \frac{x}{5}$ mol litre^{-1} = $[Cl_2]$

$[PCl_5] = 0.05$ mol litre^{-1}

$$K_C = \frac{[PCl_3][Cl_2]}{[PCl_5]} = \frac{x^2}{25 \times 0.05} = \frac{x^2}{1.25}$$

$K_P = 1.781 = K_C \times RT$

$$\therefore \quad K_C = \frac{K_P}{RT} = \frac{1.781}{0.0821 \times 523} = \frac{x^2}{1.25}$$

$$\therefore \quad x^2 = \frac{1.25 \times 1.781}{0.0821 \times 523}$$

$x^2 = 0.05185$

$\therefore \quad x = 0.2277$ moles

$\therefore$ one should have taken

$(0.05 \times 5) + (0.2277)$ mole

$= 0.4777$ moles in 5 litre

$\therefore$ no. of grams $= (0.4777 \times 208.5) = 99.6$ g

40. $K_P = 1.781 \quad K_C = \frac{1.781}{0.0821 \times 523}$

Assume that x mole of Pcl_3 is obtained by dissociation of PCl_5. Then

$$\frac{\frac{x}{5} \times \frac{0.2+x}{5}}{\frac{0.2-x}{5}} = \frac{x(0.2+x)}{5 \times (0.2-x)} = K_C = \frac{1.781}{0.0821 \times 523}$$

$$= 4.1478 \times 10^{-2}$$

$\therefore \quad x^2 + 0.2x = (0.2 - x) \times 0.2074$

$= 0.04148 - 0.2074\ x$

$\therefore \quad x^2 + 0.4074\ x - 0.04148 = 0$

Solving this quadratic

$x = 0.08435$ mol

$\therefore \quad \alpha = \dfrac{0.08435}{0.2} = 0.4217$

41. $CO_{2(g)} + C_{(s)} \rightleftharpoons 2CO_{(g)}$

$K_P = \dfrac{p_{CO}^{\ 2}}{p_{CO_2}} = 122$

$p_{CO} + p_{CO_2} = 12.2$ atm

$\therefore \quad \dfrac{p_{CO}^{\ 2}}{(12.2 - p_{CO})} = 122$

$p_{CO}^{\ 2} = 122(12.2 - p_{CO})$

$\therefore \quad p_{CO}^{\ 2} + 122\ p_{CO} - 122 \times 12.2 = 0$

Solving $p_{CO} = 11.176$ atm

mole % of CO $= \dfrac{11.176}{12.2} \times 100 = 91.6$

mole % of $CO_2 = 8.4$

42. From the detail given we take it that

$\alpha = 45\% = 0.45$

If 1 mole of $SO_{3(g)}$ is taken at the start, then at equilibrium (1 –0.45) mol of $SO_{3(g)}$ i.e., 0.55 mole,

0.45 mole of $SO_{2(g)}$ and 0.225 mole of O_2 are present.

Total no. of moles = 1.225 moles

$$p_{SO_3} = \frac{0.55}{1.225} \text{ x 1 atm}$$

$$p_{SO_2} = \frac{0.45}{1.225} \text{ atm}$$

$$p_{O_2} = \frac{0.225}{1.225} \text{ atm}$$

$$K_P = \frac{p_{SO_2} \times p_{O_2}^{1/2}}{p_{SO_3}} = \frac{\dfrac{0.45}{1.225} \times \left(\dfrac{0.225}{1.225}\right)^{1/2}}{\left(\dfrac{0.55}{1.225}\right)} \text{atm}^{1/2}$$

$= 0.3506\ \text{atm}^{1/2}$

43. 2.55 g of $H_{2(g)}$ per litre $= \dfrac{2.55}{2} = 1.275$ mol litre^{-1}

44.66 g of $N_{2(g)}$ per litre $= \dfrac{44.66}{28}$

$= 1.595$ mol litre^{-1}

4.25 g of NH_3 per litre $= \dfrac{4.25}{17} = 0.25$ mol litre^{-1}

$N_2 + 3H_2 \rightleftharpoons 2NH_3$

Under equilibrium $[N_2]$

$$= \left(1.595 - \frac{0.25}{2}\right) \text{mol litre}^{-1}$$

$[N_2] = 1.47$ mol litre^{-1}

$$[H_2] = \left[1.275 - \frac{3}{2} \times 0.25\right] \text{ mol litre}^{-1}$$

$[H_2] = 0.9$ mol litre^{-1}

$$K_C = \left\{\frac{[NH_3]^2}{[N_2][H_2]^3}\right\} = \frac{(0.25)^2}{(1.47)(0.9)^3}$$

$K_C = 0.05832\ M^{-2}$

$= 5.83 \times 10^{-2}\ M^{-2}$

44. $\Delta G° = -RT \ln K_P = -8.314 \times 523 \times \ln(1.781)$

$\simeq -2.51$ kJ

45. $\Delta G°$ is negative

46. $CO_{(g)} + Cl_{2(g)} \rightleftharpoons COCl_{2(g)} \quad \Delta G° = 718$ J

$\Delta G° = -RT \ln K_P$

$718 = -8.314 \times 300 \times 2.303 \log K_P$

$$\therefore \quad \log K_P = \left[\frac{-718}{(8.314 \times 300 \times 2.303)}\right]$$

$= -0.125 = \bar{1}.875$

$K_P = 0.75\ \text{atm}^{-1} \qquad K_P = K_C(RT)^{-1}$

$\therefore \quad K_C = K_P \times RT = 0.75 \times 0.0821 \times 300 = 18.47$

47. Molar mass $= 2 \times 65 = 130$ g

Molar mass of $PCl_5 = 208.5$ g mol^{-1}

If α is the degree of dissociation, $130(1 + \alpha)$

$= 1 \times 208$

$\therefore \quad 130\ \alpha = (208.5 - 130) = 78.5$

$\therefore \quad \alpha = \dfrac{78.5}{130} = 0.6$

48. $K_C = \dfrac{\alpha^2 C}{1-\alpha}$; $C = \left(\dfrac{3.6}{208}\right)$ mol litre^{-1}

$= 0.01731$ mol litre^{-1}

$\alpha = 0.49$

$\therefore \quad K_C = \dfrac{(0.49)^2 \times (0.01731)}{(1-0.49)}$

$\approx 8.15 \times 10^{-3}$ mol litre^{-1}

49. $\alpha = \left(\dfrac{K_P}{K_P + 4P}\right)^{1/2}$

$\therefore \quad \alpha^2 = \dfrac{K_P}{K_P + 4P}$

and $\dfrac{\alpha^2}{1-\alpha^2} = \dfrac{K_P}{4P}$

The simplest assumption is to take $(1 - \alpha^2) \simeq 1$

Then $\alpha^2 = \dfrac{K_P}{4P}$

$\therefore \quad \alpha_1^{\,2} = \dfrac{K_P}{4P_1}$; $\alpha_2^{\,2} = \dfrac{K_P}{4P_2}$

Assume $\dfrac{P_2}{P_1} = 1.1$

$\therefore \quad \dfrac{\alpha_1^{\,2}}{\alpha_2^{\,2}} = \dfrac{P_2}{P_1} = 1.1$ $\qquad$ $\dfrac{\alpha_1}{\alpha_2} = 1.0488$

$\dfrac{\alpha_2}{\alpha_1} = 0.9535$

$\therefore \quad \dfrac{\alpha_2}{\alpha_1} = \dfrac{95.35}{100}$. Thus α decreases by 4.65% by a 10% increase in pressure.

50. $(NH_4)HS_{(g)} \rightleftharpoons NH_{3(g)} + H_2S_{(g)}$.

Total pressure = 360 mm

$\therefore \quad p_{NH_3} = \left(\dfrac{180}{760}\right)$ atm

$p_{H_2S} = \left(\dfrac{180}{760}\right)$ atm

$\therefore \quad K_P = 0.0561$ atm^2

51. $2HgO_{(s)} \rightleftharpoons 2Hg_{(g)} + O_{2(g)}$.

Suppose the total pressure = p atm

$p_{Hg} + p_{O_2} = p$ but $p_{Hg} = 2p_{O_2}$

$\therefore \quad p_{O_2} = \dfrac{P}{3}$, $p_{Hg} = \dfrac{2}{3}P$

$\therefore \quad p_{Hg}^{\,2} \times p_{O_2} = 1.93 \times 10^{-2} \cdot \left(\dfrac{2}{3}P\right)^2 \times \left(\dfrac{P}{3}\right) = \dfrac{4}{27}P^3$

$= 1.93 \times 10^{-2}$

$\therefore \quad P^3 = 1.92 \times 10^{-2} \times \dfrac{27}{4} = 12.96 \times 10^{-2}$.

$P = 0.5061$ atm

$P = 0.5061 \times 760 = 384.6$ mm

52. The equation is

$$2.303 \log\left(\frac{K_2}{K_1}\right) = \frac{-\Delta H}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

$$2.303 \log\left(\frac{0.816}{0.0068}\right) = \frac{\Delta H}{1.987}\left(\frac{84}{384 \times 300}\right)$$

$$4.788 = \frac{\Delta H}{1.987} \times \frac{84}{384 \times 300}$$

$$\therefore \quad \Delta H = \frac{4.788 \times 1.987 \times 384 \times 300}{84}$$

$\Delta H = 13047$ cal ≈ 13.047 k cal mol^{-1}

53. $$2.303 \log\left(\frac{K_{P_2}}{K_{P_1}}\right) = \frac{+46.1 \times 10^3}{8.314}\left(\frac{1}{755} - \frac{1}{673}\right)$$

$$= \frac{-46100}{8.314} \times \frac{82}{673 \times 755}$$

$$\therefore \quad \log\left(\frac{K_{P_2}}{K_{P_1}}\right) = \frac{-0.8948}{2.303} = -0.3885 = \bar{1}.6115$$

$= 0.4087$

54. Making use of the equation: $2.303 \log\left(\dfrac{p_2}{p_1}\right)$

$$= -\frac{\Delta H}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

Where, $p_2 = 8.945$ mm, $p_1 = 7.125$ mm

$R \simeq 1.987$ cal deg^{-1} mol^{-1}

$T_2 = 305$ and $T_1 = 302$

The routine calculation gives

$BaCl_2.2H_2O_{(s)} \rightarrow BaCl_2.H_2O_{(s)} + H_2O_{(g)}$

$\Delta H = 13.97$ k cal

Thus for the reverse reaction $\Delta H = -13.97$ k cal

55. pH = 11

$\therefore$ $[H^+] = 10^{-11}$ mol litre^{-1}

$\therefore$ no. of $[H^+]$ ions per ml $= \dfrac{10^{-11} \times 6.02 \times 10^{23}}{10^3}$

$= 6.02 \times 10^9$

56. pH = 5

$\therefore$ $[H^+] = 10^{-5}$

$[OH^-] = \left(\dfrac{10^{-14}}{10^{-5}}\right) = 10^{-9}$ mol litre^{-1}

$\therefore$ no. of OH^- ions per ml $= \dfrac{10^{-9} \times 6.02 \times 10^{23}}{10^3}$

$= 6.02 \times 10^{11}$

57. By the principle of electroneutrality ,

$[H_3O^+] = [Cl^-] + [OH^-]$

$[H_3O^+]^2 = [H_3O^+] \times 5 \times 10^{-7} + 10^{-14}$.

This is a quadratic and may be solved.

$[H_3O^+]^2 - (5 \times 10^{-7})\,[H_3O^+] - 10^{-14} = 0$

$\therefore$ $[H_3O^+] = \dfrac{\left(5\times10^{-7}\right) \pm \sqrt{25\times10^{-14} + 4\times10^{-14}}}{2}$

$= \dfrac{5\times10^{-7} \pm 5.3852\times10^{-7}}{2}$

$[H_3O^+] = 5.1926 \times 10^{-7}$

Thus pH = 6.285

58. Wt.per litre $= 1.02 \times 10^3$ g

Wt. of $H_2SO_4 = 1.02 \times 10^3 \times \dfrac{3.24}{10^2}$

Since eqt. Wt = 49,

Normality $= \dfrac{3.24\times1.02\times10}{49} = 0.674$

pH = –log(0.674) = 0.171

59. pH = 12

$\therefore$ $p_{OH} = 2$

$[OH^-] = 10^{-2}$ litre^{-1}= concentration of NaOH

$\therefore$ $40 \times 10^{-2} = 0.4$ g NaOH to be added to 1 litre.

60. pH = 6.77 = –log$[H^+]$

$\therefore$ $\log[H^+] = -6.77 = \bar{7}.23$

$\therefore$ $[H^+] = 1.698 \times 10^{-7}$

$\therefore$ $K_w = (1.698 \times 10^{-7})^2 = 2.88 \times 10^{-14}$

61. $K_a = \dfrac{\alpha^2 C}{1-\alpha} = \dfrac{\left(3\times10^{-2}\right)^2 \times 2\times10^{-2}}{1-0.03}$

$= \dfrac{18\times10^{-6}}{0.97} = 1.856 \times 10^{-5}$

In a 0.05 M solution $1.856 \times 10^{-5} \simeq \alpha^2 \times 5 \times 10^{-2}$

$\therefore$ $\alpha^2 = 3.712 \times 10^{-4}$

$\alpha \simeq 1.93 \times 10^{-2}$ i.e., 1.93%

62. In this case the second stage of ionization is so very weak that for all practical purposes only the first stage need be considered. Thus $K_a = \dfrac{\alpha^2 C}{1-\alpha}$

$\simeq \alpha^2 C$

$\therefore$ $K_a \times C = (\alpha C)^2 = [H^+]^2 = (5.8 \times 10^{-8}) \times 0.1$

$= 5.8 \times 10^{-9}$;

$\therefore$ $[H^+] = 7.616 \times 10^{-5}$M

63. $H_2S_{(0.1M)} \rightleftharpoons 2H^+_{(aq)} + S^{2-}_{(aq)}$

$K_a = 1.1 \times 10^{-22}$; HCl = 0.1 M

$\therefore$ $[H^+] = 0.1$ M

$\therefore$ $\left\{\dfrac{\left[H^+\right]^2\left[S^{2-}\right]}{\left[H_2S\right]}\right\} = \dfrac{(0.1)^2 \times \left[S^{2-}\right]}{0.1} = 1.1 \times 10^{-22}$

$\therefore$ $[S^{2-}] = 1.1 \times 10^{-21}$ $[Cd^{2+}][S^{2-}] = 4 \times 10^{-29}$

$\therefore$ $[Cd^{2+}] = \dfrac{4\times10^{-29}}{1.1\times10^{-21}} = 3.636 \times 0^{-8}$ M

64. The $[H^+]$ in the solution is contributed by the ionization of both of the weak acids.

i.e, $[H^+] = (x + y)$M

xM from acetic acid and yM from chloroacetic acid .

$[CH_3 - COO^-] = x$, $[Cl{-}CH_2 - COO^-] = y$

$\therefore$ $\dfrac{(x+y)x}{0.02} = 1.8 \times 10^{-5}$

$\therefore$ $(x + y)x = (1.8 \times 10^{-5}) \times (2 \times 10^{-2}) = (3.6 \times 10^{-7})$

Similarly, $\dfrac{(x+y)y}{0.01} = 1.4 \times 10^{-4}$

$\therefore$ $(x + y)y = (1.4 \times 10^{-4}) \times 1 \times 10^{-2}$

$= 1.4 \times 10^{-6}$

Adding, $(x + y)^2 = (3.6 \times 10^{-7}) + (1.4 \times 10^{-6})$

$= [1.76 \times 10^{-6}]$

$\therefore$ $(x + y) = [H^+] = 1.327 \times 10^{-3}$ $\therefore$ pH = 2.877

65. $\underset{\text{weak acid}}{(CH_3)_3\overset{+}{N}H} \rightleftharpoons H^+ + N(CH_3)_3$

$[H^+] = \alpha \times C = 10^{-4} \times 1.23 \times 10^{-2}$

$[H^+] = 1.23 \times 10^{-6}$

$\therefore \quad pH = -\log(1.23 \times 10^{-6}) = 5.91$

66. $pH = \frac{1}{2} pK_w + \frac{1}{2} pK_a + \frac{1}{2} \log C$

$= 7 + \frac{4.745}{2} + \frac{1}{2} \log\left(\frac{1}{10}\right)$

$pH = (7 + 2.3725 - 0.5) = 8.8725$

67. KNO_3 is a salt of the strong acid HNO_3 and the strong base KOH.

68. $pH = \frac{1}{2} pK_w + \frac{1}{2} pK_a - \frac{1}{2} pK_b$

$= 7 + \frac{4.7447}{2} - \frac{9.3188}{2}$

$pH = 7 + 2.3724 - 4.6594 = 4.713 = \bar{5}.287$

$\therefore \quad [H^+] = 1.936 \times 10^{-5}$ M

69. $K_w = K_b K_h$

$\therefore \quad K_b = \frac{K_w}{K_h} = \frac{10^{-14}}{0.786} = 1.27 \times 10^{-14}$

70. $K_h = \frac{K_w}{K_a K_b} = \frac{10^{-14}}{7.2\times10^{-10} \times 1.8\times10^{-5}}$

$= \frac{10}{7.2\times1.8} \approx 0.7716$

$\left(\frac{h}{1-h}\right)^2 = 0.7716$

$\therefore \quad \frac{h}{1-h} = 0.8784 \quad h = 0.4676$

$pH = \frac{1}{2} pK_w + \frac{1}{2} pK_a - \frac{1}{2} pK_b$

$= 7 + 4.5713 - 2.3724$

$pH = 9.1989 \approx 9.2$

71. By mixing equal volumes, the solution gets diluted. Thus we have a mixing of 0.1 M HCl and 0.3 M sodium acetate. Thus

CH_3 –COONa + HCl → CH_3 –COOH + NaCl.

The HCl gets converted into CH_3 – COOH

i.e., 0.1 M acetic acid

Concentration of CH_3 – COONa gets reduced that much

$\therefore \quad [CH_3 - COONa] = (0.3 - 0.1)$ M = 0.2 M

We have a buffer solution.

$pH = pK_a + \log\frac{[salt]}{[acid]}. \quad pK_a = 4.75$

$pH = 4.75 + \log\left(\frac{0.2}{0.1}\right) = 4.75 + 0.3010$

$= 5.05$ (nearly)

73. $Ag_2(C_2O_4)_{(s)} \rightleftharpoons 2Ag^+ + C_2O_4^{2-}$.

If S is the solubility (mol litre^{-1})

$K_{SP} = 4S^3$

$\therefore \quad 4S^3 = 1.21 \times 10^{-11}$

$S^3 = 3.025 \times 10^{-12}$

$\therefore \quad S = 1.446 \times 10^{-4}$ mol litre^{-1}

Molar mass = 304 g mol^{-1}

$\therefore \quad S = 4.39 \times 10^{-2}$ g litre^{-1}

74. $4S^3 = 2.8 \times 10^{-11}$

$\therefore \quad S = 1.913 \times 10^{-4}$ M

Molar mass equals 78 g mol^{-1}

$\therefore \quad S = [1.913 \times 10^{-4} \times 78] = 0.015$ g litre^{-1}

75. The concentration of $KBr_{(aq)}$ = 0.1 M is so large compared to the concentration of $AgNO_3$ (0.005 M) that the influence of dilution can be ignored.

$[Ag^+][Br^-] = K_{SP} = 5 \times 10^{-3}$

$[Br^-] = 0.01$ M

$\therefore \quad [Ag^+] = \frac{5\times10^{-13}}{10^{-2}} = 5 \times 10^{-11}$ M

76. In the case of AgI, $K_{SP} = 8.3 \times 10^{-17} = S^2$

$\therefore \quad S = 9.11 \times 10^{-9}$ M = $[Ag^+]$

In the case of $Ag_3(AsO_3)$

$K_{SP} = 4.5 \times 10^{-19} = 27\ S^4$

$\therefore \quad S^4 = \frac{4.5\times10^{-19}}{27} = 1.667 \times 10^{-20}$

$\therefore \quad S = 1.136 \times 10^{-5}$ M

$[Ag^+] = 3.409 \times 10^{-5}$ M

$\therefore$ ratio of the concentrations is

$\frac{9.11\times10^{-9}}{3.409\times10^{-5}} = 2.672 \times 10^{-4}$

77. $[Ag^+]$ in the solution is contributed by the ionization of both $AgCl_{(s)}$ and $AgBr_{(s),}$ x M from AgCl and y M from AgBr.

$\therefore \quad (x + y)x = 2.25 \times 10^{-10}$ where $x = [Cl^-]$

Similarly $(x + y)y = 4.9 \times 10^{-13}$ where $y = [Br]$

$\therefore \quad (x + y)^2 = (2.25 \times 10^{-10}) + (4.9 \times 10^{-13})$

$= 2.2549 \times 10^{-10}\ (M^2)$

$\therefore \quad (x + y) = 1.5016 \times 10^{-5}$ M

78. $K_{SP} = 7.68 \times 10^{-16} = 4S^3$,

$\therefore \quad S^3 = 1.92 \times 10^{-16}$

$S = 0.5769 \times 10^{-5}$ M

$\therefore \quad [OH^-] = 2 \times 0.5769 \times 10^{-5} = 1.1538 \times 10^{-5}$ M

79. $Ag_2C_2O_4$:

$K_{SP} = 1.21 \times 10^{-11} = 4S^3$.

$S^3 = 3.025 \times 10^{-12}$

$S = 1.446 \times 10^{-4}$ M

$[Ag^+]^2\ [C_2O_4^{2-}] = 1.21 \times 10^{-11}$

$[Ag^+] = 0.1$ M $= 10^{-1}$ M

$\therefore \quad 10^{-2} \times [C_2O_4^{2-}] = 1.21 \times 10^{-11}$

$$\therefore \quad [C_2O_4^{2-}] = \frac{1.21\times10^{-11}}{10^{-2}} = 1.21 \times 10^{-9}\ M$$

80 . Dilution factor may be ignored

$[Ag^+][Cl^-] = 1.7 \times 10^{-10}$.

$[Ag^+] = 0.01$ M $= 10^{-2}$ M

$$\therefore [Cl^-] = \frac{1.7 \times 10^{-10}}{10^{-2}} = 1.7 \times 10^{-8}\ M$$

Since 1 mole of $CaCl_2$ gives two Cl^- ions;

Concentration of $CaCl_2$ in the solution

$= 0.85 \times 10^{-8}$ M

i.e, 8.5×10^{-9} M

81. According to the principle of Lechatelier.

82. Vapour pressure of water = 23.8 torr (at 25°C)

$$= \left(\frac{23.8}{760}\right) \text{atm.}$$

K_p for the $[CuSO_4.\ 5H_2O - CuSO_4.3H_2O - 2H_2O]$

$= 1.085 \times 10^{-4}\ atm^2$.

$$\therefore \quad pH_2O = \left[1.084 \times 10^{-4}\right]^{\frac{1}{2}} \text{atm} = 1.0412 \times 10^{-2} \text{ atm}$$

$\therefore$ relative humidity (%)

$$= \left(\frac{23.8}{760}\right)^{-1} \times 100 \times 1.042 \times 10^{-2} = 33\%$$

83. $2A(g) \rightleftharpoons A_2(g)$.

$$pA_2 \Big/ (pA)^2 = K_p = 3.72\ atm^{-1}$$

but $pA + pA_2 = 1.5$ atm

$$\therefore \quad 3.72 = \frac{1.5 - pA}{pA^2}$$

$\therefore \quad 3.72\ pA^2 + pA - 1.5 = 0$

$$\therefore \quad pA = \frac{-1 \pm \sqrt{1 + 4 \times 1.5 \times 3.72}}{2 \times 3.72}$$

$$= \frac{-1 \pm \sqrt{1 + 22.32}}{7.44} \text{ atm} = 0.515 \text{ atm}$$

$\therefore \quad pA_2 = (1.5 - pA) = 0.985$ atm

84. $$K_c = [C]^m \Big/ [A][B]^n = \frac{(0.35)^m}{(0.32)(0.4)^n} = 2.4$$

Then $$\frac{(0.35)^m}{(0.4)^n} = 0.768$$

Since m and n are given to be small numbers, one may use the trial method. Suppose first that m = n. Then,

$$\left(\frac{0.35}{0.4}\right)^n = 0.768$$

i.e., $(0.875)^n$

$= 0.768$. By trial if we take n = 2,

$(0.875)^2 = 0.7656$. which is $\simeq 0.768$

Thus m = n = 2

85.

	$H_{2(g)}$	+	$I_{2(g)}$	$\rightleftharpoons$	$2HI_{(g)}$	
	2 mol		1 mol		0 mol	before eqm
	$(2-0.2)$ mol		$(1-0.2)$ mol		0.4 mol	

In this case $K_p = K_c$

$$[H_2] = \frac{1.8}{V}, [I_2] = \frac{0.8}{V}.\ [HI] = \frac{0.4}{V}$$

$$\therefore \quad K_c = \frac{(0.4)^2}{1.8 \times 0.8} = \frac{0.16}{1.8 \times 0.8} = 0.11$$

86. $pN_2O_4 = 0.28$ atm $p_{NO_2} = 1.1$ atm $K_p = \dfrac{pNO_2^2}{pN_2O_4}$

$K_p = \dfrac{(1.1)^2}{(0.28)} = 4.321$ atm. We also know that if α is the degree of dissociation,

$$K_p = \frac{4\alpha^2}{(1-\alpha^2)} \times P_{total}$$

$$\therefore \quad K_p \times V = \frac{4\alpha^2}{(1-\alpha^2)} \times (nRT)$$

$$= \frac{4\alpha^2}{(1-\alpha^2)} \times (1+\alpha) \times RT = \left(\frac{4\alpha^2}{1-\alpha}\right) RT$$

Now initially $K_p \times V = \left(\dfrac{4\alpha_1^2}{1-\alpha_1}\right) \times RT.$

After doubling the volume

$$K_p \times 2V = \left(\frac{4\alpha_2^2}{1-\alpha_2}\right) \times RT$$

$$\therefore \quad 2 = \frac{\alpha^2}{(1-\alpha_2^2)} \times \frac{(1-\alpha_1)}{\alpha_1^2} \qquad \text{— (1)}$$

Initially, $K_p = 4.321 = \left(\dfrac{4\alpha_1^2}{1-\alpha_1^2}\right) \times (1.1 + 0.28)$

$$= \frac{4\alpha_1^2}{1-\alpha_1^2} \times 1.38$$

Calculation gives $\alpha_1 = 0.662$.

Substituting this in eqn (1) above given the quadratic $\alpha_2^2 + 2.592\,\alpha_2 - 2.592 = 0$.

This gives $\alpha_2 = 0.771$. Since $K_p = 4.321$

$$= \left[\frac{4\times(0.771)^2}{1-(0.771)^2}\right] \times P.$$

We get the new total pressure after doubling the volume = 0.737 atm.

$$\therefore \quad pNO_2 = \left(\frac{2\alpha_2}{1+\alpha_2}\right) \times 0.737 = 0.642 \text{ atm}$$

$pN_2O_4 = 0.095$ atm

87. $2H_2O \rightleftharpoons H_3O^+ + OH^-$ or more simplified, $H_2O \rightleftharpoons H^+ + OH^-$. From this suppose we take $K = K_w = [H^+]\,[OH^-] = 10^{-14}$.

Then $\Delta G° = -RT \ln K$

$\simeq -2.303 \times 2 \times 298 \log(10^{-14})$

$= [14 \times 2.303 \times 1.987 \times 298]$ cals

$\simeq 19091$ cal mol^{-1}

88. $\underset{(1-\alpha)}{CO_{2(g)}} + C_{(s)} + C_{(s)} \rightleftharpoons \underset{2\alpha}{2CO_{(g)}}$ under eqm

Total no. of moles = $1 + \alpha$

$\therefore \quad (1-\alpha) \times 44 + (2\alpha) \times 28 = (1+\alpha) \times 36$

$44 - 44\alpha + 56\alpha = 36 + 36\alpha.$

$\therefore \quad 44 - 36 = 36\alpha + 44\alpha - 56\alpha$

i.e., $24\alpha = 8$. $\therefore \ \alpha = \dfrac{8}{24} = \dfrac{1}{3}$

$$\therefore \quad \left.\begin{matrix}\text{Composition}\\ \text{in moles}\end{matrix}\right\} : CO_2 : CO = \frac{2}{3} : \frac{2}{3} = 1:1$$

89. Let one mole of $CO_{(g)}$ be considered with $(1-\alpha)$ mole of CO(g), α mole each of $Zn_{(g)}$ and $CO_{2(g)}$. Assuming 1 atm as the equilibrium pressure and using PM = dRT,

$M = 0.344 \times 0.0821 \times 1300 = 36.715$

$(1-\alpha)\,28 + (\alpha \times 65.4) + (\alpha \times 44)$

$= (1+\alpha) \times 36.715$ calculating

$\alpha = 0.195$

$$\therefore \quad p_{CO} = \left(\frac{1-0.195}{1+0.195}\right) \times 1\text{atm} = 0.674 \text{ atm}$$

90. Molar latent heat of vaporization = (540×18) cal

$= 9720$ cal/mo

$$2.30 \log\left(\frac{p_2}{p_1}\right) = \frac{-\Delta H}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

$p_1 = 760$ mm at 373K

$$\therefore \quad 2.303 \log\left(\frac{p_2}{760}\right) = \left(\frac{9720}{1.987}\right) \times \left(\frac{1}{368} - \frac{1}{373}\right)$$

$$= \frac{-9720}{1.987} \times \frac{5}{368 \times 373}$$

$$\therefore \quad \log\left(\frac{p_2}{760}\right) = \frac{-9720 \times 5}{2.303 \times 1.987 \times 368 \times 373}$$

$$= -0.07737$$

$$\therefore \quad \frac{p_2}{760} = 0.8368$$

$\therefore \quad p_2 = 636$ mm

92. $K_a = 1.8 \times 10^{-5} = \dfrac{\alpha^2 C}{1-\alpha} = \alpha^2 C$.

$\therefore \quad 1.8 \times 10^{-5} \times c\,(\alpha c)^2 = [H^+]^2$

$\therefore \quad [H^+] = [1.8 \times 10^{-5} \times 0.08]^{\frac{1}{2}} = 1.2 \times 10^{-3}$M. 0.05M H_2SO_4 has $[H^+] = 0.1$ M, suppose Vml are required then $V \times 0.1 = 1000 \times 1.2 \times 10^{-3} = 1.2$

$\therefore \quad V = \dfrac{1.2}{0.1} = 12$ml

93. $Ka = \dfrac{\left[H^+\right]\left[H-COO^-\right]}{\left[HCOOH\right]} = \dfrac{0.2 \times \left[HCOO^-\right]}{0.15}$

$= 1.8 \times 10^{-4}$

$\therefore \quad [HCOO^-] = \dfrac{1.8 \times 10^{-4} \times 0.15}{0.2} = 1.35 \times 10^{-4}$

The calculation is based on the assumption that (i) $[H^+]$ is practically equal to that of the strong acid HCl and (ii) $[H - COO^-]$ is so small that the weak acid HCOOH is practically non-ionized.

94. pH = 1.78 $\therefore [H^+] = 1.66 \times 10^{-2}$.
Molar mass = 94.5 g/mol

$\therefore$ Concentration of the acid $C = \dfrac{2 \times 9.45}{94.5} = 0.2$M

Taking $[H^+] = \alpha \times c = \alpha \times 0.2$.

$\alpha = \dfrac{1.66 \times 10^{-2}}{0.2} = 8.3 \times 10^{-2}$

$\therefore$ degree of dissociation, $\alpha = 8.3\%$

$K_a = \dfrac{\alpha^2 c}{(1-\alpha)} = \dfrac{\left(8.3 \times 10^{-2}\right)^2 \times 0.2}{(1-0.083)}$

$= \dfrac{13.778 \times 10^{-4}}{0.917} = 15.025 \times 10^{-4}$

$K_a = 1.503 \times 10^{-3}$

95. At the start of the titration, we have 0.1 M NaCN.

$pH = \dfrac{1}{2} pK_w + \dfrac{1}{2} pK_a + \dfrac{1}{2} \log c$

$= (7 + 4.604 - 0.5) \approx 11.1.$

At the end of the titration we have 0.05 M.HCN (besides NaCl). $K_a = 6.2 \times 10^{-10} = \dfrac{\alpha^2 c}{1-\alpha} \approx \alpha^2 c$.

$\therefore \quad K_a \times c = (\alpha c)^2 = [H^+]^2$

$\therefore \quad [H^+] = 5.568 \times 10^{-6} \quad \therefore pH = 5.25$

96. $K_h = \dfrac{K_w}{K_a} = \left[10^{-14} \Big/ 1.31 \times 10^{-10}\right]$

$= 0.76 \times 10^{-4} = \dfrac{h^2 c}{1-h}$

$\therefore \quad \dfrac{h^2}{1-h} = \dfrac{0.76 \times 10^{-4}}{0.05} = 15.267 \times 10^{-4} \simeq h^2$

$\therefore \quad h = \left[15.264 \times 10^{-4}\right]^{\frac{1}{2}} = 3.91 \times 10^{-2}$ or in %, ans $= 3.91\%$

97. $\left[\overset{+}{N}H_3 - R - COOH\right] = \dfrac{\left[NH_2 - R - COOH\right]\left[H_3O^+\right]}{K_1}$;

$[NH_2 - R - COO^-] = \dfrac{K_2[NH_2 - R - COOH]}{[H_3O^+]}$

$\therefore$ when $[NH_2 - R - COO^-]$

$= \left[\overset{+}{N}H_3 - R - COOH\right]$, we have

$[H_3O^+]^2 - K_1 K_2$. It K_a and K_b are the classical acidic and basic dissociation constant, then $K_1 = \dfrac{K_w}{K_b}$ and $K_2 = K_a$.

$\therefore \quad K_1 K_2 = \dfrac{K_w K_a}{K_b}$

$\therefore \quad [H_3O^+] = \left[\dfrac{K_w K_a}{K_b}\right]^{\frac{1}{2}}$

$\therefore \quad pH = \dfrac{1}{2} pK_w + \dfrac{1}{2} pK_a - \dfrac{1}{2} pK_b$

$= 7 + \dfrac{9.8696}{2} - \dfrac{11.6556}{2}$

$\therefore \quad pH \approx 6.11$

98. $pH = \dfrac{1}{2} pK_w + \dfrac{1}{2} pK_a - \dfrac{1}{2} pK_b = \dfrac{1}{2} pK_w = 7$

99. $K_h = \left(\dfrac{h}{1-h}\right)^2 ; \dfrac{\sqrt{K_h}}{1} = \dfrac{h}{1-h}$

$\therefore \quad h = \left(\dfrac{\sqrt{K_h}}{1+\sqrt{K_h}}\right)$

100. This is the case of a salt of a weak base and a strong acid.

$$pOH = \frac{1}{2}pK_w + \frac{1}{2}pK_b + \frac{1}{2}\log C$$

$$pOH = 7 + 4.7 + (-0.7) = 11$$

$\therefore$ $pH = 14 - 11 = 3$

101. Total volume = 250 ml 100ml of 0.2M NaOH are equivalent to $\left(\frac{100}{1000}\times 0.2\right)$ eqt = 0.02 eqt. This would form 0.02 eqt of $CH_3 - COONa$ in 250 ml. Similarly 150 ml of 0.4M. CH_3COOH

$$\equiv \frac{150}{1000}\times 0.4 = 0.06 \text{ eqt.}$$

Out of which 0.02 eqt would have been converted into $CH_3 - COONa$

$\therefore$ $(0.06 - 0.02) = 0.04$ eqt of $CH_3 - COOH$ would be present in 250 ml:

$$pH = pK_a + \log\frac{[salt]}{[acid]} = 4.74 + \log\frac{0.02}{0.04}$$

$$pH = 4.74 + \log(0.5) = 4.74 - 0.3010 \approx 4.44$$

102. At one third neutralization stage, one third of the acid initially present would have formed the salt of the acid.

$$pH = pK_a + \log\left(\frac{1/3}{2/3}\right)$$

$$4.449 = pK_a + \log\left(\frac{1}{2}\right)$$

$\therefore$ $pK_a = 4.449 + \log 2 = 4.449 + 0.3010 = 4.750$

$\therefore$ $K_a\ 1.778 \times 10^{-5}$

103. Let, α and β be the concentrations of the salt in the two solutions and let one litre of each solution be taken for mixing. Then

$$\frac{\alpha}{0.1-\alpha} = \frac{1.85\times 10^{-5}}{10^{-4}} = \frac{K_a}{10^{-4}}$$

$$\text{or } \frac{\alpha}{0.1} = \frac{K_a}{K_a + 10^{-4}}$$

$$\text{similarly, } \frac{\beta}{0.1} = \frac{K_a}{K_a + 10^{-5}}$$

$$\therefore \quad \frac{\alpha+\beta}{0.1} = K_a\left(\frac{1}{K_a+10^{-4}} + \frac{1}{K_a+10^{-5}}\right)$$

$$\therefore \quad \frac{\alpha+\beta}{0.2} = \frac{K_a}{2}\left(\frac{2K_a + 10^{-4} + 10^{-5}}{\left(K_a+10^{-4}\right)\left(K_a+10^{-5}\right)}\right)$$

$$= \frac{1.85\times 10^{-5}\left(14.7\times 10^{-5}\right)}{2\times\left(2.85\times 10^{-5}\right)\times\left(11.85\times 10^{-5}\right)}$$

$$\therefore \quad \left[\frac{(\alpha+\beta)}{0.2}\right] = \frac{1.85\times 14.7}{2\times 2.85\times 11.85} = \frac{27.195}{67.545}$$

$$\therefore \quad \frac{\alpha+\beta}{0.2-(\alpha+\beta)} = \frac{27.195}{40.35}$$

$$pH = 4.75 + \log\left(\frac{27.195}{40.35}\right) = 4.75 + (-0.17)$$

$$pH = 4.58$$

104. $pH = pK_a + \log\frac{[salt]}{[acid]}$; $\alpha = pK_a + \log\frac{x}{a}$

$$\beta = pK_a + \log\frac{y}{a}$$

$\therefore$ $\beta - \alpha = 0.6 = \log\frac{y}{x}$

$\therefore$ $\frac{y}{x} \simeq 4$

$\therefore$ $x : y = 1 : 4$

105. At the theoretical equivalence point we have NaCl and HX, each at 0.05 M. Since only an approximate value of pH is required we adopt the following procedure. By the principle of electro neutrality.

$$[Na^+] + [H^+] = [Cl^-] + [X^-] + [OH^-]\ ;\ [Na^+] = [Cl^-]$$

$\therefore$ $[H^+] = [X^-] + [OH^-]$. $[H^+]^2 = [H^+]\ [X^-] + [H^+]\ [OH^-]$

$$[H^+]^2 = (1\times 10^{-10})\ (0.05) + 10^{-14}$$

$$= 5\times 10^{-12} + 10^{-14} = 5.01\times 10^{-14}$$

$\therefore$ $[H^+] = 22.38\times 10^{-7} = 2.238\times 10^{-6}$

$\therefore$ $pH = 5.65$

106. The solubility $S = 6.8\times 10^{-2}$ g /100 g of water which is $\simeq \frac{6.8\times 10^{-1}}{461.2}$ mole / litre

$\therefore$ $K_{SP} = 1.282\times 10^{-8} = [Pb^{++}]\ [I^-]^2$. If the $[I^-]$ $= 0.01 = 10^{-2}$

then 1.282×10^{-8} $= [Pb^{++}]\ (10^{-2})^2$.

$\therefore \quad [Pb^{++}] = \dfrac{1.282\times10^{-8}}{10^{-4}}M$

$[Pb^{2+}] = 1.282 \times 10^{-4}M$

107. Even by mere inspection one can see that

$K_{SP} \simeq 4s^3$

i.e., $1.1 \times 10^{-11} = 4 \times (1.4 \times 10^{-4})^3$. $\therefore$ n = 2

108. $[Ba^{++}] = xM$ (say) $[Sr^{2+}] = yM[SO_4^{--}] = (x + y)\ M$

$\therefore \quad x\,(x + y) = 1.5 \times 10^{-9}$ $y\,(x + y) = 7.6 \times 10^{-7}$

$\therefore \quad (x + y)^2 = 7.6 \times 10^{-7} + 1.5 \times 10^{-9}$

$= 761.5 \times 10^{-9}$

or $\quad (x + y) = 8.726 \times 10^{-4}\ M = [SO_4^{2-}]$

$x = \dfrac{1.5\times10^{-9}}{8.726\times10^{-4}} = 0.1719 \times 10^{-6} = [Ba^{2+}]$

109. $4S^3 = K_{SP}$

$S = \left(\dfrac{K_{SP}}{4}\right)^{\frac{1}{3}}$

110. $pH = \dfrac{1}{2}pK_w + \dfrac{1}{2}pK_a + \dfrac{1}{2}\log c$

$= 7 + 2.372 + \dfrac{1}{2}\log(0.1)$

pH = 7 + 2.372 –0.5 = 8.872.

For the indicator pH = 8.872

$= pK_a + \log \dfrac{[\text{coloured}]}{[\text{colourless}]}$

$= 9.444 + \log \dfrac{[\text{coloured}]}{[\text{colourless}]}$

$\therefore \quad \log \dfrac{[\text{coloured}]}{[\text{colourless}]} = 8.872 - 9.444 = -0.572$

$\left[0.268 = \dfrac{[\text{coloured}]}{[\text{colourless}]}\right.$

$\therefore \quad \dfrac{[\text{coloured}]}{[\text{coloured} + \text{colourless}]} = \dfrac{0.268}{1.268} = 0.211$

111. Statement-1 wrong because K_p does not depend on P_{total} rather, the partial pressure adjust themselves so that K_p is constant at a given temperature.
Statement-2 is correct for the given system

112. Statement-1 correct ΔH is positive $\dfrac{d\ln K}{dT} = \dfrac{\Delta H}{RT^2}$ (positive)

$\therefore$ plot of ln K vs T has a positive slope

Statement-2 wrong $K_P = p^2_{H_2O}$

113. Statement-1 and Statement-2 are correct. $Ag_2S_2O_3$ dissolves AgCl while KCl in small quantities exerts a common ion effect.

114. pK_a for lactic acid = –log (2.5×10^{-4}) = 3.60

$4.30 = 3.60 + \log \dfrac{[\text{salt}]}{0.1}$

$\therefore \quad \log \dfrac{[\text{salt}]}{0.1} = 0.70$

[salt] = 0.501

$\therefore$ wt. of salt = 2 × 0.501 × 112 = 112.2 g

115. $pH = 4.74 + \log \dfrac{[\text{salt}]}{[\text{acid}]}$

$= 4.74 + \log\left[\dfrac{0.02}{0.05}\right] = 4.34$

116. $p_{OH} = pK_b + \log \dfrac{[\text{salt}]}{[\text{base}]}$

$pK_b = 4.75 \quad pOH = 14 - 9.5 = 4.5$

$\therefore \quad 4.5 = 4.75 + \log \dfrac{[\text{salt}]}{[\text{base}]}$

$\log \dfrac{[\text{salt}]}{[\text{base}]} = -0.25 = \bar{1}.75$

$\therefore \quad \dfrac{[\text{salt}]}{[\text{base}]} = 0.562$

Additional Practice Exercise

121. Required equation is,

$3\,S_{(solid)} + S_2^{2-} + S_3^{2-} \rightleftharpoons 2S_4^{2-}$

Given,

$S_2^{2-} \rightleftharpoons S_{(solid)} + S^{2-}$

$K = \dfrac{1}{12}$

$S_3^{2-} \rightleftharpoons 2S_{(solid)} + S^{2-}$

$$K = \frac{1}{132}$$

$$6S_{(solid)} + 2S^{2-} \rightleftharpoons 2S_4^{2-}$$

$$K = (528)^2$$

$$3S_{(solid)} + S_2^{2-} + S_3^{2-} \rightleftharpoons 2S_4^{2-}$$

$$K = \frac{1}{12} \times \frac{1}{132} \times 528 \times 528 = 176$$

122. $3HCHO \rightleftharpoons C_3H_6O_3$

$$K = \frac{[C_3H_6O_3]}{[HCHO]^3}$$

$$[HCHO]^3 = \frac{2}{5.4 \times 10^7} = \frac{1}{27 \times 10^6}$$

$[HCHO] = 3.33 \times 10^{-3}$ M

123. No. of moles $N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$

Initial 5 0

Eqm. 5 – 2 = 3 moles $2 \times 2 = 4$ ($\because \alpha = 0.4$)

Total pressure = 2.8 atm

$$x_{NO_2} = \frac{4}{7}; \quad x_{N_2O_4} = \frac{3}{7}$$

$$p_{NO_2} = \frac{4}{7} \times 2.8 = 1.6 \text{ atm}; \quad p_{N_2O_4} = 1.2 \text{ atm}$$

$$K_p = \frac{p^2_{NO_2}}{p_{N_2O_4}} = \frac{1.6 \times 1.6}{1.2} = 2.13 \text{ atm}$$

124. (i) $\Delta G° = -RT \ln K_C$

$\Rightarrow \Delta G° = -8.314 \times 300 \times 2.303 \log 2.1 \times 10^2$

$= -13.34$ kJ

(ii) $\Delta G = \Delta G° + RT \ln Q_c = RT \ln \frac{Q_c}{K_c}$

$$Q_c = \frac{[NO_2]^2}{[NO]^2[O_2]} = \frac{(0.005)^2}{(0.01)^2(0.0125)} = 20$$

$$\Delta G = 8.314 \times 300 \times 2.303 \log \frac{20}{2.1 \times 10^2}$$

$= -5.87$ kJ.

125. No. of moles $SO_{3(g)} \rightleftharpoons SO_{2(g)} + \frac{1}{2}O_{2(g)}$

Initial 0.25 0 0

Eqbm. 0.1375 0.1125 0.056

Total no. of moles at eqbm = 0.306

126. $A_2X.5H_2O_{(s)} \rightleftharpoons A_2X.2H_2O_{(s)} + 3H_2O_{(g)}$;

$$K_p = \left[P_{H_2O(g)} \right]^3 \Rightarrow P_{H_2O(g)} = 4 \times 10^{-4}$$

$BY_3.8H_2O_{(s)} \rightleftharpoons BY_3.3H_2O_{(s)} + 5H_2O_{(g)}$

$$K_p = \left[P_{H_2O(g)} \right]^5 \Rightarrow P_{H_2O(g)} = 2 \times 10^{-6}$$

$CZ_2.6H_2O_{(s)} \rightleftharpoons CZ_{2(s)} + 6H_2O_{(g)}$

$$K_p = \left[P_{H_2O(g)} \right]^6 \Rightarrow P_{H_2O(g)} = 2 \times 10^{-5}$$

$D_2X_3.7H_2O_{(s)} \rightleftharpoons D_2X_{3(s)} + 7H_2O_{(g)}$

$$K_p = \left[P_{H_2O(g)} \right]^7 \Rightarrow P_{H_2O(g)} = 2 \times 10^{-5}$$

$BY_3.3H_2O_{(s)}$ is the best drying agent, as it has the lowest vapour pressure of water in equilibrium with it.

127. $[H^+] = \sqrt{K_a.C} = \sqrt{3 \times 10^{-6} \times 0.12}$

$= 6 \times 10^{-4}$ moles L^{-1}

$$\Rightarrow \frac{6 \times 10^{-4} \times 6 \times 10^{23} \times 50}{1000}$$

$= 1.8 \times 10^{19}$ ions in 50 mL.

128. Weight of ethanol taken = $\frac{108 \times 85}{100} = 92$

No. of moles of ethanol = $\frac{92}{46} = 2$

Reaction : $C_2H_5OH \xrightarrow{(O)} CH_3COOH$

2 moles of ethanol give 2 moles of this acid

Conc. of acid formed = $\frac{2}{400} = 0.005$ M

$[H^+] = \sqrt{K_a.C} = \sqrt{1.74 \times 10^{-5} \times 0.005}$

$= \sqrt{8.7 \times 10^{-8}} = 2.95 \times 10^{-4}$.

129 50 mL 0.1 M HCl = 5 m.eq of HCl

50 mL 2.9 M CH_3CH_2COOH = 145 m.eq

100 mL 0.2 M CH_3CH_2COONa = 20 m.eq

$$\underset{5 \text{ m.eq}}{CH_3CH_2COONa} + \underset{5 \text{ m.eq}}{HCl} \rightarrow$$

$$\underset{5 \text{ m.eq}}{CH_3CH_2COOH} + NaCl$$

Final m. eq of HCl = 0

Final m. eq of CH_3CH_2COOH = 145 + 5 = 150

Final m. eq of CH_3CH_2COONa = 20 – 5 = 15

Final M of $CH_3CH_2COOH = \frac{150}{200} = 0.75$ M

Final M of $CH_3CH_2COONa = \frac{15}{200} = 0.075$

$$pH = pK_a + \log\frac{[Salt]}{[Acid]} = 5 + \log\frac{0.075}{0.75} = 5 - 1 = 4$$

130 Let S' be the solubility of the salt in presence of $(NH_4)_2C_2O_4$.

$\therefore \quad [Ag^+] = 2S'$ and $\left[C_2O_4^{2-}\right] = (S' + 0.1)$

$K_{sp} = (2S')^2 (S' + 0.1) = 4(S')^2 (S' + 0.1)$

Assuming that $S' << 0.1$, the above equation can be written as $0.1 \times 4(S')^2$

$\Rightarrow \quad 0.4(S')^2 = 1.21 \times 10^{-11}$;

$$(S')^2 = \frac{1.21\times10^{-11}}{0.4} \Rightarrow S' = \sqrt{\frac{1.21\times10^{-11}}{0.4}}$$

$= 5.5 \times 10^{-6}$ mol L^{-1}

131. $\frac{[AB]}{[A][B]} = K_1$

$$\frac{[A_3B]}{[A]^3[B]} = K_2 \; ; K = \frac{[A_3B]}{[AB][A]^2} = \frac{K_2}{K_1}$$

132. $\underset{\substack{1\text{ atm}\\(1-0.7)\\\text{atm}}}{2IBr_{(g)}} \rightleftharpoons \underset{\substack{\text{zero}\\0.35\\\text{atm}}}{I_2(g)} + \underset{\substack{\text{zero}\\0.35\\\text{atm}}}{Br_{2(g)}}$

for no dissociation at eqm.

$$K_p = \frac{P_{I_2} \times PBr2}{\left(P_{IBr}\right)^2} = \frac{0.35\times0.35}{(0.3)^2} = 1.361$$

133. Suppose initially the pressure (total) was 1 atm. Then $pH_2 = 0.33$ atm, $pco_2 = 0.67$ atm. At equilibrium $pH_2 = 0.33 - x$

$p_{CO_2} = 0.67 - x$, $p_{H_2O} = p_{CO} = x$ atm.

Then $K_p = 0.012 = \frac{x^2}{(0.33-x)(0.67-x)}$ we may solve this quadratic, and get $x \simeq 0.0517$ Then mole ratio of $H_{2(g)} : Co_{2(g)}$

$$= \frac{0.33-x}{0.67-x} = \frac{0.2783}{0.6183} \simeq \frac{0.28}{0.62}$$

134. 4 g $\equiv \frac{4}{92}$ moles; v = 1.5 litres T = 320K using PV = nRT,

$P = \frac{4}{92} \times \frac{0.0821 \times 320}{1.5}$ for a hypothetical undissociated state. P = 0.7615 atm But in the dissociated equilibrium state the pr $= \frac{800}{760}$

= 1.0526 atm.

$$\underset{\substack{(0.7615-x)\\\text{atm}}}{N_2O_{4(g)}} \rightleftharpoons \underset{2x\text{ atm}}{2NO_{2(g)}}$$

Total pressure = (0.7615 + x) atm = 1.0526 atm

$\therefore \quad x = 0.2911 \; \therefore \; pN_2O_4 = 0.4704$ atm

$pNO_2 = 0.5822$

$$\therefore \quad K_P = \frac{P^2_{NO2}}{pN_2O_4} \text{ atm} = \frac{(0.5822)^2}{0.4704} = 0.7206 \text{ atm}$$

135. $\underset{\substack{(2-x)\\\text{mole}}}{(PCl_{3(g)}} + \underset{\substack{1-X\\\text{mole}}}{Cl_{2(g)}} \rightleftharpoons \underset{x}{PCl_{5(g)}}$

at eqm Total number of moles (3 – x)

$$\therefore \quad p\,PCl_3 = \left(\frac{2-x}{3-x}\right)\times 1 \text{ atm } pCl_2 = \left(\frac{1-x}{3-x}\right)\times 1\text{atm}$$

$$p\,PCl_5 = \left(\frac{x}{3-x}\right)\times 1\text{atm}$$

$$K_P = 2.93 = \frac{x(3-x)}{(2-x)(1-x)}$$

we solve for x $\quad x = 0.62 \quad \therefore$ % conversion – 62%

136. $\alpha = 3\% = \frac{3}{100}$ mole per mole of $Br_{2(g)}$. Thus if we consider one litre containing 0.1 mole, the dissociated fraction (in moles) = 3×10^{-3}

$Br_2 \rightleftharpoons 2Br$

(0.1 – 0.003) 0.006 mole one litre

$$K_C = \frac{[Br]^2}{[Br_2]} = \frac{\left(6\times10^{-3}\right)^2}{(0.097)}$$

$K_C = 371.13 \times 10^{-6} \simeq 3.71 \times 10^{-4}$

137. $p_{H_2} = \left(\frac{1}{760}\right)$ atm

$p_{C_6H_6} = \left(\frac{69}{760}\right)$ atm

$$p_{C_6H_{12}} = \left(\frac{690}{760}\right) \text{ atm}$$

$$K_P = \frac{p_{C_6H_{12}}}{p_{C_6H_6 \times (p_{H_2})^3}} = \frac{690}{760} \times \frac{760}{69}\left(\frac{760}{1}\right)^3$$

$$= 10 \times (760)^3 = 10 \times (7.6)^3 \times 10^6$$

$$K_P = 43.9 \times 10^9$$

$$\Delta G = -1.987 \times 373 \times 2.303 \times (9 + 1.642)$$

$$\Delta G = -18.165 \text{ k cal}$$

$$= \Delta H - T\Delta S = -46 - T\Delta S$$

$\therefore$ $T\Delta S = (-46 + 18.165) \times 1000 = -27835$ cal

$\therefore$ $\Delta S = \dfrac{-27835}{373} = -74.6$ cal deg^{-1}

138. $2SO_{3(g)} \rightleftharpoons 2SO_{2(g)} + O_{2(g)}$

$(2 - 2x)$ $\quad 2x \quad x$

Total no. of moles = $2 + x$

Mole fractions are

$$\left(\frac{2-2x}{2+x}\right)\times P,\ \left(\frac{2x}{2+x}\right)P.\left(\frac{x}{2+x}\right)\times P$$

$$K_P = \frac{P_{SO_2}^2 \times P_{O_2}}{P_{SO_3}^2} = \frac{\left(\frac{2x}{2+x}\right)^2 \times P^2 \times \left(\frac{x}{2+x}\right)P}{\left(\frac{2-2x}{2+x}\right)^2 \times P^2}$$

$$K_P = \frac{4x^2 \times x \times P}{(2-2x)^2(2+x)} = 20 \text{ atm}$$

But $\dfrac{x}{2+x} = \dfrac{1}{5}$

$\therefore$ $5x = 2 + x$

$4x = 2 \quad x = \dfrac{2}{4} = 0.5$

$\therefore$ $\dfrac{4\times(0.5)^2 \times (0.5) \times P}{1^2 \times 2.5} = 20;\ \dfrac{4 \times 0.125 \times P}{2.5} = 20$

$\therefore$ $P = \dfrac{20 \times 2.5}{0.5} = 100$ atm

139. $C_{(s)} + O_{2(g)} \rightleftharpoons CO_{2(g)}$

$(0.2 - x) \quad (0.52 - x) \quad x$

$[CO_2] = x$ per litre

$[O_2] = (0.52 - x)$ per litre

$$K_C = K_P = \frac{[CO_2]}{[O_2]} = \left(\frac{x}{0.52 - x}\right) = 4 \times 10^{-2} = 0.04$$

$\therefore$ $\dfrac{x}{0.52} = \dfrac{0.04}{1.04}$

$\therefore$ $x = \dfrac{0.04}{1.04} \times 0.52 = 0.02$

$\therefore$ $[O_2] = 0.52 - 0.02 = 0.5$ mole in one litre

Since $PV = nRT$, $p = \dfrac{n}{V} \times RT$

$= 0.5 \times 0.0821 \times (273 + 77)$

$p_{O_2} = 14.3675$ atm

140. $K_P = p_{CO_2} = 1.642$ atm . Using $PV = nRT$

$1.642 \times 8 = n \times 0.082 \times 1000$

$\therefore$ $n = \dfrac{8 \times 1.642}{82.1} = 0.16$ mole

Now 64 g of $CaCO_3$ is $\dfrac{64}{100} = 0.64$ mole out of which 0.16 mole is decomposed

$\therefore$ % decomposed $= \dfrac{0.16}{0.64} \times 100 = 25$

$\therefore$ 25%

141. $\Delta G°$ at 27°c ie 300 K

$= -45200 + (42.82 \times 300)$ cal,

$\Delta G° = -32354$ cal. $= -RT \ln K_P$,

$= -1.987 \times 300 \ln K_P$

$\therefore$ $\ln K_P = 54.276$ $\therefore$ $K_P = 3.7305 \times 10^{23}$

$K_P^{1/2} = 6.1078 \times 10^{11}$

$\therefore$ $K_P^{-1/2} = 1.637 \times 10^{-12}$

142. $\Delta G° = -RT \ln K$

$-2.16 \times 10^3 = -1.987 \times 298 \ln K$

$\therefore$ $\ln k = 3.648$, $K = 38.4$

143. $K_P = \dfrac{[p_{I(g)}]^2}{p_{I_2(g)}}$

$$p_{I(g)} = n_I.P = \frac{20}{100} \times P = \frac{20}{100} \times 10^5 = 2 \times 10^4 \text{ Pa}$$

$$p_{I_2} = n_{I_2}.P = \frac{80}{100} \times 10^5 = 8 \times 10^4 \text{ Pa}$$

$$\therefore K_P = \frac{\left(0.2 \times 10^5\right)^2}{\left(0.8 \times 10^5\right)} = 5 \times 10^3 \text{ Pa}$$

144. By the usual method of calculation of
$\Delta G = \Delta H - T\Delta S$
reaction (i) has $\Delta G = +11874$ cal
(nonspontaneous)
reaction (ii) has $\Delta G = -8363$ cal (spontaneous)

145. (i) $\frac{1}{2}N_{2(g)} + O_{2(g)} = NO_{2(g)}$ $\Delta G = 12.27$ k cal

(ii) $N_{2(g)} + 2O_{2(g)} = N_2O_{4(g)}$ $\Delta G = 23.44$ k cal

Multiply eqn.(i) by 2, then subtract eqn (ii)

(iii) $N_2O_{4(g)} = 2NO_{2(g)}$ $\Delta G = 1.1$ k cal

Calculate K_P from this ie. $\Delta G° = -RT \ln K_P$, one gets $K_P = 0.1579$ atm. Then $K_C = \frac{K_P}{RT}$.

By calculation $K_C = 6.463 \times 10^{-3}$ mol litre $^{-1}$

146. ΔG for the forward reaction is
$[2 \times (-58.5) + (-54.6) - (-177.0)]$ k cal

$\Delta G = 5.4$ k cal $= -RT \ln K$

$\therefore \quad -\ln K = \frac{5400}{2 \times 298} = \therefore K = 1.1618 \times 10^{-4}$

147. $\Delta G = 55.64 - 0.038277 \times T$ $T = 1273$K

$\Delta G = (55.64 - 0.038277 \times 1273)$

$= 6.9134 \times 10^3$ cal

$\ln K_P = \frac{6913.4}{-1.987 \times 1273} = -2.7332$

$K_P = 0.065$

$p_{H_2} = \frac{1-x}{1+x} \times 1$, $p_{Zn} = \frac{x}{1+x} \times 1 = p_{H_2O}$

$\therefore \quad 0.065 = \frac{x^2}{1-x^2}$ $\therefore \quad \frac{0.065}{1.065} = x^2$

$x = 0.2470$

148. Ice $\rightleftharpoons$ water (contraction of vol. in forward change). By Lechalelier principle, increase of pressure shifts the equilibrium forward.

$\therefore$ ice melts and more water is formed

149.

Conc.	$COCl_2$	$\rightleftharpoons$	CO	+	Cl_2
Initial	1 mol		0		0
Equilibrium	$1-\alpha$		α		α

Total no. of moles at equilibrium $= 1 + \alpha$

$P_{CO} = P_{Cl_2} = 1$atm

i.e., $\frac{\alpha}{1+\alpha}.P = 1$

$\therefore P = \frac{1+\alpha}{\alpha}$

$\therefore P_{COCl_2} = \frac{(1-\alpha)}{(1+\alpha)}\frac{(1+\alpha)}{\alpha} = \frac{(1-\alpha)}{\alpha}$

$\therefore K_p = \frac{1 \times 1}{\left(\frac{1-\alpha}{\alpha}\right)} = 8 \times 10^{-2}$

Percentage of phosgene dissociated = 7.4%

150.

$SO_{3(g)}$	$\rightleftharpoons$	$SO_{2(g)}$	+	$\frac{1}{2}O_{2(g)}$
$1-x$		x		$\frac{x}{2}$
0.5		0.5		0.25

$\therefore$ 1.25 mole $\equiv$ 80 g

$\therefore$ molar mass $= 80 \times \frac{4}{5} = 64$ g mol^{-1}

151. p total should be divided into $\frac{2P_t}{3}$ for NH_3 and $\frac{1}{3} \times P_t$

for CO_2 $K_P = p_{NH_3}{}^2 \times pCO_2$

$= \left(\frac{2}{3}P_t\right)^2 \times \left(\frac{1}{3}P_t\right) = \frac{4}{27}P_t^3$

152. $[H^+] = 2 \times 10^{-7}$ $pH = 7 - 0.3 = 6.7$

153. In the case of extremely large dilution at 25°c
pH ≃ 7

$\therefore \quad Ka = 1.85 \times 10^{-5} = \left\{\frac{[H^+][CH_3 - COO^-]}{[CH_3 - COOH]}\right\}$

$\therefore \quad \frac{[CH_3COO^-]}{[CH_3 - COOH]}$

equals $\frac{1.85 \times 10^{-5}}{10^{-7}} = 1.85 \times 10^2$

154. By the principle of electroneutrality
$[H^+] = [Cl^-] + [OH^-]$

$\therefore \quad [H^+]^2 = [H^+] \times 10^{-7} + 10^{-14}$

Solving $[H^+] = 1.618 \times 10^{-7}$ $pH = 6.791$

155. Solution (i) and (v) have pH > 7

156. $K_h = \frac{h^2C}{1-h} = \frac{(3\times10^{-2})^2\times0.1}{1-0.03} = \frac{9\times10^{-5}}{0.97}$

$= 9.278 \times 10^{-5}$ $\quad K_w = K_a \times K_h$

$\therefore \quad K_a = \frac{1.2\times10^{-14}}{9.278\times10^{-5}} \approx 1.3 \times 10^{-10}$

157. $K_h = \frac{K_w}{K_aK_b} = \frac{1\times10^{-14}}{\left(2\times10^{-5}\right)\times\left(4\times10^{-10}\right)}$

$= \frac{10}{8} = 1.25 = \left(\frac{h}{1-h}\right)^2$

$\therefore \quad \frac{h}{1-h} = 1.118$

$h = \frac{1.118}{2.118} = 0.53$

158. Since the alkali is 10 times as strong as the acid dilution factor may be ignored 3 ml of 0.2 M alkali neutralizes 30 ml of acid.

$\therefore$ Unneutralized acid = 20ml

$pH = pK_a + \log\frac{[salt]}{[acid]} = 4.757 + \log\frac{30}{20}$

$= 4.757 + 0.1761$

$\therefore \quad pH = 4.933$

159. $p_{OH} = pK_b + \log\frac{[salt]}{[base]}$

$p_{OH} = 4.7959 + \log\left(\frac{0.4}{0.5}\right) = 4.7959 + (-0.0969)$

$= 4.699 \qquad \therefore pH = 9.301 \simeq 9.3$

160. $5.14 = pK_a + \log\frac{0.5}{0.2} = pK_a + 0.3979$

$pK_a = 4.742 \qquad K_a = 1.8 \times 10^{-5}$

161. $[H^+] = 2 \times 10^{-6}$

$\therefore \quad pH = 5.7 = pK_a + \log\frac{[salt]}{[acid]}$

$5.7 = 4.745 + \log\frac{[salt]}{[acid]}$

$\therefore \quad \log\frac{[salt]}{[acid]} = 0.955$

$\therefore \quad \frac{[salt]}{[acid]} = 9.0157$, Suppose V ml of 0.1 N alkali was added $[salt] = \left(\frac{V}{500+V}\right)\times(0.1)$

$[acid] = \frac{(500-V)}{(500+V)}\times0.1$

$\therefore \quad \frac{[salt]}{[acid]} = \frac{V}{500-V}$

$= 9.0157$

$\frac{V}{500} = \frac{9.0157}{10.0157}$

$\therefore \quad V = \frac{9.0157}{10.0157} \times 500 = 450$ ml

162. In water $4s^3 = 1.9 \times 10^{-12} \therefore s \simeq 10^{-4}$ M

In 0.1M $AgNO_3$ $(0.1)^2\ [CrO_4^{2-}] : 1.9 \times 10^{-12}$

$\therefore \quad [CrO_4^{2-}] = \frac{1.9\times10^{-12}}{10^{-2}}$

$\therefore$ Solubility $= 1.9 \times 10^{-10}$. In 0.1M $NaNO_3$, with no common ion the solubility 'S' is analogous to that of water.

In 0.1M Na_2CrO_4, $[Ag^+]^2\ [CrO_4^{2-}] = 1.9 \times 10^{-12}$

$[Ag^+]^2 = \frac{1.9\times10^{-12}}{0.1}$

$[Ag^+]^2 = 1.9 \times 10^{-11} \therefore [Ag^+] = 4.36 \times 10^{-6}$

$\therefore \quad S = 2 \times 4.36 \times 10^{-6} \qquad S = 8.72 \times 10^{-6}$ M

163. Let the solubilities of AgBr (s) and Ag(SCN) be x M and YM, Then total $[Ag^+] = (x + y)$. $(x + y)\,x = 5 \times 10^{-13}$. $(x + y)\,y = 1.1 \times 10^{-12}$

$\therefore \quad (x + y)^2 = (5 \times 10^{-13}) + (1.1 \times 10^{-12})$

$= 1.6 \times 10^{-12}$. M^2. $\therefore (x + y)$

$= [Ag^+]$ equals $\left(1.6\times10^{-12}\right)^{\frac{1}{2}}$ M

$= 1.265 \times 10^{-6}$ M

164. $H_2S \rightleftharpoons H^+ + HS^-$

$K_1 = \sim 10^{-7}$; $HS^{(-)} \rightleftharpoons H^+ + S^{2(-)}$ $K_2 = 10^{-14}$

$\therefore$ for $H_2S \rightleftharpoons 2H^+ + S^{2-}$, $K = K_1 \times K_2 = 10^{-21}$

$\therefore \quad \frac{\left[H^+\right]^2\left[S^{--}\right]}{\left[H_2S\right]} = 10^{-21}$

$$\therefore \quad [S^{--}] = \frac{10^{-21} \times [H_2S]}{[H^+]^2}$$

$$= \frac{10^{-21} \times 10^{-1}}{[H^+]^2}$$

K_{sp} of $ZnS = [Zn^{++}]\,[S^{--}] = 1.6 \times 10^{-23}$

$$= \frac{10^{-2} \times 10^{-22}}{[H^+]^2} = \frac{10^{-24}}{[H^+]^2}$$

$$\therefore \quad [H^+]^2 = \frac{10^{-24}}{1.6 \times 10^{-23}} = \frac{1}{16}$$

$$\therefore \quad [H^+] = \frac{1}{4} = 0.25M$$

$$\therefore \quad pH = -\log \frac{1}{4} = \log 4 = 0.6021$$

165. $K_{SP} = 10^{-10} = [Ba^{2+}]\,[SO_4^{2-}] = [Ba^{2+}]\,[10^{-1}]$

$\therefore \quad [Ba^{2+}]\ 10^{-9}M \ \therefore$ wt of $BaSO_4$ per litre

$= 10^{-9} \times 233.4$ g which is 2.334×10^{-7} g

166. $H_2S \rightleftharpoons 2H^+ + S^{2-} \quad K = K_1K_2$

$= 10^{-7} \times 10^{-14} = 10^{-21}$

$$\frac{[H^+]^2[S^{2-}]}{[H_2S]} = 10^{-21}\ ;\ [H_2S] = 0.1\ M$$

$$\therefore \quad [H^+]^2[S^{2-}] = 10^{-22}$$

$$\therefore \quad [S^{2-}] = \frac{10^{-22}}{[H^+]^2};\ [Mn^{2+}] = 10^{-3}$$

$$[Mn^{2+}]\,[S^{2-}] = K_{SP} = 8 \times 10^{-14}\ ;\ 10^{-3} \times \frac{10^{-22}}{[H^+]^2}$$

$$= 8 \times 10^{-14}$$

$$\therefore \quad [H^+]^2 = \frac{10^{-25}}{8 \times 10^{-14}} = 1.25 \times 10^{-12}$$

$$[H^+] = 1.118 \times 10^{-6}$$

167. $pK_a = -\log (4 \times 10^{-9}) = (9 - 0.6021)$

$= 8.3979 \simeq 8.4$

$$pH = pK_a + \log \frac{[In^-]}{[HIn]} \quad 9.4 = 8.4 + \log \frac{[In^-]}{[HIn]}$$

$$\therefore \quad \log \frac{[In^-]}{HIn} = 1 \ \therefore \ \frac{[In^-]}{[HIn]} = 10 \approx \frac{91}{9.1}$$

$\therefore$ % of HIn = 9.1

168. One half of $Na_2CO_3 \equiv (35.10 - 25.13)$ml of HCl

$= 9.97$ ml

$\therefore$ NaOH in 25 ml $\equiv 25.13 - 9.97 = 15.16$ ml of 0.0972 NHCl

This has $\left(\frac{15.16}{1000} \times 0.0972\right)$ eqt $= 1.474 \times 10^{-3}$eqt

(in 25 ml NaOH)

$\therefore$ in one litre of the solution

$(1.474 \times 10^{-3} \times 40 \times 40)$ grams NaOH which is 2.358 g

169. Normality of Na_2CO_3 solution, $\frac{26.32 \times 1.5}{10}$

$= 3.948$ N

wt/litre of $Na_2CO_3 = 3.948 \times 53 = 209.24$ g

Weight in 10 mL = 2.092 g

Weight of 10 mL $= 10 \times 1.25$

$= 12.5$ g $\rightarrow$ % of Na_2CO_3

$$= \frac{2.0924}{12.5} \times 100 = 16.7$$

170. $pH = \frac{1}{2}pK_w + \frac{1}{2}pK_a + \frac{1}{2}\log C = 7 + 2.372 - 0.5$

$$pH = 8.872 = pK_a\text{'} + \log\left(\frac{19}{81}\right) = pK_a\text{'} + (-0.63)$$

$\therefore \quad pK_a\text{'} = 8.872 + 0.63 = 9.502$

$\therefore \quad K_a\text{'} = 3.148 \times 10^{-10}$

171. Statement-1 is wrong. It should be $\left(p_{H_2O}\right)^{-2}$

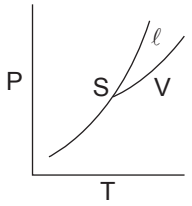
Statement-2 is correct. K_1^2 and K_2 are reciprocal

172. Statement-1 is correct. Statement-2 is wrong

$$\frac{\partial}{\partial T}\left(\frac{\Delta G^\circ}{T}\right)_p = \frac{-\Delta H^\circ}{T^2}$$

173. Both statements are correct. They are linked (since $\Delta G = 0$)

174. Statement-1 is wrong. Except in the case of the ice $\rightleftharpoons$ water system, the solid $\rightleftharpoons$ liquid equilibrium curve (in many cases) has a positive slope. Increase T at same P, solid changes to liquid (Lechatelier). But at constant T, increase P, liquid changes to solid.



Triple point diagram

Statement-2 $\Delta G^\circ = -RT \ln K_P$

$\therefore \quad K_P = \exp\left(\frac{-\Delta G^\circ}{RT}\right)$. Since $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$

$$K_P = \exp\left(\frac{-\Delta H^\circ}{RT}\right) \times \exp\left(\frac{\Delta S^\circ}{R}\right)$$

175. Statement-1 is correct.

$$\frac{d \ln K_P}{dT} = \frac{\Delta H}{RT^2} \text{ (positive)}$$

$\therefore$ $\ln K_P$ and K_P are increased when T increases

Statement-2 is wrong use of the catalyst cannot result in larger yield

176. Both statements are correct and second one the reason for first.

177. Statement-1 is wrong. Statement- 2 is correct

178. Both statements are correct. Second one the reason for first.

179. Both statements are correct but they are not linked.

180. Both statements are correct and second one is reason for first.

181. $\Delta G^\circ = -RT \ln K_P$

$= -1.987 \times 673 \times 2.303 \times \log (2.13 \times 10^{-4})$

$\Delta G^\circ = -1.987 \times 673 \times 2.303 \times (-3.6716)$ cal

$= 11306$ cal ≈ 11.31 k cal

182. $2.303 \log\left(\frac{K_{P_2}}{K_{P_1}}\right) = \frac{-\Delta H^\circ}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$

$$-2.303 \log(6) = \frac{\Delta H^\circ}{R}\left(\frac{T_2 - T_1}{T_1 T_2}\right)$$

$$= \frac{\Delta H^\circ}{1.987}\left(\frac{755 - 673}{755 \times 673}\right)$$

$$\therefore \quad \Delta H^\circ = \frac{-1.792 \times 1.987 \times 755 \times 673}{82}$$

$= -22.06$ k cal for $2NH_3$ (as given)

for $\frac{1}{2} N_{2(g)} + \frac{3}{2} H_{2(g)} \rightleftharpoons NH_3$,

$$\Delta H^\circ = \frac{-22.06}{2} \text{ k cal mol}^{-1} = -11.03 \text{ k cal mol}^{-1}$$

183. Since $\frac{d(\ln K_P)}{dT} = \frac{\Delta H^\circ}{RT^2} \cdot \frac{d(K_P)}{K_P} = \frac{\Delta H^\circ}{RT^2} dT$

$$\text{RHS} = \frac{-22.06 \times 10^3}{1.987 \times (714)^2} \times 1 = -0.0218$$

For $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$

Expressed as a %, we have the answer –2.18%

184. $pH = 3.5 = -\log[H^+]$

$\therefore \quad \log[H^+] = -(3.5) = \bar{4}.5$

$[H^+] = 3.16 \times 10^{-4}$M

Similarly for pH = 8.5

$[H^+] = 3.16 \times 10^{-9}$ M

$$\therefore \quad [OH^-] = \frac{10^{-14}}{3.16 \times 10^{-9}}$$

$$= 3.16 \times 10^{-6}$$

185. pH = 0.125

$\log [H^+] = -(0.125) = \bar{1}.875$

$[H^+] = 0.75 = N$

$$N = \frac{d\,x(\%)\,x10}{49}$$

$$\therefore \quad (\%) = \frac{0.75 x 49}{1.05 x 10} = 3.5$$

186. $[H^+] = 10^{-10}$M

$\therefore$ in 500 ml of the solution,

$[H^+] = 5 \times 10^{-11}$ mol

$\therefore$ no. of H^+ ions

$$= 5 \times 10^{-11} \times 6.023 \times 10^{23} \simeq 3 \times 10^{13}$$

$[OH^-] = 10^{-4}$ M. A similar calculation gives the value, number of OH^- ions $= 3 \times 10^{19}$

187. $K_h = \frac{K_w}{K_a} = \frac{10^{-14}}{7.2 \times 10^{-10}} = 1.39 \times 10^{-5}$

$$= \frac{h^2 C}{1 - h} \simeq h^2 C \text{ where, C} = 0.1 \text{ M}$$

Calculating, $h = 1.118 \times 10^{-2}$

$$pH = \frac{1}{2}pK_w + \frac{1}{2}pK_a + \frac{1}{2}\log C$$

$= 7 + 4.57 - 0.5 = 11.07$

188. $[OH^-] = 5.5 \times 10^{-6} = h \times C$; $C = 0.055$ M

$$\therefore \quad h = \frac{5.5\times10^{-6}}{5.5\times10^{-2}} = 10^{-4}$$

$$K_h = \frac{h^2C}{1-h} \simeq h^2C = (10^{-4})^2 \times (0.055) = 5.5 \times 10^{-10}$$

189. $$K_h = \frac{K_w}{K_aK_b} = \frac{10^{-14}}{\left(1.8\times10^{-5}\right)^2} = 3.086 \times 10^{-5}$$

$$3.086 \times 10^{-5} = \left(\frac{h}{1-h}\right)^2 ; \quad \frac{h}{1-h} = 5.556 \times 10^{-3}$$

$$\therefore \quad h = \frac{5.556\times10^{-3}}{1+\left(5.56\times10^{-3}\right)} \simeq 5.53 \times 10^{-3}$$

$$pH = \frac{1}{2}pK_w + \frac{1}{2}pK_a - \frac{1}{2}pK_b = 7$$

190. (a), (b), (c) correct

191. (a), (b), (c) correct

192. (a), (b), (c) correct

193. (b), (c) correct

194. (a), (b), (d) correct

195.(a), (b) and (c) correct

196. $K_{sp} = [Ba^{2+}][SO_4^{2-}]$

In HCl solution, there is no common ion

For (a) $x \times (0.1 + x) = K_{sp}$

(b) $(x + 0.1) \times x = K_{sp}$

(c) $(x + 0.1) \times x = K_{sp}$

197. (a), (c), (d) correct

198. (a) → r

(b) → q

(c) → q

(d) → p, r, s

(a) $CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$

$K_p = 2$ atm $\therefore P_{CO_2} = 2$ atm

(b) $P_{Cl_2} = 0.5$ atm $\qquad K_p = \frac{1}{P_{Cl_2}^6} = 64$

(c) $P_{NH_3} = P_{H_2S} = 8$ atm

$_p = P_{NH_3} \times P_{H_2S} = 64$

(d) $P_{NH_3} = \frac{2}{3} \times 6 = 4$; $P_{CO_2} = 2$

$\therefore \quad K_p = P_{NH_3}^2 \times P_{CO_2} = 32$

CHAPTER

7

DILUTE SOLUTIONS AND ELECTROCHEMISTRY

CHAPTER OUTLINE

Preview

STUDY MATERIAL

DILUTE SOLUTIONS

Introduction

Colligative Properties
- Concept Strands (1-15)

ELECTROCHEMISTRY

General

Faraday's Laws of Electrolysis

Effect of Electrodes on the Nature of Electrode Reactions During Electrolysis
- Concept Strands (16-22)

Conductance of Electrolyte Solutions
- Concept Strands (23-33)

Galvanic Cells

Reversible Electrodes

Measurement of the EMF of a Galvanic Cell

Single Electrode Potentials

Reference Electrodes

Concentration Cells
- Concept Strands (34-49)

TOPIC GRIP
- Subjective Questions (10)
- Straight Objective Type Questions (5)
- Assertion–Reason Type Questions (5)
- Linked Comprehension Type Questions (6)
- Multiple Correct Objective Type Questions (3)
- Matrix-Match Type Question (1)

IIT ASSIGNMENT EXERCISE
- Straight Objective Type Questions (80)
- Assertion–Reason Type Questions (3)
- Linked Comprehension Type Questions (3)
- Multiple Correct Objective Type Questions (3)
- Matrix-Match Type Question (1)

ADDITIONAL PRACTICE EXERCISE
- Subjective Questions (10)
- Straight Objective Type Questions (40)
- Assertion–Reason Type Questions (10)
- Linked Comprehension Type Questions (9)
- Multiple Correct Objective Type Questions (8)
- Matrix-Match Type Questions (3)

DILUTE SOLUTIONS

INTRODUCTION

A solution may broadly be defined as a homogeneous mixture of chemical species dispersed uniformly among themselves. Thus it has only a single phase, viz., gaseous, liquid or solid phase. A solution may be present in any of the above phases. *When there are two constituents in a solution, it is referred to as a binary solution. A ternary solution is one in which there are three constituents.* However, we are mainly concerned with binary solutions.

The terms solvent and solute

Generally, *the terms solvent and solute refer to the component present in larger and smaller amounts, respectively.* If a small amount of a solute is present in relatively large amount of solvent, the solution is said to be a dilute solution.

Binary solution

In a binary solution, it is possible that both the constituents are volatile or one of them is a volatile component (solvent) and the other is a non-volatile component (solute). In the former case, both the constituents of the binary solution are liquids, which have their characteristic vapour pressures at the given temperature and are completely miscible with each other or one of the components is a gas and the other component a liquid. In the second case, one of the components is a non-volatile solute (usually present in a smaller concentration) dissolved in the volatile solvent. It may be mentioned that there are other possibilities of binary solutions, wherein there are two liquid components, which are either partially miscible or completely immiscible and these cases are outside the scope of present treatment.

Binary solution in which a gas is dissolved in a liquid

Gases are completely miscible with each other. Gases also dissolve in liquids and solids. An example of dissolution of a gas in a solid is the solubility of hydrogen gas in palladium.

Solubility of a gas in a liquid is determined by several factors such as nature of the gas, nature of the liquid, temperature and pressure. **Henry's law** governs the solubility of a gas in a liquid. It states that the solubility of a gas in a liquid is directly proportional to the pressure of the gas. John Dalton also came to the conclusion that the solubility of a gas in a liquid is a function of the partial pressure of the gas.

Considering mole fraction of gas in solution as the measure of its solubility the law can be expressed as $P = K_H X$

Where, K_H is the Henry's law constant. It is evident from the relation that higher the value of K_H at a given pressure the lesser the solubility of the gas in the liquid.

Case of binary solution in which both liquid constituents are completely miscible and are volatile

A liquid mixture of this category is said to be ideal if both the constituents obey Raoult's law over the complete range of solvent composition. Raoult's law may be stated in several ways but the form which is most useful in the present context is: *The partial vapour pressure of component A is proportional to its mole fraction, x_A in the liquid phase in equilibrium with the vapour phase. The same statement is also applicable for the component B in the case of an ideal mixture.* Thus

$$p_A = \overset{\circ}{p}_A x_A ; p_B = \overset{\circ}{p}_B x_B \qquad \text{— (1)}$$

The proportionality constant in both cases is the vapour pressure of the pure constituents.

It is important to consider the conditions under which both the constituents of a liquid mixture obey Raoult's law over the complete range of solvent compositions. Generally, if the two constituents of a mixture are of similar chemical nature, it is possible to expect them to obey Raoult's law. Examples of such mixtures are (1) benzene and toluene (2) chloroform and carbon tetrachloride.

In some cases, the presence of each component tends to push the other constituent into vapour phase such that its partial pressure will be higher than what is expected from Raoult's law at that composition. Such a mixture is said to show a positive deviation from Raoult's law. Example of such mixtures (1) H_2O and ethanol (2) methanol and $CHCl_3$.

Also, there are cases where the presence of a constituent brings down the tendency of the other constituent to go into the vapour phase, i.e., the partial pressure of the second constituent is less than what is predicted by Raoult's law at the given composition. Such a mixture is said to show negative deviation from Raoult's law. This type of behaviour is observed in liquid mixtures where there is a mutual interaction between the constituents. Examples of such cases are (i) H_2O and HCl (ii) $CHCl_3$ and CH_3COCH_3 (iii) H_2O and HNO_3. The total vapour pressure 'P', for an ideal mixture is given by

$$P = p_A + p_B + p_A^0 x_A + p_B^0 x_B$$
$$= p_A^0 x_A + p_B^0 (1 - x_A)$$
$$P = p_B^0 + x_A (p_A^0 - p_B^0) \quad — (2)$$

From equation (2) it is seen that P is a linear function of mole fraction, x_A. A plot of P as a function of mole fraction of A, i.e., x_A is shown below.

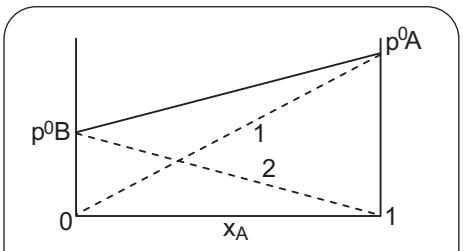


Fig. 7.1

Curves 1 and 2: Partial pressure curves of A and B, respectively.

Solid line represents the total vapour pressure curve.

It is possible to obtain a relation between the mole fraction of a given component in the liquid phase to that of the same in vapour phase by using Dalton's law. i.e.,

$\frac{p_A}{P} = x'_A$ where, x'_A is the mole fraction of A in vapour phase and substituting for p_A and P from equation (1) and (2) we get

$$\frac{p_A^0 x_A}{p_B^0 + x_A (p_A^0 - p_B^0)} = x'_A \quad — (3)$$

Variation of vapour pressure of a liquid with temperature

The vapour pressure of a liquid varies with temperature and may be quantitatively expressed by the Claypeyron – Clausius equation. This equation may be given as

$$\frac{d \ln p}{dT} = \frac{\Delta H}{RT^2} \quad — (4)$$

$\frac{d \ln p}{dT}$ (use ordinary differentiation in place partial differentiation symbol)

where, p is the vapour pressure of the liquid and ΔH is the latent heat of vaporization per mole of the liquid. R is the gas constant. General integration of the equation (4) gives

$$\ln p = -\frac{\Delta H}{RT} + C \quad — (5)$$

(Assuming ΔH to be independent of the temperature).

Between the two temperatures T_1 and T_2 where the vapour pressures are p_1 and p_2 respectively, equation (5) may be written as

$$\ln p_2 - \ln p_1 = \frac{-\Delta H}{RT_2} + \frac{\Delta H}{RT_1} \quad — (6)$$

(assuming ΔH to be constant in the temperature range T_1 and T_2)

$$\ln \left(\frac{p_2}{p_1} \right) = \frac{\Delta H}{R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right] \quad — (7)$$

$$\log \left(\frac{p_2}{p_1} \right) = \frac{\Delta H}{2.303 R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right] \quad — (8)$$

Equation (8) enables us to calculate the vapour pressure at any given temperature provided the vapour pressure at another temperature and ΔH are known.

Similarly, the enthalpy of vaporization of a liquid may be calculated if the vapour pressures at two temperatures T_1 and T_2 are known.

Vapour pressure lowering by addition of non-volatile solute

We shall now consider the case of a dilute solution containing a non-volatile solute dissolved in a solvent. Just as an ideal gas law is a limiting law, which is obeyed by real gases as the gas pressure is lowered, *a dilute solution obeys Raoult's law as the solute concentration of the solution approaches that of pure solvent. Such a solution is known as an ideal dilute solution.*

It is observed that the equilibrium vapour pressure above a solution containing a non-volatile solute is smaller than the vapour pressure of the pure solvent at the same temperature. The lowering of the vapour pressure caused by the addition of a non-volatile solute may be related to its concentration in the form of a quantitative relationship, viz., Raoult's law. *Raoult's law will be presented in a different form here, namely, the relative lowering of vapour pressure of a solvent caused by the addition of a non-volatile solute is equal to the mole fraction of the solute.* If p_A° represents the vapour pressure of the pure solvent A and p that of the solution, the relative lowering of the vapour pressure is given by $\frac{p_A^\circ - p}{p_A^\circ}$.

According to Raoult's law.

$$\frac{p_A^\circ - p}{p_A^\circ} = x_B \quad — (95)$$

where, A and B represent the solvent and solute, respectively. Rearranging the equation (9), we get

$$\frac{p}{p_A^o} = 1 - x_B = x_A \qquad \text{— (10)}$$

Expressing x_B in terms of the number of moles of solute, n_B, and number of moles of solvent, n_A, equation (9) may be written as

$$\frac{p_A^o - p}{p_A^o} = \frac{n_B}{n_A + n_B} \qquad \text{— (11)}$$

If the solution is dilute, $n_B << n_A$, and equation (11) may be simplified as

$$\frac{p_A^o - p}{p_A^o} \approx \frac{n_B}{n_A} = \frac{W_B / M_B}{W_A / M_A} = \frac{W_B \times M_A}{W_A M_B} \qquad \text{— (12)}$$

where, W_A and W_B are the weights of the solvent and solute while M_A and M_B represent their molecular weights. Equation (12) may be used to calculate the molecular weight of the solute B.

The lowering of the vapour pressure of the solvent caused by the addition of a non-volatile solute is an example of colligative property, which is defined below.

COLLIGATIVE PROPERTIES

Definition of colligative property and a list of such properties:

A colligative property is one that depends only on the tnumber of particles (concentration) present in the solution and not on their nature. Several other related properties, based on vapour pressure lowering such as (i) depression of the freezing point of a solution (ii) Elevation of boiling point of a solution (iii) osmotic pressure exerted by a solution are also colligative properties. Of these, only the first two will be discussed in detail.

Depression in freezing point

Let us consider the vapour pressure–temperature diagram for the liquid water. OA, OC, OB represent the vapour pressure curves of liquid water, ice and the effect of pressure on the freezing point of the solution. O'A' represents the vapour pressure curve of the solution containing the non-volatile solute. It can be seen that the vapour pressure curve of the solution, O'A' cuts the line BC at O' well below O i.e., well below the freezing point of water. Curve BO has a negative slope. Further, ΔT_f is measured at 1 atm, not at triple point. If T_f^o and T_f' represent the freezing points of the pure solvent and solution, respectively, then,

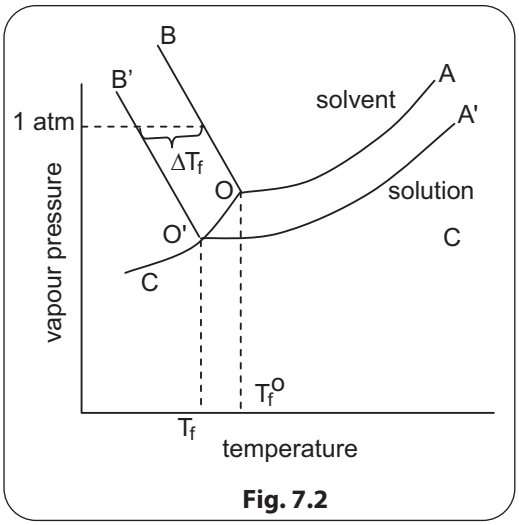


Fig. 7.2

ΔT_f = lowering of the freezing point of solution = $(T_f^o - T_f')$ — (13)

The depression in the freezing point of the solution is proportional to the molality of the solution in an ideal dilute solution. i.e.,

$$\Delta T_f \propto m \qquad \text{— (14)}$$

$$\Delta T_f = K_f \times m \qquad \text{— (15)}$$

The proportionality constant, K_f, is known as the molal freezing point depression constant and is equal to the depression in the freezing point caused by a one molal solution i.e., when m = 1. It has the unit deg kg mol^{-1}. The constant K_f depends only on the properties of the solvent and is equal to

$$K_f = \frac{RT_f^{o^2} M}{1000 \Delta H_f} \qquad \text{— (16)}$$

Note: Strictly speaking, 1 molal is not a designation for a dilute solution.

$$\therefore \quad K_f = Lt\left(\frac{\Delta T_f}{m}\right) \text{ as } m \to 0$$

where, R = gas constant, T_f^o = freezing point of the pure solvent in K, M = Molecular weight of the pure solvent, ΔH_f = enthalpy of fusion of the solvent per mol.

Equation (16) may also be written as

$$K_f = \frac{RT_f^{o^2}}{1000\frac{\Delta H_f}{M}} = \frac{RT_f^{o^2}}{1000L_f} \qquad \text{— (17)}$$

where, L_f = latent heat of fusion of the solvent per gram. The molality "m" in equation (15) may be expressed in terms of the weights of solute B (W_B) and of solvent A (W_A) as

$$m = \frac{1000W_B}{M_B W_A} \qquad \text{— (18)}$$

where, M_B is the molecular weight of the solute. Substituting equation (18) in equation (15), the equation

$$\Delta T_f = K_f \times \frac{1000W_B}{M_B W_A} \qquad \text{— (19)}$$

is obtained.

Equation (19) is used for determining the molecular weight, M_B, of the solute using known values of K_f, W_A and W_B.

Elevation of boiling point

A consequence of the lowering of vapour pressure of a solvent by the addition of a non-volatile solute is the elevation of boiling point of the solution by the added solute.

This may be observed from an examination of the vapour pressure-temperature diagram for the solvent and the solution given below.

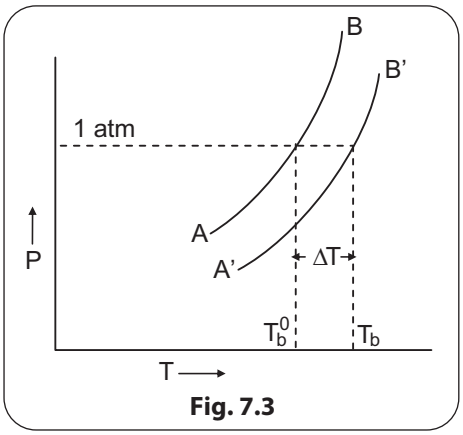


Fig. 7.3

Curves AB and A'B' represent the variation of vapour pressure with temperature for the solvent and solution, respectively. T_b^o and T_b represent the temperatures of the solvent and solution when their vapour pressures are equal to the external pressure, viz., 1 atm. In other words, they represent the boiling points of the solvent and solution, respectively.

The elevation of the boiling point of the solution, ΔT_b, i.e., $T_b - T_b^o$, is proportional to the molality of the solution (in ideal dilute solution)

or $\Delta T_b \mu\, m$ — (20)

or $\Delta T_b = K_b \times m$ — (21)

where, K_b, the proportionality constant, is known as the molal boiling point elevation constant. It is defined as the elevation in the boiling point caused by one molal solution of the non-volatile solute. It depends on the properties of the solvent and has the unit deg kg mol^{-1}. It is related to the boiling point, T_b^o of the solvent (in deg K) and the enthalpy of vapourization of the solvent (ΔH_v) and is given by

$$K_b = \frac{RT_b^{o^2} M}{1000\Delta H_V} \qquad \text{— (22)}$$

M refers to the molecular weight of the solvent in equation (22).

Equation (22) may also be expressed as

$$K_b = \frac{RT_b^{o^2}}{1000\frac{\Delta H_v}{M}} = \frac{RT_b^{o^2}}{1000L_v} \qquad \text{— (23)}$$

where L_V refers to the latent heat of vaporization per gram of the solvent.

Substituting for m from equation (18) into equation (21)

$$\Delta T_b = K_b \times \frac{1000W_B}{M_B W_A} \qquad \text{— (24)}$$

is obtained.

Abnormal behaviour of solutes in solution and their effect on colligative properties

It was mentioned earlier, that all the colligative properties depend on the total number of particles present in the solution irrespective of their nature, i.e., whether they are ionic or neutral species and whether they are single or aggregated.

If there is dissociation of an electrolytic solute in the solvent as in the case of a solute like NaCl or KCl in water, obviously there are more particles present in the solution and hence the observed colligative property is larger than the normal value. Thus there is an enhanced lowering of vapour pressure, a greater lowering of the freezing point of the solution and a greater rise in the boiling point of the solution than in the case where the solute is not dissociated.

The situation is exactly opposite in the case of aggregation of a solute in solution like, for example, acetic or benzoic acid as solute in a solvent like benzene. Because of the smaller number of particles due to association, all the

colligative properties show effects opposite to that in the case of dissociation and, correspondingly, the molecular weights calculated on the basis of the observed colligative properties will be smaller in the case of dissociated solute and higher in the case of associated solutes than normal values.

van't Hoff's factor, (i)

To take the above factors into account, van't Hoff introduced a factor, i, known as the van't Hoff's factor into the equations relating molality and ΔT_f or ΔT_b as shown below.

$$(\Delta T_f)_{actual} = iK_f m \qquad \text{— (25)}$$

$$(\Delta T_b)_{actual} = iK_b m \qquad \text{— (26)}$$

where, $$i = \frac{(\Delta T_f)_{actual}}{(\Delta T_f)_{ideal}} = \frac{(\Delta T_b)_{actual}}{(\Delta T_b)_{ideal}}$$

$$\text{'i'} = \frac{\text{observed value of colligative property}}{\text{calculated value of colligative property on the basis of ideal equation}} \qquad \text{— (27)}$$

Since the colligative property is dependent on the number of particles present in the solution, 'i' may be expressed as

$$\text{'i'} = \frac{\text{Total number of moles present in solution after dissociation or association}}{\text{actual number of moles added to the solution}} \qquad \text{— (28)}$$

Relation between degree of dissociation, α, and van't Hoff's factor, i

It is possible to relate the degree of dissociation, α, with the vant Hoff's factor, i, for an electrolyte C_pA_q giving rise to p cations and q anions on the dissociation of one molecule of the electrolyte.

The equilibrium is represented by

$$C_pA_q \rightleftharpoons pC^{q(+)}_{+}\, qA^{p(-)} \qquad \text{— (29)}$$

Let c be the initial concentration of electrolyte.

At equilibrium, we have

conc. of $C_pA_q = c(1-\alpha)$

conc. of cations, $C = \alpha pc$;

conc. of anions, $A = \alpha qc$

Total number of moles at equilibrium $= c(1-\alpha) + \alpha pc + \alpha qc = c\,[1 + \alpha(p + q - 1)]$ — (30)

$$i = \frac{c[1+\alpha(p+q-1)]}{c} \qquad \text{— (31)}$$

$$\alpha = \frac{i-1}{(p+q-1)} = \frac{i-1}{n-1} \qquad \text{— (32)}$$

[OR] $i = 1 - \alpha + n\alpha$

where, $n = p + q$.

Relation between degree of association, α, and van't Hoff's factor i

Let n molecules of a compound MA form an aggregate $(MA)_n$ according to the reaction

$$nMA \rightleftharpoons (MA)n \qquad \text{— (33)}$$

Let α be the degree of association at equilibrium and let c be the initial concentration of MA.

At equilibrium, we have

conc. of MA remaining $= c(1-\alpha)$ — (34)

conc. of associated species formed $= \frac{\alpha c}{n}$ — (35)

Total number of species formed at equilibrium

$$= c(1-\alpha) + \frac{\alpha c}{n} \qquad \text{— (36)}$$

$$i = \frac{c(1-\alpha) + \frac{\alpha c}{n}}{c} = \frac{n(1-\alpha)+\alpha}{n} \qquad \text{— (37)}$$

$$\Rightarrow \quad i = \frac{n + \alpha(1-n)}{n}$$

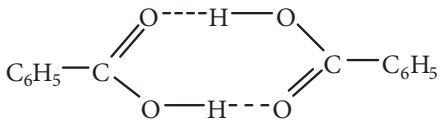
$$\Rightarrow \quad \alpha = \frac{ni - n}{1-n} = \frac{n(i-1)}{n(\frac{1}{n} - 1)}$$

$$\Rightarrow \quad \alpha = \frac{i-1}{\frac{1}{n} - 1} \qquad \text{— (38)}$$

[OR] $i = 1 - \alpha + \frac{\alpha}{n}$

As an example, let us consider the aggregate of benzoic acid in a hydrocarbon solvent according to

$$2C_6H_5COOH \rightleftharpoons (C_6H_5COOH)_2 \qquad \text{— (39)}$$



Let i = 0.6, n = 2

$$\alpha = \frac{0.6-1}{0.5-1.0} = \frac{-0.4}{-0.5} = 0.8 \qquad \text{— (40)}$$

Relation between i and abnormal molecular weights of solute

In the case of electrolytic solutes, the calculated molecular weights on the basis of the observed colligative property are smaller than their normal molecular weights. Likewise in the case of solutes, which show aggregation, the calculated molecular weights (abnormal molecular weight) on the basis of the colligative property are higher than their normal molecular weights. We can relate the abnormal molecular weights to i in the following way.

$$i = \frac{(\Delta T_f)_{obs}}{(\Delta T_f)_{Theo}} \text{ or } \frac{(\Delta T_b)_{obs}}{(\Delta T_b)_{Theo}} \qquad \text{— (41)}$$

$$\Rightarrow \quad i = \frac{(\Delta T_f)_{obs}}{(\Delta T_f)_{Theo}}$$

$$\Rightarrow \quad i = \frac{K_f 1000 W_B}{W_A (M_B)_{obs}} \Big/ \frac{K_f 1000 W_B}{W_A (M_B)_{normal}} \qquad \text{— (42)}$$

$$i = \frac{(M_B)_{normal}}{(M_B)_{Abnormal}} \qquad \text{— (43)}$$

A similar equation is applicable for i derived from $(\Delta T_b)_{obs}/(\Delta T_b)_{Theo}$ where, ΔT_b is the elevation of boiling points of the solution. Thus

$$\Delta T \alpha \frac{1}{M} \text{ (M = Molecular weight)} \qquad \text{— (44)}$$

CONCEPT STRANDS

Concept Strand 1

On bubbling N_2 gas through water at 293 K, how many grams of the gas can dissolve in 1 litre water if the partial pressure of N_2 gas is 0.987 bar and K_H for N_2 at 293K is 76.48K bar.

Solution

According to Henry's law

$$P_{N_2} = K_H X_{N_2}$$

$$X_{N_2} = \frac{0.987 \text{ bar}}{76.48 \times 10^3 \text{ bar}} = 1.29 \times 10^5$$

$$X_{N_2} = \frac{n_{N_2}}{n_{N_2} + n_{H_2O}} \approx \frac{n_{N_2}}{n_{H_2O}}$$

For 1 litre water $n = \frac{1000}{18} = 55.5$

$$n_{N_2} = K_{N_2} \times nH_2o$$

$1.29 \times 10^{-5} \times 55.55 = 7.16 \times 10^{-4}$ mol

mass of $N_2 = 7.16 \times 10^{-4}$ mol $\times$ 28 gmol^{-1}

$= 20.06 \times 10^{-3}$ g

Concept Strand 2

Calculate the mole ratio of N_2 and O_2 in water if air contains 80% N_2 and 20% O_2. K_H for $N_2 = 6.51 \times 10^7$mm Hg and that for $O_2 = 3.30 \times 10^7$ mm Hg.

Solution

Applying Henry's law

$$P_{N_2} = K_H X_{N_2}$$

$$X_{N_2} = \frac{0.8 \times 760 \text{ mmHg}}{6.5 \times 10^7 \text{ mmHg}} = 9.34 \times 10^{-6}$$

$$P_{O2} = K_H X_{O_2}$$

$$X_{O_2} = \frac{0.2 \times 760 \text{ mmHg}}{3.30 \times 10^7 \text{ mmHg}} = 4.61 \times 10^{-6}$$

Mole ratio $n_{N_2} : n_{O_2} = X_{N_2} : X_{O_2} = 2 : 1$

Concept Strand 3

The vapour pressures of benzene and toluene at 30°C are 50.7 and 32.1 mm Hg respectively. Calculate the total pressure of a mixture consisting of three moles of benzene and two moles of toluene. What will be the composition of the

vapour in equilibrium with the mixture at the same temperature?

Solution

Mole fraction of A (benzene) = $\frac{3}{3+2} = 0.6$

Mole fraction of B (toluene) = $\frac{2}{3+2} = 0.4$

$$P = p_B^o + x_A(p_A^o - p_B^o) = 32.1 + 0.6(50.7 - 32.1)$$

$$P = 32.1 + 11.16 = 43.26 \text{ mm Hg}$$

$$x_A' = \frac{p_A^o x_A}{p_B^o + x_A(p_A^o - p_B^o)}$$

$$= \frac{50.7 \times 0.6}{43.26} = \frac{30.42}{43.26} = 0.703\,,\ x_B' = 0.298$$

Concept Strand 4

The vapour pressures of pure acetone and ether are 300 mm and 650 mm Hg, respectively at 30°C. A mixture of these liquids with mole fraction of ether = 0.8 shows a vapour pressure of 700 mm Hg. Is this an ideal mixture? Explain the result.

Solution

'P' expected when mole fraction of ether = 0.8 is given by

$$P = p_A^o x_A + p_B^o x_B \text{ (A = acetone, B = ether)}$$

$$= 300 \times 0.2 + 650 \times 0.8 = 580 \text{ mm Hg}$$

Since the observed p (700 mm Hg) is higher than the predicted (580 mm Hg), the mixture shows +ve deviations from Raoult's law. Acetone is polar and ether is non-polar. Because of the dissimilarity in polarity, one component tries to push out the other component into the vapour phase.

Concept Strand 5

Calculate the vapour pressure of water at 50°C given that its boiling point is 100°C and the latent heat of vaporization of water is 9717 cal mol^{-1}.

Solution

$$\log \frac{p_2}{p_1} = \frac{\Delta H}{2.303R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$$

$$\log \frac{760}{p_1} = \frac{9717}{2.303 \times 1.987}\left[\frac{1}{323} - \frac{1}{373}\right] = 0.8812$$

$$\frac{760}{p_1} = 7.607 \text{ or}$$

$$p_1 = \frac{760}{7.607} = 99.91 \text{ mm Hg}$$

Concept Strand 6

The vapour pressure of a pure liquid at 25°C is 270 mm Hg. On dissolving 1 g of non–volatile ideal solute in one mole of the liquid, the observed vapour pressure of the solution is 268.5 mm Hg. Calculate the molecular weight of the solute assuming the solution to be ideal.

Solution

$$\frac{p_A^o - p}{p_A^o} = \frac{n_B}{n_A + n_B}$$

$$\frac{270 - 268.5}{270} = \frac{1/M}{1 + \frac{1}{M}} = \frac{1}{1+M}$$

$$\frac{1.5}{270} = \frac{1}{1+M} \text{ or } M = 179$$

Concept Strand 7

The vapour pressure of water at 30°C is 31.8 mm Hg. Calculate the vapour pressure of a solution of sodium chloride containing 0.1 moles of the salt in 25 moles of water at this temperature assuming.

(i) no dissociation of the salt
(ii) complete dissociation of the salt.

What do you conclude from the result?

Solution

$$\frac{p_A^o - p}{p_A^o} = \frac{n_B}{n_A + n_B}$$

(i) When there is no dissociation of the saltw

$$\frac{31.8 - p}{31.8} = \frac{0.1}{0.1 + 25} = \frac{0.1}{25.1}$$

$$\Rightarrow \quad \frac{p}{31.8} = \frac{25.0}{25.1} \text{ or } p = \frac{31.8 \times 25.0}{25.1} = 31.67 \text{ mm Hg}$$

(ii) When there is dissociation of the salt:
On complete dissociation, 0.1 g ion of Na^+ and 0.1 g ion Cl^- are formed in a total of 25.2 mol of solution.

$$\therefore \frac{31.8-p}{31.8}=\frac{0.1+0.1}{0.2+25}=\frac{0.2}{25.2}$$

$$\frac{p}{31.8}=\frac{25.0}{25.2};\ \text{or}\ p=\frac{31.8\times25.0}{25.2}=31.55\ \text{mm Hg}$$

The lowering of vapour pressure is a colligative property and since there are more particles on dissociation, there is a greater lowering of vapour pressure, viz., almost twice that observed in the case (i).

Concept Strand 8

In Ostwald-Walker's experiment, dry air was first passed through a solution containing 9.0 g of solute dissolved in 180 g of water and then through pure water. The loss in the weight of the solution was found to be 5.0 g while the loss in weight of water was found to be 0.025 g. Calculate the molecular weight of the solute.

Solution

The loss in weight of solution α p (vapour pressure of the solution). The loss in weight of water (solvent) α (p^o-p), lowering of vapour pressure. or p α 5g; $(p^\circ - p)\,\alpha\, 0.025$ g.

or $p^o \propto 5+0.025$g.

$$\frac{p^o-p}{p^o}=\frac{n_{solute}}{n_{solute}+n_{solvent}}$$

$$\frac{0.025}{5.025}=\frac{9.0/M}{\frac{9.0}{M}+\frac{180}{18}}=\frac{9.0/M}{\frac{9.0}{M}+10}$$

$$\frac{0.025}{5.025}=\frac{9}{9+10\,M}$$

Solving M = 180

Concept Strand 9

2.25 g of an organic compound dissolved in 250 g of benzene freezes at 5.09°C. The freezing point of benzene is 5.45°C and its enthalpy of fusion is 2.37 kcal mol^{-1}. Calculate the molecular weight of the organic compound. (R = 1.987 cal deg^{-1} mol^{-1})

Solution

$$K_f \text{ of benzene} = \frac{RT_f^{o^2}M}{1000\Delta H_f}=\frac{1.987\times(278.09)^2\times78}{1000\times2370}$$

K_f = 5.06 deg kg mol^{-1}

$$\Delta T = 5.45-5.09 = 0.36^\circ C = \frac{5.06\times1000\times2.25}{M_B\times250}$$

Molecular weight of solute

$$M_B=\frac{5.06\times1000\times2.25}{0.36\times250}=126.5\ \text{g}$$

Concept Strand 10

Ethylene glycol is used as an antifreeze. Calculate the weight of glycol which on dissolution in 2.0 kg of water prevents it from freezing at –10°C. (K_f of water = 1.85 deg kg mol^{-1}).

Solution

$$\Delta T=\frac{1000\,K_f\,W_B}{W_A\times M_B}$$

ΔT = 10 K

K_f = 1.85 K kg mol^{-1}

W_A = 2000 g

M_B = 62 g mol^{-1}

$$10=\frac{1000\times1.85\times W_B}{2000\times62}$$

Solving W_B = 670.3 g

Concept Strand 11

A solution of glucose (Mol.mass = 180) containing 18.0 g of solute per kilogram of water boils at 100.052°C. If the boiling point of water is 100°C, calculate the enthalpy of vaporization of water per gram. (R = 1.987 cal deg^{-1} mol^{-1}).

Solution

$$\Delta T_b = K_b\times m$$

$$\text{Molality of the solution} = \frac{1000\times18}{180\times1000}=0.1$$

$$0.052=\frac{RT_b^{o^2}}{1000L_v}\times0.1$$

$$=\frac{1.987\times(373)^2}{1000\times L_v}\times0.1$$

$$\therefore\quad L_v=\frac{1.987\times(373)^2\times0.1}{1000\times0.052}=532\ \text{cal g}^{-1}.$$

Concept Strand 12

6.0 gram of benzoic acid dissolved in 100 g of benzene raises its boiling point by 0.63°C. Calculate its molecular weight. Compare it with the normal molecular weight of benzoic acid and explain the result (K_b of benzene = 2.53 deg kg mol^{-1}).

Solution

$$\Delta T_b = K_b \times m$$

$$0.63 = 2.53 \times \frac{1000 \times 6.0}{M_B \times 100}; M_B = \frac{2.53 \times 1000 \times 6.0}{0.63 \times 100}$$

$$= 241$$

The normal molecular weight of benzoic acid is 122 and the observed molecular weight is close to twice the normal value. Therefore, it may be concluded that benzoic acid dimerizes in benzene.

Concept Strand 13

0.8 g of acetic acid dissolved in 50 g of benzene freezes at 4.4°C. Calculate the molecular weight of acetic acid from the given data. Compare it with normal value. What do you conclude? Calculate the van't Hoff's factor, i, and the percent extent of reaction you visualize? (K_f of benzene = 5.07 K kg mol^{-1}, freezing point of benzene = 5.45°C)

Solution

$$\text{Molality of acetic acid} = \frac{1000 \times 0.8}{M_B \times 50} = \frac{16}{M_B}$$

$$\Delta T_f = K_f m \text{ i.e., } 1.05 = \frac{5.07 \times 16}{M_B}$$

$$\text{or } M_B = \text{Molecular weight of acetic acid}$$

$$= \frac{5.07 \times 16}{1.05} = 77.3$$

Since normal molecular weight of acetic acid is 60, it is concluded that acetic acid aggregates in benzene

$$\text{i} = \text{van't Hoff's factor} = \frac{(M_B)_{\text{normal}}}{(M_B)_{\text{abnormal}}}$$

$$= \frac{60}{77.3} = 0.78$$

$$\alpha = \frac{i-1}{\left(\frac{1}{n}-1\right)} = \frac{0.78-1}{\left(\frac{1}{2}-1\right)} = 0.44$$

(assuming that n = 2 i.e., acetic acid dimerizes in benzene)

Concept Strand 14

Calculate the molecular weight of a solute when a solution containing 5 g of the substance dissolved in one kilogram of water, freezes at –0.24°C. If the normal molecular weight of the substance is 74.5, what do you conclude? Calculate the van't Hoff factor i and the extent of aggregation or dissociation as concluded? (K_f of water 1.85 deg kg mol^{-1})

Solution

$$\text{m of solute} = \frac{1000 \times 5}{M_B \times 1000}$$

$$0.24 = \frac{1.85 \times 5}{M_B} \text{ or } M_B = \frac{1.85 \times 5}{0.24} = 38.5$$

Since the observed molecular weight is less than the true value, it is concluded that there is dissociation in solution.

$$\text{van't Hoff's factor, } i = \frac{M_{\text{normal}}}{M_{\text{abnormal}}} = \frac{74.5}{38.5} = 1.93$$

$$\alpha = \frac{i-1}{n-1} = \frac{1.93-1}{2-1} = 0.93$$

$\Rightarrow$ % dissociation = 93 (assuming each solute molecule gives two ions)

Concept Strand 15

Calculate the change in freezing point and the change in boiling point of a solution of 0.02 molal solution of acetic acid given that it is 3 percent dissociated at this concentration. What will be the change in freezing point and the change in boiling point of the solution if there were no dissociation? (K_f of water = 1.85 deg kg mol^{-1}; K_b = 0.51 deg kg mol^{-1})

Solution

$$CH_3COOH \rightleftharpoons CH_3COO^- + H^+$$

Conc. eqm. 0.02 (1 – 0.03) 0.02 × 0.03 0.02 × 0.03

Total m = 0.0206

$\therefore$ $\Delta T_f = 1.85 \times 0.0206 = 0.038°C$

$\Delta T_b = 0.51 \times 0.0206 = 0.0105°C$

If there were no dissociation,

$\Delta T_f = 1.85 \times 0.02 = 0.037°C$

$\Delta T_b = 0.51 \times 0.02 = 0.0102°C$

ELECTROCHEMISTRY

GENERAL

Electrochemistry is an important topic in physical chemistry both from a fundamental as well as technological view point. It finds applications in a variety of fields such as electrometallurgy and electrorefining, electroanalysis, corrosion prevention, environmental pollution control and electrochemical energy sources, etc.

A fundamental aspect in the study of electrochemistry deals with the nature of electrical conductors and changes that occur as a result of passage of electricity through conductors. Electrical conductors may broadly be divided into two categories, viz., electronic and electrolytic conductors. There are many distinguishing features between the two classes of conductors which are given in the following table:

Table 7.1

Electronic conductors	Electrolytic conductors
Flow of current is manifested by flow of electrons.	Flow of current is manifested by flow of ions in solution.
Examples are metals and alloys	Examples are solutions of electrolytes and fused salts.
There are no visible changes occurring at electrodes during flow of current. Physical effects like heating of the conductors are observed.	Visible changes such as deposition or dissolution of material or evolution of gases occur at electrodes during electrolysis.
Specific conductivity decreases with increase of temperature.	Specific conductivity increases with increase of temperature.

The distinction between electronic and electrolytic conductors is not sharp. There are materials, which show both types of conduction and are known as mixed conductors. Examples of such substances are the inter metallic compounds formed between alkali or alkaline earth metals with metals like tin, antimony, lead, etc. Silver sulphide is another example of this type.

There is also another class of conductors known as semiconductors which exhibit a feeble electronic conductivity and in their case, the specific conductivity increases with increase of temperature. Semiconductivity appears in insulators, which are contaminated with foreign substances. Compounds like Cu(I) oxide and zinc oxide show semiconductivity due to the fact that they do not exactly contain stoichiometric amounts of metal and non metal.

A detailed account of electrolytic conductors, especially the effects of passage of charge through them and the factors affecting the conductivity of electrolytic solution is given. Electrolytic conductors are usually solutions of acids, bases or salts and thus contain ions. Ions are atoms or group of atoms having lost or gained electrons and these are charged species.

Electrolysis of an electrolytic solution

The application of an EMF from a battery through two electrodes placed in a solution of an electrolyte causes the charged particles to move with the positive ions moving towards the cathode and the negative ions moving towards the anode. At the cathode, the cations undergo reduction taking up electrons and at anode the anions undergo oxidation losing electrons.

A typical electrolysis of an electrolyte M^+A^- under the influence of an applied EMF and the resulting reactions occurring at electrodes is shown below.

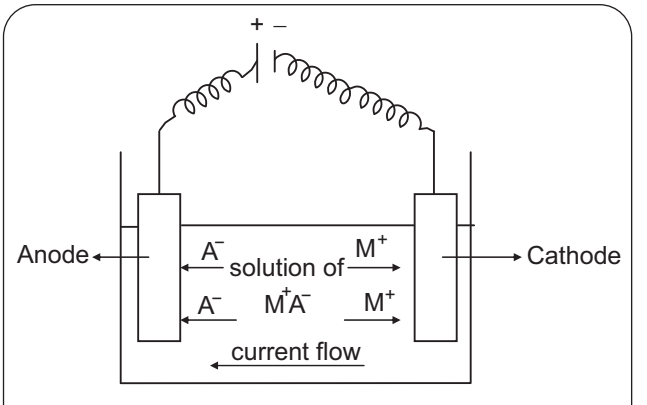


Fig. 7.4

Reactions

$$\text{Cathode}: M^+ + e^- \rightarrow M \text{ (reduction)} \quad — (1)$$

$$\text{Anode}: A^- \rightarrow A + e^- \text{ (oxidation)} \quad — (2)$$

In the external circuit, the electron flow occurs from anode to cathode while in solution current flow (movement of negative ions) occurs from cathode to anode.

It may be mentioned at this stage that in the example given above, the ions carrying the current are also the ions discharged at the electrodes. This is not always true as the discharge of an ion at an electrode depends upon the discharge potential of the ion. Further consideration of this aspect will be taken up at a later stage.

FARADAY'S LAWS OF ELECTROLYSIS

During the course of his work on the electrolysis of many salt solutions, M. Faraday observed that there is a relationship between the amount of material deposited or dissolved at an electrode and the quantity of electricity (or charge) passing through the solution. Two laws have been proposed which are known as Faraday's first and second laws of electrolysis.

Faraday's first law of electrolysis

The law states that the amount of a substance deposited (or dissolved) at an electrode on the passage of a current is proportional to the quantity of electricity passing through the solution. If "W" is the weight of a substance deposited on the passage of "C" amperes of current through a solution for "t" seconds, according to this law , $W \alpha C.\ t$, where, C.t is known as the quantity of electricity in coulombs (one coulomb = one amp × one sec)

$W = ZCt$ where Z is a proportionality constant — (3)

It is seen that W = Z when Ct = 1, i.e., one coulomb. Z is a constant for a given metal (or nonmetal) and is known as the electrochemical equivalent (also referred as e.c.e). It is equal to the weight of the metal deposited for the passage of one coulomb of electricity through the solution. This law can be verified in two ways

(i) By passing currents of different strength through a solution keeping the time of electrolysis constant, i.e.,

$W \alpha C$ (at constant time) — (4)

(ii) By passing current of a given strength through the solution for various intervals of time

$W \alpha t$ (at constant C) — (5)

Faraday's second law of electrolysis

The law states that the amounts of different substances liberated, when the same quantity of electricity is passed through different solutions, are proportional to their chemical equivalents. Let a current of "C" amperes be passed for "t" seconds through two solutions A and B and let "W_A" and "W_B" be the weights of the substances deposited at an electrode during electrolysis.

From 1st law of electrolysis, $W_A = Z_A Ct$ and $W_B = Z_B Ct$, where, Z_A and Z_B are the e.c.e's of the substances A and B.

$$\square \frac{W_A}{W_B} = \frac{Z_A}{Z_B} \quad \text{— (6)}$$

But from second law, $\frac{W_A}{W_B} = \frac{E_A}{E_B}$ — (7)

where, E_A and E_B are chemical equivalents (or equivalent weights) of A and B.

$$\therefore \quad \frac{W_A}{W_B} = \frac{E_A}{E_B} = \frac{Z_A}{Z_B} \quad \text{— (8)}$$

From equation (8) $E_A \propto Z_A$ and $E_B \propto Z_B$ i.e., the chemical equivalent is proportional to the electrochemical equivalent of a metal.

Since a given quantity of electricity (or charge) always deposits different substances (say metals) in the ratio of their equivalent weights, it is clear that a definite fixed quantity of electricity is required to deposit one gram equivalent of any substance. This quantity is known as the Faraday (F) and is equal to 96,494 coulombs. It is usually rounded off to 96,500 coulombs due to uncertainties involved in its determination. Equation (8) may, therefore, be written as

$$\therefore \quad \frac{E_A}{E_B} = \frac{F.Z_A}{F.Z_B} \quad \text{— (8a)}$$

By combining both the 1st and 2nd laws of electrolysis, a useful and important relation which may be used in calculation involving laws of electrolysis is obtained.

If "E" represents the equivalent weight of the metal and "W" its weight deposited for the passage of "I.t" coulombs of electricity through a solution.

$$W = \frac{I.t.E}{96{,}500} \quad \text{— (9)}$$

Equation (9) is useful in two ways (i) to determine the weight of any metal deposited when a known quantity of electricity, (I.t), is passed through a solution (assuming E is known) (ii) to determine the quantity of electricity and in turn the current strength "I" (knowing t) by determining the weight of the metal deposited. The apparatus used for

the latter purpose is known as a "coulometer" and there are several types of coulometer, such as silver coulometer, copper coulometer and iodine coulometer, used for this purpose. It is noteworthy that Faraday's laws are quantitative and are used to calibrate ammeters.

Significance of Faraday's laws

The charge on an electron may be determined by using Faraday's laws in the following manner.

Amount of charge contained in one gram equivalent of any ion = 1 F = 96,500 coulombs.

To deposit one mole of any ion whose valency is Z requires Z Faradays = Z × 96500 coulombs.

Since the number of ions in a mole is equal to Avogadro's number, N, charge carried by a single ion = $\frac{ZF}{N}$ coloumbs.

Since Z is an integer, being equal to 1 for a univalent ion, 2 for a divalent ion, etc., the charge carried by an ion is a multiple of $\frac{F}{N}$ or $\frac{F}{N} = \frac{96,500}{6.023 \times 10^{23}} = 1.602 \times 10^{-19}$ coulombs = $1.602 \times 10^{-19} \times 3 \times 10^{9}$ e.s.u ($\because$ 1 coulomb = 3×10^{9} e.s.u) = 4.8×10^{-10} e.s.u

Since the above calculation represents the case for Z = 1, it is clear that a univalent cation is formed by the loss of an electron. Likewise, a divalent ion is formed by the loss of two electrons, etc. The significance of these laws will be apparent by examination of the following equations, as an example,

$$Cu^{+} + e^{-} \rightarrow Cu \qquad \text{— (10)}$$

$$Cu^{2+} + 2e^{-} \rightarrow Cu \qquad \text{— (11)}$$

Equation (10) suggests that not only an electron is required to convert a cuprous ion to metallic copper but also that a Faraday of charge is required to deposit one gram ion of cuprous ion as metallic copper. Similarly, one may conclude that two Faradays of charge are required to deposit one gram ion of cupric ions to metallic copper.

Current efficiency

During the electrolysis of a solution, sometimes deviations are observed from Faraday's laws due to side reactions occurring at the given electrode. For example, in the electrolysis of copper(II) sulphate solution at high current density, the main reaction at the cathode, i.e., deposition of copper, is accompanied by evolution of hydrogen due to the reaction

$$H^{+} + e^{-} \rightarrow \frac{1}{2} H_{2(g)} \qquad \text{— (12)}$$

In other words, part of the current is used up in the side reaction involving the liberation of hydrogen. This leads us to the concept of current efficiency which is defined as the ratio of the amount of the substance actually formed at the electrode (W_{actual}) to the amount of the substance that would have been formed as calculated from Faraday's laws ($W_{theoretical}$).

$$\therefore \quad \% \text{ of current efficiency} = \frac{W_{actual}}{W_{theoretical}} \times 100 \qquad \text{— (13)}$$

EFFECT OF ELECTRODES ON THE NATURE OF ELECTRODE REACTIONS DURING ELECTROLYSIS

During the course of electrolysis, electrons are lost at the anode and are gained at the cathode. The nature of the electrode reactions occurring at electrodes depends upon the nature of electrodes which may be classified as (1) inert electrodes (2) reacting electrodes.

While inert electrodes like Pt, W do not participate in electrode reactions and merely act as an agent for transfer of electrons, reacting electrodes participate in the electrode reaction. These electrodes contribute metal ions to the solution or accept discharged ions from the solution. Examples of such electrodes are copper, silver placed in solutions of their salts. Consider for example, electrolysis of a solution of silver nitrate using silver electrodes. At cathode, s ilver ions get deposited as metallic silver, while at the anode silver dissolves from the electrode and gives rise to silver ions. The reactions are

$$Ag^{+} + e^{-} \rightarrow Ag \text{ (Cathode)} \qquad \text{— (14)}$$

$$Ag \rightarrow Ag^{+} + e^{-} \text{ (Anode)} \qquad \text{— (15)}$$

On the other hand, consider the electrolysis of an aqueous solution of HCl between platinum electrodes. The reactions at the cathode and anode are

$$2H^{+} + 2e^{-} \rightarrow H_{2} \text{ (Cathode)} \qquad \text{— (16)}$$

$$2Cl^{-} \rightarrow Cl_{2} + 2e^{-} \text{ (Anode)} \qquad \text{— (17)}$$

When there are several ions in a solution competing for discharge at an electrode, the ion having the smaller deposition potential or discharge potential will be the one which gets deposited in preference over the other.

As an example, consider the electrolysis of aqueous NaCl

(i) between Pt electrodes
(ii) between a mercury cathode and Pt anode

In the former case, the reactions at cathode and anode are

$$2H^+ + 2e^- \rightarrow H_2 \text{(cathode)}$$

$$2Cl^- \rightarrow Cl_2 + 2e^- \text{(anode)}$$

Na^+ ions are not discharged at cathode because the H^+ ions have a smaller deposition potential compared to Na^+ ions. However, if a mercury cathode is used instead of Pt cathode, Na^+ ions will be discharged at Hg cathode in preference to H^+ ions because the H^+ ions have higher discharge potential on Hg compared to Na^+ ions. The reaction at Hg cathode is, therefore,

$$Na^+ + e^- \rightarrow Na(\text{Hg cathode}) \qquad \text{— (18)}$$

$$2Cl^- \rightarrow Cl_2 + 2e^- \text{(Pt anode)}$$

The above example thus clearly illustrates the point that the ions that carry the current need not necessarily be the ones to be discharged at an electrode.

CONCEPT STRANDS

Concept Strand 16

A one molar solution of sodium sulphate is electrolysed between platinum electrode employing a current strength of one ampere for 26.8 hours.

(i) Calculate the number of coulombs passed through the solution.
(ii) Write the electrode reactions.
(iii) What are the volumes of the products formed at anode and cathode at 27°C and 1 atm?

Solution

(i) No. of coulombs = $1 \times 26.8 \times 60 \times 60 = 96480$

(ii) At cathode, $2H^+ + 2e^- \rightarrow H_2(g)$

At anode, $2OH^- \rightarrow H_2O + \frac{1}{2}O_2 + 2e^-$

(iii) 2 F are required for depositing one gram mole of H_2

1 F is required for depositing $\frac{1}{2}$ g mole of H_2

$\frac{1}{2}$ g mole of $H_2 \equiv 11.2$ litres of H_2 at N.T.P.

Vol. of H_2 liberated at cathode $= \frac{11.2 \times 300}{273} = 12.3\text{L}$

Oxygen liberated at anode $= \frac{1}{4}\text{gm mole} = \frac{22.4}{4} = 5.6\text{L}$ at S.T.P.

Oxygen liberated at 300 K at anode

$= \frac{5.6 \times 300}{273} = 6.15\text{L}$

It is assumed here that 1F = 96,480 coulombs.

Concept Strand 17

Suppose the same electrolysis as given in the above example (18.16) is carried out between a platinum anode and a mercury cathode.

(i) What are the electrode reactions?
(ii) What are the weights of the products formed at anode and cathode at 27°C and 1 atm?

Solution

(i) There is no change in the product formed at anode, i.e., oxygen. The reaction at cathode is, however, different, i.e., Na^+ ions are discharged according to $Na^+ + e^- \rightarrow Na$.

(ii) The weight of oxygen liberated at anode is also the same as obtained earlier i.e., 6.15 L at 27°C and 1 atm. Using ideal gas relation, weight of O_2 = 7.99 g. The weight of "Na" deposited for the passage of 1 F = 23 g.

Concept Strand 18

One litre of a solution of 0.1 N silver nitrate was electrolysed between silver cathode and a platinum anode for 120 minutes using a current of 0.5 amps.

(i) Write down the electrode reaction at the two electrodes.
(ii) Calculate the weight of product formed (or dissolved) at cathode.
(iii) What is the strength of the silver nitrate solution at the end of electrolysis?

(Eq. wt of Ag = 108.0) Assume that there is no change in volume of solution after electrolysis.

Solution

(i) At cathode: $Ag^+ + e^- \rightarrow Ag$

At anode: $2NO_3^- + H_2O \rightarrow 2HNO_3 + \frac{1}{2}O_2 + 2e^-$

(ii) Wt. of silver deposited = $\frac{120 \times 60 \times 0.5}{96{,}500} \times 108 = 4.03$ g

(iii) Milli equivalents of silver deposited

$= \frac{4.03}{108} \times 1000 = 37.30$

original solution contained $1000 \times 0.1 = 100$ m.eq

m.eq remaining = 100 – 37.3 = 62.7

$\Rightarrow$ $1000 \times N = 62.7$ or $N = \frac{62.7}{1000} = 0.0627$

$\therefore$ Strength of silver nitrate solution after electrolysis = 0.0627 N

Concept Strand 19

A current of 10 amperes is passed through a nickel sulphate bath for two hours to plate out nickel on the cathode. Hydrogen evolution also occurs at the cathode. The nickel deposit on cathode weighed 13.59 g.

(i) Calculate the current efficiency for deposition of nickel.

(ii) How much volume of hydrogen evolves at S.T.P. under these conditions?

(iii) Assuming that the cathode has an area of 100 cm^2, calculate the thickness of nickel obtained on the electrode. (At. wt. of Ni = 58.7; density of nickel = 8.9 g cm^{-3})

Solution

Number of coulombs passed = $10 \times 2 \times 60 \times 60 = 72000$

Wt of nickel to be deposited (on the basis of Faraday's laws)

$= \frac{72{,}000}{2 \times 96{,}500} \times 58.7 = 21.898$ g

Actual amount of Ni deposited = 13.59 g

(i) $\therefore$ % current efficiency = $\frac{13.59}{21.898} \times 100 = 62.06$

(ii) Wt. of H_2 evolved based on the efficiency for H_2 evolution

$= \frac{72{,}000}{2 \times 96{,}500} \times 2.016 \times \frac{37.94}{100} = 0.285$ g

(iii) Volume of H_2 evolved at NTP = $\frac{0.285 \times 22.4}{2.016} = 3.167$ L

Volume of nickel deposited = $\frac{13.59}{8.9} = 1.527\,cm^3$

Thickness of deposit = $\frac{1.527}{100} = 1.527 \times 10^{-2}$ cm

Concept Strand 20

During the course of the operation of a lead storage battery in an experiment, the wt % of H_2SO_4 in the battery decreased from 11.26 to 9.13 corresponding to a decrease in density from 1.075 to 1.060. Assuming that the cell volume is 5 L, calculate the charge used up by the battery in ampere hours. Also, calculate the weight of lead sulphate formed as a result of the reaction. (At. wt. of Pb = 207.2, S = 32.06, O = 16.00)

Solution

Weight of H_2SO_4 in the cell before discharge

100 g of solution contains 11.26 g of acid

5000×1.075 g solution contains

$\frac{5000 \times 1.075 \times 11.26}{100} = 605.2$ g of acid

$= \frac{605.2}{98} = 6.176$ mol H_2SO_4

Weight of H_2SO_4 in the cell after discharge

$= \frac{5000 \times 1.06 \times 9.13}{100} = 483.9$ g

$= \frac{483.9}{98} = 4.94$ mol of acid

Loss in moles of $H_2SO_4 = 6.176 - 4.940 = 1.236$

Reaction in the lead acid battery

At –ve electrode, $Pb + SO_4^{2-} \rightarrow PbSO_4 + 2e^-$

At +ve electrode, $PbO_2 + 4H^+ + SO_4^{2-} + 2e^- \rightarrow PbSO_4 + 2H_2O$

Overall reaction, $Pb + PbO_2 + 4H^+ + 2SO_4^{2-} \rightarrow 2PbSO_4 + 2H_2O$

or $Pb + PbO_2 + 2H_2SO_4 \underset{\text{charge}}{\overset{\text{discharge}}{\rightleftharpoons}} 2PbSO_4 + 2H_2O$

During discharge, 2 moles of H_2SO_4 are consumed for 2 F of electricity

1.236 moles of H_2SO_4 consumed for $\frac{1.236 \times 2}{2} = 1.236$ Faradays

Charge used up in ampere hours $= \frac{1.236 \times 96,500}{3600} = 33.13$

Also 2 Faradays deposit 2×303.26 g of $PbSO_4$

1.236 Faradays deposit $\frac{1.236 \times 2 \times 303.26}{2} = 374.8$ g $PbSO_4$

Concept Strand 21

Aniline can be produced by the reduction of nitrobenzene in acid medium. Write the relevant electrochemical reaction. Calculate the time (in hours) required for the production of 1 kg of aniline using a current of 10 amperes for electrolysis. How many electrons were used up?

Solution

Electrochemical reaction:

$$C_6H_5NO_2 + 6H^+ + 6e^- \rightarrow C_6H_5NH_2 + 2H_2O$$

6 Faradays are required to produce 93 g of aniline (one mole)

1000 g of aniline require $\frac{1000 \times 6 \times 95,500}{10 \times 3600} = 6.226 \times 10^6$ coulombs

time of electrolysis $= \frac{6.226 \times 10^6}{10 \times 3600} = 172.94$ hours

Number of electrons used up :

96,500 coulombs contains 6.023×10^{23} electrons

6.226×10^6 coulombs contain $\frac{6.226 \times 10^6 \times 6.023 \times 10^{23}}{96,500}$

$= 3.88 \times 10^{25}$ electrons

Concept Strand 22

Potassium chlorate is made by the oxidation of potassium chloride in alkaline medium according to the reaction.

$$Cl^- + 6OH^- \rightarrow ClO_3^- + 3H_2O + 6e^-.$$

In an experiment, 5 amperes of current was passed through an alkaline KCl solution for 131.3 hours and the yield of potassium chlorate was found to be 350 g. Is this in accordance with Faraday's laws? If not, what is the current efficiency for the reaction?

Solution

Number of coulombs passed = $5 \times 131.3 \times 60 \times 60 = 2363400$

$6 \times 96,500$ coulombs deposit 122.5 g $KClO_3$

2363400 coulombs deposit $\frac{2363400 \times 122.5}{6 \times 96,500} = 500$ g

But only 350 g $KClO_3$ have been deposited.

The amount of $KClO_3$ deposited is not in accordance with Faraday's laws.

$\therefore$ Current efficiency $= \frac{350}{500} \times 100 = 70\%$

CONDUCTANCE OF ELECTROLYTE SOLUTIONS

General

An important property of electrolyte solutions is their ability to conduct electricity due to the ions present in the solution on the application of an EMF from a suitable source. From conductance measurements of electrolyte solutions, a variety of physico-chemical data can be obtained such as

(i) solubility of sparingly soluble salts
(ii) degree of dissociation and dissociation constants of acids and bases, etc.

From an analytical point of view, conductance measurements can be conveniently employed to determine the equivalence points in a variety of titrations such as acid-base, precipitation, displacement reactions, etc. It is therefore important to understand the fundamental principles in this area and to apply them wherever necessary. Some of basic definitions and the relevant theory associated with them are presented below:

Specific resistance and conductivity (or specific conductance) of electrolyte solutions

Solutions of electrolytes, like metallic conductors, obey Ohm's law i.e.,

$$E \propto I \qquad \text{— (19)}$$

or $$\frac{E}{I} = R \qquad \text{— (20)}$$

where R is the resistance, E is the applied EMF and I is the current passing through conductor. In order that equation

(20) is applicable to electrolytes, the frequency has to be high (>1000 Hz) so that discharge is arrested.

Considering the resistance of an electrical conductor,

$R \propto \ell$ where, l is the length of the conductor — (21)

$R \propto \frac{1}{a}$ where, a is the cross sectional area of the conductor — (22)

Combining equations (21) and (22), we may write

$$R = \frac{\rho.\ell}{a} \quad \text{— (23)}$$

where, ρ is a proportionality constant known as specific resistance

$R = \rho$ when $\ell = 1$ cm and $a = 1$ cm^2

i.e., *the specific resistance is the resistance of a conductor of length 1 cm and cross-sectional area 1 cm^2 .*

The reciprocal of the specific resistance is known as specific conductance (or conductivity) and it has the units of ohm^{-1} cm^{-1}. Multiplication of κ in the units of ohm^{-1} cm^{-1} by 100 will give κ in the units of ohm^{-1} m^{-1}.

In the case of electronic conductors, the specific resistance is a constant for a given metal at a definite temperature. For electrolytic conductors, it is common to use specific conductance, rather than specific resistance because of the interest in the conducting power of electrolytes.

Thus one may write

$$\text{Conductivity} = \kappa = \frac{1}{\rho} = \frac{\ell}{aR} \quad \text{— (24)}$$

and the specific conductivity varies with the concentration of the electrolyte. *Specific conductivity is the conductance of a 1 cm^3 of the solution of an electrolyte.*

Equivalent and molar conductivity of electrolyte solutions

Another quantity, which is very useful in the study of the conductance of electrolytes, is known as equivalent conductance and is represented by Λ.

Λ represents the conducting power of all the ions obtainable from one gram equivalent of the electrolyte. It is defined as

$$\Lambda = \kappa V \quad \text{— (25)}$$

where, "V" is the volume in cm^3 containing 1 gram equivalent of the electrolyte. It has the units of ohm^{-1} cm^2 eq^{-1} and varies with the concentration of the electrolyte to some extent.

Equation (25) may be written in terms of the concentration C of the electrolyte when it is expressed in gram equivalents litre $^{-1}$ i.e.,

$$\Lambda = \frac{1000\kappa}{C} \quad \text{— (26)}$$

Sometimes the equivalent conductivity is expressed in the units of ohm^{-1} m^2 eq^{-1}. To convert Λ in the units of ohm^{-1} m^2 eq^{-1} from ohm^{-1} cm^2 eq^{-1}, it has to be multiplied by a factor of 10^{-4}.

To bring out the physical significance of Λ, one may state that it is equal to the conductance of a solution containing 1 gm equivalent of an electrolyte dissolved in V cm^3 of the solution and placed in a cell having a base of cross-sectional area 1 cm^2.

The molar conductivity of a solution may be defined as the conducting power of one gram mole of an electrolyte.

$$\mu = \frac{1000\kappa}{C} \quad \text{— (27)}$$

where, C = concentration of the electrolyte in g mol L^{-1}.
μ has the dimension of ohm^{-1} cm^2 mol^{-1}.

Relation between molar and equivalent conductance.

For an electrolyte which is of uni-univalent type like NaCl, KNO_3, etc, i.e., where the cation and anion carry a single charge, then $\mu = \Lambda$

If the electrolyte is of bi-bivalent type (like $NiSO_4$, $ZnSO_4$, etc.) where both cation and anion carry two negative charges each $\mu = z'\Lambda$
(where, z' = valency of cation or anion) — (28)

For an unsymmetrical electrolyte, where the cation and anion carry different charges, (like $BaCl_2$, $Fe_2(SO_4)_3$ etc.)

$$\mu = z'\Lambda \quad \text{— (29)}$$

(where, z' = product of valency and number of ions in the molecule taken either for cation or anion)

Determination of specific and equivalent conductances of an electrolyte

The determination of the specific conductivity of an electrolyte consists of two parts (i) determination of $\left(\frac{1}{a}\right)$ (ii) determination of the resistance of the electrolyte under consideration

(i) For determination of cell constant $\left(\frac{1}{a}\right)$ of the conductivity cell used in the experiments, a solution

of potassium chloride of definite concentration (either 0.01 N, or 0.02 N or 0.1 N) whose specific conductivity is already known, is taken in the conductivity cell. The conductivity cell consists of two electrodes fixed at a certain distance, say, 1 cm apart whose mean cross-sectional area is "a" cm^2, as shown below.

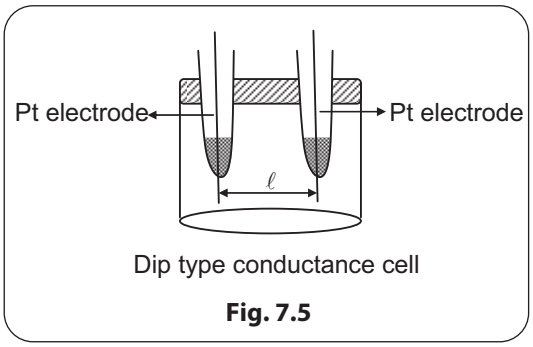


Fig. 7.5

The platinum electrodes of the cell are coated with a thin layer of platinum black to avoid undesirable electrolytic effects. The resistance of the potassium chloride solution is determined by means of Wheatstone bridge employing alternating current of low frequency (1000 C/s to 2000 C/s) and low voltage. The cell constant is calculated using equation (24) from the known values of R and κ (of the KCl solution used). The cell constant has the units of cm^{-1} .

(ii) Using the same cell whose $\frac{l}{a}$ is known, the resistance of the desired electrolyte is experimentally determined in the same manner as above. Using equation (24) again κ (of the desired electrolyte) = $\frac{l}{a}\times\frac{1}{R}$, μ of the electrolytic solution is calculated. In all conductance measurements, precise temperature control is necessary as conductivity is very sensitive to temperature changes. Also, specially purified water, known as conductivity water (having κ of the order 8 to 10×10^{-7} $ohm^{-1}cm^{-1}$) is often used for the preparation of solutions.
However, for accurate work, water purified by special methods having still lower specific conductivity compared to the above is employed. Once the specific conductivity of the desired solution is determined, its equivalent conductivity can be calculated using equation (26).

Variation of specific and equivalent conductivity with concentration

The specific conductivity of an electrolyte increases with increasing concentration because of the increase in the number of ions per unit volume with increasing concentration. In dilute solution, the specific conductivity is proportional to concentration. At higher concentration (i.e., greater than 0.1 N) deviations begin to occur due to stronger interionic attractions.

On the other hand, the equivalent conductance of an electrolyte decreases with increasing concentration or it increases with increasing dilution. The plot of (Λ) against concentration for various electrolytes reveals two distinct features.

For some electrolytes, (Λ) increases very slowly with increasing dilution (or decreasing concentration) and tends to approach a limiting value at great dilution. Such electrolytes are referred to as strong electrolytes, some examples being aqueous solutions of salts like NaCl, KCl, $NaNO_3$, KNO_3 etc.

In the case of some other electrolytes, (Λ) increases rapidly with dilution below a particular concentration. These electrolytes are known as weak electrolytes and solution of acetic acid, ammonia, benzoic acid belong to this category.

All electrolytes, whether strong or weak, tend to approach a limiting value at great dilutions. This limiting value is known as the equivalent conductivity at infinite dilution. However, this limiting value, Λ^0, cannot be determined experimentally except in some cases as shown below.

Determination of Λ_0

(i) Strong electrolytes: For strong electrolytes, a plot of Λ Vs $\sqrt{c}$ (c = conc. in gm eq $litre^{-1}$) in dilute solutions ($c \leq 0.005$ eq/litre) gives a linear relation, which on extrapolation to zero concentration yields an intercept equal to Λ^0

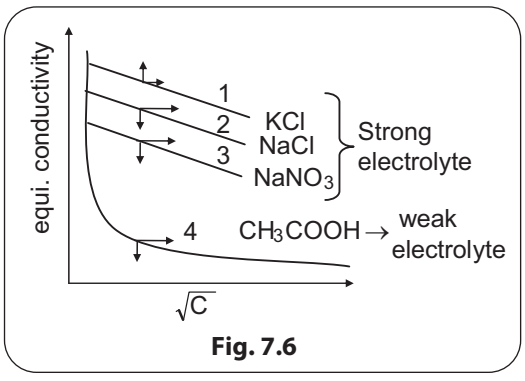


Fig. 7.6

(ii) Weak electrolytes: In the case of weak electrolytes, such extrapolation is not possible in view of the steep rise in Λ at large dilution and, therefore, an alternate procedure is adopted. It is based upon "Kohlrausch's law of independent migration of ions" which states that each ion of an electrolyte contributes a definite

amount to the equivalent conductance of an electrolyte at infinite dilution, irrespective of the nature of the other ion with which it is associated. Thus one may write

$$\Lambda^0 = \lambda^0_+ + \lambda^0_- \quad \Rightarrow (\Lambda^\infty = \lambda^\infty_+ + \lambda^\infty_-) \qquad — (30)$$

The superscripts indicate that they are at infinite dilution. λ^0_+ and λ^0_- are known as the ionic conductances at infinite dilution. They are definite constants for each ion at infinite dilution and vary only with temperature and with the nature of the solvent. Equation (30) may also be used with reasonable accuracy at other concentrations only in dilute solutions not exceeding 0.01 N, i.e.,

$$\Lambda = \lambda_+ + \lambda_- \qquad — (31)$$

Applications of ionic conductances

(i) Determination of Λ^0 of weak electrolytes: The ionic conductance data may be used to determine the Λ^0 of weak electrolytes like acetic acid, ammonium hydroxide, etc., by using the tabulated data of ion conductances from literature or by suitably combining the equivalent conductances at infinite dilution of some strong electrolytes (determined experimentally, see Λ vs $\sqrt{c}$ plot given earlier).

$$\Lambda^0 (NH_4OH) = \lambda^0_{NH_4^+} + \lambda^0_{OH^-} \qquad — (32)$$

(from tabulated data of ion conductance from literature)

or $\Lambda^0 (NH_4OH) = \lambda^0_{NH_4^+} + \lambda^0_{OH^-} = \Lambda^0(NH_4Cl) +$

$\Lambda^0(NaOH) - \Lambda^0(NaCl)$ — (33)

All electrolytes in equation (33) are strong electrolytes for which Λ^0 may be determined experimentally $\left(\text{from } \Lambda \text{ Vs} \sqrt{c}\right)$. By algebraically manipulating Λ^0 data given above,

$\Lambda^0 (NH_4OH)$ can be calculated.

$$\Lambda^0 (NH_4OH) = \lambda^0_{NH_4} + \lambda^0_{OH^-}$$
$$= \lambda^0_{NH_4^+} + \lambda^0_{Cl^-} + \lambda^0_{Na^+} + \lambda^0_{OH^-} - \lambda^0_{Na^+} - \lambda^0_{Cl^-}$$

In a similar way, it is possible to determine Λ^0 of other weak electrolytes.

(ii) Solubility of sparingly soluble salts: Solubilities of sparingly soluble salts like AgCl, $BaSO_4$ and $PbSO_4$ can be determined from conductance measurements. For this purpose, the specific conductivity of the saturated solution of the sparingly soluble salt is first determined and this can be related to its Λ by the equation.

$$\Lambda = \frac{1000\,\kappa}{C} = \frac{1000\,\kappa}{S} \qquad — (34)$$

where, the concentration C, in this case "S", is the solubility of the salt in equiv/litre. By making the assumption Λ of the salt $\approx \Lambda^0$ (35) because the solution is dilute and rewriting equation

$$\Lambda \approx \Lambda^0 = \frac{1000\,\kappa}{S} \qquad — (36)$$

$$S = \frac{1000\ \kappa}{\Lambda^0} \qquad — (37)$$

Since, both κ and Λ^0 are known, S can be calculated. The Λ^0 required in equation (37) can be calculated from the table of ionic conductances from literature. In calculating solubility, S, of the sparingly soluble salt, the specific conductivity of the saturated solution must be corrected for the conductivity of water used in the experiment.

Relation between ionic conductance and ionic speed

The equivalent conductance of an electrolyte depends or three factors (i) the number of ions obtainable from one equivalent (ii) the charge carried by the ions (iii) the velocities of the ions. Considering one gram equivalent of any electrolyte, the total charge carried by the ions (both cations and anions), i.e., the product of the number of ions and the charge carried by each ion is a constant. Thus the Λ_0 depends only on the speeds of the ions and we can write.

$$\Lambda^0 \alpha (\omega^o_+ + \omega^o_-) \qquad — (38)$$

where, ω^o_+ and ω^o_- are known as the ionic mobilities of cation and anion (under a potential gradient of 1 volt/cm)

$$\Lambda^0 = y(\omega^o_+ + \omega^o_-) \qquad — (39)$$

where y is a proportionality constant. The proportionality constant is related to the charge contained in one gram equivalent of the electrolyte which may be easily identified with the Faraday. Replacing y by F in equation (39)

$$\Lambda^0 = F(\omega^o_+ + \omega^o_-) \qquad — (40)$$

Comparing equation (40) with equation (30), the ionic conductance may be related to the ionic mobility by the equation,

$$\lambda^0_+ = F\omega^0_+;\ \lambda^0_- = F\omega^0_- \qquad — (41)$$

The ionic conductance is thus proportional to ionic mobility. It is important to note that *the ionic mobility refers to the velocity of an ion under a potential gradient of 1 volt/cm. It is also known as the absolute velocity of an ion under unit potential gradient.*

At other potential gradients, the term ionic velocity rather than ionic mobility is used.

Equations similar to (41) are applicable at other concentrations and thereby it is possible to write

$$\lambda_+ = F\omega_+ \quad \lambda_- = F\omega_- \qquad \text{— (42)}$$

In case of incomplete dissociation of an electrolyte, suitable correction must be made in the equation — (43)

The units of ionic mobility may be obtained in the following way.

$$\omega^\circ = \frac{\lambda^0}{F} = \frac{\text{ohm}^{-1}\,\text{cm}^2\,\text{eq}^{-1}}{\text{ampere}\,\text{sec}\,\text{eq}^{-1}} = \frac{\text{ohm}^{-1}\,\text{cm}^2}{\frac{\text{volts}}{\text{ohm}}\text{sec}}$$

$$= \text{cm}^2\,\text{volt}^{-1}\,\text{sec}^{-1} = \frac{\text{cm}}{\text{sec}} \times \frac{1}{\text{volt}\,\text{cm}^{-1}} \qquad \text{— (43)}$$

Relation between specific conductivity, ionic conductances in a single or mixture of electrolytes

From equation (26) the specific conductivity of an electrolyte may be expressed as $\kappa = \frac{C\Lambda}{1000}$ — (44) where, Λ is the equivalent conductivity of the electrolyte.

Equation (44) may be written in terms of ionic conductances as $\kappa = \frac{C(\lambda_+ + \lambda_-)}{1000}$ — (45) where, λ's are ionic conductances of the electrolyte.

If there is more than one electrolyte i.e., more number of ions are present, equation (45) may be generalized as $\kappa = \sum_i \frac{C_i\lambda_i}{1000}$ — (46) the summation extending over all ions. Equation (46) is very useful to calculate the specific conductivity of a solution containing a mixture of electrolytes. Even the conductance $\left(\frac{1}{R}\right)$ of the solution can be obtained by writing equation (46) as

$\frac{1}{R} = \frac{1}{1000\,\theta}\sum_i C_i\lambda_i$ — (47) where, θ = cell constant of the conductance cell.

In using equations (46) or (47), strictly the ionic conductances at the given concentration must be employed but in dilute solutions, the data at infinite dilution are often used.

Degree of dissociation, α, from conductance measurements (or conductance ratio)

Conductance measurements may be used to determine the degree of dissociation and hence the dissociation constants of weak electrolytes.

During the course of the study of the variation of Λ with concentration, Arrhenius suggested that the variation is entirely due to change in the number of ions of the electrolyte with the concentration or in other words due to a change in degree of dissociation and not due to any variation of the ionic mobilities with dilution. *Since the electrolyte is completely dissociated at infinite dilution, according to him, the ratio*

$$\frac{\Lambda}{\Lambda^0} = \alpha \qquad \text{— (48)}$$

where, Λ is the equivalent conductivity at a given concentration of electrolyte and Λ_o is the equivalent conductivity at infinite dilution. The above ratio, which is also referred as conductance ratio, is applicable to weak electrolytes and thus can be employed for determination of α of weak acid like acetic acid or weak base like NH_4OH and also their dissociation constants. Once α is known, the dissociation constant, K, of the weak acid may be calculated as follows:

$$K = \alpha^2 C \approx \left(\frac{\Lambda}{\Lambda^0}\right)^2 C \qquad \text{— (49)}$$

where, C is the stoichiometric concentration of the electrolyte.

CONCEPT STRANDS

Concept Strand 23

The specific conductance of a 0.1 N solution of potassium chloride is 0.012886 $\text{ohm}^{-1}\,\text{cm}^{-1}$ at 25°C.

(i) Calculate the resistance of the solution when measured in a cell whose electrodes are 7.5 cm apart and have a cross sectional area of 0.5 cm^2.

(ii) What will be the resistance of the solution if a cell with a cell constant half that of the earlier one is used?
(iii) What is its equivalent conductance in the unit of $ohm^{-1} cm^2 eq^{-1}$ and in $ohm^{-1} m^2 eq^{-1}$?
(iv) What is the current flowing through the cell in case (ii) if the voltage applied in the conductivity study is 5 volts?

Solution

Cell constant $= \frac{7.5}{0.5} = 15\ cm^{-1}$

(i) Resistance of solution,

$$R = \frac{l}{a} \times \frac{1}{\kappa} = \frac{15}{0.012886} = 1164 \text{ ohms}$$

(ii) When a cell with a cell constant = 7.5 cm^{-1} is used.

$$R = \frac{7.5}{0.012886} = 582.0 \text{ ohms.}$$

(iii) $$\Lambda = \frac{1000\kappa}{C} = \frac{1000 \times 0.012886}{0.1} = 128.9 ohm^{-1} cm^2 eq^{-1}$$

$$\Lambda = 128.9 \times (0.01) \times (0.01) m^2 \times eq^{-1}$$

$$= 128.9 \times 10^{-4} ohm^{-1} m^2 eq^{-1}$$

(iv) $$I = \frac{5}{582} = 8.59 \times 10^{-3} \text{ amp.}$$

Concept Strand 24

The ionic conductance at infinite dilution of Ca^{2+} and Cl^- ions are 1.19×10^{-2} $ohm^{-1} m^2 mol^{-1}$ and 7.63×10^{-3} $ohm^{-1} m^2 mol^{-1}$, respectively, at 298K.

(i) Calculate μ^0, molar conductance of $CaCl_2$ at infinite dilution in $ohm^{-1} m^2 mol^{-1}$.
(ii) Calculate molar conductance of $CaCl_2$ at infinite dilution in the unit of $ohm^{-1} cm^2 mol^{-1}$ What will be the value of Λ^0 (i.e., equivalent conductance at infinite dilution in units of $ohm^{-1} m^2 eq^{-1}$ and $ohm^{-1} cm^2 eq^{-1}$.
(iii) What will be the specific conductance of the solution in the units $ohm^{-1} cm^{-1}$ assuming that the μ^0 values in (i) are applicable at a concentration of 0.002 M.
(iv) What will be the resistance of the solution in a cell of cell constant 2.0 cm^{-1}.

Solution

(i) $$\mu^0(CaCl_2) = 1.19 \times 10^{-2} + 2 \times 7.63 \times 10^{-3}$$
$$= 1.19 \times 10^{-2} + 1.526 \times 10^{-2}$$
$$= 2.716 \times 10^{-2} ohm^{-1} m^2 mol^{-1}$$

(ii) $\mu^o(CaCl_2) = 2.716 \times 10^{-2} \times 10^4 = 271.6$ $ohm^{-1} cm^2 mol^{-1}$

$\Lambda^0 (CaCl_2)$ in $ohm^{-1} cm^2 eq^{-1}$

$$= \frac{271.6}{2} = 135.8 \ ohm^{-1} cm^2 eq^{-1}$$

$\Lambda^0 (CaCl_2)$ in $ohm^{-1} m^2 eq^{-1} = 135.8 \times 10^{-4}$ $ohm^{-1} cm^2 eq^{-1}$

(iii) $$\Lambda = \frac{1000\kappa}{C} \text{ or } \kappa = \frac{C\Lambda}{1000} = \frac{0.002 \times 135.8}{1000}$$

$$= 2.72 \times 10^{-4} ohm^{-1} cm^{-1}$$

(iv) $$R = \frac{l}{a} \times \frac{1}{\kappa} = \frac{2.0}{2.72 \times 10^{-4}} = 7352 \text{ ohm}$$

Concept Strand 25

A conductance cell had a resistance of 470 ohm when filled with 1×10^{-3} M HCl. The cell gave a resistance of 1600 ohm with a solution of 1×10^{-3} M NaCl and 1650 ohms with a solution of 1×10^{-3} M $NaNO_3$. Given $\Lambda(NaNO_3)$ =121.0 $ohm^{-1} cm^2 eq^{-1}$, calculate (i) cell constant (ii) resistance of the cell when filled with 1×10^{-3} M HNO_3 (iii) κ of a solution containing 1×10^{-3} M each of HCl, NaCl, $NaNO_3$.

Solution

(i) $$\kappa = \frac{C\Lambda}{1000} = \frac{1 \times 10^{-3} \times 121}{1000} = 1.21 \times 10^{-4} ohm^{-1} cm^{-1}$$

$$\frac{l}{a} = \kappa \times R = 1.21 \times 10^{-4} \times 1650 = 0.1996 \ cm^{-1}$$

$$\kappa \text{ of NaCl} = \frac{0.1996 \times 1}{1600} = 1.247 \times 10^{-4} ohm^{-1} cm^{-1}$$

$$\Lambda \text{ of NaCl} = \frac{1000 \times 1.247 \times 10^{-4}}{0.001} = 124.7 ohm^{-1} cm^2 eq^{-1}$$

$$\Lambda \text{ of HCl} = \frac{1000 \times 0.1996}{470 \times 10^{-3}} = 424.7 ohm^{-1} cm^2 eq^{-1}$$

$$\Lambda \text{ of } HNO_3 = \Lambda \text{ HCl} + \Lambda \text{ } NaNO_3 - \Lambda \text{ NaCl}$$

$$= 424.7 + 121 - 124.7 = 421 \ ohm^{-1} cm^{-2}$$

$$\kappa \text{ of } HNO_3 = \frac{C\Lambda}{1000}$$

$$= \frac{1 \times 10^{-3} \times 421}{1000} = 4.21 \times 10^{-4} ohm^{-1} cm^{-1}$$

$$R \text{ of } HNO_3 = \frac{l}{a\kappa} = \frac{0.1996}{4.21 \times 10^{-4}} = 474.0 \text{ ohm.}$$

(iii) κ of a mixture of 1×10^{-3} M HCl, NaCl and $NaNO_3$ each.

$$\kappa = \frac{C}{1000}\Sigma\Lambda = \frac{1\times10^{-3}}{1000}(424.7 + 124.7 + 121.0)$$
$$= 6.662 \times 10^{-4}\ \text{ohm}^{-1}\ \text{cm}^{-1}$$

Concept Strand 26

The specific conductance of a saturated solution of $BaSO_4$ in water at 25°C is 1.80×10^{-3} ohm^{-1} m^{-1} and the specific conductivity of water used in the experiment was found to be 2×10^{-4} ohm^{-1} m^{-1}. The molar ionic conductances of Ba^{2+} and SO_4^{2-} ions are 127.2 and 159.6 ohm^{-1} cm^2 mol^{-1}. Using these data, calculate the solubility and solubility product of $BaSO_4$.

Solution

Specific conductance due to $BaSO_4$ alone
$= (1.80 - 0.2) \times 10^{-3}$ ohm^{-1} m^{-1}
$= 1.60 \times 10^{-3}$ ohm^{-1}m^{-1} $= 1.60 \times 10^{-5}$ ohm^{-1} cm^{-1}

$\Lambda^0(BaSO_4) = 127.2 + 159.6 = 286.8$ ohm^{-1} cm^2 mol^{-1}

$$S = \frac{1000\kappa}{\Lambda^0} = \frac{1000\times1.60\times10^{-5}}{286.8}$$
$$= 5.58\times10^{-5}\ \text{moles/litre}$$

$\Rightarrow \quad K_{sp} = S^2 = (5.58 \times 10^{-5})^2 = 3.11 \times 10^{-9}$

Concept Strand 27

The equivalent conductivity of 0.1 N solution of acetic acid at 298K is 5.2×10^{-4} ohm^{-1} m^2 eq^{-1}. The ionic mobilities of H^+ and acetate ion at the same temperature are 36.3×10^{-8} m^2 volt^{-1} sec^{-1} and 4.23×10^{-8} m^2 volt^{-1} sec^{-1} respectively. Calculate the degree of dissociation and the dissociation constant of acetic acid?. (F = 96,500 Coulombs)

Solution

$$\Lambda^0 = F(\omega_+^o + \omega_-^o) = 96500 \times 40.53 \times 10^{-8}$$
$$= 39111.5 \times 10^{-6}\ \text{ohm}^{-1}\,\text{m}^2\,\text{eq}^{-1}$$
$$= 391.1 \times 10^{-4}\ \text{m}^2\ \text{ohm}^{-1}\,\text{eq}^{-1}$$
$$\Lambda = 5.2 \times 10^{-4}\ \text{ohm}^{-1}\ \text{m}^2\ \text{eq}^{-1}$$
$$\alpha = \frac{5.2\times10^{-4}}{391.1\times10^{-4}} = 0.0133$$
$K \cong \alpha^2 c = (0.0133)^2 \times 0.1 = 1.77 \times 10^{-5}$.

Concept Strand 28

The molar conductance of 0.01 N Na_2SO_4 solution at 298 K is 260.2 ohm^{-1} cm^2 mol^{-1}. The ionic conductance of Na^+ at this concentration is 50.0×10^{-4} ohm^{-1} m^2 mol^{-1}. (i) Calculate molar ionic conductance of SO_4^{2-} at this temperature. (ii) What is the ionic equivalent conductance of SO_4^{2-} ion and what is its mobility in the units of cm^2 volt^{-1} sec^{-1}? (iii) What will be the speed of the ion under a potential gradient of 10 volt/cm.?

Solution

$$\mu_{(Na_2SO_4)} = 260.2\ \text{ohm}^{-1}\ \text{cm}^2\ \text{mol}^{-1}$$

$$\lambda_{Na^+} = 50.0 \times 10^{-4} \times 10^4\ \text{ohm}^{-1}\ \text{cm}^2\ \text{mol}^{-1}$$
$$= 50\ \text{ohm}^{-1}\ \text{cm}^2\ \text{mol}^{-1}$$

$$\mu_{(Na_2SO_4)} = 2\lambda_{Na^+} + \lambda_{SO_4^{2-}} = 2 \times 50 + \lambda_{SO_4^{2-}}$$

(i) Molar ionic conductance of $SO_4^{2-} = \lambda_{SO_4^{2-}}$
$= 260.2 - 100 = 160.2$ ohm^{-1} cm^2 mol^{-1}

(ii) Ionic equivalent conductance of

$$SO_4^{2-} = \frac{1}{2}\lambda_{SO_4^{2-}} = \frac{160.2}{2} = 80.1\ \text{ohm}^{-1}\ \text{cm}^2\ \text{eq}^{-1}$$

Mobility of $SO_4^{2-} = \frac{80.1}{96500} = 8.30 \times 10^{-4}$ cm^2 volt^{-1} sec^{-1}

(iii) Speed of the ion $= 8.30 \times 10^{-4} \times 10 = 8.30 \times 10^{-3}$ cm^2 volt^{-1} sec^{-1}

Concept Strand 29

Calculate the transport number of OH^- ion in lithium hydroxide. Given $\Lambda_o(LiOH) = 237.2$ ohm^{-1} cm^2 mol^{-1} and the ionic conductance of OH^- ion is 198.50 ohm^{-1} cm^2 mol^{-1} at 298 K. Is this a constant for the ion? If not, give reason. [Hint: transport number is the fraction of total charge carried by an ion]

Solution

$$t^o(OH)^- = \frac{\lambda^\circ(OH^-)}{\Lambda^\circ(LiOH)} = \frac{198.50}{237.2} = 0.84$$

The t^o of an ion is not a constant unlike the case of ionic conductance. This is because, t^o depends on the nature of the other ion forming part of the electrolyte.

Concept Strand 30

100 mL of a solution of 0.001 N HCl was titrated with 0.01 N NaOH. The initial specific conductance of the acid solution was 4.260×10^{-4} ohm^{-1} cm^{-1}. After addition of 10 mL of 0.01 N NaOH, the specific conductance was found to decrease to 1.324×10^{-4} ohm^{-1} cm^{-1}. A further addition of 10 mL of 0.01 N NaOH increases the specific conductivity of the solution to 3.81×10^{-4} ohm^{-1} cm^{-1}. From these data, calculate (i) Λ of HCl (ii) Λ of NaCl (iii) Λ of NaOH (iv) Λ of ($H^+ + OH^-$). Assume that there is no volume change in these experiments.

Solution

(i) Λ of HCl $= \dfrac{1000 \times 4.26 \times 10^{-4}}{0.001} = 426.0$ ohm^{-1} cm^2

(ii) When 10 ml of 0.01 NaOH are added, the solution is exactly neutralized and 0.1 m eq of NaCl are formed in a total volume of 100 ml (no Vol. change)
C of NaCl = 0.1 m. eq/100 mL = 1 m. eq/l = 10^{-3} N

(iii) $\therefore$ Λ of NaCl $= \dfrac{1000 \times 1.324 \times 10^{-4}}{0.001}$

$= 132.4$ ohm^{-1} cm^2

(iv) When 10 ml of 0.01 N NaOH are added after the equivalence point, NaCl and NaOH are present in solution at a concentration of 1×10^{-3} each.

κ of mixture = κ due to NaCl + κ due to NaOH

$3.81 \times 10^{-4} = 1.324 \times 10^{-4} + \kappa$ due to NaOH

κ due to to NaOH $= (3.81 - 1.324) \times 10^{-4}$

$= 2.486 \times 10^{-4}$ ohm^{-1} cm^{-1}

Λ NaOH $= \dfrac{1000 \times 2.486 \times 10^{-4}}{0.001} = 248.6$ ohm^{-1} cm^{-2}

$\Lambda^0 (H^+ + OH^-) = \Lambda^0 (H^+ + Cl^-) + \Lambda^0 (Na^+ + OH^-) - \Lambda^0 (Na^+ + Cl^-)$

$= 426 + 248.6 - 132.4 = 542.2$ ohm^{-1} cm^2.

Concept Strand 31

Calculate the specific conductivity and equivalent conductivity of a 0.1 N solution of benzoic acid at 25°C given that its dissociation constant is 6.3×10^{-5} at the same temperature. The equivalent conductance at infinite dilution of HCl, NaCl and sodium benzoate are 426.2, 126.4 and 82.4 ohm^{-1} cm^2 eq^{-1} respectively.

Solution

$$\Lambda^\circ_{(C_6H_5COOH)} = \Lambda^\circ_{(HCl)} + \Lambda^\circ_{(C_6H_5COONa)} - \Lambda^\circ_{(NaCl)}$$

$= 426.2 + 82.4 - 126.4 = 382.2$ ohm^{-1} cm^2

From $\alpha^2 c \approx K$; $\alpha^2 = \dfrac{6.3 \times 10^{-5}}{0.1} = 6.3 \times 10^{-4}$; $\alpha = 2.51 \times 10^{-2}$

$\alpha = \dfrac{\Lambda}{\Lambda^0} =$; $0.0251 = \dfrac{\Lambda}{382.2}$ or $\Lambda = 9.59$ ohm^{-1} cm^2

$\kappa = \dfrac{C\Lambda}{1000} = \dfrac{0.1 \times 9.59}{1000} = 9.59 \times 10^{-4}$ ohm^{-1} cm^{-1}

Concept Strand 32

A saturated solution of magnesium hydroxide has a specific conductivity of 7.21×10^{-5} ohm^{-1} m^{-1} at 298K. Its solubility product at this temperature is 1.2×10^{-11}. Using these data calculate

(i) molar conductance μ^0 of $Mg(OH)_2$ at infinite dilution in the units of ohm^{-1} m^2 mol^{-1}.
(ii) equivalent conductance at infinite dilution in the units ohm^{-1} m^2 eq^{-1}
(iii) equivalent ionic conductance of OH^- ion given the molar ionic conductance of Mg^{2+}, $\mu^0 (Mg^{2+}) = 106.2$ ohm^{-1} cm^2 mol^{-1}

Solution

Solubility product of $Mg(OH)_2 = 4S^3 = 1.2 \times 10^{-11}$; $S^3 = 0.3 \times 10^{-11}$

$S = 1.442 \times 10^{-4}$ mole L^{-1}

(i) $\mu^\circ = \dfrac{1000 \times 7.21 \times 10^{-5}}{1.442 \times 10^{-4}} = \dfrac{7210}{14.42}$

$= 500$ ohm^{-1} cm^2 mol^{-1}

(ii) $\mu^\circ = 500 \times 10^{-4}$ ohm^{-1} m^2 mol^{-1}

$\Lambda^\circ (Mg(OH_2)) = \dfrac{500 \times 10^{-4}}{2}$

$= 250.0 \times 10^{-4}$ ohm^{-1} m^2 eq^{-1}

(iii) $\Lambda^\circ \dfrac{1}{2}(Mg^{2+}) = \dfrac{106.2}{2} = 53.1$ ohm^{-1} cm^2 eq^{-1}

$= 53.1 \times 10^{-4}$ ohm^{-1} m^2 eq^{-1}

$\therefore$ $\Lambda^\circ OH^+ = (250 - 53.1) \times 10^{-4}$

$= 196.9 \times 10^{-4}$ ohm^{-1} m^2 eq^{-1}

Concept Strand 33

The specific conductance of "ultra pure" water at 18°C is 5.0×10^{-6} ohm^{-1} m^{-1}. The ionic conductances of H^+ and OH^- ion at infinite dilution are 315.0×10^{-4} and 176.3×10^{-4} ohm^{-1} m^2 eq^{-1} respectively. Calculate the concentration of H^+ and OH^- ion in pure water and the ionic product of water at this temperature.

Solution

$$\frac{\kappa}{\Lambda^0} = \frac{5.0 \times 10^{-6}}{491.3 \times 10^{-4}} = \frac{\text{ohm}^{-1}\,\text{m}^{-1}}{\text{ohm}^{-1}\,\text{m}^2\,\text{eq}^{-1}}$$

$$= 0.0102 \times 10^{-2}\ \text{eq m}^{-3}$$

$$= 0.0102 \times 10^{-2} \times 10^{-3}\ (\text{m}^3\,\text{dm}^{-3} \times \text{eq m}^{-3})$$

$$= 1.02 \times 10^{-7}\text{eq dm}^{-3}$$

i.e., conc. of H^+ = conc. of OH^-

$$= 1.02 \times 10^{-7}\ \text{eq dm}^{-3}$$

or $K_w = [H^+][OH^-] = (1.02 \times 10^{-7})^2$

$$= 1.04 \times 10^{-14}\ \text{eq}^2\,\text{dm}^{-6}(\text{mol}^2\,\text{dm}^{-6})$$

GALVANIC CELLS

Introduction

One of the important aspects of the subject of electrochemistry is its close connection to electrical energy. Galvanic cells are sources of electrical energy in that the energy of a studied chemical reaction is converted into useful electrical energy. Although the Daniel cell is one of the earliest galvanic cells studied, rapid studies have been made in the development of galvanic cells of practical value in recent years, the most common of them being the lead storage cell, followed by the more recent developments involving various types of fuel cells. The subject is therefore, fascinating and it is pertinent to understand the fundamentals of this area which is undertaken below.

Definition and general description of galvanic cells

A galvanic cell essentially consists of two electrodes placed in a solution of an electrolyte or sometimes in different electrolytes which are separated by a porous partition such that they are in electrical contact. When they are connected by a metallic wire, a current flows. At the interface between the metal and the solution, a potential difference exists which is known as the electrode potential and the E.M.F of the cell is a composite of the two electrode potentials with their proper signs as detailed subsequently.

Galvanic cells may be broadly classified into two categories (i) chemical cells (ii), concentration cells.

In a chemical cell, there is an overall chemical reaction when the cell operates. For example, in Daniel cell, the reaction

$$Zn + CuSO_4 \rightarrow ZnSO_4 + Cu \qquad \text{— (50)}$$

takes place.

The energy of the chemical reaction provides the electrical energy delivered by the cell. In a concentration cell, on the other hand, there is no net chemical reaction but the electrical energy is derived as a result of the energy change arising out of transfer of material from one concentration to the other.

Apart from the classification given above, galvanic cells may be classified from a thermodynamic view point as (i) irreversible cells (ii) reversible cells.

In an irreversible cell a spontaneous chemical reaction takes place in the cell even though the terminals at the two half cells are not connected. Thus such cells are thermodynamically irreversible.

An example of such a cell may be given as Zn/dil. H_2SO_4/Cu where the reaction between zinc and dil.H_2SO_4 takes place spontaneously eventhough the zinc and copper terminals are not connected.

In a reversible cell, a reaction takes place in the cell only on connecting the terminals and the cell may be operated under thermodynamically reversible conditions by drawing extremely small currents through the cell. We are mainly concerned with a study of galvanic cells under reversible conditions as mentioned above.

REVERSIBLE ELECTRODES

The electrodes employed in a reversible cell must themselves be reversible and there are mainly three types of reversible electrodes that one meets with in the study of galvanic cells.

Reversible electrodes of the first kind

This type of electrode consists of a metal in contact with a solution of its ion, example. a copper electrode in contact with a solution of copper sulphate or silver electrode in contact with a solution of silver ions. Gas electrodes involving gases like hydrogen or chlorine in contact with their ions such as H^+ or Cl^- ions and in which an inert metal is placed are also examples of this category. The reaction at such electrodes, for example, Ag in contact with $AgNO_3$, H_2 gas in contact with H^+ ion may be written as

$$Ag^+ + e^- \rightarrow Ag \qquad \text{— (51)}$$

$$H^+ + e^- \rightarrow \frac{1}{2} H_{2(g)} \qquad \text{— (52)}$$

This kind of electrode is reversible to the cation.

Reversible electrodes of the second kind

Electrodes of this type involve a metal which is coated with a sparingly soluble salt of this metal and this combination is kept in a solution containing a soluble salt of this anion.

The electrode reaction in the case of Ag, AgCl electrode may be written as

$$AgCl_{(s)} + e^- \rightarrow Ag + Cl^- \qquad \text{— (53)}$$

Another example of this kind of electrode is the well known electrode, viz., calomel electrode. The electrode reaction in this case

$$Hg_2Cl_{2(s)} + 2e^- \rightarrow 2Hg_{(l)} + 2Cl^- \qquad \text{— (54)}$$

This type of electrode is reversible to the anion.

Reversible electrode of third kind

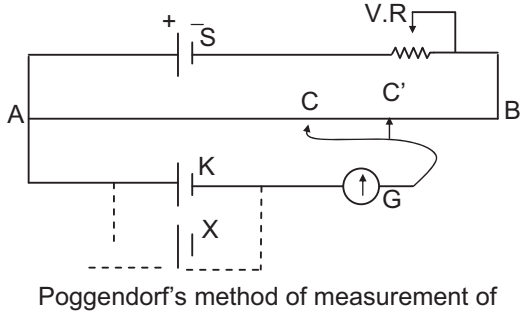
In this system, an unattackable metal like platinum (which acts as the electrode) is placed in a solution consisting of an appropriate oxidized and reduced form of a redox pair like Fe^{3+}/Fe^{2+}, Ce^{4+}/Ce^{3+}, $Fe(CN)_6^{3-}/ Fe(CN)_6^{4-}$ etc. The electrode reaction in this case is for ex. considering Fe^{3+}/Fe^{2+} electrode,

$$Fe^{3+} + e^- \rightarrow Fe^{2+} \qquad \text{— (55)}$$

Both the oxidized and reduced species are necessary for the functioning of this electrode. Most galvanic cells that we come across in the study of this area are obtained by combining any of two types of reversible electrodes, described above.

MEASUREMENT OF THE EMF OF A GALVANIC CELL

The EMF of a galvanic cell is determined by a method known as Poggendorf's compensation method. It has the advantage of giving the EMF under conditions when practically no current is drawn through the cell, i.e., virtually under open circuit conditions. The EMF so obtained is known as the "reversible EMF" of the cell because the cell may be expected to behave reversibly only when infinitesimally small currents are drawn through the cell. The following circuit is employed for measuring the EMF of a galvanic cell.



Poggendorf's method of measurement of EMF of a galvanic cell

Fig. 7.7

S: source of EMF; V.R: variable resistance AB: potentiometer wire; K: standard cell (such as a Weston cadmium cell); X: galvanic cell whose EMF is under determination; G: galvanometer; C: point of contact at zero current when standard cell is in circuit; C': point of contact at zero current when galvanic cell X is in circuit.

The important relation which enables the evaluation of the EMF, E_X, of the galvanic cell X is given by

$$\frac{E_K}{E_X} = \frac{AC}{AC'} \Rightarrow E_X = E_K \times \frac{AC'}{AC} \qquad \text{— (56)}$$

All the quantities E_K, i.e., EMF of the standard cell, AC and AC', the lengths of the potentiometric wire under zero current condition when the standard cell, K, and unknown cell X, respectively, in circuit are known. Hence, E_X may be calculated.

The following points must be borne in mind for the success of the EMF measurements by this method:

(i) The potentiometric wire is assumed to have a uniform cross sectional area.

(ii) The EMF of standard cell, E_K, must be constant and of known value at the temperature of measurement. For e.g., a standard Weston cadmium cell is commonly used for this purpose as it provides a steady and constant EMF of 1.0183 volts at 20°C and its temperature coefficient, too, is quite small.

(iii) In the measurement of EMF by this method, the standard cell or the unknown cell (E_X) must be connected in such a way that their EMF opposes the EMF of the source cell, i.e., the positive end of the potentiometer must be connected to the +ve end of the galvanic cell and the –ve end of galvanic cell should be connected to the negative end of the potentiometer.

Symbols to be used in the representation of a galvanic cell

In the representation of a galvanic cell, a metal-solution contact is indicated by a vertical line (I), metal, solid or metal, gas contact is represented by a comma (,), a liquid-liquid separation is shown by dotted lines, (⋮) the elimination of liquid junction potentials (by introducing a salt bridge) is indicated by two vertical lines (‖).

Gibbs free energy and enthalpy changes in galvanic cells

(i) From the second law of thermodynamics, it may be stated that the decrease in Gibbs free energy change, associated with a process under constant pressure and temperature conditions, is equal to all forms of reversible work excluding the P–V work, i.e.,

$$-\Delta G_{T,P} = W_{rev} - P\Delta V \qquad \text{— (57)}$$

The electrical work done by a galvanic cell operating at constant temperature and pressure under reversible condition may be related to the decrease in free energy, $\Delta G_{T,P}$, associated with the reaction in the cell as follows.

Electrical work = reversible EMF × quantity of electricity passing through the cell — (58)

$$= E_{rev} \times nF \text{ volt – coulomb or J} \qquad \text{— (59)}$$

"n" represents the number of electrons involved in the cell reaction and F = faraday. From equations (57) and (59), one may write

$$-\Delta G_{T,P} = nFE_{rev} \qquad \text{— (60)}$$

Equation (60) is consistent with the thermodynamic condition that ΔG becomes negative when E is positive for a spontaneous process involving a cell reaction.

It must be stressed that E of equation (60) is an intensive property while ΔG is extensive in nature and it is reflected by the appearance of "n", the number of electrons involved in the cell reactions, i.e., number of faradays passing through the cell.

It may also be added that it is necessary to follow a convention to write the cell reaction and give a proper sign to the EMF associated with the cell reaction. Generally, the positive sign of the cell EMF (i.e., for the spontaneous reaction) is associated with a cell representation in which the oxidation electrode (i.e., the electrode at which electrons are released) is written at the left hand side and the reduction electrode (i.e., the electrode at which electrons are consumed) is written at the right hand side.

At a later stage, a more detailed consideration of the half cell potentials of individual electrodes and the signs (+ve or –ve) associated with these electrodes is undertaken.

(ii) Gibbs-Helmholtz equation applied to galvanic cells.

The enthalpy and entropy changes associated with a cell reaction in galvanic cell may be determined by employing the Gibbs-Helmholtz equation in the following manner.

$$\Delta G = \Delta H - T\Delta S \qquad \text{— (61)}$$

$$\Delta G = \Delta H + T\frac{\partial(\Delta G)}{\partial T} \left(\because \Delta S = -\frac{\partial(\Delta G)}{\partial T} \right) \qquad \text{— (62)}$$

Considering ΔS,

$$\Delta S = -\left(\frac{\partial(\Delta G)}{\partial T}\right)_P = +nF\left(\frac{\partial E}{\partial T}\right)_P \quad (\because \Delta G = -nFE)$$

Equation (63) enables us to calculate ΔS of the reaction in the cell from the knowledge of the temperature coefficient of the EMF of the cell.

Rewriting equation (61) as ΔH = ΔG + TΔS and substituting ΔG and ΔS from equations (60) and (63), we get,

$$\Delta H = -nFE + nFT\left(\frac{\partial E}{\partial T}\right)_P \quad \text{— (64)}$$

$$\Delta H = -nF\left[E - T\left(\frac{\partial E}{\partial T}\right)_P\right] \quad \text{— (65)}$$

Equation (65) enables the determination of ΔH, provided E at the given temperature and the temperature coefficient of the cell are known.

Relation between EMF and concentrations of the constituents

The Gibbs free energy change accompanying a reaction depends on the concentration of various reactants and products according to

$$\Delta G = \Delta G^\circ + RT\ln\frac{C_M^m \times C_N^n \times \ldots}{C_A^a \times C_B^b \times \ldots} \quad \text{— (66)}$$

for the general reaction,

$$aA + bB + \ldots \rightleftharpoons mM + nN + \ldots \quad \text{— (67)}$$

where, A, B ... are reactants, M, N…. are the products and a, b, m, n are the stoichiometric coefficients. Equation (66) is known as *van't Hoff's reaction isotherm.* If reaction (67) occurs in a galvanic cell for the passage of "n" Faradays of electricity, then

$$\Delta G = -nFE \text{ and } \Delta G^\circ = -nFE^\circ \quad \text{— (68)}$$

Substituting for ΔG and ΔG° from equation (68) into equation (66), we get the equation.

$$E = E^\circ - \frac{RT}{nF}\ln\frac{C_M^m \times C_N^n \times \ldots}{C_A^a \times C_B^b \times \ldots} \quad \text{— (69)}$$

The equation (69) is known as *Nernst equation* and it relates the EMF of a cell, E, with the standard EMF, E°, and the concentration of the reactants and products. The standard EMF, E°, is the EMF of the cell when the concentration of reactants and products are unity. It is a constant for the given cell at a definite temperature. Two examples are considered as illustration of equation (69).

Examples

(i) Consider the cell

$$Pt, H_2\ (g, 1\ atm) \mid HCl\ (M) \mid AgCl\ (s), Ag \quad \text{— (70)}$$

The cell reaction for the passage of one Faraday may be written as

$$\frac{1}{2}H_{2(g)} + AgCl_{(s)} \rightarrow HCl_{(l)} + Ag_{(s)} \quad \text{— (71)}$$

and EMF of the cell is equal to

$$E = E^\circ - \frac{RT}{F}\ln\frac{C_{HCl} \times C_{Ag}}{P_{H_2}^{1/2} \times C_{AgCl}} \quad \text{— (72)}$$

The above equation may be simplified by considering (i) the concentration of all solid phases as unity (as they are constant any way) (ii) partial pressure of a gas, being a measure of the concentration, occurs in equation (71) and if its partial pressure is one atmosphere its concentration is unity. Equation (72) thus simplifies to

$$E = E^\circ - \frac{RT}{F}\ln C_{HCl} = E^\circ - \frac{RT}{F}\ln M \quad \text{— (73)}$$

In effect, the EMF of the above cell depends on the concentration of the acid taken in the cell.

(ii) Consider the cell

$$Zn/ZnSO_4(M_1)//NiSO_4(M_2)/Ni \quad \text{— (74)}$$

The reaction for the passage of two Faradays, i.e., for n = 2

$$Zn + NiSO_4(M_2) \rightarrow ZnSO_4(M_1) + Ni \quad \text{— (75)}$$

The Nernst equation for the above cell reaction is

$$E = E^\circ - \frac{RT}{2F}\ln\frac{C_{ZnSO_4}}{C_{NiSO_4}} = E^\circ - \frac{RT}{2F}\ln\frac{M_1}{M_2} \quad \text{— (76)}$$

SINGLE ELECTRODE POTENTIALS

The EMF of a cell can be written as a net contribution of the potentials of the respective half cells. For example, the EMF of the Daniel cell given by.

$$Zn \mid ZnSO_4(M_1) \parallel CuSO_4(M_2) \mid Cu \qquad — (77)$$

may be written as the difference of two half cell potentials, i.e.,

$$E = E_R - E_L = E_{Cu^{2+},Cu} - E_{Zn^{2+},Zn} \qquad — (78)$$

where, E_R and E_L are the half cell potentials of the right and left electrodes. Although E is measurable experimentally, the values of E_R and E_L are not accessible for direct measurement. Thus it is not possible to measure experimentally the potential of a single half cell like that Ag^+/Ag or Zn^{2+}/Zn etc. Only the difference in potential between two electrodes or two half cells can be measured. However, by arbitrarily assigning a value of zero to the potential of a chosen electrode system, it is possible to obtain individual half cell potentials.

The electrode commonly chosen for this purpose is known as the standard hydrogen electrode (SHE) which consists of hydrogen gas at one atmosphere pressure bubbled into a solution of hydrogen ions maintained at a concentration of one molar and into which a Pt electrode (usually coated with a thin layer of finely divided Pt black to permit rapid attainment of equilibrium) is dipped for electrical contact. By convention its potential is taken as zero at all temperatures, i.e.,

$$Pt, H_2\,(g, 1\ atm)/H^+\,(c = 1)\ E = 0\ V \qquad — (79)$$

at all temperatures and the electrode reaction may be written as

$$H^+\,(c = 1) + e^- \rightarrow \frac{1}{2} H_2\,(g, 1\ atm) \qquad — (80)$$

If any electrode, say of metal, metal ion type (M^{n+},M) is combined with a SHE to form a complete cell: SHE || M^{n+}(c) | M (81) and its EMF is measured, this is referred to as the electrode potential M^{n+},M system (w.r.t SHE) as the potential of SHE is zero volts by convention. The cell reaction in cell (equation 81) for the passage of "n" Faradays may be written as

$$\frac{n}{2} H_2\,(g, 1\ atm) + M^{n+} \rightarrow nH^+ + M \qquad — (82)$$

The E_{cell} may be written as $E_{cell} = E_R - E_L$

$$= E_{M^{n+}/M} - E^\circ_{H^+,H_2} = E_{M^{n+},M} \left(\because E_{H^+,H_2} = 0\right) \qquad — (83)$$

The Nernst equation for the above cell (equation 81) may be written as

$$E = E^o - \frac{RT}{nF} \ln \frac{C^n_{H^+} \times C_M}{P^{\frac{n}{2}}_{H_2} \times C_{M^{n+}}} = E^o - \frac{RT}{nF} \ln \frac{C_M}{C_{M^{n+}}} \qquad — (84)$$

on simplification and writing equation (84) in a general way

$$E_{el} = E^o_{el} - \frac{RT}{nF} \ln \frac{[\text{reduced state}]}{[\text{oxidised state}]} \qquad — (85)$$

Equation (85) is a general equation for the electrode potential (E_{el}) of any half cell in terms of the standard electrode potential $\left(E^o_{el}\right)$ and the concentration of the reduced and oxidized species. Since the reaction at the metal electrode is a reduction reaction, it is also referred as the reduction potential of the system. $\left(E^o_{el}\right)$ is the standard reduction potential of the system and is a characteristic of the system and varies with temperature and solvent. It corresponds to the potential of the electrode under the condition that $C_{M^{n+}}$ = 1.0 M measured versus SHE.

Sign of the electrode potential and its importance

The sign of the electrode potential (+ or –) is always the sign or polarity of the electrode when combined with SHE. It is the experimentally measured (or observed) sign of the EMF of the cell (potentiometrically) with the SHE represented on the left and the electrode in question on the right. It must be emphasized that the polarity of the electrode versus SHE measured potentiometrically is unique and not variable. It is note worthy that the sign of the electrode potential is identical with the polarity of the right hand electrode. The significance of the sign will be illustrated by some examples.

Examples

(i) The standard electrode potential of Li^+/Li half cell vs SHE is –3.04 V. This implies that the polarity of Li^+/Li half cell is negative w.r.t SHE when its e.m.f is measured potentiometrically and the cell as per the above convention must be represented as

$$Pt, H_2(g, 1\ atm)/H^+\,(C = 1)//Li^+, Li\ E = -3.04\ V$$

$$E_{el} = E_R - E_L = E_{Li^+/Li} - E^o_{SHE} = -3.04V$$

The reaction in the cell is $\frac{1}{2} H_2(g) + Li^+ \rightarrow Li + H^+$ E = –3.04 V — (86)

Which is also in agreement with the convention in writing cell reaction. Since the EMF is –ve, reaction (eqn.86) is not spontaneous but the reverse reaction actually takes place in the cell, i.e., the metal is a better reducing agent than hydrogen.

(ii) The electrode potential of Cu^{2+}/Cu electrode is +0.340 V under standard condition (i.e., when $[Cu^{2+}] = 1.0M$), i.e., for the cell

$$Pt, H_2(g, 1atm)/H^+(C = 1) // Cu^{2+}(C = 1)/Cu \quad — (87)$$

the reaction in the cell is $H_{2(g)} + Cu^{2+} \rightarrow Cu + 2H^+$ — (88)

and since its EMF (in this case $E^o = 0.340$ V) is +ve, the reaction (eqn 88) is spontaneous, i.e., hydrogen gas reduces Cu^{2+} ion to metallic copper, i.e., hydrogen gas is a better reducing agent than copper.

Standard electrode (reduction) potentials

The standard electrode potential is the potential of the given system vs SHE under the condition when the concentration of metal ions is equal to unity and it is a constant for the given system but varies with temperature and solvent. E.g., the standard electrode potential of Zn^{2+}/Zn system is the EMF of the cell

$$Pt, H_2(g, 1\ atm) / H^+(C = 1) // Zn^{2+}(C = 1) / Zn \quad — (89)$$

is given as –0.760 volts when $C_{Zn^{2+}} = 1.0$ M. The reaction in the cell (as per the convention and it is non-spontaneous) is

$$H_2(g) + Zn^{2+} \rightarrow 2H^+ + Zn \quad — (90)$$

and the expression for the electrode potential is

$$E = E^o_{el} - \frac{RT}{2F} \ln \frac{C_{Zn}}{C_{Zn^{2+}}} = E^\circ_{el} + \frac{RT}{2F} \ln C_{Zn^{2+}} \quad — (91)$$

Since $C_{Zn^{2+}} = 1$ M, $E = E^0_{Zn^{2+}/Zn}$

Although there are several methods for determining the standard potentials, this aspect will not be considered here. However, the importance of standard electrode potentials and their application will be considered. The standard electrode potentials are generally listed starting with the most negative (e.g., alkali metal system) to the most positive electrode system. (such as Cu^{2+}/Cu, Ag^+/Ag, etc.)

Important points regarding standard potential

(i) The electrode having the more negative potential is more reactive compared to the electrode lower down in the series with a less negative electrode potential. For example Na^+/Na system ($E^o = -2.70$ volts) is more reactive compared to Zn^{2+}/Zn system which has a less negative electrode potential, $E^\circ = -0.760$V, i.e., Zn^{2+}/Zn is more noble compared to Na^+/Na.

(ii) In general, a metal having a more negative standard electrode potential displaces a metal having the less negative potential down the series. E.g., zinc displaces copper from a solution of cupric ions under standard condition. This conclusion applies to any system as we go down the electrochemical series. The reaction will be more complete (such as the displacement) if there is larger difference in standard potentials. In fact, it is possible to determine equilibrium constants of many reactions using standard potential data.

REFERENCE ELECTRODES

Although the standard hydrogen electrode (SHE) is the primary choice for the determination of standard potentials of various electrodes, it is cumbersome to setup this electrode and more importantly it is susceptible to trace impurities and thereby the results will have error. Therefore, alternate reference electrodes are used and one of the most commonly employed is known as the *calomel electrode*. This electrode is an electrode of the second kind and it is reversible to anions. Depending upon the concentration of the chloride ion used, the potential of the electrode (measured vs SHE) will vary. The various types of calomel electrodes and their potential values are listed below.

Table 7.2

	Type of electrode	Representation	Electrode potential at 298 K
1	0.1 N Calomel electrode	$Hg, Hg_2Cl_{2(s)} \mid KCl$ (0.1 N)	+0.334 V
2	1 N Calomel electrode	$Hg, Hg_2Cl_{2(s)} \mid KCl$ (1 N)	+0.280V
3	Saturated Calomel electrode	$Hg, Hg_2Cl_{2(s)} \mid KCl_{(satd)}$	+0.242 V

The function of the anion reversible calomel electrode may be explained with a diagram (simplified)

(i) Electrons enter mercury through connecting wire. Hg becomes negatively charged, transfers the electrons to Hg_2Cl_2.
$Hg_2Cl_2 + 2e \rightarrow 2Hg + 2Cl^-$. Thus Cl^- ions are released into the solution, Hg adds on to Hg.

(ii) Electrons flow away from Hg through the connecting wire Hg gets positively charged, attracts Cl^- ions from the solution forming solid Hg_2Cl_2. Thus Cl^- ions are withdrawn from the solution.

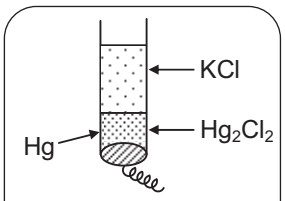


Fig. 7.8

The electrode reaction for the half cell is already given. The electrode potential for the cell reaction may be expressed as $E = E^o - \frac{RT}{2F}\ln [Cl^-]^2 = E^o \frac{-RT}{F}\ln [Cl^-]$ — (92)

Value of $\frac{RT}{nF}\ln$

It is important to use the correct value of R in calculating $\frac{RT\ln}{nF}$ so that the dimensions of this number equation are satisfied. This term must have units of "volts" which is satisfied when R is substituted in J deg^{-1} mol^{-1} i.e., V C deg^{-1} mol^{-1} which is 8.314 J deg^{-1} $mole^{-1}$

$$\therefore \frac{RT}{nF}\ln = \frac{8.314\times298\times2.303}{96{,}500\times n}\log = \frac{0.0591}{n}\log \qquad \text{— (93)}$$

The value will vary if the temperature has a value other than 298K.

Calculation of cell EMF for a typical reaction

Let us consider the cell

$$Pt/Sn^{2+}\ (0.1\ M),\ Sn^{4+}\ (0.2M)\ //\ Fe^{3+}\ (0.1\ M),\ Fe^{2+}\ (0.05\ M)\ /\ Pt \qquad \text{— (94)}$$

with $E^o_{Fe^{3+},Fe^{2+}} = 0.771$ V and $E^o_{Sn^{4+},Sn^{2+}} = 0.15$ V at 298K

1st step

$$E_{el} = E_R - E_L = E_{Fe^{3+},Fe^{2+}} - E_{Sn^{4+},Sn^{2+}} \qquad \text{— (95)}$$

2nd step

Write the electrode reaction of R.H.E. and L.H.E. in such a way that the electrons are on the reactants side (i.e., on left) Balance the number of electrons in both equation and subtract the reaction of L.H.E. from that of R.H.E to get the cell reaction.

$$2(Fe^{3+}+e^- \rightarrow Fe^{2+}) \qquad \text{— (96)}$$

$$Sn^{4+} + 2e \rightarrow Sn^{2+} \qquad \text{— (97)}$$

Eqn(96) – Eqn.(97) gives $Sn^{2+} + 2Fe^{3+} \rightarrow Sn^{4+} + 2Fe^{2+}$ — (98)

When balancing the number of electrons, the EMF of the cell (or the half cell potentials) should not be multiplied by that factor as E is an intensive property. Also the Nernst equation should not be multiplied by the same factor when expressing the EMF in terms of the concentration of reactants and products.

3rd step

Use the Nernst equation and write the expression for cell EMF employing the cell reaction (eqn (98))

$$Ecell = E^o_{cell} - \frac{RT}{2F}\ln\frac{[Sn^{4+}][Fe^{2+}]^2}{[Sn^{2+}][Fe^{3+}]^2} \qquad \text{— (99)}$$

$$= (E^o_R - E^o_L) - \frac{RT}{2F}\ln\frac{[Sn^{4+}][Fe^{2+}]^2}{[Sn^{2+}][Fe^{3+}]^2} \qquad \text{— (100)}$$

$$= (0.771-0.15) - \frac{0.0591}{2}\log\frac{(0.2)\times(0.05)^2}{(0.1)\times(0.1)^2} \qquad \text{— (101)}$$

$$= 0.621 - \frac{0.0591}{2}\log 0.5 = 0.63V$$

The same cell reaction (eqn. 98) and the expression for cell EMF (eqn. 99) can be obtained by writing the equations for electrode potentials and substituting the relevant data for the concentrations of the various species under reactants and products.

Application of electrode potentials

There are several applications of electrode potential data such as (1) in the determination of the concentration of an unknown ion in solution (2) in the determination of the equilibrium constant of a reaction (3) in the determination of solubility products of sparingly soluble salts (4) in the determination of dissociation constants of weak acids or weak bases (5) in the determination of ionic product of water (6) in the determination of equivalence point in oxidation-reduction titration, acid-base titration, precipitation titration, etc.

Some examples are given to illustrate the applications.

Determination of the concentration of an ion

(i) A galvanic cell is constructed in which one half cell consists of the unknown solution into which an electrode reversible to the ion under consideration is dipped, while the other half cell consists of a reference electrode like SCE. A salt bridge (which is an inverted U-tube containing a saturated solution of KCl or sometimes NH_4NO_3 set as a gel in agar is interposed between the two half cells not only to prevent mixing of solutions of the two half cells but also to eliminate what is known as liquid junction potential at the interface. It also serves to maintain electrical contact between the two half-cells. As an example, the determination of the concentration of silver ions in an unknown solution of silver nitrate will be considered.

The following cell may be used.

$$Ag/AgNO_3\ (C = ?)\ //SCE \qquad — (102)$$

$$E_{el} = E_R - E_L = E_{SCE} - (E^\circ_{Ag} + 0.0591 \log C_{Ag^+})$$

(Temp. of 25°C is assumed)

$$= 0.242 - (0.799 + 0.0591 \log C_{Ag^+})$$

$$= -0.557 - 0.0591 \log C_{Ag^+}$$

$$\Rightarrow \log C_{Ag^+} = \frac{-0.557 - E_{cell}}{0.0591} \qquad — (103)$$

C_{Ag^+} can be calculated by measuring E_{el} and giving it the proper sign before substituting in the above equation. It must be emphasized that an appropriate electrode must be used for the determination of the concentration of the ion under consideration. For example for determining the concentration of Cu^{2+} ion, Cu^{2+}/Cu is a suitable electrode; for H^+ ion, Pt, H_2 (g, 1 atm)/H^+ is a suitable electrode and so on.

(ii) Standard electrode potentials can be used to determine the standard Gibbs free energy changes (ΔG°) and equilibrium constants associated with cell reactions.

Consider for e.g., the reaction

$$Sn^{2+} + 2Fe^{3+} \rightarrow Sn^{4+} + 2Fe^{2+} \qquad — (104)$$

which is used to determine the concentration of Fe^{3+} by reduction with stannous chloride.

The cell in which the above reaction occurs may be formulated as

$$Pt \mid Sn^{2+}, Sn^{4+} \parallel Fe^{3+}, Fe^{2+} \mid Pt$$

$$E^\circ_{el} = E^\circ_{Fe^{3+},Fe^{2+}} - E^\circ_{Sn^{4+},Sn^{2+}} = 0.771 - (+0.15) = 0.621\ V \qquad — (105)$$

$$E^\circ_{el} = \frac{RT}{nF}\ln K = \frac{0.0591}{n}\log K = \frac{0.0591}{2}\log K \qquad — (106)$$

$$\log K = \frac{0.621 \times 2}{0.0591} = 21.015 \text{ or } K = 1.035 \times 10^{21}$$

(iii) The solubility products of sparingly soluble salts like AgCl, AgBr etc., may be determined by using standard potentials of suitable half cells. As an example, the determination of solubility product of AgCl will be considered. The solubility product equilibrium of AgCl may be represented as

$$AgCl_{(s)} \rightleftharpoons Ag^+_{(aq)} + Cl^-_{(aq)}$$

Considering the $Ag^+ \mid Ag$ and $Ag, AgCl \mid Cl^-$ half cells and combining their half cell reactions suitably the reaction pertaining to solubility equilibrium may be obtained as follows

$$AgCl_{(s)} + e \rightarrow Ag_{(s)} + Cl^-_{(aq)} \quad (a)$$

$$Ag^+_{(aq)} + e \rightarrow Ag_{(s)} \quad (b)$$

$$(a) - (b) \qquad AgCl_{(s)} \rightarrow Ag^+_{(aq)} + Cl^-_{(aq)} \qquad — (107)$$

and the corresponding cell is

$$Ag \mid Ag^+ \parallel Cl^- \mid AgCl_{(s)}, Ag$$

$$\Rightarrow E^\circ_{el} = +0.222 - 0.799 = -0.577\ V$$

$$E^\circ_{el} = \frac{RT}{nF}\ln K = \frac{0.0591}{1}\log K$$

$$\log K = -\frac{0.577}{0.0591} = -9.76 = \overline{10}.24$$

$$K = 1.74 \times 10^{-10}$$

(iv) An important application of electrode potentials or potential measurements in general lies in the determination of equivalence points in a variety of titrations, acid Vs base, oxidation-reduction titration like Fe^{++} vs $KMnO_4$ or $K_2Cr_2O_7$ and precipitation reactions involving $AgNO_3$ vs NaCl, etc. For this purpose, an electrode which is responsive to the ion being titrated (called the indicator electrode) is combined with a reference electrode like SCE to form a cell and the changes in the potential of the indicator electrode are followed as a function of the volume of the added reagent. The equivalence point is indicated by a large change in the potential of the indicator electrode as the titrant is added.

CONCENTRATION CELLS

Concentration cells have already been defined earlier. They are of two types (i) electrode type concentration cells (ii) electrolyte type concentration cells.

Electrode type concentration cells

In cells of this type, two similar electrodes at different concentrations are dipped into the same solution. Two hydrogen electrodes at different gas pressure immersed in the same solution of H^+ ion constitutes an electrode type concentration cell.

This may be represented as

Pt, $H_2(p_1)$ | solution of H^+ ion (say) HCl | $H_2(p_2)$, Pt — (108)

The EMF of such a cell can be related to the partial pressure of H_2 gas in the half cells in the following way.

$$\text{At R.H.E } H^+ + e^- \rightarrow \frac{1}{2} H_2(p_2)$$

$$\text{At L.H.E } H^+ + e^- \rightarrow \frac{1}{2} H_2(p_1)$$

The net reaction is $\frac{1}{2} H_2(p_1) \rightarrow \frac{1}{2} H_2(p_2)$ — (109)

$$E = E_R - E_L = \left(E_R^\circ - \frac{RT}{F} \ln p_2^{1/2}\right) - \left(E_L^\circ - \frac{RT}{F} \ln p_1^{1/2}\right)$$

($E_R^o = E_L^o$ because the electrode is the same)

$$E = \frac{RT}{F} \ln \frac{p_1^{1/2}}{p_2^{1/2}} = \frac{RT}{F} \ln \left(\frac{p_1}{p_2}\right)^{1/2} \quad \text{— (110)}$$

The EMF of the cell depends on the partial pressure of the gas in the two half cells as shown by equation (110).

Another example of this type of concentration cell is based on amalgams of different concentration of a given metal as indicated by

$$Zn(Hg)(C_1)/ZnSO_4/Zn(Hg)(C_2) \quad \text{— (111)}$$

Following the same procedure as in earlier case, the expression for the e.m.f of cell (eqn. 111) may be obtained. Cells of this type are sometimes referred as amalgam type concentration cells.

Electrolyte type concentration cells

Let us consider a cell composed of two electrodes of the same metal, M (or a gas electrode like the one involving H_2 gas) in contact with two solutions containing different concentrations of the metal ion (M^{n+}) say C_1 and C_2 respectively. The cell may be represented as

$$M / M^{n+}(C_1) // M^{n+}(C_2)/M \quad \text{— (112)}$$

(or Pt, H_2(g, 1atm)/$H^+(C_1)$ // $H^+(C_2)/H_2$(g, 1atm), Pt) — (113)

writing the half cell reactions in the usual manner

$M^{n+}(C_2) + ne^- \rightarrow M$ (a)

$M^{n+}(C_1) + ne^- \rightarrow M$ (b)

(a) – (b) gives $M^{n+}(C_2) \rightarrow M^{n+}(C_1)$ — (114)

or $$E = E_R - E_L = \left(E_M^o + \frac{RT}{nF} \ln C_2\right) - \left(E_M^o + \frac{RT}{nF} \ln C_1\right)$$

$$E = \frac{RT}{nF} \ln \frac{C_2}{C_1} \quad \text{— (115)}$$

The EMF of the cell thus depends on the ratio of the concentrations of the metal ion in the two half cells.

CONCEPT STRANDS

Concept Strand 34

Write the half cell reactions and the overall cell reaction for the passage of one Faraday through the following cells;

(i) Pt,H_2(g, 1 atm)/HBr(C_1)/$AgBr_{(s)}$, Ag

(ii) Hg, $Hg_2SO_{4(s)}$/$CuSO_4$(C)/Cu

(iii) Cd(Hg)(C_1) $CdSO_4$(M)/Cd(Hg)(C_2)

Solution

(i) $AgBr_{(S)} + e^- \rightarrow Ag + Br^-$

$H^+ + e^- \rightarrow \frac{1}{2} H_2$

overall reaction: $\frac{1}{2} H_2$ (g, 1 atm) + $AgBr_{(S)} \rightarrow Ag + H^+ + Br^-$

(ii) $\frac{1}{2}Cu^{2+} + e^- \rightarrow \frac{1}{2}Cu$

$\frac{1}{2}Hg_2SO_4 + e^- \rightarrow Hg + \frac{1}{2}SO_4^{2-}$

overall reaction: $\frac{1}{2}Cu^{2+} + \frac{1}{2}SO_4^{2-} + Hg \rightarrow \frac{1}{2}Cu + \frac{1}{2}Hg_2SO_4$

(iii) $\frac{1}{2}Cd^{2+}(Hg)(C_2) + e^- \rightarrow \frac{1}{2}Cd$

$\frac{1}{2}Cd^{2+}(Hg)(C_1) + e^- \rightarrow \frac{1}{2}Cd$

overall reaction: $\frac{1}{2}Cd^{2+}(C_2) \rightarrow \frac{1}{2}Cd^{2+}(C_1)$

Concept Strand 35

Formulate the galvanic cell in which the following reactions occur.

(i) $\frac{1}{2}H_{2(g)} + \frac{1}{2}I_{2(S)} \rightarrow HI\ (soln)$

(ii) $AgCl_{(S)} \rightarrow Ag^+(soln) + Cl^-(soln)$

(iii) $\frac{1}{2}Zn_{(S)} + \frac{1}{2}Hg_2Cl_{2(S)} \rightarrow \frac{1}{2}ZnCl_2(soln) + Hg$

Solution

(i) $Pt,H_2\ (g, 1\ atm) \mid HI(soln) \mid I_{2(s)}, Pt$

(ii) $Ag \mid Ag^+ \parallel Cl^- \mid AgCl_{(s)}, Ag$

(iii) $Zn/ZnCl_2/Hg_2Cl_{2(s)}, Hg$

Concept Strand 36

The EMF of the cell

Pb, $PbCl_{2(s)}$/KCl solution/$AgCl_{(s)}$, Ag is 0.490 volts at 25°C.

(i) Write the overall cell reaction for the passage of two faraday of electricity.

(ii) Calculate ΔG of the reaction.

(iii) If the entropy change for the reaction is −36.00 J deg K^{-1}, calculate ΔH of the reaction and the temperature coefficient of the EMF of the cell.

Solution

(i) $Pb_{(s)} + 2AgCl_{(s)} \rightarrow PbCl_{2(s)} + 2Ag_{(s)}$

(ii) $\Delta G = -nFE = -2 \times 96{,}500 \times 0.49 = -9.46 \times 10^4 J.$

(iii) $\Delta S = nF\left(\frac{\partial E}{\partial T}\right)_P = 2 \times 96500 \times \left(\frac{\partial E}{\partial T}\right)_P$

$\left(\frac{\partial E}{\partial T}\right)_P = \frac{-36}{2 \times 96500} = -1.865 \times 10^{-4}$ volts deg K^{-1}

$\Delta H = \Delta G + T\Delta S = -94600 - 36 \times 298 = -1.053 \times 10^5\ J$

Concept Strand 37

A cell consists of $Pt/Tl^+, Tl^{3+}$ and Pb, $PbSO_{4(s)}/SO_4^{2-}$ half cells whose standard electrode potentials are 1.25 V and −0.359 V respectively at 298K. The concentration of the different ions in the cell, $Tl^+ = 0.01$ M, $Tl^{3+} = 0.05$ M, $SO_4^{2-} = 0.1$ M.

(i) Formulate the cell in which the reaction is spontaneous.

(ii) Write the cell reaction for the passage of one faraday.

(iii) Calculate the EMF and ΔG of the reaction.

(iv) Indicate the polarities of the electrodes.

Solution

(i) $Pb, PbSO_{4(s)}/SO_4^{2-}\ (C = 0.1) \,//\, Tl^+(C = 0.01), Tl^{3+}\ (C = 0.05), /Pt$

(ii) $\frac{1}{2}(Tl^{3+} + 2\bar{e} \rightarrow Tl^+)$ (a)

$\frac{1}{2}\left(PbSO_4 + 2\bar{e} \rightarrow Pb + SO_4^{2-}\right)$ (b)

(a) − (b),

$\frac{1}{2}Tl^{3+} + \frac{1}{2}Pb_{(s)} + \frac{1}{2}SO_4^{2-} \rightarrow \frac{1}{2}Tl^+ + \frac{1}{2}PbSO_{4(s)}$

(iii) $E = E^\circ_{cell} - \frac{RT}{F}\ln\frac{(Tl^+)^{1/2}}{(Tl^{3+})^{1/2} \times (SO_4^{2-})^{\frac{1}{2}}}$ (C of $PbSO_4 = 1$, C of Pb = 1)

$E^\circ_{cell} = \left[1.25 - (-0.359)\right] - \frac{0.0591}{1}\log\frac{(0.01)^{1/2}}{(0.05)^{1/2}\ (0.1)^{1/2}}$

$= 1.61 - 0.591 \log (2)^{1/2}$

$= 1.61 - 0.0591 \times 0.1505 = 1.61 - 0.0089$

$= 1.601$ V

$\Delta G = -nFE = -1 \times 96500 \times 1.601 = -1.545 \times 10^5\ J$

(iv) Lead electrode is negative and thallium electrode is positive.

Concept Strand 38

The standard electrode potentials of Pb, $PbSO_{4(s)}/SO_4^{2-}$ and Pb, PbO_2/SO_4^{2-} electrodes are –0.310 V and +1.685 V, respectively, at 298K. Calculate the concentration of sulphuric acid given that the EMF of the cell is 2.015 V. What is the reaction in the cell during the charging?

Solution

Cell reaction:

$$Pb_{(s)} + PbO_{2(s)} + 2H_2SO_4 \rightarrow 2PbSO_4 + 2H_2O$$

$$E = E^\circ - \frac{0.0591}{2}\log\frac{[H_2O]^2}{[H_2SO_4]^2}$$

$$= 1.995 - \frac{0.0591}{2}\log\frac{1}{[H_2SO_4]^2}$$

(Assuming conc. of $H_2O = 1$)

$$\text{or } \frac{0.0591}{2}\log\frac{1}{[H_2SO_4]^2} = 1.995 - 2.015 = -0.020 \text{ V}$$

$$-0.0591 \log H_2SO_4 = -0.02$$

$$\log H_2SO_4 = \frac{0.02}{0.0591} = 0.3384$$

$$[H_2SO_4] = 2.18 \text{ M}$$

During charging, the reaction will be the opposite of what is given earlier.

Concept Strand 39

The standard electrode potentials of Ag, AgI/I^- and Ag/Ag^+ are –0.15 V and +0.800 V respectively at 25°C. Calculate the solubility product of AgI.

Solution

$$AgI + e^- \rightarrow Ag + I^- \quad \text{(a)}$$

$$Ag^+ + e^- \rightarrow Ag \quad \text{(b)}$$

(a) – (b) $AgI \rightleftharpoons Ag^+ + I^-$

$$E^\circ_{el} = E_R - E_L = -0.15 - (+0.800) = -0.95 \text{ V}$$

$$\Delta G^\circ = -nFE^\circ = -RT \ln K_{SP} = -2.303\ RT \log K_{SP}$$

$$\text{or} \quad \log K_{SP} = \frac{nFE^\circ}{2.303RT} = -\frac{0.95}{0.0591} = -16.07 = \overline{17}.93$$

$$\text{or} \quad K_{SP} = 8.5 \times 10^{-17}$$

Concept Strand 40

The EMF of the cell Pt, H_2/H^+ (C = ?) // SCE consisting of hydrogen electrode placed in a solution of H^+ ions of unknown concentration and SCE is found to be 0.656 V at 25°C. Given that the potential of saturated calomel electrode is +0.242 V, calculate the pH of this solution.

Solution

$$E_{el} = E_{SCE} - E_{H^+,\,H_2} = 0.242 - E^\circ_{H^+} - \frac{0.0591}{1}\log C_{H^+}$$

$$0.656 = 0.242 + 0.0591\ pH$$

$$0.0591\ pH = 0.414; \qquad pH = \frac{0.414}{0.0591} = 7.0$$

Concept Strand 41

The energy of combustion of the hydrocarbon fuel can be used as electrical energy in a fuel cell. Calculate the standard EMF of such a fuel cell using methane as the hydrocarbon using the standard Gibbs energy of formation of the following compounds:

$$\Delta G^\circ_f (CH_4) = -51.04 \text{ kJ};$$

$$\Delta G^\circ_f (CO_2) = -395.03 \text{ kJ}; \quad \Delta G^\circ_f (H_2O_{(\ell)}) = -237.2 \text{ kJ}$$

Solution

For the combustion of methane, the reaction is

$$CH_{4(g)} + 2O_{2(g)} \rightarrow CO_{2(g)} + 2H_2O_{(\ell)}$$

$$\Delta G^\circ \text{ of reaction} = [-395.0 + 2 \times -237.2] - [-51.04] = -818.4 \text{ kJ}.$$

Half cell reaction: $2O_{2(g)} + 8H^+ + 8e^- \rightarrow 4H_2O_{(\ell)}$

$$CO_{2(g)} + 8H^+ + 8e^- \rightarrow CH_{4(g)} + 2H_2O_{(\ell)}$$

overall reaction $CH_{4(g)} + 2O_2 \rightarrow CO_{2(g)} + H_2O_{(\ell)}$

since $8e^-$ are involved in the cell

$$E^\circ = \frac{-\Delta G^\circ}{nF} = \frac{-(-818.4 \times 1000)}{8 \times 96{,}500} = 1.06 \text{ V}$$

Concept Strand 42

The standard electrode potentials of Mn^{3+}/Mn^{2+} and Mn^{2+}/Mn electrodes at 25°C are 1.51 V and –1.18V, respectively. Calculate the standard electrode potential of $Mn^{3+} + 3e^- \rightarrow Mn$.

Solution

$Mn^{3+} + e^- \rightarrow Mn^{2+}$ $E_1^o = +1.51$ V (a)

$Mn^{2+} + 2e^- \rightarrow Mn$ $E_2^o = -1.18$ V (b)

$Mn^{3+} + 3e^- \rightarrow Mn$ $E_3^o = ?$

(a) + (b) $Mn^{3+} + 3e^- \rightarrow Mn$. That is

$n_1 E_1^o + n_2 E_2^o = n_3 E_3^o$

$1 \times 1.51 - 2 \times 1.18 = 3 \times E_3^o \Rightarrow E_3^o = -0.283$ V

Concept Strand 43

Calculate the disproportionation constant for the reaction. $2Cu^+ \rightleftharpoons Cu + Cu^{2+}$ at 298K, given that the standard potentials of Cu^{2+}/Cu and Cu^+/Cu electrodes are +0.34 V and +0.521 V, respectively.

Solution

$2Cu^+ + 2e^- \rightarrow 2Cu$ $E^o = +0.521$V (a)

$Cu^{2+} + 2e^- \rightarrow Cu$ $E^o = +0.340$ V (b)

$$(a) - (b) \log K = \frac{nE^o}{0.0591} = \frac{0.181 \times 2}{0.0591} = 6.125$$

$K = 1.334 \times 10^6$.

Concept Strand 44

The EMF of the cell

$Pt, H_2(g, 1\ atm)/HOAc\ (M = 0.01),$

$NaAc\ (M = 0.01)\ NaCl\ (M = 0.01)/AgCl_{(s)}, Ag$

(where, HOAc = acetic acid, NaOAc = sodium acetate) is 0.622 volts at 25°C. Calculate the dissociation constant of acetic acid ($E^o_{AgCl, Cl^-} = +0.222$ V at 25°C)

Solution

Cell reaction:

$$H_{2(g)} + AgCl_{(s)} \rightleftharpoons Ag_{(s)} + H^+ + Cl^-$$

or $E = E^o - 0.0591 \log\left(m_{H^+} . m_{Cl^-}\right)$

But $m_{H^+} = K_a \dfrac{[CH_3COOH]}{[CH_3COO^-]}$

$$E = E^o - 0.0591 \log\left[Ka \frac{m_{HA}}{m_{A^-}}\right] - 0.0591 \log m_{Cl^-}$$

Assuming the acid to be slightly dissociated, we can write,

$$m_{HA} = 0.01,\ m_{A^-} = 0.01,\ m_{Cl^-} = 0.01$$

substituting the data in the above equation,

$0.622 = 0.222 - 0.591 \log K_a - 0.0591 \log 0.01$

$0.0591 \log K_a = 0.222 - 0.622 + 0.1182 = -0.282$

$$\log K_a = \frac{-0.282}{0.0591} = -4.77\ ;\ K_a = 1.7 \times 10^{-5}$$

Concept Strand 45

In a hydrogen–oxygen fuel cell, the overall cell reaction is $H_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow H_2O_{(\ell)}$ Given the standard Gibbs free energy change of the reaction at 298K is –238.5 kJ, calculate E^o of the cell. Also formulate the cell in which the above reaction takes place. What is the half cell potential of the electrode?

Solution

Since the reaction

$$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)$$

occurs for the passage of 2 faraday (or $2\bar{e}$ process)

$$E^\circ = -\frac{\Delta G^\circ}{nF} = \frac{238.5 \times 1000}{96500 \times 2} = 1.236\ V$$

The cell may be formulated on the basis of the half cell reaction

$$\frac{1}{2}O_{2(g)} + 2H^+ + 2e^- \rightarrow H_2O_{(\ell)} \quad (a)$$

$$2H^+ + 2e^- \rightarrow H_{2(g)} \quad (b)$$

(a)–(b) gives the reaction

$$H_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow H_2O_{(\ell)}$$ or the cell is

$Pt, H_{2(g)} \mid$ acid solution (1 M) $\mid O_{2(g)}, Pt$

$E^o_{cell} = E^o_R - E^o_L$; or $1.236 = E^o_R - 0$

$\therefore E^o_R = 1.236$ V, i.e., the half cell potential of the given reaction is 1.236 V.

Concept Strand 46

If the temperature coefficient $\dfrac{\partial E^o}{\partial T}$ of the cell given in problem (18.43) is -8.24×10^{-4} volts deg^{-1}, calculate the standard

entropy and enthalpy changes at 298K for the formation of 1 g mole of water by the combination of hydrogen and oxygen.

Solution

$$\Delta S^\circ = nF\left(\frac{\partial E^\circ}{\partial T}\right)_p = 2\times 96500\times 8.24\times 10^{-4}$$
$$= -159.0\ \text{J/deg}$$
$$\Delta H^o = \Delta G^o + T\Delta S^o = -238.5 + 298 \times -159 \times 10^{-3}$$
$$= -238.5 - 47.4 = -285.9\ \text{kJ}$$

Concept Strand 47

Derive an equation for the EMF of the amalgam concentration cell at 298K given as Cd(Hg)(1.27 M)/$CdSO_4$ solution/Cd(Hg) (0.5 M) What is the value of the EMF?

Solution

$$\text{R.H.E} = Cd^{2+} + 2e^- \rightarrow Cd(Hg)_1$$
$$\text{L.H.E} = Cd^{2+} + 2e^- \rightarrow Cd(Hg)_2$$
$$E_R - E_L\ Cd(Hg)_2 \rightarrow Cd(Hg)_1$$
$$E_{cell} = E^o_{cell} - \frac{0.0591}{2}\log\frac{Cd(Hg)_1}{Cd(Hg)_2} = -\frac{0.0591}{2}\log$$
$$\frac{Cd(Hg)_1}{Cd(Hg)_2}\ (\because E^o_{cell} = 0)$$
$$\therefore\quad E_{cell} = \frac{0.0591}{2}\log\frac{Cd(Hg)_2}{Cd(Hg)_1}$$
$$= \frac{0.0591}{2}\log\frac{1.27}{0.5} = 0.012\ \text{V}$$

Concept Strand 48

Calculate the EMF of the concentration cell

Pt, H_2 (g, 2 atm)/H^+ (0.02 m) // H^+ (0.05 m)/H_2 (g, 5 atm), Pt at 298K.

Solution

$$\text{R.H.E: } H^+(0.05\ M) + e^- \rightarrow \frac{1}{2}H_2\,(g, 5\ atm) \qquad (a)$$

$$\text{L.H.E: } H^+(0.02\ M) + e^- \rightarrow \frac{1}{2}H_2\,(g, 2\ atm) \qquad (b)$$

Cell reaction

$$(a)-(b):\ H^+(0.05\ M) + \frac{1}{2}H_2\,(g, 2\ atm) \rightarrow$$
$$H^+(0.02\ M) + \frac{1}{2}H_2\,(g, 5\ atm)$$

(RHS, LHS represent the terms on right hand side and left hand side of cell reaction)

$$= 0 - \frac{0.0591}{1}\log\frac{0.02\times(5)^{\frac{1}{2}}}{(0.05)\times 2^{\frac{1}{2}}}$$
$$= -0.0591\ \log 0.6324 = 0.0117\ \text{volts.}$$

Concept Strand 49

The standard redox potentials of Pt/Fe^{3+}, Fe^{2+} and Pt/$Cr_2O_7^{2-}$, Cr^{3+} are 0.771 V and 1.330 V, respectively, at 298K. Calculate the equilibrium constant of the reaction which occurs in a cell formed from them. Can this reaction be used in analytical purposes for estimation of Fe^{2+}. If so, why?

Solution

$$\text{R.H.E: } Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O \qquad E^o = 1.330\ \text{V}$$
$$\text{L.H.E: } 6Fe^{3+} + 6e^- \rightarrow 6Fe^{2+} \qquad E^o = +0.771\ \text{V}$$

overall reaction:

$$Cr_2O_7^{2-} + 14H^+ + 6Fe^{2+} \rightarrow 2Cr^{3+} + 6Fe^{3+} + 7H_2O$$
$$E^o = 1.330\ \text{V} - 0.771 = 0.559\ \text{V}$$
$$\log K = \frac{nFE^o}{RT\times 2.303} = \frac{0.559\times 6}{0.0591} = 56.75$$
$$K = 5.64\times 10^{56}$$

It can be used for analytical purposes because the given reaction is practically complete.

TOPIC GRIP

Subjective Questions

1. Dry air was bubbled successively through a solution containing 0.018 kg of an ideal nonvolatile solute in 0.1 kg water and pure water for sometime. The loss of mass of solution was 2.94 g and that of the pure solvent 60 mg. What is the ratio of the molar mass of solute to that of the solvent? (Assume the solution to be a dilute one)

2. The total vapour pressure, P mm, at 30°C for mixture of Benzene and Toluene (behaving ideally) conforms the equation $P_{mm} = 32.1 + (18.6 \times x)$ where, x is the mole fraction of benzene in the liquid phase in equilibrium with vapour. Calculate the vapour pressures of pure benzene and pure toluene. Further calculate the composition of the vapour at this temperature for x = 0.5.

3. Calculate K_f, the molal freezing point constant for a certain solid solvent, given that the enthalpy of fusion per mole is 4.6 kcal, the melting point is 80°C and the molar mass = 128 g/mol.

4. A solution of m grams of glucose in a kilogram of water is cooled to –0.5°C. It is observed that approximately 250 g of ice separates out. K_f of water = 1.86 deg/molal. Calculate m (M for glucose = 180 g/mole).

5. One molal solution of benzoic acid in a solvent X boils at 81.5°C. The normal boiling point of the solvent is 80.1°C assuming that (i) the usual laws of dilute solutions apply and (ii) the solute is 89.5% dimerized, Calculate K_b for the solvent X.

6. Consider $Pb + Sn^{2+} \rightarrow Pb^{2+} + Sn$

   E°values – $Pb \rightarrow Pb^{2+} + 2e^-$; (0.13 V)

   $Sn^{2+} + 2e^- \rightarrow Sn$; (–0.14 V). Calculate the ratio concentration of the ions $\frac{[Pb^{2+}]}{[Sn^{2+}]}$ for which E = Zero at 298 K.

7. When the electric current is passed through silver nitrate solution using silver electrodes, 0.2503 g of silver is deposited (Eq.wt = 108). In series with this electrolytic cell is another cell in which dil.H_2SO_4 is electrolyzed using Pt electrodes. What volume of hydrogen would be collected over water at 20°C and 756 mm pressure. Aqueous tension = 17.56 mm at 20°C.

8. Calculate the emf of the cell at 25°C: Pt, $H_{2(g)(1\ atm)}$/0.5 M HCOOH // 1M $CH_3COOH/H_{2(g)\ 1atm,\ Pt}$. K_a values are 1.8×10^{-4} and 1.8×10^{-5} for formic and acetic acid respectively.

9. The specific conductance of 0.002 N Na_2SO_4 is 2.6×10^{-4} ohm^{-1} cm^{-1} and it increases to 7×10^{-4} ohm^{-1} cm^{-1} when the solution is saturated with $CaSO_4$. The equivalent ionic conductances of Na^+ and Ca^{2+} are 50 ohm^{-1} cm^2 and 60 ohm^{-1} cm^2 respectively. Calculate the solubility and solubility product of $CaSO_4$.

10. The standard potentials for following reactions at 298K are

    $Hg_2^{2+} + 2e^- \rightarrow 2\ Hg_{(\ell)}$ $E^\circ = +0.800$ V (a)

    $Hg^{2+} + 2e^- \rightarrow Hg_{(\ell)}$ $E^\circ = +0.850$ V (b)

    Calculate the equilibrium constant for the disproportionation reaction

    $Hg_{2(aq)}^{2+} \rightarrow Hg_{(\ell)} + Hg_{(aq)}^{2+}$

Straight Objective Type Questions

Directions: **This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

11. Vapour pressure of pure CS_2 at 50°C = 854 torr. Calculate the vapour pressure of a solution of 2 g of sulphur in 100 g of CS_2 at this temperature. Assume the formula of sulphur to be S_8. At wt. of S = 32

(a) 849 torr (b) 845 torr

(c) 8.41 torr (d) 840 torr

12. The amount of 1,2-ethane diol $\left(\begin{matrix}CH_2 & CH_2\\ | & |\\ OH & OH\end{matrix}\right)$ (w g) dissolved (an antifreeze) in 2 kg of water just prevents it from freezing at –10°C, K_f for water = 1.85 deg/molal. Calculate w.

(a) 630.7 (b) 670.3

(c) 607.3 (d) 703.6

13. 9 g of an organic compound dissolved in a kg of benzene lowers the freezing point from 5.45°C (pure benzene) to 5.09°C (for the solution). Molar enthalpy of fusion of solid benzene = 2.37 kcal/mol. The molar mass of the compound in g mol^{-1}.

(a) 167.2 (b) 172.6

(c) 127.6 (d) 216.7

14. The specific conductance of a certain electrolyte solution (0.01 M) is $1.061 \times 10^{-4}\ \Omega^{-1}\ cm^{-1}$. The molar conductance in $\Omega^{-1}\ cm^2\ mol^{-1}$.

(a) 1.061 (b) 106.1 (c) 53.05 (d) 10.61

15. The quantity of electricity in Faraday required to reduce 0.1 mole of nitrobenzene to aniline. If the process of 80% efficient, the value is

(a) 0.70 (b) 0.75 (c) 0.65 (d) 0.80

Assertion–Reason Type Questions

Directions: **Each question contains Statement-1 and Statement-2 and has the following choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False
(d) Statement-1 is False, Statement-2 is True

16. Statement 1

For a uni-univalent type of electrolyte, the equivalent conductance, $\wedge$, is the sum of the ionic conductances of the positive and negative ions λ^+ and λ^-.

and

Statement 2

Ionic conductances are rather sensitively dependent on temperature.

17. Statement 1

By thermodynamical derivation, K_b, the molal elevation constant, is (very nearly)equal to $(RT^2_B/1000\ L)$Where T_B is the normal boiling point of the solvent on the Kelvin scale and L is the latent heat of vapourisation per gram of the solvent at $T_{B.}$

and

Statement 2

The equation $\left(\frac{p_0 - p_s}{p_0}\right)$ = mole fraction of the solute, for the relative lowering of vapour pressure of a dilute solution of a non volatile solute is quite generally valid. i.e., for all non volatile solutes of varying concentration of solutions in volatile solvents.

18. Statement 1

A mixture of chloroform and acetone exhibits negative deviation from Raoult's law.

and

Statement 2

Acetone and chloroform interact through hydrogen bonding.

19. Statement 1

The magnitude of ΔG and its temperature coefficient, $\left(\frac{\partial(\Delta G)}{\partial T}\right)_p$ for a cell reaction are sufficient for evaluating the enthalpy change, ΔH of the cell reaction.

and

Statement 2

The pH of the NHE is unity.

20. Statement 1

The cell $Zn_{(s)}/Zn^{2+}_{(1M)}//H^{+}_{(1M)}/H_{2,}$ Pt an electrolytic cell.

and

Statement 2

According to the uni-univalent electrolyte in dilute solutions Λ_0 is linearly related to the square root of the concentration.

Linked Comprehension Type Questions

Directions: **This section contains 2 paragraphs. Based upon the paragraph, 3 multiple choice questions have to be anered. Each question has 4 choices (a), (b), (c)and (d), out of which ONLY ONE is correct.**

Passage I

Henry's law considers the dissolution of gas in liquids. It may be expressed as $x_2 = k'P_2$ where x_2 is the mole fraction of gas dissolving and P_2 the pressure of the gas. The equation can also be written as $P_2 = \frac{x_2}{k'}$. This form of Henry's law is preferred over the conventional form $P_2 = kx_2$. But Raoults law holds good for a solution when both solvent and solute are both volaile. Considering P_2 to be the vapour pressure of the solute (gas) dissolving in the liquid and x_2 the mole fraction of the solute $P_2 = x P_2^\circ$ where P_2° is the vapour pressure of the component. The equation is similar to $P_2 = \frac{x_2}{k'}$ given above. The two equations becomes ideal when $k' = \frac{1}{P_2^\circ}$. In that case Raoults law may be taken as a special case of Henry's law. This condition $k' = \frac{1}{P_2^\circ}$ is approached only in a few cases representing ideal solutions of gases in liquids.

21. The Henry's law constant for CO_2 is 1.25×10^6 torr at 25°C. How many grams of CO_2 dissolve in 1 kg water if the partial pressure of CO_2 over the solution is 760 torr?

(a) 3 g (b) 1.5 g (c) 0.15 g (d) 4.4 g

22. Given the Henry's law constant for gases P, Q, R, and S to be 144.97, 69.16, 76.48 and 88.84 kilobar at 298 K. The most soluble gas among at 298 K is
(a) P (b) Q (c) R (d) S

23. Heptane and octane form an ideal solution. At 298 K the vapour pressure of heptane is 105.2 kP_a and that of octane 46.8 kP_a. How much is the vapour pressure of a solution of 25 g heptane in 35 g octane at 298 K
(a) 40.7 kP_a (b) 146 kP_a (c) 100 kP_a (d) 73 kP_a

Passage II

The specific conductivity, κ ($ohm^{-1}\,cm^{-1}$) is related to the equivalent conductivity $\wedge$($ohm^{-1}\,cm^2\,eqt^{-1}$) of an electrolyte solution by the equation. $\wedge = \dfrac{1000\kappa}{c}$ where, c is the concentration of the electrolyte in g eqt/litre. For dilute solutions of uni-univalent electrolytes, $\wedge$ varies with $\sqrt{c}$: $\wedge = \wedge^0 - (A + B \wedge_0)\sqrt{c}$ where, $\wedge^0$ is the limiting value of $\wedge$ as $c \to 0$. This equation does not apply to solutions of weak electrolyte like CH_3COOH. This ratio ($\wedge/\wedge^0$) is known as the conductance ratio. For weak electrolyte solutions, the conductance ratio is taken to be the degree of dissociation, α.

24. For a certain uni-univalent strong electrolyte. Λ values are 106.3 and 101.7 ($\Omega^{-1}\,cm^2\,eqt^{-1}$) for $c = 10^{-3}$ M and 10^{-2} M respectively. The Λ^0 in $\Omega^{-1}\,cm^2\,eqt^{-1}$.
(a) 94.8 (b) 90.8 (c) 121.8 (d) 108.4

25. For a saturated solution of a very sparingly soluble salt $\wedge^0 \simeq \wedge = 124.5\ \Omega^{-1}\,cm^2\,eqt^{-1}$ and $\kappa = 1.37 \times 10^{-6}\ \Omega^{-1}\,cm^{-1}$. The solubility in eqt/litre.
(a) 1.1×10^{-5} (b) 1.3×10^{-4} (c) 1.3×10^{-5} (d) 1.42×10^{-3}

26. For a certain uni-univalent weak electrolyte at a concentration of 0.02 M, $\kappa = 3.3 \times 10^{-4}\,\Omega^{-1}\,cm^{-1}$, If $\wedge^0 = 387\ \Omega^{-1}\,cm^2\,eqt^{-1}$, the degree of dissociation, α is
(a) 3.4×10^{-2} (b) 4.3×10^{-2} (c) 3.4×10^{-3} (d) 2.7×10^{-3}



Multiple Correct Objective Type Questions

Directions: **Each question in this section has four suggested answers out of which ONE OR MORE answers will be correct.**

27. Which among the following statements are true?
(a) The electrode {$Ag/AgCl_{(s)}$, $Cl^-_{(m)}$} is an anion reversible electrode.
(b) A plot of $\wedge$ Vs $\sqrt{c}$ is linear for dilute solutions of acetic acid.
(c) A mixture of benzene and toluene (liquids) shows under equilibrium at temperature, T, a value of total vapour pressure which varies linearly with the mole fraction of benzene in the liquid phase.
(d) Trouton's constant for water is nearly (21 to 22) cal deg^{-1} mol^{-1} as for normal liquids.

28. Identify the correct statements.
(a) The pH of a solution of 10^{-6} M acetic acid is 6.
(b) The electrode potential of a bright Pt electrode dipping in a mixture of Ferrous and Ferric ions depends on the ratio of the concentrations $\left(Fe^{2+}/Fe^{3+}\right)$.
(c) The speed of a moving ion in a dilute aqueous solution depends on factors, one of which is the viscosity of the medium.
(d) $\wedge_0$ is the sum of ionic conductances of the ions in a solution.

29. Which among the following statements are true?

(a) For the electrode: $\{Cu/Cu^{2+}_{(M)}\}$ the electrode potential value decreases as M decreases.

(b) In the titration of 0.1 M HCl solution against 0.5 M NaOH solution, a plot of $(\frac{\Delta E}{\Delta V})$ against V shows a maximum at the end point (V = Volume of alkali added. $(\frac{\Delta E}{\Delta V})$ is the change of potential of the electrode per ml of alkali added).

(c) When a dilute solution of NaCl in water is cooled continuously, what separates out as solid phase is NaCl crystals along with ice.

(d) The conductance ratio $(\frac{\Lambda}{\Lambda_0})$ equals α, the degree of dissociation of solutions of both strong and weak electrolytes.

Matrix-Match Type Question

Directions: **Match the elements of Column I to elements of Column II. There can be single or multiple matches.**

30.

Column I	**Column II**
(a) Non linear variation of $\wedge$ Vs $\sqrt{c}$	(p) Comparison standard in emf measurements
(b) Calomel electrode	(q) Electrode potential varies with concentration of Cl^-
(c) $(\frac{\Lambda}{\Lambda_0}) \simeq$ degree of dissociation	(r) A concentrated solution of a bi-univalent electrolyte. E.g., $BaCl_2$
(d) $pH \simeq \frac{1}{2} pK_a + \frac{1}{2} \log c$	(s) Solutions of propanoic acid in water

IIT ASSIGNMENT EXERCISE

Straight Objective Type Questions

Directions: **This section contains multiple choice questions. Each question has 4 choices (a), (b), (c)and (d), out of which ONLY ONE is correct.**

31. What weight of NaOH is produced in the electrolysis of $NaCl_{(aq)}$ by a quantity of electricity which liberates 34 ml of gas measured at STP in electrolysis of acidified water?
(a) 8.1×10^{-2} g (b) 7.5×10^{-2} g (c) 9.5×10^{-2} g (d) 9.0×10^{-2} g

32. Anthracene can be oxidized to anthraquinone anodically with 100 % current efficiency $C_{14}H_{10} \xrightarrow{\text{oxidation}} C_{14}H_8O_2$ (anode) The weight of anthraquinone produced by a current of one amp in one hour is
(a) 2.58 g (b) 2.91 g (c) 1.29 g (d) 1.94 g

33. An alkali–chlorine cell contains 140 litres of solution obtained by operating for 10 hours with a current of 1250 amp. Analysis shows that 116.5 g of NaOH are resent per litre of the solution. The current efficiency is
(a) 0.784 (b) 0.874 (c) 0.748 (d) 0.918

34. An electrochemical cell ceases to function after some time because
(a) Anode completely gets used up
(b) Cathode is completely reduced
(c) Concentration of the solution becomes unity
(d) Electrode potential of cathode and anode becomes equal

35. $E°\ (Ag^+, Ag) = 0.7996$ volt at 25°C. K_{sp} for AgCl = 1.56×10^{-10} The $E^0_{(red)}$ for the half cell: $AgCl_{(s)} + e^- \rightleftharpoons Ag + Cl^-$
(a) 0.252 volt (b) 0.221 volt (c) 0.202 volt (d) 0.195 volt

36. $E°\ (Cu^{2+}, Cu) = 0.34$ volt. At what concentration of Cu^{2+} ion will the electrode potential $\left\{\left(\frac{Cu}{Cu^{2+}}\right)\right\}$ be zero?
(a) 2.129×10^{-11} M (b) 2.227×10^{-13} M (c) 2.982×10^{-12} M (d) 2.272×10^{-11} M

37. The Emf for the cell: $Ag/AgBr_{(s),\ a=0.34}\ Br^-, Fe^{3+}_{0.01M}, Fe^{2+}_{0.02M}/Pt$ at 25°C given.

$E°\ (AgBr + e^- \rightleftharpoons Ag + Br^-) = 0.0713$ V and $E°\ (Fe^{3+} + e^- \rightleftharpoons Fe^{2+}) = 0.77$ volt
(a) 0.6533 volt (b) 0.5631 volt (c) 0.3651 volt (d) 0.3356 volt

38. In a beaker containing 0.1 M HCl, both a silver electrode and a hydrogen electrode are placed. A little pure solid AgCl is added and well stirred. The approximate value of the emf, measured between the electrodes at 25°C.
$E°\ (Ag^+, Ag) = 0.799$ volt K_{sp} of AgCl $\simeq 1.7 \times 10^{-10}$.
(a) 0.4063 volt (b) 0.4603 volt (c) 0.6034 volt (d) 0.3406 volt

39. A current of 0.05 amp is passed through 0.1 M NaCl solution for one hour. Chlorine and hydrogen are liberated at the anode and cathode respectively. If the volume of electrolyte is 100 ml, calculate the final concentration of the OH^- ion.
(a) 1.568×10^{-2} M (b) 1.865×10^{-2} M (c) 8.65×10^{-3} M (d) 1.625×10^{-2} M

40. $E°\ (Cu^{2+}, Cu) = 0.52$ V; $E°\ (Cu^{2+}, Cu^+) = 0.153$ V (T = 298 K)
Calculate the $E°\ (Cu^{2+}, Cu)$
(a) 0.373 volt (b) 0.302 volt (c) 0.337 volt (d) 0.375 volt

41. In electroplating of gold the electrolyte preferred in $K[Au(CN)_2]$ because
(a) it is the most soluble compound of gold.
(b) the equivalent mass of gold in the same as its atomic mass in the compound.
(c) it is less soluble in water compared to other gold compounds.
(d) the equivalent mass of gold is in the compound is the least possible value.

42. K_{sp} of $Cu(OH)_{2\,(s)}$ at 25°C $\simeq 1 \times 10^{-19}$. Calculate the electrode potential of a copper electrode dipping in NaOH solution at pH = 12, containing pure precipitated $Cu(OH)_{2(s)}$ [$E°(Cu^{2+}. Cu)$] = 0.34 volt
(a) −0.1250 V (b) −0.1025 V
(c) −0.152 V (d) −0.215 V

43. The standard oxidation potential values for a sequence are

$$Mn^{2+} \xrightarrow{-1.5\,V} Mn^{3+} \xrightarrow{-1.0\,V} MnO_2.$$

Consider $2Mn^{3+} + 2H_2O \rightarrow Mn^{2+} + MnO_2 + 4\,H^+$
For unit activity of the species involved. The $\Delta G°$ for the process is
(a) −48250 J (b) −42580 J (c) −52480 J (d) −58240 J

44. Consider $Ag/AgCl_{(s)}$, 0.1 M KCl // 0.1 M $AgNO_3/Ag$. At 298 K given K_{sp} of $AgCl_{(s)} = 1.79 \times 10^{-10}$, The emf of the cell is
(a) 0.745 volt (b) 0.547 volt (c) 0.457 volt (d) 0.574 volt

45. $\frac{M}{2}$ and $\frac{M}{20}$ solutions of the metal ion x^{n+} are used to construct the cell; $x\left|\frac{M}{20}x^{n+}\right|\left|\frac{M}{2}x^{n+}\right|x$ at 18°C. The emf of the cell is 0.028 volt. Calculate the value of n given $2.303\,\frac{RT}{F} = 0.058$ volt at 18°C.
(a) 2 (b) 3 (c) 1 (d) 4

46. The process: $\underset{1atm}{H_{2(g)}} + 2AgCl_{(s)} \rightarrow 2HCl\ (1\ M) + 2Ag_{(s)}$ studied as a cell reaction at 25°C has an emf of 0.2329 V. If caried out in such a way that no electrical work is done, it gives a ΔH value of −19.4 Kcal per mole of $H_{2(g)}$ consumed. Calculate approximately ΔS for the process.
(a) −19.0 cal deg^{-1} (b) −29.0 cal deg^{-1} (c) −35.0 cal deg^{-1} (d) −15.0 cal deg^{-1}

47. The reaction taking place in the Daniell cell is the displacement of Cu^{2+} ion by Zn:
$Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$
$\{ Zn / \underset{\text{unit activity}}{Zn^{2+}} // \underset{\text{unitactivity}}{Cu^{2+}} / Cu \}$ The measured emf $\simeq$ 1.1 volt at 25°C. Calculate $\Delta G°$ and the ratio: $\frac{(Cu^{2+})}{(Zn^{2+})}$ under equilibrium. (Note: under eqm $\Delta G = 0$)
(a) 5.31×10^{-37} (b) 3.15×10^{-39} (c) 5.13×10^{-38} (d) 3.57×10^{-37}

48. A current of 60 milli amp is passed through a copper coulometer for an hour. Given (i) the current efficiency to be 95 % and (ii) the atomic weight of Cu = 63.6, calculate the weight of copper deposited on the cathode.
(a) 76.26 mg (b) 66.27 mg (c) 82.27 mg (d) 67.62 mg

49. Calculate the amount of iodine (At wt = 127) that would be liberated in the electrolysis of sodium iodide solution by a quantity of electricity that sets free 34 ml (measured at STP) of electrolytic gas in the (electrolyte) coulometer.
(a) 0.275 g (b) 0.375 g (c) 0.257 g (d) 0.175 g

50. A 50 milli amp current was passed through a silver titration coulometer for time t. At the end of the experiment 23.8 ml of 0.1 M NaCl solution were required to titrate the silver dissolved from the anode. Calculate t.
(a) 1.726 h (b) 1.627 h (c) 2.162 h (d) 1.276 h

51. A current of 0.10 amp was passed for two hours through a solution of cupra cyanide 0.3745 g of copper was deposited at the cathode. Calculate the volume of hydrogen measured at STP liberated simultaneously. (At wt of Cu = 63.6)

(a) 14.46 ml (b) 17.64 ml (c) 16.47 ml (d) 15.22 ml

52. Given (i) $Co^{3+} + e^- \rightleftharpoons Co^{2+}$; $E° = 1.82$ volt; (ii) $Cu^+ + e^- \rightleftharpoons Cu$; $E° = 0.52$ volt (iii) $Cu^{2+} + e^- \rightleftharpoons Cu^+$; $E° =$ 0.15 volt (iv) $2H_2O \rightleftharpoons O_{2(g)} + 4H^+ + 4e^-$; $E° = \frac{-1.23}{V}$. Indicate the correct stateent among the following:

(a) Both Co^{3+} and Cu^+ ions are unstable in aqueous solution.
(b) Both Co^{3+} and Cu^+ ions are stable in aqueous solution.
(c) Co^{3+} ion (but not Cu^+ ion) is stable in aqueous solution.
(d) Cu^+ ion (but not Co^{3+} ion) is stable in aqueous solution.

53. Given $\frac{1}{2}Br_2 + e^- \rightleftharpoons Br^-$; $E° = 1.09$ V; $\frac{1}{2} I_{2(s)} + e^- \rightleftharpoons I^-$; $E° = 0.535$ V.
$Fe^{3+} + e^- \rightleftharpoons Fe^{2+}$; $E° = 0.771$ volt. Indicate the correct statement among the following.

(a) Fe^{3+} ion will oxidize both Br^- and I^-ions (for unit activity)
(b) Fe^{3+} ion will not oxidize either of the two ions.
(c) Fe^{3+} ion will oxidize Br^- ion but not I^-ion.
(d) Fe^{3+} ion will oxidize I^- ion but not Br^- ion.

54. Consider the cell H_2 (Pt) (1 atm)/$H^+_{x\,M}$ ion//$Cu^{2+}_{0.1M}$/Cu. Given that $E°$ (Cu^{2+}, Cu) = 0.34 volt and $x = 10^{-6}$M. The measured emf of the cell, at 298 K is

(a) 0.586 volt (b) 0.562 volt
(c) 0.665 volt (d) 0.856 volt

55. The following data are for $\wedge$ of HCl in Ω^{-1} cm^2 mol^{-1} at two different concentrations at 298 K. Calculate $\wedge_0$. (i) $\wedge_1 = 422.74\ \Omega^{-1}$ cm^2 mol^{-1} at $C = 5 \times 10^{-4}$ mol L^{-1} (ii) $\wedge_2 = 421.36\ \Omega^{-1}$ cm^2 mol^{-1} at $C = 1 \times 10^{-3}$ mol L^{-1}

(a) 436.2 (b) 426.1 (c) 429.2 (d) 430.2

56. Concentration of solution in moles per litre is used in osmotic pressure calculation because

(a) the magnitude of osmotic pressure is large.
(b) measurement of osmotic pressure is done at constant temperature.
(c) osmosis involve decrease in concentration as the process takes place.
(d) it is not considered as a colligative property.

57. At 40°C, the total vapour pressure of a mixture of two liquids A and B is P (torr) = 119 x + 135 (x = mole fraction in the liquid phase) of liquid, A. Assume ideal behavior. What are the vapour pressures of the pure liquids. Calculate the mole fraction of A in the vapour for x = 0.5.

(a) 0.653 (b) 0.563 (c) 0.536 (d) 0.702

58. The vapour pressure of a pure liquid solvent, A = 0.80 atm at temperature TK. It drops to 0.78 atm when a non volatile solute, B is added. Calculate the mole fraction of A.

(a) 0.759 (b) 0.795 (c) 0.975 (d) 0.875

59. At 25°C, vapour pressure of pure benzene = 100 torr; while that of ethanol is 44 torr. (Observe that it has a lower vapour pressure). Assuming ideal behavior, calculate the vapour pressure of liquid mixture containing 10 g of each at 25°C.

(a) 56 torr (b) 65 torr (c) 75 torr (d) 79 torr

60. An aqueous solution boils at 100.50°C at one atm. Calculate the freezing point of the solution? $K_f = 1.86$ deg molal^{-1} $K_b = 0.51$ deg molal^{-1}.

(a) −1.28°C (b) −2.08°C (c) −2.18°C (d) −1.82°C

61. The freezing point of pure molten camphor is 178.4°C. K_f = 40 deg molal^{-1} calculate the freezing point of a solution containing 1.5 g of an organic compound of molecular wt = 125, in 35 g of camphor.
(a) 176.4°C (b) 156.4°C (c) 178.2°C (d) 164.7°C

62. The freezing point of a solution of 3.00 g of a non volatile (organic compound) in 80 g of water is −1.16°C. The compound contains only carbon, hydrogen and oxygen. % composition of the solute: C: 40%, H: 6.67%. What is the molecular formula of the solute? K_f = 1.86 deg molal^{-1}.
(a) $C_2H_4O_2$ (b) C_2H_6O (c) C_3H_6O (d) CH_2O

63. A certain solution of Benzoic acid in benzene has a freezing point of 3.1°C and boils at 82.6°C (at 1 atm pressure). Pure benzene freezes at 5.5°C. Boiling point of pure benzene is 80.1°C. (K_b and K_f for benzene are 2.53 and 5.07 K kg mol^{-1}) Indicate the correct statement among the following:
(a) Dimerization at both freezing and boiling point
(b) Dimerization only at freezing temperature
(c) Dimerization does not occur at freezing point
(d) Fraction dimerized is nearly constant at all temperatures

64. An aqueous solution of urea (mol wt = 60) freezes at −0.52°C. Calculate the osmotic pressure of the solution at 37°C. K_f = 1.86 deg molal^{-1}.
(a) 6.127 (b) 6.172 (c) 7.126 atm (d) 7.612

65. Estimate the lowering of vapour pressure of a solution (one molal) in water at 100°C.
(a) 17.4 mm (b) 13.4 mm (c) 11.4 mm (d) 15.4 mm

66. Vapour pressure of a certain liquid at 25°C is 96.0 torr. Calculate the mole fraction of the liquid in a solution (liquid mix) in which the partial vapour pressure of the liquid is 23 torr.
(a) 0.24 (b) 0.34 (c) 0.29 (d) 0.39

67. At 26°C the vapour pressure of a solution of 20g of glucose ($C_6H_{12}O_6$) in 70 g of water in 24.5 torr. Calculate the vapour pressure of pure water.
(a) 25.91 torr (b) 26.02 torr (c) 25.19 torr (d) 26.29 torr

68. At 30°C pure benzene has a vapour pressure of 121.8 torr. By dissolving 15 g of a non volatile solute in 250 g of pure benzene, the vapour pressure was lowered to 120.2 torr. Calculate approximately the molar mass of the solute (g mol^{-1}).
(a) 343 (b) 334 (c) 433 (d) 353

69. Three isomeric liquid hydrocarbons, A B, and C have vapour pressures of 109 mm, 115.7 mm and 142.6 mm at 80°C. Suggest a composition of the liquid mixture containing all of them with a vapour pressure = 130 mm.
(a) $x_A = x_B = 0.25$; $x_C = 0.5$ (b) $x_A = 0.2$ $x_B = 0.3$; $x_C = 0.5$
(c) $x_A = 0.2 = x_B$; $x_C = 0.6$ (d) $x_A = 0.207 = x_B$; $x_C = 0.586$

70. A liquid (of molar mass M_1 g mol^{-1}) has a vapour pressure of po = 440 mm at 298 K. A solution of w_2 g of a non volaile solute of molar mass = M_2 g mol^{-1} in w_1 g of the liquid has a vapour pressure, p_s = 412.7 mm. If $\left(\frac{w_2}{w_1}\right) = \left(\frac{1}{20}\right)$, calculate the ratio of the molar mass of the solute to the solvent.
(a) 0.578 (b) 0.758 (c) 0.857 (d) 0.552

71. Calculate the mole fraction of component B in the vapour phase in equilibrium with an ideal liquid mixture of A and B, with a mole fraction of 0.5 for the component A in the liquid phase. P_0 of pure A = 119 torr p_0 of pure B = 37 torr (at some temperature T)
(a) 0.237 (b) 0.327 (c) 0.372 (d) 0.273

72. Calculate the specific latent heat of vapourization of acetic acid. (given K_b = 3.1 deg molal^{-1}, normal boiling point = 118.9°C) ("Specific" means latent heat of vapourization per gram)
(a) 79 cal g^{-1} (b) 99 cal g^{-1} (c) 89 cal g^{-1} (d) 75 cal g^{-1}

73. Calculate the boiling point of a 0.5 m solution of KCl in water, given that the Vant Hoff factor, i = 1.85. K_b = 0 51 deg molal^{-1}.
(a) 100.512°C (b) 100.215°C (c) 100.327°C (d) 100.472°C

74. A solution containing 5.0 g an organic compound dissolved in 50 g of benzene shows as ΔT_f = 1.15 deg. K_f for benzene = 5.07 deg molal^{-1}. The empirical formula of the solute = C_6H_5. What is the molecular formula? (At. wt. P = 31)
(a) (C_6H_5P) (b) $(C_6H_5P)_4$ (c) $(C_6H_5P)_2$ (d) $(C_6H_5P)_3$

75. Which of the following is not true for an ideal solution?
(a) Interaction between components in the mixtures are similar to those in pure components.
(b) Vapour pressure of the mixture is a non linear function of the mole fraction.
(c) Raoult's law is obeyed.
(d) ΔH mixing = 0

76. The compound used to construct semi permeable membranes is
(a) $Fe_4[Fe(CN)_6]_3$ (b) $Cu_2[Fe(CN)_6]$ (c) CaC_2O_4 (d) BaC_2O_4

77. Which mixture among the following will exhibit positive deviation from Raoult's law?
(a) Acetone – carbon disulphide (b) Acetone – chloroform
(c) Ether – chloroform (d) Water – nitric acid

78. P°_A is the vapour pressure of pure (component) liquid A. P_A is the (partial) vapour pressure of A in solution of A and B. Then according to Raoult's law.
(a) $\frac{P_A - P^\circ_A}{P_A} = x_B$ (b) $\frac{P^\circ_A - P_A}{P_A} = x_A$ (c) $\frac{P^\circ_A - P_A}{P^\circ_A} = x_B$ (d) $\frac{P_A - P^\circ_A}{P^\circ_A} = x_B$

79. The density of 10.95 molar HNO_3 (in water) = 1.41 g ml^{-1}. Calculate the molality of the solution.
(a) 10.95 (b) 15.21 (c) 7.76 (d) 12.36

80. The vapour pressure of water at 50°C is 99.9 mm. Calculate the molar latent heat of vapourization.
(a) 9217 cal mol^{-1} (b) 9127 cal mol^{-1} (c) 9912 cal mol^{-1} (d) 9712 cal mol^{-1}

81. One molar solution of sodium sulphate is electrolyzed using Pt electrodes with a current of one ampere 6.15 litres of oxygen (dry) is collected at 27°C and one atm pressure. Calculate the time in hours for which the current is passed.
(a) 18.6 hrs (b) 26.8 hrs (c) 20.8 hrs (d) 30.8 hrs

82. A current of 10 amps is passed for two hours through a solution of Nickel sulphate to plate out Nickel on the cathode. The Nickel deposit weighed 13.59 g. At.wt. of Ni = 58.7. Calculate the current efficiency.
(a) ~ 62% (b) ~ 52% (c) ~ 72% (d) ~ 42%

83. In the cathodic plating out of Nickel in electrolysis if 22 g of Nickel are plated out on a cathode of area 100 cm^2 each on both sides, What is the thickness of the deposited Nickel if density of Nickel is 8.9 g cm^3?
(a) 2.136×10^{-2} cm (b) 2.316×10^{-3} cm (c) 1.632×10^{-2} cm (d) 1.236×10^{-2} cm

84. In the operation of a lead storage battery (in an experiment wt % of H_2SO_4 decreased from 11.26 to 9.13 corresponding) to a decrease in density from 1.075 to 1.060. The volume (total) of the liquid = 5 litres. Calculate the number of amp- hours used up.
(a) 39.13 (b) 19.33 (c) 13.93 (d) 33.19

85. Consider the electrolytic oxidation: $Cl^- + 6OH^- \rightarrow ClO_3^- + 3H_2O + 6e^-$.
5 amp current used for 131.3 hrs yielded 350 g of KCl O_3 using KCl. Calculate the current efficiency.
(a) 70% (b) 60% (c) 90% (d) 50%

86. The specific conductance for a 0.1 M solution of a uni-univalent electrolyte is K ($\Omega^{-1}cm^{-1}$). $\wedge$ = 128.9 Ω^{-1} cm^2 eqt^{-1}. Calculate K.
(a) 1.8920×10^{-2} (b) 1.2890×10^{-2} (c) 1.982×10^{-3} (d) 2.189×10^{-3}

87. Calculate the cell constant for a cell which yields an experimental resistance of 470 ohms when filled with 0.001 N MCl. K = 1.21×10^{-4} Ω^{-1} cm^{-1}. Calculate also $\wedge$ (Ω^{-1} cm^2 eqt^{-1}) (M = univalent metal)
(a) 1.692×10^{-1} cm^{-1}; 211 Ω^{-1} cm^2 mol^{-1}
(b) 1.296×10^{-2} cm^{-1}; 311 Ω^{-1} cm^2 mol^{-1}
(c) 1.996×10^{-1} cm^{-1}; 424.7 Ω^{-1} cm^2 mol^{-1}
(d) 5.687×10^{-2} cm^{-1}; 121 Ω^{-1} cm^2 mol^{-1}

88. K for saturated solution of $BaSO_4$ in water at 25°C. is 1.8×10^{-3} ohm m^{-1}.sp conductivity of water used in the experiment was 2×10^{-4} ohm^{-1} m^{-1}. Molar ionic conductance values are Ba^{2+}, and SO_4^{2-} are 127.2 Ω^{-1} cm^2 mol^{-1}, and 159.6 Ω^{-1} cm^2 mol^{-1}. Calculate the solubility and solubility product of $BaSO_4$.
(a) 6.5×10^{-4} mol L^{-1}; 3.11×10^{-8} (b) ~ 5.6×10^{-5} mol L^{-1}; 3.11×10^{-9}
(c) 6.5×10^{-6} mol L^{-1}; 3.26×10^{-9} (d) 4.9×10^{-5} mol L^{-1}; 2.62×10^{-10}

89. The ionic conductance at infinite dilution of OH^- is 198.5 S cm^2 mol^{-1} and Λ_0 of LiOH is 237.2 S cm^2 mol^{-1}. The Λ_0 of Li^+ in S cm^2 mol^{-1} is
(a) 38.7 (b) 77.4 (c) 19.35 (d) 198.5

90. Calculate the emf of the cell: $Pt/Sn^{2+}_{(0.1M)};Sn^{4+}_{(0.2M)}//Fe^{2+}_{(0.1M)};Fe^{3+}_{(0.05M)}/Pt$ $E^0_{Fe^{3+}/Fe^{2+}}$ = 0.771 V; $E^0_{Sn^{4+}/Sn^{2+}}$ = 0.15 V, T = 298 K.
(a) 0.63 volt (b) 0.59 volt (c) 0.69 volt (d) 0.73 volt

91. Standard potentials: $Ag/AgI^-,I^-$ and Ag/Ag^+ are –0.15 V and +0.800 V respectively. Calculate the solubility product of AgI (T = 298 K).
(a) 7.5×10^{-16} (b) 7.9×10^{-17} (c) 9.5×10^{-16} (d) $8.5 \times 10^{--17}$

92. Emf of the electrode: Pt (H_2) / H^+ (c) coupled with the saturated calomel electrode is 0.656 volt. The electrode potential of the saturated calomel electrode is +0.242 volt, at 298 K. Calculate the [H^+].
(a) 10^{-6} (b) 10^{-8} (c) 10^{-7} (d) 10^{-5}

93. Calculate the equilibrium constant, for the equilibrium system $2Cu^+ \rightleftharpoons Cu + Cu^{2+}$ at 298 K given. $2Cu^+ + 2e^- \rightarrow 2Cu$, E_0 = +0.521 V. $Cu^{2+} + 2e^- \rightarrow Cu$. E_0 = +0.34 V
(a) 1.012×10^5 (b) 1.334×10^6 (c) 1.012×10^6 (d) 1.492×10^6

94. The temperature coefficient of the emf for a certain cell is -8.24×10^{-4} volt deg^{-1}. The cell reaction is a two electron process. Calculate ΔS at T = 298 K.
(a) –173 J deg^{-1} (b) –159 J deg^{-1} (c) –97 J deg^{-1} (d) –112 J deg^{-1}

95. What is the emf of the cell Pt (H_2, 2 atm)/ H^+ (0.02 M)// H^+ (0.05 M) / H_2 (Pt) 5 atm?
(a) 0.0221 volt (b) 0.0302 volt (c) 0.0099 volt (d) 0.0117 volt

96. The depression of freezing point of 0.3 molal glucose solution in water is the same as that of 0.1 molal solution of (Assume complete dissociation of the salts).
(a) K_2SO_4 (b) NaCl (c) Urea (d) $MgSO_4$

97. Vapour pressure p_s of a certain solution of urea in water is 0.975 p_0 where, p_0 is the vapour pressure of water at the same temperature. In this case the amount of solute in grams added to 1 Kg of water is
(a) 96.7 (b) 72.3 (c) 85.5 (d) 69.7

98. Phenol dimerises in benzene. The degree of association is 90%. ΔT_f for the solution is 1.65 deg. K_f for the solvent = 5.2 deg $molal^{-1}$. What wt of phenol (C_6H_5OH) in Kg should be added to 2 Kg of benzene?
(a) 0.1640 (b) 0.1232 (c) 0.1460 (d) 0.1084

99. The molar enthalpy of sublimation of solid benzene is 20.21 K J mol^{-1}. Molar heat of fusion = 10.59 kJ mol^{-1}. Calculate K_b for benzene in deg $molal^{-1}$ if boiling point of benzene is 80°C.
(a) 8.446 (b) 6.446 (c) 4.648 (d) 5.468

100. Which among the following is not positive for a non ideal solution showing positive deviation?
(a) $\Delta H_{(mixing)}$ (b) $\Delta V_{(mixing)}$ (c) $\Delta G_{(mixing)}$ (d) $\Delta S_{(mixing)}$

101. 10 g of an ideal non volatile solute of molar mass 100 g mol^{-1} is dissolved in 100 g of a solvent of molar mass 20. The elevation of boiling point is 'T' K. The molal elevation constant of the solvent is
(a) 20 ΔT (b) 5 ΔT (c) 0.1 ΔT (d) ΔT

102. p_0 values of CH_3 – OH and C_2H_5 – OH at 293 K are 94 mm and 44 mm respectively. Calculate the mole fraction of each in a mixture assuming ideal behaviour if total vapour pressure is 54 mm.
(a) 0.2, 0.8 (b) 0.25, 0.75 (c) 0.15, 0.85 (d) 0.3, 0.7

103. The vapour pressure of a solvent at a certain temperature T is 71.93 mm. It is reduced to 70.5 mm if 10 grams of non-volatile solute is dissolved in 150 g of the solvent.
Calculate the ratio $\frac{\text{molar mass of solute}}{\text{molar mass of solvent}}$,
(a) 2.93 (b) 4.12 (c) 1.92 (d) 3.29

104. 1.068 g of $CoCl_3 .6H_2O$ (molar mass = 267g) dissolved in 100 g of water gave a ΔT_f value = 0.225°C. K_f for water = 1.86 deg $molal^{-1}$. Indicate the correct formula for the compound.
(a) $[Co(H_2O)_6]Cl_3$ (b) $[Co((H_2O)_5Cl)]Cl_2 . H_2O$
(c) $[Co(H_2O)_3Cl_3]3H_2O$ (d) $[Co(H_2O)_4Cl_2]Cl H_2O$

105. A 5% solution $(\frac{W}{V})$ of sucrose is isotonic with 1.14% $(\frac{W}{V})$ solution of solute A. Molar mass of A is approximately equal to
(a) 78 (b) 60 (c) 41 (d) 300

106. The vapour pressure of pure liquids A and B are 300 mm and 650 mm (of Hg) respectively. In a non ideal mixture of A and B with mole fraction of A as 0.2, the total vapour pressure is 700 mm. Indicate the nature and extent of the deviation.
(a) positive deviation, 90 mm (b) negative deviation, 80 mm
(c) positive deviation, 120 mm (d) negative deviation, 110 mm

107. In Ostwald–Walker's experiment, dry air is first passed through a solution of 9.0 g of non volatile solute dissolved in 180 g of water and then through pure water. The loss in weight of the solution was 5.0 g while loss in weight of water was 0.025 g. Calculate the molar mass of the solute (in g mol^{-1}).
(a) 160 (b) 180 (c) 210 (d) 120

108. Indicate among the following the incorrect statement.
(a) When a dilute solution of urea freezes only ice separates at first.
(b) For normal liquids the molar entropy of vapourisation is nearly constant at the respective boiling points.
(c) The ion Co^{3+} is unstable in aqueous solution.
(d) Both (Ag/AgI, I^-)and (Ag/Ag^+)are anion reversible electrodes.

109. Calculate the specific latent heat of fusion of naphthalene. Freezing point of molten naphthalene is 80.22°C. K_f = 6.85 deg $molal^{-1}$.
(a) 46.33 cal g^{-1} (b) 61.12 cal g^{-1} (c) 36.43 cal g^{-1} (d) 56.72 cal g^{-1}

110. The osmotic pressure of a certain aqueous solution of urea is 7.1 atm at 37°C. What is ΔT_f for this solution K_f for water = 1.85 deg $molal^{-1}$?
(a) 0.59 deg (b) 0.62 deg (c) 0.52 deg 0.49 deg

Assertion–Reason Type Questions

Directions: **Each question contains Statement-1 and Statement-2 and has the following choices (a), (b), (c)and (d), out of which ONLY ONE is correct.**

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False
(d) Statement-1 is False, Statement-2 is True

111. Statement 1
In the electrolysis of an aqueous solution of copper sulphate using copper electrodes
does not produce any change in concentration of solution.
and

Statement 2
Copper dissolves from anode and gets deposited on cathode.

112. Statement 1
The total vapour pressure for a liquid mixture of benzene and toluene under liquid vapour equilibrium is a linear function of the mole fraction of benzene in the liquid phase.
and

Statement 2
A liquid mixture of $CHCl_3$ and $(C_2H_5)_2O$ (of any composition) has a lower vapour pressure than that of either component at (say) 298 K.

113. Statement 1
Every Bronsted acid is also a Lewis acid and vice versa.
and

Statement 2
Adding water to an aqueous solution of acetic acid raises the pH of the solution at (say)298 K.

Linked Comprehension Type Questions

Directions: **This section contains 1 paragraph. Based upon the paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (a), (b), (c)and (d), out of which ONLY ONE is correct.**

Passage I

Tollens reagent which is ammoniacal $AgNO_3$ is used for the detection of aldehyde group. Certain carbohydrates also reduce Tollens reagent. Such carbohydrates are called reducing sugars. Glucose when added to Tollens reagent gluconic acid is formed.

$Ag^+_{(aq)} + e^- \rightarrow Ag_{(s)}$ $\quad E^0_{red} = 0.8$ V

$C_6H_{12}O_6 + H_2O \rightarrow \underset{\text{(gluconic acid)}}{C_6H_{12}O_7} + 2H^+ + 2e^-$ $\quad E^0_{oxid} = -0.05$ V

$[Ag(NH_3)_2]^{2+} + e^- \rightarrow Ag_{(s)} + 2NH_3$ $\quad E^0_{red} = 0.337$ V

Given $\dfrac{2.303\,RT}{F} = 0.0592$ and $\dfrac{F}{RT} = 38.92$ at 298K

114. The value of 'Ink' for the reaction $2Ag^{+}_{(aq)} + C_6H_{12}O_6 + H_2O \rightarrow 2Ag_{(s)} + C_6H_{12}O_7 + 2H^{+}$is calculated to be

(a) 66.13 (b) 58.38

(c) 28.30 (d) 46.29

115. When ammonia is added to the solution pH is raised to 11. Which half cell is affectedby change in pH and by how much?

(a) $E_{(\text{oxidation electrode})}$ increase by a factor 0.65 from $E^0_{(\text{oxidation electrode})}$

(b) $E_{(\text{oxidation electrode})}$ decrease by a factor 0.65 from $E^0_{(\text{oxidation electrode})}$

(c) $E_{(\text{reduction electrode})}$ increase by a factor 0.65 from $E^0_{(\text{reduction electrode})}$

(d) $E_{(\text{reduction electrode})}$ decrease by a factor 0.65 from $E^0_{(\text{reduction electrode})}$

116. Ammonia is always added to the mixture to facilitate the reaction. Which among the following statements regarding the reaction is incorrect?

(a) NH_3 combines with Ag^+ to form a complex.

(b) In absence of ammonia silver salt of gluconic acid is formed.

(c) NH_3 has affected the standard reduction potential of glucose/gluconic acid electrode.

(d) $[Ag(NH_3)_2]^+$ is a stronger oxidising agent than Ag^+.

Multiple Correct Objective Type Questions

Directions: **Each question in this section has four suggested answers out of which ONE OR MORE answers will be correct.**

117. When sodium acetate is added to dilute aqueous acetic acid [$K_a = 1.8 \times 10^{-5}$] (T = 298 K)

(a) For equal concentrations of the acid and salt pH = 4.745.

(b) An aqueous solution of sodium acetate has a pH $>$ 7 (at 298 K).

(c) In an aqueous solution of KCl, the conductance ratio $\left(\frac{\Lambda_c}{\Lambda_0}\right)$ gives α, the degree of dissociation of the salt in the solution.

(d) Comparing 0.1 M aqueous solution of (i) acetic acid (ii) formic acid and (iii) mono chloro acetic acid pH of (i) $<$ pH of (ii) = pH of (iii).

118. Which among the following statements are correct?

(a) For an aqueous solution of Propanoic acid, $[H^+] \simeq (K_a\ c)^{1/2}$ where, K_a = dissociation constant and c = concentration.

(b) The specific conductance, K of KNO_3 solution in water decreases as the solution gets diluted.

(c) A salt like NaCl is fully ionized even in the solid state.

(d) The ionic product of water is 10^{-14} i.e., constant at all temperatures.

119. Indicate among the following reactions/process involving 6 Faradays.

(a) Reduction of Fe^{3+} to Fe^{2+}

(b) Oxidation of Sn^{2+} to Sn^{4+}

(c) Reduction of $C_6H_5 - NO_2$ to $C_6H_5 - NH_2$

(d) Oxidation of Cl^- to ClO_3^-

Matrix-Match Type Question

Directions: **Match the elements of Column I to elements of Column II. There can be single or multiple matches.**

120.

	Column I		**Column II**
(a)	Relative lowering of vapour pressure	(p)	weak acids/bases
(b)	Acid base indicators	(q)	Dilute solutions of non volatile solutes in solvents
(c)	Vant Hoff factor	(r)	colour change due to structural change
(d)	Conductance ratio	(s)	degree of dissociation

ADDITIONAL PRACTICE EXERCISE

Subjective Questions

121. The normal calomel electrode has an electrode potential of +0.28 volt; E° (Cu^{++}, Cu) = +0.34 volt (at 25°C). The cell Cu/Cu^{2+} xM // KCl (1M) $Hg_2Cl_{2(s)}$/Hg has emf = Zero for a certain value of x. Calculate x.

122. Calculate the reduction potential values of the half cells

(i) Pt/Fe^{2+}(2.0 M), Fe^{3+}(0.02 M) given E°red = 0.771 volt and T = 25°C

(ii) Ag/AgCl, KCl (0.01 M), given K_{sp} of AgCl = 1.3×10^{-10} and E° (Ag^+, Ag) = 0.799volt

123. 4 L of NH_3 gas at 27°C and 0.984 atm pressure are dissolved in 89.6 g of glacial acetic acid. Calculate the freezing point of the solution assuming glacial acetic acid to be a polar solvent. Freezing point of acetic acid is 16.6°C and K_f for acetic acid is 3.9 K kg mol^{-1}.

124. During charging of a lead storage battery, a constant current of 5 A was passed for 2 hours.

(i) How many grams of lead sulphate are reduced at the cathode during this process?

(ii) How many grams of H_2SO_4 are added to the battery?

125. (i) Formulate the fuel cell using H_2 and O_2 gases as the reactants and KOH as the electrolyte and in which the overall reaction is

$2H_{2(g)} + O_{2(g)} \rightarrow 2H_2O_{(\ell).}$

(ii) Write the electrode reactions at the two half cells.

(iii) From the ΔG_f^o, ΔH_f^o and S° values of all the reactant species given below at 298K, calculate E° of the cell at 25°C and 75°C, assuming ΔH and ΔS are independent of temperature.

Themodynamic propertyt \ Species	$H_{2(g)}$	$O_{2(g)}$	$H_2O_{(l)}$
ΔG° (kJ mol^{-1})	0	0	–237.0
ΔH°(kJ mol^{-1})	0	0	–286.0
S° (J K^{-1} mol^{-1})	130.6	205.0	70.0

126. A saturated solution of silver chloride is found to have a specific conductivity of 1.12×10^{-6} ohm^{-1} cm^{-1} (corrected for the specific conductance of water used). The absolute velocities of Ag^+ and Cl^- ions under a potential gradient of unity at the same temperature are 5.6×10^{-4} and 6.8×10^{-4} cm s^{-1} respectively. Calculate the solubility product of AgCl.

[Hint: $\frac{\Lambda_o}{F}$ = ionic mobility].

127. 10 g of phenol is dissolved in 500 g of toluene. If it dimerises to the extent of 70%, what is the boiling point of the solution? ($K_{b\ (Toluene)}$ = 3.32 K kg mol^{-1}, boiling point of toluene = 110.7°C)

128. The reaction occurring in a galvanic cell is $2Ag_{(s)} + Hg_2Cl_{2\,(s)} \rightarrow 2AgCl_{(s)} + 2Hg_{(\ell)}$

(i) Formulate a suitable cell in which the above reaction occurs.

(ii) The emf of the cell is 0.0455 V at 298K and ΔH is 10.7 kJ mol^{-1}. Find the ΔG and ΔS.

129. If oxidation of methane is to be used as a fuel cell reaction, what are the anodic and cathodic reactions in a such a cell? The ΔG_f^0 values of $CH_{4(g)}$, $CO_{2(g)}$ and $H_2O_{(\ell)}$ are -50.7 kJ mol^{-1}, -394.4 kJ mol^{-1}, and -237.1 kJ mol^{-1} respectively at 298K. What is the standard emf of the cell?

130. The standard potentials of Au^{3+}/Au; Au^{+}/Au are 1.5 V and 1.69 V respectively at 298K. Will the disproportionation reaction occur spontaneously? If so, what is the equilibrium constant of the disproportionation reaction?



Straight Objective Type Questions

Directions: **This section contains multiple choice questions. Each question has 4 choices (a), (b), (c)and (d), out of which ONLY ONE is correct.**

131. Calculate Emf: Pt (H_2, 0.5 atm)/H^+ (0.1 M) // MnO_4^- (0.1 M); Mn^{2+} (1M); H^+ (0.1 M)/Pt in volts. E° (LHS) = 0; E° (RHS) = 1.51 V, T = 298 K.

(a) 1.224 V (b) 1.453 V (c) 2.142 V (d) 1.114 V

132. Calculate electrode potential of the half cell Pt/Fe^{2+} (2 M); Fe^{3+} (0.02 M).
E° = 0.771 volt T = 298 K.

(a) 0.653 volt (b) 0.356 volt (c) 0.365 volt (d) 0.536 volt

133. The mole fraction of a non-volatile solute in an aqueous solution is 0.073. What is the molality?

(a) 5.046 (b) 6.504 (c) 4.650 (d) 4.056

134. 0.1 mole of a non-volatile solute in 100 g of a solvent shows a boiling point elevation, $\Delta T_b = 0.3$°C. Calculate K_b, the molal elevation constant (in deg molal^{-1})

(a) 0.35 (b) 0.52 (c) 0.3 (d) 0.62

135. The specific conductance, K of a 0.01 M solution of a uni-univalent electrolyte is 1.061×10^{-3} Ω^{-1} cm^{-1}. The equivalent conductance Ω in Ω^{-1} cm^2 eqt^{-1} is

(a) 96.7 (b) 106.1 (c) 126.1 (d) 100.2

136. A 20 milli ampere current is used to deposit 0.1 g of silver in a silver coulometer. Atomic wt of Ag = 108. Calculate the time for which the current is passed.

(a) 1.241 hrs (b) 1.420 hrs (c) 1.121 hrs (d) 1.114 hrs

137. The freezing point of benzene is 5.45°C. $K_f = 5.06$ deg molal^{-1}. Calculate its molar enthalpy of fusion.

(a) 2930 cal mol^{-1} (b) 3290 cal mol^{-1} (c) 3920 cal mol^{-1} (d) 2390 cal mol^{-1}

138. Normal hydrogen electrode act as anode when it is connected to which of the following standard electrodes?

(a) $Cu^{2+}_{(aq)}/Cu_{(s)}$ (b) $Zn^{2+}_{(aq)}/Zn_{(s)}$ (c) $Fe^{2+}_{(aq)}/Fe_{(s)}$ (d) $Mg^{2+}_{(aq)}/Mg_{(s)}$

139. Calculate the % of NaCl in terms of weight/volume in 0.5 M solution of the salt.

(a) 2.529 (b) 3.125 (c) 2.925 (d) 2.295

140. Calculate the vapour pressure of the solution obtained by dissolving 25 g of $CuSO_4 . 5H_2O$ (M ≃ 250 g) in 360 ml of water, in the form xP° [i.e., calculate x]

(a) 0.91 (b) 0.99 (c) 0.89 (d) 0.85

141. K_b for water = 0.52 deg molal^{-1} Calculate the molar mass of a nonvolatile solute in g mole^{-1}. If 2.44 g dissolved in 75 g of water raises the boiling point by 0.413°C.

(a) 41 (b) 35 (c) 31 (d) 48

142. When a dilute solution containing the ions Na^+, K^+, Cl^- and NO_3^- is cooled to its freezing temperature the solid that separates is

(a) NaCl (b) both NaCl and KNO_3

(c) ice and the solute of lower solubility (d) ice

143. KCl is used in solution in water as a salt bridge because (i) It provides highly conducting link between the two half cells of a complete cells. (ii) it practically eliminates the liquid junction potential (iii) K^+ and Cl^- are iso electronic.

(a) (i) Only (b) (i) and (ii) (c) (iii) (d) (i) and (iii)

144. During the charging of the lead accumulator

(a) Anode layer $PbSO_4$ forms Pb. (b) Anode layer $PbSO_4$ forms PbO_2

(c) Both cathodic and anodic layers form Pb. (d) The density of the H_2SO_4 solution decreases.

145. For a certain electrolyte solution in water K= $1.75 \times 10^{-3}\ \Omega^{-1}\ cm^{-1}$, $\Lambda = 95\ \Omega^{-1}\ cm^2\ eqt^{-1}$. Calculate the normality.

(a) 1.248×10^{-3} N (b) 1.284×10^{-3} N (c) 1.842×10^{-2} N (d) 2.148×10^{-2} N

146. A current of 0.268 amp passing for 5 hours through an electrolytic (metal salt) solution deposit on the cathode one gram of the metal (At. wt = 60). What is the formula of the chloride of the metal?

(a) MCl (b) MCl_2 (c) MCl_3 (d) MCl_4

147. Which of the following exhibits at one atm (external pressure) the highest boiling point? (At.wt Na = 23; K = 39; Ba = 137.4)

(a) 1 g NaCl added to one litre of water (b) 1 g KCl added to one litre of water

(c) 1 g NH_4Cl added to one litre of water (d) 1 g $BaCl_2.2H_2O$ added to one litre of water

148. Calculate the quantity of electricity in coulombs required to reduce 0.1 mole of nitrobenzene to aniline.

(a) 70500 (b) 57900 (c) 50700 (d) 61700

149. At 298 K, $Hg_2^{2+} + 2e^- \rightarrow 2Hg_{(l)}$; E° = 0.800 V.

$Hg^{2+} + 2e^- \rightarrow Hg_{(l)}$; E° = 0.850 V.

Calculate equilibrium constant $Hg_2^{2+} \rightarrow Hg_{(l)}; + Hg_{aq}^{2+}$.

(a) 2.036×10^{-2} (b) 2.630×10^{-3} (c) 2.667×10^{-2} (d) 3.026×10^{-2}

150. To produce Aluminium by electrolysis at the rate of 50 Kg hr^{-1}, what is the current strength required? (in amps) (At. wt of Al = 27)

(a) 1.984×10^4 (b) 1.894×10^3 (c) 8.947×10^3 (d) 1.489×10^5

151. Calculate % by wt of ethyl acetate in the vapour over in a mixture of ethyl acetate and propyl acetate at 60°C which has a total vapour pressure of 234 mm. Vapour pressure of ethyl acetate at 60°C = 408 mm. Vapour pressure of propyl acetate at 60°C = 176 mm.

(a) 40 (b) 50 (c) 67 (d) 33

152. Calculate vapour pressure of a solution containing 15.4 g of cane sugar in 175 g of water at 100°C.

(a) 756.5 mm (b) 706.5 mm (c) 697.5 mm (d) 806.5 mm

153. Calculate at 20°C the vapour pressure of diethyl ether given that a solution of 5 g of a nonvolatile solute of molar mass 59.6 g mol^{-1} in 100 g of ether ($C_2H_5 - O - C_2H_5$) has a vapour pressure of 412.7 mm.

(a) 410 mm (b) 490 mm (c) 440 mm (d) 390 mm

154. Dry air was passed slowly and successively through (a) bulbs containing 10% (by wt) aqueous solution of a nonvolatile solute, x (b) bulbs containing pure water. The loss wt in the first series of bulbs was 0.836 g and the corresponding loss of wt in the second series was 0.025 g. Calculate the mol wt of x (in g mol^{-1}).

(a) 57 (b) 75 (c) 52 (d) 67

155. Given the cell $Ag/AgCl_{(s);}$ 0.1 M KCl // $AgNO_3$ (0.1 M)/Ag calculate its emf at 25°C; K_{sp} of AgCl ≃ 1.7×10^{-10} at 298 K.
(a) 0.458 volt (b) 0.584 volt (c) 0.485 volt (d) 0.628 volt

156. Consider the cell: $Cu/Cu_2O_{(s)}$, $NaOH_{(aq)}/Pt(H_2)$. The cell reaction is
$Cu_2O_{(s)} + H_{2(g)} \rightarrow 2Cu_{(s)} + H_2O$ the emf measured at 18°C is 0.461 volt. Calculate ΔG for the cell reaction (in cal).
(a) −25127 (b) −21275 (c) −27125 (d) −17525

157. Which among the following metals cannot be formed in aqueous medium by the electrolysis of their aqueous salt solutions?
(a) Copper (b) Zinc (c) Sodium (d) Magnesium

158. Consider the cell $Ag/AgNO_{3\ (0.02\,M)}$ // $AgNO_{3\ (0.1\,M)}/Ag$. Assuming that 0.1 M solution and 0.02 M solution are ionized (effectively) to 89% and 95% respectively, calculate the emf at 298 K.
(a) 0.0465 volt (b) 0.456 volt (c) 0.0544 volt (d) 0.0396 volt

159. At 100°C, the vapour pressure of a solution of a $Ca(NO_3)_2$ [6.69 g in 100 g of water] = 746.9 mm. Calculate the degree of dissociation of the solute.
(a) 0.467 (b) 0.746 (c) 0.764 (d) 0.674

160. A current passed for 6 minutes through a voltameter containing dil. H_2SO_4 liberated 40 ml of electrolytic gas measured at 15°C and 748 mm (after correction for aqueous tension). Calculate the mean value of the current in amp.
(a) 0.559 (b) 0.425 (c) 0.595 (d) 0.455

161. Ionic conductances of (H^+), (Na^+), are $\lambda^\circ_{H^+} = 318\ \Omega^{-1}cm^2\ eqt^{-1}$.

$\lambda^\circ_{Na^+} = 44\ \Omega^{-1}cm^2\ eqt^{-1}$ Specific conductivity of 0.001 M sodium acetate solution is $0.0000781\ \Omega^{-1}cm$. Assume that sodium acetate is completely ionized. Calculate Λ_0 of $CH_3 - COOH$ (in $\Omega^{-1}cm^2\ eqt^{-1}$).
(a) 317.5 (b) 351.7 (c) 175.3 (d) 513.7

162. Degree of dissociation, $\alpha = 0.0134$ at a concentration of 0.1 M for an organic monoprotic acid. Calculate α for a solution of the acid at a concentration C = 0.001 M.
(a) 0.134 (b) 0.143 (c) 0.234 (d) 0.243

163. A mixture of 0.005 N lactic acid and 0.38 N acetic acid has a $[H^+] = 3.67 \times 10^{-3}$. If K_a for acetic acid is 1.8×10^{-5}. K_a for lactic acid at T = 298 K is
(a) 3.22×10^{-5} (b) 1.87×10^{-5} (c) 1.78×10^{-5} (d) 2.23×10^{-4}

164. Consider the cell: $Cu/\underset{\text{concen.}c_1}{Cu^{2+}}$ // $\underset{\text{concen.}c_2}{Ag^+}$ /Ag. at 298 K $E°(Cu^{++}, Cu) = 0.34$ volt

$E°\ (Ag^{++}, Ag) = 0.798$ volt. At what ratio $\dfrac{\left[Ag^+\right]^2}{\left[Cu^{++}\right]}$ will the measured emf be zero?

(a) 2×10^{-15} (b) 3×10^{-16} (c) 3×10^{-14} (d) 2×10^{-14}

165. Calculate the electrochemical equivalent of Cu in gc^{-1} given that the atomic wt of Cu = 63.6.
(a) 3.295×10^{-4} (b) 2.935×10^{-3} (c) 2.593×10^{-3} (d) 3.953×10^{-2}

166. $E°\ (Ag^+, Ag) = 0.8$ volt at 25°C. K_{sp} for AgCl = 1.56×10^{-10}. Calculate E° red for the half cell: $Ag\ /\ AgCl_{(s)}, Cl^-$.
(a) 0.1214 V (b) 0.242 V (c) 0.2214 V (d) 0.312 V

167. The freezing point of a solution of $Ba(OH)_2$ containing one mole of solute in 64 litres of the solution is −0.0833°C. Asume that density of the solution is one gram per ml. For calculation take $K_f = 1.89$ deg molal $^{-1}$. Calculate approxymately the $[OH^-]$ of the solution obtained by ionization of $Ba(OH)_2$.
(a) 2.49×10^{-2} M (b) 2.94×10^{-2} M (c) 1.89×10^{-2} M (d) 9.81×10^{-2} M

168. A solution of calcium formate [0.063 mole in a Kg of water] freezes at t°C. $K_f = 1.86$ deg molal^{-1}. $\alpha = 0.85$ (deg. of dissociation) calculate t°C.
(a) −0.316°C (b) −0.260°C (c) −0.202°C (d) −0.361°C

169. 100 g of a solution of cane sugar dissolved in water. The solution just starts freezing at −0.38°C. $K_f = 1.86$ deg molal^{-1}. Calculate wt of solvent in the solution.
(a) 1134 g (b) 2114 g (c) 1431 g (d) 1034 g

170. The vapour pressure of a solution of urea in water = 736.2 mm at 100°C. Calculate approximately the osmotic pressure of the solution at 15°C. Assume the density of the solution to be 1 g ml^{-1} for the purpose of calculation.
(a) 44.10 atm (b) 54.10 atm (c) 31.41 atm (d) 41.14 atm.

Assertion–Reason Type Questions

Directions: **Each question contains Statement-1 and Statement-2 and has the following choices (a), (b), (c)and (d), out of which ONLY ONE is correct.**

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False
(d) Statement-1 is False, Statement-2 is True

171. **Statement 1**
Electrode potential of hydrogen electrode varies with pH of solution used.
and

Statement 2
NHE is a standard electrode.

172. **Statement 1**
The reaction $2Cu^+ \rightarrow Cu^{2+} + Cu$ is spontaneous.
and

Statement 2
$Cu^+ + e^- \rightarrow Cu$..$(E°)_1$ $Cu^+ \rightarrow Cu^{2+} + e^-$…$(E°)_2$ $[(E°)_1 + (E°)_2]$ has a positive value.

173. **Statement 1**
Relative lowering of vapour pressure for a dilute solution, $\left(\frac{p_o - p_s}{p_o}\right)$ equal (nearly) $\frac{mM}{1000}$ where, m = molality, M = molar mass of the solvent in g mol^{-1}.
and

Statement 2
The correct definition of the molal depression constant, K_f = is Lt m $\rightarrow$ o $\left(\frac{\Delta T_f}{m}\right)$.

174. **Statement 1**
In an aqueous solution of NaCl, Λ_m decreases as C increases, the reason being that the extent of ionization of the solute decreases at higher concentrations.
and

Statement 2
Variation of Λ_m with concentration for NaCl follows the equation $\Lambda_m = \wedge_m^0 - b\sqrt{c}$ forlow concentration.

175. Statement 1

For a dilute solution of a monoprotic weak acid, HA, with dissociation constant Ka, at a concentration C, the concentration of the A^- ion is $[A^-] = \dfrac{Ka\,C}{Ka + [H^+]}$

and

Statement 2

In the above case $[H^+] = \dfrac{Ka\,C}{Ka + [H^+]} + \dfrac{Kw}{[H^+]}$ {Kw = ionic product of water}

176. Statement 1

In a solution of molarity, M, molality, m and density, d (g ml^{-1}),

$d = M\left(\dfrac{1}{m} + \dfrac{m_2}{1000}\right)$, m_2 = molar mass of the solute (g mol^{-1})

and

Statement 2

The vapour pressure, p, of a liquid in a liquid-vapour equilibrium changes with change in temperature.

177. Statement 1

In a mixture of two miscible liquids A and B, the total vapour pressure of any temperature T is every case a linear function of the mole fraction of either A (or B).

and

Statement 2

In the distillation of an ideally behaving mixture of two liquids, A and B, the vapour is always richer in the more volatile components so that separation of the pure components, A and B practically possible by fractional distillation.

178. Statement 1

In the electrolysis of a solution containing Cu^{2+} and Bi^{3+} ions, it is possible to adjust the concentration ratio $\left[Cu^{2+}\right] \Big/ \left[Bi^{3+}\right]$ in such a way that both are deposited on the cathode deposit the difference in their E° values, i.e., E (Cu^{2+}, Cu)and E° (Bi^{3+}, Bi).

and

Statement 2

In concentrated solutions of polyvalent ions $\Lambda \vee s\ \sqrt{c}$ is nonlinear, one reason being the formation of ion pairs and other species.

179. Statement 1

Phenol and water are practically miscible liquids at not too high temperatures but any composition of the mixed liquid system becomes completely miscible (and forms one homogenous liquid phase)above a particular temperature, which depends on the composition of the liquid mixture.

and

Statement 2

Since, $\Delta G = \Delta H - T\Delta S$, it follows that at and above a particular temperature, $\Delta G < 0$ for a fore-mentioned liquid system.

180. Statement 1
Trouton's law does not hold for water at its boiling point.
and

Statement 2
Water molecules are hydrogen bonded and hence water behaves abnormally.

Linked Comprehension Type Questions

Directions: **This section contains 3 paragraphs. Based upon the paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (a), (b), (c)and (d), out of which ONLY ONE is correct.**

Passage I

In the case of strong electrolyte say NaCl, KNO_3 etc. a plot of Λ vs $\sqrt{c}$ in dilute solutions is linear with a negative slope. $\Lambda = \Lambda_0 - (\text{constant})\sqrt{c}$ at a given temperature. $\Lambda_0 \equiv \left(\underset{\sqrt{c}\to 0}{\text{Lt}\,\Lambda}\right)$. By Kohlrausch law $= \wedge^\circ = \lambda^\circ_+ + \lambda^\circ_-$ i.e. sum of the ionic conductances for the cation and anion. The above formula holds for a simple case (uni – univalent electrolyte)but can be extended to more general situations with reasonable assumptions. The ratio $\frac{\lambda^\circ_+}{\Lambda^\circ}$ and $\frac{\lambda^\circ_-}{\Lambda^\circ}$ are the limiting values of the transport numbers of the cation and anion respectively. (Note: the specific conductance, K (Ω^{-1} cm^{-1})of an electrolyte solution and the equivalent conductance, $\wedge$ (Ω^{-1} cm^2 eqt^{-1})are related as $\wedge = \frac{1000\,K}{c}$ where c = concentration in g eqt L^{-1}).

181. The conductance ratio $\frac{\Lambda}{\Lambda_0} = 0.936$ given this for a certain solution of KCl and $\Lambda = 122\ \Omega^{-1}\ cm^2\ eqt^{-1}$, and $\frac{\lambda^\circ_+}{\Lambda^\circ} = \frac{0.98}{1.98}$. Calculate the limiting values of ionic conductances of K^+ and Cl^- ions in $\Omega^{-1}\ cm^2\ eqt^{-1}$.
(a) 64.51, 65.83 (b) 74.60, 26.40 (c) 30.31, 69.69 (d) 70.12, 29.88

182. Λ_0 of a saturated solution of AgCl in water at 298 K is 124.5 $\Omega^{-1}\ cm^2\ eqt^{-1}$. Assuming this to be Λ calculate c i.e., the solubility of AgCl in water in moles L^{-1}. K is $1.37 \times 10^{-6}\ \Omega^{-1}\ cm^{-1}$.
(a) 1.6×10^{-5} (b) 1.1×10^{-5} (c) 1.9×10^{-5} (d) 1.8×10^{-5}

183. Λ_0 for water at 298 K is 497 $\Omega^{-1}\ cm^2\ eqt^{-1}$, K = $0.5 \times 10^{-7}\ \Omega^{-1}\ cm^{-1}$. Calculate the ionic product of water Kw = $[H^+]$ $[OH^-]$.
(a) 1.312×10^{-14} (b) 1.213×10^{-14}
(c) 1.012×10^{-14} (d) 1.422×10^{-14}

Passage II

According to Faraday's I law the mass of material (say a metal) liberated at an electrode during electrolysis is proportional to the number of coulombs passes i.e., m = Z it where I = mean current in amp, t = time of electrolysis in sec and Z = proportional constant known as electrochemical equivalent. By the II law of Faraday one gram eqt of any species is liberated by the passage of one Faraday (i.e., 96500 coulombs)of electricity.

184. Calculate the number of coulombs required for the production of 2 moles of $NaClO_4$ by anodic oxidation of $NaClO_3$.
(a) 4.36×10^4 (b) 3.86×10^5 (c) 3.64×10^6 (d) 6.43×10^5

185. Calculate Z, the electrochemical eqt of silver in electrolysis, in g/c.
(a) 1.912×10^{-3} (b) 1.292×10^{-2} (c) 1.119×10^{-3} (d) 1.415×10^{-4}

186. A current of 50 milliamps is passed through a solution of $AgNO_3$ for 6 hours using silver electrodes density of silver = 10.5 g/cm^3. If the cathode plate has a area of 10 cm^2 and deposition take place on both sides of the plate, what is the thickness of the deposit (cm).

(a) 7.561×10^{-3} (b) 5.756×10^{-3} (c) 6.752×10^{-2} (d) 6.257×10^{-2}

Passage III

The depression of freezing point of a dilute solution of a non-volatile solute in a solvent is a colligative property i.e., the depression ΔT_f depends on the concentration i.e., molality of the solute and not on its nature. By defination, molality = number of moles of solute dissolved in a Kg of the solvent = m = $\frac{W_2}{M_2} \times \frac{1000}{W_1}$. Where, W_2 = number of grams of solute, M_2 = molar mass of the solute, W_1 = number of grams of the solvent, ΔT_f = difference in the freezing points of the solution and the solvent. We have $\Delta T_f = K_f$ x m where, K_f is a proportionality constant known as molal depression constant (in deg molal^{-1}). The equation holds for dilute solution. K_f can be shown to be equal to $\frac{0.002 T_f^2}{l_f}$ where, T_f = freezing point of the solvent in Kelvin and l_f is the latent heat of fusion of the solid solvent in cal g^{-1}.

187. For water $l_f = 80$ cal g^{-1} and $T_f = 273$ K. Calculate K_f.

(a) 1.86 deg molal^{-1} (b) 1.80 deg molal^{-1} (c) 1.75 deg molal^{-1} (d) 1.90 deg molal^{-1}

188. Pure camphor freezes at 178.4°C. K_f = 40 deg molal^{-1}1.5 g of a compound of molar mass 125 g mol^{-1} is dissolved in 35 g of camphor. What is the freezing point?

(a) 170.4°C (b) 164.7°C (c) 160.4°C (d) 180.2°C

189. The vapour pressure p_s of a dilute solution of a nonvolatile solute in a solvent is proportional to the molality in the solution of the solute: $\frac{p_s}{p_o} = 1 - \left(\frac{mM}{1000}\right)$, p_o is the vapour pressure of the pure solvent at the temperature in question, M = molar mass of the solvent. Calculate $\frac{p_s}{p_o}$ for water at 26°C for an aqueous solution of 20 g of glucose ($C_6H_{12}O_6$) in 70 g of water.

(a) 0.9147 (b) 0.9214 (c) 0.9044 (d) 0.9714

Multiple Correct Objective Type Questions

Directions: **Each question in this section has four suggested answers of which ONE OR MORE answers will be correct.**

190. The depression of freezing point of 0.2 molal glucose solution in water is the same as that of (Assume complete dissociation of the salts).

(a) 0.1 molal KNO_3 solution (b) 0.2 molal urea solution
(c) 0.1 molal $MgSO_4$ solution (d) 0.05 molal K_2SO_4 solution

191. Phenol and water are partially miscible liquids.

(a) Each composition of the phenol–water system forms one homogenous liquid phase above a certain temperature.
(b) There is a maximum temperature above which all compositions form one homogenous liquid phase.
(c) Increase in temperature increases the mutual solubility of the liquid components.
(d) At lower temperatures the components become totally immiscible.

192. The vapour pressure of water is lowered by 2.5% (at 298 K) by adding which one of the following to one mole of water? (sucrose = $C_{12}H_{22}O_{11}$. At. mass I = 127, K = 39, Ca = 40, Na = 23, Cl = 35.5)

(a) 1.46 of NaCl (b) 1.37 of Ca $(NO_3)_2$ (c) 2.575 of KI (d) 8.55 of sucrose

193. Which among the following when dissolved in 1 kg water produce the same depression in freezing point?
(a) 6 g urea and 18 g glucose
(b) 8 g urea and 12 g glucose
(c) 10 g urea and 8 g glucose
(d) 12 g urea

194. Molar entropy of vapourization is nearly 21 to 22 cal deg^{-1} mol^{-1} for
(a) benzene
(b) water
(c) cyclohexane
(d) toluene

195. Identify the correct statements among the following.
(a) Λ_0 for acetic acid is obtained by plotting Λ vs $\sqrt{c}$ and extrapolating to the limit $c \rightarrow 0$.
(b) Λ_0 for acetic acid is = Λ_0 of $CH_3 - COO\,Na + \Lambda_0$ of $HCl - \Lambda_0$ of NaCl
(c) Λ_0 for acetic acid is obtained by plotting $(\Lambda \times c)$vs $\frac{1}{\Lambda}$ and thus from the ratio $\frac{-\text{slope}}{\text{intercept}}$ of the linear plot.
(d) An accurate value of $K_a\Lambda_0$ is obtained as the mean value of $\wedge\sqrt{c}$ measured and calculated at different concentration c.

196. Emf of an electrochemical cell varies with
(a) electrodes used
(b) temperature
(c) concentration of electrode solution
(d) size of anode and cathode used.

197. Which of the following equations are correctly represented?
(a) $\frac{p_o - p_s}{p_o}$ = mole fraction of the solute
(b) $d = M\left(\frac{1}{m} + \frac{m_2}{1000}\right)$. d = density of a solution in g mol^{-1}, M = molarity, m = molality, m_2 = molar mass of the solute.
(c) $\Delta S = nF\left(\frac{\partial E}{\partial T}\right)_p$
(d) $\Lambda = \Lambda_0 - (\text{constant}) \times c$



Matrix-Match Type Question

Directions: **Match the elements of Column I to elements of Column II. There can be single or multiple matches.**

198.

Column I	**Column II**
(a) Vapour pressures of liquid mixtures	(p) Formation of azeotrope
(b) Fractionation of ethanol–water mixture	(q) Greater mole fraction of the more volatile component in the vapour
(c) Chloroform–ether mixture	(r) Deviation from Raoults law
(d) Separation of pure components by distillation (repeated)	(s) Linear variation with mole fraction of components in the liquid phase in the case of ideal liquids

199.

Column I	**Column II**
(a) The ratio $\left(\frac{\Lambda}{\Lambda_0}\right)$	(p) linear variation with $\sqrt{c}$ for dilute solutions of uni-univalent strong electrolytes
(b) Standard used in emf measurements	(q) calomel electrode
(c) anion reversible electrode	(r) Degree of dissociation for weak electrolyte
(d) Proportional to $\sqrt{\frac{Ka}{c}}$ for weak acids	(s) $Ag/AgCl_{(s)}$, $KCl_{(M)}$

200.

Column I	**Column II**
(a) Trouton's formula	(p) Nearly constant in buffer solutions
(b) Liquid–vapour equilibrium $\left(\frac{\Delta H_v}{T}\right)$	(q) ½ pKw + ½ pK_a ½ log c in solution in water of saltsof weak acids and strong bases
(c) No specific intermolecular interaction like hydrogen bonding	(r) Molar entropy of vaporisation
(d) pH value	(s) normal liquids

HINTS AND SOLUTIONS

Topic Grip

1. $\dfrac{\text{Loss of mass of solvent}}{\text{Loss of mass of solvent} + \text{Loss of mass of solution}} = \dfrac{p^\circ - p}{p^\circ}$ $x_B = \dfrac{n_B}{n_A}$

$$\frac{60\times10^{-3}}{60\times10^{-3}+2.94} = \frac{.018\times10^{3}}{M} \Big/ \frac{0.1\times10^{3}}{18}$$

$M = 168$

$$\frac{\text{Mol. mass of solute}}{\text{Mol. mass of solvent}} = \frac{168}{18} = 9$$

2. $P = 32.1 + (18.6 \times x)$ mm. Put $x = 1$.

$P^\circ_{\text{Benzene}} = 50.7$ mm

Put $x = 0$. $P^\circ_{\text{toluene}} = 32.1$ mm.

Further put $x = 0.5$,

$P = (32.1 + 9.3)$ mm $= 41.4$ mm $= P_B + P_T$

$$\left[\left(P_0\right)_B \times x + \left(P_0\right)_T \times (1-x)\right] = 41.4 \text{ mm.}$$

But $x = 0.5$

$\therefore$ P_B mm $= (50.7 \times 0.5) = 25.35$ mm.

P_T mm $= (32.1 \times 0.5) = 16.05$ mm

$$\text{\% of Benzene in vapour} = \left(\frac{25.35}{41.4}\right) \times 100 = 61.23\%.$$

% of Toluene = 38.77%

3. $K_f = \dfrac{0.002\ T_{f^2}}{l_f}$; $T_f = 353$ K.

$$l_f = \frac{4600}{128} = 35.94 \text{ cal g}^{-1}$$

$$\therefore \quad K_f = \left(\frac{0.002\times353\times353}{35.94}\right) \text{ deg molal}^{-1}$$

$= 6.934$ deg molal^{-1}

4. When 250 g of ice have separated out the liquid phase contains 750 g of water.

$$\therefore \quad \Delta T_f = 0.5 = 1.86 \times \frac{m}{180}\ \frac{1000}{750}$$

$$\therefore \quad m = \left(\frac{0.5\times180\times750}{1.86\times1000}\right) = 36.29 \text{ g}$$

5. $\Delta T_b = (81.5^\circ - 80.1^\circ)C = 1.4^\circ C$

One molal solution contains one mole kg^{-1}. It out of this 89.5 % i.e., a fraction 0.895 is dimerized, the actual number of moles

$$= (1 - 0.895) + \frac{0.895}{2} = 0.5525$$

$\therefore$ $1.4 = K_b \times 0.5525$.

$\therefore$ $K_b = 2.534$ deg molal^{-1}

6. Adding the two given equation

$= 0.13 + (-0.14) = -0.01$ volt

$E = -0.01 - \dfrac{0.059}{2} \log \left\{\dfrac{\left[Pb^{2+}\right]}{\left[Sn^{2+}\right]}\right\}$. Take E = 0 on calculation $\dfrac{\left[Pb^{2+}\right]}{\left[Sn^{2+}\right]} = 0.5$

7. The quantity of electric charge passing through the solution is $\dfrac{0.2503}{108}$ F i.e., $\dfrac{0.2503}{108}$ eqt $\equiv \dfrac{0.2503}{2\times108}$ mole of H_2 would be liberated subtracting aqueous tension, pressure i.e., 738.44 mm. using

$PV = nRT$,

$$\frac{738.44}{760} \times V = \frac{0.2503}{216} \times 0.0821 \times 293$$

$\therefore$ V (litres) = 0.02869 litres = 28.69 ml.

8. We use $K_a = \dfrac{\alpha^2 c}{1-\alpha}$ or

$K_a C \simeq (\alpha c^2) = [H^+]^2$ $[H^+] = \sqrt{KaC}$.

For 0.5 M HCOOH, $[H^+] = (1.8 \times 10^{-4} \times 0.5)^{1/2}$

$= 0.9487 \times 10^{-2}$ M

For 1.0 M CH_3 – COOH, $[H^+] = (1.8 \times 10^{-5} \times 1)^{1/2}$

$= 4.2426 \times 10^{-3}$ M

Since this is a concentration cell,

$$\text{Emf} = \frac{0.059}{1} \log \frac{9.487\times10^{-3}}{4.2426\times10^{-3}}$$

$= 0.02062$ volt $= 20.62$ milli volt

9. Λ of $Na_2SO_4 = \dfrac{1000\times2.6\times10^{-4}}{0.002}$

$= 130$ ohm^{-1} cm^2

Assuming the solution to be dilute

$\Lambda_{eq} SO_4^{2-} = 130 - 50 = 80 \text{ ohm}^{-1} \text{ cm}^2$

κ due to $CaSO_4 = 7 \times 10^{-4} - 2.6 \times 10^{-4}$

$= 4.4 \times 10^{-4} \text{ ohm}^{-1} \text{ cm}^{-1}$

$\Lambda_{(CaSO_4)} = 60 + 80 = 140 \text{ ohm}^{-1} \text{ cm}^2$

Assuming the solution to be dilute:

$$S \text{ of } CaSO_4 = \frac{1000 \times 4.4 \times 10^{-4}}{140}$$

$= 3.14 \times 10^{-3}$ eq. L^{-1}. $= 1.57 \times 10^{-3}$ mol L^{-1}

$K_{SP} = (1.57 \times 10^{-3})^2 = 2.46 \times 10^{-6}$

10. (a) – (b) $\Rightarrow Hg_2^{2+} \rightarrow Hg_{(\lambda)} + Hg^{2+}$ $E° = -0.05$ V

$$\text{Log } K = \frac{nFE^0}{2.303RT} = \frac{2}{0.0591} \times -0.05$$

$$= -\frac{0.1}{0.0591} = -1.69$$

$K = 2.04 \times 10^{-2}$

11. We use $\frac{p_s}{p_o} = \frac{N}{N+n}$ (mole fraction solvent)

$$N = \frac{100}{76} = 1.316. \; n = \frac{2}{8 \times 32} = \frac{1}{128}$$

$$= \frac{ps}{854} = \frac{1.316}{1.316 + 0.0078}$$

$\frac{p_s}{854} = \frac{1.316}{1.3238}$. $\therefore$ ps = 849 mm. Hg

12. $\Delta T_f = 10°C = 1.85 \times \frac{m}{62} \times \frac{1000}{2000}$.

$$\therefore \quad m = \frac{10 \times 62 \times 2000}{1.85 \times 1000} \text{ g}$$

m = 670.3 g

13. W first calculate $K_f = \frac{0.002 T_f^2}{l}$

$$= \frac{0.002 \times \left((278.45)^2 \times 78\right)}{2.37 \times 1000}$$

$K_f = 5.1035$ deg mol^{-1}.

$\Delta T_f = (5.45 - 5.09)°C = 0.36°C$

$0.36 = 5.1035 \times \frac{9}{M} \times \frac{1000}{1000}$.

$$\therefore \quad M = \frac{5.1035 \times 9}{0.36} = 127.6 \text{ g mol}^{-1}$$

14. Molar conductance $= \frac{1000 \times K}{C}$

$$= \frac{1000 \times 1.061 \times 10^{-4}}{0.01}$$

$= 10.61 \; \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$

15. $(C_6H_5 - NO_2) \rightarrow (C_6H_5 - NH_2)$ is a 6 electron per mole process. i.e., one mole needs 6 Faradays.

$\therefore$ for 0.1 mole 0.6 Faradays are needed. This is ideal. The actual need will be more, say x Faraday.

$$\therefore \quad x \times \frac{80}{100} = 0.6$$

$$\therefore \quad x = 0.6 \times \frac{10}{8} = 0.75 \text{ Faraday}$$

16. Both statements are true but not linked.

17. Statement 1 is true. Statement 2 is incorrect. Laws of dilute solutions do not hold for highly concentrated solution. Moreover the solute may interact in a complex manner with the solvent.

18. Both statements are true and are interlinked.

19. Statement 2 is incorrect; the pH of NHE is zero.

20. Statement 1 incorrect. Statement 2 is correct;

$\wedge = \wedge_0 - (A + B \wedge_0) \sqrt{C}$.

21. $P_2 = k.x_2$ $\quad 760 \text{ torr} = 1.25 \times 10^6 \times \frac{n_2 x}{\left(\frac{1000}{18}\right)}$

$n_2 = .0338$ moles/kg ≈ 1.5 g/kg

22. Lesser the value of K_H more in the solubility.

23. $P_s = x_o P^\circ_{A +} x_o P^\circ_B = \frac{.25}{.557} \times 105.2 + \frac{.307}{.557} \times 468 = 73 \text{ kP}_a$

24. $106.3 = \wedge_0 - (\text{constant}) \sqrt{0.001}$,

$101.7 = \wedge_0 - (\text{constant}) \sqrt{0.001}$.

Straight forward solving procedure eliminate the second (unknown) term on the r h s, yields $\wedge_0 = 108.4$ $\Omega^{-1} \text{ cm}^2 \text{ eqt}^{-1}$

25. $\wedge = 124.5 = \frac{1000 \kappa}{C}$.

$$\therefore \quad C = \frac{1000 \times 1.37 \times 10^{-6}}{124.5} \text{ eqt L}^{-1}$$

$C = 1.1 \times 10^{-5}$ eqt L^{-1}

26. $\wedge = 124.5 = \dfrac{1000\kappa}{C}$

$= \dfrac{1000 \times 3.3 \times 10^{-4}}{0.02}\ \Omega^{-1}\ cm^2\ mol^{-1}$

$= 16.5\ \Omega^{-1}\ cm^2\ mol^{-1}$

$\alpha = \left(\dfrac{\Lambda}{\Lambda_0}\right) = \left(\dfrac{16.5}{387}\right) = 4.3 \times 10^{-2}$

27. b and d are wrong.

28. b, c, d are correct

29. a, b are correct. The curve of E measured in such cases shows an initial and final flatness with a sharp change at the end point (sigmoid variation)

$\therefore \quad \dfrac{\Delta E}{\Delta V}$ is maximum at this point. (c, d are obviously wrong)

30. (a) → (r), (s)
(b) → (p), (q)
(c) → (s)
(d)→ (s)

IIT Assignment Exercise

31. 34 ml of electrolytic gas

$2H_2O \rightarrow 2H_2 + O_2$

2 mol H_2O require 4F to liberate 67.2 L gases at STP

Hence no. of Faraday required to liberate 34 mL gas at STP $= \dfrac{34 \times 4}{67200} = 2.024 \times 10^{-3}$

Amount of NaOH produced $= 2.024 \times 10^{-3} \times 40$

$= 0.081$ g

32. $C_{14}H_{10} \xrightarrow{\text{oxidation}} C_{14}H_8O_2$ needs 6 Faraday per mole of $C_{14}H_8O_2$

one amp for one hour ≡ (1 × 3600)

coulombs ≡ $\left(\dfrac{3600}{96500}\right)$ Faraday

Molar mass of anthraquinone = 208 g mol^{-1}.

$\therefore$ mass of anthraquinone produced

$= \left(\dfrac{3600}{96500} \times \dfrac{208}{6}\right)$ g = 1.29 g

33. 116.5 g per litre ≡ $\dfrac{116.5 \times 140}{40}$ Faradays

A current of 1250amps for 10 hour

$= \dfrac{1250 \times 10 \times 3600}{96500}$ Faradays

Current efficiency is obtained by taking the ratio,

$\dfrac{116.5 \times 140}{40} = 407.75$

$\dfrac{1250 \times 10 \times 3600}{96500} = 466.32$

$\therefore$ ratio = 0.874

34. After sometime the emf reduces to zero

35. Consider $Ag/AgCl_{(s)}$, Cl^{-1} (unit activity); $a_{cl} = 1$.

$\therefore \quad [Ag^+] = 1.56 \times 10^{-10}$

$\therefore \quad E = E° + \dfrac{0.059}{1} \log (1.56 \times 10^{-10})$

$= 0.7996 + 0.059 \log (1.56 \times 10^{-10})$ volt

$E = 0.7996 + (0.059 \times \log (1.56)) - (0.059 \times 10)$

$= 0.221$ volt

36. $E = E° + \dfrac{0.059}{2} \log [Cu^{2+}] = 0.$

$\therefore \quad \dfrac{0.059}{2} \log [Cu^{2+}] = -0.34.$

$\therefore \quad \log [Cu^{2+}] = \dfrac{-0.34 \times 2}{.059}.$

$\therefore \quad Cu^{2+} = 2.982 \times 10^{-12} M$

37. On the lhs $E = 0.0713 - \dfrac{0.059}{1} \log (0.34)$

$= 0.0989$ volt. On the rhs

$E = 0.77 + \dfrac{0.059}{1} \log \dfrac{0.01}{0.02} = 0.7522$ volt.

$\therefore$ measured Emf volt = (0.7522 − 0.0989) v

=0.6533 volt

38. $[Cl^-] = 0.1$ M. $\therefore [Ag^+] = \dfrac{1.7 \times 10^{-10}}{10^{-1}} = 1.7 \times 10^{-9}$

$\therefore \quad E = 0.799 + \dfrac{0.059}{1} \log (1.7 \times 10^{-9})$ volt

$= 0.2816$ V

$[H^+] = 0.1$ M

$\therefore \quad E = \text{Zero} + \dfrac{0.059}{1} \log (10^{-1}) = -0.059$ V

$\therefore$ measured Emf = 0.3406 volt

39. 0.05 amp for one hour $\equiv \frac{0.05\times3600}{96500}$ F $\equiv \frac{0.05\times3600}{96500}$ mole of OH^- ion, in 100 ml.

$\therefore \quad [OH^-] = \left[\frac{0.05\times3600}{96500}\times10\right]$ M = 0.01865 M

40. $Cu^+ + e^- \rightleftharpoons Cu$. E° = 0.52 V.

$\therefore \quad \Delta G°_1 = -1 \times F \times 0.52$ Joules

$Cu^{2+} + e^- \rightleftharpoons Cu^+$ E° = 0.153 V.

$\therefore \quad \Delta G°_2 = -1 \times F \times 0.153$ Joules.

ΔG° values may be added. Adding the two equations:

$Cu^{2+} + 2\,e^- \rightleftharpoons Cu$.

$\Delta G° = \Delta G°_1 + \Delta G°_2 = -1 \times F \times (0.52 + 0.153)$ J,

In this case $\Delta G = -2$ F E° $= -1 \times 0.673$ Joules

$\therefore \quad E° = \frac{0.673}{2}$ 0.337 volt

41. The equivalent mass of gold is the highest hence less electricity to deposit move metal.

42. $Cu(OH)_{2(s)}{}^2 \rightleftharpoons Cu^{2+} + 2OH_{(aq)}$-.

$K_{sp} = 1 \times 10^{-19} = [Cu^{2+}][OH^-]^2$

$[OH^-] = 10^{-2}$ M. $[Cu^{2+}] = \left(\frac{1\times10^{-19}}{10^{-4}}\right) \simeq 10^{-15}$ M

E volt $= 0.34 + \frac{0.059}{2} \log [10^{-15}] = -0.1025$ V

43. $Mn^{3+} + 2H_2O \rightarrow MnO_{2(s)} + 4\,H^+ + e^-$

$\Delta G° = -1 \times 96500 \times (-1)$ J

$Mn^{3+} + e^- \rightarrow Mn^{2+}$ $\Delta G° = -1 \times 96500 \times (+1.5)$ J Adding $2Mn^{3+} + 2H_2O \rightarrow Mn^{2+} + Mn\,O_2 + 4H^+$

$\Delta G° = -1 \times 96500 \times (1.5 - 1)$

$= -1 \times 96500 \times (0.5)$ J $= -48250$ J

44. On the lhs, $[Ag^+] = \frac{1.79\times10^{-10}}{10^{-1}} = 1.79 \times 10^{-9}$ M

On the rhs $[Ag^+] = 0.1$ M.

$\therefore \quad$ Emf $= \frac{0.059}{1} \log \frac{10^{-1}}{1.79\times10^{-9}} = 0.457$ volt

45. Emf = 0.028 V $= \frac{0.058}{n} \log (10) = \frac{0.058}{n}$ V.

$\therefore \quad \underset{\text{(whole number)}}{n} = \frac{0.058}{0.028} = 2$

46. $\Delta G = -nFE = -(2 \times 96500 \times 0.2329)$ J

$= \left[\frac{-2\times96500\times0.2329}{4.18}\right]$ cal. $= -10754$ cal.

$\Delta H = -19400$ cal

$\Delta G = \Delta H - T\Delta S$. $-10754 = -19400 - T\Delta S$.

$\therefore \quad \Delta S = \frac{-19400+10754}{298}$ cal deg^{-1}

$= -29.0$ cal deg^{-1}

47. The cell reaction is a 2–electron process.

$\Delta G° = -nFE = (-2 \times 96500 \times 1.1)$ J $= -212300$ J

under equilibrium, ΔG = 0. i.e., E = 0.

$0 = E° + \frac{0.059}{2} \log \frac{[Cu^{++}]}{[Zn^{++}]}$. E° = 1.1 V.

$\therefore \quad \log \frac{[Cu^{++}]}{[Zn^{++}]} = \frac{-1.1\times2}{0.059}$

$\log \frac{[Cu^{++}]}{[Zn^{++}]} = -37.29 = \overline{38}.71$

i.e., $\frac{[Cu^{++}]}{[Zn^{++}]} = 5.13 \times 10^{-38}$

48. 60 milli amp for 1 hour $\equiv (60 \times 10^{-3} \times 3600)$ coulombs = 216 coulomb Efficiency of this = 95 %. Utilized electricity $= \left(216\times\frac{95}{100}\right)$ coulomb

Equivalent wt of copper $= \frac{63.6}{2} = 31.8$

$\therefore \quad$ 96500 coulombs would deposit 31.8 g of Cu

$\therefore \quad$ wt of copper deposited (in m g)

$= \frac{31800}{96500} \times \frac{216\times95}{100} = 67.62$ m g

49. 34ml of electrolytic gas $\equiv 34 \times \frac{2}{3}$ ml of Hydrogen which is $\frac{34}{11200} \times \frac{2}{3}$ Faraday $\equiv \frac{34\times2}{3\times11200}$ eqt of Iodine $= \frac{34\times2}{3\times11200} \times 127$ g Iodine liberated = 0.257 g

50. 23.8 ml of 0.1 M NaCl $= \left(\frac{23.8}{1000}\times0.1\right)$ eqt

$= 2.38 \times 10^{-3}$ Faraday

$50 \times 10^{-3} \times t = 2.38 \times 10^{-3} \times 96500$.

$$\therefore \quad t_{sec} = \frac{2.38 \times 96.5}{5 \times 10^{-2}}$$

$$t = \frac{2.38 \times 9650}{5 \times 3600} \text{ hrs} = 1.276 \text{ hr.}$$

51. Obviously in this case deposition of Copper and liberation of Hydrogen were competitive. 0.10 amp passed for 2 hrs ≡ 0.1 × 2 × 3600 = 720 coulombs. In cuprocyanide copper is in +1 state. ∴ Deposition of 0.3745 g of Cu ≡ $\left(\frac{0.3745}{63.6} \times 96500\right)$ coulombs ≃ 568 coulombs. Thus the remaining 152 coulombs are used in the liberation of H_2 gas.

96500 coulombs ≡ 11200 ml of H_2 gas.

∴ 152 coulombs $\frac{11200}{96500} \times 152$ ml = 17.64 ml.

52. Correct statement is (a), Both Co^{3+} and Cu^{+} ions are unstable in aqueous solution.

(i) $4\,[Co^{3+} + e^- \rightleftharpoons Co^{2+}]$

$\Delta G = -4 \times F \times (1.82)$ J

$2H_2O \rightleftharpoons O_2 + 4H^+ + 4e^-$.

$\Delta G = -4 \times F \times (-123)$ J.

Adding $4Co^{3+} + 2H_2O \rightleftharpoons 4Co^{2+} + O_2 + 4H^+$

$\Delta G = -4 \times F\,(1.82 - 1.23)$ Joule

ΔG is negative: reaction is spontaneous.

(ii) $Cu^+ + e^- \rightleftharpoons Cu$. $\Delta G = -1 \times F \times 0.52$ Joule

$Cu^+ \rightleftharpoons Cu^{2+} + e^-$,

$\Delta G = -1 \times F \times (0.15)$ Joule Adding

$2Cu^+ \rightleftharpoons Cu^{2+} + Cu$,

$\Delta G = -1 \times F\,(0.52 - 0.15)$ Joule.

ΔG is negative reaction is spontaneous.

53. $Fe^{3+} + e^- \rightleftharpoons Fe^{2+}$, $\Delta G = (-1 \times F \times 0.771)$ J

$I^- \rightleftharpoons \frac{1}{2}I_2 + e^-$ $\Delta G = (-1 \times F \times -0.535)$ J

$Fe^{3+} + I^- \rightleftharpoons Fe^{2+} + \frac{1}{2}I_2$,

$\Delta G = -1 \times F \times (0.771 - 0.535)$ J

ΔG is negative. ∴ Spontaneous

Similarly, $Fe^{3+} + Br^- \rightleftharpoons Fe^{2+} + \frac{1}{2}Br_2$,

$\Delta G = -1 \times F \times (0.771 - 1.09)$ J

ΔG is positive. ∴ not spontaneous.

54. $E_{\underset{volt}{lhs}} = 0 + \frac{0.059}{1} \log 10^{-6}$

$= -0.059 \times 6 = -0.354$ volt.

$E_{rhs} = 0.34 + \frac{0.059}{2} \log 10^{-1}$

$= 0.34 - \frac{0.059}{2} = 0.3105$ volt

∴ measured Emf volt = 0.3105 − (−0.354)

= 0.6645 volt ≃ 0.665 volt

55. $\wedge_1 = 422.74 = \wedge_0 - (\text{constant})\ (5 \times 10^{-4})^{1/2}$ (Ω^{-1} cm^2 mol^{-1})

$\wedge_2 = 421.36 = \wedge_0 - (\text{constant})\ (1 \times 10^{-3})^{1/2}$ (Ω^{-1} cm^2 mol^{-1})

Eliminating we get $\wedge_0 = 426.07\ \Omega^{-1}\ cm^2\ mol^{-1}$

56. Since osmotic pressure is measured at constant temperature. The temperature dependent concentration can be used.

57. P (torr) = 119 x + 135. x = mole fraction of A in the liquid. Put x = 1; p_A° = 254 torr,

Put x = 0 p_B° = 135 torr

mole fraction of A in the fraction

$$= \frac{p_A^\circ x}{P} = \frac{254 \times 0.5}{194.5} \text{ which is } 0.653$$

58. By Raoults law $\frac{ps}{po}$ = mole fraction of the solvent

$$= \frac{0.78}{0.80} = 0.975$$

59. 10 g of Benzene = $\frac{10}{78}$ mole = 0.1282 mole = n_1 10 g

of ethanol = $\frac{10}{46}$ = 0.2174 mole = n_2

∴ mole fraction of Benzene = $\frac{0.1282}{0.3456}$ = 0.371 mole

fraction of ethanol = $\frac{0.2174}{0.3456}$ = 0.629

∴ Total vapour pressure = $(p_A^\circ \text{ x } x_A) + (p_B^\circ \times x_B)$

which is $(100 \times 0.371) + (44 \times 0.629)$

= (37.1 + 27.676) torr = 64.776 torr ≃ 65 torr

60. $\Delta T_b = 0.5 = 0.51 \times m$. $\Delta T_f = 1.86 \times m$

$$\therefore \quad \frac{\Delta T_f}{\Delta T_b} = \frac{1.86 \times m}{0.51 \times m}.$$

$$\therefore \quad \Delta T_f = \frac{1.86}{0.51} \times 0.5 = 1.8235 \simeq 1.82$$

∴ freezing point = −1.82°C

61. $\Delta T_f = 40 \times \dfrac{1.5\times1000}{125\times35} = 13.71°C.$

$\therefore$ freezing point = (178.4 – 13.7) °C = 164.7 °C

62. $\Delta T_f = 1.16 = 1.86 \times \dfrac{3}{M}\dfrac{1000}{80}.$

$M = \dfrac{1.86\times3000}{80\times80} \simeq 60\ g\ mol^{-1}$ Empirical formula:

$\dfrac{40}{12} : \dfrac{6.67}{1} : \dfrac{53.33}{16} = 3.33 : 6.67 : 3.33.$

C: H: O = 1: 2: 1. Emp. formula = H_2O (formula wt = 30) $\therefore$ molecular formula $C_2H_4O_2$

63. $\Delta T_f = 2.4 = K_f \times m_1$ $\Delta T_b = 2.5 = K_b \times m_2$

$\therefore \quad \dfrac{\Delta T_f}{\Delta T_b} = \dfrac{2.4}{2.5} = \dfrac{K_f}{K_b} \times \dfrac{m_1}{m_2}.$

$\therefore \quad \dfrac{m_1}{m_2} = \dfrac{2.4}{2.5} \times \dfrac{2.53}{5.07}\dfrac{m_1}{m_2}$

$= 0.479 \simeq m_1$ is approximately half of m_2.

$\therefore$ At freezing point the acid is largely dimerized. At the boiling temperature however this is not the case.

64. $\Delta T_f = 0.52 = 1.86 \times m.$ $\therefore$ $m = \dfrac{0.52}{1.86} \simeq 0.28$. Thus 0.28 mole of urea is dissolved in 1000 g of water. One may (approximately) take the density as 1. Thus the volume containing 0.28 mole of solute is one litre. Using $\pi V = nRT$.

π (osm pr) = $0.28 \times 0.0821 \times 310 \equiv 7.126$ atm

65. One molal $\equiv$ one mole solute in a Kg of water

$\therefore$ number of moles of water = $\dfrac{1000}{18} = 55.56$

$\therefore$ mole fraction of solute = $\dfrac{1}{56.56} = \dfrac{po-ps}{po}.$

po = 760 mm. $\therefore$ $\dfrac{po-ps}{760} = \dfrac{1}{56.56}.$

$\therefore \quad po - ps = \dfrac{760}{56.56} = 13.44$ mm.

66. partial vapour pressure = p = po × x.

$\therefore \quad x = \dfrac{p}{po} = \dfrac{23}{96} \simeq 0.24$

67. 20 g of glucose $\equiv \dfrac{20}{180}$ mole = $n_1 = 0.11$ mole

70 g of water = $\dfrac{70}{80}$ moles = $n_2 = 3.89$ mole

mole fraction of solvent = $\dfrac{3.89}{4} = \dfrac{ps}{p_0} = \dfrac{24.5}{p_0}$

$\therefore \quad p_0 = \dfrac{24.5\times4}{3.89}$ mm $\simeq 25.19$ torr

68. $\dfrac{120.2}{121.8} = \dfrac{ps}{po}$ = mole fraction of solvent

$$= \frac{\frac{250}{78}}{\frac{250}{78}+\frac{15}{M}}.$$

$\therefore \quad \dfrac{3.205}{3.205+\frac{15}{M}} = 0.9869;$

$\dfrac{3.205}{0.9869} = 3.2475 = 3.205 + \dfrac{15}{M}.$

$\therefore \quad \dfrac{15}{M} = 0.0425.$ $\therefore$ $M \simeq 353$

69. Suppose the mole fractions of A, B and C are x_A, x_B and x_C.

Then $(x_A \times 109) + (x_B \times 115.7) + (x_C \times 142.6)$

$= 130.$

One possible solution is $x_A = 0.207 = x_B$;

$x_C = 0.586$

70. $\dfrac{ps}{po} = \dfrac{412.7}{440} = \dfrac{\frac{W_1}{M_1}}{\frac{W_1}{M_1}+\frac{W_2}{M_2}}.$

$\therefore \quad \dfrac{440}{412.7} = 1.066 = \dfrac{\frac{W_1}{M_1}+\frac{W_2}{M_2}}{\frac{W_1}{M_1}}$ 1.066

$= 1 + \dfrac{W_2}{M_2}\times\dfrac{M_1}{W_1}.$

$0.066 = \dfrac{W_2}{W_1}\times\dfrac{M_1}{M_2} = \dfrac{1}{20}\times\dfrac{M_1}{M_2}.$

$\therefore \quad = \dfrac{M_2}{M_1} = \dfrac{1}{20\times0.066} = 0.758$

71. $p_A = (p°_A \times x_A)$. $P_B = (p°_B \times x_B)$. $p_A + p_B = P$ torr $= (119 \times 0.5) + (37 \times 0.5)$. P = 78 torr. Mole fraction of B in the vapour phase is

$$\frac{p_B}{p} = \frac{p°_B \times x_B}{p} = \frac{37 \times 0.5}{78} = 0.237$$

72. $K_B = 3.1$ deg molal$^{-1} = \dfrac{0.002 T_B^2}{l}$,

$T_B = (118.9 + 273) \simeq 392$ K.

$$3.1 = \frac{0.002 \times (392)^2}{l}.$$

$$\therefore \quad l = \frac{0.002 \times (392)^2}{3.1} \text{ cal g}^{-1} \text{ which is 99 cal g}^{-1}$$

73. $\Delta T_f = K_b \times m \times z = 0.51 \times 0.5 \times 1.85 = 0.472$ deg. Boiling point = 100.472 deg.

74. $\Delta T_f = 1.15 = 5.07 \times \dfrac{5}{M} \times \dfrac{1000}{50}$.

$$\therefore \quad M = \frac{5.05 \times 5 \times 1000}{1.15 \times 50} \text{ g mol}^{-1} \quad M = 439 \text{ g mol}^{-1}$$

Empirical formula wt = 108.

$\therefore$ Molecular formula is $(C_6H_5P)_4$

79. For one litre of the solution, wt = 1410 g. This contains 10.95 moles of $HNO_3 \equiv 10.95 \times 63$ g = 689.85 g.

$\therefore$ wt of water = (1410 – 689.85) g = 720.15 g.

$\therefore$ one Kg of water contains $\left(\dfrac{10.95}{720.15} \times 1000\right)$ moles, 15.205 moles.

$\therefore$ molality $\simeq$15.21

80. $$2.303 \log \frac{760}{99.9} = \frac{-\Delta H_{vap}}{1.987}\left(\frac{1}{373} - \frac{1}{323}\right)$$

$$= \frac{\Delta H}{1.987} \times \frac{50}{373 \times 323}.$$

$$2.0285 = \frac{\Delta H \times 50}{1.987 \times 373 \times 323}.$$

$$\therefore \quad \Delta H = \frac{2.0285 \times 1.987 \times 373 \times 323}{50} \text{ cal mol}^{-1}$$

$\Delta H = 9712$ cal mol^{-1}

81. Number of moles of oxygen liberated

$$= \frac{1 \times 6.15}{0.0821 \times 300} = \frac{6.15}{24.63} \simeq \frac{1}{4} \text{ mole.}$$

$2OH^- \rightarrow H_2O + \frac{1}{2} O_{2(g)} + 2e^-$ (anode)

$\therefore$ One mole of O_2 is liberated by 4 Faradays.

$\therefore$ $\dfrac{1}{4}$ mole $\equiv$ one Faraday $\equiv$ 96480 coulombs

$= 1$ amp $\times$ t (in sec.)

$\therefore$ time required $= \dfrac{96480}{3600}$ hrs = 26.8 hrs.

82. Number of coulombs used = $10 \times 2 \times 3600$ = 72000. It there is 100% efficiency, wt. of Ni deposit expected $= \dfrac{72000}{2 \times 96500} \times 58.7 \simeq 21.9$ g

Actual deposit weighs 13.59 g

$\therefore$ current efficiency $= \dfrac{13.59}{21.9} \times 100 \simeq 62\%$

83. Volume of Nickel deposited $= \left(\dfrac{22}{8.9}\right)$ cm^3

$= 2.472$ cm^3

Area of deposit = 200 cm^2

$\therefore$ thickness of deposit $= \dfrac{2.472}{200}$ cm

$= 0.01236$ cm $= 1.236 \times 10^{-2}$ cm

84. wt of H_2SO_4 in 5 litre before discharge

$$= \frac{5000 \times 1.075 \times 11.26}{100} = 605.2 \text{ g}$$

wt of H_2SO_4 in 5 litre after discharge

$$= \frac{5000 \times 1.06 \times 9.13}{100} = 483.9 \text{ g}$$

$\therefore$ loss in moles $= \dfrac{605.2 - 483.9}{98} = 1.238$ moles.

Overall reaction:

$Pb + PbO_2 + 2H_2SO_4 \rightarrow 2PbSO_4 + 2H_2O$ which is a 2 electron process consuming 2 moles of H_2SO_4 Thus one mole of H_2SO_4 is used up per Faraday. Hence number of Faradays = 1.238 $\equiv$

$\dfrac{1.238 \times 96500}{3600}$ = amp hrs = 33.19 amp hrs

85. Number of coulombs used = $5 \times 131.3 \times 3600$.

6×96500 coulombs $\equiv$ 1 mole of $KClO_3$ (ideally)

$\equiv 122.5$ g

$\therefore$ $5 \times 131.3 \times 3600$ coulombs

$$\equiv \frac{5 \times 131.3 \times 3600}{6 \times 96500} \times 122.5 = 500 \text{ g}$$

Since the actual deposit ≡ 350 g,

current efficiency is $\frac{350}{500} \times 100 = 70\%$

86. $\wedge = \frac{1000 \times K}{c}$ $\therefore$ $K = \frac{\Lambda \times c}{1000} = \frac{128.9 \times 0.1}{1000}$

$= 1.289 \times 10^{-2}\ \Omega^{-1}cm^{-1}$

87. $R \times K$ = cell constant = $(470 \times 1.21 \times 10^{-4})\ cm^{-1}$

$= 0.05687\ cm^{-1} = 5.687 \times 10^{-2}$.

$\wedge = \frac{1000 \times K}{c} = \frac{1000 \times 1.21 \times 10^{-4}}{0.001}$

$= 121\ \Omega^{-1}\ cm^2\ eqt^{-1}$

88. Corrected K value = $(1.80 - 0.2) \times 10^{-3}\ ohm^{-1}m^{-1}$

$= 1.60 \times 10^{-5}\ ohm^{-1}\ cm^{-1}$.

$\wedge_0 = 127.2 + 159.6 = 286.8\ ohm^{-1}\ cm^2\ mol^{-1}$

$\therefore \quad S = \frac{1000K}{286.8} = \frac{1000 \times 1.6 \times 10^{-5}}{286.8}$

$= 5.58 \times 10^{-5}$ moles L^{-1}

$\therefore \quad K_{sp} = S^2 = 3.11 \times 10^{--9}$

89. $\Lambda_0\ (LiOH) = \Lambda_0\ (Li^+) + \Lambda_0\ (OH^-)$

90. $E = E° - \frac{RT}{2F} \ln \frac{\left[Sn^{4+}\right]}{\left[Sn^{2+}\right]} \times \frac{\left[Fe^{3+}\right]^2}{\left[Fe^{2+}\right]^2}$.

By Nerust equation E° = (0.771 − 0.15) V

= 0. 621 volt. Straight substitution gives

$E = 0.621 - \frac{0.0591}{2} \log \left[\frac{0.2}{0.1} \times \left(\frac{0.05}{0.1}\right)^2\right]$

= 0.63 V

91. Under standard condition E measured

= [−0.15 − (+0.80)] volt.

E measured = −0.95 volt.

$\Delta G° = -nFE° = -RT \ln K_{sp}$

$\therefore \quad \log K_{sp} = \frac{nFE^\circ}{2.303RT}$

$= \frac{-0.95}{0.0591} = -16.07$

$= -17.93$. $K_{sp} = 8.5 \times 10^{-17}$

92. E measured (0.656 V)= $0.242 - \frac{0.0591}{1} \log c$

[E° for S H E = 0], By calculation log c = −7

$[H^+] = 10^{-7}$

93. $2Cu^+ + 2e^- \rightarrow 2Cu$. E° = +0.521 V.

$Cu^{2+} + 2e^- \rightarrow Cu$; E° = +0.34 reverse the second equation, add to the first. $2Cu^+ \rightleftharpoons Cu^{2+} + Cu$

E° volt = (0.521 − 0.34) = 0.181 volt. This is a 2 electron process.

$\therefore \quad \Delta G = -nFE° = -RT \times 2.303 \log K$.

$\therefore \quad \log K = \frac{2F \times 0.181}{2.303RT}$ $\log K = \frac{2 \times 0.181}{0.0591} = 6.125$.

$\therefore \quad K = 1.334 \times 10^6$

94. $\Delta S = nF \left(\frac{dE^\circ}{dT}\right)$

$= -(2 \times 96500 \times 8.24 \times 10^{-4})\ J\ deg^{-1}$

$= -159\ J\ deg^{-1}$

95. Straight application of the Nerust equation gives

$E = -\frac{0.0591}{1} \log \left[\left(\frac{0.02}{0.05}\right) \times \left(\frac{5}{2}\right)^{1/2}\right]$

= 0.0117 volt

96. Obviously 0.1 molal salt solution behaves by dissociation like 0.3 molal i.e., one molal dissociates into 3 ions.

97. In this case $\frac{p_s}{p_o} = 0.975$ = mole fraction of solvent.

$\therefore$ mole fraction of solute = 0.025. If m grms of urea are added to one Kg of water, then

$0.025 = \frac{\frac{m}{60}}{\frac{m}{60} + \frac{1000}{18}}$

$\therefore \quad 0.025 \times 55.56 = \frac{m}{60}\ (1 - 0.025)$

$\therefore \quad m = 85.48$ g.

98. Suppose x mole of phenol is to be added 90% of the dimerises $\therefore$ Number moles actually present would be

$x\ (1 - 0.9 + \frac{0.9}{2}) = 0.55\ x$, Molar mass

$= 94\ g\ mol^{-1}$

$\therefore \quad \Delta T_f = 1.65 = 5.2 \times \frac{0.55x}{2000} \times 1000$

$= 2.6 \times 0.55\ x$

$\therefore \quad x = \left(\frac{1.65}{2.6 \times 0.55}\right)$ wt of phenol $= \frac{1.65 \times 94}{2.60 \times 0.55}$ g

wt of phenol = 108.4 g = 0.1084 Kg

99. Solid → vapour: sublimation: 20.21 K J mol^{-1}
Solid → liquid: fusion: 10.59 K J mol^{-1}
∴ liquid → vapour: (20.21 − 10.59) K J mol^{-1}
= 9.62 K J mol^{-1}
∴ Specific latent of vapourization
$= \frac{9.62\times1000}{78}$ J g^{-1}
$\therefore K_b = \frac{0.002\times353\times353\times78\times4.18}{9620}$
≃ 8.446 deg $molal^{-1}$

100. Since the process is spontaneous $\Delta G_{(mixing)}$ is negative.

101. $\Delta T = k_b.m$ = molality = 1 hence $\Delta T = k_b$

102. Let mole fraction of CH_3-OH in the liquid be x (94 × x) + [44 (1 − x)] = 54
∴ x = 0.2 of CH_3OH
(1 − x) = 0.8

103. Mole fraction of solute
$= \left(\frac{71.93-70.5}{71.93}\right) = 0.01988$
$= \frac{\frac{10}{M_2}}{\frac{10}{M_2}+\frac{150}{M_1}}$
$0.01988 = \frac{10}{10+150\times\frac{M_2}{M_1}}$.
$\therefore \frac{10+150\frac{M_2}{M_1}}{10} = 50.302$
$\therefore 10 + 150\ \frac{M_2}{M_1} = 503.$
$\therefore \frac{M_2}{M_1} = \frac{493}{150} = 3.2867 \simeq 3.29$

104. $\Delta T_f = 0.225 = 1.86 \times \frac{1.068}{M} \times \frac{1000}{100}$
$\therefore M = \frac{1.86\times1.068\times10}{0.225}$ g
M = 88.288.
$\therefore \frac{267}{88.288} = 3$

105. The effective concentration is the same in both cases
$\therefore \frac{5}{342} = \frac{1.14}{M}$.
$\therefore M = \frac{1.14\times342}{5}$ g mol^{-1}
= 78 g mol^{-1}

106. According to Raoult's law the total vapour pressure is [(0.2 × 300) + (0.8 × 650)]
= (60 + 520) mm = 580 mm
The actual vapour pressure = 700 mm
∴ positive deviation : 120 mm

107. $p_s \alpha$ 5 g $(p_o - p_s) = 0.025$ g. $\therefore p_o \alpha$ 5.025 g
$\therefore \frac{po-ps}{po} = \frac{0.025}{5.025} = \frac{\frac{9}{M}}{\frac{9}{M}+\frac{180}{18}}$.
$\therefore$ M = 180 g mol^{-1}

108. The incorrect statement is (d)

109. $K_f = 6.85 = \frac{0.002\times(353.22)^2}{l}$.
$\therefore l = \frac{0.002\times(353.22)^2}{6.85l}$ cal g^{-1}
l = 36.43 cal g^{-1}

110. π = CRT; 7.1 atm = C × 0.0821 x 310.
$\therefore C = \frac{7.1}{0.0821\times310}$ mole L^{-1} = 0.279 mole L^{-1}
This is a molarity. Since density of the solution is not given we may use the approximate calculation assuming that this is molality.
$\therefore \Delta T_f = 1.85 \times 0.279$ deg ≃ 0.52 degree

111.(b) Both statements are correct

112. (b) Ether + $CHCl_3$ mix there is hydrogen bonding volatility is reduced.

113. (d) BF_3 is Lewis acid, not a Bronsted acid $K_a \simeq \alpha^2C$. $(K_aC)^{1/2} = \alpha C = [H^+]$. Add water αC is reduced, so also $[H^+]$. ∴ pH is raised.

114. $E^0_{cell} = E^0_{cathode} - E^0_{anode}$
i.e., 0.8 − 0.05 = 0.75 V
(Θ nFE° = RT lnk)
lnk = 2 × 0.75 × 38.92
= 58.38

115. $E_{(oxidationelectrode)}$

$$= E^0_{(\text{oxidation electrode})} - \frac{0.0592}{2}\log\frac{[\text{glucose}]}{[\text{gluconic aicd}][H^+]^2}$$

When $[H^+]$ decreases to 10^{-11}

E increases by $0.0592 \times 11 = 0.6512$ V

116. NH_3 has no effect on the standard reduction potential of glucose/gluconic acid electrode.

Additional Practice Exercise

121. $E(Cu^{2+}, Cu) = \left[+0.34 + \frac{0.059}{2}\log x\right] = 0.28$ volt.

$\therefore$ 9.249×10^{-3} M approximately 10^{-2} M

122. $E = 0.771 - \frac{0.0592}{1}\log\left(\frac{[Fe^{2+}]}{[Fe^{3+}]}\right) = 0.653$ volt by calculation.

$K_{sp} = [Ag^+][Cl^-] = 1.3 \times 10^{-10}$. $[Cl^-] = 10^{-2}$ M

$\therefore$ $[Ag^+] = 1.3 \times 10^{-8}$ M

$$E = E° + \frac{0.059}{1}\log[Ag^+]$$

$$= 0.799 + 0.059 \log (1.3 \times 10^{-8})$$

$E = 0.799 + 0.059 (-7.886) = 0.334$ volt

123. Number of moles of $NH_3 = \frac{PV}{RT} = \frac{0.984 \times 4}{0.082 \times 300}$

Number of moles of salt formed = 0.16

Molality of solute $= \frac{0.16 \times 1000}{80} = 2$

Since the solvent is polar salt ionises hence i = 2

$\Delta T_f = i \times k_f m$

$= 2 \times 3.9 \times 2$

$= 15.6°$

$T = T_f - \Delta T$

$= 16.6 - 15.6 = 1°C$

$= 274$ K

124. (i) The reaction taking place in the battery during charging is

$$2PbSO_4 + 2H_2O \rightarrow Pb_{(s)} + PbO_{2(s)} + 2H_2SO_4$$

The electrons transferred in the reaction is 2

$\therefore$ weight of $PbSO_4$ reduced

$$= \frac{5 \times 2 \times 3600 \times 303}{96{,}500 \times 2} = 56.5 \text{ g}$$

(ii) Amount of H_2SO_4 added to the battery

$$= \frac{5 \times 2 \times 3600 \times 98}{96{,}500 \times 2} = 18.3 \text{ g of acid}$$

125. (i) cell:

Pt, H_2(g,1 atm)/$KOH_{(aq)}$/O_2(g,1 atm), Pt

(ii) Anode: $2H_{2(g)} + 4OH^-_{(aq)} \rightarrow 4H_2O_{(l)} + 4e^-$

Cathode: $O_{2(g)} + 2H_2O_{(l)} + 4\bar{e} \rightarrow 4OH^-_{(aq)}$

Over all reaction: $O_{2(g)} + 2H_{2(g)} \rightarrow 2H_2O_{(\lambda)}$

(iii) $\Delta G°$ of reaction $= -nFE°$

$\Delta G° = -4 \times 96{,}500 \times E° = -474{,}000$

$$E°_{25°C} = \frac{474000}{4 \times 96{,}500} = 1.23 \text{V}$$

$\Delta H°_{reaction} = -572 - 0 - 0 = -572$ kJ mol^{-1}

$\Delta S°_{reaction} = 2 \times 70 - [(2 \times 130.6) + 205]$

$= 140 - 261.2 - 205.0$

$= -326.2$ J K^{-1} mol^{-1}

$\Delta G°_{75°C} = \Delta H° - T\Delta S°$

$= -572 - [348 \times - 326.2 \times 10^{-3}]$

$= -572 + 113.5 = -458.5$ kJ mol^{-1}

$$E°_{75°C} = + \frac{458.5 \times 1000}{4 \times 96500} = 1.19 \text{ V}$$

126. $\Lambda_{0(AgCl)} = 96500 \times 10^{-4} (5.6 + 6.8)$

$= 119.7$ ohm^{-1} cm^2

$$S \text{ of AgCl} = \frac{1000 \times 1.12 \times 10^{-6}}{119.7}$$

$= 9.36 \times 10^{-6}$ eq L^{-1}

$K_{sp} = 8.75 \times 10^{-11}$

127. $\alpha = \frac{i-1}{\frac{1}{n}-1}$

or $0.70 = \frac{i-1}{0.5-1}$ or

$i - 1 = -0.35$

$i = 1 - 0.35 = 0.65$

$$\Delta T_{b(theo)} = 3.32 \times \frac{10}{94} \times \frac{1000}{500} = 0.706$$

$$i = \frac{\Delta T_{b(obs)}}{\Delta T_{b(theo)}}$$

$\Delta T_{b(obs)} = 0.65 \times 0.706 = 0.459$

Boiling point of the solution = 111.159°C

128. (i) $Ag, AgCl_{(s)}/KCl_{(aq)}/Hg_2Cl_{2(s)}, Hg$

(ii) $\Delta G = -nFE = -2 \times 96500 \times 0.0455$

$= -8.78$ kJ

$$\Delta S = \frac{\Delta H - \Delta G}{T} = \frac{(10.7 + 8.78)1000}{298}$$

$= 65.36$ J K^{-1}

129. Anode reaction:

$CH_{4(g)} + 2H_2O_{(\lambda)} \rightarrow CO_{2(g)} + 8H^+ + 8e^-$

Cathode reaction : $2O_{2(g)} + 8H^+ + 8e^- \rightarrow 4H_2O_{(\lambda)}$

Overall reaction:

$CH_{4(g)} + 2O_{2(g)} \rightarrow CO_{2(g)} + 2H_2O_{(\lambda)}$

$\Delta G°_{reactiion} = (-394.4 - 2 \times 237.1) - (-50.7)$

$= -817.9$ kJ

$$E^\circ = \frac{-\Delta G^\circ}{nF} = \frac{817.9 \times 1000}{8 \times 96500} = 1.06\,V$$

130. $3Au^+_{(aq)} \rightleftharpoons Au^{3+}_{(aq)} + 2Au_{(s)}$

$3 \times (Au^+ + e^- \rightarrow Au)$ $\quad E° = 1.69$ V — (1)

$Au^{3+} + 3e^- \rightarrow Au \quad E° = 1.50$ V — (2)

(1) – (2) $\Rightarrow$

$3Au^+ \rightleftharpoons Au^{3+} + 2Au \quad E° = 0.19$ V

Since the E° is +ve, the reaction occurs spontaneously.

$$\log K = \frac{nFE^\circ}{2.303RT} = \frac{3}{0.0591} \times 0.19 = 9.645$$

$K = 4.4 \times 10^9$

131. $5H_2 \rightarrow 10H^+ + 10e^- \quad E° = 0.00$ V

(b) $16H^+ + 2MnO_4^- + 10e^- \rightarrow 2\,Mn^{2+} + 8H_2O$

$E° = 1.51$ V

Add; $6H^+ + 2MnO_4^- + 5H_2 \rightarrow 2\,Mn^{2+} + 8H_2O$ (1.51 + 0.00) = 1.51 V $\quad E = E° - \frac{0.059}{10}$

$$\left[\log\left\{\left[Mn^{2+}\right]^2 \times a\,H_2O^8 \Big/ \left[H^+\right]^6 \left[MnO_4\right]^2 p(H_2)^5\right\}\right]$$

a H_2O = 1. p H_2 = 0.5 On substituting values =

E = 1.442 volt.

$$1.51 - \frac{0.059}{10}\log\frac{1}{(0.5)^5 \times (0.1)^2 \times (0.1)^6}$$

$$= 1.51 - \frac{0.059}{10}\log(3.2 \times 10^9) = 1.453\ V$$

132. $$E = 0.771 + \frac{0.059}{1}\log\frac{\left[Fe^{3+}\right]}{\left[Fe^{2+}\right]}$$

$= 0.771 + 0.059 \log (10^{-2}) = 0.653$ volt

133. $$\frac{n}{N+n} = \frac{m \times M}{1000};\ 0.073 = m \times \frac{18}{1000}.$$

$$\therefore \quad m = 0.073 \times \frac{1000}{18} = 4.056$$

134. $\Delta T_b = K_b \times m.\ 0.3 = K_b \times \frac{0.1}{100} \times 1000 = K_b$

$\therefore$ 0.3 deg $molal^{-1}$

135. $$\Lambda = \frac{1000\,K}{c} = \frac{1000 \times 1.061 \times 10^{-3}}{0.01}$$

$= 106.1\ \Omega^{-1}\ cm^2\ eqt^{-1}$

136. 0.1 g of Ag = $\frac{0.1}{108}$ eqt

$$= \left[\frac{0.1}{108} \times 96500\right] \text{ coloumb} = 20 \times 10^{-3} \times t_{(sec)}$$

$$\therefore \quad t = \frac{9650 \times 10^3}{108 \times 20} \text{ sec.} = 4467.59 \text{ sec} \equiv 1.241 \text{ hrs}$$

137. Freezing point = 5.45°C = 278.45 K. $K_f = 5.06$

$$= 0.002 \times \frac{T^2}{l}$$

$$5.06 = \frac{0.002 \times (278.45)^2}{l}.$$

$\therefore$ l [latent heat of fusion per gram]

$$= \frac{0.002 \times (278.45)^2}{5.06} \text{ cal}$$

$l = 30.646$ cal g^{-1}.

$\therefore$ ΔH_f (per mole) = (30.646 × 78) per mole

ΔH_f (molar) = 2390 cal mol^{-1}

138. The STD electrode potential of copper is +0.34

139. Obviously 0.5 molar solution has 0.5 mole per litre of the solution i.e., $\frac{58.5}{2}$ g in a litre. As a %

we have $\frac{58.5\times100}{2\times1000} = 2.925$

140. 25 g of $CuSO_4\,5H_2O \equiv \left(\frac{25}{250}\right) = 0.1$ mole. But due to ionization this is equivalent to 0.2 mole (in 360 ml of water. Take the density as one

$$\therefore \quad \text{molality} = \frac{0.2}{360}\times1000\ \frac{p_o - p_s}{p_o} = \frac{m\,x\,M}{1000}$$

$$= \left(\frac{0.2}{360}\times1000\right)\times\frac{18}{1000}$$

$$= \frac{0.2}{360}\times18 = 0.01$$

$$1 - \frac{p_s}{p_o} = 0.01.$$

$$\therefore \quad \frac{p_s}{p_o} = 1 - 0.01 = 0.99$$

141. $\Delta T_b = 0.413 = 0.52\times\frac{2.44}{M}\times\frac{1000}{75}$

$$\therefore \quad M = \frac{0.52\,x\,2.44\,x\,1000}{75\,x\,0.413}\ \text{g mol}^{-1}$$

$$= 40.96\ \text{g mol}^{-1}$$

142. (d)
First at the freezing point only ice separates.

143. (b)
The main purpose is to eliminate the liquid junction potential; of course, KCl bridge also provides a conducting link between the half cells.

144. Anodic oxidation takes place, $PbSO_4$ formed during discharge is converted into PbO_2.

145. $\Lambda = \frac{1000K}{c}$ where is in eqt L^{-1} i.e., normality.

$$95 = \frac{1000\times1.7510^{-3}}{c}$$

$$\therefore \quad c = \frac{1.75}{95}\ \text{N}$$

$$= 1.842\times10^{-2}\ \text{N}$$

146. (0.268 × 5 × 3600) coulombs = 4824 coulombs ≡ one gram. ∴ 96500 would liberate $\frac{96500}{4824}$ g = 20 g = eq wt. At. wt = 60 ∴ valency = $\frac{60}{20} = 3$

∴ Formula = MCl_3

147. NaCl has an effective molar mass = $\frac{58.5}{2}$

$= 29.25$

KCl has an effective molar mass = $\frac{74.5}{2}$

$= 37.25$

NH_4Cl has an effective molar mass $\frac{53.5}{2}$

$= 26.75$

$BaCl_2.\,2H_2O$ has an effective molar mass

$= \frac{244.4}{3} = 81.5$

The least is for NH_4Cl which will give the largest molality.

148. $C_6H_5 - NO_2 \xrightarrow{\text{Reduction}} C_6H_5 - NH_2$. One mole of Nitrobenzene needs 6 Faradays for reduction.

∴ 0.1 mole ≡ (0.6 × 96500) coulombs

= 57900 coulombs

149. $Hg_2^{2+} + 2e^- \rightarrow 2\,Hg_{(l)}$; 0.800 V

$Hg^{2+} + 2e^- \rightarrow Hg_{(l)}$; 0.85 V Subtract and rearrange.

$Hg_2^{2+} \rightarrow Hg^{2+} + Hg$; −0.05 V

$\Delta G° = -nFE° = -2\times96500\times(-0.05)$ J

$= 9650\ \text{J} = -RT\ln K$

$= -8.314\times298\times2.303\log K = 9650.$

∴ $K = 2.036\times10^{-2}$

150. Suppose the current strength = I amp. In one hour

$$\frac{I\times3600}{96500}\ F = \frac{50\times10^3}{9}\ F$$

$$\therefore \quad I = \frac{50\times10^3}{9}\times\frac{96500}{3600}\ \text{amp}$$

$$= 1.489\times10^5\ \text{amp}$$

151. Let the mole fraction of ethyl acetate be 'x'

$x \times 408 + (1 - x)\,176 = 234$

Solving x = 0.25 & (1 − x) = 0.75

Mole fraction of ethyl acetate in vapour state be 'x'

$x \times 234 = 0.25 \times 408$

$x = \dfrac{0.25 \times 408}{234} = 0.4359$

Mole fraction of propyl acetate = 0.5641Mass % of ethyl acetate in the vapour state

$$= \frac{0.4359 \times 88 \times 100}{0.4359 \times 88 + 0.564 \times 102} = 40\%$$

152. At 100°C, p_o = 760 mm. mole fraction of water

$$= \frac{\frac{175}{18}}{\frac{175}{18}+\frac{15.4}{342}} = \frac{9.7}{9.7+0.045}$$

$$\frac{po}{760} = \frac{9.7}{9.7+0.045} = \frac{9.7}{9.745}.$$

$$\therefore \quad ps = 760 \times \frac{9.7}{9.745} = 756.5 \text{ mm}$$

153. Number of moles of solute = $\left(\dfrac{5}{59.6}\right)$

number of moles of solvent = $\dfrac{100}{74}$

$$\therefore \quad \frac{p_s}{p_o} = \frac{412.7}{p_o} = \frac{\frac{100}{74}}{\frac{100}{74}+\frac{5}{59.6}} \simeq 440 \text{ mm}$$

154. $p_s \alpha$ 0.836 g; $(p_o - p_s) \alpha$ 0.025 g

$$\therefore \quad p_o \alpha = 0.861 \text{ g} \therefore \frac{p_o - p_s}{p_o} = \frac{0.025}{0.861}$$

$$= \frac{\frac{10}{M}}{\frac{10}{M}+\frac{90}{18}} \cdot \frac{0.861}{0.025} = 1 + \frac{90}{18} \times \frac{M}{10}$$

$$\therefore \quad M = 67 \text{ g mol}^{-1}$$

155. $[Ag^+]$ on the lhs = $\dfrac{1.7 \times 10^{-10}}{10^{-1}}$

$= 1.7 \times 10^{-9}$ M. $[Ag^+]$ rhs = 10^{-1} M

$\therefore$ as a concentration cell,

$$\text{Emf} = \frac{0.059}{1} \log \left(\frac{10^{-1}}{1.7 \times 10^{-9}}\right)$$

$= (0.059 \times 7.77)$ V = 0.458 volt

156. This is a two electron process.

$$\therefore \quad \Delta G = -nFE = \frac{-2 \times 96500 \times 0.461}{4.182} \text{ cal}$$

$\Delta G = -21275$ cal

157. Magnesium cannot be formed in aqueous medium an electrolysis since E° is much less than that of water.

158. $E = \dfrac{0.059}{1} \log \left(\dfrac{0.1 \times 0.89}{0.02 \times 0.95}\right) = 0.0396$ volt

159. $\dfrac{p_o - p_s}{p_o}$ = mole fraction of solute

$$= \frac{mM}{1000} = \frac{760 - 746.9}{760} = \frac{m \times 18}{1000}$$

$$\therefore \quad m = \frac{1000}{18} \times 0.01724 \text{ m} = 0.958.$$

But $\dfrac{6.69}{M} \times \dfrac{1000}{100} = \dfrac{6.69 \times 10}{M}$. Suppose we use the theoretical molar mass m = 164 g in this $\dfrac{66.9}{164}$ = 0.408. Suppose the degree of dissociation is x

$(0.408 - x) + 3x = 0.958.$

$\therefore \quad x = 0.275.$

$\therefore$ degree of dissociation is $\dfrac{0.275}{0.408} = 0.674$

160. 40 ml of electrolytic gas $\equiv 40 \times \dfrac{2}{3}$ ml of H_2 gas

$= 26.67$ ml.

Number of moles of $H_{2(gas)}$ is calculated =

$$\frac{748}{760} \times \frac{26.67}{1000} = n \times 0.0821 \times 288$$

$\therefore \quad n = 1.11 \times 10^{-3}$ moles of $H_{2(g)}$

$= 2.22 \times 10^{-3}$ Faraday. If the current is I,

then $\dfrac{I \times 360}{96500} = 2.22 \times 10^{-3}$.

$\therefore \quad I = 0.595$ amp

161. $\wedge$ for sodium acetate

$$= \frac{1000K}{c} = \frac{1000 \times 7.81 \times 10^{-5}}{0.001} \ \Omega^{-1} \text{ cm}^2 \text{ eqt}^{-1}$$

$\wedge = 78.1\ \Omega^{-1} \text{ cm}^2 \text{ eqt}^{-1} = \wedge_0 . \lambda^\circ_{Na+} = 44.4\ \Omega^{-1} \text{ cm}^2 \text{ eqt}^{-1}$

$$\therefore \quad \lambda^\circ_{CH_3-COO} = 33.7\ \Omega^{-1} \text{ cm}^2 \text{ eqt}^{-1}.$$

$\therefore \quad \Lambda_{0\,CH_3-COOH} = (318 + 33.7)\ \Omega^{-1}\ cm^2\ eqt^{-1}$

$\Lambda_0 = 351.7\ \Omega^{-1}\ cm^2\ eqt^{-1}$

162. $Ka \simeq \alpha^2 c = (0.0134)^2 \times 0.1 = \alpha^2 \times 0.001.$

$\therefore \quad \alpha^2 = (0.0134)^2 \times \dfrac{0.1}{0.001} \quad \alpha^2 = (0.0134)^2 \times 10^2.$

$\therefore \quad \alpha = 0.0134 \times 10 = 0.134$

163. We shall use the formula $[H^+] = (Ka\ C)^{1/2}$. Both acid contribute to the $[H^+]$ with (of course) a common ion effect

$\therefore \quad (Ka \times 5 \times 10^{-3})^{1/2} + (1.80 \times 10^{-5} \times 0.38)^{1/2}$

$= 3.67 \times 10^{-3}$

$\therefore \quad (Ka \times 5 \times 10^{-3})^{1/2} = 3.67 \times 10^{-3} - 2.615 \times 10^{-3}$

$= 1.055 \times 10^{-3}$

$\therefore \quad Ka \times 5 \times 10^{-3} = 1.113 \times 10^{-6}$

$\therefore \quad Ka = 0.226 \times 10^{-3}$

$Ka = 2.23 \times 10^{-4}$

Note: This is approximate but quiet reasonable. A more accurate calculation would give a result of the same order of magnitude.

164. On the lhs. $E_l = 0.34 + \dfrac{0.059}{2} \log c_1$

On the rhs $E_r = 0.798 + \dfrac{0.059}{1} \log c_2$

when the measured emf is zero $E_l = E_r$

$\therefore \quad 0.34 + \dfrac{0.059}{2} \log c_1 = 0.798 + \dfrac{0.059}{1} \log c_2^2$

$\therefore \quad 0.798 - 0.34 = \dfrac{0.059}{2} \log \left(\dfrac{c_1}{c_{2^1}^2} \right)$

$15.525 = \log \left(\dfrac{c_1}{c_{2^1}^2} \right) \ \therefore\ 3.350 \times 10^{15}$

$\therefore \quad \dfrac{c_{2^1}^2}{c_1} = 2.985 \times 10^{-16}$

165. According to Faraday's laws m (mass liberated) $= Z \times I \times t$. $Z = \dfrac{m}{i \times t}$. Since 31.8 g are liberated for one Faraday = 96500 coulombs.

$Z = \dfrac{31.8}{96500}$ g coulomb$^{-1.}$

$Z = 3.295 \times 10^{-4}$ g coulomb$^{-1.}$

166. For E°, $[Cl^-] = 1 \ \therefore\ [Ag^+] = 1.56 \times 10^{-10}$

$\therefore \quad = 0.8 + \dfrac{0.059}{1} \log (1.56 \times 10^{-10}) = 0.2214$ volt

167. $\Delta T_f = K_f \text{ x m}. \ \therefore\ m = \dfrac{0.0833}{1.89} = 0.04407$. But one mole in 64 litres ≡ 0.015625 M

Ratio $= \dfrac{0.04407}{0015625} = 2.82$. Thus one mole behaves like 2.82 moles in ionization.

$\therefore$ Concentration of OH^- ion

$= 0.015625 \times 2.82 \times \dfrac{2}{3} = 2.9375 \times 10^{-2}$ M

[factor $\dfrac{2}{3}$ because one "molecule" of $Ba(OH)_2$ gives 3 ions of which 2 ions are OH [Note: we are considering here only the ionization of $Ba(OH)_2$ and not including the ionization of water]

168. $\alpha = 0.85 = \dfrac{i-1}{3-1}$.

$\therefore \quad I = 1 + 1.70 = 2.7.\ \Delta T_f = i \times K_f \times m$

$\Delta T_f = 2.7 \times 1.86 \times 0.063 = 0.316°$.

$\therefore$ freezing point = –0.316°C

169. $\Delta T_f = 0.38 = 1.86 \times \dfrac{100}{342} \times \dfrac{1000}{W_1}$.

$W_1 = 1.86 \times \dfrac{100}{342} \times \dfrac{1000}{0.38}$ g

$W_1 = 1431$ g

170. $\dfrac{p_o - p_s}{p_o} = \dfrac{760 - 736.2}{760} = 0.03132$

$= \dfrac{m\,M}{1000} = \dfrac{m \times 18}{1000}$. M = 1.74

i.e., 1.74 mole of solute in approximately one litre,

$\therefore \quad \pi = 1.74 \times 0.0821 \text{ x } 288$ atm = 41.14 atm

171. (b)

Both statements are true statement 2 is not and explain to statement 1.

172. (a)

When E° (total) is positive, $\Delta G° = -nFE°$ is negative hence spontaneous reaction.

173. (b)

Both statements are correct. They are independent.

174. (d) In a solution of NaCl, decrease of Λ with increase in C is not due to varying extent of ionization but due to interionic attractions.

175. (a)

The two statements are related,

$$Ka = \frac{[H^+][A^-]}{[HA]}.$$

$$\therefore \quad \frac{Ka}{[H^+]} = \frac{[A^-]}{[HA]};$$

$$\frac{Ka}{Ka + [H^+]} = \frac{[A^-]}{C}.$$

$\therefore \quad [A^-] = \dfrac{KaC}{Ka + [H^+]}$. Also by the principle of electroneutrality

$$[H^+] = [A^-] + [OH^-] = \left[\frac{KaC}{Ka + [H^+]} + \frac{Kw}{[H^+]}\right]$$

176. Suppose molarity is M i.e., m moles per litre of the solution.

Statement-1

If density is d, then $[1000 \times d - M\,m_2]$ = wt of solvent.

$$\therefore \quad \left[\frac{M}{1000d - Mm_2} \times 1000\right] = m.$$

$$\therefore \quad \frac{1000d - Mm_2}{M} = \frac{1000}{m}$$

$$\therefore \quad 1000\,d - M\,m_2 = 1000\frac{M}{m}$$

$$\therefore \quad d - \frac{Mm_2}{1000} = \frac{M}{m}$$

$$\therefore \quad d = \frac{M}{m} + \frac{Mm_2}{1000} = M\left(\frac{1}{m} + \frac{m_2}{1000}\right)$$

Statement 2 is correct

177. (d) linearity is not observed in every case (i.e., +ve and –ve deviations from Raoult's law)

Statement 2 is correct.

178. (b) Both statements are correct but are not linked.

179. (a) linked correct statements

180. (a) linked correct statements

181. $\dfrac{\Lambda}{\Lambda_0} = \dfrac{122}{\Lambda_0} = 0.936.$

$$\therefore \quad \Lambda_0 = \frac{122}{0.936}$$

$$= 130.34\ \Omega^{-1}\ cm^2\ eqt^{-1}\ \frac{\lambda^0_+}{130.34} = \frac{0.98}{1.98}$$

$$\therefore \quad \lambda^0_+ = 64.51\ \Omega^{-1}\ cm^2\ eqt^{-1}.\ \frac{\lambda^0_-}{130.34} = \frac{1}{1.98}$$

$$\therefore \quad \lambda^0_- = \frac{130.34}{1.98}\ \Omega^{-1}\ cm^2\ eqt^{-1}$$

$$= 65.83\ \Omega^{-1}\ cm^2\ eqt^{-1}$$

182. $\dfrac{1000K}{C} = \Lambda.$

$$\therefore \quad C = \frac{1000K}{\Lambda} = \frac{1000 \times 1.37 \times 10^{-6}}{124.5}\ M$$

$$= 1.1 \times 10^{-5}\ M$$

183. Using $\Lambda = \dfrac{1000K}{C}$ $C = \dfrac{1000 \times 0.5 \times 10^{-7}}{497}$ eqt L^{-1}

$= 1.100 \times 10^{-7}$. This is $[H^+]$.

$\therefore$ Ionic product $= (1.006 \times 10^{-7})^2 = 1.012 \times 10^{-14}$

184. $NaClO_3 \xrightarrow{oxid} NaClO_4$ i.e., 2 Faradays per mole of $NaClO_4$,

$\therefore$ for 2 moles 4×96500 coulombs

$= 3.86 \times 10^5$ coulomb

185. $m = Zit \ \therefore Z = \dfrac{m}{it} = \dfrac{108}{96500}$ in g coulomb $^{-1}$

$= 1.119 \times 10^{-3}$ g coulomb $^{-1}$

186. $\dfrac{50 \times 10^{-3} \times 6 \times 3600}{96500} \times 108$ g deposited $= 1.2087$

Volume $= \dfrac{1.2087}{10.5}\ cm^3 = 0.1151\ cm^3$

Total area of deposit $= 20\ cm^2$

$\therefore$ thickness of the deposit $= \dfrac{0.1151}{20}$ cm

$= 5.756 \times 10^{-3}$

187. $K_f = \dfrac{0.002T_{f^2}}{l_f} = \dfrac{0.002 \times (273)^2}{80}$ deg molal $^{-1}$

1.86 deg molal $^{-1}$

188. $\Delta T_f = K_f \times m = 40 \times \frac{1.5}{125} \times \frac{1000}{35} = 13.714°C$

$\therefore$ melting point = (178.4 – 13.714)°C = 164.686

$\simeq$ 164.7°C

189. $\frac{p_s}{p_o} = 1 - \frac{mM}{1000}$. $m = \frac{20 \times 1000}{180 \times 70} = 1.5873$

$\therefore \quad \frac{p_s}{p_o} = 1 - \frac{1.5873 \times 18}{1000} = 0.9714$

190. a, b, c are correct

191. a, b, c are correct

192. b, d correct

193. a, b, d are correct

194. a, c, d are correct

195. b and c correct

196. a, b, c are correct.
The size of electrode does not affect emf of cell.

197. a, b are correct

This page is intentionally left blank.

CHAPTER

8

CHEMICAL KINETICS AND SOLID STATE

CHAPTER OUTLINE

Preview

STUDY MATERIAL

CHEMICAL KINETICS

Introduction and Importance of the Subject

Definition of "Rate"

Order of a Reaction

Molecularity of a Reaction

Evaluation of Rate Constant, k

Methods for Following the Progress of a Reaction

Methods of Determination of Order of a Reaction

Collision Theory

Activated Complex Theory of Reaction Rates

Kinetics of Consecutive, Parallel and Opposing Reactions

- Concept Strands (1-22)

SOLID STATE

Classification of Solids

Types of Unit Cells

Closest Packing of Constituent Particles (Spheres) in Crystals

Number of Atoms Per Unit Cell in a Cubic Lattice

Packing in Ionic Compounds

Defects in Solids

- Concept Strands (23-34)

TOPIC GRIP

- Subjective Questions (10)
- Straight Objective Type Questions (5)
- Assertion–Reason Type Questions (5)
- Linked Comprehension Type Questions (6)
- Multiple Correct Objective Type Questions (3)
- Matrix-Match Type Question (1)

IIT ASSIGNMENT EXERCISE

- Straight Objective Type Questions (80)
- Assertion–Reason Type Questions (3)
- Linked Comprehension Type Questions (3)
- Multiple Correct Objective Type Questions (3)
- Matrix-Match Type Question (1)

ADDITIONAL PRACTICE EXERCISE

- Subjective Questions (10)
- Straight Objective Type Questions (45)
- Assertion–Reason Type Questions (8)
- Linked Comprehension Type Questions (9)
- Multiple Correct Objective Type Questions (5)
- Matrix-Match Type Questions (3)

CHEMICAL KINETICS

INTRODUCTION AND IMPORTANCE OF THE SUBJECT

In the study of chemical equilibria, time is not a variable factor because we are concerned with the initial and final states only in this topic. However, the subject of chemical kinetics deals with the rates of reactions i.e., with the time required for a chemical system to reach the equilibrium state. This aspect is important from a commercial or preparative point of view. In kinetic studies, it is also pertinent to consider the effect of various factors such as concentration, temperature, pressure on the rates of reactions. Such rate studies provide all important information about the "mechanism" of the reactions. Thus chemical kinetics is basically concerned with the details of the process whereby a chosen reaction proceeds from one state to another and the time required for this change.

The main interest in the study of rate of reactions is that it offers the most general method of determining the mechanism of the reaction. By mechanism of a reaction, we mean the various individual steps or elementary processes involving different species (say atom, molecules or ions) that occur simultaneously or otherwise in producing the overall observed chemical reaction.

Factors affecting the rate of a reaction

Any chemical reaction proceeds at a definite rate depending upon the chosen experimental conditions. The most important of them are (i) concentration or pressure (in case of gaseous reactants) (ii) temperature (iii) presence of a catalyst, if any (iv) light of suitable wavelength to activate the reaction in case of photochemical reactions. For example, acetone decomposes to give ethane and carbon monoxide when irradiated with UV radiation in the wavelength range 200 – 300 mμ (v) nature of the solvent medium (in case of reactions in solutions).

Rates of reactions varying over wide range

The rates of reactions vary over a wide range. Some reactions are very rapid that they occur almost instantaneously. An example of such a reaction is the neutralization reaction given by

$$H^+_{(aq)} + OH_{(aq)} \rightarrow H_2O \qquad — (1)$$

At the other extreme, there are reactions, which are so slow at ordinary temperatures that they appear not to take place at all. The reaction between hydrogen and oxygen to give water according to the equation

$$2H_{2(g)} + O_{2(g)} \rightarrow 2H_2O_{(\ell)} \qquad — (2)$$

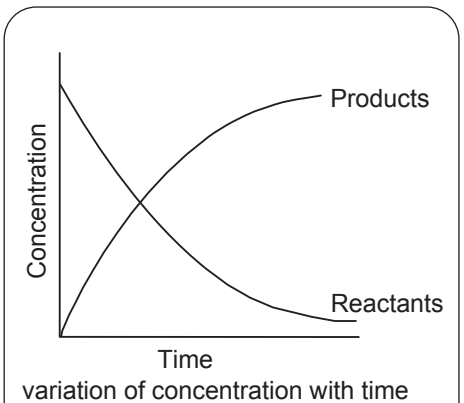
in the absence of a catalyst is an example of this category. In between these extreme cases, there are many reactions involving both inorganic and organic compounds, which occur at measurable rates at ordinary temperatures, such as, decomposition of N_2O_5 in gaseous state, hydrolysis of esters like methyl acetate in presence of mineral acids etc.

In this chapter we are mainly concerned with the study of such reactions.

Formulation of the kinetics of a reaction

From the law of mass action proposed by Guldberg and Waage, the rate of a chemical reaction is proportional to the "active masses" or the concentrations of the reacting substances.

Since the reacting substances are continuously consumed during the course of the reaction, their concentrations must decrease steadily. It follows, therefore, that the rate of the reaction (expressed as $\frac{-dc}{dt}$) varies continuously with time, becoming slower and slower as the reactants are consumed (Fig 8.1).



variation of concentration with time

Fig. 8.1

DEFINITION OF "RATE"

The rate of a reaction is generally defined as the rate of change of concentration of a substance (pressure in the case of gas) taking part in the reaction with time with a negative or positive sign attached to it depending upon whether it is a reactant or a product. Consider a general reaction such as

$$aA + bB + \ldots \rightarrow gG + hH + \ldots \qquad — (3)$$

where a,b,g,h are stoichiometric coefficients, the rate may be expressed in terms of the reactants as $\frac{-d[A]}{dt}$, $\frac{-d[B]}{dt}$, the negative sign indicating that their concentrations are decreasing with time. For the product, the rate is given as $\frac{d[G]}{dt}$, $\frac{d[H]}{dt}$, with the +ve sign (not shown) showing that the concentration of the products is increasing with time. The above equations for rate are, however, incomplete. In giving the numerical values of rate (i) units of time and concentration must be specified (ii) the particular substance whose concentration is under consideration must also be specified, the reason being that the reaction may be using different number of moles of various reactants. Considering $H_2 - O_2$ reaction (equation (2)) as an example, it is seen that hydrogen is reacting at twice the rate of oxygen in a given interval of time dt i.e.,

$$\frac{-d[H_2]}{dt} = -2\frac{d[O_2]}{dt} \text{ or } -\frac{1}{2}\frac{d[H_2]}{dt} = -\frac{d[O_2]}{dt} = \frac{1}{2}\frac{d[H_2O]}{dt} \qquad — (4)$$

This equation may be extended to the general reaction (equation (3)) by writing

$$-\frac{1}{a}\frac{d[A]}{dt} = -\frac{1}{b}\frac{d[B]}{dt} = +\frac{1}{g}\frac{d[G]}{dt} = +\frac{1}{h}\frac{d[H]}{dt} \qquad — (5)$$

In other words, the rate consumption or formation of various reactants and products divided by their respective stoichiometric coefficients are equal.

ORDER OF A REACTION

An important parameter in understanding the rates of reactions is the "order of the reaction". *Reactions are classified on the basis of the "order" ie by the number of atoms or molecules whose concentrations determine the rate of the reaction.* A few examples are considered to illustrate the significance of the term.

Examples

(i) The rate of thermal decomposition of N_2O_5 in gas phase, given by the reaction,

$$N_2O_{5(g)} \rightarrow N_2O_{4(g)} + \frac{1}{2}O_{2(g)} \qquad — (6)$$

is observed to be proportional to the concentration of N_2O_5 i.e.,

$$\frac{-d[N_2O_5]}{dt} = k[N_2O_5] \qquad — (7)$$

The reaction is, therefore said to be of first order.

(ii) The hydrolysis of an ester like methyl acetate in dilute aqueous solution is experimentally found to be of first order although the reaction involves a molecule of ester and a molecule of water given as

$$CH_3COOCH_3 + H_2O \overset{H^+}{\rightleftharpoons} CH_3COOH + CH_3OH \qquad — (8)$$

The rate of hydrolysis is given by

$$\frac{-d[CH_3COOCH_3]}{dt} = k'[CH_3COOCH_3][H_2O] \qquad — (9)$$

$$= k[CH_3COOCH_3] \text{ where } k = k'[H_2O] \qquad — (10)$$

The amount of water consumed in the reaction is quite small and hence it may be assumed to be practically constant (see equation (10)). As the reaction proceeds quite slowly at normal temperatures, a mineral acid like HCl at definite concentration is used as a catalyst whose concentration remains unchanged at the end of the reaction. The complete rate equation may, therefore, be written as

$$\frac{-d[CH_3COOCH_3]}{dt} = k''[CH_3COOCH_3][H_2O][H_3O^+] \qquad — (11)$$

$= k[CH_3COOCH_3]$ ($\because$ both $[H_2O]$ and $[H_3O^+]$ are constants) — (12)

Here again, experimentally, it behaves as a reaction of first order. Hence it is often referred as a pseudo first order reaction.

(iii) The decomposition of HI (in the gas phase) is a second order reaction ie the experimentally observed rate is proportional to the square of the concentration of HI i.e., for the reaction

$$2HI_{(g)} \rightleftharpoons H_{2(g)} + I_{2(g)} \qquad \text{— (13)}$$

$$\frac{-d[HI]}{dt} = k[HI]^2 \qquad \text{— (14)}$$

It is not possible to predict the rate equation or the order of the reaction from the stoichiometric equation (from equations like (6), (8), (13)) as they are determined only by experiment.

Consider a general reaction

$$A + B + C \rightarrow \text{Products} \qquad \text{— (15)}$$

Assuming that the experimentally observed rate of the reaction is proportional to n_1th power of concentration of A, n_2th power of the concentration of B and n_3th power of the concentration of C, the rate equation may be written as

$$\text{Rate} = kC_A^{n_1} C_B^{n_2} C_C^{n_3} \qquad \text{— (16)}$$

From equation (16), the total order of the reaction = $n_1 + n_2 + n_3$ — (17)

Further, the order with respect to A is n_1, order with respect to B is n_2 and the order with respect to C is n_3.

The powers n_1, n_2, n_3 are often positive integers but they may be fractional or even negative depending on the complexity of the reaction. The order of a reaction thus describes the mathematical form in which the observed results are best presented in the form of rate equation.

MOLECULARITY OF A REACTION

The term molecularity of a reaction is also often used in the study of the mechanisms of reactions and sometimes mentioned as synonymous with the order of the reaction. It is to be emphasized that the terms "order" and "molecularity" are not the same although under certain circumstances they are identical. The molecularity of a reaction refers to number of molecules in the step leading to chemical reaction or to the number of molecules participating in an elementary step of reaction.

The concept is, therefore, theoretical in nature and linked to the mechanism of the reaction.

The distinguishing features between the two terms are summarized in the table 8.1.

Table. 8.1: Differences between order and molecularity

Order	Molecularity
1) It is determined from experiments	1) It is a theoretical concept
2) Order may be integral, fractional or even a negative number	2) It cannot be fractional or negative or zero.
3) For a reaction taking place in a single step, order may be the same as the molecularity of the reaction. The decomposition of HI is an example of this category.	3) For a reaction taking place in a single step, the molecularity is usually the same as its order as in the case of hydrogen iodide decomposition. For a multi step reaction, each step has its own molecularity.

Units of rate constant

The proportionality constant, k (equation (16)) is known as the rate constant, velocity constant or specific reaction rate of the reaction. For a nth order reaction,

$$\frac{-dC}{dt} = kC^n$$

$$k = \frac{-dC}{dt} / C^n \qquad \text{— (18)}$$

k has the units $(\text{conc})^{1-n}\ \text{time}^{-1}$. If the concentration is expressed in moles / litre and time is expressed in seconds, k has the units $\left(\frac{\text{mol}}{\text{litre}}\right)^{1-n}\ s^{-1}$. From this general expression,

the units for the rate constant of reactions of various orders may be derived as shown in Table 8.2.

Table 8.2: Units of reactions of various orders.

Order of the reaction	Units
First order reaction, n = 1	sec^{-1}
Second order reaction, n = 2	$L\ mol^{-1}\ sec^{-1}$
Third order reaction, n = 3	$L^2\ mol^{-2}\ sec^{-1}$
Half order reaction, n = $\frac{1}{2}$	$mol^{½}\ L^{-½}\ sec^{-1}$
Zero order reaction, n = 0	$mol\ L^{-1}\ sec^{-1}$

Reactions of zero order

When the rate of a reaction is not affected by changes in the concentration of one or more reactants ie the rate is independent of the concentration of a reactant, the reaction is said to be zero order with respect to that reactant.

In general for a zero order reaction, we can write

$$\frac{-dC_A}{dt} = kC_A^0 = k \text{ (where A is the reactant)}$$

$$\frac{-dC}{dt} = k$$

On general integration, $-\int dC_A = k\int dt$

$$-C_A = kt + \text{constant} \qquad \text{— (81)}$$

At t = 0, constant = $-C_A^0$

Substituting for the "constant" in equation (81)

$$-C_A = kt - C_A^0 \text{ or } kt = C_A^0 - C_A$$

$$\text{or } k = \frac{1}{t}(C_A^0 - C_A) \qquad \text{— (82)}$$

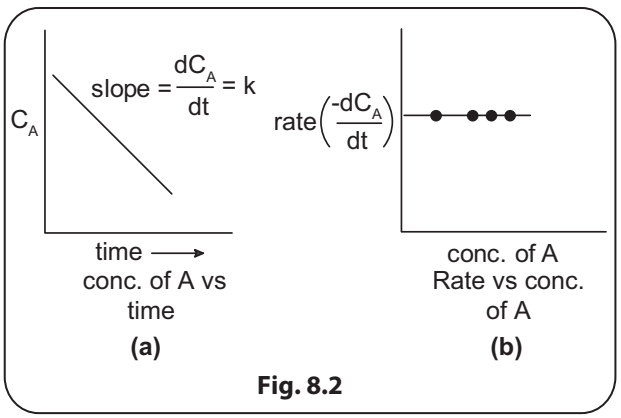


Fig. 8.2

The rate constant may be evaluated from a plot of C_A vs t as shown in Fig. 8.2(a)

A plot of the rate vs concentration of the reactant is linear and is horizontal to the concentration axis as shown in Fig. (8.2(b)).

An example of such a reaction is the bromination or iodination of acetone which is first order with respect to ketone but zero order with respect to halogen.

Reactions of first order

In a reaction of 1st order, the rate at any instant is proportional to the first power of the concentration of the reactant at that instant. Mathematically, this may be expressed as

$$\frac{-dC}{dt} = kC \qquad \text{— (19)}$$

From the above equation, it is seen that

$$\frac{-dC}{dt} = k, \text{ when } C = 1 \qquad \text{— (20)}$$

The rate constant of a reaction is a constant for the reaction and depends on temperature and solvent.

Any kinetic investigation of a reaction involves the determination of the rate constant by a suitable analytical method, the effect of temperature and catalyst on the rate etc. and ultimately the formulation of a mechanism for the reaction.

General integration of equation (19) gives

$$-\int \frac{dC}{C} = k\int dt \qquad \text{— (21)}$$

$$\text{or } -\ln C = kt + \text{constant} \qquad \text{— (22)}$$

To evaluate the integration constant, consider t = 0, constant = $-\ln C_0$ where, C_0 is the initial concentration of the reactant.

Substituting for the "constant" in equation (22) and rewriting the equation, we get

$$kt = \ln \frac{C_0}{C} \qquad \text{— (23)}$$

$$\text{or } k = \frac{1}{t}\ln\frac{C_0}{C} = \frac{2.303}{t}\log_{10}\frac{C_0}{C} \qquad \text{— (24)}$$

Equation (23) may also be expressed in the exponential form as $C = C_0 e^{-kt}$. (24 a)

In equation (24) C is the concentration of the reactant remaining at the time t. An alternative form of equation (24) is often used for evaluating k.

If "a" represents the initial concentration of the reactant, "x" is the amount of the reactant (in concentration

units) reacting in time "t", then amount of reactant remaining after time, t = (a-x)

The rate equation may be written as

$$\frac{-d(a-x)}{dt} = k(a-x) \qquad \text{— (25)}$$

General integration of equation (25) gives,

$$\int \frac{dx}{a-x} = k\int dt \qquad \text{— (26)}$$

$$-\ln (a-x) = kt + \text{constant} \qquad \text{— (27)}$$

$$\text{At } t = 0, \times = 0, \therefore \text{ constant} = -\ln a \qquad \text{— (28)}$$

Equation (27) may now be written as

$$k = \frac{1}{t}\ln\frac{a}{a-x} = \frac{2.303}{t}\log\left(\frac{a}{a-x}\right) \qquad \text{— (29)}$$

where a = initial concentration of the reactant.
(a – x) = concentration of reactant remaining at time t.

From equation (29), it is seen that for a first order reaction, the rate constant, k, is independent of concentration units and thus has the units of time^{-1}.

Time taken to complete a definite fraction of the reaction

Let us consider the time of half reaction, $t_{½}$, the time required for the initial concentration to become half its value.

Substituting $\times = \frac{a}{2}$ in equation (29), we get

$$k = \frac{2.303}{t_{½}}\log\frac{a}{\frac{a}{2}} = \frac{2.303}{t_{½}}\log 2$$

$$\text{or} \quad t_{½} = \frac{2.303}{k}\log 2 = \frac{0.693}{k} \qquad \text{— (30)}$$

Equation (30) shows that $t_{½}$ is a constant and is independent of the initial concentration. This is a very important characteristic of first order reactions. Thus for a first order reaction, it will take the same time to reduce the concentration from one molar to 0.5 molar or from 1×10^{-3} M to 5×10^{-4} M

In general, the time taken to complete any definite fraction of the reaction (say, $\frac{1}{4}$th of the reaction or $\frac{3}{4}$th of the reaction) is independent of the initial concentration.

This result may be arrived at by substituting

x = (¼ a) or ($^3/_4$ a) in equation (29).

EVALUATION OF RATE CONSTANT, K

The rate constant k may be evaluated by

(i) analytical method
(ii) graphical method.

Analytical method

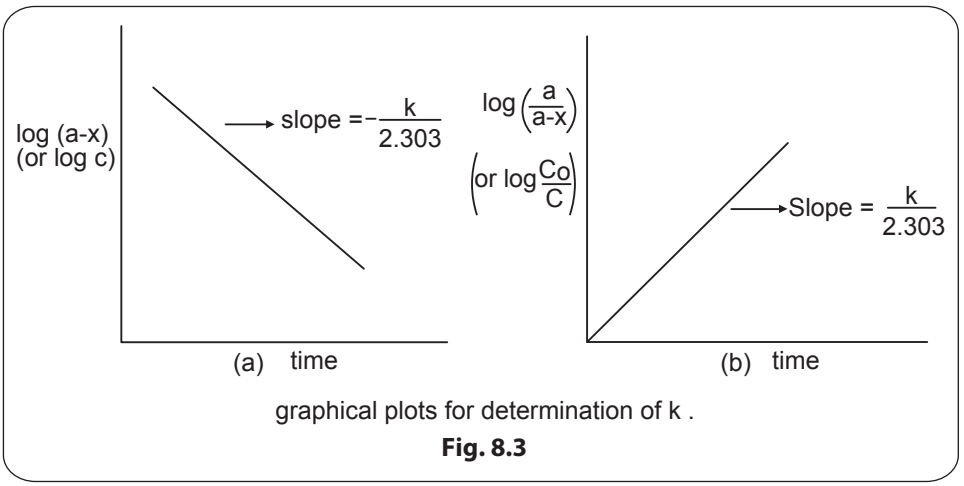
After initiating the reaction, the reaction mixture is analysed at various intervals of time and the resulting data are substituted in the equation for k to evaluate the same. Some examples will be discussed subsequently.

Graphical method

For a first order reaction, the plot of log (a – x) (or log C) against t should give a linear relation and from the slope, k can be calculated.

Alternatively, a plot of $\log\left(\frac{a}{a-x}\right)\left(\text{or } \log\frac{C_0}{C}\right)$ versus t is a straight line passing through the origin and from the slope, k can be calculated. The nature of these plots is shown in Fig. 8.3.

The graphical method involving the plot of log (a – x) vs time (Fig. 8.3(a)), for determination of k has the advantage that it does not require a knowledge of the initial concentration of the reactant.



graphical plots for determination of k .

Fig. 8.3

METHODS FOR FOLLOWING THE PROGRESS OF A REACTION

Depending upon the nature of the reaction under consideration, various methods may be adopted for following the course of a reaction. Some typical reactions and the methods adopted for studying their rates are given in Table 8.3.

Table 8.3 Various method for following rates

Nature of reaction with example	Method for following the rate
1. Reaction in which a species is formed or consumed and can be determined by a titrimetric method. Example Acid hydrolysis of an ester	The reaction can be followed by titrating the acid formed at various intervals of time with a standard base.
2. Reaction involving the formation or disappearance of a coloured species. In this case, a colorimetric or spectro-photometric method may be employed for following the reaction. Example Bromination or iodination of acetone	The reaction can be followed by following the disappearance of the colour of halogen at a suitable wavelength at different time intervals.
3. In the case of gaseous reactions which involve a change in the number of molecules, the reaction can be followed by noting the change in pressure at constant volume (or by following the change in volume at constant pressure) at various time intervals. Example $N_2O_5(g) \rightarrow N_2O_4(g) + ½\ O_2(g)$	The reaction can be followed by following the change in pressure under constant volume conditions.
4. Sometimes, reactions occur with the formation or consumption of ions which results in a large change of conductivity. Example, Alkaline hydrolysis of an ester	In this case, as the reaction occurs, there is a decrease of conductivity due to the fact that highly mobile OH^- ions are replaced by the organic acid anions (such as CH_3COO^-) which are slow moving.
5. Any specific difference in properties of reactants and products is sometimes used effectively. Example Hydrolysis of sucrose	Here, the rotation of plane of polarization of plane polarisd light changes from right to left and this change is followed in a polarimeter.

There are several other possibilities for following the kinetics of reactions and the above listed methods are only representative.

Some examples of first order reactions

Hydrolysis of methyl acetate (or any other ester) in presence of a mineral acid

The reaction is given by

$$CH_3COOCH_3 + H_2O \rightarrow CH_3COOH + CH_3OH \quad \text{— (31)}$$

and it proceeds slowly at room temperature but is catalysed by acids like HCl, $HClO_4$ etc. The reaction is followed by titrating the acetic acid formed (+ the catalyst acid at a fixed concentration) at various intervals of time against a standard base like NaOH.

It is also referred as a pseudo unimolecular reaction indicating that although there are two reacting molecules effectively, it is a first order reaction because of the negligible change in the concentration of water as a reactant.

The equation for evaluating the rate constant using the titre values of the base is obtained as follows.

If the titre value at the beginning of the experiment = V^o
If the titre values at any time, t = V_t
If the titre value at the end of the experiment = V_∞
(i.e., when all the ester is hydrolysed)
one may write, $a \propto (V_\infty - V_o)$;
$x \propto (V_t - V_o)$ — (32)

$(a - x) \propto (V_\infty - V_t)$

substituting for a and (a – x) in equation (29)

$$k = \frac{2.303}{t}\log\frac{(V_\infty - V_o)}{(V_\infty - V_t)} \quad \text{— (33)}$$

k is calculated using equation (33) at various intervals of time and it is a constant at the given temperature.

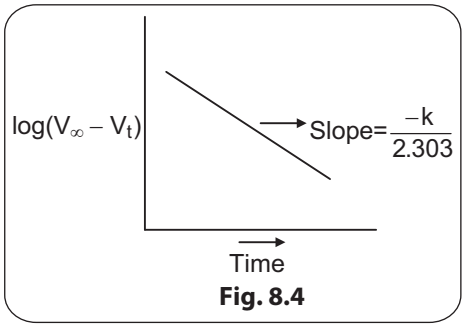


Fig. 8.4

k can also be evaluated graphically by plotting log $(V_\infty - V_t)$ vs t.

The resulting graph is a straight line as shown in Fig. 8.3 and k may be evaluated from the slope given by $\frac{-k}{2.303}$.

This method has the advantage that it eliminates the use of $(V_\infty - V_o)$ which is associated with some error in the determination of V_o at the beginning of experiment.

By considering the results at two time intervals t_1 and t_2, it is possible to calculate k without using $(V_\infty - V_o)$ in equation (33). At the time intervals t_1 and t_2, we may write

$$t_1 = \frac{2.303}{k}\log\left(\frac{V_\infty - V_o}{V_\infty - V_{t_1}}\right)$$

$$t_2 = \frac{2.303}{k}\log\left(\frac{V_\infty - V_o}{V_\infty - V_{t_2}}\right) \quad \text{— (34)}$$

$$\text{or} \quad (t_2 - t_1) = \frac{2.303}{k}\log\left(\frac{V_\infty - V_{t_1}}{V_\infty - V_{t_2}}\right) \quad \text{— (35)}$$

Equation (35) enables evaluation of k from data at two time intervals t_1 and t_2.

Hydrolysis of sucrose

Sucrose hydrolyses in aqueous solution in presence of an acid (which acts as a catalyst) to give glucose and fructose. This is also a pseudo unimolecular reaction and the reaction is given by

$$C_{12}H_{22}O_{11} + H_2O \xrightarrow[\text{Catalyst}]{H^+} \underset{\text{glucose}}{C_6H_{12}O_6} + \underset{\text{fructose}}{C_6H_{12}O_6} \quad \text{— (36)}$$

The rate of hydrolysis can be conveniently followed by measuring the angle of rotation of polarized light in a polarimeter. Sucrose rotates the plane of polarized light to the right (called dextrorotatory) while the mixture of glucose and fructose rotates it to the left (called leavorotatory). By measuring the angle of rotation at different time intervals, it is possible to follow the kinetics of the reaction.

Let α_o, α_t and α_∞ be the angles of rotation including the sign at zero time, at time t and at the end of the experiment (∞ time) respectively. Then

$$a \propto (\alpha_o - \alpha_\infty);\ x \propto (\alpha_o - \alpha_t);\ (a - x) \propto (\alpha_t - \alpha_\infty)$$

$$\text{or} \quad k = \frac{2.303}{t}\log\frac{(\alpha_o - \alpha_\infty)}{(\alpha_t - \alpha_\infty)} \quad \text{— (37)}$$

k can be evaluated using equation (37) or by a graphical method similar to that in the case of ester hydrolysis reaction. It may be noted that k evaluated in the two reactions

discussed above is applicable at the particular concentration of the acid used and is proportional to it.

$$\text{i.e., } k \propto [H^+]_{expt} \text{ or } k = k_o[H^+] \qquad \text{— (38)}$$

where k_o is independent of acid concentration and is sometimes referred as the true rate constant of the reaction. Equation (37) is often used to determine $[H^+]$ concentration i.e., the catalyst, using a calibration plot of data of k in solution of known $[H^+]$.

It is noteworthy that in the case of first order reaction, the rate constant is independent of concentration unit. This is an advantage and facilitates evaluation of rate constants by substituting any physical property proportional to the concentration (such as optical rotation, absorbance etc.,) in the equation for k. Another important class of reactions belonging to first order, is the radio active disintegration. In this case, the rate constant is referred as "decay constant". More details on the radioactive decay are given in the chapter on nuclear chemistry.

General expression for the rate constant of nth order reaction

A reaction in which the order with respect to the reactant "x" is n may be represented as

$$nx \rightarrow \text{products} \qquad \text{— (39)}$$

The rate equation may be written as

$$\frac{-d[x]}{dt} = k[x]^n \qquad \text{— (40)}$$

Rewriting equation (40) and making a general integration of the resulting equation

$$-\int \frac{d[x]}{[x]^n} = k\int dt \qquad \text{— (41)}$$

$$\frac{-[x]^{(-n+1)}}{(-n+1)} = kt + \text{constant} \qquad \text{— (42)}$$

To evaluate the integration constant, let t = 0, Then

$$\text{Constant} = \frac{-[x_o]^{(-n+1)}}{(-n+1)} = \frac{1}{(n-1)}[x_o]^{(-n+1)} \qquad \text{— (43)}$$

Substituting for the constant from equation (43) in equation (42) and rewriting the resulting equation.

$$k = \frac{1}{t(n-1)}\left[\frac{1}{[x]^{n-1}} - \frac{1}{[x_o]^{n-1}}\right] \qquad \text{— (44)}$$

$[x_o]$ represents the concentration of × at zero time and [x] is the concentration remaining at time t. Equation (44) is a general equation applicable when $n \neq 1$ and the concentration terms apply to the same reactant (or two reactants at the same initial concentration).

METHODS OF DETERMINATION OF ORDER OF A REACTION

There are five methods commonly adopted for determining the order of a reaction and they are detailed below:

(i) Trial and error method

The simplest method is to substitute the analytical data into the rate equation for a first, second or third order reaction. The equation, which gives the most constant values for the rate constant gives the order of the reaction.

(ii) Graphical method

If a reaction is 1st order, the plot of log (a – x) vs t is linear with a slope $= -\frac{k}{2.303}$

For a reaction of order greater than one, the general equation obtained earlier (equation 44) can be rearranged as

$$t = \frac{1}{k(n-1)}\frac{1}{[x]^{n-1}} - \frac{1}{k(n-1)}\frac{1}{[x_o]^{n-1}} \qquad \text{— (45)}$$

$$= \frac{1}{k(n-1)}\frac{1}{[x]^{n-1}} \text{ - constant} \qquad \text{— (46)}$$

From equation (46), it is seen that a plot of $\frac{1}{[x]^{n-1}}$ vs t is linear where n is the order of the reaction. For example, for a second order reaction plot of $\frac{1}{[x]}$ vs t is linear.

(iii) Method based on $t_{½}$ of the reaction

By determining the time required to complete a definite fraction of the reaction, say one half of the reaction, at different initial concentrations, the order of the reaction can be determined.

Substituting $[x] = \frac{[x_o]}{2}$ in equation (44), we get

$$k = \frac{1}{t_{1/2}(n-1)}\left[\frac{1}{\left[\frac{x_0}{2}\right]^{n-1}} - \frac{1}{[x_o]^{n-1}}\right] = \frac{1}{t_{1/2}(n-1)}\frac{2^{n-1}-1}{[x_o]^{n-1}} \quad \text{— (47)}$$

Rearranging equation (47)

$$t_{1/2} = \frac{1}{k(n-1)}\frac{2^{n-1}-1}{[x_o]^{n-1}} = \text{ constant} \times \frac{1}{[x_o]^{n-1}} \quad \text{— (48)}$$

$$\text{or } t_{1/2} \times [x_o]^{n-1} = \text{constant} \quad \text{— (49)}$$

$$\text{i.e., For a reaction of nth order, } t_{1/2} \propto \frac{1}{[x_o]^{n-1}} \quad \text{— (50)}$$

Thus by varying the initial concentration and measuring $t_{½}$ (or another fraction of $[x_0]$) and determining the product for various values of "n", the order "n" can be determined. However, it must be stated that this method is not applicable when n = 1.

(iv) Differential method

Consider the reaction

$$A + B \rightarrow \text{Products} \quad \text{— (51)}$$

The order with respect to the reactants A and B may be determined in the following manner. Initially the concentration of B is kept constant and that of A is varied. For two such concentrations C_{A_1} and C_{A_2}, we have for the initial rates (rates in the initial stages of the reaction when it is not more than 10% complete)

$$\frac{-dC_{A_1}}{dt} = k'C_{A_1}^{n_A}C_B^{n_B} = kC_{A_1}^{n_A} \; ; \frac{-dC_{A_2}}{dt} = k'C_{A_2}^{n_A}C_B^{n_B} = kC_{A_2}^{n_A} \quad \text{— (52)}$$

where, $k = k'C_B^{nB}$. It is assumed that C_B is virtually constant because the reaction has not progressed much. Taking logarithm

$$\log\left(\frac{-dC_{A_1}}{dt}\right) = \log k + n_A \log C_{A_1} \quad \text{— (53(a))}$$

$$\log\left(\frac{-dC_{A_2}}{dt}\right) = \log k + n_A \log C_{A_2} \quad \text{— (53(b))}$$

Subtracting equation (53(b)) from (53(a))

$$\frac{\log\left(\frac{-dC_{A_1}}{dt}\right) - \log\left(\frac{-dC_{A2}}{dt}\right)}{\log\left(C_{A_1}\right) - \log\left(C_{A_2}\right)} = n_A \quad \text{— (54)}$$

Repeating the above procedure with concentration of A being kept constant and varying B, the order with respect to B is obtained. The initial rates can be found from a graph of concentration vs time at different time intervals (when the reaction has not progressed beyond about 10%).

(v) Ostwald's isolation method

Considering the same reaction (equation (51)) as an example, the rate equation may be written as

$$\frac{dx}{dt} = kC_A^{n_A}C_B^{n_B} \quad \text{— (55)}$$

In one experiment the concentration of B is kept at a relatively large value compared to A so that the order with respect to A can be determined from the equation.

$$\frac{dx}{dt} = k'C_A^{n_A} \text{ where } k' = kC_B^{n_B} \quad \text{— (56)}$$

In a second experiment, the concentration of A is kept large so that the rate equation becomes

$$\frac{dx}{dt} = k'C_B^{n_B} \text{ where } k' = kC_A^{n_A} \quad \text{— (57)}$$

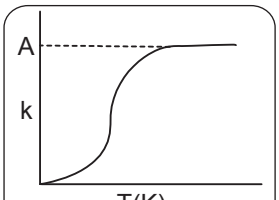
and the order with respect to B can be determined

Effect of temperature on the rate

Increase of temperature generally increases the rate of a reaction markedly. For homogeneous reactions, the rate constant increases by a factor of two or three for a ten degree rise in temperature. The effect of temperature on the rate of a reaction can be expressed quantitatively through an equation proposed by Arrhenius and is given by

$$k = Ae^{-E/RT} \quad \text{— (58)}$$

where k is the rate constant, E_a is the energy of activation, A is the frequency factor and T is the temperature in K, A and E are constants for a given reaction (strictly, A and E_a depend on the temperature to a small extent.) A plot of k against temperature (K), according to Arrhenius equation, has the following form (Fig. 8.4).



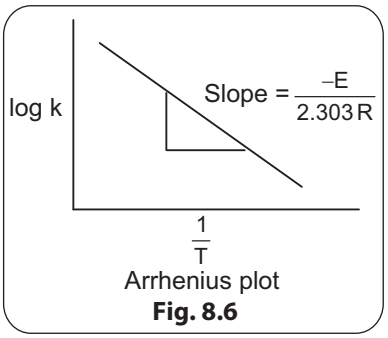
Effect of temperature on rate

Fig. 8.5

Experimentally, the lower rising part of the curve is the region accessible for study. By taking logarithms, equation (58) may be written as

$$\log_{10} k = \log_{10} A - \frac{E_a}{2.303RT} \quad \text{— (59)}$$

From equation (59), it is seen that a plot of log k Vs $\frac{1}{T}$ is linear, the slope being equal to $-\frac{E}{2.303R}$ as shown in Fig. 8.6.



Arrhenius plot
Fig. 8.6

Thus the energy of activation for a reaction can be calculated from the slope of the plot of rate constant data obtained at various temperatures. It may be mentioned that the value of the frequency factor (A) cannot be obtained from the above graph accurately as the linearity of the plot is restricted only to a small temperature range.

Alternatively, E_a can be determined by considering the rate constants k_1 and k_2 at two temperatures T_1 and T_2. Writing the Arrhenius equation at the two temperatures T_1 and T_2,

$$\log k_1 = \log A - \frac{E_a}{2.303RT_1} \quad \text{— (60(a))}$$

$$\log k_2 = \log A - \frac{E_a}{2.303RT_2} \quad \text{— (60(b))}$$

And subtracting equation (60(a)) from equation (60(b))

$$\log \frac{k_2}{k_1} = \frac{E_a}{2.303\ R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right] = \frac{E_a}{2.303\ R}\left[\frac{T_2 - T_1}{T_1T_2}\right] \quad \text{— (61)}$$

Significance of E_a and A

The energy of activation of a reaction is an important parameter in kinetic studies as it provides information on the energetics of the reaction and also about its mechanism. The explanation of the terms, energy of activation (E_a) and frequency factor (A) is possible on considering of the theories of reaction rates. There are two important theories which are widely employed (1) collision theory (2) transition state theory or activated complex theory. Both theories aim at explaining the large differences in rates of immeasurably slow reactions on the one hand and very fast reactions on the other by giving an interpretation to the terms A and E_a in the Arrhenius equation. A brief account of the collision theory of reaction rates is given to highlight the significance of the terms E_a and A.

COLLISION THEORY

Consider a bimolecular reaction between two gases A and B. *According to collision theory, the primary requirement for a reaction between the two species is that the two molecules must collide.* The total number of collisions that occur between the molecules of A and B at a given temperature and pressure, Z_{AB} (which is required for calculating the rate) can be obtained from the kinetic theory of gases as

$$Z_{AB} = n_A \times n_B\ \sigma^2 \left(8\pi RT \times \frac{M_A + M_B}{M_A M_B}\right)^{1/2} \quad \text{— (62)}$$

where, Z_{AB} = number of collisions between A and B molecules per mL per second.

n_A, n_B = number of A molecules (n_A) and number of B molecules (n_B) per unit volume.

σ = mean collision diameter of A and B molecules = $\left(\frac{\sigma_A + \sigma_B}{2}\right)$

M_A, M_B = molecular masses of A and B

If the reaction involves only a single gas, represented by A, the number of collisions between A molecules Z_{AA} is given by

$$Z_{AA} = 2n_A^2\sigma_A^2 \left(\frac{8\pi RT}{M}\right)^{1/2} \quad \text{— (63)}$$

where, n_A, σ_A have the same significance given earlier.

Z is known as the collision frequency. However, the number of collisions calculated from kinetic theory expressions (eqn 62 or eqn 63) is much higher than the number of molecules reacting to give the products at any given temperature. Secondly, although the number of collisions increases with increase of temperature, to some extent, the rate of the reaction increases by a much larger factor.

To explain the above observations, it was suggested that when two molecules collide, a reaction occurs only if they possess energy in excess of a certain minimum value. Thus collisions between molecules having energy less than the minimum value are ineffective and do not lead to reaction. Thus one arrives at the concept of activation energy, E, which is defined as the minimum energy (over and above

a certain average value) which the molecules must acquire before they can react on collision. The proportion of collisions between molecules possessing an energy E in excess of the average value is known as Boltzmann's factor and is given as $e^{-E/RT}$.

∴ The actual number of collisions leading to reaction is given by the product of Z and $e^{-E/RT}$ i.e.,

$$k = Ze^{-E/RT} \qquad \text{— (64)}$$

The rate constant in equation (64) has the units of number of molecules reacting per mL per sec. lt is seen that the Arrhenius equation (eqn 58) and equation (64) are very similar with the frequency factor (A) being identical with the collision frequency (Z).

The collision theory is found to be satisfactory in the case of simple reactions (like decomposition of HI) but in many cases, the rates calculated from kinetic theory expression (eqn (64)) are much higher than the experimentally observed values. Thus this theory has certain drawbacks. An alternate theory, viz., the transition state theory which aims to overcome the drawbacks of the collision theory, is proposed. Details of the theory are given under "additional reference".

Additional reference

Reactions of second order

(i) *For a second order, the rate depends on two concentration terms, both of them may refer to the same reactant or to two different reactants.*

If the two reacting molecules are the same, the reaction may be written as

$$2X \rightarrow \text{Products} \qquad \text{— (65)}$$

for which the rate equation is

$$\frac{-d[X]}{dt} = k[X]^2 \qquad \text{— (66)}$$

Using the equation (44) viz., the expression for the rate constant of a nth order reaction, with n = 2 (i.e., substitute n = 2 in equation (44) we get

$$k = \frac{1}{t(2-1)}\left[\frac{1}{[X]^1} - \frac{1}{[X_o]^1}\right] = \frac{1}{t}\left[\frac{1}{[X]} - \frac{1}{[X_o]}\right] \qquad \text{— (67)}$$

Employing the nomenclature $[X_o] = a$ and $[X] = (a - x)$ in equation (67),

$$k = \frac{1}{t}\left[\frac{1}{(a-x)} - \frac{1}{a}\right] = \frac{1}{t}\frac{x}{a(a-x)} \qquad \text{— (68)}$$

(ii) For a third order reaction under similar conditions

$$k = \frac{1}{2t}\left[\frac{1}{(a-x)^2} - \frac{1}{a^2}\right] \qquad \text{— (69)}$$

(iii) Considering a more general case of a second order reaction involving two different reactants A and B

$$A + B \rightarrow \text{products} \qquad \text{— (70)}$$

The rate equation may be written as

$$\frac{-dC_A}{dt} = \frac{-dC_B}{dt} = kC_AC_B \qquad \text{— (71)}$$

Writing for the initial conc of A = a
and initial conc of B = b
Conc of A and B reacting at the time t = x — (72)
Equation (71) may be written as

$$\frac{-d(a-x)}{dt} = \frac{-d(b-x)}{dt} = k(a-x)(b-x) \qquad \text{— (73)}$$

When a = b, i.e., when the reactants A and B are at the same concentrations, equation (73) reduces to

$$\frac{dx}{dt} = k(a-x)^2 = k(b-x)^2 \qquad \text{— (74)}$$

which on integration leads to equation (68)

Integration of equation (73) by the method of partial fractions gives

$$k = \frac{2.303}{t(b-a)}\log\frac{a(b-x)}{b(a-x)} \qquad \text{— (75)}$$

The dimensions of a second order reaction is $conc^{-1}$ $time^{-1}$ or $litres\ mole^{-1}\ sec^{-1}$ when concentrations are expressed in the units of moles $litre^{-1}$ and time is expressed in sec.

(iv) $t_{½}$ of a second order reaction

When there is a single reactant, or when two reactants are at the same initial concentration, equation (68) may be simplified as

$$t_{½} = \frac{1}{ak} \qquad \text{— (76)}$$

The $t_{½}$ depends on the initial concentration (a) and is inversely proportional to it.

(v) Examples of second order reactions

Among reactions occurring in the gas phase, mention may be made of

(a) combination of ethylene and hydrogen to give ethane

$$C_2H_{4(g)} + H_{2(g)} \rightarrow C_2H_{6(g)} \qquad \text{— (77)}$$

(b) decomposition of nitrosyl chloride to give nitric oxide and chlorine

$$2NOCl_{(g)} \rightarrow 2NO_{(g)} + Cl_{2(g)} \qquad \text{— (78)}$$

Some examples of second order reactions in solution phase are

(i) The hydrolysis of an ester in alkaline medium for example, the hydrolysis of ethyl acetate given by

$$CH_3COOC_2H_5 + OH^- \rightarrow CH_3COO^- + C_2H_5OH \quad \text{— (79)}$$

which is first order in the concentration of ester and first order in the concentration of alkali.

(ii) The reaction between triethylamine and ethyl bromide in benzene as solvent to form the quaternary ammonium bromide according to

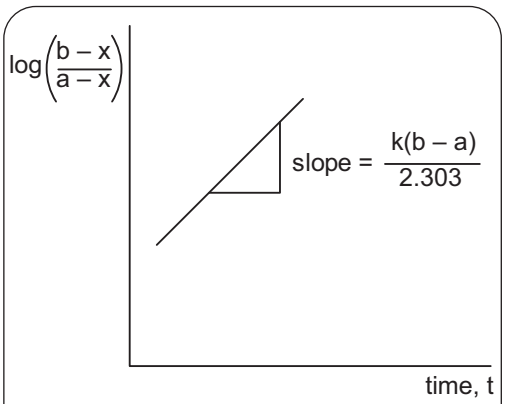
$$(C_2H_5)_3N + C_2H_5Br \rightarrow (C_2H_5)_4NBr \quad \text{— (80)}$$

Considering the alkaline hydrolysis of an ester as an example, the rate of the reaction can be followed by two ways (a) analytically by following the decrease in the concentration of the base at various intervals of time.

This is done by arresting the reaction at any instant by adding an excess of a known volume of a strong acid of given strength and then back titrating the remaining acid with a base of known strength or (b) by following the decrease in conductivity arising as a result of replacement of highly mobile OH^- ions by slow moving acetate ions at various intervals of time.

Evaluation of rate constant of 2nd order reaction

The rate constant can be calculated by substituting the experimental data at various intervals of time in the 2nd order rate equation (see for example, equation (68) or equation (75)). Alternatively, a graphical method may be employed by plotting $\log \frac{(b-x)}{(a-x)}$ vs t and from the slope of the linear graph, k can be calculated as shown in Fig. 8.7.



Evaluation of k of a second order reaction

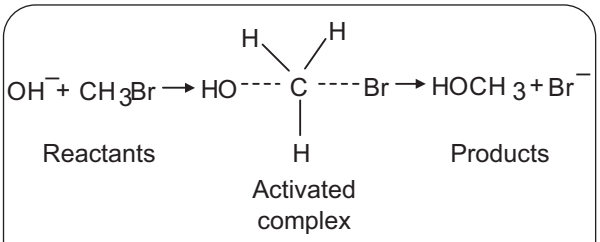
Fig. 8.7

ACTIVATED COMPLEX THEORY OF REACTION RATES

To overcome the drawbacks of the collision theory, another theory was proposed which is known as the transition state theory or the activated complex theory. This theory developed by H. Eyring and M. Polanyi in 1935, postulates the existence of an activated complex as an intermediate in all reactions.

When two (or more) molecules with the requisite energy come very close they react to give products. Before the products are formed, there is a rearrangement of energy as well as the valence bonds to form the activated complex, which has only a transient existence. The activated complex decomposes at a definite rate to form the products.

The nature of the activated complex can be understood by considering an example such as the hydrolysis of methyl bromide, as shown in Fig. 8.8.



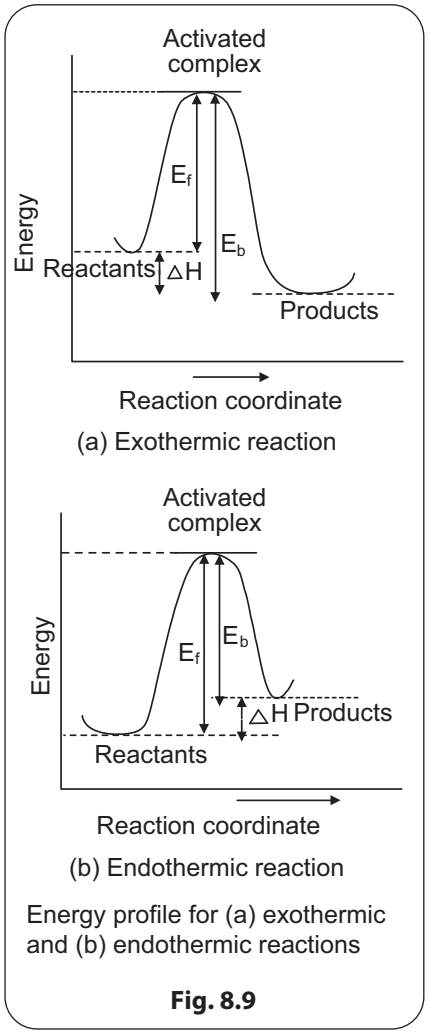
Nature of the activated complex in hydrolysis of methyl bromide

Fig. 8.8

According to this theory, the activation energy of a reaction (E) is defined as the additional energy which the

reactant molecules must acquire to form the activated complex. Thus E is the difference in energy between the activated complex and the reactants.

The formation of the activated complex for exothermic and endothermic reactions is shown in the following diagram, Fig. 8.9.



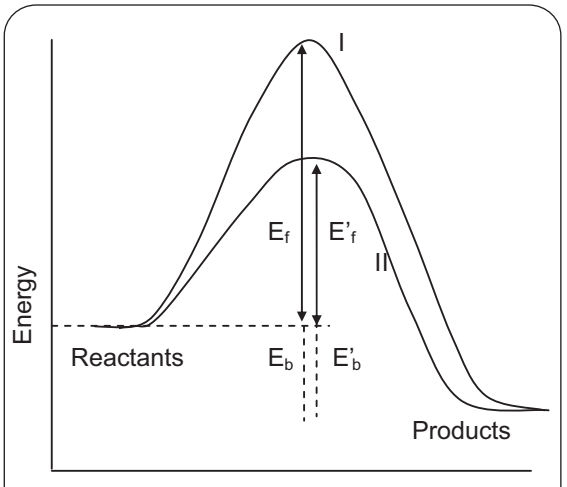
Energy profile for (a) exothermic and (b) endothermic reactions

Fig. 8.9

The reaction coordinate in Fig. 8.9 is a measure of the progress of the reaction. From the above figure, it is seen that for an exothermic reaction $\Delta H = E_f - E_b$ = –ve for an endothermic reaction $\Delta H = E_f - E_b$ = +ve.

Effect of a catalyst

If a catalyst is used in the reaction, the height of the maximum in the energy barrier is lowered as shown in Fig. 8.10.



Energy diagram for an exothermic reaction in presence of a catalyst.

Fig. 8.10

The catalyst merely provides an alternative reaction channel II with a lower activation curves it does not remove channel I

Curve I → Energy diagram in the absence of a catalyst.
Curve II → Energy diagram in presence of a catalyst.

E_f → Energy of activation for forward reaction in the absence of catalyst.

E_b → Energy of activation for reverse reaction in the absence of catalyst.

E'_f → Energy of activation for forward reaction in presence of catalyst.

E'_b →Energy of activation for backward reaction in presence of catalyst.

The following points regarding a catalyst and the phenomenon of catalysis need to be emphasized.

(i) A catalyst (positive catalyst) increases the rate of a reaction but its composition remains unchanged at the end of the reaction.

(ii) The phenomenon of the increase in the rate of a reaction in presence of a catalyst is known as catalysis.

(iii) The addition of a catalyst lowers the activation energies of the forward and backward reactions to the same extent. Thus the rates of both forward and reverse reactions are accelerated to the same extent.

(iv) The catalyst does not affect (or alter) the position of the equilibrium in a reversible reaction. It merely speeds up the approach to equilibrium of both forward and backward reactions.

The phenomenon of negative catalysis (or a negative catalyst) is also important and needs mention. When a catalyst reduces the rate of a reaction, it is referred as a negative catalyst (or inhibitor) and this property is known as negative catalysis. Two examples will be cited as an illustration of this phenomenon.

(i) The decomposition of H_2O_2 into water and oxygen according to the reaction

$$2H_2O_2 \rightarrow 2H_2O + O_2 \qquad \text{— (83)}$$

is retarded in the presence of glycerol or dilute acids. So, these substances act as negative catalysts.

(ii) Tetraethyl lead, $Pb(C_2H_5)_4$, acts as an antiknock agent when it is added to petrol. It prevents or slows down the rapid combustion of petrol which is responsible for the "knocking" of the engine.

Relation between the rate constant and activation energy

It is assumed that an equilibrium exists between the reactants and the activated complex which subsequently decomposes to products at a definite rate i.e.,

$$\text{Reactants} \overset{K^{\#}}{\rightleftharpoons} \text{Activated complex} \xrightarrow{k} \text{Products} \qquad \text{— (84)}$$

The rate constant k of the reaction is related to the equilibrium constant $K^{\#}$ for the formation of the activated complex according to

$$k = \frac{RT}{Nh}K^{\#} \qquad \text{— (85)}$$

where, R is the gas constant, N is the Avogadro's number, h is the Planck's constant, $K^{\#}$ is the equilibrium constant for the formation of activated complex. But we can write

$$\Delta G^{\#} = \Delta H^{\#} - T\Delta S^{\#} = -RT\ln K^{\#} \qquad \text{— (86)}$$

where, $\Delta G^{\#}$ is the Gibbs free energy change for the formation of activated complex, $\Delta H^{\#}$ and $\Delta S^{\#}$ are the enthalpy and entropy of activation or

$$K^{\#} = e^{-\Delta H^{\#}/RT}\, e^{\Delta S^{\#}/R} \qquad \text{— (87)}$$

Substituting for $K^{\#}$ in (85), we get

$$k = \frac{RT}{Nh}e^{-\Delta H^{\#}/RT} \times e^{\Delta S^{\#}/R} \qquad \text{— (88)}$$

If we identify $\Delta H^{\#}$ with the energy of activation E, of the reaction equation (88) may be written as

$$k = \frac{RT}{Nh}e^{\Delta S^{\#}/R} \times e^{-E/RT} \qquad \text{—(89)}$$

$$k = A.e^{-E/RT} \qquad \text{— (90)}$$

where, A, the frequency factor of the Arrhenius equation is equal to $\frac{RT}{Nh}e^{\Delta S^{\#}/R}$ or in other words it is related to the entropy of activation of the reaction. Although the enthalpy of activation and energy of activation of a reaction are not equal, it may be stated that in the case of unimolecular reaction in gas phase and for solution reactions $\Delta H^{\#} \approx E$. The transition state theory is able to explain the low rates in the case of some complex reactions, as in such cases $\Delta S^{\#}$ is quite or even negative small. However, collision theory fails in such cases.

KINETICS OF CONSECUTIVE, PARALLEL AND OPPOSING REACTIONS

Hitherto, simple reactions or isolated reactions only have been considered. Reactions can proceed in a complex way and a few such cases are considered briefly.

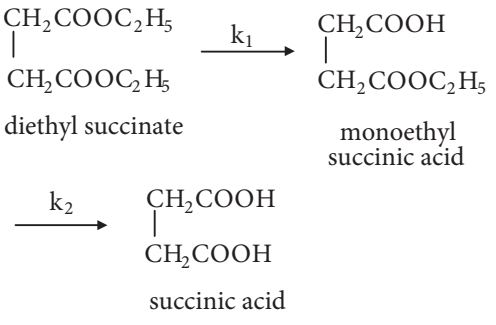
(i) Consecutive reactions

These reactions proceed from reactants to products through one or more intermediate steps. A reaction of the type

$$A \xrightarrow{k_1} B \xrightarrow{k_2} C \qquad \text{— (91)}$$

where, k_1 and k_2 are rate constants of the two stages is an example of a consecutive reaction. An example of such a reaction is the acid hydrolysis of an ester of a dibasic acid like diethyl succinate.

Its hydrolysis may be represented as



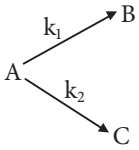
diethyl succinate

monoethyl succinic acid

succinic acid

(ii) Parallel reactions

In some cases, the reactant may be removed by two or more reactions occurring simultaneously according to



— (93)

where, A is the reactant. The rate of the reaction may be expressed as

$$\frac{-dC_A}{dt} = \frac{-d(a-x)}{dt}$$
$$= k_1(a-x) + k_2(a-x) = (k_1+k_2)(a-x) \quad \text{— (94)}$$

assuming that both parallel reactions are first order. On integration

$$-\int \frac{d(a-x)}{(a-x)} = (k_1+k_2)\int dt$$

$$-\ln (a-x) = (k_1+k_2)t + \text{constant} \quad \text{— (95)}$$

At t = 0, constant = $-\ln a$ ($\therefore$ x = 0)

On substitution in equation (95), we get $(k_1 + k_2) =$

$$\frac{1}{t}\ln\left(\frac{a}{a-x}\right) \quad \text{— (96)}$$

The reaction, as a whole, behaves as a first order reaction.

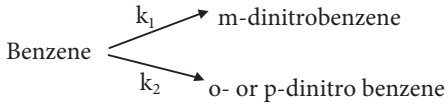
To determine k_1 and k_2 separately

Rate of formation of B = $k_1 (a - x)$

Rate of formation of C = $k_2 (a - x)$

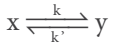
Thus at any time, $\frac{[B]}{[C]} = \frac{k_1}{k_2}$ — (97)

If the concentrations of B and C are determined at any stage of the reactions (or even at the end), the ratio $\frac{k_1}{k_2}$ will be known and combining with the earlier determined values of $(k_1 + k_2)$, individual k_1 and k_2 values may be calculated. An example of such a reaction is the nitration of benzene which gives m-dinitro and o– or p– dinitrobenzenes.



(iii) Opposing reactions

A reversible reaction in which a first order forward reaction is opposed by a first order backward reaction will be considered as an example



(109)

$$\text{Rate} = \frac{dx}{dt} = k(a-x) - k'x \quad \text{— (100)}$$

($\because$ conc. of Y formed at any instant = x)

where, x is the amount of X reacted at time t, a is the initial concentration of X. At equilibrium, the rates of the forward and reverse reactions are equal.

$$\therefore \quad k(a - x_e) = k'x_e \quad \text{— (101)}$$

where x_e is the conc of X reacted or Y formed at equilibrium

$$\text{or} \quad k' = \frac{k(a-x_e)}{x_e} = \left(\frac{ka}{x_e} - k\right) \quad \text{— (102)}$$

substituting for k′ from equation (102) into equation (100)

$$\frac{dx}{dt} = k(a-x) - x\left(\frac{ka}{x_e} - k\right) \quad \text{— (103)}$$

$$\frac{dx}{dt} = \frac{ka}{x_e}(x_e - x)$$

Rewriting equation (103) as

$$-\int \frac{dx}{(x_e - x)} = \frac{ka}{x_e}\int dt$$

In general integration and evaluation of the integration constant the equation

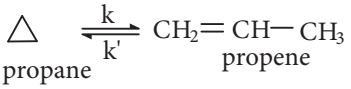
$$\frac{ka}{x_e} = \frac{1}{t}\ln\left(\frac{x_e}{x_e - x}\right) \quad \text{— (104)}$$

is obtained. By following the reaction at various intervals of time, x can be determined. Also, x_e, the amount of Y formed at equilibrium may be determined. Thus k can be calculated. Since

$$k + k' = \frac{ka}{x_e} \quad \text{— (105)}$$

k′ can also be determined.

An example of such a reaction is the isomerisation of cyclopropane to give propene.



Mechanism of reactions

The basic objective of a kinetic study is to understand the various stages by which the reaction proceeds or in other words, understand the mechanism of the reaction. Many reactions, which are simple from a chemical viewpoint, do not necessarily proceed in a simple manner kinetically. Such reactions are said to be complex in contrast to elementary

reactions, which take place in a single step, for example, the reactions between hydrogen and iodine to give HI.

Especially reactions in which the stoichiometric equation contains three or more molecules occur in many steps and all the steps together constitute the mechanism of the reaction. The slowest step among them is said to be the rate determining stage of the reactions. It may also be mentioned that even reactions which are extremely simple may be occurring by a complex mechanism (example, reaction between H_2 and Br_2 to give HBr).

To illustrate the mechanism, the reaction between hydrogen peroxide and hydrogen iodide will be considered. The stoichiometric equation for the reaction may be expressed as

$$H_2O_2 + 2H_3O^+ + 2I^- \rightarrow 4H_2O + I_2 \qquad \text{— (107)}$$

A kinetic study of the reaction shows that the overall order of the reaction is two, being first order with respect to concentration of H_2O_2 and first order with respect to concentration of iodide ion. This observation may be explained by considering the reaction to take place in stages which are given as

$$\left.\begin{aligned} &H_2O_2 + I^- \rightarrow H_2O + IO^- \text{ (slow)} \\ &IO^- + H_3O^+ \rightarrow HIO + H_2O \text{ (fast)} \\ &HIO + H_3O^+ + I^- \rightarrow 2H_2O + I_2 \text{ (fast)} \end{aligned}\right\} \qquad \text{— (108)}$$

The experimentally observed order of the reaction corresponds to the slow step of the reaction which governs the rate. It must be mentioned, however, that there is no unique mechanism for a reaction and the observed order may be explained by more than one mechanism. Thus further experiments have to be devised to choose the most probable one among the various possible mechanisms.

Deriving the rate law from the mechanism of a complex reaction

Case (i)

Suppose we have a complex reaction involving the conversion of reactants to products through a series of intermediates given by

$$\underset{\text{Reactants}}{X} \xrightarrow{k_1} X_1 \xrightarrow{k_2} X_2 \xrightarrow{k_3} X_3 \cdots\cdots \xrightarrow{k_{n-1}} X_n \xrightarrow{k_n} \underset{\text{Products}}{Y} \qquad \text{—(109)}$$

where, X_1, X_2, X_3 … X_n are intermediates and k_1, k_2, k_3, … k_n are rate constants of the various steps. If one of the steps is the slowest and hence rate determining, the derivation of rate law is simplified by assuming all steps preceeding rate determining step are fast equilibria. An example illustrating the above method is given below.

The experimentally observed rate of the reaction between NO and H_2 to give nitrogen and water according to the reaction.

$$2NO + 2H_2 \rightarrow N_2 + 2H_2O \qquad \text{— (110)}$$

is rate = $k[NO]^2[H_2]$. One of the mechanisms proposed for the reaction is

$$NO + H_2 \underset{k_2}{\overset{k_1}{\rightleftharpoons}} NOH_2 \quad \text{(fast equilibrium)}$$

$$NOH_2 + NO \xrightarrow{k_3} N_2 + H_2O_2 \text{ (slow)} \qquad \text{— (111)}$$

$$H_2O_2 + H_2 \xrightarrow{k_4} 2H_2O \text{ (fast)}$$

To derive the rate law, one can write

$$\text{Rate} = k_3 [NOH_2][NO] \qquad \text{— (112)}$$

The concentration of the intermediate NOH_2 is given by

$$\frac{[NOH_2]}{[NO][H_2]} = K = \frac{k_1}{k_2} \qquad \text{— (113)}$$

$$\text{or} \quad [NOH_2] = K[NO][H_2] \qquad \text{— (114)}$$

Equation (114) on substitution in equation (112) gives

$$\text{Rate} = k_3 K [NO]^2[H_2] = k[NO]^2[H_2] \qquad \text{— (115)}$$

The derived rate law (equation (115)) is in agreement with the experimental rate equation.

Case (2)

Sometimes the derivation of the rate law is simplified by applying what is known as "steady state approximation" for a transient species or short lived reactive intermediate. The principle of steady state approximation is based on the observation that when a short lived reaction intermediate exists at low concentration in a system, its rate of formation may be considered to be equal to its rate of disappearance. If "I" is such an intermediate, we can write,

$$\frac{d[I]}{dt} = 0 \qquad \text{— (116)}$$

The following example will illustrate the application of "steady state approximation". The gaseous decomposition of ozone according to

$$O_3(g) \underset{k'}{\overset{k}{\rightleftharpoons}} O_2(g) + O$$

$$O_3 + O \xrightarrow{k''} 2O_2 \qquad \text{— (117)}$$

obeys the rate law $\frac{-d[O_3]}{dt} = k_1 \frac{[O_3]^2}{[O_2]}$ — (118)

It is possible to obtain the above rate law by applying steady state approximation to the species "oxygen atom".

$$\text{Rate} = \frac{-d[O_3]}{dt} = k[O_3] - k'[O_2][O] + k''O_3][O] \quad \text{— (119)}$$

Applying steady state approximation to "oxygen atom" concentration

i.e., $\frac{d[O]}{dt} = 0,$

$$\frac{d[O]}{dt} = k[O_3] - k'[O_2][O] - k''[O_3][O] \quad \text{— (120)}$$

$$[O] = \frac{k[O_3]}{k'[O_2]+k''[O_3]} \quad \text{— (121)}$$

Substituting equation (121) in equation (119), we get

$$\frac{-d[O_3]}{dt} = k[O_3] + \frac{k[O_3]}{k'[O_2]+k''[O_3]}\{k''[O_3] - k'[O_2]\} \quad \text{— (122)}$$

$$\frac{-d[O_3]}{dt} = \frac{k[O_3]k'[O_2] + kk''[O_3]^2 - kk'[O_2][O_3]}{k'[O_2]+k''[O_3]} \quad \text{— (123)}$$

$$= \frac{2kk''[O_3]^2}{k'[O_2]+k''[O_3]} \quad \text{— (124)}$$

If we make the additional assumption that $k'[O_2] >> k''[O_3]$, equation (125) simplifies to

$$\frac{-d[O_3]}{dt} = \frac{2kk''[O_3]^2}{k'[O_2]} = \frac{k_1[O_3]^2}{[O_2]} \quad \text{— (125)}$$

where, $k_1 = \frac{2kk''}{k'}$

Thus rate laws can be derived for complex mechanisms by involving steady state principle.

CONCEPT STRANDS

Concept Strand 1

The hydrolysis of methyl bromide in aqueous solution can be followed by titrating the bromide ion formed at various intervals of time with standard silver nitrate solution. The following data were obtained at 25°C in an experiment.

Time (min)	0	75	100	250	400	∞
Bromide ion formed (mols/litre)	0	0.031	0.040	0.090	0.131	0.300

Calculate the rate constant in sec^{-1}. What are the values of $t_{½}$ and $t_{_}$ in seconds.

Solution

a, the initial conc of methyl bromide α 0.300
(a – x) ∝ (0.300 – x values at various times)
(a – x) represents the concentration of CH_3Br remaining unreacted.

1st data: $k = \frac{2.303}{75\times60}\log\frac{0.300}{0.269}$

$= 2.424 \times 10^{-5}\,sec^{-1}$

2nd data: $k = \frac{2.303}{100\times60}\log\frac{0.300}{0.260} = 2.385 \times 10^{-5}\,sec^{-1}$

3rd data: $k = \frac{2.303}{250\times60}\log\frac{0.300}{0.210} = 2.378 \times 10^{-5} sec^{-1}$

4th data: $k = \frac{2.303}{400\times60}\log\frac{0.300}{0.169} = 2.392 \times 10^{-5}\,sec^{-1}$

Ave. $k = 2.395 \times 10^{-5}\,sec^{-1}$

$$t_{½} = \frac{0.693}{2.395\times10^{-5}} = 2.894 \times 10^4 \text{ sec}$$

$t_{3/4}$ will be the twice the above value i.e., 5.788×10^4 sec

Concept Strand 2

The rate constant for the thermal decomposition of acetone is 0.516 min^{-1} at 873K. Starting with an initial pressure of 360 mm Hg of acetone, how many half lives does it take to bring down the pressure of acetone to 45 mm Hg. What is the partial pressure of acetone after 840 sec.

Solution

$$t_{½} = \frac{0.693\times60}{0.516} = 80.6\,\text{sec}$$

$$t = \frac{2.303}{0.516}\times\log\frac{360}{45} = \frac{2.303}{0.516}\times\log 8 = 4.03\,\text{min} = 241.8\ \text{sec}$$

$$\text{No. of half lives} = \frac{4.03\times60}{80.6} = 3$$

Partial pressure at the end of 840 sec.

$$\frac{0.516}{60} = \frac{2.303}{840}\log\left(\frac{360}{a-x}\right)$$

$$\log\left(\frac{360}{a-x}\right) = \frac{0.516\times840}{60\times2.303} = 3.1368$$

$$\left(\frac{360}{a-x}\right) = 1370.2 \text{ or } (a-x) = 0.263 \text{ mm Hg}$$

Concept Strand 3

The hydrolysis of methyl acetate in 1 N HCl at 25°C gave the following data on titration of 20 mL of reaction mixture with 0.1 N NaOH at various intervals of time.

Time (minutes)	5.7	20.70	45.80	79.80	∞
Total strength of acid in reaction mixture (mol/l)	0.132	0.139	0.148	0.159	0.199

Calculate the rate constant and $t_{3/4}$ of the reaction.

Solution

The initial titre value at zero time is not given. Therefore, the conc data at two time intervals are to be taken and k evaluated. $(C_\infty - C_t)$ at time t is proportional to the amount of ester remaining at time t.

Taking the first two data, we can write

$$k = \frac{2.303}{(t_2-t_1)}\log\left(\frac{C_\infty - C_{t_1}}{C_\infty - C_{t_2}}\right) = \frac{2.303}{(20.7-5.7)}$$

$$\log\left(\frac{0.199-0.132}{0.199-0.139}\right)$$

$$= \frac{2.303}{15.0}\log\frac{0.067}{0.060} = \frac{2.303\times0.0479}{15} = 7.35\times10^{-3}\ \text{min}^{-1}$$

Using the 3rd and 2nd sets of data,

$$K = \frac{2.303}{(45.8-20.70)}\log\left(\frac{0.199-0.139}{0.199-0.148}\right) = \frac{2.303}{25.1}\log\frac{0.060}{0.051}$$

$$= \frac{2.303}{25.1}\times0.0706 = 6.48\times10^{-3}\ \text{min}^{-1}$$

Using 4th and 3rd sets of data

$$k = \frac{2.303}{79.8-45.8}\log\left(\frac{0.199-0.148}{0.199-0.159}\right) = \frac{2.303}{34}\log\left(\frac{0.051}{0.04}\right)$$

$$= \frac{2.303\times0.1055}{34} = 7.15\times10^{-3}\ \text{min}^{-1}$$

Ave. $k = 6.99\times10^{-3}\ \text{min}^{-1}$

$t_{3/4}$ = time taken for 3/4 th of the reaction to be completed $= \frac{2.303}{k}\log 4$

$$= \frac{2.303}{6.99\times10^{-3}}\log4 = \frac{2.303\times0.6021}{6.99\times10^{-3}} = 198.4\,\text{min}$$

Concept Strand 4

The decomposition of benzene diazonium chloride in water at 323K according to the reaction.

$C_6H_5-{}^{\oplus}N\equiv NCl^{\ominus} + H_2O \rightarrow C_6H_5OH + HCl + N_2(g)$ was followed by measuring the volume of nitrogen in a gas burette at various intervals of time.

Time (min)	0	7.5	10.0	15.0	20	∞
Volume of N_2 evolved (mL)	0	20.7	28.0	33.8	42.1	56.2

Show that this reaction is 1st order and calculate the rate constant?

Solution

The initial concentration of diazonium chloride, a, is proportional to $(V_\infty - V_0) = (56.2 - 0)$ of nitrogen. At any time t, the concentration of diazonium chloride remaining, i.e., (a – x) is proportional to $(V_\infty - V_t)$.

$$k \text{ from second data} = \frac{2.303}{7.5}\log\frac{56.2}{(56.2-20.7)}$$

$$= \frac{2.303}{7.5}\log\frac{56.2}{35.5} = 0.06126\ \text{min}^{-1}$$

$$k \text{ from third data} = \frac{2.303}{10}\log\frac{56.2}{28.2} = 0.06897\ \text{min}^{-1}$$

k from fourth data = $\frac{2.303}{15}\log\frac{56.2}{22.4} = 0.06133\ \text{min}^{-1}$

k from fifth data = $\frac{2.303}{20}\log\frac{56.2}{14.1} = 0.06914\ \text{min}^{-1}$

Since k at various times is nearly constant, it may be concluded that it is a first order reaction.

Ave.k = $0.0652 = 6.52 \times 10^{-2}\ \text{min}^{-1}$

Concept Strand 5

For the reaction, $2NO_{(g)} + 2H_{2(g)} \rightarrow N_{2(g)} + 2H_2O_{(g)}$ at 1100K the following initial rates were measured once by keeping the pressure of hydrogen constant and next by keeping the pressure of "NO" constant. The data are given

$\left[P_{H_2}\right]_o$ = 400 mm Hg

$\left[P_{NO}\right]_o$ /mm Hg	290	152
$\left(\frac{-dp}{dt}\right)_o$ /mm Hg	1.03	0.25

$\left[P_{NO}\right]_o$ = 400 mm Hg

$\left[P_{H_2}\right]_o$ / mm Hg	205	147
$\left(\frac{-dp}{dt}\right)_o$ /mm Hg sec^{-1}	1.10	0.79

Determine the order with respect to "NO" and also H_2 and the total order of the reaction.

Solution

The order w.r.t "NO" is obtained from the relation

$$\frac{\log\left(\frac{-dp}{dt}\right)_1 - \log\left(\frac{-dp}{dt}\right)_2}{\log(P_1) - \log(P_2)} = n_{NO}$$

$$\frac{\log 1.03 - \log 0.25}{\log 290 - \log 152} = \frac{0.6149}{0.2806} = 2.19$$

The order with respect to "NO" is = 2.

Similarly, the order w.r.t H_2 is obtained from

$$\frac{\log 1.10 - \log 0.79}{\log 205 - \log 147} = \frac{0.1438}{0.1444} = 0.996$$

$\therefore$ The order w.r.t H_2 is = 1.0

Total order = 2 + 1 = 3

Concept Strand 6

The decomposition of ethylene oxide in vapour phase into methane and carbon monoxide was followed at 687K at constant volume by measuring the increase in pressure at various intervals of time.

Time (sec)	0	240	600	900
Pressure (atm)	0.153	0.161	0.171	0.181

Show that the reaction is first order and calculate the rate constant. Calculate the time required for the pressure to reach 0.200 atm. What is the decrease in the pressure of ethylene oxide at the above time?

Solution

Let the initial pressure of ethylene oxide be P_i atm. Let the decrease in its pressure be x atm at time t.

$\therefore$ Partial pressure of ethylene oxide at time t = $(P_i - x)$

Partial pressure of methane = partial pressure of CO = x

Total pressure $P = P_{\text{Ethylene oxide}} + P_{CH_4} + P_{CO}$

$= P_i - x + x + x = (P_i + x)$

or $x = (P - P_i)$

and $(a - x)\ \alpha\ P_i - P + P_i = (2P_i - P)$; $a\ \alpha\ P_i$

using the first order equation $k = \frac{2.303}{t}\log\frac{P_i}{(2P_i - P)}$

At t = 240 sec, k =

$$\frac{2.303}{240}\log\left(\frac{0.153}{0.306-0.161}\right) = 2.23 \times 10^{-4}\ \text{sec}^{-1}$$

At t = 600 sec, k =

$$\frac{2.303}{600}\log\left(\frac{0.153}{0.306-0.171}\right) = 2.08 \times 10^{-4}\ \text{sec}^{-1}$$

At t = 900 sec, k =

$$\frac{2.303}{900}\log\left(\frac{0.153}{0.306-0.181}\right) = 2.25 \times 10^{-4}\ \text{sec}^{-1}$$

Average $k = 2.19 \times 10^{-4}\ \text{sec}^{-1}$

When P = 0.200 atm, t = ?

$$2.19 \times 10^{-4} = \frac{2.303}{t}\log\left(\frac{0.153}{0.306 - 0.200}\right) = \frac{2.303}{t}\log\frac{0.153}{0.106}$$

or $t = \frac{2.303}{2.19 \times 10^{-4}} \times 0.1594 = 1676.1 \text{ sec}$

$x = P - P_i = 0.200 - 0.153 = 0.047 \text{ atm}$

Concept Strand 7

The following data are obtained in the decomposition reaction of a compound at 323K.

Initial conc (moles/l)	0.10	0.20	0.30
$t_{½}$ (min)	30	7.5	3.3

What is the order of the reaction and what is the rate constant?

Solution

It is not a zero order reaction because, in that case, $t_{½} \alpha a$ which is not observed.

It is not a first order reaction since in such case $t_{½}$ is independent of initial concentration which is not observed. It is also not a second order reaction, since in that case, $t_{½} \times a$ = constant which is not observed.

For third order reaction, $t_{½} \alpha \frac{1}{a^2}$

$t_{½} \times a^2$ for all the three data given is found to be constant and equal to 0.3 … The reaction is third order.

For third order reaction

$$k = \frac{1}{2t}\left[\frac{1}{(a-x)^2} - \frac{1}{a^2}\right] \text{ or } t_{½} = \frac{1}{2k}\frac{4-1}{a^2} = \frac{1}{2k}\frac{3}{a^2}$$

or $$k = \frac{1 \times 3}{30 \times 2(0.1)^2} = \frac{3}{60 \times 0.01} = 5 \text{ litre}^2 \text{ mol}^{-2} \text{ sec}^{-1}$$

Concept Strand 8

The acid catalysed hydrolysis of sucrose proceeds according to the rate law

$$\frac{-d[\text{sucrose}]}{dt} = k'[\text{Sucrose}]$$

k′ (expressed in terms of $t_{½}$), however, is found to vary with H_3O^+ concentration and may be expressed as

$$k' = k[H_3O^+]^x$$

The $t_{½}$ of the reaction at a pH = 1.0 is found to be 96 sec and at pH = 2.0 it increases to 960 sec. What is the order with respect to $[H_3O^+]$. What is the total order of the reaction.

Solution

$$k' \text{ (at pH = 1.0)} = \frac{0.693}{96} = 7.219 \times 10^{-3} \text{ min}^{-1}$$

$$k'' \text{(at pH = 2.0)} = \frac{0.693}{960} = 7.219 \times 10^{-4} \text{ min}^{-1}$$

$$\frac{k'(\text{pH}=1.0)}{k''(\text{pH}=2.0)} = \frac{[H_3O^+]_1^x}{[H_3O^+]_2^x} \text{ i.e., } \frac{7.219 \times 10^{-3}}{7.219 \times 10^{-4}} = \left(\frac{0.1}{0.01}\right)^x$$

Taking logarithm, log 10 = x log10 or x = 1

The order in H_3O^+ species is 1 and we know that it is also of first order w.r.t sucrose 1.

The total order is 2.

Concept Strand 9

For the reaction, xA + yB + zC → products, the rate law may be expressed as

$$\text{rate} = k = [A]^x [B]^y [C]^z.$$

The following table gives the initial rates of the reaction under varying concentration conditions of A, B and C. Determine the orders x, y, z with respect to the reactants A, B and C. Also calculate the rate constant of the reaction.

Expt. No.	[A] (moles litre^{-1})	[B] (moles litre^{-1})	[C] (moles litre^{-1})	initial rate (moles litre^{-1} sec^{-1})
1.	0.002	0.005	0.001	8×10^{-4}
2.	0.004	0.005	0.001	1.6×10^{-3}
3.	0.002	0.010	0.001	4.0×10^{-4}
4.	0.002	0.005	0.002	8.0×10^{-4}

Solution

$$\text{Rate} = k[A]^x [B]^y [C]^z$$

$$\frac{(\text{Rate})_1}{(\text{Rate})_2} = \frac{k[0.002]^x[0.005]^y[0.001]^z}{k[0.004]^x[0.005]^y[0.001]^z} = \frac{[0.002]^x}{[0.004]^x}$$

$$\text{i.e., } \frac{8\times10^{-4}}{1.6\times10^{-3}} = 0.5 = \left(\frac{1}{2}\right)^x; \log 0.5 = x \log 0.5;$$

$\therefore$ x=1

$$\frac{(\text{Rate})_1}{(\text{Rate})_3} = \frac{k[0.002]^1[0.005]^y[0.001]^z}{k[0.002]^1[0.01]^y[0.001]^z} = \frac{[0.005]^y}{[0.01]^y}$$

$$\frac{8\times10^{-4}}{4\times10^{-4}} = 2 = \left(\frac{1}{2}\right)^y; \log 2 = -y \log 2; \quad y = -1$$

$$\frac{(Rate)_1}{(Rate)_4} = \frac{k[0.002]^1[0.005]^{-1}[0.001]^z}{k[0.002]^1[0.005]^{-1}[0.002]^z} = \frac{[0.005]^z}{[0.002]^z}$$

$$\frac{8\times10^{-4}}{8.0\times10^{-4}} = \left(\frac{1}{2}\right)^z ; \log 1 = z\log 1 - z\log 2 = 0 - z\log 2;$$

$$-z \log 2 = 0; z = -\frac{0}{\log 2} = 0$$

Total order = 1 – 1 + 0 = 0

Rate constant

Data 1: $8 \times 10^{-4} = k[0.002]^1[0.005]^{-1}[0.001]^0$; $= k[0.002]/[0.005]$

$$k = \frac{8\times10^{-4}\times0.005}{0.002} = 20\times10^{-4} = 2\times10^{-3} \text{ mole litre}^{-1}\text{ sec}^{-1}$$

Data 2: $2 \times 8 \times 10^{-4} = k[0.004]^1[0.005]^{-1}[0.001]^0$ or

$$k = \frac{8\times10^{-4}\times2\times0.005}{0.004} = 20\times10^{-4} = 2\times10^{-3} \text{ mole litre}^{-1}\text{ sec}^{-1}$$

Data 3: $4 \times 10^{-4} = k[0.002]^1[0.01]^{-1}[0.001]^0$ or

$$k = \frac{4\times10^{-4}\times0.01}{0.002} = 20\times10^{-4} = 2\times10^{-3} \text{ mole litre}^{-1}\text{ sec}^{-1}$$

Data 4: $8 \times 10^{-4} = k[0.002]^1[0.005]^{-1}[0.002]^0$ or

$$k = \frac{8\times10^{-4}\times0.005}{0.002} = 20\times10^{-4} = 2\times10^{-3} \text{ mole litre}^{-1}\text{ sec}^{-1}$$

$\therefore$ Ave.k = 2×10^{-3} mole litre^{-1} sec^{-1}

Concept Strand 10

A second order reaction given by, A + B $\rightarrow$ products is 60% complete in 150 minutes, when the initial concentration of A and B are each equal to 0.025 M. Calculate

(i) the rate constant of the reaction
(ii) half life of the reaction
(iii) the time required for the reaction to be 60% complete when the initial concentration of A and B are each equal to 0.1 M. What is $t_{½}$ under these conditions?

Solution

$$a = 0.025 \text{ M}, x = \frac{0.025\times60}{100} = 0.0150\text{M}, \; a - x = 0.01 \text{ M}$$

(i) $$k = \frac{1}{150}\times\frac{0.015}{0.025\times0.01} = 0.4 \text{ litres mol}^{-1}\text{ min}^{-1}$$

(ii) $$t_{½} = \frac{1}{ak} = \frac{1}{0.025\times0.4} = 100\text{min}$$

(iii) when a = 0.1 M, $x = \frac{0.1\times60}{100} = 0.06$ M , a – x = 0.1 – 0.06 = 0.04 M

$$t = \frac{1}{k}\frac{x}{a(a-x)} = \frac{1}{0.4}\times\frac{0.06}{0.1\times0.04} = 37.5 \text{ min}$$

$$\frac{\left(t_{½}\right)_1}{\left(t_{½}\right)_2} = \frac{(a)_2}{(a)_1} \Rightarrow \frac{100}{\left(t_{½}\right)_2} = \frac{0.1}{0.025} = 4;$$

$$\left(t_{½}\right)_2 = \frac{100}{4} = 25 \text{ min}$$

Concept Strand 11

A reaction is 50 per cent complete in 40 minutes and 75 per cent complete in 60 minutes. What is the order of the reaction and what is the rate constant? How long will it take for the reaction to be 90 per cent complete?

Solution

The reaction is not first order because $t_{½}$ is not independent of initial concentration as seen from the given data.

When the equation for a second order reaction is applied to the given data, k is not constant.

Considering it to be zero order, and taking C_o = one mole/litre

$$k_1 = \frac{C_o - C_1}{t_1} = \frac{1-0.5}{40} = 0.0125 \text{ moles litre}^{-1}\text{ min}^{-1}$$

$$k_2 = \frac{C_o - C_2}{t_2} = \frac{1-0.25}{60}$$

$$= \frac{0.75}{60} = 0.0125 \text{ moles litre}^{-1}\text{ min}^{-1}$$

(note that C_1 and C_2 are the concentration remaining at times t_1 and t_2)

$\therefore$ Reaction is zero order. k = 0.0125 moles litre^{-1} min^{-1}

Concept Strand 12

The rate constant k of a second order reaction is 6.6 litres mole^{-1} min^{-1}. Express k in the units of

(i) cm^3 mole^{-1}min^{-1}.
(ii) cm^3 molecule^{-1} min^{-1}
(iii) cm^3 mole^{-1} sec^{-1}
(iv) cm^3 molecule^{-1}sec^{-1}

Solution

(i) $6.6 \text{ litres mole}^{-1} \text{ min}^{-1} \times \dfrac{1000 \text{ cm}^3}{\text{litre}} = 6600 \text{ cm}^3 \text{ mole}^{-1} \text{ min}^{-1}$

(ii) $6.6 \dfrac{1000 \text{cm}^3}{\text{litre}} \times \dfrac{1}{6.023 \times 10^{23}}$ moles molecule^{-1} litres mole^{-1} min^{-1}

$= \dfrac{6600}{6.023 \times 10^{23}} = 1.096 \times 10^{-20}$ cm^3 molecule^{-1} min^{-1}

(iii) $6.6 \times \dfrac{1000 \text{cm}^3}{\text{litre}} \times \text{litre mole}^{-1} \text{ min}^{-1} \times \dfrac{1}{60} \text{min sec}^{-1} =$ 110 cm^3 mole^{-1} sec^{-1}

(iv) $6.6 \times \dfrac{1000 \text{cm}^3}{\text{litre}} \times \dfrac{1}{6.023 \times 10^{23}}$ moles molecule^{-1} $\times \dfrac{1}{60}$ min sec^{-1} litres moles^{-1} min^{-1}

$= 1.826 \times 10^{-22}$ cm^3 molecule^{-1} sec^{-1}

Concept Strand 13

The reaction, $RCH_2NO_2 + OH^- \rightarrow RCHNO_2^- + H_2O$ is overall second order being first order with respect to each of the reactants. The rate constant at 273K is 6.5×10^{-1} litres mole^{-1} sec^{-1}. An aqueous solution is made up of 5×10^{-3} moles of nitroalkane and 6×10^{-3} moles of NaOH per litre. Calculate the time required for 75 percent of nitroalkane to react?

Solution

Let concentration of NaOH = a; Let conc of nitro alkane = b

$$k = \frac{2.303}{t(a-b)} \log \frac{b(a-x)}{a(b-x)}$$

$$x = \frac{5 \times 10^{-3} \times 75}{100} = 3.75 \times 10^{-3}$$

$$\Rightarrow \quad t = \frac{2.303}{0.65 \times 1 \times 10^{-3}} \log \frac{5 \times 10^{-3} \times 2.25 \times 10^{-3}}{6 \times 10^{-3} \times 1.25 \times 10^{-3}}$$

$$= \frac{2.303}{0.00065} \log 1.5 = 623.9 \text{ sec}$$

Concept Strand 14

A compound decomposes to give two parallel products A and B, both processes being first order reaction. If the rates of decomposition have frequency factors of 5×10^{10} sec^{-1} and 5×10^{11} sec^{-1} and activation energies of 13.0 kcal and 15.0 kcal for the process giving products A and B respectively, calculate the temperature at which both the products are formed at equal rates.

Solution

For the reaction giving product A, $k_1 = 5 \times 10^{10} \, e^{-13,000/1.987T}$
For the reaction giving product B, $k_2 = 5 \times 10^{11} \, e^{-15,000/1.987T}$

$$\frac{k_2}{k_1} = \frac{5 \times 10^{11} \, e^{-15,000/1.987T}}{5 \times 10^{10} \, e^{-13,000/1.987T}} = 10 \times \frac{e^{-15,000/1.987T}}{e^{-13,000/1.987T}}$$

But $\dfrac{k_2}{k_1} = 1$

$\therefore \quad 0.1 = \times e^{-15,000/1.987T} \times e^{+13,000 \,/\, 1.987T}$

$$\ln 0.1 = \ln e^{-15,000/1.987T} + \ln e^{+13,000/1.987T}$$

$$= \frac{-15,000}{1.987T} + \frac{13,000}{1.987T} = -\frac{2000}{1.987T}$$

$$-2.303 = -\frac{2000}{1.987T}$$

or $\quad T = \dfrac{2000}{2.303 \times 1.987} = 437.1\text{K} = 164.1°\text{C}$

Concept Strand 15

The frequency factor and the activation energy for the decomposition of ethyl chloride are 2.8×10^{13} sec^{-1} and 230.0 kJ/mole respectively. Calculate the temperature at which the half life of the reaction is 150 minutes.

Solution

$$\log \; k = \log A - \frac{E}{2.303 \, RT} = 13.45 - \frac{230 \times 1000}{8.314 \times T \times 2.303}$$

$$k = \frac{0.693}{150 \times 60} = 7.7 \times 10^{-5} \text{ sec}^{-1}$$

$$\Rightarrow \quad -4.113 = 13.45 - \frac{12012.2}{T}$$

$$\frac{12012.2}{T} = 13.45 + 4.113 = 17.563$$

$$T = 683.9\text{K} = 410.9°\text{C}$$

Concept Strand 16

The rate constants for the decomposition of N_2O_5 at two temperatures are given below:

T/deg K	308	318
k/min^{-1}	8.1×10^{-3}	3.0×10^{-2}

Calculate the energy of activation, the pre-exponential factor and the entropy of activation of the reaction in this temperature range.

Solution

$$\log \frac{k_2}{k_1} = \frac{E}{2.303R}\left[\frac{T_2 - T_1}{T_2.T_1}\right]$$

$$\log \frac{3.0\times10^{-2}}{8.1\times10^{-3}} = \frac{E}{2.303\times8.314}\left[\frac{318-308}{318\times308}\right]$$

$$0.569 = \frac{E}{19.15}\times\frac{10}{318\times308};$$

$$E = \frac{0.569\times19.15\times318\times308}{10\times1000} = 106.7 \text{ kJ / mole}$$

$$\log k = \log A - \frac{E}{2.303RT}$$

$$\log (3.0\times10^{-2}) = \log A - \frac{106.7}{2.303\times8.314\times10^{-3}\times318}$$

$$-1.523 = \log A - 17.524;$$

$$\log A = 16.00 \text{ or } A = 10^{16} \text{ min}^{-1}$$

A = Frequency factor = 10^{16} min^{-1}

Entropy of activation

$$A = \frac{RT}{Nh}e^{\Delta S^{\#}/R}; \log A = \log\frac{RT}{Nh} + \frac{\Delta S^{\#}}{2.303R}$$

$$\log 1.67\times10^{14} (\text{sec}^{-1}) = \log\left(\frac{8.314\times313}{6.023\times10^{23}\times6.626\times10^{-34}}\right) + \frac{\Delta S^{\#}}{2.303\times8.314}$$

$$14.22 = \log\left(\frac{2602.3}{39.91\times10^{-11}}\right) + \frac{\Delta S^{\#}}{19.15} = 12.81 + \frac{\Delta S^{\#}}{19.15}$$

$$\frac{\Delta S^{\#}}{19.15} = 1.41; \ \Delta S^{\#} = 19.15\times1.41 = 27.0 \text{ J K}^{-1}$$

Concept Strand 17

In the kinetic study of a certain reaction, it was observed that 10 percent of the reactant disappeared in 15 minutes at 37°C where as at 47°C the same amount of the reactant reacted in 5 minutes. Calculate the activation energy of the reaction.

Solution

rate α $\frac{1}{\text{time}}$ ($\because$ rate decreases with time)

Rate α k (ie the rate constant)

$$\frac{r_1}{r_2} = \frac{k_1}{k_2} = \frac{t_2}{t_1}$$

$$\Rightarrow \quad \frac{k_2}{k_1} = \frac{t_1}{t_2} = \frac{15}{5} = 3$$

$$\log 3 = \frac{E}{2.303\times8.314}\left[\frac{10}{310\times320}\right]$$

$$\Rightarrow \quad E = \frac{0.477\times2.303\times8.314\times310\times320}{10}$$

$$= 90.6 \text{ kJ mol}^{-1} = 21.6 \text{ kcal mol}^{-1}$$

Concept Strand 18

For the iodination of acetone, the following mechanism has been suggested.

$$CH_3COCH_3 + OH^- \underset{k_2}{\overset{k_1}{\rightleftharpoons}} CH_3COCH_2^- + H_2O$$

$$CH_3COCH_2^- + I_2 \xrightarrow{k_3} CH_3COCH_2I + I^-$$

Applying steady state approximation for the species $CH_3COCH_2^-$ and assuming that $k_2 >> k_3 [I_2]$, derive the rate law for the reaction.

Solution

Rate = $k_3 [CH_3CO\,CH_2^{\ominus}][I_2]$ (1) ($\because$ Product is formed in 2nd step)

$$\frac{d[CH_3COCH_2^-]}{dt}$$

= 0 according to steady state assumption

$$\frac{d[CH_3COCH_2^-]}{dt}$$

$= k_1[CH_3COCH_3][OH^-] - k_2[CH_3COCH_2^-] - k_3[CH_3COCH_2^-][I_2]$

Using the previous step,

$$[CH_3COCH_2^-] = \frac{k_1[CH_3COCH_3][OH^-]}{k_2 + k_3[I_2]} \quad (2)$$

Substituting for $[CH_3COCH_2]$ in equation (1) we get

$$\text{Rate} = k_3[I_2]\ \frac{k_1[CH_3COCH_3][OH^-]}{k_2 + k_3[I_2]} \qquad (3)$$

When $k_2 >> k_3\,[I_2]$, equation (3) simplifies to

$$\text{Rate} = k_1k_3\ \frac{[CH_3COCH_3][I_2][OH^-]}{k_2}$$

$$= k'\,[CH_3COCH_3][I_2][OH^-]$$

where $k' = \dfrac{k_1k_3}{k_2}$

Concept Strand 19

The entropies of activation of two first order reactions differ by 12 entropy units. What will be the ratio of the rate constants at 320K if the two reactions have the same activation energy.

(1 eu = 1 cal deg K^{-1} mol^{-1})

Solution

Let the rate constants of two reactions be k_1 and k_2

For reaction 1, $k_1 = \dfrac{RT}{Nh}\,e^{-E/RT}\,e^{(\Delta S^{\#}),/R}$

For reaction 2, $k_2 = \dfrac{RT}{Nh}\,e^{-E/RT}\,e^{(\Delta S^{\#})_2/R}$

Or $\dfrac{k_2}{k_1} = \dfrac{e^{(\Delta S^{\#})_2/R}}{e^{(\Delta S^{\#})_1/R}}$; $\ln\dfrac{k_2}{k_1} = \ln e^{(\Delta S^{\#})_2/R} - \ln e^{(\Delta S^{\#})/R}$

(Note that T is the same for the two reactions)

i.e., $\ln\dfrac{k_2}{k_1} = \dfrac{(\Delta S^{\#})_2}{R} - \dfrac{(\Delta S^{\#})_1}{R}$

let $(\Delta S^{\#})_2 - (\Delta S^{\#})_1 = 12$ e.u

Thus $\log\dfrac{k_2}{k_1} = \dfrac{1}{2.303R}\left[\left(\Delta S^{\#}\right)_2 - \left(\Delta S^{\#}\right)_1\right]$

$$= \frac{12}{2.303 \times 1.987} = 2.6223$$

$$\Rightarrow \quad \frac{k_2}{k_1} = 419.1$$

Concept Strand 20

Calculate the ratio of the rate constants of a reaction occurring at 320 K in the absence and in the presence of a catalyst, given that the activation energies of the uncatalysed and catalysed reactions are 70 kJ and 50 kJ respectively.

Solution

$$\frac{k_{cat}}{k_{uncat}} = \frac{Ae^{-Ecat/RT}}{Ae^{-Euncat/RT}} = \frac{e^{-50000/R\times320}}{e^{-70000/R\times320}}$$

$$\ln\ \frac{k_{cat}}{k_{uncat}} = \frac{-50,000}{8.314\times320} + \frac{70,000}{8.314\times320} = \frac{20,000}{8.314\times320}$$

$$\log\frac{k_{cat}}{k_{uncat}} = \frac{20,000}{2.303\times8.314\times320}$$

$$\frac{k_{cat}}{k_{uncat}} = 1837.4$$

Concept Strand 21

Derive the rate law for the mechanism of the reaction given below:

$$A + B \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} C \text{ (fast equilibrium)}$$

$$C \xrightarrow{k} D \text{ (slow)}$$

$$D \rightarrow \text{Products (fast)}$$

Solution

$$\text{Rate} = \frac{-dc_c}{dt} = k\,[C]$$

But $\dfrac{[C]}{[A][B]} = K$ (K = Equilibrium constant)

$$\therefore \quad \text{Rate} = \frac{-dc_c}{dt} = Kk[A][B] = k'[A][B]$$

The reaction behaves like a second order reaction.

Concept Strand 22

The first order rate constant for the decomposition of azomethane is 1.50×10^{-2} min^{-1} at 27°C. Calculate the partial pressures of azomethane, nitrogen and ethane when azomethane at an initial pressure of 300 mm Hg decomposes for sixty minutes. The reaction for the decomposition is given by $CH_3 - N = N - CH_3 \rightarrow C_2H_{6(g)} + N_{2(g)}$.

Solution

Let initial pressure of azomethane be P_i and let x be the decrease in its pressure at time t.

$\therefore$ Partial pressures of azomethane at time t = $(P_i - x)$;
Partial pressure of C_2H_6 = Partial pressure of N_2 = x.

Total pressure,

$P = P_i - x + x + x = (P_i + x)$

$\therefore \quad x = (P - P_i);$

$(a - x) = P_i - P + P_i = (2P_i - P)$ $(\because\ a \propto P_i)$

i.e., $1.5\times10^{-2} = \frac{2.303}{60} \log \frac{300}{600 - P};$

$$\log\left(\frac{300}{600-P}\right) = \frac{1.5\times10^{-2}\times 60}{2.303} = 0.391$$

$$\left(\frac{300}{600-P}\right) = 2.459;\ 1475.4 - 2.459P = 300;$$

$P = 478$ mm

$x = 478 - 300 = 178$ mm

$\therefore$ Partial pressure of azomethane $= 600 - 478 = 122$ mm

$P_{C_2H_6} = P_{N_2} = 178$ mm

SOLID STATE

Introduction

Solid materials are characterized by some properties which may be stated as (i) they are rigid and have a definite shape (ii) they occupy a certain volume which is independent of the shape and size of the container (iii) they are incompressible (iv) diffusion in solids is quite slow compared to that in liquids and gases. The particles in a solid viz., molecules, atoms or ions are held together by strong cohesive forces and thus they cannot move in a random manner. On the basis of these properties, it may be inferred that the particles of solids occupy fixed positions in their spatial arrangements.

CLASSIFICATION OF SOLIDS

Solids are generally classified as either crystalline or amorphous. *In a crystalline solid, the constituent structural entities are arranged in a regular geometrical configuration characteristic of the substance. This definite and ordered arrangement extends over a large distance in crystalline solids and is known as long range order.* Crystals of substances like sodium chloride, copper sulphate etc have a sharp melting point and a finite heat of fusion. The atoms in a crystal are fixed within a definite set of surfaces known as planes.Crystals may be classified into four types and Table 8.4 summarizes their characteristics and some examples are provided for each class.

Table 8.4: Various types of crystals

Type of crystal	Nature of particles	Forces operating in crystals	Properties	Examples
Molecular crystals	Small molecules	van der Waals type	Volatile, soft, low melting point, poor conductivity (thermal/electrical), low heat of fusion	Ice, solid CO_2, sulphur, iodine.
Ionic crystals	Positive and negative ions arranged regularly in lattice	Electrostatic plus short range repulsion	High melting point, large heats of fusion, poor conductivity (thermal and electrical)	NaCl, CsF
Covalent crystals	Network of chemically bound atoms	Covalent	Hard, high melting point, poor conductivity of heat and electricity, high heats of fusion	Silicon, diamond
Metallic crystals	+ve ions amidst electrons	Electrical attraction	Soft to hard, low to high melting points, good thermal and electrical conductivity, malleable and ductile	Metals like Cu, Zn, and also alloys

Substances like glass, rubber and high polymers are amorphous solids. Although they possess some of the properties of crystalline solids such as rigidity and hardness, they do not possess regular internal structure. They do not have a sharp melting point but melt over a range of temperatures. However, amorphous solids exhibit the same value of a property. (For example, thermal conductivity, electrical conductivity, refractive index) in all directions and are said to be isotropic. On the other hand, crystalline solids are anisotropic i.e., in these cases the magnitude of the property depends upon the direction along which it is measured.

Crystal lattice and unit cell

A crystal lattice may be defined as an ordered three dimensional structure formed by the constituent particles like ions, atoms or molecules, whose locations are shown by points in the following Fig. 8.11. This arrangement is also called a space lattice. It is possible to consider a one dimensional, two dimensional or three dimensional lattice. *In a one dimensional lattice, a set of points are repeated at regular intervals along a line. A two dimensional lattice results when a set of points are repeated in a regular manner on a plane along a two coordinate system. The two dimensional lattice can be extended in three dimensions by arranging the points in a regular repetitive manner along three co-ordinate axes.* The crystals lattices are summarized in Fig. 8.11.

For every crystal lattice, it is possible to select a group of lattice points and connect them by sets of parallel lines along the three co-ordinate axes. The space lattice will be then divided into a large number of small symmetric units, which are basic to the whole lattice. They are known as unit cells. The unit cell is characterized by the distances a, b, c along the three edges and the angles α, β and γ between the pairs of edges bc, ca and ab respectively. These are known as unit cell parameters.

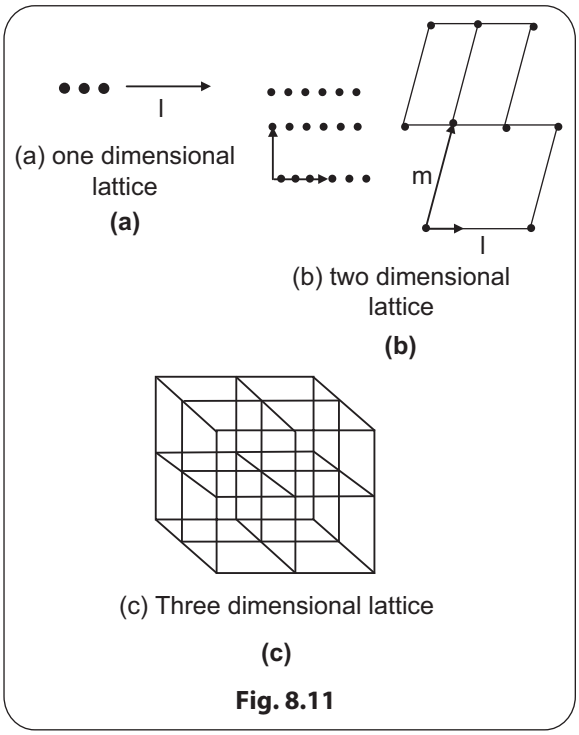


Fig. 8.11

TYPES OF UNIT CELLS

Seven types of basic or primitive unit cells have been recognized from these parameters and in these cases the unit cell has points only at the corners.

However, there are other possible arrangements of points which fulfill the requirements of a space lattice i.e., each point in space has identical surroundings. Such arrangements are

(i) when there are lattice points at the corners and at the centre of a unit cell in which case it is called a body centred unit cell.

(ii) when there are lattice points at the corners and at the centres of all faces, this case being described as a face centred unit cell. A cubic lattice has three types of unit cells, known as primitive, body centered (bcc) and face centered (fcc) unit cells.

The seven types of primitive cells are shown in Fig. 8.12.

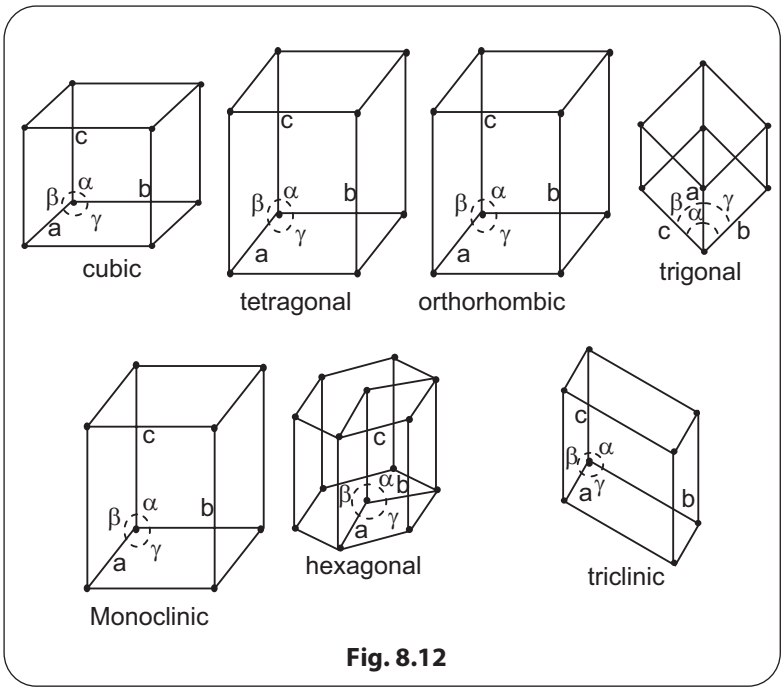


Fig. 8.12

Seven types primitive cells described under Table 8.5

Table 8.5: Axial distances, angles in various crystal systems

Name of the system	Number and names of Bravais lattices	Axial distances	Axial angles	Examples
Cubic	3 (primitive, face centred, body centred)	a=b=c	$\alpha=\beta=\gamma=90^{o}$	NaCl, CaF_2, zinc blende
Tetragonal	2 (Primitive, body centred)	a=b≠c	$\alpha=\beta=\gamma=90^{0}$	TiO_2, SnO_2, white tin
Orthorhombic	4 (Primitive, face centred, body centred, end-centred)	a≠b≠c	$\alpha=\beta=\gamma=90^{0}$	Rhombic S, KNO_3, $BaSO_4$, TiO_2
Rhombohedral (or Trigonal)	1 (Primitive)	a=b=c	$\alpha=\beta=\gamma\neq90^{0}$	Calcite, ice
Monoclinic	2 (primitive, end-centred)	a≠b≠c	$\alpha=\gamma=90^{0}$ $\beta\neq90^{0}$	Monoclinic S, $CaSO_4.2H_2O$, $Na_2SO_4.10H_2O$
Hexagonal	1 (Primitive)	a=b≠c	$\alpha=\beta=90^{0}$ $\gamma=120^{0}$	Graphite, Mg, PbI_2
Triclinic	1 (primitive)	a≠b≠c	$\alpha\neq\beta\neq\gamma\neq90^{0}$	$K_2Cr_2O_7$, $CuSO_4.5H_2O$

Apart from these seven types, there are seven other types of unit cells i.e., a total of fourteen different unit cells also referred as *Bravais lattices.* Table 8.5 summarises the various types of unit cells along with the typical characteristics of the unit cell.

CLOSEST PACKING OF CONSTITUENT PARTICLES (SPHERES) IN CRYSTALS

When crystals are formed the constituent particles (assumed to be spherical, for simplicity) get closely packed and the available space is used in a most economical manner.

A close packed three dimensional structure may be obtained by stacking such layers one on top of the other. However, there are different ways of doing this stacking and thereby different types of close packed structures result.

There are two ways of achieving efficient packing in the closest possible way which are known as (i) hexagonal close packing (hcp) and cubic close packing (ccp).

Let us consider the hexagonal close packing. At first a number of spheres (equal sized) are arranged on a flat surface such that they touch each other and are as close as possible as shown below:

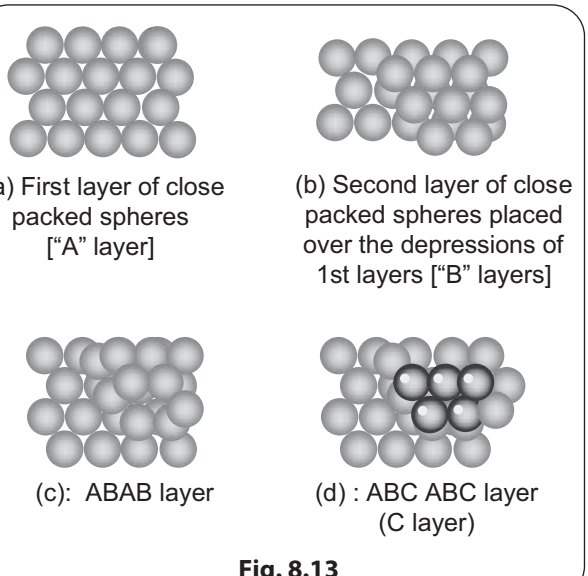


Fig. 8.13

Let us call this as "A" layer. Now a second layer of spheres is arranged over the first layer in such a way that they lie in the depressions of the first layer. Let this be "B" layer. A third layer may be added over the second layer (B layer) in either of two ways.

In the first case, the spheres are placed in such a way that they are exactly over the spheres of A layer Fig. 8.13 (c) or in other words, they reproduce the first layer to give a pattern called the ABA type layers. If the ABAB pattern is repeated to give a series of layers ABABAB, the resulting arrangement is called hexagonal close packed.

In the second case, the spheres are placed over the gaps in the first layer. In the case, neither the A layer spheres nor B layer spheres are eclipsed and therefore, the layer is a typically different one and may be called C layer [Fig. 8.13(d)]. Thus this pattern may be referred as ABCABC type and the spheres are cubic close packed (ccp) The ccp structure results in a face centred cubic unit cell (or fcc).

The compactness of close packed structures, discussed above, may be assessed in two ways (i) by their co-ordination number and (ii) by calculating their packing fraction, i.e., the fraction of the space occupied by the spheres.

The coordination number i.e., the number of nearest neighbours in both hcp and fcc arrangements is twelve viz., six in the same layer, three in the layer above and three in the layer below.

Before the calculation of packing fraction in various close packed structures, it is necessary to find the number of lattice points (or atoms) per unit cell in these structures.

NUMBER OF ATOMS PER UNIT CELL IN A CUBIC LATTICE

Simple cubic lattice (or primitive cubic unit cell)

In a primitive or simple cubic lattice the atoms are present at the corners only, each atom is shared equally by eight unit cells. The contribution of each atom to the unit cell is, therefore, $\frac{1}{8}$ and the total number of atoms per unit cell is thus $8 \times \frac{1}{8} = 1$.

The packing fraction in this case may be calculated in the following manner. Along the edge of the unit cell, two spheres (or atoms) touch each other. Let "r" and "l" represent the radius of the sphere (or atom) and the length of the unit cell edge. Thus 2r = l

Fraction of the total volume occupied = Packing fraction (P.F.)

$$\text{P.F.} = \frac{\text{Volume of the sphere}}{\text{Unit cell volume}}$$

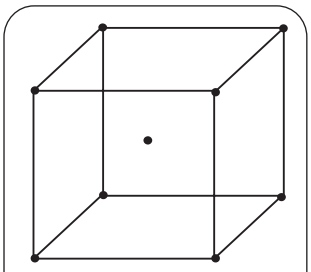
$$= \frac{\frac{4}{3}\pi r^3}{(2r)^3} = 0.524$$

This implies that effectively 52.4 percent of a unit cell volume is occupied by the spheres.

$\therefore$ void fraction (V.F) = 1 − 0.524 = 0.476

Body centred cubic unit cell (bcc)

In this lattice, there is an atom at the body centre exclusively to the unit cell in addition to the eight atoms at the corners.



Body centred cubic unit cell

Fig. 8.14

Therefore the total number of atoms per unit cell works to be

Contribution from corner atoms = $8 \times \frac{1}{8} = 1.$

Contribution from body centre = 1

i.e., there are a total of two atoms per unit cell in this case.

To calculate the packing fraction it may be noted that the spheres touch one another along the body diagonal.

If "l" is the length of the unit cell edge and "r" is the radius of the sphere, it is seen that

(body diagonal)^2 = (face diagonal)^2 + (unit cell edge)^2

$$= \left(\sqrt{2}l\right)^2 + l^2 = 3l^2$$

or body diagonal = $\sqrt{3}l$ and is equal to r + 2r + r = 4r.

$\Rightarrow l = 4r / \sqrt{3}$

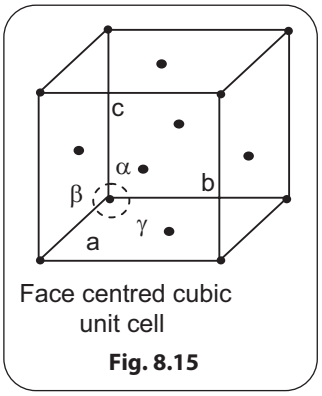
Since it is known that there are two atoms per unit cell in this case, the packing fraction is

$$\text{P.F} = \frac{2 \times \frac{4}{3}\pi r^3}{\left(\frac{4r}{\sqrt{3}}\right)^3} = \frac{\sqrt{3}\pi}{8} = 0.680$$

Or void fraction V.F. = 1 − 0.680 = 0.32

Face centred cubic unit cell (fcc)

In a fcc lattice, not only are the corners occupied, there are atoms occupying the six faces.



Face centred cubic unit cell

Fig. 8.15

A face atom is shared equally between two unit cells and therefore, such an atom contributes one half to unit cell.

Therefore, the total number of atoms per unit cell in a fcc lattice is given by

Contribution of six face atoms = $6 \times \frac{1}{2} = 3$

Contribution of 8 corner atoms = $8 \times \frac{1}{8} = 1.$

Thus making a total of four. The packing fraction (P.F) in this case may be calculated by noting that the spheres touch along the face diagonal.

Face diagonal = $\sqrt{2}l$ where, l = length of unit cell edge

$\sqrt{2}l = r + 2r + r = 4r$ (where, r = radius of the sphere)

$l = \frac{4r}{\sqrt{2}}$

$\therefore$ Fraction of the volume of unit cell occupied

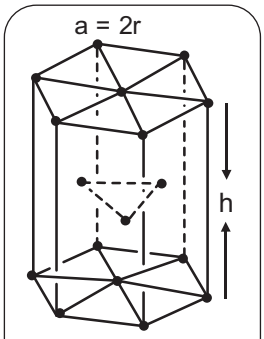
$$\text{P.F.} = \frac{4 \times \frac{4}{3}\pi r^3}{\left(\frac{4r}{\sqrt{2}}\right)^3} = \frac{4 \times \frac{4}{3} \times \pi \times r^3 \times 2\sqrt{2}}{64\, r^3}$$

$$= \frac{\sqrt{2} \times \pi}{6} = 0.741$$

$\therefore$ void fraction = 1 – 0.741 = 0.259

Hexagonal primitive unit cell

In a hexagonal primitive unit cell, the atoms are arranged at corners, face centers and body centers of the unit cell



Hexagonal primitive unit cell

Fig. 8.16

Each corner atom will be common to six other unit cells and therefore its contribution to the unit cell is $\frac{1}{6}$. However, there are twelve corner atoms, three body centre atoms and two face centre atoms.

Total number atoms per unit cell

$$= \left(12 \times \frac{1}{6}\right) + 3 + \left(2 \times \frac{1}{2}\right) = 6$$

The packing fraction is given by

$$\text{P.F} = \frac{6 \times \frac{4}{3}\pi r^3}{\text{Base area} \times \text{height}} = \frac{6 \times \frac{4}{3}\pi r^3}{6 \times \frac{\sqrt{3}}{4}(2r)^2 \times 4r\sqrt{\frac{2}{3}}}$$

$$= \frac{\frac{4}{3}\pi r^3}{4r^2 \times \sqrt{3} \times \frac{\sqrt{2}}{\sqrt{3}}r}$$

$$= \frac{\pi}{3\sqrt{2}} = 0.741$$

Edge centred cubic lattice

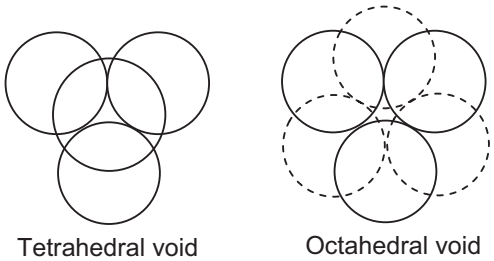
In this type of lattice, an edge atom is shared by four unit cells and there are twelve edges of the unit cell. The contribution of the edge atoms to the unit cell is ¼ and the twelve edges contribute 12 × ¼ atoms to the unit cell. Apart from this the eight corners contribute one atom to the unit cell. Total no of atoms per unit cell = 4.

Density of a unit cell

The density of a unit cell (ρ) (which is the same as the density of the crystal lattice) may be calculated from the relation $\rho = \frac{n \times M}{V \times N}$ where n is the number of molecules or atoms in the unit cell, M is the molecular weight of the substance, V is the volume of the unit cell and N is the Avogardo's number.

Tetrahedral and octahedral voids

Two kinds of voids occur in all closest packed structures and they are known as tetrahedral voids and octahedral voids. A tetrahedral void is surrounded by four spheres while



Different types of voids

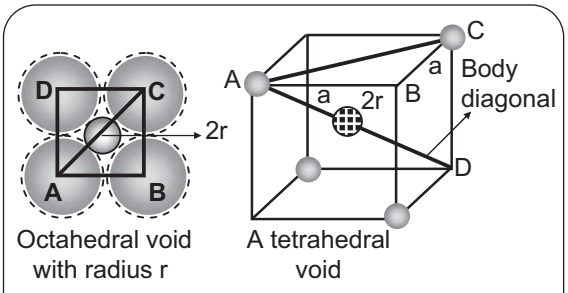
Fig. 8.17

an octahedral hole is surrounded by six spheres. These voids are shown in the following diagram (Fig. 8.17)

For a tetrahedral void, four spheres are arranged at the corners of a regular tetrahedron whereas in the case of an octahedral void, six spheres are arranged at the corners of a regular octahedron.

These arrangements are shown in the Fig. 8.18.

In a fcc structure, there are eight tetrahedral holes and four octahedral holes per unit cell. In a hexagonal primitive unit cell, there are twelve tetrahedral and six octahedral voids.



Arrangement of tetrahedral and octahedral voids

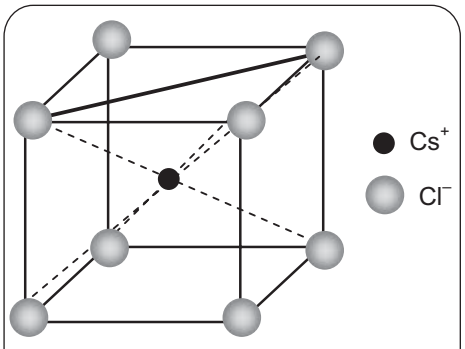
Fig. 8.18

PACKING IN IONIC COMPOUNDS

In ionic compounds, the packing of spheres involves cations and anions. In a case where the cations and anions carry equal but opposite charges, a stable electrically neutral structure requires the number of cations surrounding an anion to be the same as the number of anions surrounding a cation. When both cation and anion are of the same size, it is not possible to have an electrically neutral structure with co-ordination number 12. However a bcc or fcc close packed structure is possible.

Cesium chloride structure

A bcc type arrangement of such a type involves like charged ions in which an ion at the body centre is surrounded by eight oppositely charged ions at the corners of a cube.



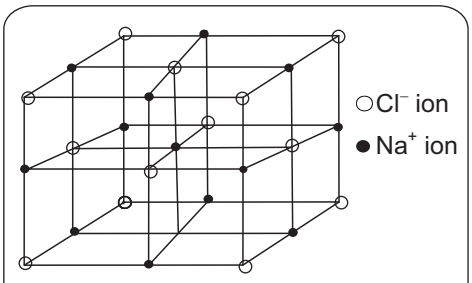
Cs^+ ion surrounded by $8Cl^-$ ions
bcc lattice of CsCl

Fig. 8.19

Cesium chloride is an example of the type in which the co-ordination number of cesium ion is eight. The unit cell of cesium chloride structure contains one molecule of CsCl because the eight corner Cl^- ions contribute one Cl^- to the unit cell ($8 \times \frac{1}{8} = 1$) while the body centred Cs^+ ion contributes one Cs^+ ion to the unit cell.

Rock salt structure

Sodium chloride has a fcc structure in which the corners are occupied by Cl^- ions and they alternate with Na^+ ions in such a way that each Cl^- ion is surrounded by six Na^+ ions and inturn each Na^+ ion is surrounded by six chloride ion. Thus a coordination number of six is achieved by each ion.



fcc lattice of NaCl

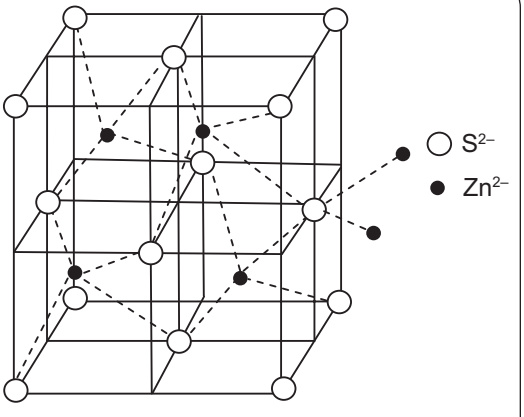
Fig 8.20

In this arrangement all cations occupy octahedral holes. It is also possible to treat the unit cell of sodium chloride such that all anions occupy octahedral holes in an array of fcc cations.

It is seen from the unit cell structure of NaCl given in Fig. 8.20 there are 14 Cl^- and 13 Na^+ ions. The eight Cl^- ions at the corners contribute one Cl^- ion to the unit cell while the six Cl^- ions at the face centre contribute three Cl^- to the unit cell making a total of four Cl^- ions per unit cell. A similar calculation shows that there is one Na^+ ion in body centre contributing to unit cell. The first twelve edge centred Na^+ ions contribute three Na^+ ions to the unit cell making a total of four Na^+ ions per unit cell. Thus there are four "NaCl" molecules per unit cell.

(iii) Zinc blende (ZnS) structure

This structure may be considered as an fcc arrangement (Fig. 8.21) of sulphide ions in which half of the tetrahedral holes are alternately occupied by zinc ions.



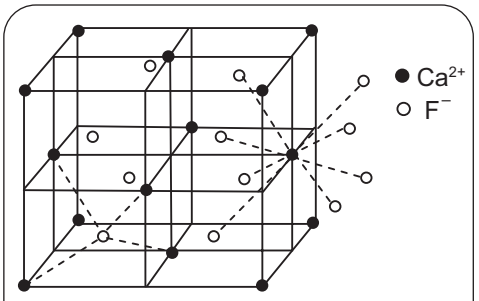
Zinc blende unit cell

Fig. 8.21

Each sulphide ion is surrounded by four zinc ions tetrahedrally and like wise each Zn^{2+} ion is surrounded by four sulphide ion at the corners of a tetrahedron. Thus a 4:4 coordination number is achieved in this case. Ex. CuCl, AgI

(iv) Fluorite (CaF_2) structure

In this structure, calcium ions are arranged in a fcc type structure while the fluoride ions occupy all tetrahedral holes



Calcium fluorite unit cell

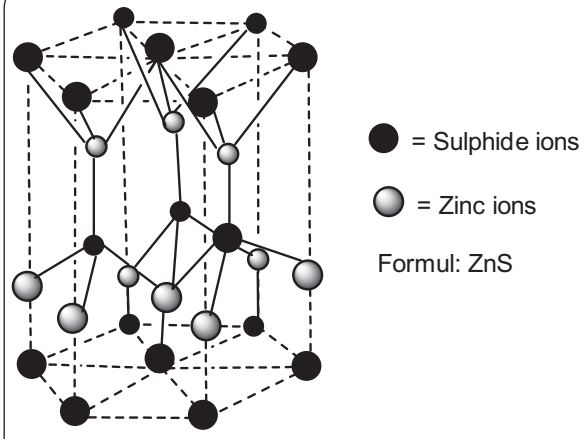
Fig. 8.22

Each fluoride ion is tetrahedrally co-ordinated to four Ca^{2+} ions and each calcium ion is surrounded by eight fluoride ions at the corners of a cube. Thus a 8:4 coordination number is attained in this case. UO_2 and ThO_2 are other compounds that exhibit this structure.

(v) Antifluorite structure

Oxides like Li_2O and other alkali oxides Na_2O, K_2O and Rb_2O exhibit this structure. This unit cell is exactly the reverse of fluorite structure. Here, the oxide ions exhibit fcc structure while Li^+ ions occupy tetrahedral holes or voids. There are four oxide ions and eight lithium ions per unit cell. So a 4:8 corodination number is attained.

(vi) Wurtzite structure



Wurtzite structure

Fig. 8.23

This structure differs from that of zinc blende structure in that it is derived from an expanded hexagonally close packed array of anions rather than fcc arrangement.

Thus the S^{2-} ions are arranged in hexagonally close packed layers. However, as in zinc blende structure the cations occupy one type of tetrahedral holes. This structure has a 4:4 coordination.

(vii) Perovskite structure

Examples of this class are $CaTiO_3$ and $BaTiO_3$ in which one of the cations is bivalent and the other is tetravalent. The bivalent ions are present in primitive cubic lattice with oxide ions being present on the six face centres. The tetravalent cation is in the centre of unit cell occupying octahedral void.

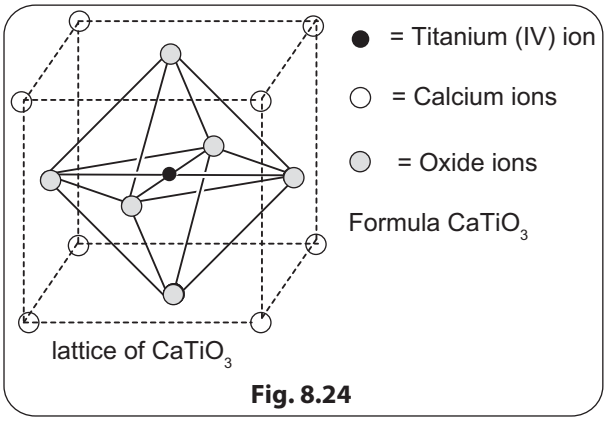


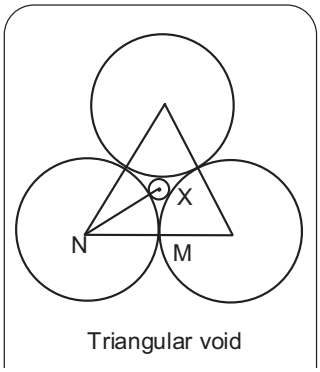
Fig. 8.24

Effect of ion size on crystal structure

In close packed systems, oppositely charged ions touch each other. The difference in crystal structure depends on the size of the ions. The geometrical requirement for a given structure is expressed by radius ratio of the smaller to the larger ion ie $R = \frac{r_s}{r_\ell} = \frac{r_+}{r_-}$ where, usually the smaller size ion is the cation (r_s or r_+ and larger size ion is the anion (r_ℓ or r_-).

Calculations involving radius ratio

(i) Triangular voids (Co-ordination no = 3)



Let (r_+) be the radius of the cations (smaller ion, central sphere) and (r_-) be the radius of the anions (larger ion, coordinating sphere) from the Fig. 8.25

$$\frac{NM}{NX} = \frac{r_-}{r_- + r_+} = \cos 30 \quad (\because \text{ the angle XNM} = 30°)$$

$$1 + \frac{r_+}{r_-} = \frac{1}{\cos 30} = \frac{1}{0.866} = 1.155$$

$$\frac{r_+}{r_-} = 0.155$$

(ii) Tetrahedral voids (Co-ordination no = 4)

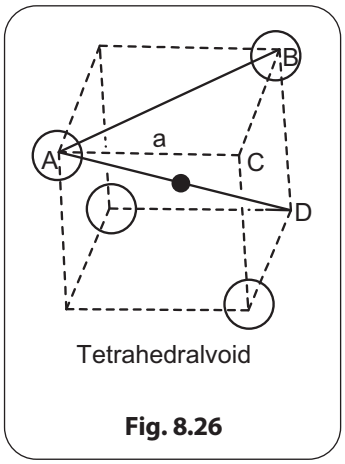


Fig. 8.26

To simplify calculations, a tetrahedral void may be represented in a cube as shown in the Fig. 8.26, in which three spheres form the triangular base, the fourth lies at the top and the shaded sphere occupies the tetrahedral void.

Suppose the length of the side of the cube = a

From right angled triangle ABC, face diagonal

$$AB = \sqrt{AC^2 + BC^2} = \sqrt{a^2 + a^2} = \sqrt{2}\,a$$

Let r_+ be the radius of the cation (shaded sphere) and r_- be the radius of the anion (co-ordinating sphere).

As spheres A and B are actually touching each other face diagonal $AB = 2r_-$

Hence $2r_- = \sqrt{2}\,a$ or $r_- = \frac{1}{\sqrt{2}}a$

Again from the right angled triangle ABD, body diagonal $AD = \sqrt{AB^2 + BD^2} = \sqrt{\left(\sqrt{2}\,a\right)^2 + a^2} = \sqrt{3}\,a$

But as shaded sphere touches others spheres, evidently, body diagonal $AD = 2(r_- + r_+)$

$$\therefore \quad 2(r_- + r_+) = \sqrt{3}a \text{ or } r_- + r_+ = \frac{\sqrt{3}}{2}a$$

We have got $r_- = \frac{1}{\sqrt{2}}a$

$$\therefore \quad \frac{r_- + r_+}{r_-} = \frac{\sqrt{3}a/2}{a/\sqrt{2}} = \frac{\sqrt{3}}{\sqrt{2}}$$

$$1 + \frac{r_+}{r_-} = \frac{\sqrt{3}}{\sqrt{2}}$$

$$\frac{r_+}{r_-} = \frac{\sqrt{3}}{\sqrt{2}} - 1 = 0.225$$

(iii) Octahedral voids (Co-ordination no = 6)

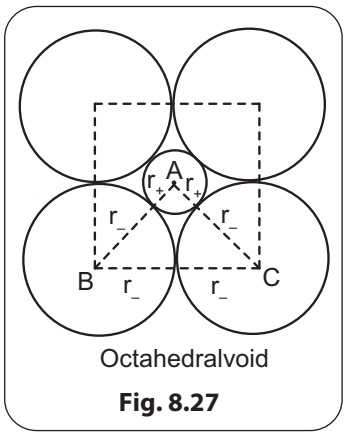


Fig. 8.27

In the Fig. 8.27 a cross section of the octahedral site is shown. The small central circle represents the cation (r_+) which is surrounded by six negative ions (r_-). However, only four negative ions are depicted in the above figure.

Obviously ABC is a right angled triangle. Applying Pythagoras theorem,

$$BC^2 = AB^2 + AC^2$$

$$(2r_-)^2 = (r_- + r_+)^2 + (r_- + r_+)^2 = 2(r_- + r_+)^2$$

$$\frac{(2r_-)^2}{2} = (r_- + r_+)^2$$

$$\left(\sqrt{2}\,r_-\right)^2 = (r_- + r_+)^2$$

$$\sqrt{2}\,r_- = r_- + r_+$$

$$r_+ = \sqrt{2}\,r_- - r_- = r_-\left(\sqrt{2} - 1\right)$$

$$\frac{r_+}{r_-} = \sqrt{2} - 1 = 0.414$$

The following table gives the limiting ratios of $\frac{r_+}{r_-}$, the corresponding co-ordination number, shape of the compound expected along with some examples.

Table 8.6

Limiting radius ratio, $\frac{r_+}{r_-}$	Co-ordination no	Shape of species	Example
< 0.155	2	Linear	$BeCl_2$
Between 0.155 and 0.225	3	Triangular	B_2O_3
Between 0.225 and 0.414	4	Tetrahedral	ZnO
Between 0.414 and 0.732	6	Octahedral	NaF
> greater than 0.732	8	BCC	CsBr

DEFECTS IN SOLIDS

An ideal crystalline solid, in which every lattice point is occupied, is a concept strictly valid at absolute zero of temperature. At any definite temperature, a configurational disorder occurs in the structure of solids. Apart from the presence of such intrinsic defects, it is possible to introduce defects mechanically known as extrinsic or non–equilibrium defects, as for example, by incorporation of an impurity atom. Thus any deviation from a chemically pure stoichiometric crystal or compound (defined as a compound in which the number of positive and negative ions are exactly in the ratios indicated by their stoichiometric formulae) is called an imperfection or a defect. The existence of such defects strongly influences many physical properties such as mechanical strength of the crystal, optical absorption and electrical conductivity.

The various types of defects may be classified into three categories (i) point defects (ii) line defects and (iii) plane defects. We shall deal with the point defect in some detail.

Various types of point defects

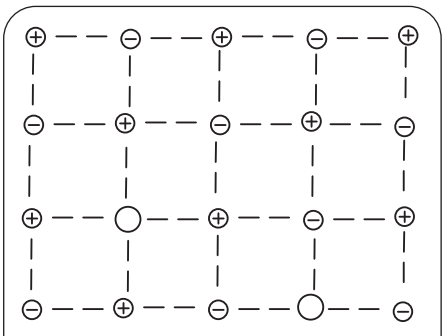
The point defects may belong to several categories such as (i) those arising due to a vacancy (ii) Schottky/ Frenkel type as in interstitial impurity (iii) non-stoichiometry etc. A vacancy defect, for example, arises due to the fact that some lattice points normally occupied by metal ions (or atoms) in metals like Cu, are missing from their lattice positions.

Schottky defects

In this type of defect, a vacancy is created in a metal by the migration of metal atom (or ion) to the surface. If a Schottky defect occurs in an ionic crystal, both anion and cation migrate. The number of Schottky defects in a crystal depends on several factors

(i) The mass of the ionic crystal
(ii) The average energy required to produce a Schottky defect
(iii) Temperature. The fraction of the defects increases exponentially with temperature.

Fig. 8.28 shows the Schottky defect in a crystal.



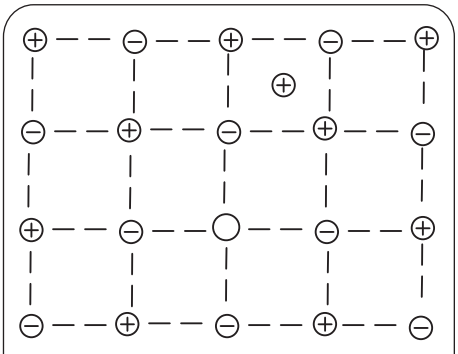
Schottky defect

Fig. 8.28

Schottky defects are dominant in closest packed structures like that of NaCl. This type of defect is shown by crystals like NaCl, CsCl

Frenkel defects

These defects arise when an atom or an ion becomes displaced from its proper position in the crystal lattice and takes up an interstitial position along side. This is possible when the lattice contains large enough interstices. Frenkel defects are common in open structures like those of silver halides. Frenkel defects also increase with temperature.



Frenkel defect

Fig. 8.29

This type of defect is shown by crystals like ZnS, AgCl, AgI

Metal excess and metal deficiency defects

Among the point defects observed, these types of defects are less common. A brief description of these defects is given below.

(i) Metal excess defects

When a crystal of NaCl is heated in sodium vapours, it acquires a yellow colour. This yellow colour results from the formation of a non stoichiometric compound of NaCl in which there is a slight excess of Na^+ ions which are formed as a result of their ionization. While the cation possibly occupies the normal cation site in NaCl, the electron occupies an anion vacancy. The electron is loosely bound to the vacant site and leads to absorption in the visible region.

(ii) Metal deficiency defects

This deficiency arises when one of the metallic cations (ie positive ion) is missing from its lattice site. The extra negative charge there by created is balanced by another metal ion near it acquiring an additional +ve charge. A deficiency of metal ions thus occurs. Compounds of transition metals like NiO, FeO exhibit this type of defect.

It is thus seen that all types of point defects arise due to the creation of vacancies or holes in the lattices of crystals.

Non–stoichiometry

The formulae ZnO and FeO for the oxides of zinc and iron suggest that they are stoichiometric compounds. However, use of sophisticated instrumental techniques has shown some small departures from stoichiometry in such compounds. Elements exhibiting variable valency usually form compounds with variable composition. For example, copper sulphide exists in the range $Cu_{1.7}S$ to Cu_2S, since appropriate number of Cu^{2+} ions can maintain charge neutrality.

Semiconductivity

When impurities are deliberately added into silicon and germanium, the materials develop semi-conducting properties. The semi-conducting property of silicon and germanium is greatly enhanced by introduction of impurity atoms (from group III or group V). When a trivalent atom like indium is introduced in silicon or germanium there arises a deficiency of one electron in the filled zone of valency electrons. When an electrical potential is applied, the electron deficiency may be considered to move from one atom to the next and the conductivity of such crystals is said to be due to positive carriers or holes and they are known as p-type semiconductors.

On the other hand, when a group V atom like antimony atom is introduced in silicon or germanium, an extra electron is introduced into an empty zone above valency zone. The added electrons move in an electric field and the conductivity of the crystal is due to negative carriers. Such conductors are called n-type semiconductors.

Both types of semiconductors are used in the manufacture of transistors, rectifiers and other semiconducting devices. Lattice defects of various types are responsible for the high catalytic activity of some metal oxides.

CONCEPT STRANDS

Concept Strand 23

A fcc lattice contains "X" type atoms at corners and "Y" type atoms in the face centre of the unit cell.

(i) What is the formula of the compound?
(ii) Suppose two "X" atoms are missing from two corners. What is the formula?

Solution

(i) Eight corners contribute $8 \times \frac{1}{8} = 1$ atom of X to unit cell

6 faces contribute $6 \times ½ = 3$ atoms of Y to unit cell

$\therefore$ formula of compound – XY_3

(ii) 6 atoms at the corners contribute $\frac{6}{8} = \frac{3}{4}$ atoms to unit cell.

6 faces contribute . $6 \times ½ = 3$ atoms to unit cell.

Formula : $X_{3/4}Y_3 = X_3Y_{12} = XY_4$

Concept Strand 24

Silver crystallizes in fcc structure. Given its edge length as 4.1 A^0, calculate its radius. Also calculate the number of unit cells per mole and the Avogadro's number (ρ = 10.6 gcm^{-3}, At wt = 108.0)

Solution

In an fcc lattice, the atoms touch along a face diagonal. The face diagonal "a" is related to edge length "ℓ" by

$$a = \sqrt{2}\ell$$

"a" is also related to the radius of the atom (r) by the equation

$$a = 4r$$

$$4r = \sqrt{2}\ell\ ;$$

$$r = \frac{\sqrt{2}}{4}\ell = \frac{\sqrt{2}}{4} \times 4.1 = 1.45\ A^0$$

Unit cell volume = $(4.1 \times 10^{-8})^3 = 68.92 \times 10^{-24}$ cm^3
Molar volume of Ag = 108/10.6 = 10.19 cm^3mol^{-1}

$$\text{No of unit cells in one mole} = \frac{10.19}{68.92 \times 10^{-24}}$$

$$= 1.48 \times 10^{23}$$

From fcc system, no. of atoms per unit cell = 4

$\therefore$ Total no of atoms per mole = $1.48 \times 10^{23} \times 4 = 5.92 \times 10^{23}$

Concept Strand 25

Calculate the number of molecules of lead sulphide in the unit cell from the following data. Edge length = 5.0 A°. Molecular weight = 239.2, N = 6.023×10^{23}. density = 12.7 gm cm^{-3}. What is its structure?

Solution

ρ = density = $\frac{n \times M}{V \times N}$

n = no of molecules in unit cell

M = mol. wt.

V = unit cell volume

and N = Avogardo number

Unit cell volume = $(5\times 10^{-8})^3$

$= 125 \times 10^{-24} cm^3$

$n = \frac{\rho \times V \times N}{M} = \frac{12.7 \times 125 \times 10^{-24} \times 6.023 \times 10^{23}}{239.2}$

$= 4.0$

since there are 4 molecules per unit cell, the structure of PbS is fcc.

Concept Strand 26

Assuming that sodium has a density of 1.0 gm cm^{-3} at 293K and that its edge length in cubic lattice is 4.24 A°, calculate the number of atoms per unit cell. What is its crystal structure, simple cubic, fcc or bcc? What is the radius of the atom? (At wt of Na = 23)

Solution

Unit cell volume = $(4.24 \times 10^{-8})^3 = 76.22 \times 10^{-24} cm^3$

$n = \frac{\rho \times V \times N}{M} = \frac{76.22 \times 10^{-24} \times 1 \times 6.023 \times 10^{23}}{23} = 2$

Since there are two atoms per unit cell, Na crystallizes in bcc lattice. In a body centred cubic lattice, the atoms touch along a body diagonal. The body diagonal "a" is related to edge length "ℓ" by the relation.

$a = \sqrt{3}\ell$ and

a = 4r where r = radius of the atom

$\therefore \quad r = \frac{\sqrt{3}\ell}{4} = \frac{\sqrt{3} \times 4.24}{4} = 1.835A°$

Concept Strand 27

When the same metal ie Na (given in the previous problem) is cooled to –195°C, its density increases by 4 percent and unit cell length increases by 24.8 percent. What type of unit cell does it have at the lower temperature. (Use density and unit cell length given in problem 4)

Solution

Density at –195°C = $1.00 + \frac{1 \times 4}{100} = 1.04 \text{ g cm}^{-3}$

Unit cell length at –195°C = $4.24 + 4.24 \times \frac{24.8}{100}$

$= 4.24 + 1.051 = 5.291 \text{ A}^0$

n, when cooled to -195°C = $\frac{\rho \times V \times N}{M}$

$= \frac{1.04 \times (5.291 \times 10^{-8})^3 \times 6.023 \times 10^{23}}{23} = 4.03 = 4$

Since the number of atoms per unit cell is close to 4, it has a face centred cubic lattice under the low temperature conditions.

Concept Strand 28

Cesium chloride forms a bcc type lattice in which there is cesium ion at the body centre and chloride ions at the corners of the unit cell. The Cs^+ and Cl^- ions contact each other along the body diagonal. If the density of cesium chloride is 4.0 g cm^{-3} and the radius of the Cs^+ ion is 1.76A° , calculate the radius of chloride ion. (Mol. Wt of CsCl = 168.4)

Solution

The volume of the unit cell, V can be calculated from the equation $\rho = \frac{n \times M}{V \times N}$

where n = no of formula units per unit cell,

M = Mol wt, N = Avogardo number

In the case CsCl n = 1

($\therefore$ one Cs^+ ion and one Cl^- are present in unit cell)

or $V = \frac{1 \times 168.4}{4.0 \times 6.023 \times 10^{23}} = 7.0 \times 10^{-23} \text{ cm}^3$

But $V = a^3$ when a = unit cell length

$\therefore \quad a = \sqrt[3]{7.0 \times 10^{-23}} = 4.12 \times 10^{-8} \text{ cm}$

Since the Cs^+ and Cl^- ions are in contact along body diagonal

$(\text{body diagonal})^2 = (\text{face diagonal})^2 + (\text{edge length})^2$

$$\{2(r_c + r_a)\}^2 = \left(\sqrt{2}a\right)^2 + a^2 = 3a^2$$

$$r_c + r_a = \sqrt{\frac{3a^2}{4}} = \frac{\sqrt{3}}{2}a = \frac{\sqrt{3}}{2} \times 4.12 = 3.57$$

or $1.76 + r_a = 3.57 \quad \therefore \; r_a = 1.81 A°$

Concept Strand 29

A metal crystallizes in bcc lattice and its density is 7.86 g cm^{-3} and its unit cell length is 2.88 A°. Calculate the number of unit cells and the number of atoms in 200 grams of the metal.

Solution

Vol of 200 g of metal $= \frac{200}{7.86} = 25.445 \text{ cm}^3$

Vol of unit cell $= \left(2.88 \times 10^{-8}\right)^3 = 23.89 \times 10^{-24} \text{ cm}^3$

$\therefore$ no of unit cells in 200g $= \frac{25.445}{23.89 \times 10^{-24}} = 1.065 \times 10^{24}$

since the metal crystallizes in bcc, no of atoms per unit cell is equal to 2.

$\therefore$ number of atoms $= 1.065 \times 10^{24} \times 2 = 2.130 \times 10^{24}$

Concept Strand 30

Calculate the unit cell volume of a compound crystallizing in ortho rhombic fcc lattice given the density of the compound as 0.67 g cm^{-3} and its molecular weight as 21.8.

Solution

$$\rho = \frac{n \times M}{V \times N}$$

$$V = \frac{n \times M}{\rho \times N} = \frac{4 \times 21.8}{0.67 \times 6.023 \times 10^{23}} = 21.61 \times 10^{-23} \text{ cm}^3$$

Concept Strand 31

Copper has a unit cell of fcc type with the edge length of 3.68 A°. Calculate its density if 100 g of copper contains 9.5×10^{23} atoms.

Solution

Unit cell volume $= (3.68 \times 10^{-8})^3 = 49.84 \times 10^{-24} \text{ cm}^3$

There are 4 atoms per unit cell

$$\text{Number of unit cells in 100 g} = \frac{9.5 \times 10^{23}}{4} = 2.37 \times 10^{23}$$

Total unit cell volume in 100 g $= 49.84 \times 10^{-24} \times 2.37 \times 10^{23} = 11.81 \text{ cm}^3$

$$\rho = \frac{\text{mass}}{\text{vol}} = \frac{100}{11.81} = 8.46 \text{ g cm}^{-3}$$

Concept Strand 32

Sodium fluoride forms a cubic close packed arrangement with fluoride ions occupying the corners of the lattice and sodium ions occupying the octahedral voids. If the edge length of the unit cell is 4.7 A°, calculate the limiting ionic radii of Na^+ and F^- ions.

Solution

Edge length $= 2(r_+ + r_-) = 4.7$ A°

Or $r_+ + r_- = 2.35$ A°

For an octahedral void, $\frac{r_+}{r_-} = 0.414$

Or $0.414\, r_- + r_- = 2.35$

$$r_- = \frac{2.35}{1.414} = 1.66 A°$$

$\therefore \; r_+ = 2.35 - 1.66 = 0.69$ A°

Concept Strand 33

Calculate the packing fraction and void fraction in antifluorite structure per unit cell.

Solution

In antifluorite structure (example: Na_2O), oxide ions occupy the corners and face centres of fcc ($\therefore$ this is the closest packing). Na^+ ions occupy the tetrahedral holes. There are four oxide ions and 8 Na^+ ions per unit cell. For the face diagonal in this case.

$$4r_- = \sqrt{2}a \text{ , } a = 2\sqrt{2}\; r_-$$

a is the unit cell length and r_- is the radius of the oxide ion.

Volume of unit cell $= a^3 = \left(2\sqrt{2}r_-\right)^3 = 16\sqrt{2}r_-^3$

$$\text{P.F} = \frac{4 \times \frac{4}{3}\pi r_-^3 + 8 \times \frac{4}{3}\pi r_+^3}{16\sqrt{2}r_-^3} = \frac{\pi}{3\sqrt{2}}\left[1 + \frac{2r_+^3}{r_-^3}\right]$$

In antifluorite structure, $\frac{r_+}{r_-} = 0.225$.

Substituting this value in the above equation,

$$P.F = \frac{\pi}{3\sqrt{2}}\left[1+2(0.225)^3\right] = 0.758$$

$$V.F = 1 - 0.758 = 0.242$$

Concept Strand 34

The unit cell length of lithium is 3.5 A°and it crystallizes in bcc lattice. Calculate its theoretical density. If its experimental density is 0.53 g cm^{-3}, what is the percentage of vacant sites.
(At wt of Li = 6.9)

Solution

$$\rho = \frac{nM}{V \times N} = \frac{2 \times 6.9}{\left(3.5 \times 10^{-8}\right)^3 \times 6.023 \times 10^{23}}$$

$$= 0.534 \text{ g cm}^3$$

$$\% \text{ occupied sites} = \frac{\text{Experimental density}}{(\text{ideal density})} \times 100$$

$$= \frac{0.53}{0.534} \times 100 = 99.25$$

$$\% \text{ vacant sites} = 100 - 99.25$$

$$= 0.75$$

TOPIC GRIP

Subjective Questions

1. Decomposition of NH_3 gas on heated tungsten surface gave the following data. Calculate the order and the mean value of the rate constant.

initial pressure p_o (in mm Hg)	65	105	150	185
$T_{1/2}$ (sec)	290	460	670	820

2. Decomposition of $N_2O_{(g)}$ conforms to I order kinetics. The rate constant gets quadrupled over the temperature range 298 K to 308 K. Calculate an approximate value of the activation energy, E_a. Further, assuming constancy of activation energy calculate the temperature at which, the rate constant again gets quadrupled.

3. A first order reaction is 30% complete in one hour. How much more time will elapse to reach the stage when 30% of the initial concentration will remain?

4. For the reaction, $A_{(g)} \rightarrow 2\ B_{(g)} + C_{(g)}$ at 100°C, I order kinetics is observed starting with pure A, the total pressure after 10 minutes is 176 mm. After a long time, the total pressure had the limiting value of 270 mm. Calculate an approximate value of the rate constant.

5. The rate of a chemical reaction increases two times for every 20°C rise of temperature. If the rate constant at 30°C is 'K' what is the value of rate constant at90°C?

6. Metallic lead crystallizes in the fcc system. The side length of the unit cell = 4.92 Å. Atomic weight of lead = 207.2 g. Calculate the density. $N_0 = 6.02 \times 10^{23}\ mol^{-1}$.

7. (a) In an fcc lattice of atoms what is the number of nearest neighbours and next nearest neighbours for each atom.

(b) In an fcc lattice of anions what is the cation–anion radius ratio, $\left(\frac{r_+}{r_-}\right)$ if the cation is of the right size, just to fit the octahedral hole.

8. CsI lattice consists of two interpenetrating simple cubic lattices side length of the unit cell = 4.562 Å. Calculate density. At. wt of Cs = 133, I = 127.

9. Silver (at. wt = 107.88 g) crystallizes in the fcc system density = 10.53 g/cm^3.
$N_0 = 6.02 \times 10^{23}$. Calculate the side length of the unit cell.

10. (i) Barium titanate has the perovskite structure i.e., cubic with Ba^{2+} ions at the corners, O^{2-}ions at the face centers Ti^{4+} at the centre of the unit cell. If Ti^{4+} ion is taken to occupy voids (or holes) in the oxide lattice, what type of void is occupied?

(ii) Metallic gold crystallizes in the fcc lattice. Side length a, of the unit cell = 4.07 Å. What is the closest distance between the atoms of gold?

Straight Objective Type Questions

Directions: **This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

11. The $T_{1/2}$ for a certain reaction equals $\frac{\left(2-\sqrt{2}\right)C_0^{1/2}}{K}$. What is the order of the reaction?

(a) 1.0 (b) 0.5 (c) 1.5 (d) 2.0

12. For a I order reaction, the initial rate = 0.84% per minute. Calculate $T_{1/2}$.

(a) 85.2 (b) 52.8

(c) 82.5 (d) 58.2

13. For an elementary chemical reaction A + B $\rightarrow$ C + D the rate of reaction 'r' becomes equal to rate constant at which of the following values of concentration of A and B

	[A]/mol L^{-1}	**[B]/mol L^{-1}**
(a)	0	1
(b)	0.5	0.5
(c)	0.5	2
(d)	1	1

14. For PbS, the edge length of the unit cell is 5.0Å. Molar mass = 239.2 g mol^{-1}density = 12.7 g cm^{-3} calculate the number of "molecules" of PbS present in the unit cell. What is the type of lattice?

(a) 1.0 sc (b) 2.0 bcc (c) 3.0 mixed fcc and sc (d) 4.0 fcc

15. In the crystal structure of AB, assume that we have a close–packed structure of B^- ions forming an fcc lattice. A^+ ions just fit into the octahedral voids. Edge length of the unit cell = 4.7 Å. Calculate the radii of A^+ and B^- ions in Å.

(a) 0.59, 1.76 (b) 0.69, 1.66 (c) 0.72, 1.63 (d) 0.45, 1.90

Assertion–Reason Type Questions

Directions: **Each question contains Statement-1 and Statement-2 and has the following choices (a), (b), (c)and (d), out of which ONLY ONE is correct.**

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False
(d) Statement-1 is False, Statement-2 is True

16. Statement 1

Rate constant for the process $^{238}_{92}U \rightarrow {}^{234}_{90}Th + {}^{4}_{2}$ does not change with temperature.

and

Statement 2

It is a nuclear process.

17. **Statement 1**
In a reaction sequence A $\xrightarrow{K_1}$ B $\xrightarrow{K_2}$ C, if $K_1 << K_2$ then after starting the reaction with only A, [B] may reach a maximum and remain steady over a period of time.
and

Statement 2
In the above case [C] may also remain steady over a period of time.

18. **Statement 1**
All first order reactions are unimolecular in nature and are not influenced by positive catalysts.
and

Statement 2
A simplified explanation for positive catalysis is that such a catalyst provides an alternative channel of low activation energy for reaction to take place.

19. **Statement 1**
Unit cell of a crystal exists but cannot be isolated.
and

Statement 2
Unit cells can be primitive or centred.

20. **Statement 1**
In the Zinc blende lattice, sulphide ions form an fcc structure while Zn ion occupy all the octahedral voids.
and

Statement 2
In the fluorite lattice (CaF_2)calcium ions form an fcc structure while fluoride ions occupy all the tetrahedral holes.

Linked Comprehension Type Questions

Directions: **This section contains 2 paragraphs. Based upon the paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (a), (b), (c)and (d), out of which ONLY ONE is correct.**

Passage I

During the course of a chemical reaction, the concentrations of the reactants as well as the rate of the reaction decrease. An equation that relates the rate of the reaction to the concentration of the reactants is known as the rate equation. This may have a complicated form. For convenience however one adopts the form: rate = $K\, c_1^{n_1}\, c_2^{n_2} \ldots$ Where, c_1, c_2 etc are the concentrations of the reactants designated, 1, 2…K is known as the specific rate constant. n_1, n_2…are the partial orders of the reactants 1, 2 …and ($n_1 + n_2 + \ldots$)is known as the total order of the reaction. The rate equation has only an empirical (experimental) basis and the orders n_1, n_2…may be whole numbers, fraction zero and in some cases even negative. In a simple case of one reactant the equation takes the form $-\frac{dc}{dt} = K\, c^n$ which may be integrated. n = 0, 1, 2, correspond to orders0, 1, and 2. The correctness of the integrated form, C as a function of t may be checked

against experimental data for various orders. $C = C_0 - Kt$ for zero order, $K = \left(\frac{2.303}{t}\right) \log \left(\frac{c_0}{c}\right)$ for I order and $\frac{1}{c} - \frac{1}{c_0} = Kt$ for II order.

21. Azomethane decomposes conforming to I order kinetics.
$CH_3 - N = N - CH_{3(g)} \rightarrow C_2H_{6(g)} + N_{2(g)}$ Initial pressures of reactants and products are $CH_3 - N = N - CH_3$: 300 mm, C_2H_6: zero, N_2: zero $K = 1.50 \times 10^{-2}$ min at 300 K. Calculate at the end of 60 minutes the partial pressures of the reactant and the products.

(a) 122 mm, 178 mm, 178 mm
(b) 152 mm, 148 mm, 148 mm
(c) 145 mm, 155 mm, 155 mm
(d) 133 mm, 167 mm, 167 mm

22. The activation energies for the uncatalyzed and catalyzed channels for a certain elementary step in a reaction are 70 kJ and 50 kJ respectively at 320 K. Calculate the ratio of the rate constants $\left[\frac{K_{catalysed}}{K_{uncatalysed}}\right]$.

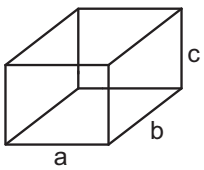
(a) 1480 (b) 1360 (c) 1290 (d) 1840

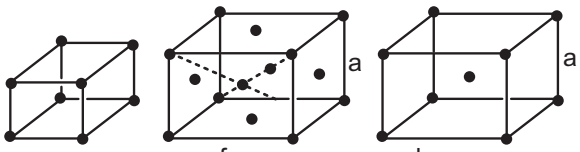
23. A reaction is 50% complete in 40 minutes and 75% complete in 60 minutes. Calculate the order of the reaction and the rate constant. How long will it take for the reaction to be 90% complete.

(a) zero, 0.0152 mol L^{-1} min^{-1}, 64 min
(b) zero, 0.0165 mol L^{-1} min^{-1}, 56 min
(c) zero, K = 0.0125 mol L^{-1} min^{-1}, 72 min
(d) zero, 0.0172 mol L^{-1} min^{-1}, 48 min

Passage II

One may regard the bulk form of a crystalline solid as made up certain basis structures known as unit cells. Each unit cell is a geometric structure, a parallelopiped (fig) with side lengths a, b, c and angles α, β, γ between sides b and c, c and a and a and b. The special case where a = b = c and $\alpha = \beta = \gamma = \frac{\pi}{2}$ is known as a cubic structure or lattice. There are three basic cubic lattices, simple cubic (sc), face centred cubic (fcc) and body centred cubic (bcc). Each corner is a lattice point. (see Fig).





simplecubic Vol = a^3 has an average one lattice point

f.c.c vol = a^3 has an average four lattice points

b.c.c vol=a^3 has an average two lattice points

24. The unit cell of NaCl has Cl^- ions forming an fcc lattice and a Na^+ ion in between pairs of non touching Cl^- ions, i.e., (along and) at the mid point of edges and at the centre of the unit cell. What is the number of nearest neighbours and next nearest neighbours of Na^+ for each Na^+ and Cl^- for each Cl^- ions?

(a) 12, 6 (b) 8, 8 (c) 6, 8 (d) 8, 12

25. A crystalline substance has molar mass of 75 g mol^{-1} and density 2 g cm^{-3}. The unit cell has edge length of 500 pm. It structure is

(a) sc (b) bcc (c) fcc (d) mixed sc and bcc

26. MX_4 (solid) crystallizes with a cubic lattice, the edge length of the unit cell is 0.5 n. m. density 0.874 g cm^{-3}. 5 g of MX_4 (non electrolyte) dissolved in 100 g of water results in boiling point elevation of 0.291°C. K_f = 0.51 deg $molal^{-1}$ what is the type of lattice?

(a) sc (b) fcc

(c) bcc (d) mixed type

Multiple Correct Objective Type Questions

Directions: **Each question in this section has four suggested answers of which ONE OR MORE answers will be correct.**

27. Which among the following statements are correct?

(a) The molecularity of a reaction is a concept relevant to an elementary step in the mechanism of the reaction.

(b) The order of a reaction can be fractional.

(c) A reaction may have its order equal to its molecularity.

(d) The stoichiometry of a reaction, in most cases yields its order.

28. Identify the correct statement among the following.

(a) The cation–anion radius ratio gives an insight into the coordination numbers of ions in (assumed)purely ionic structures. But cannot be applied to covalent structures.

(b) The Ceasium halide structure with 8–coordination is a bcc structure.

(c) The hcp and ccp structures are the closest packed structures with minimal void space.

(d) The colour of PbI_2 (solid)is due to its structural defect.

29. Which of the following statements are correct?

(a) In a reaction sequence $A \xrightarrow{K_1} B \xrightarrow{K_2} C$, steady state approximation for [B] may be useful if $A \rightarrow B$ is a slow step while $B \rightarrow C$ is a fast step.

(b) In an elementary step of a reaction sequence, the rate constant $K = \frac{kT}{h} \times exp\left(\frac{-\Delta G^{\neq}}{RT}\right)$ where, K = rate constant, k = Boltmann constant, T = temperature (Kelvin), DG# is the free energy of activation.

(c) Schottky defect are dominant in ZnS, silver halides

(d) Impure semiconductors have either excess electrons above the valence band or electron holes or vacancies.

Matrix-Match Type Questions

Directions: **Match the elements of Column I to elements of Column II. There can be single or multiple matches.**

30.

Column I	**Column II**
(a) Interstitial occupation of atoms in a crystal lattice.	(p) Schottky defect
(b) defective crystal structure	(q) Rate constant $K = \frac{abp}{1+ap}$ where, p is reactant pressure, a, b = constants
(c) Surface catalysis with partial surface coverage	(r) Reaction channel with lower activation energy
(d) order zero, one or fractional	(s) Frenkel defect

IIT ASSIGNMENT EXERCISE

Straight Objective Type Questions

Directions: **This section contains multiple choice questions. Each question has 4 choices (a), (b), (c)and (d), out of which ONLY ONE is correct.**

31. A $\rightarrow$ products; K = 1×10^{-4} s^{-1}at 283 K. E_a = 60 kJ mol^{-1}. Calculate an approximate value of K, at 0°C in s^{-1}.

(a) 5×10^{-5} (b) 1×10^{-5} (c) 9×10^{-6} (d) 4×10^{-5}

32. A + B $\rightarrow$ C. The rate of the reaction becomes fourfold when [A] is doubled. The rate is independent of [B]. What are the partial orders?

(a) (2, 0) (b) (1, 0) (c) (2, 1) (d) (1, 1)

33. For a reaction of order, 3 how is $T_{1/2}$ related to the initial concentration, C_0.

(a) $T_{1/2} \alpha C_0^2$ (b) $T_{1/2} \alpha \frac{1}{C_0^2}$ (c) $T_{1/2} \alpha \frac{1}{C_0^{1/2}}$ (d) $T_{1/2} \alpha C_0$

34. Decomposition of H_2O_2 in aqueous solution conforms to I order kinetics. The data given below give the titre value, V ml, of standard $KMnO_4$ needed to react with a definite volume of the H_2O_2 solution (at regular time intervals. Calculate the rate constant (s^{-1}).

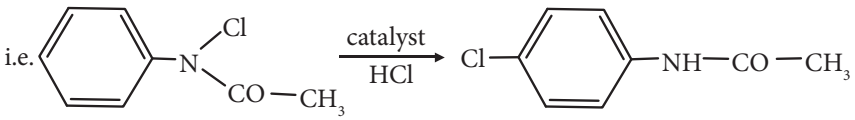
Time (minutes)	0	10	20
V ml	22.8	13.8	8.25

(a) 5.084×10^{-2} (b) 4.85×10^{-2} (c) 5.84×10^{-3} (d) 4.58×10^{-3}

35. For a certain I order reaction, $\log K\ (min^{-1}) = 33.91 - \frac{7933}{T}$. Calculate the activation energy in kcal mol^{-1}.

(a) 33.60 (b) 36.30 (c) 30.63 (d) 30.36

36. The conversion of N–chloroacetanilide into p–chloroacetanilide



conforms to the I order and may be followed by removing a measured quantity of the solution from time to time, adding to a solution of KI and titrating the liberated Iodine against standard thiosulphate. Volume of thiosulphate (V ml)consumed is proportional to the concentraion of the reactant given. Calculate V_0 at t = 0 (ml).

t (hours)	1	4
V (ml)	35.6	13.8

(a) 48.85 (b) 45.88
(c) 54.82 (d) 58.24

37. If a I order reaction is completed to the extent of 60% and x% in time intervals, t_1 and t_2 (min) calculate x given that $(\frac{t_1}{t_2}) = 4.106$.

(a) 20 (b) 35 (c) 10 (d) 40

38. A solution of sucrose is hydrolyzed to the extent of 20% after 30 minutes, from the start. After what time will the reaction to be 90% complete?

(a) 369.0 min (b) 309.6 min (c) 296.3 min (d) 269.3 min

39. Determine the order of the reaction: $(NH_4)CNO_{(aq)} \longrightarrow O{=}C\begin{matrix}NH_2\\NH_2\,(aq)\end{matrix}$ given the following data

Initial reactant concentration (molar)	Time for half conversion (h)
0.05	37.03
0.10	19.15
0.20	9.45

(a) 1.5 (b) 1 (c) 0 (d) 2

40. For the reaction A $\rightarrow$ Products of order 1.5, by what factor (approximately) is the rate increased when [A] is tripled?

(a) 5.196 (b) 5.690 (c) 6.15 (d) 6.52

41. In the acid catalyzed hydrolysis of an ester (in aqueous solution at 25°C) calculate $T_{1/2}$ given the data that 69.12% of the ester was hydrolyzed in 350 min.

(a) 206.4 min (b) 246 min (c) 264 min (d) 220 min

42. NH_3 gas decomposes on the surface of tungsten conforming to zero order reaction. The mean value of the rate constant is 0.1127 mm sec^{-1}. The initial pressure was found to be 65 mm. Calculate (i) total pressure after 15 sec (ii) partial pressure of H_2 after 50 sec $2NH_{3(g)} \rightarrow N_{2(g)} + 3H_2$.

(a) 69.73 mm, 9.26 mm (b) 66.69 mm, 8.45 mm (c) 69.22 mm, 9.62 mm (d) 68.32 mm, 8.98 mm

43. Consider the reaction $A_{(g)} \rightarrow 3\ B_{(g)}$. At t = 0, P°_A= 312 mm and P°_B = 0. The reaction conforms to I order. $T_{1/2}$ = 27 minutes calculate the time interval over which the total pressure rises from its initial value to 468 mm.

(a) 13.70 min (b) 11.21 min (c) 10.11 min (d) 12.75 min

44. $T_{1/2}$ of a reaction is doubled as the initial reactant concentration is doubled. The order of the reaction is

(a) 1 (b) 0.5 (c) 2 (d) zero

45. A first order reaction is 50% complete in 100 minutes at 300 K and in 10 minutes at 400 K, calculate the mean value of the activation energy in kJ mol^{-1}.

(a) 23 (b) 40 (c) 60 (d) 11.5

46. 50 bacteria are placed inside a flask containing nutrients at 35°C and their multiplication is studied as per data given below.

Time in minutes	0	15	30	45	60
Number of baceria	50	100	200	400	800

How many bacteria will be there after 3 hours?

(a) 2.4×10^3 (b) 2.04×10^5 (c) 1.02×10^5 (d) 6.4×10^4

47. For a first order reaction the value of 'A' and 'E_a' in the Arrhenius equation are$4 \times 10^{13}\ s^{-1}$ and 98.6 kJ mol^{-1} respectively. The temperature at which its half life period will be 10 minutes is

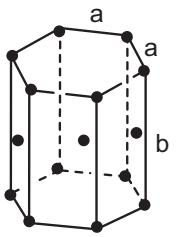
(a) 311.8 K (b) 623.6 K (c) 313.8 K (d) 295.15 K

48. Derive the rate law for the given mechanism $A + B \underset{K_1}{\overset{K_1}{\rightleftharpoons}} C \xrightarrow{K} D \xrightarrow{fast}$ product. (K_1 and K_{-1}) >>K

(a) rate = K' [A] [B] (b) rate = $K'\frac{[A]}{[C]}$ (c) rate = $K'\frac{[B]}{[C]}$ (d) rate = $K'\frac{[A][B]}{[C]}$

49. What is the % of void space in a bcc structure Assume that atoms are in contact along the body diagonal of the cube?
(a) 48 (b) 26 (c) 52 (d) 32

50. Shown here is a unit cell of the hcp structure how many lattice points may be assigned to the volume of the unit cell?



(a) 3
(b) 6
(c) 4
(d) 8

51. A fcc lattice has atoms, x, at the corners and atoms, y, at the face centres. What is the formula of the compound?
(a) xy_2 (b) xy (c) xy_3 (d) x_2y

52. A metal crystallizing in the bcc lattice has density 7.86 g cm^{-3}. The unit cell has a side length = 2.88 A°. How many unit cells are present per gram of the metal?
(a) 5.62×10^{23} (b) 5.22×10^{22} (c) 5.32×10^{21} (d) 6.25×10^{21}

53. Calculate the unit cell volume for a crystal having the fcc structure. Molar mass of the compound = 22 g mol^{-1} density = 0.67 g cm^{-3}.
(a) 28.22×10^{-23} cm^3 (b) 26.82×10^{-23} cm^3 (c) 20.24×10^{-23} cm^3 (d) 21.82×10^{-23} cm^3

54. Metallic silver crystallizes in the fcc lattice. Radius of the atom = 1.45A°. Assuming close packing calculate the density in g cm^{-3}.At. wt = 108
(a) 10.9 (b) 10.4 (c) 11.12 (d) 9.9

55. Sodium metal at moderately high temperatures has the bcc structure. Assume the density to be nearly 1 g cm^{-3}. Calculate the atomic radius, making reasonable assumptions about packing of atoms.
(a) 1.732A° (b) 1.892 A° (c) 1.837 A° (d) 1.720 A°

56. The face centred unit cell of nickel has an edge length of 352.39 pm. If the density of nickel is 8.9 g/cc. The calculated alue of Avogadro constant is
(at mass of nickel = 58.7 u)
(a) 5.028×10^{23} (b) 6.028×10^{23} (c) 6.82×10^{23} (d) 6.203×10^{24}

57. An organic compound crystallizes with an orthorhombic lattice; a = 6.8 A°, b = 4.4 A°, c = 7.2 A°; density = 1.42 g cm^{-3}. The unit cell has 2 molecules. Calculate molar mass in g mol^{-1}.
(a) 85 (b) 112 (c) 78 (d) 92

58. A certain crystal has (i) density = 0.456 g cm^{-3}, (ii) a fourfold axis of symmetry and (iii) 4 molecules per unit cell. Side lengths of the unit cell are a = b = 658 pm and c = 593 pm. Calculate the molar mass in g mol^{-1}.
(a) 16.20 (b) 17.62 (c) 19.25 (d) 23

59. A certain compound has a molar mass = 155 g mol^{-1} and density of 4 g cm^{-3}. The crystal structure orthorhombic and has 4 molecules in the unit cell, given a = 634 pm, b = 784 pm. Calculate c.
(a) 581 pm (b) 652 pm (c) 518 pm (d) 625 pm

60. CsCl lattice is cubic with say Cl^- ions at the corners and with just one Cs^+ ion at the centre of the unit cell. Assume that the cation and the anions are in contact along the body diagonal of the cube. Density = 4 g cm^{-3}. Molar mass of CsCl = 168.4 g mol^{-1} given that Cs^+ ion has a radius, r^+ = 1.76 A°. Calculate the radius of the Cl^- ion (i.e., r^-) in A°.
(a) 1.708 (b) 1.920 (c) 1.625 (d) 1.807

61. In a mixed oxide, O^{2-} ions form a close packed structure. If X^{3+} occupy one-fourth of the tetrahedral voids and one-fourth of the octahedral voids are occupied by Y^{2+} ions what is the formula of the oxide?
(a) $X_2(YO_4)$ (b) $(XO)Y_4$ (c) X_4OY_4 (d) XY_4O_4

62. Lithium metal crystallizes in the bcc lattice. If the experimental density= 4.5 x 10^2 kg m^3 and the vacancy i.e., unocupied volume (due to defect structure) is 10% calculate the edge length (cm) given at mass of lithium = 6.99 g cm^{-3}.
(a) 3.59×10^{-8} (b) 4.12×10^{-8} (c) 3.24×10^{-8} (d) 4.27×10^{-8}

63. A certain metal crystallizes in the bcc lattice; the side length of the unit cell = 4 A°. Calculate the molar volume in cm^3 mol^{-1}.
(a) 16.429 (b) 14.296 (c) 12.964 (d) 19.264

64. A compound AB of molar mass 80 g mol^{-1} has a tetragonal lattice: a = 4.5 A°, b = c = 3 A° at 25°C. Assume that there are two molecules in the unit cell calculate the density in g ml^{-1}
(a) 5.266 (b) 6.222 (c) 6.562 (d) 5.222

65. Given the following data for the unit cell of a crystalline solid: a = 11.2 A°, b = 10.8 A°, c = 6.2 A° $\alpha = 92°$, $\beta = 82°$, $\gamma = 91°$, state the crystallographic system to which the solid belongs.
(a) Rhombic (b) Triclinic (c) Monoclinic (d) Cubic

66. A certain metal x crystallizes with fcc lattice. The side length of the unit cell is 5 A°. The closest distance between two atoms of x is.
(a) 3.535 A° (b) 2.5 A° (c) 4.33 A° (d) 1.76 A°

67. For a certain ionic compound AB, the radius ratio of A^+ to B^- = 0.524. The side length of the unit cell is 2.76 A° radius of the anion is.
(a) 1.01 A° (b) 0.906 A° (c) 0.65 A° (d) 0.85 A°

68. The unit cell of ZnS (Zinc blende) contains 4 "molecules" of ZnS. Calculate the number of unit cells per gram. Molar mass = 97.3 g mol^{-1}.
(a) 1.547×10^{21} (b) 6.02×10^{23} (c) 1.61×10^{22} (d) 3.2×10^{22}

69. In a close packed crystal structure of the compound, AB where in the anions B^- are assumed to be hard spheres, r^- = 0.150 nm. Calculate the largest radius of the cation A^+ that will fit into an octahedral hole.
(a) 0.274 nm (b) 0.109 nm (c) 0.0620 nm (d) 0.205 nm

70. The compound AB has 'NaCl' structure. The shortest distance between the cation and the anion is 300 pm. Calculate the volume of the unit cell.
(a) 2.16×10^{-23} cm^3 (b) 2.16×10^{-22} cm^3 (c) 0.216×10^{-23} cm^3 (d) 21.6×10^{-22} cm^3

71. In the reaction, $2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)}$, the rate of reaction is expressed as $\frac{-d[O_2]}{dt} = 2.5 \times 10^{-4}$ mol L^{-1} s^{-1}. The rate of disappearance of SO_2 is
(a) 2.5×10^{-4} mol L^{-1} s^{-1} (b) 5.0×10^{-4} mol L^{-1} s^{-1} (c) 7.5×10^{-4} mol L^{-1} s^{-1} (d) 1.25×10^{-4} mol L^{-1} s^{-1}

72. A reaction A $\rightarrow$ B follows second order kinetics. Doubling the concentration of A will increase the rate of formation of B by a factor of
(a) 2 (b) $\frac{1}{2}$ (c) 4 (d) $\frac{1}{4}$

73. Rate law for a reaction is given as $r = k\,[A]^{1/3}$. The unit of rate constant is
(a) s^{-1} (b) $mol^{1/3}\ L^{-1/3}\ s^{-1}$ (c) $mol^{2/3}\ L^{-2/3}\ s^{-1}$ (d) $mol^{-2/3}\ L^{2/3}\ s^{-1}$

74. For a first order reaction, time required to complete $\frac{1}{3}$ of a reaction is
(a) $\frac{2.303}{k}\log\frac{1}{3}$ (b) $\frac{2.303}{k}\log 3$ (c) $\frac{2.303}{k}\log\frac{2}{3}$ (d) $\frac{2.303}{k}\log\frac{3}{2}$

75. The rate of a certain reaction increases 1.9 times, when temperature is raised from 300 K to 310 K. If k is the rate constant at 300 K, then the energy of activation of the reaction (in kJ mol^{-1}) is
(a) 74.8 (b) 25.1 (c) 41.8 (d) 49.7

76. For a certain reaction, $2A + 4B \rightarrow 3C$, if $\frac{-d[A]}{dt} = k_1[2B]^2$, $\frac{-d[B]}{dt} = k_2[B]^2$, $\frac{d[C]}{dt} = k_3[B]^2$ then,
(a) $6k_1 = 3k_2 = 5k_3$ (b) $12k_1 = 2k_2 = 3k_3$ (c) $k_1 = 3k_2 = 4k_3$ (d) $24k_1 = 3k_2 = 4k_3$

77. The phenomenon in which polar substances when heated produce electric current is called
(a) Piezoelectricity (b) Ferroelectricity (c) Pyroelectricity (d) Antiferroelectricity

78. Among the following, the one with antifluorite structure is
(a) NaCl (b) ZnS (c) CaF_2 (d) Na_2O

79. Volume of space occupied in face centred close packing is
(a) 52% (b) 74% (c) 68% (d) 48%

80. Formula of a compound in which atoms Q are present at the corners of a cube and atoms P at the body centre is
(a) P_2Q (b) PQ_2 (c) PQ (d) PQ_3

81. The rate constant of a reaction is 2×10^{-3} mol L^{-1} sec^{-1}. What is the order of the reaction?
(a) 1 (b) 1.5 (c) zero (d) 2

82. $A_{(g)} \rightarrow B_{(g)} + C_{(g)} + D_{(g)}$. Initial pressure for A = 250 mm; for B, C, D, initial pressures are zero. After 20 minutes, the total pressure is 550 mm calculate the rate constant in min^{-1}.
(a) 1.51×10^{-2} min^{-1} (b) 4.582×10^{-2} min^{-1} (c) 3.3×10^{-2} min^{-1} (d) 2.27×10^{-2} min^{-1}

83. The souring of milks takes 70 hrs at 20°C and 350 hrs at 5°C (one may assume I order kinetics). Calculate the activation energy.
(a) 6.772 kJ (b) 67.72 kJ (c) 7.267 kJ (d) 75.289 kJ

84. $T_{1/2}$ = 360 sec for a reaction conforming to I order kinetics. Calculate the time taken for the concentration to reduce from 0.8 M to 0.25 M.
(a) 400 sec (b) 720 sec (c) 1152 sec (d) 604 sec

85. The activation energy of a I order reaction is 90 kJ mol^{-1} calculate the magnitude of the slope of the graph of log k vs $\frac{1}{T}$.
(a) -4.7×10^3 (b) -7.4×10^3 (c) -3.7×10^3 (d) -2.35×10^3

86. The isomerization of diisopropyl ether in vapour phase to allyl acetone conforms to I order kinetics $\log k\ (s^{-1}) = 11.73 - \frac{6.36 \times 10^3}{T}$. Calculate the activation energy in kcal mol^{-1}.
(a) 21.490 (b) 29.104 (c) 19.422 (d) 41.21

87. In the thermal decomposition of a gas the following data were observed

$T_{1/2}$ (sec)	400	160	114.3
Initial pressure p° mm Hg	200	500	700

Determine the order of the reaction.
(a) 2 (b) 1 (c) 1.5 (d) zero

88. For the 1st order decomposition of azomethane at 400 mm Hg
$CH_3 - N = N - CH_3 \rightarrow C_2H_6 + N_2$, $k = 1.0 \times 10^{-2}$ min^{-1} at a certain temperature. The time taken for pressure of nitrogen to build up to 100 mm Hg is
(a) 28.7 min (b) 138 min (c) 100 min (d) 13.8 min

89. A certain catalyzed reaction at 300K proceeds at the same rate as the uncatalyzed one at 600K. The ratio of activation energy of catalyzed reaction to uncatalzyed one is

(a) 0.2 (b) 0.5 (c) 0.33 (d) 0.25

90. Derive the rate law for the following reaction $2NO_{(g)} + Br_{2(g)} \rightarrow 2NOBr_{(g)}$

Mechanism (i) $NO_{(g)} + Br_{2(g)} \underset{k^{-1}\ \text{fast}}{\overset{k_1}{\rightleftharpoons}} NOBr_{2(g)}$; (ii) $NOBr_2 + NO \xrightarrow[k_2]{\text{slow}} 2NOBr_{(g)}$

(a) Rate = K [NO] [Br_2] (b) Rate = K^1 [NO] [Br_2]^2 (c) Rate = K^1 [NO]^2 [Br_2] (d) Rate = K^1 [NO]^2 [Br_2]^2

91. The rate of the chemical reaction $C_2H_5ONa + CH_3I \rightarrow C_2H_5OCH_3 + NaI$ increases by 2.5 times for every 10°C rise of temperature. How many times will the rate of reaction increase when the temperature is increased from 25°C to 80°C?

(a) 55 (b) 77.2 (c) 154.4 (d) 308.8

92. For an elementary step of a certain reaction $A + 3B \rightarrow 2C + D$. Indicate the correct rate equation.

(a) $-\frac{d[B]}{dt} = \frac{d[C]}{dt}$ (b) $\frac{d[B]}{dt} = \frac{d[C]}{dt}$ (c) $-\frac{d[A]}{dt} = +\frac{d[D]}{dt}$ (d) $-\frac{d[A]}{dt} = +\frac{1}{3}\frac{d[B]}{dt}$

93. Which of the following statements is true?

(a) The reaction $NaOH + HCl \rightarrow NaCl + H_2O$ can be studied by the usual methods and is bimolecular.

(b) The reaction $S_2O_8^{2-} + 2I^- \rightarrow 2\,SO_4^{2-} + I_2$ is a reaction of the third order.

(c) Halogenation of methane say $CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$ etc in the presence of light is a chain reaction.

(d) The formation of quarternary ammonium halides say $(CH_3)_3\,N + CH_3X \rightarrow (CH_3)_4\,N^+\,X^-$ is a first reaction.

94. Which of the following statements is correct?

(a) NaCl, KCl and CsCl all have the same structure.

(b) NaCl has the fluorite structure.

(c) The coordination no. of NaCl and CsCl is 6:6.

(d) Number of formula units in the unit cells of NaCl and CsCl are 4 and 1 respectively.

95. In hydrated copper sulphate, the number of water molecules attached to Cu^{2+} and SO_4^{2-} ions are respectively.

(a) 4 and 1 (b) 1 and 4 (c) 5 and zero (d) 3 and 2

96. A crystalline substance has molar mass of 75 g mol^{-1}, and it belongs to bcc structure and edge length 500 pm. Its density in g cm^{-3} is its

(a) 5 (b) 2 (c) 3.5 (d) 4.8

97. r_+ = 45 pm in a compound with zinc blende structure. The ideal magnitude of r_- is

(a) 400 pm (b) 200 pm (c) 50 pm (d) 20 pm

98. For a certain ionic compound AB, radius ratio $\left(\frac{r_+}{r_-}\right)$ = 0.414. The edge length of the unit cell is 2.21 Å. What is the value of r_-?

(a) 0.78 Å (b) 0.68 Å (c) 0.87 Å (d) 0.59 Å

99. Silver (at wt 108) crystallizes in the cubic system. The number of unit cells per mole = 1.48×10^{23}. Calculate the side length of the unit cell density = 10.6 g cm^{-3}.

(a) 3.5 Å (b) 5.1 Å (c) 4.1Å (d) 4.9 Å

100. In a hypothetical isolated unit cell of NaCl the number of Na^+ and Cl^- ions present are

(a) 4 and 4 (b) 6 and 6 (c) 13 and 14 (d) 27 and 27

101. In a bcc lattice what is (i) the nearest and (ii) the next nearest number of atoms for a given atom?

(a) 8, 8 (b) 8, 6 (c) 6, 8 (d) 6, 6

102. Aluminium crystallizes in the cubic system. a = 4.05 Å density = 2.70 g cm^{-3} [At. wt = 27 g] what is the unit cell type.?
(a) fcc (b) sc (c) bcc (d) rhombohedral

103. What is the ratio of void space percentage in bcc and fcc unit cells.
(a) 1.231 (b) 1.414 (c) 1.33 (d) 1.5

104. AgI has the same structure as ZnS, density of silver iodide = 5.67 g cm^{-3}[At. wt: Ag:108 g, I:127 g] calculate the length of the unit cell edge.
(a) 5.16 Å (b) 6.02 Å (c) 6.51 Å (d) 7.22 Å

105. At room temperature sodium has a bcc structure a = 4.2906 Å. At −195°C, the density is only 4% larger but a is now 5.35 Å. What is the unit cell type at −195°C.?
(a) sc (b) trigonal (c) bcc (i.e., no change) (d) fcc

106. A first order reaction is 50% completed in 30 minutes at 27°C and in 10 minutes at 47°C. Calculate the rate constant of the reaction at 27°C and the activation energy of the reaction in kJ mol^{-1}.
(a) $k = 0.0231\ min^{-1}$, $E_a = 43.85\ kJ\ mol^{-1}$
(b) $k = 0.04385\ min^{-1}$, $E_a = 23.1\ kJ\ mol^{-1}$
(c) $k = 0.0462\ min^{-1}$, $E_a = 43.85\ kJ\ mol^{-1}$
(d) $k = 0.0693\ mol^{-1}$, $E_a = 12.5\ kJ\ mol^{-1}$

107. For the reaction $2A + B \rightarrow D + E$, the following mechanism is proposed. $A + B \rightarrow C + D$ (slow)
$A + C \rightarrow E$ (fast).
The rate law expression for the reaction is
(a) $r = k[A]^2[B]$ (b) $r = k[A][B]$ (c) $r = k[A]^2$ (d) $r = k[A][C]$

108. 50% of a first order reaction is complete in 346 s. The time required to complete 90% of the reaction in seconds is
(a) 796.8 (b) 499.2 (c) 115.15 (d) 1151.5

109. A metallic conductor among the following is
(a) ReO_3 (b) CrO_2 (c) VO (d) all the above

110. Even though, noble gas crystals and certain metals are crystalline in cubic closed pack structure, noble gases are low melting because
(a) they are held by van der Waals forces.
(b) they have low molecular weight.
(c) it is a nonmetal.
(d) gases have low melting points.



Assertion–Reason Type Questions

Directions: **Each question contains Statement-1 and Statement-2 and has the following choices (a), (b), (c)and (d), out of which ONLY ONE is correct.**

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False
(d) Statement-1 is False, Statement-2 is True

111. Statement 1
A reaction with simple stoichiometry may have a complex mechanism.

and

Statement 2
A reaction of the second order involving in its rate equation the concentrations of two different reactants A and B may show first order kinetics if one of the reactants is taken in large excess.

112. Statement 1
The number of atoms per unit cell is a primitive cubic structure is two.
and

Statement 2
The packing efficiency in simple cubic structure is 52.4%.

113. Statement 1
CrO_2 is used in making magnetic tapes.
and

Statement 2
CrO_2 is a ferromagnetic substance.

Linked Comprehension Type Questions

Directions: **This section contains 1 paragraph. Based upon the paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

Passage I
It is a rather striking observation that reaction rates are quiet sensitive to changes in temperature. Thus a change from 30°C to 40°C (about 3% change on the Kelvin scale)may produce a 100% change in the rate constant (or more) to explain this Arrhenius proposed a simple equation for the rate constant of a gaseous reaction: viz., $K = A \times \exp\left(\frac{-Ea}{RT}\right)$. A is sometimes called the frequency factor but is best described as a pre exponential factor. Ea is called an experimentally observed activation energy. It is given the explanation that molecules must posses at least this much of energy for colliding molecules to undergo the reaction. Thus $\exp\left(\frac{-Ea}{RT}\right)$ is a Boltzmann factor or probability factor for the reaction. One may formulate the above equation in the form, $\log K = \frac{-Ea}{2.303RT}$.+ log A and use it in investigation of reactions.

114. The rate constant K for a I order reaction increases from 8.1×10^{-3} min^{-1} to 3.0×10^{-2} min^{-1} when the temperature increase from 308 K to 318 K. Calculate the mean activation energy for the given temperature range.
(a) 106.6 kJ mol^{-1} (b) 160.6 kJ mol^{-1} (c) 96.6 kJ mol^{-1} (d) 126.6 kJ mol^{-1}

115. The activation energy for a reaction of the I order is 21.6 kcal mol^{-1}. Calculate the slope of the plot of log K (min) vs $\frac{1}{T}$.
(a) −5710 (b) −4720 (c) −7240 (d) −6335

116. The activation energies of two different reactions of the same order differ b25 kJ mol^{-1}. The ratio between their rate constants at 27°C is
(a) 2.198×10^2 (b) 2.25×10^4 (c) 4.44 (d) 2.5×10^3

Multiple Correct Objective Type Questions

Directions: **Each question in this section has four suggested answers of which ONE OR MORE answers will be correct.**

117. Which among the following statements are correct?

(a) For a complex reaction, showing a simple kinetic order there are many elementary steps, one of which is the slowest and rate determining.

(b) For a I order reaction K has the dimensions of a frequency but is often characterized by a frequency factor which is $\frac{RT}{h} \times e^{\frac{\Delta S^{\neq}}{R}}$.

(c) Order of reaction is in many cases not defined such as Rate = $\frac{k_1 k_2 c}{1+k_1 c}$.

(d) The true order of the reaction of the thermal decomposition of CH_3 – CHO vapour is integral.

118. Identify the correct statements among the following.

(a) CsCl should be correctly regarded as having a bcc structure.

(b) The unit cell of KCl is actually fcc but shown by x-ray diffraction to be simple cubic.

(c) The hydrated copper ion in $CuSO_4$; $5H_2O$ is 4 coordinate with a tetrahedral disposition of water molecules.

(d) Both ccp and hcp structures have the same % of void space in the unit cell.

119. The incorrect statement among the following is.

(a) The complex formed in a surface catalyzed reaction the order may depend on the reactant concentration. E.g., Rate = $\frac{ab[A]}{1+a[A]}$ where, [A] is the reactant concentration, (a, b constants).

(b) In a reaction sequence A $\xrightarrow{K_1}$ B $\xrightarrow{K_2}$ C where $K_2 >> K_1$ if one starts with [A] = C_0 at t = 0, [B] = [C] = 0 at t = 0 the concentration of B shows a maximum (and remain almost steady over a period)slowly varying with the rate constant K_1.

(c) A negative catalyst raises the activation energy of an uncatalyzed reaction.

(d) Titration of warm acidified oxalic acid against $KMnO_4$ is an auto catalyzed reaction.

Matrix-Match Type Question

Directions: **Match the elements of Column I to elements of Column II. There can be single or multiple matches.**

120.

Column-I	**Column-II**
(a) Thermal decomposition of N_2O_5 (studied by Hinshel wood)	(p) Rotation of plane of polarization of polarized light
(b) Acid catalyzed inversion of sucrose	(q) II order kinetics
(c) H_2 – Br_2 reaction yielding HBr	(r) Complex multi step mechanism
(d) Rate constant × initial reactant concentration is constant kC_0 = constant	(s) Kinetically I order

ADDITIONAL PRACTICE EXERCISE

121. The following data were obtained in the kinetic study of acid hydrolysis of ethyl acetate in presence of 0.5 M HCl.

Time in s	600	1500	2700	4200	∞
Vol. of NaOH consumed (mL)	25.7	26.9	29.7	32.1	41.4

(i) Determine the pseudo first order rate constant of the reaction.
(ii) If the rate is also first order in $[H^+]$, calculate the second order rate constant.

122. Nitrous oxide undergoes catalytic decomposition at 900°C by gold. At an initial pressure of 200 torr 50% of the reaction was completed in 53 minutes and 73% of the reaction in 100 minutes. How much of it will decompose in 100 minutes at 600 torr at the same temperature?
(a) 53% (b) 50% (c) 73% (d) 33.3%

123. The slope of the plot of log k versus $\frac{1}{T}$ for a certain reation is –9900. Calculate the activation energy of the reaction. If the rate constant of the reaction is 4×10^{-2} L mol^{-1} s^{-1} at 730K, what is the frequency factor?

124. The $t_{½}$ of the reaction A → products is 1200 s. If its initial concentration is 1 M and if the reaction is zero order in A,
(i) What is the value of the rate constant?
(ii) What concentration of A is left after 30 minutes?
(iii) What is the value of k in the units of molecules cm^{-3} min^{-1}?

125. A half order reaction is 36% complete in 128 minutes. How long will it take for the reaction to be 75% complete?

126. The unit cell dimensions of an organic solid crystallizing in orthorhombic lattice are a = 6.8A°, b= 4.4 A°, c = 7.2 A°. Its density is 1.42 g cm^{-3}. If it has two molecules per unit cell, how many moles of the compound are there in 828 g of the compound.

127. A crystal MX (rock salt structure) has molecular weight $y \times 10^{-9}$ amu, where, y is an arbitrary number in amu. If the minimum distance between cation and anion is $y^{1/3}$ pm and the observed density is 20 kg m^{-3} find the theoretical density and indicate the type of crystal defect.

128. KCl crystallizes in NaCl structure if $\frac{r^+_{Na}}{r^-_{Cl}} = 0.5$ and $\frac{r^+_{K}}{r^-_{Cl}} = 0.7$ the ratio of the edge length of a unit cell of KCl to that of NaCl is
(a) 1.205 (b) 1.937 (c) 1.172 (d) 1.133

129. NaCl is doped with 10^{-3} mol % $SrCl_2$. What is the concentration of cation vacancy?
(a) 6.02×10^{18} mol^{-1} (b) 3.01×10^{20} mol^{-1} (c) 1.56×10^{24} mol^{-1} (d) 6.02×10^{13} mol^{-1}

130. A metal crystallizes like NaCl at a particular temperature and crystallizes like CsCl at another temperature. If the unit cell lengths are 3 A° and 4 A° respectively, find the ratio of density of fcc type to bcc type of the metal.

Straight Objective Type Questions

131. A compound, AB, has a = 4.5 A°, b = c = 3 A° at 25°C. The unit cell has two particles, what is the molar volume? (in cm^3 mol^{-1})
(a) 12.19 (b) 19.21 (c) 11.92 (d) 11.29

132. E_a for a I order reaction = 105 kJ mol^{-1}. Pre exponential factor = 10^{12} sec^{-1} calculate the temperature at which $T_{1/2}$ = 2 minutes.

(a) 580 K (b) 385 K (c) 480 K (d) 512 K

133. A face centred cubic unit cell of a non-ionic non-volatile solute soluble in water has atoms A at corners, atoms B at face centres and atom C at body centre. A solution of 10 g of the solute in 100 gram water gives $\Delta T_b = 0.17°C$ K_b for water = 0.51 deg $molal^{-1}$. Calculate density in g/cm^3 if a = 6Å.

(a) 3.27 (b) 3.11 (c) 2.307 (d) 2.23

134. The unit of rate of a reaction is the same as the rate constant if the order of the reaction is

(a) zero (b) one (c) two (d) 1.5

135. For a I order reaction $\frac{K_1}{K_2} = 2$ when T_2 = 308 K and T_1 =298 K calculate the activation energy (kcal mol^{-1}).

(a) 17.22 (b) 11.27 (c) 12.72 (d) 15.32

136. $T_{1/2}$ of I order reaction is one hour. What would be the fraction of the reactant left after 30 minutes from the start of the reaction?

(a) 78.2% (b) 70.7% (c) 64.1% (d) 82.1%

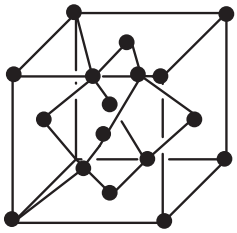
137. If 2% of initial concentration C_0 decomposes per minute at the start of a I order reaction what is $T_{1/2}$ of the reaction?

(a) 36.72 (b) 43.2 (c) 30.27 (d) 34.65

138. In the decomposition $A_{(g)} \rightarrow$ gaseous products conforming to I order kinetics, initial pressure of A = 308.2 mm. The pressure of $A_{(g)}$ after 6000 sec is 137.2 mm. Calculate the rate constant K. (t = 35°C) in sec^{-1}.

(a) 1.35×10^{-4} (b) 1.53×10^{-4} (c) 2.12×10^{-5} (d) 2.72×10^{-3}

139. Diamond lattice is shown in the figure. Carbon atoms at the corners and face centres and four atoms of carbon (shown in black)inside the unit cell at alternate tetrahedral holes. The solid is covalent. C-C bond length is 1.54 A°, what is the side length of the unit cell in A°.



(a) 3.556
(b) 3.122
(c) 2.78
(d) 2.23

140. Which of the following crystalline solids have $\left(\frac{r_+}{r_-}\right)$ ratio close to 0.75.?

(i) Rb^+Br^- (ii) Na^+F^- (iii) K^+Cl^- (iv) Cs^+I^-

(a) (i), (ii) (b) (ii), (iv) (c) (ii), (iii), (iv) (d) (i), (iii), (iv)

141. For the bimolecular dissociation of hydrogen iodide. $K = 10^8 \, T \exp\left(\frac{-E}{RT}\right)$.

Take 556 K = T. E = +44.3 kcal. Calculate K (magntude).

(a) 3.12×10^{-7} (b) 2.78×10^{-7} (c) 3.17×10^{-6} (d) 2.32×10^{-8}

142. For the reaction $2HI_{(g)} \rightarrow H_{2(g)} + I_{2(g)}$. E_a = 44.3 K cal.

ΔH = –1.35 K cal for ½ H_2 + ½ I_2 = HI. What is E_a for the reverse process (in kcal)

(a) 47.0 (b) 45.65 (c) 41.6 (d) 42.95

143. In the thermal decomposition at 518°C of a certain gaseous reactant, $T_{1/2}$ was 410 sec, for an initial pressure of 363 mm and 880 sec for an initial pressure of 169 mm. What is the apparent order of the reaction?

(a) 1.5 (b) 2 (c) 1 (d) zero

144. For the reaction system: $CH_3\text{-}COOH + C_2H_5OH \rightleftharpoons CH_3\text{–}COOC_2H_5 + H_2O$ at room temperature, the equilibrium constant K = 2.92. If the rate constant for the forward process is 2.38×10^{-4} (in magnitude), what is the rate constant for the reverse reaction in magnitude?

(a) 1.72×10^{-4} (b) 1.25×10^{-4} (c) 7.30×10^{-5} (d) 8.15×10^{-5}

145. Mechanism for the thermal decomposition of $CH_3\text{-}CHO$ is given below:

(I) $CH_3\text{-}CHO \rightarrow \dot{C}H_3 + \dot{C}HO$ (initiation)

(II) $\dot{C}H_3 + CH_3\text{-}CHO \rightarrow CH_4 + CH_3\text{-}CO$

(III) $CH_3\text{-}\dot{C}O \rightarrow \dot{C}H_3 + CO$ (propagation steps)

(IV) $\dot{C}H_3 + \dot{C}H_3 \rightarrow C_2H_6$ (termination step)

Apply the steady state treatment for $\dot{C}H_3$ radicals and arrive at the rate equation.

(a) Rate = $k\,[CH_3\text{-}CHO]^{3/2}$ (b) Rate = $k\,[CH_3\text{-}CHO]$

(c) Rate = $k\,[CH_3\text{-}CHO]^2$ (d) Rate = k

146. For a certain I order thermal decomposition of a gaseous substance; the rate constant increases in the ratio 35 : 3 for an increase in temperature from 45°C to 65°C. what is the approximate magnitude of E_a (mean value) kcal mol^{-1}?

(a) 14.62 (b) 16.42 (c) 26.41 (d) 21.46

147. Calculate the factor by which the reaction rate of a I order reaction may increase for a difference of 20 kcal mol^{-1} in the activation energies in the uncatalyzed and catalyzed reactions at 800 K.

(a) 323185 (b) 413712 (c) 192287 (d) 268337

148. Which of the following is/are examples of "negative" catalysis?

(a) Oxidation of $C_6H_5\text{-}CHO$ in the presence of anthracene.

(b) Reaction of $2SO_{2(g)} \xrightarrow{O_2} 2SO_{3(g)}$ in the presence of platinum.

(c) Acid catalysed ester hydrolysis.

(d) Haber process of synthesis of NH_3.

149. Derive the rate law: $A \underset{k_2}{\overset{k_1}{\rightleftharpoons}} B$; (fast) $B \xrightarrow{k_3} C$ (slow).

(a) $\frac{k_1 k_3}{k_2}[A]$ (b) $\left(\frac{k_1}{k_2}\right)^{\frac{1}{2}} k_3[A]$ (c) $\frac{k_2 k_3}{k_1}[A]$ (d) $\left(\frac{k_2}{k_1}\right)^{\frac{1}{2}} k_3[A]$

150. Which among the following is photovoltaic?

(a) amorphous sodium silicate (b) amorphous silicon

(c) Lead zirconate (d) Barium citrate

151. Which of the following have a unit cell, neither cubic nor hexagonal?

(a) ZnS (b) SiO_2 (c) CaF_2 (d) $CaCO_3$

152. In the fluorite (CaF_2) unit cell (i) where are the fluoride ions located? (ii) how many formula units (i.e., "molecules") of CaF_2 are present per unit cell?

(a) tetrahedral holes 8 molecules (b) octahedral sites 4 molecules

(c) tetrahedral sites or holes, 4 molecules (d) octahedral sites 6 molecules

153. White tin crystallizes in the tetragonal system. a = 5.82 A°; b = c = 3.175 A°. Given the atomic wt = 118.7 and density = 7.29 g cm^3, determine the number of atoms in the unit cell.

(a) 4 (b) 2 (c) 1 (d) 8

154. Thermal decomposition of 2-nitropropane yields propene as a major product: $\frac{d}{dt}[C_3H_6] = k\,[C_3H_7NO_2]$ $k = 1.11 \times 10^{11} \exp\left(\frac{-39300}{RT}\right)$ (sec^{-1}). Determine $T_{1/2}$ at 300°C approximately. [Take R = 2 cal deg^{-1} mol^{-1}, T = 573 K]

(a) 4884 sec (b) 4024 sec (c) 8424 sec (d) 6328 sec

155. For a certain I order reaction, $\log K\,(min^{-1}) = 33.91 - \frac{18150}{T}$. Calculate E_a (in kcal mol^{-1}).

(a) 68.4 (b) 72.6 (c) 83.6 (d) 89.9

156. When ammonia was decomposed by heating to 856°C by the passage of electricity through a tungsten wire it was found that after 100 sec, the ammonia content decreased by 13.5 mm when the initial pressure was 100 mm and 14.0 mm when it was 200 mm. What is the order of the reaction?

(a) one (b) 1.5 (c) 2 (d) zero

157. The molar mass and density of a crystalline solid $A^+ B^-$ are 235 g mol^{-1} and 5.67 g cm^3. The solid crystallizes in the cubic system, the side length of the unit cell = 6.50 A°. Determine the number of formula units AB in the unit cell.

(a) one (b) two
(c) four (d) mixed type with point defects

158. The catalytic decomposition of N_2O on a gold surface at 900°C with initial pressure = 200 mm was 50% complete in 53 min and 73% complete in 100 min. What is the order of the reaction?

(a) 0.5 (b) 1 (c) 1.5 (d) 2

159. For the reaction: $C_6H_5–N_2Cl \rightarrow C_6H_5Cl + N_2$ (I order) The rate constant increases by a factor of 10 when the temperature increases from 30°C to 50°C. At 60°C k = 0.251 min^{-1}. Calculate the pre exponential factor A in the Arrhenius rate constant equation,

(a) 1.25×10^{14} (b) 2.62×10^{15} (c) 2.26×10^{14} (d) 1.52×10^{15}

160. Hydrogenation of vegetable oils at 25°C reduces the pressure from 2 atm to 1 atm in 6.84 min. What is the mean rate of the reaction in mol L^{-1} per sec.

(a) 1×10^{-4} (b) 2×10^{-4} (c) 2×10^{-5} (d) 3×10^{-5}

161. For the reaction $2A + 2B \rightarrow C$, when the concentration of A is doubled by keeping the concentration of B constant, the rate increases four times. Keeping the concentration of A constant, when the concentration of B is doubled rate is also doubled. The order of the reaction is

(a) 4 (b) 3 (c) 2.5 (d) 1.5

162. The initial concentration of a zero order reaction is decreased from 4 mol L^{-1} to 1 mol L^{-1}. Time for half change

(a) remains the same (b) is reduced to one fourth
(c) becomes four times (d) is doubled

163. The rate constant for the reaction $2A \rightarrow$ products is 3.0×10^{-2} s^{-1}. If the rate is 2.40×10^{-2} mol L^{-1} s^{-1}, the concentration of A in mol L^{-1} is

(a) 2.4 (b) 1.4 (c) 6 (d) 0.8

164. Example of a crystal containing both Schottky and Frenkel defect is

(a) NaCl (b) KCl (c) CaF_2 (d) AgBr

165. Decomposition of $N_2O_{5(g)}$ into $NO_{2(g)}$ and $O_{2(g)}$ is a first order reaction. If the initial concentration of $N_2O_{5(g)}$ is 0.03 mol L^{-1}. Calculate the concentration of $N_2O_{5(g)}$ after 30 minutes. Given rate constant of the reaction is 1.35×10^{-4} s^{-1}.

(a) 0.003 mol L^{-1} (b) 0.0235 mol L^{-1} (c) 0.01175 mol L^{-1} (d) 0.0175 mol L^{-1}

166. For a certain reaction, when the initial concentration is 2 mol L^{-1}, the $T_{½}$ of the reaction is 0.5 and when the initial concentration is 4 mol L^{-1}, $T_{½}$ decreases to 0.25. The order of the reaction is
(a) 1 (b) 2 (c) 3 (d) 0

167. The specific reaction rate of a first order reaction is 2.3×10^{-3} s^{-1}. If the reaction started with 2 mol L^{-1}, calculate the amount remaining after 50 seconds.
(a) 1.36 mol L^{-1} (b) 0.632 mol L^{-1} (c) 0.22 mol L^{-1} (d) 1.78 mol L^{-1}

168. For the reaction, $2NO_2^- + 4H^+ + 2I^- \rightarrow I_2 + 2NO + 2H_2O$, rate $= k[NO_2][H^+]^2[I^-]$. Keeping all concentrations constant, the pH of solution was reduced by 1 unit. The rate will
(a) increase by two times (b) decrease by two times
(c) increase by four times (d) increase by hundred times

169. The activation energy of a first order reaction is 90 kJ mol^{-1}. What will be the slope of the line in the graph of log k versus $\frac{1}{T}$?
(a) −4700K (b) −4819K (c) +4819K (d) 1082K

170. What will be the rate constant of a reaction whose 2/3 life ($t_{2/3}$) is 2.3×10^5 min?
(a) 4.77×10^{-6} min^{-1} (b) 1.76×10^{-6} min^{-1} (c) 4.77×10^{-5} min^{-1} (d) 1.76×10^{-5} min^{-1}

171. For a certain ionic compound AB, the radius ratio of A^+ to B^- is 0.524. If the length of an edge is 5.52A°, the radius of the cation is
(a) 1.81 A° (b) 0.95 A° (c) 1.9 A° (d) 0.65 A°

172. The rate constant of a first order reaction which takes 40 min for 30% completion is
(a) 0.85×10^{-3} min^{-1} (b) 8.9×10^{-3} min^{-1} (c) 3.01×10^{-2} min^{-1} (d) 2.1×10^{-2} min^{-1}

173. For a reaction A → B, the activation energies for the forward and backward reactions are found to be 40 kJ mol^{-1} and 52 kJ mol^{-1}. The ΔH for the reaction is
(a) −12 kJ mol^{-1} (b) 40 kJ mol^{-1} (c) 12 kJ mol^{-1} (d) 52 kJ mol^{-1}

174. The rate constant of a reaction is 2×10^{-3} mol L^{-1} s^{-1}. If the concentration of the reactant is doubled the rate of the reaction is
(a) 2×10^{-3} mol L^{-1} s^{-1} (b) 4×10^{-3} mol L^{-1} s^{-1} (c) 5×10^{-2} mol L^{-1} s^{-1} (d) 8×10^{-3} mol L^{-1} s^{-1}

175. The rate of a first order reaction increases by 4000 times at 27°C in presence of a catalyst. The change in activation energy in presence of the catalyst
(a) 20.7 kJ mol^{-1} (b) 41.4 kJ mol^{-1} (c) 10.35 kJ mol^{-1} (d) 12.5 kJ mol^{-1}

Assertion–Reason Type Questions

Directions: **Each question contains Statement-1 and Statement-2 and has the following choices (a), (b), (c)and (d), out of which ONLY ONE is correct.**

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False
(d) Statement-1 is False, Statement-2 is True

176. Statement 1
The void space per unit cell is a minimum in the hcp and ccp structures.

and

Statement 2

The closest packing of identical spheres is observed in ccp and hcp structures.

177. Statement 1

Coordination number of the cation (or anion)in CsCl is 8.

and

Statement 2

CsCl crystal structure is made up of two interpenetrating simple cubic arrangements, one of cations and the other of anions.

178. Statement 1

The side length of a cubic unit cell is related to the molar volume of an ionic crystal.

and

Statement 2

The radius ratio, $\frac{r_+}{r_-}$ for ionic crystals provides a useful guide for the coordination number of cations and anions.

179. Statement 1

A heterogeneous (solid)positive catalyst used in a gaseous reaction is supposed to provide on its surface an alternative channel (for the same reaction)with a lower activation energy.

and

Statement 2

The mechanistic steps in a catalyzed reaction are in general different from those of the corresponding uncatalyzed reaction.

180. Statement 1

Most of the known reactions have integral orders.

and

Statement 2

The order exhibited at the start of a reaction is justifiably the 'true' order of the reaction also called order with respect to concentration.

181. Statement 1

Acid catalyzed ester hydrolysis and acid catalysed inversion of sucrose are examples of pseudo first order reactions.

and

Statement 2

The equation $K = A \exp\left(\frac{-E_a}{RT}\right)$ yields a value of A, which is temperature independent (i.e., constant)over a wide range of temperature.

182. Statement 1

In a reaction sequence $A \xrightarrow{K_1} B \xrightarrow{K_2} C \xrightarrow{K_3} \ldots$ if $k_1 << (k_2, k_3 \ldots)$ then a steady state or secular equilibrium may be observed after a fairly long time interval from the start of the reaction.

and

Statement 2

In a sequential reaction of many elementary steps every step has its own molecularity.

183. Statement 1

In parallel reactions one branch or reaction channel may be exclusively favoured at low temperatures.

and

Statement 2

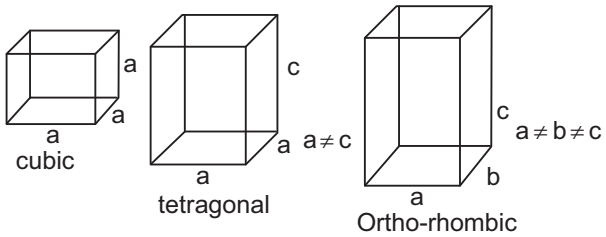
Rates of different branches of parallel reactions are always the same.

Linked Comprehension Type Questions

Directions: **This section contains 3 paragraphs. Based upon the paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

Passage I

The structure of a crystalline solid may be regarded as made up of unit cells belonging to various types of crystal system. In the cubic system one has the simple cubic, face centred cubic and body centred unit cells. In the cube all sides are equal. In the tetragonal system two sides are equal. In the orthorhombic unit cell, the three sides are unequal. In the unit cell contains n formula units of the substance in questions (atoms or "molecules") then we have $\frac{(\text{unit cell volume})}{n} \times$ Avogadro number = molar volume of the solid.



184. In the Zinc blende and fluorite lattices which ions and how many of these occupy the tetrahedral holes.
(a) Zn^{2+} 4 ; F^- 8 (b) F^- 8 ; Zn^{2+} 8
(c) Zn^{2+} 6 ; F^- 6 (d) F^- 6 ; Zn^{2+} 8

185. In a crystal AB, of the fcc type, distance between cation and anion is 300 pm, what is the unit cell volume in cm^3?
(a) 2.61×10^{-23} (b) 2.16×10^{-22}
(c) 2.68×10^{-21} (d) 3.12×10^{-22}

186. Number of tetrahedral voids present in a unit cell of NaCl are
(a) 4 (b) 12
(c) 8 (d) zero

Passage II

In order to account for the abnormal increase in rate of reaction with temperature Arrhenius proposed the equation $k = A\bar{e}^{E_a/RT}$. The pre exponential factor A is only slightly influenced by temperature. $\bar{e}^{E_a/RT}$ is known as boltzmann factor and E_a is considered to be a constant in a narrow range of temperature the logarithmic form of the equation or $\log k = \log A - \frac{E_a}{2.303RT}$ plot of log K vs $\frac{1}{T}$ would be linear with the slope $\left(-\frac{E_a}{2.303R}\right)$. For two different temperatures we have $\log\left(\frac{K_2}{K_1}\right) = -\frac{E_a}{2.303R}\left(\frac{1}{T_2}-\frac{1}{T_1}\right)$. In differential form we have $\frac{d\ln K}{dT} = \frac{E_a}{RT^2}$.

187. For a certain reaction $\log K = \text{constant} - \frac{2090}{T}$. Calculate E_a in kJ mol^{-1}.
(a) 45 (b) 50 (c) 35 (d) 40

188. For a certain I order reaction $E_a = 52$ kJ $mole^{-1}$. Calculate the % change of K per degree rise of temperature at 300 K.
(a) 6.95 (b) 5.6 (c) 4.9 (d) 3.7

189. If t_1 and t_2 are the half lives of the reaction at temperatures T_1 and T_2, then activation energy of the reaction can be given by

(a) $\frac{0.693R(T_2-T_1)}{T_1T_2}\ln\left(\frac{t_1}{t_2}\right)$ (b) $\frac{RT_1T_2}{(T_2-T_1)}\text{In}\left(\frac{t_1}{t_2}\right)$

(c) $\frac{-2.303\,RT_1T_2}{T_2-T_1}\log\frac{t_1}{t_2}$ (d) $\frac{+2.303RT_1T_2}{T_2-T_1}\ln\frac{t_1}{t_2}$

Passage III

For a majority of chemical reactions, the rate constant and hence the rate of chemical reaction increases with temperature. The temperature dependence is given by the Arrhenius equation.

A catalyst increases the rate of reaction by providing an alternate reaction pathway with lower activation energy.

The given temperature due to lower activation energy greater proportion of collision have energy greater than the activation energy leading to faster reaction rate.

190. In the kinetic study of a certain I order reaction the rate constant K, changes by a factor 3 for a temperature change from 37°C to 47°C. i.e., $\frac{K_2}{K_1} = 3$. Calculate the temperature at which $K = 2K_1$.

(a) 40.2°C (b) 44.7°C (c) 43.3°C (d) 45.2°C

191. By what factor does the rate constant K change at 800 K if the use of a catalyst lowers the activation energy by 10 K cal $mole^{-1}$.

(a) 518 (b) 622 (c) 477 (d) 562

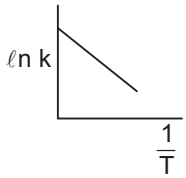
192. For the scheme of parallel I order reactions, $A \xrightarrow{K_1} B(E_a = E_1)$, $A \xrightarrow{K_2} C(E_a = E_2)$, Express the overall activation energy E in terms of E_1 and E_2.

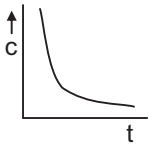
(a) $E_1 + E_2$ (b) $\frac{E_1}{K_1}+\frac{E_2}{K_2}$ (c) $\frac{(E_1E_2)^{1/2}}{K_1+K_2}$ (d) $\frac{K_1E_1+K_2E_2}{K_1+K_2}$

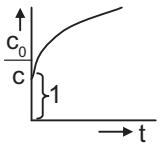
Multiple Correct Objective Type Questions

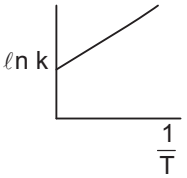
Directions: **Each question in this section has four suggested answers of which ONE OR MORE answers will be correct.**

193. For a reaction for which K = 0.25 min, which of the following graphs are constant.

(a) 

(b) 

(c) 

(d) 

194. The organic compound, RCl reacts with NaI in acetone to yield a precipitate.

(a) The reaction is irreversible under the conditions it is conducted.

(b) The reaction is a bimolecular displacement of 'Cl' by 'I'.

(c) The precipitate has an fcc lattice.

(d) RCl but not NaI in soluble in acetone.

195. Indicate the reactions among the following which are kinetically of the same order.

(a) Thermal decomposition of N_2O_5
(b) Thermal decomposition of Arsine
(c) The H_2-Br_2 reaction
(d) The catalysed decomposition of H_2O_2 in solution

196. Radius ratio $\left(\frac{r_+}{r_-}\right)$ calculations

(a) assume 100% ionicity for crystals to which they apply.
(b) indicates coordination number of ions 8, 6 etc.
(c) are only approximate since many factors such as polarizabilities of ions are ignored.
(d) can decide the crystal structure of ionic solids.

197. Which of the following statements are correct?

(a) Maximum packing of identical sized spheres is achievable in hcp structures.
(b) Scattering of x-rays by electrons in diffraction work is analogous to Compton scattering.
(c) x-ray diffraction methods are not useful in fixing hydrogen atoms in crystals.
(d) A change of temperature may result in a change of structure say bcc $\rightleftharpoons$ fcc.

Matrix-Match Type Question

Directions: **Match the elements of Column I to elements of Column II. There can be single or multiple matches.**

198.

Column I	**Column II**
(a) Madelung constant	(p) solvation energy
(b) Type of coordination in ionic structures	(q) Radius ratio $\frac{r_+}{r_-}$
(c) (Related to) solubility in polar solvents	(r) Lattice energy
(d) Stability of crystals	(s) Temperature of thermal dissociation

199.

Column I	**Column II**
(a) Pseudo I order reaction	(p) $T_{1/2} \alpha\ c_0^{1-n}$ (c_0 = initial concentration)
(b) Order of reaction = $n \neq 1$	(q) Acid catalyzed ester hydrolysis
(c) Rate constant proportional to concentration of H^+ ion.	(r) Reactant concentration decreases in geometric progression as time measured increases in arithmetic progression.
(d) Rate constant has dimensions $\left(\frac{\text{mole}}{\text{litre}}\right)^{1-n}$ (time)$^{-1}$ $n \neq 1$	(s) $\left[\frac{1-f^{n-1}}{f^{n-1}}\right] = (n-1)\ c_0^{1-n}\ Kt\ (n \neq 1)\ \left[\frac{c}{c_0}=f\right]$

200.

Column I	**Column II**
(a) Close packing of equal sized spheres	(p) void space approximately 26%
(b) Bond structure with filled and empty bands at $T \rightarrow 0$	(q) Insulators
(c) Atom-layers ..ABC ABC…..	(r) ccp structure
(d) Narrow energy gap between valence and conduction bands	(s) Semiconductors

SOLUTIONS

ANSWER KEYS

Topic Grip

1. 0.1127 mm s^1
2. 318.7 K
3. 2.376 hr
4. 6.498×10^{-2} mm^{-1}
5. 8
6. 11.56 g cm^{-3}
7. a, 12, 6
b, 0.414
8. 4.549 g cm^{-3}
9. 4.083 Å
10. octahedral voids
2.878 Å

11. (b)	**12.** (c)	**13.** (d)
14. (d)	**15.** (b)	**16.** (a)
17. (c)	**18.** (d)	**19.** (b)
20. (d)	**21.** (a)	**22.** (d)
23. (c)	**24.** (a)	**25.** (b)

26. (a)
27. (a), (b), (c)
28. (a), (c), (d)
29. (a), (b), (d)
30. (a) → (s)
(b) → (p), (s)
(c) → (q), (r)
(d) → (q)

IIT Assignment Exercise

31. (d)	**32.** (a)	**33.** (b)
34. (a)	**35.** (b)	**36.** (a)
37. (a)	**38.** (b)	**39.** (d)
40. (a)	**41.** (a)	**42.** (b)
43. (b)	**44.** (d)	**45.** (a)
46. (b)	**47.** (c)	**48.** (a)
49. (d)	**50.** (b)	**51.** (c)
52. (c)	**53.** (d)	**54.** (b)
55. (c)	**56.** (a)	**57.** (d)
58. (b)	**59.** (c)	**60.** (d)
61. (a)	**62.** (a)	**63.** (d)
64. (c)	**65.** (b)	**66.** (a)

67. (b)	**68.** (a)	**69.** (c)
70. (b)	**71.** (b)	**72.** (c)
73. (c)	**74.** (d)	**75.** (**d**)
76. (d)	**77.** (c)	**78.** (d)
79. (b)	**80.** (c)	**81.** (c)
82. (b)	**83.** (d)	**84.** (d)
85. (a)	**86.** (b)	**87.** (a)
88. (a)	**89.** (b)	**90.** (c)
91. (d)	**92.** (c)	**93.** (c)
94. (d)	**95.** (a)	**96.** (b)
97. (b)	**98.** (a)	**99.** (c)
100. (c)	**101.** (b)	**102.** (a)
103. (a)	**104.** (c)	**105.** (d)
106. (a)	**107.** (b)	**108.** (d)
109. (d)	**110.** (a)	**111.** (b)
112. (d)	**113.** (a)	**114.** (a)
115. (b)	**116.** (b)	

117. (a), (b), (c)
118. (b), (d)
119. (a), (b), (d)
120. (a) → (r), (s)
(b) → (p), (r), (s)
(c) → (r)
(d) → (q)

Additional Practice Exercise

121. $K = 1.66 \times 10^{-4}$ s^{-1},
$K_2 = 3.32 \times 10^{-4}$ mol^{-1} s^{-1}
122. c
123. $A = 1.44 \times 10^{12}$ L mol^{-1} s^{-1}
124. (i) $K = 4.17 \times 10^{-4}$ mol L^{-1} s^{-1}
(ii) C = 0.25 M
(iii) $K = 1.5 \times 10^{19}$ molecules cm^{-3} min^{-1}
125. 320 minutes
126. 9
127. Metal excess defect
128. (d)
129. (a)
130. 4.74

131. (a)	**132.** (b)	**133.** (c)

134. (a)	**135.** (c)	**136.** (b)
137. (d)	**138.** (a)	**139.** (a)
140. (d)	**141.** (b)	**142.** (c)
143. (b)	**144.** (d)	**145.** (a)
146. (c)	**147.** (d)	**148.** (a)
149. (a)	**150.** (b)	**151.** (d)
152. (c)	**153.** (b)	**154.** (a)
155. (c)	**156.** (d)	**157.** (c)
158. (b)	**159.** (a)	**160.** (a)
161. (b)	**162.** (b)	**163.** (d)
164. (d)	**165.** (b)	**166.** (b)
167. (d)	**168.** (d)	**169.** (a)
170. (a)	**171.** (b)	**172.** (b)
173. (a)	**174.** (a)	**175.** (a)
176. (a)	**177.** (a)	**178.** (b)
179. (b)	**180.** (d)	**181.** (c)
182. (b)	**183.** (c)	**184.** (a)
185. (b)	**186.** (c)	**187.** (d)
188. (a)	**189.** (b)	**190.** (c)

191. (a)
192. (d)
193. (a), (c)
194. (a), (b), (c)
195. (a), (b), (d)
196. (a), (b), (c), (d)
197. (a), (c), (d)
198. (a) → (r)
(b) → (q)
(c) → (p), (r)
(d)→ (q), (r), (s)
199. (a) → (p), (q), (r)
(b) → (p), (s)
(c) → (q)
(d) → (p), (s)
200. (a) → (p), (r)
(b) → (q), (s)
(c) → (p), (r)
(d) → (s)

HINTS AND EXPLANATIONS

Topic Grip

1. For a reaction of order, n, $T^{1/2} \times C_0^{n-1}$ = constant. In this case $T^{1/2}$ increase with p_o. In fact

$T^{1/2} \times p_0^{n-1}$ $\frac{T_{1/2}}{p_o}$ is nearly constant

i.e., $\frac{290}{65} = 4.46$, $\frac{460}{105} = 4.38$,

$\frac{670}{150} = 4.47$, $\frac{820}{185} = 4.43$

$\therefore$ n = 0 Mean value of $\frac{T_{1/2}}{p_o} = 4.435$

$\therefore$ $\frac{p_o}{T_{1/2}} = 0.2255$ mm sec^{-1}.

since $\frac{dp}{dt} = -K$.

$P = p_0 - Kt$.

Put $P = \frac{p_0}{2}$, Then $K\,T^{1/2} = \frac{p_0}{2}$.

$\therefore$ $K = \frac{p_o}{2T_{1/2}} = 0.1127$ mm sec^{-1}

2. $\therefore$ $2.303 \log\left(\frac{K_2}{K_1}\right) = \frac{E_0}{R}\left(\frac{(T_2 - T_1)}{T_1 T_2}\right)$

i.e., $2.303 \log 4 = \frac{E_a}{R}\left(\frac{10}{298 \times 308}\right)$

$\therefore$ $\frac{2.303 \times 0.6021 \times 298 \times 308}{10} = \frac{E_0}{R}$

= 12727. Take R = 1.987.

E_a = 25286 cal $mol^{-1} \simeq$ 25.3 K cal mol^{-1}For the second part we have $\frac{2.303 \times 0.6021 \times 308 \times T}{T - 308}$

= 12727

i.e., $\frac{(427.1)T}{T - 308} = 12727$.

$12727\,T - 308 \times 12727 = 427\,T$

$\therefore$ $T = \frac{3919916}{12300} = 318.7$ K

3. $K = \frac{2.303}{1}\log\frac{100}{70} = \frac{2.303}{t}\log\frac{100}{30}$

$\therefore$ $t \log \frac{100}{70} = \log \frac{100}{30}$. $t \times 0.1549 = 0.5229$

$\therefore$ t = 3.376 hrs. The question is how much more time.

$\therefore$ Ans. (3.376 − 1) hr = 2.376 h

4. After a long time A would have all been decomposed

i.e., one mole $\rightarrow$ 3 mole (pr =270 mm).

$\therefore$ initial presure of A = 90 mm

$K = \frac{2.303}{10} \log \frac{90}{p}$. Now suppose after 10 minute the partial pressure of A = p. Then total pressure = p + 3 (90 − p)

i.e., 176 = 270 − 2 p.

$\therefore$ p = 47 mm. $\therefore$ $K = \frac{2.303}{10} \log\left(\frac{90}{47}\right)$.

$K = 6.498 \times 10^{-2}$ min^{-1}

5. Rate becomes twice for every 20°C rise

30°C − K 50°C − 2 K

70°C − 4 K 90°C − 8 K

6. $\frac{a^3}{4} \times 6.02 \times 10^{23} = \frac{207.2}{d}$

$\therefore$ $\frac{(4.92)^3}{4} \times 10^{-24} \times 6.02 \times 10^{23} = \frac{207.2}{d}$

$\therefore$ $17.924 = \frac{207.2}{d}$ $\therefore$ d ≃ 11.56 g cm^{-3}

7. (a) Take for example the NaCl structure; the Na^+ atoms alone form an fcc lattice. Now consider a face centre, in one plane there are 4 nearest neighbours but there are two more planes perpendicular to this. Hence 12 nearest neighbours. By an extension of the same argument the next nearest neighbors are 6 in number.

(b) Assume that the anions are of larger size. In an fcc lattice anions are in contact along a face diagonal. $\therefore$ if a is the side length, then $4(r^-) = \sqrt{2}\,a$ and $2(r^+ + r^-) = a$ $\therefore \frac{(r^+ + r^-)}{2r^-} = \frac{1}{\sqrt{2}} = 0.707$. $\therefore$ $\frac{(r^+ + r^-)}{r^-} = 1.414$.

$\therefore$ $\frac{r^+}{r^-} = 0.414$

8. The unit cell contains one $Cs^+ I^-$ pair

$\therefore \quad \frac{a^3}{1} \times 6.02 \times 10^{23} = \frac{260}{d} = 57.156 = \frac{260}{d}$

$\therefore \quad d = 4.549 \text{ g cm}^{-3}$

9. $\frac{a^3}{4} \times 6.02 \times 10^{23} = \frac{107.88}{10.53}$ If a is in A°

Then $\frac{107.88}{10.53} \times \frac{10^{24} \times 4}{6.02 \times 10^{23}} = a^3 = 68.073$ A°.

$a = 4.083$ A°

10. (i) $BaTiO_3$; octahedral voids of the oxide lattice.

(ii) In the fcc lattice atoms touch along a face diagonal in close packing.

$\therefore$ closest distance between centres of atoms is one half of the face diagonal length

i.e., $\frac{1}{2} \times \sqrt{2} \times 4.07$ A° = 2.878 A°

11. $T_{1/2}$ is proportional to $C_0^{(1-n)}$. In this case $1 - n = ½$

$\therefore \quad n = 0.5$

12. $\frac{-dc}{dt} = Kc$

$\therefore$ very nearly $\frac{dc}{cdt} = K$,

Now $\frac{dc}{c} = \frac{0.84}{100}$. Taking dt = 1 min. we have

$K = \frac{0.84}{100} \text{ min}^{-1}$

$\therefore \quad T_{1/2} = \frac{0.693}{K} = \frac{0.693 \times 100}{0.84}$ min

$= 82.5$ min

13. Rate becomes rate constant when concentration of each of the reactants are independently unity.

14. $\frac{a^3}{n} \times N_0 = \frac{M}{P} \cdot \frac{5^3 \times 10^{-24}}{n} \times 6.02 \times 10^{23}$

$= \frac{239.2}{12.7} \quad \therefore \quad \frac{125 \times 6.02}{10 \times n} = 18.835.$

$\therefore \quad n \simeq 4. \therefore$ fcc

15. $4r^- = \sqrt{2} \times 4.7$ A. $\therefore \; r^- = 1.66$ A°

$2(r^+ + r^-) = 4.7$

$\therefore \quad r^+ + r^- = 2.35$ A°. $\therefore \; r^+ = 0.69$ A°

16. Both are true statements 2 is the reason for statement 1

17. Statement 1 is true. Statement 2 is wrong [C] Continuously increases.

18. Statement 1 not true. Statement 2 is true

19. Both statements are correct. But statement 2 is not an explanation to statement 1

20. Statement 1 wrong, statement 2 correct.

21. $K = 1.50 \times 10^{-2} = \frac{2.303}{60} \log\left(\frac{300}{p}\right)$

$\therefore \quad \log\left(\frac{300}{p}\right) = 0.3908.$

$\therefore \quad \frac{300}{p} = 2.4592 \therefore \; p = 122$ mm

$\therefore \quad p_{N_2} = (300 - 122)$ mm = 178 mm

$pC_2H_6 = 178$ mm

22. Ratio of rate constant = $\exp\left(\frac{(70-50)10^3}{8.314 \times 320}\right)$

$= \exp(7.5174) \simeq 1840$

23. The data fit into order zero

$C = Co - Kt$, $50 = 100 - K(40)$.

$\therefore \quad K = \frac{50}{40}$ % per min = 1.25% per min

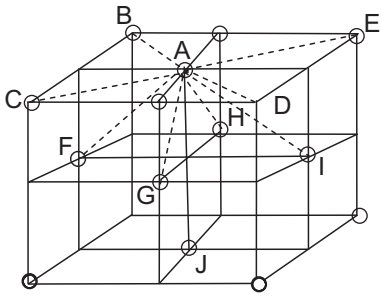
$25 = 100 - K\,60$, $75 = K \times 60$

$\therefore \quad K = \frac{75}{60} = 1.25$ % per min

$K = \frac{1.25}{100}$ per min $= 0.0125$ mol L^{-1} min^{-1}

Since 10% of the reaction complete itself in 8 min, 90% will be completed in 72 min

24.



Sphere A is equidistant from spheres B, C, D and E in the same layer. There are 4 more spheres in the next down layer. F, G, H, and I at the same distance. In

the same way there will be four spheres at the same distance in the upper layer. Hence the no. of nearest neighbours is 12. Next nearest means sphere in position identical to J. There will be six of them. Hence the no. of nearest neighbours = 12 and next nearest neighbours 6.

25. $\frac{5^3\times10^{-24}\times6\times10^{23}}{n}=\frac{75}{2},\frac{125\times6}{10n}=\frac{75}{2}$

$$n=\frac{(125\times62)}{10\times75}=2$$

BCC

26. $\Delta T_b = K_b \times m;\ 0.291 = 0.51 \times \frac{5}{M}\times\frac{1000}{100}$

$0.291 = \frac{0.51}{M}\times 50$

$\therefore \quad M=\frac{0.51\times50}{0.291}\simeq 87.6\ g\ mol^{-1}$

$\frac{5^3\times10^{-24}}{n}\times6\times10^{23}=\frac{87.6}{0.874}\therefore n\simeq0.75$

The value is close to n = 1. The deviation is due to experimental error in boiling point determination $\therefore$ structure is sc

27. a, b, c correct

28. a, c, d correct. Ceasium halide structure is correctly described as two interpenetrating SC structures (oberve) solution of PbI_2 in hot water is colourless

29. a, b, d correct

IIT Assignment Exercise

31. $2.303\log\left(\frac{1\times10^{-4}}{K}\right)$

$$=\frac{60\times10^3}{8.314}\times\frac{10}{273\times283}$$

$\therefore \quad \log\left(\frac{1\times10^{-4}}{K}\right)=0.4056.$

$\therefore \quad \frac{1\times10^{-4}}{K}=2.544$

$\therefore \quad K=3.93\times10^{-5}\ s^{-1}=4\times10^{-5}\ s^{-1}$

32. Obviously the rate is proportional to $[A]^2$ but ineendent of [B].

$\therefore$ Partial orders are (2, 0)

33. $T^{1/2}\ \alpha\ C_0^{1-n}$. In this case n = 3.

$\therefore \quad T^{1/2}\ \alpha\ \frac{1}{C_0^2}$. Actually one may give the proportionality constant also $\frac{-dC}{dt}=Kc^3$

$\therefore \quad \frac{-dC}{C^3}=K\,dt.$

By integration $\frac{1}{2C^2}=Kt+\text{constant}$

Constant $=\frac{1}{2C_0^2}$. $\therefore$ $Kt=\frac{1}{2}\left(\frac{1}{C^2}\frac{1}{C_0^2}\right)$.

Put $t=T_{1/2}$ and $C=\frac{C_0}{2}$.

$$K\,T_{1/2}=\frac{1}{2}\left(\frac{4}{C_0^2}\frac{1}{C_0^2}\right)=\frac{1}{2}\times\frac{3}{C_0^2}$$

$\therefore \quad T_{1/2}=\frac{3}{2K}\times\frac{1}{C_0^2}$

34. $K\ (s^{-1})=\frac{2.303}{10}\log\frac{22.8}{13.8}$

$$=5.022\times10^{-2}\ s^{-1}$$

Also $K\ (s^{-1})=\frac{2.303}{20}\log\left(\frac{22.8}{8.25}\right)$

$$=5.084\times10^{-2}\ s^{-1}$$

Also $K\ (s^{-1})=\frac{2.303}{10}\log\left(\frac{13.8}{8.25}\right)$

$$=5.145\times10^{-2}\ s^{-1}$$

The drift in the values is due to experimental error, one may take an average value

$$=5.084\times10^{-2}\ s^{-1}$$

35. Since $-Ea/RT\ K = A\ e\ \ln K = \ln A\ -\frac{Ea}{RT}$.

$$2.303\log K=2.303\log A\ -\frac{Ea}{RT}$$

$\therefore \quad \log K=\log A\ -\frac{Ea}{2.303R}\times\frac{1}{T}$ compare

$$\log K=33.91\ -\frac{7933}{T}$$

we have $\frac{Ea}{2.303R}=7933$

$\therefore \quad Ea = \dfrac{7933 \times 2.303 \times 1.987}{1000}$

$= 36.30 \text{ K cal mol}^{-1}$

36. Suppose V at t = 0 = V_0, then V after 1 hr = $V_{0\alpha} = 35.6$ and V after 4 hrs = $V_{0\alpha}{}^4 = 13.8$.

Dividing $\alpha^3 = \dfrac{13.8}{35.6}$

$\therefore \quad \alpha = 0.729$

$\therefore \quad V_0 = \dfrac{V_{0\alpha}}{\alpha} = \dfrac{35.6}{0.729} = 48.83$ ml.

Also $V_0 = \dfrac{V_{0\alpha^4}}{\alpha^4} = \dfrac{13.8}{(0.729)^4}$ $V_0 = 48.86$ ml.

Mean value 48.85 ml

37. $t_1 = \dfrac{2.303}{K}\log\dfrac{100}{40}$ $t_2 = \dfrac{2.303}{K}\log\left(\dfrac{100}{100-x}\right)$

$\therefore \quad \dfrac{t_1}{t_2} = 4.106 = \dfrac{0.3979}{\log\left(\dfrac{100}{100-x}\right)}$

$\therefore \quad \log\left(\dfrac{100}{100-x}\right) = \dfrac{0.3979}{4.106}$

$\log\left(\dfrac{100}{100-x}\right) = 0.09691,$

$\therefore \quad \dfrac{100}{100-x} = 1.25 = 1.25\,(100 - x) = 100$

$\therefore \quad x = 20\%$

38. $K = \dfrac{2.303}{30}\log\dfrac{100}{80} = \dfrac{2.303}{t}\log\dfrac{100}{10} = \dfrac{2.303}{t}$

$3.23 \times 10^{-3} = \dfrac{1}{t}$ $\therefore$ t = 309.6 min

39. Clearly $37.03 \times 0.05 = 1.8515$; 19.15×0.1

$= 1.915$; $9.45 \times 0.2 = 1.89$.

i.e., $T_{1/2} \times C_0$ is nearly constant

$\therefore$ order = 2

40. Rate = $K c^{1.5}$, Rate $R_1 = K c^{1.5}$,

Rate $R_2 = K (3c)^{1.5}$

$\therefore \quad \dfrac{R_2}{R_1} = 3^{1.5} = 5.196$

41. $K = \dfrac{2.303}{350}\log\left(\dfrac{100}{30.88}\right) = 3.3579 \times 10^{-3} \text{ min}^{-1}$

$\therefore \quad T_{1/2} = \dfrac{0.693}{K} = 206.4$ min

42. For a zero order reaction; $\dfrac{-dp}{dt} = K$

$\therefore \quad p = Kt + p_0$. In this case po = 65 mm

$\therefore \quad p_{15\,sec} = (-0.1127 \times 15 + 65)$ mm = 63.31 mm

But $\underset{2\,moles}{2NH_3} \rightarrow \underset{4\,mole}{N_2} + 3H_2$.

$\therefore$ if p = (po − x)

Total pressure = (po − x) + 2x = po + x. in this case x = (65 − 63.31) mm = 1.69 mm

$\therefore$ Total pressure = 65 + 1.69 = 66.69 mm

Then equation shows that 2 moles of NH_3 yields 3 moles of H_2

$\therefore$ after 50 sec decrease in pressure of NH_3

= (0.1127 × 50) mm = 5.635 mm

$\therefore$ partial pressure of H_2 = 8.45 mm

43. $T_{1/2} = 27$ min.

$\therefore \quad K = 2.567 \times 10^{-2} \text{ min}^{-1}$ If after time t, $p_A = 312 - x$,

Total pressure = (312 − x) + 3x

= 312 + 2x

312 + 2x mm = 468 mm.

$\therefore \quad x = 78$ mm. $p_A = (312 - 78)$ mm $p_A = 234$ mm

$\therefore \quad 2.567 \times 10^{-2} = \dfrac{2.303}{t}\log\left(\dfrac{312}{234}\right)$

$\therefore \quad \dfrac{2.303}{2.567 \times 10^{-2}}\log\left(\dfrac{312}{234}\right) = 11.21$ min

44. $T_{1/2} \alpha C_0^{1-n}$, in the case $T_{1/2} \alpha C_0$.

$\therefore \quad n = 0$

45. $T_{1/2} = \dfrac{0.693}{K}$ $\therefore$ $T_{1/2}$ is inversely proportional to K

$\therefore \quad \dfrac{K_2 \text{ at } 400\,K}{K_1 \text{ at } 300\,K} = 10.$

$2.303 \log\left(\dfrac{K_2}{K_1}\right) = \dfrac{Ea}{R}\left(\dfrac{100}{300 \times 400}\right)$

$\therefore \quad \dfrac{Ea}{R} = \dfrac{2.303 \times 300 \times 400}{100}$

$$E_{a\ kJ\ mol^{-1}} \simeq \frac{2.303 \times 300 \times 400 \times 8.314}{100 \times 1000}$$

$Ea \simeq 23\ KJ\ mol^{-1}$

46. The process is first order since the time taken for doubling the amount is the same. Equation for the process is

$$t = \frac{2.303}{k}\log\left(\frac{a}{a+x}\right)$$

Since the rate increases with progress of the process

$$k = \frac{0.693}{t_{1/2}} = \frac{0.693}{15} = 0.0462\ min^{-1}$$

$$3 \times 60 = \frac{2.303}{0.0462}\log\left(\frac{n}{50}\right)$$

Solving $n = 2.04 \times 10^5$

47. When $t_{1/2} = 10$ minutes

$$k = \frac{0.693}{10 \times 60} = 1.155 \times 10^{-3}$$

$$\log 1.155 \times 10^{-3}$$

$$= \log 4 \times 10^{13} - \frac{98.6 \times 10^3}{2.303 \times 83.4 \times T}$$

$$\frac{5149.6}{T} = 16.412$$

$T = 313.8\ K$

48. Since K_1 and K_{-1} are both a greater than K, one may assume at equilibrium

i.e., $\frac{[C]}{[A][B]} = \frac{K_1}{K_{-1}} = (\text{constant})$

$\therefore$ [C] = (constant) x [A] [B] = K [A] [B]

rate = K[C] = K K [A] [B] = K^1 [A] [B]

49. In a bcc structure, the unit cell has 2 atoms along the body diagonal = $4r = \sqrt{3}\,a$ where a = side length

$\therefore$ $r = \frac{\sqrt{3}a}{4}$. Volume of the unit cell = a^3

volume of two atoms = $2 \times \frac{4}{3}\pi r^3$

$\therefore$ occupied volume is $\frac{\left(2 \times \frac{4}{3}\pi r^3\right)}{a^3}$

$$= \frac{\left(2 \times \frac{4}{3}\pi\right)r^3}{\left(\frac{4}{\sqrt{3}}\right)^3 r^3} = \frac{8}{3} \times \frac{\pi \times 3\sqrt{3}}{64} = \frac{\pi\sqrt{3}}{8} = 0.68$$

$\simeq 68\%$ $\therefore$ void space = 32%

50. Each corner of the hexagon is shared by 3 hexagons. $\therefore$ Corners contribute $\frac{6}{3} = 2$ point to each hexagon with one more at the centre each hexagon has 2 + 1 point. Since hexagon is common to 2 unit cells and there are 2 hexagonal faces we have $2 \times \frac{3}{2} = 3$ points. Inside the hexagon there are 3 more points.

$\therefore$ each unit cell shown in the fig. has 6 lattice points.

51. x atoms = $8 \times \frac{1}{8}$ = one x. 6 face centres

$= \frac{6}{2} = 3$ atoms of y

$\therefore$ formula = xy_3

52. One mole has 6.02×10^{23} atoms.

$\therefore$ one gram has $\frac{6.02 \times 10^{23}}{M}$ atoms where, M is the molar mass

$\therefore$ in bcc lattice number of unit cell per gram $\frac{6.02 \times 10^{23}}{2M} = n$ (say)

volume per mole, $V = \frac{M}{7.86}\ cm^3$. volume per gram is $\frac{1}{7.86}\ cm^3$

volume per unit cell = $(2.88 \times 10^{-8})\ cm^3$

$= (23.89 \times 10^{-24})\ cm^3$

$\therefore$ $n = \frac{10^{24}}{7.86 \times 23.89} = 5.3255 \times 10^{21}$

53. $\frac{a^3}{4} \times 6.02 \times 10^{23} = \left(\frac{22}{0.67}\right)$

$\therefore$ $a^3 = \frac{22}{0.67} \times \frac{4}{6.02 \times 10^{23}}$

$a^3 = 21.82 \times 10^{-23}\ cm^3$

54. In the fcc lattice atoms in close packing are in contact along the face diagonal. $\therefore$ $4r = \sqrt{2} \times a$ where, a side length.

$\therefore$ $a\ cm = \frac{4 \times 1.45 \times 10^{-8}}{\sqrt{2}} = 4.1018 \times 10^{-8}\ cm$

$$\frac{a^3}{4} \times 6.02 \times 10^{23} = \frac{M}{\rho}$$

$$\frac{(4.1018)^3 \times 10^{-24} \times 6.02 \times 10^{23}}{4} = \frac{108}{\rho}$$

$$10.386 = \frac{108}{\rho}.$$

$\therefore \quad \rho = 10.399 \text{ g cm}^{-3}$

55. $\frac{a^3}{2} \times 6.02 \times 10^{23} = \frac{23}{1} = 23 \text{ cm}^3 \text{ mol}^{-1}$

$a^3 = \frac{23 \times 2}{6.02 \times 10^{23}} = 7.64 \times 10^{-23} = 76.4 \times 10^{-24}$,

$a = 4.243 \times 10^{-8}$

$a = 4.243 \times 10^{-8}$ Å, in a bcc lattice. $4r = \sqrt{3}\, a$

$\therefore \quad r = \frac{\sqrt{3}a}{4} = \left(\frac{1.732}{4} \times 4.243\right)$ Å $= 1.837$ Å

56. $D = \frac{ZM}{a^3 N}$

$$8.9 = \frac{4 \times 58.7}{(352.39 \times 10^{-10} \text{ cm})^3 \times N}$$

57. $\frac{abc}{2} \times 6.02 \times 10^{23} = \frac{M}{1.42}$

$\frac{6.8 \times 4.4 \times 7.2}{2 \times 10} \times 6.02 \times 1.42 = M$

$M = 92 \text{ g mol}^{-1}$

58. $\frac{6.58 \times 6.58 \times 5.93 \times 6.02}{40} \times 0.456$

$= M = 17.62 \text{ g mol}^{-1}$

59. $\frac{6.34 \times 7.84 \times c \times 6.02}{40} = \frac{155}{4}$

$\therefore \quad c = \frac{155}{4} \times \frac{40}{6.34 \times 7.84 \times 6.02}$ Å

$= 5.18$ Å $= 518$ pm

60. $\frac{a^3}{1} \times 6.02 \times 10^{23} = \frac{168.4}{4}$

$\therefore \quad a^3 = \frac{168.4}{4 \times 6.02 \times 10^{23}} = 6.99 \times 10^{-23} \text{ cm}^3$

$a^3 = 69.9 \times 10^{-24} \text{ cm}^3$. $a = 4.119 \times 10^{-8}$ cm

$= 4.119$ A°

length of body diagonal $= 2(r_+ + r_-)$

$= 4.119 \times \sqrt{3} = 7.134$ A°

$\therefore \quad r^+ + r^- = 3.567$ A° $r_+ = 1.76$ A°.

$\therefore \quad r_- = 1.807$ A°

61. Assuming fcc structure four O^{2-} ions, two X^{3+} ions and one Y^{2+} ion. Formula X_2YO_4 Charge neutrality $= (2) \times (+3) + (1) \times (+2) + (4)\ (-2) = 0$

62. $\frac{a^3}{2} \times 6.02 \times 10^{23} = \frac{M}{p} = \frac{6.99}{p}$. In this P is the ideal density. Observed density $= 0.45 \text{ g cm}^{-3}$ is only 90% of P.

$\therefore \quad 0.45 = 0.9 \times p$

$\therefore \quad p = \frac{0.45}{0.90} = 0.5 \text{ g cm}^{-1}$

$\therefore \quad \frac{a^3}{2} \times 6.02 \times 10^{23} = \frac{6.99}{0.5} = 13.98$

$$a^3 = \frac{13.98 \times 2}{6.02 \times 10^{23}} \text{ cm}^3$$

$\therefore \quad a^3 = 4.6445 \times 10^{-23}$

$\therefore \quad a = 3.59 \times 10^{-8}$ cm

63. $\frac{a^3}{2} \times 6.02 \times 10^{23} = \frac{M}{p} =$ molar volume, V

$\therefore \quad V = \frac{64 \times 10^{-24} \times 6.02 \times 10^{23}}{2} \text{ cm}^3$

$= 19.264 \text{ cm}^3 \text{ mol}^{-1}$

64. $\frac{4.5 \times 3 \times 3 \times 10^{-24}}{2} \times 6.02 \times 10^{23}$

$= \frac{80}{p} = \frac{40.5 \times 6.02}{20}$

$\therefore \ p = \frac{80 \times 20}{40.5 \times 6.02} \text{ g cm}^{-3} = 6.562 \text{ g cm}^{-3}$

65. It is Triclinic.

66. In the fcc lattice atoms (in a close-packed structure) are in contact along a face diagonal.

$\therefore \quad 4r = 5 \times \sqrt{2}$ distance of closest approach is

$2r = \frac{5 \times \sqrt{2}}{2} = \frac{7.07}{2} = 3.535$ Å

67. (i) For the radius ratio of 0.524. We have an fcc lattice. Side length = 2.76 Å

$\therefore$ $(r_+ + r_-) = 1.38$ Å. $r_+ = 0.524\ r_-$ (Assume cation anion contact along the side)

$$\therefore \quad \left(r_+ + \frac{r_+}{0.524}\right) = r_+\left(1 + \frac{1}{0.524}\right) = 1.38$$

$$= \frac{1.524}{0.524}\ r_+$$

$$\therefore \quad r_+ = 1.38 \times \frac{0.524}{1.524}\ \text{Å} = 0.474\ \text{Å}$$

$$r_- = 1.38 \times \frac{1}{1.524} = 0.906\ \text{Å}$$

(ii) On the contrary one may assume that the cation is too small and the anions are in contact along a face diagonal.

Then $4\,r_- = \sqrt{2} \times 2.76$ Å

$\therefore$ $r^- = 0.976$ Å Then $r_+ = 0.524 \times 0.976$

$= 0.511$ Å.

The two approaches give different values. Actually the first approach is more reasonable.

68. Number of molecules of ZnS per gram

$$= \frac{6.02\times10^{23}}{97.3}$$

$\therefore$ number of unit cells per gram $= \dfrac{6.02\times10^{23}}{97.3\times4}$

$= 1.547 \times 10^{21}$

69. For a octahedron with close packed anion side length of the octahedron $= 2 \times 0.15$ nm

$= 0.3$ nm

$\therefore$ centre to centre distance of cation and anion is $\dfrac{0.3}{\sqrt{2}} = 0.212$ nm

$\therefore$ cation radius $= (0.212 - 0.15)$ nm which is 0.062 nm.

70. Shortest distance = 300 pm = 3A°.

$\therefore$ side length = 6 A°. volume $= (6 \times 10^{-8})^3$ cm^3

$= 216 \times 10^{-24}$ cm^3 $= 2.16 \times 10^{-22}$ cm^3

71. $$-\frac{1}{2}\frac{d[SO_2]}{dt} = -\frac{d[O_2]}{dt} = 2.5 \times 10^{-4}\ \text{mol L}^{-1}\ \text{s}^{-1}$$

$$-\frac{d[SO_2]}{dt} = 5.0 \times 10^{-4}\ \text{mol L}^{-1}\ \text{s}^{-1}$$

72. $$-\frac{d[A]}{dt} = \frac{d[B]}{dt} = k\,[A]^2$$

when the conc. of A is doubled,

$$\left[-\frac{d[A]}{dt}\right]_1 = k[2A]^2 = k \times 4[A]^2$$

$$\frac{\left(-\dfrac{d[A]}{dt}\right)_1}{\left(-\dfrac{d[A]}{dt}\right)} = \frac{k \times 4 \times [A]^2}{k[A]^2} = 4$$

73. $$k = \frac{\text{rate}}{[A]^{1/3}} \Rightarrow \frac{\text{mol litre}^{-1}\ \text{s}^{-1}}{\text{mol}^{1/3}\ \text{litre}^{-1/3}}$$

$$= \text{mol}^{2/3}\ \text{L}^{-2/3}\ \text{s}^{-1}$$

74. $$t = \frac{2.303}{k}\log\frac{a}{\frac{2a}{3}} = \frac{2.303}{k}\log\frac{3a}{2a}$$

$$= \frac{2.303}{k}\log\frac{3}{2}$$

75. $$\log\frac{k_2}{k_1} = \frac{E_a}{2.303\ R}\frac{T_2 - T_1}{T_1T_2}$$

$$\log 1.9 = \frac{E_a}{2.303\times8.314} \times \frac{310-300}{310\times300}$$

$$E_a = 49.7\ \text{kJ mol}^{-1}$$

76. $$\frac{-1}{2}\frac{d[A]}{dt} = \frac{-1}{4}\frac{dB}{dt} = +\frac{1}{3}\frac{dC}{dt}$$

substituting $\dfrac{1}{2} \times 4k_1[B]^2 = \dfrac{1}{4} \times k_2[B]^2 = \dfrac{1}{3} \times k_3[B]^2$

$$2k_1 = \frac{1}{4}k_2 = \frac{1}{3}k_3$$

$$24k_1 = 3k_2 = 4k_3$$

77. Pyroelectricity is the phenomenon in which polar substances produce electric current when heated.

78. Na_2O has antifluorite structure.

79. Since fcc is the closest packed arrangement, the volume of space occupied is 74%.

80. Q atoms contribute one atom to unit cell and P atoms also contribute one atom to the unit cell.

$\therefore$ formula is PQ.

81. The unit of the rate constant are in (mol L^{-1}) per sec. Obviously the order is zero.

82. After 20 minutes partial pressure of A = (250 – p) mm. Partial pressures of B, C and D are p mm each.

$\therefore$ Total pressure = (250 – p) + 3p

= 250 + 2p = 550 mm.

$\therefore$ p = 150 mm.

$\therefore$ partial pressure of A after 20 min is (250 – 150) mm. The question indicates in its wording that the reaction conforms to I order.

$$\therefore \quad k = \frac{2.303}{20}\log\left(\frac{250}{100}\right) = 4.582 \times 10^{-2}\ \text{min}^{-1}$$

83. One may take 70 hours and 350 hours as representative of $T_{1/2}$ values.

$$\therefore \quad 2.303\ \log\left(\frac{350}{70}\right) = \frac{Ea}{R} = \left[\frac{15}{293 \times 288}\right]$$

Taking R = 8.314 J mol deg^{-1} and calculating, one gets Ea = 75.289 kJ mol^{-1}

84. $T_{1/2}$ = 360 sec

$$\therefore \quad K = \left(\frac{0.693}{360}\right)\ \text{sec}^{-1}$$

$$= \frac{2.303}{t}\log\left(\frac{0.8}{0.25}\right)$$

$\therefore$ t = 604 sec

85. $\log K = (\text{constant}) - \left[\frac{Ea}{2.303R}\right] x \frac{1}{T}$

$$\text{Slope} = -\frac{90 \times 10^3}{2.303 \times 8.314} = -4.7 \times 10^3$$

86. $\frac{-Ea}{2.303R} = -6.36 \times 10^3$

$$\therefore \quad Ea = \frac{6.36 \times 10^3 \times 2.303 \times 1.987}{10^3}\ \text{K cal mol}^{-1}$$

Ea = (6.36 × 2.303 × 1.987) K cal mol^{-1}

= 29.104 K cal mol^{-1}

87. It is clear that $T_{1/2} \times po$ = constant

$$\therefore \quad T_{1/2} \alpha \frac{1}{po}$$

$\therefore$ order = 2

88. $CH_3N = N - CH_3 \rightarrow C_2H_6 + N_2$

t = 0	400 mm Hg	0	0
t = t	400 – p	p	p

given p = 100

$$t = \frac{2.303}{k}\log\frac{400}{400-100} = \frac{20303}{10^{-2}}\log 4/3$$

= 28.7 min

89. $k_{cat} = k_{uncat}$

$$e^{\frac{-E_{a(cat)}}{RT}} = e^{\frac{-E_{a(uncat)}}{RT}}$$

$$\frac{E_{a(cat)}}{300} = \frac{E_{a(uncat)}}{600}$$

$$\frac{E_{a(cat)}}{E_{a(uncat)}} = \frac{300}{600} = 0.5$$

90. By steady state approximation.

$$K_1[NO][Br_2] = K_{-1}[NOBr_2] + K_2[NOBr_2]\ [NO]$$

$$\therefore \quad [NOBr_2] = \frac{K_1\ [NO]\ [Br_2]}{K_{-1} + K_2[NO]}$$

$$\therefore \quad \text{Rate} = K_2\ [NOBr_2]\ [NO]$$

$$= \frac{K_1K_2[NO]^2[Br_2]}{K_{-1} + K_2[NO]}$$

$$= \frac{\left(\frac{K_1K_2}{K_{-1}}\right)[NO]^2[Br_2]}{1 + \frac{K_2}{K_{-1}}[NO]} \quad \text{since } K_2 << K_{-1}$$

$\therefore$ rate = $K^1\ [NO]^2\ [Br_2]$

91. Since rate becomes 2.5 times for 10°C rise.
The increase is by $(2.5)^{5.5}$ when temperature increase from 25°C to 80°C.

92. For the reaction the rate of reaction is $-\frac{d[A]}{dt} = \frac{d[D]}{dt}$

93. Chlorination of methane in presence of light is a chain reaction.

94. Unit cell of NaCl has 4 units and that of CsCl has only one unit.

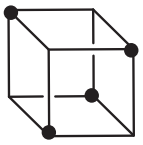
95. In $CuSO_4.5H_2O$ four H_2O molecules are co-ordinated to Cu^{2+} and one H_2O molecule is hydrogen bonded to SO_4^{2-}.

96. $p = \dfrac{z.M}{a^3N} = \dfrac{2\times 75\,g\,mol^{-1}}{\left(5\times10^{-8}\,cm\right)^3\times 6.02\times10^{23}\,mol^{-1}}$

$= 2\ g\ cm^{-3}$

97. The Zn^{2+} ions are in tetrahedral holes of small cubes. If side length is a then face diagonal is $\sqrt{2}\times a = 2\ r^-$ but

$\sqrt{3}\ a = 2\ (r^+ + r^-)$



a

$\therefore \quad \dfrac{\sqrt{3}}{\sqrt{2}} = \dfrac{r^+ + r^-}{r^-} = \dfrac{r^+}{r^-} + 1 = \dfrac{1.732}{1.414} = 1.2249$

$\therefore \quad \dfrac{r^+}{r^-} = 0.2249$

$\therefore \quad r^- = \dfrac{45}{0.2249} = 200\ pm.$

98. The ratio, 0.414 is ideal for octahedral coordination. Edge length = 2.21 A° = 2 ($r_+ + r_-$). Since $\dfrac{r_+}{r^-} = 0.414$

$r_+ = 0.414\ r^-$

$\therefore \quad 2.21 = 2\ (0.414\ r^- + r_-) = 2\ (1.414\ r^-)$

$\therefore \quad r^- = \dfrac{2.21}{2\times1.414}\ A° = 0.78\ A°$

99. Molar volume $= \dfrac{108}{10.6}\ cm^3 = 10.1887\ cm^3$

$\therefore$ volume of the unit cell =

$a^3 = \dfrac{10.1887\times10^{24}\ A^3}{1.48\times10^{23}}$

$a^3 = 68.84\ A°;\ a = 4.098\ A° \simeq 4.1\ A°$

100. Solution Cl^- ions are present at the corners and face centres of the cube. Na^+ ions are present at the edge centres and body centre.

101. Ans. 8 and 6 as a figure would show.

102. $\dfrac{\left(4.05\times10^{-8}\right)^3\times6.02\times10^{23}}{n} = \dfrac{27}{2.7} = 10$

$n = \dfrac{66.43\times6.02}{10\times10} \simeq 4.$ Thus fcc type.

103. The void space % in bcc = 32

void space % in fcc (also ccp) = 26

ratio = 1.231

104. In ZnS the S^{2-} ions form an fcc lattice. Zn^{2+} ion occupy four of the tetrahedral holes. In AgI, we have iodide ion forming fcc lattice. The unit cell has 4 formula units of AgI.

$\therefore \quad \dfrac{a^3}{4}\times6.02\times10^{23} = \dfrac{235}{5.67}$

$\therefore \quad a^3 = \dfrac{41.4462\times4}{6.02\times10^{23}} = 27.539\times10^{-23}$

$= 275.39\times10^{-24}\ cm^3$

$a = 6.51\times10^{-8}\ cm = 6.51\ A°$

105. $\dfrac{\left(4.2906\times10^{-8}\right)\times6.02\times10^{23}}{2} = \dfrac{23}{p_1};\ \dfrac{\left(5.35\times10^{-8}\right)^3}{n}$

$\times6.02\times10^{23} = \dfrac{23}{p_2}\left(\dfrac{5.35}{4.2906}\right)^3\times\dfrac{2}{n} = \dfrac{p_1}{p_2} = \dfrac{1}{1.04}.$

$\dfrac{1.9396\times2}{n} = \dfrac{1}{1.04}$

$\therefore \quad n = 4 \qquad \therefore$ fcc type.

106. $k_{27°C} = \dfrac{0.693}{t_{1/2}} = 0.0231\ min^{-1}$

$K_{47°C} = \dfrac{0.693}{10} = 0.0693\ min^{-1}$

$\log\dfrac{0.0693}{0.0231} = \dfrac{E_a}{2.303\times8.314}\times\left(\dfrac{1}{300} - \dfrac{1}{320}\right)$

$E_a = 43.85\ kJ\ mol^{-1}$

107. $r = k[A][B]$

Rate determining step is the slow step

108. $k = \dfrac{0.693}{t_{1/2}} = \dfrac{0.693}{346} = 2\times10^{-3}$

$t = \dfrac{2.303}{2\times10^{-3}}\log\dfrac{100}{10} = 1151.5\ s$

109. All ReO_3, CrO_2 and VO exhibit good conducting property.

110. The atoms in noble gas crystals are held together by van der Waals forces and so they show lower melting points.

111. Both statements are correct but second statement is not the explanation to first statement.

112. Statement 1 is wrong and statement 2 is correct.

113. Both statements are correct. Statement 2 is the correct explanation to statement 1.

114. $2.303 \log\left(\frac{3.0\times10^{-2}}{8.1\times10^{-3}}\right)=\frac{Ea}{8.314}\left[\frac{10}{308\times318}\right]$

By calculation Ea ≃ 106.64 kJ mol^{-1}

115. Slope $= \frac{-21.6\times10^{3}}{2.303\times1.987} = -4720.$

$$\left[\log K=\frac{-Ea}{2.303R}\times\frac{1}{T}+\text{constant}\right]$$

116. $\log\left(\frac{k_1}{k_2}\right)=\frac{E_2-E_1}{2.303\,RT}$

$= 4.35$

$\frac{k_1}{k_2} = 2.25\times10^{4}$

117. The order of the reaction $CH_3CHO \rightarrow CH_4 + CO$ is $\frac{3}{2}$

118. CsCl contain Cs^+ in a simple cubic structure of Cl^-. $[Cu(H_2O)_4]^{2+}$ is square planar.

119. a, b, d correct. Catalyst does not alter the activation energy of a reaction, but it provides alternate reaction pathway.

Additional Practice Exercise

121. (i) For two times t_1 and t_2, we write

$$k=\frac{2.303}{t_1}\log\left(\frac{a}{a-x_1}\right)$$

$$k=\frac{2.303}{t_2}\log\left(\frac{a}{a-x_2}\right)$$

$$(t_2-t_1)=\frac{2.303}{k}\left[\log\left(\frac{a-x_1}{a-x_2}\right)\right]$$

$$=\frac{2.303}{k}\log\left(\frac{T_\alpha-T_{t_1}}{T_\alpha-T_{t_2}}\right)$$

Since the titre value at the beginning of the reaction is not given, we can use data at two different time intervals and calculate k.

Taking data at 600 s and 1500 s we get

$$k=\frac{2.303}{900}\log\left(\frac{15.7}{14.5}\right)=8.84\times10^{-5}\ s^{-1}$$

Taking data at 1500 s and 2700 s we get

$$k=\frac{2.303}{1200}\log\left(\frac{14.5}{11.7}\right)=1.8\times10^{-4}\ s^{-1}$$

Taking data at 2700 s and 4200 s we get

$$k=\frac{2.303}{1500}\log\left(\frac{11.7}{9.3}\right)=1.53\times10^{-4}\ s^{-1}$$

Taking 2nd and 4th data

$$k=\frac{2.303}{2700}\log\left(\frac{14.5}{9.3}\right)=1.64\times10^{-4}\ s^{-1}$$

Average of last three calculation = $k = 1.66\times10^{-4}\ s^{-1}$

The reaction is first order because of constancy of k.

(ii) The observed rate constant k' may be expressed as

$k' = k_2[H^+]$

or $k_2=\frac{k'}{[H^+]}=\frac{1.66\times10^{-4}}{0.5}$

$= 3.32\times10^{-4}\ L\ mol^{-1}\ s^{-1}$

122. The reaction is first order since k calculated for 50% and 73% are the same on using first order rate equation.

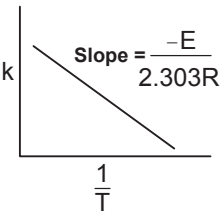
Since the time for the same fraction of reaction to take place is the same for a first order reaction at constant temperature

$t_{73\%}$ = 100 minutes

123. $k = Ae^{-E/RT}$

$$\log k = \log A - \frac{E}{2.303RT}$$

$$\text{slope} = -9900 = -\frac{E}{2.303R}$$



Or $E = 9900\times2.303\times8.314\times10^{-3} = 189.5$ kJ

$$\log(4\times10^{-2}) = \log A - \frac{E}{2.303\,RT}$$

$$= \log A - \frac{189.5}{2.303\times8.314\times10^{-3}\times730}$$

$-1.4 = \log A - 13.56$

$\log A = 12.16$

$A = 1.44 \times 10^{12}\ L\ mol^{-1}\ s^{-1}$

124. (i) For a zero order reaction,

C_O = initial concentration.

C = conc. at time t

$$k = \frac{C_o - C}{t} = \frac{0.5}{1200} = 4.17 \times 10^{-4}\ mol\ L^{-1}\ s^{-1}.$$

(ii) After 30 min, $C_o - C = kt$

$= 4.17 \times 10^{-4} \times 30 \times 60$

$= 75.1 \times 10^{-2}$

$C = 1 - 0.751 = 0.25\ M$

(iii) $k \Rightarrow 4.17 \times 10^{-4} \times 6 \times 10^{23}$ molecules $\times 0.001\ cm^{-3} \times 60\ min^{-1}$

$= 4.17 \times 6 \times 6 \times 10^{17}$ molecules $cm^{-3}\ min^{-1}$

$= 1.5 \times 10^{19}$ molecules $cm^{-3}\ min^{-1}$

125. For a nth order reaction,

$$k = \frac{1}{t(n-1)}\left[\frac{1}{C^{n-1}} - \frac{1}{C_o^{n-1}}\right]$$

$$k = \frac{1}{128} \times -2\left[\frac{1}{(0.64)^{-1/2}} - \frac{1}{(1)^{-1/2}}\right]$$

$$= \frac{1}{128} \times -2\,(0.8-1)$$

$= 3.125 \times 10^{-3}\ mol^{½}\ L^{-½}\ min^{-1}$

$$t = -\frac{2}{3.125 \times 10^{-3}}[0.5-1] = 320 \text{ minutes}$$

126. $$\rho = \frac{\text{mass of molecule} \times \text{no. of molecules}}{\text{volume}}$$

$1.42\ g\ cm^{-3} =$

$$\frac{M \times 2}{N_A\ mol^{-1} \times 6.8 \times 10^{-8}\ cm \times 4.4 \times 10^{-8}\ cm \times 7.2 \times 10^{-8}\ cm}$$

$M = 92\ g\ mol^{-1}$

$$\text{Number of moles} = \frac{828}{92} = 9 \text{ moles}$$

127. Since crystal belongs to NaCl type

$n = 4$

$M = y \times 10^{-9}\ g$

$a = 2y^{1/3}$ pm

$$\rho = \frac{nM}{N_0 a^3}$$

$$\rho = \frac{4 \times y \times 10^{-9} \times 10^{-3}\ Kg\ mol^{-1}}{6.02 \times 10^{23}\ mol^{-1} \times \left(2y^{1/3} \times 10^{-12}\ m\right)^3}$$

$= 0.83\ Kg\ m^{-3}$

The calculated density is less than the observed density, the crystal has metal excess defect.

128. Edge length of NaCl $= 2\left(r^+_{Na} + r^-_{Cl}\right)$

Edge length of KCl $= 2(r^+_k + r^-_{Cl})$

$$\frac{\text{Edge length of KCl}}{\text{Edge length of NaCl}} = \frac{r^+_k + r^-_{Cl}}{r^+_{Na} + r^-_{Cl}}$$

$$= \frac{\frac{r^+_K}{r^-_{Cl}} + 1}{\frac{r^+_{Na}}{r^-_{Cl}} + 1} = \frac{1.7}{1.5}$$

$= 1.133$

129. No. of vacancy per mole

$$= \frac{6.02 \times 10^{23} \times 10^{-3}}{100}$$

$= 6.02 \times 10^{18}$

Since one $SrCl_2$ will make a vacancy of one Na^+ ion.

130. For fcc crystal $\rho(fcc) = \frac{4M}{N_0 a^3}$

For bcc crystal $\rho(bcc) = \frac{2M}{N_0 a^3}$

$$\frac{\rho(fcc)}{\rho(bcc)} = \left(\frac{4}{2}\right)\left(\frac{4}{3}\right)^3 = 4.74$$

131. $\frac{abc}{2} = 6.02 \times 10^{23}$ = molar volume $(cm^3\ mol^{-1})$

$$= \frac{(4.5 \times 3 \times 3) \times 10^{-24}}{2} \times 6.02 \times 10^{23}\ ]$$

$$= \frac{40.5 \times 6.02}{20}\ cm^3\ mol^{-1}$$

$= 12.1905\ cm^3\ mol^{-1}$

132. $K = A \exp\left(\frac{-E_a}{RT}\right) = 10^{12} \times e^{\frac{-105\times10^3}{8.314\times T}}$

$T_{1/2} = 2 \text{ min} = 120 \text{ sec}. \; K = \frac{0.693}{120}.$

$\therefore \quad \frac{0.693}{120} = 10^{12} \times \exp\left(\frac{-105\times10^3}{8.314\,T}\right)$

$\therefore \quad \exp \frac{105\times10^3}{8.314\times T} = \frac{10^{12}\times120}{0.693}$

$= 1.7316 \times 10^{14}$

$\therefore \quad \frac{105\times10^3}{8.314\times T} = 14.2384 \times 2.303$

$T = \frac{105\times10^3}{8.314\times14.2384\times2.303}$

$= 385 \text{ K}$

133. $\Delta T_b = K_b \times m, \; 0.17 = 0.51 \times \frac{10}{M}\times\frac{1000}{100}.$

$M = \frac{0.51\times100}{0.17}$ g. $M = 300 \text{ g mol}^{-1}.$

The formula unit = AB_3C. One unit cell contains one formula unit.

$\therefore \quad \frac{a^3}{1} 6.02 \times 10^{23} = \frac{300}{P}$

$216 \times 10^{-24} \times 6.02 \times 10^{23} = \frac{M}{\rho}$

$= \frac{216\times6.02}{10} = \frac{300}{\rho}$

$\therefore \quad \rho = \frac{300\times10}{216\times6.02} \text{ g cm}^3 = 2.307 \text{ g cm}^3$

134. $\frac{-dc}{dt} = K$. i.e., zero order

135. $2.303 \log\left(\frac{K_1}{K_2}\right) = \frac{E_a}{R}\left(\frac{10}{298\times308}\right)$

$\therefore \quad E_a = 2.303 \times (\log 2) \times \frac{2\times298\times308}{10\times10^3} \text{ K cal mol}^{-1}$

$E_a = \frac{0.693\times2\times298\times308}{10^4} \text{ K cal mol}^{-1}$

$= 12.72 \text{ K cal mol}^{-1}$

136. Suppose the initial concentration is c_0.

c_1 after 30 min = $c_0\,\alpha$. c_2 after one hour = $c_0\,\alpha^2$

$c_0\,\alpha^2 = \frac{c_0}{2},$

$\therefore \quad d^2 = \frac{1}{2}, \; \alpha = \frac{1}{\sqrt{2}}.$

$\therefore \quad c_0\,\alpha = \frac{c_0}{\sqrt{2}} = c_1$

$\frac{c_1}{c_0} = \frac{1}{\sqrt{2}} = 0.707 = 70.7\%$

137. $\frac{-dc}{dt} = \frac{\Delta c}{\Delta t} = kc.$

$\therefore \quad \frac{\Delta c}{c\Delta t} = k; \; \frac{\Delta c}{c} = \frac{2}{100}. \; \Delta t = 1 \text{ min}$

$\therefore \quad \frac{2}{100\times1} = k = 2 \times 10^{-2} \text{ min}^{-1}.$

$\therefore \quad T_{1/2} = \frac{0.693}{k} = \frac{0.693\times100}{2} \text{ min}$

$T_{1/2} = \frac{0.693}{2} \text{ min} = 34.65 \text{ min}$

138. $K = \frac{2.303}{6000} \log\left(\frac{308.2}{137.2}\right)$

$= 1.349 \times 10^{-4} \text{ sec}^{-1}$

139. distance between a corner and face centre is ℓ, A tetrahedral angle = $\theta = 109°28'$. $\cos\theta = -\frac{1}{3}$

$\therefore \quad \ell^2 = (1.54)^2 + (1.54)^2 + 2\,(1.54)\,(1.54) \times \frac{1}{3}. \; (\text{Å}^2)$

$\ell^2 = 2 \times (1.54)^2 \left[1 + \frac{1}{3}\right] = (1.54)^2 \times \frac{8}{3} \text{ Å}^2$

$\therefore \quad \ell = (1.54 \times 1.633) \text{ A°}$

$\therefore$ length of face diagonal

$= 2 \times 1.54 \times 1.633 = \sqrt{2}\,a$

$\therefore$ side length $a = \frac{1.54\times1.633\times2}{\sqrt{2}}$

$= (1.54 \times 1.633 \times \sqrt{2}) \text{ A°} = 3.556 \text{ A°}$

140. The radius ratio for Na^+F^- is close to 0.7.

141. $K = 10^8 \times 556 \times \exp\left(\frac{-44.3\times10^3}{2\times556}\right)$

$= 10^8 \times 556 \times e^{-39.838} = 2.78 \times 10^{-7}$

142. $H_{2(g)} + I_{2(g)} \rightarrow 2HI_{(g)}\ \Delta H_b = -2 \times 1.35$ K cal $= -2.7$ K cal

$\therefore \quad \Delta H_f = +2.7$ K cal

$\therefore \quad E_f = E_b + \Delta H;\ 44.3 = E_b + 2.7$ (K cal)

$\therefore \quad E_b = (44.3 - 2.7)$ K cal $= 41.6$ K cal

143. It is clear that $T_{1/2}\ a_0^{n-1}$ = constant when n = 2

i.e., $410 \times 363 = 148830$

$880 \times 169 = 148720$

$\therefore$ second order

144. $\dfrac{2.38\times10^{-4}}{K_b} = 2.92$

$\therefore \quad K_b = \dfrac{2.38\times10^{-4}}{2.92} = 8.1507 \times 10^{-5}$

in magnitude

145. The steady state approximation gives

$k_1\ [CH_3\text{–}CHO] = k_4\ [\dot{C}H_3]_2$

The overall rate $= k_2\ [\dot{C}H_3]\ [CH_3\text{–}CHO]$

$= [\text{constant}]\ [CH_3\text{–}CHO]^{3/2}$

146. $2.303 \log\left(\dfrac{35}{3}\right) = \dfrac{E_a}{R}\left(\dfrac{20}{318\times338}\right)$

$\therefore \quad E_a = 2.303 \log\left(\dfrac{35}{3}\right) \times \dfrac{318\times338\times2}{20\times1000}$

$= 26.41$ K cal mol^{-1}

147. $k = Ae^{-E_a/RT}$ taking ratio for the uncatalysed and catalyzed reaction. $\dfrac{k_1}{k_2} = \exp\left(\dfrac{20\times10^3}{2\times800}\right) = \exp\left(\dfrac{20000}{1600}\right)$

$\dfrac{k_1}{k_2} = \exp\left(\dfrac{25}{2}\right) = 268337$

148. Anthracene act as a negative catalyst in the oxidation of C_6H_5CHO.

149. $A \underset{K_2}{\overset{K_1}{\rightleftharpoons}} B$ (Fast). There is a pre equilibrium: $\dfrac{k_1}{k_2} = \dfrac{[B]}{[A]}$

Rate $= k_3\ [B] = \dfrac{k_1}{k_2} \times k_3 [A]$

150. Amorphous silicon is photovoltaic.

151. $CaCO_3$ exist in orthorhombic structure.

152. In fluorite structure $\bar{F}$ ions occupied tetrahedral voids. There are four CaF_2 formula units per unit cell.

153. $\dfrac{abc}{n} \times 6.02 \times 10^{23} = \dfrac{118.7}{7.29}$

$\dfrac{5.82\times3.175\times3.175\times6.02}{10\times n} = \dfrac{118.7}{7.29}$. $\therefore$ n $\simeq$ 2

154. K at 300°C $= 1.11 \times 10^{11} \exp\left(\dfrac{-39300}{2\times573}\right)$

$= 1.11 \times 10^{11} \times 1.2784 \times 10^{-15}$

$= 1.419 \times 10^{-4}$ sec^{-1}

$\therefore \quad T_{1/2} = \dfrac{0.693}{K} = 4884$ sec

155. $\log K = 33.91 - \dfrac{18150}{T} = \log A - \dfrac{E_a}{2.303R}\left(\dfrac{1}{T}\right)$

$\therefore \quad \dfrac{E_a}{2.303R} = 18150$

$\therefore \quad E_a$ (K cal mol^{-1}) $\simeq \dfrac{18150\times2.303\times2}{1000}$

$= 83.6$ K cal mol^{-1}

156. $\dfrac{\Delta p}{\Delta t} = \dfrac{13.5}{100}$ and $\dfrac{14.0}{100}$ nearly constant

$\therefore$ order zero.

157. $\dfrac{(6.50)^3 \times10^{-24} \times6.02\times10^{23}}{n} = \dfrac{235}{5.67}$

$n = \dfrac{(6.50)^3 \times16.02}{10\times235} \times 5.67 = 4$

158. Since $T_{1/2} = 53$ min, 75% completion would be observed in 106 min if the reaction is of I order. This is observed $\therefore$ order = 1

159. $2.303 \times \log 10 = \dfrac{E_a}{R}\left(\dfrac{20}{323\times303}\right)$

$\therefore \quad E_a = \left[\dfrac{2.303\times2\times323\times303}{20}\right]$ cal mol^{-1}

$= 22539$ cal mol^{-1}

$K = A \times \exp\left(\dfrac{-E_a}{RT}\right)$

$0.251 = A \times \exp\left(\dfrac{-22539}{2\times333}\right)$

$\therefore \quad A = 1.25 \times 10^{14}$

160. $p = c \times RT$ where, c is in mol L^{-1}

$\frac{(2-1)}{(0.0821\times298)6.84\times60}$ = mean rate of reaction in mole L^{-1} per sec.

Ans. 9.959×10^{-5} mole L^{-1} per sec

$= 1 \times 10^{-4}$ mole L^{-1} per sec

161. $\frac{dx}{dt} = k[A]^2[B]^1$

order = 2 + 1 = 3

162. For a zero order reaction $t_{1/2} \mu a$.

When the initial concentration is reduced to ¼th, $t_{½}$ becomes ¼th

163. As the value of k is 3×10^{-2} s^{-1} the reaction is of 1st order

$[A] = \frac{r}{k} = \frac{2.4\times10^{-2}}{3\times10^{-2}} = 0.8$ mol L^{-1}

164. AgBr contains both Schottky and Frenkel defects.

165. $t = \frac{2.303}{k}\log\left(\frac{a}{a-x}\right)$

$30 \times 60 = \frac{2.303}{1.35\times10^{-4}}\log\frac{(0.03)}{(a-x)}$

$(a - x) = 0.0235$ mol L^{-1}

166. $t_{1/2} \alpha \frac{1}{a^{n-1}}$

From the given data, $\frac{0.5}{0.25} = \left(\frac{4}{2}\right)^{n-1}$

$2^1 = 2^{n-1}$

$\Rightarrow$ $n - 1 = 1$ or $n = 2$

167. $k = \frac{2.303}{t}\log\frac{a}{a-x}$

$2.3\times10^{-3} = \frac{2.303}{50}\log\frac{2}{2-x}$

$\log\frac{2}{2-x} = \frac{50\times2.3\times10^{-3}}{2.303} = 0.05$

$\frac{2}{a-x} = 1.122$

$a - x = \frac{2}{1.122} = 1.78$ mol L^{-1}

168. If pH reduces by one unit, $[H^+]$ increases by 10 times

rate $\mu [H^+]^2 = [10]^2 = 100$

169. Slope $= -\frac{E_a}{2.303R}$

$= \frac{-90\times10^3 \text{ J mol}^{-1}}{2.303\times8.314 \text{ J K}^{-1}\text{ mol}^{-1}} = -4700$K

170. Let initial moles = a

No. of moles remaining = $a - \frac{2}{3}a = \frac{1}{3}a$

$k = \frac{2.303}{t}\log\frac{a}{a-x}$

$k = \frac{2.303}{2.3\times10^5}\log\frac{a}{a/3}$

$= 10^{-5} \log 3 = 4.771 \times 10^{-6}$ min^{-1}

171. Given $\frac{r_{A^+}}{r_{B^-}} = 0.524$

This lies between 0.414 and 0.732

$\therefore$ it has coordination number 6.

i.e., edge length = $2(r_{A^+} + r_{B^-})$

$2(r_{A^+} + r_{B^-}) = 5.52$ A°

$0.524\ r_{B^-} + r_{B^-} = 2.76$ A°

$\Rightarrow$ $r_{B^-} = \frac{2.76}{1.524} = 1.81$ A°

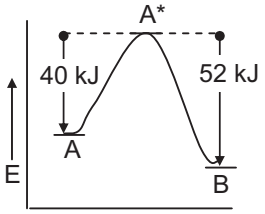
$r_{A^+} = 2.76 - 1.81 = 0.95$ A°

172. $k = \frac{1}{t} \times 2.303 \log \frac{a}{0.7a}$

$= \frac{1}{40} \times 2.303 \log 1.43$

$= 8.91 \times 10^{-3}$ min^{-1}

173. To reach the same point i.e., A* 'A' requires 40 kJ and 'B' requires 52 kJ. So B is below A by 52 − 40 = 12 kJ so $\Delta H = -12$ kJ mol^{-1}



174. Since the unit of the rate of constant reaction is mol L^{-1} s^{-1} it is a zero order reaction.

$\therefore$ If concentration is doubled, the rate of reaction remain same

175. $$\frac{k_{(catalyzed)}}{k_{(uncatalyzed)}} = \frac{e^{\frac{-E_{a(catalysed)}}{RT}}}{e^{\frac{-E_{a(uncatalysed)}}{RT}}}$$

$$\log\left(\frac{k_{catalyzed}}{k_{uncatalyzed}}\right) = \frac{E_{a(uncatalyzed)} - E_{a(catalyzed)}}{2.303RT}$$

$$\log 4000 = \frac{\Delta E_a}{2.303 \times 8.314 \times 300}$$

$$\Delta E_a = 20.7 \text{ kJ mol}^{-1}$$

176. Statement 1 is true, statement 2 is true, statement 2 is a correct explanation for statement 1.

177. Statement 1 is true, statement 2 is true, statement 2 is a correct explanation for statement 1.

178. Statement 1 is true, statement 2 is true, statement 2 is not a correct explanation for statement 1.

179. Statement 1 is true, statemen t 2 is true, statement 2 is not a correct explanation for statement 1.

180. Statement 1 is false and statement 2 is true

181. Statement 1 is true and statement 2 is false

182. Statement 1 is true, statement 2 is true, statement 2 is not a correct explanation for statement 1.

183. Statement 1 is true and statement 2 is false

184. In Zinc blende structure Zn^{2+} occupy alternate tetrahedral voids and in fluorite structure all the tetrahedral voids are occupied by F.

185. Cation-anion contact along the edge of the unit cell just like NaCl fcc type.

$\therefore$ side length a = 2 × 3 = 6 A°

$\therefore$ unit cell volume = $(6 \times 10^{-8})^3$ $cm^3 = 216 \times 10^{-24}$ cm^3

i.e., 2.16×10^{-22} cm^3

186. There are eight tetrahedral voids in a unit cell of NaCl.

187. $$2090 = \frac{E_a}{2.303R}$$

$$\therefore \quad E_a = \frac{2.303 \times 8.314 \times 2090}{1000} \text{ kJ mol}^{-1}$$

$$E_a = 40 \text{ kJ mol}^{-1}$$

188. $\frac{d\ell nk}{dT} = \frac{E_a}{RT^2} \cdot \frac{dk}{kdT} = \frac{52 \times 1000}{8.314 \times (300)^2}$ The lhs multiplied by 100 $\frac{100 \times dk}{kdT}$ = % change in k per degree rise of temperature.

$\therefore$ Ans. = $\frac{52 \times 1000}{8.314 \times 300 \times 300} \times 100 = 6.949$

189. $t_{1/2} = \frac{0.693}{k}$

We know that $\log\frac{k_2}{k_1} = \frac{E_a.(T_2 - T_1)}{2.303\, R\, T_1 T_2}$

190. $$2.303 \log 3 = \frac{E_a}{R}\left(\frac{10}{310 \times 320}\right)$$

$$\therefore \quad \frac{E_a}{R} = \frac{1.0988 \times 310 \times 320}{10}$$

$$2.303 \log 2 = \frac{E_a}{R}\left(\frac{T-310}{310 \times T}\right) \quad \frac{E_a}{R} = \frac{0.693 \times 310T}{T-310}$$

$$\frac{1.0988 \times 310 \times 320}{10} = \frac{0.693 \times 310 \times T}{T-310}$$

$$\therefore \quad \frac{T-310}{T} = \frac{0.693 \times 310 \times 10}{1.0988 \times 310 \times 320} = 0.019709$$

$$1 - \frac{310}{T} = 0.019709. \quad \frac{310}{T} = 0.9803$$

$\therefore$ T = 316.23 K = 43.3°C

191. $\exp\left(\frac{10 \times 1000}{2 \times 800}\right) = \exp(6.25) = 518$

192. $\frac{d\ell nk_1}{dT} = \frac{E_1}{RT^2}$

$$\therefore \quad \frac{dk_1}{kdT} = \frac{k_1 E_1}{RT^2}$$

similarly, $\frac{dk_2}{kdT} = \frac{k_2 E_2}{RT^2}$

$k_1 + k_2 = k$

$$\therefore \quad \frac{dk}{kdT} = \frac{k_1 E_1 + k_2 E_2}{RT^2} = \frac{kE}{RT^2} \cdot$$

$$\frac{d\ell nk}{dT} = \frac{E}{RT^2} \quad \therefore E = \left(\frac{k_1 E_1 + k_2 E_2}{k_1 + k_2}\right)$$

This page is intentionally left blank.

CHAPTER

9

SURFACE CHEMISTRY AND NUCLEAR CHEMISTRY

CHAPTER OUTLINE

Preview

STUDY MATERIAL

SURFACE CHEMISTRY

Adsorption and Absorption

Physical Adsorption and Chemisorption

- Concept Strands (1-5)

Colloids

Types of Colloidal Systems

Classification of Sols

Methods of Preparation of Colloidal Solutions

Purification of Colloidal Solutions

Properties of Colloidal Solutions

Applications of Colloids

Emulsifiers

NUCLEAR CHEMISTRY

α, β, γ-Rays and Their Properties

Kinetics of Radioactive Decay

Units of Radioactivity

Isotopes, Isobars and Isotones

Nuclear Fission and Fusion Reactions

Artificial Disintegration or Transmutation of Elements

- Concept Strands (6-19)

TOPIC GRIP

- Subjective Questions (10)
- Straight Objective Type Questions (5)
- Assertion–Reason Type Questions (5)
- Linked Comprehension Type Questions (6)
- Multiple Correct Objective Type Questions (3)
- Matrix-Match Type Question (1)

IIT ASSIGNMENT EXERCISE

- Straight Objective Type Questions (46)
- Assertion–Reason Type Questions (4)
- Linked Comprehension Type Questions (6)
- Multiple Correct Objective Type Questions (3)
- Matrix-Match Type Question (1)

ADDITIONAL PRACTICE EXERCISE

- Subjective Questions (6)
- Straight Objective Type Questions (18)
- Assertion–Reason Type Questions (12)
- Linked Comprehension Type Questions (6)
- Multiple Correct Objective Type Questions (6)
- Matrix-Match Type Questions (2)

SURFACE CHEMISTRY

Introduction

The subject of surface chemistry deals with a study of solid and liquid surfaces, particularly with reference to the type of forces operating on the surfaces and the consequences of these forces on the reactivity of various types of surfaces. We shall consider a liquid surface, as an example, to understand the type of forces operating at any surface.

Properties of liquid surfaces

A molecule in the interior of a liquid is uniformly surrounded by other molecules and so, on an average, is attracted equally in all directions. However, when a molecule at the surface is considered, there is a net attraction inwards because the number of molecules per cm^3 in the liquid is greater than in the vapour. Therefore, there is a strong inward pull from the molecules in the bulk on the surface molecules due to the intermolecular forces in the liquid. As a result, the surface of a liquid is in a state of strain or tension because of the net attraction of the molecules in the bulk to the molecules on surface layer. Thus its surface always tends to occupy the smallest area and work has to be done to increase the surface area of a liquid against the inward attractive force. Since a decrease of the surface area is associated with a decrease of free energy, the surface of a liquid always tends to contract.

ADSORPTION AND ABSORPTION

The same phenomenon is also applicable to a solid surface, which is associated with a residual field of force. The tendency for the free energy of a surface to decrease, results in the phenomenon of adsorption. *Adsorption may be defined as the presence of a higher concentration of a substance at the surface of a liquid or solid as compared to the bulk concentration.*

It is important to distinguish adsorption from absorption, which refers to the presence of a uniform concentration of a substance throughout the bulk of the medium. The reason for adsorption at a surface is that it leads to a decrease of interfacial area and thereby decreases the interfacial tension. This is a spontaneous process associated with a decrease of free energy.

For example, dissolution of acetic acid or oxalic acid by water is a case of absorption of these acids by water resulting in uniform concentration of the acids in water. The same acids are, however, adsorbed from solution by finely divided charcoal.

Adsorption of gases on solids

(i) Adsorption occurs on all surfaces but the effects are very much evident when the adsorbent (i.e., the substance which adsorbs) is porous and has a large surface area for a given mass. The substance adsorbed is known as adsorbate. In the example cited in the previous section, finely divided charcoal is the adsorbent and the acids are the absorbates.

Among the very commonly used adsorbents, mention may be made of silica gel, alumina, clay, wood charcoal or charcoal obtained by burning coconut shell activated by heating in vacuum at temperatures of 350°C – 400°C. Molecular sieves is another important adsorbent used extensively in drying of gases and separation of components from a mixture by preferential adsorption.

(ii) Effect of pressure and temperature on adsorption of gases: The extent of adsorption of a gas increases with increase of pressure and a decrease of temperature of the gas. The general nature of adsorption isotherms of a gas is shown in Fig. 9.1 given below.

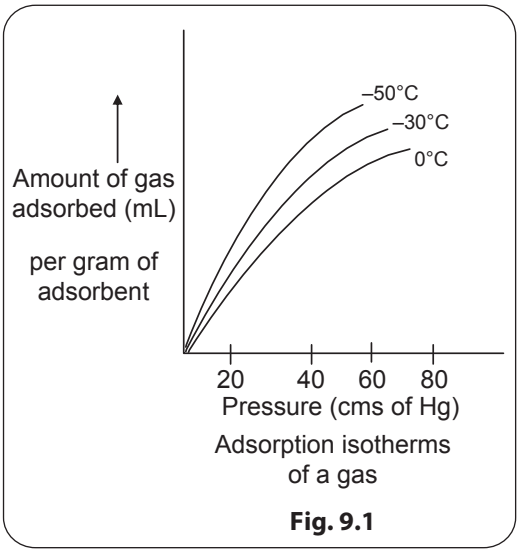


Fig. 9.1

It is seen from Fig. 9.1, that adsorption increases rapidly as the temperature is lowered. This observation finds practical application in the production of high vacuum. Since adsorption decreases with increase of temperature, it follows from Le-Chatelier's principle that adsorption is an exothermic process.

PHYSICAL ADSORPTION AND CHEMISORPTION

Depending on the type of forces involved in the adsorption of gases on solids, two distinct types of adsorption may be recognized.

(i) Physical adsorption or van der Waals adsorption.
(ii) Chemisorption or activated adsorption.

The distinguishing features between the two types of adsorption are given in the following table

Table 9.1

Physical adsorption	Chemisorption
1) It is exhibited by all gases at low to moderately low temperatures.	1) It occurs at moderately high temperatures.
2) The heat of adsorption is of the order of 20 to 40 kJ mol^{-1} which is of the same order as the heats of condensation of gases to liquids. Thus the forces involved in this adsorption are similar to cohesive forces in the liquid state and hence are of van der Waals type.	2) Heat of adsorption is much higher ranging from 40 to 400 kJ mol^{-1}. The adsorbate molecules are held by comparatively strong valence forces.
3) The extent of adsorption correlates well with the ease of liquefaction of the gas among a number of gases. The most easily liquefiable gases are thus, the most strongly adsorbed	3) Chemisorption is highly specific in nature depending on the properties of the gas and the adsorbent.
4) This type of adsorption is reversible i.e., if the pressure of the gas is reduced, desorption occurs and the same isotherm is followed as in adsorption process.	4) Chemisorption is not reversible and usually it is necessary to raise the temperature significantly to remove the adsorbed gas.
5) In this case, usually a multimolecular layer adsorption occurs.	5) Chemisorption generally occurs with the formation of a monomolecular layer.

Freundlich adsorption isotherm

The variation of adsorption with pressure at a given temperature can be represented over a small range of pressures by Freundlich adsorption isotherm given by

$$\frac{x}{m} = kp^{1/n} \qquad \text{—(1)}$$

where, "x" is the amount of gas adsorbed by "m" grams of the adsorbent, "p" is the equilibrium pressure, "k" and "n" are constants for the given system at the temperature of study. $\frac{1}{n}$ is less than unity and thus the amount of gas adsorbed increases less rapidly with pressure.

On taking logarithms, equation (1) may be written as

$$\log\left(\frac{x}{m}\right) = \log k + \frac{1}{n}\log p \qquad \text{— (2)}$$

A plot of $\log\left(\frac{x}{m}\right)$ Vs log p should be linear and the slope of such a graph is equal to $\frac{1}{n}$ and it is shown in Fig. 9.2.

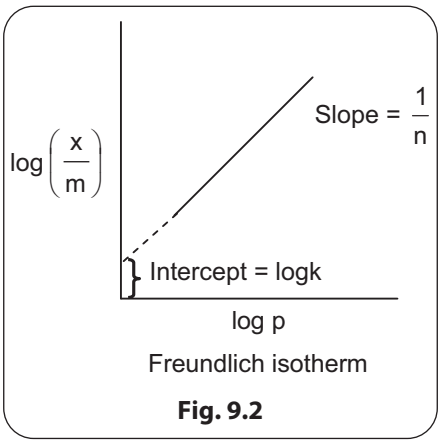


Fig. 9.2

There are however, deviations observed from the linearity especially at low temperatures.

Langmuir adsorption isotherm

Langmuir (1916) first proposed the formation of an adsorbed layer which is unimolecular thick on the surface of the adsorbent. This hypothesis is based on the fact that chemical forces decrease rapidly with distance and hence chemisorption does not extend beyond a single molecular layer in thickness on the surface of the adsorbent.

We consider the dynamics of adsorption by considering an adsorbing surface exposed to a gas. Gas molecules will strike the surface, adhere for an appreciable time while other molecules tend to evaporate from the surface. After a certain time, adsorption equilibrium is established and then the rate at which gas molecules are adsorbed is equal to the rate of evaporation of the molecules from the surface i.e., at adsorption equilibrium, the rate of adsorption is equal to the rate of desorption. On this basis, Langmuir derived an equation known as Langmuir adsorption isotherm which is given by

$$a = \frac{k_1 p}{k_2 p + 1} \qquad \text{— (3)}$$

where, "a" is the amount of gas adsorbed by unit mass of adsorbent, "k_1" and "k_2" are constants and "p" is the equilibrium pressure. Two special cases of equation (3) are of interest.

(i) When p is very low, $k_2 p < 1$. Equation (3) simplifies to

$$a = k_1 p \qquad \text{— (4)}$$

i.e., the amount of gas adsorbed is proportional to the gas pressure which has been experimentally verified.

(ii) When p is quite high, $k_2 p >> 1$, in that case, $a = \frac{k_1}{k_2}$ — (5)

i.e., the amount of gas adsorbed at any temperature should reach a constant limiting value at high pressures. This situation arises when the surface is covered completely by a uniform layer of gas molecules. At intermediate pressures the relation

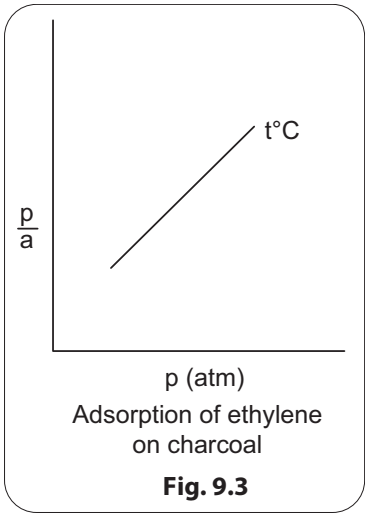
$$a = k p^{1/n} \qquad \text{— (6)}$$

is valid where, "$\frac{1}{n}$" lies between 0 and 1.

On rearranging the equation (3), we get

$$\frac{p}{a} = \frac{k_2}{k_1} p + \frac{1}{k_1} \qquad \text{— (7)}$$

A plot of $\frac{p}{a}$ against p at temperature t^0C should be linear as shown in Fig. 9.3, in the case of adsorption of ethylene as charcoal.



Adsorption of ethylene on charcoal

Fig. 9.3

A number of examples are available to support the validity of this isotherm as, for example, in the case of the adsorption of ethylene on charcoal. Deviations from Langmuir isotherm are known to occur in cases where there is a simultaneous occurrence of physical adsorption and chemisorption.

An interesting form of Langmuir's adsorption isotherm is obtained by considering, θ, the fraction of the surface of adsorbent covered by a unimolecular layer in terms of the volume of the gas adsorbed.

Let $\theta = \frac{v}{v_m}$ — (8)

where, v is the volume of the gas adsorbed at any pressure p, v_m = volume of the gas adsorbed in the unimolecular layer. We can write

$$\theta = \frac{v}{v_m} = \frac{kp}{kp + k'} \qquad \text{— (9)}$$

k and k' are constants and p is the equilibrium presure. Rewriting equation (9) as

$$\frac{v_m}{v} = \frac{k'}{k}\frac{1}{p} + 1 \qquad \text{— (9a)}$$

or $$\frac{1}{v} = \frac{1}{v_m} + \frac{k'}{k}\frac{1}{v_m}\frac{1}{p} \qquad \text{— (10)}$$

From equation (10), it is seen that a plot of $\frac{1}{v}$ against $\frac{1}{p}$ should be linear with an intercept equal to $\frac{1}{v_m}$. It is thus possible to calculate, v_m, the volume of gas adsorbed in the unimolecular layer and hence also the surface area of the adsorbent.

Adsorption from solution

As a result of the tendency of the free energy of a surface to decrease, the concentration of a solute at the surface differs from that in the bulk of the solution. The component with the lower surface tension tends to concentrate at the surface because this process is associated with a decrease of free energy. This process is, however, opposed by the thermal motion and the intermolecular forces among the molecules in solution and ultimately adsorption equilibrium is reached.

In general, it may be stated that if a solute lowers the interfacial tension at an interface, then it will be present in greater proportion at the interface compared to that in the bulk of the solution. Under these conditions, the solute is said to be adsorbed from the solution.

Substances, which produce a marked reduction in surface tension are said to be surface active. Examples of such substances are long chain fatty acids like palmitic acid, oleic acid and long chain alcohols like hexyl alcohol and they reduce the interfacial tension when placed in water.

There are also substances which produce an increase in interfacial tension i.e., their concentration at the interface is less than that in the bulk of the solution. Electrolytes belong to this category and hence they are not surface active.

Gibbs equation

J. W. Gibbs (1878) derived a quantitative relation between the excess concentration of a solute near the surface and the concentration of the same in the bulk of the solution. The equation known as Gibbs adsorption isotherm is given by

$$S = -\frac{C}{RT}\frac{\partial \gamma}{\partial C} \qquad \text{— (11)}$$

In this equation, S = excess concentration of the solute per sq cm of the surface as compared to that in the bulk of the solution. If n_2 is the number of moles of solute associated with n_1 moles of the solvent in the surface phase and n_2^0 is the number of moles of solute in n_1^0 moles of solvent in the bulk phase, $S = \frac{\left(n_2 - \frac{n_1 n_2^0}{n_1^o}\right)}{s}$ where, "s" is the area of the interface.

S has the units of mole cm^{-2},

$\frac{\partial \gamma}{\partial C}$ = rate of increase of interfacial tension with the concentration of the solute,

C = equilibrium concentration of solute,

T = temperature in deg K,

R = gas constant.

From equation (11) it is seen that if $\frac{\partial \gamma}{\partial C}$ is –ve, S is +ve. This implies that the concentration of the solute on the surface is higher than in the bulk. In this situation when the interfacial tension is reduced by the added solute.

However, if $\frac{\partial \gamma}{\partial C}$ is +ve, S is –ve implying that the concentration of the solute on the surface is less than that in the bulk. In this case, the interfacial tension increases by the addition of the solute. This is known as negative adsorption.

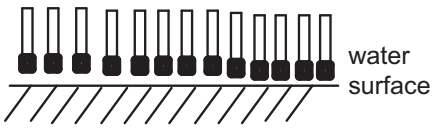
Unimolecular films

An interesting surface phenomenon related to adsorption is the spreading of films of certain insoluble substances on the surface of water. This has important application in preventing extensive loss of water by evaporation from lakes and rivers.

Addition of a small amount of a hydrocarbon oil to water results in the formation of a lens like drop of oil on water. But if we add a substance containing polar groups like –COOH, $-CH_2OH$ etc. the oil spreads uniformly on the surface as a thin unimolecular film.

Since such substances are usually solid, they are added to water as a solution in benzene. By suitable experiments it is possible to calculate the surface area of the film and also the area occupied by a single molecule in the closely packed monolayer. The area of the surface occupied per molecule is found to be constant i.e., 21.0 sq A^0 = 21.0 × 10^{-16} cm^2 for a large number of compounds containing polar end groups like –COOH, $-CH_2OH$. This value is independent of chain length and this observation suggests that the orientation of the molecules on the surface is such that the polar heads are attached to water molecules while the

hydrocarbon chains point upwards away from water as shown in the figure.



Applications of adsorption

The principle of adsorption is widely employed in industry as well as in analyti cal practice. Some of its important applications are presented below.

(i) Chromatographic adsorption: The separation, purification and analysis of a variety of substances is possible by the principle of chromatographic adsorption. *The term "chromatography" derives its name as it was first employed in the separation and purification of coloured substances.*
Chromatography is used in several forms.

(a) Gas-liquid chromatography or gas chromatography is a powerful analytical tool widely used in the analysis of hydrocarbon mixtures required in petroleum industry.
(b) Solid-liquid chromatography (also known as column chromatography) is based on the principle that a given adsorbent will adsorb to varying extents the constituents of a mixture present in the solution. Another form of this method is known as paper chromatography where, a sheet of filter paper is the adsorbing material and this method is widely used in the analysis of amino acids, and other important biochemicals.

(ii) Ion exchange separation: Soils and certain minerals like zeolites are capable of exchanging cations they contain with other cations that are present in solution. They are known as cation exchangers. Similarly, we have anion exchangers. The ion exchangers are used for softening of hard water. Boiler feed water in boilers is also obtained in a similar manner.
(iii) Adsorption finds application in many industrial processes such as manufacture of ammonia, synthetic petrol etc.
(iv) Removal of unwanted colouring matter from coloured solutions is another important application of adsorption principle.
(v) Detergent action involving use of surface active agents like soaps is another useful aspect of adsorption principle
(vi) Dyeing of fabrics using mordant dyes, tanning of leather are other examples illustrating the principle of adsorption.

CONCEPT STRANDS

Concept Strand 1

The adsorption of nitrogen on silica gel at 0^0C follows Langmuir isotherm and a plot of $\frac{1}{v}$ vs $\frac{1}{p}$ gave an intercept equal to 0.008 mL^{-1}. Given that the nitrogen molecule occupies 16.2 sq. A^0 on the surface of adsorbent, calculate the surface area of silica gel. Assume that the amount of silica gel in the experiment is one gram.

Solution

$$\text{Intercept} = \frac{1}{v_m} = 0.008 \text{ or}$$

v_m = vol adsorbed in unimolecular layer

$$= \frac{1}{0.008} = 125.0 \text{ mL.}$$

At N.T.P, 125 mL corresponds to $\frac{125}{22,400} = 0.0056$ mol

$\therefore$ Number of molecules $= 0.0056 \times 6.023 \times 10^{23}$
$= 337.3 \times 10^{19}$

Total area occupied by these molecules = $337.3 \times 10^{19} \times 16.2 \times 10^{-16} = 5.465 \times 10^6$ $cm^2 g^{-1}$

Concept Strand 2

1.06×10^{-4} g of stearic acid (mol wt = 284) is found to occupy 500 cm^2 of the surface of water. Calculate the cross sectional area of stearic acid molecule and the thickness of the unimolecular film. (density of stearic acid = 0.85 g cm^{-3}).

Solution

Number of molecules of stearic acid

$$= \frac{1.06 \times 10^{-4}}{284} \times 6.023 \times 10^{23}$$

Cross sectional area per molecule

$$= \frac{500 \times 284}{1.06 \times 6.023 \times 10^{19}}$$

$$= 22.24 \times 10^{-16}\ cm^2 = 22.24\ (A°)^2.$$

Volume occupied by 1.06×10^{-4} g stearic acid

$$= \frac{1.06 \times 10^{-4}}{0.85} = 1.25 \times 10^{-4}\ cm^3$$

Thickness of the film

$$= \frac{1.25 \times 10^{-4}\ cm^3}{500\ cm^2} = 25.0\ A^0$$

Concept Strand 3

The rate of change of surface tension with concentration for a solution of n-butanol in water at 293K is –165.6 dynes cm^{-1} L mol^{-1}. Calculate the excess concentration of butanol at a concentration of 0.2 mol L^{-1}.

Solution

$$S = -\frac{C}{RT}\frac{\partial\gamma}{\partial C} = -\frac{0.2}{8.314 \times 10^7 \times 293} \times -165.6$$

$$= 13.6 \times 10^{-10}\ \text{moles}\ cm^2$$

Concept Strand 4

For the adsorption of carbon monoxide on wood charcoal at 273K, the values of $\frac{x}{m}$ ($cm^3 g^{-1}$) as a function of pressure (mm Hg) are given below.

$\frac{x}{m}$ ($cm^3 gm^{-1}$)	P (mm Hg)
6.93	200
13.22	450

Calculate the values of the constants k and n in the Freundlich isotherm. What is the volume of gas adsorbed per g at a pressure of 300 mm Hg?

Solution

Taking logarithm of the Freundlich isotherm $\log \frac{x}{m} = \log k + \frac{1}{n}\log P$

And substituting the two data, we get

(a) $\log(13.22) = \log k + \frac{1}{n}\log 450$

(b) $\log(6.93) = \log k + \frac{1}{n}\log 200$

(a) – (b) gives $1.122 - 0.8407 = 0 + \frac{1}{n}(2.6532 - 2.3010)$

$$\frac{1}{n} \times 0.3522 = 0.2813$$

$$\frac{1}{n} = 1.7986;\ n = 1.252$$

To evaluate k,

$1.122 = \log k + 0.7986 \times 2.6532 = \log k + 2.118$

$\log k = -0.996$; $k = 0.1009$

At 300 mm Hg

$$\log \frac{x}{m} = \log 0.1009 + 0.7986 \log 300 = -0.9961 + 1.9782$$

$$= 0.9821$$

$$\frac{x}{m} = 9.59\ cm^3 g^{-1}$$

Concept Strand 5

In an adsorption study of acetic acid from aqueous solution by charcoal, 200 mL of 0.20 N acetic acid was thoroughly shaken with 10.0 g of charcoal at room temperature. After adsorption equilibrium was established, 50 mL of acetic acid solution required 25 mL of 0.2 N NaOH for neutralization. Calculate the weight of acetic acid adsorbed per gram of adsorbent. What is the equilibrium concentration of acetic acid in solution?

Solution

Number of m.eq of acetic acid initially present $= 200 \times 0.2 = 40$

At adsorption equilibrium,

normality of acetic acid remaining $= \frac{25 \times 0.2}{50} = 0.1$

Or m.eq of acetic acid present in 200 mL after adsorption $= 200 \times 0.1 = 20$ m.eq.

$\therefore$ m.eq of acid adsorbed $= 40 - 20 = 20$

wt of adsorbed acid $= 20 \times 10^{-3} \times 60 = 1.2$ g

wt. of acid adsorbed per gm of adsorbent $= \frac{1.2}{10} = 0.12$ g

Equilibrium conc. of acetic acid = 0.1 N

COLLOIDS

Introduction

True solutions such as solutions of urea, sugar, sodium chloride in water consist of simple molecules or ions. But a suspension like that of chalk or finely powdered carbon in water contains particles which can be freely seen by the eye or in a microscope. Between these two extremes occur the colloidal systems which consist of particles larger than molecules but smaller in size than the particles in suspensions.

Colloids and approximate size of colloidal particles

A colloidal solution may be said to be intermediate in size between molecular dimensions and suspensions. The range in which the size of colloidal particles lies may be seen from the table 9.1.

Table 9.2 Size of colloidal particles

Coarse suspension	Colloidal particles	Molecules
$10^{-3} - 2 \times 10^{-5}$ cm	$2 \times 10^{-5} - 5 \times 10^{-7}$ cm	$10^{-7} - 10^{-8}$
(10,000 mμ – 200 mμ)	(200 mμ – 5 mμ)	(1 mμ – 0.1 mμ)

(1 mm = 10–7cm)

The properties of colloidal systems arise due to two important factors

(i) the ratio of their surface area to volume is large.
(ii) colloids are two phase systems with a definite surface of separation between the colloidal particles and the surrounding medium.

TYPES OF COLLOIDAL SYSTEMS

The two phases in colloidal systems are known as

(i) the dispersed phase i.e., the phase forming the particles.
(ii) the dispersion medium i.e., the medium in which the particles are dispersed.

Either the dispersed phase or the dispersion medium may be solid, liquid or gas. Depending upon the nature of the two phases, the colloid is known by different names.

Table 9.3 gives the list of different colloidal systems with examples.

Table 9.3 Different types of colloidal systems

Dispersed phase	Dispersion medium	Name of colloidal system	Examples
Solid	Liquid	Sol or colloidal solution	Metal sols like silver sol, gold sol etc.
Liquid	Liquid	Emulsion	Milk, mayonnaise
Gas	Liquid	Foam	Whipped cream, froth
Solid	Solid	Solid sol	Ruby glass, gems
Solid	Gas	Aerosol	Smoke, dust
Liquid	Gas	Aerosol	Mist, cloud, fog
Gas	Solid	Solid foam	Pumice stone, foam rubber
Liquid	Solid	Solid emulsion (Gel)	Jellies, butter, cheese

In this chapter, colloids involving liquid as a dispersion medium such as sols and emulsions will be considered.

CLASSIFICATION OF SOLS

Sols are of two types

(i) *lyophobic sols i.e., sols which do not have affinity for the solvent.* Examples of such sols are metal sols, sulphur sol, sols of silver halides etc.

(ii) *lyophilic sols i.e., sols which have affinity for the solvent.* Examples being substances of high molecular weight such as starch, gums, proteins dispersed in water.

Some of the characteristic differences between lyophobic and lyophilic sols are given in the table 9.4.

Table 9.4 Distinguishing features of lyophobic and lyophilic sols

Lyophobic sols	Lyophilic sols
1) Generally unstable	1) Stable
2) Small quantities of electrolytes cause coagulation and precipitation of colloidal particles.	2) Large quantities of electrolytes are required to cause precipitation.
3) On warming a lyophobic sol, a solid is often obtained which cannot be brought back to colloidal state again reversibly.	3) Lyophilic sols can be reversibly brought back to colloidal state easily by the addition of dispersion medium.
4) The colloidal particles travel towards the anode or cathode depending on their charge, under the influence of an electric field.	4) The particles may migrate in either direction (towards anode or cathode) or not at all under the influence of an electric field.
5) Viscosity of these sols is nearly the same as that of the dispersion medium.	5) These sols have higher viscosities compared to the dispersion medium.
6) Lyophobic sols are generally prepared by indirect methods	6) Lyophilic sols are prepared by direct mixing with dispersion medium.

Lyophilic sols

Lyophilic sols, also referred as intrinsic sols, are substances of high molecular weight such as gelatin, starch and they can be brought into colloidal form by warming them with the dispersion medium.

Lyophobic sols

These sols, also referred as extrinsic colloids are of substances of low molecular weight.

METHODS OF PREPARATION OF COLLOIDAL DISPERSIONS

There are two general methods of preparation of these colloids viz., (i) condensation methods and (ii) dispersion methods.

Condensation methods

The principle of this method is that the materials from which the sol is prepared are originally true solution and as a result of interaction between them, insoluble particles of colloidal size are obtained. The experimental conditions such as the concentration of electrolyte must be properly controlled to prevent precipitation of the colloidal particles. Various chemical reactions are employed to obtain sols under this category The various approaches by which sols are prepared using condensation methods are described below.

(a) Reduction

Sols of metals like gold sol, silver sol have been prepared by this method by reduction of their soluble salts using suitable non-electrolytic reducing agents such as hydrogen, formaldehyde, hydrazine etc. For example, gold sol may be

prepared by reduction of auric chloride with formaldehyde according to the reaction.

$$2AuCl_3 + 3HCHO + 3H_2O \rightarrow \underset{\text{(gold sol)}}{2Au} + 3HCOOH + 6HCl \quad \text{— (12)}$$

(b) Double decomposition

Examples of this method include preparation of silver halide sols like that of silver chloride, for example, which is prepared by reacting a dilute solution of silver nitrate with a dilute solution of potassium chloride in nearly equivalent proportions. This sol is usually stabilised by the presence of Ag^+ or Cl^- ions.

$$AgNO_3 + KCl \rightarrow AgCl(sol) + KNO_3 \quad \text{— (13)}$$

Another example of this category is the arsenious sulphide sol which is prepared by passing a slow stream of hydrogen sulphide gas through a solution of arsenious oxide. The reaction is

$$As_2O_3 + 3H_2S \rightarrow \underset{\text{yellow sol}}{As_2S_3} + 3H_2O \quad \text{— (14)}$$

Sulphur sol may be prepared by oxidation of hydrogen sulphide by bromine according to the reaction

$$H_2S + Br_2 \rightarrow \underset{\text{sulphur sol}}{S} + 2HBr \quad \text{— (15)}$$

The reaction with SO_2 is

$$2H_2S + SO_2 \rightarrow 2H_2O + \underset{\text{sulphur sol}}{3S} \quad \text{— (16)}$$

(c) Hydrolysis

Hydroxides of weakly electropositive metals like Al, Fe, Cr using salts $AlCl_3$, $FeCl_3$ are prepared by hydrolyzing them in water. The reaction in the case of $FeCl_3$ is

$$FeCl_3 + 3H_2O \rightarrow Fe(OH)_3 + 3HCl \quad \text{— (17)}$$

Dispersion methods

In this method, the substance containing particles of a large size is brought down to colloidal dimensions by disintegrating it in two ways. (a) Mechanical disintegration (b) Bredig's arc method

(a) Mechanical disintegration

The mechanical disintegration of a substance into particles of colloidal size is carried in a colloidal mill which consists of two steel discs each rotating in opposite direction at a certain speed. The dispersion medium and the substance to be dispersed are passed through the mill, then a colloidal solution results, because of the breaking down of the large particles by the shearing forces.

(b) Bredig's arc method

Sols of metals like platinum, silver and gold are prepared by this method. The method consists in striking an electric arc between the electrodes of the desired metal in water. The high temperature of the arc evaporates the metal and its vapours are condensed in water to give particles of colloidal range. Colloidal systems obtained in this manner (either method (a) and (b)) have to be stabilised by adding an agent known as a peptizing agent, which is generally a reagent having an ion in common with the colloidal particles.

For example, hydroxides of metals like Fe, Al are peptized by small amounts of alkali hydroxide. Metal sols prepared by Bredig's method are stabilised by the addition of trace amounts of alkali hydroxide.

In the case of lyophilic sols, the dispersion medium itself acts as the peptizing agent

PURIFICATION OF COLLOIDAL DISPERSIONS

The colloids prepared by the above methods are purified to remove any ionic or other impurities as they may cause coagulation of the colloidal particles. Two methods are generally employed for this purpose and are known as "dialysis" and "ultrafiltration".

Dialysis

Particles in true solution can diffuse through animal membranes or through parchment paper or cellophane sheet containing fine pores. However, colloidal particles, be-

ing of much larger dimensions cannot pass through such pores. If a colloidal solution containing small amounts of ionic or other impurities is placed in such a parchment bag and this bag is placed in a vessel containing pure water, the ions diffuse out through the membrane leaving the colloidal particles behind. The water in the vessel is frequently replaced.

Generally the process of dialysis is slow but can be hastened by the application of electric field if the colloidal solution contains electrolytic impurities. This method is known as electrodialysis. Under the influence of the field, the ionic impurities pass out through the membrane to the respective electrodes.

Ultrafiltration

Ordinary filter papers have pore sizes greater than about 1000 μ and since the dimensions (size) of the colloidal particles are much smaller, they pass through the filter paper easily. However, if the filter papers are soaked in a solution of gelatin or collodion, the pore sizes are much reduced. Such modified filter papers are known as ultrafilters. When a colloidal solution is passed through ultrafilters it is possible to retain the colloidal particles on the ultrafilter while the still smaller ionic species or other small sized solutes pass out. This process of separation of colloidal particles from the solution containing such impurities as above by filtration is known as "ultrafiltration". Ultrafiltration is, however, a slow process.

PROPERTIES OF COLLOIDAL DISPERSIONS

The properties of colloidal systems may, broadly, be divided into three categories.

(i) Optical properties.
(ii) Kinetic properties
(iii) Electrical properties.

They are briefly discussed below:

Optical properties (Tyndall effect)

Colloidal particles are too small to be seen in an ordinary microscope but their presence can be detected by optical means because of the scattering they produce when a beam of light falls on them. If a beam of light passes through a true solution, viewed perpendicular to the direction of the incident light, the beam is invisible. *But in presence of a colloidal solution, the beam of light is clearly seen under the same conditions (i.e., perpendicular to the incident light beam). This is known as Tyndall effect. Zsigmondy (1923) developed an apparatus known as "Ultramicroscope" which utilizes the scattering of light by colloidal particles for their observation. By means of this apparatus, it is possible to observe the various spots of light as bright halos around the colloidal particles.* No information about the size or shape of colloidal particles can be obtained from the ultramicroscopic study of colloidal particles.

Kinetic properties (Brownian movement)

The invention of the ultra microscope has enabled a study of the movement of the colloidal particles in the dispersion medium. The particles are observed to be in a state of chaotic and zig zag movement randomly and this is known as the Brownian motion. This name is derived after Robert Brown who observed the movement of pollen grains in aqueous suspension under a microscope as a ceaseless, choatic and random motion. The Brownian movement is essentially due to the bombardment of the colloidal particles by the molecules of the dispersion medium.

Electrical properties

When a hydrophobic sol is placed in an electric field, the colloidal particles move either towards anode or cathode. This suggests that colloidal particles are charged with respect to the dispersion medium. The phenomenon of migration of colloidal particles under the influence of an applied EMF is known as electrophoresis. The charge on the colloidal particles may be determined by the direction of movement of colloidal particles under the influence of applied field.

The velocity of colloidal particles in cm s^{-1} per unit potential gradient is known as the electrophoretic mobility which is similar to the ionic mobility. The order of the electrophoretic mobility is about 10^{-4} cm s^{-1}.

Sols of metals, sulphur, acidic hydroxides usually carry a negative charge while sols of hydrous oxides of metals like Fe, Al carry a +ve charge. The charge carried by colloidal particles depends on the method of their preparation.

The electric charge carried by colloidal particles is important for the stability of hydrophobic sols. The charge on the colloidal particles is due to the presence of traces of electrolytes present in the sol.

Electrical double layer in colloids

The surface of a colloidal particle acquires a +ve or –ve charge by selective adsorption of a layer of ions around it. This may be explained by the following example. A silver iodide sol with a –ve charge may be prepared by the addition of a slight excess of potassium iodide to a solution of silver nitrate. The same sol with a +ve charge can be prepared by the addition of potassium iodide to excess silver nitrate. These sols may be pictorially represented as shown in fig. 9.4

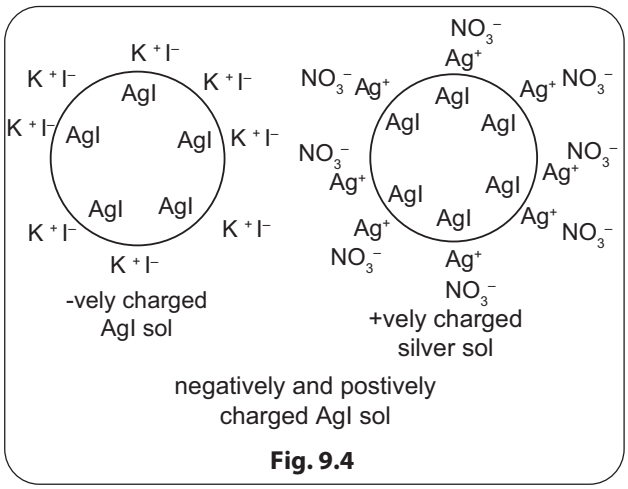


Fig. 9.4

The negatively charged AgI sol arises due to the adsorption of I^- ions and the +vely charged AgI sol is formed due to the adsorption of silver ions on the colloidal particles. In these cases, ions of opposite charge i.e., potassium ions in the first case and NO_3^- ions in the second case remain in the proximity to the negatively charged and positively charged AgI sol due to electrostatic attraction. The presence of a charge adsorbed on colloidal particles and the oppositely charged ions surrounding them in the vicinity gives rise to what is known as electrical double layer and the potential difference across the double layer is known as zeta potential.

This double layer consists of a fixed part i.e., due to the charge on the colloidal particles and the mobile layer i.e., the diffuse part due to the oppositely charged ions in the vicinity of colloidal particles in solution.

The electrical double layer present in a colloidal solution is schematically shown in Fig. 9.5 in the case of a +vely charged sol.

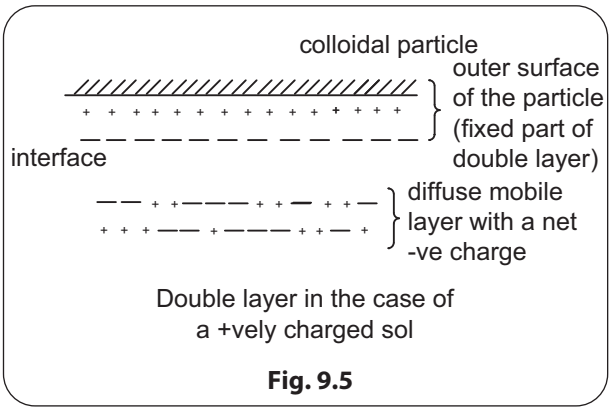


Fig. 9.5

The existence of electrical double layer in colloidal solution gives rise to interesting phenomena, (known as electrokinetic phenomena) arising out of the relative movement of the two layers with respect to each other and they will not be considered further.

APPLICATIONS OF COLLOIDS

Colloids find a wide variety of applications in diverse fields such as agriculture, biology, medicine as well as in day to day life. The applications of some colloidal solutions are enumerated below:

(i) Foods: Whipped cream, jellies, egg yolk

(ii) Medicines: Colloidal silver protected by gelatin is used as eye lotion. Colloidal calcium, colloidal gold are used as tonics. Milk of magnesia is used to cure stomach disorders.

(iii) Agriculture: Colloidal sulphur is an effective germ and weed killer.

(iv) Purification of water: Well water [or municipal water] contains impurities of colloidal nature and hence is turbid. It is clarified by the addition of alum. The Al^{3+} ions present in alum coagulate the colloidal particles and the particles settle down as a precipitate.

(v) Purification of air from smoke: Smoke from industrial plants is mainly a colloidal dispersion of carbon particles in air. The pollution due to smoke may be eliminated by passing through an apparatus, known as Cottrell precipitator.

The last two of the above point to the deleterious effects of colloidal systems and the methods of eliminating the same.

Emulsions, surfactants and micelles

Emulsions

An emulsion is a colloid in which the dispersed phase and the dispersion medium are liquids. The droplets or globules of the dispersed liquid are of fairly large size ranging from 10^{-5} to 10^{-4} cm in diameter. Common examples of emulsions are milk and cod liver oil emulsion (used as a source of vitamin A and D supplement). Milk is an emulsion in which small drops of liquid fat are dispersed in water.

Different types of emulsions

Emulsions are of two types (i) oil in water type (o/w) (ii) water in oil type (w/o). In the first case, oil is the dispersed phase and water is the dispersion medium. The milky liquid obtained during steam distillation of nitrobenzene is an example of this category. In the second case, water is the dispersed phase and oil is the dispersion medium. Cod liver oil emulsion is an example of this category.

Methods of distinguishing between the two types of emulsions

Two methods are suggested to distinguish between the two types of emulsions. A small amount of an oil soluble dye is added to the emulsions under study. If it is an oil in water type emulsion, the dye will not dissolve or mix and the emulsion remains colourless. If it is water in oil type emulsion, the dye is taken up by the oil and becomes coloured.

A second method is based on addition of water or oil to the emulsion under study. If water is added to an oil in water type emulsion, the emulsion retains its emulsion character. Similarly, when oil is added to a water in oil type emulsion, the emulsion character is not disturbed.

EMULSIFIERS

Emulsions are generally unstable but can be stabilised in the presence of an emulsifying agent. By using an emulsifying agent, it is possible to prepare emulsions containing a high percentage of the dispersed phase in the chosen medium.

The emulsifiers consist of ionic as well as non-ionic surfactants. They generally act by reducing the interfacial tension at oil-water interface. There are three categories of emulsifying agents (or emulsifiers) listed below along with the nature of their action.

Emulsifiers of 1st category

These are long chain compounds with polar groups. Soaps, long chain sulphonic acids and long chain compounds containing sulphate end groups belong to this category. They are also referred as detergents or surfactants. They get adsorbed at the interface between dispersed oil droplets and the dispersion medium as a unimolecular layer and lower the interfacial tension.

Thus the mixing of the two liquids is facilitated. The cleaning action exhibited by soaps or other detergents is ascribed to their capacity to emulsify the grease or oil along with the dirt present in the material to be cleaned. Alkali metal soaps favour the formation of oil in water type emulsions while alkaline earth metal soaps like those of Zn, Fe favour water in oil type emulsions.

Emulsifiers of 2nd category

Lyophilic substances of high molecular weight such as proteins, gums, agar are grouped in this category. These substances are believed to form a multimolecular strong film round droplets of dispersed phase and thereby prevent them from coalescing. Casein, which is a protein, forms a strong multimolecular film around liquid fat in milk and prevents coagulation of fat globules.

Emulsifiers of 3rd category

In this category, some insoluble substances like the basic sulphates of iron, copper, nickel may be cited. They stabilize oil in water type emulsions. Carbon black, for example stabilizes water in oil emulsions.

Breaking of emulsions

Emulsions can be broken or de-emulsified in two ways (i) by using an agent which destroys the emulsifying agent. For

example an acid converts a soap into a fatty acid which has no emulsifying action. Thus the emulsion breaks.

Physical methods like heating, freezing or centrifuging also results in de-emulsification.

Applications of emulsions

Emulsions find wide applications. Asphalt emulsified with water mechanically is used in road construction. In mining industry the finely ground ore is treated with an oil emulsion and foam such that the metallic part of the ore is carried to the surface which is then separated.

Surface active agents or surfactants

Surface active agents or surfactants are substances which get adsorbed preferentially at various interfaces such as air-water, oil-water or solid-water interfaces. They form monolayers at the interface in which the hydrophilic groups or the polar groups point towards the aqueous phase while the hydrocarbon chains point towards oil or air.

Surfactants may be divided into three groups

(i) cationic surfactants
(ii) anionic surfactants
(iii) non-ionic surfactants.

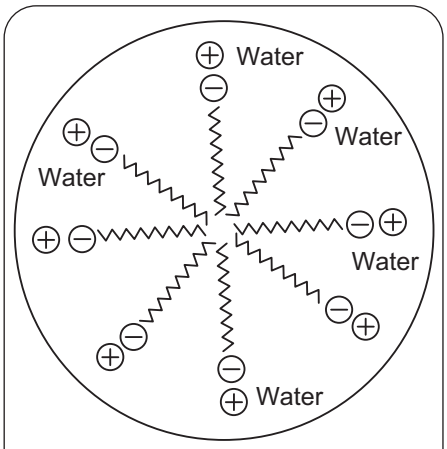
As example of cationic surfactants, we may cite salts such as cetyl trimethyl ammonium chloride, $C_{16}H_{33}(CH_3)_3N^+Cl^-$ and octadecylammonium chloride, $C_{18}H_{37}NH_3^+Cl^-$. They dissociate in water to give positively charged ions. Anionic surfactants include sodium salts of higher fatty acids like sodium palmitate $C_{15}H_{31}COONa$, sodium stearate, $C_{17}H_{35}COONa$, and salts of sulphonic acids of high molecular weight having the general formula $C_nH_{2n+1}SO_3^-M^+$ or $C_nH_{2n+1}C_6H_4SO_3^-M^+$ (where $M^+ = Na^+$, K^+ or NH_4^+). They give rise to anions in solutions.

Non-ionic surfactants also are known (which do not give rise to ions) and an example of such surfactant may be cited as the compound resulting from the reaction of a high molecular weight alcohol, $C_nH_{2n+1}OH$ with many molecules of ethylene oxide, $C_nH_{2n+1}(OCH_2CH_2)_mOH$. These surfactants are capable of converting hydrophobic groups into hydrophilic ones when they are adsorbed from aqueous solution.

Micelles

Micelles are aggregates of organic molecules which contain both hydrophobic as well as hydrophilic parts. Soaps and detergents (see under surfactants) give rise to micelles when hundred or more molecules aggregate in solution. In micelles, the hydrophobic parts (i.e., the hydrocarbon part) point inwards while the hydrophilic parts (or ionic parts) point outwards in the case where the dispersion medium is polar like water.

A schematic diagram of a micelle is shown in Fig. 9.6.



An ionic surfactant forming a micelle in aqueous medium

Fig. 9.6

where, ⟿⊖ represents the hydrocarbon chain with the negatively charged group pointing outwards; ⊕ represents the counter cation.

The surfactant micelles randomly shift between single molecules and aggregates. Actually, the micelle formation begins to occur when the concentration of surfactant reaches a certain value. The lowest concentration at which micelles begin to form is known as critical micelle concentration (CMC) and it is possible to recognize the onset of micelle formation from a study of the variation of the conductivity of the solution with surfactant concentration. All the compounds cited under surfactants are capable of giving rise to micelles.

The cleaning action of soap or detergent in removing grease and dirt from a surface results from the formation of micelles.

NUCLEAR CHEMISTRY

Radioactivity

The phenomenon of radioactivity was discovered by H. A. Becquerel in 1896 when he observed that minerals of uranium like pitchblende continuously emitted a kind of radiation having the following properties

(i) they can penetrate thin sheets of metal.
(ii) they can affect a photographic plate in the dark.
(iii) they can ionize a gas through which they pass i.e., the gas becomes an electrical conductor.

This property of emitting active radiation is called "Radioactivity" and it was observed that not only uranium but other elements like radium and compounds of thorium possess radioactive property. The rate of radioactive change is not influenced by external conditions like temperature or pressure or by the state of combination of the compound under study. It was, therefore, concluded that radioactivity is a nuclear property and is associated with nuclear transformations.

Nuclear reactions differ from chemical reactions in the following respects, as summarized in table 9.5.

Table 9.5 Differences between chemical reations and nuclear reactions

Chemical reactions	Nuclear reactions
1. There is rearrangement of electrons in the atom but the nuclei are unaffected. Thus there is no change in the atomic species taking part in the reactions.	1. There is a change in the number of protons or neutrons or both with the result that a new atomic species is formed. Thus nuclear reactions lead to atomic changes.
2. External factors like temperature, pressure affect the chemical reactions.	2. Temperature or pressure change has no effect on these reactions.
3. Energy changes involved in chemical reactions are of the order of a few kilocalories.	3. Energy changes involved in nuclear reactions are about a million times larger than in chemical reactions.
4. Electrons in atomic orbitals are involved in breaking or forming of bonds.	4. Protons, neutrons, electrons and other elementary particles are involved in the reactions.

α, β, γ-RAYS AND THEIR PROPERTIES

The emitted radiation from radioactive materials is found to be composite in nature as shown from studies by the application of electric or magnetic field. Three types of rays can be distinguished when these radiations are subjected to electric or magnetic field.

The first type are known as "α-rays" which behave like positively charged particles. The second type are referred as "β-rays" and they behave like negatively charged particles. The third type are known as "γ-rays" which are unaffected by an electric or magnetic field.

The nature of the emitted rays under the influence of an applied field is illustrated in Fig. 9.7.

Some important characteristics of these radiations are summarized in Table 9.6.

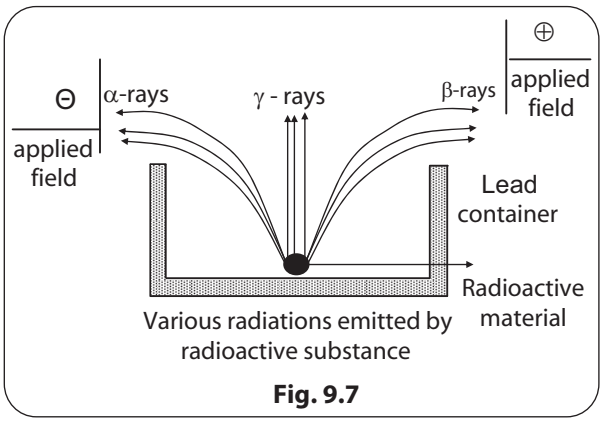


Fig. 9.7

Table 9.6 Charactersticits of α-, β- and γ-rays

Name of radiation	Composition of particles	Charge and mass of particles	Other information
α-rays	Rutherford proved by spectrographic methods that they are helium nuclei	Have a charge of +2 and mass 4	Have velocities in the range of $1.5 - 2 \times 10^9$ cm sec^{-1}. Possess considerable kinetic energy and can cause ionization of gas. Have a range of about 2.8 cm and cannot penetrate a thin sheet of paper or Al foil.
β-rays	Electrons	Have a unit –ve charge	Have velocities approaching that of light and have energies of the order of 1 MeV. Because of their small mass, they can be easily deflected. More penetrating than α-particles. They can be stopped by Al foil of 2 mm thickness. They can cause ionization of the medium they pass through.
γ-rays	Electromagnetic radiation having very short wavelengths in the range $10^{-8} - 10^{-11}$ cm.	Zero	Have velocities close to that of light. Rays are analogous to x-rays. Energies range from 1×10^{-2} to 10 MeV. They can also be easily deflected because of their small mass. Penetrating power is 100 times more than α-particles.

Detection of ionizing radiation

The presence of α-, β-rays and γ-rays can be detected by the following methods (i) Wilson's cloud chamber method (ii) Bubble chamber (iii) Ionization chamber.

Other instruments known as proportional counters, Geiger Muller counter and scintillation counters are also used for detecting and estimating the number of particles. The Geiger – Muller counter consists of a copper cylinder closed with a mica window at one end. In the middle of the cylinder, is another small sealed cylinder containing a thin wire of steel or aluminium fixed along its axis. The wire is held at a potential of 1000 V relative to the walls of the cylinder. The entry of any charged particle, like α-rays, produces ionization of the air kept at low pressure inside. This ionization is transmitted to a counter and the count is registered by a deflection of the needle in the apparatus.

Radioactive decay

Radioactive elements undergo spontaneous change with the atoms of the element giving rise to new atoms of a different element. In such changes the radiation of one type or the other are emitted and these changes are referred to as radioactive decay.

Rutherford and Soddy proposed their theory of radioactive disintegration in 1902 which is elaborated further. According to them, the nucleus of a radioactive species undergoes a spontaneous change accompanied by the emission of an α or β-particle and the formation of the nucleus of a different element. The new element which is physically and chemically different from its parent may itself be unstable and emit a particle with the formation of still another element. Such changes occur in a series till the final product is a non-radioactive element.

KINETICS OF RADIOACTIVE DECAY

The rate of decay of a radioactive element is proportional to the number of atoms of the element present at that instant. Radioactive decay is thus a first order process and may be expressed mathematically as

$$\frac{-dN}{dt} = \lambda N \qquad \text{— (1)}$$

Where, N = number of radioactive atoms present at any instant and λ = decay constant or disintegration constant. Rewriting equation (1) and on general integration,

$$\frac{-dN}{N} = \lambda dt \qquad -\int \frac{dN}{N} = \lambda \int dt \qquad \text{— (2)}$$

$$-\ln N = \lambda t + \text{constant} \qquad \text{— (3)}$$

$$\text{At } t = 0, \text{ constant} = -\ln N_o \qquad \text{— (4)}$$

Or substituting for the constant from equation (4) in equation (3), we get

$$\ln\left(\frac{N_0}{N}\right) = \lambda t$$

or $$N = N0e^{-\lambda t} \qquad \text{— (5)}$$

From equation (5) it is seen that radioactive decay is an exponential process. If the ratio $\frac{N}{N_o}$ is plotted as a function of time, the following graph is obtained, as shown in Fig. 9.8

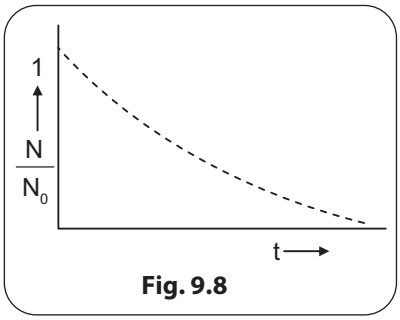


Fig. 9.8

At $t = 0$

$$\frac{N}{N_0} = 1$$

The equation for the decay constant may be given as

$$\lambda = \frac{1}{t}\ln\left(\frac{N_o}{N}\right) = \frac{2.303}{t}\log\left(\frac{N_o}{N}\right) \qquad \text{— (6)}$$

It is possible to determine λ analytically from equation (5) or equation (6). Alternatively, a plot of log N V$_s$ time is made which is linear with a slope equal to $\frac{-\lambda}{2.303}$

Average or mean life of a radioactive element (τ)

The average life of a radioactive substance is defined as the ratio of the total lifetime of all the radioactive elements to the total number of atoms present in it. It is given by

$$\tau = \frac{1}{\lambda} \qquad \text{— (6a)}$$

i.e., it is equal to the reciprocal of the decay constant. The half life period $t_{½}$ may be shown to be equal to

$t_{½} = 0.693 \times \tau$ — (6b)

Half life period of a radioactive element

The half life period ($t_{½}$) is defined as the time required for the amount (or the activity) of a radioactive substance to be reduced to half its initial value i.e., $\left(\frac{N}{N_0}\right) = 0.5$

From equation (6), $t_{½} = \frac{2.303}{\lambda}\log 2 = \frac{0.693}{\lambda}$ — (7)

It is seen from equation (7) that $t_{½}$ is independent of the initial amount taken just as in the case of a 1st order reaction. In fact, this is true for the time taken for any fraction of the initial amount to disintegrate. If "n" is the number of half life periods (where, $n = \frac{t}{t_{1/2}}$), the number of atoms N left after a period of time "t" is given by

$$N = N_o\left(\frac{1}{2}\right)^n = \frac{N_o}{2^n} \qquad \text{— (8)}$$

The half life periods of radioactive elements vary widely from few minutes to several years as seen for some elements given below:

Protactinium, Pa^{234} = 1.14 minutes,
Radon, Rn^{222} = 3.82 days,
Radium, Ra^{226} = 1540 years,
Uranium, $U^{238} = 4.5 \times 10^9$ years.

Radioactive equilibrium

If a radioactive parent element, like uranium, gives rise to another daughter element which is also radioactive, a state of equilibrium is reached when the new element decays at the same rate as it is being formed from its parent. This is called radioactive equilibrium. If A represents the parent element and B the daughter element, the condition of radioactive equilibrium is given by

$$\lambda_A N_A = \lambda_B N_B \qquad \text{— (9)}$$

Assuming that there are several decay stages, at equilibrium.

$$\lambda_A N_A = \lambda_B N_B = \lambda_C N_C \ldots\ldots \qquad \text{— (10)}$$

where, N_A, N_B, N_C are the number of atoms of various radioactive species present at equilibrium. Substituting equation (7) in equation (10), we get

$$\frac{0.693}{\left(t_{1/2}\right)_A}\times N_A = \frac{0.693}{\left(t_{1/2}\right)_B}\times N_B = \ldots\ldots = \text{Constant} \quad — (11)$$

$$\text{or } \frac{N_A}{\left(t_{1/2}\right)_A} = \frac{N_B}{\left(t_{1/2}\right)_B} = \ldots. \quad — (12)$$

It is seen from equation (12) that the number of atoms of any member of a decay series is proportional to its half life. Equation (12) can be used to calculate the half life of a radioactive species.

UNITS OF RADIOACTIVITY

Radioactivity is expressed in the units of disintegrations (or disintegrating atoms) per second (dps). The commonly used unit is the curie (Ci) which is equal to 3.7×10^{10}dps. Another unit also finds use i.e., Rutherford (Rd) which equal to 10^6 dps. The curie and "Rd" units are related by the fact

$$\text{One curie} = 3.7 \times 10^4\,\text{Rd} \quad — (13)$$

The specific activity of a radioactive substance is expressed in many ways like millicuries per gram (D_g) or millicuries per ml (D_v) or millicuries per millimole (D_M). The amount of a radioactive isotope is measured in the units of curies (Ci) or millicuries (mCi).

It can be converted into weight, for example, we can calculate the number of milligrams of the isotope corresponding to one millicurie of activity as shown by the following calculation.

$$1\ \text{mCi} = 3.7 \times 10^7 \text{ disintegrations per sec}$$

$$= \frac{3.7\times10^7}{6.023\times10^{23}}\ \text{gm atoms sec}^{-1}$$

$$= \frac{3.7\times10^7}{6.023\times10^{23}} \times \text{atomic mass of the isotope}$$

(A.M) gm sec^{-1}

To remove sec^{-1}, we multiply the above by $t_{½}/0.693$ (which is equal to $\frac{1}{\lambda}$ which has the units of "sec").

$$\text{or } 1\ \text{mCi} = \frac{3.7\times10^7}{6.023\times10^{23}} \times \text{A.M} \times \frac{t_{1/2}}{0.693} \times 1000\ \text{mg}$$

$$= 8.864 \times 10^{-14} \times \text{A.M.} \times t_{½}\ \text{mg} \quad — (14)$$

(Note: $t_{½}$ is to be given in "sec")

Radioactive series

When a radioactive element loses an α- particle, its atomic mass decreases by four units and the atomic number by two units than its parent. The product (or the daughter element) occupies two places to the left of the parent element in the periodic table. Expulsion of a β-particle does not result in any change of the atomic mass but its atomic number goes up by one unit and hence the product occupies one position to the right in the periodic table (compared to parent element). The relative displacement of the products formed in the periodic table is known as Fajans and Soddy group displacement law.

Some examples illustrating the law are given below:

Loss of α-particle

Ex1: $_{92}U^{238} \rightarrow {}_{90}Th^{234} + {}_{2}He^{4}$ — (15)

Ex2: $_{84}Po^{218} \rightarrow {}_{82}Pb^{214} + {}_{2}He^{4}$ —(16)

Loss of β-particle

Ex1: $_{90}Th^{234} \rightarrow {}_{91}Pa^{234} + {}_{-1}e^{0}$ — (17)

Ex2: $_{88}Ra^{228} \rightarrow {}_{89}Ac^{228} + {}_{-1}e^{0}$ — (18)

In the above equation, the superscripts are mass numbers and subscripts are atomic numbers.

β decay does not include only the expulsion of a fast moving electron from the nucleus. It includes also the expulsion of a positron and the orbital electron (usually K shell electron) capture.

There are three distinct natural radioactive series, namely Thorium series (4n series), Uranium series (4n + 2 series), and Actinium series (4n + 3 series) and one artificial series, namely Neptunium series (4n + 1 series).

After a number of successive radioactive changes, all of them involving emission of α- or β-particles, in each series a stable, non-radioactive end product is formed which is a form of lead in the case of natural radioactive series and a form of bismuth in the case of artificial series. Each radioactive series is complete with a definite number of stages from the initial to the final product. More details on each of the products will not be given here.

ISOTOPES, ISOBARS AND ISOTONES

Isotopes

Suppose an element loses an α-particle and two β-particles successively, it will then revert to its original position forming a new element in the periodic table but its mass will be less by four units. Thus the two elements have the same atomic number and thus have similar chemical properties. Such elements differing only in mass (ie have different mass numbers) but have the same atomic number are known as isotopes. For example, Uranium – I and Uranium – II are isotopes having an atomic number of 92 but having atomic weights 238 and 234 respectively. The atomic number of an element is equal to the number of extra nuclear electrons in the atom. Since the electroneutrality of the atom must be maintained, the number of protons in the nucleus must be equal to the number of extra-nuclear electrons.

The different mass numbers of the isotopes may then be accounted by the difference in the number of neutrons. Isotopes are not necessarily radioactive. For example, lead, the ultimate product derived by disintegration from uranium, thorium and actinium series has an atomic number 82 but the mass number varies from 206 to 208 and is not radioactive.

Since isotopes have same number of extra-nuclear electrons and hence have identical chemical properties, separation of isotopes by chemical methods is very difficult. The total number of protons + neutrons = mass number (integer closest to atomic weight).

Isobars

Different substances having the same mass number but different atomic numbers are referred as isobars. It is readily apparent that in isobars, the total number of protons and neutrons is constant but they differ in the individual numbers of the two quantities.

For example,

$_{82}Pb^{214}$ has 82 protons + 132 neutrons

$_{83}Bi^{214}$ has 83 protons + 131 neutrons

$_{84}Po^{214}$ has 84 protons + 130 neutrons

are isobars and similarly $_{18}Ar^{40}$, $_{19}K^{40}$, and $_{20}Ca^{40}$ are also examples of isobars.

Isotones

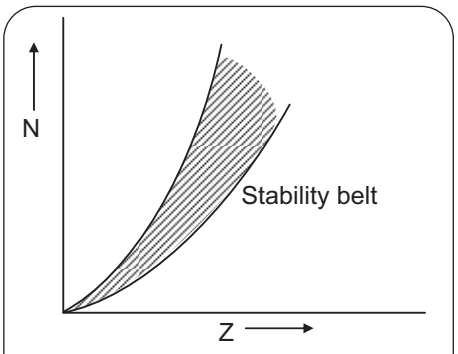
Isotones are nuclides (a term used to describe an atomic species that is identified by the composition of its nucleus) having different atomic numbers and having different atomic masses but having the same number of neutrons.

For example,

$_{14}Si^{30}$, $_{15}P^{31}$, and $_{16}S^{32}$ are all isotones, each having 16 neutrons.

Stability of nuclei

Stable nuclei are those, which do not undergo spontaneous disintegration. Useful information about the stability of nuclei can be obtained from a graph of the number of neutrons (N) Vs the number of protons (Z) for various nuclei as shown in Fig. 9.9



N Vs Z plot indicating zone of stability

Fig. 9.9

The graph has an initial slope of unity but curves upwards as we go to higher atomic numbers. For nuclides lighter than $_{20}Ca^{40}$, $\frac{N}{Z}$ ratio is close to unity. However, when the nuclear charge increases beyond 20, the ratio $\frac{N}{Z}$ deviates from unity and $\frac{N}{Z} > 1$. The reason for the upward curvature of the graph is due to the increase in the number of protons,

which causes electrostatic repulsion and lowers the stability of the nucleus.

There are two ways in which the $\frac{N}{Z}$ ratio could be brought close to and inside the stability belt depending upon the $\frac{N}{Z}$ value of the nuclides under study. For a nucleus with too many neutrons e.g., C–14, emission of a negative β-particle decreases the number of neutrons, increases the number of protons and thus the product comes within the stability belt conversion of a neutron into a proton is shown by a downward arrow (i.e., decrease in neutron number) and a left to right arrow to denote the increase in the number of protons. Actually these arrows are imaginary and do not indicate a time sequence.

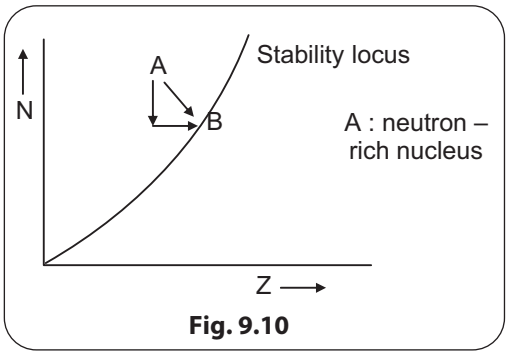


Fig. 9.10

They are simultaneous processes. The result is the product B on the stability locus.

The reverse process of positron decay i.e., conversion of a proton into a neutron (in a proton rich nucleus) is shows here

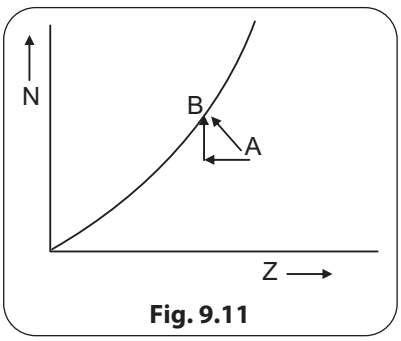


Fig. 9.11

In the latter case the loss of a positron decreases the atomic number by one unit and this implies that an orbital electron is also lost. i.e., the total loss of mass energy is about 1 MeV.

In an energy poor nucleus, this loss may not be probable. In such a case an alternative process to change a proton into a neutron is that the nucleus swallows up an orbital electron usually a K-shell electron. This is also a kind of β-decay.

Similarly, $\frac{N}{Z}$ can be increased for nuclides lying below the belt of stability by the loss of an α-particle from the same as it increases $\frac{N}{Z}$. Sometimes, the $\frac{N}{Z}$ ratio can be increased by a positron emission ($_1e^0$) also.

It is observed that nuclides in which the number of protons and neutrons are equal to 2, 8, 20, 28, 50, 82 and 126 are more stable and more abundant in nature. These numbers are thus referred as magic numbers.

Use of Radioactive isotopes

The manifold uses of radio isotopes are due to the fact that they decay with a characteristic half life and that their presence can be detected by suitable techniques. Generally, one uses "labelled compounds" containing the desired isotope and follows its decay in the process under consideration.

Some important applications of radioisotopes are discussed below:

Radio carbon dating

Radioactive $_6C^{14}$ is a β-ray emitter. Any living matter such as a tree is in equilibrium with it during photo synthesis and thereby contains a definite ratio of C^{14}/C^{12}. This ratio changes when the tree dies and using these observations it is possible to estimate the age of the tree (or of the wood).

This method was discovered by W. F. Libby in 1960 for which he was awarded the Nobel prize.

The CO_2 taken by plants from atmosphere contains trace of radioactive C^{14} through the reaction

$$_7N^{14}\text{ (from atmosphere)} + {}_0n^1 \rightarrow {}_6C^{14} + {}_1H^1 \qquad \text{— (20)}$$

The C^{14} isotope so formed, mixes with atmospheric CO_2 containing $_6C^{12}$, the stable isotope of carbon and is absorbed by living plants or animals. When the plants (or the tree or the wood of the tree) are alive, $_6C^{14}$ decays continuously according to the reaction,

$$_6C^{14} \rightarrow {}_7N^{14} + {}_{-1}e^o \ (_6C^{14}\text{ has a } t_{½} = 5600 \text{ years}) \qquad \text{— (21)}$$

and is replaced from atmosphere steadily so that C^{14}/C^{12} ratio remains constant. When the plant (or the wood of the plant) dies, the replenishment of C^{14} does not occur and thus the ratio C^{14}/C^{12} decreases continuously. It is therefore, possible to determine the age of plant or wood from the equation.

$$t = \frac{2.303}{\lambda} \log \frac{a}{a-x} = \frac{2.303 \times t_{½}}{0.693}$$

$$\log \frac{C^{14}/C^{12}\text{ ratio in living plants or freshwood}}{C^{14}/C^{12}\text{ in the dead plants or dead wood}} \qquad \text{— (22)}$$

There are other applications of radio isotopes such as in

(i) Isotope dilution analysis: It is possible to determine the solubility of sparingly soluble salts using this method. For example, the solubility of $SrSO_4$ can be determined using the isotope $_{34}Sr^{90}$.

(ii) Elucidation of reaction mechanism: Mechanisms of reactions like the hydrolysis of esters has been elucidated by using oxygen isotope, O^{18}.

(iii) Construction of atomic clocks as standards of time measurement.

(iv) Medical applications like treatment of disorders of thyroid gland and location of brain tumours etc.

NUCLEAR FISSION AND FUSION REACTIONS

Nuclear fission reactions

In nuclear fission reactions, an atomic nucleus is fissioned or broken into two or more products having a lower mass and also considerable amount of energy is released. The fission is achieved by the bombardment of the atomic nucleus with various projectiles like α-particles, ($_2He^4$), neutron ($_0n^1$), proton ($_1H^1$), deuteron ($_1D^2$) etc. For example, when uranium nuclei $_{92}U^{235}$ are bombarded with slow neutrons, the products formed are found to have a lower mass and lower atomic number than the parent nucleus according to the reaction

$$_{92}U^{235} + {}_0n^1 \rightarrow {}_{56}Ba^{144} + {}_{36}Kr^{90} + 2{}_0n^1 \qquad \text{— (23)}$$

This reaction was first observed by O. Hahn and F. Strassman in 1939 and it is also associated with a release of about 205 MeV of energy. Neutrons are particularly useful as bombarding particles because they are electrically neutral.

Fission reactions can be classified into various types according to the nature of the projectile used and the particles that are released. The following Table 9.7 gives the various reaction types with examples for each category.

Table 9.7
Classification of fission reaction based on nature of projectile and particle released

Nature of projectile	Nature of particles emitted	Reaction type	Example
α-particles	Neutron (n)	α, n reaction	$_4Be^9 + {}_2He^4 \rightarrow {}_6C^{12} + {}_0n^1$
Protons (p)	Neutron (n)	p, n reaction	$_{20}Ca^{44} + {}_1H^1 \rightarrow {}_{21}Sc^{44} + {}_0n^1$
Deuterons (d)	Protons (p)	d, p reaction	$_4Be^9 + {}_1D^2 \rightarrow {}_4Be^{10} + {}_1H^1$
Neutrons (n)	Protons (p)	n, p reaction	$_{13}Al^{27} + {}_0n^1 \rightarrow {}_{12}Mg^{27} + {}_1H^1$

When neutrons produced in fission reactions initiate any new fission processes, they are referred as chain reactions. Chain reactions are utilized in the manufacture of atom bombs and nuclear power generators.

Nuclear reactor: *A nuclear reactor is a device in which the chain reaction is allowed to run in a controlled way to use the liberated energy. In a nuclear reactor, a material known as moderator is used to absorb or slow down the fast neutrons produced during fission. In the absence of the moderator, the fission chain cannot be sustained.* Graphite is another material used as moderator.

Nuclear fusion reaction

Nuclear fusion reaction is opposite to that of a fission reaction. *In nuclear fusion reactions, atoms of light elements combine to form heavier elements. They generally occur at very high temperatures where the kinetic energy of the nuclei outweighs the electrical repulsions between them. They then combine to give an unstable species which decomposes to give a stable nucleus and release a particle like protons, neutrons etc.* The simplest fusion reaction involves

$$_1D^2 + {}_1D^2 \xrightarrow{\text{fusion}} {}_2He^3 + {}_0n^1 + \text{energy (3.2 MeV)} \qquad \text{— (24)}$$

Another reaction is

$$_1D^2 + {}_1D^2 \xrightarrow{\text{fusion}} {}_1T^3 + {}_1H^1 + \text{energy (4.0 MeV)} \qquad \text{— (25)}$$

However, this reaction does not take place under ordinary condition because of the requirement of high activation energy. Very high temperatures of the order of 3.5×10^8 K are required for the reaction to occur.

Other reactions include the fusion of two hydrogen nuclei to produce a deuterium nucleus according to

$$_1H^1 + {}_1H^1 \xrightarrow{\text{fusion}} \rightarrow {}_1D^2 + {}_1e^0 \quad \text{(unstable)} \qquad \text{— (26)}$$

The principle of fusion reaction is used in making hydrogen bombs.

Additional reference

Mass defect, packing fraction and nuclear binding energy

For all atoms (except H atom) the atomic mass is less than the sum of the constituent protons, neutrons and electrons. The difference between the two numbers is referred as the "mass defect". Let A be the mass number and Z be the atomic number of an element. The difference (A–Z) corresponds to the number of neutrons in the element. If m_p and m_n represent the masses of proton and neutron respectively and M(x) denotes the mass of the nucleus of the element (i.e., excluding the electrons), the mass defect is given by

$$\Delta Y = Zm_p + (A–Z)m_n – M(x) \qquad \text{— (27)}$$

where, $M(x) = M – Z\ m_{e^-}$ since the atom contains Z electrons of mass m_{e^-} and M is its atomic mass.

Equation (27) may be written as

$$\Delta Y = Zm_H – Z\,m_{e^-} + (A–Z)m_n – M + Z\,m_{e^-} \qquad \text{— (28)}$$

where, m_H is the atomic mass of hydrogen (which contains one proton and one electron)

or $$\Delta Y = Zm_H + (A–Z)\,m_n – M \qquad \text{— (29)}$$

It is expected that ΔY is a positive quantity for stable nuclei because a stable nucleus must have less energy than its constituent particles, As an example, let us consider the mass defect in the case of $_2He^6$ whose mass is 6.01889 a.m.u.

$$\Delta Y = 2 \times 1.00783 + 4 \times 1.00867 – 6.01889$$
$$= 2.01566 + 4.03468 – 6.01889$$

(where $m_H = 1.00783$ a.m.u., $m_n = 1.00867$ a.m.u)

$$\Delta Y = 6.05034 – 6.01889 = 0.03145 \text{ a.m.u} \qquad \text{— (30)}$$

But 1 a.m.u. = 931.5 MeV — (31)

The mass defect, ΔY, obtained above can be converted into binding energy of the nucleus using the conversion factor (eqn 31). The binding energy of the nucleus may be defined as the energy released in the formation of a nucleus from its constituent nucleons.

Binding energy of $_2He^6 = 0.03145 \times 931.5$
$= 29.29$ MeV — (32)

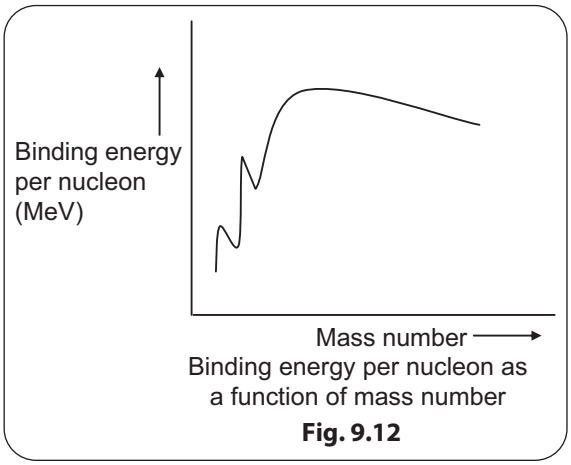
A large mass defect indicates large binding energy and it indicates a greater stability for the nucleus. The total binding energy of the nucleus increases with the number of the nucleons. A comparison of the stability of nuclei of different elements is done by calculating the average binding energy per nucleon. For $_2He^6$, this value is

Binding energy per nucleon for

$$_2He^6 = \frac{\text{Total binding energy}}{\text{Number of nucleons}} = \frac{29.29}{6}$$

$$= 4.882 \text{ MeV} \qquad \text{— (33)}$$

A plot of the binding energy per nucleon against mass number is shown in Fig 9.10



Binding energy per nucleon as a function of mass number

Fig. 9.12

It is seen from Fig 9.10: that the binding energy per nucleon rises sharply for lighter nuclei and subsequently it changes slightly and attains a fairly constant value of about 8 MeV for stable nuclei. This constancy occurs around iron.

Packing fraction: It is defined as

$$\text{packing fraction} = \frac{\text{isotopic mass} – \text{mass number}}{\text{mass number}} \qquad \text{— (34)}$$

Packing fraction may have a positive or negative sign. A negative value of packing fraction indicates stability of the

nucleus and a positive value implies instability of the nucleus. For example, the packing fraction of $_2He^4$ is given by

$$\text{Packing fraction of } {}_2He^4 = \frac{4.0026 - 4.0000}{4} = 0.00065 \quad \text{— (35)}$$

Since the packing fractions are usually expressed in parts per 10,000 the packing fraction of $_2He^4$ becomes 6.5.

Transuranium elements

Elements with atomic number greater than 93 are not found in nature (presumably because they have very short half lives and any of them present when earth was formed, have decayed). However, they have been prepared in the laboratory and so far elements upto atomic number 104 (name : Rutherfordium, Rf) have been reported. The genesis of their existence is, however, of interest. When $_{92}U^{238}$ is irradiated with slow neutrons, a neutron is captured by the nucleus resulting in a (n, γ) reaction

i. e.,

$$_{92}U^{238} + {}_0n^1 \rightarrow {}_{92}U^{239} + \gamma$$

$$_{92}U^{239} \rightarrow {}_{93}Np^{239} + {}_{-1}e^0 \{\beta\text{-particle}\} \quad \text{— (36)}$$

The element with atomic number 93 is neptunium (Np) and its half life is 2.3 days. This also decays with the emission of β-particles forming the new element plutonium, $_{94}Pu^{239}$ according to

$$_{93}Np^{239} \rightarrow {}_{94}Pu^{239} + {}_{-1}e^0 \quad \text{— (37)}$$

Plutonium has a long half life of 24,400 years and it decays giving rise to naturally occurring $_{92}U^{235}$. Several other isotopes of Np and Pu have been obtained as a result of interaction of accelerated deuterons and α-particles with isotopes of uranium. Americium, another transuranium element having an atomic number 95, is obtained according to the reaction

$$_{94}Pu^{241} \rightarrow {}_{95}Am^{241} + {}_{-1}e^0 \quad \text{— (38)}$$

$_{94}Pu^{241}$ is an α-emitter with $t_{½}$ =500 years.

ARTIFICIAL DISINTEGRATION OR TRANSMUTATION OF ELEMENTS

The spontaneous disintegration or transmutation of an element like radium continuously into two other elements like helium and radon led scientists like Rutherford and Curie to find new ways to bring about disintegration of naturally stable elements by using high speed charged particles like α-rays, protons, deuterons etc. The first such artificial transmutation reaction achieved by Rutherford in 1918, was by bombardment of $_7N^{14}$ by α-particles.

The reaction is

$$_7N^{14} + {}_2He^4 \rightarrow {}_8O^{17} + {}_1H^1 \quad \text{— (39)}$$

In due course, newer disintegration reactions were observed using high energy protons, deuterons etc. In order to give them the high energy required to cause disintegrations, they are accelerated by the application of high voltage in devices known as "linear accelerators". Other types of apparatus called "cyclotron" or "synchrotron" are used for the same purpose. A few examples of disintegration are given below.

Examples for disintegration by protons are

$$_9F^{19} + {}_1H^1 \rightarrow {}_8O^{16} + {}_2He^4 \quad \text{— (40a)}$$

$$_4Be^9 + {}_1H^1 \rightarrow {}_4Be^8 + {}_1D^2 \quad \text{— (40b)}$$

Charged particles like α-particles, have the disadvantage of requiring high energy to overcome the repulsion between +vely charged nucleus and the +vely charged bombarding particle, hence uncharged particles like neutrons are preferred.

Neutrons cannot be accelerated and hence those produced in other nuclear reactions are used for this purpose.

An example for neutron induced disintegration is

$$_{48}Cd^{113} + {}_0n^1 \rightarrow {}_{48}Cd^{114} + \gamma \text{ rays} \quad \text{— (41)}$$

Many transuranium elements are prepared by neutron bombardment.

Artificial radioactivity

Artificial radioactivity was discovered by I. Joliet Curie and F. Joliet in 1934. When light elements like Al, Mg and B were bombarded with α-particles, it was observed that both neutrons and positrons were produced. When the source of α-particles was removed, neutron emission ceased but positron emission continued for some time.

These observations were explained as arising due to the formation of unstable nuclide, which decays with the release of positrons.

Some examples are given below to illustrate the above observations.

An example with α-particle as the bombarding particle is given by

Ex1: $_{13}Al^{27} + _{2}He^{4} \rightarrow {}_{15}{}^{*}P^{30} + {}_{0}n^{1}$
(unstable)

$${}_{15}{}^{*}P^{30} \rightarrow {}_{14}Si^{30} + {}_{1}e^{o} \quad \text{— (42)}$$

Another example involves deuterium as the bombarding particle.

$$_{12}Mg^{26} + {}_{1}D^{2} \rightarrow {}_{11}{}^{*}Na^{24} + {}_{2}He^{4}$$
(unstable)

$${}_{11}{}^{*}Na^{24} \rightarrow {}_{12}Mg^{24} + {}_{-1}e^{0} \quad \text{— (43)}$$

Another set of reactions using neutrons is given by

$$_{15}P^{31} + {}_{o}n^{1} \rightarrow {}_{14}{}^{*}Si^{31} + {}_{1}H^{1}$$
(unstable)

$${}_{14}{}^{*}Si^{31} \rightarrow {}_{15}P^{31} + {}_{-1}e^{0} \text{— (44)}$$

CONCEPT STRANDS

Concept Strand 6

Calculate the mass of radium (atomic mass = 226) having an activity of 0.5 curie.

($t_{½}$ of Ra^{226} = 1600 yrs)

Solution

$$\lambda = \frac{0.693}{1600 \times 365 \times 24 \times 60 \times 60} = 1.37 \times 10^{-11} \text{ sec}^{-1}$$

$0.5 \text{ curie} = 0.5 \times 3.7 \times 10^{10} \text{ dps}$

$$\frac{-dN}{dt} = \lambda N; \text{ or } N = \frac{-dN/dt}{\lambda}$$

$$= \frac{1.85 \times 10^{10}}{1.37 \times 10^{-11}} = 1.35 \times 10^{21} \text{ atoms}$$

$$\text{or } N = \frac{1.35 \times 10^{21}}{6.023 \times 10^{23}} = 0.224 \times 10^{-2} \text{ moles}$$

$$= 0.224 \times 10^{-2} \times 226 = 0.51 \text{ g}$$

Concept Strand 7

Calculate the α-ray activity in millicuries of one gram of Th^{232} whose $t_{½} = 1.4 \times 10^{10}$ years.

Solution

$$\lambda = \frac{0.693}{1.4 \times 10^{10} \times 365 \times 24 \times 60 \times 60} = 1.569 \times 10^{-18} \text{ sec}^{-1}$$

$$1 \text{ g of } Th^{232} = \frac{1}{232} \times 6.023 \times 10^{23} = 2.6 \times 10^{21} \text{ atoms}$$

$$\Rightarrow \frac{-dN}{dt} = \lambda N; \text{ ie } \frac{-dN}{dt} = 1.569 \times 10^{-18} \times 2.6 \times 10^{21}$$

$$= 4.08 \times 10^{3} \text{ dps}$$

3.7×10^{7} dps = 1 millicurie

$$4.08 \times 10^{3} \text{ dps} = \frac{4.08 \times 10^{3}}{3.7 \times 10^{7}}$$

$$= 1.103 \times 10^{-4} \text{ millicurie}$$

Concept Strand 8

Calculate the time required in half-life units for the fraction $\frac{N}{N_o}$ to reach 0.01565 of a radioactive element.

Solution

$$\frac{N}{N_o} = \frac{1}{2^n} \text{ where, } n = \frac{t}{t_{½}}$$

$$0.01565 = \frac{1}{2^n}$$

$$\log 0.01565 = \log 1 - n \log 2$$

$$-1.8055 = 0 - (n \times 0.3010)$$

$$n = \frac{1.8055}{0.3010} = 6$$

Concept Strand 9

$_{83}Bi^{210}$ has a half life of 5.0 days. Calculate (i) the time required for one gram of the material to become 0.001 g (ii) starting with an isotope of one curie activity, what is the activity in units of dps at the end of 30 days.

Solution

$$\lambda = \frac{0.693}{t_{1/2}} = \frac{0.693}{5 \times 24 \times 60 \times 60} = 1.604 \times 10^{-6}\ \text{sec}^{-1}$$

(i) $$t = \frac{2.303}{1.604 \times 10^{-6}} \times \log 1000$$

$$= \frac{2.303 \times 3}{1.604 \times 10^{-6}} = 4.307 \times 10^{6}\ \text{sec}$$

$$\frac{4.306 \times 10^{6}}{3600 \times 24} = 49.84\ \text{days}$$

(ii) $$\lambda = \frac{2.303}{t} \log \frac{N_0}{N}\ ;\ \log\left(\frac{N_0}{N}\right)$$

$$= \frac{1.604 \times 10^{-6} \times 30 \times 24 \times 3600}{2.303} = 1.805$$

$$\left(\frac{N_0}{N}\right) = 63.87;\ \frac{N}{N_0} = 0.0156 \text{ or}$$

$$N = 0.0156 \times 1 = 0.0156\ \text{curies}$$

1 curie = 3.7×10^{10} dps

1.56×10^{-2} curies = $1.56 \times 3.7 \times 10^{8} = 5.77 \times 10^{8}$ dps

Concept Strand 10

The $_6C^{14}$ activity of a fresh piece of wood is 14300 counts per sec per kg while an old sample of wood gives 6000 counts per sec per kg for its $_6C^{14}$ content. Calculate the age of the wood sample. $t_{½}$ of $_6C^{14}$ is 5600 years.

Solution

$$\lambda = \frac{0.693}{t_{1/2}} = \frac{0.693}{5600} = 1.238 \times 10^{-4}\ \text{yr}^{-1}$$

using the first order equation

$$t = \frac{2.303}{1.238 \times 10^{-4}} \log\left(\frac{14300}{6000}\right)$$

$$= \frac{2.303}{1.238 \times 10^{-4}} \times 0.3772 = 7016.6\ \text{years}$$

Concept Strand 11

$_{90}Th^{232}$ has a half life of 1.4×10^{10} years and $_{88}Ra^{224}$ has a half life of 3.65 days. Calculate the mass of $_{88}Ra^{224}$ in equilibrium with 5 gms of $_{90}Th^{232}$.

Solution

$$\frac{N_{Th}}{N_{Ra}} = \frac{\lambda_{Ra}}{\lambda_{Th}} = \frac{0.693}{t_{1/2\,Ra}} \times \frac{t_{1/2\,(Th)}}{0.693}$$

$$\left(\frac{5.0}{232}\right) \times \frac{6.023 \times 10^{23}}{N_{Ra}} = \frac{1.4 \times 10^{10} \times 365}{3.65}$$

$$\text{or} N_{Ra} = \frac{5.0}{232} \times \frac{6.023 \times 10^{23} \times 3.65}{1.4 \times 10^{10} \times 365} = 9.27 \times 10^{9}\ \text{atoms}$$

$$= \frac{9.27 \times 10^{9}}{6.023 \times 10^{23}} \times 224 = 3.45 \times 10^{-12}\ \text{g}$$

Concept Strand 12

The half life period, $t_{½}$, of $_{19}K^{40}$ is 1.2×10^{9} years. If a lunar rock sample contains 4×10^{18} atoms per gram, how long would the sample have decayed? Assume that the original rock contains 25% of $_{19}K^{40}$ only.

Solution

$$\lambda = \frac{0.693}{1.2 \times 10^{9}} = 5.8 \times 10^{-10}\ \text{years}$$

one gram of original lunar sample contains

$$\frac{0.25}{40} \times 6.023 \times 10^{23}$$

$= 3.764 \times 10^{21}$ atoms ($\therefore \frac{1 \times 25}{100} = 0.25$ g)

or

$$t = \frac{2.303}{\lambda} \log\left(\frac{3.764 \times 10^{21}}{4 \times 10^{18}}\right)$$

$$= \frac{2.303}{5.8 \times 10^{-10}} \log(0.94 \times 10^{3})$$

$$= \frac{2.303}{5.8 \times 10^{-10}} \times 2.974 = 1.18 \times 10^{10}\ \text{years}$$

Concept Strand 13

The nucleidic mass of $_{14}Si^{29}$ is 28.9765 a.m.u. Calculate the binding energy of this nuclide, binding energy per nucleon and its packing fraction. (mass of H atom = 1.00783 a.m.u; mass of neutron = 1.00867 amu).

Solution

$_{14}Si^{29}$ contains 14 protons and 15 neutrons.

ΔY = Binding energy = $14 \times 1.00783 + 15 \times 1.00867 - 28.9765$

$= 14.10962 + 15.13005 - 28.9765 = 0.26317$ a.m.u.

Binding energy in MeV = 0.26317×931.5

$= 245.14$ MeV.

Binding energy per nucleon = $\frac{245.14}{29} = 8.453$ MeV

$$\text{Packing fraction} = \frac{28.9765 - 29.0000}{29} \times 10000 = -8.10$$

Concept Strand 14

Natural antimony has two isotopes of masses 121 and 123 respectively. Given that the atomic weight of natural antimony is 121.77, calculate the percentage amounts of the two isotopes in natural antimony.

Solution

Let the percentage of isotope of mass 121 be x

Percentage of isotope of mass 123 is (100–x)

We now write

$$\frac{121 \times x + (100 - x)123}{100} = 121.77$$

$2x = 123; \quad x = 61.5$

$\therefore$ % of isotope of mass 123 = 38.5

Concept Strand 15

A sample of uranium ore, on analysis showed 0.075 g of Pb^{206} and 0.58 gms of U^{238}. Given that the decay constant of $U^{238} = 1.54 \times 10^{-10}$ yr^{-1}, calculate the age of the ore assuming that the Pb^{206} is entirely formed out of disintegration of U^{238}

Solution

$$\lambda = \frac{2.303}{t} \log \frac{N_0}{N} = \frac{2.303}{t} \log \frac{U^{238}_{\text{remaining}} + Pb^{206}_{\text{formed}}}{U^{238}_{\text{remaining}}}$$

$$1.54 \times 10^{-10} = \frac{2.303}{t} \log \left(\frac{0.655}{0.58} \right)$$

or

$$t = \frac{2.303}{1.54 \times 10^{-10}} \log 1.129$$

$$= \frac{2.303 \times 0.0528}{1.54 \times 10^{-10}} = 7.9 \times 10^8 \text{ years}$$

Concept Strand 16

Calculate the energy released in the fusion reaction, $_3Li^6 + {}_1D^2 \rightarrow 2\,{}_2He^4$ given that the masses of $_3Li^6$ $_1D^2$ and $_2He^4$ are 6.01512 a.m.u, 2.0141 a.m.u, 4.0026 a.m.u. respectively.

Solution

Total mass on the reactants side = 6.01512 + 2.01410

= 8.02922 a.m.u

Total mass on the product side = 2×4.0026

= 8.0052 a.m.u

Loss in mass = 0.02402 a.m.u.

Energy released = 0.02402×931.5

= 22.37 MeV

Concept Strand 17

Based on neutron, proton stability ratio predict whether a β-emission, α-emission or positron emission occurs. Fill up the product.

(a) $_{16}S^{35} \rightarrow$ ……..+……

(b) $_7N^{16} \rightarrow$ ……..+……

(c) $_{11}Na^{24} \rightarrow$ ……..+……

(d) $_6C^{14} \rightarrow$ ……..+……

(e) $_5B^8 \rightarrow$ ……..+……

Solution

(a) $_{16}S^{35} \rightarrow {}_{17}Cl^{35} + {}_{-1}e^0$

(b) $_7N^{16} \rightarrow {}_8O^{16} + {}_{-1}e^0$

(c) $_{11}Na^{24} \rightarrow {}_{12}Mg^{24} + {}_{-1}e^0$

(d) $_6C^{14} \rightarrow {}_7N^{14} + {}_{-1}e^0$

(e) $_5B^8 \rightarrow {}_4Be^8 + {}_1e^0$

In first four cases a β particle is emitted. In the last case, a positron emission occurs.

Concept Strand 18

Complete the following nuclear reactions

(a) $_{92}U^{235} + {}_0n^1 \rightarrow {}_{56}Ba^{144} + \ldots\ldots + 2\,{}_0n^1$

(b) $_{92}U^{238} + {}_6C^{12} \rightarrow \ldots\ldots\ldots\ldots + 4\,{}_0n^1$

(c) $_{86}Rn^{222} \rightarrow \ldots\ldots + {}_2He^4$

(d) $_{83}Bi^{209} + {}_1D^2 \rightarrow \ldots\ldots\ldots + {}_1H^1$

(e) $_{90}Th^{232} + {}_0n^1 \rightarrow \ldots\ldots\ldots\ldots + 2\,{}_{-1}e^0$

(f) $_{92}U^{238} + {}_0n^1 \rightarrow \ldots. + 2\,{}_{-1}e^0$

Solution

Equate the mass numbers and atomic numbers on the L.H.S and R.H.S of the given equation to arrive at the result.

(a) $_{92}U^{235} + {}_{0}n^{1} \rightarrow {}_{56}Ba^{144} + {}_{b}\times^{a} + 2\,{}_{0}n^{1}$

Equating mass numbers

$235 + 1 = 144 + a + 2 \Rightarrow a = 90$

Equating atomic numbers

$92 + 0 = 56 + b + 0 \Rightarrow b = 36$

${}_{b}\times^{a} = {}_{36}x^{90}$ i.e., ${}_{36}Kr^{90}$

$_{92}U^{235} + {}_{0}n^{1} \rightarrow {}_{56}Ba^{144} + {}_{36}Kr^{90} + 2\,{}_{0}n^{1}$

(b) Similarly, solving as in the above case

$_{92}U^{238} + {}_{6}C^{12} \rightarrow {}_{98}Cf^{246} + 4\,{}_{0}n^{1}$

(c) $_{86}Rn^{222} \rightarrow {}_{84}Po^{218} + {}_{2}He^{4}$

(d) $_{83}Bi^{209} + {}_{1}D^{2} \rightarrow {}_{83}Bi^{210} + {}_{1}H^{1}$

(e) $_{90}Th^{232} + {}_{0}n^{1} \rightarrow {}_{92}U^{233} + 2\,{}_{-1}e^{0}$

(f) $_{92}U^{238} + {}_{0}n^{1} \rightarrow {}_{94}Pu^{239} + 2\,{}_{-1}e^{0}$

Concept Strand 19

$_{89}Ac^{231}$ disintegrates to $_{82}Pb^{207}$ in the actinium series. How many α and β particles are emitted in this transformation?

Solution

Let the equation be

$$_{89}Ac^{231} \rightarrow {}_{82}Pb^{207} + x\,{}_{2}He^{4} + y\,{}_{-1}e^{0}$$

Comparing mass number

$231 = 207 + 4x + y \times 0$

or $4x = 24;\ x = 6$

comparing atomic number

$89 = 82 + 2x - y = 82 + 12 - y$

or $y = 94 - 89 = 5$

ie 6α-particles and 5β-particles are emitted.

$$_{89}Ac^{231} \rightarrow {}_{82}Pb^{207} + 6\,{}_{2}He^{4} + 5\,{}_{-1}e^{0}$$

TOPIC GRIP

Subjective Questions

1. The minimum concentration of an electrolyte required to cause coagulation of a sol is called its flocculation value. It is expressed in millimoles per litre. If the flocculation value of $MgSO_4$ for standard As_2S_3 sol is 0.833. How many milligrams of $MgSO_4$ is to be added to 20 mL standard As_2S_3 sol so that flocculation just starts?
2. 0.12 g of dextrin is required to prevent the colour change from red to blue of 200 mL standard Gold sol on adding 2 mL 10% NaCl. Calculate the gold number of dextrin.
3. The formula of a Zeolite used as molecular seive has the formula $M_{56}[(AlO_2)_{56}(SiO_2)_{136}]$. 250 H_2O. What is the valency of the metal M?
4. Two alpha particles with energies 7.908 MeV and 7.710 MeV are emitted by an atom of radium. Calculate the wavelength of gamma ray accompanying the decay and express the value in pico meter as a single digit number.
   ($h = 6.6 \times 10^{-34}$ Js Electronic charge = 1.6×10^{-19} coulomb)
5. Carbon from freshly cut piece of wood gives 16000 β emissions per minute per kg and that from an old wooden bowl gives 500 β emission per minutes per kg. Considering $t_{½}$ of C – 14 to be 6000 y, calculate the age of the wooden bowl and express the value as a multiple of 10^4 years
6. Calculate the corrected whole number ratio for the time taken for 50% of a radioactive substance to decay to time taken by it to undergo 9.09% decay
7. Burning of 1 kg coal produce 3.2×10^6 J of heat. Fission of each $^{241}_{94}Pu$ nucleus produce 200 MeV energy. Calculate the number of 10 g pellets of $^{241}_{94}Pu$ required to produce the same amount of energy by nuclear fission as by burning 1000 tons of coal.
8. Calculate the ratio of the number of β particles produced to number of neutrons captured per atom of $^{232}_{90}Th$ when $^{232}_{90}Th$ is converted to $^{233}_{92}U$ in a breeder reactor.
9. Assuming no other emission occur in the nucleus, calculate maximum kinetic energy in MeV possessed by β particles emitted in the process $^{6}_{2}He \rightarrow {}^{6}_{3}Li + \overline{\beta}$ if the mass defect is calculated to be 0.0043 u (energy equivalent of1 u = 931 MeV)
10. A positron and an electron each having mass 0.537 mg mol^{-1} annihilate to form two 'ϒ' rays. Taking energy of 1 u = 931 MeV calculate the energy of each gamma ray produced in MeV and express the value as a multiple of 10^5.

Straight Objective Type Questions

Directions: **This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

11. Which among the following statement(s) is (are) correct?
    (a) Silica gel adsorbs and anhydrous calcium chloride absorbs moisture.
    (b) Both silica gel and anhydrous calcium chloride adsorb moisture.
    (c) Both silica gel and anhydrous calcium chloride absorb moisture.
    (d) Silica gel absorbs moisture and anhydrous calcium chloride adsorbs moisture.

12. The formation of ferric hydroxide sol by adding ferric chloride to freshly formed ferric hydroxide precipitate is called
(a) Flocculation (b) Dialysis
(c) Peptisation (d) Pyrolysis

13. In the transition ${}^{y}_{x}A \rightarrow {}^{y-16}_{x-4}B$, the number of α and β particles emitted are
(a) 4 and 6 (b) 4 and 4
(c) 8 and 4 (d) 4 and 8

14. Which among the following is formed when ${}^{14}_{7}N$ undergoes (α, p) reaction?
(a) ${}^{16}_{8}O$ (b) ${}^{17}_{8}O$
(c) ${}^{12}_{6}C$ (d) ${}^{11}_{6}C$

15. The average life of a radioactive element with half life 50 s is
(a) 144 s (b) 1.44 s (c) 14.4 s (d) 72 s



Assertion–Reason Type Questions

Directions: **Each question contains Statement-1 and Statement-2 and has the following choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False
(d) Statement-1 is False, Statement-2 is True

16. Statement 1
The accumulation of molecular species on the surface rather than the bulk of a solid or liquid is termed adsorption.
and

Statement 2
Adsorption is essentially a surface phenomenon.

17. Statement 1
Aqueous solution of coloured raw sugar when passed over beds of animal charcoal becomes colourless.
and

Statement 2
The colouring substances are adsorbed by animal charcoal.

18. Statement 1
The phenomenon of both adsorption and absorption taking place is called sorption.
and

Statement 2
The process of removing adsorbed substance from surface is called desorption.

19. Statement 1
Radioactivity of an element is independent of its physical state, its chemical environment or temperature.
and

Statement 2
Radioactivity is a property of the nucleus.

20. Statement 1
β particles are fast moving electrons.
and

Statement 2
Orbital electrons in a β active nuclide is emitted as β particles.

Linked Comprehension Type Questions

Directions: **This section contains 2 paragraphs. Based upon the paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

Passage I

The mathematical statement of Freundlich adsorption isotherm is $\frac{x}{m} = kP^{\frac{1}{n}}$. At low pressure n = 1 and the expression becomes $\frac{x}{m} = kP$ and at high pressure the expression becomes $\frac{x}{m} = k$.

21. For a plot of log $\frac{x}{m}$ Vs log (P) for adsorption of a gas on solid surface, the slope is given by.
(a) log P (b) log k (c) $\frac{1}{n}$ (d) $\frac{1}{k}$

22. Freundlich adsorption isotherm better explains.
(a) physisorption (b) chemisorption (c) occlusion (d) absorption

23. The curve showing the variation of extent of adsorption with pressure at constant temperature is called
(a) adsorption isobar (b) adsorption isotherm (c) adsorption isochore (d) adsorption isostere

Passage II

Instability of an atomic nucleus is due to unfavourable $\frac{N}{P}$ ratio. The range of $\frac{N}{P}$ value for stability is 1 to 1.56. If the $\frac{N}{P}$ ratio of a nuclide is greater than 1.56 it is radioactive. There is no naturally existing nuclide other than hydrogen having $\frac{N}{P}$ ratio less than 1. The neutrons to protons ratio of stable nuclei are plotted to get a graph indicating stability belt. If the $\frac{N}{P}$ ratio of a nuclide is greater than or less the stability belt it is radioactive.

24. $^{60}_{27}Co$ is radioactive because
(a) it has high $\frac{N}{P}$ ratio (b) it has less $\frac{N}{P}$ ratio
(c) its atomic number is high (d) the size of atom is small

25. Unstable nuclei of atomic number less than 20 with $\frac{N}{P}$ ratio falling below the stability belt decay by
(a) α decay (b) β decay (c) positron emission (d) K-electron capture

26. In which of the following case $\frac{N}{P}$ ratio increases
(a) $^{238}_{92}U$ undergoes α emission (b) $^{3}_{1}H$ undergoes β emission
(b) $^{14}_{6}C$ undergoes β emission (d) $^{18}_{8}O$ undergoes α emission

Multiple Correct Objective Type Questions

Directions: **Each question in this section has four suggested answers out of which ONE OR MORE answers will be correct.**

27. Which among the following statements regarding physisorption is not correct?

(a) It is reversible in nature
(b) It requires activation energy
(c) It takes place at high temperature
(d) It has low heat of adsorption

28. Which among the following contain material particles?

(a) α-rays (b) β-rays (c) ϒ-rays (d) X-rays

29. Atomic number of a radioactive nucleus decreases when

(a) α-particle is emitted
(b) β-particle is emitted
(c) positron is emitted
(d) K-electron capture takes place

Matrix-Match Type Question

Directions: **Match the elements of Column I to elements of Column II. There can be single or multiple matches.**

30.

Column l (Property)	**Column ll (Reason)**
(a) Tyndall effect	(p) Unequal bombardment of solvent molecules
(b) Coagulation	(q) Scattering of light
(c) Electrophoresis	(r) Neutralization of charge
(d) Brownian movement	(s) Presence of charge
	(t) particle size in the range 100 Å

IIT ASSIGNMENT EXERCISE

Straight Objective Type Questions

Directions: **This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

31. When As_2S_3 sol and $Fe(OH)_3$ sol are mixed
(a) both become dilute
(b) both become more stable
(c) both become true solutions
(d) mutual coagulation takes place

32. Which of the following is an example of heterogeneous catalysis?
(a) Decomposition of H_2O_2 solution in presence of KI
(b) Lead chamber process for the manufacture of H_2SO_4
(c) MnO_2 in the decomposition of $KClO_3$
(d) Hydrolysis of ethyl acetate in presence of dilute H_2SO_4

33. Example of a shape selective catalyst is
(a) Zeolite (b) Spongy iron (c) V_2O_5 (d) Pyrolusite

34. The greater the Brownian motion in a sol, greater the stability because
(a) entropy decreases
(b) aggregation is prevented
(c) dispersed phase and the medium are stable
(d) kinetic energy of molecules are high

35. NH_3 is adsorbed more quickly than N_2 or H_2 because
(a) it has pyramidal structure
(b) NH_3 is a compound and others are elements
(c) NH_3 has higher van der Waals forces of attraction
(d) NH_3 is basic and reactive while N_2 and H_2 are comparatively inert.

36. Rate of the process does not show any significant change with time in the case of
(a) absorption (b) chemisorption (c) physisorption (d) all sorption processes

37. Among the following, the compound capable of forming micelles in water is
(a) sodium palmitate (b) sodium benzoate (c) glucose (d) potassium oxalate

38. The quantity of gas adsorbed by a finely divided adsorbent like charcoal increases with
(a) increase in temperature and decrease in pressure
(b) decrease in temperature and increase in pressure
(c) decrease in temperature and decrease in pressure
(d) increase in temperature and increase in pressure

39. The critical temperature of SO2, CO2 and NH3 are 157°C, 31°C and 133°C respectively. The volumes of the gases (V) adsorbed by a given mass of adsorbent under similar conditions of temperature and pressure are in the order
(a) $V_{SO_2} > V_{NH_3} > V_{CO_2}$ (b) $V_{CO_2} > V_{NH_3} > V_{SO_2}$ (c) $V_{NH_3} > V_{SO_2} > V_{CO_2}$ (d) $V_{NH_3} > V_{CO_2} > V_{SO_2}$

40. The activation energy requirement for physisorption is much less than that for chemisorption because physisorption
(a) occurs with decrease in free energy
(b) does not involve chemical bond formation
(c) is an exothermic process
(d) proceeds with decrease in entropy

41. During an adsorption process, gas molecules are adsorbed appreciably at all temperatures. This is expected to be
(a) reversible
(b) highly specific
(c) trimolecular layered
(d) having low activation energy for desorption

42. Among the following colloidal solutions, more scattering of light is observed for
(a) gum (b) soap (c) proteins (d) ferric hydroxide

43. CMC of a given soap in water is 10^{-3} mol L^{-1}. A 10^{-5} mol L^{-1} solution of this soap in water forms
(a) true solution (b) colloidal solution (c) heterogeneous solution (d) suspension.

44. If a nuclide of an element of group II undergoes radioactive decay giving out only one type of particle and become a noble gas, the particle may be
(a) 2 positrons (b) 3 Antiprotons (d) 4 β particles (d) 2 α particles.

45. Which among the following is incorrect?
(a) As low a concentration as one mole colloidal platinum in 10^8 litres can catatlyze decomposition of H_2O_2.
(b) Rate of inversion of Cane sugar catalysed by hydrogen ions vary with concentration of hydrogen ions present in solution.
(c) Efficiency of a solid catalyst such as platinum increases when it is present in finely divided state or deposited on some inert material.
(d) The composition and amount of catalyst is found to vary at the end of the reaction.

46. A fossil sample has C^{14} activity 25% that of fresh sample of carbon. If half-life period of C^{14} is 5750 years, the life of the fossil (in years) is
(a) 5750 (b) 2 × 5750 (c) 3 × 5750 (d) 4 × 5760

47. Energy equivalent to one proton is 938 MeV. Energy equivalent to one electron in MeV is
(a) 938 (b) 0.938 (c) 0.470 (d) 0.51

48. 4 g of a radioactive substance decays to 1 g in 40 s. Its rate constant is
(a) 3.5×10^{-2} L mol^{-1} s^{-1} (b) 0.035 L^2 mol^{-2} s^{-1} (c) 35 mol L^{-1} s^{-1} (d) 0.035 s^{-1}

49. Curie point is the temperature at which
(a) an element has maximum radioactivity (b) a non-radioactive element becomes radioactive
(c) a radioactive element becomes non-radioactive (d) a ferromagnetic substance becomes paramagnetic

50. Which of the following are isotones?
(a) $_{92}U^{238}$ and $_{92}U^{235}$ (b) $_{19}K^{40}$ and $_{18}Ar^{40}$ (c) $_{20}Ca^{40}$ and $_{12}Mg^{24}$ (d) $_{15}P^{31}$ and $_{16}S^{32}$

51. An old wooden statue weighing 2.1 kg shows a C^{14} activity of 21000 disintegrations per minute. A fresh C^{14} sample shows 15.3 disintegrations per gram per minute. The age of the wooden statue (in years) is (Given $t_{1/2}$ of C^{14} = 5600 years)
(a) 4875 (b) 3438 (c) 5159 (d) 1720

52. The activity of 1.14 micrograms of $_{20}Ca^{45}$ is 20 milli curie. The decay constant in min^{-1} is
(a) 48.7×10^{-9} (b) 24.5×10^{-6} (c) 14.7×10^{-8} (d) 29.2×10^{-7}

53. The fusion reaction involving two deuterium atoms is given by $_1D^2 + {}_1D^2 \rightarrow {}_2He^3 + {}_0n^1$. Given nuclidic masses D^2 = 2.01410 amu, He^3 = 3.01605 amu and neutron = 1.00867 amu, the energy released in kcals per gram atom of $_1D^2$ is
(a) 1.87×10^8 (b) 7.5×10^5 (c) 3.73×10^7 (d) 14.9×10^4

54. The formation of C^{14} in upper atmosphere is due to the action of neutrons on
(a) $^{14}_{7}N$ (b) $^{12}_{6}C$ (c) $^{18}_{8}O$ (d) CO_2

55. The approximate radius of a nucleide (in Fermi) with mass number 125 is
(a) 1.75 (b) 3.5 (c) 0.7 (d) 7

56. A nucleus is unstable with n/p ratio lying below the stability line. The nucleus will become stable by emitting
(a) α particle (b) β particle (c) γ rays (d) both α and β particles.

57. Identify the missing nuclide, $_{11}Na^{23} + {}_1H^2 \rightarrow (?) + {}_1H^1$.
(a) $_{12}Mg^{24}$ (b) $_{10}Ne^{20}$ (c) $_{11}Na^{24}$ (d) $_{13}Al^{27}$

58. The isotope that does not occur naturally is
(a) $_6C^{14}$ (b) $_{16}S^{32}$ (c) $_{35}Br^{80}$ (d) $_{13}Al^{27}$

59. Which one of the following is deflected in a magnetic field?
(a) X-rays (b) γ-rays (c) β-rays (d) neutrons

60. A first order radioactive decay is represented as $_ZA^M \rightarrow {}_{Z-4}B^{M-8} + 2\ {}_2He^4$. If $t_{½}$ of A is 10 days, volume of He collected at S.T.P after 20 days with 1 mole of A, will be
(a) 11.2 L (b) 22.4 L (c) 33.6 L (d) 44.8 L

61. For which of the following reactions a change in catalyst can bring about a change in the products formed?
(a) $N_2 + 3H_2 \rightarrow NH_3$ (b) $2KClO_3 \rightarrow 2KCl + 3O$
(c) $CO + 3H_2 \rightarrow CH_4 + H_2O$ (d) $2SO_2 + O_2 \rightarrow 2SO_3$

62. Conversion of glucose to ethyl alcohol is bought about by the enzyme.
(a) diastase (b) maltase (c) zymase (d) Invertase

63. The rate of an enzyme catalyzed reaction increases upto 45°C and then rapidly decrease with further increase of temperature. This is because
(a) The rate of backward reaction increases more than that for forward reactio.
(b) Enzyme alters the products formed.
(c) The enzyme undergoes denaturation.
(d) The reactants are deactivated at higher temperatures.

64. Which among the following is a process that is not occurring at the interface?
(a) dissolution (b) heterogeneous catalysis
(c) Corrosion (d) Neutralization

65. Air in a closed chamber can be dried by
(a) heating (b) introducing ice crystals
(c) introducing silica gel (d) Cooling

66. Which among the following regarding adsorption is incorrect? During adsorption there is a
(a) a decrease in residual force on surface. (b) a decrease in surface energy.
(c) a restriction to movement of adsorbed molecules. (d) absorption of heat.

67. Which among the following is incorrect?
(a) Oxygen is adsorbed on metals by virtue of oxide formation.
(b) Hydrogen is adsorbed by transition metals due to hydride formation.
(c) Physisorption of a gas adsorbed at low temperature may change into chemisorption at high temperature.
(d) Usually less pressure is favorable for chemisorption.

68. Collodion is
(a) 4% solution of nitro cellulose in a mixture of alcohol and ether.
(b) 10% solution of formaldehyde in water.
(c) 2% solution of cellulose acetate in acetone.
(d) 5% solution of nitro cellulose mixed with 50% TNT and dispersed in petroleum jelly.

69. Identify the product 'X' of the reaction ^9_4Be (α, n) X.
(a) $^{14}_6C$ (b) $^{14}_7N$ (c) $^{12}_6C$ (d) $^{16}_8O$

70. $^{27}_{13}Al$ is a stable isotope and $^{29}_{13}Al$ is radioactive. What type of radiation is expected to be produced by $^{29}_{13}Al$?
(a) α rays (b) β rays (c) γ rays (d) positrons

71. $^{238}_{92}U$ belonging to group 3 of periodic table undergoes α emission to form a daughter element which belongs to

(a) group 3 (b) group 2 (c) group 1 (d) group 4

72. How many neutrons is present in the atom formed from $^{238}_{92}X$ on emission of an alpha particle?

(a) 146 (b) 144 (c) 142 (d) 138

73. Each atom of $^{235}_{92}U$ absorb a slow moving neutron and undergoes fission to form $^{139}_{54}Xe$ and $^{94}_{38}Sr$. How many neutrons are emitted per uranium atom?

(a) 2 (b) 3 (c) 4 (d) 0

74. The number of β particles emitted by a radioactive nucleus $^{b}_{a}X$ when it is transformed to $^{d}_{c}Y$ is

(a) $\left(\frac{b-d}{4}\right)+(a-c)$ (b) $(b-d)\times 2$

(c) $c+\frac{(b-d)}{2}-a$ (d) $\left(\frac{a-b}{2}\right)+c-d$

75. $\frac{N}{P}$ ratio is the highest for

(a) $^{4}_{2}He$ (b) $^{208}_{82}Pb$ (c) $^{3}_{1}H$ (d) $^{238}_{92}U$

76. Which among the following belong to the same radioactive series as $^{219}_{86}Rn$?

(a) $^{206}_{82}Pb$ (b) $^{209}_{83}Bi$ (c) $^{238}_{92}U$ (d) $^{207}_{82}Pb$

Assertion–Reason Type Questions

Directions: **Each question contains Statement-1 and Statement-2 and has the following choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False
(d) Statement-1 is False, Statement-2 is True

77. **Statement 1**
Enthalpy of chemisorption is high as it involves chemical bond formation.
and

Statement 2
Chemisorption involves absorption of adsorbate into the adsorbent.

78. **Statement 1**

$\frac{x}{m}=K.P^{n}$ the mathematical statement of Freundlich adsorption isotherm where n is a non zero integer.
and

Statement 2
Freundlich isotherm fails at high pressure, equation holds good over a limited range of pressure.

79. Statement 1
A minimum mass of a fissionable material required to undergo nuclear fission.
and

Statement 2
Nuclear fission is brought about by detonation of fissionable material with high explosive such as TNT.

80. Statement 1
Nuclear fusion reaction does not produce harmful material byproducts.
and

Statement 2
Hydrogen is the fuel in nuclear fusion reaction.

Linked Comprehension Type Questions

Directions: **This section contains 2 paragraphs. Based upon the paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

Passage I

Coagulation of a colloid is the process of dispersed phase forming aggregation and separating from the dispersed medium. The minimum concentration of an electrolyte in millimoles per litre to required to bring about coagulation or flocculation is called flocculation value. It is inversely proportional to coagulating power flocculation values of different electrolytes are different. This behaviour is easily understood from Hardy-Schulze rule.

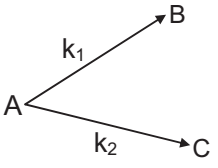
81. Which among the following ions has the higher coagulating power for As_2S_3 sol?
(a) Na^+ (b) Ca^{2+} (c) Al^{3+} (d) NH_4^+

82. $Fe(OH)_3$ is a positive sol, the highest capacity to precipitate it is shown by
(a) $FeSO_4$ (b) $K_4[Fe(CN)_6]$ (c) Na_3PO_4 (d) $AlCl_3$

83. Addition of which of the following can bring about the coagulation of standard aqua sol of gold?
(a) NaCl (b) Starch sol (c) water (d) gelatin

Passage II

Radioactive decay follows first order kinetics. There are many radioactive processes involving parallel reactions. The rate constant for nucleus A decomposing to give B and C by parallel paths is related to the individual rate constants of production of daughter nuclei B and C by $k = k_1 + k_2$.



The ratio of k_1 to k_2 is given by the percentage of the two daughter nuclei formed. An important application of ^{14}C nucleus is in radio carbon dating. It is possible to find the age of carbonaceous fossils by radio carbon dating. The relevant equations are

$$^{14}_{7}N + ^{1}_{0}n \rightarrow ^{14}_{6}C + ^{1}_{1}H$$

$^{14}_{6}C$ decay as $^{14}_{6}C \rightarrow ^{0}_{-1}\beta + ^{14}_{7}N$

^{14}C activity in living organisms remains a constant. But in dead matter, ^{14}C activity decreases with time. From the activity details, it is possible to find the age of articles. Answer the following questions on parallel reaction, radio carbon dating and radioactivity.

84. ^{227}Ac has a half life of 21.8 years with respect to radioactive decay. The decay follows two parallel paths, one leading to ^{227}Th and another leading to ^{223}Fr. The percentage yields of the two daughter nuclides are 2 and 98. The rate constants for production of ^{227}Th and ^{223}Fr are

(a) $6 \times 10^{-4}\ y^{-1}$, $3.11 \times 10^{-2}\ y^{-1}$ (b) $3.8 \times 10^{-2}\ y^{-1}$, $3.14 \times 10^{-4}\ y^{-1}$

(c) $3.8 \times 10^{-1}\ y^{-1}$, $3.14 \times 10^{-5}\ y^{-1}$ (d) $3.8 \times 10^{-4}\ min^{-1}$, $3.14 \times 10^{-2}\ min^{-1}$

85. A small quantity of a solution containing ^{24}Na [$t_{1/2}$ = 693 minutes] of activity 3.7×10^4 dps is injected into the blood of a person. A sample of blood of volume 1 cc taken after 231 minutes shows an activity of 296 dpm. The total volume of the blood of the person is

(a) 5953 litres (b) 595.3 cc (c) 5967 cc (d) 59.53 litres

86. The half life of ^{64}Cu is 12.80 hours. Cu in copper acetate is used to study brain tumour. If we start with 15 mg of copper acetate how much of it will remain after 64 hours?

(a) 4.74 mg (b) 0.474 mg (c) 0.0474 mg (d) 0.00474 mg

Multiple Correct Objective Type Questions

Directions: **Each question in this section has four suggested answers out of which ONE OR MORE answers will be correct.**

87. Which among the following is a biological catalyst?

(a) glucose (b) zymase (c) glycine (d) maltase

88. Which among the following is not a term related to a reaction in which one of the products catalyze the reaction?

(a) acid base catalyzis (b) enzyme catalyzis (c) autocatalyzis (d) negative catalyzis

89. Which among the following atomic nuclei belong to 4n + 2 series?

(a) $^{206}_{82}Pb$ (b) $^{234}_{90}Th$ (c) $^{228}_{89}Ac$ (d) $^{222}_{86}Rn$

Matrix-Match Type Question

Directions: **Match the elements of Column I to elements of Column II. There can be single or multiple matches.**

90.

Column I	**Column II**
(a) α-emission	(p) X-ray
(b) β-emission	(q) ϒ-ray
(c) Positron emission	(r) increase in no. of protons
(d) K-electron capture	(s) increase in no. of neutrons
	(t) decrease in mass no.

ADDITIONAL PRACTICE EXERCISE

Subjective Questions

91. 5 g finely powered charcoal of density 1.25 g/cc was taken in an evacuated 1004 mL glass bulb and Argon gas at 273 K and 750 torr was filled in it. Due to adsorption of gas on charcoal the pressure reduced to 560 torr. How many centilitres of the gas at STP are absorbed per gram of charcoal?

92. The tritium activity of an old sample of champane was found to be 3.9×10^{-3} times that of a recent sample. What is the estimated age of the old champane in centuries if the half life period of tritium is 12.33 years?

93. In a nuclear fusion reaction, protons interact to form ${}_{2}^{4}\text{He}$ as the final product releasing 2 positrons along with each helium nucleus produced. How many protons are required to form a helium nucleus in such a fusion reaction?

94. How many neutrons are emitted when ${}_{92}^{235}\text{U}$ undergoes fission by slow moving neutrons to produce ${}_{42}^{97}\text{Mo}$ and ${}_{50}^{137}\text{Sn}$?

95. Half life period for the process ${}_{79}^{198}\text{Au} \rightarrow {}_{80}^{198}\text{Hg} + {}_{-1}^{0}\beta$ is 65 hours. How many grams of Mercury will be present after 260 hours if the experiments starts with 3.2 g gold?

96. It is observed that a uranium mineral sample contained 1.667g ${}_{92}^{238}\text{U}$ associated with every 0.206 g ${}_{82}^{206}\text{Pb}$. Considering the half life period of ${}_{92}^{238}\text{U} \rightarrow {}_{82}^{206}\text{Pb}$ to be 4.851×10^{9} yrs and that all lead has come from decay of ${}_{92}^{238}\text{U}$ what is the age of the mineral. Express the value as multiple of 10^{9} years.

Straight Objective Type Questions

Directions: **This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

97. A true solution and colloidal dispersion can be distinguished
(a) by measuring the density
(b) from the colour
(c) observing Tyndall effect
(d) by osmotic pressure measurement

98. The minimum concentration of an electrolyte in millimoles per litre required to cause precipitation of a sol in two hours is called
(a) Flocculation value
(b) Hardy-Schulze number
(c) Stability ratio
(d) gold number

99. The combination of two layers of opposite charges around colloidal particles are called
(a) Electro kinetic double layer
(b) zeta multilayer
(c) Helmholtz electrical double layer
(d) Electromagnetic double layer

100. Electro osmosis occurs when electrical potential is applied to
(a) colloidal dispersion where movement of particles i.e., electrophoresis is prevented by suitable means.
(b) the colloidal dispersion and pure solvent are separated by a porous partition.
(c) electrodes consists of semi permeable conducting materials.
(d) osmosis is prevented by impregnating the pores of semi permeable membrane using inert insoluble ionic substances.

101. Fertile soils are colloidal in nature. Which among the following act as protective colloid in fertile soil?
(a) Gelatin (b) Starch (c) Humus (d) Soil bacteria

102. Animal hides on soaking in tannin forms hardened leather. This is due to
(a) Mutual coagulation of positively charged colloidal particles of animal hide by negative charged particles of tannin.
(b) Exchange of colloidal particles in animal hide by colloidal tannin particles.
(c) Scattering of colloidal particles in animal hide by colloidal particles in tannin.
(d) Exchange of dispersion medium between animal hide and tannin solution.

103. Whipped cream is an example of
(a) associated colloid (b) macromolecular colloid
(c) multimolecular colloid (d) a gel

104. Two isotopes X and Y of atomic masses 25 and 50 respectively in the mass ratio 1 : 1 are taken in a container. The half life of the first isotope is 25 days. After 50 days the ratio by mass of X and Y was found to be 1 : 4 the half life period of Y is
(a) 25 days (b) 50 days (c) infinity (d) zero

105. 25 mL of 2 M oxalic acid solution is shaken with 0.2 g of wood charcoal. After adsorption, the concentration of the final solution was found to be 1.5 M. The amount of hydrated oxalic acid adsorbed per gram of charcoal is
(a) 6 g (b) 8 g (c) 4.7 g (d) 0.32 g

106. $_{26}Fe^{55}$ can be converted into $_{25}Mn^{55}$ by
(a) α-Emission (b) β-Emission
(c) Proton emission (d) Electron capture

107. What will be binding energy of ^{16}O, if its mass defect is 0.210 amu?
(a) 1.89×10^{16} J mol^{-1} (b) 1.89×10^{15} J mol^{-1}
(c) 1.89×10^{14} J mol^{-1} (d) 1.89×10^{13} J mol^{-1}

108. The radioactive isotope which is used in cardiac pacemaker is
(a) I^{131} (b) P^{32} (c) Co^{60} (d) Pu^{238}

109. The time taken by a radioactive element to undergo 90% decay of the initial amount is ($t_{1/2}$ = 100 sec)
(a) 332.3 sec (b) 90 sec (c) 26 days (d) 40.5 min

110. A radioactive isotope has a half life of 2.3 days. A certain quantity of it was sealed in a container and sent to a place where it was received after 9.2 days. If 300 mg of the isotope was left in the container, the initial quantity of it was
(a) 2.4 g (b) 1.2 g (c) 3.6 g (d) 4.8 g

111. If radium and chlorine combine to form radium chloride, the compound is
(a) As radioactive as radium
(b) Half as radioactive as radium
(c) No longer radio active
(d) Twice as radio active as radium

112. An example for fertile nuclide is/are
1. ^{235}U 2. ^{238}U 3. ^{232}Th 4. ^{208}Pb
(a) 1 and 2 (b) 2 and 3 (c) 3 and 4 (d) 1, 2 and 3

113. A radioactive substance shows a current strength of 37.5 units at a certain time and 12.5 units after 300 hrs. The disintegration constant of the substance is
(a) 3.6×10^{-2} hr^{-1} (b) 3.66×10^{-3} hr^{-1} (c) 3.66 hr^{-1} (d) 4.6×10^{-3} hr^{-1}

114. The wrong statement among the following is
(a) Greater the Binding energy per nucleon, greater is the stability of the nucleus.
(b) Nucleides with negative packing fraction are less stable.
(c) Nucleides with positive packing fraction are unstable.
(d) Longer half-life period indicates that the nucleide is stable.



Assertion–Reason Type Questions

Directions: **Each question contains Statement-1 and Statement-2 and has the following choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False
(d) Statement-1 is False, Statement-2 is True

115. **Statement 1**
MnO_2 added to $KClO_3$ during its thermal decomposition increases the decomposition temperature.
and

Statement 2
MnO_2 in this process act as a positive catalyst.

116. **Statement 1**
Mixture of CO and H_2 can form CH_4, CH_3OH or HCHO depending on the catalyst used.
and

Statement 2
The ability of a catalyst to make the reactants active towards a chemical reaction is known as activity.

117. **Statement 1**
Acidity in the stomach produces indigestion.
and

Statement 2
Decrease in pH decreases the enzyme activity.

118. **Statement 1**
Electrodialysis is possible only when the impurities in colloidal dispersion are electrolytes.
and

Statement 2
Ultra filtration is preferred over peptisation in preparing colloidal dispersions.

119. **Statement 1**
Milk is easily assimilated by young animals since it is an emulsion.
and

Statement 2
Boiling of milk make the dispersed fat separate from the dispersion medium.

120. Statement 1

K-electron capture is a nuclear process involving orbital electron.

and

Statement 2

K-capture produce the same effect as positron emission.

121. Statement 1

The value of 'n' for the end products of different decay series are either 51 or 52.

and

Statement 2

The value of 'n' in the decay series correspond to atomic number of the parent element.

122. Statement 1

The lowest $\frac{N}{P}$ ratio for naturally existing elements is for $^{1}_{1}H$ and the highest value is for $^{238}_{92}U$.

and

Statement 2

$^{1}_{1}H$ is not radioactive but $^{238}_{92}U$ is an alpha emitter.

123. Statement 1

One curie is the amount of radioactive isotope that gives 3.7×10^{10} disintegrations per second.

and

Statement 2

It is the activity associated with 1 g radium – 225 with half life 1600 years.

124. Statement 1

Bombardment of $_{82}Pb$ – nucleus with $_{30}Zn$ nucleus produce $_{112}Cn$ nucleus.

and

Statement 2

Trans uranium elements are all radioactive.

125. Statement 1

In a breeder reactor an alloy of sodium and potassium is usually used as coolant.

and

Statement 2

$^{233}_{92}U$ produced from $^{232}_{90}Th$ or $^{239}_{94}Pa$ obtained from $^{238}_{92}U$ undergoes fission in breeder reactors.

126. Statement 1

The absorption of neutrons by any nucleus produces an activated species that decays by a process characteristic of the nucleus involved.

and

Statement 2

The various isotopes of elements differ considerably in their ability to absorb neutrons.

Linked Comprehension Type Questions

Directions: **This section contains 3 paragraphs. Based upon the paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

Passage I

Adsorption is an important concept when we discuss about chemistry related to surfaces. The variation of extent of adsorption with adsorption pressure at definite temperature is called isotherm. Some important adsorption isotherms are

(1) Freundlich isotherm
(2) Gibbs adsorption isotherm and
(3) Langmuir adsorption isotherm

In Freundlich isotherm we have the equation $\frac{x}{m} = kc^{1/n}$, where, x is mass adsorbed on mass 'm' of absorbent, k and n are constants and c is concentration.

In Langmuir isotherm, θ, the fraction of surface covered by adsorbent is given by $\theta = \frac{bp}{1+bp}$, where b is constant and p is pressure.

In Gibbs isotherm we have $S = -\frac{C}{RT}\left(\frac{\partial\gamma}{\partial c}\right)_T$

where, S is surface excess, C is concentration, $\frac{\partial\gamma}{\partial C}$ is the variation of surface tension with concentration. Answer the following questions on adsorption.

127. An adsorption process is
(a) exothermic
(b) always spontaneous
(c) having small activation energy
(d) has negative ΔG

128. If the constant b in the expression $\frac{bp}{1+bp}$ is 1 at 1 atmosphere pressure then the fraction of surface covered is
(a) 0.25 (b) 0.5 (c) 0.75 (d) 0.9

129. The surface excess for a 10^{-4} M solution of n-butanoic acid is 3.2×10^{-6} mol m^{-2} at 298K. The rate of variation of surface tension with concentration for n-butanoic acid is therefore,
(a) –0.04 (b) –0.08 (c) 0.04 (d) 0.08

Passage III

The spontaneous emission of invisible penetrating rays from potassium uranyl sulphate was first observed in 1891. This phenomenon was named radioactivity and it was a characteristic of an unstable nucleus. Two types of rays, α and β were identified first, later γ rays were identified. α was compared to helium atom, β to the electron and γ is a high frequency electromagnetic radiation.

130. The deflection of β particles is greater than that of α, when placed in a magnetic or electric field because
(a) β particles are negatively charged.
(b) β particles has lower mass than α particles.
(c) α particles have least penetrating power.
(d) b particles travel in wavy lines unlike a particles.

131. Which of the following emissions from an unstable atom will give an isotope of the parent nucleus?
(a) γ ray emission
(b) γ ray + β emission
(c) γ ray and 1α emission
(d) 1α and 2β particles emission.

132. U-238 is disintegrated by emitting 8 α and 6 β particles to form a stable element. The atomic number and mass number of the stable product are respectively
(a) 83 and 206 (b) 83 and 202 (c) 82 and 202 (d) 82 and 206

Multiple Correct Objective Type Questions

Directions: **Each question in this section has four suggested answers out of which ONE OR MORE answers will be correct.**

133. In homogenous catalysis the rate of reaction depends on
(a) concentration of catalyst
(b) free energy change of reaction
(c) physical state of catalyst
(d) temperature

134. Which among the following is effective in coagulation of As_2S_3 sol
(a) KCl (b) $Al_2(SO_4)_3$ (c) $Fe(NO_3)_3$ (d) $K_4[Fe(CN)_6]$

135. Gold number does not express
(a) the amount of gold present in standard gold sol.
(b) number of particles of gold sol to be present in 1 mL of it to give red colour to it.
(c) the protective action of a lyophillic colloid.
(d) the sedimenting capacity of a lyophobic colloid.

136. Half life period of pure $^{14}_{6}C$ is observed to 5770 years. In which among the following compounds ^{14}C exhibit the same half life period
(a) $^{14}_{6}CO_2$ (b) $^{14}C_6H_{12}O_6$ (c) $^{14}CH_4$ (d) ^{14}C (fullerene)

137. Which among the following is used for absorbing neutrons in nuclear reactions?
(a) Graphite (b) Boron steel (c) heavy water (d) Cadmium

138. Which among the following are employed for producing fissionable nuclei in Breeder reactors
(a) $^{1}_{1}H$ (b) $^{238}_{92}U$ (c) $^{232}_{90}Th$ (d) $^{233}_{91}Pa$

Matrix-Match Type Questions

Directions: **Match the elements of Column I to elements of Column II. There can be single or multiple matches.**

139.

Column I	**Column II**
(a) Moderator	(p) Uranium
(b) Control rods	(q) Heavy water
(c) Fuel rods	(r) Cadmium
(d) Coolant	(s) Graphite
	(t) Sodium

140.

Column I	**Column II**
(a) Action of phosphoric acid on aqueous Hydrogen peroxide	(p) Homogenous catalysis
(b) Ethanol formation from dilute glucose solution by yeast	(q) Heterogeneous catalysis
(c) Influence of MnO_2 on thermal decomposition of $KClO_3$	(r) Enzyme catalysis
(d) Influence of $MnSO_4$ formed from $KMnO_4$ during reaction of oxalic acid on acidified permanganate on the same reaction.	(s) Auto catalysis
	(t) Negative catalysis

SOLUTIONS

ANSWERS KEYS

Topic Grip

1. (2) **2.** (6) **3.** (1)
4. (6) **5.** (3) **6.** (7)
7. (4) **8.** (2) **9.** (4)
10. (5)
11. (a) **12.** (c) **13.** (b)
14. (b) **15.** (d) **16.** (b)
17. (a) **18.** (b) **19.** (a)
20. (c) **21.** (c) **22.** (a)
23. (b) **24.** (a) **25.** (c)
26. (a)
27. (b), (c)
28. (a), (b)
29. (a), (c), (d)
30. (a) → (q), (t)
(b) → (r)
(c) → (r), (s), (t)
(d) → (p), (s), (t)

IIT Assignment Exercise

31. (d) **32.** (c) **33.** (a)
34. (b) **35.** (c) **36.** (a)
37. (a) **38.** (b) **39.** (a)
40. (b) **41.** (b) **42.** (d)
43. (a) **44.** (a) **45.** d)
46. (b) **47.** (d) **48.** (d)
49. (d) **50.** (d) **51.** (b)
52. (d) **53.** (c) **54.** (a)
55. (d) **56.** (a) **57.** (c)
58. (c) **59.** (c) **60.** (c)
61. (c) **62.** (c) **63.** (c)
64. (d) **65.** (c) **66.** (d)
67. (d) **68.** (a) **69.** (c)
70. (b) **71.** (a) **72.** (b)
73. (b) **74.** (c) **75.** (c)
76. (d) **77.** (c) **78.** (d)
79. (c) **80.** (b) **81.** (c)
82. (b) **83.** (a) **84.** (a)
85. (c)
86. (b)
87. (b), (d)
88. (a), (b), (d)
89. (a), (b), (d)
90. (a) → (q), (t)
(b) → (q), (r)
(c) → (p), (s)
(d) → (d), (p), (s)

Additional Practice Exercise

91. (5) **92.** (100) **93.** (4)
94. (2) **95.** (3) **96.** (1)
97. (c) **98.** (a) **99.** (c)
100. (b) **101.** (c) **102.** (a)
103. (c) **104.** (c) **105.** (b)
106. (d) **107.** (d) **108.** (d)
109. (a) **110.** (d) **111.** (a)
112. (b) **113.** (b) **114.** (b)
115. (d) **116.** (b) **117.** (a)
118. (c) **119.** (b) **120.** (b)
121. (c) **122.** (d) **123.** (a)
124. (b) **125.** (b) **126.** (b)
127. (a) **128.** (b) **129.** (b)
130. (b) **131.** (d) **132.** (d)
133. (a), (d)
134. (a), (b), (c), (d)
135. (a), (b), (d)
136. (a), (b), (c), (d)
137. (b), (d)
138. (b), (c)
139. (a) → (q), (s)
(b) → (r)
(c) → (p)
(d) → (q), (t)
140. (a) → (p), (t)
(b) → (p), (r)
(c) → (q)
(d) → (p), (s)

HINTS AND EXPLANATIONS

1. 1 lit As_2S_3 sol require 0.833 m mol
$= 0.833 \times 120$ mg $MgSO_4$
20 ml require $\frac{0.833 \times 120 \times 20}{1000} = 2$ mg

2. 120 mg dextrin is required by 200 ml gold sol
10 ml require 6 mg dextrin i.e., gold number = 6

3. solution formula of Zeolite is
$M_{x/n}\left[\left(AlO_2\right)_x\left(SiO_2\right)_y\right] zH_2O$
where 'n' is the valency of the metal M
Valency of metal = 1

4. The lowering of energy of second α particle emitted is due to emission of γ-ray
$\Delta E = E_2 - E_1 = 7.908$ MeV $- 7.710$ MeV
$= 0.198$ MeV
wavelength of γ ray $= \frac{hc}{\Delta E}$
$= \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{0.198 \times 10^6 \times 1.6 \times 10^{-19}}$
$= 6.25 \times 10^{-12}$ m ≈ 6 Pm

5. $T_{½} = 6000$ y $\lambda = \frac{.693}{6000} y^{-1} = 1.155 \times 10^{-4} y^{-1}$
$t = \frac{2.303}{\lambda} \log \frac{No}{N} = \frac{2.303}{1.155 \times 10^{-4}} \log \frac{16000}{500}$
$= 3 \times 10^4$ yrs

6. $\frac{t_{1/2}}{t_{9.09}} = \frac{\log 2}{\log \frac{100}{90.909}} = \frac{.301}{.0414} \approx 7$
$\frac{\log 2 \times 2.303}{\frac{1}{10}} = .693 \times 10 = 6.93 \approx 7$

7. 1000 ton coal produce $3.2 \times 10^6 \times 10^6$ J
$= 3.2 \times 10^{12}$ J
10 g $^{241}_{94}Pu$ produce
$= \frac{10 \times 200 \times 10^6 \times 1.6 \times 10^{-19} \times 6.02 \times 10^{23}}{241}$ J
$= 8 \times 10^{11}$
Number of 10 g pellets required $= \frac{3.2 \times 10^{12}}{8 \times 10^{11}} = 4$

8. The reaction is
$^{232}_{90}Th + {}^{1}_{0}n \rightarrow {}^{233}_{90}Th \xrightarrow{-\beta} {}^{233}_{91}Pa \xrightarrow{-\beta} {}^{233}_{92}U$
no. of β particles = 2

9. Energy $= .00434 \times 931$ MeV $u^{-1} = 4$ MeV

10. Mass of electron = 0.000537 u
Energy of electron $= 0.000537 \times 931 \times 10^6$ eV
$= 5 \times 10^5$ eV

11. Silica gel adsorbs and anhydrous $CaCl_2$ absorbs moisture.

12. Peptisation. During peptisation, the suitable ions from the added electrolyte are adsorbed by the particles of the precipitate. The charged particles repel one another and form colloidal solution.

13. 4 α and 4 β
No. of α-particles $= \frac{\text{Mass difference}}{4}$
No. of β-particles = 2 × No. of α-particles – difference in atomic no.

14. $^{14}_{7}N + {}^{4}_{2}He \rightarrow {}^{17}_{8}O + {}^{1}_{1}H$

15. Average life $= 1.44 \times t_{½} = 1.44 \times 50 = 72$

16. Both statements are correct but statement 2 is not the explanation to statement 1

17. Both statements are correct and statement 2 is the reason for statement 1

18. Both statements are correct statement 2 is not the correct explanation to statement 1

19. Both statements are correct and statement 2 is the reason for statement 1

20. Statement 1 is correct but
Statement 2 is wrong.

21. $\log \frac{x}{m} = \log k + \frac{1}{n} \log P$

22. It better explains physisorption

23. It is adsorption isotherm

24. The $\frac{n}{P}$ ratio of Co-60 is higher. It is higher than that of the stable nucleus of cobalt.

25. Higher nuclei with low $\frac{n}{P}$ ratio undergo positron emission

26. $\frac{n}{P}$ ratio for $^{238}_{92}U = \frac{146}{92} = 1.587$

$\frac{n}{P}$ ratio for $^{234}_{90}Th = \frac{144}{90} = 1.6$

27. (b) and (c)
It does not resume activation energy and it takes place at low temperature.

28. (a) and (b)
α rays streams of rays are streams of α particles and β electrons.

29. (a), (c)and (d)
in all these cases the atomic number of the atomic nucleus undergoes a decrease.

30. (a) → (q), (t)
(b) → (r)
(c) → (r), (s), (t)
(d) → (p), (s), (t)

IIT Assignment Exercise

31. When As_2S_3 and $Fe(OH)_3$ sols are mixed, mutual coagulation takes place; As_2S_3 is negatively charged, while $Fe(OH)_3$ is positively charged sol.

32. MnO_2 and $KClO_3$ are in different phases

33. Zeolites are microporous alumino silicates. The reactions in zeolites depend upon the size of cavities and pores present in them; shape selectivity of catalyst depends on the pore structure; the pore size in zeolites generally varies between 260 pm and 740 pm.

34. Due to constant rapid motion, it is difficult for the particles to aggregate, hence they are more stable.

35. NH_3 has higher molecular interactions and is easily liquefiable, hence easily adsorbed.

36. Rate of absorption remains constant with time.

37. Sodium palmitate is the salt of long chain carboxylic acid, palmitic acid and NaOH. It is capable of forming micelles in water with the hydrocarbon chain pointing inwards (i.e., hydrophobic part) and the ionic part (hydrophilic part) pointing outwards. Others have low molecular weights and do not form micelles.

38. Adsorption is an exothermic process. It increases with decrease of temperature. Also according to Freundlich isotherm, it is proportional to pressure.

39. The ease of liquefaction of a gas affects the adsorbability of a gas. Since the ease of liquefaction of the gases lie in the order $SO_2 > NH_3 > CO_2$, the volumes adsorbed are also in the same order. i.e., SO_2 is the most adsorbed, followed by NH_3 and CO_2.

40. As there is only intermolecular force of attraction, E_a is not required to start the reaction.

41. Adsorption that takes place appreciably at all temperatures is chemisorption.
Chemisorption is irreversible, monomolecular layered and highly specific.

42. Larger the size of the dispersed phase particles, more the scattering (i.e.,) lyophobic ($Fe(OH)_3$; (colloidal Au) sols show more scattering than lyophilic colloids (gum, starch, soap, proteins etc).

43. CMC is critical micellisation concentration (i.e.,) the minimum concentration required for micelle formation. For soaps CMC is 10^{-3} mol $L^{-1.}$
$10^{-5} < 10^{-3}$
$\therefore$ forms true solution.

44. Element of group II must either lose two electrons or gain 16 electrons to become a noble gas. This is possible only on emitting two positrons.

45. The amount and composition of catalyst is found to be the same as that at the beginning

46. $t = \frac{2.303}{k}\log\frac{N_0}{N}$

$\frac{N_0}{N} = \frac{100}{25}$; $t_{1/2} = 5750$ yrs

$t = \frac{2.303 \times 5750}{0.693}\log\frac{100}{25}$

$\therefore t = 2 \times 5750\,\text{yrs} = 11{,}500$ years

47. 1.67×10^{-27} kg have 938 MeV

9.1×10^{-31} kg have × MeV

$x = \frac{9.1 \times 10^{-31}}{1.67 \times 10^{-27}} \times 938 = 0.51$ MeV.

48. $k = \frac{2.303}{t} \log \frac{C_0}{C}$

$k = \frac{2.303}{40} \log \frac{4}{1} = 0.035\ s^{-1}$

49. At curie point ferromagnetic substance becomes paramagnetic.

50. Isotones are nuclides having the same number of neutrons. Since $_{15}P^{31}$ and $_{16}S^{32}$ are the species having the same number of neutrons, they are isotones.

51. C^{14} activity of statue per gram = $\frac{21000}{2100}$ = 10 disintegrations per minute.

$$\lambda = \frac{0.693}{t_{1/2}} = \frac{0.693}{5600} = 1.237 \times 10^{-4}\ yr^{-1}$$

$$t = \frac{2.303}{\lambda} \log \frac{N_0}{N}$$

$$t = \frac{2.303}{1.237 \times 10^{-4}} \log \frac{15.3}{10}$$

$$= \frac{2.303}{1.237 \times 10^{-4}} \times 0.1847 = 3438 \text{ years}$$

52. 20 milli curies = $20 \times 3.7 \times 10^7$ disintegrations per second

$1.14\mu g = 1.14 \times 10^{-6}$ g

i.e., $\frac{1.14 \times 10^{-6} \times 6 \times 10^{23}}{45} = 1.52 \times 10^{16}$ atoms

$$-\frac{dN}{dt} = \lambda N$$

or $20 \times 3.7 \times 10^7 = 1.52 \times 10^{16} \times \lambda$

$$\lambda = \frac{20 \times 3.7 \times 10^7}{1.52 \times 10^{16}} = 48.7 \times 10^{-9}\ s^{-1}$$

$$= 29.2 \times 10^{-7}\ min^{-1}$$

53. Change in mass is

$2 \times 2.0141 - (3.01605 + 1.00867) = 0.00348$ amu

Energy released = $0.00348 \times 931 = 3.24$ MeV = $3.24 \times 1.6 \times 10^{-13}$ J = $3.24 \times 1.6 \times 10^{-6}$ ergs

$= 5.184 \times 10^{-6}$ ergs atom^{-1}

(1 amu = 931 MeV)

(1 eV = $1.6 \times 10^{-19} \times 1$ V = 1.6×10^{-19} coulomb)

(1 MeV = $1.6 \times 10^{-19} \times 10^6$ J = 1.6×10^{-13}J)

Energy released per gram atom =

$$\frac{5.184 \times 10^{-6} \times 6.023 \times 10^{23}}{10^7 \times 4.184 \times 1000 \times 2} = 3.73 \times 10^7 \text{ kcal}$$

(The factor 2 in denominator is due to the fact that in the equation we have two $_1D^2$ atoms.).

54. The reaction is $_7N^{14} + {}_0n^1 \rightarrow {}_6C^{14} + {}_1H^1$

55. $r_n = 1.4 \times 10^{-15} (125)^{1/3}$

$= 7 \times 10^{-15}$ m

56. $\frac{n}{p}$ ratio lying below the stability line means $\frac{n}{p} < 1$

disintegration should increase the $\frac{n}{p}$ value, this happens by the emission of α particles. So the nucleus has lower charge and lower mass.

57. $_{11}Na^{23} + {}_1H^2 \rightarrow {}_{11}Na^{24} + {}_1H^1$

58. Nuclides with odd number of proton and neutron are least likely to occur naturally ($_1H^2$ is exception).

(a) $_6C^{14}$ 6p; 8n

(b) $_{16}S^{32}$ 16p; 16n

(c) $_{35}Br^{80}$ 35p; 45n

(d) $_{13}Al^{27}$ 13p; 14n

$^{80}_{35}Br$ does not occur naturally. Naturally occurring Bromine consists of $_{35}Br^{79}$ (50.54%) and $_{35}Br^{81}$ (49.46%)

59. β-rays are deflected by magnetic field as they have negative charge

60. $k = \frac{0.693}{10} = 0.0693\ day^{-1}$

$$0.0693 = \frac{2.303}{20} \log\left(\frac{1}{a-x}\right)$$

$$\log \frac{1}{(a-x)} = \frac{0.0693 \times 20}{2.303} = 0.60$$

$a - x = 0.25$

$\therefore$ 0.75 moles of A is reacted to form 1.5 moles of He which corresponds to 33.6 L at S.T.P

61. CO and H_2 can react to form CH_4, CH_3OH or HCHO depending on the Catalyst used

62. Zymase convert glucose to ethyl alcohol and CO_2.

63. At temperature above 45° c. The enzyme undergoes thermal denaturation.

64. Neutralization is not a surface phenomenon.

65. Silica gel adsorb moisture and make air dry.

66. adsorption is an exothermic process

67. Usually high pressure is favourable for chemisorption

68. Collodion is a 4% solution of nitro cellulose in alcohol ether mixture

69. $^{9}_{4}Be + {}^{4}_{2}He \rightarrow {}^{12}_{6}C + {}^{1}_{0}n$

70. $\frac{N}{P}$ ratio of $^{29}_{13}Al$ is higher than that is $^{29}_{13}Al$. Hence it decays by β emission

71. Both $_{92}U$ and $_{90}Th$ belong to actinide series which is in group 3

72. No. neutrons in $^{234}_{90}Y$ = 234 – 90 = 144

73. $^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{139}_{54}Xe + {}^{94}_{38}Sr + 3\,{}^{1}_{0}n$

74. No. of α particles = $\frac{b-d}{4}$

Decrease of At. Number due to α particles

$= \frac{2\times(b-d)}{4}$

actual decrease = (a – c)

The difference is due to β particles.

75. For $^{3}_{1}H$ $\frac{n}{P}$ ratio is 2 which is the highest

76. Both $^{219}_{86}Rn$ and $^{207}_{82}Pb$ belong to 4n + 3 series

77. Statement 2 is wrong

78. Statement 1 is wrong and statement 2 is correct

79. Statement 1 is correct but Statement 2 is wrong

80. Both statements are correct

81. Hardy-Schulze rule

82. Hardy-Schulze rule

83. NaCl is an electrolyte. It brings about coagulation of gold sol.

84. $k_1 + k_2 = \frac{0.693}{t_{1/2}}$

$= \frac{0.693}{21.8} = k\ (\text{overall})$

$k_1 = 0.02 \times k_{\text{overall}}$

$= 0.02 \times \frac{0.693}{21.8} = 6\times10^{-4}\ y^{-1}$

$k_2 = 0.98 \times \frac{0.693}{21.8} = 3.11\times10^{-2}\ y^{-1}$

85. Let the volume of blood be VmL

Initial activity = $3.7 \times 10^4 \times 60$ dpm

final activity = (296 × V) dpm

$T_{1/2} = 693$ min

$\lambda = 10^{-3}\ min^{-1}$

$\frac{N_0}{N} = \frac{\text{initial activity}}{\text{final activity}}$

$t = \frac{2.303}{\lambda}\log\frac{No}{N}$

$231\ \text{min} = \frac{2.303}{10^{-3}\ \text{min}^{-1}}\log\frac{3.7\times10^4\times60}{296\times V}$

$0.1003 = \log\frac{7500}{V}$

Antilog (0.1003) = 1.259

$= \frac{7500}{V}$

V = 5967 mL

86. $\lambda = \frac{0.693}{t_{1/2}} = \frac{0.693}{12.8}\text{hour}^{-1}$

$\frac{0.693}{12.8} = \left(\frac{1}{64}\right)2.303\log\frac{15}{N}$

$\log 15 - \log N = \frac{0.693}{12.8}\times\frac{64}{2.303}$

log N = 1.176 – 1.50 = –0.324

N = 0.474 mg

87. (b) and (d)

Maltase and zymase are enzymes

88. (a), (b), (d)

The process referred is essentially autocatalysis.

89. (a), (b) and (d)

Mass number of there elements gives a remainder two when divided by 4

90. (a) → (q), (t)

(b) → (q), (r)

(c) → (q), (s)

(d) → (p), (s)

Emission of α and β particles are associated with ϒ–ray emission.

Additional Practice Exercise

91. Vol of charcoal = $\frac{5g}{1.25g/a} = 4cc$

Vol of gas = 1004cc − 4cc = 1000 cc = 1L

Pressure drop = 750 torr − 560 torr = 190 torr

= 0.25 atom

Moles of Argon adsorbed

$$= \frac{0.25atm \times 1L}{0.0821 L atm K^{-1} mol^{-1} \times 273K}$$

Moles of Argon adsorbed per gram of charcoal

$$= \frac{1.115 \times 10^{-3}}{5}$$

$= 2.23 \times 10^{-3}$ mL

Volume of gas at STP adsorbed per gram of charcoal.

$= 2.23 \times 10^{-3}$ mol × 22400 mL mol^{-1}

= 50 mL = 5 centilitre

92. $t_{½} = 12.33$ yrs

3.9×10^{-3} times the activity means $\frac{1}{256}$ times hence age of the sample = $8 \times t_{½} = 100$ yrs.

93. The reaction is

$$4\,{}^{1}_{1}H \rightarrow {}^{4}_{2}He + 2\,{}^{0}_{1}e + 24.7MeV$$

94. The reaction is

$${}^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{97}_{42}Mo + {}^{137}_{50}Sn + 2\,{}^{1}_{0}n$$

95. $T = 4\,T_{½}$

Amount of gold remaining = 0.2 g

Amount of Mercury formed = 3 g

96. $t = \frac{2.303}{\lambda}\log\frac{a}{a-x}$

$$= \frac{2.303 \times 4.851 \times 10^{9}}{.693} \times \log\frac{8 \times 10^{-3}}{7 \times 10^{-3}}$$

$$\approx 1 \times 10^{9}\ yr$$

97. Only colloidal dispersions give Tyndall effect

98. It is the definition for flocculation value

99. It is known as Helmholtz double layer

100. When electro phorosis prevented electro osmosis take place.

101. Humus acts as protective colloid in fertile soil

102. Mutual coagulation results in formation of hardened leather

103. It is gas dispersed in liquid. It is an example of multi-molecular colloid.

104. For radioactive material $a = \frac{a_0}{2^n}$

Initial mass ratio of $\frac{x}{y} = \frac{1}{1}$

Let $T_{1/2}$ of y = t

After 50 days amount of x remaining = $\frac{1}{2^2} = \frac{1}{4}$

After 50 days amount of y remaining = $\frac{1}{2^{50/t}}$

$\frac{1}{4} : \frac{1}{2^{50/t}} = 1 : 4$

here $\frac{50}{t} = 0 \quad t = \infty$

105. Amount of oxalic acid initially present

$$= \frac{2 \times 126 \times 25}{1000} = 6.3\ g.$$

Amount of oxalic acid left after adsorption

$$= \frac{1.5 \times 126 \times 25}{1000} = 4.725\ g$$

Amount of oxalic acid adsorbed

= 6.3 – 4.725 = 1.575 g

Amount of oxalic acid adsorbed per gram of charcoal.

$$= \frac{1.575}{0.2} = 7.9\ g = 8\ g$$

106. ${}^{55}_{26}Fe \pm {}_{-1}e^{o} \rightarrow {}^{55}_{25}Mn$

107. 1 amu ≡ 931.5 MeV $atom^{-1} \approx 8.98 \times 10^{13}$ $Jmol^{-1}$

$0.210\ u = 0.21 \times 8.98 \times 10^{13} \approx 1.89 \times 10^{13}$ mol^{-1}

108. In cardiac pace makers ^{238}Pu is used

109. $\lambda = \frac{0.693}{t_{1/2}} = \frac{0.693}{100} = 0.00693$

$t = \frac{2.303}{\lambda} \log \frac{No}{N}$ (i.e.,) $\frac{2.303}{0.00693}\log\frac{100}{10}$

= 332.3 s

110. Number of half lives = $\frac{9.2}{2.3} = 4$

$N_0 = 2n \times N = 24 \times 300$ mg = $\frac{16 \times 300}{1000}$ g = 4.8g

111. Chemical combination does not affect radioactive property

112. A nuclide that can be converted into fissionable nuclides are called fertile nuclides. ^{238}U and ^{232}Th are examples.

113. $\lambda = \frac{2.303}{300} \log \frac{37.5}{12.5} = 3.66 \times 10^{-3}\ hr^{-1}$

114. Nucleides with negative packing factors are more stable.

115. Statement 1 is wrong and statement 2 is correct

116. Both statements are true

117. Both statements are true. Statement 2 is the correct explanation to statement 1

118. Statement 2 is wrong
Statement 1 is correct

119. Both statements are correct

120. Both statements are correct

121. Statement 1 is correct
Statement 2 is wrong

122. Statement 1 is wrong
Statement 2 is correct

123. Both statements are correct. Statement 2 is the correct explanation for statement 1

124. Both Statement 1 and Statement 2 are correct

125. Both Statements are true

126. Both Statements are correct

127. Adsorption is an exothermic process. It is not always spontaneous.

128. $\frac{bp}{1+bp} = \theta$

$\frac{1\times1}{1+1} = \frac{1}{2} = 0.5$

$\theta = 0.5$

129. $s = \frac{-C}{RT}\left(\frac{\partial\gamma}{\partial c}\right)_T$

$3.2 \times 10^{-6}\ mol\ m^{-2} = \frac{-10^{-4} \times 10^{3}\ mol\ m^{-3}}{(8.314\ J\ mol^{-1}\ K^{-1})(298K)}\left(\frac{\partial\gamma}{\partial C}\right)_T$

$\therefore \quad \left(\frac{\partial\gamma}{\partial C}\right)_T = -0.08\ N\ m^2\ mol^{-1}$

130. Due to lower mass, greater the deflection

131. $M_z^A \xrightarrow{-\alpha} M_{Z-2}^{A-4} \xrightarrow{-\beta} M_{Z-1}^{A-4} \xrightarrow{-\beta} \underset{\text{isotope of } M_Z^A}{M_Z^{A-4}}$

132. $_{92}U^{238}$ $\begin{matrix} -32 & 0 \\ 8\alpha & 6\beta \\ -16 & +6 \end{matrix}$ $\begin{matrix} = 206 \\ = 82 \end{matrix}$

133. (a) and (d)
n homogenous catalysis the amount of catalyst and temperature decide the rate of reaction

134. (a), (b), (c), (d)
All are electrolytes and can coagulate As_2S_3 sol

135. (a), (b), and (d)
These statements are not true regarding gold number

136. (a), (b), (c), and (d)
The half life period of C – 14 is not affected by compound formation.

137. (b) and (d)
Both Cadmium and Boron steel are used for making control rods in nuclear reactors.

138. (b) and (c)

$^{238}_{92}U$ and $^{232}_{90}Th$ are used for the production of fissionable nuclei in breeder reactors.

139. a – q, s, b – r, c – p, d – q, t

140. a – p, t, b – p, r, c – q, d – p, s